Design

As a cloud architect or VI administrator, understand the supported design options for VCF, and review a set of decision points, justifications, implications, and considerations for designing each component.

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview and VCF Release Notes. For information about deploying, converging to or upgrading VCF, see Overview of VCF Deploy, Converge, and Upgrade.

Organization of This Design Guidance

The design content consists of the following sections:

Section	Description
Architecture Options in VMware Cloud Foundation	Contains an overview of each component of a VCF private cloud with high level comparisons of the benefits and implications of the options available within each component.
Design Blueprints for VMware Cloud Foundation	A collection of design requirements and recommendations based on a defined design profile. It can be used as a full end-to-end design for a VMware Cloud Foundation platform or as a starting point and adjusted to suit your specific objectives by altering the design choices made.
Design Library for VMware Cloud Foundation	Contains detailed design for each component of the VCF private cloud.

Design Elements

The *VMware Cloud Foundation Design Guide* contains requirements and recommendations for the design of each component of the VCF private cloud. In situations where a configuration choice exists, requirements and recommendations are available for each choice. Implement only those that are relevant to your target configuration.

Design Element	Description
Design Requirement	Required for the operation of VCF. Deviations are not permitted.
Design Recommendation	Recommended as a best practice. Deviations are permitted.
Design Choice	You must choose one of a number of options. Could be simple configuration options, or a choice between lower level (more elemental models) from the design library.

Scope of This Design Guidance

The VMware Cloud Foundation Design contains:

- An overview of each component of a VCF private cloud with high-level comparisons of the benefits and implications of the available architectural options
- A library of designs for each of the architectural options
- A number of blueprints (defined architecture topologies, based on specific design profiles) to help accelerate time to value when building a new VMware Cloud Foundation platform

The following groups of design blueprints are available:

- Infrastructure Modernization
- Application Modernization
- Security Modernization

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Architectural Options in VMware Cloud Foundation

To design a VMware Cloud Foundation platform, you must understand the supported architectures and deployment models surrounding the components that deliver the capabilities of the platform. This section contains an overview of each component of a VCF private cloud with high-level comparisons of the benefits and implications of the options available within each component. For detailed design, including requirements and recommendations for each of the component models that are discussed in this section, refer to the Design Library for VMware Cloud Foundation section of this document.

With the release of VMware Cloud Foundation 9.x, new constructs, have been introduced into the architecture making it important to understand them, see the VCF Taxonomy section of the Product Overview for more details.

Functional Area	Models
VCF Fleet	• VCF Fleet Deployment Models • VCF Fleet Sizing Models
Consumption	• VCF Automation Models • vSphere Supervisor Models • Network Consumption Models • Workload Connectivity Models • Load Balancer Models • VCF Edge Models • Private AI Foundation Platform Models • Private AI Foundation Compute Models
Operations	• VCF Management Services Models • VCF Management Network Models • VCF Operations Models • Log Management Models • VCF Operations for Networks Models • VCF Recovery Options
Security Governance and Compliance	• Identity Broker Models • VCF Single Sign-On Models • Lateral Security with vDefend Models
Virtual Infrastructure	• VCF Domain Models • vSphere Cluster Models • Distributed Switch Models • Storage Models • NSX Manager and Control Plane Models • NSX Edge Cluster Models • Virtual Network Appliance Cluster Models
Physical Infrastructure	• Network Fabric Models

VCF Fleet Deployment Models

This section outlines the *VCF Fleet Deployment Models* and available design options, offering guidance on selecting the most appropriate deployment model to meet your specific requirements.

Understanding the VCF Fleet Construct

A VCF fleet represents a unified operational framework that enables organizations to deploy, manage, and operate one or more VCF Instances as a cohesive infrastructure ecosystem. The VCF fleet construct serves as the foundational organizational model that provides:

- **Centralized Management Plane**: A single control point for managing multiple VCF Instances across geographic locations.
- **Unified Operations Framework**: Consistent operational practices, policies, and governance across all instances.
- **Standardized Lifecycle Management**: Coordinated deployment, upgrade, and maintenance procedures across the fleet.
- **Integrated Monitoring and Analytics**: Comprehensive visibility into performance, capacity, and health across all instances.
- **Consistent Automation Capabilities**: Standardized automation templates and workflows deployable across the entire fleet.

Common Elements Across VCF Fleet Deployment Designs

This section highlights the common elements that are shared across the four fleet deployment designs. Each design is built on a unified framework for operating and managing a VCF fleet across single or multiple instances, leveraging centralized management and automation capabilities. These designs provide a consistent approach to deploying, consuming, and automating virtual machines, Kubernetes, and other infrastructure resources, ensuring scalability and operational efficiency.

Table 11: Deployment Design Considerations

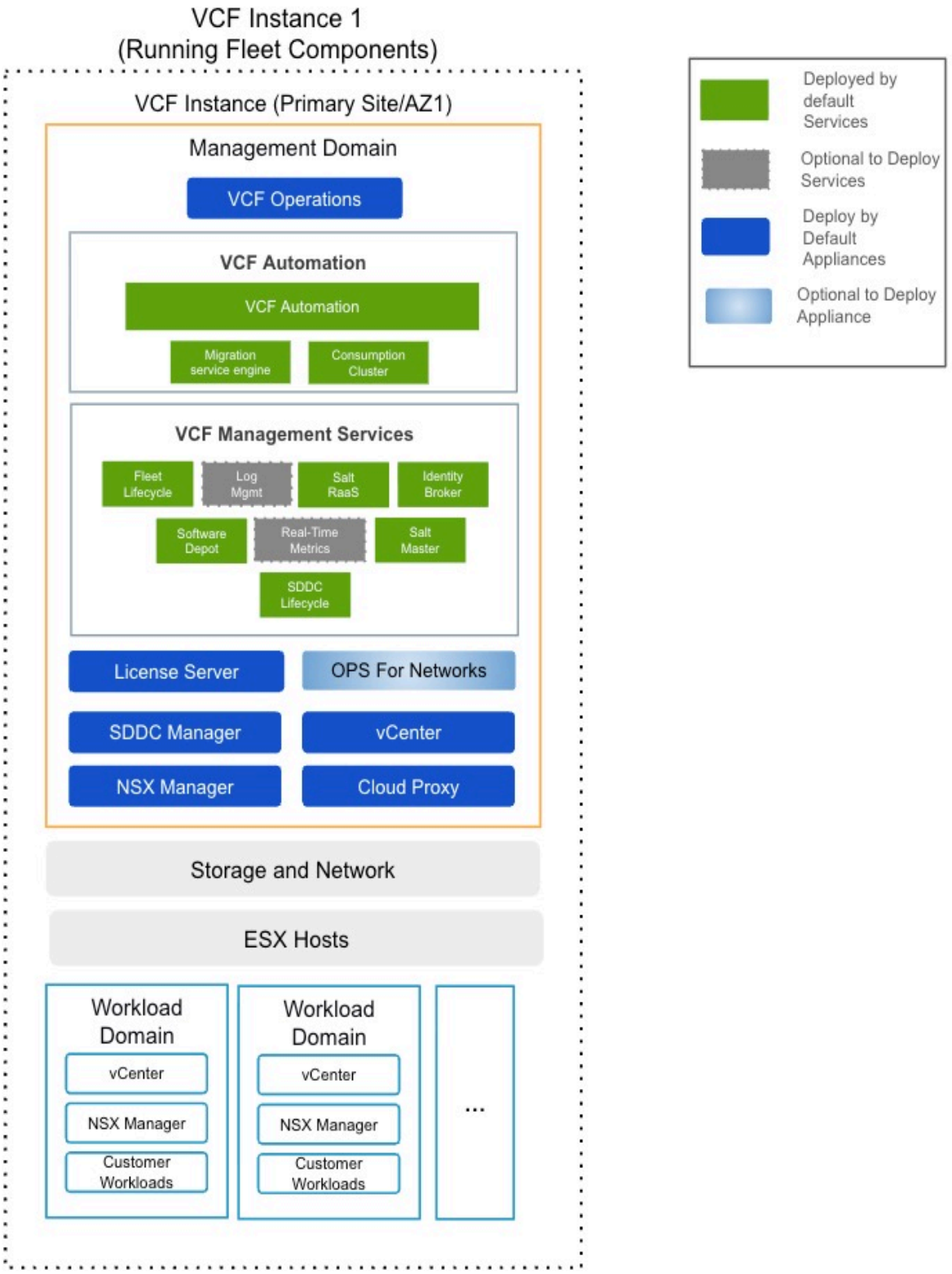
Attributes	Details
External Services	External Services Detailed Design
Distributed Switch Models	Distributed Switch Detailed Design
Storage Models	Storage Detailed Design
VCF Operations Models	VCF Operations Detailed Design
VCF Automation Models	VCF Automation Detailed Design
Log Management Models	Log Management Detailed Design
VCF Operations for Networks Models	VCF Operations for Networks Detailed Design
VCF Management Services	VCF Management Services Detailed Design
NSX Manager Models	NSX Manager and Control Plane Models
NSX Edge Models	NSX Edge Cluster Models
Identity Broker Models	Identity Broker Detailed Design

There are common attributes across all the designs, below we cover all the common attributes and details.

Table 12: Common Key Provisions Across Designs

Attribute	Benefits	Implications
Centralized Management	• Provides a centralized management framework for VCF Instances and vCenters. • Uses a single VCF Operations instance to manage the environment. • Uses a single VCF Automation instance to manage the environment.	Centralized management reduces operational complexity and ensures consistency across deployments.
Foundational Deployment Pattern	• The Basic Design serves as the architectural foundation for all fleet deployments. • Supports incremental expansion from single to multi-instance fleets. • Enables geographic distribution while maintaining unified operations.	• Initial deployment establishes operational baselines and standards. • Expansion follows consistent patterns reducing complexity. • Enables seamless scaling based on business growth.

Attribute	Benefits	Implications
Architecture Flexibility	• Single-domain or multi-domain architectures per instance. • Mixed cluster types (single-rack, multi-rack, stretched) within domains. • Flexible workload segmentation (VDI, Kubernetes, general purpose).	• Organizations tailor architecture to specific use cases. • Supports diverse workload requirements and isolation policies. • Enables gradual evolution from simple to complex topologies.
Latency	• Review Ports and Protocol Site to get all latency requirements.	• Verify that all latency before finalizing design.
Scale	• Organizations may choose to deploy multiple VCF fleets to account for the scale in their environment.	More management overhead of environment and needs to be considered.
Network Connectivity	The placement of VCF fleet should account for the latency between VCF Operations and all the components of the fleet. Ideally, placed in a low-latency, high-bandwidth network segment. This ensures efficient communication between VCF components and other monitored components.	Multiple VCF fleets could need to be created, one scenario one fleet per region (for example, AMER, EMEA, APAC).



VCF Fleet Deployment Basic Design

The *VCF Fleet Deployment Basic Design* provides centralized management for VCF Instances within a single availability zone or region. Organizations deploy one management domain instance to control multiple workload domain instances through a unified framework.

Start with a single VCF Instance, then scale to multiple VCF Instances while maintaining centralized operations and automation. This design establishes the foundation for fleet-based deployments where consistent management and operational efficiency are priorities.

Table 13: Key Provisions of the VCF Fleet Deployment Basic Design

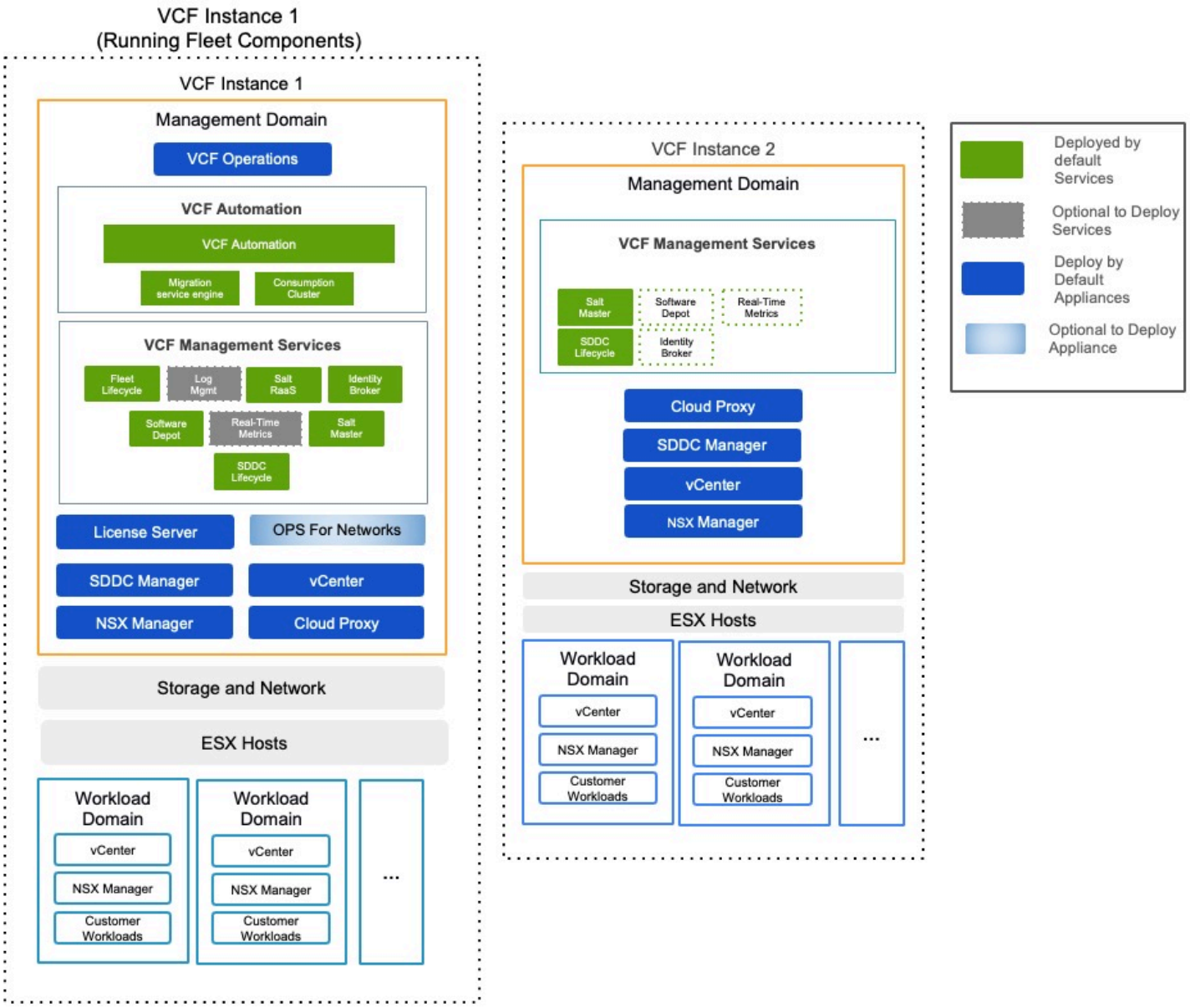
Attribute	Details	Implications
Design Deployment Model	• Designed for a single VCF fleet that is built in one availability zone or region. • Does not establish infrastructure in multiple availability zones for high availability or fault tolerance. • Uses the best practices around vSphere HA and physical hardware design constructs.	The deployment is limited to a single availability zone or region, which simplifies the architecture but does not provide site-level high availability or disaster recovery.
Cluster Models	• Supports Single-Rack vSphere Cluster Model for management domain. • Supports Single-Rack vSphere Cluster Model and Multi-Rack vSphere Cluster Detailed Design for workload domains. • Provides flexibility in choosing the appropriate cluster model based on the environment's needs.	• Any failure at the availability zone or region level could result in downtime for the entire VCF fleet. • Scaling beyond the single site would require transitioning to a multi-site deployment model.
High Availability Approach	• vSphere HA for compute resiliency. • vSAN or external storage redundancy. • N+1 host redundancy within clusters. • No cross-zone fault tolerance.	• Protection against single host or storage device failures. • Entire fleet vulnerable to zone-level outages. • Relies on physical.

Note:

Infrastructure Blueprint Selection

This fleet design corresponds with the following infrastructure blueprints for implementation:

- VCF Fleet in a Single Site with Minimal Footprint
- VCF Fleet in a Single Site



Note:

Log management and Real-time metrics are optional in fleet deployment.

VCF Fleet Deployment with Site High Availability (Across Zones) Design

The *VCF Fleet Deployment with Site High Availability (Across Zones) Design* enhances the *VCF Fleet Deployment Basic Design* by introducing fault domains that distribute resources across multiple physical hardware groups. This architecture protects workloads from localized failures—single availability zone power outages, network disruptions, cooling system issues, and hardware breakdowns.

The design maintains operational continuity when these failures occur by ensuring workloads remain available across zones. Built on vSphere HA best practices, it supports three cluster models: Single-Rack, Multi-Rack, and Stretched configurations. Organizations can select the appropriate model based on their infrastructure requirements and availability targets.

Deploy this design when minimizing downtime and maintaining continuous availability are operational priorities.

Table 14: Key Provisions of the VCF Fleet Deployment with Site High Availability (Across Zones) Design

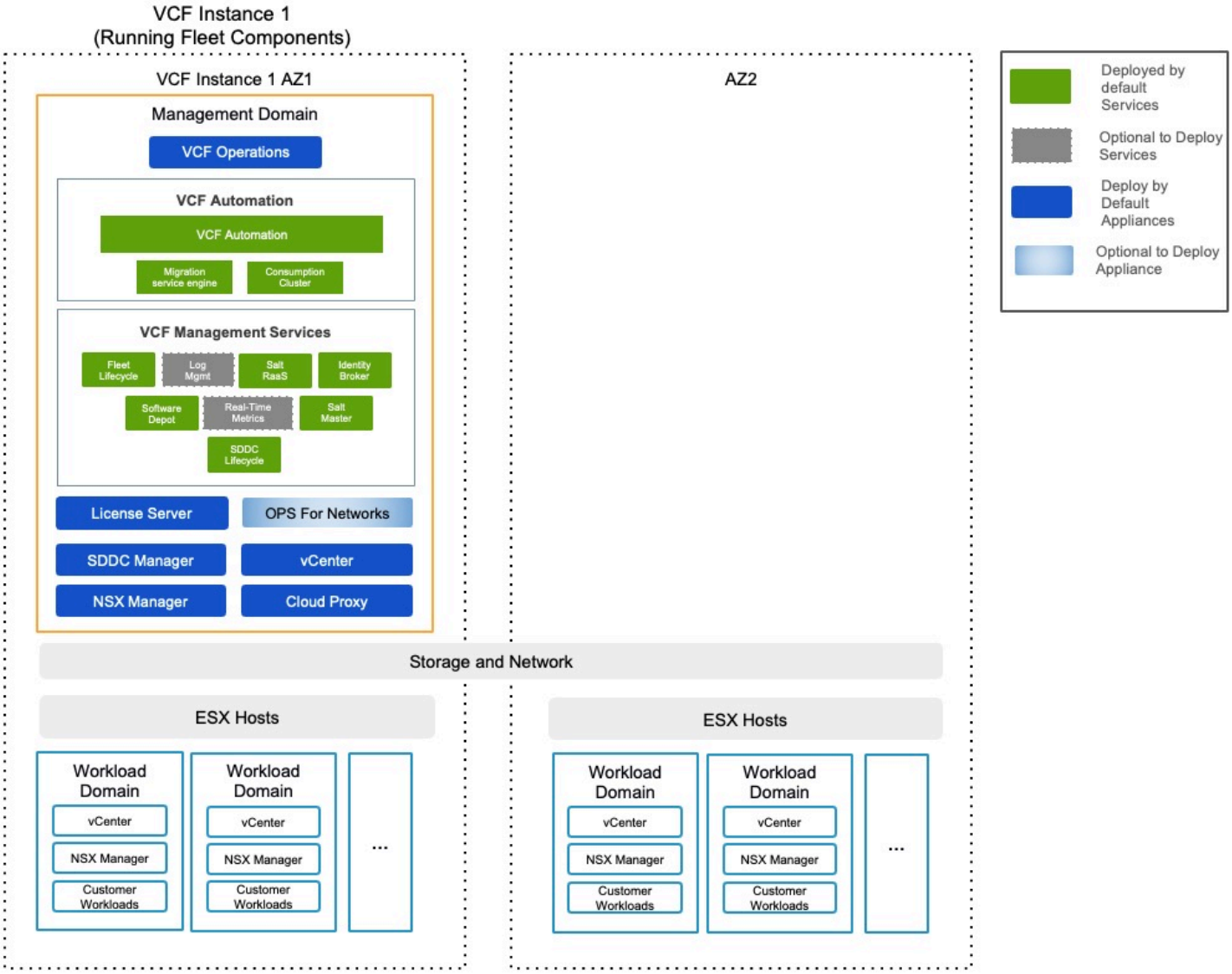
Attribute	Details	Implications
Events to Consider	• Single availability zone power outage. • Network failure in a single availability zone. • Datacenter cooling issues.	High availability is limited to the site level, meaning that while hardware failures are mitigated, site-wide outages (e.g., power or network failures) cannot be addressed without additional disaster recovery mechanisms.
Fault Domain Implementation	• Fault domains map to physical availability zones. • vSphere cluster spans both zones. • Resources balanced across fault domains. • Automated fail over between availability zones.	• Zone failure triggers automatic virtual machine restart in the surviving zone. • No data loss from synchronous storage replication. • Requires stretched Layer 2 networks or advanced routing.
Design Deployment Model	• Built on the single VCF fleet deployment. • Uses the best practices around vSphere HA and physical hardware design constructs. • Designed for a single VCF fleet instance across two availability zones. • Incorporates fault domains within the two availability zones to enhance availability.	Provides active-active availability across two availability zones, ensuring resilience to zone-level failures but requires low-latency, high-bandwidth network connectivity between zones.
Cluster Models	• Supports Single-Rack vSphere Cluster Modeland Stretched vSphere Cluster Model for management domain. • Supports Single-Rack vSphere Cluster Model, Multi-Rack Layer 2 vSphere Cluster Model, Multi-Rack Layer 3 vSphere Cluster Model, and Stretched vSphere Cluster Model for workload domains. • Provides flexibility in choosing the appropriate cluster model based on the environment's needs.	
High Availability Mechanisms	• vSphere HA with zone awareness. • vSAN stretched cluster across zones. • NSX stretched segments. • Witness node for split-brain avoidance.	• Near-instantaneous fail-over for protected workloads. • Storage remains accessible during zone failure. • Network connectivity maintained via stretched segments.
Network Portability	The design requires IP addresses to be stretched between the two availability zones to ensure seamless communication and workload mobility across zones.	• Stretched IP addresses enable seamless failover and workload mobility between availability zones, ensuring high availability and minimizing downtime during failures. • Configuring stretched IPs requires advanced network setup, such as Layer 2 extension or routing solutions, which can increase operational complexity. • Communication between availability zones must meet latency requirements to avoid performance degradation for workloads relying on stretched IPs.

Note:

Infrastructure Blueprint Selection

This fleet design corresponds with the following infrastructure blueprint for implementation:

- VCF Fleet with Multiple Sites in a Single Region



Note:

Log management and Real-time metrics are optional in fleet deployment.

VCF Fleet Deployment with Disaster Recovery (Across Regions) Design

The *VCF Fleet Deployment with Disaster Recovery (Across Regions) Design* protects organizations from site-wide failures through integrated disaster recovery capabilities. Building on the *VCF Fleet Deployment Basic Design*, this architecture uses VMware Live Recovery to enable seamless failover and failback between sites or regions.

The design replicates critical workloads across locations, reducing downtime and data loss during outages. Both Single-Rack and Multi-Rack configurations are supported, providing deployment flexibility based on infrastructure requirements. Organizations gain a resilient foundation that is built on fault tolerance and disaster recovery best practices—essential for maintaining business operations when disruptions occur.

Table 15: Key Provisions of the VCF Fleet Deployment with Disaster Recovery (Across Regions) Design

Attribute	Details	Implications
Events to Consider	• Natural disasters. • Cyber-Attacks. • Data center disaster. • Regional disaster. • Application failures.	A robust disaster recovery (DR) strategy is essential to ensure business continuity and minimize the impact of such events.
Disaster Recovery Capabilities	• Provides disaster recovery mechanisms to ensure business continuity in the event of a disaster. • Includes strategies for data replication, failover, and recovery.	• Organizations must plan and test disaster recovery strategies regularly to ensure that they meet recovery time objectives (RTO) and recovery point objectives (RPO). • Disaster recovery capabilities require additional infrastructure and operational overhead, such as maintaining a secondary site or region.
Design Deployment Model	• Built on the single VCF fleet deployment. • Uses the best practices around vSphere HA and physical hardware design constructs. • Designed for a single VCF fleet deployment built in one availability zone or region with a second instance that is deployed in a separate availability zone or region for disaster recovery. • Incorporates disaster recovery plans to protect against site-wide failures.	Network latency and bandwidth between the primary and secondary sites must meet VMware's requirements to ensure seamless replication and failover.
Cluster Models	• Supports both Single-Rack vSphere Cluster Model for management domain. • Supports Single-Rack vSphere Cluster Model, Multi-Rack Layer 2 vSphere Cluster Model, and Multi-Rack Layer 3 vSphere Cluster Model for workload domains.	Organizations must balance the cost of implementing disaster recovery capabilities with the potential impact of downtime or data loss.
Application Disaster Recovery	• Site-wide protection protects against site-wide failures, such as natural disasters, power outages, or cyber attacks. • Data replication ensures that data is replicated to a secondary site, providing recovery options if there is catastrophic failure. • Business continuity enables failover to a secondary site to maintain operations during major disruptions.	• Longer recovery time disaster recovery typically involves manual or semi-automated failover processes, leading to longer recovery times compared to high availability. • Data loss risk depending on the Recovery Point Objective (RPO), there may be some data loss during failover. • Higher complexity requires careful planning, testing, and maintenance of replication and failover mechanisms. • Significant investment in a secondary site, including hardware, software, and network connectivity.

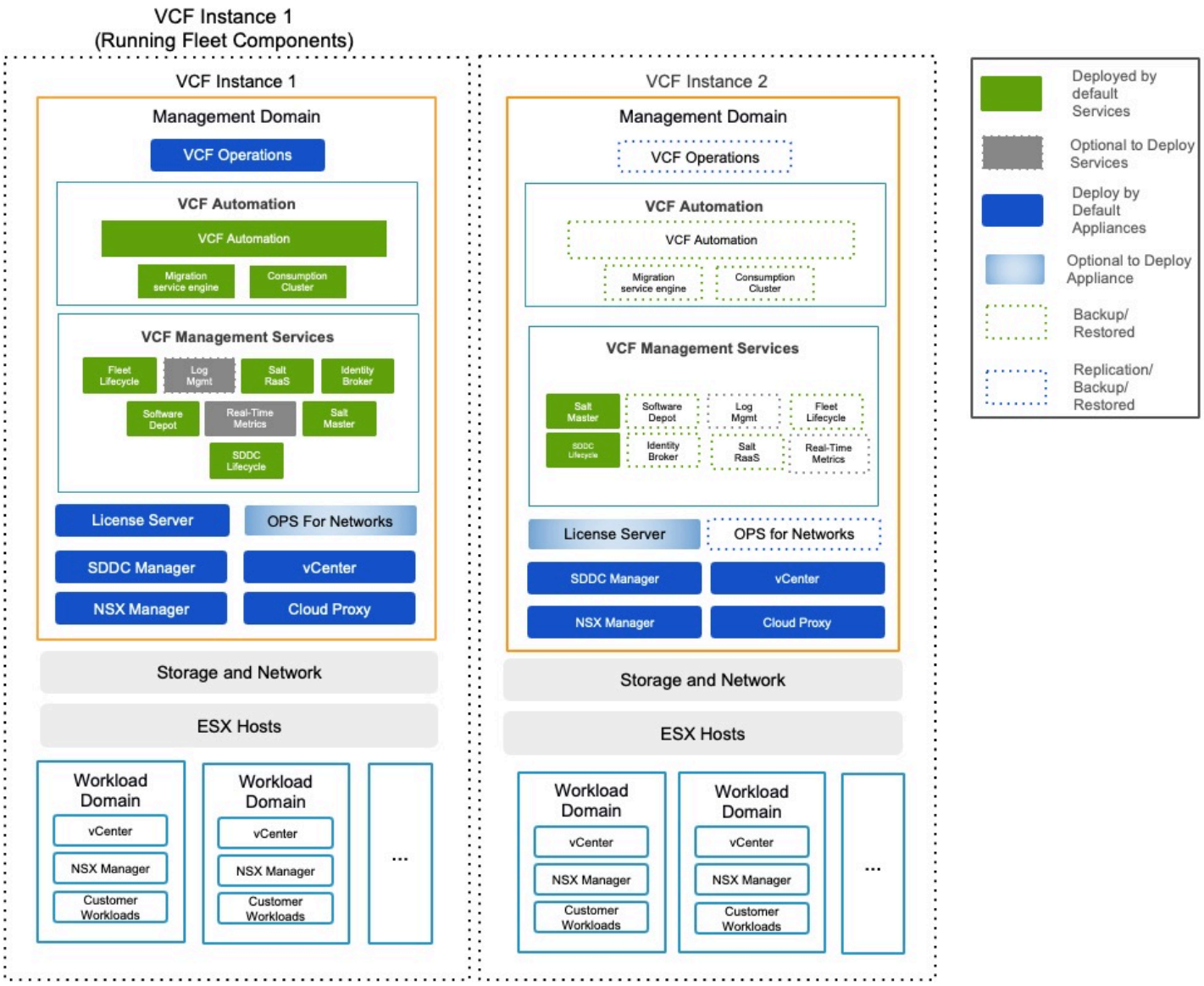
Attribute	Details	Implications
Network Portability	The design requires IP addresses to be stretched between the two availability zones to ensure seamless communication and workload mobility across zones.	• Stretched IP addresses enable seamless failover and workload mobility between availability zones, ensuring high availability and minimizing downtime during failures. • Configuring stretched IPs requires advanced network setup, such as Layer 2 extension or routing solutions, which can increase operational complexity. • Communication between availability zones must meet latency requirements to avoid performance degradation for workloads relying on stretched IPs.

Note:

Infrastructure Blueprint Selection

This fleet design corresponds with the following infrastructure blueprint for implementation:

- VCF Fleet with Multiple Sites Across Multiple Regions



Note:

Log management and Real-time metrics are optional in fleet deployment.

VCF Fleet Deployment with Fault Domains and Disaster Recovery Design

The *VCF Fleet Deployment with Fault Domains and Disaster Recovery Design* combines site-level high availability with cross-region disaster recovery protection. This architecture integrates capabilities from both the `VCF Fleet Deployment with Site High Availability (Across Zones)` and *VCF Fleet Deployment with Disaster Recovery (Across Regions)* designs.

Fault domains distribute resources across physical hardware groups to minimize the impact of localized failures—power outages, hardware breakdowns, or network disruptions within a single zone. Disaster recovery features, including replication and failover, protect workloads against site-wide outages by enabling recovery across regions.

The design supports Single-Rack, Stretched, and Multi-Rack configurations, and follows vSphere HA, fault tolerance, and disaster recovery best practices. Deploy this architecture when critical operations require protection from both localized hardware failures and large-scale site disruptions.

Table 16: Key Provisions of the VCF Fleet with Fault Domains and Disaster Recovery Design

Attribute	Details	Implications
Events to Consider	• Single availability zone power outage. • Network failure in single availability zone. • Natural disasters. • Cyber-Attacks. • Datacenter disaster. • Regional disaster. • Application failures.	The design must account for these risks by incorporating high availability and disaster recovery mechanisms to ensure business continuity and minimize the impact of such events.
Enhanced High Availability	• Provides fault domains to ensure site high availability. • Distributes resources across fault domains to minimize the impact of hardware failures.	High availability is limited to the site level, meaning that while hardware failures are mitigated, site-wide outages (e.g., power or network failures) require disaster recovery mechanisms to maintain operations.
Disaster Recovery Capabilities	• Provides disaster recovery mechanisms to ensure business continuity in the event of a disaster. • Includes strategies for data replication, failover, and recovery.	Proper configuration of fault domains is critical to ensure resilience within the availability zone.
Design Deployment Model	• Built on the *VCF Fleet Deployment with Disaster Recovery (Across Regions) Design*. • Uses the best practices around vSphere HA and physical hardware design constructs. • Designed for a single fleet instance across two availability zones with a second instance that is deployed in a separate availability zone or region for disaster recovery. • Incorporates fault domains and disaster recovery plans to protect against site-wide failures.	Organizations must plan and test disaster recovery strategies regularly to meet Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO).
Cluster Models	• Supports Single-Rack vSphere Cluster Modeland Stretched vSphere Cluster Model for management domain. • Supports Single-Rack vSphere Cluster Model, Multi-Rack Layer 2 vSphere Cluster Model, Multi-Rack Layer 3 vSphere Cluster Model, and Stretched vSphere Cluster Model for workload domains.	Network latency and bandwidth between the primary and secondary sites must meet VMware's requirements to ensure seamless replication and failover.
Application High Availability	• Minimized downtime ensures that the application remains operational during hardware or software failures within the same site. • Automatic failover resources are automatically shifted to healthy nodes in the cluster without manual intervention. • Real-time availability provides continuous availability for critical applications. • Simplified management centralized management within the same site.	• Single-site dependency high availability is limited to the same site, so it does not protect against site-wide disasters like power outages or natural disasters. • Resource overhead requires additional resources (e.g., redundant nodes) to maintain failover capacity. • Higher infrastructure costs due to the need for redundant hardware and licensing.

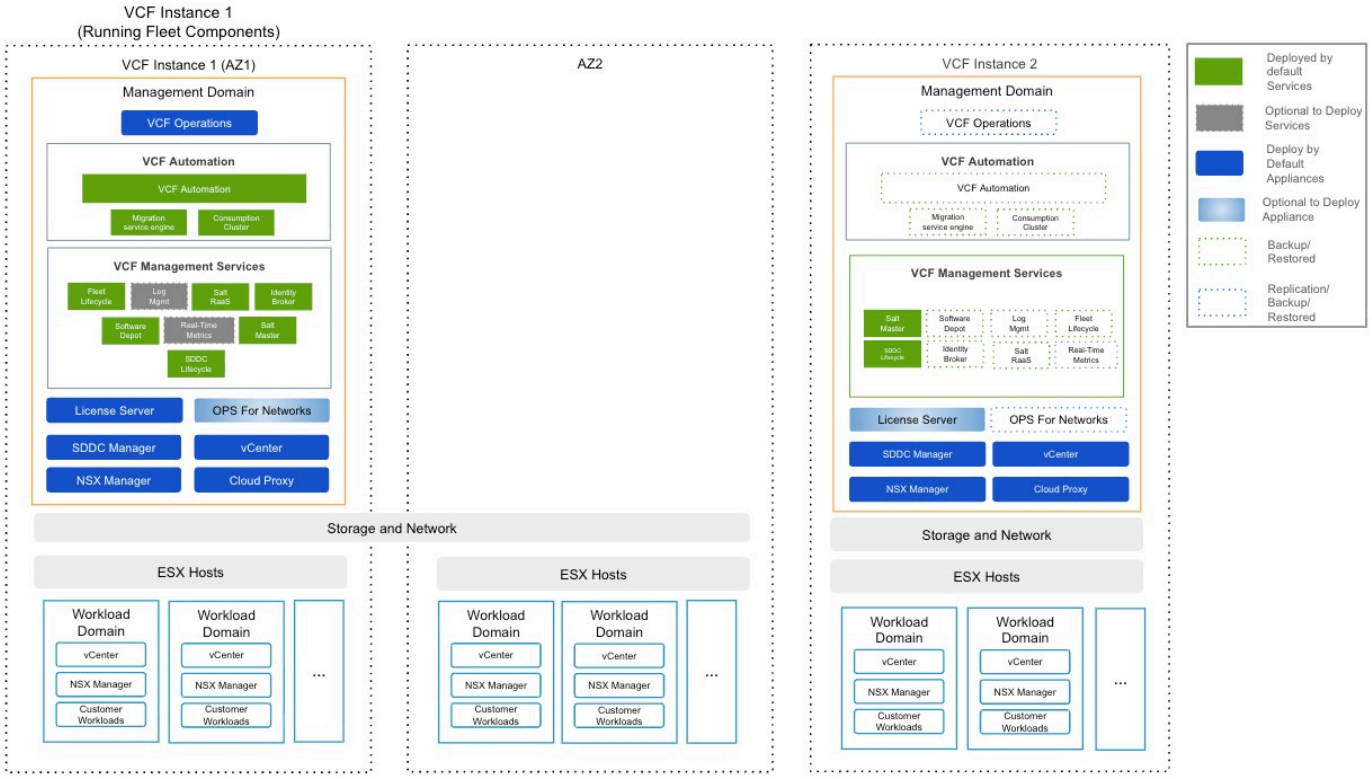
Attribute	Details	Implications
Application Disaster Recovery	• Data replication ensures that data is replicated to a secondary site, providing recovery options if there is catastrophic failure. • Business continuity enables failover to a secondary site to maintain operations during major disruptions.	• Data Loss Risk depending on the Recovery Point Objective (RPO), there may be some data loss during failover. • Higher complexity requires careful planning, testing, and maintenance of replication and failover mechanisms. • Significant investment in a secondary site, including hardware, software, and network connectivity.
Application High Availability and Disaster Recovery	• Comprehensive protection combines real-time availability (HA) with site-wide disaster recovery for maximum resilience. • Minimal downtime ensures high availability within the primary site and rapid recovery if there was site-wide failure. • Data integrity protects against both localized and large-scale failures.	• Requires significant investment in both redundant infrastructure (HA) and a secondary site (DR). • The increased complexity managing both high availability and disaster recovery setups requires advanced planning, monitoring, and testing. • Resource requirements demand more hardware, software, and network resources to support both configurations.
Network Portability	The design requires IP addresses to be stretched between the two availability zones to ensure seamless communication and workload mobility across zones.	• Stretched IP addresses enable seamless failover and workload mobility between availability zones, ensuring high availability and minimizing downtime during failures. • Configuring stretched IPs requires advanced network setup, such as Layer 2 extension or routing solutions, which can increase operational complexity. • Communication between availability zones must meet latency requirements to avoid performance degradation for workloads relying on stretched IPs.

Note:

Infrastructure Blueprint Selection

This fleet design corresponds with the following infrastructure blueprint for implementation:

- VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)



Note:

Log management and Real-time metrics are optional in fleet deployment.

VCF Fleet Sizing Models

This section outlines the *VCF Fleet Sizing Models* and available sizing options within VCF Installer. Scale limits

First VCF Instance

Table 17: VMware Cloud Foundation Installer Deployment Profiles

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for the initial VCF instance within a VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	1x Small • 1x Control Plane Nodes • 2x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Large • 3x Control Plane Nodes • 4x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Medium	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
VCF Operations	1x Small	2x Small	3x Medium	3x Large
License Server	1x Default	1x Default	1x Default	1x Default
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard
VCF Automation	1x Small	1x Medium	3x Medium	3x Large

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
Resource Requirements	• vCPU: 76 • RAM: 251 GB • DISK: 7448 GB	• vCPU: 106 • RAM: 335 GB • DISK: 8422 GB	• vCPU: 184 • RAM: 656 GB • DISK: 11445 GB	• vCPU: 298 • RAM: 949 GB • DISK: 15357 GB

Additional VCF Instance

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for any additional VCF instance joining an existing VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Table 18: VMware Cloud Foundation Installer Deployment Profiles

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	• 1x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Small	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard
Resource Requirements	• vCPU: 34 • RAM: 111 GB • DISK: 4562GB	• vCPU: 62 • RAM: 187 GB • DISK: 5462 GB	• vCPU: 74 • RAM: 244 GB • DISK: 5813 GB	• vCPU 120 • RAM 353 GB • DISK 8321 GB

Resource Requirements

Download the Planning and Preparation Workbook and populate the **Management Domain Sizing** tab to determine resource requirements based on your desired design.

VCF Automation Models

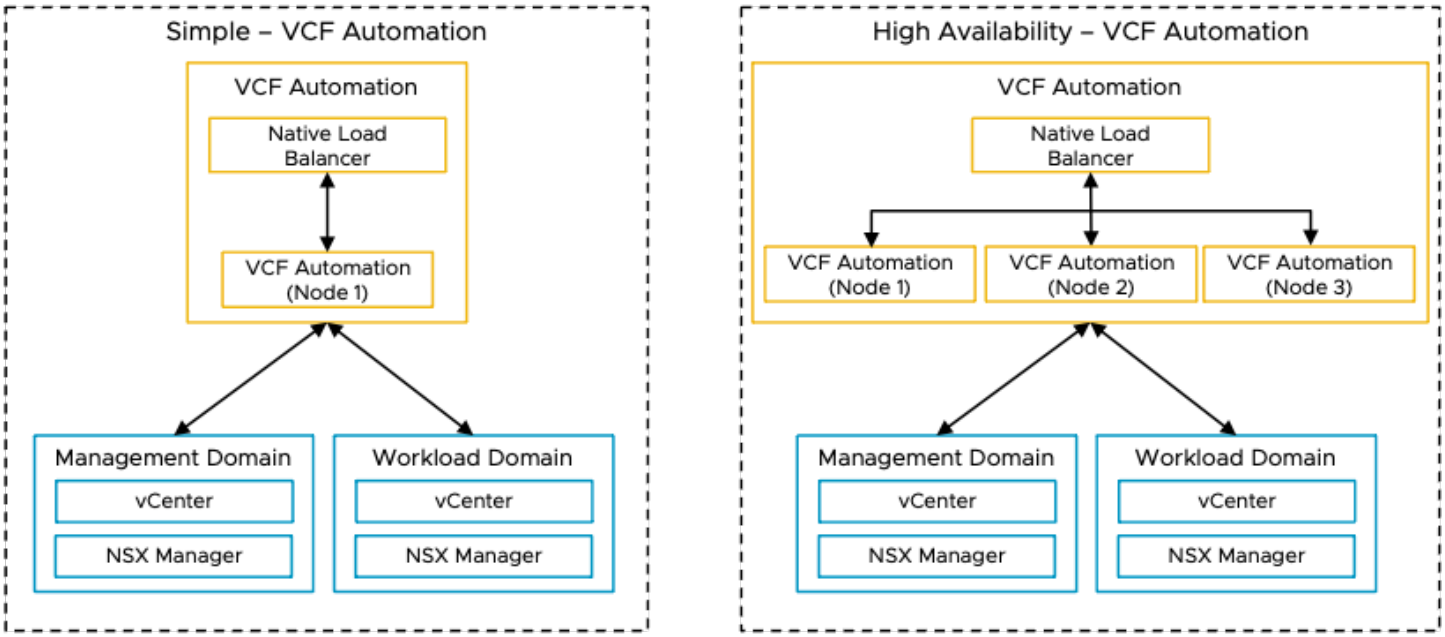
A VMware Cloud Foundation platform uses VCF Automation as the central provisioning console for your VCF fleet to provide users self-service consumption of Kubernetes and cloud infrastructure resources "as a Service", while maintaining control, security, and compliance, all available in a self-service portal.

VCF Automation runs on containers within a Kubernetes cluster, which is deployed on dedicated virtual appliances. The VCF Automation infrastructure is visible in the vCenter inventory as individual nodes. During installation the dedicated Kubernetes cluster is to orchestrate the deployment of the VCF Automation component containers and services.

Table 19: VCF Automation Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Simple VCF Automation Model	• Single node. • Applies to small appliance size. • Can be scaled out to the high availability model by resizing the node to Medium or Large, which automatically scales the deployment out to 3 nodes.	• Recommended deployment scenarios: – Proof of Concept (POC) or evaluation. – Small-scale, non-critical workloads. – Development and testing environments.	No application-level high availability.
High Availability VCF Automation Model	• Three node cluster. • Applies to medium or large node sizes. • Supports use of an external load balancer.	• Recommended deployment scenarios: – Production environments. – Mission-critical applications. – Enterprise deployments. – Scalable infrastructure, ideal for organizations that anticipate future growth and need the ability to scale their infrastructure resources. • Can be scaled up or down in size as necessary (for example, from medium to large nodes or the reverse).	Provides application-level availability.

Figure 13: VCF Automation Models in VMware Cloud Foundation



vSphere Supervisor Models

vSphere Supervisor is a workload domain capability that, when activated, enables a self-service consumption experience that uses a single declarative API to create resources such as Virtual Machines, Kubernetes Guest Clusters, Containers, Storage Volumes, and other advanced services.

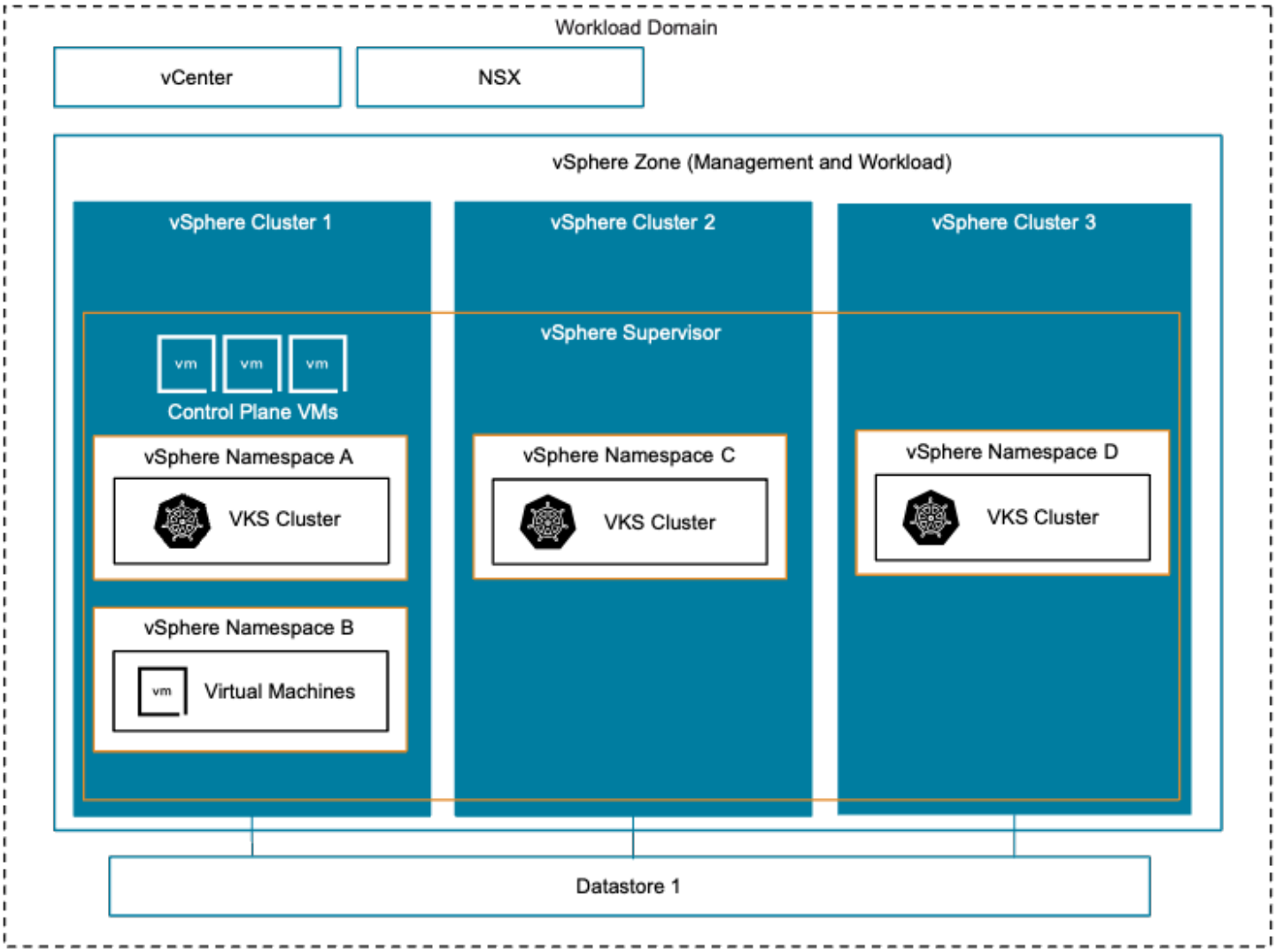
vSphere Zones

A vSphere Zone is the boundary that can map to a single or to multiple vSphere clusters.

vSphere Zones Type	Attributes	Benefits	Implications
Single vSphere Cluster Zone	Maps to a single vSphere cluster	Simplest deployment model.	• If a vSphere Supervisor needs more resources, it is achieved by adding a vSphere cluster as an additional workload zone. • Removing the only vSphere cluster from a vSphere zone requires draining all the workloads from that zone.
Multi vSphere Cluster Zones	Maps to multiple vSphere clusters	• Allows scaling the compute capacity of a zone by aggregating multiple vSphere clusters. • Easier capacity expansion by adding new clusters to existing zones. • Enhanced lifecycle handling. Seamless cluster replacements, upgrades, or retirements with no downtime or rework.	• Requires shared storage and shared VDS to support cluster decommissioning. • Storage Policies should be identical between clusters. • Networking Segments must be uniformly available between clusters. • vSphere Clusters that make up a vSphere Zones should have similar performance characteristics to maintain consistent and predictable performance. • A vSphere namespace can only allocate resources from a single vSphere Cluster from a Multi vSphere Cluster Zone.

Figure 14: Multi Cluster vSphere Zone Diagram

This diagram illustrates an example of a *Multi Cluster vSphere Zone* deployment.



Note:

A vSphere Zone will always be the boundary for high availability or mobility operations.

vSphere Zones can be tagged as vSphere Supervisor Management zones and/or assigned to one or more vSphere Namespaces as Workload Zones. A vSphere Zone can only be associated to one vSphere Supervisor instance.

Supervisor Zone Type	Attributes
Management Zone	• One vSphere Zone in a Single Management Zone configuration. • Three vSphere Zones in a Three Management Zone configuration. • Where vSphere Supervisor Control Plane nodes and Foundational Load Balancer (FLB) (optional component) appliances are deployed. • Must be selected at vSphere Supervisor activation.
Workload Zone	• Can be added or marked for removal from vSphere Namespaces post vSphere Supervisor activation. • Can be added to one or more vSphere Namespaces.

Note:

ESX hosts that make up a vSphere Zone within each zone type should have similar performance characteristics to maintain consistent and predictable performance.

vSphere Supervisor Control Plane Availability Models

For each vSphere Supervisor instance, one or three control plane nodes are deployed to the selected Management Zones. The vSphere Supervisor control plane appliance can be deployed with one control plane node (Simple) and scale to three control plane (High-availability) nodes post activation for *Single Management Zone* model deployments.

Table 20: vSphere Supervisor Control Plane Availability Models in VMware Cloud Foundation

Control Plane Availability	Attributes	Benefits	Implications
Simple vSphere Supervisor Control Plane Model	• Single control plane node. • Default option when activating vSphere Supervisor as part of workload domain creation.	• Simplified vSphere Supervisor activation. • Minimal resources required.	• Single point of failure. • Downtime during upgrade. • Not an option for Three Zone Management deployments.
High Availability vSphere Supervisor Control Plane Model	• Three control plane nodes deployed for each vSphere Supervisor instance. • In a Three Management Zone model, one control plane node is deployed per zone. • Only option for Three Management Zone model.	Protects against single control plane node failure.	Additional resources required.

vSphere Supervisor Zone Models

vSphere Supervisor can be deployed with a combination of Single or Three Management Zones and Combined or Isolated Workload Zones. A hybrid approach to these models is possible but not recommended.

Table 21: vSphere Supervisor Zone Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Simplified vSphere Supervisor Model	• Single control plane VM. • Single vNIC control plane VM. • No load balancer. • Single vSphere Zone that is used for vSphere Supervisor management and workloads.	• Minimal resources required. • Supports running VM Service.	• Does not support running vSphere Pods and other Supervisor services. • Not supported for use by VCF Automation All Apps Organization.
Single Management Zone with Combined Workload Zones Model	• A single vSphere Zone that is used for vSphere Supervisor management and application workloads. • Simple or high-availability control plane option.	No additional vSphere clusters required for application workloads zones.	vSphere cluster availability impacts both Supervisor management and application workloads zones.
Single Management Zone with Isolated Workload Zones Model	• One vSphere Zone used for vSphere Supervisor management workloads. • vSphere Zone that is used for management workloads are separate to application workloads. • Simple or high-availability control plane option.	vSphere Supervisor management and application workload availability impacts are independent of each other.	Additional vSphere clusters required for application workloads zones.

Model	Attributes	Benefits	Implications
Three Management Zones with Combined Workload Zones Model	Three vSphere Zones that are used for vSphere Supervisor management and application workloads.	• vSphere Supervisor management workload is protected from single vSphere cluster outages. • No additional vSphere clusters required for application workloads zones.	• High-availability control plane option only. • Activation by API only.
Three Management Zones with Isolated Workload Zones Model	• Three vSphere Zones used for vSphere Supervisor management workloads. • vSphere Zones that are used for management workloads are separate to application workloads.	vSphere Supervisor management and application workload availability impacts are independent of each other.	• High-availability control plane option only. • Additional vSphere clusters required for application workloads zones. • Activation by API only.

Load Balancer Models and Network Connectivity Model Support Matrix for vSphere Supervisor

Load Balancer		Distributed VLAN Connectivity Model	Centralized Connectivity Model	NSX Segment Connectivity Model	VLAN Networking Connectivity Model	VLAN Backed VPC Connectivity Model	Distributed VXLAN Connectivity Model for Version 9.1.0	Distributed VXLAN Connectivity Model for Version 9.1.1
Avi Load Balancer Model		✓	✓	✓	✓	✓		✓
NSX Edge VPC Load Balancer Model			✓					
NSX Edge Tier-1 Load Balancer Model				✓	✓			
Virtual Network Appliance Load Balancer Model		✓				✓		✓
Foundation Load Balancer Model					✓			

vSphere Zones and Storage

vSphere Supervisor uses storage policies to integrate with storage available to vSphere. The policies represent datastores and manage storage placement of vSphere Supervisor objects. Storage policies support vSphere cluster principal and supplemental datastores configured.

When assigning storage for vSphere Supervisor, availability and performance considerations should be made in relation to the underlying storage that supports the vSphere Zones that are used by vSphere Supervisor.

For detailed information, refer to the Storage for Supervisor and VKS Clusters in the *vSphere Supervisor Platform* documentation.

	Considerations	Examples
Availability	Storage availability constructs should align with vSphere Zones. There is no replication of data between zones on the infrastructure layer. Each zone should be treated as an independent failure domain.	• If a datastore is a failure domain, each zone should use an independent datastore. • If a zone has physical failure domain boundaries such as ESX hosts and physical networking, physical storage placement should be co-located.
Performance	Storage consumed by the zone type and vSphere Namespace should have the same performance characteristics to maintain consistent and predictable performance.	• Storage for management zones should have the same performance characteristics. • Storage for all workload zones within the same vSphere namespace should have the same performance characteristics.

Storage Topologies for vSphere Zones

vSphere Supervisor supports different datastore topologies for vSphere Zones. These topologies define how storage is presented to hosts in each zone and how workloads behave during a zone failure.

Figure 15: Zonal datastore

A **zonal datastore** is local to a single vSphere Zone and is only available to workloads in that zone.

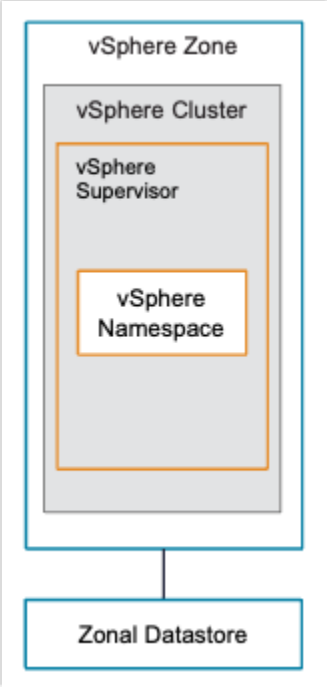


Figure 16: Multi-zone datastore

A **multi-zone datastore topology** uses separate datastores per zone; each zone has its own datastore, with no storage-level replication or automatic failover between zones, so application-level resilience is required.

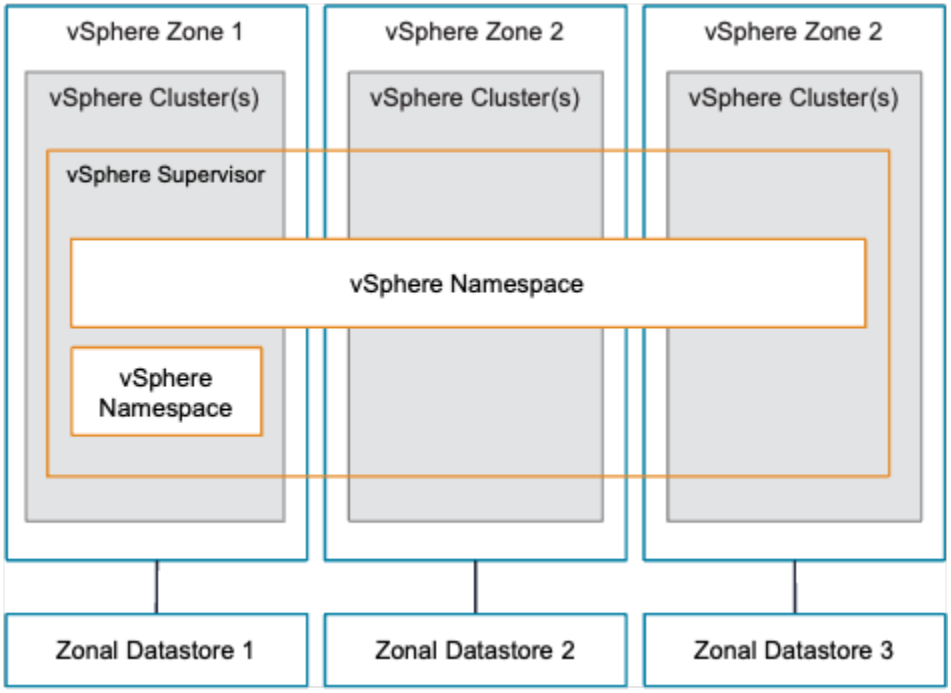
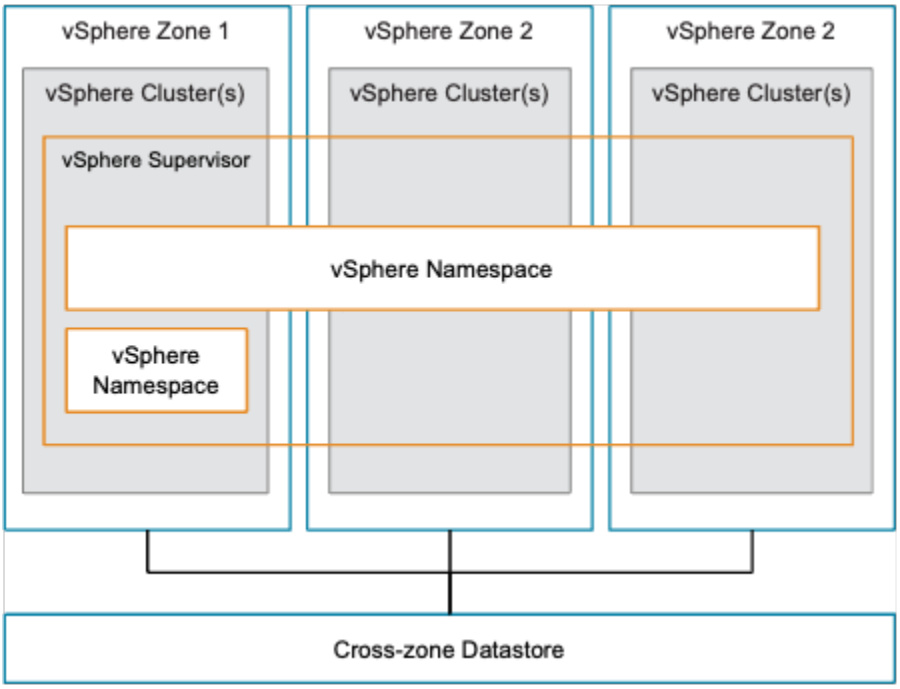


Figure 17: Cross-zone datastore

A **cross-zone datastore** is a single logical datastore shared across multiple zones, providing infrastructure-level availability and protection against the failure of an entire zone.



Zone Datastore Type	Description
Zonal datastore	• The Zonal datastore is local to a single vSphere Zone. • Datastore is attached to all the hosts that are part of the underlying vSphere cluster. • Datastore is only accessible by workloads running within that vSphere Zone. • This configuration provides high availability at the vSphere cluster level.
Multi-zone datastore	• Multiple vSphere Zones, each with its own availability zone paired with a dedicated datastore. • Datastore is attached to all the hosts that are part of the underlying vSphere cluster. • Datastore is only accessible by workloads running within that vSphere Zone. • Does not provide replication or failover capability. The application must provide HA functionality
Cross-zone datastore	• A single logical datastore shared across multiple vSphere Zones, with the datastore spanning multiple availability zones. • This configuration provides high availability and fault tolerance at the infrastructure level, protecting workloads from the failure of an entire availability zone.

Supervisor vSAN Compute and Storage Model Support

Storage	Zonal datastore	Multi-zone datastore	Cross-zone datastore
vSAN Cluster	✓	✓	
vSAN Cluster and vSAN File Services	✓	✓	
Stretched vSAN Cluster	✓		
Stretched vSAN Cluster and vSAN File Services	✓		
vSAN Storage Cluster	✓	✓	✓
vSAN Storage Cluster and vSAN File Services	✓	✓	✓
Stretched vSAN Storage Cluster	✓		✓
Stretched vSAN Storage Cluster and vSAN File Services	✓		✓

Table 22: Zonal datastore

Topology	Attributes	Benefits	Implications
vSAN Cluster or vSAN Storage	• A zonal datastore is local to a single vSphere Zone • Compute and storage reside within the same vSphere zone.	Simplest configuration.	Failure of a single vSphere zone impacts workloads in that zone.
vSAN Cluster or vSAN Storage Cluster with File Services	• A zonal datastore is local to a single vSphere Zone • Compute and storage reside within the same vSphere zone. • vSAN File Services provide RWX file shares for shared persistent volumes.	Volume Access Modes.	Failure of a single vSphere zone impacts workloads in that zone.
Stretched vSAN Cluster	• A zonal datastore is local to a single vSphere Zone. • Compute and storage reside within the same vSphere zone. • Storage is configured in a multi-availability zone configuration.	• Highly resilient and available storage architecture. • During availability zone failure, workloads are automatically restarted in the surviving availability zone.	Failure of a single vSphere zone impacts workloads in that zone.
Stretched vSAN Cluster and vSAN File Services	• A zonal datastore is local to a single vSphere Zone. • vSAN File Services provide RWX file shares for shared persistent volumes. • Compute and storage reside within the same vSphere zone. • Storage is configured in a multi-availability zone configuration.	• Highly resilient and available storage architecture. • During availability zone failure, workloads are automatically restarted in the surviving availability zone. • Volume Access Modes.	Failure of a single vSphere zone impacts workloads in that zone.
Stretched vSAN Storage Cluster	• A zonal datastore is local to a single vSphere Zone. • Compute and storage resides within the same vSphere zone. • Storage is configured in a multi-availability zone configuration.	• Highly resilient and available storage architecture. • During compute availability Zone failure, workloads are automatically restarted in the surviving availability zone.	• Failure of a single vSphere zone impacts workloads in that zone. • vSAN Compute Site Coupling must be configured. vSAN Storage Cluster must be stretched before mounting.
Stretched vSAN Storage Cluster and File Services	• A zonal datastore is local to a single vSphere Zone. • vSAN File Services provide RWX file shares for shared persistent volumes. • Storage is configured in a multi-availability zone configuration.	• Highly resilient and available storage architecture. • During compute availability Zone failure, workloads are automatically restarted in the surviving availability zone. • Volume Access Modes.	• Failure of a single vSphere zone impacts workloads in that zone. • vSAN Compute Site Coupling must be configured. vSAN Storage Cluster must be stretched before mounting. • vSAN datastore is mounted and File Services is enabled per zone.

Figure 18: Stretched vSAN Storage Cluster

Single Zone storage configuration with Stretched vSAN Storage Cluster

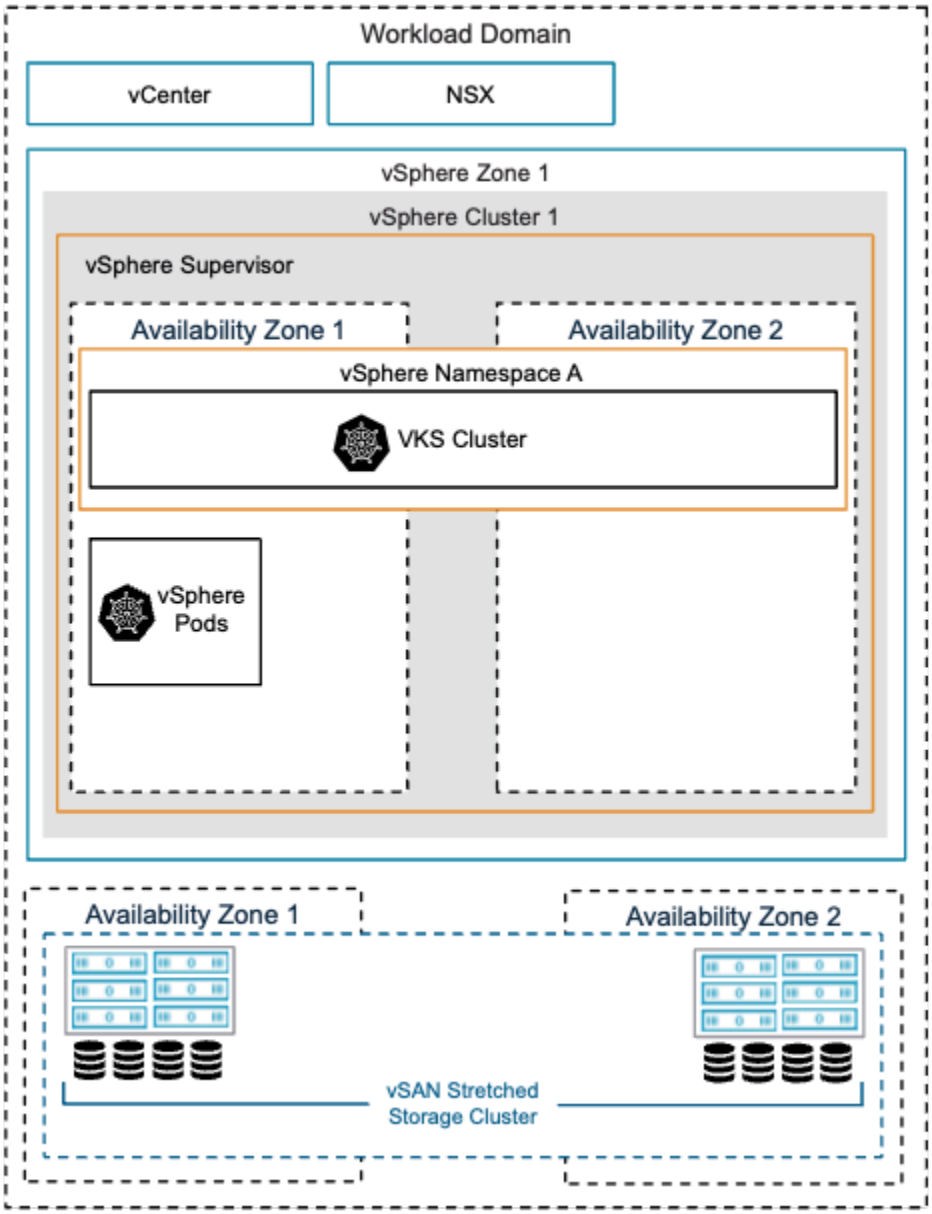


Table 23: Multi-zone datastore

Topology	Attributes	Benefits	Implications
vSAN Cluster	• Spans across multiple vSphere Zones. • Each cluster within the vSphere Zone has its own dedicated vSAN datastore. • Zone-aware storage policies ensure that pods and volumes stay co-located.	• Zonal independence, using app level replicas. • During Single Zone failure, the application remains available via replicas in other zones.	• Application requires its own app-level replication. • StatefulSet replicas distributed across zones for high availability.
vSAN Cluster and vSAN File Services	• Spans across multiple vSphere Zones. • Each cluster within the vSphere Zone has its own dedicated vSAN datastore. • Zone-aware storage policies ensure that pods and volumes stay co-located. • vSAN File Services provide RWX file shares for shared persistent volumes.	• Zonal independence, using app level replicas. • During vSphere Zone failure, the application remains available via replicas in other zones. • Volume Access Modes.	• Application requires its own app-level replication. • StatefulSet replicas distributed across zones for high availability. • Zonal failure RWX file volumes are unavailable. • RWX volumes are local to the zone.
vSAN Storage Cluster	• Spans across multiple vSphere Zones. • Each Cluster within the vSphere Zone has its own dedicated vSAN Storage Cluster Datastore. • Zone-aware storage policies ensure that pods and volumes stay co-located.	• Zonal independence, using app level replicas. • During Single Zone failure, the application remains available via replicas in other zones.	• Application requires its own app-level replication. • StatefulSet replicas distributed across zones for high availability.
vSAN Storage Cluster and vSAN File Services	• Spans across multiple vSphere Zones. • Each cluster within the vSphere Zone has its own dedicated vSAN Storage Cluster Datastore. • Zone-aware storage policies ensure that pods and volumes stay co-located. • vSAN File Services provide RWX file shares for shared persistent volumes.	• Zonal independence, using app level replicas. • During Single Zone failure, the application remains available via replicas in other zones. • Volume Access Modes.	• Application requires its own app-level replication. • StatefulSet replicas distributed across zones for high availability. • During Zonal failure RWX file volumes are unavailable. • RWX volumes are local to the zone.

Figure 19: vSAN Cluster

Multi Zone storage configuration with vSAN Cluster

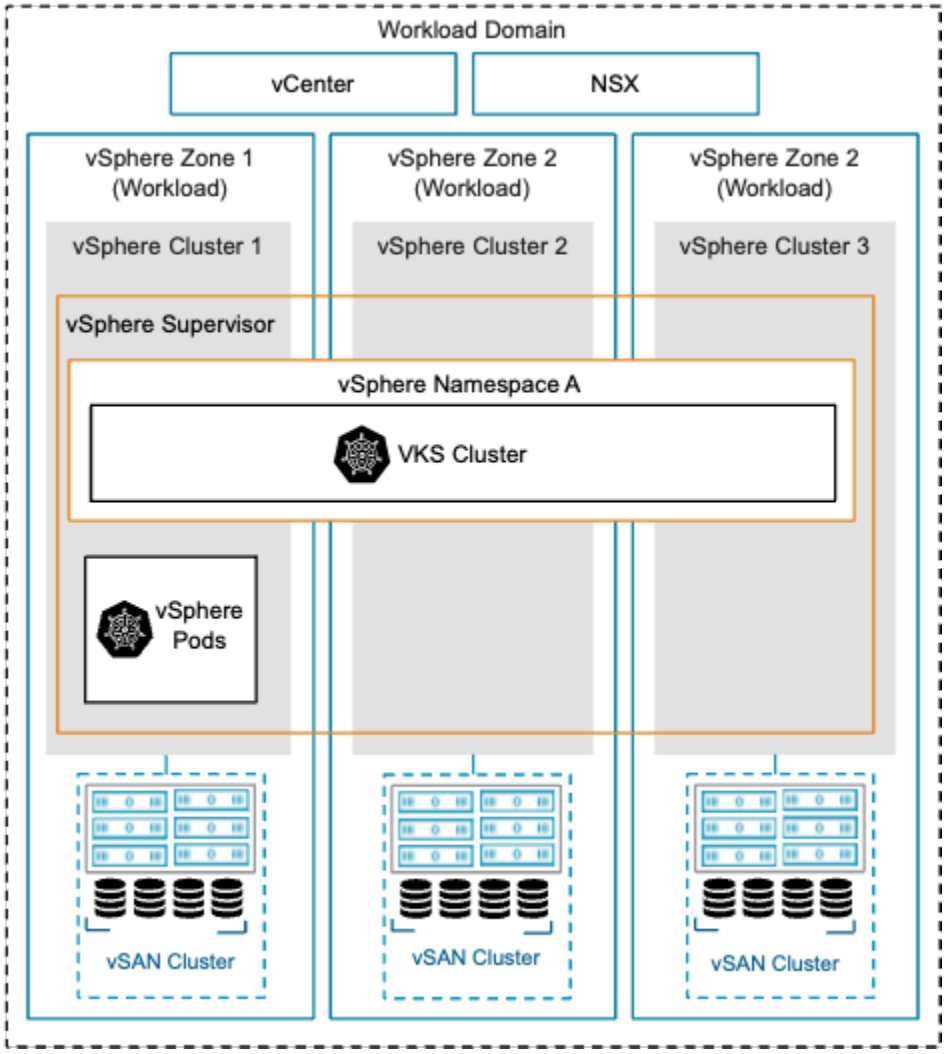


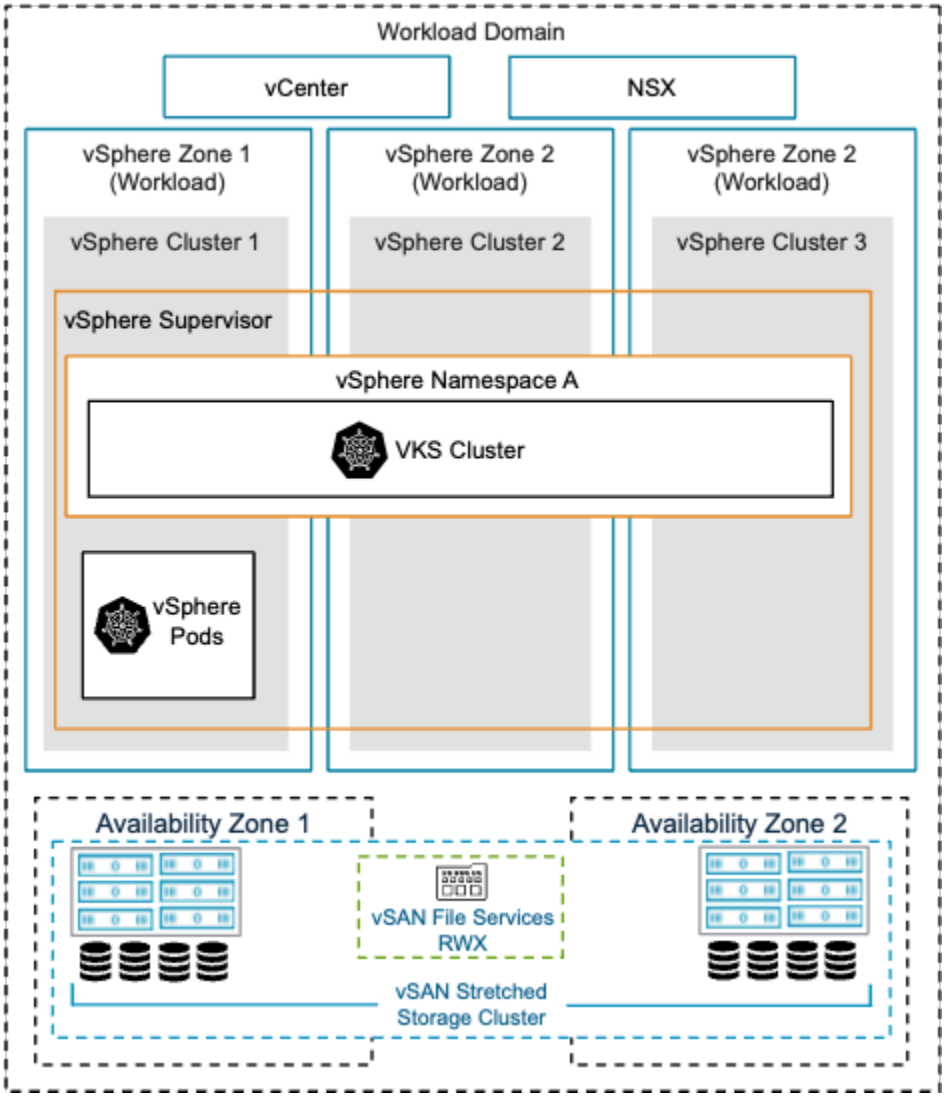
Table 24: Cross-zone datastore

Topology	Attributes	Benefits	Implications
vSAN Storage Cluster	• Spans across multiple vSphere Zones. • vSAN Storage datastore shared across the vSphere zones. • Zone-aware storage policies ensure that pods and volumes stay co-located.	• Zonal independence, using with app-level replicas. • During single vSpherezone failure, the application remains available via replicas in other zones.	• Application requires its own app-level replication. • StatefulSet replicas distributed across zones for high availability.
vSAN Storage Cluster and vSAN File Services	• Spans across multiple vSphere Zones. • vSAN Storage datastore shared across the vSphere zones • vSAN File Services provide RWX file shares for shared persistent volumes.	• During single vSpherezone failure, the application remains available via replicas in other zones. • During Single Zone failure, workloads maintain access to RWX volumes. • Volume Access Modes.	• Application requires its own app-level replication. • StatefulSet replicas distributed across zones for high availability. • vSAN datastore is mounted and File Services is enabled per zone.

Topology	Attributes	Benefits	Implications
Stretched vSAN Storage Cluster	• Spans across multiple vSphere Zones. • Stretched vSAN Storage datastore shared across the vSphere Zones.	• During Single Zone failure, workloads are automatically restarted in the surviving zone. • Highly resilient and available storage architecture.	• vSAN Compute Site Coupling must be configured. • vSAN Storage Cluster must be stretched before mounting.
Stretched vSAN Storage Cluster + File Services	• Spans across multiple vSphere Zones. • Stretched vSAN Storage datastore shared across the vSphere Zones • vSAN File Services provide RWX file shares for shared persistent volumes.	• During Single Zone failure, workloads are automatically restarted in the surviving zone. • During Single Zone failure, workloads maintain access to RWX volumes. • Volume Access Modes.	• vSAN Compute Site Coupling must be configured. • vSAN Datastore is mounted and File Services is enabled per zone. • vSAN Storage Cluster must be stretched before mounting.

Figure 20: vSAN Storage Cluster and vSAN File Services

Cross Zone storage configuration with vSAN Storage Cluster and vSAN File Services



Network Consumption Models

Network Consumption Models are the methods cloud users use to deploy and manage networking and security capabilities withVMware Cloud Foundation. Different capabilities are supported according to the selected *Network Consumption Model*.

VCF Automation All Apps Orgs Consumption Model

When the goal is to use the VMware Cloud Foundation infrastructure using VCF Automation via an All Apps Orgs, where the management of compute, storage, and network is centralized via single UI/API, the cloud user should not interact directly with the NSX UI/API other than for troubleshooting purposes. The cloud user should create and administer Virtual Private Clouds (VPCs) via the VCF Automation API/UI.

Virtual Private Cloud (VPC) Consumption Model

A VPC is a cloud networking construct, a secure, isolated virtual network within a cloud environment. VPCs bring the same characteristics and consumption model of the public cloud to VMware Cloud Foundation.

The VPC is a foundational and required component for designs requiring VCF Automation All Apps Orgs, but it is also the recommended network consumption model for any network topology aimed at supporting the manual deployment of workloads, VCF Automation VM Only Orgs, or third-party automation tools.

VPCs can be configured and managed from the vCenter making the vSphere client the single user interface for managing compute, network, and storage for simple single-tenant deployments. Multi-tenant deployments that do not require the full capabilities of VCF Automation All Apps Org can leverage the NSX multi-tenancy framework to provision self-service network and security services via the NSX UI or API.

NSX Segment Consumption Model

This model leverages NSX objects such as Tier-0/Tier-1 gateways and segments to deploy network topologies. This model allows the network administrator to leverage all of the advanced NSX network capabilities but requires more extensive networking and NSX-specific knowledge.

This model does not support VCF Automation All Apps Orgs, and while it does support VMware vSphere Kubernetes Service (VKS), it is recommended that new deployments of VKS embrace the VPC-based topology.

The *NSX Segment Consumption Model* should be considered when the capabilities it provides compared to the VPC Consumption Model are critical to meeting the design requirements. Among those, the ability to extend network topologies across workload domains respective NSXinstances in different VMware Cloud Foundation instances via NSX Federation is a primary use case.

VLAN Networking Consumption Model

In this model, the only VMware Cloud Foundation component participating in the networking functionalities is the vSphere Distributed Switch, responsible for switching the traffic between the VMs running in the hypervisor and the physical uplinks. Once the packets are delivered to the upstream physical switch, it is the responsibility of the physical network fabric to deliver them to the appropriate destination. Virtual machines are connected to vSphere Distributed Port Groups or NSX VLAN segments.

This model lacks the benefits provided by the NSX network virtualization which decouples the physical network from the topology requirements of the applications and services that run on top of it. With NSX network virtualization, the underlaying physical network fabric can be designed and operated based on a simpler and more stable model that does not require following the application's lifecycle. The role of adapting to the ever-changing application requirements is offloaded to the NSX network virtualization layer while the physical network fabric design can be centered on maximizing throughput and high availability, minimizing latency, and optimizing operational efficiency.

The *VLAN Networking Consumption Model* is supported in VMware Cloud Foundation and may be appropriate for specific use cases, but it is not recommended for the majority of scenarios as it lacks the benefits provided by network virtualization.

Network Consumption Models Comparison

This table provides a comparison between the supported *Network Consumption Models*.

Model	Attributes	Benefits	Implications
VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model	• Consumption via VCF Automation • Prescriptive workload network design based on the VPC model. • Supports the following Workload Connectivity models: — Centralized Connectivity Model- — Distributed VLAN Connectivity Model- — Distributed VXLAN Connectivity Model for Version 9.1.1-	• Supports vSphere Supervisor and Supervisor services (e.g., VKS). • Consistent cloud-like experience for networking. • The physical network is abstracted. No changes are required to support the application lifecycle. • Easy to achieve rack availability via vSphere HA for compute workloads regardless of the underlaying network fabric topology. • Software-defined allows scaling of the networking and security stack.	• Administrator should not interact directly with NSX UI/API except for troubleshooting. • Best suited for new workloads.
Virtual Private Cloud (VPC) Consumption Model	• Administrator utilizes vCenter or NSX UI/API to create and use NSX VPC • Cloud networking constructs. • Supports NSX projects as part of the NSX multi-tenancy framework. • Supports the following Workload Connectivity models: — Centralized Connectivity Model- — Distributed VLAN Connectivity Model- — Distributed VXLAN Connectivity Model for Version 9.1.0- — Distributed VXLAN Connectivity Model for Version 9.1.1-	• Simple consumption for non network specialists. • For single tenant, unified UI/API for managing compute, network, and storage (vCenter). • Self-service network and security services. • Embedded IPAM simplifies the creation of new networks. • Supports NAT with these connectivity models: — Centralized Connectivity Model — Distributed VLAN Connectivity Model • Supports Supervisor services (e.g., VKS) with these connectivity models: • Centralized Connectivity Model • Distributed VLAN Connectivity Model • VM mobility across Layer 3 boundaries with these models: — Centralized Connectivity Model — Distributed VXLAN Connectivity Model • The physical network is abstracted. No changes are required to support the application lifecycle.	• VPC model cannot be consumed via NSX Global Manager. • Without VCF Automation All Apps Orgs, IaaS tenancy is provided at the networking and security level only.

Model	Attributes	Benefits	Implications
		• Easy to achieve rack availability via vSphere HA for compute workloads regardless of the underlaying network fabric topology. • Software-defined allows scaling of the networking and security stack.	
NSX Segment Consumption Model	• Administrator utilizes NSX UI/API to create logical network topologies. • Uses policy NSX objects (Tier-0/Tier-1 gateways and segments). • Provides advanced NSX network and visibility capabilities. • Full support for NSX Federation, refer to NSX Segment Connectivity Model with NSX Federation. • Supports NSX projects as part of the NSX multi-tenancy framework (Not in conjunction with NSX Federation) • Supports the following Workload Connectivity model: – NSX Segment Connectivity Model	• Allows extension of network topologies across NSX instances. • Full access to advanced NSX features. • Supports Supervisor services (e.g., VKS). • VM mobility across Layer 3 boundaries – any topology. • The physical network is abstracted. No changes are required to support the application lifecycle. • Easy disaster recovery without re-IP via NSXFederation or multi availability zone design. • Software-defined allows scaling of the networking and security stack.	• Requires networking and NSX-specific knowledge. • Does not support VCF Automation All Apps Orgs. • No embedded IPAM for networks (DHCP is available).
VLAN Networking Consumption Model	• Administrator utilizes vCenter or NSX UI/API to create and VLANs • VMs connect to distributed port-groups or NSX VLAN segments. • Physical fabric handles network routing and delivery.	• Familiar traditional networking model for vSphere admins. • Supports vSphere Supervisor and Supervisor services (e.g., VKS).	• Lacks network virtualization benefits. • Tight coupling between physical network and application topologies. • Requires physical fabric changes to adapt to application changes. • VM mobility is limited to the span of a VLAN (usually one rack). • Slow application deployment dependent on physical network reconfigurations. • Achieving rack availability via vSphere HA requires extending VLANs between racks. • Disaster recovery requires re-IP or extending networks across sites via the physical network.

NSX Federation Capabilities in the Different Network Consumption Models

NSX Federation is compatible with all the network consumption models, but the level of integration and the benefits it can provide vary across them. The table below summarize its capabilities in the context of each of the available network consumption models.

Table 25: NSX Federation Capabilities in the Different Network Consumption Models

Network Consumption Model	NSX Federation Capabilities	
	NSX Enterprise Administrator centrally manages security policy and groups across NSX instances	**Layer 2 and Layer 3 network topology extension across NSX instances**
VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model	Yes. Tenant or VPC scoped groups and security policies are local to a single NSX instance.	No
Virtual Private Cloud (VPC) Consumption Model	Yes. Tenant or VPC scoped groups and security policies are local to a single NSX instance.	No
NSX Segment Consumption Model	Yes	Yes
VLAN Networking Consumption Model	Yes	No

Workload Connectivity Models

In this topic we will introduce the different network connection types and will provide the associated *Workload Connectivity Models*.

Workload Connectivity Models

The *Workload Connectivity Models* define methods of providing layer 2/3 connectivity to workload virtual machines and network services. Each *Workload Connectivity Model* presents different capabilities regarding simplicity, performance, service availability, and scalability.

Each *Workload Connectivity Model* may be deployed and consumed via one or more *Network Consumption Models*.

Workload Connectivity Model	Supported Network Consumption Models	Attributes	Benefits	Implications
Centralized Connectivity Model	• VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model • Virtual Private Cloud (VPC) Consumption Model	• Requires NSX Edge nodes. • Active/active or active/standby Tier-0 gateway. • Active/active or active/standby Centralized Transit Gateway – Active/Standby required for VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model	• Support forVCF Automation All Apps Orgs and VKS. • Support for VPC network and stateful services i.e., Default Outbound NAT (Auto-SNAT), custom NAT, Gateway Firewall, External IPs (stateless 1:1 NAT) and load balancing using NSX Edge or Avi Load Balancer. • Support for DHCP relay for external DHCP servers. • Support for VPN. • Support for vCenter UI.	North-south throughput is determined by the number and type of NSX Edge nodes deployed.
Distributed VLAN Connectivity Model	• VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model • Virtual Private Cloud (VPC) Consumption Model	• Requires VNA nodes for stateful services. • North-south direct connectivity from the ESX host with the physical network fabric	• Support forVCF Automation All Apps Orgs and VKS. • Support for VPC network and stateful services i.e., Default Outbound NAT (Auto-SNAT), custom NAT, External IPs (stateless 1:1 NAT) and load balancing using VNA or Avi Load Balancer. • Support for DHCP relay for external DHCP servers (9.1.1). • Can achieve near host pNIC line rate for traffic not being redirected to the VNA cluster. • Support for vCenter UI. • Does not require NSX Edge nodes.	• No support for VPN. • No support for Multi-Rack Layer 3 vSphere Cluster Model.

Workload Connectivity Model	Supported Network Consumption Models	Attributes	Benefits	Implications
VLAN Backed VPC Connectivity Model	• VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model • Virtual Private Cloud (VPC) Consumption Model	• Requires VNA load balancer or Avi Load Balancer for load balancing capabilities. • North-south direct connectivity from the ESX host with the physical network fabric • No NSX Host TEPs required on the ESX hosts. • No NSX Overlay traffic required between ESX hosts.	• Support forVCF Automation All Apps Orgs and VKS. • With VCF Automation All Apps Orgs multiple VPC public subnets can share a single external VLAN connection. • Support for VPC public subnets and load balancing using VNA or Avi Load Balancer. • Can achieve near host pNIC line rate for traffic not being redirected to the VNA cluster. • Support for vCenter UI. • Does not require VNA nodes if no load balancer is required. • Does not require NSX Edge nodes.	• No NAT support • No support for VPN. • No support for Multi-Rack Layer 3 vSphere Cluster Model. • No support for Private VPC subnets. • When using vCenter UI or NSX UI every new VPC public subnet requires a new VLAN.
Distributed VXLAN Connectivity Model for Version 9.1.0	Virtual Private Cloud (VPC) Consumption Model	• Requires EVPN Route Controller node(s) to exchange MP-BGP -EVPN NLRI. • Requires a standard based EVPN-VXLAN network fabric.	• Integration with physical EVPN-VXLAN network fabric. • Support for VPC network services. • Support for External IPs (stateless 1:1 NAT). • Can achieve near host pNIC line rate for traffic. • Supports Multi-Rack Layer 3 vSphere Cluster Model. • Support for vCenter UI for consumption. • Does not require NSX Edge nodes.	• No support ofVCF Automation All Apps Orgs and VKS. • No support for stateful services including VPN. • No support for DHCP relay for external DHCP servers.

Workload Connectivity Model	Supported Network Consumption Models	Attributes	Benefits	Implications
Distributed VXLAN Connectivity Model for Version 9.1.1	• VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model • Virtual Private Cloud (VPC) Consumption Model	• Requires VNA nodes for stateful services. • Requires EVPN Route Controller node(s) to exchange Multiprotocol BGP-EVPN NLRI. • Requires a standard based EVPN-VXLAN network fabric.	• Integration with physical EVPN-VXLAN network fabric. • Support forVCF Automation All Apps Orgs and VKS. • Support for VPC network and stateful services i.e., Default Outbound NAT (Auto-SNAT), custom NAT, External IPs (stateless 1:1 NAT) and load balancing using VNA or Avi Load Balancer. • Support for DHCP relay for external DHCP servers. • Can achieve near host pNIC line rate for traffic not being redirected to the VNA cluster. • Support of Direct Path Optimizations. • Supports Multi-Rack Layer 3 vSphere Cluster Model. • Support for vCenter UI for consumption. • Does not require NSX Edge nodes.	• No support for VPN.
NSX Segment Connectivity Model	NSX Segment Consumption Model	• Requires NSX Edge nodes. • Active/active or active/standby Tier-0 gateway. • Tier-1 gateway provides network and stateful services. • Full NSX Federation support.	• Supports vSphere Supervisor and VKS. • Scalable north-south throughput. • Supports VM Mobility across NSX instances. • Supports single point of management for security policies across NSX instances. • Supports VPN. • Supports advanced security features on Tier-1 gateway. • Supports Multi-Rack Layer 3 vSphere Cluster Model. • Supports IPv6 workloads.	• No support ofVCF Automation All Apps Orgs. • No simplified VPC consumption. • North-south throughput is determined by the number and type of NSX Edge nodes deployed. • No support for vCenter UI consumption. NSX segments are presented as distributed virtual port-groups in vCenter. The VI Admin can attach VMs to them, but no CRUD operation is available for NSX networking object.

Workload Connectivity Model	Supported Network Consumption Models	Attributes	Benefits	Implications
VLAN Networking Connectivity Model	VLAN Networking Consumption Model	• Virtual machines connect to vCenter distributed port-groups or NSX VLAN segments. • Requires VLANs created on the physical switches and configured on the switch trunk ports connecting to ESX hosts. • Physical network fabric handles routing. • Network admin is heavily involved for the creation and lifecycle of networks. • More suited to a static network configuration.	• Familiar traditional networking model for vSphere admins. • Supports vSphere Supervisor and VKS. • Supports IPv6 workloads.	• Network changes require the reconfiguration of multiple devices. • Limited to 4094 VLANs in the layer 2 domain. • Network admin required to add VLANs on the physical switches for new networks. • Slow application deployment dependent on physical network reconfigurations • Single point of management for the entire logical network. The network can't completely embrace an Infrastructure as Code (IaC) approach, where network constructs follow the application lifecycle.

Load Balancer Models

VMware Cloud Foundation supports multiple *Load Balancer Models* that provide different levels of network services availability, performance, resilience, and scale. In the section, we cover the various models and the highlights of each model.

Load Balancer Models

VMware Cloud Foundation has multiple options to provide load-balancing capabilities for many types of workloads, including virtual machine, vSphere Supervisor based workloads, and the VCF management components including VCF Operations.

Table 26: Load Balancer Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Avi Load Balancer Model	• Avi Controllers act as the Control plane. • Avi Service Engines are in the datapath. • Avi Service Engine automatic scale and self healing. • Supports both Active/Active, Active/Standby, and N+M high availability models. • Single or three Avi controllers can be deployed. • Supports the following connectivity models for workloads: – Centralized Connectivity Model – Distributed VLAN Connectivity Model – Distributed VXLAN Connectivity Model for Version 9.1.1 – VLAN Backed VPC Connectivity Model – NSX Segment Connectivity Model – VLAN Networking Connectivity Model	• Layer 7 load-balancing support. • VCF Automation All Apps and VM Apps support. • Suitable for vSphere Supervisor and VKS-based workloads. • Supports VPCs • Required for general-purpose load balancing without Supervisor deployment. • Includes built-in Web Application Firewall (WAF) security that protects against web attacks for traditional workloads and VKS workloads. • Intelligent traffic management across multiple locations for enhanced disaster recovery.	Additional entitlement required.
Virtual Network Appliance Load Balancer Model	• Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services). • Virtual Network Appliance cluster required, with two Virtual Network Appliance nodes for high availability. • Active/Standby mode only. • The size of the Virtual Network Appliance VM determines the load balancer size and the number of load balancers. • Distributed Transit Gateway required. • Supports the following connectivity models for workloads: – Distributed VLAN Connectivity Model – Distributed VXLAN Connectivity Model for Version 9.1.1 – VLAN Backed VPC Connectivity Model	• Suitable for vSphere Supervisor and VKS-based workloads. • VCF Automation All Apps support. • Supports VPCs. • One can Virtual Network Appliancehold multiple load balancer instances for different VPCs.	• General-purpose load-balancing features and layer 7 application rules are not included in the VCF entitlement. • VPC load-balancer instance configuration using NSX API only. • Can only support VCF Operations load balancer if deployed on a VPC subnet. • Does not support the NSX Segment Connectivity Model

Model	Attributes	Benefits	Implications
NSX Edge VPC Load Balancer Model	• Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services). • NSX Edge cluster required. • Two NSX Edge nodes are deployed in an NSX Edge cluster to provide high availability. • Active/Standby mode only. • The size of the edge VM determines the load balancer size and number of load balancers. • Supports the Centralized Connectivity Model	• VCF Automation All Apps support. • Suitable for vSphere Supervisor and VKS-based workloads. • Supports VPCs.	• General-purpose load-balancing features and layer 7 application rules are not included in the VCF entitlement. • VPC load-balancer instance configuration using NSX API only. • Does not support the NSX Segment Connectivity Model
NSX Edge Tier-1 Load Balancer Model	• Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services) • NSX Edge cluster required. • Two NSX Edge nodes are deployed in an NSX Edge cluster to provide high availability. • Active/Standby mode only. • Tier-1 gateway required for load balancer when connecting to NSX segment or vlan networking. • Supports the following Connectivity Models: – NSX Segment Connectivity Model – VLAN Networking Connectivity Model	NSX UI can be used to set up Tier-1 load balancer for VCF Operations.	• General-purpose load-balancing features and layer 7 application rules are not included in the VCF entitlement. • Does not support VCF Automation All Apps. • Does not support VPCs
Foundation Load Balancer Model	• One or two foundational load balancer VMs. • Active/Standby mode only if two VMs deployed. • Can be deployed in One-arm or Two-arm modes. • vSphere Distributed Switch Port Groups required for connectivity. • Deployed from vCenter as part of vSphere Supervisor activation. • Supports VLAN Networking Connectivity Model	Supports vSphere Supervisor and VKS workloads with vSphere Distributed Switch networking option.	• Not supported for VCF Automation All Apps. • Only supports Layer 4 services. • Not supported with VPC Networking. • Not supported with NSX segment networking. • In a multi-zone Supervisor, Foundation Load Balancer VMs are only placed on the first Management zone.
External Load Balancer Model	• Third-party solution. • Third-party qualification needed.	Existing load balancer infrastructure could be reused.	• Extra hardware may be needed. • Third-party support needed.

VCF Management Services Models

VMware Cloud Foundation supports multiple *VCF Management Services Models* that provide different levels of availability for VCF management components.

VCF Management Services Component Models

VCF 9.1 introduces VCF management services to provide a unified architecture for centralized lifecycle and operational functionalities. The VCF services runtime instance of the first VCF Instance hosts the fleet-level VCF management services components that perform global operations and the instance-level components for that VCF Instance. Every VCF Instance has a VCF services runtime instance that hosts the instance-level VCF management services that run local tasks. This foundation allows for a unified architecture, simplified lifecycle management, and streamlined backup and recovery of all hosted components.

VMware Cloud Foundation supports multiple *VCF Management Services Models* that provide different levels of availability.

When deploying a new VCF Instance, you choose a component model and a deployment model for VCF management services based on the profiles and needs of that instance.

Table 27: VCF Management Services Component Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
VCF Management Services in the First VCF Instance Model	Runs the following fleet level components: • Fleet lifecycle • Salt RaaS • Log management Runs the following VCF Instance level components: • SDDC lifecycle • Telemetry • Salt master • Identity broker • Real-time metrics • Software depot	Central instance for running VCF fleet and VCF Instance level components.	Deploying the log managementand real-time metricscomponents is a day-N operation and requires adding more worker nodes.
VCF Management Services in an Additional VCF Instance Model	Runs the following VCF Instance level components: • SDDC lifecycle • Telemetry • Salt master • Identity broker • Real-time metrics • Software depot	Smaller VCF management services instance for running VCFInstance level components only.	Deploying the real-time metrics,identity broker, and Software depot components are a day-N operation and requires adding more worker nodes.

Figure 21: VCF Management Services Models in a VCF fleet

The diagram below shows the components for each model. The dotted line components are deployed as a day-N operation.

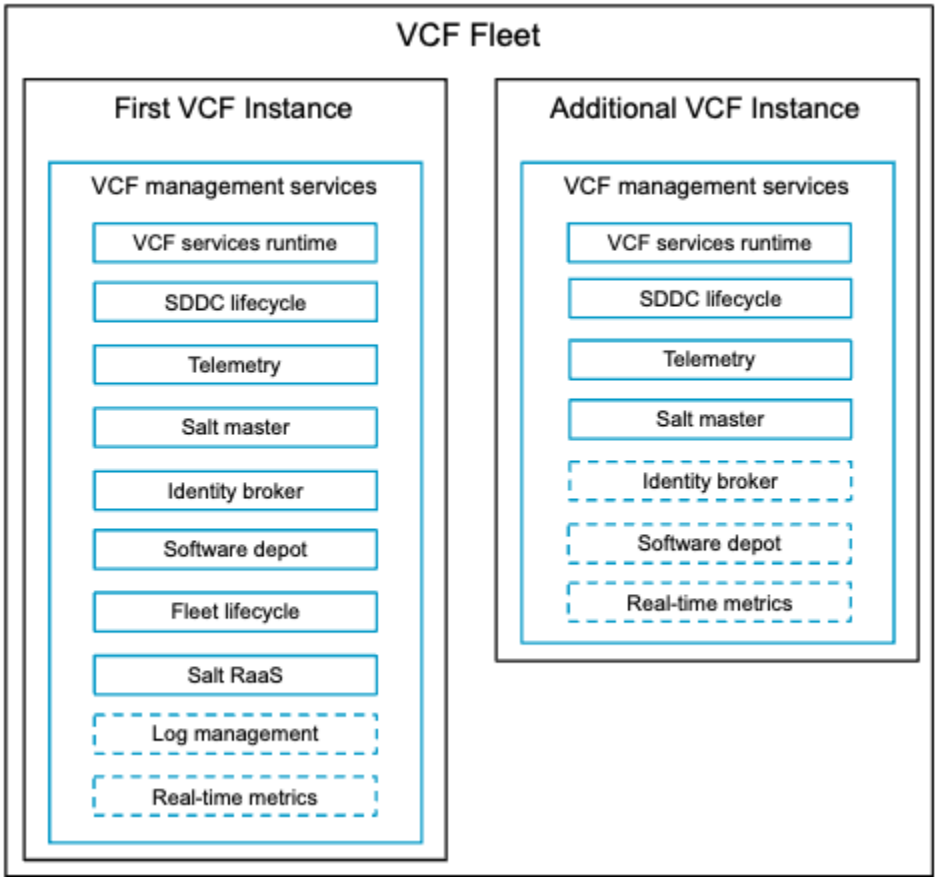
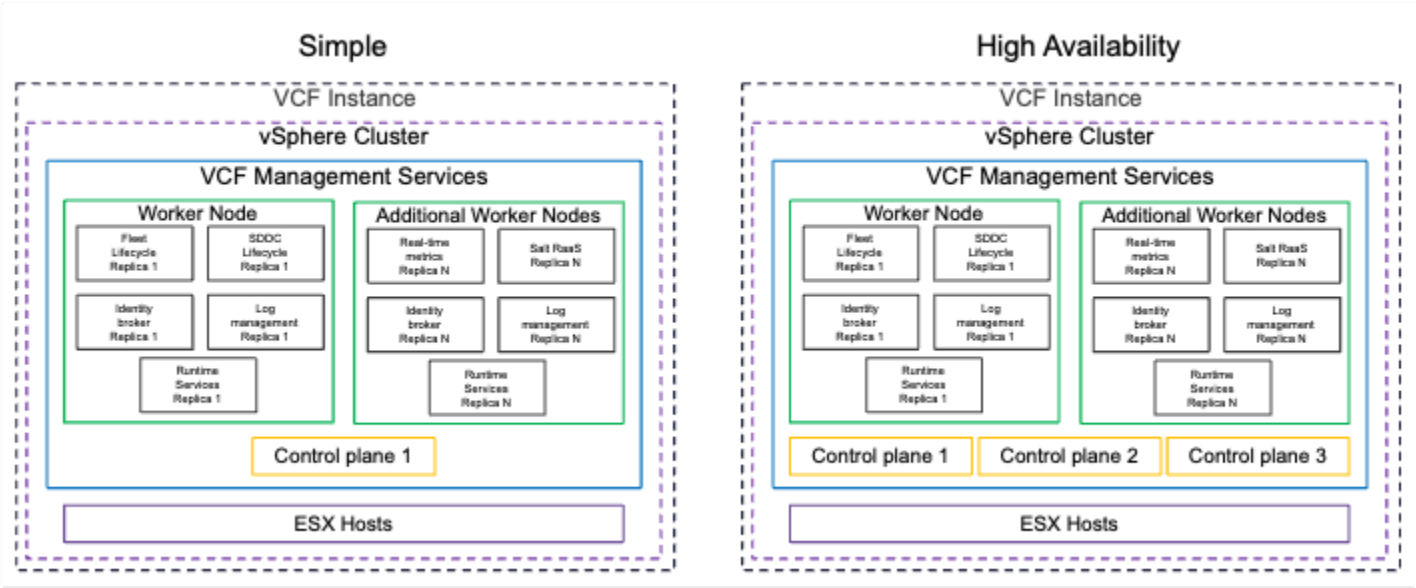


Table 28: VCF Management Services Deployment Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Simple VCF Management Services Model	• Single control plane node. • Three worker nodes.	Smaller footprint for testing and proof of concept deployments.	Loss of the single control plane node would require redeploying the cluster and restoring from backup.
High Availability VCF Management Services Model	• Three control plane nodes. • Worker node specification and count depend on the chosen deployment size.	High availability for the control plane and worker nodes.	Requires additional resources.

Figure 22: VCF Management Services Deployment Models for a VCF Instance



VCF Management Network Models

VMware Cloud Foundation provides flexibility for the deployment of the VCF management components including (VCF management services, VCF Operations, and VCF Automation) to different network types. Either vSphere distributed port groups or NSX segments can be used.

Using VCF Installer by default the VCF management components are deployed to the **VM management network**, the same network that is used to deploy vCenter, NSX Manager, and SDDC Manager. Or they can be deployed to a dedicated distributed port group on a dedicated VLAN, known as the **VCF management network**.

Alternatively, custom networking can be selected which defers the deployment of VCF Operations and VCF Automation so that an NSX segment or different vSphere distributed port group can be used. You must configure these networks after the initial VCF Installer deployment is complete and before deploying the deferred VCF management components with VCF Installer.

> **Note:**
>
> The VCF management services must always be deployed to a vSphere distributed port group. The full list of VCF management services are described here VCF Management Services Models

NSX segments are required or if there are plans to stretch the network between VCF Instances using NSX Federation. This might be the case if disaster recovery is implemented for the VCF management components.

> **Note:**
>
> The cloud proxy is a VCF Instance level management component and is always deployed to the **VM management network** by VCF Installer.

Table 29: VCF Management Network Models

Model	Attributes	Benefits	Implications
VCF Management Shared VLAN Network Model	VCF management components are connected to the **VM management network** a vSphere distributed port group also used by vCenter, NSX Manager, and SDDC Manager.	• Deployed using VCF Installer during initial deployment. • Simple deployment model done day 0 • Single subnet and VLAN	No logical isolation of VCF fleet level components and VCF Instance components.
VCF Management Dedicated VLAN Network Model	VCF management components are connected to the **VCF management network** a dedicated vSphere distributed port group. VCF Operations cloud proxy is still connected to the default **VM management network**.	Provides logical isolation from other networks, can be used to have separate security for the VCF management components while using a VLAN backed vSphere distributed port group.	• An additional VLAN is required for a separate network. • Stretched cluster deployments require additional VLAN to be stretched between sites.
VCF Management Dedicated VLAN and NSX Overlay Segment Network Model	VCF management components are connected to an NSX overlay segment and a vSphere distributed port group. The.VCF management services are deployed day 0 and connect to the vSphere distributed port group. NSX Overlay supported for: • VCF Operations • VCF Automation • License Server • VCF Operations for networks	• Provides isolation from other networks, can be used to have separate security for the VCF management components while using an NSX overlay segment. • The NSX overlay segment can be used to connect an NSX Tier-1 one-arm load balancer for VCF Operations.	• Additional steps are required to deploy NSX networking. • A Day N operation is required to deploy deferred components using VCF Installer. • Additional steps required to deploy an NSX Edge cluster. • BGP routing required for NSX Edge cluster Tier-0 gateway if NSX Federation is used to stretch the network.
VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model	VCF management components are connected to an NSX stretched overlay segment and a vSphere distributed port group. The.VCF management services are deployed day 0 and connect to the vSphere distributed port group. NSX stretched overlay is only supported for: • VCF Operations • VCF Automation • License Server • VCF Operations for networks	• NSX Overlay segment provides IP mobility for the VCF management components connected to the Overlay. • For Backup and Restore of VCF management services in the case of disaster recovery a new network can be used at the recovery site without having to stretch the VLAN between sites. • The NSX stretched overlay segment can be used to connect an NSX Tier-1 one-arm load balancer for VCF Operations.	• NSX Federation Global Mangers must be deployed as Active/Standby. • Additional steps are required to deploy NSX networking. • A Day N operation is required to deploy deferred components using VCF Installer • Additional steps required to deploy an NSX Edge cluster. • BGP routing required for NSX Edge cluster T0 gateway.

VCF Operations Models

VMware Cloud Foundation utilizes VCF Operations as the central console for operating your VCF fleet. It provided dashboards, log analysis, issues reporting, and licensing. VCF Operations also provides fleet management capabilities such as fleet extension, lifecycle, certificate, and password management.

VCF Operations Models

Use the VCF Installer to deploy your new VCF fleet, which includes vCenter, NSX, VCF Operations, VCF Automation, and SDDC Manager. At deployment time you must select one of the *VCF Operations Models* based on your organizational objectives.

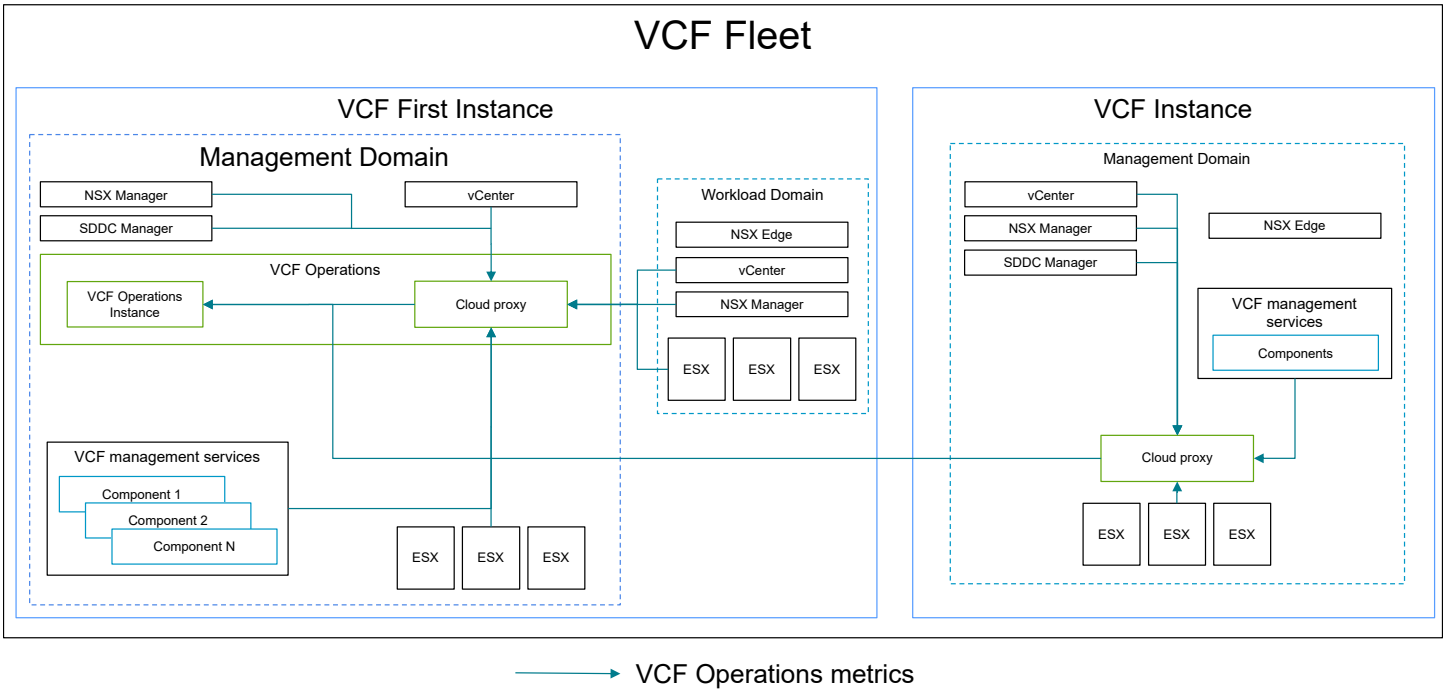
vCenter and SDDC Manager are always deployed as single node. You have the flexibility to choose whether to deploy NSX Manager, VCF Operations, VCF Automation, and VCF Management Services as single node deployments, or as three node deployments to provide high availability capabilities.

Table 30: VCF Operations Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Simple VCF Operations Model	• Single node forVCF Operations. • Additional appliances deployed: — License server — Cloud proxy — SDDC Manager • Supports scale-up and scale-out.	• Smallest footprint. • Can be scaled-out to be highly available.	• Slower recovery from failure. • Relies on vSphere HA to restart the nodes in the event of an ESX host failure. • Service interruption may occur for: — monitoring and alerting — fleet management
High Availability VCF Operations Model	• Three node cluster for VCF Operations: — Primary node — Replica node — Data node • Provides greater service availability. • Additional appliances deployed: — License server — Cloud proxy — SDDC Manager • Supports an optional external load balancer. • Supports scale-up for all nodes and scale-out with additional data nodes.	• Highly available cluster with data redundancy. • Rapid recovery of a single node failure in the cluster.	• Cloud proxys expanded to a collector group manually. • When using an external load balancer: — Additional IP address and FQDN are required. — FQDNs for all cluster nodes and the FQDN for the load balancer must be included in the certificate's SAN.

Model	Attributes	Benefits	Implications
Continuous Availability VCF Operations Model	• A cluster with a pair of nodes, across two availability zones: — Primary node <-> Primary replica node — Data node <-> Data Node • Provides best service availability. • Additional appliance, deployed in the primary availability zone: — License server • Additional appliances, deployed in both availability zones: — Cloud proxy — SDDC Manager • Supports an optional external load balancer. • Supports scale-up for all nodes and scale-out with additional data nodes.	• Highly available cluster with data redundancy across different availability zones. • No service interruption in case of a single availability zone failure.	• Manual operations after the initial installation • Two availability zones that hosts the replica appliances and a third availability zone for the witness. • It is recommended to have a separate VCF Instance in the second availability zone that could be used as a starting point for new "primary management domain". • Maximum network latency between availability zones 10ms.

Figure 23: VCF Operations and Real-Time Metrics in VMware Cloud Foundation



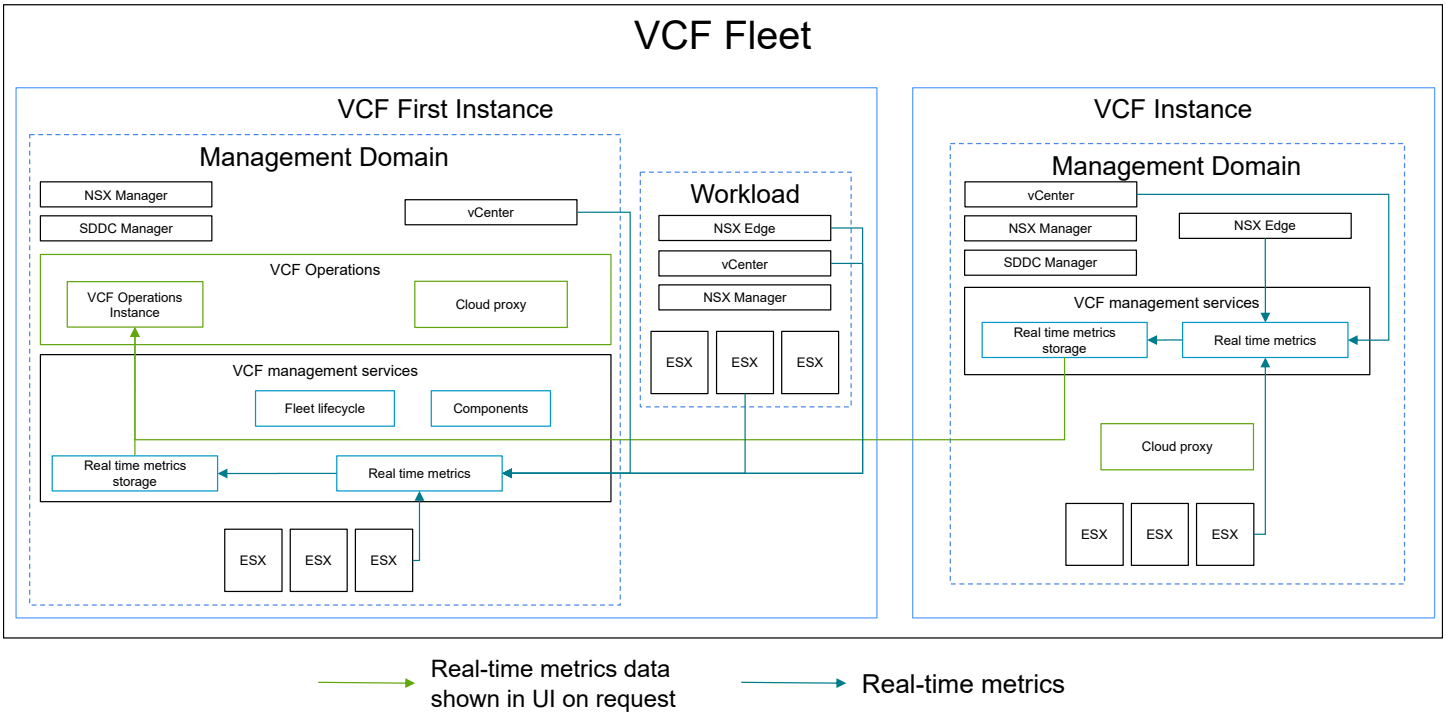
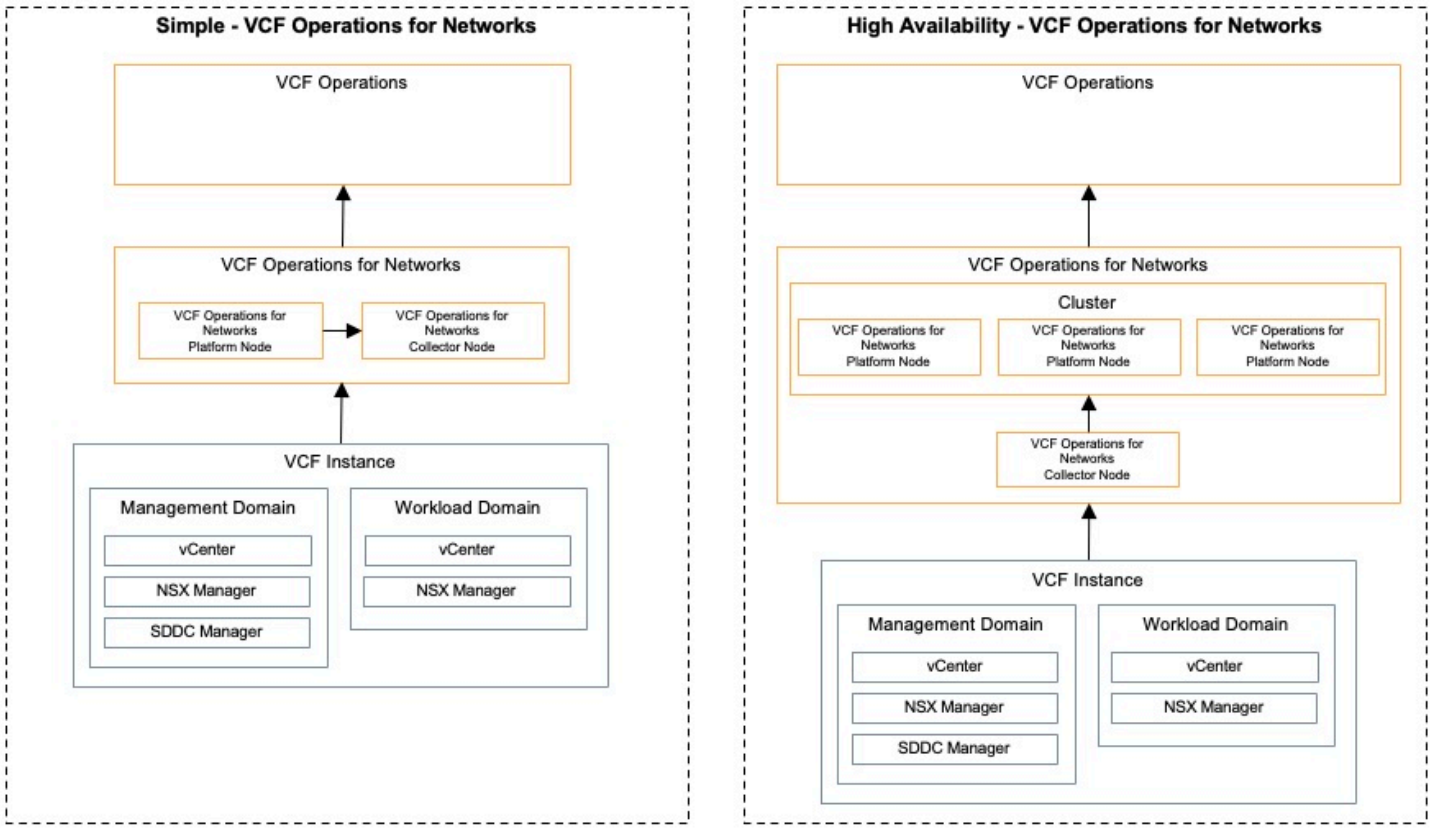

VCF Fleet
VCF First Instance
Management Domain
NSX Manager
SDDC Manager
vCenter
VCF Operations
VCF Operations Instance
Cloud proxy
VCF management services
Fleet lifecycle
Components
Real time metrics storage
Real time metrics
ESX
ESX
ESX
Workload
NSX Edge
vCenter
NSX Manager
ESX
ESX
ESX
VCF Instance
Management Domain
vCenter
NSX Manager
SDDC Manager
NSX Edge
VCF management services
Real time metrics storage
Real time metrics
Cloud proxy
ESX
ESX
ESX
Real-time metrics data shown in UI on request
Real-time metrics

Figure 24: VCF Operations for Networks Models in VMware Cloud Foundation



Log Management Models

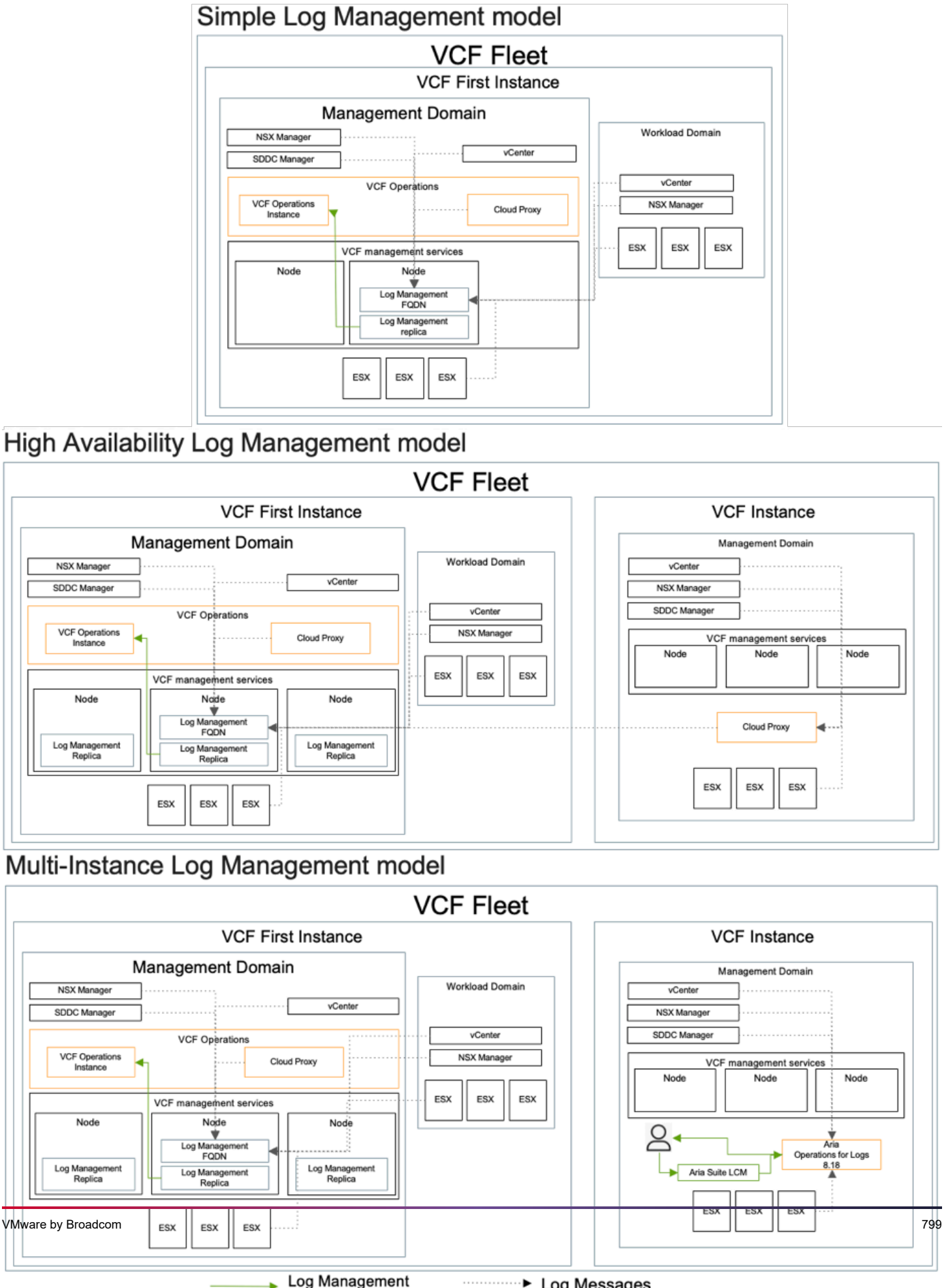
Log management can be added to your VCF fleet to extend the capabilities of VCF Operations by providing a centralized log management framework to collect, analyze, and visualize logs from across your VMware Cloud Foundation platform.

You deploy log management as a Day-2 operation using VCF Operations, see Deploy Log Management. At deployment time you must select one of the *Log Management Models* based on your organizational objectives.

Table 31: Log Management Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Simple Log Management Model	• Single replica. • Supports scale-up and scale out.	• Smallest footprint. • Can be extended to be highly available.	• Failures have a more significant impact. • Relies on VCF management services and vSphere HA to restart nodes in the event of an ESX host or log management component failure. • No application high availability. • Deployed by usingVCF Operations as a Day-2 operation. • VIP IP address and FQDN are required during deployment. You can add VIP IP addresses and FQDNs later.
High Availability Log Management Model	• Three or more replica clusters. • Additional options on deployment: — Number of replicas (3-19) • Supports scale-up and scale out.	• Distributed cluster with integrated load balancer. • Full to partial service maintained in the event of a node failure.	• Deployed by using VCF Operations as a Day-2 operation. • VIP IP address and FQDN are required during deployment. You can add VIP IP addresses and FQDNs later.
Multi-Instance Log Management Model	• Individually managed and deployed instances, each scaled independently • Each instance supports one of the other deployment models (simple or High Availability) • One instance integrated with VCF Operations, the others operate standalone, running Aria Operations for Logs 8.18.	• Supports large-scale environments • Supports full isolation between instances for: — Remote sites with limited bandwidth to the VCF fleet. — Organizationally separate environments.	• Deployed through VCF Operations as a Day-2 operation for primary instance, and manually for secondary instances. • VIP IP address and FQDN are required during deployment. You can add VIP IP addresses and FQDNs later.

Figure 25: Log Management Models in VMware Cloud Foundation



VCF Operations for Networks Models

VCF Operations for Networks Models

VCF Operations for networks extends the capabilities of VCF Operations providing network visibility and analytics to minimize risk during application migration, optimize network performance, manage, and scale your network across the VMware Cloud Foundation platform.

You deploy VCF Operations for networks as a Day-2 operation using VCF Operations, see Deploying VCF Operations for networks. At deployment time you must select one of the *VCF Operations for Networks Models* based on your organizational objectives.

Note:

For the detailed design relating to *VCF Operations for Networks Models* in the table below see VCF Operations for Networks Detailed Design.

Table 32: VCF Operations for Networks Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Simple VCF Operations for Networks Model	• Single nodes for — VCF Operations for networks platform node — VCF Operations for networks collector node • Supports scale-up and scale out.	• Smallest footprint • Can be scaled-out to a platform cluster. • Supports 10,000 virtual machines or objects.	• Slower recovery from failure. • Relies on vSphere HA to restart the nodes in the event of an ESX host failure. • No application high availability. • Requires a CPU and RAM reservation of 100% of the CPU and RAM assigned. • Requires the single node be deployed as an Extra Large size. • Deployed through VCF Operations as a Day 2 operation.
High Availability VCF Operations for Networks Model	• Three node cluster. • Single VCF Operations for networkscollector node. • Provides greater application scale and performance. • Supports scale-up and scale out.	• Highly performant • Highly scalable	• Cluster does not provide high availability. It is for performance. • Requires higher CPU and memory reservations to support additional nodes. • Deployed through VCF Operations as a Day 2 operation.

VCF Recovery Options

VMware Cloud Foundation supports multiple *VCF Recovery Options* depending on the scenario.

VCF Recovery Options

Table 33: Recovery Options in VMware Cloud Foundation

Option	Attributes	Benefits	Implications
VCF Component Backup and Restore	• Backup and restore at an individual component level. • Applicable when individual VCF components fail and must be restored from backup.	Supports recovery of individual components without the need to recover the entire VCF Instance.	• Requires an SFTP server. • Requires a third-party image backup tool.
VCF Instance Backup and Restore	• Full VCF Instance level recovery process. • Applicable when an entire VCF Instance has failed and must be recovered in the same location. • Accounts for sequencing and dependencies.	• Supports full VCF Instance restoration. • Satisfies regulatory requirements such as the Digital Operational Resilience Act (DORA). • Powershell automation to assist in the recovery process.	Recovery replicates the identity of the source environment exactly (VLANs, IPs, and hostnames). Test restorations require an isolated lab environment to prevent IP conflicts with the production source.
VCF Fleet Disaster Recovery	• Recovery of fleet level components to another VCF Instance or physical site. • Applicable when an entire VCF Instance has failed and must be recovered in another location.	Supports restoration of key VCF management components across two physical sites. • VCF Operations • Fleet lifecycle • Salt RaaS • Log management • VCF Automation	• Requires more than one VCF Instance and physical site. • IP mobility is required between the sites.
Cyber Recovery for VMware Cloud Foundation	Applicable when an entire or parts of a VCF Instance are affected by a ransomware attack and must be recovered in another location.	Supports seamless recovery of on-premises business-critical workloads after a cyber event by using an isolated clean room with push-button network isolation and integrated Endpoint Detection and Response (EDR).	Requires more than one VCF Instance to host the isolated clean room.

Identity Broker Models

VMware Cloud Foundation provides identity and access management services to VCF fleet components. You build your authentication design by using three main components - your external Identity Provider (IdP), identity broker, and VCF Single Sign-On.

Your corporate identity provider is the centralized source of digital identities for your enterprise, which provides authentication and identity assertion. You connect the corporate identity provider to your VCF fleet components by using identity broker. Identity broker enables a VCF Single Sign-On implementation and streamlines authentication across different VCF fleet components, except for SDDC Manager. The third component of this design is VCF Single Sign-On, which provides centralized authentication, identity federation, and user management.

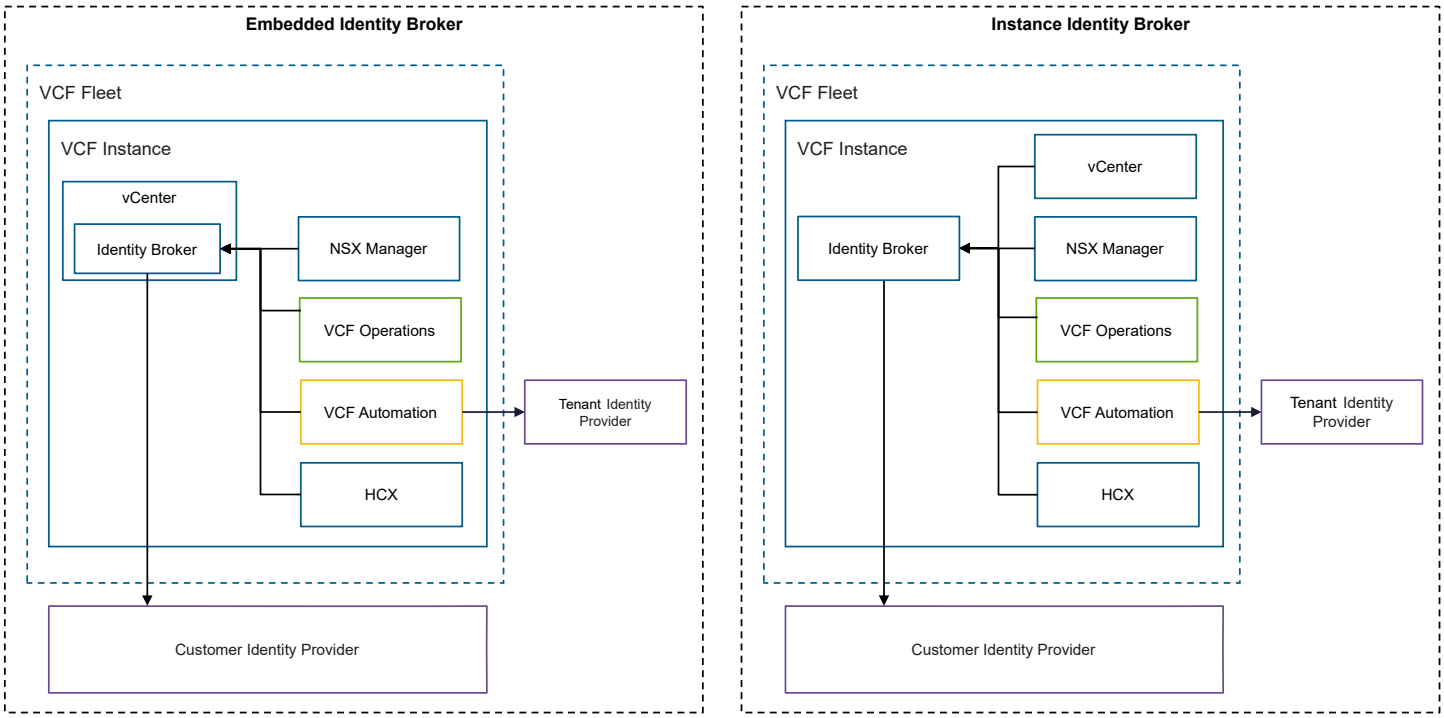
Identity Broker Models

The identity broker component acts as the single broker for VCF Single Sign-On that all components in a VMware Cloud Foundation platform can authenticate to. It connects to your identity provider or directory, maintains Single Sign-On tokens and validates users as they log into VCF fleet components. Identity broker is consumed as an embedded service, running on vCenter, or deployed as an instance. Instance model is the recommended deployment for production environments.

Table 34: Identity Broker Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Embedded Identity Broker Model	• Service deployed as part of vCenter. • Resiliency is provided through vSphere HA.	No additional appliance required.	Limited scale and availability supported by embedded model.
Instance Identity Broker Model	• Deployed as a component on VCF management services. • Resiliency is provided through a multi-node deployment.	Ensures availability in the event of ESX host failure events.	Additional resource footprint.

Figure 26: Identity Broker Models for VMware Cloud Foundation



VCF Single Sign-On Models

To accommodate multiple VCF Instances as your VMware Cloud Foundation platform grows, you make design decisions regarding the placement and size of the identity broker component. These decisions impact the scale, availability, blast radius, and span of your VCF Single Sign-On environment.

Table 35: VCF Single Sign-On Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
VCF Fleet-Wide Single Sign-On Model	Single identity broker instance per VCF fleet.	Provides a large Single Sign-On realm that is across all components in a VCF fleet.	Offers the lowest resource footprint but results in a large blast radius and reduced availability.
Cross VCF Instance Single Sign-On Model	Single identity broker instance for multiple VCF instances in VCF fleet.	Balances scale, blast radius, and availability by extending SSO realm across a subset of VCF instances.	Achieves a balanced resource footprint by grouping multiple instances under a shared Single Sign-On realm.
Single VCF Instance Single Sign-On Model	Single identity broker instance per VCF Instance.	Limits the impact of an authentication failure to a single VCF instance.	Provides high availability and scale at the cost of a large resource footprint and limited Single Sign-On realm.

VCF Domain Models

When deploying a VCF Instance within a VCF fleet, you determine the *VCF Domain Models* according to your objectives for availability, networking, and security.

A VCF domain represents a logical unit of application-ready infrastructure that groups ESX hosts managed by a vCenter instance with specific characteristics according to VMware recommended practices. A VCF domain can have one or more *vSphere Cluster Models*.

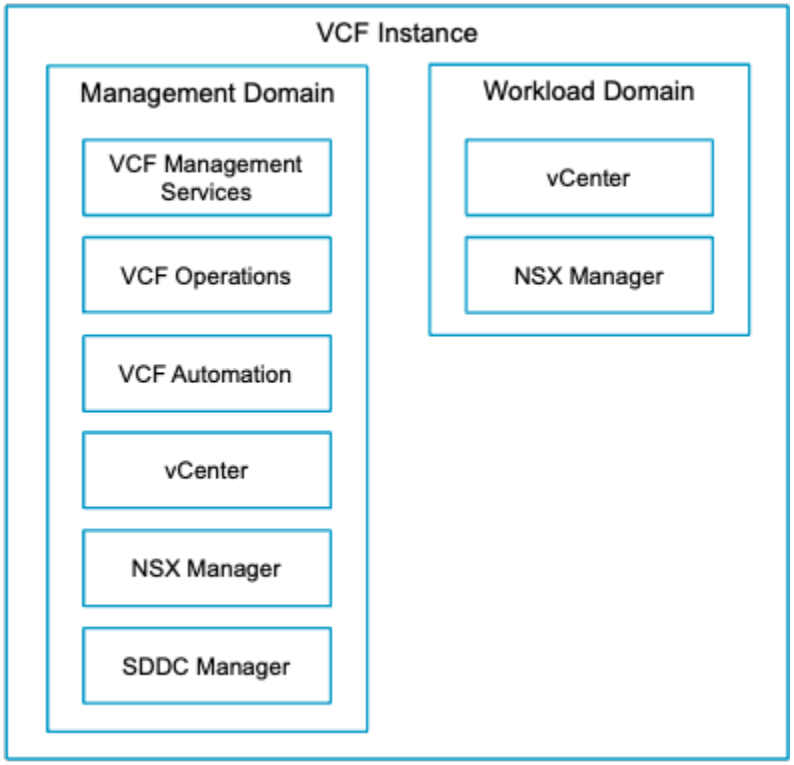
Note:

VCF Instances that are upgraded from an earlier version may have shared vCenter Single Sign-On domains.

Table 36: VCF Domain Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Management Domain Model	• First workload domain deployed in a VCF Instance. • Contains the following management components for all workload domains: – vCenter – NSX Manager – SDDC Manager • The first management domain contains the VCF fleet management components and instance level components. • Additional management domains contain instance level components only. • Can also run business workloads. • Can contain NSX Edgenodes.	• Enables the use of specific hardware to meet the requirements of the management components. • Enables the use of dedicated physical compute, network, and storage, separate from resources for additional workloads. • Enables separate lifecycle management of management and workload components. • Can share an NSXinstance with other workload domains.	You may need to expand the resources in the management domain to accommodate future deployments of additional workload domains and management components.
Workload Domain Model	• Represents an additional dedicated set of resources for running customer workloads using dedicated ESX hosts. • Has a dedicated vCenter and vCenter Single Sign-On domain. • Provides support for configurations with a dedicated identity provider.	• Can share an NSXinstance with other VCF domains. • Can share NSX Edge clusters when sharing NSX instances. • Scales up to 40 workload domains per VCF Instance. • Enables independent lifecycle management.	Requires additional hardware for running business workloads.

Figure 27: VCF Domain Models in VMware Cloud Foundation



Comparison between dedicated NSX Instance per Workload Domain and Shared NSX Instance across Workload Domains

Attribute	Dedicated NSX Instance per Workload Domain	Shared NSX Instance across Workload Domains
Footprint	Higher footprint as each workload domain requires a dedicated NSX Manager cluster.	Reduced footprint as a single NSX Manager cluster is shared across workload domains.
Availability	Independent NSX management and control plane per workload domain increases overall availability.	A single NSX management and control plane shared across workload domains increases the blast radius in the event of a failure.
Scalability	Each workload domain can be scaled independently based on its own requirements.	Scalability can be constrained as the NSX instance is shared across workload domains.
Lifecycle	Upgrades and patching can be performed independently per workload domain.	Upgrades and patching impact all included workload domains.

vSphere Cluster Models

VMware Cloud Foundation supports multiple *vSphere Cluster Models* that provide different levels of availability to the workloads running in a workload domain.

vSphere Cluster Models

A *vSphere Cluster Model* is defined and implemented per vSphere cluster and describes the maximum level of availability that can be provided to virtual machines on that vSphere cluster. Each workload domain or VCF Instance can contain a combination of the defined *vSphere Cluster Models*. You can also select to deploy a single *vSphere Cluster Model* in a workload domain or VCF Instance according to your availability objectives.

If a vSphere cluster is part of a workload domain with vSphere Supervisor, the vSphere cluster is known as a vSphere zone that can be tagged as a vSphere Supervisor management and workload zone.

Table 37: vSphere Cluster Models in VMware Cloud Foundation

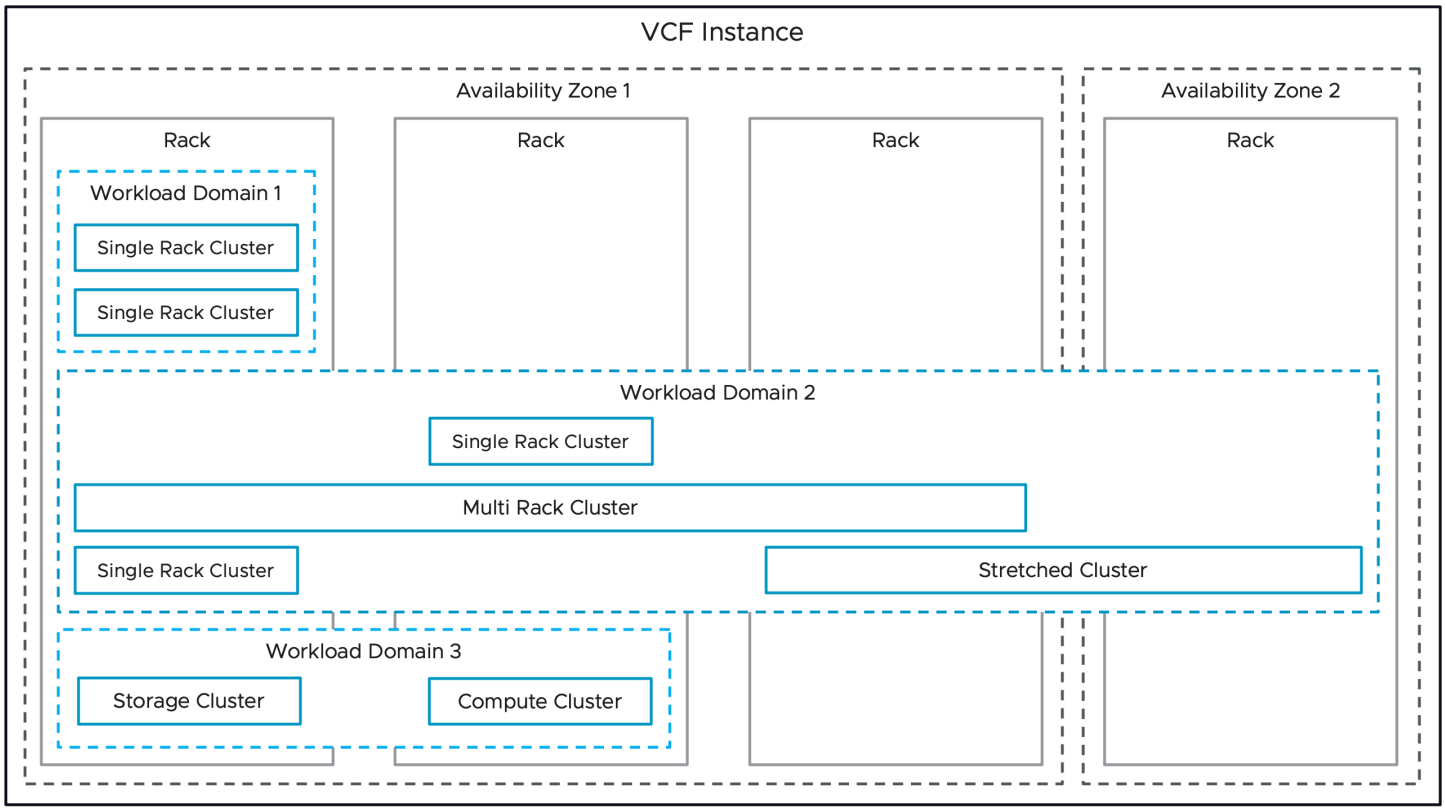
Model	Attributes	Benefits	Implications
Single-Rack vSphere Cluster Model	• Single storage fault domain. • Supports Layer 2 networking between ESX hosts. • Protects VMs using vSphere HA. • When using vSAN, can be a: — HCI Cluster. ** — Storage Custer. — Compute Cluster.	• Supports a minimal footprint. • Support for recovery after an ESX host failure events. • Supported in all VCFdomains.	No protection against a rack failure.
Multi-Rack Layer 2 vSphere Cluster Model	• Multiple storage fault domains. • Supports Layer 2 networking between ESX hosts. • Protects VMs using vSphere HA. • When using vSAN, can be a: — HCI Cluster. ** — Storage Cluster. — Compute Cluster.	Support for recovery after ESX host or physical rack failure events.	• Requires multiple physical racks. • Management domain default cluster supported.
Multi-Rack Layer 3 vSphere Cluster Model	• Multiple storage fault domains. • Supports Layer 3 networking between ESX hosts. • Protects VMs using vSphere HA. • When using vSAN, can be a: — HCI Cluster. ** — Compute Cluster.	• Support for recovery after ESX host or physical rack failure events. • Provides isolated layer 2 network broadcast domains per rack.	• Requires multiple physical racks. • More complex networking requirements with extra subnets and VLANs per rack. • Management domain default cluster not supported.
Stretched vSphere Cluster Model	• Dual storage fault domains. • Protects VMs using vSphere HA. • Can stretch single-rack vSphere clusters.*	Support for recovery after host and availability zone failure events.	Layer 3 multi-rack vSphere clusters cannot be stretched.

Important:

* If the cluster uses vSAN Storage Client traffic separation, you cannot stretch a single-rack Storage Cluster.

** An HCI Cluster refers to a Storage Model that uses vSAN ESA or vSAN OSA architectures.

Figure 28: vSphere Cluster Models in VMware Cloud Foundation



Tip:

You can implement disaster recovery between VCF Instances on any of the above *vSphere Cluster Models* by following the VCF Fleet Disaster Recovery Protection Blueprint.

Distributed Switch Models

VMware Cloud Foundation supports multiple *Distributed Switch Models* that provide different levels of network availability, performance, resilience, and scale.

Distributed Switch Models

VMware Cloud Foundation uses vSphere Distributed Switches to provide many network services, infrastructure networks, virtual networking, etc. A vSphere Distributed Switch acts as a single switch across all associated ESX hosts in a vSphere cluster to provide centralized provisioning, administration, and monitoring of networks.

vSphere Distributed Switches provide several services to ESX hosts and virtual machines. You can enable two types of network services in ESX:

- Connecting virtual machines to the physical network and to each other.
- Connecting VMkernel services (such as management, vSphere vMotion, vSAN, and NSX host TEP) to the physical network.

When you design vSphere cluster networking, consider the configuration of the vSphere Distributed Switches, distributed port groups, and VMkernel adapters in the VMware Cloud Foundation platform. VMware Cloud Foundation provides a flexible approach to vSphere Distributed Switch deployment options, the *Distributed Switch Models* are chosen based on the design qualities, availability, manageability, performance, recoverability and security. The following table shows some of the most common *Distributed Switch Models*, each with its benefits and implications listed.

Note:

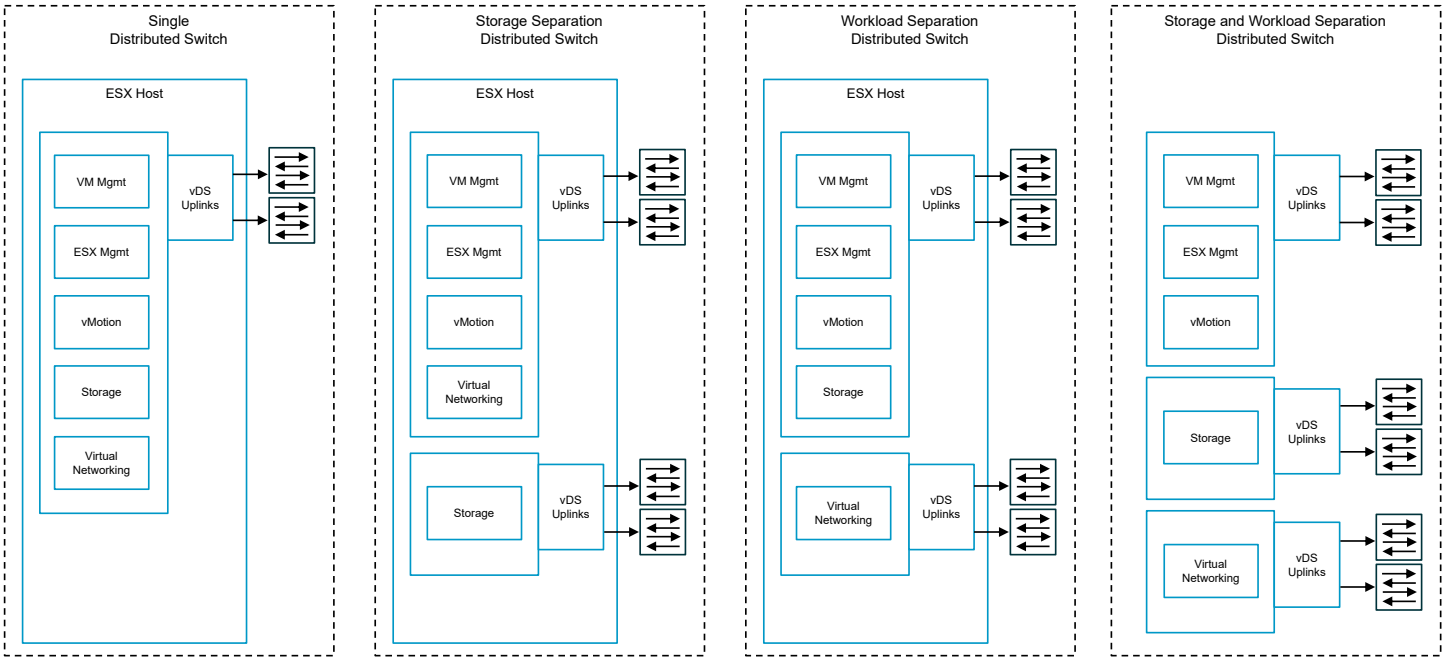
Starting with VCF 9.1.1 for greenfield deployments it is now supported to share a vSphere Distributed Switch between clusters in a VCF domain.

Table 38: Distributed Switch Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Single Distributed Switch Model	Option for minimal deployment.	• Easier to manage with least number of vSphere Distributed Switches. • Can be deployed with least number of ESX host physical NICs (one ESX host physical NIC using API). • Lowest physical switch port density required.	Does not provide network traffic separation to another physical network fabric.
Storage Separation Distributed Switch Model	Option for larger-scale deployments where bandwidth and quality of service is a concern, especially storage traffic bandwidth requirement.	• Separates Storage traffic onto dedicated physical ESX host NICs and provides the option to isolate the storage traffic to a different network fabric. • In secure environments, you can provide physical air-gapped separation of network traffic. • Can provide maximum bandwidth and performance for storage traffic with no contention from other traffic.	• Additional vSphere Distributed Switches increase management overhead. • Can be a more complex network configuration. • Higher number of ESX host physical NICs required than using single vSphere Distributed Switch with two physical NICs. • Higher physical switch port density required.
Workload Separation Distributed Switch Model	Option for larger-scale deployments where bandwidth and quality of service is a concern, especially for workload traffic.	• Separates virtual networking traffic onto different physical ESX host NICs and provides the option to isolate the workload traffic to a different network fabric. • In secure environments, you can provide physical air-gapped separation of network traffic. • Can provide maximum bandwidth and performance for workload traffic with no contention from other traffic on the ESX hosts physical NICs.	• Additional vSphere Distributed Switches increase management overhead. • Can be a more complex network configuration. • Higher number of ESX host physical NICs required than using single vSphere Distributed Switch with two physical NICs. • Higher physical switch port density required.

Model	Attributes	Benefits	Implications
Storage and Workload Separation Distributed Switch Model	Option for larger-scale deployments where isolation to different fabrics is a requirement, also offers quality-of-service benefits.	• Separates virtual networking and storage traffic onto different physical ESX host NICs and provides the option to isolate the storage and virtual networking traffic to different network fabrics. • In secure environments, you can provide physical air-gapped separation of network traffic • Can provide maximum bandwidth and performance for virtual networking traffic and Storage traffic with no contention from other traffic on the ESX hosts physical NICs.	• Additional vSphere Distributed Switches increase management overhead. • Most complex network configuration. • Highest number of ESX host physical NICs required, minimum of six (6) with redundancy. • Highest switch port density required.
Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model	Option for the deployment of NSX Edge nodes to vSphere clusters dedicated to only host edges, this design can achieve maximum bandwidth and performance for the edge VMs.	• Provides maximum bandwidth for NSX Edge node traffic. • NSX Edge utilizes all physical NICs. • Built on workload traffic separation model for simpler deployment.	• Network I/O control on both vSphere Distributed Switches must be disabled manually post deployment for best performance. • Additional vSphere Distributed Switches required increasing management overhead.
vSAN Storage Cluster Separation Distributed Switch Model	Option for vSAN Storage Clusters where vSAN traffic for the cluster is separated from client vSAN traffic flowing to the compute clusters.	• Provides maximum bandwidth for vSAN storage cluster server traffic. • Isolates client and server traffic onto two different network fabrics if necessary.	• Two (2) vSphere Distributed Switches increase management overhead • Higher physical switch port density required.

Figure 29: Distributed Switch Models in VMware Cloud Foundation



Storage Models

VMware Cloud Foundation supports multiple *Storage Models* that provide different levels of availability to workloads running within a VCFdomain.

Principal and Supplemental Storage

- Principal storage is used during the creation of a VCF domain or a vSphere cluster and is capable of running workloads.
- Supplemental storage can be added after the creation of a VCF domain or a vSphere cluster and can run workloads or be used for data at rest storage (such as virtual machine templates, backup data, and ISO images).

Table 39: Principal and Supplemental Storage in VMware Cloud Foundation

Storage Model	Management Domain - Default Cluster	Management Domain - Additional Clusters	Workload Domain
Single-Rack vSAN ESA Storage Model	Principal	Principal	Principal
Single-Rack vSAN OSA Storage Model	Principal	Principal	Principal
Single-Rack vSAN Storage Cluster Storage Model	Non Supported	Principal	Principal
Single-Rack vSAN Compute Cluster Storage Model	Non Supported	Principal	Principal
Multi-Rack vSAN Storage Model	Non Supported	Principal	Principal
Multi-Rack vSAN Compute Cluster Storage Model	Non Supported	Principal	Principal
Fibre Channel Storage Model	• Principal • Supplemental	• Principal • Supplemental	• Principal • Supplemental
NFS Storage Model (NFSv3)	• Principal • Supplemental	• Principal • Supplemental	• Principal • Supplemental
NFSv4.1	• Principal* • Supplemental	• Principal* • Supplemental	• Principal* • Supplemental
iSCSI	• Principal* • Supplemental	Supplemental	• Principal* • Supplemental
NVMe/TCP	• Principal* • Supplemental	Supplemental	• Principal* • Supplemental
NVMe/FC	• Principal* • Supplemental	Supplemental	• Principal* • Supplemental
NVMe/RDMA	• Principal* • Supplemental	Supplemental	• Principal* • Supplemental

Important:

* Principal Storage marked with an asterisk can be achieved by converging existing virtual infrastructure to a VMware Cloud Foundation instance for a management domain or importing an existing virtual infrastructure to a VMware Cloud Foundation for a workload domain. See KB 416270

Table 40: Principal Storage Models Feature Comparison in VMware Cloud Foundation

Function	vSAN	Fibre Channel	NFS	iSCSI	NVMe
Provides a Full SDDC Solution	✓	X	X	X	X
Storage Policy Based Management	✓	X	X	X	X
Automated Deployment and Scale	✓	X	X	X	X
Automated LCM, Patching, and Upgrades	✓	X	X	X	X
Stretched Clusters*	✓	✓	✓	X	X
Remote Clusters	✓	✓	✓	✓	✓
Compute Only Clusters	✓	X	X	X	X

Important:

*Fibre Channel and NFS stretched cluster can be achieved using vSphere Metro Storage Cluster (vMSC). See KB 417356

Table 41: Storage Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Single-Rack vSAN ESA Storage Model	• Single-tier storage architecture. • Storage devices are NVMe drives only. • Can be stretched across two availability zones. • Supports data services.	• Easy to manage. • Highly scalable. • Storage is configured and provisioned as part of automated workflows. • Allows for data re-protection after ESX host and availability zone failure events. • vSphere handles lifecycle operations. • Supports Virtual Machine Storage Policies. • Supports Host Rebuild Reserve. • Uses Auto-Policy Management.	• Requires additional network configuration. • CPU and memory overhead needs to be considered for vSAN. • Additional RAW storage capacity is required to ensure data is reported during failure scenarios.
Single-Rack vSAN OSA Storage Model	• Two-tier architecture. • Storage devices can be SATA, SAS, and NVMe. • Can be stretched across two availability zones. • Supports data services.	• Ease of management. • Highly scalable. • Storage is configured and provisioned as part of automated workflows. • Allows for data re-protection after ESX host and availability zone failure events. • vSphere handles lifecycle operations. • Support for Virtual Machine Storage Policies. • Supports Host Rebuild Reserve.	• Requires additional network configuration. • CPU and memory overhead needs to be considered for vSAN. • Additional RAW storage capacity is required to ensure data is reported during failure scenarios.

Model	Attributes	Benefits	Implications
Single-Rack vSAN Storage Cluster Storage Model	• vSAN storage-based cluster. • Storage Pool consists of NVMe drives only. • Can be achieved with — Non-stretched vSAN — Stretched vSAN	• Provides disaggregated storage for vSphere HCI enabled clusters. • Allows recovery after host and availability zone failure events. • Allows use of storage dense nodes.	• Not recommended running virtual machine workloads directly on the vSAN Storage Clusters. • Requires vSAN ESA. • Requires additional network configuration.
Single-Rack vSAN Compute Cluster Storage Model	• vSAN Client Compute Cluster. • Requires vSAN Storage Cluster. • No local vSAN storage required. • Can be stretched across two availability zones. • Allows for VM storage policies. • Allows for Site Coupling	• Consumes vSAN storage from a vSAN storage cluster. • Allows ESX host compute and network to scale independently of storage. • Allows for data re-protection after ESX host and availability zone failure events.	• Must be co-located in the same workload domain as its corresponding vSAN Storage cluster. • Requires additional network configuration.
Multi-Rack vSAN Storage Model	• Can be configured with both vSAN ESA and vSAN OSA • Can be configured with both Layer 2 cluster type and Layer 3 cluster type. • Layer 2 cluster type can be stretched across availability zones. • Supports data services.	• Highly scalable. • Storage is configured and provisioned as part of automated workflows. • Allows for data re-protection after ESX host and availability zone failure events. • vSphere handles lifecycle operations. • Supports Virtual Machine Storage Policies. • Supports Host Rebuild Reserve. • Uses Auto-Policy Management.	• Requires additional network configuration. • CPU and memory overhead needs to be considered for vSAN. • Additional RAW storage capacity is required to ensure data is reported during failure scenarios.
Multi-Rack vSAN Compute Cluster Storage Model	• vSAN Client Compute Cluster. • Requires vSAN Storage Cluster. • No local vSAN storage required. • Single availability zone only.	• Consumes vSAN storage from a vSAN storage cluster. • Allows ESX host compute and network to scale independently of storage. • Allows for data re-protection after ESX host and availability zone failure events.	• Must be co-located in the same workload domain as its corresponding vSAN Storage cluster. • Requires additional network configuration.
Fibre Channel Storage Model	• Supports data services. • Supports Datastore clusters.	Can leverage existing storage arrays.	• Requires dedicated fiber channel fabric configuration. • Storage provisioning and configuration are not completed by VMware Cloud Foundation automated workflows. • Requires additional management tools to configure.

Model	Attributes	Benefits	Implications
NFS Storage Model	• Distributed file system protocol. • Supports data services. • Supports Datastore clusters.	Can leverage existing storage arrays.	• Storage provisioning and configuration are not completed by VMware Cloud Foundation automated workflows. • Requires additional management tools to configure.
iSCSI	Supports data services.	Can leverage existing storage arrays.	• Storage provisioning and configuration are not completed by VMware Cloud Foundation automated workflows. • Requires additional management tools to configure. • Can only be configured as Primary Storage through use of Import/Converge Workflows
NVMe over Fabric (FC/TCP/RDMA)	Supports data services.	Can leverage existing storage arrays.	• Storage provisioning and configuration are not completed by VMware Cloud Foundation automated workflows. • Requires additional management tools to configure. • Can only be configured as Primary Storage through use of Import/Converge Workflows

Note:

For the detailed design related to *Storage Models*, see Storage Detailed Design.

Important:

vSAN Datastore Sharing (also known as "HCI Mesh") is permitted on all vSAN storage models within a VCF domain. This allows multiple independent vSAN HCI Clusters to consume storage of adjacent vSAN storage resources. This represents a flexible cross-cluster capacity sharing capability between vSAN clusters.

Figure 30: vSAN Storage

vSAN aggregates local, direct-attached storage devices across a vSphere cluster to create a single datastore that all hosts and virtual machines in a vSAN cluster can access.

vSAN can be deployed in multiple different topologies such as single rack, multi-rack, stretched or as separate storage and compute clusters.

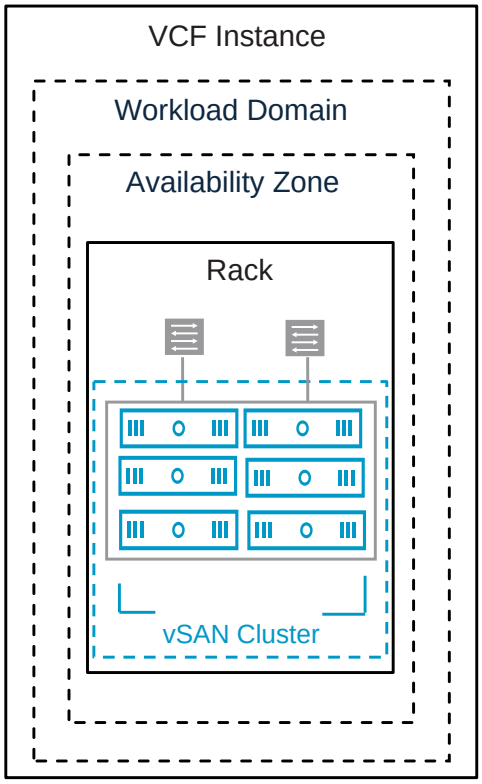


Table 42: vSAN Storage Model Attributes

Attribute	Detail
vSAN HCI Cluster	A cluster that provides both compute and storage to virtual machines.
vSAN Storage Cluster	A vSAN cluster that provides disaggregated storage for vSAN and vSphere clusters.
vSAN Compute Cluster	A vSAN Compute Cluster is a vSphere cluster that does not have local vSAN storage, but instead, it mounts a remote vSAN datastore.
vSAN Stretched Cluster	A vSAN Stretched Cluster is a vSAN cluster topology that extends a single logical vSAN cluster across two availability zones. A vSAN Stretched Cluster can tolerate a site failure.
vSAN Multi-Rack Cluster	A vSAN multi-rack vSAN cluster is a vSAN cluster topology spread over different physical racks, enhancing resilience and scalability. A multi-rack cluster can use vSAN fault domains to tolerate a rack failure.
vSAN Datastore	A distributed datastore that supports features that require shared storage.
vSAN Physical Networking	Host network interfaces can be dedicated or shared with other traffic types.
vSAN Availability	Workloads are protected by synchronously replicating objects in a vSAN cluster.
Data Services	• Deduplication and/or compression. • Encryption. • Integrated file services. • vSAN iSCSI Target Service. • Kubernetes persistent storage. services (block and file). • Cloud Native Services (CNS).

Table 43: vSAN Cluster Model Sizing Considerations

vSAN Storage Model	Minimum Number of vSAN Nodes	vSAN Node Failure response
Management domain default cluster	Three (3) (Simple Install)	• Tolerates one host failure. No automatic rebuild of vSAN components. • Three nodes do not support rebuild of vSAN objects to storage policy compliance.
	Four (4) (Highly Available Install)	Tolerates one host failure and supports automatic rebuild of vSAN objects to return vSAN objects to storage policy compliance.
All workload domain clusters **Note:** Includes additional clusters on management domains.	Three (3)	Tolerates one host failure. Three nodes do not support rebuild of vSAN objects to storage policy compliance.

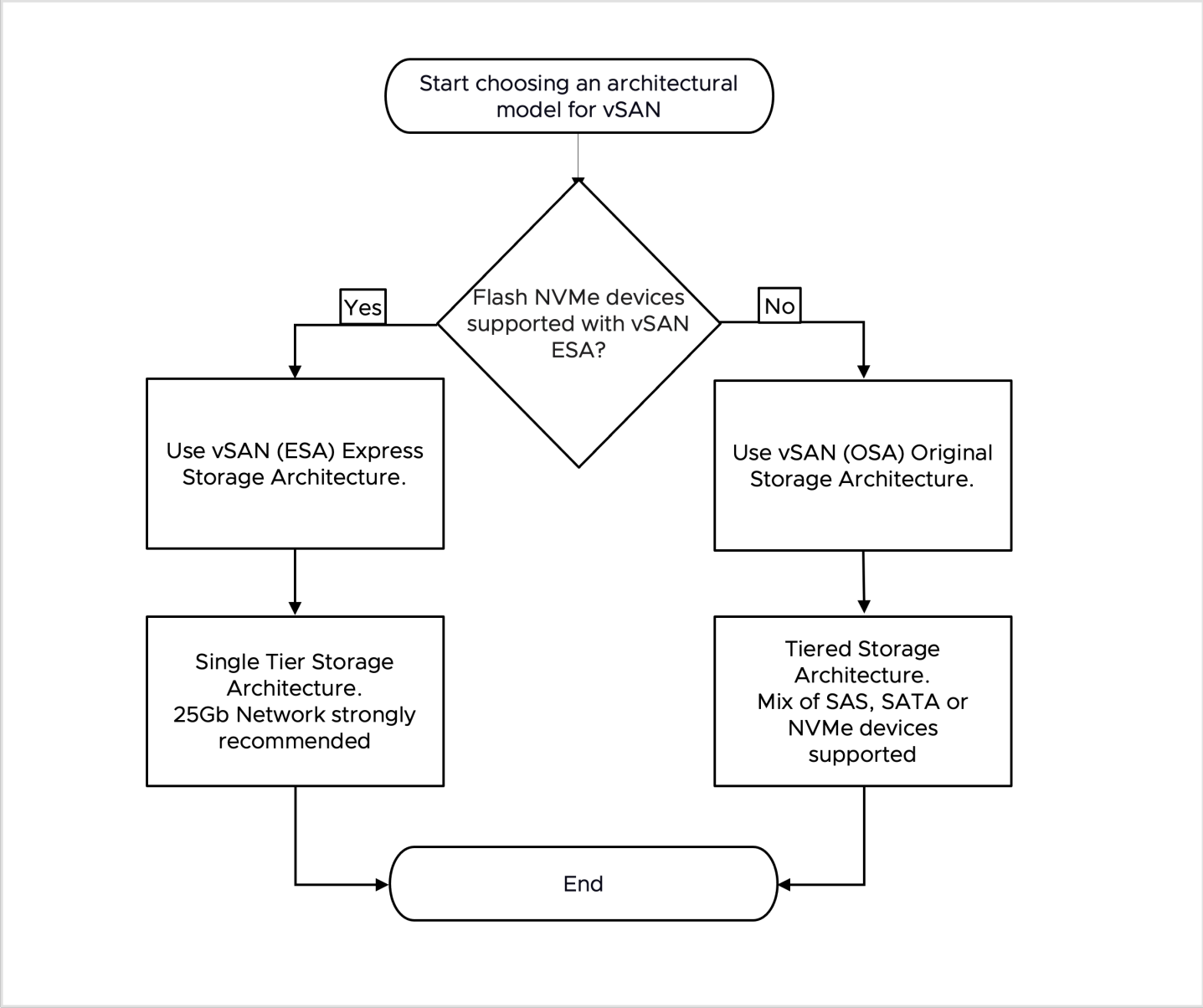
Table 44: vSAN Storage Options

Design Area	Options
vSAN Storage Architecture	• vSAN ESA uses a single-tier architecture that utilizes high-performance flash storage devices. • vSAN ESA supports only NVMe high-performance storage devices. • vSAN ESA can be deployed as a disaggregated vSAN Storage Cluster.
	• vSAN OSA uses a two-tier storage architecture that utilizes distinct cache and capacity tier storage devices. • vSAN OSA can support both flash and magnetic disks that can be a combination of SATA, SAS, and NVMe storage devices to configure an all-flash or hybrid vSAN cluster.
vSAN Host Networking	• 10GbE • 25GbE or higher **Important:** vSAN ESA Ready node profile, vSAN-ESA-AF-0, is the only vSAN ESA ready node that supports 10GbE. All other vSAN ESA nodes require 25GbE or greater. For more detail see https://partnerweb.vmware.com/comp_guide2/vsanesa_profile.php

Figure 31: Choosing a vSAN Storage Architecture Model

For high-performance workloads requiring low latency and high IOPS, vSAN ESA should be chosen as the desired vSAN Storage Architecture. vSAN ESA supports vSAN HCI storage as well as vSAN disaggregated storage clusters.

If newer hardware is not available, vSAN OSA maybe chosen as the vSAN architecture.



For detailed design models for vSAN storage see vSAN Storage Models.

NSX Manager and Control Plane Models

NSX Manager and Control Plane is contained within the NSX Manager nodes. These manage and configure virtual networking, including logical switches, Virtual Private Clouds (VPCs), and transit gateways through a user interface and API.

The NSX Manager nodes serve as the control plane for VMware Cloud Foundation networking by managing traffic flow and network topologies.

Table 45: NSX Manager Models in VMware Cloud Foundation

Model	Details	Benefits	Implications
Simple NSX Manager Model	• Single node with a Virtual IP (VIP) address for future expansion. • vSphere HA protects the node applying high restart priority.	Smallest footprint.	• Slower recovery from node failure. • Management and control plane impacted during upgrade operations. • Supports limited scale.
High Availability NSX Manager Model	• Three node cluster with a Virtual IP (VIP) address for UI and API management. • Highly available management and control plane. • Anti-affinity rule to keep cluster nodes on different ESX hosts. • vSphere HA protects the cluster nodes applying high restart priority.	• Highly available management and control plane with rapid recovery of single node failure. • No impact to control plane during upgrade operations.	All API-based management traffic utilizing the VIP is targeted to a single node.
High Availability NSX Manager With External Load Balancer Model	• Three node cluster with a Virtual IP (VIP) address for UI management. • Highly available management and control plane. • Anti-affinity rule to keep the cluster nodes on different ESX hosts. • vSphere HA protects the cluster nodes applying high restart priority. • External load balancer with additional VIP to distribute API requests across all NSX Manager cluster nodes.	• Highly available management and control plane with rapid recovery of single node failure. • No impact to control plane during upgrade operations. • Increase API-based management scale for high-churn environments.	Requires an external load balancer. See Load Balancer Models

NSX Global Manager Models

NSX Global Managers establish a centralized management plane to orchestrate consistent networking and security policies across an NSX Federation. In contrast, NSX Local Managers provide management and control plane services unique to a single VCF Instance or networking workload domain. While local managers maintain localized objects and traffic flows, global managers enable the extension of segments and security groups across workload domains residing within the same or different VCF Instances.

Table 46: NSX Global Manager Models in VMware Cloud Foundation

Models	Details	Benefits	Implications
High Availability NSX Global Manager Model	• Three node cluster with a Virtual IP (VIP) address for UI and API management. • Highly available global management plane. • Anti-affinity rule to keep cluster nodes on different ESX hosts. • vSphere HA protects the cluster nodes applying high restart priority.	• Highly available global management plane with rapid recovery of single node failure. • No impact to management plane during upgrade operations.	VIP for API-based management that is targeted to a single node.
High Availability with Standby NSX Global Manager Model	• Three node cluster with a Virtual IP (VIP) address for UI and API management. • Highly available global management plane. • Anti-affinity rule to keep cluster nodes on different ESX hosts. • vSphere HA protects the cluster nodes applying high restart priority. • One active and one standby cluster.	• Highly available management plane with rapid recovery of single node failure. • No impact to management plane during upgrade operations. • Standby cluster ready for manual failover.	• Additional global manager cluster needed. • Manual failover to standby cluster. • VIP for API-based management that is targeted to a single node.

NSX Edge Cluster Models

VMware Cloud Foundation supports multiple *NSX Edge Cluster Models*, you determine the form factor, number of nodes and placement according to the requirements for network services in a VCF instance. NSX Edge nodes can be shared across workload domains where such workload domain shares an NSX Manager cluster.

An NSX Edge node is an appliance that provides centralized networking services which cannot be distributed to hypervisors. Some services, such as Tier-0 gateways, are limited to a single instance per NSX Edge node. However, multiple instances of most services can coexist in these nodes.

NSX Edge Form Factor

An NSX Edge node can be deployed as a virtual appliance, or installed on bare-metal hardware. The NSX Edge node on bare-metal hardware can have better performance capabilities at the expense of more challenging deployment and limited deployment topology use cases.

Table 47: NSX Edge Form Factors in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
NSX Edge Virtual Appliance	• Can be deployed by using vCenter or NSX Manager. • Centralized password management through VCF Operations. • Supports four (4) data path virtual NICs. • Xlarge size supports the SECURITY Core Allocation Profile, see details Broadcom knowledge base article 441953 (requires version 9.1.1).	• Automated password management by using VCF Operations. • Protected by vSphere HA. • Can be recovered across availability zones via vSphere HA. • Easy to scale up by modifying the specification of the virtual appliance. • Provides sufficient performance for most scenarios.	Workloads with small packet size might not have the best performance.
NSX Edge Bare-Metal Appliance	• Physical server where the NSX Edge software is installed by mounting the installation ISO file or using PXE. • After the image installation, it requires manual registration to NSX Manager. • Workload domain upgrades include NSX Edge bare metal nodes. • Supports up to 16 data path physical interfaces. • Supports the L4FORWARDING and SECURITY Core Allocation Profiles, see details Broadcom knowledge base article 441953 (requires version 9.1.1). • Centralized password management through VCF Operations.	• Automated password management by using VCF Operations. • Ability to consume the bandwidth of multiple 100G interfaces. • Workloads with small packet size might have better performance. • Independent from ESX lifecycle management.	• Has hardware compatibility requirements (network interface cards and CPUs). • Requires individual hardware lifecycle management and monitoring of failures, firmware, and drivers. • Must be manually deployed and registered to NSX Manager.

NSX Edge Cluster Models

NSX Edge Cluster Models are a grouping of one or more NSX Edge nodes, representing a pool of capacity for NSX network services.

Important:

During NSX Edge node deployment, the NSX Edge node supports either centralized password management via VCF Operations or locally configured passwords. VCF Operations supports the password management task 'update', which requires the current password before a new one can be configured. When centralized password management is selected during a fresh NSX Edge node deployment, randomized passwords are generated. These passwords can be retrieved, for example, via the SDDC Manager API (GET /v1/credentials), and subsequently used to update the passwords through VCF Operations.

Table 48: NSX Edge Cluster Models in VMware Cloud Foundation

Deployment Model	Attributes	Benefits	Implications
Host Fault Tolerant NSX Edge Cluster Model	• Suited to single availability zone configurations. • Uses the Single-Rack vSphere Cluster Model. • Supports virtual appliance or bare-metal form factor.	• Allows for the minimal footprint. • Allows for recovery after host failure events. • Supports all VCF domains.	• No protection against a rack failure. • North-south throughput is limited to the physical network fabric top of rack switch uplinks in a single rack.
Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters	• Suited to single availability zone configurations. • Uses multiple instances of the Single-Rack vSphere Cluster Model. • Supports virtual appliance or bare-metal form factor.	• Supports protection against rack failures. • Independent storage (vSAN) for each rack. • Higher north-south throughput scalability across multiple racks.	Requires additional resources to support multiple vSphere clusters that are deployed across different racks with independent storage (vSAN).
Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster	• Suited to single availability zone configurations. • Uses Multi-Rack Layer 2 vSphere Cluster Model. • Supports virtual appliance or bare-metal form factor.	• Supports protection against rack failures. • Reduced resources requirements with a single vSphere cluster compared to Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters. • Higher north-south throughput scalability across multiple racks.	• Limited availability with a single shared storage across racks. • Dependency on physical network fabric with the requirement to support Layer 2 adjacencies for VLANs spanning racks, for example, using VXLAN encapsulation.
Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster	• Suited to single availability zone configurations. • Uses Multi-Rack Layer 3 vSphere Cluster Model. • Supports virtual appliance or bare-metal form factor.	• Supports protection against rack failures. • Reduced resources requirements with a single vSphere cluster compared to Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters. • Higher north-south throughput scalability across multiple racks.	Limited availability with a single shared storage across racks.
Dual Availability Zone Based on NSX Edge Node HA Model	• Suited to dual availability zone configurations. • Uses: – Multiple instances of the Single-Rack vSphere Cluster Model. Or – Stretched vSphere Cluster Model. • Supports virtual appliance or bare-metal form factor.	• Fast network failover across availability zones. • No dependency on physical network fabric when using multiple instances of the Single-Rack vSphere Cluster Model. • Maintain optimal fault isolation between availability zones when using multiple instances of the Single-Rack vSphere Cluster Model.	• Dependency on physical network fabric BGP configuration, if maintaining symmetric routing is required. • Stretched vSphere Cluster Model usage implies common datastore supporting NSX Edge cluster across availability zones.

Deployment Model	Attributes	Benefits	Implications
Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model	• Suited to dual availability zone configurations. • Uses Stretched vSphere Cluster Model. • Supports virtual appliance form factor only.	Symmetric routing in and out of the virtual network.	• Slow network failover across availability zones. • Dependency on physical network fabric to extend NSX Edge uplink VLANs, NSX Edge TEP VLAN, and NSX Edge management VLANs. • Stretched vSphere Cluster Model usage implies common datastore supporting NSX Edge cluster across availability zones.

Virtual Network Appliance Cluster Models

VMware Cloud Foundation supports multiple *Virtual Network Appliance Cluster Models*, you determine the form factor, node count, and placement based on the requirements for stateful network services within distributed VPC environments running in a VCF domain.

A *Virtual Network Appliance* node is an appliance that provides stateful network services such NAT or VNA load balancer which cannot be distributed to hypervisors running the Distributed Transit Gateway. North-South traffic that is sourced from or destined to the hypervisor that requires stateful network services such as NAT are redirected to the *Virtual Network Appliance* node.

Virtual Network Appliance Form Factor

Virtual Network Appliance nodes can only be deployed as virtual appliances; other form factors are not available.

Table 49: Virtual Network Appliance Form Factor in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Virtual Appliance	• Can be deployed by using vCenter or NSX Manager. • Centralized password management through VCF Operations. • Supports four (4) data path virtual NICs.	• Automated password management by using VCF Operations. • Protected by vSphere HA. • Easy to scale up by modifying the specification of the virtual appliance.	None.

Virtual Network Appliance Cluster Models

Virtual Network Appliance nodes are grouped in one or more *Virtual Network Appliance* clusters, representing a pool of capacity for stateful network services consumed by VPCs connected to a Distributed Transit Gateway.

Important:

Centralized password management via VCF Operations is the only supported option; local credentials cannot be configured during the Virtual Network Appliance node deployment. VCF Operations supports the password management task 'update', which requires the current password before a new one can be configured. A fresh Virtual Network Appliance node deployment generates randomized passwords, which can be, for example, retrieved via the SDDC Manager API (GET /v1/credentials) and subsequently used to update the passwords using VCF Operations.

Note:

When using the Distributed VLAN Connectivity Model and a Virtual Network Appliance cluster is required, the Multi-Rack Layer 3 vSphere Cluster Model is not supported. The external VLAN is, by design, stretched across racks, which inherently implies a Multi-Rack Layer 2 vSphere Cluster Model. This limitation does not apply when using the Distributed VXLAN Connectivity Model for Version 9.1.1.

Note:

The Virtual Network Appliance cluster provides a pool of capacity and runs one or more service instances for stateful network services. One service instance is automatically configured per VPC when a stateful network service is configured. When two or more Virtual Network Appliance nodes are part of the Virtual Network Appliance cluster, the service instance is deployed in active/standby High Availability mode; active/active mode is not supported. Each service instance is connected to exactly one of the four data plane fastpath interfaces, and multiple service instances are distributed across all four data plane fastpath interfaces.

Note:

This section is exclusively for the use of Virtual Network Appliance nodes for stateful network services such as NAT and not when the Virtual Network Appliance node is used as an EVPN Route Controller. The design section for the EVPN Route Controller including the Virtual Network Appliance nodes is covered within the Workload Connectivity Models section for VPC with EVPN integration.

Table 50: Virtual Network Appliance Cluster Models in VMware Cloud Foundation

Deployment Model	Attributes	Benefits	Implications
Host Fault Tolerant Virtual Network Appliance Cluster Model	• Suited to single availability zone configurations. • Uses the Single-Rack vSphere Cluster Model. • Supports virtual appliance form factor only.	• Allows for the minimal footprint. • Allows for recovery after ESX host failure events. • Supports all workload domains.	No protection against a rack failure.
Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters	• Suited to single availability zone configurations. • Uses multiple instances of the Single-Rack vSphere Cluster Model. • Supports virtual appliance form factor only.	Independent storage (vSAN) for each rack.	• When using the Distributed VLAN Connectivity Model Layer 2 adjacency for the Distributed Transit Gateway external VLAN is required across all racks where VNA nodes from the same VNA cluster reside. • Requires additional resources to support multiple vSphere clusters deployed in or across different racks with independent storage (vSAN).
Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster	• Suited to single availability zone configurations. • Uses Multi-Rack Layer 2 vSphere Cluster Model. • Supports virtual appliance form factor only.	Reduced resources requirements with a single vSphere cluster compared to Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters.	• Layer 2 adjacency for all VLANs must be extended across all racks. • Limited availability with a single shared storage across racks.

Deployment Model	Attributes	Benefits	Implications
Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster	• Suited to single availability zone configurations. • Uses Multi-Rack Layer 3 vSphere Cluster Model. • Supports virtual appliance form factor only.	Reduced resources requirements with a single vSphere cluster compared to Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters.	• This model is only supported when using the Distributed VXLAN Connectivity Model for Version 9.1.1. • Limited availability with a single shared storage across racks.
Dual Availability Zone Based on Virtual Network Appliance Node HA Model	• Suited to dual availability zone configurations. • Uses: – Multiple instances of the Single-Rack vSphere Cluster Model. Or – Stretched vSphere Cluster Model. • Supports virtual appliance form factor only.	• Fast network failover across availability zones. • No dependency on physical network fabric when using multiple instances of the Single-Rack vSphere Cluster Model. • Maintain optimal fault isolation between availability zones when using multiple instances of the Single-Rack vSphere Cluster Model.	• When using the Distributed VLAN Connectivity Model Layer 2 adjacency for the Distributed Transit Gateway external VLAN is required across all racks where VNA nodes from the same VNA cluster reside. • Stretched vSphere Cluster Modelusage implies common datastore supporting VNA cluster across availability zones. • Traffic between availability zones may occur.
Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model	• Suited to dual availability zone configurations. • Uses Stretched vSphere Cluster Model. • Supports virtual appliance form factor only.	Cross-availability zone traffic can be avoided by deploying workload VMs exclusively in the primary availability zone.	• Layer 2 adjacency for all VLANs must be extended across the availability zones. • Stretched vSphere Cluster Model usage implies common datastore supporting VNA cluster across availability zones. • Slow network failover across availability zones.

Network Fabric Models

The data center network fabric facilitates communication between ESX hosts and VMware Cloud Foundation components within and across VCF Instances. It also provides networking services to workload virtual machines via VLANs and Virtual Networking.

Data Center Network Fabric Options

The *Network Fabric Models* define the physical and logical topologies that present Layer 2 and Layer 3 networking to ESX hosts. There are four primary network topologies, each with implementation-specific advantages and integration requirements:

- Campus-style Network Fabric (Core–Aggregation–Access)
- CLOS Network Fabric (Leaf-Spine/Superspine)
- Hardware SDN - Standards Based BGP-EVPN
- Hardware SDN - Proprietary Fabric

Each deployment has unique performance requirements based on design objectives, vSphere Cluster Models, Network Consumption Models, and Storage Models. Key considerations include latency, bandwidth, MTU settings, gateway and switch availability, and inter-site networking capacity, all of which must be properly sized for the specific design. To fully leverage the VMware Cloud Foundation platform, the fabric must be highly performant data center-grade ethernet fabric that meets the design availability goals.

VMware Cloud Foundation is generally agnostic to physical networking configurations, offering customers flexibility in selecting their preferred topology. However, certain data center network fabric options and vendor-specific

implementations may affect the deployment of supported topologies and configurations. A thorough review of network fabric requirements detailed in the design library section should be conducted in collaboration with the network fabric vendor.

VMware Cloud Foundation supports optional integration with standards based EVPN-VXLAN fabrics to connect the Virtual private clouds (VPCs) to the physical EVPN-VXLAN network fabric. In this model, VCF components setup BGP-EVPN exchange between EVPN Route Controller and the physical fabric EVPN gateway to advertise internal prefixes directly into the fabric's control plane. This integration utilizes BGP-EVPN Type-5 routes (IP Prefix routes) to advertise workload prefixes associated with tenant VPCs. Utilizing Type-5 routes allows the physical fabric to dynamically learn the location of workloads without requiring broad Layer 2 extension or manual static route configuration on the ToR switches. See Distributed VXLAN Connectivity Model for Version 9.1.0 for more details.

Network Fabric Models

The Network Fabric Models describe the physical and logical topologies to present Layer 2 and Layer 3 networking to your ESX hosts. You must select the appropriate Network Fabric Model based on your specific requirements for vSphere cluster scale, physical fault isolation, and availability spanning single racks, multiple racks, or multiple data center availability zones.

Table 51: Network Fabric Models in VMware Cloud Foundation

Model	Supported Cluster Models	Details	Benefits	Implications
Single-Rack Network Fabric Model	Single-Rack vSphere Cluster Model	• Fabric designed to support vSphere clusters isolated to a single rack. • All virtual machines within a vSphere cluster reside within an individual rack. • Building block approach for datacenters with multiple instances of Single-Rack vSphere Cluster Model.	• Simplest network fabric model. • In data centers with multiple racks, each may be independent of the other.	Does not support vSphere clusters which span multiple racks.
Multi-Rack Network Fabric Model	• Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model	• Fabric designed to support vSphere clusters which span across multiple racks. • Virtual machines within a vSphere cluster may move between racks to provide resilience for a rack outage. • Fabric may be: – Layer 2 (Management and workload domain). – Layer 3 (workload domain only).	Supports *vSphere Cluster Models* and *Storage Models* to tolerate rack-level outages.	• More complex data center fabric needed to span VLANs between racks when using the Layer 2 multi-rack model. • Routing peering between racks may be more complex when using NSX Edge Node deployed without BER model.

Model	Supported Cluster Models	Details	Benefits	Implications
Multi Availability Zone Network Fabric Model	Stretched vSphere Cluster Model	• Fabric designed to support vSphere clusters which span across multiple data center availability zones. • Virtual machines within a vSphere cluster may move between availability zones to provide resilience for an availability zone outage. • Fabric may be: – Layer 2 (management and workload domain – Layer 3 (workload domain) • Availability zones are defined by customer - may be separate racks in the same data center hall, or separate physical buildings within 5ms RTT latency.	Supports *vSphere Cluster Models* and *Storage Models* to tolerate availability zone-level outages.	• More complex data center fabric needed to span VLAN Segments between availability zones. • Routing peering between availability zones may be more complex.

Infrastructure IP Addressing Models

The *Infrastructure IP Addressing Models* define the IP address family—IPv4 and IPv6—allocated to your management components and infrastructure VMkernel (VMK) interfaces. Depending on your organization's network readiness, IP space constraints, and operational requirements, you can choose to maintain a traditional single-stack IPv4 environment or implement a dual-stack configuration.

Model	Description	Benefits	Implications
Single-Stack - IPv4 Only	Uses the traditional IPv4 addressing scheme exclusively for all VCF components.	• Simplifies network planning and management • Uses existing IPv4 infrastructure and expertise. • Lower operational complexity.	May face IP address exhaustion in large environments.
Dual-stack - IPv4 & IPv6	Configures many VCF components to use both IPv4 and IPv6 addresses concurrently. Some components and traffic types remain exclusively IPv4, while others may be designated as IPv6 only. **Note:** This model does not eliminate the need for IPv4 addressing.	Can support some components with IPv6 only, therefore limiting IPv4 address space requirements.	• Increases network planning and operational complexity (managing two address families, routing, security). • Not all VCF components support dual-stack or IPv6-only, requiring some to remain IPv4-only. • Required IPv6 capable fabric configuration and data center services (DNS, NTP)

Note:

Addressing of workload virtual machines is separate from the addressing of management and workload domain infrastructure components.

VCF Edge Models

VCF Edge (formerly Remote Clusters) is an optimized configuration of VMware Cloud Foundation, tailored for edge use cases. VCF Edge provides a private cloud infrastructure platform for edge locations, with integrated enterprise-class compute, storage, networking, management, and security capabilities.

Table 52: VCF Edge Requirements

Category	Requirement	Justification
Site distribution	Minimum 10 sites or more.	Supports geographically diverse edge locations.
CPU per host	Minimum 8 CPU cores per host.	Ensures enough resources for each Edge host to qualify for VCF Edge deployment.
Maximum CPU cores per site	Maximum of 256 CPU cores per site.	Maximum host and CPU capacity to scale of individual edge locations.
Host Placement	VCF Edge hosts or clusters must be deployed in a physically distinct location from your data center workloads, unless you have strict physical segregation measures in place.	Place Edge hosts or clusters in a separate rack or separate switch than your data center workload hosts.
Edge vCenter	• Edge vCenter can be hosted in a Data Center or a Co-location facility • Dedicate separate vCenter for the Edge workload	You can use an existing vCenter or Deploy a net new vCenter for the Edge workload.
VCF Operations	VCF Operations is the mandatory requirement for licensing of your Edge environment.	None

Important:

If you are deploying a VCF Edge compute cluster without a full VCF stack, then you have to deploy VCF Operations along with a vCenter manually.

Please refer to the VCF Edge Design section for all the design patterns.

Use Cases

In order to enable businesses to provide edge infrastructure everywhere, edge compute, network technologies, and architectures will be required as business applications and workloads develop and spread beyond data centers and the cloud to the sites where data is produced and consumed.

For customers looking to deploy edge solutions for use cases within retail, banking, energy, healthcare, and utilities, VMware Cloud Foundation is a scalable solution that enables the ability to support secure, edge solutions using the full stack or required elements of VMware Cloud Foundation.

Private AI Foundation Platform Models

There are multiple ways an organization can consume the Private AI capabilities of VMware Cloud Foundation. Use cases and the level of abstraction of the underlying virtual infrastructure are the key factors in selecting the most appropriate *Private AI Foundation Platform Model*.

Private AI Foundation Platform Models

Table 53: Private AI Foundation Platform Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
Private AI Services Model	• Fully managed AI application platform. • UI-driven deployment of model endpoints, knowledge bases, and agents. • No Kubernetes/VM knowledge required for the Developers and ML Operation persona. • OpenAI-compatible API. • Built-in Model Runtime, Data Indexing & Retrieval, Agent Builder. • Supports vLLM, Infinity, LlamaCPP inference engines.	• Ultimate simplicity for LLM-based applications. • End-to-end RAG workflow (model → knowledge base → agent). • Self-service for application developers. • Abstract most of the infrastructure complexity. • Integrated with Harbor model gallery and VMware Data Services Manager.	• Only supports LLM use cases: Conversational AI, Data Indexing and retrieval, MCP servers, and content generation. • Cannot run: Computer vision, time-series, optimization, audio/speech models. • Requires one-time namespace activation by admin. • Depends on supported inference engines (e.g., cannot use Triton for Computer Vision models). • OpenAI API not suited for structured data applications.
AI and ML Platform Based on VMware Kubernetes Services Model	• Kubernetes cluster with access to vGPU or full GPU via DirectPath. • Can be deployed via VCF Automation catalog or manually (kubectl). • Best suited for ML engineers and platform teams. • Can integrate with VMware Data Services Manager vector databases.	• Suitable for production inference and training/ • Supports any AI workload (Computer Vision, time-series, LLM, audio, optimization). • Suitable for Distributed Inference and Training across multiple physical hosts.	• Requires understanding of Kubernetes concepts. • More complex than DLVM for single-user scenarios. • Catalog deployment limited to vGPU before VCF 9.1 (DirectPath only available via manual deployment in VCF 9.0).
AI and ML Platform Based on Deep Learning Virtual Machines Model	• Pre-built Ubuntu image with Docker, ML Package Managers, ML frameworks, NVIDIA Container Toolkit. • Can be deployed via VCF Automation catalog, vSphere Client, or kubectl. • Standardized, validated software stack (PyTorch, TensorFlow, Triton, DCGM).	• Fastest time to productivity for individual data scientists. • Automated vGPU driver installation matching the host driver version. • Automated NVIDIA Driver installation when using DirectPath. • Pre-integrated with VCF CLI for model management.	• Manual deployment requires understanding OVF properties and base64 encoding. • Catalog deployment limited to vGPU before VCF 9.1 (DirectPath only available via manual deployment in VCF 9.0).

Private AI Foundation Compute Models

In VCF, AI/ML workloads may run on top of different compute infrastructure, which may be equipped with a GPU or CPU only. Each option presents different tradeoffs between performance, cost, and operational flexibility.

Private AI Foundation Compute Models

Table 54: Private AI Foundation Compute Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
vGPU Compute Model	• GPU sharing: multiple VMs can share a single physical GPU. • Two sharing modes: Time-sliced (serial execution) or MIG (hardware isolation). • Requires NVIDIA vGPU Manager (ESX VIBs) installed on the ESX host. • Profile-based allocation (frame buffer size, compute allocation). • Supports live migration (vMotion) when enabled. • Multi-GPU profiles for large models. • DRS integration for automated placement and vSphere HA restart.	• vMotion support: live migration for maintenance and load balancing. • Flexible capacity management and resource pooling. • Consistent operations with other virtualized workloads thanks to live vMotion support. • Cost efficiency through GPU sharing. • Native support for NVLink/ NVSwitch device groups enabling direct high-speed GPU-to-GPU communication for multi-GPU configurations.	• Requires NVIDIA vGPU licensing. • Additional software layer: vGPU Manager updates as part of the vLCM cluster image. • Unified Virtual Memory (UVM) disables vMotion when enabled.
Direct Path GPUs Compute Model	• Direct hardware access: GPU exclusively assigned to a single VM. • Bare-metal performance with minimal virtualization overhead. • Native GPU drivers run in the guest OS. • Assignment using Dynamic DirectPath (device-independent labels) or legacy DirectPath (fixed PCI address). • DRS integration for automated initial placement and vSphere HA restart.	• Maximum performance for single GPU inference. • Simpler troubleshooting. • No hypervisor-level GPU driver dependencies. • No vGPU license requirements.	• No live vMotion support: VMs must be powered off to be migrated. • No GPU sharing: one or more GPUs per VM limits density. • Reduced flexibility for capacity management. • Multi-GPU configurations must be manually assigned and require GPU fabric manager software on a workload VM.

Model	Attributes	Benefits	Implications
CPU Only Compute Model	• Applicable to the Private AI Services Model. • No GPU hardware required: inference runs on CPU cores using llama.cpp runtime. • Standard VM configuration with memory and vCPU allocation. • llama.cpp optimized for CPU inference with SIMD instructions (AVX, AVX2, AVX-512, ARM NEON). • Suitable for smaller models (7B-13B parameters) and embedding models.	• Zero GPU infrastructure cost: no specialized hardware procurement or licensing. • Full vSphere operational flexibility. vMotion, DRS, and HA work without restrictions. • llama.cpp's efficient CPU utilization enables reasonable inference performance on commodity hardware. • Simplified capacity planning: uses standard compute resources. • Lower power consumption compared to GPU deployments.	• Significantly lower inference throughput compared to GPU solutions. • llama.cpp performance depends on available CPU instructions. AVX/AVX2 (universal on modern server CPUs from 2013+) provides baseline performance. AVX-512 can provide up to 2x better inference performance. • Limited to smaller quantized models due to CPU memory bandwidth constraints. • Higher per-token latency makes it unsuitable for real-time interactive applications. • Not viable for training workloads.

Lateral Security with vDefend Models

The *Lateral Security with vDefend Models* allows you to secure a VMware Cloud Foundation platform by using VMware vDefend and the Security Services Platform.

Lateral Security with vDefend Models in VMware Cloud Foundation

Model	Attributes	Benefits	Implications
License Hub for vDefend and Avi Load Balancer	• Central vDefend license management and reporting. • Support for connected and disconnected modes of operations. • Automated reactivation on renewal (for connected environments). • License endpoints: NSX Manager, vDefend Security Services Platform, Avi Controller.	• Simplifies license operations. • Reduces complexity and risk. • Easy to stay compliant with usage data updated every 180 days. • Centralized license usage analytics across multiple vDefend and Avi installations in a single pane of glass.	Single License HUB support - up to 120 endpoints.
Security Services Platform for VCF Model	• Security Services Platform employs an appliance-based deployment model. • Delivered and life cycled viaSecurity Services Platform Installer • Single Security Services Platform Installer deploys and manages a single Security Services Platform instance. • A single Security Services Platform instance onboards only one NSX Manager cluster.	• Enables the use of advanced vDefend security services, including Security Intelligence, Network Detection and Response, Malware Prevention, and more. • Supports management domain and workload domain deployment models. • Simplifies customer Security Journey via the vDefend DFW 1-2-3-4 prescriptive workflow. • Enables customers to assess their security posture via analytics, dashboards, and reports - Security Segmentation Score and Report. • Firewall Rule Analysis: support customer firewall rule management and optimization. • Enables Distributed Firewall for Bare Metal Workloads. • Security Metrics: Collects and stores metrics • Enables License Hub for VMware vDefend.	You may need to expand the resources in the management domain or the workload domain to accommodate Security Services Platform deployments.
Management Domain Security Model	• Employs a prescriptive methodology to secure the VCF management domain components with vDefend DFW. • Follows a prescriptive staged approach with Infrastructure, Environments, and Application level protection. • Provides a Terraform automation script for straightforward implementation.	• Enables VCF management component protection. • Simplifies design, planning, and implementation.	N/A

Model	Attributes	Benefits	Implications
Workload Domain Security Model	• Facilitates DFW 1-2-3-4: Four Stage Prescriptive Segmentation Deployment Journey. • Segmentation Planning Employees import customer CSV metadata files to tag VMs and create groups, which are foundational for defining security policies in subsequent stages. • Supports essential enterprise functions, including Active Directory, DNS, NTP, logging, and monitoring utilities. • Delivers cross-zone traffic monitoring, either the list of traffic or a set of recommended rules, so that exceptions can then be granted. • Employs VM-to-application mapping with auto-tagging and grouping for granular application microsegmentation.	• Speed up microsegmentation deployment without guesswork. • Improve efficiency through an automated multi-stage segmentation workflow. • Secure VCF workloads quickly and easily.	Requires Security Services Platform deployment.
Distributed Virtual Patching	• Covers design details and decisions to implement vDefend Intrusion Detection and Prevention System (IDPS) to protect unpatched workloads in VMware Cloud Foundation (VCF). • Follows a 4-stage prescriptive workflow that takes an organization from risk identification through to control retirement. • Defines virtual patching rules and policy placement. • Utilize per-signature IDPS profiles by selecting relevant signatures based on CVE or specific threats. • Covers centralized event collection, proactive alerting, and optional context enrichment through vDefend Network Detection and Response (NDR)	• Compensating control for rapid identification of novel vulnerabilities via Frontier AI Models. • By applying a network-level signature tailored to detect and block the specific exploit mechanism at the vNIC boundary, the threat is neutralized at the earliest possible stage.	• Requires cross-team input (NetSec, SOC, application owners) and access to assessment tooling such as Security Intelligence or BAS tools. • Requires commercial scanning tooling.

Design Blueprints for VMware Cloud Foundation

This section contains an overview of the design blueprints that can be consumed when designing your VCF private cloud with VMware Cloud Foundation.

- Infrastructure Modernization

- Application Modernization
- Security Modernization

Infrastructure Modernization

Build and Deploy Private Cloud infrastructure On-Premises	
VCF Fleet in a Single Site with Minimal Footprint	The *VCF Fleet in a Single Site with Minimal Footprint* is an infrastructure blueprint where the management domain of the first VCF Instance is deployed with a minimal footprint. All management and workload components are deployed in a single vSphere cluster.
VCF Fleet in a Single Site	The *VCF Fleet in a Single Site* is an infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components. Workloads are isolated to a separate workload domain. All components are located within a single data center and a single availability zone.
VCF Fleet with Multiple Sites in a Single Region	The *VCF Fleet with Multiple Sites in a Single Region* is an infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components. Workloads are isolated to a separate workload domain. Multiple sites exist and stretched clusters are used to increase availability across sites.
VCF Fleet with Multiple Sites Across Multiple Regions	The *VCF Fleet with Multiple Sites Across Multiple Regions* is an infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components. Workloads are isolated to a separate workload domain. Multiple sites exist in different regions.
VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)	The *VCF Fleet with Multiple Sites in a Single Region Plus Additional Region(s)* is an infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components. Workloads are isolated to a separate workload domain. Multiple sites exist, where some sites co-exist in the same region, and other sites exist in a different region.
Optimize Infrastructure and Operations	
VCF Fleet Management Blueprint	VCF Operations helps build, manage, and scale your private cloud infrastructure while providing that consistency across each of its components.
VCF Monitoring and Alerting Blueprint	This blueprint provides comprehensive guidance for monitoring a VMware Cloud Foundation platform by using the monitoring components Real-time metrics, Log management, and VCF Operations for networks. Effective monitoring is crucial for maintaining the stability, performance, and health of the SDDC infrastructure.
VCF Troubleshooting Blueprint	This blueprint is intended to help troubleshooting efforts for typical scenarios experienced in an enterprise environment. The blueprint focuses on incident management - identifying the cause of an occurring problem, and determining potential solutions.
Extend Data Center to the Edge, Public, and Sovereign Clouds	
VCF Edge: Single Node with Argo CD Design Blueprint	A single-node VCF Edge deployment consolidates the ESX hypervisor, vSphere Supervisor, and workload plane onto one physical host.
VCF Edge Design Blueprint for Government and Defense Use Case	This blueprint focuses on federal use cases for a VCF Edge to address the unique operational, security, and compliance needs of federal agencies on and off the battlefield.
VCF Edge Design Blueprint for Manufacturing Use Case	This blueprint focuses on manufacturing use cases for a VCF Edge to address the unique operational, security, and compliance needs.

Application Modernization

Enable Self-Service Infrastructure and Cloud Services for App Teams	
Self-Service Multi-Tenant Private Cloud Consumption Design Blueprint	The *Self-Service Multi-Tenant Private Cloud Consumption* blueprint provides guidance for implementing a self-service, multi-tenant private cloud. The blueprint delivers a robust, scalable, and automated platform that offers agility and efficiency in resource consumption, while adhering to commonly adopted design patterns.

Build, Run, and Manage Kubernetes and Other Modern Workloads	
VMware vSphere Kubernetes Service Consumption Blueprint	The *VMware vSphere Kubernetes Service Consumption Blueprint* provides architectural guidance for selecting and implementing the appropriate VMware vSphere Kubernetes Service (VKS) consumption model within VMware Cloud Foundation platforms.
Build, Run, and Manage Private AI Apps	
Private AI Services Consumption Blueprint	The *Private AI Services Consumption Blueprint* describes the design selections and architectural components required to deploy and operate a GenAI platform on VMware Cloud Foundation. It is intended to be used in conjunction with an underlying infrastructure blueprint.

Security Modernization

Bolster Lateral Security and Intrusion Detection and Prevention	
Lateral Security with vDefend	The *Lateral Security with vDefend* blueprints describes the architectural components and design selections required to provide centralized, enterprise-grade security services for a VMware Cloud Foundation platform using VMware vDefend.
Accelerate Recovery from Ransomware and Other Disasters	
VCF Component Backup and Restore	The *VCF Component Backup and Restore* blueprint provides guidance on ensuring the resilience and continuity of your VCF private cloud. In the event of data loss or system failure, having up-to-date backups allows for efficient restoration of critical services, minimizing downtime and maintaining operational integrity. Implementing a consistent backup strategy protects your infrastructure against unforeseen disruptions.
VCF Instance Backup and Restore	The *VCF Instance Backup and Restore* blueprint provides guidance on the recovery of an entire VCF Instance that has failed, and must be recovered in the same site.
VCF Fleet Disaster Recovery	This *VCF Fleet Disaster Recovery* protection blueprint provides the ability to handle a VCF Instance, or site failure, and recover components in another location, where that location is typically outside metro radius.
Cyber Recovery for VMware Cloud Foundation	This *Cyber Recovery for VMware Cloud Foundation* protection blueprint provides a clean room environment on a vCenter instance that allows you to inspect, analyze, and recover infected virtual machines before restoring them to a production environment.

Infrastructure Modernization

Broadcom, through its portfolio of VMware products, delivers a proven path to infrastructure modernization — helping enterprises replace fragmented, legacy environments with a unified, software-defined platform. VMware solutions consolidate compute, storage, and networking into a streamlined architecture that supports private cloud, hybrid deployments, and modern application frameworks. The result: reduced operational overhead, stronger security posture, and the agility to scale on demand — all without disrupting business continuity.

How to select the right VCF Fleet Blueprint

Decision Criteria	Blueprint 1: Minimal Footprint	Blueprint 2: Single Site	Blueprint 3: Multi-Site Single Region	Blueprint 4: Multi-Region	Blueprint 5: Hybrid Multi-Region
Site Config	**Single Site (3 hosts)**	**Single Site (4+ hosts)**	**Multiple Sites Same Region**	**Multiple Sites Multiple Regions**	**Multiple Sites Region + Additional**
Availability	ESX host, Network path, Single rack, Management	ESX host, Network path, Single rack, Management	ESX host, Network path, Single rack, Management, Availability zone	ESX host, Network path, Single rack, vSphere cluster, Management	ESX host, Network path, Single rack, vSphere cluster, Management, Availability zone
Isolation	Hypervisor-based, Tenant isolation	Hypervisor-based, Network traffic, Cluster-based, Tenant isolation	Hypervisor-based, Network traffic, Cluster-based, Tenant isolation	Hypervisor-based, Network traffic, Cluster-based, Tenant isolation	Hypervisor-based, Network traffic, Cluster-based, Tenant isolation
DR Capability	**Limited (Backup only)**	**Limited (Backup only)**	**Enhanced (Site-to-site)**	**Full (Cross-region)**	**Comprehensive (Multi-option)**

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Expansion after Infrastructure Blueprint Deployment

The included infrastructure blueprints are starting points for the initial deployment and cover either a management domain only, or a management domain and the first workload domain. All VCF deployments may be expanded in the following ways and are subject to the limits defined in VMware Configuration Maximums.

Note:

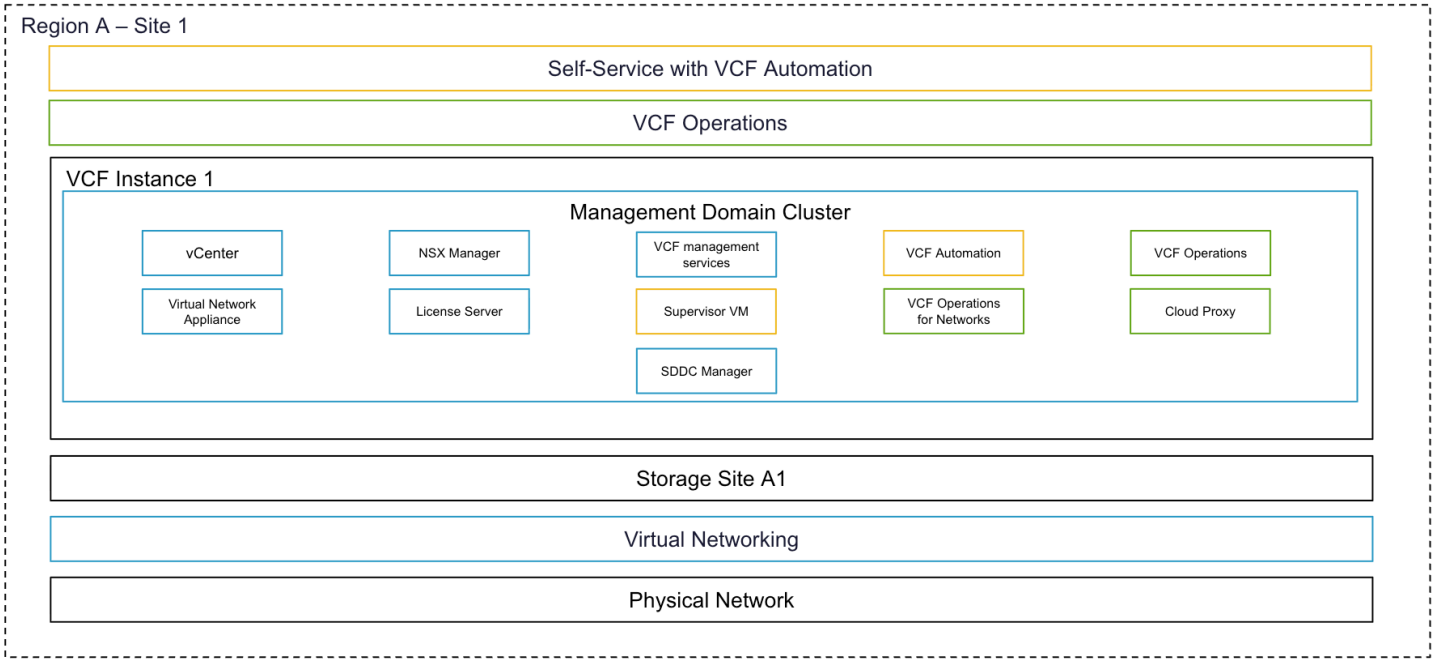
Type of Expansion	Method of Expansion	Description
Resource Expansion	Add hosts to an existing vSphere cluster.	Adds compute capacity (and storage if using vSAN) with automated workload balancing and failover within the scope of the entire cluster. All ESX hosts in a cluster must have the same configuration and access to resources (networking, storage).
	Add vSphere clusters to an existing VCF domain.	Adds compute capacity (and storage if using vSAN) in an isolated scope from workloads in other clusters. Can have different configuration, networking and/or storage from other clusters. Automated failover and workload balancing does not span across clusters.
	Add workload domains to an existing VCF Instance.	Provides separate management and control plane (vCenter) from other workload domains, allowing for different configurations or to support separate locations or access.
	Add VCF Instances to an existing VCF fleet.	Provides separate management and networking from other instances, allowing for different configurations or to support separate locations, while maintaining a common consumption instance (VCF Operations, and VCF Automation.
Availability Expansion	Add Interconnected Site (in the same region)	Steps are required to create additional availability by deploying new or modifying the configuration of existing vSphere clusters in that site.
	Add Interconnected Site (in another region)	Steps are required to create additional availability by deploying new or modifying the configuration of existing vSphere clusters in that site.

VCF Fleet in a Single Site with Minimal Footprint

The *VCF Fleet in a Single Site with Minimal Footprint* is an infrastructure blueprint where the management domain of the first VCF Instance is deployed with a minimal footprint. All management and workload components are deployed in a single vSphere cluster.

Architecture Overview

Figure 32: VCF Fleet in a Single Site with Minimal Footprint



Design Profile

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint is suited to customers who are looking for the following design profile for their VMware Cloud Foundation platform.

Attribute	Value
Consumption	• Self-service consumption via VCF Automation supporting modern workloads like VKS, Data Services, AI Services, and so on. with multi-tenancy. • Direct vCenter consumption supporting modern workloads like VKS, Data Services, AI Services, and so on. This infrastructure blueprint can use the following consumption blueprint: • Self-Service Multi-Tenant Private Cloud Consumption
Physical Site Configuration	Single Site (One VCF instance)
Availability	• Tolerates failure of a single ESX host. • Tolerates failure of an individual network path.
Isolation	• Hypervisor-based (logical isolation). • Tenant/Tenant isolation.
Recoverability	This infrastructure blueprint can use the following protection blueprints: • VCF Component Backup and Restore • VCF Instance Backup and Restore
Expansion	For available resource and availability expansions, see Expansion after Infrastructure Blueprint Deployment.

Management Domain Design Selections

Layer	Selected Model	Additional Information
Consumption	Simple VCF Automation Model	Consists of a single node that is deployed in a vSphere cluster. Availability is provided by the native vSphere HA functionality within the vSphere cluster.
	Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability is provided using vSphere HA.
	Simple vSphere Supervisor Control Plane Model	The *Simple vSphere Supervisor Control Plane Model* deploys a single vSphere Supervisor control plane VM into the selected Supervisor management zone.
	Tenancy Deployment Model 1: Consolidated Management and Organization Workloads in a VCF Instance	This tenancy deployment model consists of one VCF domain that hosts the management components and organization workloads.
	Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for vSphere Supervisor and VKS workloads, it supports VPC-based networking.
Operations	Simple VCF Operations Model	Deploys a single node of the VCF Operations components. Availability is provided by the vSphere HA. Nodes: • Primary appliance • Cloud Proxy appliance
	VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services instance with a combination of fleet, and instance level components for managing all VCF Instances in the fleet.

Layer	Selected Model	Additional Information
	Simple VCF Management Services Model	The VMware Cloud Foundation platform includes a comprehensive suite of management services that are designed to automate and orchestrate private cloud operations at scale.
	Single License Server Model	The *Single License Server Model* is the default configuration for the VMware Cloud Foundation platform, designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.
	Simple VCF Operations for Networks Model	Single VCF Operations for networks nodes deployed in a vSphere cluster. Availability is provided by the native vSphere HA functionality within the vSphere cluster. Nodes: • Platform • Collector
	Simple Log Management Model	A simple log management deployment in the VCF fleet's primary VCF Instance management domain. Availability is provided by vSphere HA within a single vSphere cluster. The goal of the simple model is to consume minimal resources and require minimal configuration.
Security Governance and Compliance	Instance Identity Broker Model	Deploys the Identity Broker component on theVCF Management Services instance in the management domain.
	VCF Fleet-Wide Single Sign-On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all supported components across all VCF Instances in the same VCF fleet.
Virtual Infrastructure	Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.
	VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.
	vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.
	ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCFdomain vSphere cluster.
	Single-Rack vSphere Cluster Model	The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.
	Single Distributed Switch Model	The *Single Distributed Switch Model* provides a minimal deployment of networking using a single vSphere distributed switch for all network services such as management, vMotion, IP-based storage and, NSX.

Layer	Selected Model	Additional Information
	Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture that utilizes high-performance flash storage devices. vSAN ESAsupports only NVMe high-performance storage devices.
	Simple NSX Manager Model	This model addresses use cases requiring a minimal infrastructure footprint, lower scale, and lower availability requirements.
Physical Infrastructure	Single-Rack Network Fabric Model	The data center network fabric provides connectivity to the vSphere clusters. The *vSphere Cluster Models* and *Storage Models* chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a *Single-Rack vSphere Cluster Model*. Depending upon the storage type chosen, additional VLAN requirements may apply.

Minimum Requirements for VCF Fleet in a Single Site with Minimal Footprint

The *VCF Fleet in a Single Site with Minimal Footprint* outlines the minimum requirements to stand up this infrastructure blueprint.

VCF Installer Deployment Options

Deployment Model	Deployment Size
Simple	Small

Minimal IP Requirements

Requirement	Additional Information
Physical Networks	• ESX Management (3 IPs) • VM Management (24 IPs) • vMotion (3 IPs) • vSAN (3 IPs) • Host TEP (6 IPs) • (Optional) Edge TEP Network (4 IPs) — 2 IPs per Edge VM (x2 Edge VMs) • (Optional) Edge Uplink Network (2 IPs) — 2 IP per Edge VM (x2 Edge VMs)
DNS Entries	• vCenter • ESX — 3 Nodes • SDDC Manager • NSX — VIP — Node A • VCF Operations — Primary — Cloud Proxy • VCF Management Services — VCF Services Runtime — VCF Management Instance Service — VCF Fleet Component • VCF Automation • VCF Service Runtime for VCF Automation • License server • Identity Broker • Log Management • VCF Operations for Networks — Platform — Collector
Virtual Networks	• vSphere Supervisor Service CIDR — e.g. 10.96.0.0/23 • VPC Private IP Block (Control Plane) • Distributed Transit Gateway Private IP Block (Control Plane) • External IP Block (Control Plane and Workload/Organizations) • Private VPC Block for each Organization's VPCs • Private TGW IP Block for each Organization

First VCF Instance

Table 55: VMware Cloud Foundation Installer Deployment Profiles

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for the initial VCF instance within a VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	1x Small • 1x Control Plane Nodes • 2x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Large • 3x Control Plane Nodes • 4x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Medium	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
VCF Operations	1x Small	2x Small	3x Medium	3x Large
License Server	1x Default	1x Default	1x Default	1x Default
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard
VCF Automation	1x Small	1x Medium	3x Medium	3x Large
Resource Requirements	• vCPU: 76 • RAM: 251 GB • DISK: 7448 GB	• vCPU: 106 • RAM: 335 GB • DISK: 8422 GB	• vCPU: 184 • RAM: 656 GB • DISK: 11445 GB	• vCPU: 298 • RAM: 949 GB • DISK: 15357 GB

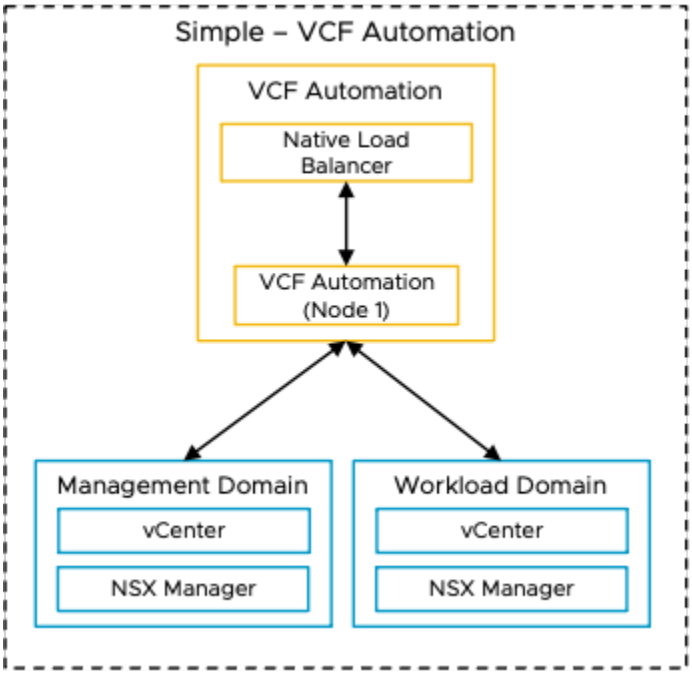
Management Domain Design Selections

The following models in the design align with the objectives for deploying this *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint.

Consumption Layer for VCF Fleet in a Single Site with Minimal Footprint
The consumption layer for the *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

Simple VCF Automation Model
The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Simple VCF Automation Model*.

Selected Model	Additional Information
Simple VCF Automation Model	Consists of a single node that is deployed in a vSphere cluster. Availability is provided by the native vSphere HA functionality within the vSphere cluster.



Simple VCF Automation Model Attributes

A *Simple VCF Automation Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out to high the availability model. **Note:** Scale-up to a larger node size and scale-out to three nodes at the same time.
Recoverability	Restore through a backup.

Simple VCF Automation Model Design Requirements

Table 56: Simple VCF Automation Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AUTO-REQD-SIM-001	Deploy VCF Automation as a single node in the default management domain vSphere cluster.	• Supports the simple deployment objective. • VCF Automation supports scale-out to a high availability model.	• The simple deployment model defaults to the small appliance size. • Administrators must be aware of the scale and performance implications of a single small-size node to their design.
VCF-AUTO-REQD-SIM-002	Protect the VCF Automation node using vSphere HA.	Ensures availability of VCF Automation without requiring manual intervention during an ESX host failure event.	None.

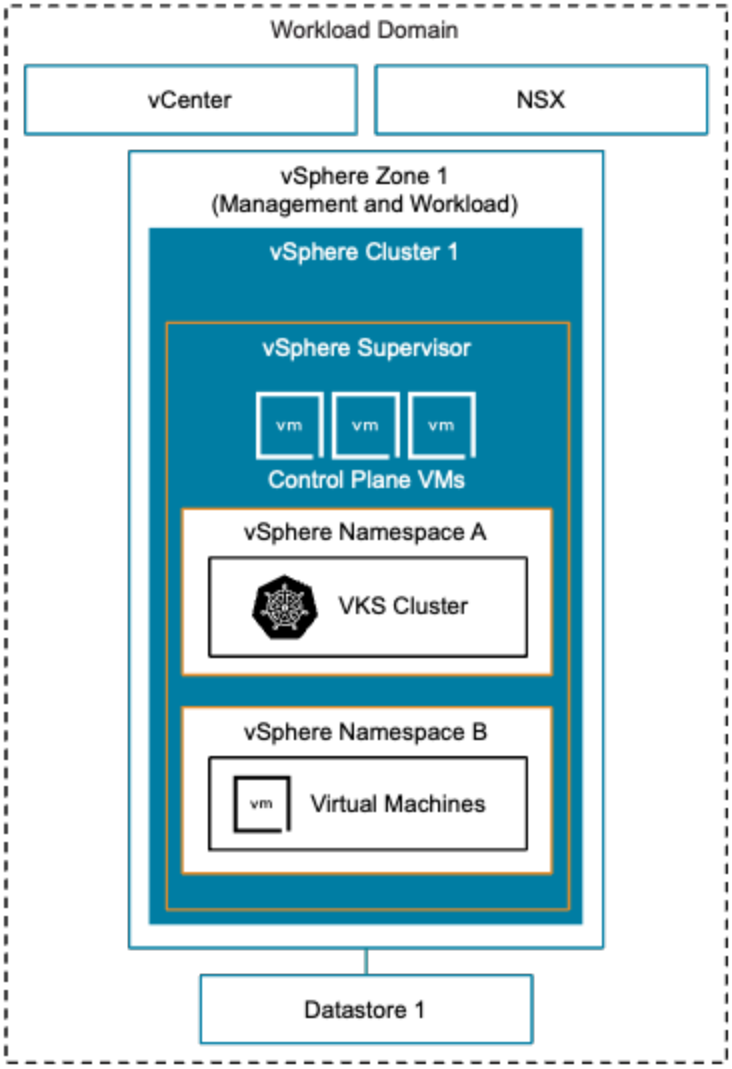
Single Management Zone with Combined Workload Zones Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Single Management Zone with Combined Workload Zones Model*.

Selected Model	Additional Information
Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* uses resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.

Figure 33: Single Management Zone with Combined Workload Zones Model Diagram

This diagram illustrates an example of a *Single Management Zone with Combined Workload Zones Model* deployment.



Single Management Zone with Combined Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single - Combined with management
vSphere Zone Datastore Type	Zonal datastore

Attribute	Detail
Activation	• At workload domain creation. OR • From vCenter.

vSphere Supervisor Design Requirements for All Zone Models

Table 57: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

vSphere Supervisor Single Management Zone Design Requirements

Table 58: vSphere Supervisor Single Management Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-SMZ-REQD-CFG-001	One available vSphere Zone that is backed by a supported vSphere cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	vSphere Zone for Supervisor management workload use.	The selected vSphere Zone is dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.

vSphere Supervisor Common Networking Design Recommendations

Table 59: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Design Recommendations

Table 60: vSphere Supervisor Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

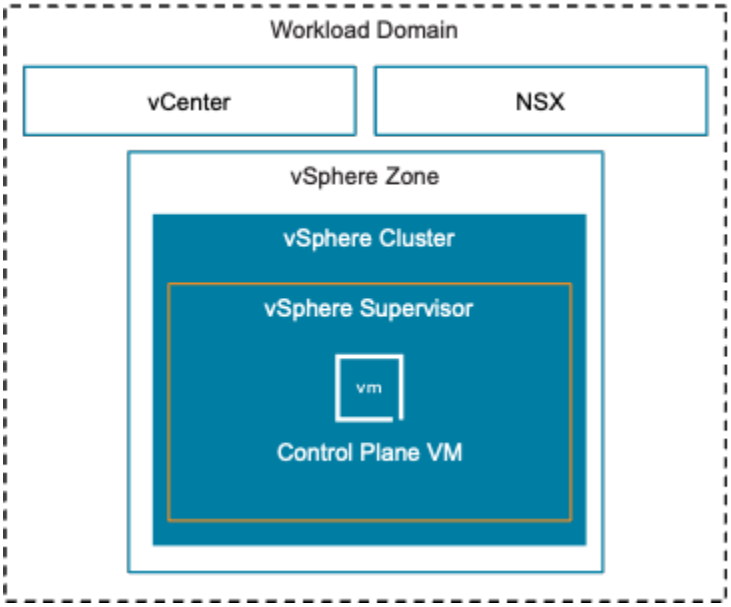
Simple vSphere Supervisor Control Plane Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Simple vSphere Supervisor Control Plane Model*.

Selected Model	Additional Information
Simple vSphere Supervisor Control Plane Model	The *Simple vSphere Supervisor Control Plane Model* deploys a single vSphere Supervisor control plane VM into the selected vSphere Supervisor management zone.

Figure 34: Simple vSphere Supervisor Control Plane Model

This diagram illustrates an example of a *Simple vSphere Supervisor Control Plane Model* deployment.



Simple vSphere Supervisor Control Plane Model Attributes

Attribute	Detail
Control Planes Nodes	One
Availability	Relies on vSphere HA
Scalability	Scale up and scale out
Placement	Hosted in a single vSphere Supervisor management zone.
Applicable Zone Models	• Simplified vSphere Supervisor Model • Single Management Zone with Combined Workload Zones Model • Single Management Zone with Isolated Workload Zones Model

vSphere Supervisor Control Plane Model Design Requirements

Table 61: vSphere Supervisor Control Plane Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-CP-REQD-CFG-001	Five consecutive management IP addresses routable to vCenter NSX, and optionally Avi management networks if Avi Load Balancer are used.	Required for integration and management communications.	Requires precise IP address management.
VCF-SUP-CP-REQD-CFG-002	Three vSphere Zones are required for the High-Availability Model in a Three Management Zone Model.	Control pane nodes are evenly placed with one per zone to protect against a single vSphere cluster failure.	Additional vSphere zones each back by separate supported vSphere clusters are required.

Tenancy Deployment Model 1: Consolidated Management and Organization Workloads in a VCF Instance

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Tenancy Deployment Model 1: Consolidated Management and Organization Workloads* in a VCF Instance.

Selected Model	Additional Information
Tenancy Deployment Model 1: Consolidated Management and Organization Workloads in a VCF Instance	This tenancy deployment model consists of one VCF domain that hosts the management components and organization workloads.

Management components and workloads can share one vSphere cluster, or can be deployed on different vSphere clusters.

When adopting this model, carefully evaluate the potential implications, including resource contention, performance impact, security risks, and reduced availability that may arise from hosting organization workloads within the same cluster as management components.

Figure 35: Management and Organization Workload Sharing One vSphere Cluster within a vCenter

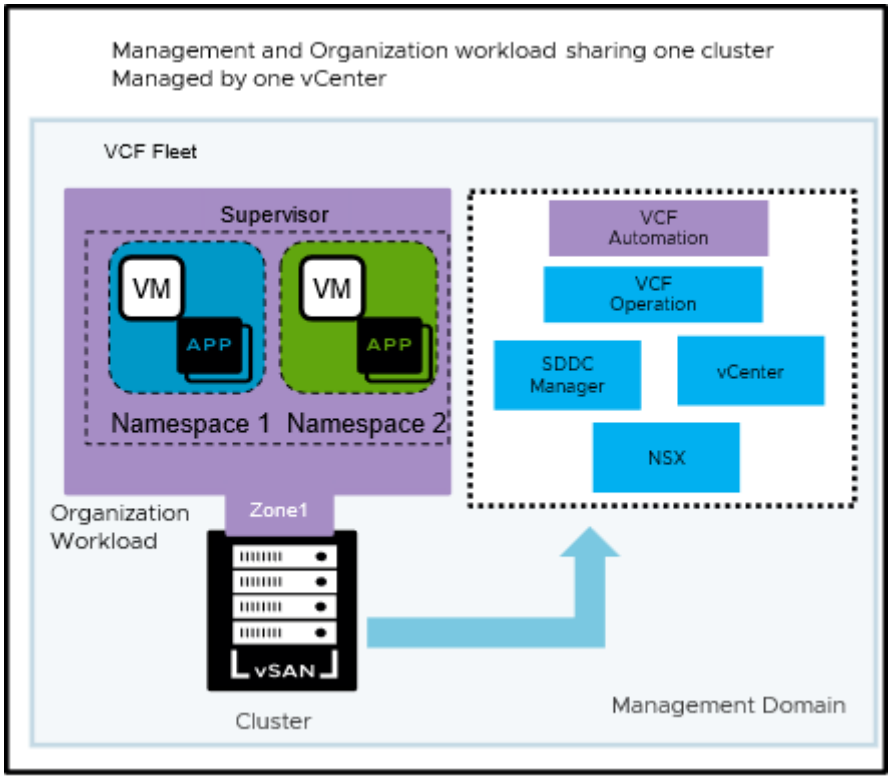
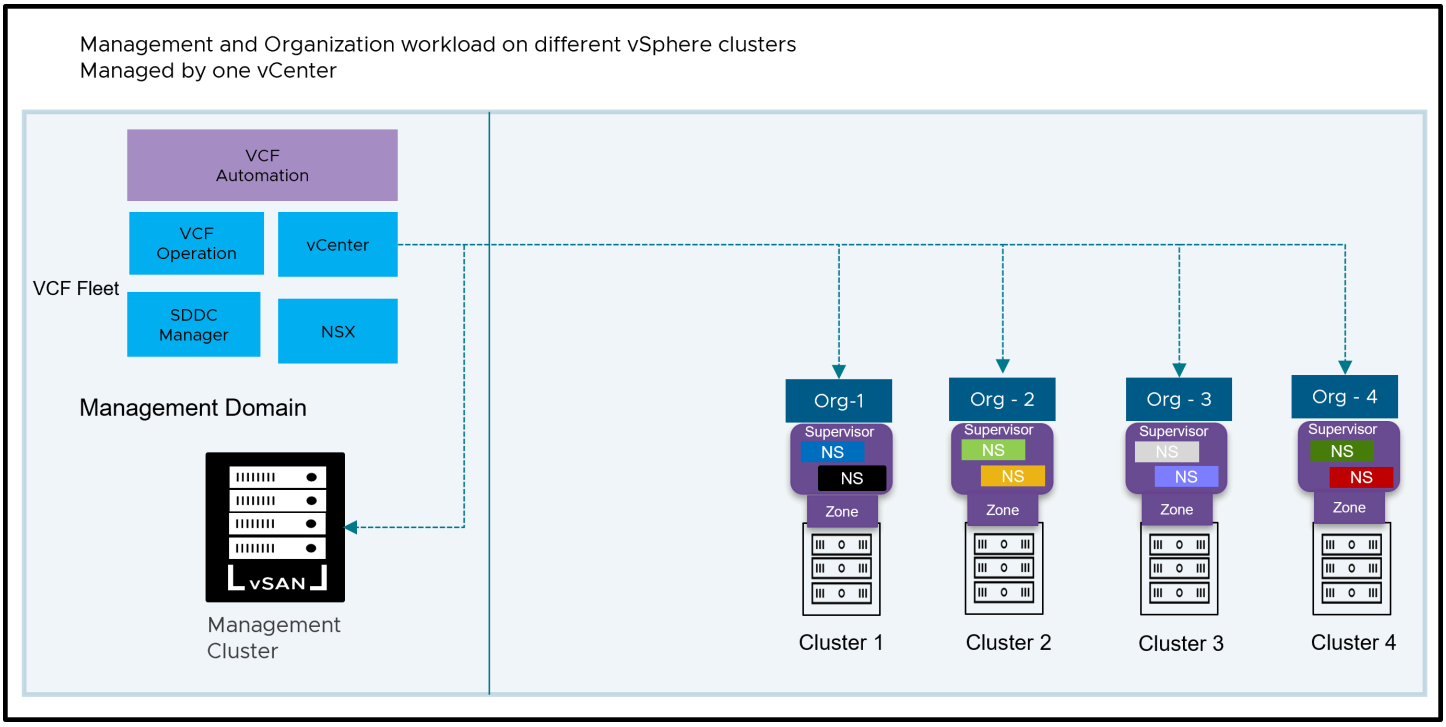


Figure 36: Management and Organization Workload on Different vSphere Clusters within in a vCenter



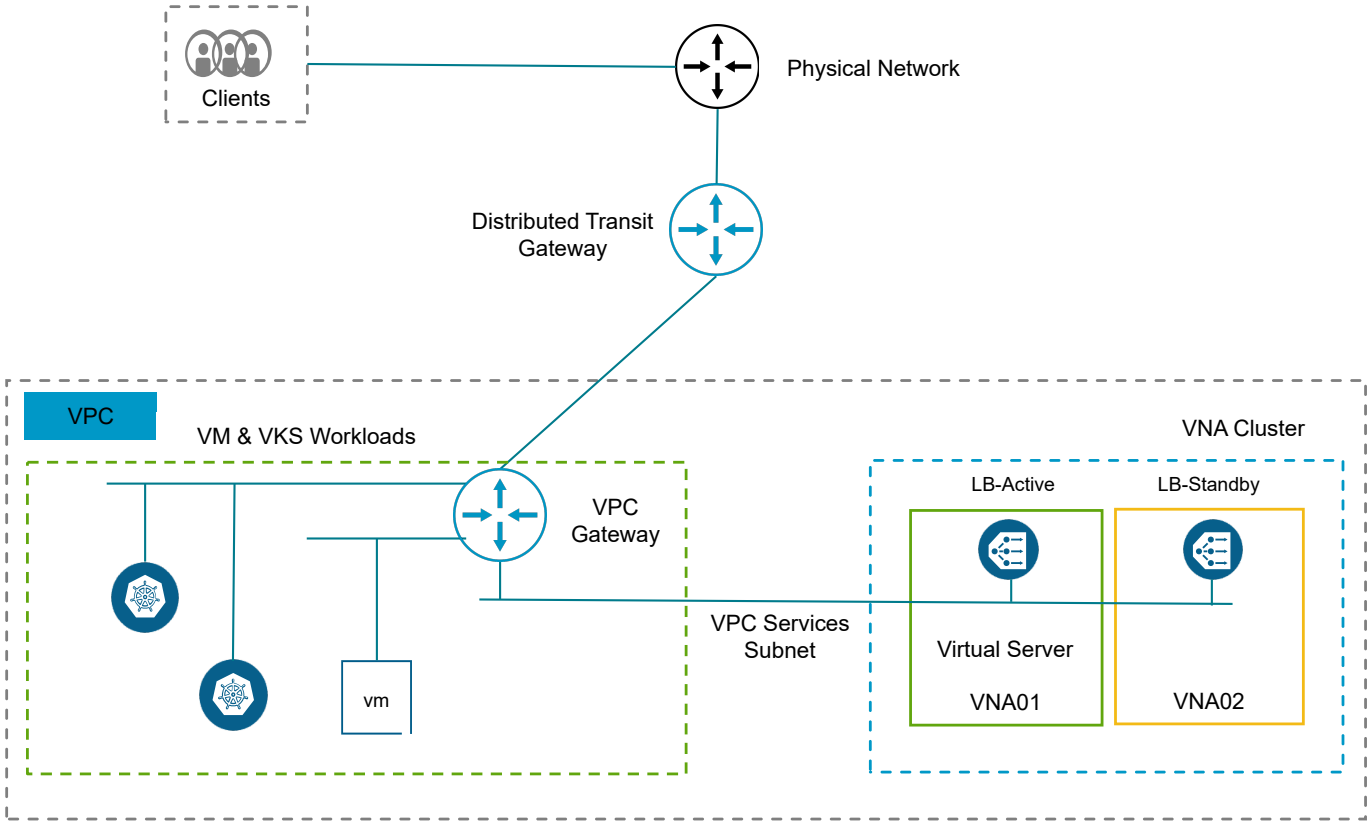
Benefits	Best Fit Scenarios
• Consolidating workloads can streamline the initial setup of a VCF Instance, which results in fewer physical hosts and clusters to configure, which speeds up the deployment process. • Using a single cluster lowers the overall hardware requirements that lead to cost savings in terms of server hardware, networking equipment, and data center space.	For organizations with smaller environments and limited resource requirements, consolidating workloads can provide a cost-effective and manageable solution.

Virtual Network Appliance Load Balancer Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Virtual Network Appliance Load Balancer Model*.

Selected Model	Additional Information
Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for vSphere Supervisor and VKS workloads. It supports VPC-based networking.

Figure 37: Virtual Network Appliance Load Balancer Model in VMware Cloud Foundation



Virtual Network Appliance Load Balancer Model Attributes

A *Virtual Network Appliance Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Load Balancing for VCF infrastructure components (i.e. VCF appliances) • Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services)
Networking Models Supported	VPC Networking Required
Entitlement	Included with VMware Cloud Foundation
Availability	Active/Standby
Scalability	• Scale up Virtual Network Appliance nodes in the Virtual Network Appliance cluster • Scale out a single Virtual Network Appliance Cluster • Create additional Virtual Network Appliance clusters
Deployment Model	Virtual Network Appliance Cluster with two or more Virtual Network Appliances. See Virtual Network Appliance Cluster Detailed Design for further details.

Virtual Network Appliance Load Balancer Model Options

Design Area	Options
Number of Virtual Network Appliances per cluster	1 to 10
Deployment Options	• Deploy with a new workload domain • Deploy Day 2 from NSX Manager • Deploy Day 2 from vCenter
Virtual Network Appliance Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger scale depoloyments where a large number of load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

Virtual Network Appliance Load Balancer Design Requirements and Recommendations

Table 62: Virtual Network Appliance Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-LB-REQD-CFG-01	Create a Virtual Network Appliance cluster with two or more Virtual Network Appliance nodes.	Provides availability for a load balancer instance.	None.
VCF-VNA-LB-REQD-CFG-02	Configure the VPC connectivity profile to use Virtual Network Appliance cluster in the VPC Service Gateway Configurations.	Enables the VPC to use the Virtual Network Appliance cluster for load balancer instances.	Virtual Network Appliance cluster can be shared between VPCs.
VCF-VNA-LB-REQD-CFG-03	Create a load balancer instance for the VPC using the API	Supervisor and VKS-based workloads, load balancer creation is automated.	No UI available to configure load balancer instances.
VCF-VNA-LB-REQD-CFG-04	Create a load balancer configuration for VPC using the API.	Supervisor and VKS-based workloads, load balancer configuration is automated.	No UI available to configure load balancer instances.

Table 63: Virtual Network Appliance Load Balancer Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-LB-RCMD-CFG-01	Configure failures domain for Virtual Network Appliance clusters with more than two Virtual Network Appliance nodes in the cluster.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-VNA-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from vCenter UI.

Operations Layer for VCF Fleet in a Single Site with Minimal Footprint

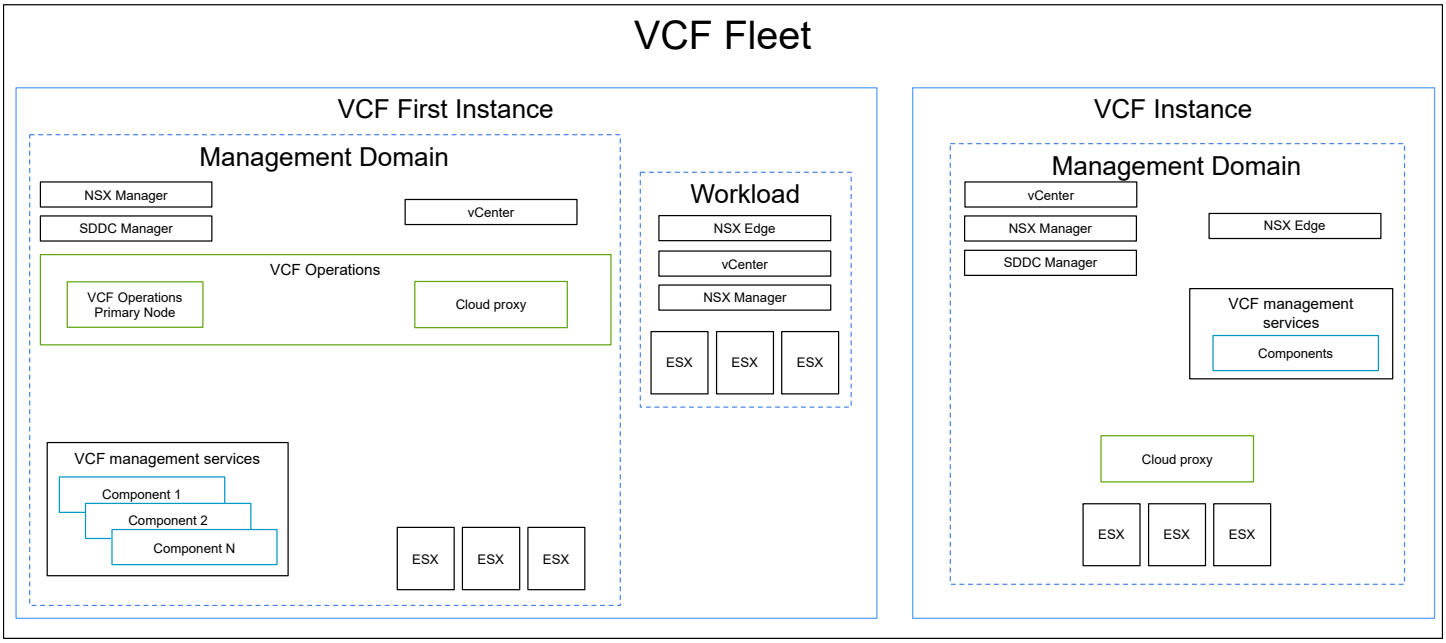
The operations layer serves as the central nervous system of the enterprise infrastructure and applications, providing comprehensive visibility, management, and control capabilities across the entire technology stack. This layer ensures that systems remain available, performant, secure, and compliant with organizational standards and regulatory requirements.

Simple VCF Operations Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Simple VCF Operations Model*.

Selected Model	Additional Information
Simple VCF Operations Model	Deploys the following VCF Operations components and uses vSphere HA as an availability mechanism: • VCF Operations Primary Appliance • Cloud Proxy Appliance

Figure 38: Simple VCF Operations Model in VMware Cloud Foundation



Simple VCF Operations Model Attributes

A *Simple VCF Operations Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out. **Note:** Scale-out is supported for the VCF Operations data nodes and cloud proxy nodes.

Attribute	Detail
Recoverability	Restore through a backup.

Common VCF Operations Design Elements for All Models

Table 64: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

Table 65: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

Simple VCF Operations Model Design Requirements

Table 66: Simple VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-SIM-001	Deploy VCF Operations as a single node in the default management vSphere cluster.	• Supports the simple deployment objective. • Provides the capacity that is required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out to a high availability model.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • No application level high availability.

VCF Management Services in the First VCF Instance Model

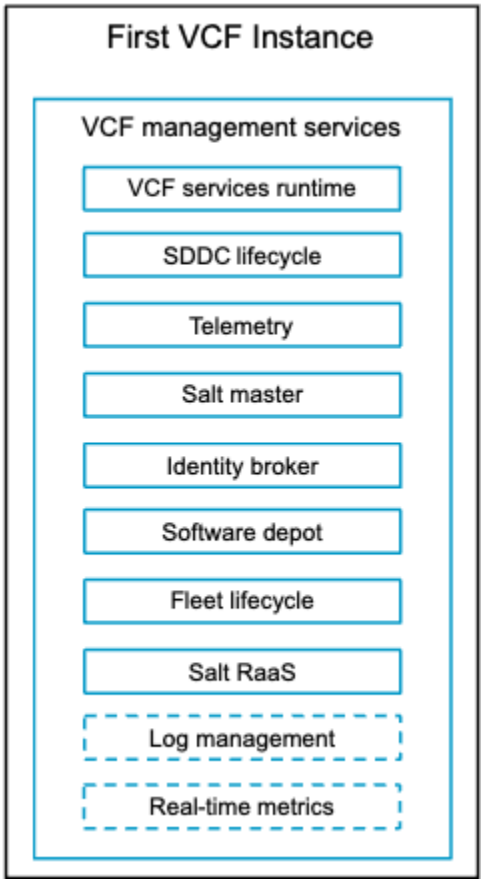
The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *VCF Management Services in the First VCF Instance Model*.

Selected Model	Additional Information
VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.

VCF Management Services in the First VCF Instance Model

Figure 39: VCF Management Services in the First VCF Instance Model in VMware Cloud Foundation

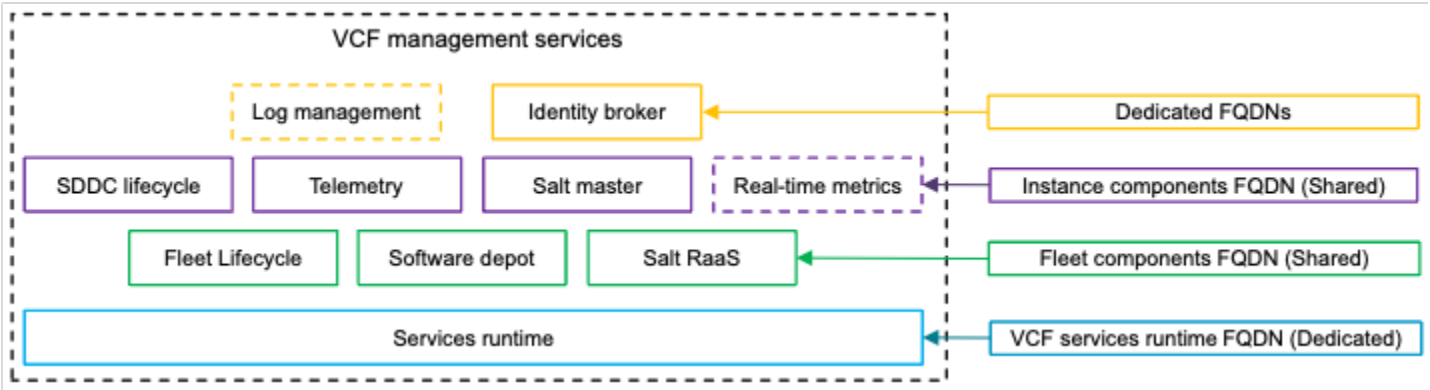
The diagram below shows the various components for this model. The dotted line components are deployed day-N.



VCF Management Services in the First VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 40:



VCF Management Services in the First VCF Instance Model Design Requirements

Table 67: VCF Management Services in the First VCF Instance Model Design Requirements

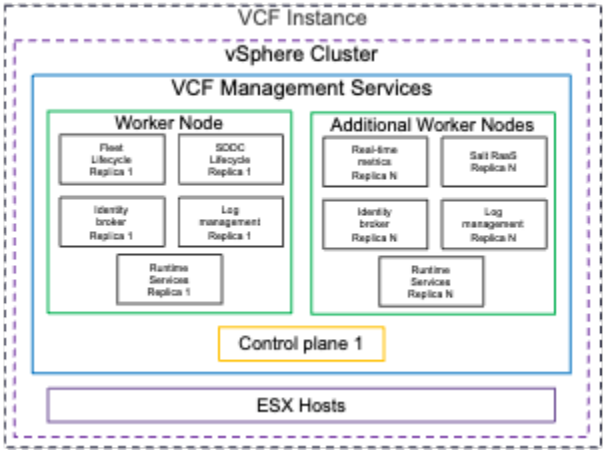
Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-FIRST-001	Create DNS entries for the following shared FQDNs: • Fleet components FQDN • Instance components FQDN	Required for deploying VCF management services fleet and instance level components in the first VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-FIRST-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN • Identity broker FQDN	Required for deploying VCF management services in the first VCF Instance.	You must create the DNS entries before starting the deployment.

Simple VCF Management Services Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Simple VCF Management Services Model*.

Selected Model	Additional Information
Simple VCF Management Services Model	The *Simple VCF Management Services Model* deploys a single control plane node and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Figure 41: Simple VCF Management Services Model in VMware Cloud Foundation



Simple VCF Management Services Model Attributes

Attribute	Detail
Recommended Usage	• Proof of Concept • Lab deployments
Deployment Size	Small **Note:** Refer to VCF Fleet Sizing Models for more details on sizing.
Availability	• Relies on vSphere HA. • Loss of the single control plane node would result in a need to redeploy the cluster and restore from backup.
Scalability	Scale out to a medium or large high availability deployment model is supported.

Attribute	Detail
Placement	Hosted on a single vSphere cluster.

Common VCF Management Services Model Design Requirements

Table 68: Common VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-001	Allocate a sufficient IP range for the required number of control plane and worker nodes for your chosen deployment type.	Required for initial deployment.	Adding additional components may require auto-scale out of the VCF management services.
VCF-MS-REQD-COM-002	Use `Kubernetes Cluster API Provider vSphere (CAPV)` node anti-affinity to ensure the nodes run on different ESX hosts.	Using `CAPV` prevents the VCF management services nodes from running on the same ESX host and risking the high availability of the cluster.	Distributed Resource Scheduler (DRS) anti-affinity rules are not required.
VCF-MS-REQD-COM-003	Protect the VCF management services nodes by using vSphere HA.	Ensures availability of the VCF management services nodes without requiring manual intervention during an ESX host failure event.	None.

Simple VCF Management Services Model Design Requirements

Table 69: Simple VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-SIM-001	Deploy VCF management services with a single control plane node and three worker nodes in the default management vSphere cluster.	• Supports the simple deployment objective. • VCF management services supports scale-out and up to a high availability model.	Loss of the single control plane node would result in a need to redeploy and restore from backup.

Single License Server Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Single License Server Model*.

Selected Model	Additional Information
Single License Server Model	The *Single License Server Model* is the default configuration for the VMware Cloud Foundation platform. The *Single License Server Model* is designed to provide centralized license management for the entire fleet and is suitable for deployments where reduced operational overhead is prioritized.

Table 70: Single License Server Model Attributes

Attribute	Detail
Use Cases	This is the typical license server deployment model for VMware Cloud Foundation and is suitable for most VCF deployments.
Deployment mechanism	One license server per VCF fleet. A license server is automatically deployed as part of the installation or upgrade to VMware Cloud Foundation 9.1.
Backup and restore	Backup is not required. If the original license server becomes unavailable, deploy a new license server instance and migrate the licenses within 90 days.

Attribute	Detail
Lifecycle	The upgrade and patch management of the license server is managed through VCF Operations.
High Availability	High availability is provided by vSphere HA.
License Scope	A single license server provides licensing management for the whole VCF fleet.

Single License Server Model Design Elements

Table 71: Single License Server Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LIC-REQD-LS-001	Register the default license server in VCF Business Services console.	Registration allows the VCF Business Services console to add license allocations to the license server.	None.

Connected Mode Licensing for VCF Operations Design Recommendations

Table 72: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Disconnected Mode Licensing for VCF Operations Design Recommendations

Table 73: Licensing for VCF Operations Design Recommendations

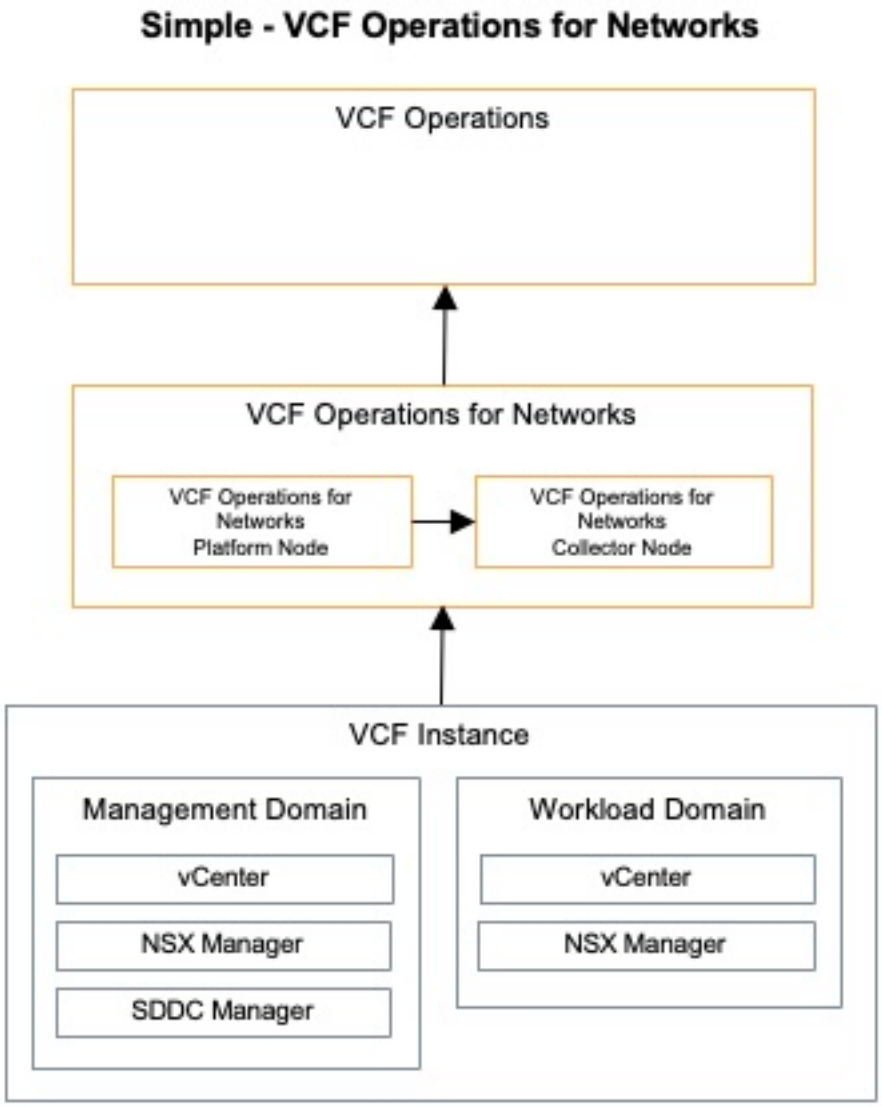
Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

Simple VCF Operations for Networks Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Simple VCF Operations for Networks Model*.

Selected Model	Additional Information
Simple VCF Operations for Networks Model	A single VCF Operations for networks nodes is deployed in a vSphere cluster. Availability is provided by vSphere HA.

Figure 42: Simple VCF Operations for Networks Model in VMware Cloud Foundation



Simple VCF Operations for Networks Model Attributes

A *Simple VCF Operations for Networks Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out.
Recoverability	Restore through a backup.

Simple VCF Operations for Networks Model Options

A *Simple VCF Operations for Networks Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Medium • Large • Extra Large

Simple VCF Operations for Networks Model Design Elements

Table 74: Simple VCF Operations for Networks Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-SIM-001	Deploy VCF Operations for networks as a non-clustered, single platform node in the default management vSphere cluster.	• Provides capacity for monitoring of up to 10,000 virtual machine or objects. • Supports scale-out with additional VCF Operations for networks platform nodes to a platform cluster.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks platform node will be applied.
VCF-NET-REQD-SIM-002	Deploy a single VCF Operations for networks collector node in the default management vSphere cluster.	A minimum of one VCF Operations for networks collector node is required to provide the VCF Operations for networks platform node with network monitoring data.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks collector node will be applied.

Table 75: Simple VCF Operations for Networks Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-SIM-001	Deploy an appropriately sized VCF Operations for networksplatform node for your needs.	Ensures resource availability and usage efficiency.	None.
VCF-NET-RCMD-SIM-002	Protect all VCF Operations for networks nodes by using vSphere High Availability.	Supports the availability objective for VCF Operations for networks without requiring manual intervention during an ESX host failure event.	You must ensure sufficient spare capacity for vSphere High Availability failover operations.
VCF-NET-RCMD-SIM-003	In an environment with multiple VCF Instances, deploy the VCF Operations for networks collector node(s) in the default management vSphere cluster in each VCF Instance by usingVCF Operations fleet management.	• Ensures a consistent deployment in each VCF Instances regarding network data collection. • It is assumed that each VCF Instances introduces a sufficiently high number of virtual machines, so that the deployment of dedicated VCF Operations for networks collector nodes is justified.	Each VCF Operations for networks collector node must be registered with a VCF Operations for networks platform node.
VCF-NET-RCMD-SIM-004	If the number of virtual machines exceeds 10,000 or if any other object scale limitations are reached, scale out to a VCF Operations for networks platform cluster deployment by adding additional platform nodes by usingVCF Operations fleet management.	Ensures that the VCF Operations for networks platform cluster has enough capacity to meet the SDDC object growth.	You must consider the additional resources required by VCF Operations for networks platform nodes in the management vSphere cluster when scaling out to a platform cluster.
VCF-NET-RCMD-SIM-005	For each vCenter data source, enable NetFlow on each vSphere Distributed Switch within the workload domain.	Provides the collection of network flows via the IPFIX protocol.	For the VCF domains, VCF Operations for networks will automatically update the NetFlow settings for each cluster's vSphere Distributed Switch.

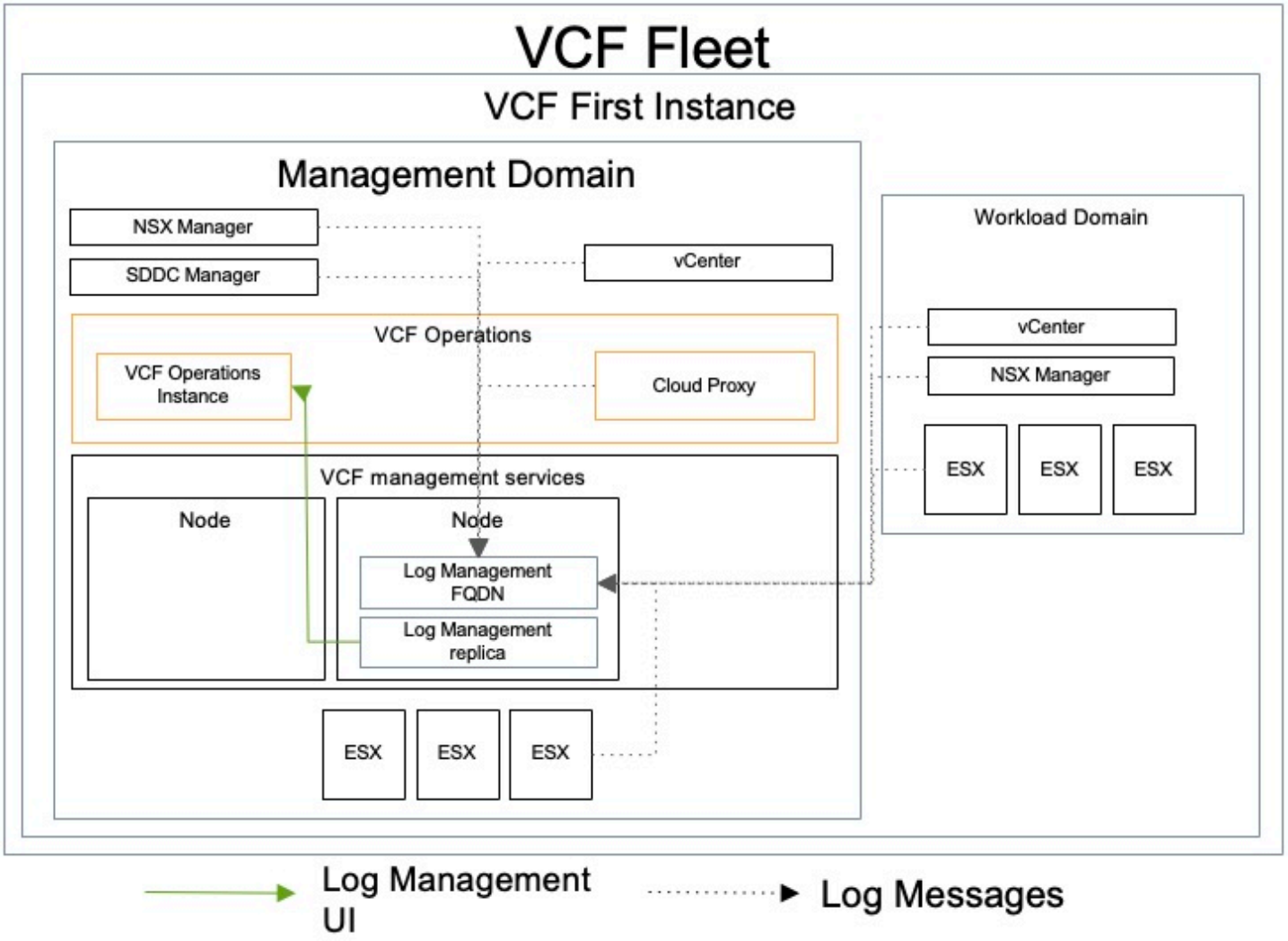
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-SIM-006	For each NSX data source, enable IPFIX for the distributed firewall.	Provides the collection of network flows via the IPFIX protocol.	The distributed firewall service must be enabled on the NSX Local Manager for the VCF workload domain.
VCF-NET-RCMD-SIM-007	For each NSX Manager data source, enable latency metric collection.	Provides the collection of latency metrics from NSX Transport Nodes.	Any firewall rule sets from all ESX hosts to the VCF Operations for networks collector must allow traffic on TCP 1991.
VCF-NET-RCMD-SIM-008	Use the default retention period of one month for the VCF Operations for networks platform node.	Keeping the default value for retention minimizes required storage resources administrative efforts.	None.
VCF-NET-RCMD-SIM-009	Use a CA-Signed certificate containing the fully qualified domain names (FQDNs) of each VCF Operations for networks platform and collector node in the SAN attributes, when deploying VCF Operations for networks.	Configuring a CA-Signed certificate ensures that the communication to the externally facing web UI and API VCF Operations for networks for is encrypted.	• Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered. • Each time a node is added the certificate must be replaced to include the fully qualified domain name of the additional node.
VCF-NET-RCMD-SIM-010	Use an SHA-2 or higher algorithm when signing certificates.	The SHA-1 algorithm is considered less secure and has been deprecated.	Not all certificate authorities support SHA-2.

Simple Log Management Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Simple Log Management Model*.

Selected Model	Additional Information
Simple Log Management Model	A basic log management deployment in the VCF fleet's primary VCF Instance management domain. Availability is provided by vSphere HA within a single vSphere cluster. The goal of the simple model is to consume minimal resources and require minimal configuration.

Figure 43: Simple Log Management Model in VMware Cloud Foundation



Simple Log Management Model Attributes

A *Simple Log Management Model* has the following attributes.

Attribute	Detail
Centralized Management	Provides a centralized log management framework for vCenter, ESX, NSX, VCF Operations, VCF Automation, and identity broker.
Availability	Relies on VCF management services and vSphere HA to restart nodes in the event of an ESX host or log management component failure, resulting in potential outages for up to 5 minutes depending on the failure.

Attribute	Detail
Scalability	Supports scale-up, storage scale up, and scale out. **CAUTION:** Once a quantity (size or number of replicas) is scaled up, it cannot be scaled back down.
Recoverability	Restore configuration through a backup, restore logs from archives.

Simple Log Management Model Design Requirements

Table 76: Simple Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-SIM-001	Deploy log management as a single small-size replica.	• Supports the simple deployment objective. • Supports scale-out to a high availability model.	None.

Table 77: Common Log Management Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

Security Governance and Compliance Layer for VCF Fleet in a Single Site with Minimal Footprint

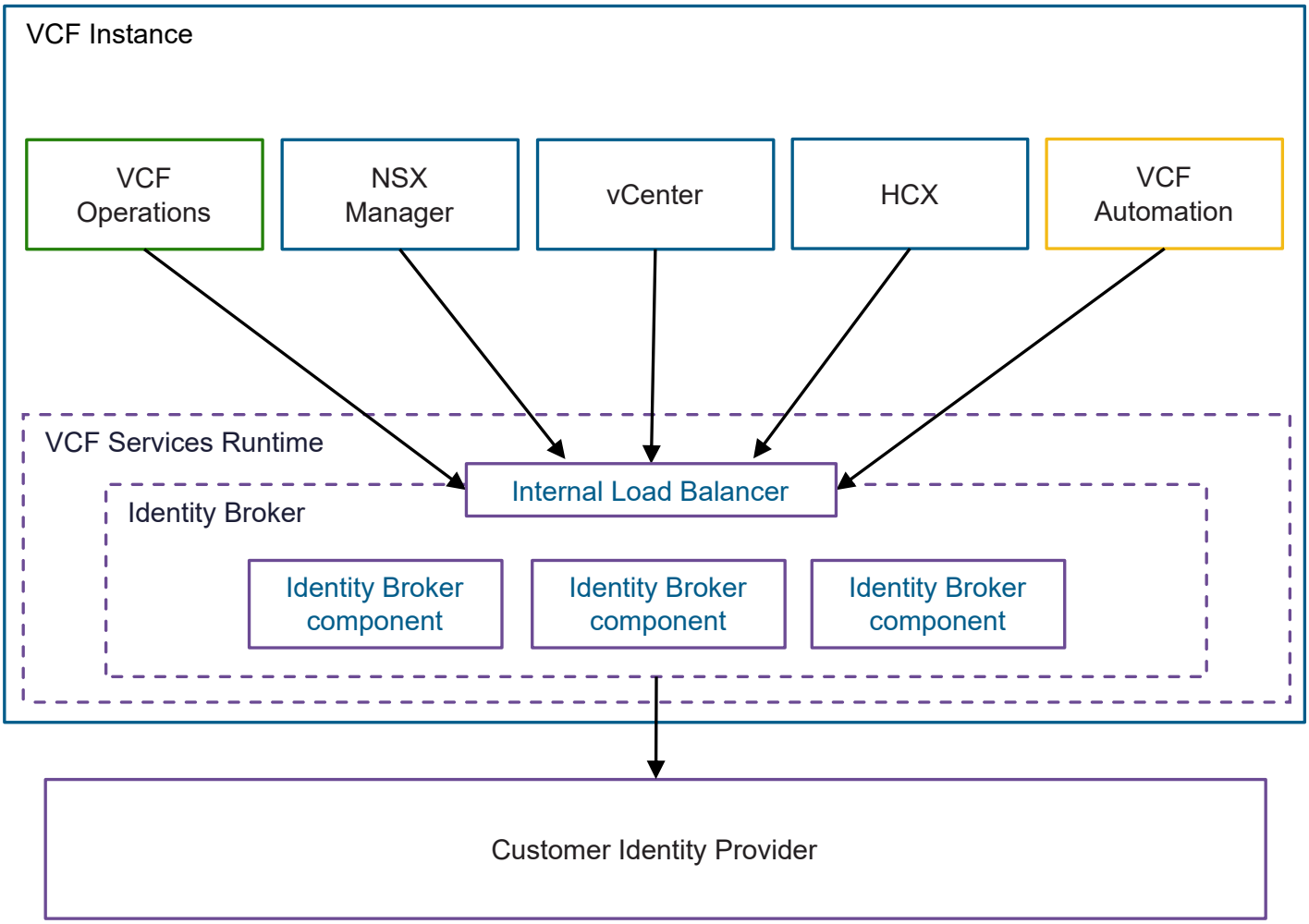
The security governance and compliance layer refers to the framework of policies, procedures, and controls that organizations establish to manage security risks while meeting regulatory, legal, and industry standards related to authentication.

Instance Identity Broker Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Instance Identity Broker Model*.

Selected Model	Additional Information
Instance Identity Broker Model	Deploys the Identity Broker component on the VCF Management Services instance in the management domain.

Figure 44: Instance Identity Broker Model in VMware Cloud Foundation



Instance Identity Broker Model Attributes

The *Instance Identity Broker Model* has the following attributes.

Attribute	Detail
Service Location	Identity broker runs as a service on VCF Management Services.
Deployment mechanism	An identity broker is automatically deployed during the installation or upgrade to VMware Cloud Foundation 9.1 in the first management domain. Additional identity brokers can be deployed in each VCF Instance as part of a day 2 operation.
Number of Nodes	multi-node cluster.
Scale considerations	• Scale up: Identity broker supports small, medium, and large sizes to match different configuration demands. • Scale out: Support for multiple identity broker instances to increase redundancy and throughput. • Recommendations: – Scaling up is preferred over scaling out when a single large-size identity broker meets security, resilience, and performance standards, as it reduces operational complexity and simplifies SSO management. – Schedule scale-up operations within a maintenance window, as these activities can impact active services.
Supported services	For support directories and IdPs, see Identity Providers and Protocols Supported for Single Sign-On.
Backup and restore	Use VCF management services backup and restore process.
Resiliency	Resiliency is provided by a distributed multiple-node deployment running each node on separate vSphere hosts.
Operational Impacts	As an isolated, external service, operational tasks performed on vCenter do not impact the authentication stack of the VMware Cloud Foundation components.
Migration path	Migrating from the appliance to embedded requires a reset of the VCF Single Sign-On configuration. Users must be re-added to components.

Instance Identity Broker Model Design Requirements

Table 78: Instance Identity Broker Model Design Requirements

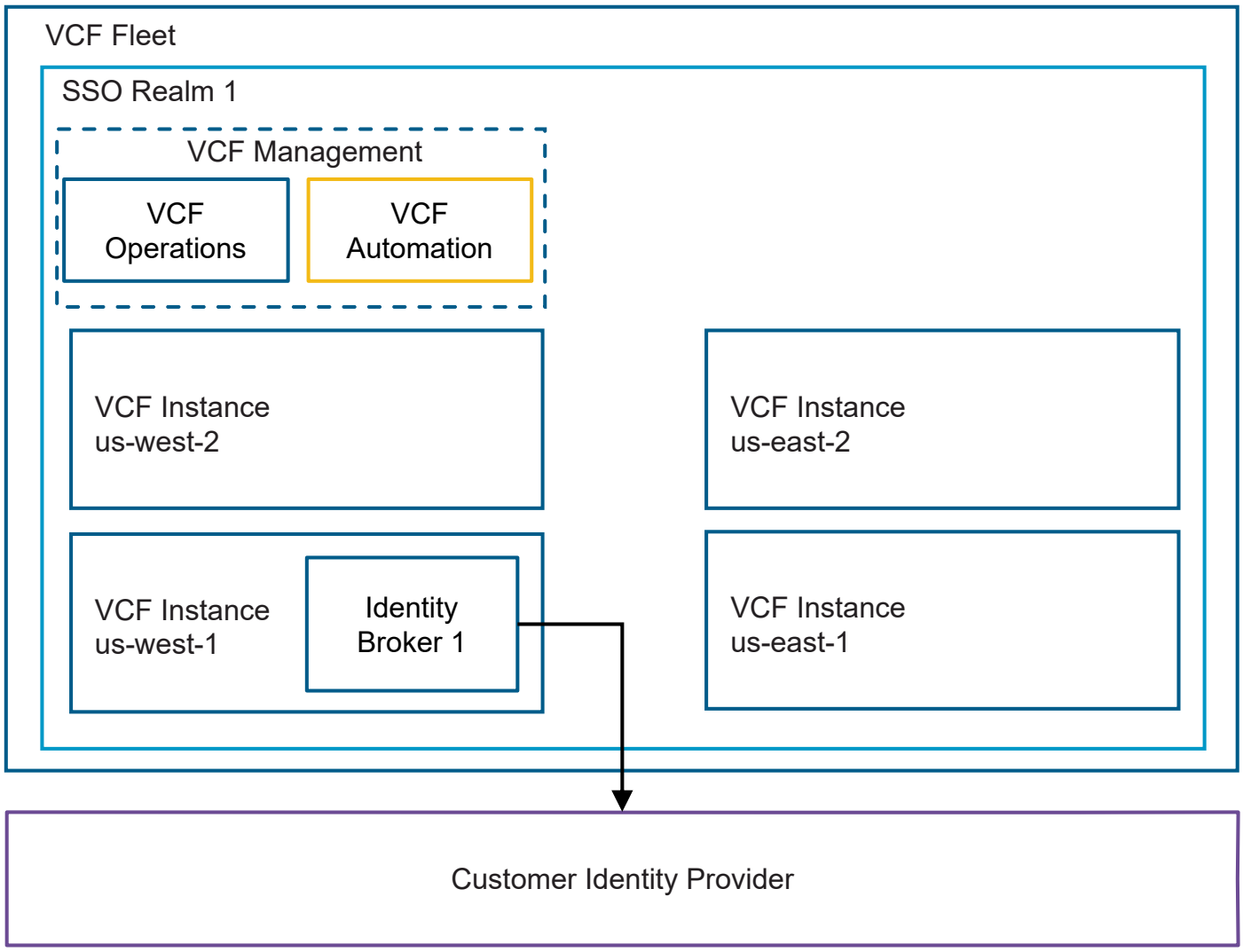
Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-EXT-001	Deploy additional identity broker instances as a Day N operation as needed.	• Provides high availability to the authentication source for a VCF Instance. • Provides additional scale.	Increases the operational overhead of managing the identity broker.
VCF-SSO-REQD-EXT-002	Connect the identity broker service to your corporate IdP.	Provides directory services (users and group) and user authentication service to the identity broker service.	None.

VCF Fleet-Wide Single Sign-On Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *VCF Fleet-Wide Single Sign-On Model*.

Selected Model	Additional Information
VCF Fleet-Wide Single Sign-On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all components across all VCF Instances in the same VCF fleet.

Figure 45: VCF Fleet-Wide Single Sign-On Model in VMware Cloud Foundation



VCF Fleet-Wide Single Sign-On Model Attributes

The *VCF Fleet-Wide Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**Low Scale -** A single identity broker services all VCF Instances in your VCF fleet.
Resources requirements	**Low Resource Footprint** - A single identity broker deployment is required for one VCF fleet.
Operational Overhead	**Low Operational Overhead** - An identity broker per VCF fleet will need to be managed separately for operational tasks such as backup and restore.
Availability	**Low Availability per Private Cloud** - A single identity broker servicing all VCF Instances in your VCF fleet.
Blast Radius	**Large Blast Radius** - An outage of one identity broker instance will impact logins for a single VCF fleet.
Single Sign-On Realm	**Large SSO Realm** - Users do not need to re-authenticate when moving between different VCF Instances within the same VCF fleet.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a only): Because the identity broker is hosted on the primary VCF instance, VCF instance restoration is required; Option 2b is not applicable. • Failure Scenario 3

VCF Fleet-Wide Single Sign-On Model Design Requirements

Table 79: VCF Fleet-Wide Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VFL-001	Configure additional VCF Instances in the VCF fleet to consume the same identity broker configured in the first VCF Instance.	• Provides an authentication source for all components in the VCF fleet. • Provides VCF Single Sign-On capabilities to all components in the VCF fleet.	An outage to the identity broker service impacts the VCF fleet.

Virtual Infrastructure Layer for VCF Fleet in a Single Site with Minimal Footprint

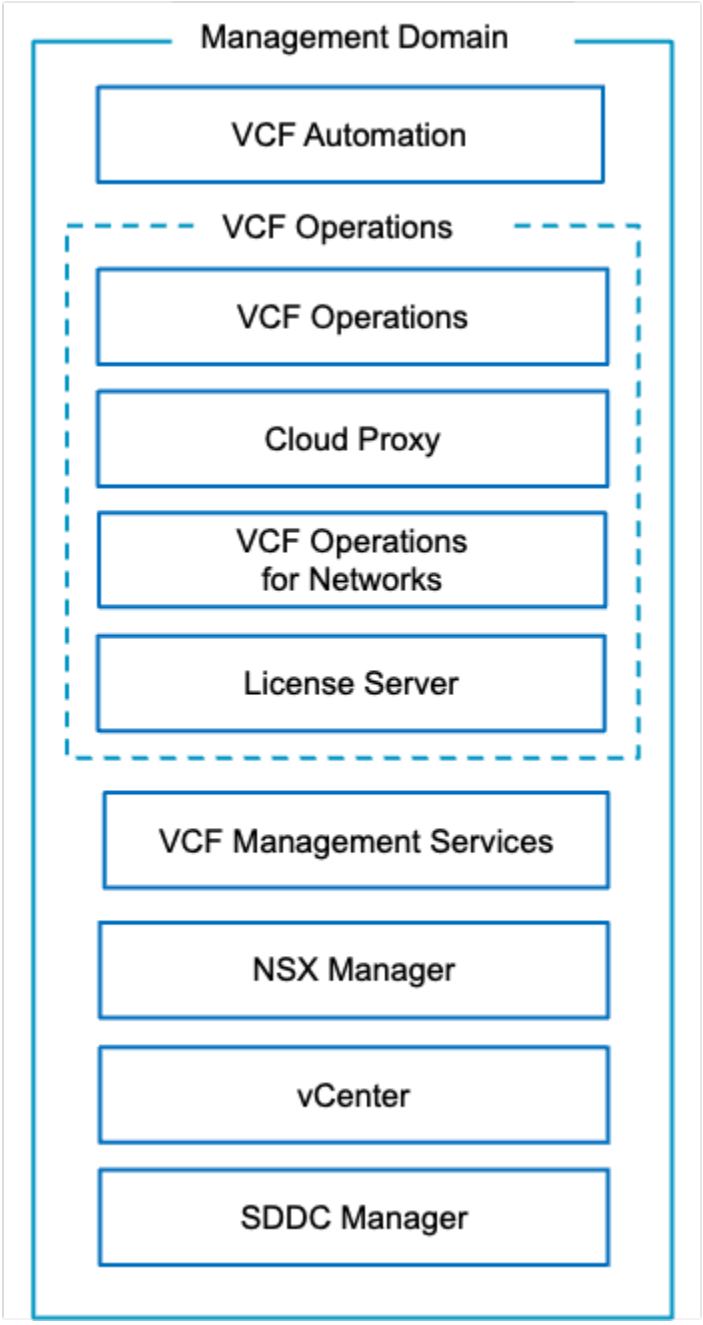
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's Software-Defined Data Center. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Management Domain Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Management Domain Model*.

Selected Model	Additional Information
Management Domain Model	The management domain hosts both the fleet-level and instance-level management components.

Figure 46: Management Domain Model in VMware Cloud Foundation



Management Domain Model Attributes

The *Management Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• VCF fleet management components. • VCF Instance management components. • NSX Edge nodes. • Can also run business workloads.
Management Domain in the First VCF Instance	• The management domain in the first VCF Instance contains the following management components: – VCF management services (fleet and instance level components) – VCF Operations – Cloud Proxy – VCF Operations for Networks – VCF Automation – ESX hosts – vCenter – NSX Manager – SDDC Manager
Management Domains in Additional VCF Instances	• Management domains in additional VCF Instances contain the following instance-level management components: – VCF management services (instance level components) – Cloud Proxy – ESX hosts – vCenter – NSX Manager – SDDC Manager

Management Domain Model Design Requirements

Table 80: Management Domain Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLD-REQD-CFG-001	Deploy a management domain to host all management components for the VCF Instance.	• Guarantees sufficient resources for management components. • Enables the use of specific hardware to meet only the needs of the management components. • Enables the use of dedicated physical compute, network, and storage separately from those used for additional workloads. • Enables separate lifecycle management of management and workload components.	• You must carefully size the management domain to accommodate planned deployments of additional workload domains and management components. • Hardware might not be fully utilized until full-scale deployment has been reached. • The management components for all greenfield VCF domains are deployed to the same cluster, and share the same subnet.

Management Domain Model Design Recommendations

Table 81: Management Domain Model Design Recommendations

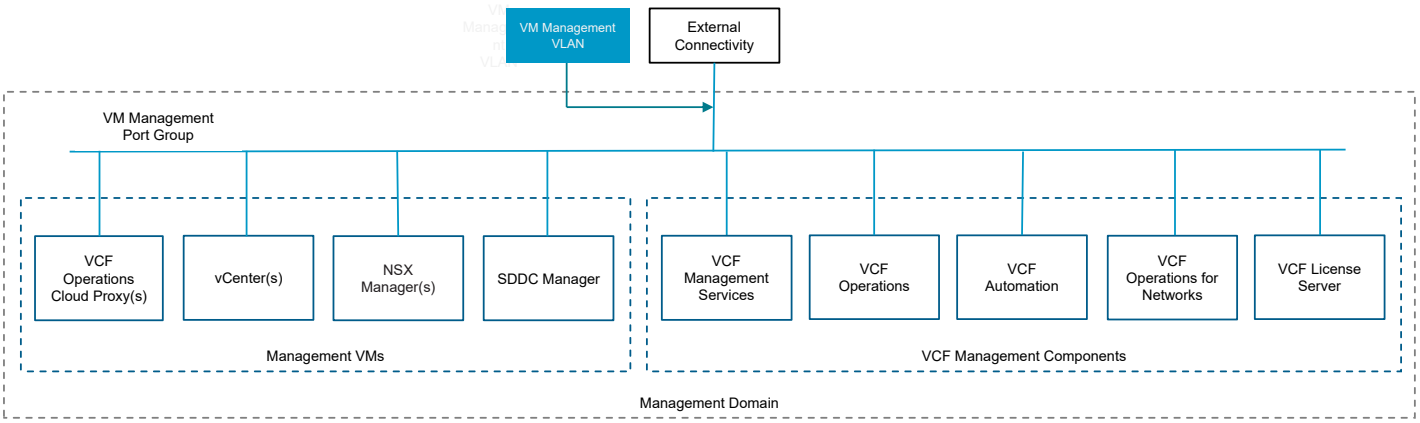
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-RCMD-NSX-001	Use a dedicated NSX Manager instance for the management domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Managers to manage and patch.

VCF Management Shared VLAN Network Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *VCF Management Shared VLAN Network Model*.

Selected Model	Additional Information
VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Figure 47: VCF Management Shared VLAN Network Model in VMware Cloud Foundation



VCF Management Shared VLAN Network Model Attributes

A *VCF Management Single VLAN Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	VLAN-backed vSphere distributed port group.
Network Isolation	No; Sharing VLAN with the VCF management domain components.
NSX Edge Cluster Required	No; Optional to provide NSX load balancer for VCF Operations. (requires additional NSX VLAN segment for Load balancer connectivity)
Stretching Network Between Availability Zones	• **VM management network** must be stretched in the physical network fabric between availability zones. • Provides automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• Not suited for a disaster recovery use case as the network is shared with management domain components. • Failover testing is not supported.

VCF Management Shared VLAN Network Model Design Requirements

Table 82: VCF Management Shared VLAN Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-SV-NET-REQD-001	Use the management domain vCenter VM management distributed port group for the VCF fleet level components.	No additional network required.	• No logical network isolation. • Done automatically during VCF Installer deployment.
VCF-MGMT-SV-NET-REQD-002	Use the same VLAN for VCF management components and the VCF Instance management virtual machines. e.g. vCenter and NSX Managers.	No additional VLAN required.	• Separate physical firewall rules may not be possible with a single VLAN. NSX Distributed Firewall may be an alternative solution. • Not suitable for Fleet Disaster Recovery use case with single VLAN.

VCF Management Shareed VLAN Network Model Design Recommendations

Table 83: VCF Management Shared VLAN Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-SV-NET-RCMD-001	Deploy the VCF management components during initial deployment using VCF Installer.	Simplified deployment using VCF Installer UI.	None.

vCenter Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *vCenter Model*.

Selected Model	Additional Information
vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

vCenter Model Attributes

A *vCenter Model* has the following attributes.

Design Area	Attributes
vCenter Deployment Type	• One vCenter instance for the management domain that manages the management components of the VCF Instance. • Optionally, additional vCenter instances for workload domains to support workloads. • vSphere HA protecting all vCenter appliances.
vCenter Single Sign-On	• Each vCenter has a distinct vCenter Single Sign-On domain.

Methods for Protecting vCenter

VMware Cloud Foundation supports multi-rack management vSphere clusters, vSAN stretched clusters, or 3rd party synchronous replication solutions such as metro clusters that are designed to provide disaster recovery and/or high availability for the entire management stack (vCenter, NSX Manager, etc.) as opposed to a single vCenter.

Table 84: Methods for Protecting vCenter in VMware Cloud Foundation

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere High Availability	Yes	You must ensure sufficient resources in the cluster to allow virtual machines to restart on another ESX host in the event of an ESX host failure.
vCenter High Availability (vCenter HA)	No	• vCenter services must fail over to the passive node so there is no continuous availability. • Recovery time can be up to 15 mins. • You must meet additional networking requirements for the private network. • vCenter HA requires additional resources for the passive and witness nodes. • Lifecycle management is complicated because you must manually delete and recreate the standby virtual machines during a lifecycle management operation.
vSphere Fault Tolerance (vSphere FT)	No	• The vCPU limit of vSphere FT vCPU would limit vCenter appliance size to medium. • You must provide a dedicated network.

vCenter Model Design Requirements and Recommendations

Table 85: vCenter Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-CFG-001	Deploy a dedicated vCenter appliance for the management domain of the VCF Instance.	• Isolates vCenter failures to management or customer workloads. • Isolates vCenter operations between management and business workloads. • Supports a scalable vSphere cluster design where you can reuse the management components as more customer workloads are added to the VCF Instance. • Simplifies capacity planning for customer workloads because you do not consider management workloads for the workload domain vCenter. • Improves the ability to upgrade the vSphere environment and related components by enabling for explicit separation of maintenance windows: — Management workloads remain available while you are upgrading the tenant workloads. — Workloads remain available while you are upgrading the management nodes. • Supports clear separation of roles and responsibilities to ensure that only administrators with granted authorization can control the management workloads. • Facilitates quicker troubleshooting and problem resolution. • Simplifies disaster recovery operations by supporting a clear separation between recovery of the management components and tenant workloads. • Provides isolation of potential network issues by introducing network separation of the vSphere clusters in the VCF Instance.	The vSphere cluster must have sufficient resources.
VCF-VCS-REQD-NET-001	Place all workload domain vCenter appliances on the VM management network in the management domain.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-SSO-001	Create all vCenter instances within a VCF Instance in their own unique vCenter Single Sign-On domain.	• Sharing vCenter Single Sign-On domains is not supported. • Enables isolation at the vCenter Single Sign-On domain layer for increased security separation.	• Each vCenter instance is managed through its own pane of glass using a different set of administrative credentials. • You must manage password rotation for each vCenter Single Sign-On domain separately.

Table 86: vCenter Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VCS-RCMD-CFG-001	Deploy an appropriately sized vCenter appliance for each workload domain.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate compute resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-002	Deploy a vCenter appliance with the appropriate storage size.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate storage resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-003	Protect workload domain vCenter appliances by using vSphere HA.	vSphere HA is the only supported method to protect vCenter availability in VCF.	vCenter becomes unavailable during a vSphere HA failover.
VCF-VCS-RCMD-CFG-004	In vSphere HA, set the restart priority policy for the vCenter appliance to high.	vCenter is the management and control plane for physical and virtual infrastructure. In a vSphere HA event, to ensure that the rest of the SDDC management stack comes up faultlessly, the workload domain vCenter must be available first, before the other management components come online.	If the restart priority for another virtual machine is set to highest, the connectivity delays for the management components are longer.

ESX Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *ESX Model*.

Selected Model	Additional Information
ESX Model	In the design of the ESX host configuration for your VCF platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCF domain and vSphere cluster.

ESX Hardware and Sizing Considerations

For more information on host sizing, refer to the VCF Planning and Preparation Workbook.

Table 87: ESX Hardware and Sizing Considerations

Hardware Element	Considerations
BIOS/Firmware	Ensure the ESX host hardware is on the Hardware Compatability List (HCL) and that you are running the latest supported BIOS and firmware.
CPU	• Total CPU requirements for the workloads that are running in the cluster. • Host failure and maintenance scenarios. • CPU over-commitment ratio guidelines – Management Domain • To ensure optimal performance and maintain resilience during host failures or maintenance, keep the management domain vCPU-to-pCPU over-commitment ratio at or below 2:1. **Note:** While higher ratios such as 3:1 or 4:1 are supported, you must closely monitor CPU ready times to ensure that the management cluster has enough capacity to run the workloads. – Workload domains • CPU over-commitment ratios for workload domains should be calculated on a case by case basis depending on the CPU intensity of the workloads. You must closely monitor CPU ready times to ensure optimal performance. • Number of physical cores, not logical cores. – Simultaneous multi-threading (SMT) in CPUs improve CPU performance however the performance enhancement will not be equivalent to 100% more CPU power.
Memory	• Total memory requirements for the workloads that are running in the cluster. • When sizing memory for the ESX hosts in a cluster, to reserve the resources of one host for failover or maintenance, set the admission control setting to N+1, which reserves the resources of one host for failover or maintenance. • *vSAN* OSA (Original Storage Architecture). – To support the maximum number of disk groups, you must provide 32 GB of RAM. • *vSAN* ESA (Express Storage Architecture). – You must provide a minimum of 128GB of RAM.
Storage	• Use high-endurance devices such as a hard drive or SSD for boot device. • Use a 128GB or greater boot device to maximize the space available for ESX-OS Data. • Use hosts with homogeneous configuration.
Network	• Use multiple physical NICs connected to multiple ToR switches to ensure redundancy. • 25GbE is recommended • Use physical NICs of the same speed, vendor, and model across all host within a cluster.
Power	• Use redundant power supplies, preferably connected to separate circuits to ensure availability. • Consider the impact of enabling Power Management on your ESX hosts. In order to allow ESX to control CPU power-saving features you need to set power management in the BIOS to *OS Controlled Mode*.

ESX Model Design Requirements and Recommendations

In your ESX design, you apply certain best practices.

Table 88: ESX Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-ESX-REQD-CFG-001	Install no less than the minimum number of ESX hosts required for the vSphere cluster model being deployed.	• Ensures that availability requirements are met. • If one of the ESX hosts is not available because of a failure or maintenance event, the CPU over-commitment ratio becomes 2:1.	None.
VCF-ESX-REQD-CFG-002	Ensure each ESX host matches the required CPU, memory, and storage specification.	Ensures that workloads will run without contention even during failure and maintenance conditions.	Assemble the server specification and number according to the sizing in the VCF Planning and Preparation Workbook which is based on projected deployment size.
VCF-ESX-REQD-SEC-001	Regenerate the certificate of each ESX host after assigning the ESX host an FQDN.	Establishes a secure connection with the VCF Installer during the initial deployment and prevents man-in-the-middle (MiTM) attacks.	You must manually regenerate the certificates of the ESX hosts before the deployment.

Table 89: ESX Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-001	When using vSAN as primary storage in the management domain, use vSAN ReadyNodes for each ESX host.	Your management domain is fully compatible with vSAN at deployment. **Note:** For information about the models of physical servers that are vSAN-ready, see vSAN Compatibility Guide for vSAN ReadyNodes.	• Hardware choices might be limited. • If you plan to use a server configuration that is not a vSAN ReadyNode, your CPU, disks, and I/O modules must be listed on the VMware Compatibility Guide under *CPU Series* and *vSAN Compatibility List* aligned to the ESX version.
VCF-ESX-RCMD-CFG-002	Allocate ESX hosts with uniform configuration across the default management vSphere cluster.	A balanced vSphere cluster has these advantages: • Predictable performance even during hardware failures. • Minimal impact of resynchronization or rebuild operations on performance.	You must apply vendor sourcing, budgeting, and procurement considerations for uniform server nodes on a per vSphere cluster basis.
VCF-ESX-RCMD-CFG-003	When sizing CPU, do not consider multi-threading technology and associated performance gains.	Although multi-threading technologies increase CPU performance, the performance gain depends on running workloads and differs from one case to another.	Because you must provide more physical CPU cores, costs increase and hardware choices become limited.
VCF-ESX-RCMD-CFG-004	Install and configure all ESX hosts in the default management vSphere cluster to boot using a 128-GB device or larger.	Provides ESX hosts that have large memory, that is, greater than 512 GB, with enough space for the scratch partition when using vSAN.	None.

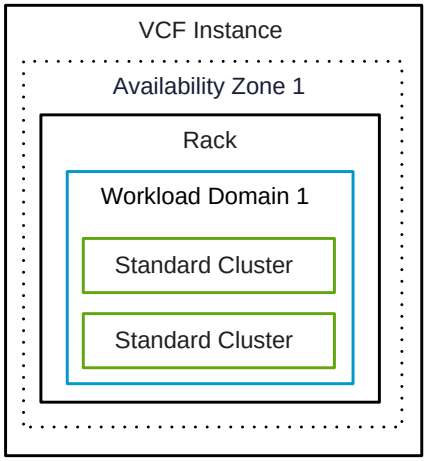
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-005	Use the default configuration for the scratch partition on all ESX hosts in the default management vSphere cluster.	• If a failure in the vSAN cluster occurs, the ESX hosts remain responsive and log information is still accessible. • If using vSAN storage its not possible to use a vSAN datastore for the scratch partition.	None.
VCF-ESX-RCMD-CFG-006	For workloads running in the default management vSphere cluster, save the virtual machine swap file at the default location.	Simplifies the configuration process.	Increases the amount of replication traffic for management workloads that are recovered as part of the disaster recovery process.
VCF-ESX-RCMD-NET-001	Place the ESX hosts in each management domain vSphere cluster on the ESX management network that is separate from the VM management network.	• Enables the separation of the physical VLAN between ESX hosts and the other management components for security reasons. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Increases the number of VLANs required.
VCF-ESX-RCMD-NET-002	Place the ESX hosts in each workload domain on a separate ESX host management VLAN-backed network.	Enables the separation of the physical VLAN between the ESX hosts in different workload domains for security reasons.	Increases the number of VLANs required. For each workload domain, you must allocate a separate management subnet.
VCF-ESX-RCMD-NET-003	Use dedicated VLANs for each traffic type.	Enables the separation of the physical VLAN for vSAN, vMotion, NFS & backup traffic.	Increases the number of VLANs required. For each traffic type you must allocate a separate VLAN and subnet.
VCF-ESX-RCMD-SEC-001	Set the advanced setting `UserVars.SuppressShellWarning` to `0` across all ESX hosts in the management domain.	• Ensures compliance with the *vSphere Security Configuration Guide* and with security best practices • Enables the warning message that appears in the vSphere client every time SSH access is activated on an ESX host.	You must turn off SSH enablement warning messages manually when performing troubleshooting or support activities.

Single-Rack vSphere Cluster Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Single-Rack vSphere Cluster Model*.

Selected Model	Additional Information
Single-Rack vSphere Cluster Model	The *Single-Rack vSphere Cluster Model* is the simplest form of protection, where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.

Figure 48: Single-Rack vSphere Cluster Model in VMware Cloud Foundation



Single-Rack vSphere Cluster Model Attributes

A *Single-Rack vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter.
Physical Rack Mapping	All cluster hosts exist in one physical rack.
Resilience	vSphere HA protects against host failures.

Single-Rack vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere which supports your storage type and VCF domain considerations.

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
Minimum number of ESX hosts	vSAN	Simple	Three (3)	• 33% CPU and memory • Tolerates one host failure	Three (3)	• 33% CPU and memory • Tolerates one host failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
	NFS or FC	Simple	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Important:

vSphere clusters are configured allowing for the failure of a single ESX host while still having enough resources to run all vSphere cluster workloads.

Attention:

* Designs should be completed in consultation with your storage vendor.

Common vSphere Cluster Design Requirements

Table 90: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Table 91: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Common vSphere Cluster Design Recommendations

Table 92: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Table 93: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

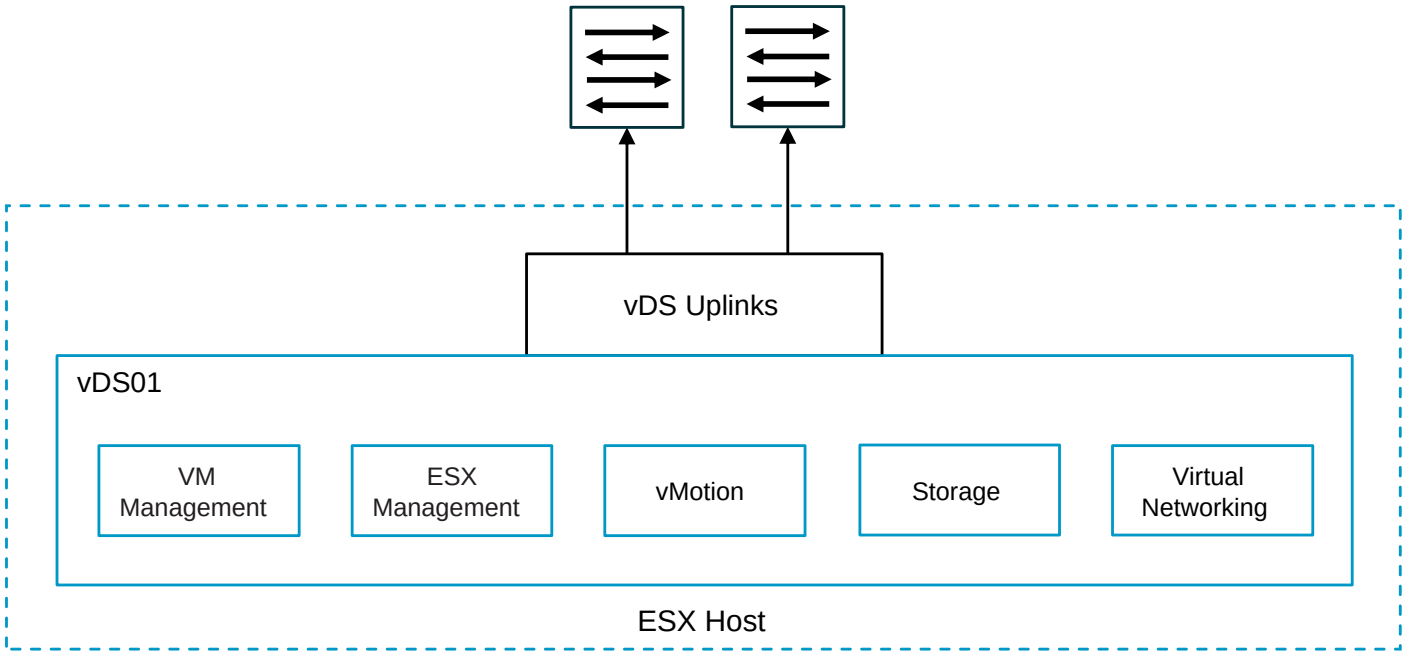
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Single Distributed Switch Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Single Distributed Switch Model*.

Selected Model	Additional Information
Single Distributed Switch Model	A *Single Distributed Switch Model* in VMware Cloud Foundationprovides a minimal deployment of networking using a single vSphere distributed switch for all network services such as management, vMotion, IP-based storage and, NSX.

Figure 49: Single Distributed Switch Model in VMware Cloud Foundation



Single Distributed Switch Model Attributes

A *Single Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Single
Number of Physical NICs per ESX Host	Two (2)
Bandwidth	Shared for all network traffic
Network I/O Control	Enabled by default

Single Distributed Switch Model Options

A *Single Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Physical NIC Level Redundancy (NLR)	• Use a single 2-port NIC (no NLR). • Use two separate 2-port NICs. – Each traffic type is spread across two separate NIC to avoid single point of failure. – Results in unused ports on NIC.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Common Distributed Switch Design Requirements

Single Distributed Switch Model Design Requirements

Table 94: Single Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-SDS-REQD-CFG-001	Create the following distributed port groups on the vSphere Distributed Switch: • VM management (mgmt domain only) • ESX management • vMotion • Storage	Provides the required network services for the vSphere cluster.	VM management distributed port group must be manually created on additional vSphere clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-SDS-REQD-CFG-002	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switch for each additional rack: • ESX management • vMotion • Storage (vSAN/NFS)	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack	Additional port groups are created for each additional rack.
VCF-VDS-SDS-REQD-CFG-003	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switch for availability zone 2 • ESX management • vMotion • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2.

Common Distributed Switch Design Recommendations

Table 95: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Single Distributed Switch Model Design Recommendations

Table 96: Single Distributed Switch Model Design Recommendations

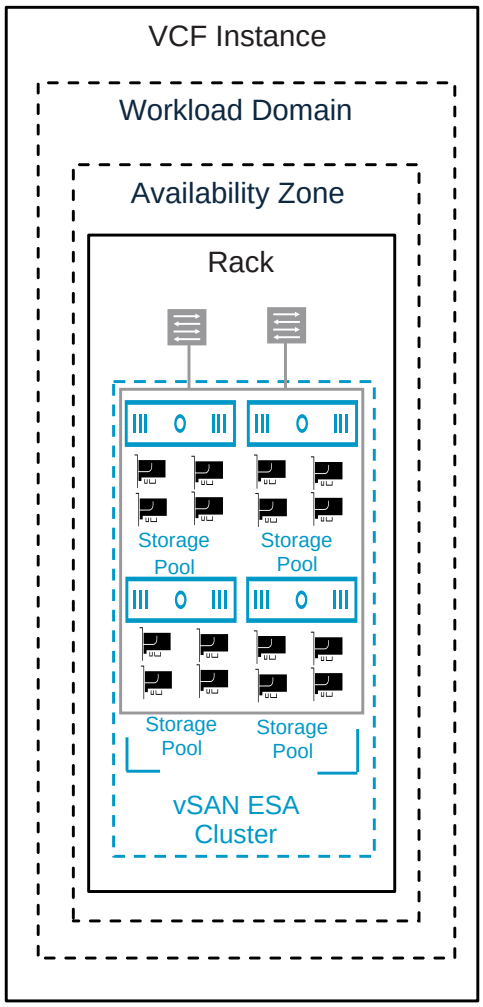
Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SDS-RCMD-DPG-001	Use the **Route based on physical NIC load** teaming algorithm for any storage distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-SDS-RCMD-NIO-001	Enable Network I/O Control on vSphere Distributed Switch (default setting).	Increases resiliency and performance when network traffic shares uplinks.	Network I/O Control might impact network performance for critical traffic types if misconfigured.
VCF-VDS-SDS-RCMD-NIO-002	Set the share value for management traffic to Normal (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than vSAN traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-SDS-RCMD-NIO-003	Set the share value for vMotion traffic to Low (default setting).	During times of network contention, vMotion traffic is not as important as virtual machine or storage traffic.	During times of network contention, vMotion takes longer than usual to complete.
VCF-VDS-SDS-RCMD-NIO-004	Set the share value for virtual machines to High (default setting).	Virtual machines are the most important asset in a VCF Instance. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-SDS-RCMD-NIO-005	Set the share value for principal IP-based storage traffic to High (default setting).	During times of network contention, storage traffic needs guaranteed bandwidth to support virtual machine performance.	None.
VCF-VDS-SDS-RCMD-NIO-006	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

Single-Rack vSAN ESA Storage Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Single-Rack vSAN ESA Storage Model*.

Selected Model	Additional Information
Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture with high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.

Figure 50: Single-Rack vSAN ESA Storage Model in VMware Cloud Foundation



Single-Rack vSAN ESA Storage Model Attributes

A *Single-Rack vSAN ESA Storage Model* has the following attributes.

Attribute	Detail
Single Tier	All data is stored on the same type of storage device, regardless of access frequency or importance.
Storage Pool	All NVMe Flash Storage.
Data Services	• Compression (enabled by default). • Encryption. (in transit and at rest) • vSAN Global Deduplication **Restriction:** vSAN Global Deduplication is not currently compatible with vSAN Stretched Clusters and Two Node vSAN Clusters. • Integrated vSAN File Services. • vSAN iSCSI Target Service. • vSAN Storage Clusters. • Stretched Cluster. • Fault domains. • Kubernetes persistent storage services (block and file). • Native scalable snapshots

Attribute	Detail
Virtual Machine Storage policies	vSAN storage policies (also known as Virtual Machine Storage Policies or VM Storage Policies) are rules defining virtual machine (VM) storage requirements that vSAN automatically enforces across a cluster. vSAN 9.1 automatically configures vSAN Auto-RAID policy on new vSAN ESA clusters. vSAN Auto-RAID policy manages the resilience for data residing in a cluster, regardless of the size or topology of the vSAN cluster. vSAN Auto-RAID creates a single policy within the vCenter server,

Common vSAN Storage Model Design Requirements

Table 97: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Common vSAN Host Network Model Design Requirements

Table 98: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

vSAN ESA Storage Model Design Requirements

Table 99: vSAN ESA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-ESA-REQD-CFG-001	Ensure vSAN certified Drives used for vSAN ESA clusters.	• When using VMware Cloud Foundation automated workflows, vSAN ESA cluster creation maybe blocked unless vSAN certified drives are detected. • vSAN Auto-RAID will be enabled when vSAN certified components are detected. This allows vSAN to generate an optimal default storage policy, based on the cluster topology and the number of ESX hosts.	None.
VCF-VSAN-ESA-REQD-CFG-002	1.6 TB drive size or higher for vSAN ESA devices for storage pools.	Minimum supported vSAN ESA drive size is 1.6TB.	vSAN ESA supports up to 24 drives per node. Smaller capacity drives may constrain raw capacity of a vSAN node.
VCF-VSAN-ESA-REQD-CFG-003	16 Cores or higher per vSAN node. vSAN ESA has a minimum value of 16 cores per vSAN Node	Minimum vSAN ESA core count per node is 16 for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	Smaller number of cores per node may require more vSAN nodes to support targeted workloads.
VCF-VSAN-ESA-REQD-CFG-004	Memory size per vSAN Ready node 128GB or higher	Minimum vSAN ESA memory is 128GB for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	• Smaller memory footprint per node will limit number of vSAN drives per node. • Smaller memory footprint per node for vSAN ESA will limit VM workloads being placed on vSAN nodes.

Common vSAN Storage Model Design Recommendations

Table 100: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Common vSAN Host Network Model Design Recommendations

Table 101: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

vSAN ESA Storage Model Design Recommendations

Table 102: vSAN ESA Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-001	• Select a specific size of drives used according to the requirements of target workloads. • Use multiple storage devices per vSAN node • NVMe TLC devices with a performance Class F (100000 - 349999) or higher with endurance class 3 DWPD or higher.	• The number of storage devices has an impact on the performance of vSAN ESA. • NVMe drives with mixed-use endurance and performance class typically classified on the vSAN HCL as "3DWPD" (3 Drive Writes per Day) will give a balance between performance and durability.	• Smaller drive sizes will impact overall capacity of vSAN ESA. Large drive sizes will introduce larger capacity fault domain in case of failure. • Sizing vSAN Nodes requires determining the workload capacity requirements prior to deployment. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-002	Use vSAN ready nodes with high core count.	vSAN Express Storage Architecture (ESA) is designed to be highly parallel and leverages multiple CPU cores for improved performance, making it multithreaded, and optimized for NVMe- TLC based flash devices.	There will be a correlation between target workload, number of cores and NVMe devices per vSAN Node. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

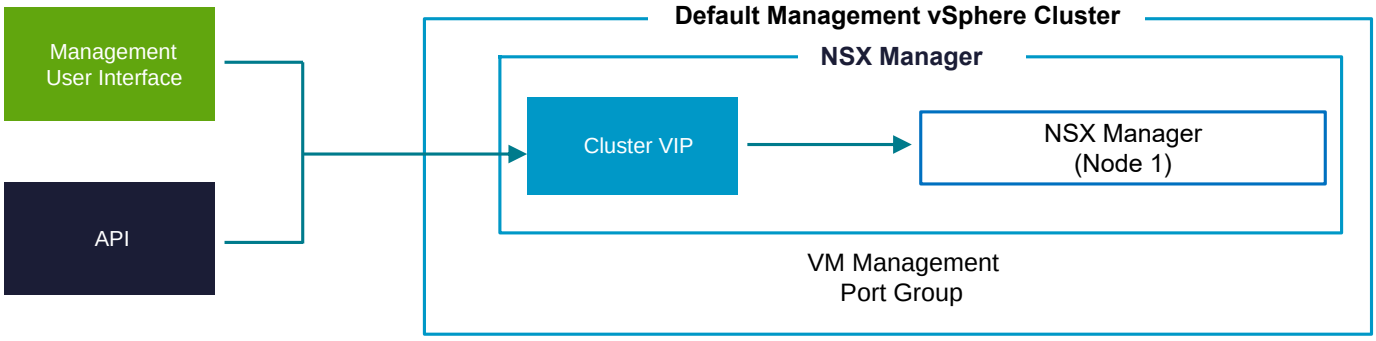
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-003	Use vSAN ready nodes with high memory capacity per vSAN node.	When enabled vSAN has required memory overheads and must have adequate resources account for VM workloads.	Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsanizer/home.
VCF-VSAN-ESA-RCMD-CFG-004	At least four (4) nodes for vSAN cluster size.	This ensures in the event of a fault domain failure, vSAN objects can be re-protected to maintain policy compliance. In conjunction with auto policy management, four (4) node vSAN ESA clusters will allow vSAN to automatically create a default storage policy that is capacity optimized.	None. **Note:** New vSAN clusters in VCF 9.1 will use clusters will automatically be configured to use Auto-RAID.
VCF-VSAN-ESA-RCMD-CFG-005	• Select the same minimum requirements for vSAN Storage cluster nodes as vSAN ESA HCI nodes. • Review vSAN Storage minimum node requirements	• vSAN Storage Cluster Nodes use the same architecture as regular vSAN ESA nodes. • vSAN ESA nodes can be re-purposed for vSAN Storage Clusters or vSAN HCI Clusters	None.

Simple NSX Manager Model

The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Simple NSX Manager Model*.

Selected Model	Additional Information
Simple NSX Manager Model	This model addresses use cases requiring a minimal infrastructure footprint, lower scale, and lower availability requirements.

Figure 51: Simple NSX Manager Model in VMware Cloud Foundation



Simple NSX Manager Model Attributes

The *Simple NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Single NSX Manager Appliance with Internal VIP.
Availability	• vSphere HA protects the node applying high restart priority. • Slower recovery from node failure.
Infrastructure Impact during Upgrade	• Management and control plane impacted during upgrade operations. • Deployment/Power-on/Restart/vMotion of VMs may be impacted.
Virtual Networking Scale	Reduced scale
API Request Scale	Limited scale

Common NSX Manager Design Requirements for All Models

Table 103: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

NSX Manager Simple Deployment Model Design Requirements

Table 104: NSX Manager Simple Deployment Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-004	Deploy a single NSX Manager node in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports the objective of a small-scale deployment.	• Slower recovery from node failure. • Management and control plane is impacted during upgrade operations. • Supports limited scale.

Common NSX Manager Model Design Recommendations for All Models

Table 105: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

Physical Infrastructure Layer for VCF Fleet in a Single Site with Minimal Footprint
The physical infrastructure layer constitutes the hardware and facilities that provide the foundational computing, storage, and networking resources.

Single-Rack Network Fabric Model
The *VCF Fleet in a Single Site with Minimal Footprint* infrastructure blueprint uses the *Single-Rack Network Fabric Model*.

Selected Model	Additional Information
Single-Rack Network Fabric Model	The data center network fabric provides connectivity to the vSphereclusters. The selected *vSphere Cluster Models* and *Storage Models* determine which data center availability model can support the connectivity and availability in the cluster. This section describes the requirements to support a *Single-Rack vSphere Cluster Model.* Depending upon the storage type chosen, additional VLAN requirements may apply.

The data center network fabric provides connectivity to the vSphere clusters. The vSphere Cluster Models and Storage Models chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a Single-Rack vSphere Cluster Model. Depending upon the storage type chosen, additional VLAN requirements may apply.

Note:

If this rack will contain NSX Edge nodes, additional VLAN Requirements will apply. NSX Edge Cluster Models for Single Availability Zone

Table 106: Infrastructure VLAN Scope and First Hop Gateway Availability for Single Rack

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management VM Appliances*		Rack Local	Highly Available	1500	1500
ESX host	ESX management VMK *				1500
	IP Storage VMKs (vSAN, NFS, iSCSI)				9000
	vMotion VMK				
	Host TEP			1700	
NSX Edge node	Refer to Host Fault Tolerant NSX Edge Cluster Model or Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters				
	NSX Edge Node management	Rack Local	Highly Available	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available Static Routing with BFD - Optional BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
	BGP-EVPN Peering	Across Racks	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* When using separate VLAN segments for management component VMs and ESX VMKs.					

Table 107: Workload VLAN Scope and First Hop Gateway Availability for Single Rack

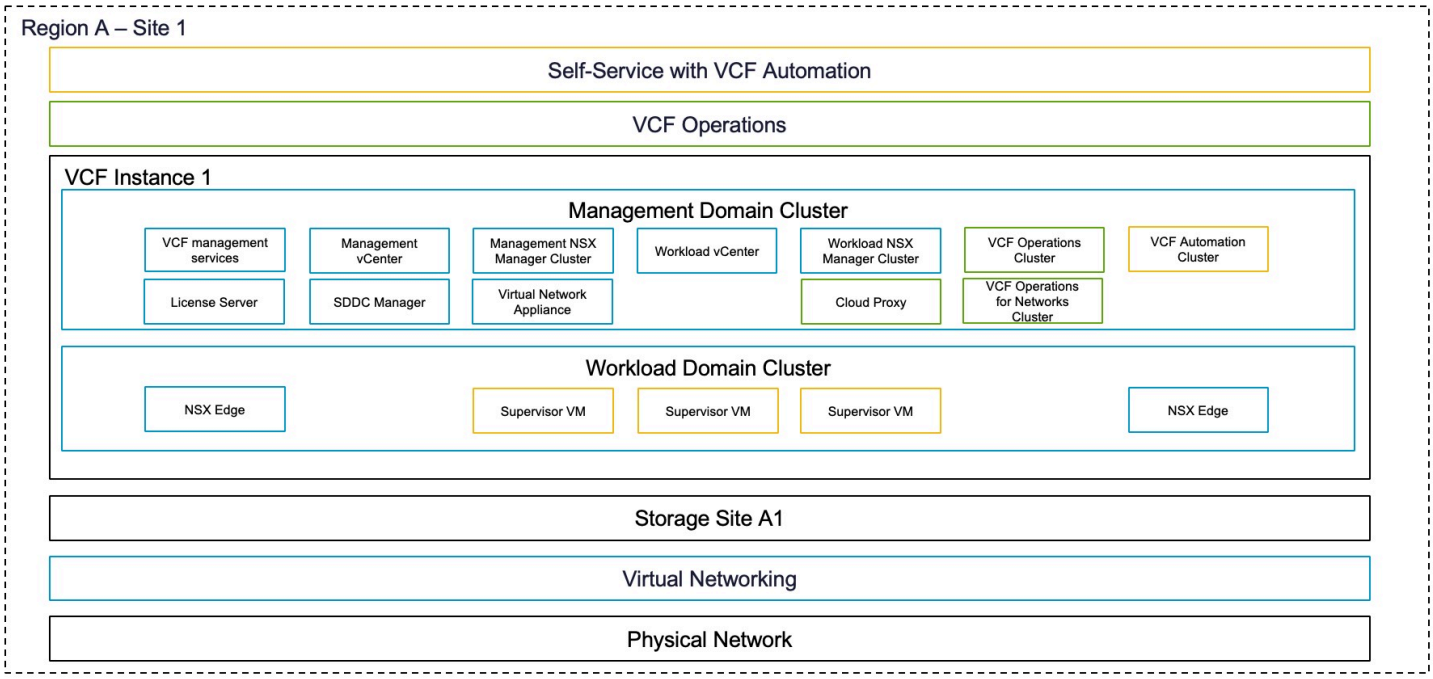
Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VLAN Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model Attributes	Refer to Edges			

VCF Fleet in a Single Site

The *VCF Fleet in a Single Site* is an infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components and workloads are isolated to a separate workload domain. All components are located within a single data center and a single availability zone.

Architecture Overview

Figure 52: VCF Fleet in a Single Site



Design Profile

The *VCF Fleet in a Single Site* infrastructure blueprint is well suited to customers who are looking for the following design profile for their VMware Cloud Foundation platform.

Attribute	Value
Consumption	• Self-service consumption via VCF Automation supporting modern workloads like VKS, Data Services, AI Services, and so on. with multi-tenancy. • Direct vCenter consumption supporting modern workloads like VKS, Data Services, AI Services, and so on. This infrastructure blueprint can use the following Consumption blueprint: • Self-Service Multi-Tenant Private Cloud Consumption
Physical Site Configuration	Single Site (Single instance)

Attribute	Value
Availability	• Tolerates failure of a single ESX host • Tolerates failure of an individual network path • Tolerates failure of a single rack • Tolerates failure of a single management component
Isolation	• Hypervisor-based (logical isolation) • Network traffic type isolation • Cluster-based (physical isolation) • Tenant / Tenant isolation
Recoverability	This infrastructure blueprint can use the following protection blueprints: • VCF Component Backup and Restore • VCF Instance Backup and Restore
Expansion	For available resource and availability expansions, see Expansion after Infrastructure Blueprint Deployment.

Management Domain Design Selection

The following models were selected for the management domain to align with the design profile of the *VCF Fleet in a Single Site* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	High Availability VCF Automation Model	The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.
	Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.
Operations	High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single cloud proxy appliance. To achieve high availability after the initial deployment, an additional cloud proxy appliance should be deployed and then combined in a collector group.
	VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.
	High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.
	Single License Server Model	The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, designed to provide centralized license management for the entire fleet. It is suitable for most deployments where reduced operational overhead is prioritized.
	High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a three-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Layer	Selected Model	Additional Information
	High Availability Log Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind an internal load balancer. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.
Security Governance and Compliance	Instance Identity Broker Model	Deploys the Identity Broker component on the VCF Management Services instance in the management domain.
	VCF Fleet Level Single Sign On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all components across all VCF Instances in the same VCF fleet.
Virtual Infrastructure	Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.
	VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Workload Domain Design Selections

The following models were selected for the workload domain to align with the design profile of the *VCF Fleet in a Single Site* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.
	High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected Supervisor management zones.
	Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for vSphere Supervisor and VKS workloads, it supports VPC-based networking.
	Multi Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone	This approach provides robust resource isolation for different lines of business (LOBs) within a single vSphere Supervisor Zone.
Virtual Infrastructure	Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.
	Host Fault Tolerant NSX Edge Cluster Model	A *Single-Rack vSphere Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Shared Design Selection Across Management Domain and Workload Domain

The following models are shared across both the management domain and workload domain in the *VCF Fleet in a Single Site* infrastructure blueprint.

Layer	Selected Model	Additional Information
Virtual Infrastructure	vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.
	ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCFdomain vSphere cluster.
	Single-Rack vSphere Cluster Model	The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.
	Storage Separation Distributed Switch Model	A *Storage Separation Distributed Switch Model* in VMware Cloud Foundation provides separation of storage traffic from all other network services running on the vSphere cluster.
	Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture that utilizes high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.
	High Availability NSX Manager Model	This model addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.
Physical Infrastructure	Single-Rack Network Fabric Model	The data center network fabric provides connectivity to the vSphere clusters. The *vSphere Cluster Models* and *Storage Models* chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a *Single-Rack Cluster Model*. Depending upon the storage type chosen, additional VLAN requirements may apply.

Minimum Requirements for VCF Fleet in a Single Site

VCF Fleet in a Single Site minimal requirements to stand up this infrastructure blueprint.

VCF Installer Deployment Options

Deployment Model	Deployment Size
High Availability	Medium

Minimal IP Requirements

Requirement	Additional Information
Physical Networks	• ESX Management (4 IPs) • VM Management (35 IPs) • vMotion (4 IPs) • vSAN (4 IPs) • Host TEP (8 IPs) • (Optional) Edge TEP Network (4 IPs) — 2 IPs per Edge VM (x2 Edge VMs) • (Optional) Edge Uplink Network (2 IPs) — 2 IP per Edge VM (x2 Edge VMs)
DNS Entries	• vCenter • ESX — 4 Nodes • SDDC Manager • NSX Manager — VIP — Node A — Node B — Node C • VCF Operations — (Optional) VIP — Primary — Replica — Data — Cloud Proxy • VCF Management Services — VCF Services Runtime — VCF Management Instance Service — VCF Fleet Component • VCF Automation • VCF Service Runtime for VCF Automation • License Server • Identity Broker • Log Management • VCF Operations for Networks — Platform 1 — Platform 2 — Platform 3 — Collector
Virtual Networks	• vSphere Supervisor Service CIDR — e.g. 10.96.0.0/23 • VPC Private IP Block (Control Plane) • Distributed Transit Gateway Private IP Block (Control Plane) • External IP Block (Control Plane & Workload/Organizations) • Private VPC Block for each Organization's VPCs • Private TGW IP Block for each Organization

First VCF Instance

Table 108: VMware Cloud Foundation Installer Deployment Profiles

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for the initial VCF instance within a VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	1x Small • 1x Control Plane Nodes • 2x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Large • 3x Control Plane Nodes • 4x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Medium	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
VCF Operations	1x Small	2x Small	3x Medium	3x Large
License Server	1x Default	1x Default	1x Default	1x Default
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard
VCF Automation	1x Small	1x Medium	3x Medium	3x Large
Resource Requirements	• vCPU: 76 • RAM: 251 GB • DISK: 7448 GB	• vCPU: 106 • RAM: 335 GB • DISK: 8422 GB	• vCPU: 184 • RAM: 656 GB • DISK: 11445 GB	• vCPU: 298 • RAM: 949 GB • DISK: 15357 GB

Management Domain Design Selections

The following models in the design align with the objectives for deploying this *VCF Fleet in a Single Site* infrastructure blueprint.

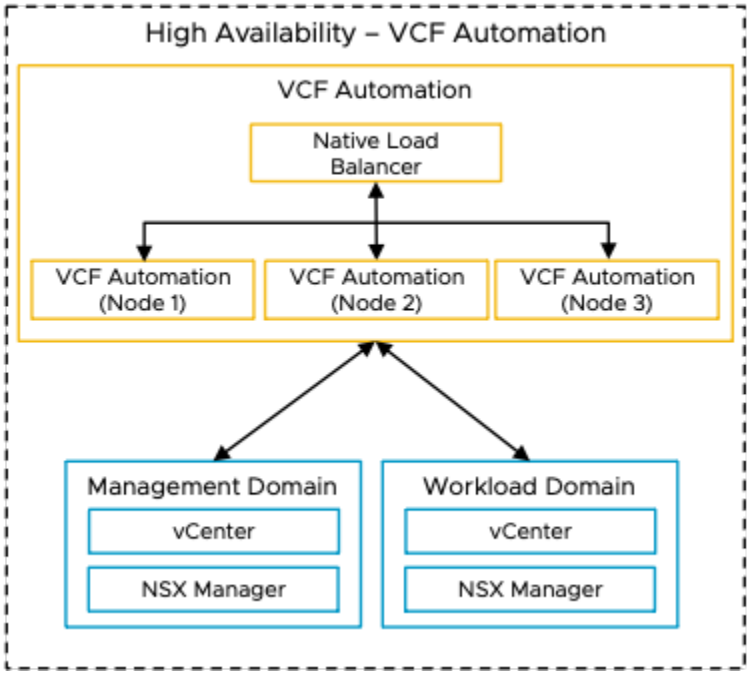
Consumption Layer for VCF Fleet in a Single Site

The consumption layer for the VCF Fleet in a Single Site infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

High Availability VCF Automation Model

The *High Availability VCF Automation Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Automation Model	The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.



High Availability VCF Automation Model Attributes

A *High Availability VCF Automation Model* has the following attributes.

Attribute	Detail
Availability	• Application level availability provided by the three node cluster • Appliance level level availability provided by vSphere HA
Scalability	Supports scale-up and scale-down of node sizes.
Recoverability	Restore through a backup.

High Availability VCF Automation Model Design Requirements

Table 109: High Availability VCF Automation Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AUTO-REQD-HA-001	Deploy VCF Automation as a three node cluster.	Supports high availability objectives.	• Analysis is required to determine which node size is appropriate for your deployment. • Node can be resized after deployment as necessary.
VCF-AUTO-REQD-HA-003	Protect all VCF Automation nodes using vSphere HA.	Ensures availability of VCF Automation without requiring manual intervention during an ESX host failure event.	None

VCF Automation Tenant Authentication Requirements

Table 110: VCF Automation Tenant Authentication Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-TEN-001	Configure an identity provider for VCF Automation organizations.	• Allows tenants to use enterprise directory. • Enables named access control to tenant level constructs. • Required to provision administrative or user access to the Organization portal for any account other than the first user account established during Organization creation.	If not configured, the only access to the Organization Portal will be through the first user account established during Organization creation.

VCF Automation Enterprise Authentication Design Requirements

Table 111: VCF Automation Enterprise Authentication Design Requirements

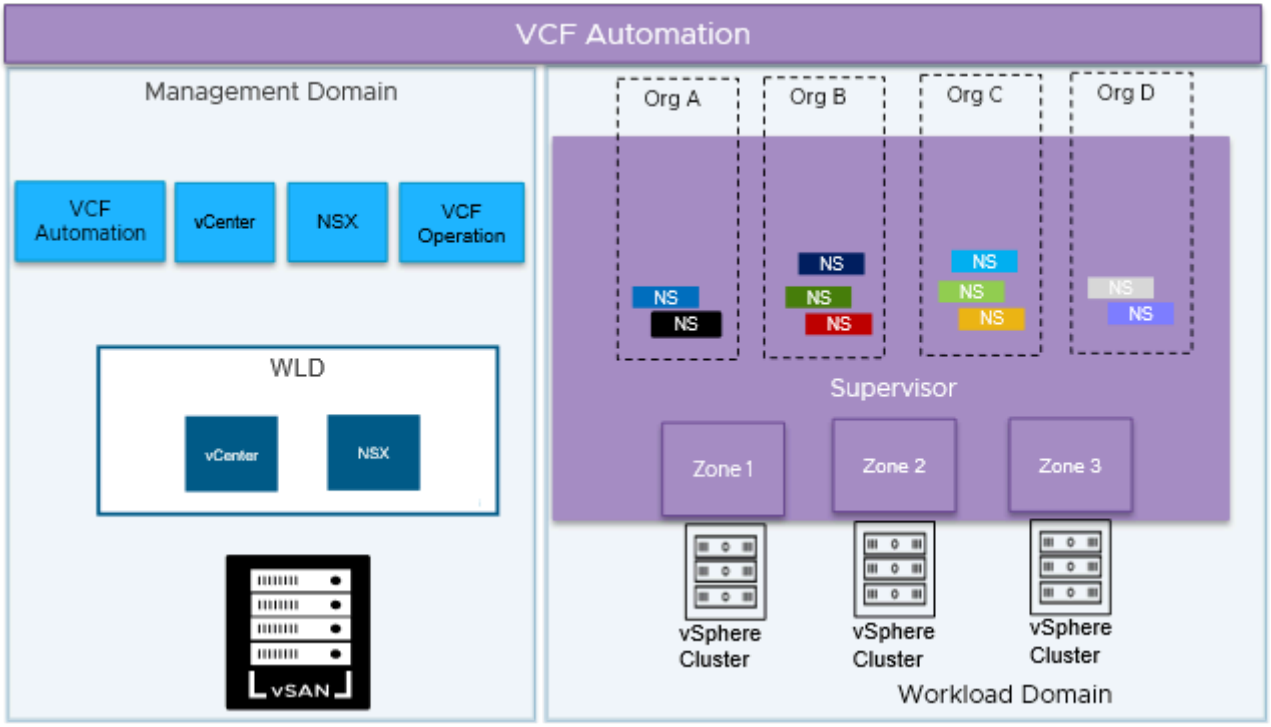
Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-IDP-001	The provider uses the same IdP as the tenant.	• Allows consistent authentication across provider and tenant	None

Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance

Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.

Figure 53: Shared Workload Domain for Multiple Organizations in a VCF Instance



Benefits	Best fit scenarios
• Resource Sharing: Tenants share the underlying infrastructure (workload domain, vCenter, NSX), leading to cost savings and efficient resource utilization. • Simplified Management and Operational Consistency: While providing tenant isolation, the centralized management of the VCF Management Domain simplifies overall infrastructure operations, including patching, upgrades, monitoring, and capacity planning. Consistent policies can be applied across all tenants. • Scalability and Flexibility: The model is highly scalable. As the number of tenants or their resource demands grow, the underlying workload domain can be scaled up by adding more hosts. Resource allocations within Namespaces can also be adjusted dynamically. If the requirement comes, the service provider can add more workload domains for multiple other organizations to share or can dedicate a workload domain per tenant like explained in Model 2. • vSphere Namespace Isolation: vSphere Namespaces provide logical resource isolation for tenant workloads, preventing interference. • Network Segmentation: Network policies and security groups can be used to further segment tenant traffic and can enhance security.	• Cloud Service Providers (CSPs) Offering Multi-Tenant Cloud Services: This model allows the CSP to host multiple customers within a single VCF instance, maximizing resource utilization across their infrastructure. The Supervisor Cluster ensures strong isolation between each customer's workloads and data. Each customer gets their own vSphere Namespace, providing them with a secure and logically separated environment for deploying VMs and containers. Network segmentation (via NSX) further guarantees that customer traffic remains isolated. • Large Enterprises with Multiple Business Units or Departments: A large corporation has several independent business units (e.g., Finance, Marketing, R&D) that require their own isolated IT environments but can share underlying infrastructure. • Organizations Offering Internal PaaS or Developer Platforms: An organization wants to provide a self-service platform for its internal development teams to deploy and manage their applications (including containerized microservices) in an isolated and governed manner. • Educational Institutions or Research Organizations with Multiple Projects or Teams: A university hosts various research projects or has different departments with distinct IT requirements and security needs.

Operations Layer for VCF Fleet in a Single Site

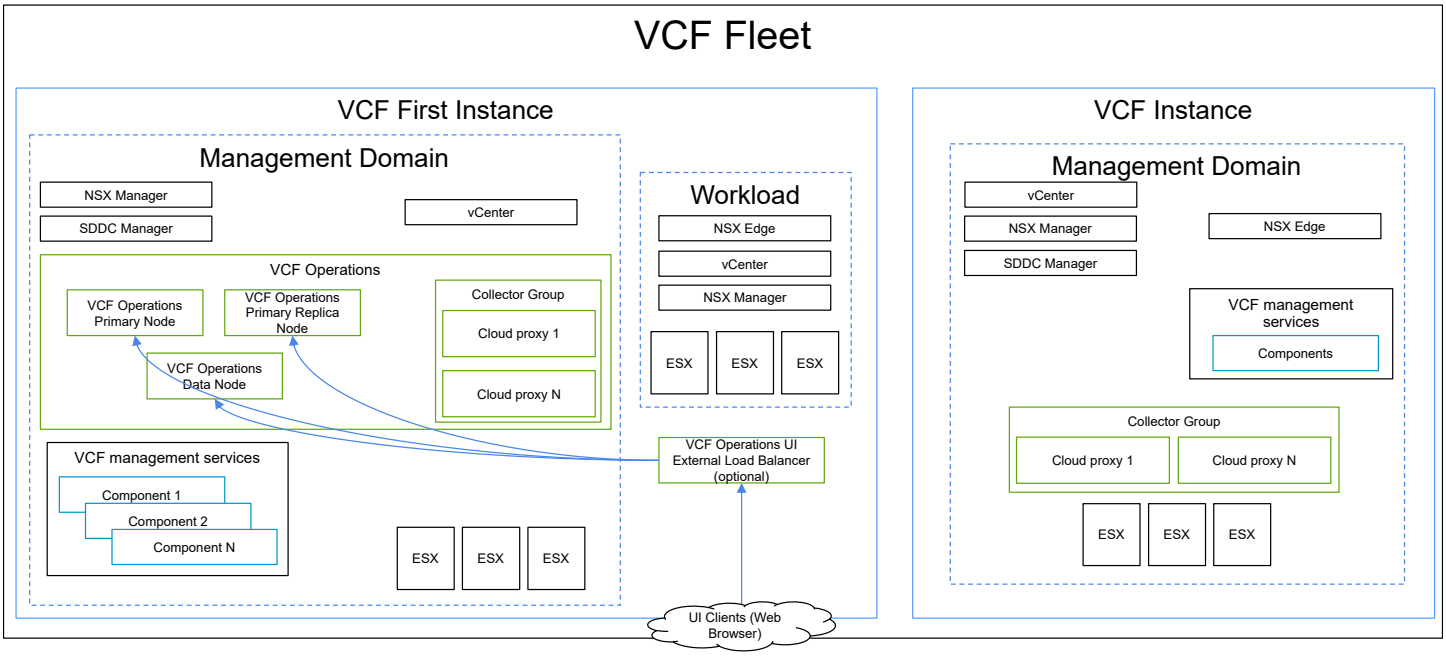
The operations layer serves as the central nervous system of enterprise infrastructure and applications, providing comprehensive visibility, management, and control capabilities across the entire technology stack. This layer ensures that systems remain available, performant, secure, and compliant with organizational standards and regulatory requirements.

High Availability VCF Operations Model

The *High Availability VCF Operations Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single cloud proxy appliance. To achieve high availability after the initial deployment, an additional cloud proxy appliance should be deployed and then combined in a collector group.

Figure 54: High Availability VCF Operations Model in VMware Cloud Foundation



High Availability VCF Operations Model Attributes

A *High Availability VCF Operations Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure. Additional level of availability for the components in: • VCF Operations cluster • VCF Operations collector groups
Scalability	Supports scale-up and scale-out. **Note:** Scale-out is supported for the VCF Operations data nodes and cloud proxynodes.
Recovery	Restore through a backup.
Load balancing	Optional external load balancer for the VCF Operations cluster UI.

Common VCF Operations Design Requirements for All Models

Table 112: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Operations Model Design Requirements

Table 113: High Availability VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-001	Deploy VCF Operations as a cluster of three nodes in the default management vSphere cluster.	• Supports the high availability objectives by deploying VCF Operations as a cluster. • Provides the capacity that is required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out with additional data nodes.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • Create an anti affinity rule (Separate Virtual Machines) for the VMs in the cluster after the deployment.
VCF-OPS-REQD-HA-002	Deploy a single Unified VCF Operations collector node in the default management vSphere cluster.	Removes the load from the VCF Operations cluster from collecting metrics from local-instance applications.	You may need to scale-up the VCF Operations collector nodes to a greater size based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-HA-003	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations cluster nodes.	Using vSphere DRS prevents the VCF Operations cluster nodes from running on the same ESX host and risking the high availability of the cluster.	• You must perform an additional configuration to set up an anti-affinity rule. • You must have a minimum of four ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode. • If additional data nodes are added, you must update the anti-affinity rule and ensure there are enough ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode.
VCF-OPS-REQD-HA-004	The certificate for VCF Operations must include the FQDNs for all cluster nodes in the SAN (subject alternative name) attribute. **Note:** The SAN does not need to include the FQDN of VCF Operations collector nodes.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted.	Before scale out, the certificate for VCF Operations must be changed to also include the FQDN of the additional node in the SAN (subject alternative name) attribute.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-005	If using an optional external load balancer in the high availability deployment model, the certificate for VCF Operations **must** also include the FQDN of the load balancer in the SAN (subject alternative name) attribute.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted when using an external load balancer with the high availability model.	None.

Common VCF Operations Design Recommendations for All Models

Table 114: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

High Availability VCF Operations Model Design Recommendations

Table 115: High Availability VCF Operations Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-HA-001	Deploy VCF Operations with an external load balancer to load balance the connections across the VCF Operations cluster nodes.	Required to deploy a VCF Operations cluster deployment with distributed user interface access across members.	You must configure the external load balancer configuration to support this VCF Operations configuration.
VCF-OPS-RCMD-HA-002	Create a collector group in VCF Operations for each VCF Instance and assign the default VCF Operations collector appliance.	Provides increased availability to metric collection across the VCF Operations collector appliances that are assigned to the collector group.	Manual reconfiguration of the VMware Cloud Foundation adapter required to utilize multiple VCF Operations collector appliances with the group.
VCF-OPS-RCMD-HA-003	Deploy an additional VCF Operations collector node in the default management vSphere cluster.	Extends the availability of the VCF Operations collectors by balancing the load across the collectors when added to a collector group.	• You must deploy the collector node after the deployment of the VCF Operations. • Additional collector node should be deployed for all VCF Instances that requires a collector group.
VCF-OPS-RCMD-HA-004	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations collector nodes.	Using vSphere DRS prevents the VCF Operations collectors from running on the same ESX host and risking the high availability of the collectors.	You must perform manual configuration to set up an anti-affinity rule.
VCF-OPS-RCMD-HA-005	Reconfigure the default VMware Cloud Foundation Adapter in VCF Operations to use a local-instance collector group for a VCF Instance.	Increases the availability of metric collection.	Must be reconfigured post-deployment.

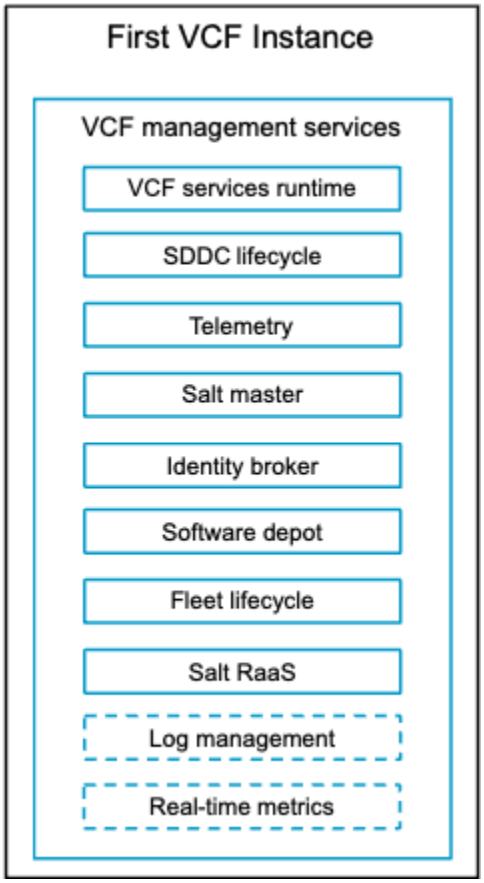
VCF Management Services in the First VCF Instance Model

The *VCF Management Services in the First VCF Instance Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.

Figure 55: VCF Management Services in the First VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.



VCF Management Services in the First VCF Instance Model Components

Table 116: VCF Management Services in the First VCF Instance Model Components

The following components run in the first VCF Instance in the VCF fleet.

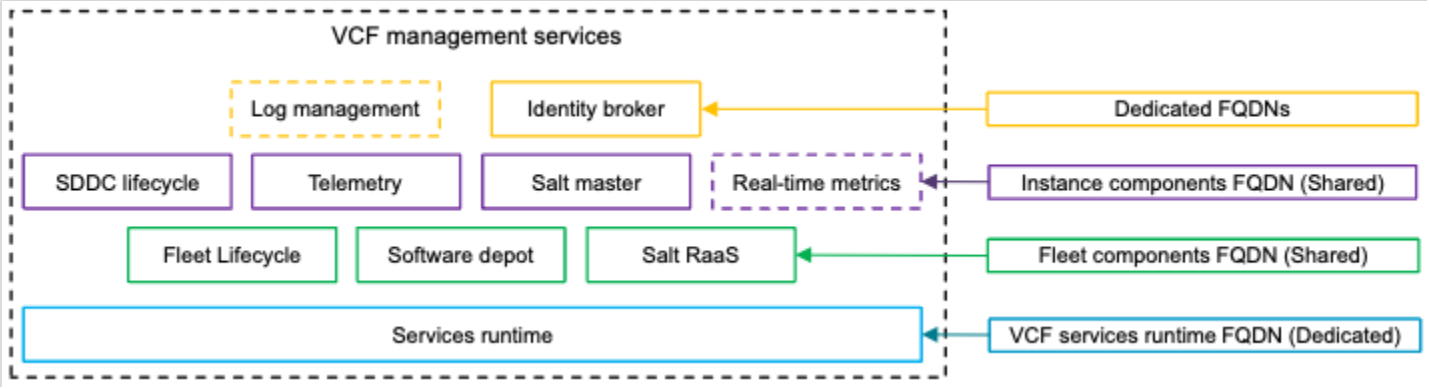
Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations across the fleet.	
Fleet lifecycle	Orchestrates fleet-wide life-cycle operations, including the planning and sequencing of upgrades and patches across multiple VCF Instances.	

Service	Description	Deployment Time
Salt RaaS	Defines and manages desired state configurations and policies (Recipe as a Service).	
Log management	Collects logs from all fleet components	Day-N Components
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

VCF Management Services in the First VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 56:



VCF Management Services in the First VCF Instance Model Design Requirements

Table 117: VCF Management Services in the First VCF Instance Model Design Requirements

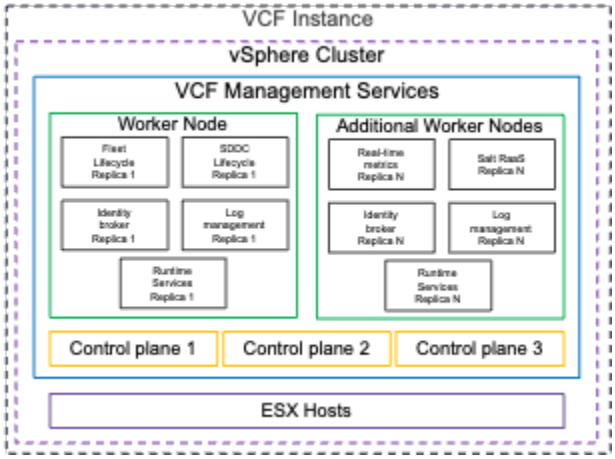
Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-FIRST-001	Create DNS entries for the following shared FQDNs: • Fleet components FQDN • Instance components FQDN	Required for deploying VCF management services fleet and instance level components in the first VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-FIRST-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN • Identity broker FQDN	Required for deploying VCF management services in the first VCF Instance.	You must create the DNS entries before starting the deployment.

High Availability VCF Management Services Model

The *High Availability VCF Management Services Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Figure 57: High Availability VCF Management Services Model in VMware Cloud Foundation



High Availability VCF Management Services Model Attributes

Attribute	Detail
Availability	Relies on vSphere HA
Scalability	Scale out/up worker nodes
Placement	Hosted in a single vSphere cluster

High Availability VCF Management Services Model Options

According to your objectives, the *High Availability VCF Management Services Model* can have options from the following design areas.

Design Area	Options
Infrastructure IP Address Model	• IPv4 Addressing Model • Dual Stack IP Addressing Model
Deployment Size	• Medium • Large Refer to VCF Fleet Sizing Models for more details on sizing.
VCF Management Network Model	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model

Common VCF Management Services Model Design Requirements

Table 118: Common VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-001	Allocate a sufficient IP range for the required number of control plane and worker nodes for your chosen deployment type.	Required for initial deployment.	Adding additional components may require auto-scale out of the VCF management services.
VCF-MS-REQD-COM-002	Use *Kubernetes Cluster API Provider vSphere (CAPV)* node anti-affinity to ensure the nodes run on different ESX hosts.	Using *CAPV* prevents the VCF management services nodes from running on the same ESX host and risking the high availability of the cluster.	Distributed Resource Scheduler (DRS) anti-affinity rules are not required.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-003	Protect the VCF management services nodes by using vSphere HA.	Ensures availability of the VCF management services nodes without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Management Services Model Design Requirements

Table 119: High Availability VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-HA-001	Deploy VCF management services in high availability mode with three (3) control plane nodes and the corresponding number of worker nodes in the default management vSphere cluster.	• Supports the high availability deployment objective. • VCF management services supports scale-out and up from small to medium or large.	Requires additional resources.

Single License Server Model

The *Single License Server Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Single License Server Model	The *Single License Server Model* is the default configuration for the VMware Cloud Foundation platform, designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.

Table 120: Single License Server Model Attributes

Attribute	Detail
Use Cases	This is the typical license server deployment model for VMware Cloud Foundation and is suitable for most VCF deployments.
Deployment mechanism	One license server per VCF fleet. A license server is automatically deployed as part of the installation or upgrade to VMware Cloud Foundation 9.1.
Backup and restore	Backup is not required. If the original license server becomes unavailable, deploy a new license server instance and migrate the licenses within 90 days.
Lifecycle	The upgrade and patch management of the license server is managed through VCF Operations.
High Availability	High availability is provided by vSphere HA.
License Scope	A single license server provides licensing management for the whole VCF fleet.

Single License Server Model Design Elements

Table 121: Single License Server Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LIC-REQD-LS-001	Register the default license server in VCF Business Services console.	Registration allows the VCF Business Services console to add license allocations to the license server.	None.

Licensing for VCF Operations Design Recommendations Connected Mode

Table 122: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Licensing for VCF Operations Design Recommendations Disconnected Mode

Table 123: Licensing for VCF Operations Design Recommendations

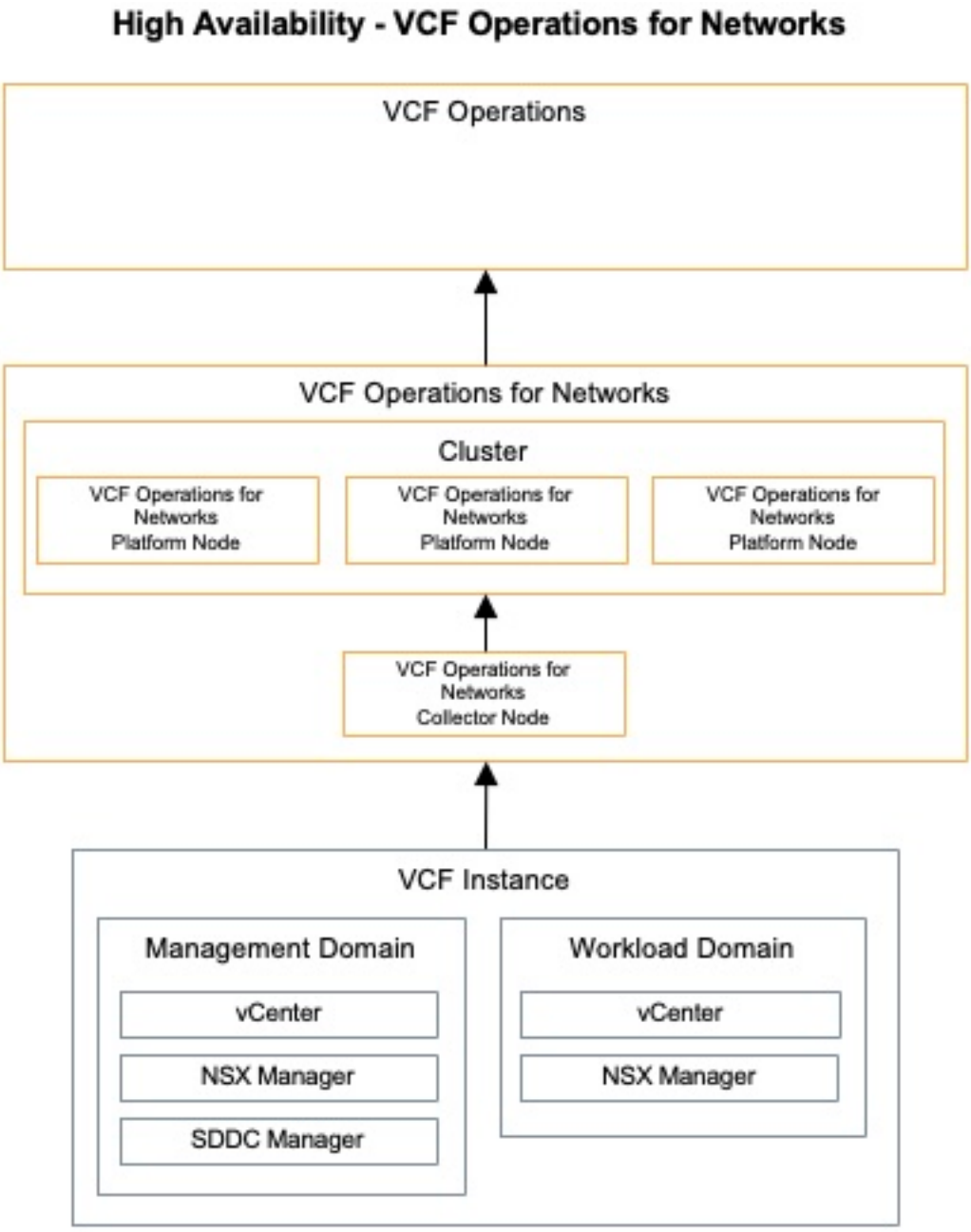
Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

High Availability VCF Operations for Networks Model

The *High Availability VCF Operations for Networks Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a four-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Figure 58: High Availability VCF Operations for Network Model in VMware Cloud Foundation



High Availability VCF Operations for Networks Model Attributes

A *High Availability VCF Operations for Networks Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out.
Recoverability	Restore through a backup.

High Availability VCF Operations for Networks Model Options

A *High Availability VCF Operations for Networks Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Large • Extra Large

High Availability VCF Operations for Networks Model Design Elements

Table 124: High Availability VCF Operations for Networks Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-HA-001	Deploy threeVCF Operations for networksplatform nodes in a cluster.	• Highly Performant. • Highly scalable, can scale up to 15 Extra Large nodes.	• Cluster does not provide high availability. It is for performance. • Requires higher CPU and memory reservations to support additional nodes.
VCF-NET-REQD-HA-002	Deploy a single VCF Operations for networks collector node in the default management vSphere cluster.	A minimum of one VCF Operations for networks collector node is required to provide the VCF Operations for networks platform node with network monitoring data.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks collector node will be applied.

Table 125: High Availability VCF Operations for Networks Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-001	Deploy appropriately sized VCF Operations for networksplatform nodes for your needs.	Ensures resource availability and usage efficiency.	None.
VCF-NET-RCMD-HA-002	Protect all VCF Operations for networks nodes by using vSphere High Availability.	Supports the availability objective for VCF Operations for networks without requiring manual intervention during an ESX host failure event.	You must ensure sufficient spare capacity for vSphere High Availability failover operations.
VCF-NET-RCMD-HA-003	In an environment with multiple VCF Instances, deploy the VCF Operations for networkscollector node(s) in the default management vSphere cluster in each VCF Instance by using VCF Operations fleet management.	• Ensures a consistent deployment in each VCF Instances regarding network data collection. • It is assumed that each VCF Instances introduces a sufficiently high number of virtual machines, so that the deployment of dedicated VCF Operations for networks collector nodes is justified.	Each VCF Operations for networks collector node must be registered with a VCF Operations for networks platform node.
VCF-NET-RCMD-HA-004	For each vCenter data source, enable NetFlow on each vSphere Distributed Switch within the workload domain.	Provides the collection of network flows via the IPFIX protocol.	For the VCF domains, VCF Operations for networks will automatically update the NetFlow settings for each cluster's vSphere Distributed Switch.

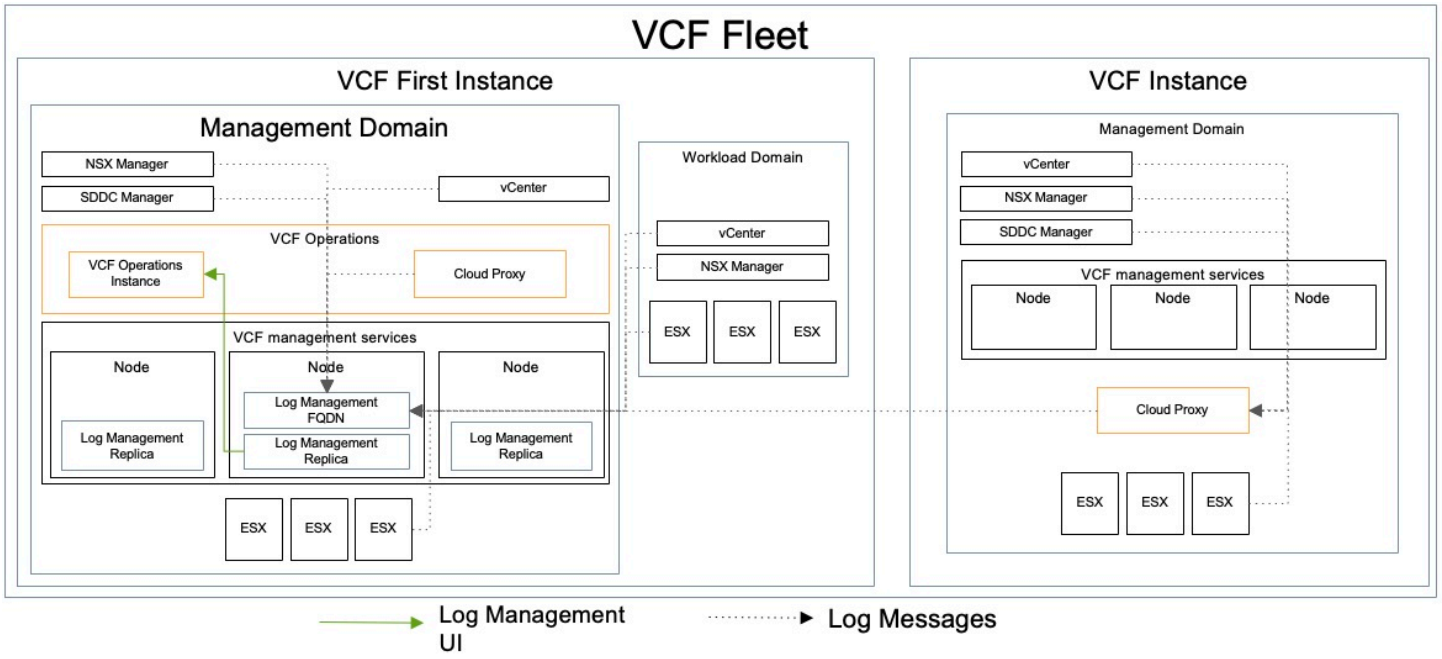
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-005	For each NSX data source, enable IPFIX for the distributed firewall.	Provides the collection of network flows via the IPFIX protocol.	The distributed firewall service must be enabled on the NSX Local Manager for the VCF workload domain.
VCF-NET-RCMD-HA-006	For each NSX Manager data source, enable latency metric collection.	Provides the collection of latency metrics from NSX Transport Nodes.	Any firewall rule sets from all ESX hosts to the VCF Operations for networks collector must allow traffic on TCP 1991.
VCF-NET-RCMD-HA-007	Use the default retention period of one month for the VCF Operations for networks platform node.	Keeping the default value for retention minimizes required storage resources administrative efforts.	None.
VCF-NET-RCMD-HA-008	Use a CA-Signed certificate containing the fully qualified domain names (FQDNs) of each VCF Operations for networks platform and collector node in the SAN attributes, when deploying VCF Operations for networks.	Configuring a CA-Signed certificate ensures that the communication to the externally facing web UI and API VCF Operations for networks for is encrypted.	• Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered. • Each time a node is added the certificate must be replaced to include the fully qualified domain name of the additional node.
VCF-NET-RCMD-HA-009	Use an SHA-2 or higher algorithm when signing certificates.	The SHA-1 algorithm is considered less secure and has been deprecated.	Not all certificate authorities support SHA-2.

High Availability Log Management Model

The *High Availability Log Management Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
High Availability Log Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind and internal load balancer. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.

Figure 59: High Availability Log Management Model in VMware Cloud Foundation



High Availability Log Management Model Attributes

A *High Availability Log Management Model* has the following attributes.

Attribute	Detail
Centralized management	Provides a centralized log management framework for vCenter, ESX, NSX, VCF Operations, VCF Automation, and identity broker.
Availability	Relies on cluster technology in the event of an outage or VCF management services and vSphere HA to restart nodes in the event of an ESX host failure.

Attribute	Detail
Scalability	Supports scale-up, storage scale up, and scale out.
Recoverability	Restore configuration through a backup, restore logs from archives.

High Availability Log Management Model Options

A *High Availability Log Management Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Small • Medium • Large

High Availability Log Management Model Design Requirements

Table 126: High Availability Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-HA-001	Deploy log management with at least 3 replicas.	• Provides high availability. • Using the integrated load balancer prevents a single point of failure. • Using the integrated load balancer simplifies the log management deployment and subsequent integration.	• You must deploy a minimum of three appropriately sized replicas. • All replicas are sized identically. • If the capacity of your log management cluster must expand, identical capacity is added to each node.
VCF-LOGS-REQD-HA-002	Protect all VCF Management service nodes by using vSphere High Availability.	Supports the availability objectives for log management without requiring manual intervention during an ESX host failure event.	None.

Common Log Management Design Recommendations for All Models

Table 127: Common Log Management Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

High Availability Log Management Model Design Recommendations

Table 128: High Availability Log Management Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-HA-001	Enable HA on the VCF management services control plane. (Instance deployed with the High Availability deployment model, or scale up VCF services runtime to medium or large.)	To support the high availability objective, the control plane for the VCF management services should be configured for HA, as log management depends upon VCF management services.	Requires additional nodes and capacity in the vSphere management cluster.

Security Governance and Compliance Layer for VCF Fleet in a Single Site

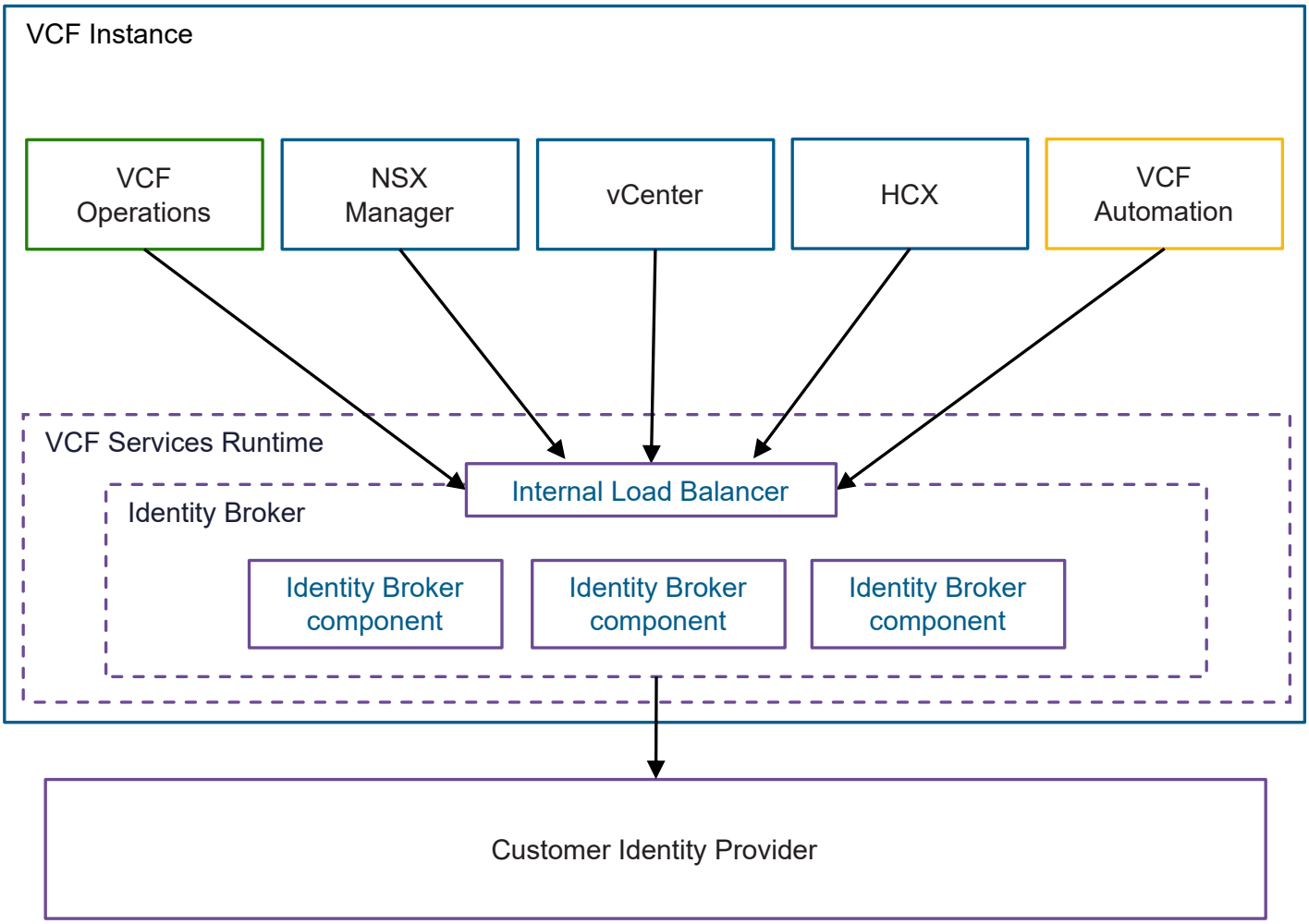
The security governance and compliance layer refers to the framework of policies, procedures, and controls that organizations establish to manage security risks while meeting regulatory, legal, and industry standards around authentication.

Instance Identity Broker Model

The *Instance Identity Broker Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Instance Identity Broker Model	Deploys the Identity Broker component on the VCF Management Services instance in the management domain.

Figure 60: Instance Identity Broker Model in VMware Cloud Foundation



Instance Identity Broker Model Attributes

The *Instance Identity Broker Model* has the following attributes.

Attribute	Detail
Service Location	Identity broker runs as a service on VCF Management Services.
Deployment mechanism	An identity broker is automatically deployed during the installation or upgrade to VMware Cloud Foundation 9.1 in the first management domain. Additional identity brokers can be deployed in each VCF Instance as part of a day 2 operation.
Number of Nodes	multi-node cluster.
Scale considerations	• Scale up: Identity broker supports small, medium, and large sizes to match different configuration demands. • Scale out: Support for multiple identity broker instances to increase redundancy and throughput. • Recommendations: – Scaling up is preferred over scaling out when a single large-size identity broker meets security, resilience, and performance standards, as it reduces operational complexity and simplifies SSO management. – Schedule scale-up operations within a maintenance window, as these activities can impact active services.
Supported services	For support directories and IdPs, see Identity Providers and Protocols Supported for Single Sign-On.
Backup and restore	Use VCF management services backup and restore process.
Resiliency	Resiliency is provided by a distributed multiple-node deployment running each node on separate vSphere hosts.
Operational Impacts	As an isolated, external service, operational tasks performed on vCenter do not impact the authentication stack of the VMware Cloud Foundation components.
Migration path	Migrating from the appliance to embedded requires a reset of the VCF Single Sign-On configuration. Users must be re-added to components.

Instance Identity Broker Model Design Requirements

Table 129: Instance Identity Broker Model Design Requirements

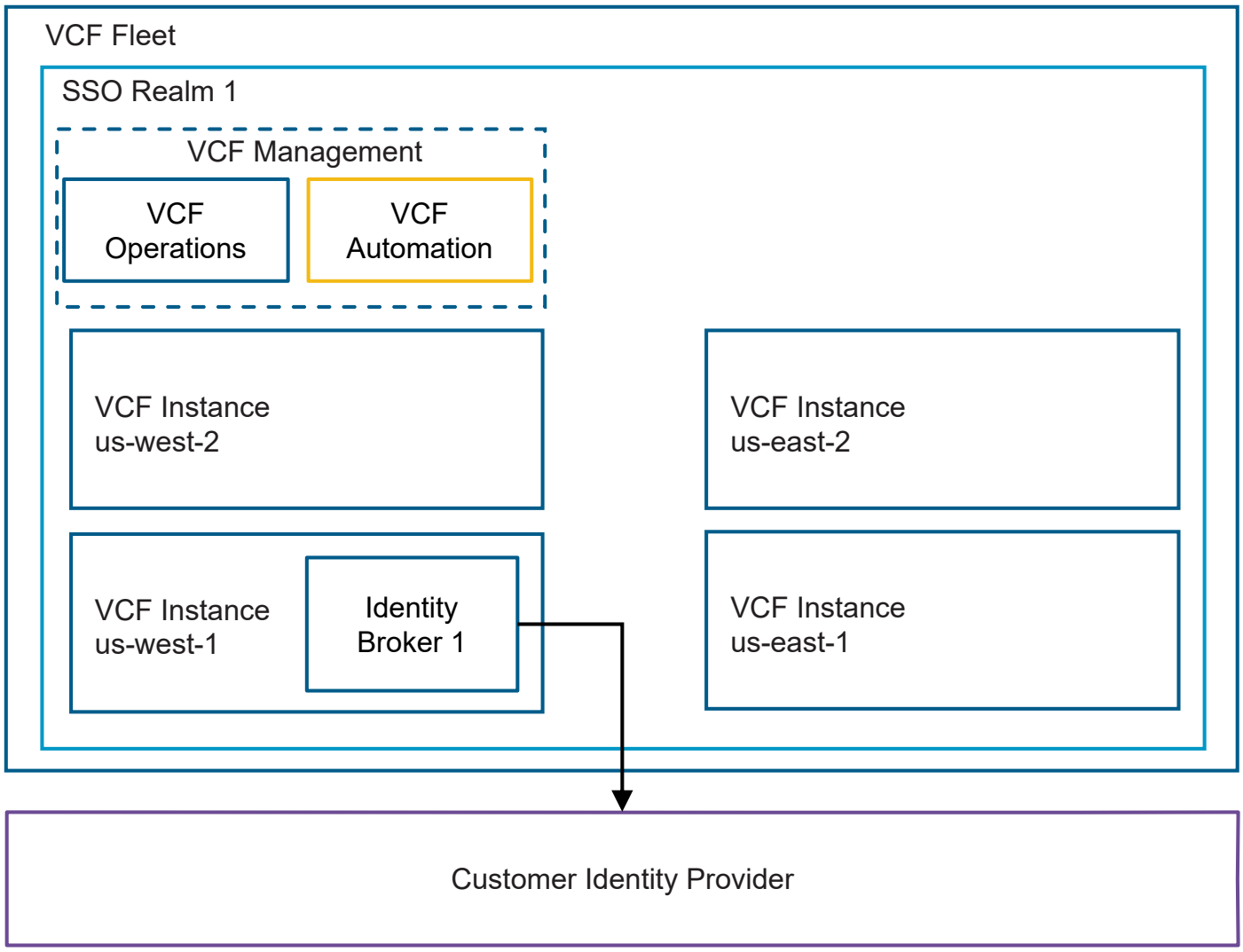
Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-EXT-001	Deploy additional identity broker instances as a Day N operation as needed.	• Provides high availability to the authentication source for a VCF Instance. • Provides additional scale.	Increases the operational overhead of managing the identity broker.
VCF-SSO-REQD-EXT-002	Connect the identity broker service to your corporate IdP.	Provides directory services (users and group) and user authentication service to the identity broker service.	None.

VCF Fleet Level Single Sign On Model

The *VCF Fleet Level Single Sign On Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
VCF Fleet Level Single Sign-On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all components across all VCFinstances in the same VCF fleet.

Figure 61: VCF Fleet-Wide Single Sign-On Model in VMware Cloud Foundation



VCF Fleet-Wide Single Sign-On Model Attributes

The *VCF Fleet-Wide Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**Low Scale -** A single identity broker services all VCF Instances in your VCF fleet.
Resources requirements	**Low Resource Footprint** - A single identity broker deployment is required for one VCF fleet.
Operational Overhead	**Low Operational Overhead** - An identity broker per VCF fleet will need to be managed separately for operational tasks such as backup and restore.
Availability	**Low Availability per Private Cloud** - A single identity broker servicing all VCF Instances in your VCF fleet.
Blast Radius	**Large Blast Radius** - An outage of one identity broker instance will impact logins for a single VCF fleet.
Single Sign-On Realm	**Large SSO Realm** - Users do not need to re-authenticate when moving between different VCF Instances within the same VCF fleet.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a only): Because the identity broker is hosted on the primary VCF instance, VCF instance restoration is required; Option 2b is not applicable. • Failure Scenario 3

VCF Fleet-Wide Single Sign-On Model Design Requirements

Table 130: VCF Fleet-Wide Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VFL-001	Configure additional VCF Instances in the VCF fleet to consume the same identity broker configured in the first VCF Instance.	• Provides an authentication source for all components in the VCF fleet. • Provides VCF Single Sign-On capabilities to all components in the VCF fleet.	An outage to the identity broker service impacts the VCF fleet.

Virtual Infrastructure Layer for VCF Fleet in a Single Site

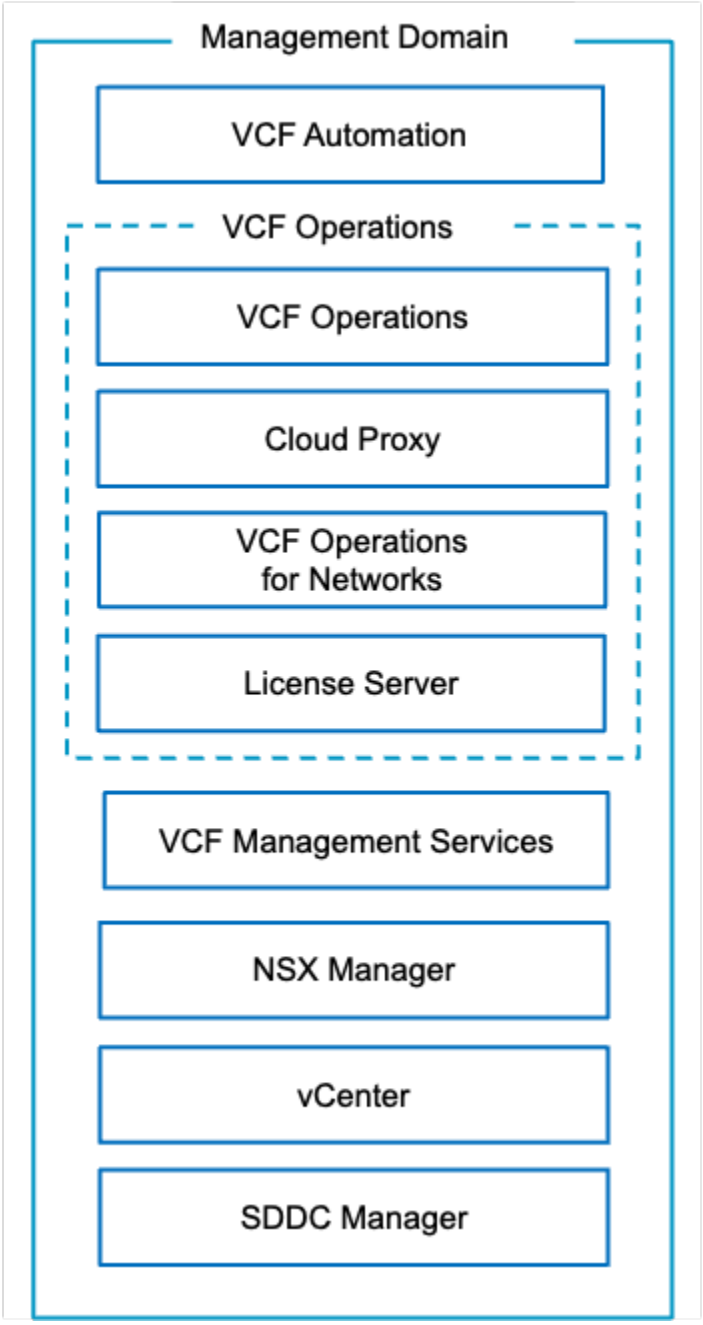
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Management Domain Model

The *Management Domain Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.

Figure 62: Management Domain Model in VMware Cloud Foundation



Management Domain Model Attributes

The *Management Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• VCF fleet management components. • VCF Instance management components. • NSX Edge nodes. • Can also run business workloads.
Management Domain in the First VCF Instance	• The management domain in the first VCF Instance contains the following management components: – VCF management services (fleet and instance level components) – VCF Operations – Cloud Proxy – VCF Operations for Networks – VCF Automation – ESX hosts – vCenter – NSX Manager – SDDC Manager
Management Domains in Additional VCF Instances	• Management domains in additional VCF Instances contain the following instance-level management components: – VCF management services (instance level components) – Cloud Proxy – ESX hosts – vCenter – NSX Manager – SDDC Manager

Management Domain Model Design Requirements

Table 131: Management Domain Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLD-REQD-CFG-001	Deploy a management domain to host all management components for the VCF Instance.	• Guarantees sufficient resources for management components. • Enables the use of specific hardware to meet only the needs of the management components. • Enables the use of dedicated physical compute, network, and storage separately from those used for additional workloads. • Enables separate lifecycle management of management and workload components.	• You must carefully size the management domain to accommodate planned deployments of additional workload domains and management components. • Hardware might not be fully utilized until full-scale deployment has been reached. • The management components for all greenfield VCF domains are deployed to the same cluster, and share the same subnet.

Management Domain Model Design Recommendations

Table 132: Management Domain Model Design Recommendations

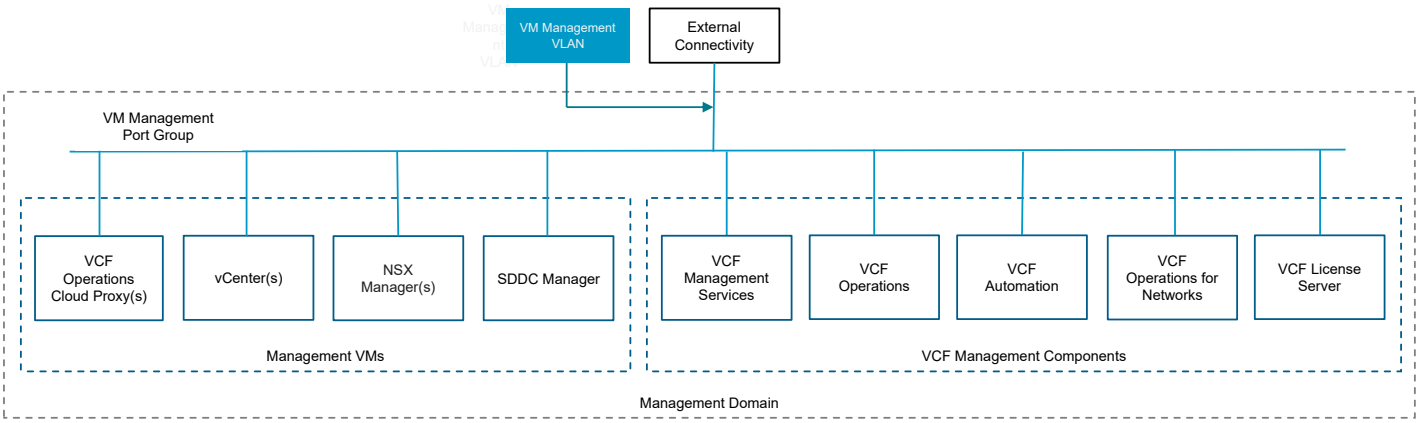
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-RCMD-NSX-001	Use a dedicated NSX Manager instance for the management domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Managers to manage and patch.

VCF Management Shared VLAN Network Model

The *VCF Management Shared VLAN Network Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Figure 63: VCF Management Shared VLAN Network Model in VMware Cloud Foundation



VCF Management Shared VLAN Network Model Attributes

A *VCF Management Single VLAN Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	VLAN-backed vSphere distributed port group.
Network Isolation	No; Sharing VLAN with the VCF management domain components.
NSX Edge Cluster Required	No; Optional to provide NSX load balancer for VCF Operations. (requires additional NSX VLAN segment for Load balancer connectivity)
Stretching Network Between Availability Zones	• **VM management network** must be stretched in the physical network fabric between availability zones. • Provides automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• Not suited for a disaster recovery use case as the network is shared with management domain components. • Failover testing is not supported.

VCF Management Shared VLAN Network Model Design Requirements and Recommendations

Table 133: VCF Management Shared VLAN Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-SV-NET-REQD-001	Use the management domain vCenter VM management distributed port group for the VCF fleet level components.	No additional network required.	• No logical network isolation. • Done automatically during VCF Installer deployment.
VCF-MGMT-SV-NET-REQD-002	Use the same VLAN for VCF management components and the VCF Instance management virtual machines. e.g. vCenter and NSX Managers.	No additional VLAN required.	• Separate physical firewall rules may not be possible with a single VLAN. NSX Distributed Firewall may be an alternative solution. • Not suitable for Fleet Disaster Recovery use case with single VLAN.

Table 134: VCF Management Shared VLAN Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-SV-NET-RCMD-001	Deploy the VCF management components during initial deployment using VCF Installer.	Simplified deployment using VCF Installer UI.	None.

Workload Domain Design Selections

The following models in the design align with the objectives for deploying the *VCF Fleet in a Single Site* infrastructure blueprint.

Consumption Layer for VCF Fleet in a Single Site

The consumption layer for the *VCF Fleet in a Single Site* infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

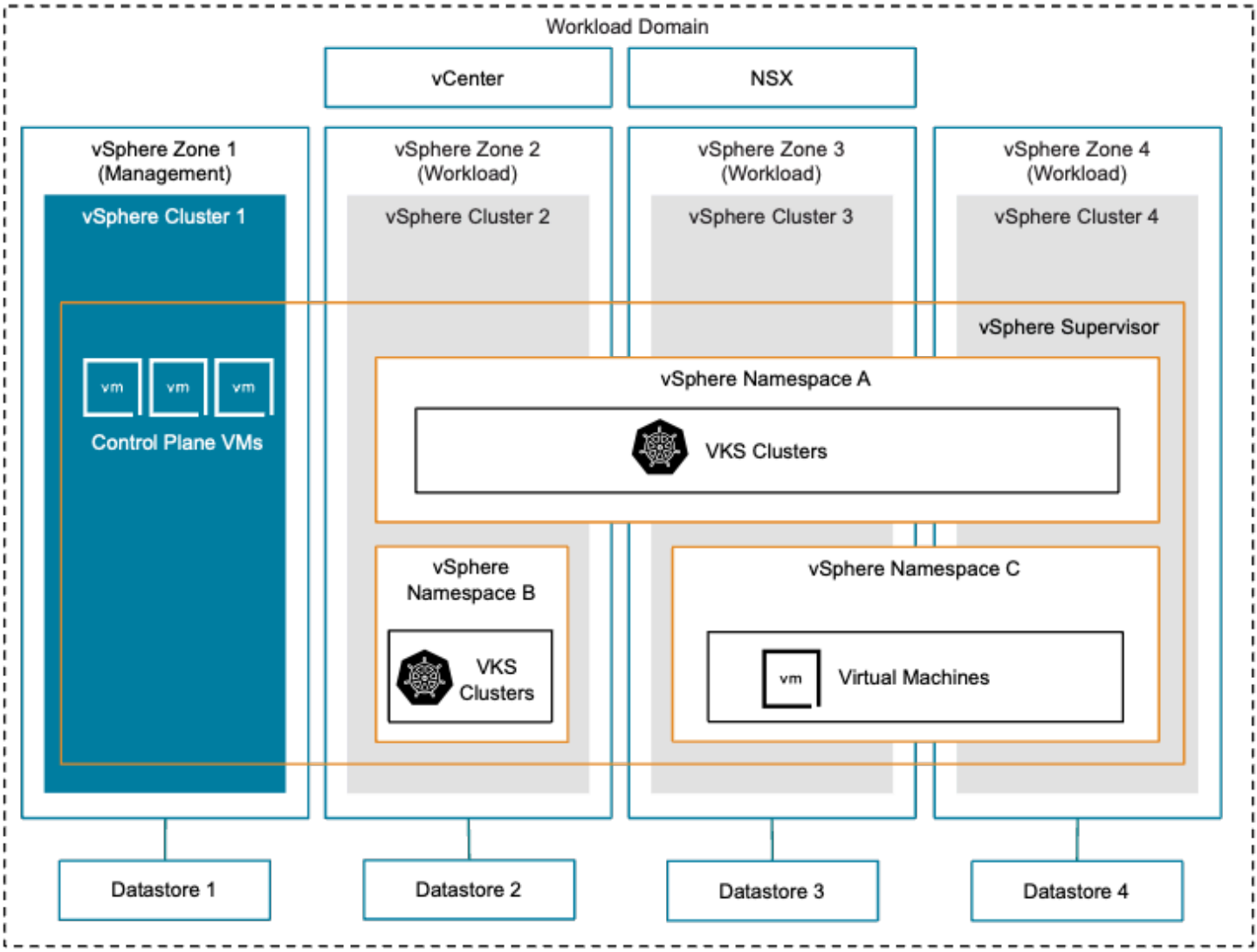
Single Management Zone with Combined Workload Zones Model

The *Single Management Zone with Combined Workload Zones Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.

Figure 64: Single Management Zone with Isolated Workload Zones Model

This diagram illustrates an example of a *Single Management Zone with Isolated Workload Zones Model* deployment.



Single Management Zone with Combined Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single - Combined with management
vSphere Zone Datastore Type	Zonal datastore

Attribute	Detail
Activation	• At workload domain creation. OR • From vCenter.

vSphere Supervisor Design Requirements for all Zone Models

Table 135: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

vSphere Supervisor Single Management Zone Design Requirements

Table 136: vSphere Supervisor Single Management Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-SMZ-REQD-CFG-001	One available vSphere Zone that is backed by a supported vSphere cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	vSphere Zone for Supervisor management workload use.	The selected vSphere Zone is dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.

vSphere Supervisor Common Networking Design Recommendations

Table 137: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Design Recommendations

Table 138: vSphere Supervisor Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

High Availability vSphere Supervisor Control Plane Model

The *High Availability vSphere Supervisor Control Plane Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected vSphere Supervisor management zones.

Figure 65: *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model deployment.

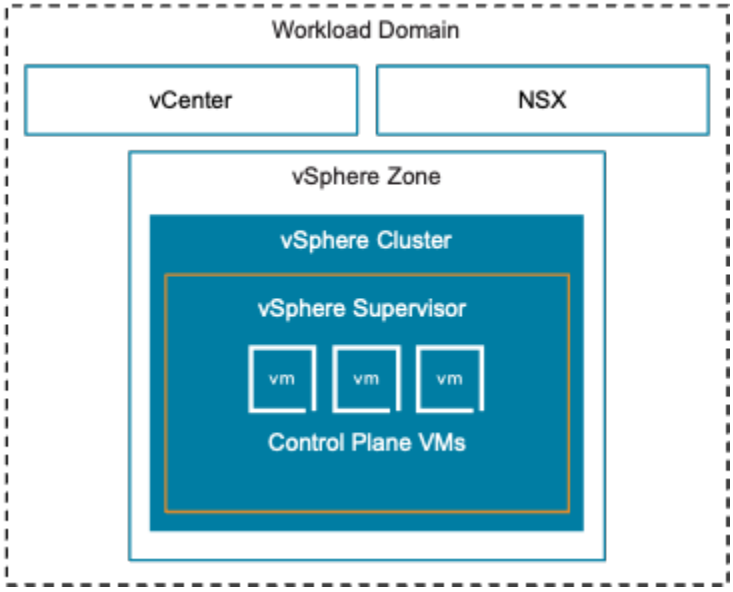
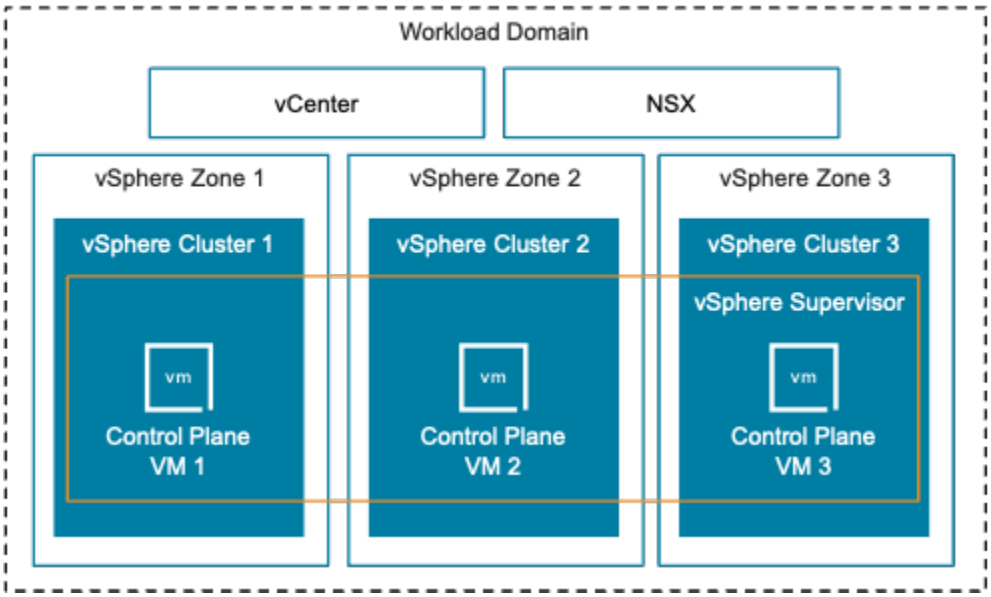


Figure 66: *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model deployment.



High Availability vSphere Supervisor Control Plane Model Attributes

Attribute	Detail
Control Planes Nodes	Three
Availability	• Relies on vSphere HA. • Three management zones, which are protected from single cluster failure.
Scalability	Scale up only
Placement	• Single management zone, all nodes in a single management zone. • Three management zones, one node per management zone.

Attribute	Detail
Applicable Zone Models	• Single Management Zone with Combined Workload Zones Model • Single Management Zone with Isolated Workload Zones Model • Three Management Zones with Combined Workload Zones Model • Three Management Zones with Isolated Workload Zones Model

Table 139: vSphere Supervisor Control Plane Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-CP-REQD-CFG-001	Five consecutive management IP addresses routable to vCenter NSX, and optionally Avi management networks if Avi Load Balancer are used.	Required for integration and management communications.	Requires precise IP address management.
VCF-SUP-CP-REQD-CFG-002	Three vSphere Zones are required for the High-Availability Model in a Three Management Zone Model.	Control pane nodes are evenly placed with one per zone to protect against a single vSphere cluster failure.	Additional vSphere zones each back by separate supported vSphere clusters are required.

vSphere Supervisor Control Plane Design Recommendations

Table 140: vSphere Supervisor Control Plane Design Recommendations

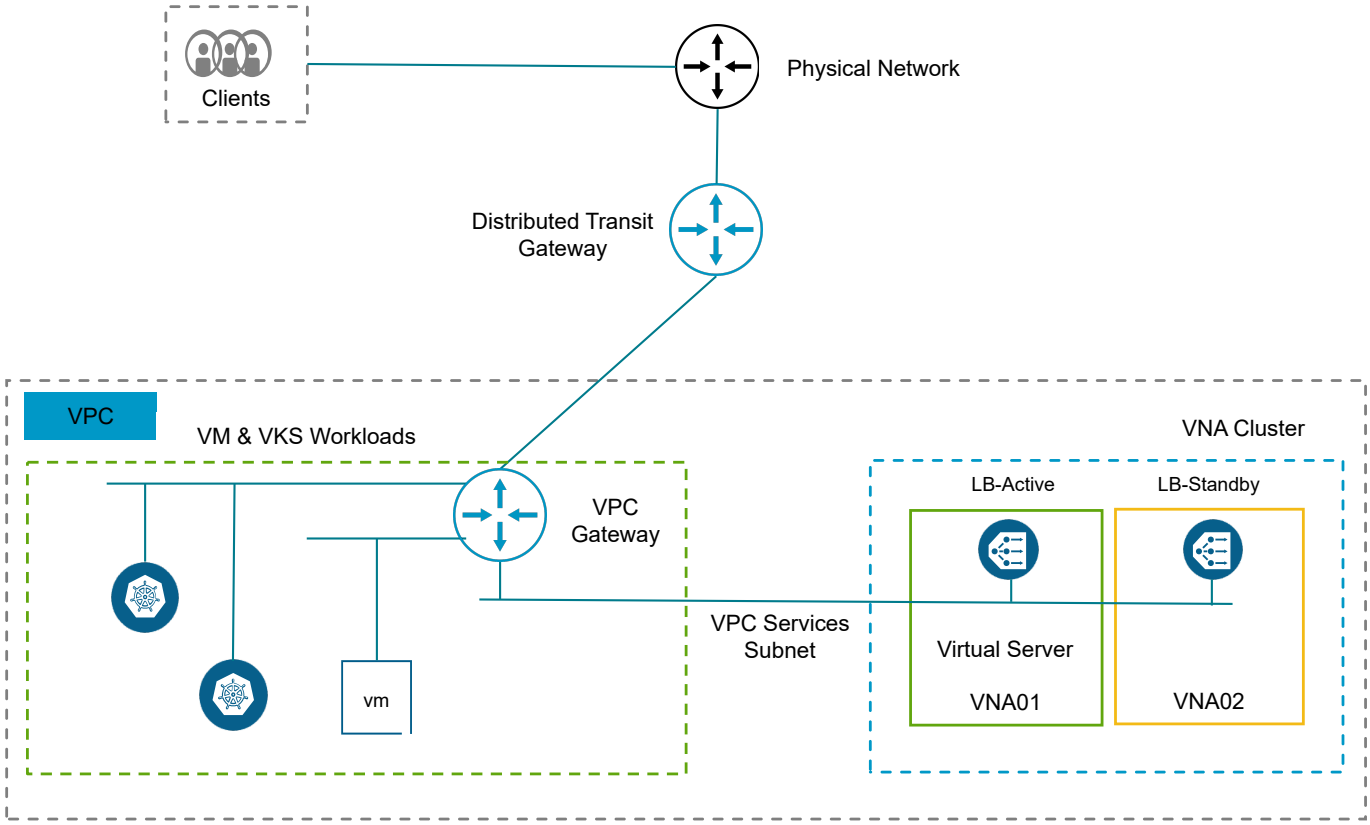
Design Requirement ID	Design Recommendation	Justification	Implication
VCF-SUP-CP-RCMD-CFG-001	Deploy the supervisor control plane in high availability mode with three control plane VMs. *Applicable to Non-Simplified Supervisor models only.	Three control plane VMs allow for continuous availability of the Supervisor control plane during rolling upgrades and single node outages.	Additional resources are required to support the extra control plane VMs.

Virtual Network Appliance Load Balancer Model

The *Virtual Network Appliance Load Balancer Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for VCF Supervisor and VKS workloads, it supports VPC-based networking.

Figure 67: Virtual Network Appliance Load Balancer Model in VMware Cloud Foundation



Virtual Network Appliance Load Balancer Model Attributes

A *Virtual Network Appliance Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Load Balancing for VCF infrastructure components (i.e. VCF appliances) • Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services)
Networking Models Supported	VPC Networking Required
Entitlement	Included with VMware Cloud Foundation
Availability	Active/Standby
Scalability	• Scale up Virtual Network Appliance nodes in the Virtual Network Appliance cluster • Scale out a single Virtual Network Appliance Cluster • Create additional Virtual Network Appliance clusters
Deployment Model	Virtual Network Appliance Cluster with two or more Virtual Network Appliances. See Virtual Network Appliance Cluster Detailed Design for further details.

Virtual Network Appliance Load Balancer Model Options

Design Area	Options
Number of Virtual Network Appliances per cluster	1 to 10
Deployment Options	• Deploy with a new workload domain • Deploy Day 2 from NSX Manager • Deploy Day 2 from vCenter
Virtual Network Appliance Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger scale depoloyments where a large number of load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

Virtual Network Appliance Load Balancer Design Requirements

Table 141: Virtual Network Appliance Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-LB-REQD-CFG-01	Create a Virtual Network Appliance cluster with two or more Virtual Network Appliance nodes.	Provides availability for a load balancer instance.	None.
VCF-VNA-LB-REQD-CFG-02	Configure the VPC connectivity profile to use Virtual Network Appliance cluster in the VPC Service Gateway Configurations.	Enables the VPC to use the Virtual Network Appliance cluster for load balancer instances.	Virtual Network Appliance cluster can be shared between VPCs.
VCF-VNA-LB-REQD-CFG-03	Create a load balancer instance for the VPC using the API	Supervisor and VKS-based workloads, load balancer creation is automated.	No UI available to configure load balancer instances.
VCF-VNA-LB-REQD-CFG-04	Create a load balancer configuration for VPC using the API.	Supervisor and VKS-based workloads, load balancer configuration is automated.	No UI available to configure load balancer instances.

Virtual Network Appliance Load Balancer Design Recommendations

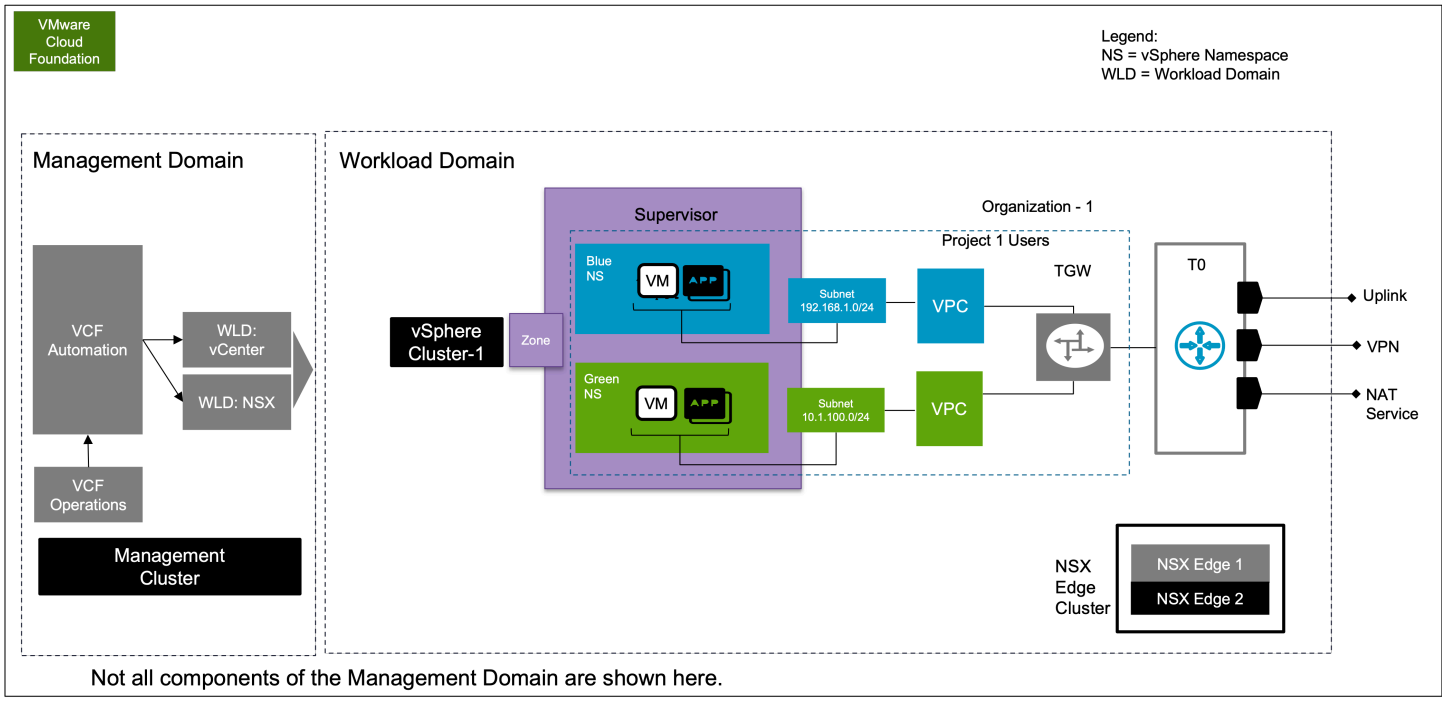
Table 142: Virtual Network Appliance Load Balancer Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-LB-RCMD-CFG-01	Configure failures domain for Virtual Network Appliance clusters with more than two Virtual Network Appliance nodes in the cluster.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-VNA-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from vCenter UI.

Multi Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone
The *Multi Tenancy Consumption Pattern 2 Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone	This approach provides robust resource isolation for different lines of business (LOBs) within a single vSphere Supervisor Zone.

Figure 68: Pattern 2: Resource Isolation with multiple VPCs for multiple lines of business in one Supervisor zone - Centralized Gateway (CTGW) Design



Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone Design Attributes

Design Attributes	Details	Diagram Representation
Organization	One	Represents as one Organization
Line of Business	Two	Describes Blue and Green as separate LOB
Number of vSphere Clusters	One	A compute and vSAN storage cluster used in a workload domain construct
Number of vSphere Supervisors	One	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	One	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Projects	One	Represented by "Project 1 - Users". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One	The T0 gateway provides external connectivity. This is where NAT and VPN services are configured.
vSphere Namespaces	Two	Blue and Green LOB has dedicated vSphere Namespace for workload isolation.
VPCs	Two	Each LOB/ tenant has their own dedicated VPC for network isolation.
TGWs (Transit Gateway)	One	A single TGW connects all the VPCs to the Provider Gateway.
Tenant Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Virtual Infrastructure Layer for VCF Fleet in a Single Site

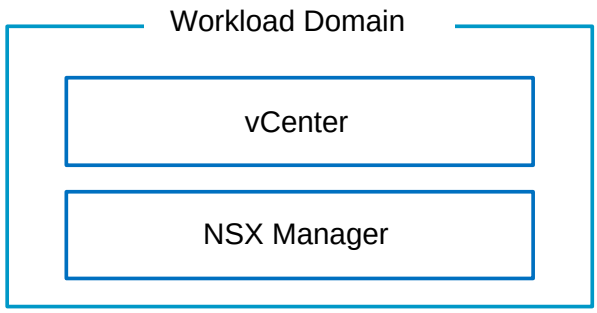
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Workload Domain Model

The *Workload Domain Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.

Figure 69: Workload Domain Model in VMware Cloud Foundation



Workload Domain Model Attributes

A *Workload Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• NSX Edge nodes. • Business workloads.
vCenter Single Sign-On Domain	Dedicated.

Workload Domain Model Design Recommendations

Table 143: Workload Domain Model Design Recommendations

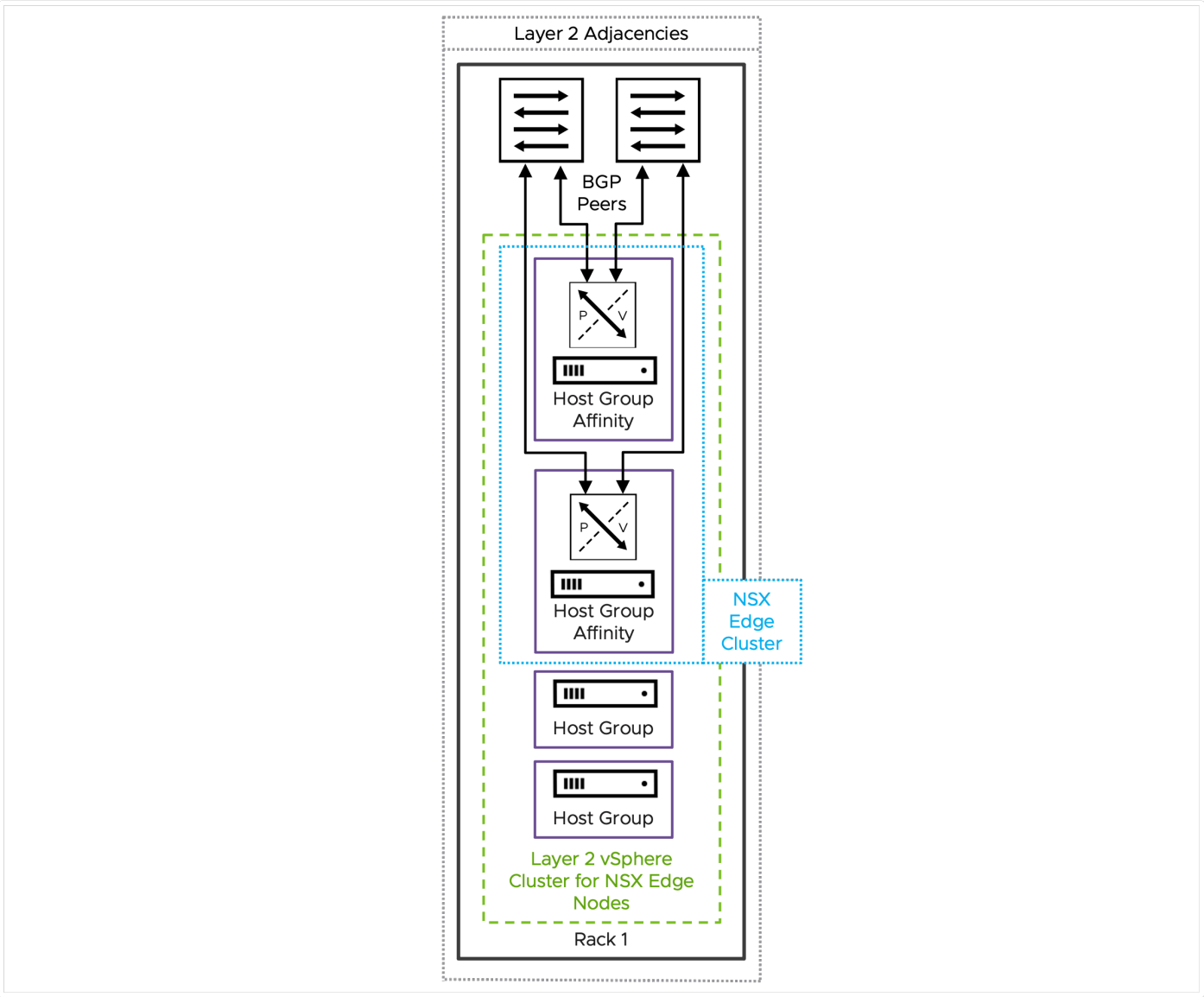
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLD-RCMD-CFG-002	Use workload domains for customer workloads.	• Aligns with the VMware best practice of separating management workloads from customer workloads. • Provides better long-term flexibility and expansion options.	Requires additional hardware.
VCF-WLD-RCMD-NSX-002	Use a dedicated or shared NSX Manager instance for the workload domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Manager to manage and patch.

Host Fault Tolerant NSX Edge Cluster Model

The *Host Fault Tolerant NSX Edge Cluster Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Host Fault Tolerant NSX Edge Cluster Model	A *Single-Rack Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Figure 70: Host Fault Tolerant NSX Edge Cluster Model in VMware Cloud Foundation



Host Fault Tolerant NSX Edge Cluster Model Attributes

The *Host Fault Tolerant NSX Edge Cluster Model* has the following attributes.

Attribute	Details
Availability Model	• Host level redundancy only. • Single shared storage. • This design cannot ensure continuous service availability for north-south communication and centralized network services during a complete rack failure.
NSX Edge Node Mobility	Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same rack that hosts the NSX Edge node.
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.

Attribute	Details
North-South Bandwidth Growth	• Add hosts to the single existing vSphere cluster or preferably transition to a rack fault tolerant design with two or more vertical Layer 2 clusters in separate racks. • When deploying more NSX Edge nodes on existing hosts, ensure each host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Common NSX Edge Cluster for Single Availability Zone Design Requirements

Table 144: Common NSX Edge Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMSAZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric Settings.

Host Fault Tolerant NSX Edge Cluster Model Design Requirements

Table 145: Host Fault Tolerant NSX Edge Cluster Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-HFTSR-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Table 146: Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMSAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMSAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMSAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs. • When deploying two NSX Edge nodes per host, each NSX Edge node requires two dedicated physical NICs configured for optimal performance.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMSAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface already exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMSAZ-008	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow.
VCF-NSX-EDGE-RCMD-COMSAZ-009	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-NSX-EDGE-RCMD-COMSAZ-010	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESXhost.	If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-011	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-NSX-EDGE-RCMD-COMSAZ-012	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually.
VCF-NSX-EDGE-RCMD-COMSAZ-013	Create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-014	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload virtual machines and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.
VCF-NSX-EDGE-RCMD-COMSAZ-015	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMSAZ-016	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-017	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSX on dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMSAZ-018	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two data plane fastpath interfaces gets an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP)	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMSAZ-019	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (Physical link state up but not forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-020	Limit the span of NSX Edge node networks to the local racks.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two (2) NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP session.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.
VCF-NSX-EDGE-RCMD-COMSAZ-021	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard for the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.
VCF-NSX-EDGE-RCMD-COMSAZ-022	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Host Fault Tolerant NSX Edge Cluster Model Design Recommendations

Table 147: Host Fault Tolerant NSX Edge Cluster Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-HFTSR-001	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across different hosts.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-NSX-EDGE-RCMD-HFTSR-002	Configure an NSX failure domain for each host where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability if there was a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Shared Design Selection between Management and Workload Domain

Shared Design Selections between Management and Workload Domain are the models that are the same across the *VCF Fleet in a Single Site* infrastructure blueprint.

Virtual Infrastructure Layer for VCF Fleet in a Single Site
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

vCenter Model
The *vCenter Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

vCenter Model Attributes

A *vCenter Model* has the following attributes.

Design Area	Attributes
vCenter Deployment Type	• One vCenter instance for the management domain that manages the management components of the VCF Instance. • Optionally, additional vCenter instances for workload domains to support workloads. • vSphere HA protecting all vCenter appliances.
vCenter Single Sign-On	• Each vCenter has a distinct vCenter Single Sign-On domain.

Methods for Protecting vCenter

VMware Cloud Foundation supports multi-rack management vSphere clusters, vSAN stretched clusters, or 3rd party synchronous replication solutions such as metro clusters that are designed to provide disaster recovery and/or high availability for the entire management stack (vCenter, NSX Manager, etc.) as opposed to a single vCenter.

Table 148: Methods for Protecting vCenter in VMware Cloud Foundation

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere High Availability	Yes	You must ensure sufficient resources in the cluster to allow virtual machines to restart on another ESX host in the event of an ESX host failure.
vCenter High Availability (vCenter HA)	No	• vCenter services must fail over to the passive node so there is no continuous availability. • Recovery time can be up to 15 mins. • You must meet additional networking requirements for the private network. • vCenter HA requires additional resources for the passive and witness nodes. • Lifecycle management is complicated because you must manually delete and recreate the standby virtual machines during a lifecycle management operation.
vSphere Fault Tolerance (vSphere FT)	No	• The vCPU limit of vSphere FT vCPU would limit vCenter appliance size to medium. • You must provide a dedicated network.

vCenter Model Design Requirements and Recommendations

Table 149: vCenter Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-CFG-001	Deploy a dedicated vCenter appliance for the management domain of the VCF Instance.	• Isolates vCenter failures to management or customer workloads. • Isolates vCenter operations between management and business workloads. • Supports a scalable vSphere cluster design where you can reuse the management components as more customer workloads are added to the VCF Instance. • Simplifies capacity planning for customer workloads because you do not consider management workloads for the workload domain vCenter. • Improves the ability to upgrade the vSphere environment and related components by enabling for explicit separation of maintenance windows: — Management workloads remain available while you are upgrading the tenant workloads. — Workloads remain available while you are upgrading the management nodes. • Supports clear separation of roles and responsibilities to ensure that only administrators with granted authorization can control the management workloads. • Facilitates quicker troubleshooting and problem resolution. • Simplifies disaster recovery operations by supporting a clear separation between recovery of the management components and tenant workloads. • Provides isolation of potential network issues by introducing network separation of the vSphere clusters in the VCF Instance.	The vSphere cluster must have sufficient resources.
VCF-VCS-REQD-NET-001	Place all workload domain vCenter appliances on the VM management network in the management domain.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-SSO-001	Create all vCenter instances within a VCF Instance in their own unique vCenter Single Sign-On domain.	• Sharing vCenter Single Sign-On domains is not supported. • Enables isolation at the vCenter Single Sign-On domain layer for increased security separation.	• Each vCenter instance is managed through its own pane of glass using a different set of administrative credentials. • You must manage password rotation for each vCenter Single Sign-On domain separately.

Table 150: vCenter Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VCS-RCMD-CFG-001	Deploy an appropriately sized vCenter appliance for each workload domain.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate compute resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-002	Deploy a vCenter appliance with the appropriate storage size.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate storage resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-003	Protect workload domain vCenter appliances by using vSphere HA.	vSphere HA is the only supported method to protect vCenter availability in VCF.	vCenter becomes unavailable during a vSphere HA failover.
VCF-VCS-RCMD-CFG-004	In vSphere HA, set the restart priority policy for the vCenter appliance to high.	vCenter is the management and control plane for physical and virtual infrastructure. In a vSphere HA event, to ensure that the rest of the SDDC management stack comes up faultlessly, the workload domain vCenter must be available first, before the other management components come online.	If the restart priority for another virtual machine is set to highest, the connectivity delays for the management components are longer.

ESX Model

The *ESX Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCF domain vSphere cluster.

ESX Hardware and Sizing Considerations

For more information on host sizing, refer to the VCF Planning and Preparation Workbook.

Table 151: ESX Hardware and Sizing Considerations

Hardware Element	Considerations
BIOS/Firmware	Ensure the ESX host hardware is on the Hardware Compatability List (HCL) and that you are running the latest supported BIOS and firmware.
CPU	• Total CPU requirements for the workloads that are running in the cluster. • Host failure and maintenance scenarios. • CPU over-commitment ratio guidelines – Management Domain • To ensure optimal performance and maintain resilience during host failures or maintenance, keep the management domain vCPU-to-pCPU over-commitment ratio at or below 2:1. **Note:** While higher ratios such as 3:1 or 4:1 are supported, you must closely monitor CPU ready times to ensure that the management cluster has enough capacity to run the workloads. – Workload domains • CPU over-commitment ratios for workload domains should be calculated on a case by case basis depending on the CPU intensity of the workloads. You must closely monitor CPU ready times to ensure optimal performance. • Number of physical cores, not logical cores. – Simultaneous multi-threading (SMT) in CPUs improve CPU performance however the performance enhancement will not be equivalent to 100% more CPU power.
Memory	• Total memory requirements for the workloads that are running in the cluster. • When sizing memory for the ESX hosts in a cluster, to reserve the resources of one host for failover or maintenance, set the admission control setting to N+1, which reserves the resources of one host for failover or maintenance. • *vSAN* OSA (Original Storage Architecture). – To support the maximum number of disk groups, you must provide 32 GB of RAM. • *vSAN* ESA (Express Storage Architecture). – You must provide a minimum of 128GB of RAM.
Storage	• Use high-endurance devices such as a hard drive or SSD for boot device. • Use a 128GB or greater boot device to maximize the space available for ESX-OS Data. • Use hosts with homogeneous configuration.
Network	• Use multiple physical NICs connected to multiple ToR switches to ensure redundancy. • 25GbE is recommended • Use physical NICs of the same speed, vendor, and model across all host within a cluster.
Power	• Use redundant power supplies, preferably connected to separate circuits to ensure availability. • Consider the impact of enabling Power Management on your ESX hosts. In order to allow ESX to control CPU power-saving features you need to set power management in the BIOS to *OS Controlled Mode*.

ESX Model Design Requirements and Recommendations

In your ESX design, you apply certain best practices.

Table 152: ESX Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-ESX-REQD-CFG-001	Install no less than the minimum number of ESX hosts required for the vSphere cluster model being deployed.	• Ensures that availability requirements are met. • If one of the ESX hosts is not available because of a failure or maintenance event, the CPU over-commitment ratio becomes 2:1.	None.
VCF-ESX-REQD-CFG-002	Ensure each ESX host matches the required CPU, memory, and storage specification.	Ensures that workloads will run without contention even during failure and maintenance conditions.	Assemble the server specification and number according to the sizing in the VCF Planning and Preparation Workbook which is based on projected deployment size.
VCF-ESX-REQD-SEC-001	Regenerate the certificate of each ESX host after assigning the ESX host an FQDN.	Establishes a secure connection with the VCF Installer during the initial deployment and prevents man-in-the-middle (MiTM) attacks.	You must manually regenerate the certificates of the ESX hosts before the deployment.

Table 153: ESX Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-001	When using vSAN as primary storage in the management domain, use vSAN ReadyNodes for each ESX host.	Your management domain is fully compatible with vSAN at deployment. **Note:** For information about the models of physical servers that are vSAN-ready, see vSAN Compatibility Guide for vSAN ReadyNodes.	• Hardware choices might be limited. • If you plan to use a server configuration that is not a vSAN ReadyNode, your CPU, disks, and I/O modules must be listed on the VMware Compatibility Guide under *CPU Series* and *vSAN Compatibility List* aligned to the ESX version.
VCF-ESX-RCMD-CFG-002	Allocate ESX hosts with uniform configuration across the default management vSphere cluster.	A balanced vSphere cluster has these advantages: • Predictable performance even during hardware failures. • Minimal impact of resynchronization or rebuild operations on performance.	You must apply vendor sourcing, budgeting, and procurement considerations for uniform server nodes on a per vSphere cluster basis.
VCF-ESX-RCMD-CFG-003	When sizing CPU, do not consider multi-threading technology and associated performance gains.	Although multi-threading technologies increase CPU performance, the performance gain depends on running workloads and differs from one case to another.	Because you must provide more physical CPU cores, costs increase and hardware choices become limited.
VCF-ESX-RCMD-CFG-004	Install and configure all ESX hosts in the default management vSphere cluster to boot using a 128-GB device or larger.	Provides ESX hosts that have large memory, that is, greater than 512 GB, with enough space for the scratch partition when using vSAN.	None.

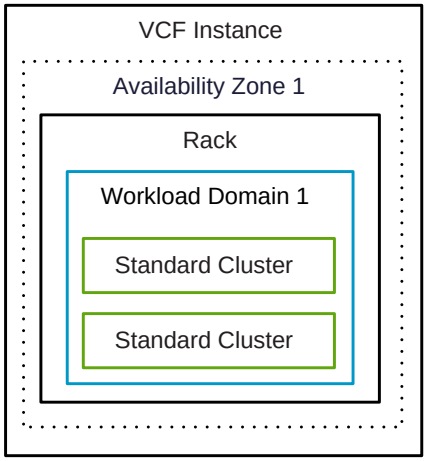
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-005	Use the default configuration for the scratch partition on all ESX hosts in the default management vSphere cluster.	• If a failure in the vSAN cluster occurs, the ESX hosts remain responsive and log information is still accessible. • If using vSAN storage its not possible to use a vSAN datastore for the scratch partition.	None.
VCF-ESX-RCMD-CFG-006	For workloads running in the default management vSphere cluster, save the virtual machine swap file at the default location.	Simplifies the configuration process.	Increases the amount of replication traffic for management workloads that are recovered as part of the disaster recovery process.
VCF-ESX-RCMD-NET-001	Place the ESX hosts in each management domain vSphere cluster on the ESX management network that is separate from the VM management network.	• Enables the separation of the physical VLAN between ESX hosts and the other management components for security reasons. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Increases the number of VLANs required.
VCF-ESX-RCMD-NET-002	Place the ESX hosts in each workload domain on a separate ESX host management VLAN-backed network.	Enables the separation of the physical VLAN between the ESX hosts in different workload domains for security reasons.	Increases the number of VLANs required. For each workload domain, you must allocate a separate management subnet.
VCF-ESX-RCMD-NET-003	Use dedicated VLANs for each traffic type.	Enables the separation of the physical VLAN for vSAN, vMotion, NFS & backup traffic.	Increases the number of VLANs required. For each traffic type you must allocate a separate VLAN and subnet.
VCF-ESX-RCMD-SEC-001	Set the advanced setting `UserVars.SuppressShellWarning` to `0` across all ESX hosts in the management domain.	• Ensures compliance with the *vSphere Security Configuration Guide* and with security best practices • Enables the warning message that appears in the vSphere client every time SSH access is activated on an ESX host.	You must turn off SSH enablement warning messages manually when performing troubleshooting or support activities.

Single-Rack vSphere Cluster Model

The *Single-Rack vSphere Cluster Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack vSphere Cluster Model	The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.

Figure 71: Single-Rack vSphere Cluster Model in VMware Cloud Foundation



Single-Rack vSphere Cluster Model Attributes

A *Single-Rack vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter.
Physical Rack Mapping	All cluster hosts exist in one physical rack.
Resilience	vSphere HA protects against host failures.

Single-Rack vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere which supports your storage type and VCF domain considerations.

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
Minimum number of ESX hosts	vSAN	Simple	Three (3)	• 33% CPU and memory • Tolerates one host failure	Three (3)	• 33% CPU and memory • Tolerates one host failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
	NFS or FC	Simple	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Important:

vSphere clusters are configured allowing for the failure of a single ESX host while still having enough resources to run all vSphere cluster workloads.

Attention:

* Designs should be completed in consultation with your storage vendor.

Common vSphere Cluster Design Requirements

Table 154: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Table 155: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Common vSphere Cluster Design Recommendations

Table 156: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Table 157: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

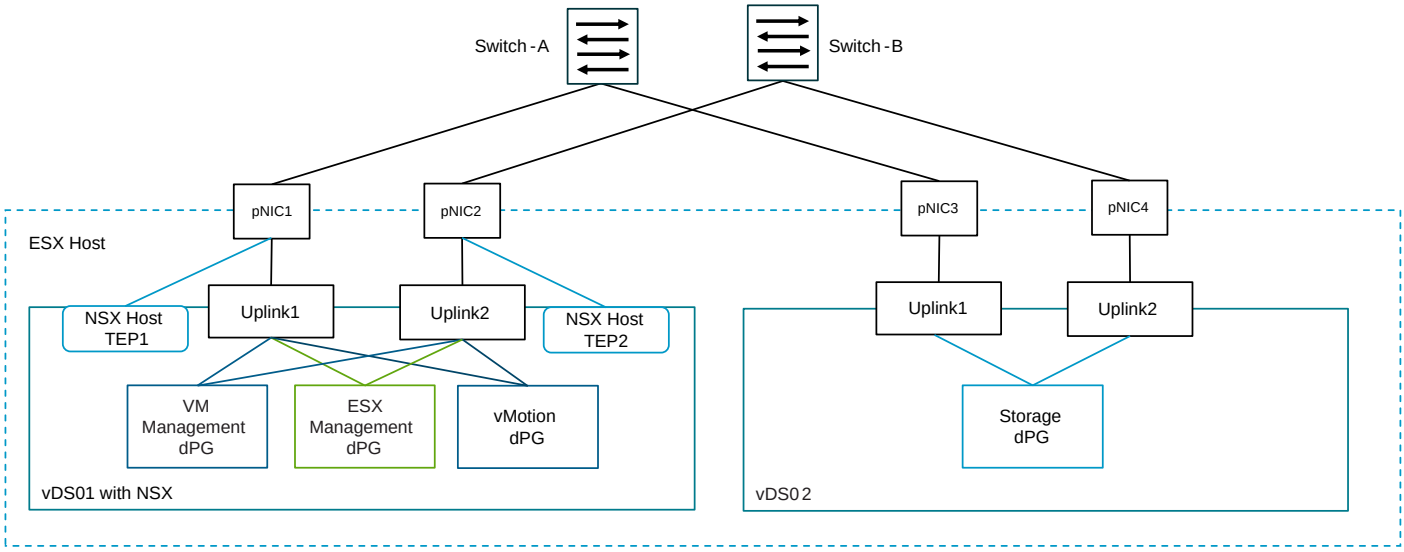
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Storage Separation Distributed Switch Model

The *Storage Separation Distributed Switch Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Storage Separation Distributed Switch Model	A *Storage Separation Distributed Switch Model* in VMware Cloud Foundation provides separation of storage traffic from all other network services running on the vSphere cluster.

Figure 72: Storage Separation Distributed Switch Model With Single Network Fabric in VMware Cloud Foundation



Storage Separation Distributed Switch Model Attributes

A *Storage Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Shared for Management, vSphere vMotion, NSX workloads. • Dedicated for IP storage.

Storage Separation Distributed Switch Model Options

A *Storage Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (storage traffic isolated to dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two or more network cards. • Each traffic type is spread across separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Common Distributed Switch Design Requirements

Storage Separation Distributed Switch Model Design Requirements

Table 158: Storage Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-SS-REQD-CFG-001	Use two vSphere Distributed Switch per vSphere cluster: • vDS01 — Management — vMotion — NSX • vDS02 — Storage	• Separates storage traffic onto a dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for storage traffic (NFS or vSAN) with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switch required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than the simple model.
VCF-VDS-SS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion	Provides the required network services for the vSphere cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-SS-REQD-CFG-003	Create the following distributed port groups on vDS02: • Storage	Provides connectivity for the hosts in the cluster to the storage network.	vSphere Distributed Switch storage traffic separation profile provides the required configuration.
VCF-VDS-SS-REQD-CFG-004	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switches for each additional rack: vDS01: • ESX management • vMotion vDS02: • Storage	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack
VCF-VDS-SS-REQD-CFG-005	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switches for availability zone 2 vDS01: • ESX management • vMotion vDS02: • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2

Common Distributed Switch Design Recommendations

Table 159: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Storage Separation Distributed Switch Model Design Recommendations

Table 160: Storage Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-DPG-001	Use the **Failover Order** teaming algorithm for the storage port group.	Provides a consistent traffic flow through a single physical ToR switch without needing to traverse Inter Switch Link during normal operations.	It must be manually configured during deployment using vSphere Distributed Switche custom profile option.
VCF-VDS-SS-RCMD-NIO-001	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than Principal storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-SS-RCMD-NIO-002	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as virtual machine traffic.	During times of network contention, vMotion takes longer than usual to complete.

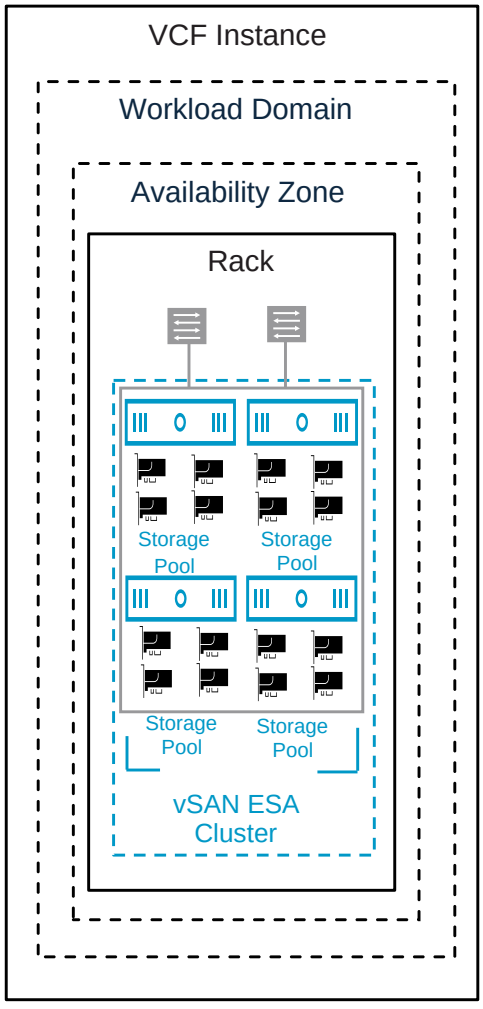
Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-NIO-003	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in the SDDC. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-SS-RCMD-NIO-004	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

Single-Rack vSAN ESA Storage Model

The *Single-Rack vSAN ESA Storage Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture that utilizes high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.

Figure 73: Single-Rack vSAN ESA Storage Model in VMware Cloud Foundation



Single-Rack vSAN ESA Storage Model Attributes

A *Single-Rack vSAN ESA Storage Model* has the following attributes.

Attribute	Detail
Single Tier	All data is stored on the same type of storage device, regardless of access frequency or importance.
Storage Pool	All NVMe Flash Storage.
Data Services	• Compression (enabled by default). • Encryption. (in transit and at rest) • vSAN Global Deduplication **Restriction:** vSAN Global Deduplication is not currently compatible with vSAN Stretched Clusters and Two Node vSAN Clusters. • Integrated vSAN File Services. • vSAN iSCSI Target Service. • vSAN Storage Clusters. • Stretched Cluster. • Fault domains. • Kubernetes persistent storage services (block and file). • Native scalable snapshots
Virtual Machine Storage policies	vSAN storage policies (also known as Virtual Machine Storage Policies or VM Storage Policies) are rules defining virtual machine (VM) storage requirements that vSAN automatically enforces across a cluster. vSAN 9.1 automatically configures vSAN Auto-RAID policy on new vSAN ESA clusters. vSAN Auto-RAID policy manages the resilience for data residing in a cluster, regardless of the size or topology of the vSAN cluster. vSAN Auto-RAID creates a single policy within the vCenter server,

Common vSAN Storage Model Design Requirements

Table 161: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Common vSAN Host Network Model Design Requirements

Table 162: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

vSAN ESA Storage Model Design Requirements

Table 163: vSAN ESA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-ESA-REQD-CFG-001	Ensure vSAN certified Drives used for vSAN ESA clusters.	• When using VMware Cloud Foundation automated workflows, vSAN ESA cluster creation maybe blocked unless vSAN certified drives are detected. • vSAN Auto-RAID will be enabled when vSAN certified components are detected. This allows vSAN to generate an optimal default storage policy, based on the cluster topology and the number of ESX hosts.	None.
VCF-VSAN-ESA-REQD-CFG-002	1.6 TB drive size or higher for vSAN ESA devices for storage pools.	Minimum supported vSAN ESA drive size is 1.6TB.	vSAN ESA supports up to 24 drives per node. Smaller capacity drives may constrain raw capacity of a vSAN node.
VCF-VSAN-ESA-REQD-CFG-003	16 Cores or higher per vSAN node. vSAN ESA has a minimum value of 16 cores per vSAN Node	Minimum vSAN ESA core count per node is 16 for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	Smaller number of cores per node may require more vSAN nodes to support targeted workloads.
VCF-VSAN-ESA-REQD-CFG-004	Memory size per vSAN Ready node 128GB or higher	Minimum vSAN ESA memory is 128GB for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	• Smaller memory footprint per node will limit number of vSAN drives per node. • Smaller memory footprint per node for vSAN ESA will limit VM workloads being placed on vSAN nodes.

Common vSAN Storage Model Design Recommendations

Table 164: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Common vSAN Host Network Model Design Recommendations

Table 165: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

vSAN ESA Storage Model Design Recommendations

Table 166: vSAN ESA Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-001	• Select a specific size of drives used according to the requirements of target workloads. • Use multiple storage devices per vSAN node • NVMe TLC devices with a performance Class F (100000 - 349999) or higher with endurance class 3 DWPD or higher.	• The number of storage devices has an impact on the performance of vSAN ESA. • NVMe drives with mixed-use endurance and performance class typically classified on the vSAN HCL as "3DWPD" (3 Drive Writes per Day) will give a balance between performance and durability.	• Smaller drive sizes will impact overall capacity of vSAN ESA. Large drive sizes will introduce larger capacity fault domain in case of failure. • Sizing vSAN Nodes requires determining the workload capacity requirements prior to deployment. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-002	Use vSAN ready nodes with high core count.	vSAN Express Storage Architecture (ESA) is designed to be highly parallel and leverages multiple CPU cores for improved performance, making it multithreaded, and optimized for NVMe- TLC based flash devices.	There will be a correlation between target workload, number of cores and NVMe devices per vSAN Node. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

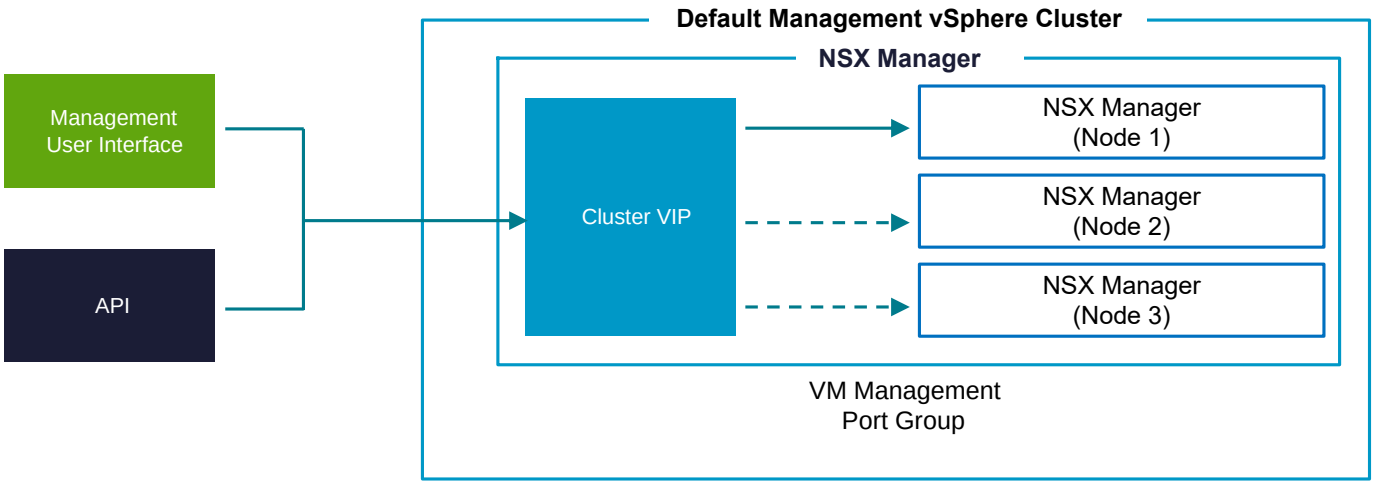
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-003	Use vSAN ready nodes with high memory capacity per vSAN node.	When enabled vSAN has required memory overheads and must have adequate resources account for VM workloads.	Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsanizer/home.
VCF-VSAN-ESA-RCMD-CFG-004	At least four (4) nodes for vSAN cluster size.	This ensures in the event of a fault domain failure, vSAN objects can be re-protected to maintain policy compliance. In conjunction with auto policy management, four (4) node vSAN ESA clusters will allow vSAN to automatically create a default storage policy that is capacity optimized.	None. **Note:** New vSAN clusters in VCF 9.1 will use clusters will automatically be configured to use Auto-RAID.
VCF-VSAN-ESA-RCMD-CFG-005	• Select the same minimum requirements for vSAN Storage cluster nodes as vSAN ESA HCI nodes. • Review vSAN Storage minimum node requirements	• vSAN Storage Cluster Nodes use the same architecture as regular vSAN ESA nodes. • vSAN ESA nodes can be re-purposed for vSAN Storage Clusters or vSAN HCI Clusters	None.

High Availability NSX Manager Model

The *High Availability NSX Manager Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
High Availability NSX Manager Model	This model addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.

Figure 74: High Availability NSX Manager Model in VMware Cloud Foundation



High Availability NSX Manager Model Attributes

The *High Availability NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Three-node cluster of NSX Manager appliances with internal virtual IP
Availability	• Highly available cluster of virtual IP NSX Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.
API Request Scale	Internal virtual IP directs API requests to a single NSX Manager appliance.

Common NSX Manager Design Requirements for All Models

Table 167: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

High Availability NSX Manager Model Design Requirements

Table 168: High Availability NSX Manager Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-005	Deploy three (3) NSX Manager nodes in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports high availability of the NSX Manager cluster.	You must have sufficient resources in the default cluster of the management domain to run three NSX Manager nodes.

Common NSX Manager Model Design Recommendations for All Models

Table 169: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

High Availability NSX Manager Model Design Recommendations

Table 170: High Availability NSX Manager Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-003	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Manager appliances.	Keeps the NSX Manager appliances running on different ESX hosts for high availability.	You must allocate at least four physical hosts so that the three NSX Manager appliances continue running if an ESX host failure occurs.

Physical Infrastructure Layer for VCF Fleet in a Single Site

The physical infrastructure constitutes the tangible hardware and facilities layer that provides the foundational computing, storage, and networking resources upon which all digital services depend. This layer encompasses the actual physical assets—servers, storage arrays, network equipment, and supporting facilities—that form the bedrock of modern IT operations.

Single-Rack Network Fabric Model

The *Single-Rack Network Fabric Model* was chosen for the *VCF Fleet in a Single Site* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack Network Fabric Model	The data center network fabric provides connectivity to the vSphere clusters. The *vSphere Cluster Models* and *Storage Models* chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a *Single-Rack Cluster Model*. Depending upon the Storage type chosen, additional VLAN requirements may apply.

The data center network fabric provides connectivity to the vSphere clusters. The vSphere Cluster Models and Storage Models chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a Single-Rack vSphere Cluster Model. Depending upon the storage type chosen, additional VLAN requirements may apply.

Note:

If this rack will contain NSX Edge nodes, additional VLAN Requirements will apply. NSX Edge Cluster Models for Single Availability Zone

Table 171: Infrastructure VLAN Scope and First Hop Gateway Availability for Single Rack

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management VM Appliances*		Rack Local	Highly Available	1500	1500
ESX host	ESX management VMK *				1500
	IP Storage VMKs (vSAN, NFS, iSCSI)				9000
	vMotion VMK				
	Host TEP			1700	
NSX Edge node	Refer to Host Fault Tolerant NSX Edge Cluster Model or Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters				
	NSX Edge Node management	Rack Local	Highly Available	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available Static Routing with BFD - Optional BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
	BGP-EVPN Peering	Across Racks	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* When using separate VLAN segments for management component VMs and ESX VMKs.					

Table 172: Workload VLAN Scope and First Hop Gateway Availability for Single Rack

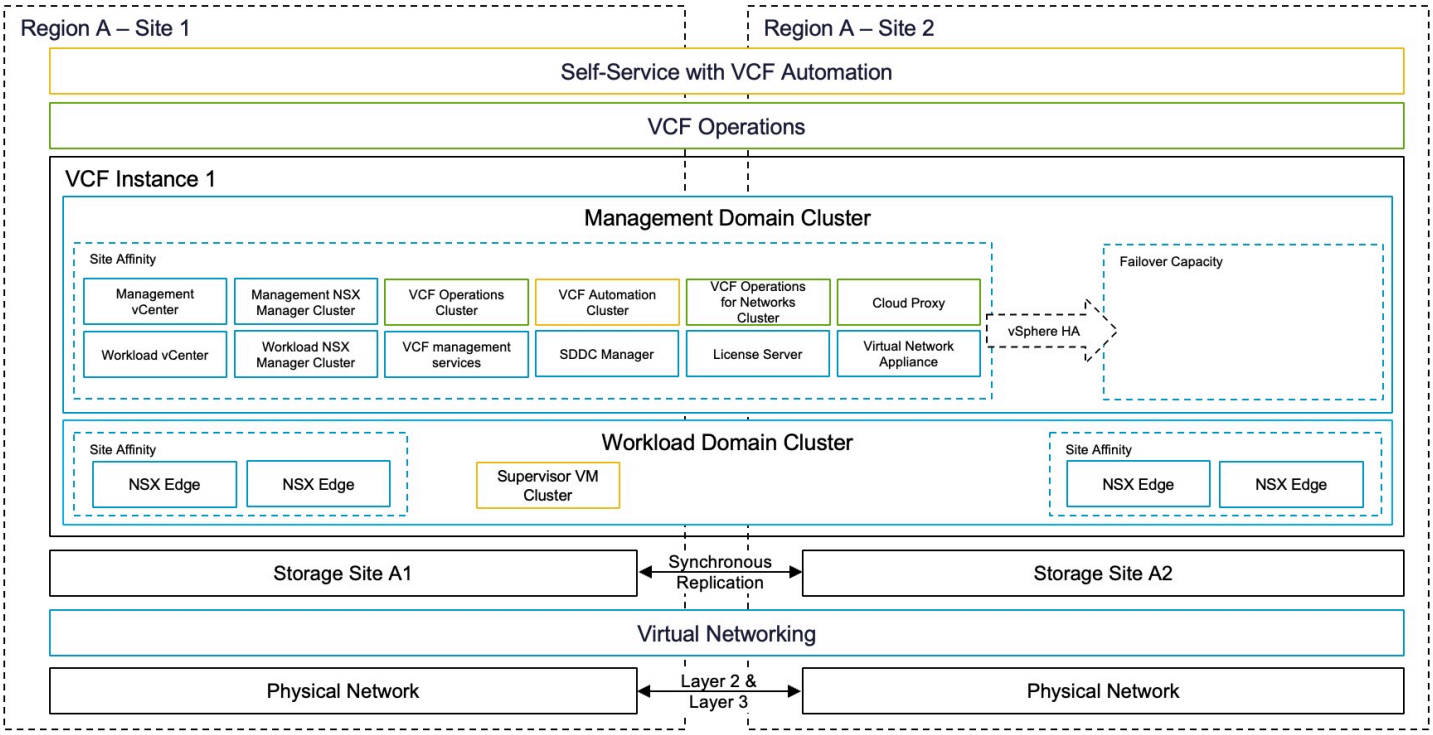
Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VLAN Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model Attributes	Refer to Edges			

VCF Fleet with Multiple Sites in a Single Region

The *VCF Fleet with Multiple Sites in a Single Region* is an infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components and workloads are isolated to a separate workload domain. Multiple sites exist and stretched clusters are used to increase availability across sites.

Architecture Overview

Figure 75: VCF Fleet for Multiple Sites in a Single Region



Design Profile

The *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint is well suited to customers who are looking for the following design profile for their VMware Cloud Foundation platform.

Attribute	Value
Consumption	• Self-service consumption via VCF Automation supporting modern workloads like VKS, Data Services, AI Services, and so on. with multi-tenancy. • Direct vCenter consumption supporting modern workloads like VKS, Data Services, AI Services, and so on. This infrastructure blueprint can use the following consumption blueprint: • Self-Service Multi-Tenant Private Cloud Consumption
Physical Site Configuration	Multiple sites within a single region (single instance, stretched cluster between the sites for maximum availability)

Attribute	Value
Availability	• Tolerates failure of a single ESX host • Tolerates failure of an individual network path • Tolerates failure of a single rack • Tolerates failure of a single management component • Tolerates failure of a single availability zone
Isolation	• Hypervisor-based (logical isolation) • Network traffic type isolation • Cluster-based (physical isolation) • Tenant / Tenant isolation
Recoverability	This infrastructure blueprint can use the following protection blueprints: • VCF Component Backup and Restore • VCF Instance Backup and Restore
Expansion	For available resource and availability expansions, see Expansion after Infrastructure Blueprint Deployment.

Management Domain Design Selections

The following models in the design align with the objectives for deploying the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	High Availability VCF Automation Model	The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.
	Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.
Operations	High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single cloud proxy appliance. To achieve high availability after the initial deployment, an additional cloud proxy appliance should be deployed and then combined in a collector group.
	VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.
	High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.
	Single License Server Model	The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.

Layer	Selected Model	Additional Information
	High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a three-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.
	High Availability Log Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind an internal load balancer. This model provides resiliency at both the application level through the multi-replica deployment, VCF management services to restart replicas, and vSphere HA to restart nodes in the event of an ESX host or log management component failure.
Security Governance and Compliance	Instance Identity Broker Model	Deploys the Identity Broker component on the VCF management services instance in the management domain.
	VCF Fleet-Wide Single Sign-On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all components across all VCF Instances in the same VCF fleet.
Virtual Infrastructure	Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.
	VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Workload Domain Design Selections

The following models in the design align with the objectives for deploying the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.
	High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected vSphere Supervisor management zones.
	Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for VCF Supervisor and VKS workloads, it supports VPC-based networking.
	Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone	This approach provides robust resource isolation for different lines of business (LOBs) within a single vSphere Supervisor Zone.
Virtual Infrastructure	Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.

Layer	Selected Model	Additional Information
	Dual Availability Zone Design Based on NSX Edge Node HA Model	A *Single-Rack Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Shared Design Selection between Management and Workload Domain

Shared Design Selections between Management and Workload Domain are the models that are the same across the *VCF Fleet in a Single Site* infrastructure blueprint.

Layer	Selected Model	Additional Information
Virtual Infrastructure	vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.
	ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCFdomain vSphere cluster.
	Stretched vSphere Cluster Model	The *Stretched vSphere Cluster Model* allows you to protect workloads on vSphere clusters in your environment against a failure of a single fault domain by implementing a stretched vSphere cluster across multiple availability zones.
	Storage Separation Distributed Switch Model	A *Storage Separation Distributed Switch Model* in VMware Cloud Foundation provides separation of storage traffic from all other network services running on the vSphere cluster.
	Stretched vSAN Storage Model	A *Stretched vSAN Storage Model* is a cluster configuration where the ESX hosts, that comprise a single vSAN cluster, reside in two availability zones.
	High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.
Physical Infrastructure	Multi Availability Zone Network Fabric Model	This section describes the VLAN and gateway requirements to support a vSphere cluster within multiple availability zones.

Minimum Requirements for VCF Fleet with Multiple Sites in a Single Region

VCF Fleet with Multiple Sites in a Single Region requirement to stand up this blueprint.

VCF Installer Deployment Options

Deployment Model	Deployment Size
High Availability	Medium

Minimal IP Requirements

Requirement	Additional Information
Physical Networks	• AZ 1 ESX Management (4 IPs) • VM Management (35 IPs) • AZ 1 vMotion (4 IPs) • AZ 1 vSAN (4 IPs) • AZ 1 Host TEP (8 IPs) • AZ 2 ESX Management (4 IPs) • AZ 2 vMotion (4 IPs) • AZ 2 vSAN (4 IPs) • AZ 2 Host TEP (8 IPs) • (Optional) Edge TEP Network (4 IPs) — 2 IPs per Edge VM (x2 Edge VMs) • (Optional) Edge Uplink Network (2 IPs) — 1 IP per Edge VM (x2 Edge VMs)
DNS Entries	• vCenter • AZ 1 ESX — 4 Nodes • AZ 2 ESX — 4 Nodes • SDDC Manager • NSX Manager — VIP — Node A — Node B — Node C • VCF Operations — Primary — Replica — Data — Cloud Proxy • VCF Management Services — VCF Services Runtime — VCF Management Instance Service — VCF Fleet Component • VCF Automation • VCF Service Runtime for VCF Automation • License Server • Identity Broker • Log Management • VCF Operations for Networks — Platform 1 — Platform 2 — Platform 3 — Collector
Virtual Networks	• vSphere Supervisor Service CIDR — e.g. 10.96.0.0/23 • VPC Private IP Block (Control Plane) • Distributed Transit Gateway Private IP Block (Control Plane) • External IP Block (Control Plane & Workload/Organizations) • Private VPC Block for each Organization's VPCs • Private TGW IP Block for each Organization

First VCF Instance

Table 173: VMware Cloud Foundation Installer Deployment Profiles

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for the initial VCF instance within a VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	1x Small • 1x Control Plane Nodes • 2x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Large • 3x Control Plane Nodes • 4x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Medium	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
VCF Operations	1x Small	2x Small	3x Medium	3x Large
License Server	1x Default	1x Default	1x Default	1x Default
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard
VCF Automation	1x Small	1x Medium	3x Medium	3x Large
Resource Requirements	• vCPU: 76 • RAM: 251 GB • DISK: 7448 GB	• vCPU: 106 • RAM: 335 GB • DISK: 8422 GB	• vCPU: 184 • RAM: 656 GB • DISK: 11445 GB	• vCPU: 298 • RAM: 949 GB • DISK: 15357 GB

Management Domain Design Selections

The following models in the design align with the objectives for deploying the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

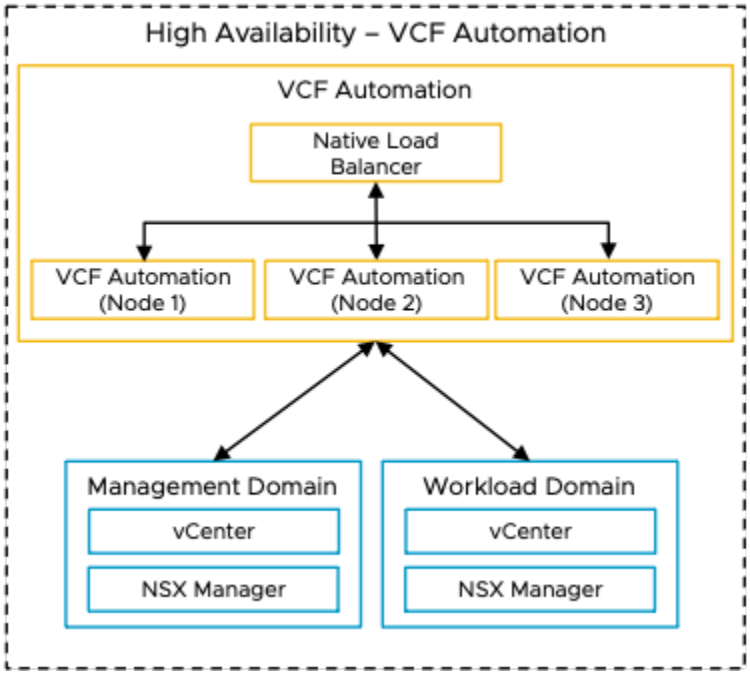
Consumption Layer for VCF Fleet with Multiple Sites in a Single Region

The consumption layer for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

High Availability VCF Automation Model

The *High Availability VCF Automation Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Automation Model	The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.



High Availability VCF Automation Model Attributes

A *High Availability VCF Automation Model* has the following attributes.

Attribute	Detail
Availability	• Application level availability provided by the three node cluster • Appliance level level availability provided by vSphere HA
Scalability	Supports scale-up and scale-down of node sizes.
Recoverability	Restore through a backup.

High Availability VCF Automation Model Design Requirements

Table 174: High Availability VCF Automation Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AUTO-REQD-HA-001	Deploy VCF Automation as a three node cluster.	Supports high availability objectives.	• Analysis is required to determine which node size is appropriate for your deployment. • Node can be resized after deployment as necessary.
VCF-AUTO-REQD-HA-003	Protect all VCF Automation nodes using vSphere HA.	Ensures availability of VCF Automation without requiring manual intervention during an ESX host failure event.	None

VCF Automation Tenant Authentication Requirements

Table 175: VCF Automation Tenant Authentication Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-TEN-001	Configure an identity provider for VCF Automation organizations.	• Allows tenants to use enterprise directory. • Enables named access control to tenant level constructs. • Required to provision administrative or user access to the Organization portal for any account other than the first user account established during Organization creation.	If not configured, the only access to the Organization Portal will be through the first user account established during Organization creation.

VCF Automation Enterprise Authentication Design Requirements

Table 176: VCF Automation Enterprise Authentication Design Requirements

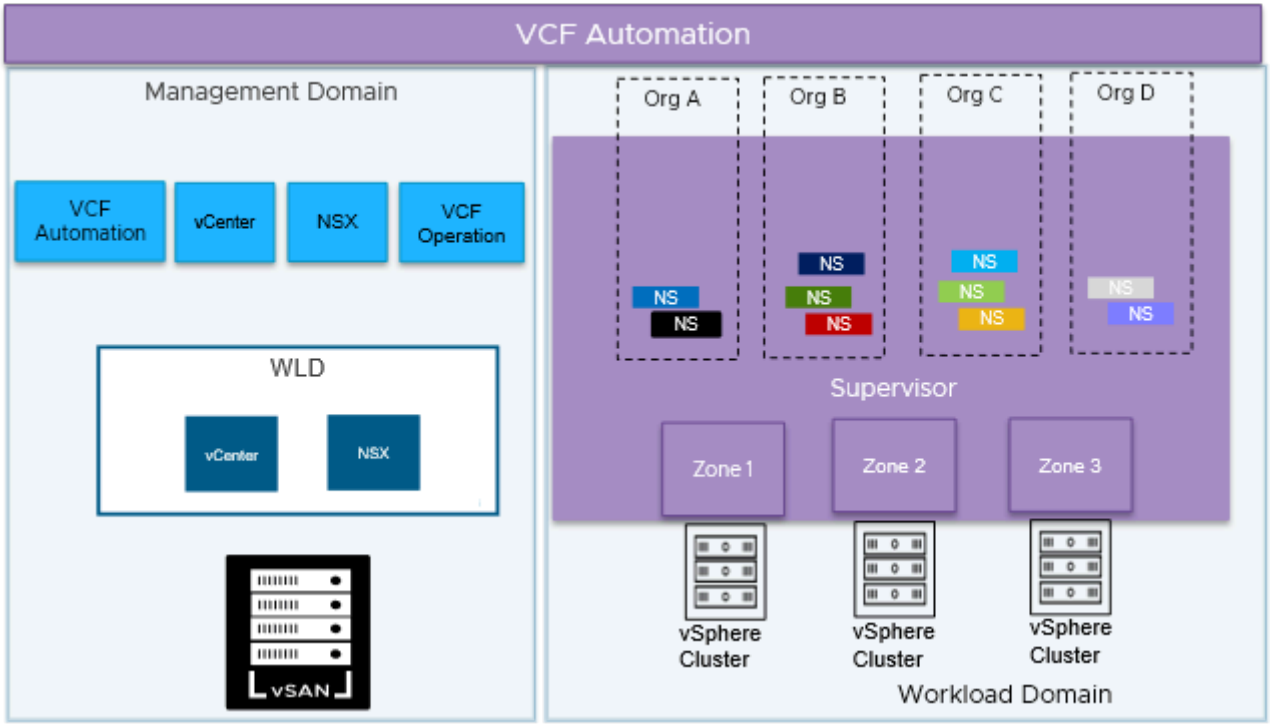
Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-IDP-001	The provider uses the same IdP as the tenant.	• Allows consistent authentication across provider and tenant	None

Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance

The*tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.

Figure 76: Shared Workload Domain for Multiple Organizations in a VCF Instance



Benefits	Best fit scenarios
• Resource Sharing: Tenants share the underlying infrastructure (workload domain, vCenter, NSX), leading to cost savings and efficient resource utilization. • Simplified Management and Operational Consistency: While providing tenant isolation, the centralized management of the VCF Management Domain simplifies overall infrastructure operations, including patching, upgrades, monitoring, and capacity planning. Consistent policies can be applied across all tenants. • Scalability and Flexibility: The model is highly scalable. As the number of tenants or their resource demands grow, the underlying workload domain can be scaled up by adding more hosts. Resource allocations within Namespaces can also be adjusted dynamically. If the requirement comes, the service provider can add more workload domains for multiple other organizations to share or can dedicate a workload domain per tenant like explained in Model 2. • vSphere Namespace Isolation: vSphere Namespaces provide logical resource isolation for tenant workloads, preventing interference. • Network Segmentation: Network policies and security groups can be used to further segment tenant traffic and can enhance security.	• Cloud Service Providers (CSPs) Offering Multi-Tenant Cloud Services: This model allows the CSP to host multiple customers within a single VCF instance, maximizing resource utilization across their infrastructure. The Supervisor Cluster ensures strong isolation between each customer's workloads and data. Each customer gets their own vSphere Namespace, providing them with a secure and logically separated environment for deploying VMs and containers. Network segmentation (via NSX) further guarantees that customer traffic remains isolated. • Large Enterprises with Multiple Business Units or Departments: A large corporation has several independent business units (e.g., Finance, Marketing, R&D) that require their own isolated IT environments but can share underlying infrastructure. • Organizations Offering Internal PaaS or Developer Platforms: An organization wants to provide a self-service platform for its internal development teams to deploy and manage their applications (including containerized microservices) in an isolated and governed manner. • Educational Institutions or Research Organizations with Multiple Projects or Teams: A university hosts various research projects or has different departments with distinct IT requirements and security needs.

Operations Layer for VCF Fleet with Multiple Sites in a Single Region

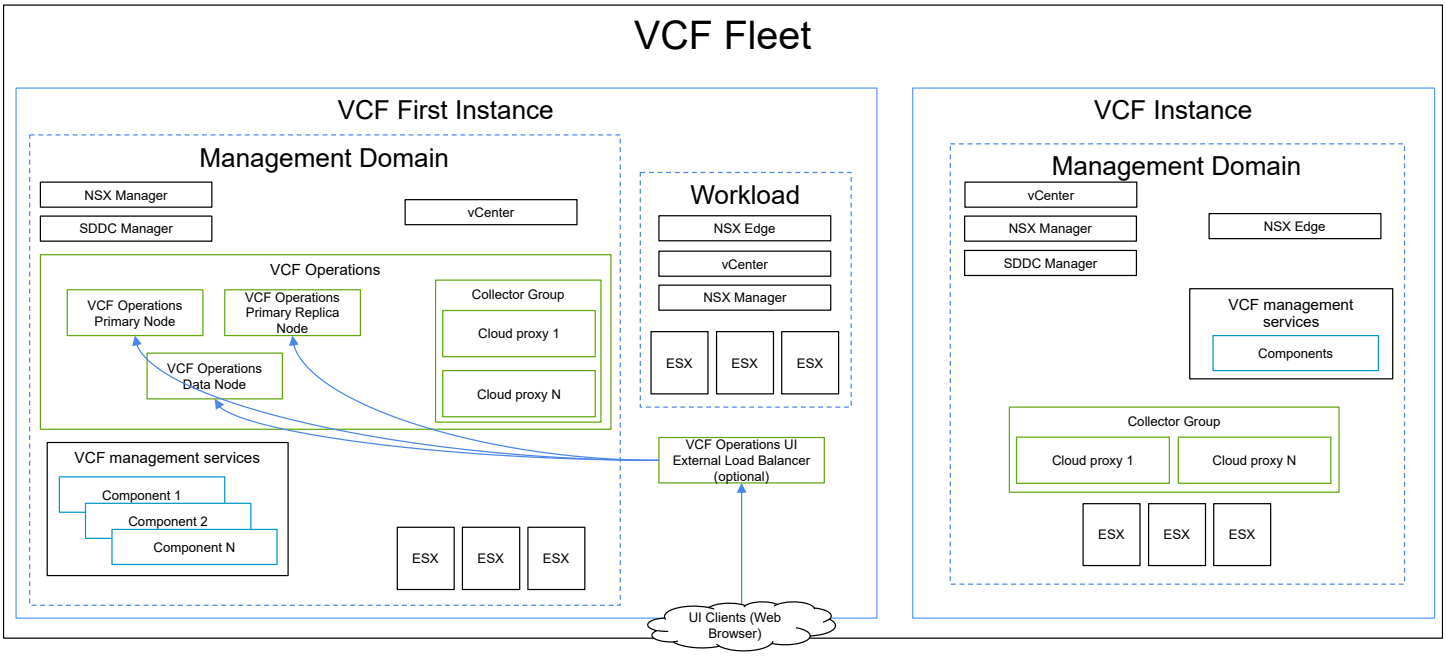
The operations layer serves as the central nervous system of enterprise infrastructure and applications, providing comprehensive visibility, management, and control capabilities across the entire technology stack. This layer ensures that systems remain available, performant, secure, and compliant with organizational standards and regulatory requirements.

High Availability VCF Operations Model

The *High Availability VCF Operations Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single cloud proxy appliance. To achieve high availability after the initial deployment, an additional cloud proxy appliance should be deployed and then combined in a collector group.

Figure 77: High Availability VCF Operations Model in VMware Cloud Foundation



High Availability VCF Operations Model Attributes

A *High Availability VCF Operations Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure. Additional level of availability for the components in: • VCF Operations cluster • VCF Operations collector groups
Scalability	Supports scale-up and scale-out. **Note:** Scale-out is supported for the VCF Operations data nodes and cloud proxynodes.
Recovery	Restore through a backup.
Load balancing	Optional external load balancer for the VCF Operations cluster UI.

Common VCF Operations Design Requirements for All Models

Table 177: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Operations Model Design Requirements

Table 178: High Availability VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-001	Deploy VCF Operations as a cluster of three nodes in the default management vSphere cluster.	• Supports the high availability objectives by deploying VCF Operations as a cluster. • Provides the capacity that is required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out with additional data nodes.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • Create an anti affinity rule (Separate Virtual Machines) for the VMs in the cluster after the deployment.
VCF-OPS-REQD-HA-002	Deploy a single Unified VCF Operations collector node in the default management vSphere cluster.	Removes the load from the VCF Operations cluster from collecting metrics from local-instance applications.	You may need to scale-up the VCF Operations collector nodes to a greater size based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-HA-003	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations cluster nodes.	Using vSphere DRS prevents the VCF Operations cluster nodes from running on the same ESX host and risking the high availability of the cluster.	• You must perform an additional configuration to set up an anti-affinity rule. • You must have a minimum of four ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode. • If additional data nodes are added, you must update the anti-affinity rule and ensure there are enough ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode.
VCF-OPS-REQD-HA-004	The certificate for VCF Operations must include the FQDNs for all cluster nodes in the SAN (subject alternative name) attribute. **Note:** The SAN does not need to include the FQDN of VCF Operations collector nodes.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted.	Before scale out, the certificate for VCF Operations must be changed to also include the FQDN of the additional node in the SAN (subject alternative name) attribute.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-005	If using an optional external load balancer in the high availability deployment model, the certificate for VCF Operations **must** also include the FQDN of the load balancer in the SAN (subject alternative name) attribute.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted when using an external load balancer with the high availability model.	None.

Common VCF Operations Design Recommendations for All Models

Table 179: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

High Availability VCF Operations Model Design Recommendations

Table 180: High Availability VCF Operations Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-HA-001	Deploy VCF Operations with an external load balancer to load balance the connections across the VCF Operations cluster nodes.	Required to deploy a VCF Operations cluster deployment with distributed user interface access across members.	You must configure the external load balancer configuration to support this VCF Operations configuration.
VCF-OPS-RCMD-HA-002	Create a collector group in VCF Operations for each VCF Instance and assign the default VCF Operations collector appliance.	Provides increased availability to metric collection across the VCF Operations collector appliances that are assigned to the collector group.	Manual reconfiguration of the VMware Cloud Foundation adapter required to utilize multiple VCF Operations collector appliances with the group.
VCF-OPS-RCMD-HA-003	Deploy an additional VCF Operations collector node in the default management vSphere cluster.	Extends the availability of the VCF Operations collectors by balancing the load across the collectors when added to a collector group.	• You must deploy the collector node after the deployment of the VCF Operations. • Additional collector node should be deployed for all VCF Instances that requires a collector group.
VCF-OPS-RCMD-HA-004	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations collector nodes.	Using vSphere DRS prevents the VCF Operations collectors from running on the same ESX host and risking the high availability of the collectors.	You must perform manual configuration to set up an anti-affinity rule.
VCF-OPS-RCMD-HA-005	Reconfigure the default VMware Cloud Foundation Adapter in VCF Operations to use a local-instance collector group for a VCF Instance.	Increases the availability of metric collection.	Must be reconfigured post-deployment.

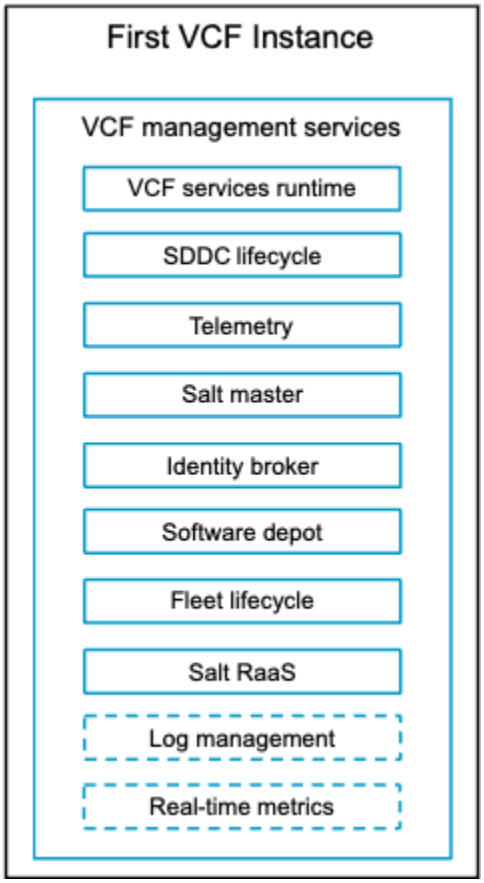
VCF Management Services in the First VCF Instance Model

The *VCF Management Services in the First VCF Instance Model* chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.

Figure 78: VCF Management Services in the First VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.



VCF Management Services in the First VCF Instance Model Components

Table 181: VCF Management Services in the First VCF Instance Model Components

The following components run in the first VCF Instance in the VCF fleet.

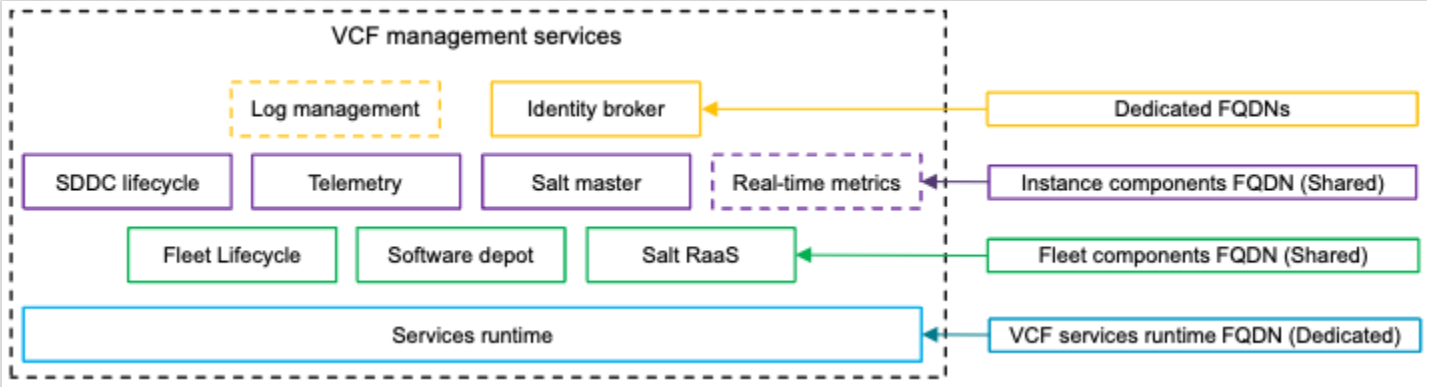
Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations across the fleet.	
Fleet lifecycle	Orchestrates fleet-wide life-cycle operations, including the planning and sequencing of upgrades and patches across multiple VCF Instances.	

Service	Description	Deployment Time
Salt RaaS	Defines and manages desired state configurations and policies (Recipe as a Service).	
Log management	Collects logs from all fleet components	Day-N Components
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

VCF Management Services in the First VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 79:



VCF Management Services in the First VCF Instance Model Design Requirements

Table 182: VCF Management Services in the First VCF Instance Model Design Requirements

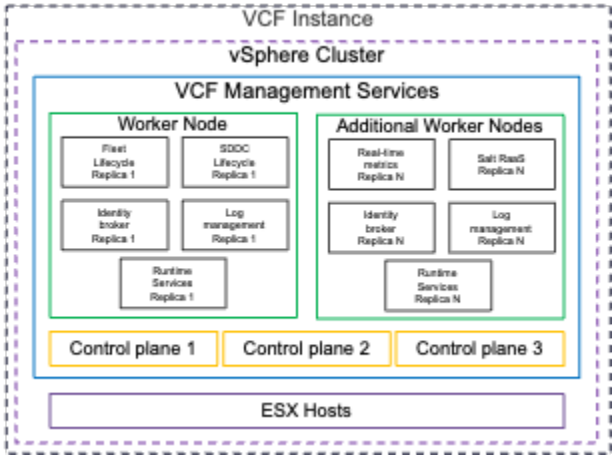
Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-FIRST-001	Create DNS entries for the following shared FQDNs: • Fleet components FQDN • Instance components FQDN	Required for deploying VCF management services fleet and instance level components in the first VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-FIRST-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN • Identity broker FQDN	Required for deploying VCF management services in the first VCF Instance.	You must create the DNS entries before starting the deployment.

High Availability VCF Management Services Model

The *High Availability VCF Management Services Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Figure 80: High Availability VCF Management Services Model in VMware Cloud Foundation



High Availability VCF Management Services Model Attributes

Attribute	Detail
Availability	Relies on vSphere HA
Scalability	Scale out/up worker nodes
Placement	Hosted in a single vSphere cluster

Common VCF Management Services Model Design Requirements

Table 183: Common VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-001	Allocate a sufficient IP range for the required number of control plane and worker nodes for your chosen deployment type.	Required for initial deployment.	Adding additional components may require auto-scale out of the VCF management services.
VCF-MS-REQD-COM-002	Use *Kubernetes Cluster API Provider vSphere (CAPV)* node anti-affinity to ensure the nodes run on different ESX hosts.	Using *CAPV* prevents the VCF management services nodes from running on the same ESX host and risking the high availability of the cluster.	Distributed Resource Scheduler (DRS) anti-affinity rules are not required.
VCF-MS-REQD-COM-003	Protect the VCF management services nodes by using vSphere HA.	Ensures availability of the VCF management services nodes without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Management Services Model Design Requirements

Table 184: High Availability VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-HA-001	Deploy VCF management services in high availability mode with three (3) control plane nodes and the corresponding number of worker nodes in the default management vSphere cluster.	• Supports the high availability deployment objective. • VCF management services supports scale-out and up from small to medium or large.	Requires additional resources.

Single License Server Model

The *Single License Server Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Single License Server Model	The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.

Table 185: Single License Server Model Attributes

Attribute	Detail
Use Cases	This is the typical license server deployment model for VMware Cloud Foundation and is suitable for most VCF deployments.
Deployment mechanism	One license server per VCF fleet. A license server is automatically deployed as part of the installation or upgrade to VMware Cloud Foundation 9.1.
Backup and restore	Backup is not required. If the original license server becomes unavailable, deploy a new license server instance and migrate the licenses within 90 days**.**
Lifecycle	The upgrade and patch management of the license server is managed through VCF Operations.
High Availability	High availability is provided by vSphere HA.
License Scope	A single license server provides licensing management for the whole VCF fleet.

Single License Server Model Design Elements

Table 186: Single License Server Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LIC-REQD-LS-001	Register the default license server in VCF Business Services console.	Registration allows the VCF Business Services console to add license allocations to the license server.	None.

Licensing for VCF Operations Design Recommendations Connected Mode

Table 187: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Licensing for VCF Operations Design Recommendations Disconnected Mode

Table 188: Licensing for VCF Operations Design Recommendations

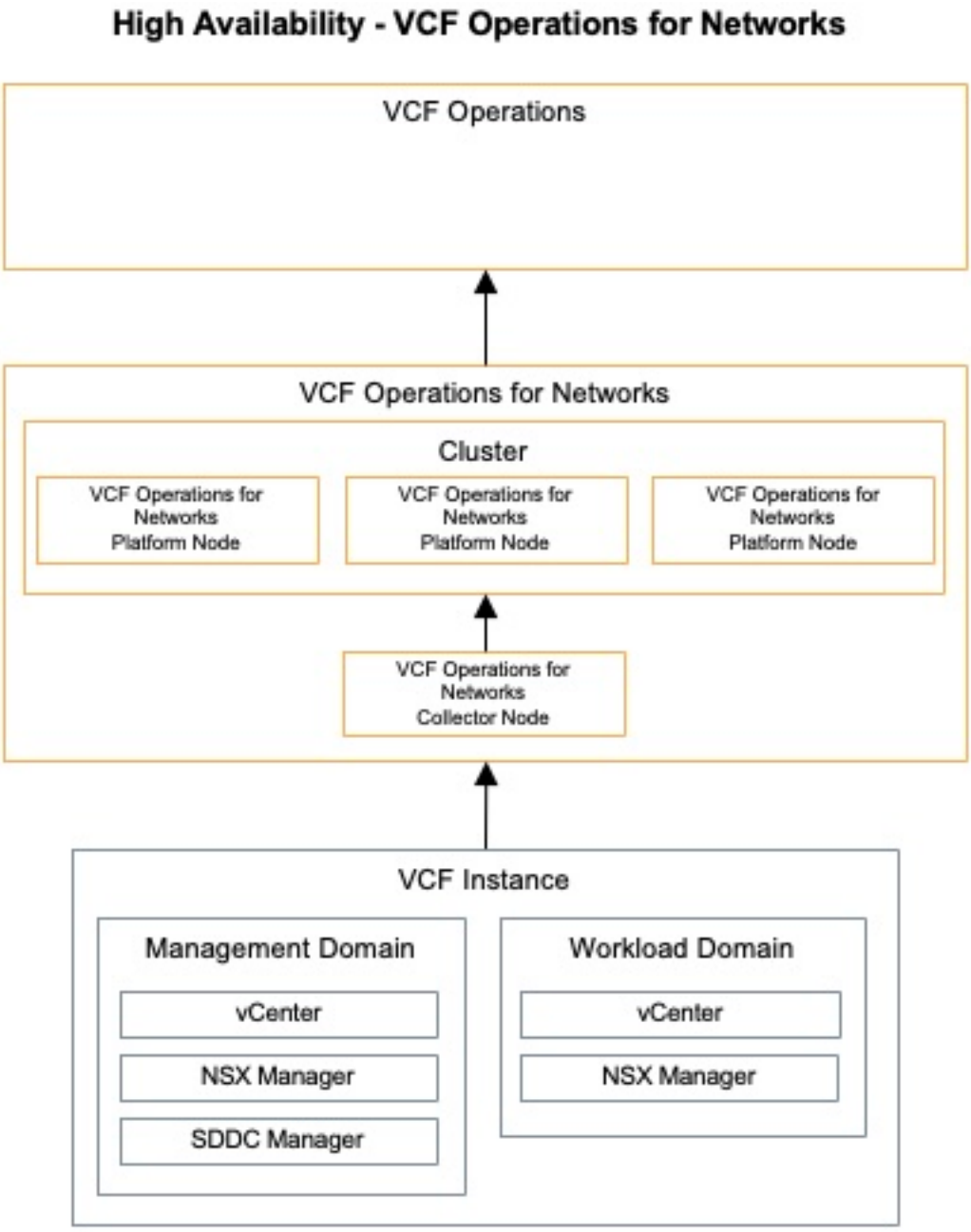
Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

High Availability VCF Operations for Networks Model

The *High Availability VCF Operations for Networks Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a three-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Figure 81: High Availability VCF Operations for Network Model in VMware Cloud Foundation



High Availability VCF Operations for Networks Model Attributes

A *High Availability VCF Operations for Networks Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out.
Recoverability	Restore through a backup.

High Availability VCF Operations for Networks Model Options

A *High Availability VCF Operations for Networks Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Large • Extra Large

High Availability VCF Operations for Networks Model Design Elements

Table 189: High Availability VCF Operations for Networks Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-HA-001	Deploy threeVCF Operations for networksplatform nodes in a cluster.	• Highly Performant. • Highly scalable, can scale up to 15 Extra Large nodes.	• Cluster does not provide high availability. It is for performance. • Requires higher CPU and memory reservations to support additional nodes.
VCF-NET-REQD-HA-002	Deploy a single VCF Operations for networks collector node in the default management vSphere cluster.	A minimum of one VCF Operations for networks collector node is required to provide the VCF Operations for networks platform node with network monitoring data.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks collector node will be applied.

Table 190: High Availability VCF Operations for Networks Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-001	Deploy appropriately sized VCF Operations for networksplatform nodes for your needs.	Ensures resource availability and usage efficiency.	None.
VCF-NET-RCMD-HA-002	Protect all VCF Operations for networks nodes by using vSphere High Availability.	Supports the availability objective for VCF Operations for networks without requiring manual intervention during an ESX host failure event.	You must ensure sufficient spare capacity for vSphere High Availability failover operations.
VCF-NET-RCMD-HA-003	In an environment with multiple VCF Instances, deploy the VCF Operations for networkscollector node(s) in the default management vSphere cluster in each VCF Instance by using VCF Operations fleet management.	• Ensures a consistent deployment in each VCF Instances regarding network data collection. • It is assumed that each VCF Instances introduces a sufficiently high number of virtual machines, so that the deployment of dedicated VCF Operations for networks collector nodes is justified.	Each VCF Operations for networks collector node must be registered with a VCF Operations for networks platform node.
VCF-NET-RCMD-HA-004	For each vCenter data source, enable NetFlow on each vSphere Distributed Switch within the workload domain.	Provides the collection of network flows via the IPFIX protocol.	For the VCF domains, VCF Operations for networks will automatically update the NetFlow settings for each cluster's vSphere Distributed Switch.

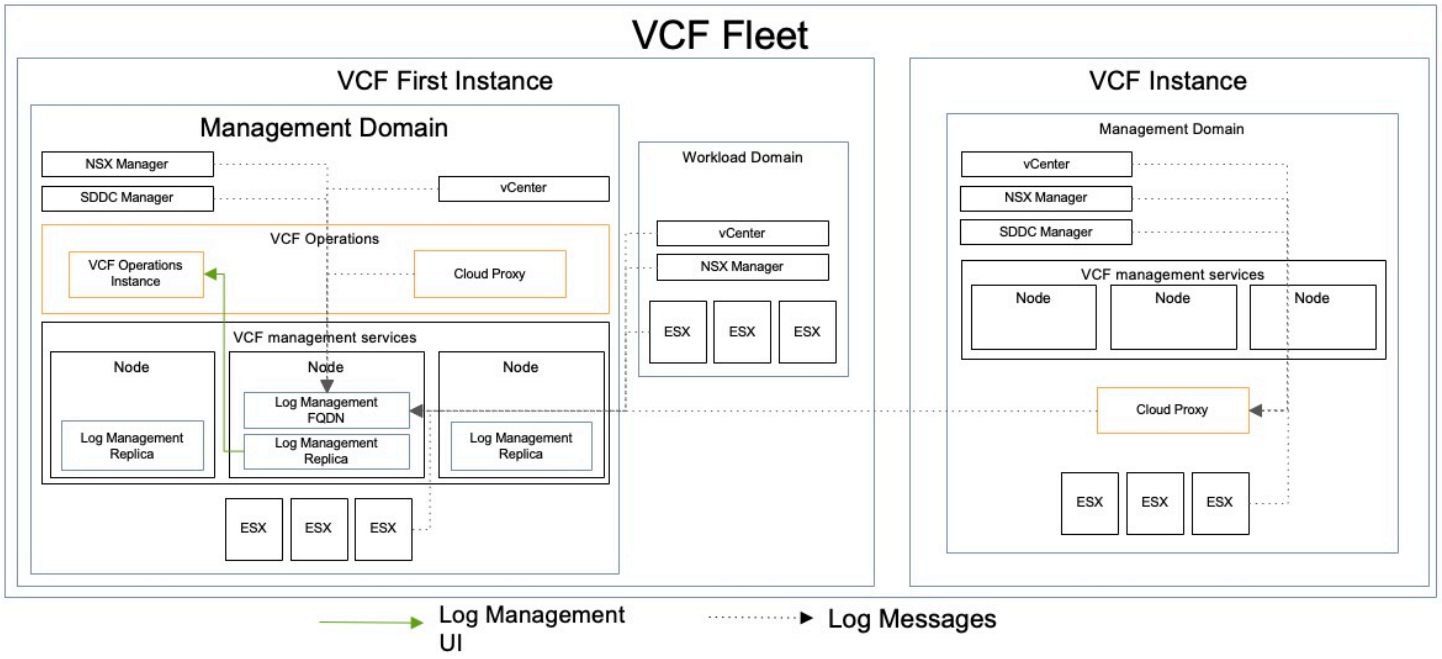
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-005	For each NSX data source, enable IPFIX for the distributed firewall.	Provides the collection of network flows via the IPFIX protocol.	The distributed firewall service must be enabled on the NSX Local Manager for the VCF workload domain.
VCF-NET-RCMD-HA-006	For each NSX Manager data source, enable latency metric collection.	Provides the collection of latency metrics from NSX Transport Nodes.	Any firewall rule sets from all ESX hosts to the VCF Operations for networks collector must allow traffic on TCP 1991.
VCF-NET-RCMD-HA-007	Use the default retention period of one month for the VCF Operations for networks platform node.	Keeping the default value for retention minimizes required storage resources administrative efforts.	None.
VCF-NET-RCMD-HA-008	Use a CA-Signed certificate containing the fully qualified domain names (FQDNs) of each VCF Operations for networks platform and collector node in the SAN attributes, when deploying VCF Operations for networks.	Configuring a CA-Signed certificate ensures that the communication to the externally facing web UI and API VCF Operations for networks for is encrypted.	• Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered. • Each time a node is added the certificate must be replaced to include the fully qualified domain name of the additional node.
VCF-NET-RCMD-HA-009	Use an SHA-2 or higher algorithm when signing certificates.	The SHA-1 algorithm is considered less secure and has been deprecated.	Not all certificate authorities support SHA-2.

High Availability Log Management Model

The *High Availability Log Management Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
High Availability Log Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind and internal load balancer. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.

Figure 82: High Availability Log Management Model in VMware Cloud Foundation



High Availability Log Management Model Attributes

A *High Availability Log Management Model* has the following attributes.

Attribute	Detail
Centralized management	Provides a centralized log management framework for vCenter, ESX, NSX, VCF Operations, VCF Automation, and identity broker.
Availability	Relies on cluster technology in the event of an outage or VCF management services and vSphere HA to restart nodes in the event of an ESX host failure.

Attribute	Detail
Scalability	Supports scale-up, storage scale up, and scale out.
Recoverability	Restore configuration through a backup, restore logs from archives.

High Availability Log Management Model Design Requirements

Table 191: High Availability Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-HA-001	Deploy log management with at least 3 replicas.	• Provides high availability. • Using the integrated load balancer prevents a single point of failure. • Using the integrated load balancer simplifies the log management deployment and subsequent integration.	• You must deploy a minimum of three appropriately sized replicas. • All replicas are sized identically. • If the capacity of your log management cluster must expand, identical capacity is added to each node.
VCF-LOGS-REQD-HA-002	Protect all VCF Management service nodes by using vSphere High Availability.	Supports the availability objectives for log management without requiring manual intervention during an ESX host failure event.	None.

Common Log Management Design Recommendations for All Models

Table 192: Common Log Management Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

High Availability Log Management Model Design Recommendations

Table 193: High Availability Log Management Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-HA-001	Enable HA on the VCF management services control plane. (Instance deployed with the High Availability deployment model, or scale up VCF services runtime to medium or large.)	To support the high availability objective, the control plane for the VCF management services should be configured for HA, as log management depends upon VCF management services.	Requires additional nodes and capacity in the vSphere management cluster.

Security Governance and Compliance Layer for VCF Fleet with Multiple Sites in a Single Region

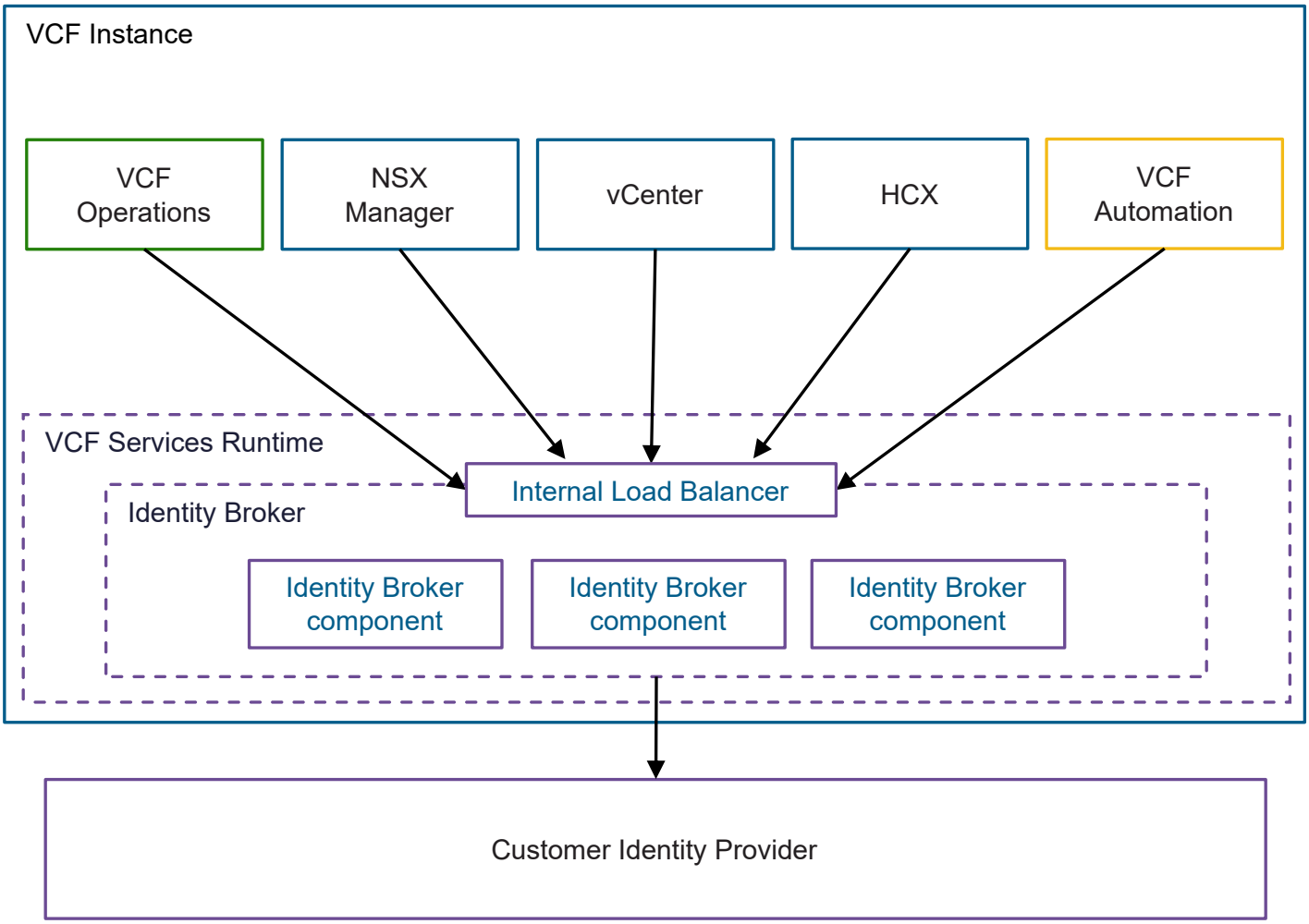
The security governance and compliance layer refers to the framework of policies, procedures, and controls that organizations establish to manage security risks while meeting regulatory, legal, and industry standards around authentication.

Instance Identity Broker Model

The *Instance Identity Broker Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Instance Identity Broker Model	Deploys the Identity Broker component on the VCF Management Services instance in the management domain.

Figure 83: Instance Identity Broker Model in VMware Cloud Foundation



Instance Identity Broker Model Attributes

The *Instance Identity Broker Model* has the following attributes.

Attribute	Detail
Service Location	Identity broker runs as a service on VCF Management Services.
Deployment mechanism	An identity broker is automatically deployed during the installation or upgrade to VMware Cloud Foundation 9.1 in the first management domain. Additional identity brokers can be deployed in each VCF Instance as part of a day 2 operation.
Number of Nodes	multi-node cluster.
Scale considerations	• Scale up: Identity broker supports small, medium, and large sizes to match different configuration demands. • Scale out: Support for multiple identity broker instances to increase redundancy and throughput. • Recommendations: – Scaling up is preferred over scaling out when a single large-size identity broker meets security, resilience, and performance standards, as it reduces operational complexity and simplifies SSO management. – Schedule scale-up operations within a maintenance window, as these activities can impact active services.
Supported services	For support directories and IdPs, see Identity Providers and Protocols Supported for Single Sign-On.
Backup and restore	Use VCF management services backup and restore process.
Resiliency	Resiliency is provided by a distributed multiple-node deployment running each node on separate vSphere hosts.
Operational Impacts	As an isolated, external service, operational tasks performed on vCenter do not impact the authentication stack of the VMware Cloud Foundation components.
Migration path	Migrating from the appliance to embedded requires a reset of the VCF Single Sign-On configuration. Users must be re-added to components.

Instance Identity Broker Model Design Requirements

Table 194: Instance Identity Broker Model Design Requirements

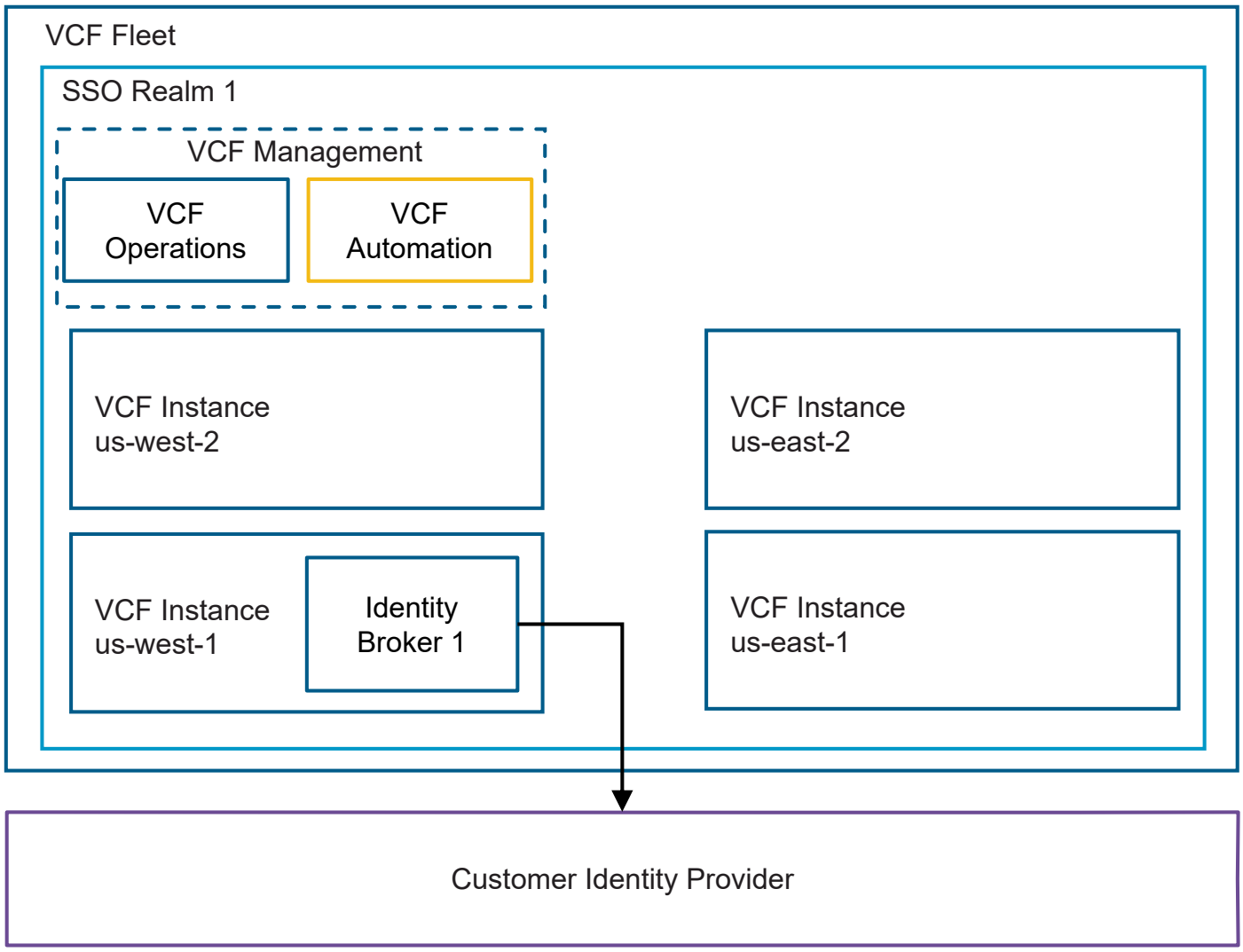
Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-EXT-001	Deploy additional identity broker instances as a Day N operation as needed.	• Provides high availability to the authentication source for a VCF Instance. • Provides additional scale.	Increases the operational overhead of managing the identity broker.
VCF-SSO-REQD-EXT-002	Connect the identity broker service to your corporate IdP.	Provides directory services (users and group) and user authentication service to the identity broker service.	None.

VCF Fleet-Wide Single Sign-On Model

The *VCF Fleet-Wide Single Sign-On Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
VCF Fleet-Wide Single Sign-On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all components across all VCF instances in the same VCF fleet.

Figure 84: VCF Fleet-Wide Single Sign-On Model in VMware Cloud Foundation



VCF Fleet-Wide Single Sign-On Model Attributes

The *VCF Fleet-Wide Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**Low Scale -** A single identity broker services all VCF Instances in your VCF fleet.
Resources requirements	**Low Resource Footprint** - A single identity broker deployment is required for one VCF fleet.
Operational Overhead	**Low Operational Overhead** - An identity broker per VCF fleet will need to be managed separately for operational tasks such as backup and restore.
Availability	**Low Availability per Private Cloud** - A single identity broker servicing all VCF Instances in your VCF fleet.
Blast Radius	**Large Blast Radius** - An outage of one identity broker instance will impact logins for a single VCF fleet.
Single Sign-On Realm	**Large SSO Realm** - Users do not need to re-authenticate when moving between different VCF Instances within the same VCF fleet.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a only): Because the identity broker is hosted on the primary VCF instance, VCF instance restoration is required; Option 2b is not applicable. • Failure Scenario 3

VCF Fleet-Wide Single Sign-On Model Design Requirements

Table 195: VCF Fleet-Wide Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VFL-001	Configure additional VCF Instances in the VCF fleet to consume the same identity broker configured in the first VCF Instance.	• Provides an authentication source for all components in the VCF fleet. • Provides VCF Single Sign-On capabilities to all components in the VCF fleet.	An outage to the identity broker service impacts the VCF fleet.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region

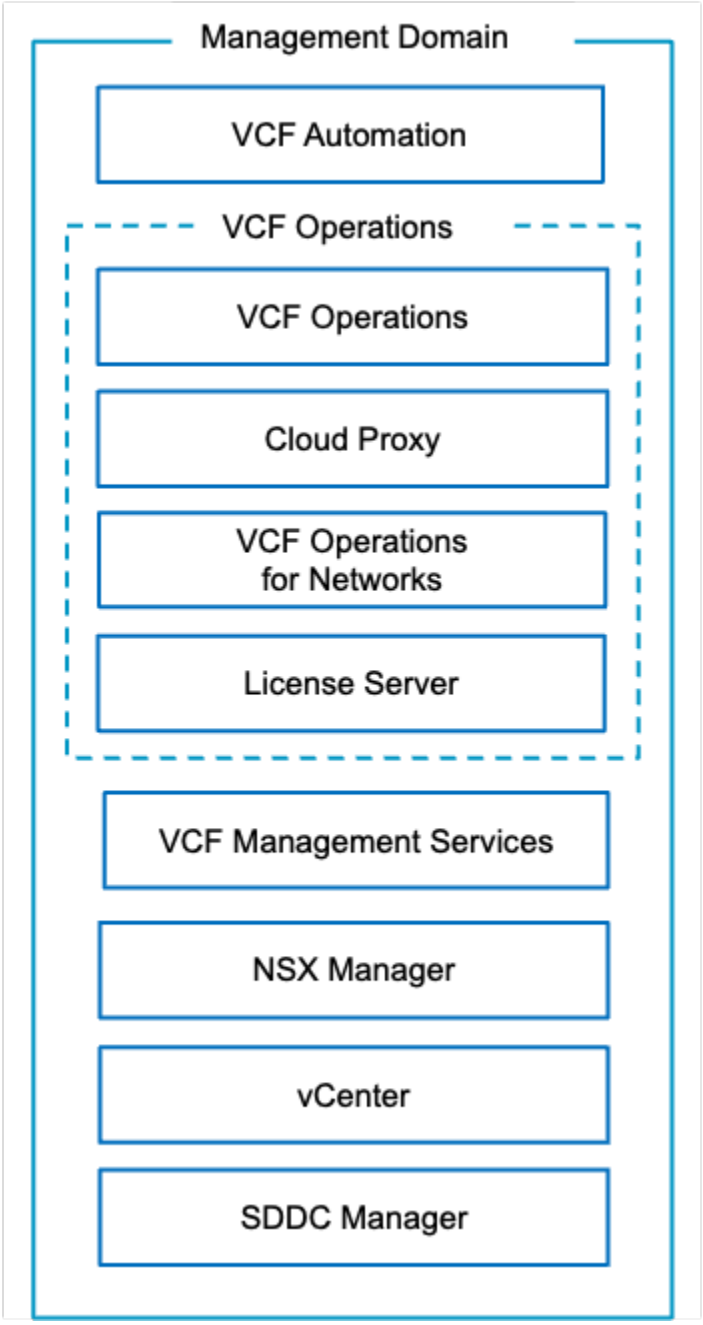
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Management Domain Model

The *Management Domain Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.

Figure 85: Management Domain Model in VMware Cloud Foundation



Management Domain Model Attributes

The *Management Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• VCF fleet management components. • VCF Instance management components. • NSX Edge nodes. • Can also run business workloads.
Management Domain in the First VCF Instance	• The management domain in the first VCF Instance contains the following management components: – VCF management services (fleet and instance level components) – VCF Operations – Cloud Proxy – VCF Operations for Networks – VCF Automation – ESX hosts – vCenter – NSX Manager – SDDC Manager
Management Domains in Additional VCF Instances	• Management domains in additional VCF Instances contain the following instance-level management components: – VCF management services (instance level components) – Cloud Proxy – ESX hosts – vCenter – NSX Manager – SDDC Manager

Management Domain Model Design Requirements

Table 196: Management Domain Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLD-REQD-CFG-001	Deploy a management domain to host all management components for the VCF Instance.	• Guarantees sufficient resources for management components. • Enables the use of specific hardware to meet only the needs of the management components. • Enables the use of dedicated physical compute, network, and storage separately from those used for additional workloads. • Enables separate lifecycle management of management and workload components.	• You must carefully size the management domain to accommodate planned deployments of additional workload domains and management components. • Hardware might not be fully utilized until full-scale deployment has been reached. • The management components for all greenfield VCF domains are deployed to the same cluster, and share the same subnet.

Management Domain Model Design Recommendations

Table 197: Management Domain Model Design Recommendations

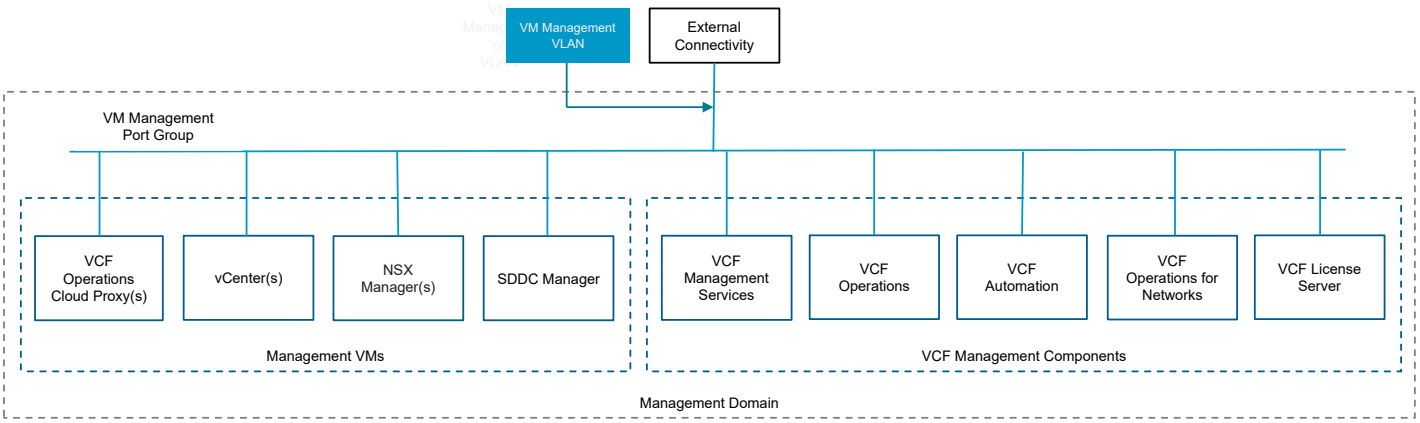
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-RCMD-NSX-001	Use a dedicated NSX Manager instance for the management domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Managers to manage and patch.

VCF Management Shared VLAN Network Model

The *VCF Management Shared VLAN Network Model* chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Figure 86: VCF Management Shared VLAN Network Model in VMware Cloud Foundation



VCF Management Shared VLAN Network Model Attributes

A *VCF Management Single VLAN Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	VLAN-backed vSphere distributed port group.
Network Isolation	No; Sharing VLAN with the VCF management domain components.
NSX Edge Cluster Required	No; Optional to provide NSX load balancer for VCF Operations. (requires additional NSX VLAN segment for Load balancer connectivity)
Stretching Network Between Availability Zones	• **VM management network** must be stretched in the physical network fabric between availability zones. • Provides automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• Not suited for a disaster recovery use case as the network is shared with management domain components. • Failover testing is not supported.

VCF Management Shared VLAN Network Model Design Requirements and Recommendations

Table 198: VCF Management Shared VLAN Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-SV-NET-REQD-001	Use the management domain vCenter VM management distributed port group for the VCF fleet level components.	No additional network required.	• No logical network isolation. • Done automatically during VCF Installer deployment.
VCF-MGMT-SV-NET-REQD-002	Use the same VLAN for VCF management components and the VCF Instance management virtual machines. e.g. vCenter and NSX Managers.	No additional VLAN required.	• Separate physical firewall rules may not be possible with a single VLAN. NSX Distributed Firewall may be an alternative solution. • Not suitable for Fleet Disaster Recovery use case with single VLAN.

Table 199: VCF Management Shared VLAN Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-SV-NET-RCMD-001	Deploy the VCF management components during initial deployment using VCF Installer.	Simplified deployment using VCF Installer UI.	None.

Workload Domain Design Selections

The following models in the design align with the objectives for deploying the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Consumption Layer for VCF Fleet with Multiple Sites in a Single Region

The consumption layer for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

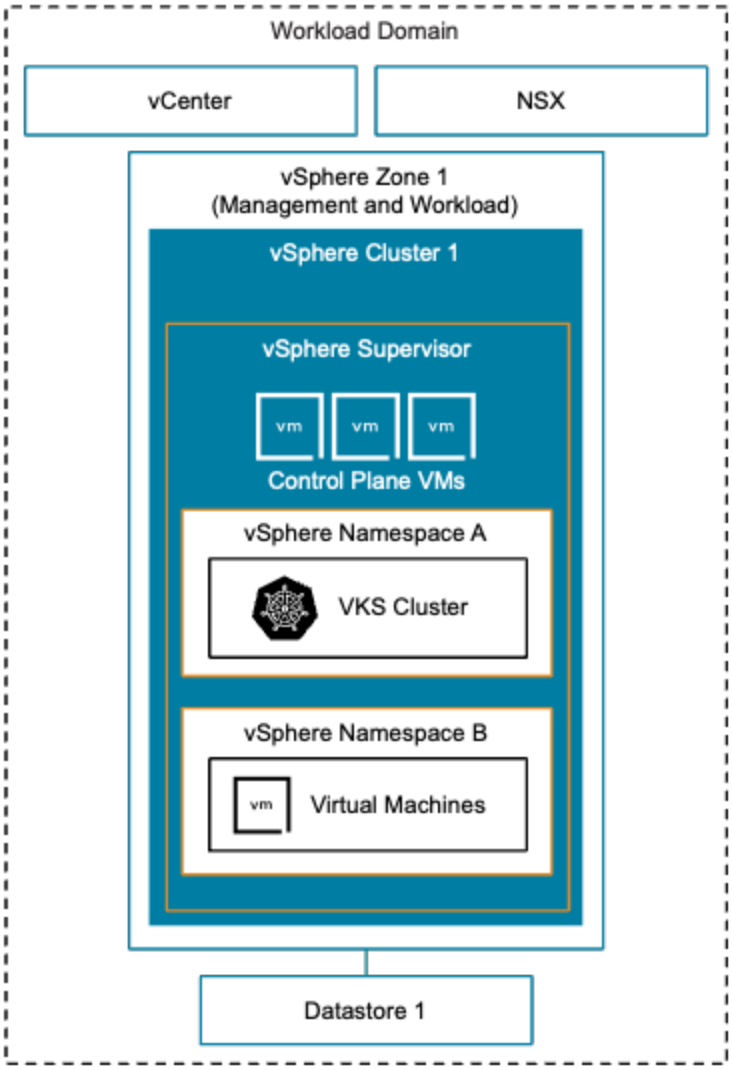
Single Management Zone with Combined Workload Zones Model

The *Single Management Zone with Combined Workload Zones Model* chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.

Figure 87: Single Management Zone with Combined Workload Zones Model Diagram

This diagram illustrates an example of a *Single Management Zone with Combined Workload Zones Model* deployment.



Single Management Zone with Combined Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single - Combined with management
vSphere Zone Datastore Type	Zonal datastore

Attribute	Detail
Activation	• At workload domain creation. OR • From vCenter.

vSphere Supervisor Design Requirements for all Zone Models

Table 200: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

vSphere Supervisor Design Requirements

Table 201: vSphere Supervisor Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-REQD-CFG-001	Topology aware storage policies created for control plane nodes when deploying in a Multi Zone Deployment model.	Storage policies should have placement considerations that are aligned with the availability of control plane nodes.	Additional storage policies are required.
VCF-SUP-REQD-CFG-002	A single vSAN stretched cluster that is not already assigned to another vSphere Supervisor instance must be available at initial deployment in a vSAN stretched cluster deployment model.	When deploying supervisor on vSAN stretched cluster, the vSANcluster must be stretched before supervisor activation.	vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Common Networking Design Recommendations

Table 202: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Design Recommendations

Table 203: vSphere Supervisor Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

High Availability vSphere Supervisor Control Plane Model

The *High Availability vSphere Supervisor Control Plane Model* chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected vSphere Supervisor management zones.

Figure 88: *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model deployment.

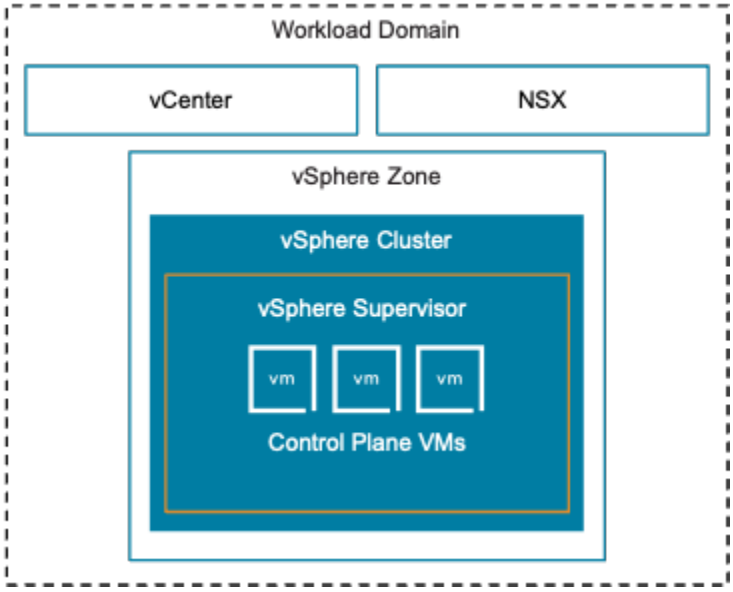
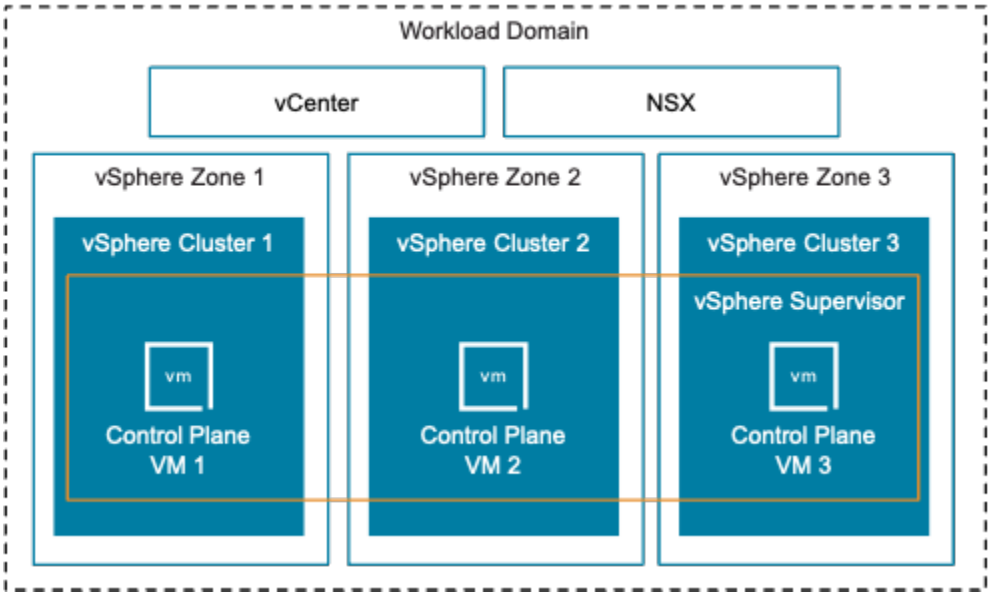


Figure 89: *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model deployment.



High Availability vSphere Supervisor Control Plane Model Attributes

Attribute	Detail
Control Planes Nodes	Three
Availability	• Relies on vSphere HA. • Three management zones, which are protected from single cluster failure.
Scalability	Scale up only

Attribute	Detail
Placement	• Single management zone, all nodes in a single management zone. • Three management zones, one node per management zone.
Applicable Zone Models	• Single Management Zone with Combined Workload Zones Model • Single Management Zone with Isolated Workload Zones Model • Three Management Zones with Combined Workload Zones Model • Three Management Zones with Isolated Workload Zones Model

vSphere Supervisor Control Plane Model Design Requirements

Table 204: vSphere Supervisor Control Plane Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-CP-REQD-CFG-001	Five consecutive management IP addresses routable to vCenter NSX, and optionally Avi management networks if Avi Load Balancer are used.	Required for integration and management communications.	Requires precise IP address management.
VCF-SUP-CP-REQD-CFG-002	Three vSphere Zones are required for the High-Availability Model in a Three Management Zone Model.	Control pane nodes are evenly placed with one per zone to protect against a single vSphere cluster failure.	Additional vSphere zones each back by separate supported vSphere clusters are required.

vSphere Supervisor Control Plane Design Recommendations

Table 205: vSphere Supervisor Control Plane Design Recommendations

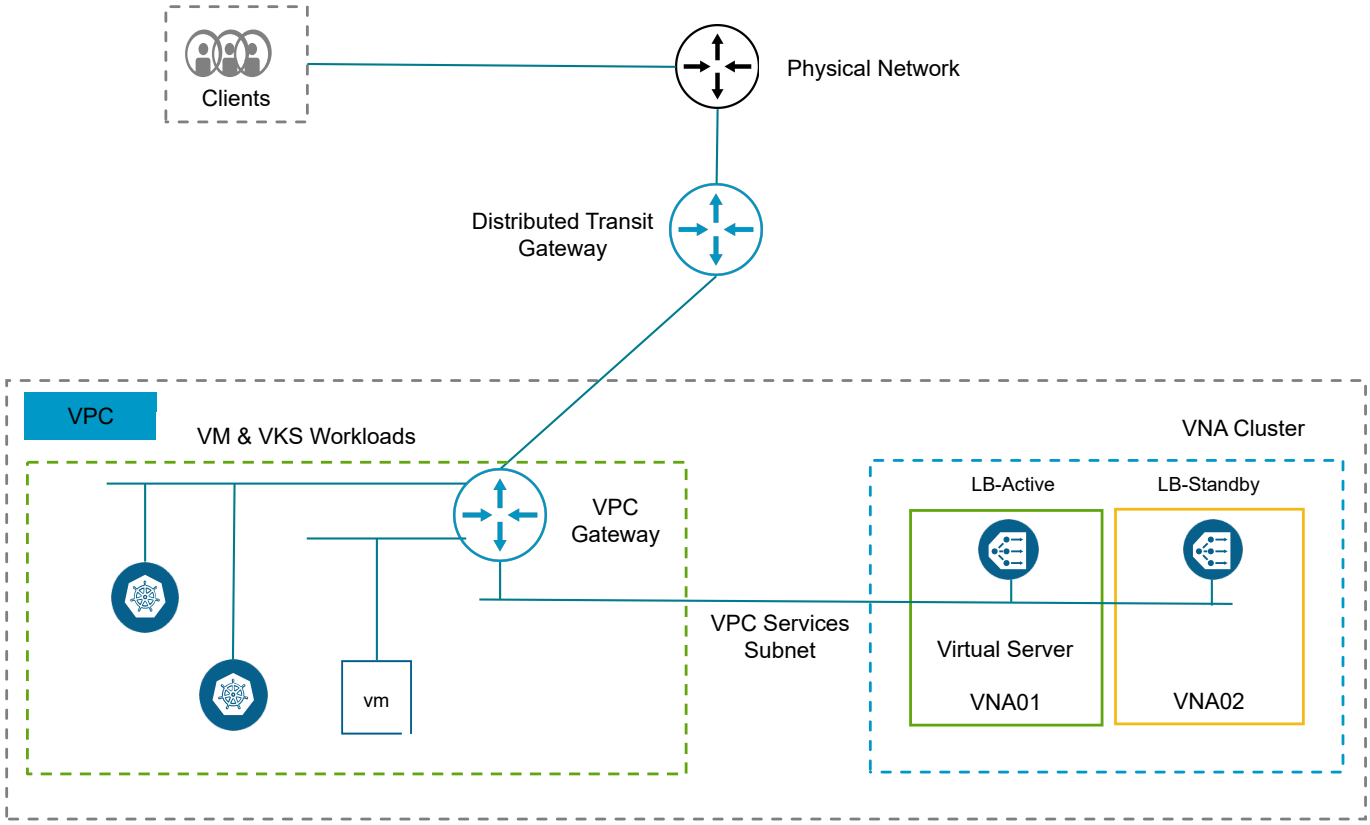
Design Requirement ID	Design Recommendation	Justification	Implication
VCF-SUP-CP-RCMD-CFG-001	Deploy the supervisor control plane in high availability mode with three control plane VMs. *Applicable to Non-Simplified Supervisor models only.	Three control plane VMs allow for continuous availability of the Supervisor control plane during rolling upgrades and single node outages.	Additional resources are required to support the extra control plane VMs.

Virtual Network Appliance Load Balancer Model

The *Virtual Network Appliance Load Balancer Model* chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for vSphere Supervisor and VKS workloads, it supports VPC-based networking.

Figure 90: Virtual Network Appliance Load Balancer Model in VMware Cloud Foundation



Virtual Network Appliance Load Balancer Model Attributes

A *Virtual Network Appliance Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Load Balancing for VCF infrastructure components (i.e. VCF appliances) • Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services)
Networking Models Supported	VPC Networking Required
Entitlement	Included with VMware Cloud Foundation
Availability	Active/Standby
Scalability	• Scale up Virtual Network Appliance nodes in the Virtual Network Appliance cluster • Scale out a single Virtual Network Appliance Cluster • Create additional Virtual Network Appliance clusters
Deployment Model	Virtual Network Appliance Cluster with two or more Virtual Network Appliances. See Virtual Network Appliance Cluster Detailed Design for further details.

Virtual Network Appliance Load Balancer Model Options

Design Area	Options
Number of Virtual Network Appliances per cluster	1 to 10
Deployment Options	• Deploy with a new workload domain • Deploy Day 2 from NSX Manager • Deploy Day 2 from vCenter
Virtual Network Appliance Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger scale depoloyments where a large number of load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

Virtual Network Appliance Load Balancer Design Requirements

Table 206: Virtual Network Appliance Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-LB-REQD-CFG-01	Create a Virtual Network Appliance cluster with two or more Virtual Network Appliance nodes.	Provides availability for a load balancer instance.	None.
VCF-VNA-LB-REQD-CFG-02	Configure the VPC connectivity profile to use Virtual Network Appliance cluster in the VPC Service Gateway Configurations.	Enables the VPC to use the Virtual Network Appliance cluster for load balancer instances.	Virtual Network Appliance cluster can be shared between VPCs.
VCF-VNA-LB-REQD-CFG-03	Create a load balancer instance for the VPC using the API	Supervisor and VKS-based workloads, load balancer creation is automated.	No UI available to configure load balancer instances.
VCF-VNA-LB-REQD-CFG-04	Create a load balancer configuration for VPC using the API.	Supervisor and VKS-based workloads, load balancer configuration is automated.	No UI available to configure load balancer instances.

Virtual Network Appliance Load Balancer Design Recommendations

Table 207: Virtual Network Appliance Load Balancer Design Recommendations

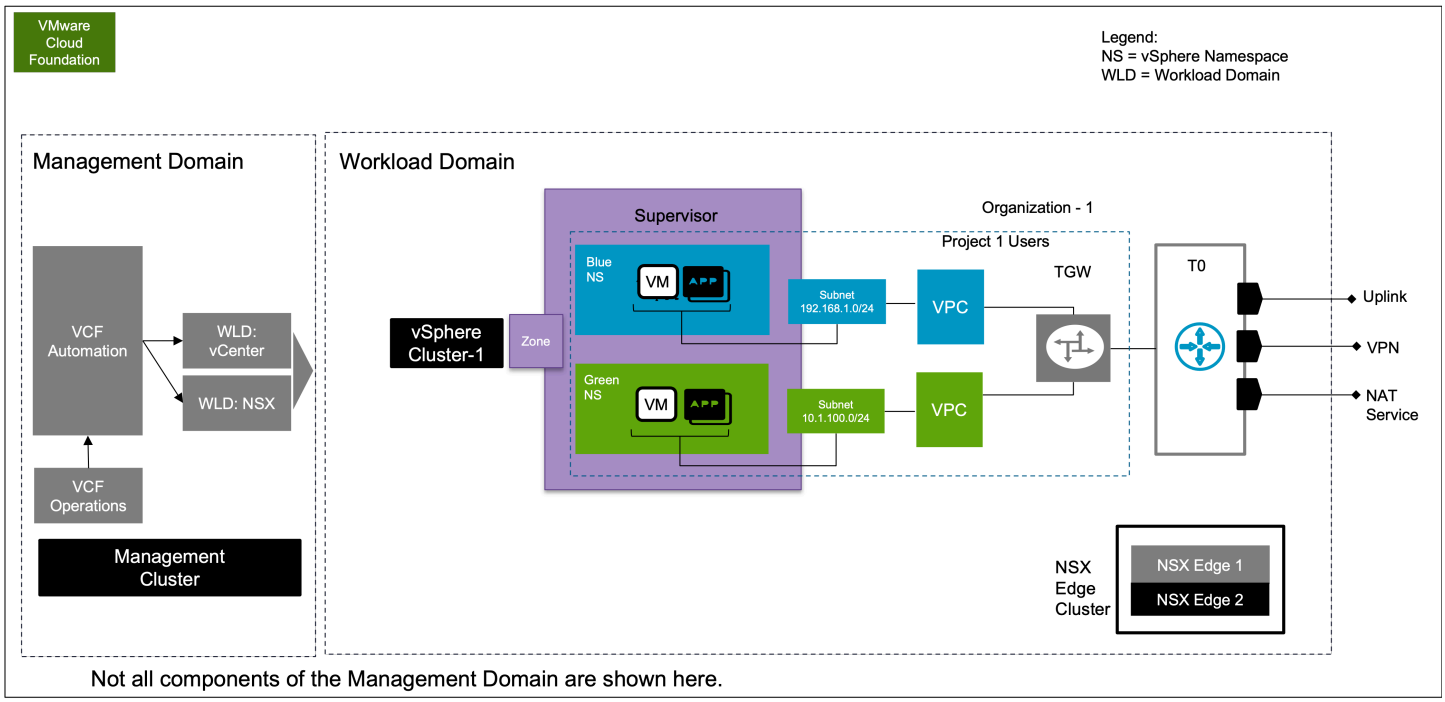
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-LB-RCMD-CFG-01	Configure failures domain for Virtual Network Appliance clusters with more than two Virtual Network Appliance nodes in the cluster.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-VNA-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from vCenter UI.

Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone
Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone	This approach provides robust resource isolation for different lines of business (LOBs) within a single vSphere Supervisor Zone.

Pattern 2: Resource Isolation with Multiple VPCs for Multiple lines of Business in One Supervisor Zone - Centralized Gateway (CTGW) Design

Figure 91: Pattern 2: Resource Isolation with multiple VPCs for multiple lines of business in one Supervisor zone - Centralized Gateway (CTGW) Design



Pattern 2: Resource Isolation with Multiple VPCs for Multiple lines of Business in One Supervisor Zone - Centralized Gateway (CTGW) Design Attributes

Design Attributes	Details	Diagram Representation
Organization	One	Represents as one Organization
Line of Business	Two	Describes Blue and Green as separate LOB
Number of vSphere Clusters	One	A compute and vSAN storage cluster used in a workload domain construct
Number of vSphere Supervisors	One	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	One	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Projects	One	Represented by "Project 1 - Users". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One	The T0 gateway provides external connectivity. This is where NAT and VPN services are configured.
vSphere Namespaces	Two	Blue and Green LOB has dedicated vSphere Namespace for workload isolation.
VPCs	Two	Each LOB/ tenant has their own dedicated VPC for network isolation.
TGWs (Transit Gateway)	One	A single TGW connects all the VPCs to the Provider Gateway.
Tenant Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region

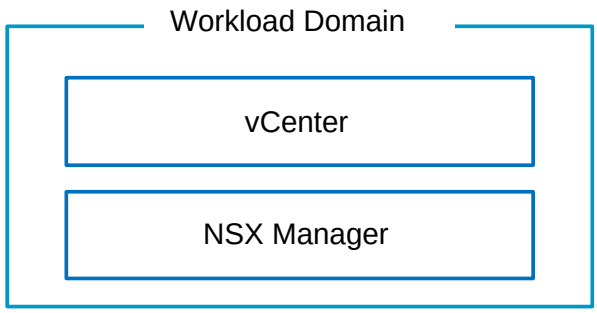
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Workload Domain Model

The *Workload Domain Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.

Figure 92: Workload Domain Model in VMware Cloud Foundation



Workload Domain Model Attributes

A *Workload Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• NSX Edge nodes. • Business workloads.
vCenter Single Sign-On Domain	Dedicated.

Workoad Domain Model Design Recommendations

Table 208: Workload Domain Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLD-RCMD-CFG-002	Use workload domains for customer workloads.	• Aligns with the VMware best practice of separating management workloads from customer workloads. • Provides better long-term flexibility and expansion options.	Requires additional hardware.
VCF-WLD-RCMD-NSX-002	Use a dedicated or shared NSX Manager instance for the workload domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Manager to manage and patch.

Dual Availability Zone Design Based on NSX Edge Node HA Model

The *Dual Availability Zone Design Based on NSX Edge Node HA Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Dual Availability Zone Design Based on NSX Edge Node HA Model	A *Single-Rack Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Figure 93: Dual Availability Zone Design Based on NSX Edge Node HA Model (Dedicated vSphere Cluster) in VMware Cloud Foundation

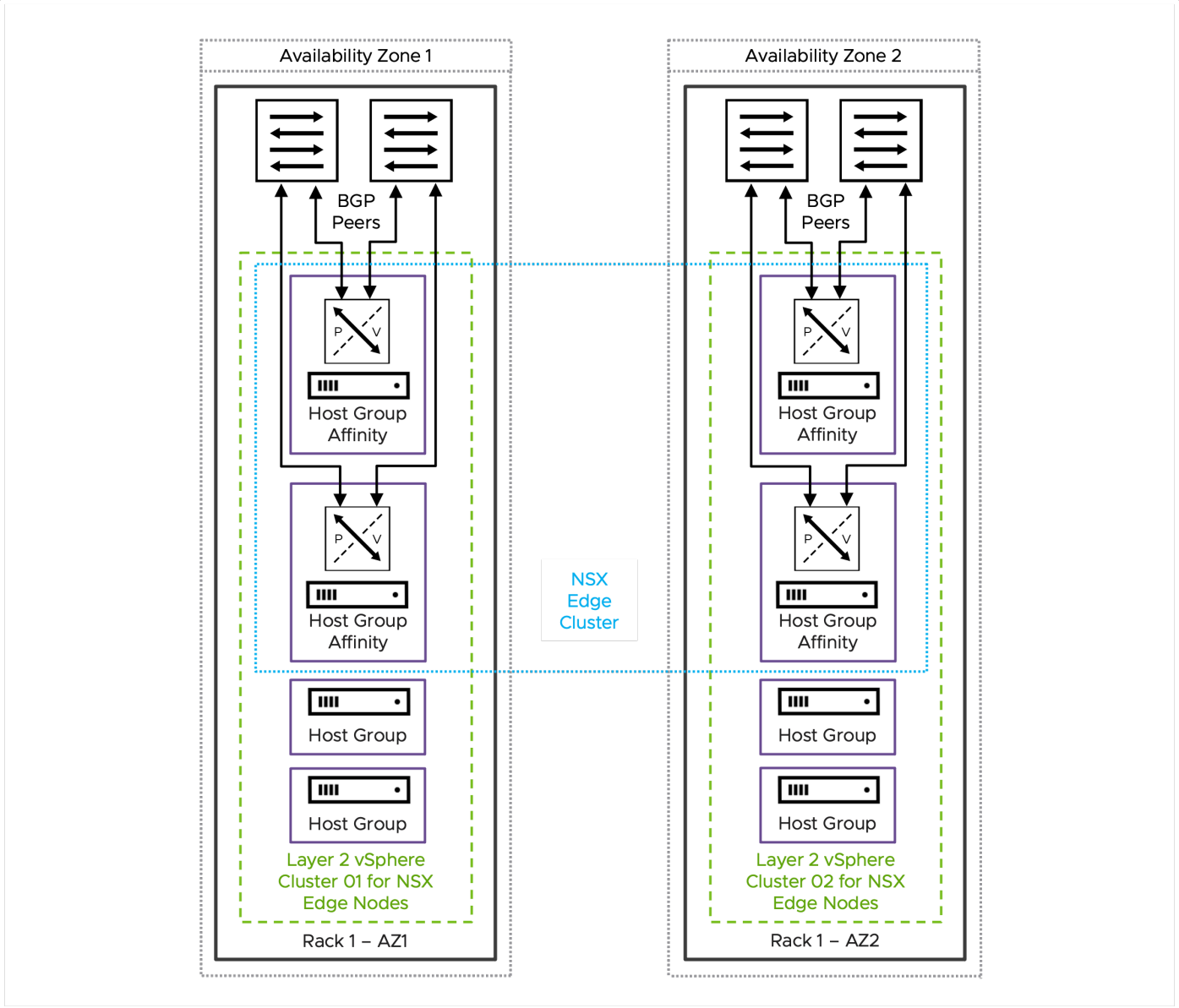
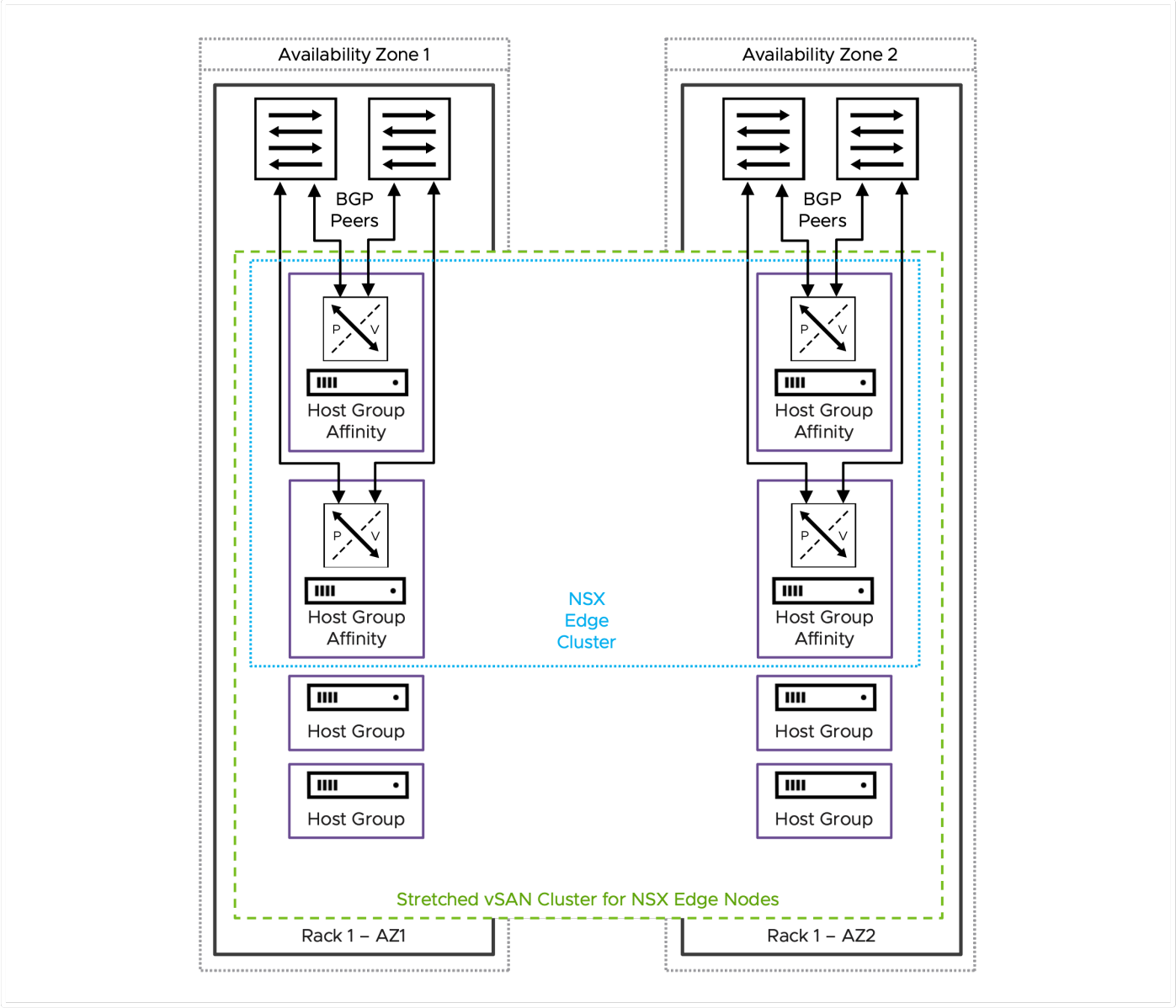


Figure 94: Dual Availability Zone Design Based on NSX Edge Node HA Model (Stretched vSAN) in VMware Cloud Foundation



Dual Availability Zone Design Based on NSX Edge Node HA Model Attributes

The *Dual Availability Zone Design Based on NSX Edge Node HA Model* has the following attributes.

Attribute	Details
Availability Model	• Host and availability zone redundancy. • Independent shared storage per vSphere cluster when using vSAN or single shared storage. • This design ensures continuous north-south communication and centralized network services even if one availability zone completely fails.
NSX Edge Node Mobility	• Mobility is not supported across racks/availabilty zones when using multiple instances of the Single-Rack vSphere Cluster Model. Mobility is not supported across availability zones when using the Stretched vSphere Cluster Model. • Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same availability zones that hosts the NSX Edge node.
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Add more hosts to both vSphere clusters simultaneously or scale out the stretched vSphere cluster with additional hosts in each availability zone. • When deploying more NSX Edge nodes on existing ESX hosts, ensure each ESX host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Common NSX Edge Cluster for Dual Availability Zones Design Requirements

Table 209: Common NSX Edge Cluster for Dual Availability Zones Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMD AZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric settings.

Dual Availability Zone Design based on NSX Edge Node HA Model Design Requirements

Table 210: Dual Availability Zone Design based on NSX Edge Node HA Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-DAZE HA-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High-Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Dual Availability Zone Design Based on NSX Edge Node HA Model BGP Routing Design Requirements

Table 211: Dual Availability Zone Design Based on NSX Edge Node HA Model BGP Routing Design Requirements

These BGP routing design requirements are specific requirements for the *Dual Availability Zone Design Based on NSX Edge Node HA Model* for routing option B. Routing option B implements a High Availability design using Tier-0 gateways configured in active/active mode and ensures symmetric traffic in and out of the virtual network. For routing options A, C, and D, these specific recommendations do not apply; only the general BGP routing design requirements and rrecommendations that are foundin the Workload Connectivity Modelssection are relevant.

Design requirement ID	Design Requirement	Justification	Implication
VCF-NSX-BGP-REQD-DAZEHA-001	Create a route-map that prepends the BGP ASN path using the local Tier-0 gateway ASN and matches a new configured IP prefix list set to "Any". Apply this route-map as an outbound filter to all BGP neighbors configured with the top of rack switches in the second availability zone.	The Tier-0 gateway uses BGP ASN path prepending in BGP route advertisements to the second availability zone's top of rack switches, ensuring ingress traffic preferentially flows through the primary availability zone, as BGP selects paths with a shorted AS path.	• Additional manual configuration. • The Tier-0 gateway will route north-south traffic through the second availability zone only if the connection to their BGP neighbors in the primary availability zone is lost, for example, if a failure of the top of the rack switch pair or in the availability zone occurs.
VCF-NSX-BGP-REQD-DAZEHA-002	The top of rack switches in the primary availability zone must advertise 0.0.0.0/1 and 128.0.0.0/1 to the NSX Edge nodes in the primary availability zone and the top of rack switches in the second availability zone must advertise 0.0.0.0/0 to the NSX Edge nodes in the second availability zone.	The NSX Edge nodes in the primary availability zone will receive more specific routes compared to the NSX Edge nodes in the second availability zone, ensuring the egress traffic preferentially flows through the primary availability zone.	• Additional manual configuration on the top of rack switches. • The origin of 0.0.0.0/1 and 128.0.0.0/1 depends on the routing topology, but in many cases it is the same device where the default route is injected — for example, the internet-facing firewall.

Common NSX Edge Cluster for Dual Availability Zones Design Recommendations

Table 212: Common NSX Edge Cluster for Dual Availability Zones Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node ESX hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMDAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or ESX hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload virtual machines have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart after ESX host failures if it cannot meet their memory reservations on surviving ESX hosts. • Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMDAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are generally less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMDAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.
VCF-NSX-EDGE-RCMD-COMDAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMDAZ-008	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across two dedicated vSphere clusters respective across two availability zones.	Maintain High Availability and prevent single points of failure.	• Additional servers are required. • This design recommendation is not applicable to the design model based on vSphere HA Recovery
VCF-NSX-EDGE-RCMD-COMDAZ-009	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload VMs and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-010	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMDAZ-011	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMDAZ-012	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge nodes and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge nodes, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSXon dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMDAZ-013	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two (2) data plane fastpath interfaces get an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP).	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMDAZ-014	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (physical link state up but no forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-015	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard or the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-016	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Dual Availability Zone Design based on NSX Edge Node HA Model Design Recommendations

Table 213: Dual Availability Zone Design based on NSX Edge Node HA Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZEHA-001	Limit the span of NSX Edge node networks to the local rack.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP sessions.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.
VCF-NSX-EDGE-RCMD-DAZEHA-002	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	• Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow. • This design recommendation is not applicable with routing option C - Active/Standby Tier-0 gateway with symmetric routing.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZE HA-003	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	• The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-004	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESX host.	• Increased configuration effort. • If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-005	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-006	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZE HA-007	Create a compute policy type Best Effort Restart (BER) and assign the previously created vCenter tag and category to it.	• The compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-008	Configure an NSX failure domain for each availability zone where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability in case of a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Shared Design Selection between Management and Workload Domain

Shared Design Selections between Management and Workload Domain are the models that are the same across the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region

The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

vCenter Model

The *vCenter Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

vCenter Model Attributes

A *vCenter Model* has the following attributes.

Design Area	Attributes
vCenter Deployment Type	• One vCenter instance for the management domain that manages the management components of the VCF Instance. • Optionally, additional vCenter instances for workload domains to support workloads. • vSphere HA protecting all vCenter appliances.
vCenter Single Sign-On	• Each vCenter has a distinct vCenter Single Sign-On domain.

Methods for Protecting vCenter

VMware Cloud Foundation supports multi-rack management vSphere clusters, vSAN stretched clusters, or 3rd party synchronous replication solutions such as metro clusters that are designed to provide disaster recovery and/or high availability for the entire management stack (vCenter, NSX Manager, etc.) as opposed to a single vCenter.

Table 214: Methods for Protecting vCenter in VMware Cloud Foundation

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere High Availability	Yes	You must ensure sufficient resources in the cluster to allow virtual machines to restart on another ESX host in the event of an ESX host failure.
vCenter High Availability (vCenter HA)	No	• vCenter services must fail over to the passive node so there is no continuous availability. • Recovery time can be up to 15 mins. • You must meet additional networking requirements for the private network. • vCenter HA requires additional resources for the passive and witness nodes. • Lifecycle management is complicated because you must manually delete and recreate the standby virtual machines during a lifecycle management operation.

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere Fault Tolerance (vSphere FT)	No	• The vCPU limit of vSphere FT vCPU would limit vCenter appliance size to medium. • You must provide a dedicated network.

vCenter Model Design Requirements and Recommendations

Table 215: vCenter Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-CFG-001	Deploy a dedicated vCenter appliance for the management domain of the VCF Instance.	• Isolates vCenter failures to management or customer workloads. • Isolates vCenter operations between management and business workloads. • Supports a scalable vSphere cluster design where you can reuse the management components as more customer workloads are added to the VCF Instance. • Simplifies capacity planning for customer workloads because you do not consider management workloads for the workload domain vCenter. • Improves the ability to upgrade the vSphere environment and related components by enabling for explicit separation of maintenance windows: — Management workloads remain available while you are upgrading the tenant workloads. — Workloads remain available while you are upgrading the management nodes. • Supports clear separation of roles and responsibilities to ensure that only administrators with granted authorization can control the management workloads. • Facilitates quicker troubleshooting and problem resolution. • Simplifies disaster recovery operations by supporting a clear separation between recovery of the management components and tenant workloads. • Provides isolation of potential network issues by introducing network separation of the vSphere clusters in the VCF Instance.	The vSphere cluster must have sufficient resources.
VCF-VCS-REQD-NET-001	Place all workload domain vCenter appliances on the VM management network in the management domain.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-SSO-001	Create all vCenter instances within a VCF Instance in their own unique vCenter Single Sign-On domain.	• Sharing vCenter Single Sign-On domains is not supported. • Enables isolation at the vCenter Single Sign-On domain layer for increased security separation.	• Each vCenter instance is managed through its own pane of glass using a different set of administrative credentials. • You must manage password rotation for each vCenter Single Sign-On domain separately.

Table 216: vCenter Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VCS-RCMD-CFG-001	Deploy an appropriately sized vCenter appliance for each workload domain.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate compute resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-002	Deploy a vCenter appliance with the appropriate storage size.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate storage resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-003	Protect workload domain vCenter appliances by using vSphere HA.	vSphere HA is the only supported method to protect vCenter availability in VCF.	vCenter becomes unavailable during a vSphere HA failover.
VCF-VCS-RCMD-CFG-004	In vSphere HA, set the restart priority policy for the vCenter appliance to high.	vCenter is the management and control plane for physical and virtual infrastructure. In a vSphere HA event, to ensure that the rest of the SDDC management stack comes up faultlessly, the workload domain vCenter must be available first, before the other management components come online.	If the restart priority for another virtual machine is set to highest, the connectivity delays for the management components are longer.

ESX Model

The *ESX Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCF domain vSphere cluster.

ESX Hardware and Sizing Considerations

For more information on host sizing, refer to the VCF Planning and Preparation Workbook.

Table 217: ESX Hardware and Sizing Considerations

Hardware Element	Considerations
BIOS/Firmware	Ensure the ESX host hardware is on the Hardware Compatability List (HCL) and that you are running the latest supported BIOS and firmware.
CPU	• Total CPU requirements for the workloads that are running in the cluster. • Host failure and maintenance scenarios. • CPU over-commitment ratio guidelines – Management Domain • To ensure optimal performance and maintain resilience during host failures or maintenance, keep the management domain vCPU-to-pCPU over-commitment ratio at or below 2:1. **Note:** While higher ratios such as 3:1 or 4:1 are supported, you must closely monitor CPU ready times to ensure that the management cluster has enough capacity to run the workloads. – Workload domains • CPU over-commitment ratios for workload domains should be calculated on a case by case basis depending on the CPU intensity of the workloads. You must closely monitor CPU ready times to ensure optimal performance. • Number of physical cores, not logical cores. – Simultaneous multi-threading (SMT) in CPUs improve CPU performance however the performance enhancement will not be equivalent to 100% more CPU power.
Memory	• Total memory requirements for the workloads that are running in the cluster. • When sizing memory for the ESX hosts in a cluster, to reserve the resources of one host for failover or maintenance, set the admission control setting to N+1, which reserves the resources of one host for failover or maintenance. • *vSAN* OSA (Original Storage Architecture). – To support the maximum number of disk groups, you must provide 32 GB of RAM. • *vSAN* ESA (Express Storage Architecture). – You must provide a minimum of 128GB of RAM.
Storage	• Use high-endurance devices such as a hard drive or SSD for boot device. • Use a 128GB or greater boot device to maximize the space available for ESX-OS Data. • Use hosts with homogeneous configuration.
Network	• Use multiple physical NICs connected to multiple ToR switches to ensure redundancy. • 25GbE is recommended • Use physical NICs of the same speed, vendor, and model across all host within a cluster.
Power	• Use redundant power supplies, preferably connected to separate circuits to ensure availability. • Consider the impact of enabling Power Management on your ESX hosts. In order to allow ESX to control CPU power-saving features you need to set power management in the BIOS to *OS Controlled Mode*.

ESX Model Design Requirements and Recommendations

In your ESX design, you apply certain best practices.

Table 218: ESX Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-ESX-REQD-CFG-001	Install no less than the minimum number of ESX hosts required for the vSphere cluster model being deployed.	• Ensures that availability requirements are met. • If one of the ESX hosts is not available because of a failure or maintenance event, the CPU over-commitment ratio becomes 2:1.	None.
VCF-ESX-REQD-CFG-002	Ensure each ESX host matches the required CPU, memory, and storage specification.	Ensures that workloads will run without contention even during failure and maintenance conditions.	Assemble the server specification and number according to the sizing in the VCF Planning and Preparation Workbook which is based on projected deployment size.
VCF-ESX-REQD-SEC-001	Regenerate the certificate of each ESX host after assigning the ESX host an FQDN.	Establishes a secure connection with the VCF Installer during the initial deployment and prevents man-in-the-middle (MiTM) attacks.	You must manually regenerate the certificates of the ESX hosts before the deployment.

Table 219: ESX Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-001	When using vSAN as primary storage in the management domain, use vSAN ReadyNodes for each ESX host.	Your management domain is fully compatible with vSAN at deployment. **Note:** For information about the models of physical servers that are vSAN-ready, see vSAN Compatibility Guide for vSAN ReadyNodes.	• Hardware choices might be limited. • If you plan to use a server configuration that is not a vSAN ReadyNode, your CPU, disks, and I/O modules must be listed on the VMware Compatibility Guide under *CPU Series* and *vSAN Compatibility List* aligned to the ESX version.
VCF-ESX-RCMD-CFG-002	Allocate ESX hosts with uniform configuration across the default management vSphere cluster.	A balanced vSphere cluster has these advantages: • Predictable performance even during hardware failures. • Minimal impact of resynchronization or rebuild operations on performance.	You must apply vendor sourcing, budgeting, and procurement considerations for uniform server nodes on a per vSphere cluster basis.
VCF-ESX-RCMD-CFG-003	When sizing CPU, do not consider multi-threading technology and associated performance gains.	Although multi-threading technologies increase CPU performance, the performance gain depends on running workloads and differs from one case to another.	Because you must provide more physical CPU cores, costs increase and hardware choices become limited.
VCF-ESX-RCMD-CFG-004	Install and configure all ESX hosts in the default management vSphere cluster to boot using a 128-GB device or larger.	Provides ESX hosts that have large memory, that is, greater than 512 GB, with enough space for the scratch partition when using vSAN.	None.

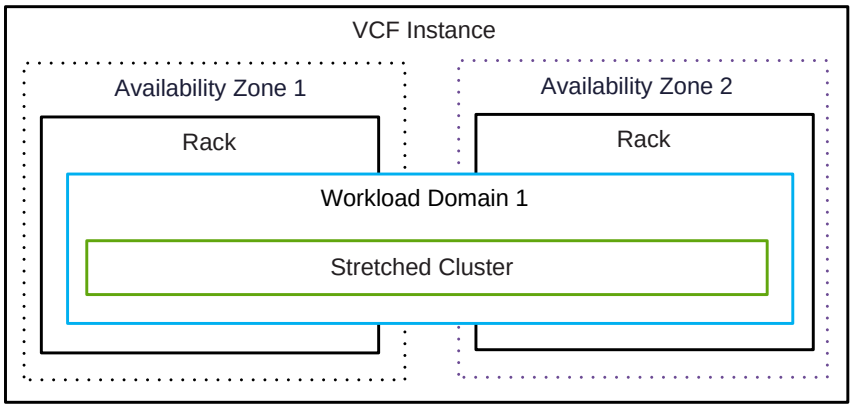
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-005	Use the default configuration for the scratch partition on all ESX hosts in the default management vSphere cluster.	• If a failure in the vSAN cluster occurs, the ESX hosts remain responsive and log information is still accessible. • If using vSAN storage its not possible to use a vSAN datastore for the scratch partition.	None.
VCF-ESX-RCMD-CFG-006	For workloads running in the default management vSphere cluster, save the virtual machine swap file at the default location.	Simplifies the configuration process.	Increases the amount of replication traffic for management workloads that are recovered as part of the disaster recovery process.
VCF-ESX-RCMD-NET-001	Place the ESX hosts in each management domain vSphere cluster on the ESX management network that is separate from the VM management network.	• Enables the separation of the physical VLAN between ESX hosts and the other management components for security reasons. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Increases the number of VLANs required.
VCF-ESX-RCMD-NET-002	Place the ESX hosts in each workload domain on a separate ESX host management VLAN-backed network.	Enables the separation of the physical VLAN between the ESX hosts in different workload domains for security reasons.	Increases the number of VLANs required. For each workload domain, you must allocate a separate management subnet.
VCF-ESX-RCMD-NET-003	Use dedicated VLANs for each traffic type.	Enables the separation of the physical VLAN for vSAN, vMotion, NFS & backup traffic.	Increases the number of VLANs required. For each traffic type you must allocate a separate VLAN and subnet.
VCF-ESX-RCMD-SEC-001	Set the advanced setting `UserVars.SuppressShellWarning` to `0` across all ESX hosts in the management domain.	• Ensures compliance with the *vSphere Security Configuration Guide* and with security best practices • Enables the warning message that appears in the vSphere client every time SSH access is activated on an ESX host.	You must turn off SSH enablement warning messages manually when performing troubleshooting or support activities.

Stretched vSphere Cluster Model

The *Stretched vSphere Cluster Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Stretched vSphere Cluster Model	The *Stretched vSphere Cluster Model* allows you to protect workloads on vSphere clusters in your environment against a failure of a single fault domain by implementing a stretched vSphere cluster across multiple availability zones.

Figure 95: Stretched vSphere Cluster Model in VMware Cloud Foundation



Stretched vSphere Cluster Model Attributes

A *Stretched vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Multiple datacenters
Cluster Rack Mapping	• VCF domain cluster with multiple availability zones, each zone in a single-rack vSphere cluster. • VCF domain cluster with multiple availability zones, with Layer 2 multi-rack vSphere clusters in each zone.
Resilience	• vSphere HA protects workloads against host failures. • Multiple availability zones protect against availability zone failures.
Bandwidth	• When using vSAN stretched clusters, the bandwidth between the zones must be at least 10 Gbps and the round-trip latency must be less than 5 ms. • When using storage array based stretched clusters, you must work with your storage vendor to define the bandwidth and latency requirements.
Management Domain Requirements	Having the management domain on a stretched cluster is a prerequisite to configure and implement stretched cluster in a workload domain. In this way, the critical management components managing the workload domains are still available if an availability zone failure occurs.
Availability Zones	• You can have up to two availability zones associated with a given stretched vSphere cluster. – When using vSAN stretched clusters, a third independent fault domain is required to host the vSAN witness appliance.

Stretched vSphere Cluster Model VLANs and Subnets

Function	VLANs and Subnets
VM management	Shared across availability zones
VCF management	Shared across availability zones
ESX management	Unique per availability zone
vSphere vMotion	Unique per availability zone
vSAN*	Unique per availability zone
Host overlay	Unique per availability zone
NFS*	Depends on stretched storage design

Note:

*Optional based on the principal storage choice for the cluster.

Stretched vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere cluster which supports your storage type and workload domain considerations.

Storage Type	VCF Installer Deployment Model	Management Domain (First Cluster)	Management Domain (Additional Clusters) Workload Domain (All Clusters)
vSAN	Simple	Six (6)	Six (6)
	Highly Available	Eight (8)	
NFS, FC *	Simple	Refer to KB417356	**Minimum:** • One (1) host in Site A • One (1) host in Site B **Total: 2 ESX hosts** **Recommended:** • Two (2) hosts in Site A • Two (2) hosts in Site B **Total: 4 ESX hosts**
	Highly Available	Refer to KB417356	

Important:

vSphere clusters are configured allowing for the failure of a single availability zone while still having enough resources to run all vSphere cluster workloads.

Attention:

* Storage and networking design requirements should be completed in consultation with your storage vendor.

Common vSphere Cluster Design Requirements

Table 220: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Table 221: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Stretched vSphere Cluster Model Design Requirements

Table 222: Stretched vSphere Cluster Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-005	Enable the **Override default gateway for this adapter** setting on the vSAN VMkernel adapters on all ESX hosts.	Enables routing the vSAN data traffic through the vSAN network gateway rather than through the management gateway.	vSAN networks across availability zones must have a route to each other.
VCF-CLS-REQD-CFG-006	Create a host group for each availability zone and add the ESX hosts in the zone to the respective group.	Makes it easier to manage which virtual machines run in which availability zone.	None.

Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Requirements

Table 223: Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements for a stretched cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-SC-REQD-CFG-001	Create an uplink profile for each availability zone.	Enables different VLAN ids for Host TEP in each availability zone.	Requires additional Host TEP VLAN in availability zone 2.
VCF-VNI-SC-REQD-CFG-002	Create NSX host sub transport node profile for second availability zone.	Enables NSX host transport node configuration per availability zone.	None.

Common vSphere Cluster Design Recommendations

Table 224: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Table 225: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Stretched vSphere Cluster Model Design Recommendations

Table 226: Stretched vSphere Cluster Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-008	Increase admission control percentage to half of the ESX hosts in the vSphere cluster.	Allocating only half of a stretched cluster ensures that all VMs have enough resources if an availability zone outage occurs.	• In a vSphere cluster of eight (8) ESX hosts, the resources of only four (4) ESX hosts are available for use. • If you add more ESX hosts to the default management vSphere cluster, add them in pairs, one per availability zone.
VCF-CLS-RCMD-CFG-009	Create a virtual machine group for each availability zone and add the virtual machines in the zone to the respective group.	Ensures that virtual machines are located only in the assigned availability zone to avoid unnecessary vSphere vMotion migrations.	You must add virtual machines to the allocated group manually.
VCF-CLS-RCMD-CFG-010	Create a should-run-on-hosts-in-group VM-Host affinity rule to run each group of virtual machines on the respective group of hosts in the same availability zone.	Ensures that virtual machines are located only in the assigned availability zone to avoid unnecessary vSphere vMotion migrations.	You must manually create the rules.

Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Recommendations

Table 227: Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations for a stretched cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

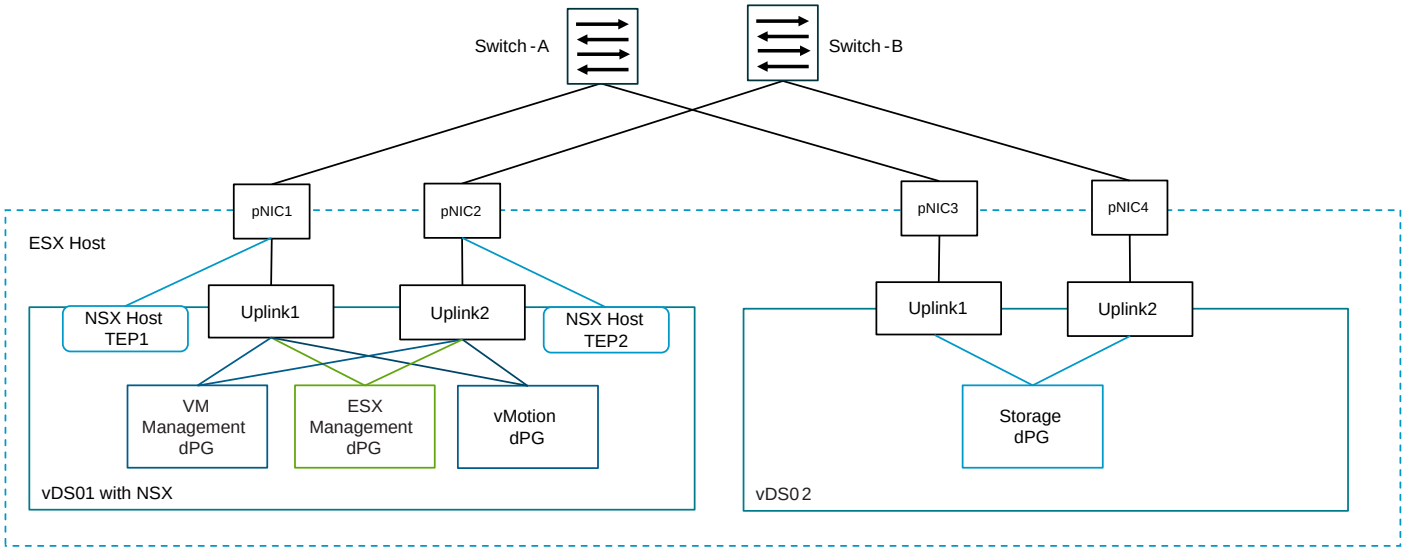
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-SC-RCMD-CFG-001	Create a host TEP IP pool for each availability zone with a subnet that is allocated per availability zone and a gateway that is configured for the network.	• Provides the host TEP IP addressing for each availability zone. • Enables a Layer 3 routed TEP traffic between availability zones. • Using an external DHCP server for the host overlay VLANs in both availability zones is not required.	Requires additional subnet per availability zone.
VCF-VNI-SC-RCMD-CFG-002	Configure an NSX sub-transport node profile for hosts in availability zone 2.	• The NSX transport node profile can remain attached when using two separate VLANs for host TEPs at each availability zone. • You can use static IP pools for the host TEPs in each availability zone.	Changes to the host transport node configuration are done at the vSphere cluster level.

Storage Separation Distributed Switch Model

The *Storage Separation Distributed Switch Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Storage Separation Distributed Switch Model	A *Storage Separation Distributed Switch Model* in VMware Cloud Foundation provides separation of storage traffic from all other network services running on the vSphere cluster.

Figure 96: Storage Separation Distributed Switch Model With Single Network Fabric in VMware Cloud Foundation



Storage Separation Distributed Switch Model Attributes

A *Storage Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Shared for Management, vSphere vMotion, NSX workloads. • Dedicated for IP storage.

Storage Separation Distributed Switch Model Options

A *Storage Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (storage traffic isolated to dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two or more network cards. • Each traffic type is spread across separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Common Distributed Switch Design Requirements

Storage Separation Distributed Switch Model Design Requirements

Table 228: Storage Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-SS-REQD-CFG-001	Use two vSphere Distributed Switch per vSphere cluster: • vDS01 — Management — vMotion — NSX • vDS02 — Storage	• Separates storage traffic onto a dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for storage traffic (NFS or vSAN) with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switch required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than the simple model.
VCF-VDS-SS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion	Provides the required network services for the vSphere cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-SS-REQD-CFG-003	Create the following distributed port groups on vDS02: • Storage	Provides connectivity for the hosts in the cluster to the storage network.	vSphere Distributed Switch storage traffic separation profile provides the required configuration.
VCF-VDS-SS-REQD-CFG-004	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switches for each additional rack: vDS01: • ESX management • vMotion vDS02: • Storage	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack
VCF-VDS-SS-REQD-CFG-005	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switches for availability zone 2 vDS01: • ESX management • vMotion vDS02: • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2

Common Distributed Switch Design Recommendations

Table 229: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Storage Separation Distributed Switch Model Design Recommendations

Table 230: Storage Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-DPG-001	Use the **Failover Order** teaming algorithm for the storage port group.	Provides a consistent traffic flow through a single physical ToR switch without needing to traverse Inter Switch Link during normal operations.	It must be manually configured during deployment using vSphere Distributed Switche custom profile option.
VCF-VDS-SS-RCMD-NIO-001	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than Principal storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-SS-RCMD-NIO-002	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as virtual machine traffic.	During times of network contention, vMotion takes longer than usual to complete.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-NIO-003	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in the SDDC. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-SS-RCMD-NIO-004	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

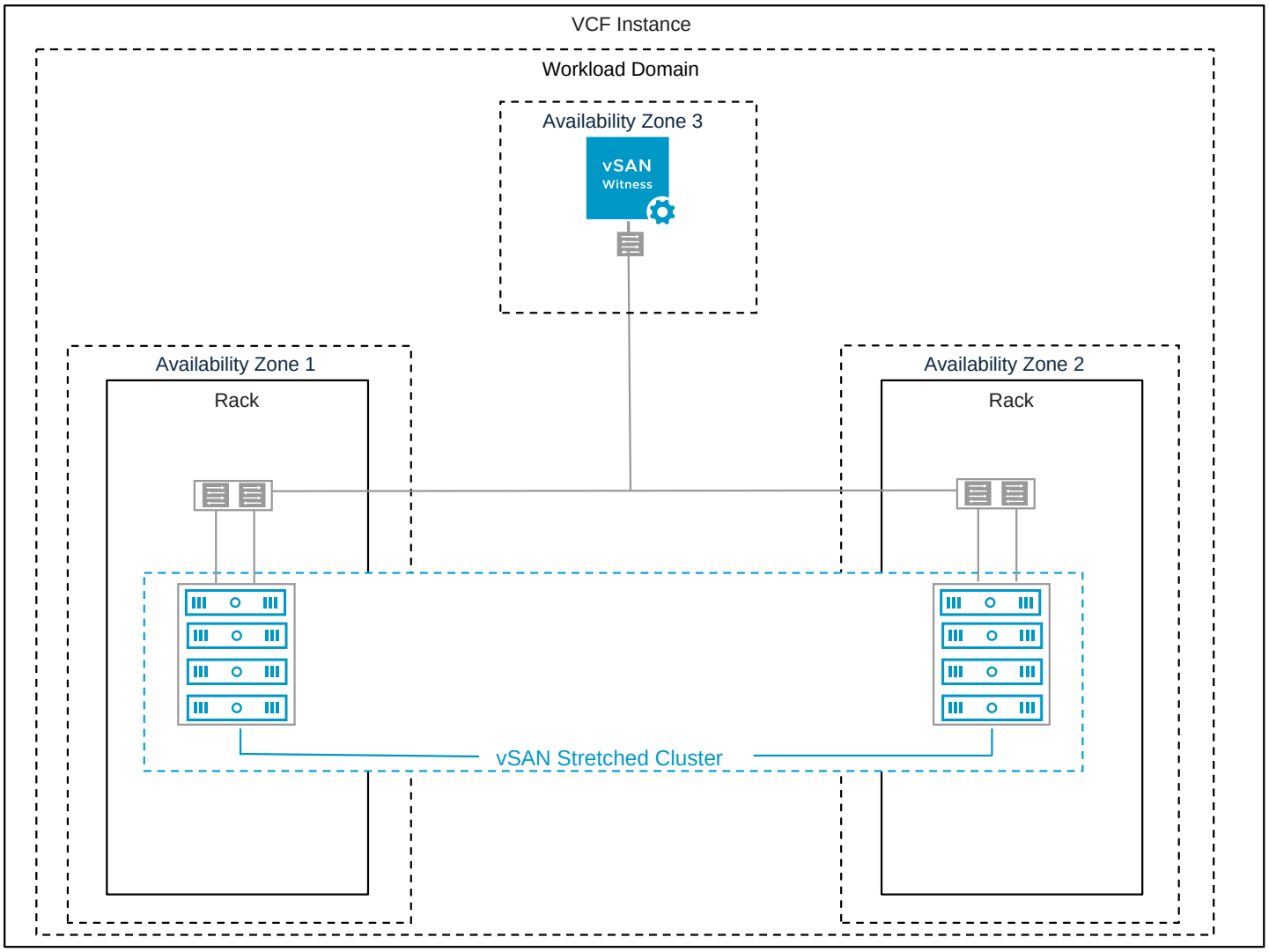
Stretched vSAN Storage Model

The *Stretched vSAN Storage Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Stretched vSAN Storage Model	A *Stretched vSAN Storage Model* is a cluster configuration where the ESX hosts, that comprise a single vSAN cluster, reside in two availability zones.

Figure 97: Stretched vSAN Storage Model in VMware Cloud Foundation

Stretched vSAN Storage Model depicting three availability zones for vSAN. Two availability zones are designated for vSAN data nodes and a third availability zone is designated for a vSAN witness node.


VCF Instance
Workload Domain
Availability Zone 3
vSAN
Witness
Availability Zone 1
Rack
Availability Zone 2
Rack
vSAN Stretched Cluster

Stretched vSAN Storage Model Attributes

A *Stretched vSAN Storage Model* has the following attributes.

Attribute	Detail
Fault Domain / Avilability zone	• Stretched vSAN clusters comprise of three (3) Fault domains. A fault domain comprises of one or more vSAN nodes. • A fault domain or availability zone can represent a physical location. Fault domains / availability zones can either be two distinct data centers in a metro distance, or two safety or fire sectors (data halls) in the same large-scale datacenter.
vSAN Traffic Type	• Stretched vSAN clusters utilize vSAN Traffic type for data traffic. • Stretched vSAN clusters utilize vSAN Witness Traffic type for vSAN meta-data traffic. **Note:** Stretched vSAN Storage clusters (ESA) can be configured to use vSAN Storage Cluster Client traffic for vSAN compute client clusters. See Stretched vSAN Storage Cluster Storage Model.
vSAN Witness Appliance	A witness node is a pre-configured appliance that runs ESX and is distributed as an OVA file.

Common vSAN Storage Model Design Requirements

Table 231: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Common vSAN Host Network Model Design Requirements

Table 232: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

vSAN ESA Storage Model Design Requirements

Table 233: vSAN ESA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-ESA-REQD-CFG-001	Ensure vSAN certified Drives used for vSAN ESA clusters.	• When using VMware Cloud Foundation automated workflows, vSAN ESA cluster creation maybe blocked unless vSAN certified drives are detected. • vSAN Auto-RAID will be enabled when vSAN certified components are detected. This allows vSAN to generate an optimal default storage policy, based on the cluster topology and the number of ESX hosts.	None.
VCF-VSAN-ESA-REQD-CFG-002	1.6 TB drive size or higher for vSAN ESA devices for storage pools.	Minimum supported vSAN ESA drive size is 1.6TB.	vSAN ESA supports up to 24 drives per node. Smaller capacity drives may constrain raw capacity of a vSAN node.
VCF-VSAN-ESA-REQD-CFG-003	16 Cores or higher per vSAN node. vSAN ESA has a minimum value of 16 cores per vSAN Node	Minimum vSAN ESA core count per node is 16 for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	Smaller number of cores per node may require more vSAN nodes to support targeted workloads.
VCF-VSAN-ESA-REQD-CFG-004	Memory size per vSAN Ready node 128GB or higher	Minimum vSAN ESA memory is 128GB for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	• Smaller memory footprint per node will limit number of vSAN drives per node. • Smaller memory footprint per node for vSAN ESA will limit VM workloads being placed on vSAN nodes.

Stretched vSAN Storage Model Design Requirements

Table 234: Stretched vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-001	Two (2) availability zones and a third site for the witness appliance location.	Fault domains are mapped to availability zones to provide logical and physical ESX host separation and ensure a copy of vSAN data is always available even when an availability zone goes offline.	• This requires both availability zones to have an equal number of ESX hosts residing in each site, and are connected using a high bandwidth/low latency connection known as an inter-site link, or ISL. • The third site should host the witness appliance and must not have a logical or physical dependency on either availability zone.
VCF-VSAN-SC-REQD-CFG-002	The minimum latency required between the data sites hosting virtual machine objects should not exceed 5 milliseconds (ms) in round trip time (RTT), or 2.5ms one way.	The latency minimums exist largely to ensure that applications can write data resiliently across two sites in a timely manner.	Stretched cluster network latency requirements will dictate the realistic geographic distance between sites.
VCF-VSAN-SC-REQD-CFG-003	The minimum latency required between witness site and the data sites should not exceed 200ms RTT, or 100ms one way.	The witness site only stores small amounts of metadata, and is not a part of the vSAN data path.	None.
VCF-VSAN-SC-REQD-CFG-004	vSAN Stretched Cluster Data Site to Data Site Bandwidth minimum is no less than 10GbE. • The vSAN Stretched Cluster Bandwidth Sizing document can be used for general guidance in sizing for vSAN OSA and ESA architectures.	The network requirements for the data-site to data site in a stretch cluster remain the same for both vSAN ESA and OSA.	• VM workloads will ultimately determine the network bandwidth demand between data-sites. • Please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing
VCF-VSAN-SC-REQD-CFG-005	vSAN Stretched Cluster Data Site to Witness Site Bandwidth is based on number of VMs that is protected by a stretch cluster storage policy.	• Network communication from the data sites to the witness host appliance is comprised entirely of metadata and this is wholly dependent on the number of VM and components protected by vSAN Stretch Cluster polices. • There is no minimum bandwidth defined as estimating the bandwidth required for the witness site is based on the number of VM components.	For accurate sizing please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-006	Management Domain initial cluster Minimum of eight (8) nodes for vSAN stretched cluster for high availability (HA) install	Management domain initial clusters deployed in high availability (HA) mode have a requirement for 4 nodes in a non-stretched deployment. Stretching a cluster is a day two operation and will require 4 additional hosts on the second availability zone.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure
VCF-VSAN-SC-REQD-CFG-007	Workload domain clusters (except for management Domain initial cluster) Minimum of six (6) nodes for vSAN stretched clusters	This will provide minimum requirements for site availability and site local availability, tolerating a site failure and one failure on each availability zone. This will give a balance between a host failure on a local availability zone and a full availability zone failure.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure with minimum three (3) nodes per site. However with three nodes per site, vSAN will not automatically rebuild objects that are impacted by a local failure.
VCF-VSAN-SC-REQD-CFG-008	For vSAN OSA architecture Modify the default Storage Policy as follows • **Site disaster tolerance:** Site mirroring • **Failure to tolerate :** One (1) failure RAID-1 (Mirroring)	Provides the minimum necessary protection for VMs in each availability zone, with the ability to recover from an availability zone outage. **Remember:** • For vSAN ESA , Auto Policy configuration will automatically alter the default storage policy to reflect the number of hosts and underlying vSAN topology, when the cluster is converted to a stretch cluster. • With 9.1 vSAN Auto-RAID will be configured by default to all new vSAN ESA clusters and automticlly configure the optimal policy based on the cluster topology	• Additional Storage is necessary to satisfy the storage policy requirements in both availability zones. • Additional policies can be configured for specific workloads depending on their individual performance or availability requirements which may differ from what the default vSAN stretched cluster policy.
VCF-VSAN-SC-REQD-CFG-009	Deploy a vSAN witness appliance in a location that is not co-located to the ESX hosts in any of the availability zones.	Ensures availability of vSAN witness appliance in the event of a failure of one of the availability zones.	You must provide a third availability zone that is logically and physically independent of the data sites The third site must be able to provide a environment capable to host the witness appliance.
VCF-VSAN-SC-REQD-CFG-010	Deploy a witness appliance that corresponds to vSAN Architecture, (ESA or OSA).	Ensures the witness appliance deployed is compatible to the correct vSAN Storage architecture	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-011	Connect the first VMkernel (vmk0) adapter of the vSAN witness appliance to the management network in the witness site.	The ESX host management networks in both availability zones must be routed to the vSAN witness host management network in the third witness site.	When using witness traffic separation, the second interface on vSAN witness host (vmk1) can be removed.
VCF-VSAN-SC-REQD-CFG-012	Configure vSAN Witness host identity as per standard ESX host recommendations.	• Allows vSAN Witness host to be managed identically as a ESX host. • Ensures host identity, DNS and NTP is consistently configured	Apply the requirements for External Services Design
VCF-VSAN-SC-REQD-CFG-013	Enable vSphere Lifecycle Manager images on the vSAN witness host.	• VMware Cloud Foundation 9.0 requires vSphere Lifecycle Manager images for ESX hosts. • This will prevent warnings or vSAN health check alarms.	This requires either deploying or converting a vSAN witness host as a vSphere Lifecycle Manager image managed entity.

Common vSAN Storage Model Design Recommendations

Table 235: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Common vSAN Host Network Model Design Recommendations

Table 236: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

vSAN ESA Storage Model Design Recommendations

Table 237: vSAN ESA Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-001	• Select a specific size of drives used according to the requirements of target workloads. • Use multiple storage devices per vSAN node • NVMe TLC devices with a performance Class F (100000 - 349999) or higher with endurance class 3 DWPD or higher.	• The number of storage devices has an impact on the performance of vSAN ESA. • NVMe drives with mixed-use endurance and performance class typically classified on the vSAN HCL as "3DWPD" (3 Drive Writes per Day) will give a balance between performance and durability.	• Smaller drive sizes will impact overall capacity of vSAN ESA. Large drive sizes will introduce larger capacity fault domain in case of failure. • Sizing vSAN Nodes requires determining the workload capacity requirements prior to deployment. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-002	Use vSAN ready nodes with high core count.	vSAN Express Storage Architecture (ESA) is designed to be highly parallel and leverages multiple CPU cores for improved performance, making it multithreaded, and optimized for NVMe- TLC based flash devices.	There will be a correlation between target workload, number of cores and NVMe devices per vSAN Node. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-003	Use vSAN ready nodes with high memory capacity per vSAN node.	When enabled vSAN has required memory overheads and must have adequate resources account for VM workloads.	Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsanizer/home.
VCF-VSAN-ESA-RCMD-CFG-004	At least four (4) nodes for vSAN cluster size.	This ensures in the event of a fault domain failure, vSAN objects can be re-protected to maintain policy compliance. In conjunction with auto policy management, four (4) node vSAN ESA clusters will allow vSAN to automatically create a default storage policy that is capacity optimized.	None. **Note:** New vSAN clusters in VCF 9.1 will use clusters will automatically be configured to use Auto-RAID.
VCF-VSAN-ESA-RCMD-CFG-005	• Select the same minimum requirements for vSAN Storage cluster nodes as vSAN ESA HCI nodes. • Review vSAN Storage minimum node requirements	• vSAN Storage Cluster Nodes use the same architecture as regular vSAN ESA nodes. • vSAN ESA nodes can be re-purposed for vSAN Storage Clusters or vSAN HCI Clusters	None.

Stretched vSAN Storage Model Design Recommendations

Table 238: Stretched vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-RCMD-001	Use Layer 3 (routed) networking between the two data sites, and the witness site. • The data sites will only be able to communicate with each other using the ISL • The witness host should only be able to communicate with hosts in each data site directly, and not through another data site.	Using Layer 3 helps avoid spanning tree protocol (STP) from redirecting traffic across an undesirable link, such as the more bandwidth constrained link to the witness site	Ensure Layer 3 gateways are highly available such that it tolerates the failure of an entire availability zone.
VCF-VSAN-SC-RCMD-002	Use witness traffic separation when configuring a stretched vSAN cluster.	• Network Flexibility. You can configure different VLANs or subnets for vSAN data and witness traffic, allowing for better network segmentation and resource allocation. • Mixed MTU. This will allow vSAN witness traffic type to use a different MTU size than the vSAN data traffic. This avoids configuring static routes between vSAN networks. • Simplified Troubleshooting. Identifying and resolving network issues related to witness traffic becomes easier when it is isolated from the main vSAN data traffic.	This requires the vSAN witness network to be routable to both the ESX host management networks on either availability zone as well to VM management network if configured.
VCF-VSAN-SC-RCMD-CFG-003	Configure the vSAN witness host to use the first VMkernel adapter (vmk0), that is the management interface, for vSAN witness traffic.	• Removes the requirement to have two IP addresses allocated to the vSAN witness host. • Removes the requirement to have static routes in the witness as the witness traffic is routed over the management networks	• The management networks in both availability zones must be routed to the management witness site. • Ensure to use the same MTU size as the host management interfaces. **Note:** Default MTU size is 1500 for management interfaces

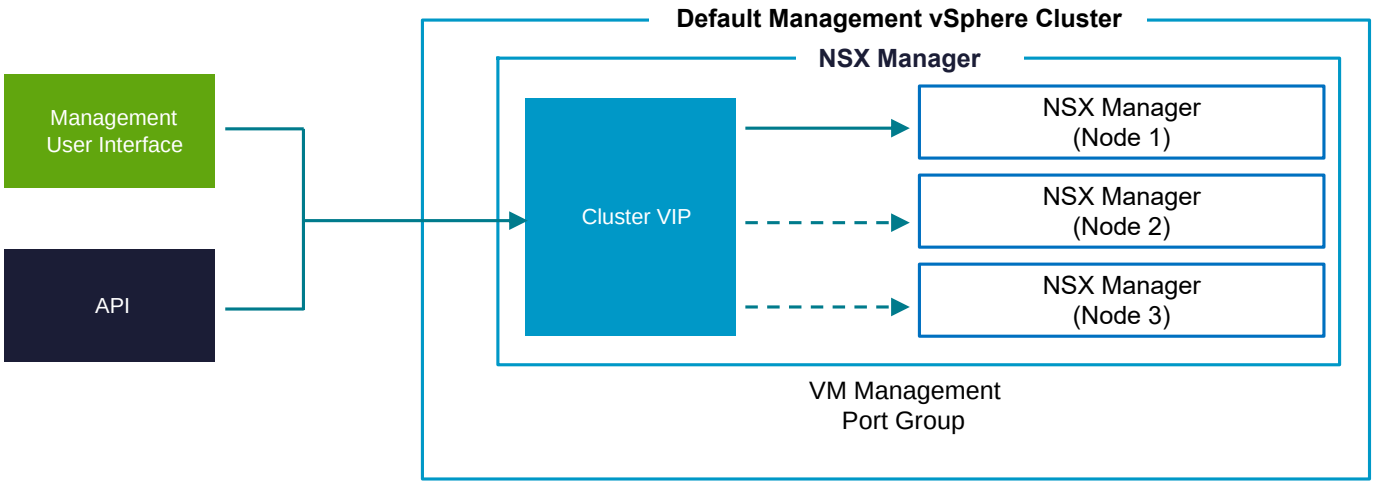
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-RCMD-CFG-004	For vSAN OSA clusters, configure vSAN storage policies with **Force Provisioning**.	vSAN objects are created if an availability zone becomes unavailable because of a site maintenance operation or a failure.	This would require modifying vSAN OSA policies used as part of the stretched cluster. **Tip:** **Force Provisioning** is no longer an applicable storage rule in a stretched cluster using Auto-RAID. Objects will continue to be created in the event of site mainteance.
VCF-VSAN-SC-RCMD-CFG-005	Ensure DRS Host Groups, VM Groups, and VM/Host Rules are in conjunction with vSAN storage policies.	• VM and vSAN data resilience desired outcomes are satisfied. • Suggests or forces a behavior of VM placement in a vSAN stretched cluster while in steady state. • vSphere HA will trigger the desired behavior in response to a failure.	• Requires creation of additional DRS Host groups, VM Groups, and VM/Host DRS Rules. • The placement rules should match the desired failure response.
VCF-VSAN-SC-RCMD-CFG-006	vSphere HA Admission control fail-over capacity is configured to reserve 50% of the CPU and memory capacity of the entire stretched cluster.	This ensures that VMs can fail between AZs and have adequate resources.	Requires adequate capacity guarantee successful fail-over of protected workloads in the event of failure of an entire availability zone.
VCF-VSAN-SC-RCMD-CFG-007	Do not enable reserved Capacity in a stretched vSAN cluster.	vSAN capacity reserve is not supported in conjunction with the following vSAN cluster topologies: • Three (3) node clusters. • Stretched vSAN clusters. • vSAN clusters using fault domains.	This will require manually calculating vSAN slack space. Refer to the vSAN sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

High Availability NSX Manager Model

The *High Availability NSX Manager Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.

Figure 98: High Availability NSX Manager Model in VMware Cloud Foundation



High Availability NSX Manager Model Attributes

The *High Availability NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Three-node cluster of NSX Manager appliances with internal virtual IP
Availability	• Highly available cluster of virtual IP NSX Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.
API Request Scale	Internal virtual IP directs API requests to a single NSX Manager appliance.

Common NSX Manager Design Requirements for All Models

Table 239: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

High Availability NSX Manager Model Design Requirements

Table 240: High Availability NSX Manager Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-005	Deploy three (3) NSX Manager nodes in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports high availability of the NSX Manager cluster.	You must have sufficient resources in the default cluster of the management domain to run three NSX Manager nodes.

Common NSX Manager Model Design Recommendations for All Models

Table 241: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

High Availability NSX Manager Model Design Recommendations

Table 242: High Availability NSX Manager Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-003	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Manager appliances.	Keeps the NSX Manager appliances running on different ESX hosts for high availability.	You must allocate at least four physical hosts so that the three NSX Manager appliances continue running if an ESX host failure occurs.

Common NSX Manager Design Recommendations for Stretched Clusters for All Models

Table 243: Common NSX Manager Design Recommendations for Stretched Clusters for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-002	Add the NSX Manager appliances to the virtual machine group for the first availability zone.	Ensures that the NSX Manager appliances are powered on an ESX host in the primary availability zone.	None.

Physical Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region

Physical Infrastructure constitutes the tangible hardware and facilities layer that provides the foundational computing, storage, and networking resources upon which all digital services depend. This layer encompasses the actual physical

assets—servers, storage arrays, network equipment, and supporting facilities—that form the bedrock of modern IT operations.

Multi Availability Zone Network Fabric Model

The *Multi Availability Zone Network Fabric Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region* infrastructure blueprint.

Selected Model	Additional Information
Multi Availability Zone Network Fabric Model	This section describes the VLAN and gateway requirements to support a vSphere cluster within multiple availability zones.

Table 244: Infrastructure VLAN Scope and First Hop Gateway Availability for Multiple AZ

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management component VMs*		Global across AZs	Highly Available - Across AZs **	1500	9000
ESX host	ESX management VMK	AZ Local***	Highly Available - Within AZ**		1500
	IP Storage VMKs (VSAN, NFS, iSCSI)		Highly Available - Within AZ **		9000
	vMotion VMK		Highly Available - Within AZ **		
	Host TEP		Highly Available - Within AZ **	1700	
NSX Edge node	Refer to Dual Availability Zone Based on NSX Edge Node HA Model				
	NSX Edge Node Management	AZ Local	Highly Available - Within AZ **	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available - Within AZ** BGP - None	1700	
	Refer to Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model				
	NSX Edge Node Management	Global across AZs	Highly Available - Across AZs *	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available - Across AZs ** BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Dual Availability Zones				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
	BGP-EVPN Peering	Across AZs	Single-Hop BGP - None Multi-Hop BGP - Highly Available - Across AZs **	1500	1500

* When using separate VLAN segments for management component virtual machines and ESX VMkernels.
** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.
**In this table, ESX host infrastructure VLANs are limited to an individual availability zone. Stretching of these VLANs between availability zones is supported, however the first hop gateway must also be highly available between availability zones.

Table 245: Workload VLAN Scope and First Hop Gateway Availability for Multiple Availability Zones

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Global between AZs	Highly Available - Across AZs**	1500	9000
Distributed VLAN Connectivity Model	Global between AZs*	Highly Available - Across AZs**	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model	Refer to Edges			
NSX Segment Connectivity Model	Refer to Edges			
* All Node of a VNA cluster must share a common L2 network. Optional stretching of this VLAN between racks will enable seamless maintenance and recovery across racks. ** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.				

This design requires stretching a minimum number of VLAN segments between each availability zone. The physical network fabric stretches these segments to provide a single Layer 2 domain across zones, enabling virtual machine mobility during an availability zone outage. For these fabric-stretched VLAN segments, the first-hop gateway must remain highly available across both availability zones. You must give special consideration to VLAN segments for NSX Edge uplinks when using the Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model, as BGP peering from the NSX Edge node to the fabric occurs over these networks.

Note:

VLAN IDs are switch-level numerical labels identifying a Layer 2 domain. In some fabric configurations, the same VLAN ID may identify different domains, or different VLAN IDs may identify the same domain. A distributed port group can be associated with a single VLAN ID. Any distributed port groups supporting virtual machines must have the same VLAN ID and Layer 2 domain mapping across the entire vSphere cluster. You may use different VLAN IDs for separate vSphere clusters that do not share a common vSphere Distributed Switch.

Table 246: Infrastructure VLAN Scope and First Hop Gateway Availability for Multiple AZ

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management component VMs*		Global across AZs	Highly Available - Across AZs **	1500	9000
ESX host	ESX management VMK	AZ Local***	Highly Available - Within AZ**		1500
	IP Storage VMKs (VSAN, NFS, iSCSI)		Highly Available - Within AZ **		9000
	vMotion VMK		Highly Available - Within AZ **		
	Host TEP		Highly Available - Within AZ **	1700	
NSX Edge node	Refer to Dual Availability Zone Based on NSX Edge Node HA Model				
	NSX Edge Node Management	AZ Local	Highly Available - Within AZ **	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available - Within AZ** BGP - None	1700	
	Refer to Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model				
	NSX Edge Node Management	Global across AZs	Highly Available - Across AZs *	1500	9000

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available - Across AZs ** BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Dual Availability Zones				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
	BGP-EVPN Peering	Across AZs	Single-Hop BGP - None Multi-Hop BGP - Highly Available - Across AZs **	1500	1500

* When using separate VLAN segments for management component virtual machines and ESX VMkernels.
** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.
**In this table, ESX host infrastructure VLANs are limited to an individual availability zone. Stretching of these VLANs between availability zones is supported, however the first hop gateway must also be highly available between availability zones.

Table 247: Workload VLAN Scope and First Hop Gateway Availability for Multiple Availability Zones

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Global between AZs	Highly Available - Across AZs**	1500	9000
Distributed VLAN Connectivity Model	Global between AZs*	Highly Available - Across AZs**	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model	Refer to Edges			
NSX Segment Connectivity Model	Refer to Edges			

* All Node of a VNA cluster must share a common L2 network. Optional stretching of this VLAN between racks will enable seamless maintenance and recovery across racks.
** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.

IP Connectivity and Symmetry

Traffic must be routed between availability zones for all VLAN segments that require cross-site communication. For example, vSAN networks in each availability zone must have a route to each other to support synchronous data replication.

For VLAN segments that stretch between availability zones, the physical network fabric determines the ingress and egress paths. In most cases, to simplify troubleshooting, symmetrical routing is recommended.

For workloads, the ingress and egress path depends upon the Network Consumption Models chosen. For the Centralized Connectivity Modelingress and egress paths depend on the specific NSX Edge Cluster Models for Dual Availability Zonesand routing model:

- **Static Routing**: The physical network fabric determines the ingress and egress paths.
- **eBGP**: A combination of the physical fabric and the NSX Edge gateway determines the ingress and egress paths.

For all stretched VLAN network segments (infrastructure networks, VLAN Networking Connectivity Model or Distributed VLAN Connectivity Model), or Distributed VXLAN Connectivity Model for Version 9.1.0, the physical network fabric must provide a highly available Layer 3 gateway at the first hop. This ensures that virtual machines maintain connectivity on the same IP address when moving between availability zones.

Witness Location

For vSAN stretched cluster, an independent witness site is required. This requires independent network paths from each availability zone to the witness site, such that the failure of an individual availability zone does not impact the traffic path from the other availability zones to the witness site.

When configured with vSAN Witness Traffic Separation (WTS), the vSAN witness traffic flows between the ESX management VMkernels and the vSAN Witness node. This traffic uses the MTU of the ESX management VMkernels.

Availability Zone Data Center Interconnect

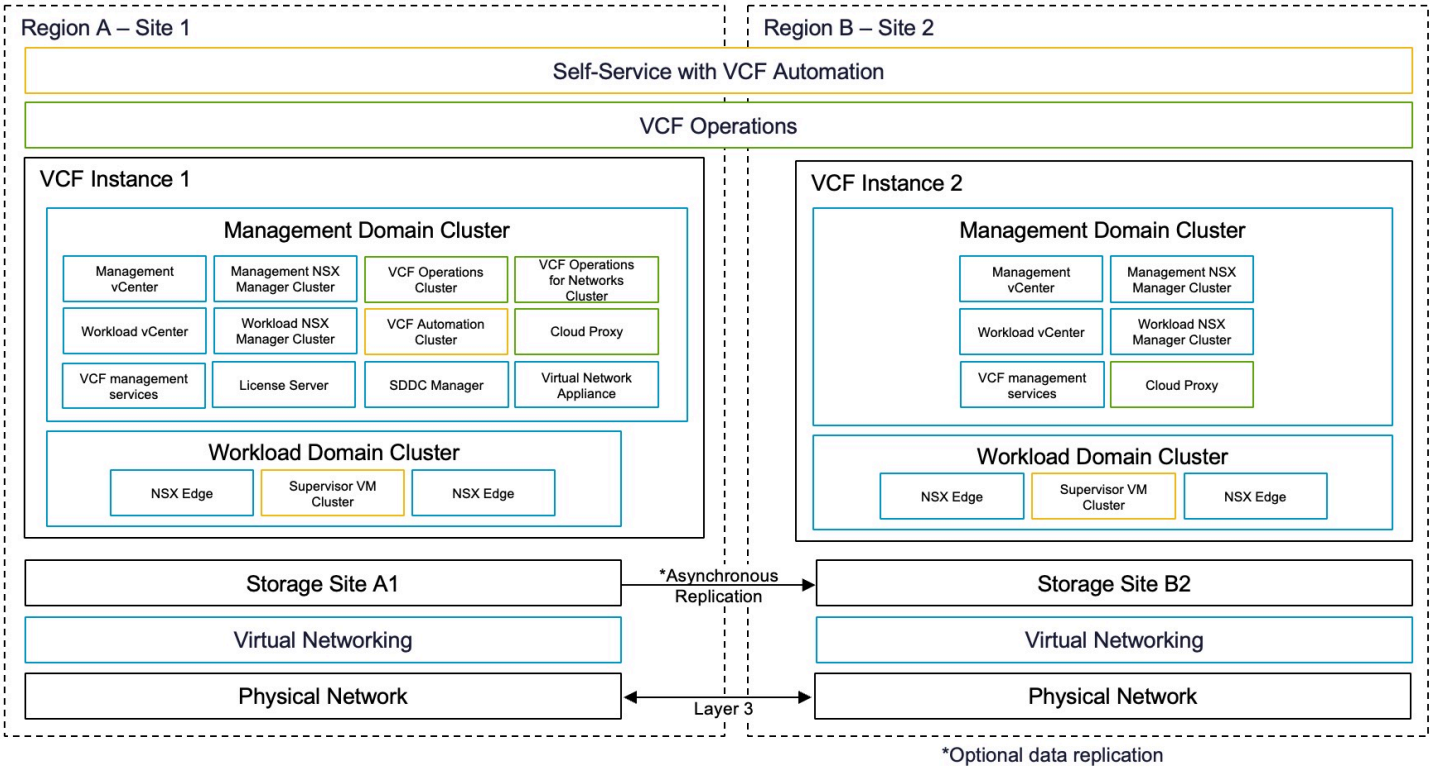
The data center inter-connect between the availability zones must be appropriately sized to handle all infrastructure, workload, and storage replication traffic. The interconnect is important to the operations of the vSAN Cluster. It should have the appropriate level of resilience to achieve business objectives.

VCF Fleet with Multiple Sites Across Multiple Regions

The *VCF Fleet with Multiple Sites Across Multiple Regions* an infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components and workloads are isolated to a separate workload domain. Multiple sites exist in different regions.

Architecture Overview

Figure 99: VCF Fleet with Multiple Sites Across Multiple Regions



Design Profile

The *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint is well suited to customers who are looking for the following design profile for their VMware Cloud Foundation platform.

Attribute	Value
Consumption	• Self-service consumption via VCF Automation supporting modern workloads like VKS, Data Services, AI Services, and so on. with multi-tenancy. • Direct vCenter consumption supporting modern workloads like VKS, Data Services, AI Services, and so on. This infrastructure blueprint can use the following Consumption blueprint: • Self-Service Multi-Tenant Private Cloud Consumption
Physical Site Configuration	Multiple sites (across multiple regions)

Attribute	Value
Availability	• Tolerates failure of a single ESX host • Tolerates failure of an individual network path • Tolerates failure of a single rack • Tolerates failure of a vSphere cluster • Tolerates failure of a single management component
Isolation	• Hypervisor-based (logical isolation) • Network traffic type isolation • Cluster-based (physical isolation) • Tenant / Tenant isolation
Recoverability	This infrastructure blueprint can use the following protection blueprints: • VCF Component Backup and Restore • VCF Instance Backup and Restore • VCF Fleet Disaster Recovery
Expansion	For available resource and availability expansions, see Expansion after Infrastructure Blueprint Deployment.

Management Domain Design Selections

The following models in the design align with the objectives for deploying in this *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	High Availability VCF Automation Model	The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.
	Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.
Operations	High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single Cloud Proxy appliance. To achieve high availability after the initial deployment, an additional Cloud Proxyappliance should be deployed and then combined in a collector group.
	VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.
	VCF Management Services in an Additional VCF Instance Model	Additional VCF Instances in a fleet run a VCF management services cluster with instance level components for managing that instance only.
	High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.
	Single License Server Model	The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.

Layer	Selected Model	Additional Information
	High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a four-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.
	High Availability Log Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind and internal load balancer. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.
Security Governance and Compliance	Identity Broker Model	Deploys the Identity Broker component on the VCF management services instance in the management domain.
	VCF Fleet-Wide Single Sign-On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all components across all VCF instances in the same VCFfleet.
Virtual Infrastructure	Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.
	VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Workload Domain Design Selections

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.
	High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected vSphere Supervisor management zones.
	Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for VCF Supervisor and VKS workloads, it supports VPC-based networking.
	Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways	This design provides multi-tenant resource isolation across multiple regions by combining dedicated gateways for strict tenant separation with shared gateways for efficient common services.
Virtual Infrastructure	Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.
	Host Fault Tolerant NSX Edge Cluster Model	A *Single-Rack Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Shared Design Selection between Management and Workload Domain

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

	Selected Model	Additional Information
Virtual Infrastructure	vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.
	ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCFdomain vSphere cluster.
	Single-Rack vSphere Cluster Model	The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.
	Workload Separation Distributed Switch Model	A *Workload Separation Distributed Switch Model* in VMware Cloud Foundation provides a separation of workload traffic running on NSX segments from all other network services running on the vSphere cluster.
	Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture that uses high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.
	High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.
Physical Infrastructure	Single-Rack Network Fabric Model	The *Single-Rack Network Fabric Model* describes the VLAN and gateway requirements to support the Single-Rack vSphere Cluster Model.

Minimum Requirements for VCF Fleet with Multiple Sites Across Multiple Regions

VCF Fleet with Multiple Sites Across Multiple Regions requirements to stand up this infrastructure blueprint.

VCF Installer Deployment Options

Deployment Model	Deployment Size
High Availability	Medium

Minimal IP Requirements

Requirement	Additional Information
Physical Networks	• Region 1 ESX Management (4 IPs) • Region 1 VM Management (35 IPs) • Region 1 vMotion (4 IPs) • Region 1 vSAN (4 IPs) • Region 1 Host TEP (8 IPs) • Region 1 (Optional) Edge TEP Network (4 IPs) – 2 IPs per Edge VM (x2 Edge VMs) • Region 1 (Optional) Edge Uplink Network (2 IPs) – 1 IP per Edge VM (x2 Edge VMs) • Region 2 ESX Management (4 IPs) • Region 2 VM Management (15 IPs) • Region 2 vMotion (4 IPs) • Region 2 vSAN (4 IPs) • Region 2 Host TEP (8 IPs) • Region 2 (Optional) Edge TEP Network (4 IPs) – 2 IPs per Edge VM (x2 Edge VMs) • Region 2 (Optional) Edge Uplink Network (2 IPs) – 1 IP per Edge VM (x2 Edge VMs)
DNS Entries	• Region 1 vCenter • Region 1 ESX – 4 Nodes • Region 1 SDDC Manager • Region 1 NSX Manager – VIP – Node A – Node B – Node C • Region 1 VCF Operations – Primary – Replica – Data – Cloud Proxy • Region 1 VCF Management Services – VCF Services Runtime – VCF Management Instance Service – VCF Fleet Component • Region 1 VCF Automation • Region 1 VCF Automation Service Runtime • Region 1 License Server • Region 1 Identity Broker • Region 1 Log Management • Region 1 VCF Operations for Networks – Platform 1 – Platform 2 – Platform 3 – Collector • Region 2 vCenter • Region 2 ESX – 4 Nodes • Region 2 SDDC Manager • Region 2 NSX Manager – VIP – Node A – Node B – Node C • Region 2 VCF Operations – Cloud Proxy • Region 2 VCF Management Services

Requirement	Additional Information
Virtual Networks	• Supervisor Service CIDR — e.g. 10.96.0.0/23 • VPC Private IP Block (Control Plane) • Distributed Transit Gateway Private IP Block (Control Plane) • External IP Block (Control Plane & Workload/Organizations) • Private VPC Block for each Organization's VPCs • Private TGW IP Block for each Organization

First VCF Instance

Table 248: VMware Cloud Foundation Installer Deployment Profiles

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for the initial VCF instance within a VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	1x Small • 1x Control Plane Nodes • 2x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Large • 3x Control Plane Nodes • 4x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Medium	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
VCF Operations	1x Small	2x Small	3x Medium	3x Large
License Server	1x Default	1x Default	1x Default	1x Default
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard
VCF Automation	1x Small	1x Medium	3x Medium	3x Large
Resource Requirements	• vCPU: 76 • RAM: 251 GB • DISK: 7448 GB	• vCPU: 106 • RAM: 335 GB • DISK: 8422 GB	• vCPU: 184 • RAM: 656 GB • DISK: 11445 GB	• vCPU: 298 • RAM: 949 GB • DISK: 15357 GB

Additional VCF Instance

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for any additional VCF instance joining an existing VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Table 249: VMware Cloud Foundation Installer Deployment Profiles

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	• 1x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Small	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
Resource Requirements	• vCPU: 34 • RAM: 111 GB • DISK: 4562GB	• vCPU: 62 • RAM: 187 GB • DISK: 5462 GB	• vCPU: 74 • RAM: 244 GB • DISK: 5813 GB	• vCPU 120 • RAM 353 GB • DISK 8321 GB

Management Domain Design Selections

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

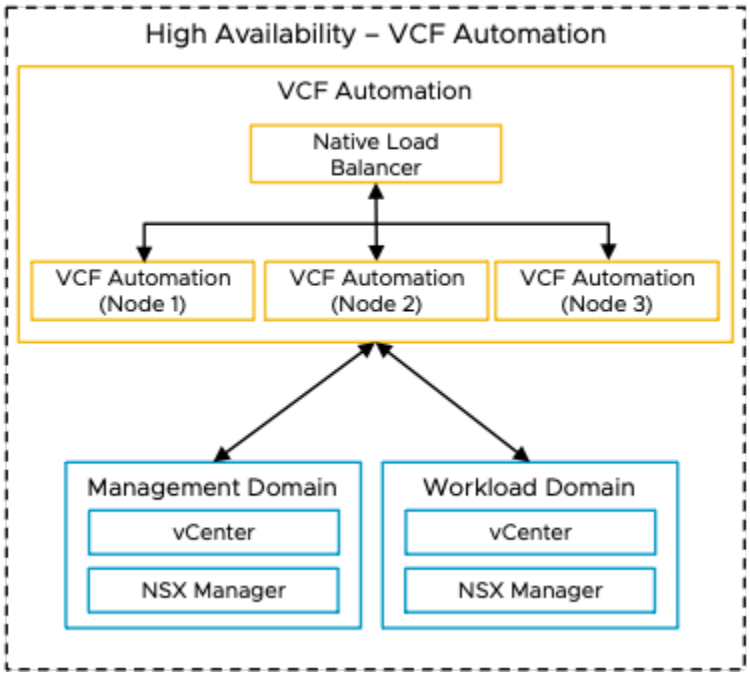
Consumption Layer for VCF Fleet with Multiple Sites Across Multiple Regions

The consumption layer for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

High Availability VCF Automation Model

The *High Availability VCF Automation Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Automation Model	The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.



High Availability VCF Automation Model Attributes

A *High Availability VCF Automation Model* has the following attributes.

Attribute	Detail
Availability	• Application level availability provided by the three node cluster • Appliance level level availability provided by vSphere HA
Scalability	Supports scale-up and scale-down of node sizes.
Recoverability	Restore through a backup.

High Availability VCF Automation Model Design Requirements

Table 250: High Availability VCF Automation Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AUTO-REQD-HA-001	Deploy VCF Automation as a three node cluster.	Supports high availability objectives.	• Analysis is required to determine which node size is appropriate for your deployment. • Node can be resized after deployment as necessary.
VCF-AUTO-REQD-HA-003	Protect all VCF Automation nodes using vSphere HA.	Ensures availability of VCF Automation without requiring manual intervention during an ESX host failure event.	None

VCF Automation Tenant Authentication Requirements

Table 251: VCF Automation Tenant Authentication Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-TEN-001	Configure an identity provider for VCF Automation organizations.	• Allows tenants to use enterprise directory. • Enables named access control to tenant level constructs. • Required to provision administrative or user access to the Organization portal for any account other than the first user account established during Organization creation.	If not configured, the only access to the Organization Portal will be through the first user account established during Organization creation.

VCF Automation Enterprise Authentication Design Requirements

Table 252: VCF Automation Enterprise Authentication Design Requirements

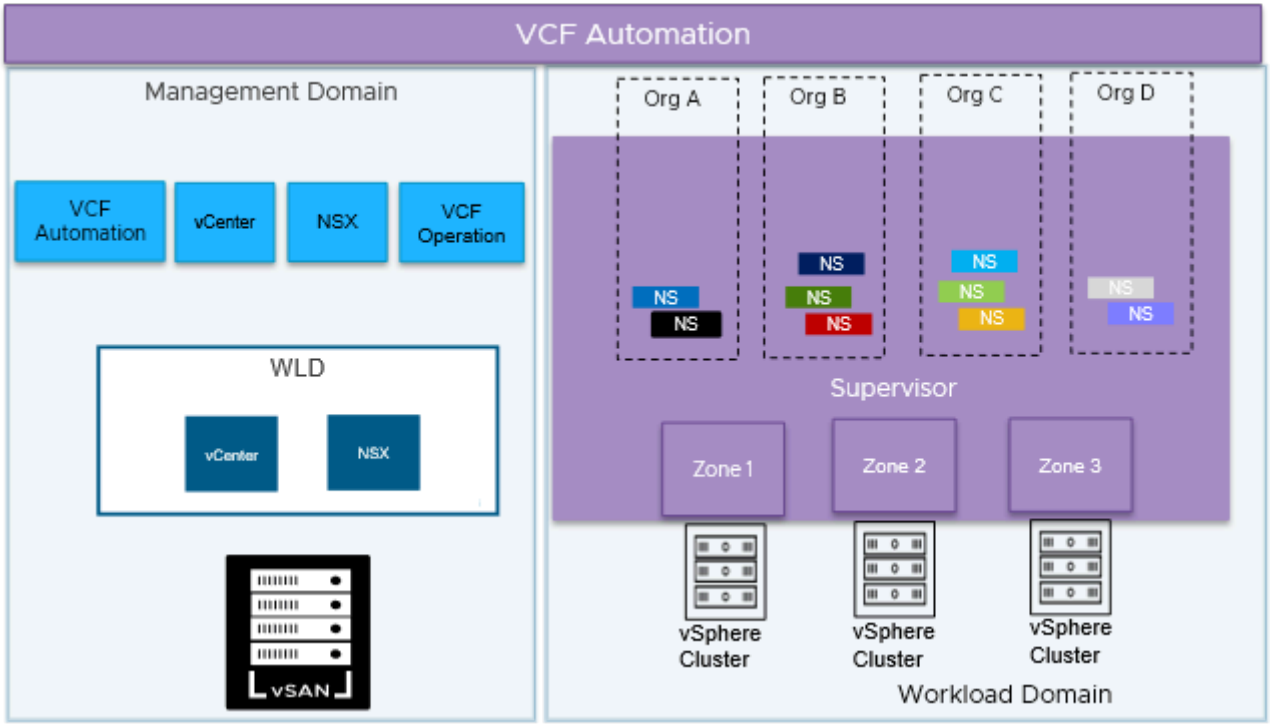
Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-IDP-001	The provider uses the same IdP as the tenant.	• Allows consistent authentication across provider and tenant	None

Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance

The *Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.

Figure 100: Shared Workload Domain for Multiple Organizations in a VCF Instance



Benefits	Best fit scenarios
• Resource Sharing: Tenants share the underlying infrastructure (workload domain, vCenter, NSX), leading to cost savings and efficient resource utilization. • Simplified Management and Operational Consistency: While providing tenant isolation, the centralized management of the VCF Management Domain simplifies overall infrastructure operations, including patching, upgrades, monitoring, and capacity planning. Consistent policies can be applied across all tenants. • Scalability and Flexibility: The model is highly scalable. As the number of tenants or their resource demands grow, the underlying workload domain can be scaled up by adding more hosts. Resource allocations within Namespaces can also be adjusted dynamically. If the requirement comes, the service provider can add more workload domains for multiple other organizations to share or can dedicate a workload domain per tenant like explained in Model 2. • vSphere Namespace Isolation: vSphere Namespaces provide logical resource isolation for tenant workloads, preventing interference. • Network Segmentation: Network policies and security groups can be used to further segment tenant traffic and can enhance security.	• Cloud Service Providers (CSPs) Offering Multi-Tenant Cloud Services: This model allows the CSP to host multiple customers within a single VCF instance, maximizing resource utilization across their infrastructure. The Supervisor Cluster ensures strong isolation between each customer's workloads and data. Each customer gets their own vSphere Namespace, providing them with a secure and logically separated environment for deploying VMs and containers. Network segmentation (via NSX) further guarantees that customer traffic remains isolated. • Large Enterprises with Multiple Business Units or Departments: A large corporation has several independent business units (e.g., Finance, Marketing, R&D) that require their own isolated IT environments but can share underlying infrastructure. • Organizations Offering Internal PaaS or Developer Platforms: An organization wants to provide a self-service platform for its internal development teams to deploy and manage their applications (including containerized microservices) in an isolated and governed manner. • Educational Institutions or Research Organizations with Multiple Projects or Teams: A university hosts various research projects or has different departments with distinct IT requirements and security needs.

Operations Layer for VCF Fleet with Multiple Sites Across Multiple Regions

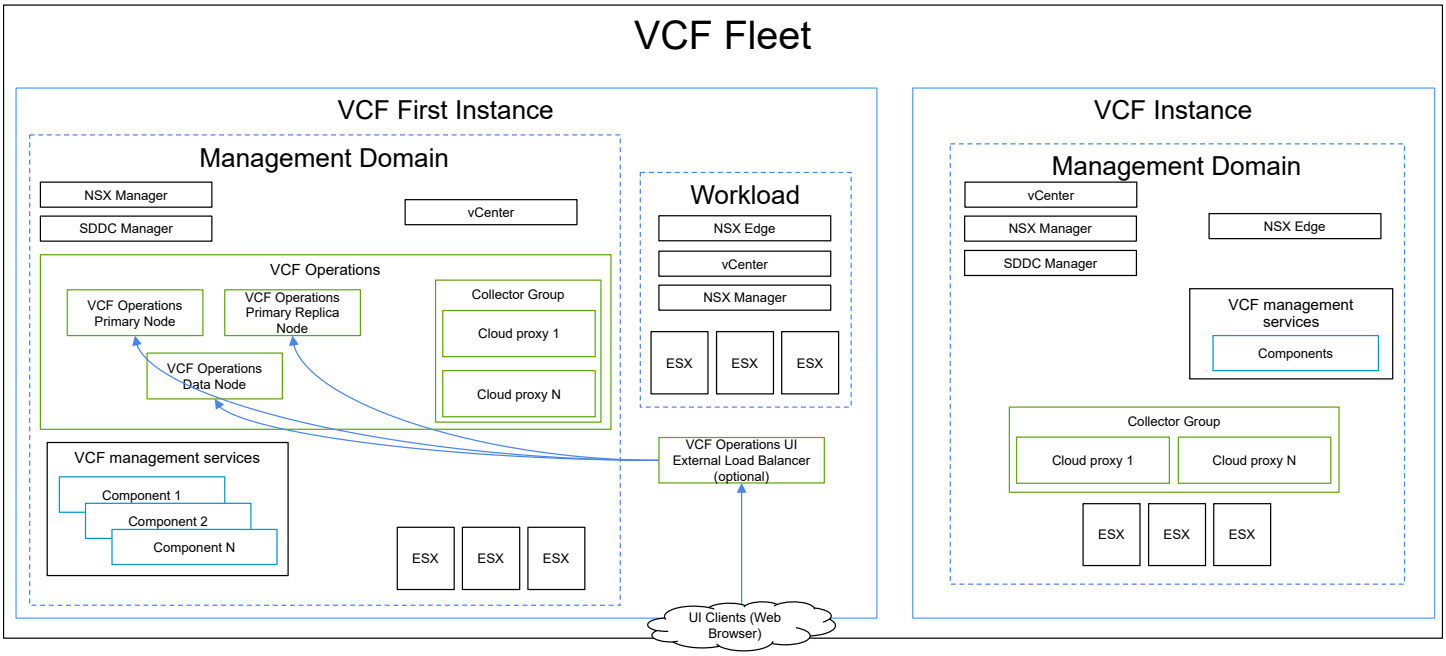
The operations layer serves as the central nervous system of enterprise infrastructure and applications, providing comprehensive visibility, management, and control capabilities across the entire technology stack. This layer ensures that systems remain available, performant, secure, and compliant with organizational standards and regulatory requirements.

High Availability VCF Operations Model

The *High Availability VCF Operations Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single Cloud Proxy appliance. To achieve high availability after the initial deployment, an additional Cloud Proxy appliance should be deployed and then combined in a collector group.

Figure 101: High Availability VCF Operations Model in VMware Cloud Foundation



High Availability VCF Operations Model Attributes

A *High Availability VCF Operations Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure. Additional level of availability for the components in: • VCF Operations cluster • VCF Operations collector groups
Scalability	Supports scale-up and scale-out. **Note:** Scale-out is supported for the VCF Operations data nodes and cloud proxynodes.
Recovery	Restore through a backup.
Load balancing	Optional external load balancer for the VCF Operations cluster UI.

Common VCF Operations Design Requirements for All Models

Table 253: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Operations Model Design Requirements

Table 254: High Availability VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-001	Deploy VCF Operations as a cluster of three nodes in the default management vSphere cluster.	• Supports the high availability objectives by deploying VCF Operations as a cluster. • Provides the capacity that is required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out with additional data nodes.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • Create an anti affinity rule (Separate Virtual Machines) for the VMs in the cluster after the deployment.
VCF-OPS-REQD-HA-002	Deploy a single Unified VCF Operations collector node in the default management vSphere cluster.	Removes the load from the VCF Operations cluster from collecting metrics from local-instance applications.	You may need to scale-up the VCF Operations collector nodes to a greater size based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-HA-003	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations cluster nodes.	Using vSphere DRS prevents the VCF Operations cluster nodes from running on the same ESX host and risking the high availability of the cluster.	• You must perform an additional configuration to set up an anti-affinity rule. • You must have a minimum of four ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode. • If additional data nodes are added, you must update the anti-affinity rule and ensure there are enough ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode.
VCF-OPS-REQD-HA-004	The certificate for VCF Operations must include the FQDNs for all cluster nodes in the SAN (subject alternative name) attribute. **Note:** The SAN does not need to include the FQDN of VCF Operations collector nodes.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted.	Before scale out, the certificate for VCF Operations must be changed to also include the FQDN of the additional node in the SAN (subject alternative name) attribute.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-005	If using an optional external load balancer in the high availability deployment model, the certificate for VCF Operations **must** also include the FQDN of the load balancer in the SAN (subject alternative name) attribute.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted when using an external load balancer with the high availability model.	None.

Common VCF Operations Design Recommendations for All Models

Table 255: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

High Availability VCF Operations Model Design Recommendations

Table 256: High Availability VCF Operations Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-HA-001	Deploy VCF Operations with an external load balancer to load balance the connections across the VCF Operations cluster nodes.	Required to deploy a VCF Operations cluster deployment with distributed user interface access across members.	You must configure the external load balancer configuration to support this VCF Operations configuration.
VCF-OPS-RCMD-HA-002	Create a collector group in VCF Operations for each VCF Instance and assign the default VCF Operations collector appliance.	Provides increased availability to metric collection across the VCF Operations collector appliances that are assigned to the collector group.	Manual reconfiguration of the VMware Cloud Foundation adapter required to utilize multiple VCF Operations collector appliances with the group.
VCF-OPS-RCMD-HA-003	Deploy an additional VCF Operations collector node in the default management vSphere cluster.	Extends the availability of the VCF Operations collectors by balancing the load across the collectors when added to a collector group.	• You must deploy the collector node after the deployment of the VCF Operations. • Additional collector node should be deployed for all VCF Instances that requires a collector group.
VCF-OPS-RCMD-HA-004	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations collector nodes.	Using vSphere DRS prevents the VCF Operations collectors from running on the same ESX host and risking the high availability of the collectors.	You must perform manual configuration to set up an anti-affinity rule.
VCF-OPS-RCMD-HA-005	Reconfigure the default VMware Cloud Foundation Adapter in VCF Operations to use a local-instance collector group for a VCF Instance.	Increases the availability of metric collection.	Must be reconfigured post-deployment.

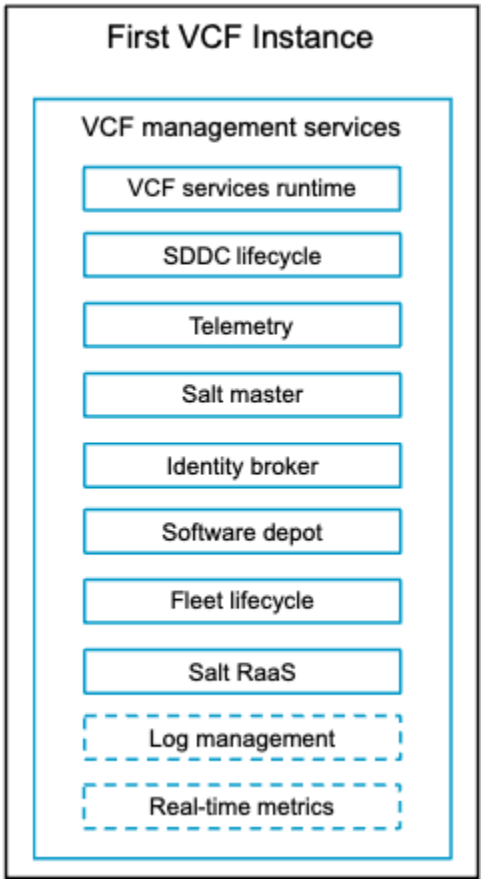
VCF Management Services in the First VCF Instance Model

The *VCF Management Services in the First VCF Instance Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.

Figure 102: VCF Management Services in the First VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.



VCF Management Services in the First VCF Instance Model Components

Table 257: VCF Management Services in the First VCF Instance Model Components

The following components run in the first VCF Instance in the VCF fleet.

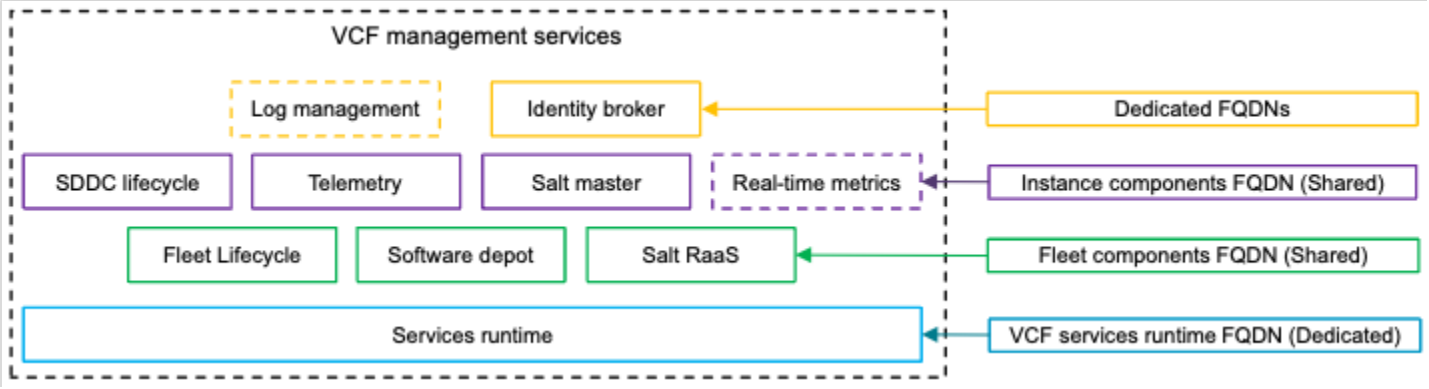
Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations across the fleet.	
Fleet lifecycle	Orchestrates fleet-wide life-cycle operations, including the planning and sequencing of upgrades and patches across multiple VCF Instances.	

Service	Description	Deployment Time
Salt RaaS	Defines and manages desired state configurations and policies (Recipe as a Service).	
Log management	Collects logs from all fleet components	Day-N Components
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

VCF Management Services in the First VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 103:



VCF Management Services in the First VCF Instance Model Design Requirements

Table 258: VCF Management Services in the First VCF Instance Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-FIRST-001	Create DNS entries for the following shared FQDNs: • Fleet components FQDN • Instance components FQDN	Required for deploying VCF management services fleet and instance level components in the first VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-FIRST-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN • Identity broker FQDN	Required for deploying VCF management services in the first VCF Instance.	You must create the DNS entries before starting the deployment.

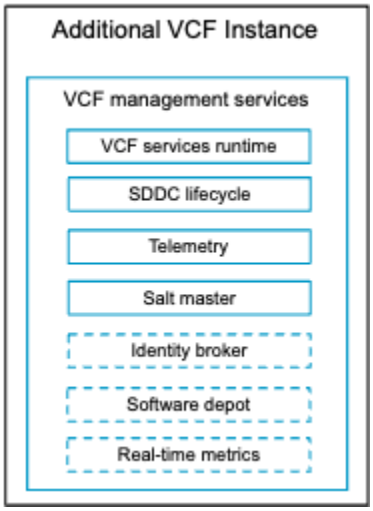
VCF Management Services in an Additional VCF Instance Model

The *VCF Management Services in an Additional VCF Instance Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Services in an Additional VCF Instance Model	Additional VCF Instances in a fleet run a VCF management services cluster with instance level components for managing that instance only.

Figure 104: VCF Management Services in an Additional VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.



VCF Management Services in an Additional VCF Instance Model Components

Table 259: VCF Management Services in an Additional VCF Instance Model Components

The following components run in each additional VCF Instance in the VCF fleet.

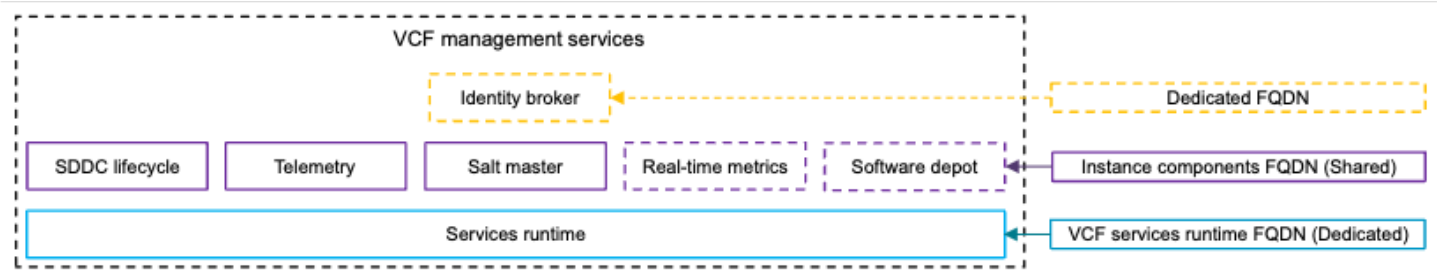
Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	Day-N Components
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations within this VCF Instance.	
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

VCF Management Services in an Additional VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 105: VCF Management Services FQDN Mapping in an Additional VCF Instance

The diagram below shows the various components for this model. The dotted line components are optional, and are deployed day-N.



VCF Management Services in an Additional VCF Instance Model Design Requirements

Table 260: VCF Management Services in an Additional VCF Instance Model Design Requirements

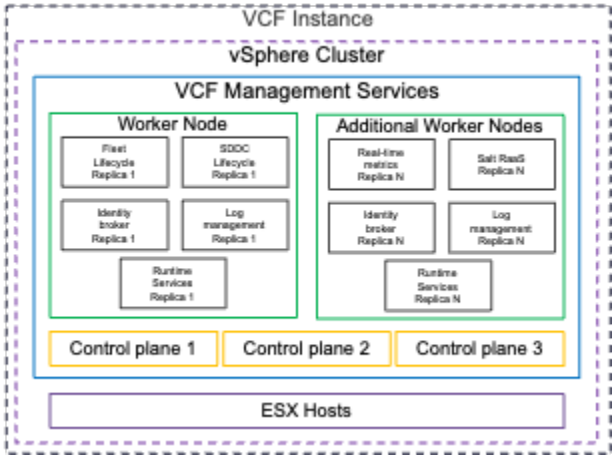
Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-ADTL-001	Create DNS entries for the following shared FQDNs: • Instance components FQDN	Required for deploying VCF management services in the additional VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-ADTL-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN	Required for deploying VCF management services in the additional VCF Instance.	You must create the DNS entries before starting the deployment.

High Availability VCF Management Services Model

The *High Availability VCF Management Services Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Figure 106: High Availability VCF Management Services Model in VMware Cloud Foundation



High Availability VCF Management Services Model Attributes

Attribute	Detail
Availability	Relies on vSphere HA
Scalability	Scale out/up worker nodes
Placement	Hosted in a single vSphere cluster

Common VCF Management Services Model Design Requirements

Table 261: Common VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-001	Allocate a sufficient IP range for the required number of control plane and worker nodes for your chosen deployment type.	Required for initial deployment.	Adding additional components may require auto-scale out of the VCF management services.
VCF-MS-REQD-COM-002	Use *`Kubernetes Cluster API Provider vSphere (CAPV)`* node anti-affinity to ensure the nodes run on different ESX hosts.	Using *`CAPV`* prevents the VCF management services nodes from running on the same ESX host and risking the high availability of the cluster.	Distributed Resource Scheduler (DRS) anti-affinity rules are not required.
VCF-MS-REQD-COM-003	Protect the VCF management services nodes by using vSphere HA.	Ensures availability of the VCF management services nodes without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Management Services Model Design Requirements

Table 262: High Availability VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-HA-001	Deploy VCF management services in high availability mode with three (3) control plane nodes and the corresponding number of worker nodes in the default management vSphere cluster.	• Supports the high availability deployment objective. • VCF management services supports scale-out and up from small to medium or large.	Requires additional resources.

Single License Server Model

The *Single License Server Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Single License Server Model	The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, which is designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.

Table 263: Single License Server Model Attributes

Attribute	Detail
Use Cases	This is the typical license server deployment model for VMware Cloud Foundation and is suitable for most VCF deployments.
Deployment mechanism	One license server per VCF fleet. A license server is automatically deployed as part of the installation or upgrade to VMware Cloud Foundation 9.1.
Backup and restore	Backup is not required. If the original license server becomes unavailable, deploy a new license server instance and migrate the licenses within 90 days**.**
Lifecycle	The upgrade and patch management of the license server is managed through VCF Operations.
High Availability	High availability is provided by vSphere HA.
License Scope	A single license server provides licensing management for the whole VCF fleet.

Single License Server Model Design Elements

Table 264: Single License Server Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LIC-REQD-LS-001	Register the default license server in VCF Business Services console.	Registration allows the VCF Business Services console to add license allocations to the license server.	None.

Licensing for VCF Operations Design Recommendations Connected Mode

Table 265: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Licensing for VCF Operations Design Recommendations Disconnected Mode

Table 266: Licensing for VCF Operations Design Recommendations

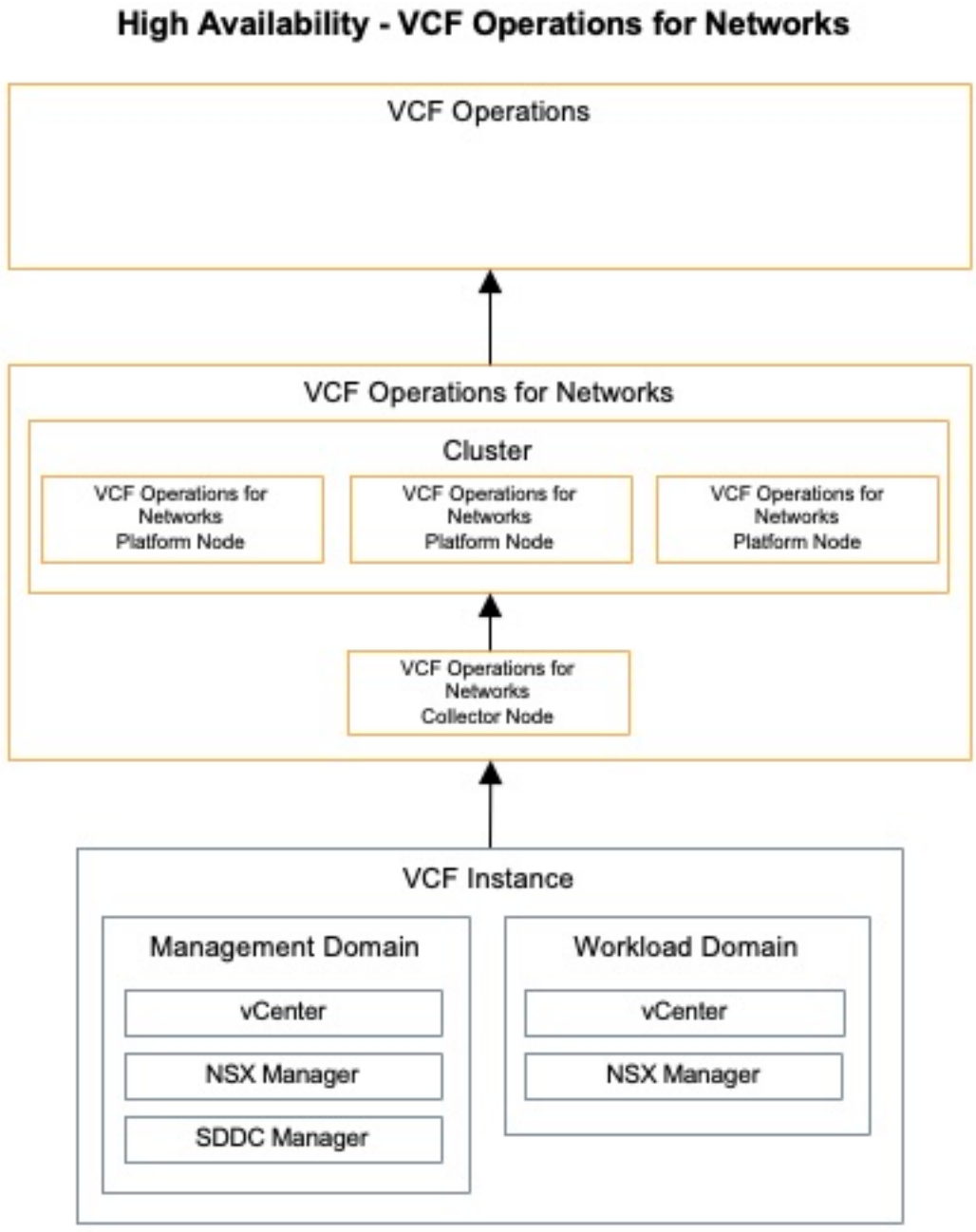
Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

High Availability VCF Operations for Networks Model

The *High Availability VCF Operations for Networks Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a four-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Figure 107: High Availability VCF Operations for Network Model in VMware Cloud Foundation



High Availability VCF Operations for Networks Model Attributes

A *High Availability VCF Operations for Networks Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out.
Recoverability	Restore through a backup.

High Availability VCF Operations for Networks Model Options

A *High Availability VCF Operations for Networks Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Large • Extra Large

High Availability VCF Operations for Networks Model Design Elements

Table 267: High Availability VCF Operations for Networks Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-HA-001	Deploy threeVCF Operations for networksplatform nodes in a cluster.	• Highly Performant. • Highly scalable, can scale up to 15 Extra Large nodes.	• Cluster does not provide high availability. It is for performance. • Requires higher CPU and memory reservations to support additional nodes.
VCF-NET-REQD-HA-002	Deploy a single VCF Operations for networks collector node in the default management vSphere cluster.	A minimum of one VCF Operations for networks collector node is required to provide the VCF Operations for networks platform node with network monitoring data.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks collector node will be applied.

Table 268: High Availability VCF Operations for Networks Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-001	Deploy appropriately sized VCF Operations for networksplatform nodes for your needs.	Ensures resource availability and usage efficiency.	None.
VCF-NET-RCMD-HA-002	Protect all VCF Operations for networks nodes by using vSphere High Availability.	Supports the availability objective for VCF Operations for networks without requiring manual intervention during an ESX host failure event.	You must ensure sufficient spare capacity for vSphere High Availability failover operations.

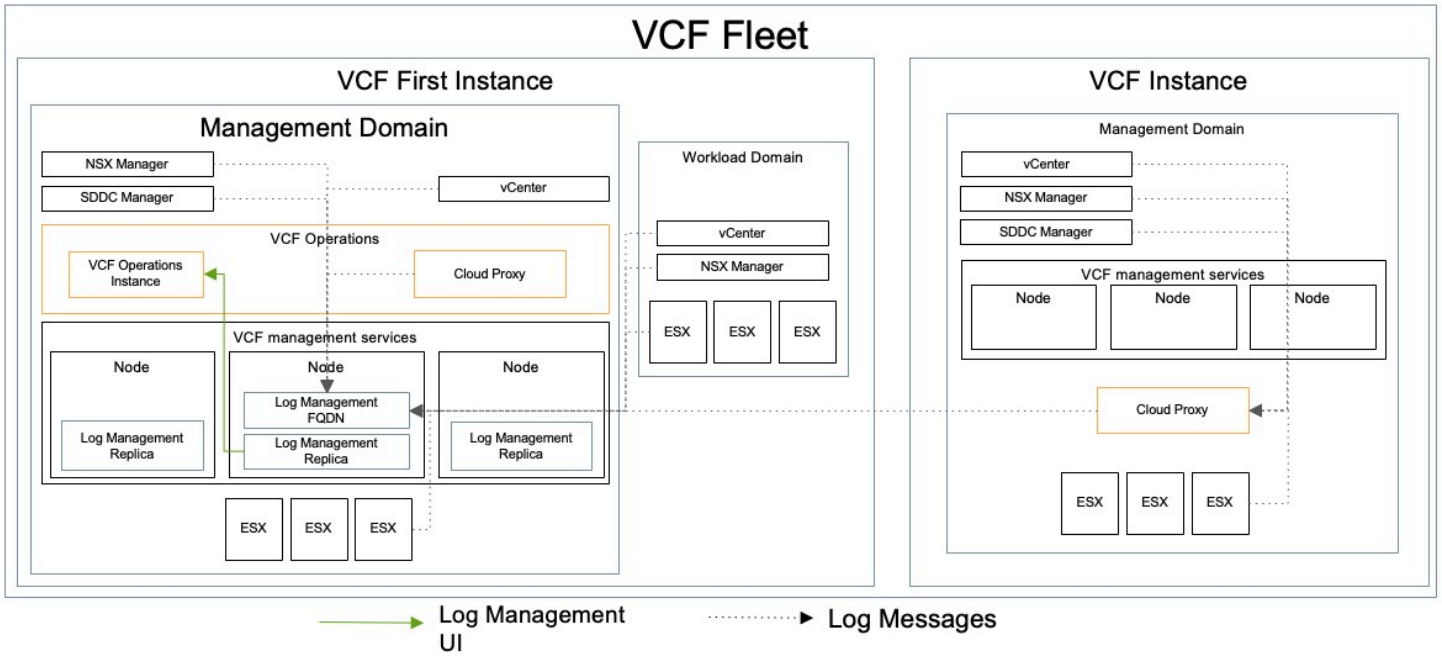
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-003	In an environment with multiple VCF Instances, deploy the VCF Operations for networkscollector node(s) in the default management vSphere cluster in each VCF Instance by using VCF Operations fleet management.	• Ensures a consistent deployment in each VCF Instances regarding network data collection. • It is assumed that each VCF Instances introduces a sufficiently high number of virtual machines, so that the deployment of dedicated VCF Operations for networks collector nodes is justified.	Each VCF Operations for networks collector node must be registered with a VCF Operations for networks platform node.
VCF-NET-RCMD-HA-004	For each vCenter data source, enable NetFlow on each vSphere Distributed Switch within the workload domain.	Provides the collection of network flows via the IPFIX protocol.	For the VCF domains, VCF Operations for networks will automatically update the NetFlow settings for each cluster's vSphere Distributed Switch.
VCF-NET-RCMD-HA-005	For each NSX data source, enable IPFIX for the distributed firewall.	Provides the collection of network flows via the IPFIX protocol.	The distributed firewall service must be enabled on the NSX Local Manager for the VCF workload domain.
VCF-NET-RCMD-HA-006	For each NSX Manager data source, enable latency metric collection.	Provides the collection of latency metrics from NSX Transport Nodes.	Any firewall rule sets from all ESX hosts to the VCF Operations for networks collector must allow traffic on TCP 1991.
VCF-NET-RCMD-HA-007	Use the default retention period of one month for the VCF Operations for networks platform node.	Keeping the default value for retention minimizes required storage resources administrative efforts.	None.
VCF-NET-RCMD-HA-008	Use a CA-Signed certificate containing the fully qualified domain names (FQDNs) of each VCF Operations for networks platform and collector node in the SAN attributes, when deploying VCF Operations for networks.	Configuring a CA-Signed certificate ensures that the communication to the externally facing web UI and API VCF Operations for networks for is encrypted.	• Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered. • Each time a node is added the certificate must be replaced to include the fully qualified domain name of the additional node.
VCF-NET-RCMD-HA-009	Use an SHA-2 or higher algorithm when signing certificates.	The SHA-1 algorithm is considered less secure and has been deprecated.	Not all certificate authorities support SHA-2.

High Availability Log Management Model

The *High Availability Log Management Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
High Availability Log Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind and internal load balancer. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.

Figure 108: High Availability Log Management Model in VMware Cloud Foundation



High Availability Log Management Model Attributes

A *High Availability Log Management Model* has the following attributes.

Attribute	Detail
Centralized management	Provides a centralized log management framework for vCenter, ESX, NSX, VCF Operations, VCF Automation, and identity broker.
Availability	Relies on cluster technology in the event of an outage or VCF management services and vSphere HA to restart nodes in the event of an ESX host failure.

Attribute	Detail
Scalability	Supports scale-up, storage scale up, and scale out.
Recoverability	Restore configuration through a backup, restore logs from archives.

High Availability Log Management Model Options

A *High Availability Log Management Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Small • Medium • Large

High Availability Log Management Model Design Requirements

Table 269: High Availability Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-HA-001	Deploy log management with at least 3 replicas.	• Provides high availability. • Using the integrated load balancer prevents a single point of failure. • Using the integrated load balancer simplifies the log management deployment and subsequent integration.	• You must deploy a minimum of three appropriately sized replicas. • All replicas are sized identically. • If the capacity of your log management cluster must expand, identical capacity is added to each node.
VCF-LOGS-REQD-HA-002	Protect all VCF Management service nodes by using vSphere High Availability.	Supports the availability objectives for log management without requiring manual intervention during an ESX host failure event.	None.

Common Log Management Design Recommendations for All Models

Table 270: Common Log Management Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

High Availability Log Management Model Design Recommendations

Table 271: High Availability Log Management Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-HA-001	Enable HA on the VCF management services control plane. (Instance deployed with the High Availability deployment model, or scale up VCF services runtime to medium or large.)	To support the high availability objective, the control plane for the VCF management services should be configured for HA, as log management depends upon VCF management services.	Requires additional nodes and capacity in the vSphere management cluster.

Security Governance and Compliance Layer for VCF Fleet with Multiple Sites Across Multiple Regions

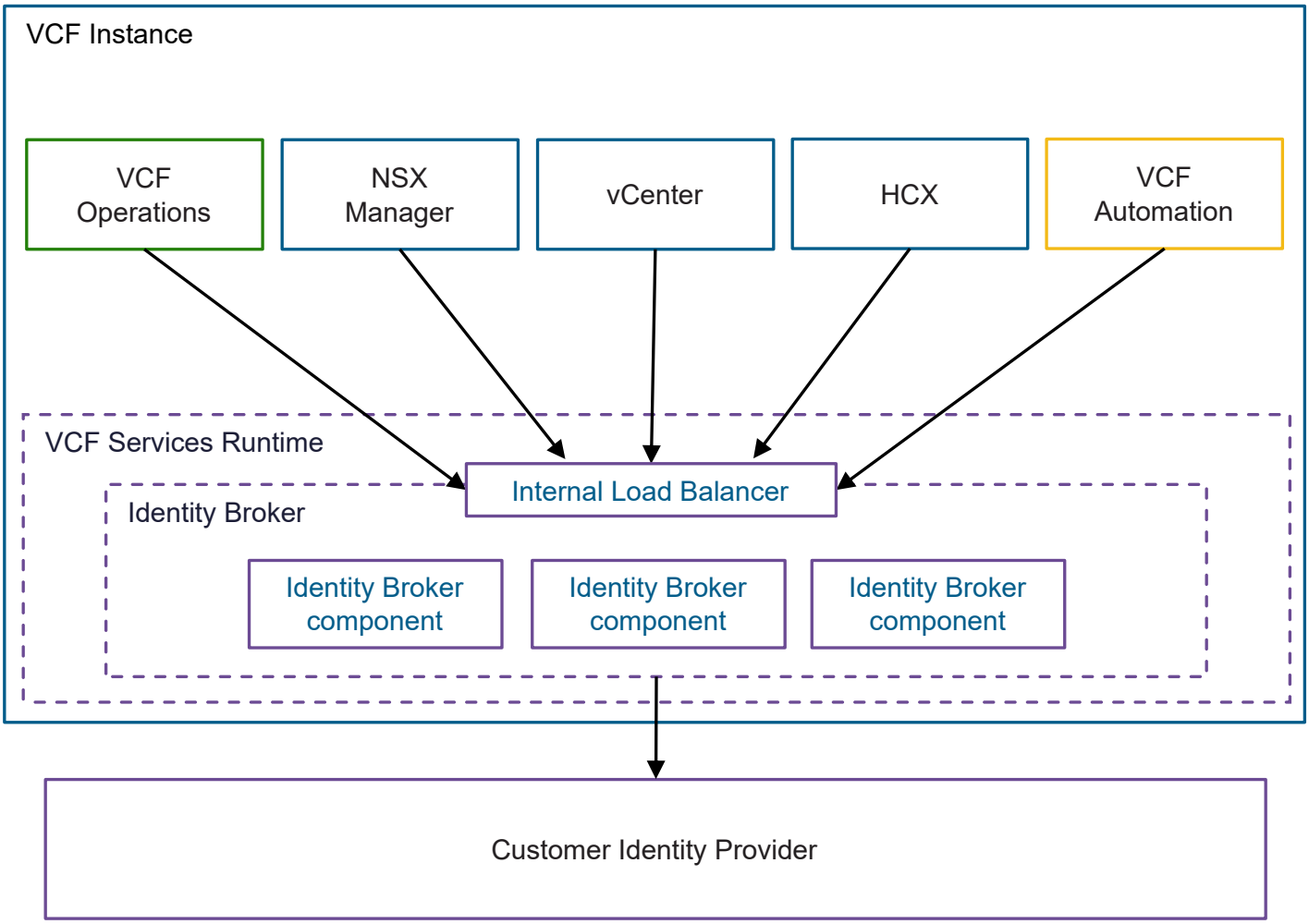
The security governance and compliance layer refers to the framework of policies, procedures, and controls that organizations establish to manage security risks while meeting regulatory, legal, and industry standards around authentication.

Identity Broker Model

The *Identity Broker Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Identity Broker Model	Deploys the Identity Broker component on the VCF Management Services instance in the management domain.

Figure 109: Instance Identity Broker Model in VMware Cloud Foundation



Instance Identity Broker Model Attributes

The *Instance Identity Broker Model* has the following attributes.

Attribute	Detail
Service Location	Identity broker runs as a service on VCF Management Services.
Deployment mechanism	An identity broker is automatically deployed during the installation or upgrade to VMware Cloud Foundation 9.1 in the first management domain. Additional identity brokers can be deployed in each VCF Instance as part of a day 2 operation.
Number of Nodes	multi-node cluster.
Scale considerations	• Scale up: Identity broker supports small, medium, and large sizes to match different configuration demands. • Scale out: Support for multiple identity broker instances to increase redundancy and throughput. • Recommendations: – Scaling up is preferred over scaling out when a single large-size identity broker meets security, resilience, and performance standards, as it reduces operational complexity and simplifies SSO management. – Schedule scale-up operations within a maintenance window, as these activities can impact active services.
Supported services	For support directories and IdPs, see Identity Providers and Protocols Supported for Single Sign-On.
Backup and restore	Use VCF management services backup and restore process.
Resiliency	Resiliency is provided by a distributed multiple-node deployment running each node on separate vSphere hosts.
Operational Impacts	As an isolated, external service, operational tasks performed on vCenter do not impact the authentication stack of the VMware Cloud Foundation components.
Migration path	Migrating from the appliance to embedded requires a reset of the VCF Single Sign-On configuration. Users must be re-added to components.

Instance Identity Broker Model Design Requirements

Table 272: Instance Identity Broker Model Design Requirements

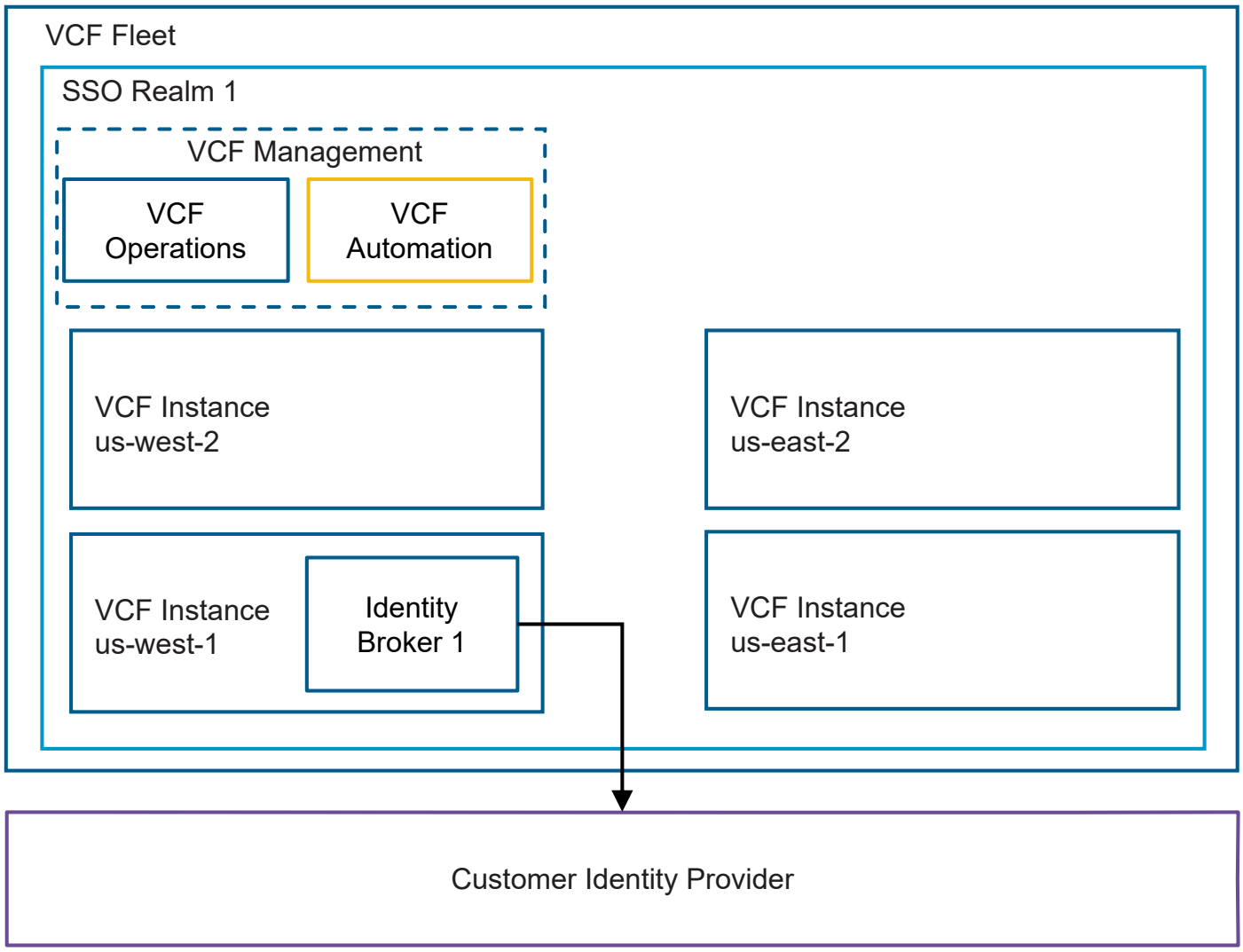
Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-EXT-001	Deploy additional identity broker instances as a Day N operation as needed.	• Provides high availability to the authentication source for a VCF Instance. • Provides additional scale.	Increases the operational overhead of managing the identity broker.
VCF-SSO-REQD-EXT-002	Connect the identity broker service to your corporate IdP.	Provides directory services (users and group) and user authentication service to the identity broker service.	None.

VCF Fleet-Wide Single Sign-On Model

The *VCF Fleet-Wide Single Sign-On Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
VCF Fleet-Wide Single Sign-On Model	Single Identity Broker deployment. A user can log in once and can gain VCF Single Sign-On capabilities for all components across all VCF instances in the same VCF fleet.

Figure 110: VCF Fleet-Wide Single Sign-On Model in VMware Cloud Foundation



VCF Fleet-Wide Single Sign-On Model Attributes

The *VCF Fleet-Wide Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**Low Scale -** A single identity broker services all VCF Instances in your VCF fleet.
Resources requirements	**Low Resource Footprint** - A single identity broker deployment is required for one VCF fleet.
Operational Overhead	**Low Operational Overhead** - An identity broker per VCF fleet will need to be managed separately for operational tasks such as backup and restore.
Availability	**Low Availability per Private Cloud** - A single identity broker servicing all VCF Instances in your VCF fleet.
Blast Radius	**Large Blast Radius** - An outage of one identity broker instance will impact logins for a single VCF fleet.
Single Sign-On Realm	**Large SSO Realm** - Users do not need to re-authenticate when moving between different VCF Instances within the same VCF fleet.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a only): Because the identity broker is hosted on the primary VCF instance, VCF instance restoration is required; Option 2b is not applicable. • Failure Scenario 3

Table 273: VCF Fleet-Wide Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VFL-001	Configure additional VCF Instances in the VCF fleet to consume the same identity broker configured in the first VCF Instance.	• Provides an authentication source for all components in the VCF fleet. • Provides VCF Single Sign-On capabilities to all components in the VCF fleet.	An outage to the identity broker service impacts the VCF fleet.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites Across Multiple Regions

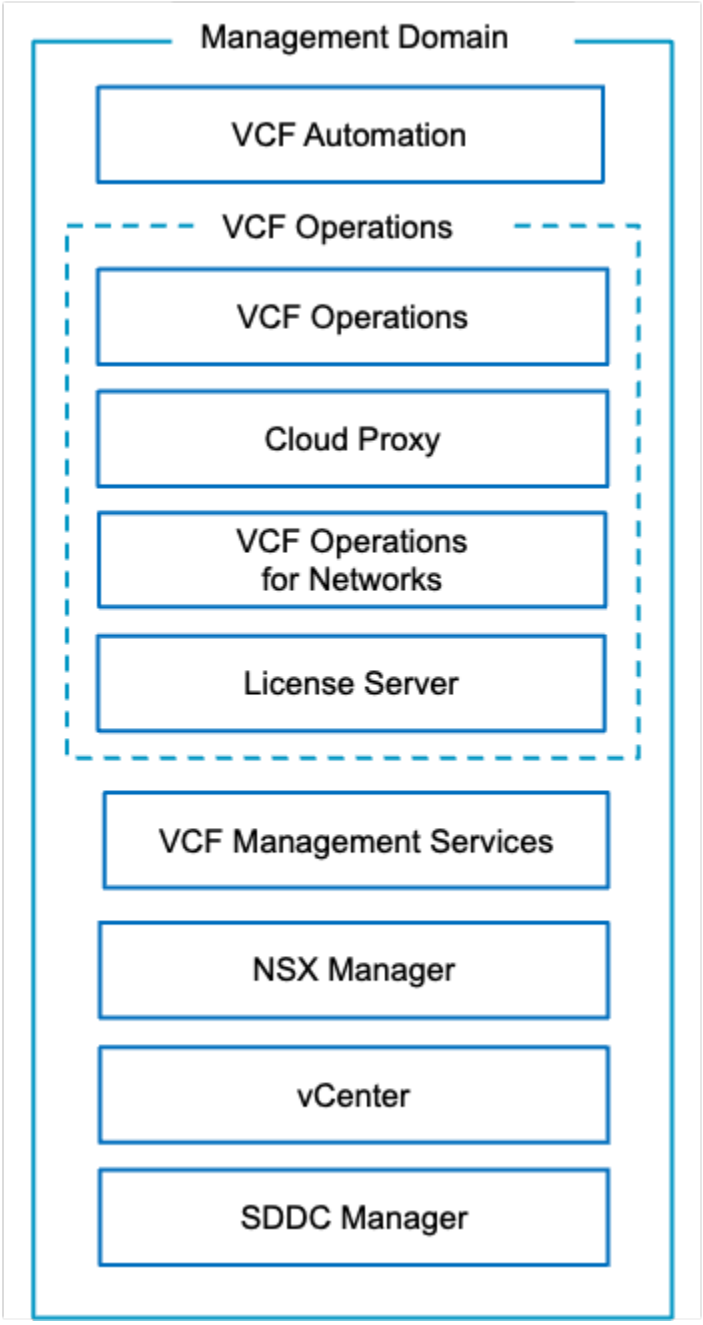
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Management Domain Model

The *Management Domain Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.

Figure 111: Management Domain Model in VMware Cloud Foundation



Management Domain Model Attributes

The *Management Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• VCF fleet management components. • VCF Instance management components. • NSX Edge nodes. • Can also run business workloads.
Management Domain in the First VCF Instance	• The management domain in the first VCF Instance contains the following management components: – VCF management services (fleet and instance level components) – VCF Operations – Cloud Proxy – VCF Operations for Networks – VCF Automation – ESX hosts – vCenter – NSX Manager – SDDC Manager
Management Domains in Additional VCF Instances	• Management domains in additional VCF Instances contain the following instance-level management components: – VCF management services (instance level components) – Cloud Proxy – ESX hosts – vCenter – NSX Manager – SDDC Manager

Management Domain Model Design Requirements

Table 274: Management Domain Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLD-REQD-CFG-001	Deploy a management domain to host all management components for the VCF Instance.	• Guarantees sufficient resources for management components. • Enables the use of specific hardware to meet only the needs of the management components. • Enables the use of dedicated physical compute, network, and storage separately from those used for additional workloads. • Enables separate lifecycle management of management and workload components.	• You must carefully size the management domain to accommodate planned deployments of additional workload domains and management components. • Hardware might not be fully utilized until full-scale deployment has been reached. • The management components for all greenfield VCF domains are deployed to the same cluster, and share the same subnet.

Management Domain Model Design Recommendations

Table 275: Management Domain Model Design Recommendations

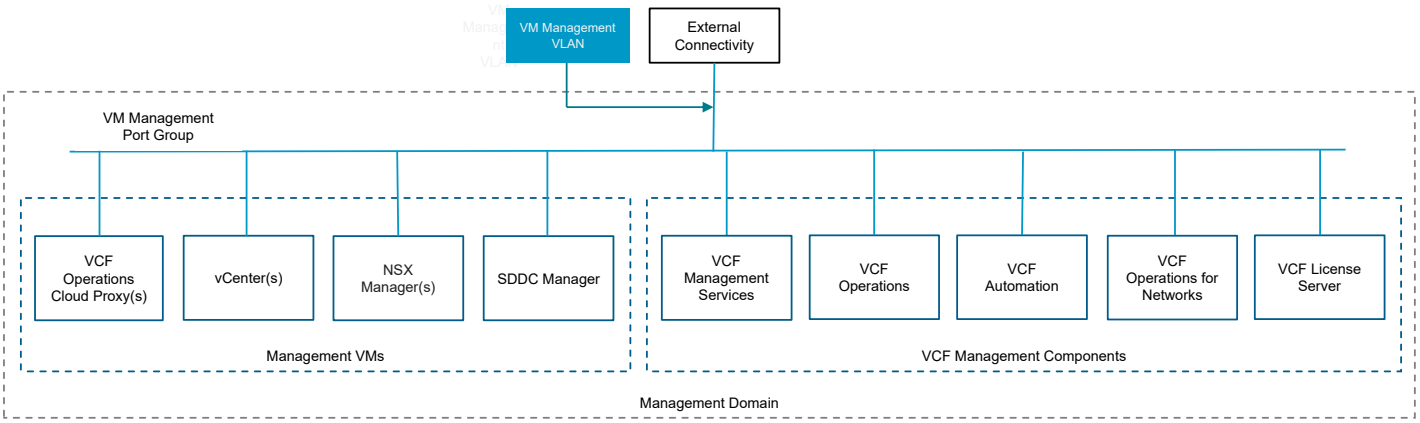
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-RCMD-NSX-001	Use a dedicated NSX Manager instance for the management domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Managers to manage and patch.

VCF Management Shared VLAN Network Model

The *VCF Management Shared VLAN Network Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Figure 112: VCF Management Shared VLAN Network Model in VMware Cloud Foundation



VCF Management Shared VLAN Network Model Attributes

A *VCF Management Single VLAN Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	VLAN-backed vSphere distributed port group.
Network Isolation	No; Sharing VLAN with the VCF management domain components.
NSX Edge Cluster Required	No; Optional to provide NSX load balancer for VCF Operations. (requires additional NSX VLAN segment for Load balancer connectivity)
Stretching Network Between Availability Zones	• **VM management network** must be stretched in the physical network fabric between availability zones. • Provides automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• Not suited for a disaster recovery use case as the network is shared with management domain components. • Failover testing is not supported.

VCF Management Shared VLAN Network Model Design Requirements and Recommendations

Table 276: VCF Management Shared VLAN Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-SV-NET-REQD-001	Use the management domain vCenter VM management distributed port group for the VCF fleet level components.	No additional network required.	• No logical network isolation. • Done automatically during VCF Installer deployment.
VCF-MGMT-SV-NET-REQD-002	Use the same VLAN for VCF management components and the VCF Instance management virtual machines. e.g. vCenter and NSX Managers.	No additional VLAN required.	• Separate physical firewall rules may not be possible with a single VLAN. NSX Distributed Firewall may be an alternative solution. • Not suitable for Fleet Disaster Recovery use case with single VLAN.

Table 277: VCF Management Shared VLAN Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-SV-NET-RCMD-001	Deploy the VCF management components during initial deployment using VCF Installer.	Simplified deployment using VCF Installer UI.	None.

Workload Domain Design Selections

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Consumption Layer for VCF Fleet with Multiple Sites Across Multiple Regions
The consumption layer for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

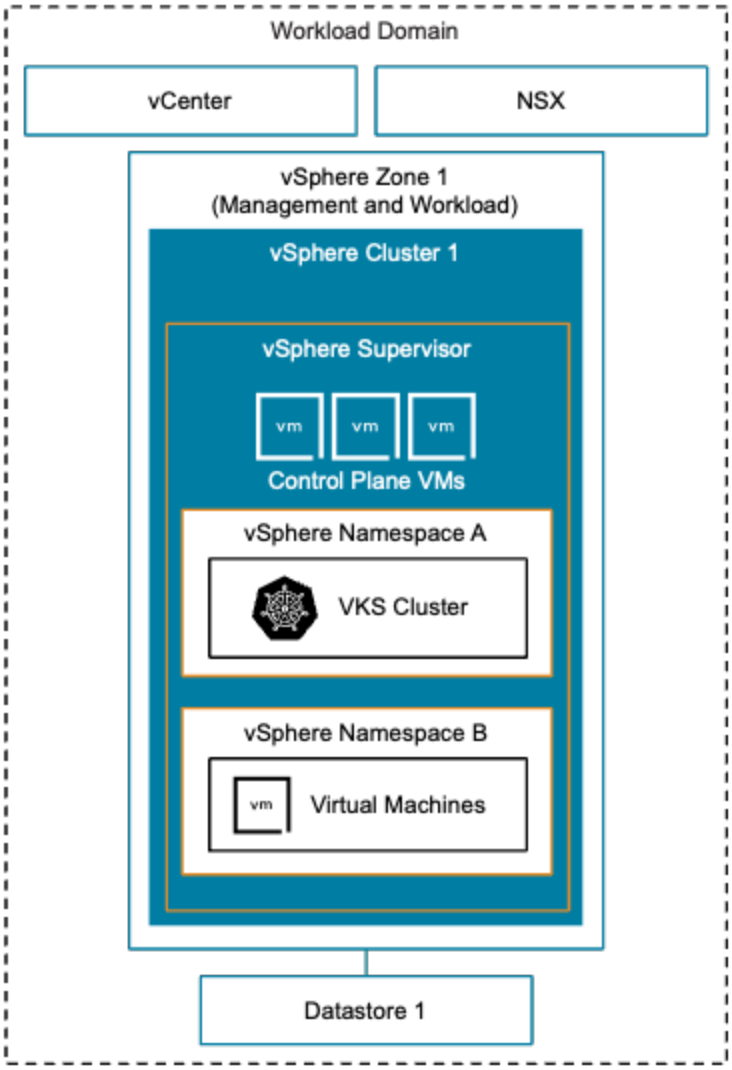
Single Management Zone with Combined Workload Zones Model

The *Single Management Zone with Combined Workload Zones Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.

Figure 113: Single Management Zone with Combined Workload Zones Model Diagram

This diagram illustrates an example of a *Single Management Zone with Combined Workload Zones Model* deployment.



Single Management Zone with Combined Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single - Combined with management
vSphere Zone Datastore Type	Zonal datastore

Attribute	Detail
Activation	• At workload domain creation. OR • From vCenter.

vSphere Supervisor Design Requirements for all Zone Models

Table 278: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

vSphere Supervisor Single Management Zone Design Requirements

Table 279: vSphere Supervisor Single Management Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-SMZ-REQD-CFG-001	One available vSphere Zone that is backed by a supported vSphere cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	vSphere Zone for Supervisor management workload use.	The selected vSphere Zone is dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.

vSphere Supervisor Design Requirements

Table 280: vSphere Supervisor Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-REQD-CFG-001	Topology aware storage policies created for control plane nodes when deploying in a Multi Zone Deployment model.	Storage policies should have placement considerations that are aligned with the availability of control plane nodes.	Additional storage policies are required.
VCF-SUP-REQD-CFG-002	A single vSAN stretched cluster that is not already assigned to another vSphere Supervisor instance must be available at initial deployment in a vSAN stretched cluster deployment model.	When deploying supervisor on vSAN stretched cluster, the vSANcluster must be stretched before supervisor activation.	vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Common Networking Design Recommendations

Table 281: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Design Recommendations

Table 282: vSphere Supervisor Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

High Availability vSphere Supervisor Control Plane Model

The *High Availability vSphere Supervisor Control Plane Model* was chosen for the V*CF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected vSphere Supervisor management zones.

Figure 114: *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model deployment.

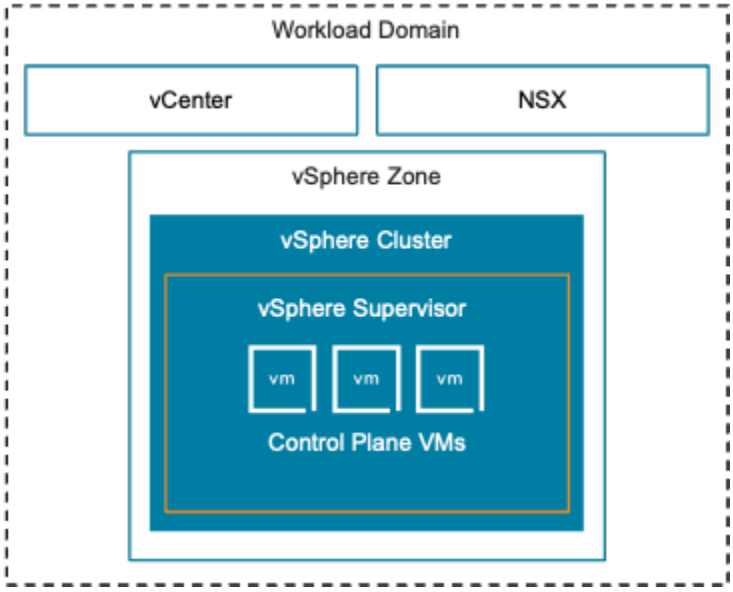
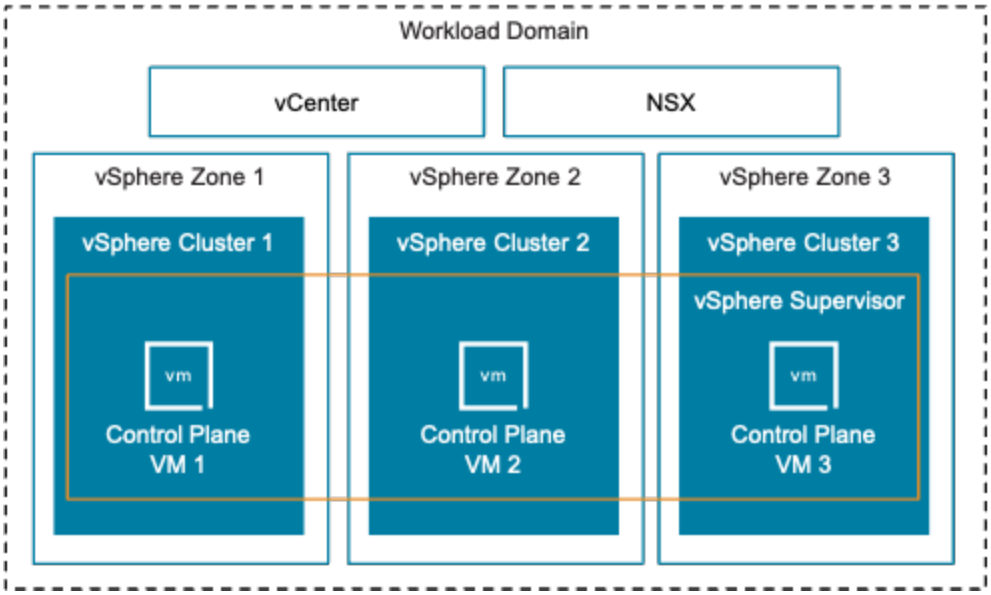


Figure 115: *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model deployment.



High Availability vSphere Supervisor Control Plane Model Attributes

Attribute	Detail
Control Planes Nodes	Three
Availability	• Relies on vSphere HA. • Three management zones, which are protected from single cluster failure.
Scalability	Scale up only
Placement	• Single management zone, all nodes in a single management zone. • Three management zones, one node per management zone.
Applicable Zone Models	• Single Management Zone with Combined Workload Zones Model • Single Management Zone with Isolated Workload Zones Model • Three Management Zones with Combined Workload Zones Model • Three Management Zones with Isolated Workload Zones Model

vSphere Supervisor Control Plane Model Design Requirements

Table 283: vSphere Supervisor Control Plane Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-CP-REQD-CFG-001	Five consecutive management IP addresses routable to vCenter NSX, and optionally Avi management networks if Avi Load Balancer are used.	Required for integration and management communications.	Requires precise IP address management.
VCF-SUP-CP-REQD-CFG-002	Three vSphere Zones are required for the High-Availability Model in a Three Management Zone Model.	Control pane nodes are evenly placed with one per zone to protect against a single vSphere cluster failure.	Additional vSphere zones each back by separate supported vSphere clusters are required.

vSphere Supervisor Control Plane Design Recommendations

Table 284: vSphere Supervisor Control Plane Design Recommendations

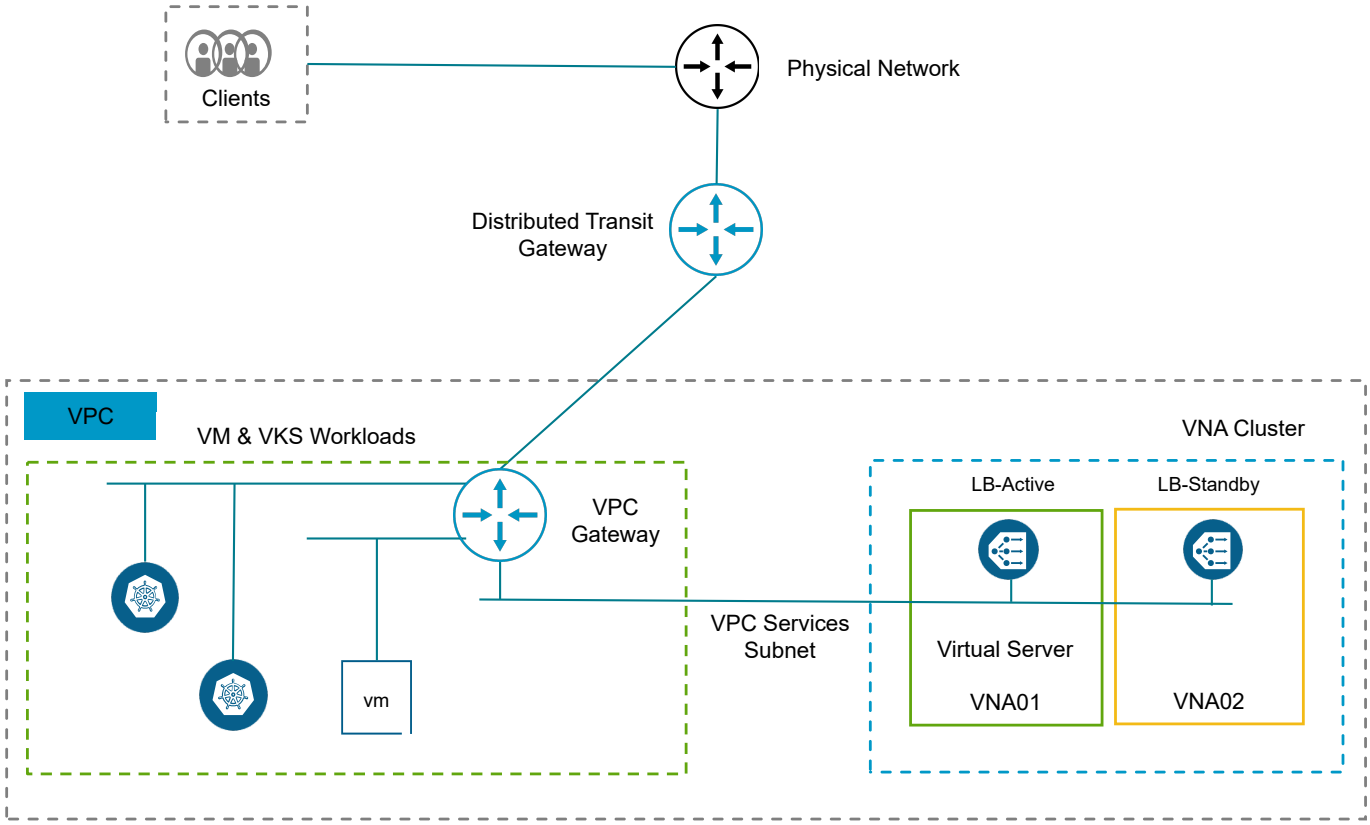
Design Requirement ID	Design Recommendation	Justification	Implication
VCF-SUP-CP-RCMD-CFG-001	Deploy the supervisor control plane in high availability mode with three control plane VMs. *Applicable to Non-Simplified Supervisor models only.	Three control plane VMs allow for continuous availability of the Supervisor control plane during rolling upgrades and single node outages.	Additional resources are required to support the extra control plane VMs.

Virtual Network Appliance Load Balancer Model

The *Virtual Network Appliance Load Balancer Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for VCF vSphere Supervisor and VKS workloads, it supports VPC-based networking.

Figure 116: Virtual Network Appliance Load Balancer Model in VMware Cloud Foundation



Virtual Network Appliance Load Balancer Model Attributes

A *Virtual Network Appliance Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Load Balancing for VCF infrastructure components (i.e. VCF appliances) • Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services)
Networking Models Supported	VPC Networking Required
Entitlement	Included with VMware Cloud Foundation
Availability	Active/Standby
Scalability	• Scale up Virtual Network Appliance nodes in the Virtual Network Appliance cluster • Scale out a single Virtual Network Appliance Cluster • Create additional Virtual Network Appliance clusters
Deployment Model	Virtual Network Appliance Cluster with two or more Virtual Network Appliances. See Virtual Network Appliance Cluster Detailed Design for further details.

Virtual Network Appliance Load Balancer Model Options

Design Area	Options
Number of Virtual Network Appliances per cluster	1 to 10
Deployment Options	• Deploy with a new workload domain • Deploy Day 2 from NSX Manager • Deploy Day 2 from vCenter
Virtual Network Appliance Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger scale depoloyments where a large number of load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

Virtual Network Appliance Load Balancer Design Requirements

Table 285: Virtual Network Appliance Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-LB-REQD-CFG-01	Create a Virtual Network Appliance cluster with two or more Virtual Network Appliance nodes.	Provides availability for a load balancer instance.	None.
VCF-VNA-LB-REQD-CFG-02	Configure the VPC connectivity profile to use Virtual Network Appliance cluster in the VPC Service Gateway Configurations.	Enables the VPC to use the Virtual Network Appliance cluster for load balancer instances.	Virtual Network Appliance cluster can be shared between VPCs.
VCF-VNA-LB-REQD-CFG-03	Create a load balancer instance for the VPC using the API	Supervisor and VKS-based workloads, load balancer creation is automated.	No UI available to configure load balancer instances.
VCF-VNA-LB-REQD-CFG-04	Create a load balancer configuration for VPC using the API.	Supervisor and VKS-based workloads, load balancer configuration is automated.	No UI available to configure load balancer instances.

Virtual Network Appliance Load Balancer Design Recommendations

Table 286: Virtual Network Appliance Load Balancer Design Recommendations

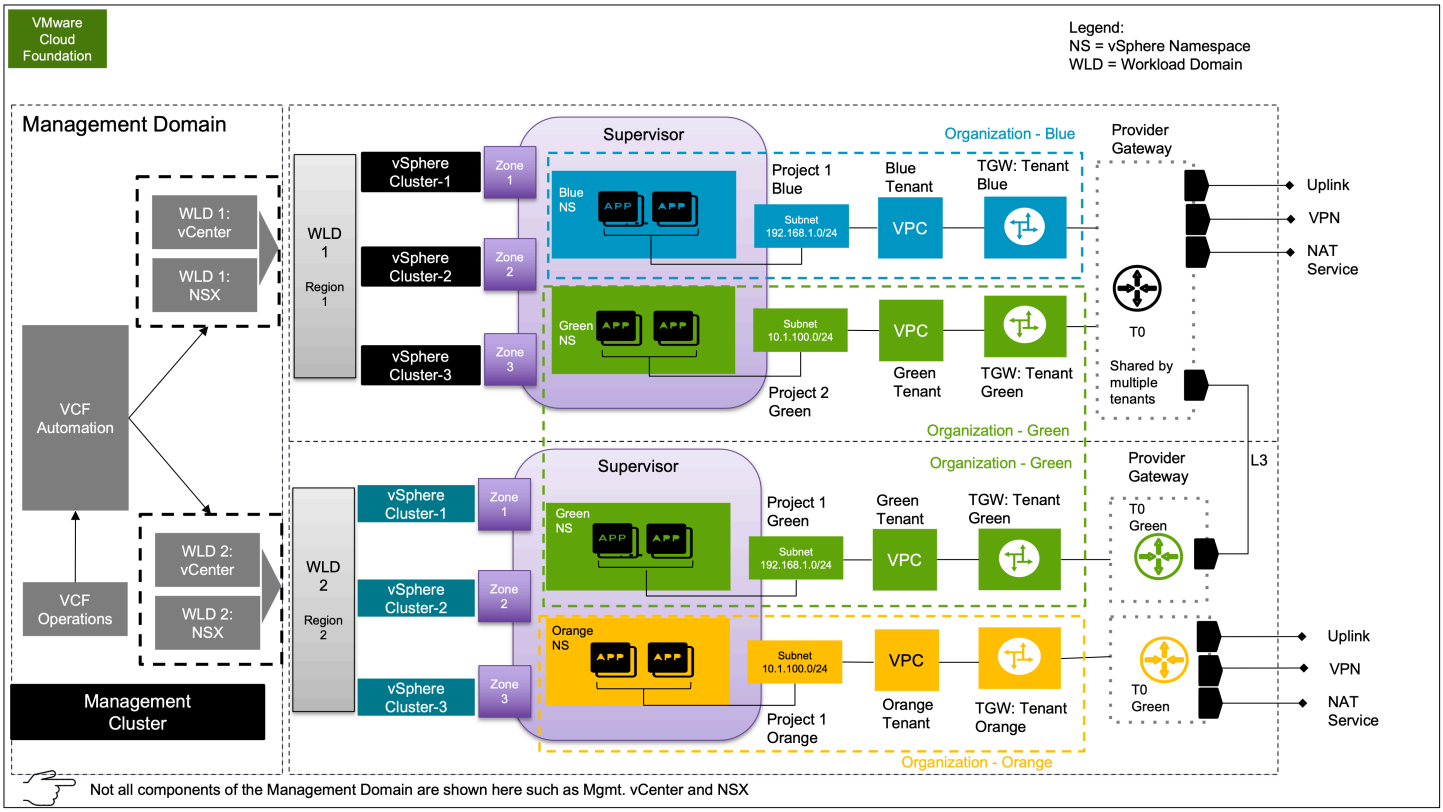
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-LB-RCMD-CFG-01	Configure failures domain for Virtual Network Appliance clusters with more than two Virtual Network Appliance nodes in the cluster.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-VNA-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from vCenter UI.

Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways
The *Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways	This design provides multi-tenant resource isolation across multiple regions by combining dedicated gateways for strict tenant separation with shared gateways for efficient common services.

Multi-Tenancy Consumption Pattern 5 Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways - Centralized Gateway (CTGW) Design

Figure 117: Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways - Centralized Gateway (CTGW) Design



Multi-Tenancy Consumption Pattern 5 Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways - Centralized Gateway (CTGW) Design Attributes

Design Attributes	Details	Diagram Representation
Management Domain	One	Centralized Management: Includes VCF Automation, workload domain vCenter, NSX, and VCF Operations. Consist of a vSphere Cluster -the underlying hardware for the management domain.
Workload Domains	Two	A workload domain with three compute vSphere clusters for tenant workloads.
NSX Local Managers	Two	Each workload domain has one NSX local manager
Organizations	Three	Represents three separate Organizations. Such as Blue, Green and Orange
Number of vSphere Clusters	Six	Six compute and vSAN storage cluster used in two workload domain construct.
Number of vSphere Supervisors	Six	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	Six	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Projects	Three	Represented by "Project 1 - User" and "Project 2- User". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One in each region	A single T0 gateway provides external connectivity, including Uplink, VPN, VRF (Virtual Routing and Forwarding), and NAT services.
VRFs	Two	Tenant "Green" and "Orange" has a dedicated VRF instance in a T0. Both tenant sharing a T0 but still they have network isolation.
vSphere Namespaces	Two	Blue and Green tenant has dedicated vSphere Namespace for workload isolation.
VPCs	One	Each LOB/ tenant sharing a single VPC
TGWs (Transit Gateway)	Two	Each tenant has a dedicated TGW, "TGW: Tenant Blue" and "TGW: Tenant Green", for connecting their respective VPCs.
Tenant Networks	Four	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites Across Multiple Regions

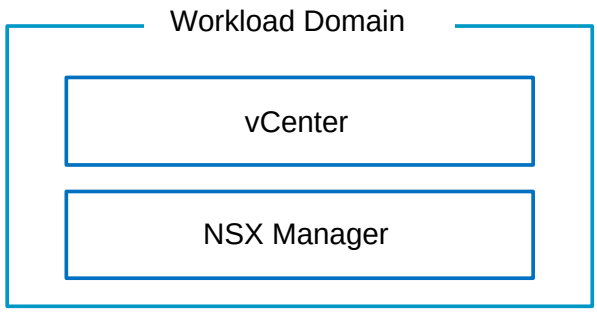
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Workload Domain Model

The *Workload Domain Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.

Figure 118: Workload Domain Model in VMware Cloud Foundation



Workload Domain Model Attributes

A *Workload Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• NSX Edge nodes. • Business workloads.
vCenter Single Sign-On Domain	Dedicated.

Workload Domain Model Design Recommendations

Table 287: Workload Domain Model Design Recommendations

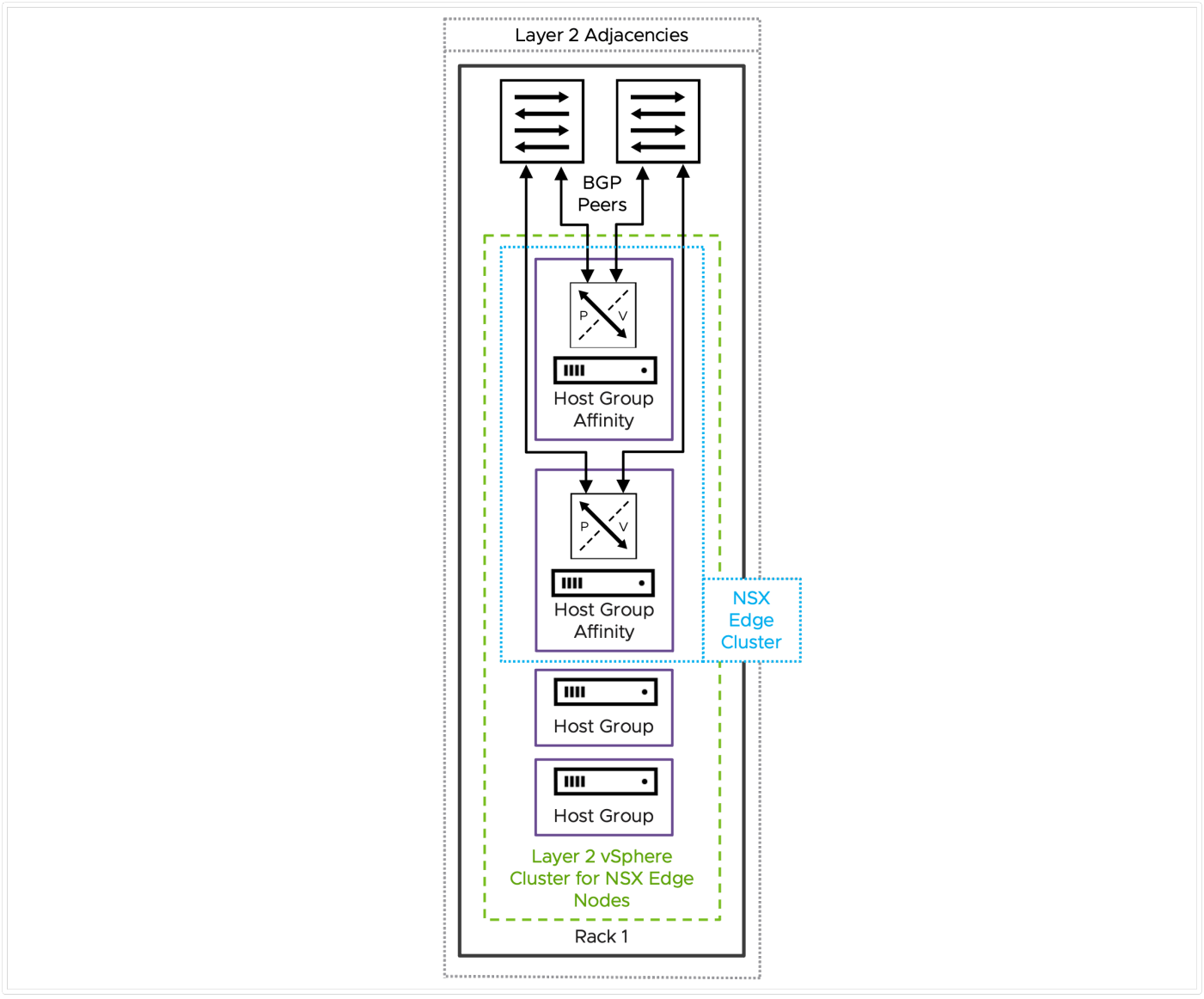
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLD-RCMD-CFG-002	Use workload domains for customer workloads.	• Aligns with the VMware best practice of separating management workloads from customer workloads. • Provides better long-term flexibility and expansion options.	Requires additional hardware.
VCF-WLD-RCMD-NSX-002	Use a dedicated or shared NSX Manager instance for the workload domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Manager to manage and patch.

Host Fault Tolerant NSX Edge Cluster Model

The *Host Fault Tolerant NSX Edge Cluster Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Host Fault Tolerant NSX Edge Cluster Model	A *Single-Rack vSphere Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Figure 119: Host Fault Tolerant NSX Edge Cluster Model in VMware Cloud Foundation



Host Fault Tolerant NSX Edge Cluster Model Attributes

The *Host Fault Tolerant NSX Edge Cluster Model* has the following attributes.

Attribute	Details
Availability Model	• Host level redundancy only. • Single shared storage. • This design cannot ensure continuous service availability for north-south communication and centralized network services during a complete rack failure.
NSX Edge Node Mobility	Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same rack that hosts the NSX Edge node.
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Add hosts to the single existing vSphere cluster or preferably transition to a rack fault tolerant design with two or more vertical Layer 2 clusters in separate racks. • When deploying more NSX Edge nodes on existing hosts, ensure each host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Common NSX Edge Cluster for Single Availability Zone Design Requirements

Table 288: Common NSX Edge Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMSAZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric Settings.

Host Fault Tolerant NSX Edge Cluster Model Design Requirements

Table 289: Host Fault Tolerant NSX Edge Cluster Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-HFTSR-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Table 290: Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMSAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMSAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMSAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs. • When deploying two NSX Edge nodes per host, each NSX Edge node requires two dedicated physical NICs configured for optimal performance.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMSAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface already exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMSAZ-008	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow.
VCF-NSX-EDGE-RCMD-COMSAZ-009	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-NSX-EDGE-RCMD-COMSAZ-010	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESXhost.	If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-011	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-NSX-EDGE-RCMD-COMSAZ-012	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually.
VCF-NSX-EDGE-RCMD-COMSAZ-013	Create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-014	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload virtual machines and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.
VCF-NSX-EDGE-RCMD-COMSAZ-015	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMSAZ-016	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-017	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSX on dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMSAZ-018	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two data plane fastpath interfaces gets an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP)	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMSAZ-019	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (Physical link state up but not forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-020	Limit the span of NSX Edge node networks to the local racks.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two (2) NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP session.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.
VCF-NSX-EDGE-RCMD-COMSAZ-021	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard for the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.
VCF-NSX-EDGE-RCMD-COMSAZ-022	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Host Fault Tolerant NSX Edge Cluster Model Design Recommendations

Table 291: Host Fault Tolerant NSX Edge Cluster Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-HFTSR-001	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across different hosts.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-NSX-EDGE-RCMD-HFTSR-002	Configure an NSX failure domain for each host where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability if there was a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Shared Design Selection between Management and Workload Domain

Shared Design Selections between Management and Workload Domain are the models that are the same across the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites Across Multiple Regions
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

vCenter Model
The *vCenter Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

vCenter Model Attributes

A *vCenter Model* has the following attributes.

Design Area	Attributes
vCenter Deployment Type	• One vCenter instance for the management domain that manages the management components of the VCF Instance. • Optionally, additional vCenter instances for workload domains to support workloads. • vSphere HA protecting all vCenter appliances.
vCenter Single Sign-On	• Each vCenter has a distinct vCenter Single Sign-On domain.

Methods for Protecting vCenter

VMware Cloud Foundation supports multi-rack management vSphere clusters, vSAN stretched clusters, or 3rd party synchronous replication solutions such as metro clusters that are designed to provide disaster recovery and/or high availability for the entire management stack (vCenter, NSX Manager, etc.) as opposed to a single vCenter.

Table 292: Methods for Protecting vCenter in VMware Cloud Foundation

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere High Availability	Yes	You must ensure sufficient resources in the cluster to allow virtual machines to restart on another ESX host in the event of an ESX host failure.
vCenter High Availability (vCenter HA)	No	• vCenter services must fail over to the passive node so there is no continuous availability. • Recovery time can be up to 15 mins. • You must meet additional networking requirements for the private network. • vCenter HA requires additional resources for the passive and witness nodes. • Lifecycle management is complicated because you must manually delete and recreate the standby virtual machines during a lifecycle management operation.
vSphere Fault Tolerance (vSphere FT)	No	• The vCPU limit of vSphere FT vCPU would limit vCenter appliance size to medium. • You must provide a dedicated network.

vCenter Model Design Requirements and Recommendations

Table 293: vCenter Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-CFG-001	Deploy a dedicated vCenter appliance for the management domain of the VCF Instance.	• Isolates vCenter failures to management or customer workloads. • Isolates vCenter operations between management and business workloads. • Supports a scalable vSphere cluster design where you can reuse the management components as more customer workloads are added to the VCF Instance. • Simplifies capacity planning for customer workloads because you do not consider management workloads for the workload domain vCenter. • Improves the ability to upgrade the vSphere environment and related components by enabling for explicit separation of maintenance windows: — Management workloads remain available while you are upgrading the tenant workloads. — Workloads remain available while you are upgrading the management nodes. • Supports clear separation of roles and responsibilities to ensure that only administrators with granted authorization can control the management workloads. • Facilitates quicker troubleshooting and problem resolution. • Simplifies disaster recovery operations by supporting a clear separation between recovery of the management components and tenant workloads. • Provides isolation of potential network issues by introducing network separation of the vSphere clusters in the VCF Instance.	The vSphere cluster must have sufficient resources.
VCF-VCS-REQD-NET-001	Place all workload domain vCenter appliances on the VM management network in the management domain.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-SSO-001	Create all vCenter instances within a VCF Instance in their own unique vCenter Single Sign-On domain.	• Sharing vCenter Single Sign-On domains is not supported. • Enables isolation at the vCenter Single Sign-On domain layer for increased security separation.	• Each vCenter instance is managed through its own pane of glass using a different set of administrative credentials. • You must manage password rotation for each vCenter Single Sign-On domain separately.

Table 294: vCenter Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VCS-RCMD-CFG-001	Deploy an appropriately sized vCenter appliance for each workload domain.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate compute resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-002	Deploy a vCenter appliance with the appropriate storage size.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate storage resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-003	Protect workload domain vCenter appliances by using vSphere HA.	vSphere HA is the only supported method to protect vCenter availability in VCF.	vCenter becomes unavailable during a vSphere HA failover.
VCF-VCS-RCMD-CFG-004	In vSphere HA, set the restart priority policy for the vCenter appliance to high.	vCenter is the management and control plane for physical and virtual infrastructure. In a vSphere HA event, to ensure that the rest of the SDDC management stack comes up faultlessly, the workload domain vCenter must be available first, before the other management components come online.	If the restart priority for another virtual machine is set to highest, the connectivity delays for the management components are longer.

ESX Model

The *ESX Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCF domain vSphere cluster.

ESX Hardware and Sizing Considerations

For more information on host sizing, refer to the VCF Planning and Preparation Workbook.

Table 295: ESX Hardware and Sizing Considerations

Hardware Element	Considerations
BIOS/Firmware	Ensure the ESX host hardware is on the Hardware Compatability List (HCL) and that you are running the latest supported BIOS and firmware.
CPU	• Total CPU requirements for the workloads that are running in the cluster. • Host failure and maintenance scenarios. • CPU over-commitment ratio guidelines – Management Domain • To ensure optimal performance and maintain resilience during host failures or maintenance, keep the management domain vCPU-to-pCPU over-commitment ratio at or below 2:1. **Note:** While higher ratios such as 3:1 or 4:1 are supported, you must closely monitor CPU ready times to ensure that the management cluster has enough capacity to run the workloads. – Workload domains • CPU over-commitment ratios for workload domains should be calculated on a case by case basis depending on the CPU intensity of the workloads. You must closely monitor CPU ready times to ensure optimal performance. • Number of physical cores, not logical cores. – Simultaneous multi-threading (SMT) in CPUs improve CPU performance however the performance enhancement will not be equivalent to 100% more CPU power.
Memory	• Total memory requirements for the workloads that are running in the cluster. • When sizing memory for the ESX hosts in a cluster, to reserve the resources of one host for failover or maintenance, set the admission control setting to N+1, which reserves the resources of one host for failover or maintenance. • *vSAN* OSA (Original Storage Architecture). – To support the maximum number of disk groups, you must provide 32 GB of RAM. • *vSAN* ESA (Express Storage Architecture). – You must provide a minimum of 128GB of RAM.
Storage	• Use high-endurance devices such as a hard drive or SSD for boot device. • Use a 128GB or greater boot device to maximize the space available for ESX-OS Data. • Use hosts with homogeneous configuration.
Network	• Use multiple physical NICs connected to multiple ToR switches to ensure redundancy. • 25GbE is recommended • Use physical NICs of the same speed, vendor, and model across all host within a cluster.
Power	• Use redundant power supplies, preferably connected to separate circuits to ensure availability. • Consider the impact of enabling Power Management on your ESX hosts. In order to allow ESX to control CPU power-saving features you need to set power management in the BIOS to *OS Controlled Mode*.

ESX Model Design Requirements and Recommendations

In your ESX design, you apply certain best practices.

Table 296: ESX Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-ESX-REQD-CFG-001	Install no less than the minimum number of ESX hosts required for the vSphere cluster model being deployed.	• Ensures that availability requirements are met. • If one of the ESX hosts is not available because of a failure or maintenance event, the CPU over-commitment ratio becomes 2:1.	None.
VCF-ESX-REQD-CFG-002	Ensure each ESX host matches the required CPU, memory, and storage specification.	Ensures that workloads will run without contention even during failure and maintenance conditions.	Assemble the server specification and number according to the sizing in the VCF Planning and Preparation Workbook which is based on projected deployment size.
VCF-ESX-REQD-SEC-001	Regenerate the certificate of each ESX host after assigning the ESX host an FQDN.	Establishes a secure connection with the VCF Installer during the initial deployment and prevents man-in-the-middle (MiTM) attacks.	You must manually regenerate the certificates of the ESX hosts before the deployment.

Table 297: ESX Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-001	When using vSAN as primary storage in the management domain, use vSAN ReadyNodes for each ESX host.	Your management domain is fully compatible with vSAN at deployment. **Note:** For information about the models of physical servers that are vSAN-ready, see vSAN Compatibility Guide for vSAN ReadyNodes.	• Hardware choices might be limited. • If you plan to use a server configuration that is not a vSAN ReadyNode, your CPU, disks, and I/O modules must be listed on the VMware Compatibility Guide under *CPU Series* and *vSAN Compatibility List* aligned to the ESX version.
VCF-ESX-RCMD-CFG-002	Allocate ESX hosts with uniform configuration across the default management vSphere cluster.	A balanced vSphere cluster has these advantages: • Predictable performance even during hardware failures. • Minimal impact of resynchronization or rebuild operations on performance.	You must apply vendor sourcing, budgeting, and procurement considerations for uniform server nodes on a per vSphere cluster basis.
VCF-ESX-RCMD-CFG-003	When sizing CPU, do not consider multi-threading technology and associated performance gains.	Although multi-threading technologies increase CPU performance, the performance gain depends on running workloads and differs from one case to another.	Because you must provide more physical CPU cores, costs increase and hardware choices become limited.
VCF-ESX-RCMD-CFG-004	Install and configure all ESX hosts in the default management vSphere cluster to boot using a 128-GB device or larger.	Provides ESX hosts that have large memory, that is, greater than 512 GB, with enough space for the scratch partition when using vSAN.	None.

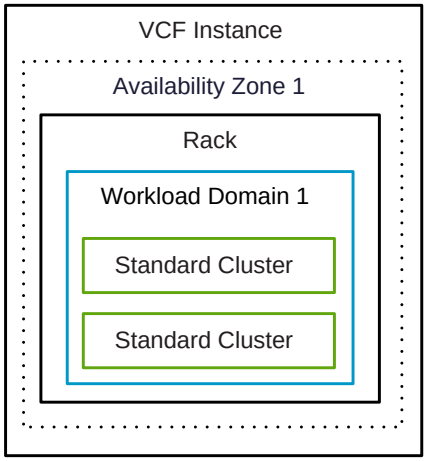
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-005	Use the default configuration for the scratch partition on all ESX hosts in the default management vSphere cluster.	• If a failure in the vSAN cluster occurs, the ESX hosts remain responsive and log information is still accessible. • If using vSAN storage its not possible to use a vSAN datastore for the scratch partition.	None.
VCF-ESX-RCMD-CFG-006	For workloads running in the default management vSphere cluster, save the virtual machine swap file at the default location.	Simplifies the configuration process.	Increases the amount of replication traffic for management workloads that are recovered as part of the disaster recovery process.
VCF-ESX-RCMD-NET-001	Place the ESX hosts in each management domain vSphere cluster on the ESX management network that is separate from the VM management network.	• Enables the separation of the physical VLAN between ESX hosts and the other management components for security reasons. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Increases the number of VLANs required.
VCF-ESX-RCMD-NET-002	Place the ESX hosts in each workload domain on a separate ESX host management VLAN-backed network.	Enables the separation of the physical VLAN between the ESX hosts in different workload domains for security reasons.	Increases the number of VLANs required. For each workload domain, you must allocate a separate management subnet.
VCF-ESX-RCMD-NET-003	Use dedicated VLANs for each traffic type.	Enables the separation of the physical VLAN for vSAN, vMotion, NFS & backup traffic.	Increases the number of VLANs required. For each traffic type you must allocate a separate VLAN and subnet.
VCF-ESX-RCMD-SEC-001	Set the advanced setting `UserVars.SuppressShellWarning` to `0` across all ESX hosts in the management domain.	• Ensures compliance with the *vSphere Security Configuration Guide* and with security best practices • Enables the warning message that appears in the vSphere client every time SSH access is activated on an ESX host.	You must turn off SSH enablement warning messages manually when performing troubleshooting or support activities.

Single-Rack vSphere Cluster Model

The *Single-Rack vSphere Cluster Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack vSphere Cluster Model	The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.

Figure 120: Single-Rack vSphere Cluster Model in VMware Cloud Foundation



Single-Rack vSphere Cluster Model Attributes

A *Single-Rack vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter.
Physical Rack Mapping	All cluster hosts exist in one physical rack.
Resilience	vSphere HA protects against host failures.

Single-Rack vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere which supports your storage type and VCF domain considerations.

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
Minimum number of ESX hosts	vSAN	Simple	Three (3)	• 33% CPU and memory • Tolerates one host failure	Three (3)	• 33% CPU and memory • Tolerates one host failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
	NFS or FC	Simple	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Important:

vSphere clusters are configured allowing for the failure of a single ESX host while still having enough resources to run all vSphere cluster workloads.

Attention:

* Designs should be completed in consultation with your storage vendor.

Common vSphere Cluster Design Requirements

Table 298: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Table 299: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Common vSphere Cluster Design Recommendations

Table 300: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Table 301: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

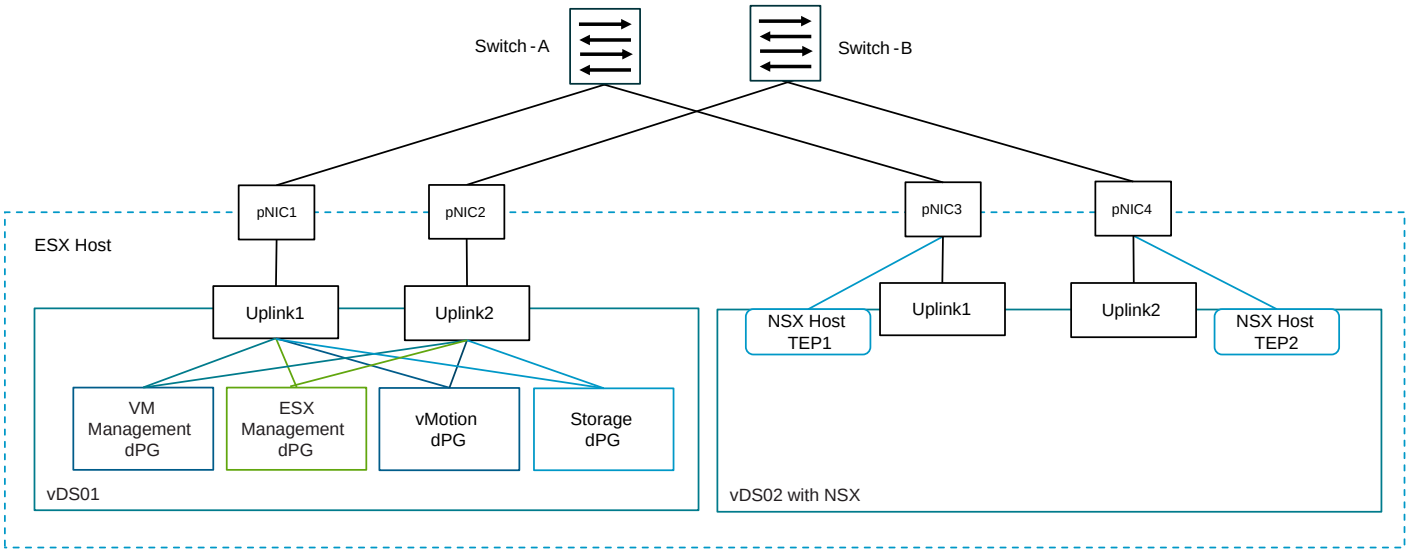
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Workload Separation Distributed Switch Model

The *Workload Separation Distributed Switch Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Workload Separation Distributed Switch Model	A *Workload Separation Distributed Switch Model* in VMware Cloud Foundation provides a separation of workload traffic running on NSX segments from all other network services running on the vSphere cluster.

Figure 121: Workload Separation Distributed Switch Model in VMware Cloud Foundation



Workload Separation Distributed Switch Model Attributes

A *Workload Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Shared for Management, vSphere vMotion, vSAN. • Dedicated for workloads on NSX.

Workload Separation Distributed Switch Model Options

A *Workload Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (workload traffic using NSX and VPC networks isolated to a dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two separate 2-port network cards. • Each traffic type is spread across two separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Common Distributed Switch Design Requirements

Workload Separation Distributed Switch Model Design Requirements

Table 302: Workload Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-WS-REQD-CFG-001	Use two (2) vSphere Distributed Switches per vSphere cluster: • vDS01 — ESX Management — vMotion — Storage • vDS02 — NSX	• Separates NSX workload traffic onto dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for workload traffic with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switches required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than the simple model.
VCF-VDS-WS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion • Storage	Provides the required network services for the cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain clusters.
VCF-VDS-WS-REQD-CFG-003	Configure vDS02 for NSX.	Provides NSX networking capabilities for the workloads on the cluster.	None.
VCF-VDS-WS-REQD-CFG-004	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switch for each additional rack: • ESX management • vMotion • Storage (vSAN/NFS)	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack.
VCF-VDS-WS-REQD-CFG-005	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switch for availability zone 2: • ESX management • vMotion • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2.

Common Distributed Switch Design Recommendations

Table 303: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Workload Separation Distributed Switch Model Design Recommendations

Table 304: Workload Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-WS-RCMD-DPG-001	Use the **Route based on physical NIC load** teaming algorithm for the storage distributed port group on vDS01 (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-WS-RCMD-NIO-001	Enable Network I/O Control on vSphere Distributed Switch vDS01 (default setting).	Increases resiliency and performance when network traffic shares uplinks.	Network I/O Control might impact network performance for critical traffic types if misconfigured.
VCF-VDS-WS-RCMD-NIO-002	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than principal storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.

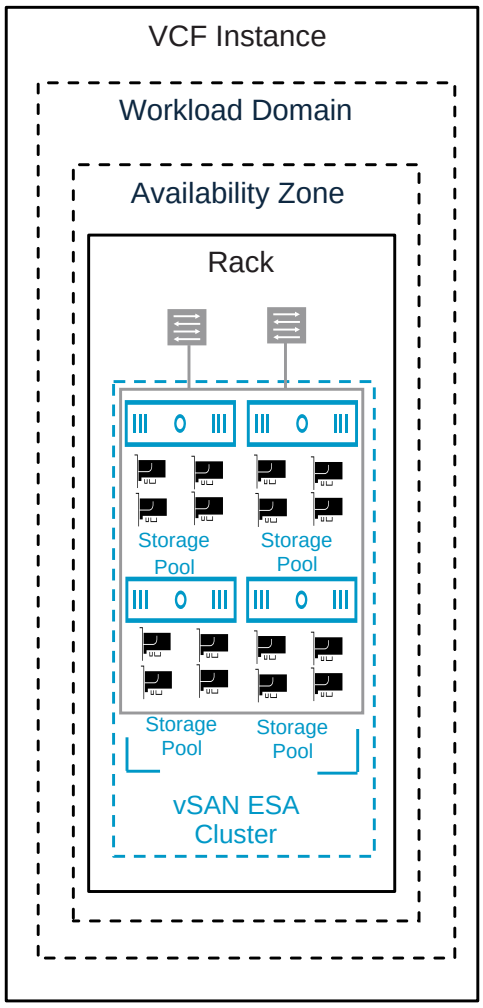
Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-WS-RCMD-NIO-003	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as storage traffic.	During times of network contention, vMotion takes longer than usual to complete.
VCF-VDS-WS-RCMD-NIO-004	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in an VCF Instance. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-WS-RCMD-NIO-005	Set the share value for principal IP-based Storage traffic to High for vDS01 (default setting).	During times of network contention, storage traffic needs guaranteed bandwidth to support virtual machine performance.	None.
VCF-VDS-WS-RCMD-NIO-006	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

Single-Rack vSAN ESA Storage Model

The *Single-Rack vSAN ESA Storage Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture that uses high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.

Figure 122: Single-Rack vSAN ESA Storage Model in VMware Cloud Foundation



Single-Rack vSAN ESA Storage Model Attributes

A *Single-Rack vSAN ESA Storage Model* has the following attributes.

Attribute	Detail
Single Tier	All data is stored on the same type of storage device, regardless of access frequency or importance.
Storage Pool	All NVMe Flash Storage.
Data Services	• Compression (enabled by default). • Encryption. (in transit and at rest) • vSAN Global Deduplication **Restriction:** vSAN Global Deduplication is not currently compatible with vSAN Stretched Clusters and Two Node vSAN Clusters. • Integrated vSAN File Services. • vSAN iSCSI Target Service. • vSAN Storage Clusters. • Stretched Cluster. • Fault domains. • Kubernetes persistent storage services (block and file). • Native scalable snapshots

Attribute	Detail
Virtual Machine Storage policies	vSAN storage policies (also known as Virtual Machine Storage Policies or VM Storage Policies) are rules defining virtual machine (VM) storage requirements that vSAN automatically enforces across a cluster. vSAN 9.1 automatically configures vSAN Auto-RAID policy on new vSAN ESA clusters. vSAN Auto-RAID policy manages the resilience for data residing in a cluster, regardless of the size or topology of the vSAN cluster. vSAN Auto-RAID creates a single policy within the vCenter server,

Common vSAN Storage Model Design Requirements

Table 305: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Common vSAN Host Network Model Design Requirements

Table 306: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

vSAN ESA Storage Model Design Requirements

Table 307: vSAN ESA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-ESA-REQD-CFG-001	Ensure vSAN certified Drives used for vSAN ESA clusters.	• When using VMware Cloud Foundation automated workflows, vSAN ESA cluster creation maybe blocked unless vSAN certified drives are detected. • vSAN Auto-RAID will be enabled when vSAN certified components are detected. This allows vSAN to generate an optimal default storage policy, based on the cluster topology and the number of ESX hosts.	None.
VCF-VSAN-ESA-REQD-CFG-002	1.6 TB drive size or higher for vSAN ESA devices for storage pools.	Minimum supported vSAN ESA drive size is 1.6TB.	vSAN ESA supports up to 24 drives per node. Smaller capacity drives may constrain raw capacity of a vSAN node.
VCF-VSAN-ESA-REQD-CFG-003	16 Cores or higher per vSAN node. vSAN ESA has a minimum value of 16 cores per vSAN Node	Minimum vSAN ESA core count per node is 16 for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	Smaller number of cores per node may require more vSAN nodes to support targeted workloads.
VCF-VSAN-ESA-REQD-CFG-004	Memory size per vSAN Ready node 128GB or higher	Minimum vSAN ESA memory is 128GB for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	• Smaller memory footprint per node will limit number of vSAN drives per node. • Smaller memory footprint per node for vSAN ESA will limit VM workloads being placed on vSAN nodes.

Common vSAN Storage Model Design Recommendations

Table 308: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Common vSAN Host Network Model Design Recommendations

Table 309: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

vSAN ESA Storage Model Design Recommendations

Table 310: vSAN ESA Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-001	• Select a specific size of drives used according to the requirements of target workloads. • Use multiple storage devices per vSAN node • NVMe TLC devices with a performance Class F (100000 - 349999) or higher with endurance class 3 DWPD or higher.	• The number of storage devices has an impact on the performance of vSAN ESA. • NVMe drives with mixed-use endurance and performance class typically classified on the vSAN HCL as "3DWPD" (3 Drive Writes per Day) will give a balance between performance and durability.	• Smaller drive sizes will impact overall capacity of vSAN ESA. Large drive sizes will introduce larger capacity fault domain in case of failure. • Sizing vSAN Nodes requires determining the workload capacity requirements prior to deployment. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-002	Use vSAN ready nodes with high core count.	vSAN Express Storage Architecture (ESA) is designed to be highly parallel and leverages multiple CPU cores for improved performance, making it multithreaded, and optimized for NVMe- TLC based flash devices.	There will be a correlation between target workload, number of cores and NVMe devices per vSAN Node. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

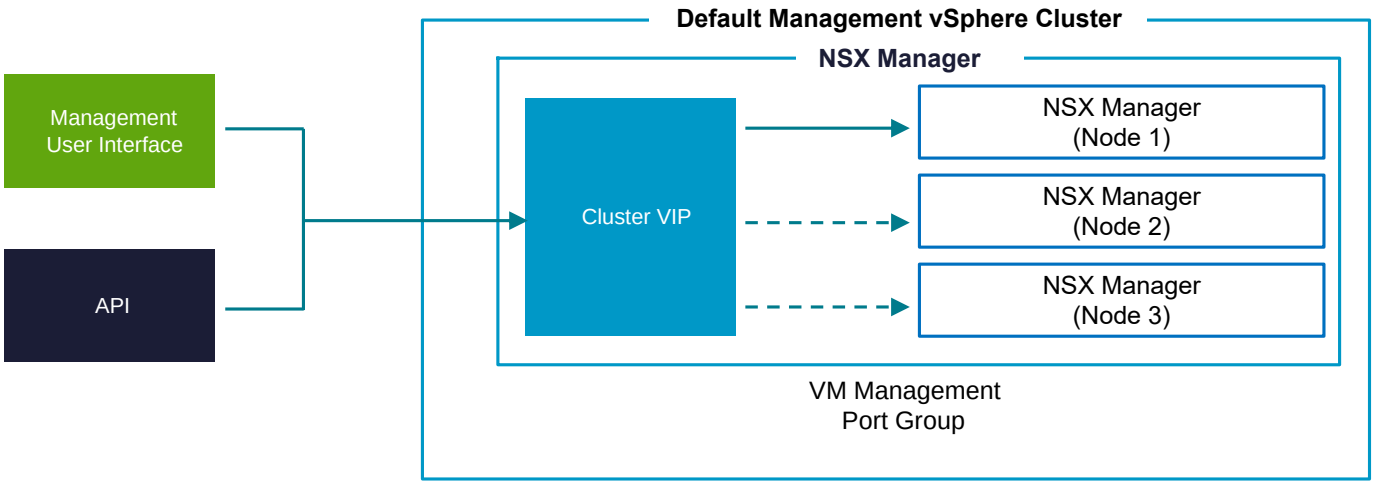
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-003	Use vSAN ready nodes with high memory capacity per vSAN node.	When enabled vSAN has required memory overheads and must have adequate resources account for VM workloads.	Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsanizer/home.
VCF-VSAN-ESA-RCMD-CFG-004	At least four (4) nodes for vSAN cluster size.	This ensures in the event of a fault domain failure, vSAN objects can be re-protected to maintain policy compliance. In conjunction with auto policy management, four (4) node vSAN ESA clusters will allow vSAN to automatically create a default storage policy that is capacity optimized.	None. **Note:** New vSAN clusters in VCF 9.1 will use clusters will automatically be configured to use Auto-RAID.
VCF-VSAN-ESA-RCMD-CFG-005	• Select the same minimum requirements for vSAN Storage cluster nodes as vSAN ESA HCI nodes. • Review vSAN Storage minimum node requirements	• vSAN Storage Cluster Nodes use the same architecture as regular vSAN ESA nodes. • vSAN ESA nodes can be re-purposed for vSAN Storage Clusters or vSAN HCI Clusters	None.

High Availability NSX Manager Model

The *High Availability NSX Manager Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* blueprint.

Selected Model	Additional Information
High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.

Figure 123: High Availability NSX Manager Model in VMware Cloud Foundation



High Availability NSX Manager Model Attributes

The *High Availability NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Three-node cluster of NSX Manager appliances with internal virtual IP
Availability	• Highly available cluster of virtual IP NSX Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.
API Request Scale	Internal virtual IP directs API requests to a single NSX Manager appliance.

Common NSX Manager Design Requirements for All Models

Table 311: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

High Availability NSX Manager Model Design Requirements

Table 312: High Availability NSX Manager Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-005	Deploy three (3) NSX Manager nodes in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports high availability of the NSX Manager cluster.	You must have sufficient resources in the default cluster of the management domain to run three NSX Manager nodes.

Common NSX Manager Model Design Recommendations for All Models

Table 313: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

High Availability NSX Manager Model Design Recommendations

Table 314: High Availability NSX Manager Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-003	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Manager appliances.	Keeps the NSX Manager appliances running on different ESX hosts for high availability.	You must allocate at least four physical hosts so that the three NSX Manager appliances continue running if an ESX host failure occurs.

Physical Infrastructure Layer for VCF Fleet with Multiple Sites Across Multiple Regions
The physical infrastructure layer constitutes the tangible hardware and facilities layer that provides the foundational computing, storage, and networking resources upon which all digital services depend. This layer encompasses the actual physical assets—servers, storage arrays, network equipment, and supporting facilities—that form the bedrock of modern IT operations.

Single-Rack Network Fabric Model

The *Single-Rack Network Fabric Model* was chosen for the *VCF Fleet with Multiple Sites Across Multiple Regions* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack Network Fabric Model	The *Single-Rack Network Fabric Model* describes the VLAN and gateway requirements to support the Single-Rack vSphere Cluster Model.

The data center network fabric provides connectivity to the vSphere clusters. The vSphere Cluster Models and Storage Models chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a Single-Rack vSphere Cluster Model. Depending upon the storage type chosen, additional VLAN requirements may apply.

Note:

If this rack will contain NSX Edge nodes, additional VLAN Requirements will apply. NSX Edge Cluster Models for Single Availability Zone

Table 315: Infrastructure VLAN Scope and First Hop Gateway Availability for Single Rack

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management VM Appliances*		Rack Local	Highly Available	1500	1500
ESX host	ESX management VMK *				1500
	IP Storage VMKs (vSAN, NFS, iSCSI)				9000
	vMotion VMK				
	Host TEP			1700	
NSX Edge node	Refer to Host Fault Tolerant NSX Edge Cluster Model or Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters				
	NSX Edge Node management	Rack Local	Highly Available	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available Static Routing with BFD - Optional BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
	BGP-EVPN Peering	Across Racks	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* When using separate VLAN segments for management component VMs and ESX VMKs.					

Table 316: Workload VLAN Scope and First Hop Gateway Availability for Single Rack

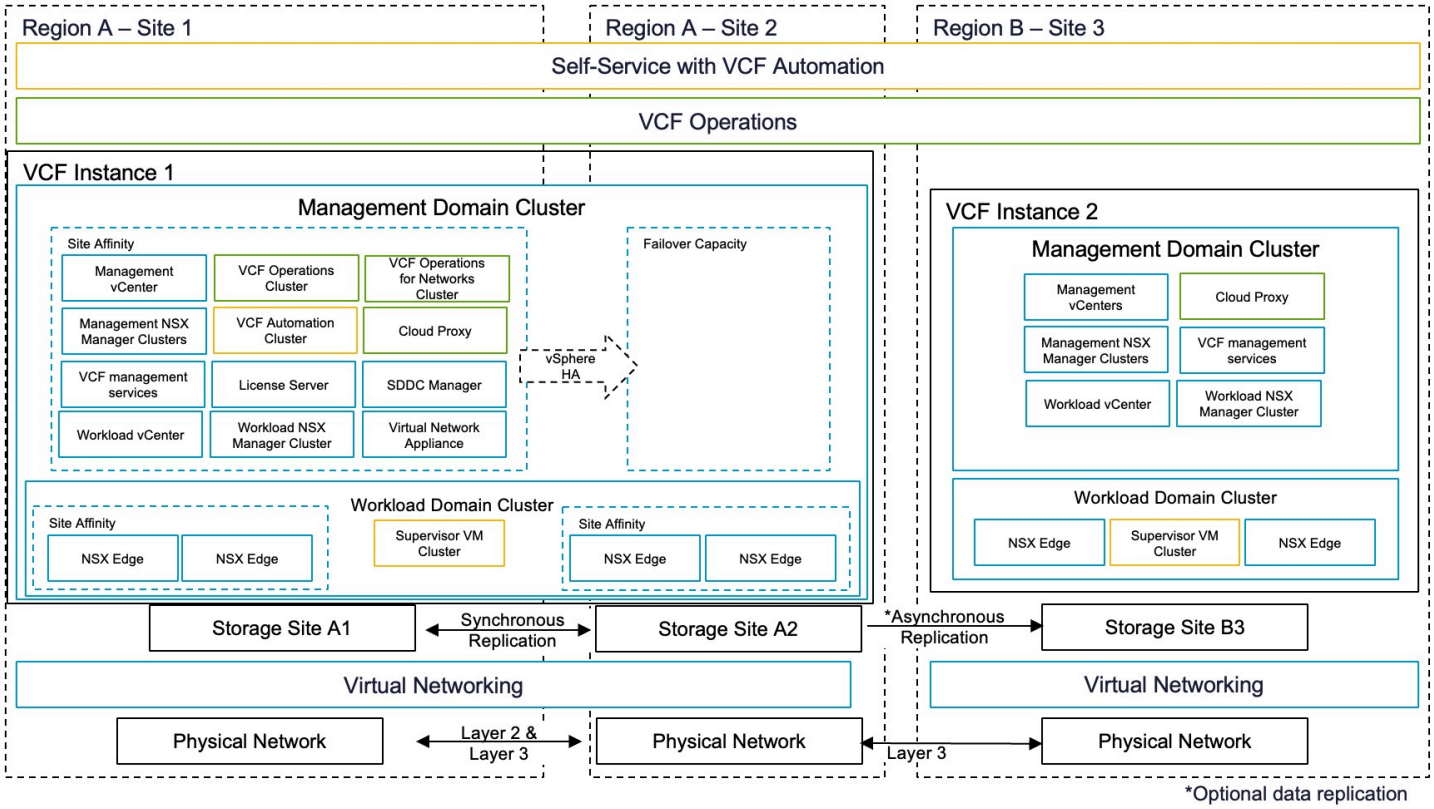
Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VLAN Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model Attributes	Refer to Edges			

VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)

The *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint, where the management domain of the first VCF Instance is deployed with high availability for management components and workloads are isolated to a separate workload domain. Multiple sites exist, where some sites co-exist in the same region, and other site(s) exist in a different region.

Architecture Overview

Figure 124: VCF Fleet with Multiple Sites in a Single Region Addtional Region(s)



Design Profile

The *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint is well suited to customers who are looking for the following design profile for their VMware Cloud FoundationVMware Cloud Foundation platform.

Attribute	Value
Consumption	• Self-service consumption via VCF Automation supporting modern workloads like VKS, Data Services, AI Services, and so on. with multi-tenancy. • Direct vCenter consumption supporting modern workloads like VKS, Data Services, AI Services, and so on. This infrastructure blueprint can use the following Consumption blueprint: • Self-Service Multi-Tenant Private Cloud Consumption
Physical Site Configuration	Multiple sites within a single region (single instance, stretched cluster between the sites for maximum availability)

Attribute	Value
Availability	• Tolerates failure of a single ESX host • Tolerates failure of an individual network path • Tolerates failure of a single rack • Tolerates failure of a single management component • Tolerates failure of a single availability zone
Isolation	• Hypervisor-based (logical isolation) • Network traffic type isolation • Cluster-based (physical isolation) • Tenant / Tenant isolation
Recoverability	Can provide recovery within a region for single site loss providing both sites have sufficient external network connectivity and compute resources to support all workloads to be recovered. This infrastructure blueprint can use the following protection blueprints: • VCF Component Backup and Restore • VCF Instance Backup and Restore • VCF Fleet Disaster Recovery
Expansion	For available resource and availability expansions, see Expansion after Infrastructure Blueprint Deployment.

Management Domain Design Selections Region

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	High Availability VCF Automation Model	The High Availability VCF Automation Model consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.
	Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.
Operations	High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single Cloud Proxy appliance. To achieve high availability after the initial deployment, an additional Cloud Proxyappliance should be deployed and then combined in a collector group.
	VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.
	VCF Management Services in an Additional VCF Instance Model	Additional VCF Instances in a fleet run a VCF management services cluster with instance level components for managing that instance only.
	High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Layer	Selected Model	Additional Information
	Single License Server Model	The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.
	High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a four-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.
	High Availability Logs Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind and internal load balancer. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.
Security Governance and Compliance	Identity Broker Model	Deploys the Identity Broker component on the VCF management services instance in the management domain.
	Single VCF Instance Single Sign On Model	With the *Single VCF Instance Single Sign-On Model* each VCF Instance will utilize its own dedicated identity broker. The VCF Single Sign-On scope will be limited to the VCF Instance. VCF management components, such as VCF Operations and VCF Automation, can connect to only one of identity broker instances for Single Sign-On.
Virtual Infrastructure	Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.
	VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Workload Domain Design Selections

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Layer	Selected Model	Additional Information
Consumption	Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.
	High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected vSphere Supervisor management zones.
	Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for vSphere Supervisor and VKS workloads, it supports VPC-based networking.

Layer	Selected Model	Additional Information
	Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways	This design provides multi-tenant resource isolation across multiple regions by combining dedicated gateways for strict tenant separation with shared gateways for efficient common services.
Virtual Infrastructure	Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.
	Dual Availability Zone Design Based on NSX Edge Node HA Model	A *Single-Rack vSphere Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Shared Design Selection between Management and Workload Domain Region A

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Layer	Selected Model	Additional Information
Virtual Infrastructure	vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.
	ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCFdomain vSphere cluster.
	Stretched vSphere Cluster Model	The *Stretched vSphere Cluster Model* allows you to protect workloads on vSphere clusters in your environment against a failure of a single fault domain by implementing a stretched vSphere cluster across multiple availability zones.
	Stretched vSAN Storage Model	A *Storage Separation Distributed Switch Model* in VMware Cloud Foundation provides separation of storage traffic from all other network services running on the vSphere cluster.
	Storage Separation Distributed Switch Model	A *Stretched vSAN Storage Model* is a cluster configuration where the ESX hosts, that comprise a single vSAN cluster, reside in two availability zones.
	High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.
Physical Infrastructure	Multi Availability Zone Network Fabric Model	This section describes the VLAN and gateway requirements to support a vSphere cluster within multiple availability zones.

Shared Design Selection between Management and Workload Domain Region B

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

	Selected Model	Additional Information
Virtual Infrastructure	vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

	Selected Model	Additional Information
	ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCFdomain vSphere cluster.
	Single-Rack vSphere Cluster Model	The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.
	Workload Separation Distributed Switch Model	A *Workload Separation Distributed Switch Model* in VMware Cloud Foundation provides a separation of workload traffic running on NSX segments from all other network services running on the vSphere cluster.
	Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture that uses high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.
	High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.
Physical Infrastructure	Single-Rack Network Fabric Model	The *Single-Rack Network Fabric Model* describes the VLAN and gateway requirements to support the Single-Rack vSphere Cluster Model.

Minimum Requirements for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)

VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s) requirements to stand up this infrastructure blueprint.

VCF Installer Deployment Options

Deployment Model	Deployment Size
High Availability	Medium

Minimal IP Requirements

Requirement	Additional Information
Physical Networks	• Region 1 AZ1 ESX Management (4 IPs) • Region 1 VM Management (35 IPs) • Region 1 AZ1 vMotion (4 IPs) • Region 1 AZ1 vSAN (4 IPs) • Region 1 AZ1 Host TEP (8 IPs) • Region 1 (Optional) Edge TEP Network (4 IPs) — 2 IPs per Edge VM (x2 Edge VMs) • Region 1 (Optional) Edge Uplink Network (2 IPs) — 1 IP per Edge VM (x2 Edge VMs) • Region 1 AZ2 ESX Management (4 IPs) • Region 1 AZ2 vMotion (4 IPs) • Region 1 AZ2 vSAN (4 IPs) • Region 1 AZ2 Host TEP (8 IPs) • Region 2 ESX Management (4 IPs) • Region 2 VM Management (15 IPs) • Region 2 vMotion (4 IPs) • Region 2 vSAN (4 IPs) • Region 2 Host TEP (8 IPs) • Region 2 (Optional) Edge TEP Network (4 IPs) — 2 IPs per Edge VM (x2 Edge VMs) • Region 2 (Optional) Edge Uplink Network (2 IPs) — 1 IP per Edge VM (x2 Edge VMs)
DNS Entries	• Region 1 vCenter • Region 1 AZ1 ESX — 4 Nodes • Region 1 SDDC Manager • Region 1 NSX Manager — VIP — Node A — Node B — Node C • Region 1 VCF Operations — Primary — Replica — Data — Cloud Proxy • Region 1 VCF Management Services — VCF Services Runtime — VCF Management Instance Service — VCF Fleet Component • Region 1 VCF Automation • Region 1 VCF Service Runtime for VCF Automation • Region 1 License Server • Region 1 Identity Broker • Region 1 Log Management • Region 1 VCF Operations for Networks — Platform 1 — Platform 2 — Platform 3 — Collector • Region 1 AZ2 ESX — 4 Nodes • Region 2 vCenter • Region 2 ESX — 4 Nodes • Region 2 SDDC Manager • Region 2 NSX Manager — VIP

Requirement	Additional Information
Virtual Networks	• vSphere Supervisor Service CIDR — e.g. 10.96.0.0/23 • VPC Private IP Block (Control Plane) • Distributed Transit Gateway Private IP Block (Control Plane) • External IP Block (Control Plane & Workload/Organizations) • Private VPC Block for each Organization's VPCs • Private TGW IP Block for each Organization

First VCF Instance

Table 317: VMware Cloud Foundation Installer Deployment Profiles

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for the initial VCF instance within a VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	1x Small • 1x Control Plane Nodes • 2x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Medium • 3x Control Plane Nodes • 3x Worker Nodes	1x Large • 3x Control Plane Nodes • 4x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Medium	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
VCF Operations	1x Small	2x Small	3x Medium	3x Large
License Server	1x Default	1x Default	1x Default	1x Default
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard
VCF Automation	1x Small	1x Medium	3x Medium	3x Large
Resource Requirements	• vCPU: 76 • RAM: 251 GB • DISK: 7448 GB	• vCPU: 106 • RAM: 335 GB • DISK: 8422 GB	• vCPU: 184 • RAM: 656 GB • DISK: 11445 GB	• vCPU: 298 • RAM: 949 GB • DISK: 15357 GB

Additional VCF Instance

This table details the default sizing profiles in the VMware Cloud Foundation Installer and lists the underlying component sizes for any additional VCF instance joining an existing VCF Fleet. Custom sizes can be configured using a JSON deployment specification file.

Table 318: VMware Cloud Foundation Installer Deployment Profiles

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
VCF Management Services	• 1x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes	• 3x Control Plane Nodes • 2x Worker Nodes
SDDC Manager	1x Default	1x Default	1x Default	1x Default
vCenter	1x Small	1x Small	1x Medium	1x Large
NSX	1x Medium	3x Medium	3x Medium	3x Large
Cloud Proxy	1x Small	1x Small	1x Standard	1x Standard

Component	Simple	High Availability - Small	High Availability - Medium	High Availability - Large
Resource Requirements	• vCPU: 34 • RAM: 111 GB • DISK: 4562GB	• vCPU: 62 • RAM: 187 GB • DISK: 5462 GB	• vCPU: 74 • RAM: 244 GB • DISK: 5813 GB	• vCPU 120 • RAM 353 GB • DISK 8321 GB

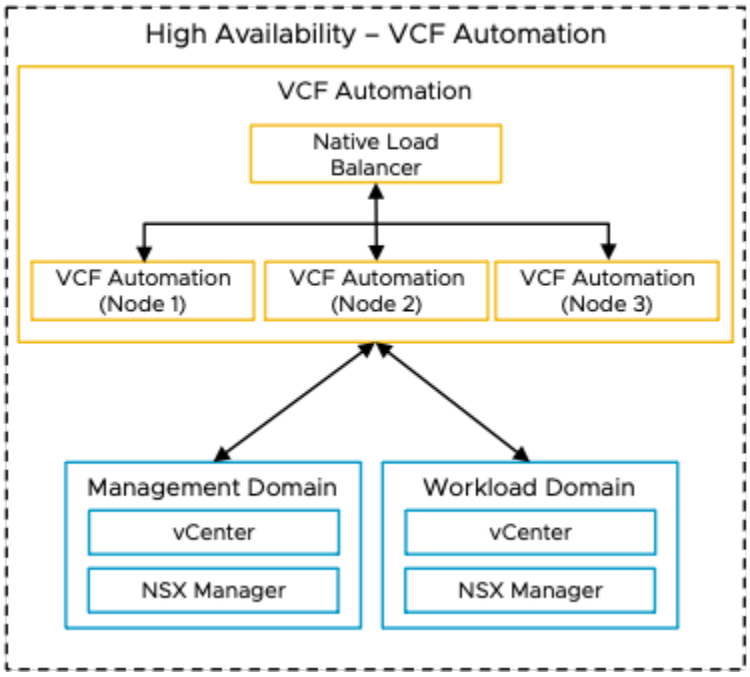
Management Domain Design Selections

The following models in the design align with the objectives for deploying this *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Consumption Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
The consumption layer for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint represents the top-most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

High Availability VCF Automation Model
The *High Availability VCF Automation Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Automation Model	The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.



High Availability VCF Automation Model Attributes

A *High Availability VCF Automation Model* has the following attributes.

Attribute	Detail
Availability	• Application level availability provided by the three node cluster • Appliance level level availability provided by vSphere HA
Scalability	Supports scale-up and scale-down of node sizes.
Recoverability	Restore through a backup.

High Availability VCF Automation Model Design Requirements

Table 319: High Availability VCF Automation Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AUTO-REQD-HA-001	Deploy VCF Automation as a three node cluster.	Supports high availability objectives.	• Analysis is required to determine which node size is appropriate for your deployment. • Node can be resized after deployment as necessary.
VCF-AUTO-REQD-HA-003	Protect all VCF Automation nodes using vSphere HA.	Ensures availability of VCF Automation without requiring manual intervention during an ESX host failure event.	None

VCF Automation Tenant Authentication Requirements

Table 320: VCF Automation Tenant Authentication Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-TEN-001	Configure an identity provider for VCF Automation organizations.	• Allows tenants to use enterprise directory. • Enables named access control to tenant level constructs. • Required to provision administrative or user access to the Organization portal for any account other than the first user account established during Organization creation.	If not configured, the only access to the Organization Portal will be through the first user account established during Organization creation.

VCF Automation Enterprise Authentication Design Requirements

Table 321: VCF Automation Enterprise Authentication Design Requirements

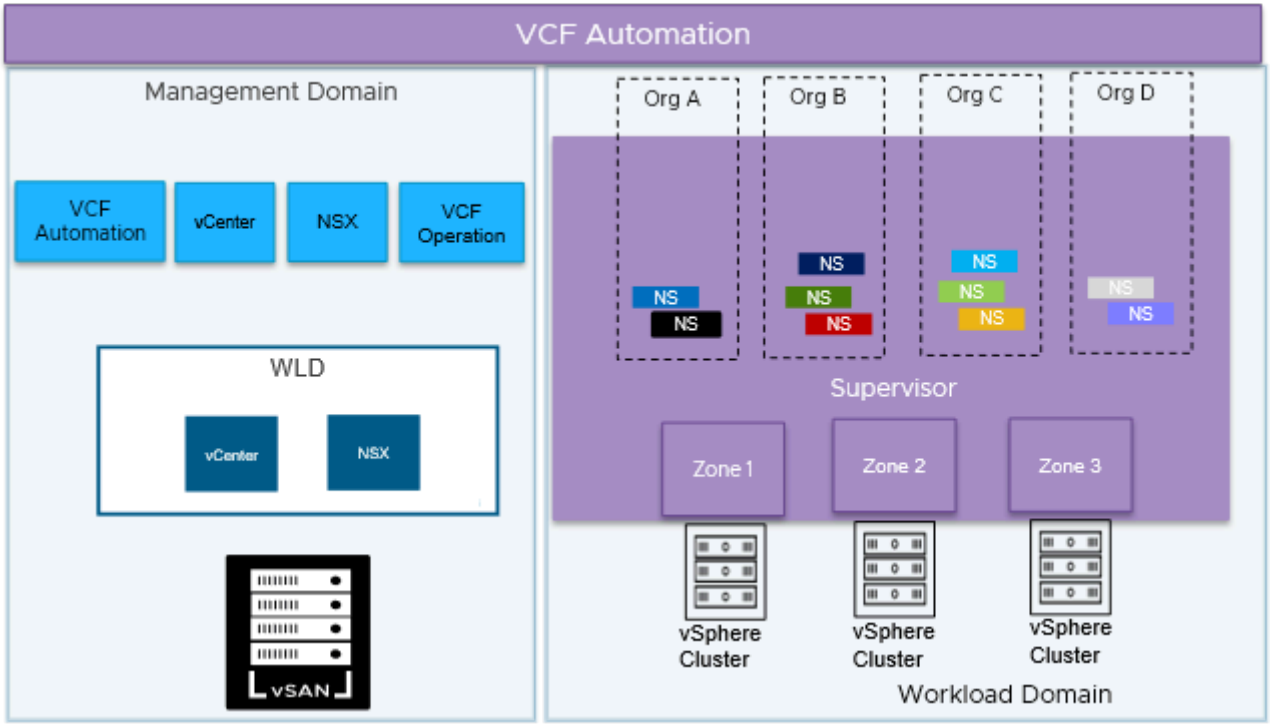
Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-IDP-001	The provider uses the same IdP as the tenant.	• Allows consistent authentication across provider and tenant	None

Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance

The *Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Tenancy Deployment Model 2: Shared workload domain for multiple organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.

Figure 125: Shared Workload Domain for Multiple Organizations in a VCF Instance



Benefits	Best fit scenarios
• Resource Sharing: Tenants share the underlying infrastructure (workload domain, vCenter, NSX), leading to cost savings and efficient resource utilization. • Simplified Management and Operational Consistency: While providing tenant isolation, the centralized management of the VCF Management Domain simplifies overall infrastructure operations, including patching, upgrades, monitoring, and capacity planning. Consistent policies can be applied across all tenants. • Scalability and Flexibility: The model is highly scalable. As the number of tenants or their resource demands grow, the underlying workload domain can be scaled up by adding more hosts. Resource allocations within Namespaces can also be adjusted dynamically. If the requirement comes, the service provider can add more workload domains for multiple other organizations to share or can dedicate a workload domain per tenant like explained in Model 2. • vSphere Namespace Isolation: vSphere Namespaces provide logical resource isolation for tenant workloads, preventing interference. • Network Segmentation: Network policies and security groups can be used to further segment tenant traffic and can enhance security.	• Cloud Service Providers (CSPs) Offering Multi-Tenant Cloud Services: This model allows the CSP to host multiple customers within a single VCF instance, maximizing resource utilization across their infrastructure. The Supervisor Cluster ensures strong isolation between each customer's workloads and data. Each customer gets their own vSphere Namespace, providing them with a secure and logically separated environment for deploying VMs and containers. Network segmentation (via NSX) further guarantees that customer traffic remains isolated. • Large Enterprises with Multiple Business Units or Departments: A large corporation has several independent business units (e.g., Finance, Marketing, R&D) that require their own isolated IT environments but can share underlying infrastructure. • Organizations Offering Internal PaaS or Developer Platforms: An organization wants to provide a self-service platform for its internal development teams to deploy and manage their applications (including containerized microservices) in an isolated and governed manner. • Educational Institutions or Research Organizations with Multiple Projects or Teams: A university hosts various research projects or has different departments with distinct IT requirements and security needs.

Operations Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)

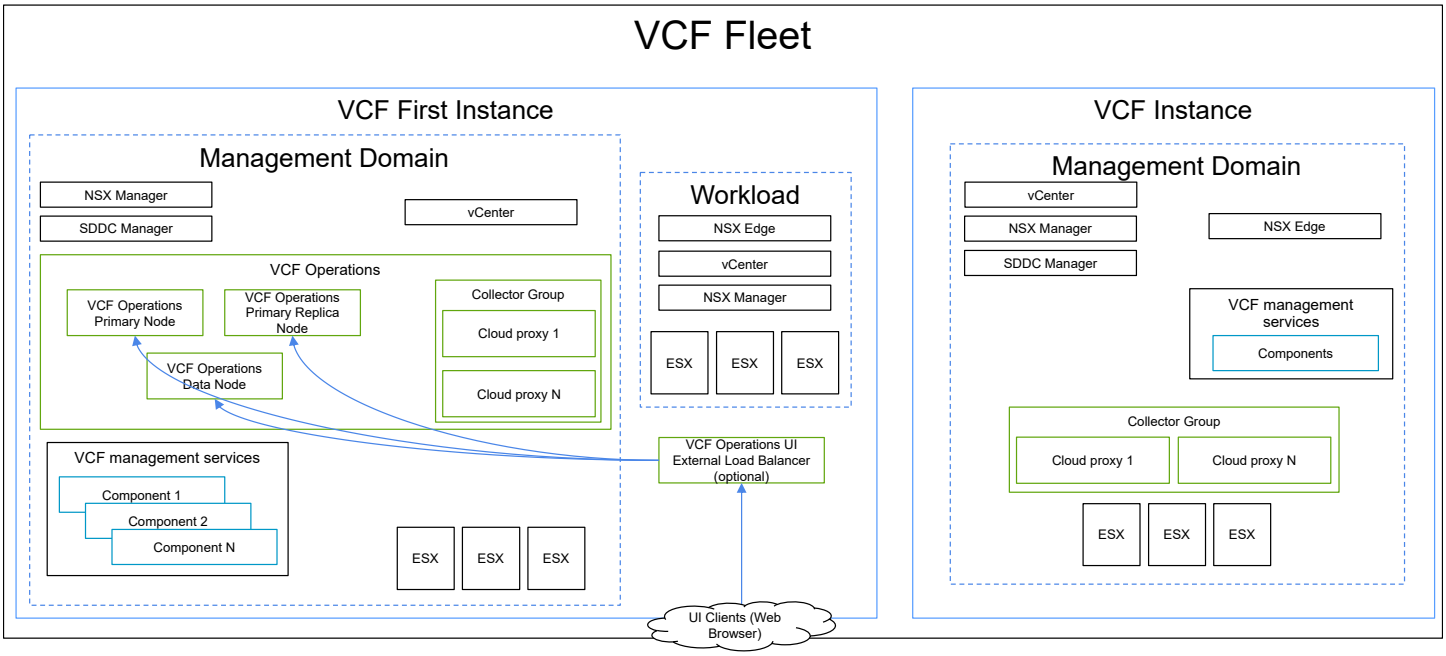
The operations layer serves as the central nervous system of enterprise infrastructure and applications, providing comprehensive visibility, management, and control capabilities across the entire technology stack. This layer ensures that systems remain available, performant, secure, and compliant with organizational standards and regulatory requirements.

High Availability VCF Operations Model

The *High Availability VCF Operations Model* was chosen for the V*CF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations Model	The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and a single Cloud Proxy appliance. To achieve high availability after the initial deployment, an additional Cloud Proxy appliance should be deployed and then combined in a collector group.

Figure 126: High Availability VCF Operations Model in VMware Cloud Foundation



High Availability VCF Operations Model Attributes

A *High Availability VCF Operations Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure. Additional level of availability for the components in: • VCF Operations cluster • VCF Operations collector groups
Scalability	Supports scale-up and scale-out. **Note:** Scale-out is supported for the VCF Operations data nodes and cloud proxynodes.
Recovery	Restore through a backup.
Load balancing	Optional external load balancer for the VCF Operations cluster UI.

Common VCF Operations Design Requirements for All Models

Table 322: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Operations Model Design Requirements

Table 323: High Availability VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-001	Deploy VCF Operations as a cluster of three nodes in the default management vSphere cluster.	• Supports the high availability objectives by deploying VCF Operations as a cluster. • Provides the capacity that is required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out with additional data nodes.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • Create an anti affinity rule (Separate Virtual Machines) for the VMs in the cluster after the deployment.
VCF-OPS-REQD-HA-002	Deploy a single Unified VCF Operations collector node in the default management vSphere cluster.	Removes the load from the VCF Operations cluster from collecting metrics from local-instance applications.	You may need to scale-up the VCF Operations collector nodes to a greater size based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-HA-003	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations cluster nodes.	Using vSphere DRS prevents the VCF Operations cluster nodes from running on the same ESX host and risking the high availability of the cluster.	• You must perform an additional configuration to set up an anti-affinity rule. • You must have a minimum of four ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode. • If additional data nodes are added, you must update the anti-affinity rule and ensure there are enough ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode.
VCF-OPS-REQD-HA-004	The certificate for VCF Operations must include the FQDNs for all cluster nodes in the SAN (subject alternative name) attribute. **Note:** The SAN does not need to include the FQDN of VCF Operations collector nodes.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted.	Before scale out, the certificate for VCF Operations must be changed to also include the FQDN of the additional node in the SAN (subject alternative name) attribute.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-005	If using an optional external load balancer in the high availability deployment model, the certificate for VCF Operations **must** also include the FQDN of the load balancer in the SAN (subject alternative name) attribute.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted when using an external load balancer with the high availability model.	None.

Common VCF Operations Design Recommendations for All Models

Table 324: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

High Availability VCF Operations Model Design Recommendations

Table 325: High Availability VCF Operations Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-HA-001	Deploy VCF Operations with an external load balancer to load balance the connections across the VCF Operations cluster nodes.	Required to deploy a VCF Operations cluster deployment with distributed user interface access across members.	You must configure the external load balancer configuration to support this VCF Operations configuration.
VCF-OPS-RCMD-HA-002	Create a collector group in VCF Operations for each VCF Instance and assign the default VCF Operations collector appliance.	Provides increased availability to metric collection across the VCF Operations collector appliances that are assigned to the collector group.	Manual reconfiguration of the VMware Cloud Foundation adapter required to utilize multiple VCF Operations collector appliances with the group.
VCF-OPS-RCMD-HA-003	Deploy an additional VCF Operations collector node in the default management vSphere cluster.	Extends the availability of the VCF Operations collectors by balancing the load across the collectors when added to a collector group.	• You must deploy the collector node after the deployment of the VCF Operations. • Additional collector node should be deployed for all VCF Instances that requires a collector group.
VCF-OPS-RCMD-HA-004	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations collector nodes.	Using vSphere DRS prevents the VCF Operations collectors from running on the same ESX host and risking the high availability of the collectors.	You must perform manual configuration to set up an anti-affinity rule.
VCF-OPS-RCMD-HA-005	Reconfigure the default VMware Cloud Foundation Adapter in VCF Operations to use a local-instance collector group for a VCF Instance.	Increases the availability of metric collection.	Must be reconfigured post-deployment.

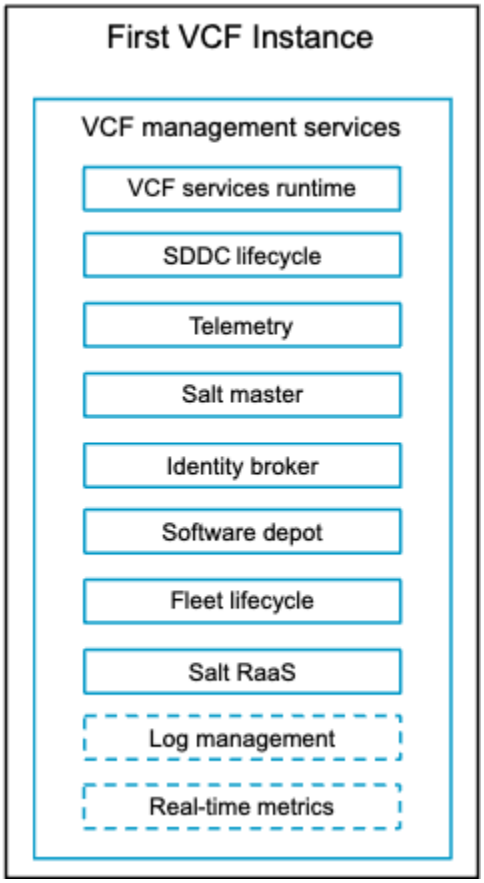
VCF Management Services in the First VCF Instance Model

The *VCF Management Services in the First VCF Instance Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Services in the First VCF Instance Model	The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.

Figure 127: VCF Management Services in the First VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.



VCF Management Services in the First VCF Instance Model Components

Table 326: VCF Management Services in the First VCF Instance Model Components

The following components run in the first VCF Instance in the VCF fleet.

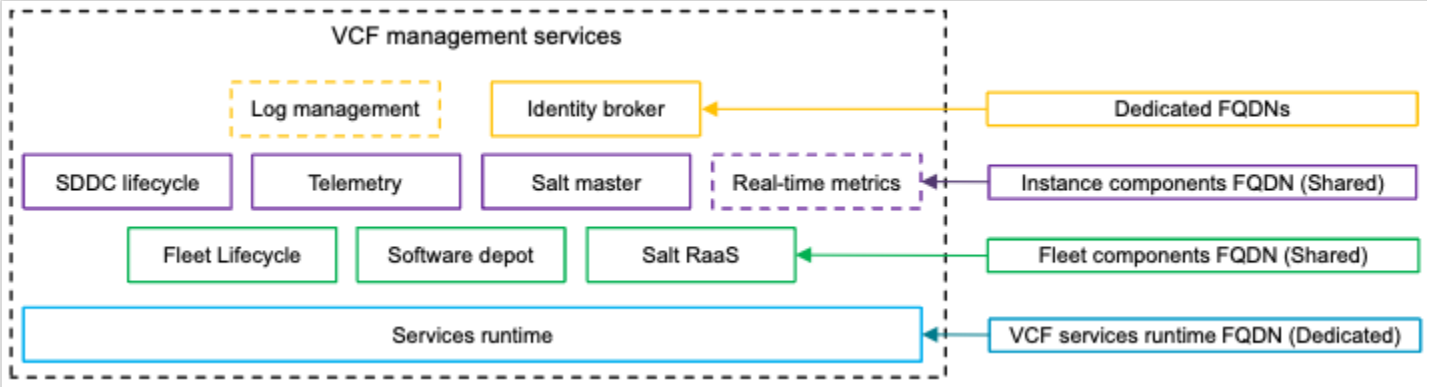
Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations across the fleet.	
Fleet lifecycle	Orchestrates fleet-wide life-cycle operations, including the planning and sequencing of upgrades and patches across multiple VCF Instances.	

Service	Description	Deployment Time
Salt RaaS	Defines and manages desired state configurations and policies (Recipe as a Service).	
Log management	Collects logs from all fleet components	Day-N Components
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

VCF Management Services in the First VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 128:



VCF Management Services in the First VCF Instance Model Design Requirements

Table 327: VCF Management Services in the First VCF Instance Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-FIRST-001	Create DNS entries for the following shared FQDNs: • Fleet components FQDN • Instance components FQDN	Required for deploying VCF management services fleet and instance level components in the first VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-FIRST-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN • Identity broker FQDN	Required for deploying VCF management services in the first VCF Instance.	You must create the DNS entries before starting the deployment.

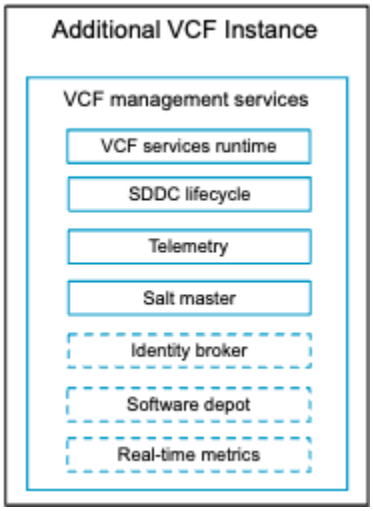
VCF Management Services in an Additional VCF Instance Model

The *VCF Management Services in an Additional VCF Instance Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
VCF Management Services in an Additional VCF Instance Model	Additional VCF Instances in a fleet run a VCF management services cluster with instance level components for managing that instance only.

Figure 129: VCF Management Services in an Additional VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.



VCF Management Services in an Additional VCF Instance Model Components

Table 328: VCF Management Services in an Additional VCF Instance Model Components

The following components run in each additional VCF Instance in the VCF fleet.

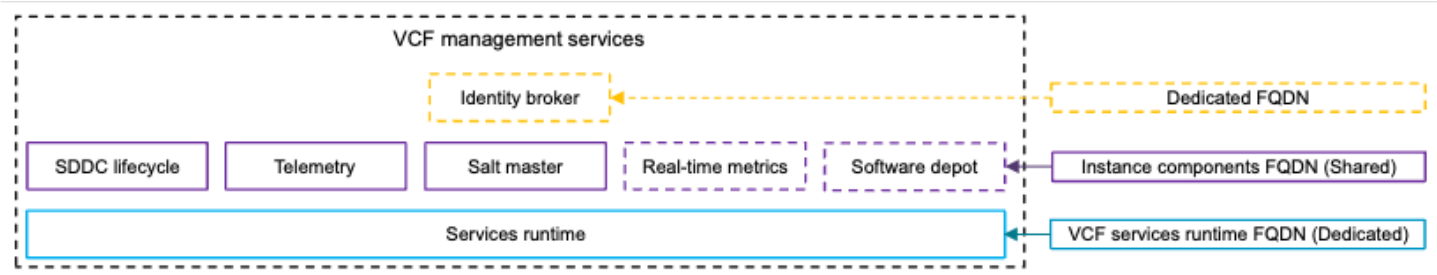
Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	Day-N Components
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations within this VCF Instance.	
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

VCF Management Services in an Additional VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 130: VCF Management Services FQDN Mapping in an Additional VCF Instance

The diagram below shows the various components for this model. The dotted line components are optional, and are deployed day-N.



VCF Management Services in an Additional VCF Instance Model Design Requirements

Table 329: VCF Management Services in an Additional VCF Instance Model Design Requirements

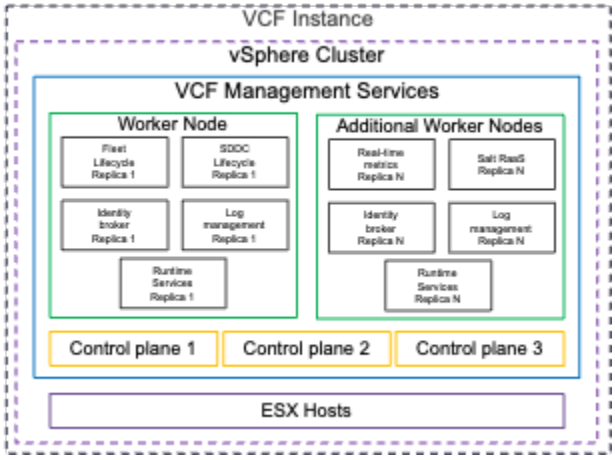
Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-ADTL-001	Create DNS entries for the following shared FQDNs: • Instance components FQDN	Required for deploying VCF management services in the additional VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-ADTL-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN	Required for deploying VCF management services in the additional VCF Instance.	You must create the DNS entries before starting the deployment.

High Availability VCF Management Services Model

The *High Availability VCF Management Services Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructue blueprint.

Selected Model	Additional Information
High Availability VCF Management Services Model	The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Figure 131: High Availability VCF Management Services Model in VMware Cloud Foundation



High Availability VCF Management Services Model Attributes

Attribute	Detail
Availability	Relies on vSphere HA
Scalability	Scale out/up worker nodes
Placement	Hosted in a single vSphere cluster

Common VCF Management Services Model Design Requirements

Table 330: Common VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-001	Allocate a sufficient IP range for the required number of control plane and worker nodes for your chosen deployment type.	Required for initial deployment.	Adding additional components may require auto-scale out of the VCF management services.
VCF-MS-REQD-COM-002	Use *`Kubernetes Cluster API Provider vSphere (CAPV)`* node anti-affinity to ensure the nodes run on different ESX hosts.	Using *`CAPV`* prevents the VCF management services nodes from running on the same ESX host and risking the high availability of the cluster.	Distributed Resource Scheduler (DRS) anti-affinity rules are not required.
VCF-MS-REQD-COM-003	Protect the VCF management services nodes by using vSphere HA.	Ensures availability of the VCF management services nodes without requiring manual intervention during an ESX host failure event.	None.

High Availability VCF Management Services Model Design Requirements

Table 331: High Availability VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-HA-001	Deploy VCF management services in high availability mode with three (3) control plane nodes and the corresponding number of worker nodes in the default management vSphere cluster.	• Supports the high availability deployment objective. • VCF management services supports scale-out and up from small to medium or large.	Requires additional resources.

Single License Server Model

The *Single License Server Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Single License Server Model	The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, which is designed to provide centralized license management for the entire fleet. It is suitable for most typical deployments where reduced operational overhead is prioritized.

Table 332: Single License Server Model Attributes

Attribute	Detail
Use Cases	This is the typical license server deployment model for VMware Cloud Foundation and is suitable for most VCF deployments.
Deployment mechanism	One license server per VCF fleet. A license server is automatically deployed as part of the installation or upgrade to VMware Cloud Foundation 9.1.
Backup and restore	Backup is not required. If the original license server becomes unavailable, deploy a new license server instance and migrate the licenses within 90 days**.**
Lifecycle	The upgrade and patch management of the license server is managed through VCF Operations.
High Availability	High availability is provided by vSphere HA.
License Scope	A single license server provides licensing management for the whole VCF fleet.

Single License Server Model Design Elements

Table 333: Single License Server Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LIC-REQD-LS-001	Register the default license server in VCF Business Services console.	Registration allows the VCF Business Services console to add license allocations to the license server.	None.

Licensing for VCF Operations Design Recommendations Connected Mode

Table 334: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Licensing for VCF Operations Design Recommendations Disconnected Mode

Table 335: Licensing for VCF Operations Design Recommendations

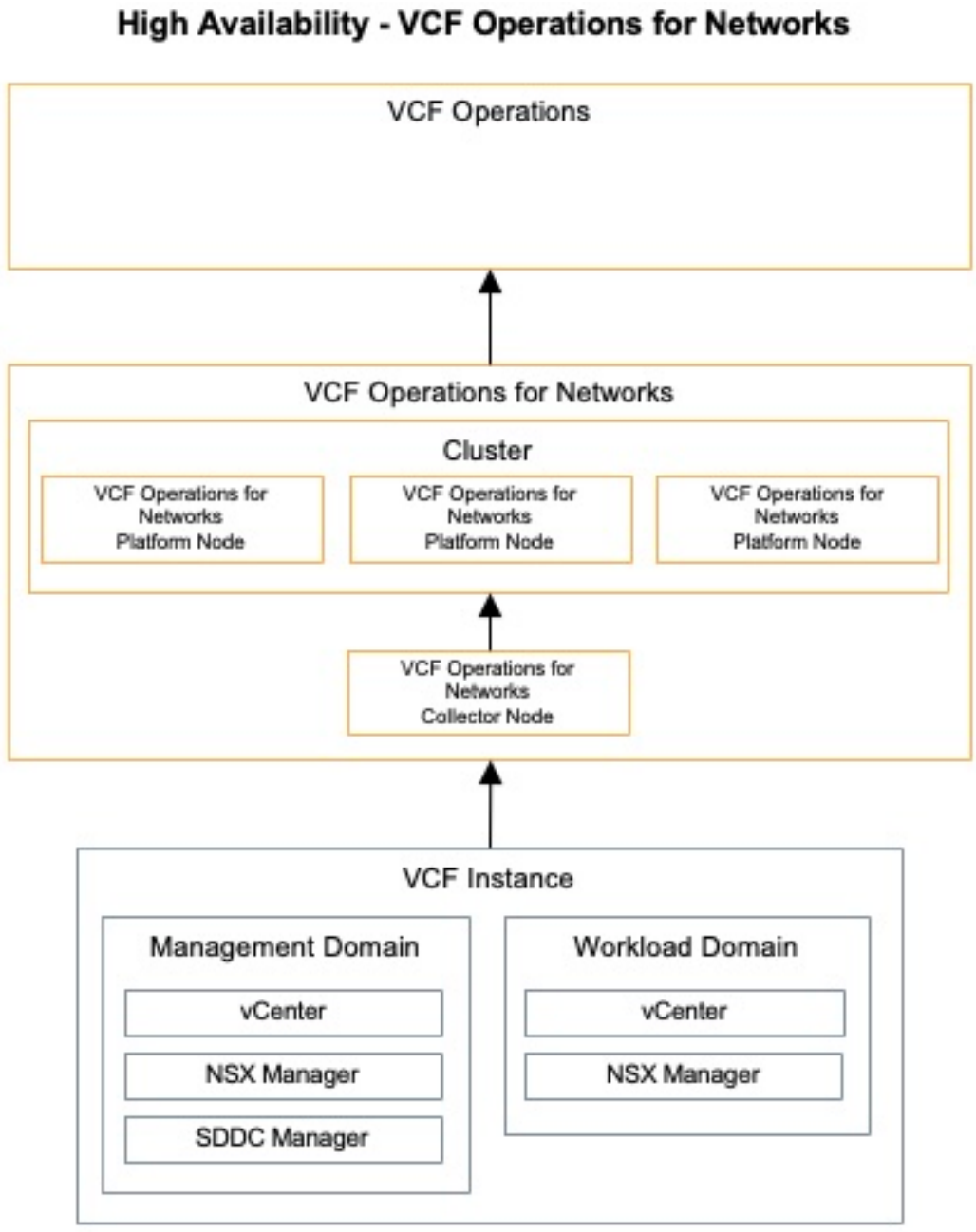
Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

High Availability VCF Operations for Networks Model

The *High Availability VCF Operations for Networks Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
High Availability VCF Operations for Networks Model	The *High Availability VCF Operations for Networks Model* consists of a four-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Figure 132: High Availability VCF Operations for Network Model in VMware Cloud Foundation



High Availability VCF Operations for Networks Model Attributes

A *High Availability VCF Operations for Networks Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out.
Recoverability	Restore through a backup.

High Availability VCF Operations for Networks Model Options

A *High Availability VCF Operations for Networks Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Large • Extra Large

High Availability VCF Operations for Networks Model Design Elements

Table 336: High Availability VCF Operations for Networks Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-HA-001	Deploy threeVCF Operations for networksplatform nodes in a cluster.	• Highly Performant. • Highly scalable, can scale up to 15 Extra Large nodes.	• Cluster does not provide high availability. It is for performance. • Requires higher CPU and memory reservations to support additional nodes.
VCF-NET-REQD-HA-002	Deploy a single VCF Operations for networks collector node in the default management vSphere cluster.	A minimum of one VCF Operations for networks collector node is required to provide the VCF Operations for networks platform node with network monitoring data.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks collector node will be applied.

Table 337: High Availability VCF Operations for Networks Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-001	Deploy appropriately sized VCF Operations for networksplatform nodes for your needs.	Ensures resource availability and usage efficiency.	None.
VCF-NET-RCMD-HA-002	Protect all VCF Operations for networks nodes by using vSphere High Availability.	Supports the availability objective for VCF Operations for networks without requiring manual intervention during an ESX host failure event.	You must ensure sufficient spare capacity for vSphere High Availability failover operations.

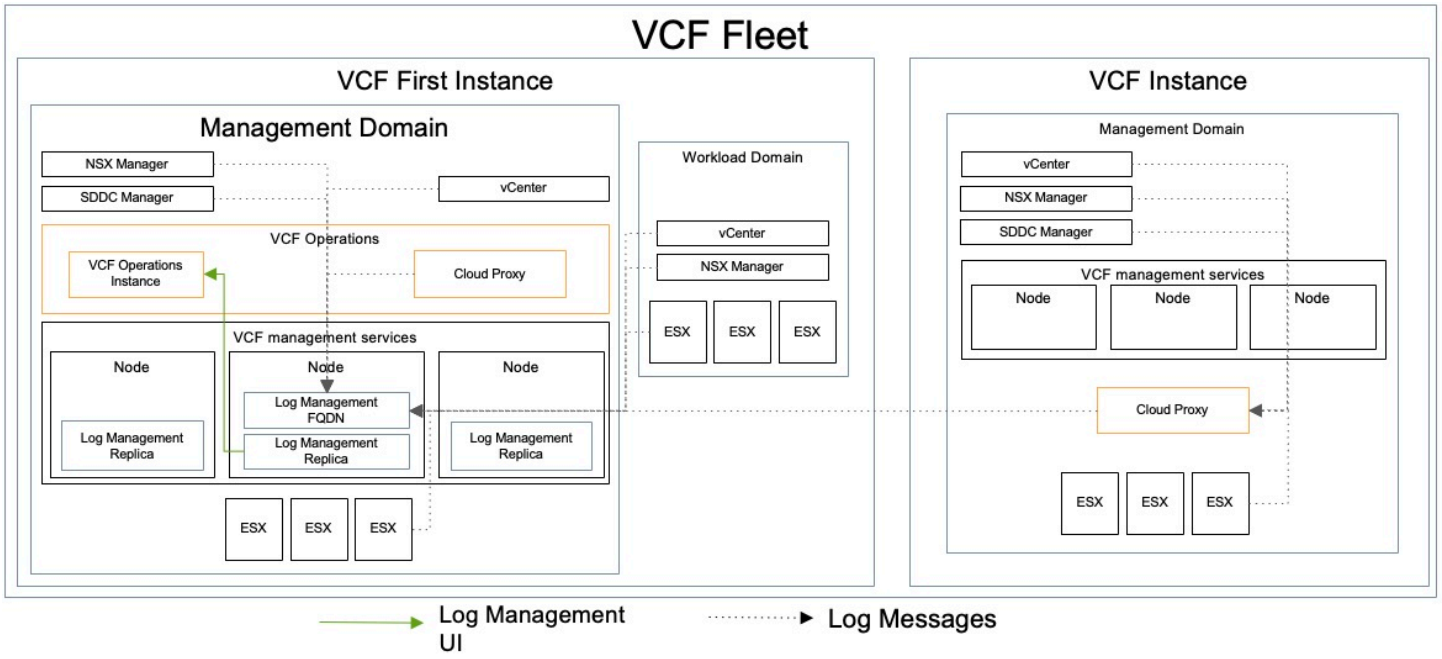
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-003	In an environment with multiple VCF Instances, deploy the VCF Operations for networkscollector node(s) in the default management vSphere cluster in each VCF Instance by using VCF Operations fleet management.	• Ensures a consistent deployment in each VCF Instances regarding network data collection. • It is assumed that each VCF Instances introduces a sufficiently high number of virtual machines, so that the deployment of dedicated VCF Operations for networks collector nodes is justified.	Each VCF Operations for networks collector node must be registered with a VCF Operations for networks platform node.
VCF-NET-RCMD-HA-004	For each vCenter data source, enable NetFlow on each vSphere Distributed Switch within the workload domain.	Provides the collection of network flows via the IPFIX protocol.	For the VCF domains, VCF Operations for networks will automatically update the NetFlow settings for each cluster's vSphere Distributed Switch.
VCF-NET-RCMD-HA-005	For each NSX data source, enable IPFIX for the distributed firewall.	Provides the collection of network flows via the IPFIX protocol.	The distributed firewall service must be enabled on the NSX Local Manager for the VCF workload domain.
VCF-NET-RCMD-HA-006	For each NSX Manager data source, enable latency metric collection.	Provides the collection of latency metrics from NSX Transport Nodes.	Any firewall rule sets from all ESX hosts to the VCF Operations for networks collector must allow traffic on TCP 1991.
VCF-NET-RCMD-HA-007	Use the default retention period of one month for the VCF Operations for networks platform node.	Keeping the default value for retention minimizes required storage resources administrative efforts.	None.
VCF-NET-RCMD-HA-008	Use a CA-Signed certificate containing the fully qualified domain names (FQDNs) of each VCF Operations for networks platform and collector node in the SAN attributes, when deploying VCF Operations for networks.	Configuring a CA-Signed certificate ensures that the communication to the externally facing web UI and API VCF Operations for networks for is encrypted.	• Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered. • Each time a node is added the certificate must be replaced to include the fully qualified domain name of the additional node.
VCF-NET-RCMD-HA-009	Use an SHA-2 or higher algorithm when signing certificates.	The SHA-1 algorithm is considered less secure and has been deprecated.	Not all certificate authorities support SHA-2.

High Availability Logs Management Model

The *High Availability Logs Management Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
High Availability Logs Management Model	The *High Availability Log Management Model* consists of a three-node deployment behind and internal load balancer. This model provides resiliency at both the application level through the clustered deployment and the vSphere cluster level through vSphere HA and vSphere DRS functionality.

Figure 133: High Availability Log Management Model in VMware Cloud Foundation



High Availability Log Management Model Attributes

A *High Availability Log Management Model* has the following attributes.

Attribute	Detail
Centralized management	Provides a centralized log management framework for vCenter, ESX, NSX, VCF Operations, VCF Automation, and identity broker.
Availability	Relies on cluster technology in the event of an outage or VCF management services and vSphere HA to restart nodes in the event of an ESX host failure.

Attribute	Detail
Scalability	Supports scale-up, storage scale up, and scale out.
Recoverability	Restore configuration through a backup, restore logs from archives.

High Availability Log Management Model Options

A *High Availability Log Management Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Small • Medium • Large

High Availability Log Management Model Design Requirements

Table 338: High Availability Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-HA-001	Deploy log management with at least 3 replicas.	• Provides high availability. • Using the integrated load balancer prevents a single point of failure. • Using the integrated load balancer simplifies the log management deployment and subsequent integration.	• You must deploy a minimum of three appropriately sized replicas. • All replicas are sized identically. • If the capacity of your log management cluster must expand, identical capacity is added to each node.
VCF-LOGS-REQD-HA-002	Protect all VCF Management service nodes by using vSphere High Availability.	Supports the availability objectives for log management without requiring manual intervention during an ESX host failure event.	None.

Common Log Management Design Recommendations for All Models

Table 339: Common Log Management Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

High Availability Log Management Model Design Recommendations

Table 340: High Availability Log Management Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-HA-001	Enable HA on the VCF management services control plane. (Instance deployed with the High Availability deployment model, or scale up VCF services runtime to medium or large.)	To support the high availability objective, the control plane for the VCF management services should be configured for HA, as log management depends upon VCF management services.	Requires additional nodes and capacity in the vSphere management cluster.

Security Governance and Compliance Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)

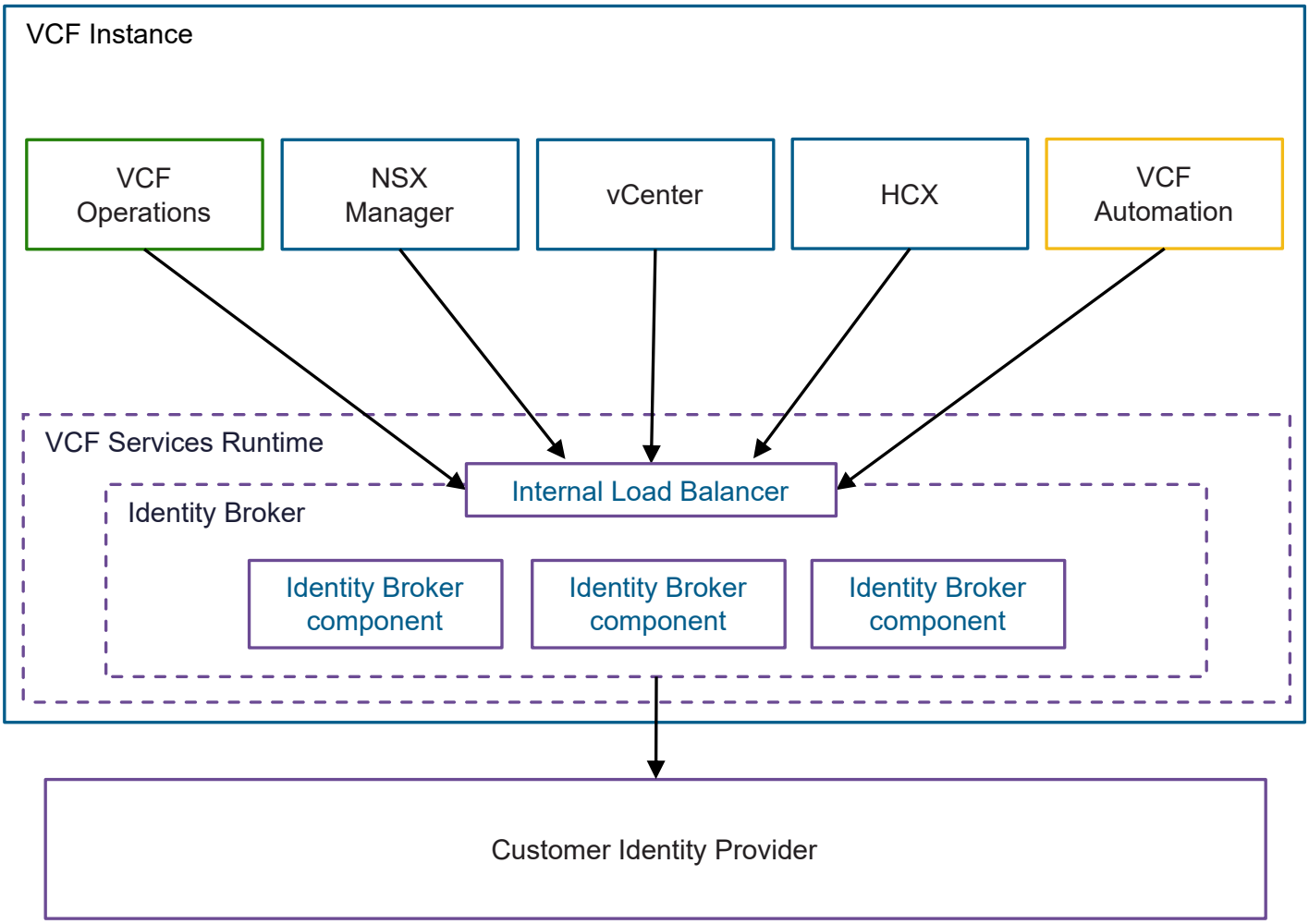
The security governance and compliance layer refers to the framework of policies, procedures, and controls that organizations establish to manage security risks while meeting regulatory, legal, and industry standards around authentication.

Identity Broker Model

The *Identity Broker Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Identity Broker Model	Deploys the Identity Broker component on the VCF management services instance in the management domain.

Figure 134: Instance Identity Broker Model in VMware Cloud Foundation



Instance Identity Broker Model Attributes

The *Instance Identity Broker Model* has the following attributes.

Attribute	Detail
Service Location	Identity broker runs as a service on VCF Management Services.
Deployment mechanism	An identity broker is automatically deployed during the installation or upgrade to VMware Cloud Foundation 9.1 in the first management domain. Additional identity brokers can be deployed in each VCF Instance as part of a day 2 operation.
Number of Nodes	multi-node cluster.
Scale considerations	• Scale up: Identity broker supports small, medium, and large sizes to match different configuration demands. • Scale out: Support for multiple identity broker instances to increase redundancy and throughput. • Recommendations: – Scaling up is preferred over scaling out when a single large-size identity broker meets security, resilience, and performance standards, as it reduces operational complexity and simplifies SSO management. – Schedule scale-up operations within a maintenance window, as these activities can impact active services.
Supported services	For support directories and IdPs, see Identity Providers and Protocols Supported for Single Sign-On.
Backup and restore	Use VCF management services backup and restore process.
Resiliency	Resiliency is provided by a distributed multiple-node deployment running each node on separate vSphere hosts.
Operational Impacts	As an isolated, external service, operational tasks performed on vCenter do not impact the authentication stack of the VMware Cloud Foundation components.
Migration path	Migrating from the appliance to embedded requires a reset of the VCF Single Sign-On configuration. Users must be re-added to components.

Instance Identity Broker Model Design Requirements

Table 341: Instance Identity Broker Model Design Requirements

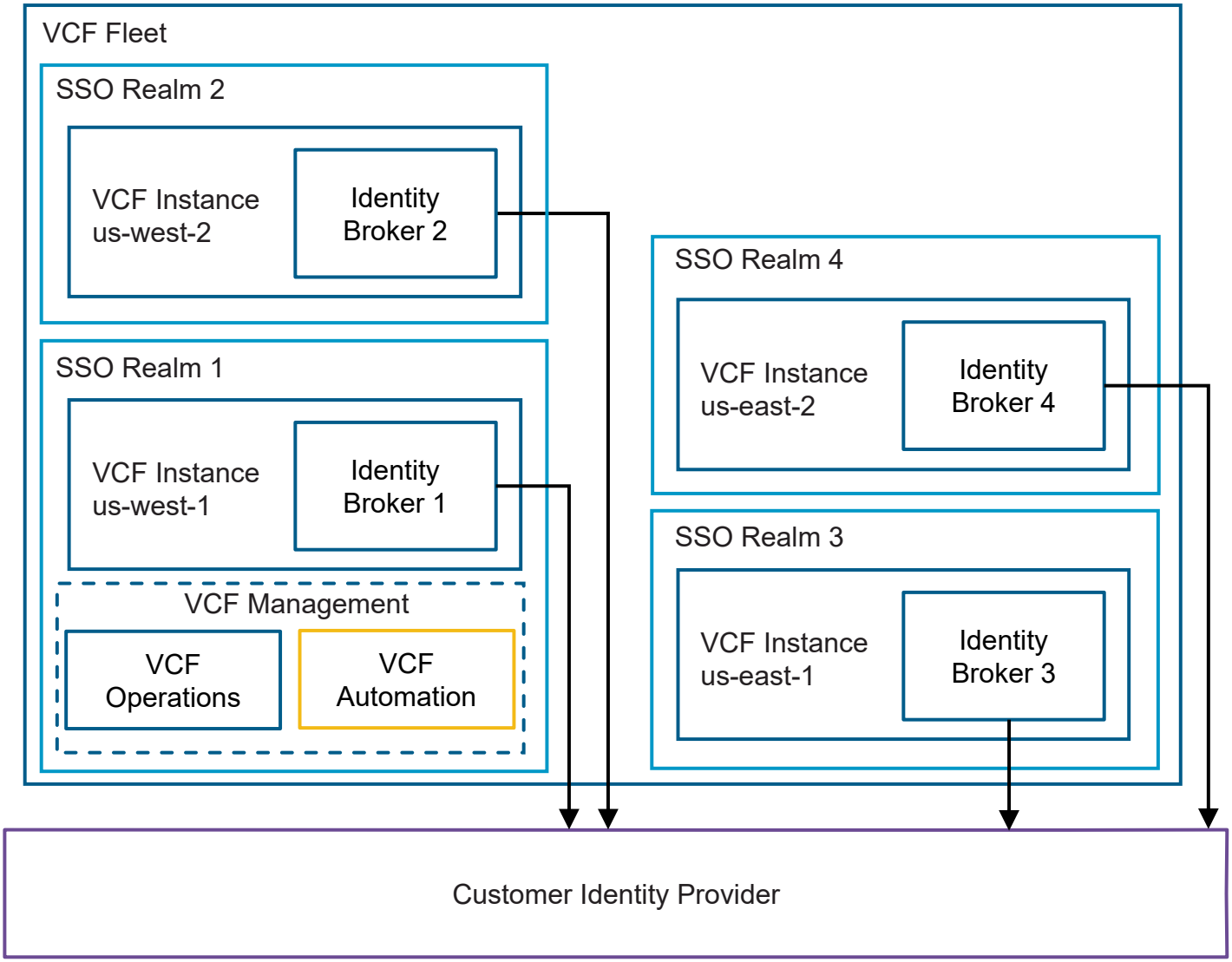
Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-EXT-001	Deploy additional identity broker instances as a Day N operation as needed.	• Provides high availability to the authentication source for a VCF Instance. • Provides additional scale.	Increases the operational overhead of managing the identity broker.
VCF-SSO-REQD-EXT-002	Connect the identity broker service to your corporate IdP.	Provides directory services (users and group) and user authentication service to the identity broker service.	None.

Single VCF Instance Single Sign On Model

The *Single VCF Instance Single Sign On Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Single VCF Instance Single Sign-On Model	With the *Single VCF Instance Single Sign-On Model* each VCF Instance will utilize its own dedicated identity broker. The VCF Single Sign-On scope will be limited to the VCF Instance. VCF management components, such as VCF Operations and VCF Automation, can connect to only one of identity broker instances for Single Sign-On.

Figure 135: Single VCF Instance Single Sign-On Model in VMware Cloud Foundation



Single VCF Instance Single Sign-On Model Attributes

The *Single VCF Instance Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**High Scale -** A single identity broker service for each VCF Instance.
Resources requirements	**Large Resource Footprint** - Each identity broker deployment will require additional compute / memory / storage resources.
Operational Overhead	**High Operational Overhead** - Each identity broker will need to be managed separately for operational tasks such as backup and restore.
Availability	**High Availability per Private Cloud** - A single identity broker servicing each VCF Instance.
Blast Radius	**Small Blast Radius** - An outage of one identity broker instance will only impact logins for the VCF Instance associated with it.
Single Sign-On Realm	**Small SSO Realm** - For each identity broker deployment users will have an additional Single Sign-On realm. Moving across instances will require re-entering credentials.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a only): Option 2b is not a viable option because restoring across different SSO realms is not supported. • Failure Scenario 3 Note: local accounts and VCF SSO emergency access client (for API access) continue to work during outages.

Single VCF Instance Single Sign-On Model Design Requirements

Table 342: Single VCF Instance Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VSI-001	Deploy identity broker in the management domain of the first VCF Instance.	• Provides an authentication service for the first VCF Instance. • Provides VCF Single Sign-On capabilities to the VCF Instance.	All workload domains in a VCF Instance use the same authentication source as the management domain.
VCF-SSO-REQD-VSI-002	Configure VCF Instances to consume the identity broker installed in its management domain.	• Provides an authentication source for all components in the configured VCF Instance. • Provides VCF Single Sign-On capabilities to all components in the configured VCF Instance.	Any outage to the identity broker service impacts all components in the VCF Instance.

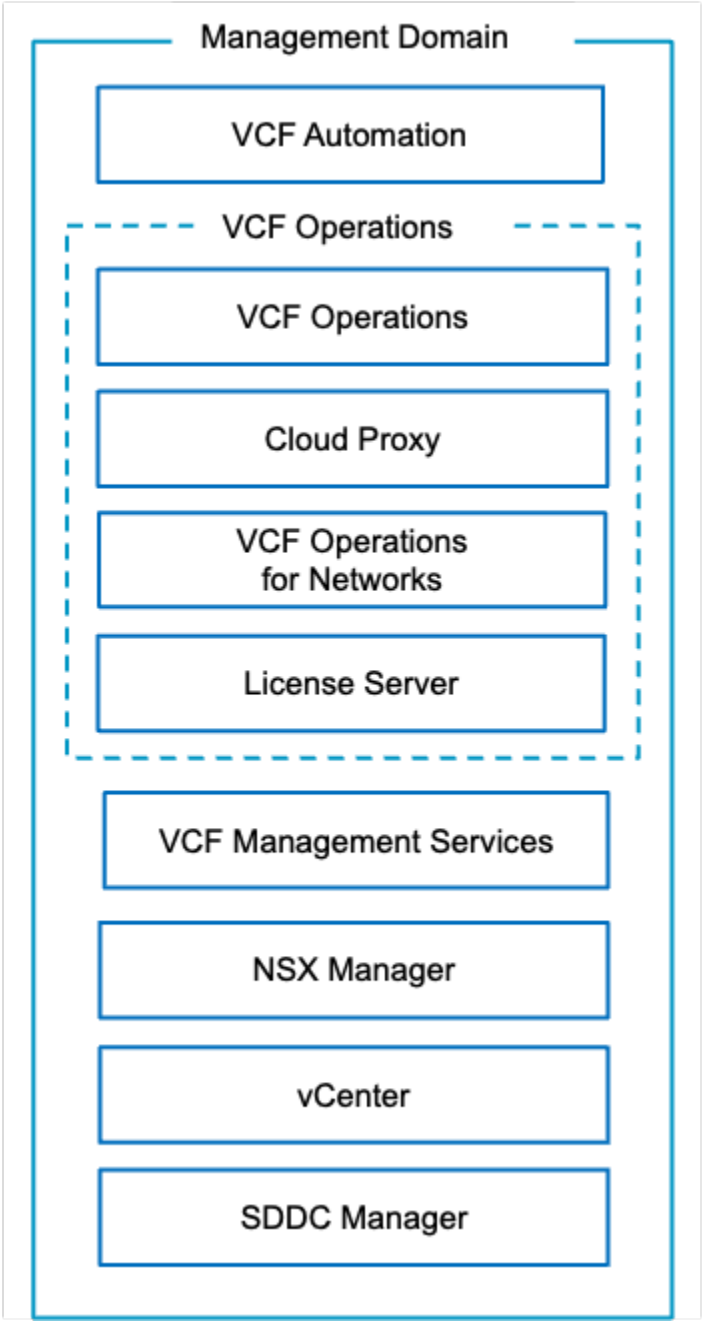
Virtual Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Management Domain Model

The *Management Domain Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Management Domain Model	The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.

Figure 136: Management Domain Model in VMware Cloud Foundation



Management Domain Model Attributes

The *Management Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• VCF fleet management components. • VCF Instance management components. • NSX Edge nodes. • Can also run business workloads.
Management Domain in the First VCF Instance	• The management domain in the first VCF Instance contains the following management components: – VCF management services (fleet and instance level components) – VCF Operations – Cloud Proxy – VCF Operations for Networks – VCF Automation – ESX hosts – vCenter – NSX Manager – SDDC Manager
Management Domains in Additional VCF Instances	• Management domains in additional VCF Instances contain the following instance-level management components: – VCF management services (instance level components) – Cloud Proxy – ESX hosts – vCenter – NSX Manager – SDDC Manager

Management Domain Model Design Requirements

Table 343: Management Domain Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLD-REQD-CFG-001	Deploy a management domain to host all management components for the VCF Instance.	• Guarantees sufficient resources for management components. • Enables the use of specific hardware to meet only the needs of the management components. • Enables the use of dedicated physical compute, network, and storage separately from those used for additional workloads. • Enables separate lifecycle management of management and workload components.	• You must carefully size the management domain to accommodate planned deployments of additional workload domains and management components. • Hardware might not be fully utilized until full-scale deployment has been reached. • The management components for all greenfield VCF domains are deployed to the same cluster, and share the same subnet.

Management Domain Model Design Recommendations

Table 344: Management Domain Model Design Recommendations

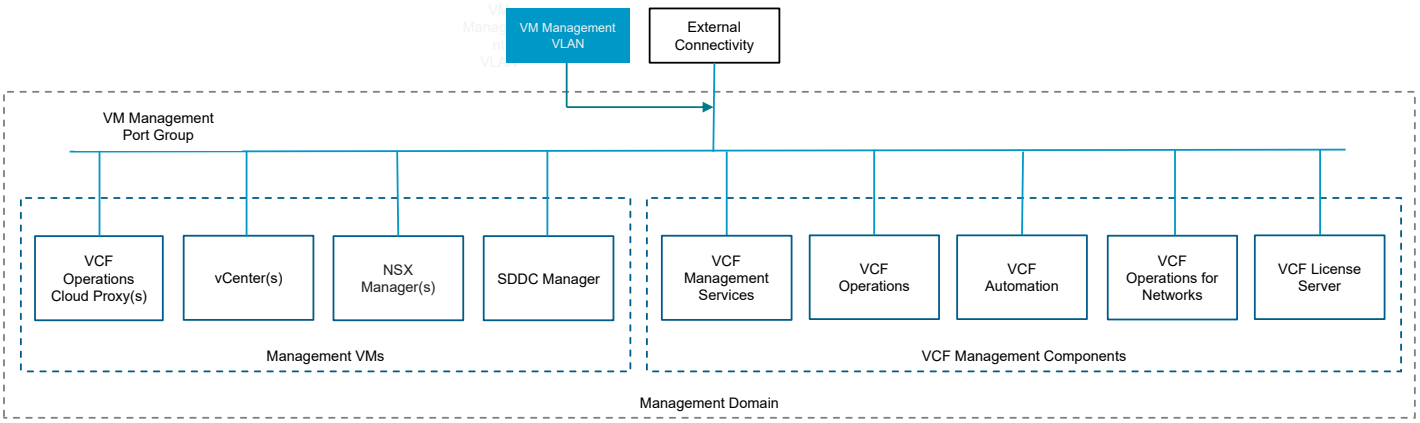
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-RCMD-NSX-001	Use a dedicated NSX Manager instance for the management domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Managers to manage and patch.

VCF Management Shared VLAN Network Model

The *VCF Management Shared VLAN Network Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

VCF Management Shared VLAN Network Model	VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Figure 137: VCF Management Shared VLAN Network Model in VMware Cloud Foundation



VCF Management Shared VLAN Network Model Attributes

A *VCF Management Single VLAN Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	VLAN-backed vSphere distributed port group.
Network Isolation	No; Sharing VLAN with the VCF management domain components.
NSX Edge Cluster Required	No; Optional to provide NSX load balancer for VCF Operations. (requires additional NSX VLAN segment for Load balancer connectivity)
Stretching Network Between Availability Zones	• **VM management network** must be stretched in the physical network fabric between availability zones. • Provides automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• Not suited for a disaster recovery use case as the network is shared with management domain components. • Failover testing is not supported.

VCF Management Shared VLAN Network Model Design Requirements and Recommendations

Table 345: VCF Management Shared VLAN Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-SV-NET-REQD-001	Use the management domain vCenter VM management distributed port group for the VCF fleet level components.	No additional network required.	• No logical network isolation. • Done automatically during VCF Installer deployment.
VCF-MGMT-SV-NET-REQD-002	Use the same VLAN for VCF management components and the VCF Instance management virtual machines. e.g. vCenter and NSX Managers.	No additional VLAN required.	• Separate physical firewall rules may not be possible with a single VLAN. NSX Distributed Firewall may be an alternative solution. • Not suitable for Fleet Disaster Recovery use case with single VLAN.

Table 346: VCF Management Shared VLAN Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-SV-NET-RCMD-001	Deploy the VCF management components during initial deployment using VCF Installer.	Simplified deployment using VCF Installer UI.	None.

Workload Domain Design Selections

The following models in the design align with the objectives for deploying the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Consumption Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
The consumption layer for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint represents the top most tier of a modern data and application architecture, serving as the primary interface between end users and underlying systems. This layer is where business value is realized through direct interaction with processed data, applications, and services.

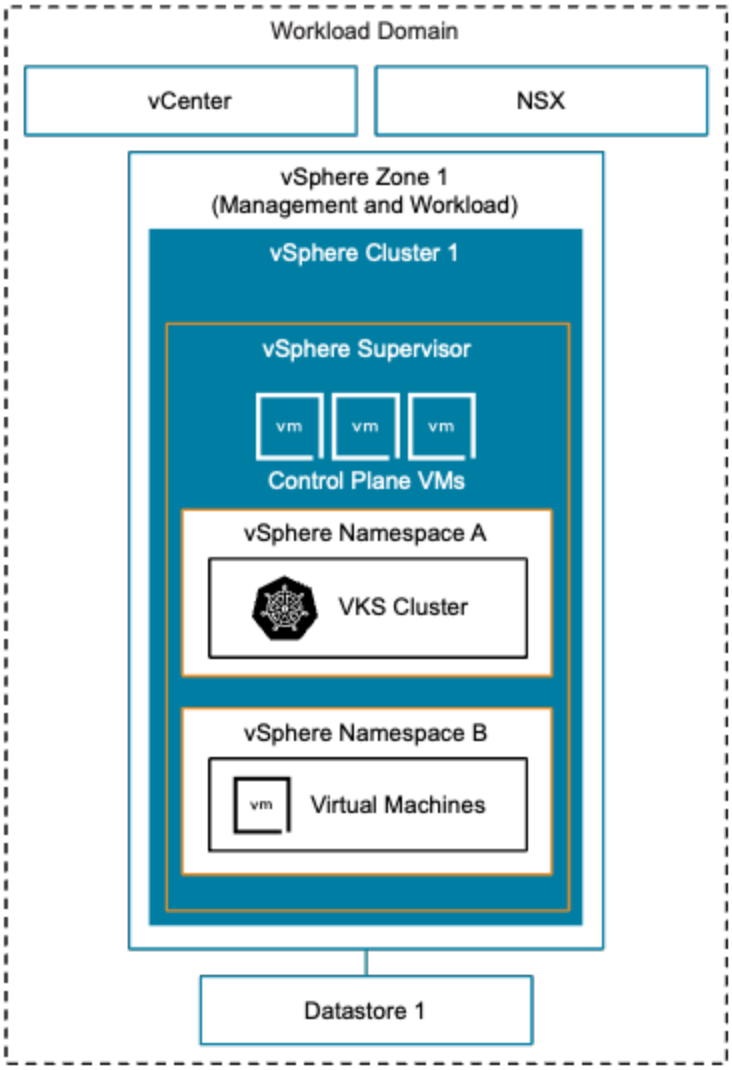
Single Management Zone with Combined Workload Zones Model

The *Single Management Zone with Combined Workload Zones Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Single Management Zone with Combined Workload Zones Model	A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.

Figure 138: Single Management Zone with Combined Workload Zones Model Diagram

This diagram illustrates an example of a *Single Management Zone with Combined Workload Zones Model* deployment.



Single Management Zone with Combined Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single - Combined with management
vSphere Zone Datastore Type	Zonal datastore

Attribute	Detail
Activation	• At workload domain creation. OR • From vCenter.

vSphere Supervisor Design Requirement for all Zone Models

Table 347: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

vSphere Supervisor Single Management Zone Design Requirements

Table 348: vSphere Supervisor Single Management Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-SMZ-REQD-CFG-001	One available vSphere Zone that is backed by a supported vSphere cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	vSphere Zone for Supervisor management workload use.	The selected vSphere Zone is dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.

vSphere Supervisor Design Requirements

Table 349: vSphere Supervisor Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-REQD-CFG-001	Topology aware storage policies created for control plane nodes when deploying in a Multi Zone Deployment model.	Storage policies should have placement considerations that are aligned with the availability of control plane nodes.	Additional storage policies are required.
VCF-SUP-REQD-CFG-002	A single vSAN stretched cluster that is not already assigned to another vSphere Supervisor instance must be available at initial deployment in a vSAN stretched cluster deployment model.	When deploying supervisor on vSAN stretched cluster, the vSANcluster must be stretched before supervisor activation.	vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Common Networking Design Recommendations

Table 350: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

vSphere Supervisor Design Recommendations

Table 351: vSphere Supervisor Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

High Availability vSphere Supervisor Control Plane Model

The *High Availability vSphere Supervisor Control Plane Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
High Availability vSphere Supervisor Control Plane Model	The *High Availability vSphere Supervisor Control Plane Model* deploys three vSphere Supervisor control plane VMs into the selected vSphere Supervisor management zones.

Figure 139: *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model deployment.

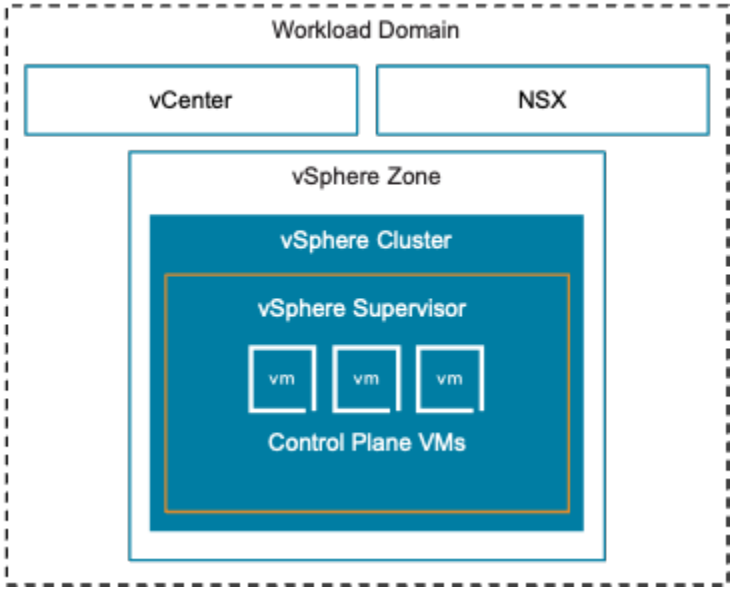
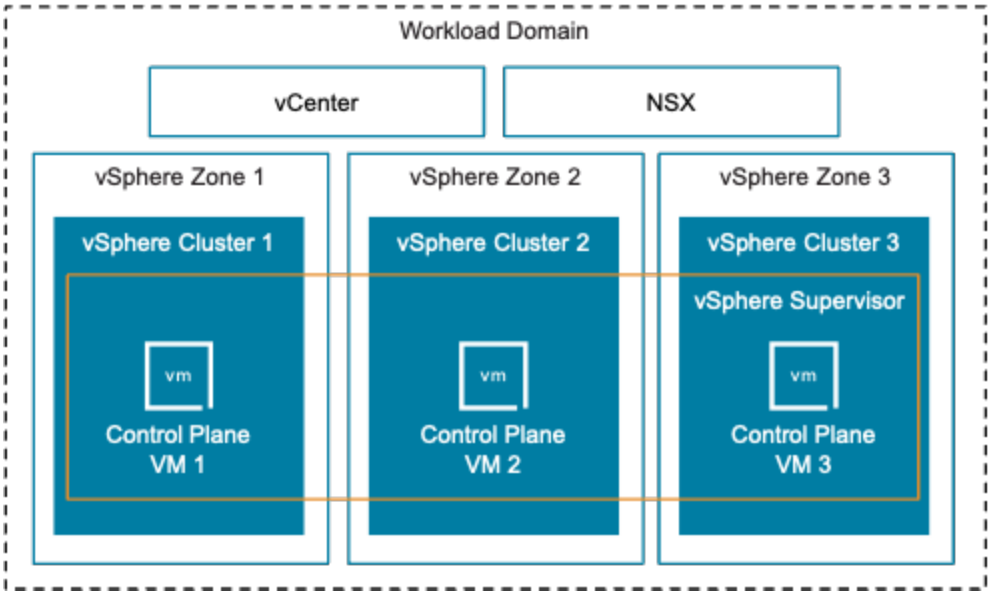


Figure 140: *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model deployment.



High Availability vSphere Supervisor Control Plane Model Attributes

Attribute	Detail
Control Planes Nodes	Three
Availability	• Relies on vSphere HA. • Three management zones, which are protected from single cluster failure.
Scalability	Scale up only
Placement	• Single management zone, all nodes in a single management zone. • Three management zones, one node per management zone.
Applicable Zone Models	• Single Management Zone with Combined Workload Zones Model • Single Management Zone with Isolated Workload Zones Model • Three Management Zones with Combined Workload Zones Model • Three Management Zones with Isolated Workload Zones Model

vSphere Supervisor Control Plane Model Design Requirements

Table 352: vSphere Supervisor Control Plane Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-CP-REQD-CFG-001	Five consecutive management IP addresses routable to vCenter NSX, and optionally Avi management networks if Avi Load Balancer are used.	Required for integration and management communications.	Requires precise IP address management.
VCF-SUP-CP-REQD-CFG-002	Three vSphere Zones are required for the High-Availability Model in a Three Management Zone Model.	Control pane nodes are evenly placed with one per zone to protect against a single vSphere cluster failure.	Additional vSphere zones each back by separate supported vSphere clusters are required.

vSphere Supervisor Control Plane Design Recommendations

Table 353: vSphere Supervisor Control Plane Design Recommendations

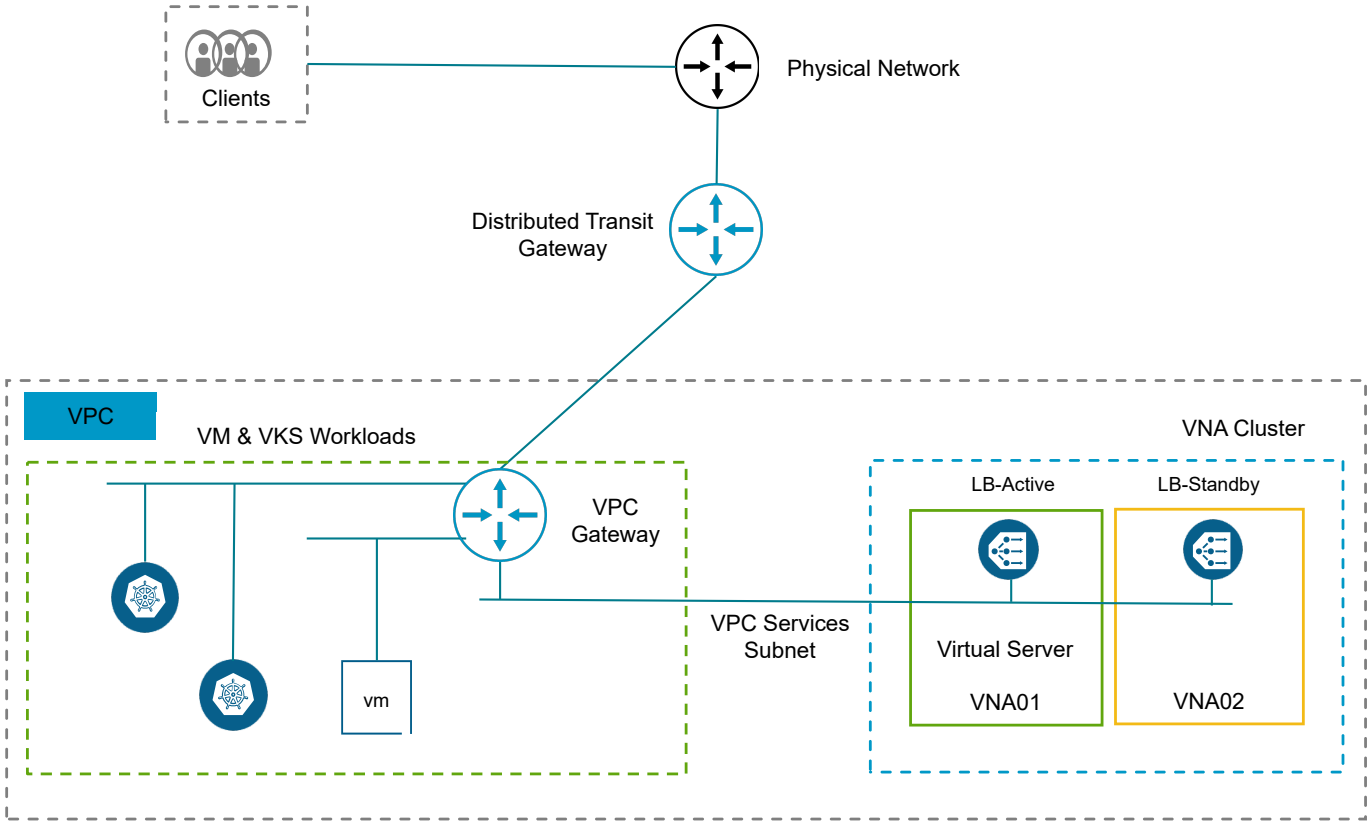
Design Requirement ID	Design Recommendation	Justification	Implication
VCF-SUP-CP-RCMD-CFG-001	Deploy the supervisor control plane in high availability mode with three control plane VMs. *Applicable to Non-Simplified Supervisor models only.	Three control plane VMs allow for continuous availability of the Supervisor control plane during rolling upgrades and single node outages.	Additional resources are required to support the extra control plane VMs.

Virtual Network Appliance Load Balancer Model

The *Virtual Network Appliance Load Balancer Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Virtual Network Appliance Load Balancer Model	The Virtual Network Appliance (VNA) load balancer can be used to provide load balancing for vSphere Supervisor and VKS workloads, it supports VPC-based networking.

Figure 141: Virtual Network Appliance Load Balancer Model in VMware Cloud Foundation



Virtual Network Appliance Load Balancer Model Attributes

A *Virtual Network Appliance Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Load Balancing for VCF infrastructure components (i.e. VCF appliances) • Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services)
Networking Models Supported	VPC Networking Required
Entitlement	Included with VMware Cloud Foundation
Availability	Active/Standby
Scalability	• Scale up Virtual Network Appliance nodes in the Virtual Network Appliance cluster • Scale out a single Virtual Network Appliance Cluster • Create additional Virtual Network Appliance clusters
Deployment Model	Virtual Network Appliance Cluster with two or more Virtual Network Appliances. See Virtual Network Appliance Cluster Detailed Design for further details.

Virtual Network Appliance Load Balancer Model Options

Design Area	Options
Number of Virtual Network Appliances per cluster	1 to 10
Deployment Options	• Deploy with a new workload domain • Deploy Day 2 from NSX Manager • Deploy Day 2 from vCenter
Virtual Network Appliance Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger scale depoloyments where a large number of load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

Virtual Network Appliance Load Balancer Design Requirements

Table 354: Virtual Network Appliance Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-LB-REQD-CFG-01	Create a Virtual Network Appliance cluster with two or more Virtual Network Appliance nodes.	Provides availability for a load balancer instance.	None.
VCF-VNA-LB-REQD-CFG-02	Configure the VPC connectivity profile to use Virtual Network Appliance cluster in the VPC Service Gateway Configurations.	Enables the VPC to use the Virtual Network Appliance cluster for load balancer instances.	Virtual Network Appliance cluster can be shared between VPCs.
VCF-VNA-LB-REQD-CFG-03	Create a load balancer instance for the VPC using the API	Supervisor and VKS-based workloads, load balancer creation is automated.	No UI available to configure load balancer instances.
VCF-VNA-LB-REQD-CFG-04	Create a load balancer configuration for VPC using the API.	Supervisor and VKS-based workloads, load balancer configuration is automated.	No UI available to configure load balancer instances.

Virtual Network Appliance Load Balancer Design Recommendations

Table 355: Virtual Network Appliance Load Balancer Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-LB-RCMD-CFG-01	Configure failures domain for Virtual Network Appliance clusters with more than two Virtual Network Appliance nodes in the cluster.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-VNA-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure VPC load balancer availability by controlling Actve/ Standby placement.	Manually configured from vCenter UI.

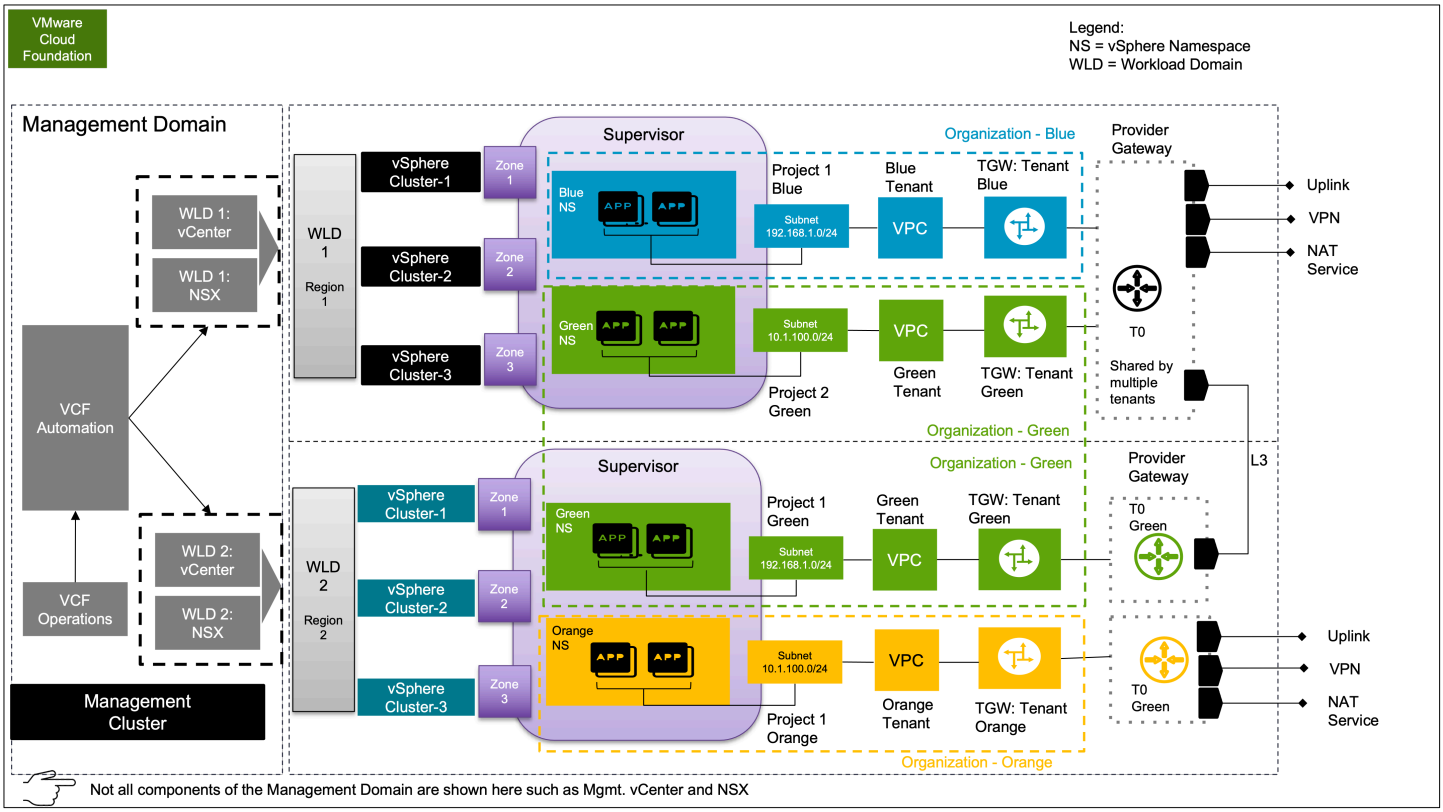
Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways

The *Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways	This design provides multi-tenant resource isolation across multiple regions by combining dedicated gateways for strict tenant separation with shared gateways for efficient common services.

Multi-Tenancy Consumption Pattern 5 Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways - Centralized Gateway (CTGW) Design

Figure 142: Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways - Centralized Gateway (CTGW) Design



Multi-Tenancy Consumption Pattern 5 Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways - Centralized Gateway (CTGW) Design Attributes

Design Attributes	Details	Diagram Representation
Management Domain	One	Centralized Management: Includes VCF Automation, workload domain vCenter, NSX, and VCF Operations. Consist of a vSphere Cluster -the underlying hardware for the management domain.
Workload Domains	Two	A workload domain with three compute vSphere clusters for tenant workloads.
NSX Local Managers	Two	Each workload domain has one NSX local manager
Organizations	Three	Represents three separate Organizations. Such as Blue, Green and Orange
Number of vSphere Clusters	Six	Six compute and vSAN storage cluster used in two workload domain construct.
Number of vSphere Supervisors	Six	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	Six	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Projects	Three	Represented by "Project 1 - User" and "Project 2- User". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One in each region	A single T0 gateway provides external connectivity, including Uplink, VPN, VRF (Virtual Routing and Forwarding), and NAT services.
VRFs	Two	Tenant "Green" and "Orange" has a dedicated VRF instance in a T0. Both tenant sharing a T0 but still they have network isolation.
vSphere Namespaces	Two	Blue and Green tenant has dedicated vSphere Namespace for workload isolation.
VPCs	One	Each LOB/ tenant sharing a single VPC
TGWs (Transit Gateway)	Two	Each tenant has a dedicated TGW, "TGW: Tenant Blue" and "TGW: Tenant Green", for connecting their respective VPCs.
Tenant Networks	Four	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

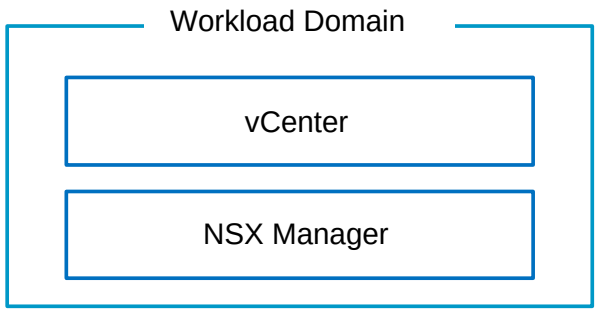
Virtual Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

Workload Domain Model

The *Workload Domain Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Workload Domain Model	As you expand your VCF fleet, you can add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.

Figure 143: Workload Domain Model in VMware Cloud Foundation



Workload Domain Model Attributes

A *Workload Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• NSX Edge nodes. • Business workloads.
vCenter Single Sign-On Domain	Dedicated.

Workload Domain Model Design Recommendations

Table 356: Workload Domain Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLD-RCMD-CFG-002	Use workload domains for customer workloads.	• Aligns with the VMware best practice of separating management workloads from customer workloads. • Provides better long-term flexibility and expansion options.	Requires additional hardware.
VCF-WLD-RCMD-NSX-002	Use a dedicated or shared NSX Manager instance for the workload domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Manager to manage and patch.

Dual Availability Zone Design Based on NSX Edge Node HA Model

The *Dual Availability Zone Design Based on NSX Edge Node HA Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Dual Availability Zone Design Based on NSX Edge Node HA Model	A *Single-Rack Cluster Model* for hosting NSX Edge nodes offers a minimalist footprint while maintaining basic host-level availability.

Figure 144: Dual Availability Zone Design Based on NSX Edge Node HA Model (Dedicated vSphere Cluster) in VMware Cloud Foundation

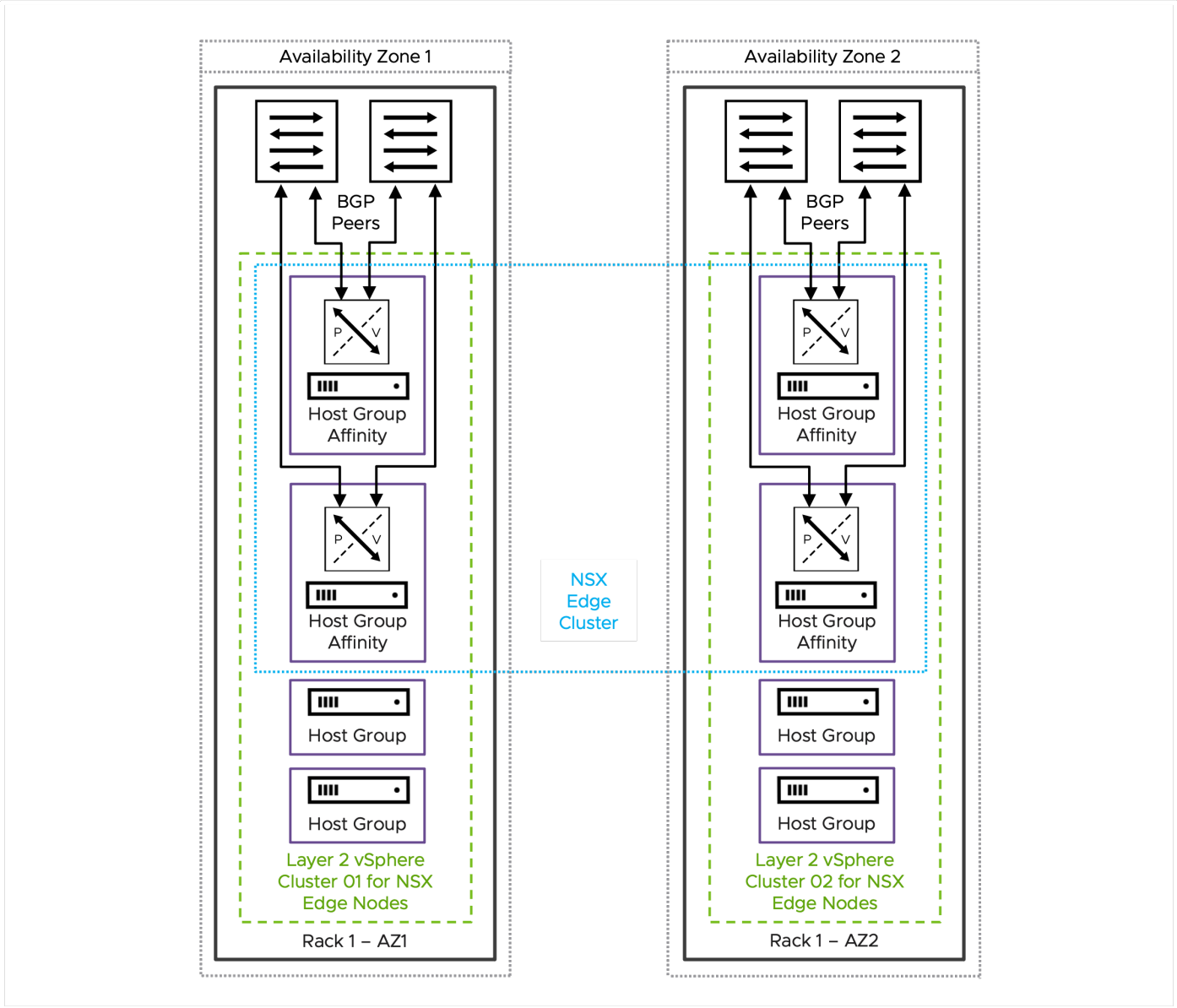
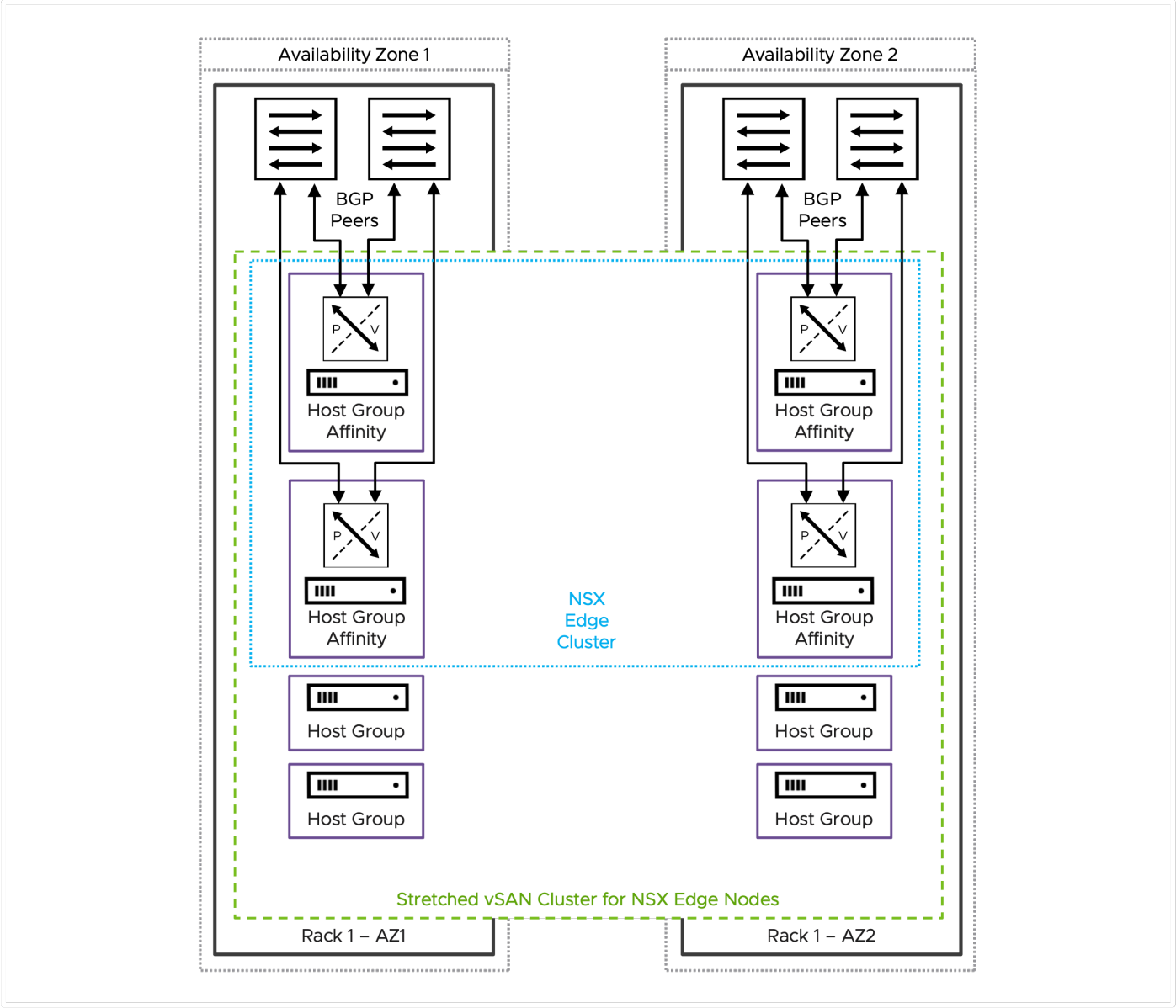


Figure 145: Dual Availability Zone Design Based on NSX Edge Node HA Model (Stretched vSAN) in VMware Cloud Foundation



Dual Availability Zone Design Based on NSX Edge Node HA Model Attributes

The *Dual Availability Zone Design Based on NSX Edge Node HA Model* has the following attributes.

Attribute	Details
Availability Model	• Host and availability zone redundancy. • Independent shared storage per vSphere cluster when using vSAN or single shared storage. • This design ensures continuous north-south communication and centralized network services even if one availability zone completely fails.
NSX Edge Node Mobility	• Mobility is not supported across racks/availabilty zones when using multiple instances of the Single-Rack vSphere Cluster Model. Mobility is not supported across availability zones when using the Stretched vSphere Cluster Model. • Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same availability zones that hosts the NSX Edge node.
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Add more hosts to both vSphere clusters simultaneously or scale out the stretched vSphere cluster with additional hosts in each availability zone. • When deploying more NSX Edge nodes on existing ESX hosts, ensure each ESX host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Common NSX Edge Cluster for Dual Availablity Zones Design Requirements

Table 357: Common NSX Edge Cluster for Dual Availability Zones Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMD AZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric settings.

Dual Availability Zone Design based on NSX Edge Node HA Model Design Requirements

Table 358: Dual Availability Zone Design based on NSX Edge Node HA Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-DAZE HA-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High-Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Common NSX Edge Cluster for Dual Availability Zones Design Recommendations

Table 359: Common NSX Edge Cluster for Dual Availability Zones Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node ESX hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMDAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or ESX hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload virtual machines have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart after ESX host failures if it cannot meet their memory reservations on surviving ESX hosts. • Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMDAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are generally less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMDAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.
VCF-NSX-EDGE-RCMD-COMDAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMDAZ-008	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across two dedicated vSphere clusters respective across two availability zones.	Maintain High Availability and prevent single points of failure.	• Additional servers are required. • This design recommendation is not applicable to the design model based on vSphere HA Recovery
VCF-NSX-EDGE-RCMD-COMDAZ-009	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload VMs and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-010	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMDAZ-011	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMDAZ-012	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge nodes and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge nodes, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSXon dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMDAZ-013	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two (2) data plane fastpath interfaces get an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP).	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMDAZ-014	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (physical link state up but no forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-015	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard or the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-016	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Dual Availability Zone Design Based on NSX Edge Node HA Model Design Recommendations

Table 360: Dual Availability Zone Design based on NSX Edge Node HA Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZEHA-001	Limit the span of NSX Edge node networks to the local rack.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP sessions.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.
VCF-NSX-EDGE-RCMD-DAZEHA-002	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	• Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow. • This design recommendation is not applicable with routing option C - Active/Standby Tier-0 gateway with symmetric routing.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZE HA-003	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	• The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-004	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESX host.	• Increased configuration effort. • If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-005	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-006	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZE HA-007	Create a compute policy type Best Effort Restart (BER) and assign the previously created vCenter tag and category to it.	• The compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-008	Configure an NSX failure domain for each availability zone where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability in case of a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Shared Design Selection between Management and Workload Domain Region A

Shared Design Selections between Management and Workload Domain are the models that are the same across the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
The virtual infrastructure layer represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

vCenter Model
The *vCenter Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

vCenter Model Attributes

A *vCenter Model* has the following attributes.

Design Area	Attributes
vCenter Deployment Type	• One vCenter instance for the management domain that manages the management components of the VCF Instance. • Optionally, additional vCenter instances for workload domains to support workloads. • vSphere HA protecting all vCenter appliances.
vCenter Single Sign-On	• Each vCenter has a distinct vCenter Single Sign-On domain.

Methods for Protecting vCenter

VMware Cloud Foundation supports multi-rack management vSphere clusters, vSAN stretched clusters, or 3rd party synchronous replication solutions such as metro clusters that are designed to provide disaster recovery and/or high availability for the entire management stack (vCenter, NSX Manager, etc.) as opposed to a single vCenter.

Table 361: Methods for Protecting vCenter in VMware Cloud Foundation

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere High Availability	Yes	You must ensure sufficient resources in the cluster to allow virtual machines to restart on another ESX host in the event of an ESX host failure.
vCenter High Availability (vCenter HA)	No	• vCenter services must fail over to the passive node so there is no continuous availability. • Recovery time can be up to 15 mins. • You must meet additional networking requirements for the private network. • vCenter HA requires additional resources for the passive and witness nodes. • Lifecycle management is complicated because you must manually delete and recreate the standby virtual machines during a lifecycle management operation.
vSphere Fault Tolerance (vSphere FT)	No	• The vCPU limit of vSphere FT vCPU would limit vCenter appliance size to medium. • You must provide a dedicated network.

vCenter Model Design Requirements and Recommendations

Table 362: vCenter Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-CFG-001	Deploy a dedicated vCenter appliance for the management domain of the VCF Instance.	• Isolates vCenter failures to management or customer workloads. • Isolates vCenter operations between management and business workloads. • Supports a scalable vSphere cluster design where you can reuse the management components as more customer workloads are added to the VCF Instance. • Simplifies capacity planning for customer workloads because you do not consider management workloads for the workload domain vCenter. • Improves the ability to upgrade the vSphere environment and related components by enabling for explicit separation of maintenance windows: — Management workloads remain available while you are upgrading the tenant workloads. — Workloads remain available while you are upgrading the management nodes. • Supports clear separation of roles and responsibilities to ensure that only administrators with granted authorization can control the management workloads. • Facilitates quicker troubleshooting and problem resolution. • Simplifies disaster recovery operations by supporting a clear separation between recovery of the management components and tenant workloads. • Provides isolation of potential network issues by introducing network separation of the vSphere clusters in the VCF Instance.	The vSphere cluster must have sufficient resources.
VCF-VCS-REQD-NET-001	Place all workload domain vCenter appliances on the VM management network in the management domain.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-SSO-001	Create all vCenter instances within a VCF Instance in their own unique vCenter Single Sign-On domain.	• Sharing vCenter Single Sign-On domains is not supported. • Enables isolation at the vCenter Single Sign-On domain layer for increased security separation.	• Each vCenter instance is managed through its own pane of glass using a different set of administrative credentials. • You must manage password rotation for each vCenter Single Sign-On domain separately.

Table 363: vCenter Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VCS-RCMD-CFG-001	Deploy an appropriately sized vCenter appliance for each workload domain.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate compute resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-002	Deploy a vCenter appliance with the appropriate storage size.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate storage resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-003	Protect workload domain vCenter appliances by using vSphere HA.	vSphere HA is the only supported method to protect vCenter availability in VCF.	vCenter becomes unavailable during a vSphere HA failover.
VCF-VCS-RCMD-CFG-004	In vSphere HA, set the restart priority policy for the vCenter appliance to high.	vCenter is the management and control plane for physical and virtual infrastructure. In a vSphere HA event, to ensure that the rest of the SDDC management stack comes up faultlessly, the workload domain vCenter must be available first, before the other management components come online.	If the restart priority for another virtual machine is set to highest, the connectivity delays for the management components are longer.

ESX Model

The *ESX Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCF domain vSphere cluster.

ESX Hardware and Sizing Considerations

For more information on host sizing, refer to the VCF Planning and Preparation Workbook.

Table 364: ESX Hardware and Sizing Considerations

Hardware Element	Considerations
BIOS/Firmware	Ensure the ESX host hardware is on the Hardware Compatability List (HCL) and that you are running the latest supported BIOS and firmware.
CPU	• Total CPU requirements for the workloads that are running in the cluster. • Host failure and maintenance scenarios. • CPU over-commitment ratio guidelines – Management Domain • To ensure optimal performance and maintain resilience during host failures or maintenance, keep the management domain vCPU-to-pCPU over-commitment ratio at or below 2:1. **Note:** While higher ratios such as 3:1 or 4:1 are supported, you must closely monitor CPU ready times to ensure that the management cluster has enough capacity to run the workloads. – Workload domains • CPU over-commitment ratios for workload domains should be calculated on a case by case basis depending on the CPU intensity of the workloads. You must closely monitor CPU ready times to ensure optimal performance. • Number of physical cores, not logical cores. – Simultaneous multi-threading (SMT) in CPUs improve CPU performance however the performance enhancement will not be equivalent to 100% more CPU power.
Memory	• Total memory requirements for the workloads that are running in the cluster. • When sizing memory for the ESX hosts in a cluster, to reserve the resources of one host for failover or maintenance, set the admission control setting to N+1, which reserves the resources of one host for failover or maintenance. • *vSAN* OSA (Original Storage Architecture). – To support the maximum number of disk groups, you must provide 32 GB of RAM. • *vSAN* ESA (Express Storage Architecture). – You must provide a minimum of 128GB of RAM.
Storage	• Use high-endurance devices such as a hard drive or SSD for boot device. • Use a 128GB or greater boot device to maximize the space available for ESX-OS Data. • Use hosts with homogeneous configuration.
Network	• Use multiple physical NICs connected to multiple ToR switches to ensure redundancy. • 25GbE is recommended • Use physical NICs of the same speed, vendor, and model across all host within a cluster.
Power	• Use redundant power supplies, preferably connected to separate circuits to ensure availability. • Consider the impact of enabling Power Management on your ESX hosts. In order to allow ESX to control CPU power-saving features you need to set power management in the BIOS to *OS Controlled Mode*.

ESX Model Design Requirements and Recommendations

In your ESX design, you apply certain best practices.

Table 365: ESX Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-ESX-REQD-CFG-001	Install no less than the minimum number of ESX hosts required for the vSphere cluster model being deployed.	• Ensures that availability requirements are met. • If one of the ESX hosts is not available because of a failure or maintenance event, the CPU over-commitment ratio becomes 2:1.	None.
VCF-ESX-REQD-CFG-002	Ensure each ESX host matches the required CPU, memory, and storage specification.	Ensures that workloads will run without contention even during failure and maintenance conditions.	Assemble the server specification and number according to the sizing in the VCF Planning and Preparation Workbook which is based on projected deployment size.
VCF-ESX-REQD-SEC-001	Regenerate the certificate of each ESX host after assigning the ESX host an FQDN.	Establishes a secure connection with the VCF Installer during the initial deployment and prevents man-in-the-middle (MiTM) attacks.	You must manually regenerate the certificates of the ESX hosts before the deployment.

Table 366: ESX Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-001	When using vSAN as primary storage in the management domain, use vSAN ReadyNodes for each ESX host.	Your management domain is fully compatible with vSAN at deployment. **Note:** For information about the models of physical servers that are vSAN-ready, see vSAN Compatibility Guide for vSAN ReadyNodes.	• Hardware choices might be limited. • If you plan to use a server configuration that is not a vSAN ReadyNode, your CPU, disks, and I/O modules must be listed on the VMware Compatibility Guide under *CPU Series* and *vSAN Compatibility List* aligned to the ESX version.
VCF-ESX-RCMD-CFG-002	Allocate ESX hosts with uniform configuration across the default management vSphere cluster.	A balanced vSphere cluster has these advantages: • Predictable performance even during hardware failures. • Minimal impact of resynchronization or rebuild operations on performance.	You must apply vendor sourcing, budgeting, and procurement considerations for uniform server nodes on a per vSphere cluster basis.
VCF-ESX-RCMD-CFG-003	When sizing CPU, do not consider multi-threading technology and associated performance gains.	Although multi-threading technologies increase CPU performance, the performance gain depends on running workloads and differs from one case to another.	Because you must provide more physical CPU cores, costs increase and hardware choices become limited.
VCF-ESX-RCMD-CFG-004	Install and configure all ESX hosts in the default management vSphere cluster to boot using a 128-GB device or larger.	Provides ESX hosts that have large memory, that is, greater than 512 GB, with enough space for the scratch partition when using vSAN.	None.

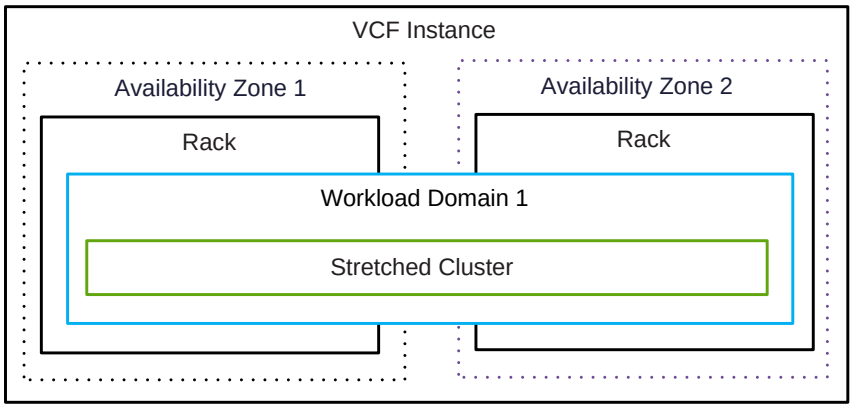
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-005	Use the default configuration for the scratch partition on all ESX hosts in the default management vSphere cluster.	• If a failure in the vSAN cluster occurs, the ESX hosts remain responsive and log information is still accessible. • If using vSAN storage its not possible to use a vSAN datastore for the scratch partition.	None.
VCF-ESX-RCMD-CFG-006	For workloads running in the default management vSphere cluster, save the virtual machine swap file at the default location.	Simplifies the configuration process.	Increases the amount of replication traffic for management workloads that are recovered as part of the disaster recovery process.
VCF-ESX-RCMD-NET-001	Place the ESX hosts in each management domain vSphere cluster on the ESX management network that is separate from the VM management network.	• Enables the separation of the physical VLAN between ESX hosts and the other management components for security reasons. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Increases the number of VLANs required.
VCF-ESX-RCMD-NET-002	Place the ESX hosts in each workload domain on a separate ESX host management VLAN-backed network.	Enables the separation of the physical VLAN between the ESX hosts in different workload domains for security reasons.	Increases the number of VLANs required. For each workload domain, you must allocate a separate management subnet.
VCF-ESX-RCMD-NET-003	Use dedicated VLANs for each traffic type.	Enables the separation of the physical VLAN for vSAN, vMotion, NFS & backup traffic.	Increases the number of VLANs required. For each traffic type you must allocate a separate VLAN and subnet.
VCF-ESX-RCMD-SEC-001	Set the advanced setting `UserVars.SuppressShellWarning` to `0` across all ESX hosts in the management domain.	• Ensures compliance with the *vSphere Security Configuration Guide* and with security best practices • Enables the warning message that appears in the vSphere client every time SSH access is activated on an ESX host.	You must turn off SSH enablement warning messages manually when performing troubleshooting or support activities.

Stretched vSphere Cluster Model

The *Stretched vSphere Cluster Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Stretched vSphere Cluster Model	The *Stretched vSphere Cluster Model* allows you to protect workloads on vSphere clusters in your environment against a failure of a single fault domain by implementing a stretched vSphere cluster across multiple availability zones.

Figure 146: Stretched vSphere Cluster Model in VMware Cloud Foundation



Stretched vSphere Cluster Model Attributes

A *Stretched vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Multiple datacenters
Cluster Rack Mapping	• VCF domain cluster with multiple availability zones, each zone in a single-rack vSphere cluster. • VCF domain cluster with multiple availability zones, with Layer 2 multi-rack vSphere clusters in each zone.
Resilience	• vSphere HA protects workloads against host failures. • Multiple availability zones protect against availability zone failures.
Bandwidth	• When using vSAN stretched clusters, the bandwidth between the zones must be at least 10 Gbps and the round-trip latency must be less than 5 ms. • When using storage array based stretched clusters, you must work with your storage vendor to define the bandwidth and latency requirements.
Management Domain Requirements	Having the management domain on a stretched cluster is a prerequisite to configure and implement stretched cluster in a workload domain. In this way, the critical management components managing the workload domains are still available if an availability zone failure occurs.
Availability Zones	• You can have up to two availability zones associated with a given stretched vSphere cluster. – When using vSAN stretched clusters, a third independent fault domain is required to host the vSAN witness appliance.

Stretched vSphere Cluster Model VLANs and Subnets

Function	VLANs and Subnets
VM management	Shared across availability zones
VCF management	Shared across availability zones
ESX management	Unique per availability zone
vSphere vMotion	Unique per availability zone
vSAN*	Unique per availability zone
Host overlay	Unique per availability zone
NFS*	Depends on stretched storage design

Note:

*Optional based on the principal storage choice for the cluster.

Stretched vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere cluster which supports your storage type and workload domain considerations.

Storage Type	VCF Installer Deployment Model	Management Domain (First Cluster)	Management Domain (Additional Clusters) Workload Domain (All Clusters)
vSAN	Simple	Six (6)	Six (6)
	Highly Available	Eight (8)	
NFS, FC *	Simple	Refer to KB417356	**Minimum:** • One (1) host in Site A • One (1) host in Site B **Total: 2 ESX hosts** **Recommended:** • Two (2) hosts in Site A • Two (2) hosts in Site B **Total: 4 ESX hosts**
	Highly Available	Refer to KB417356	

Important:

vSphere clusters are configured allowing for the failure of a single availability zone while still having enough resources to run all vSphere cluster workloads.

Attention:

* Storage and networking design requirements should be completed in consultation with your storage vendor.

Stretched vSphere Cluster Model Options

A *Stretched vSphere Cluster Model* can apply options from the design areas that are listed below, depending on your objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Storage Models (principal)	• Stretched vSAN Storage Model • Stretched vSAN Storage Cluster Storage Model • Stretched vSAN Compute Cluster Storage Model • Fibre Channel Storage Model

Common vSphere Cluster Design Requirements

Table 367: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Table 368: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Stretched vSphere Cluster Model Design Requirements

Table 369: Stretched vSphere Cluster Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-005	Enable the **Override default gateway for this adapter** setting on the vSAN VMkernel adapters on all ESX hosts.	Enables routing the vSAN data traffic through the vSAN network gateway rather than through the management gateway.	vSAN networks across availability zones must have a route to each other.
VCF-CLS-REQD-CFG-006	Create a host group for each availability zone and add the ESX hosts in the zone to the respective group.	Makes it easier to manage which virtual machines run in which availability zone.	None.

Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Requirements

Table 370: Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements for a stretched cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-SC-REQD-CFG-001	Create an uplink profile for each availability zone.	Enables different VLAN ids for Host TEP in each availability zone.	Requires additional Host TEP VLAN in availability zone 2.
VCF-VNI-SC-REQD-CFG-002	Create NSX host sub transport node profile for second availability zone.	Enables NSX host transport node configuration per availability zone.	None.

Common vSphere Cluster Design Recommendations

Table 371: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Table 372: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Stretched vSphere Cluster Model Design Recommendations

Table 373: Stretched vSphere Cluster Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-008	Increase admission control percentage to half of the ESX hosts in the vSphere cluster.	Allocating only half of a stretched cluster ensures that all VMs have enough resources if an availability zone outage occurs.	• In a vSphere cluster of eight (8) ESX hosts, the resources of only four (4) ESX hosts are available for use. • If you add more ESX hosts to the default management vSphere cluster, add them in pairs, one per availability zone.
VCF-CLS-RCMD-CFG-009	Create a virtual machine group for each availability zone and add the virtual machines in the zone to the respective group.	Ensures that virtual machines are located only in the assigned availability zone to avoid unnecessary vSphere vMotion migrations.	You must add virtual machines to the allocated group manually.
VCF-CLS-RCMD-CFG-010	Create a should-run-on-hosts-in-group VM-Host affinity rule to run each group of virtual machines on the respective group of hosts in the same availability zone.	Ensures that virtual machines are located only in the assigned availability zone to avoid unnecessary vSphere vMotion migrations.	You must manually create the rules.

Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Recommendations

Table 374: Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations for a stretched cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

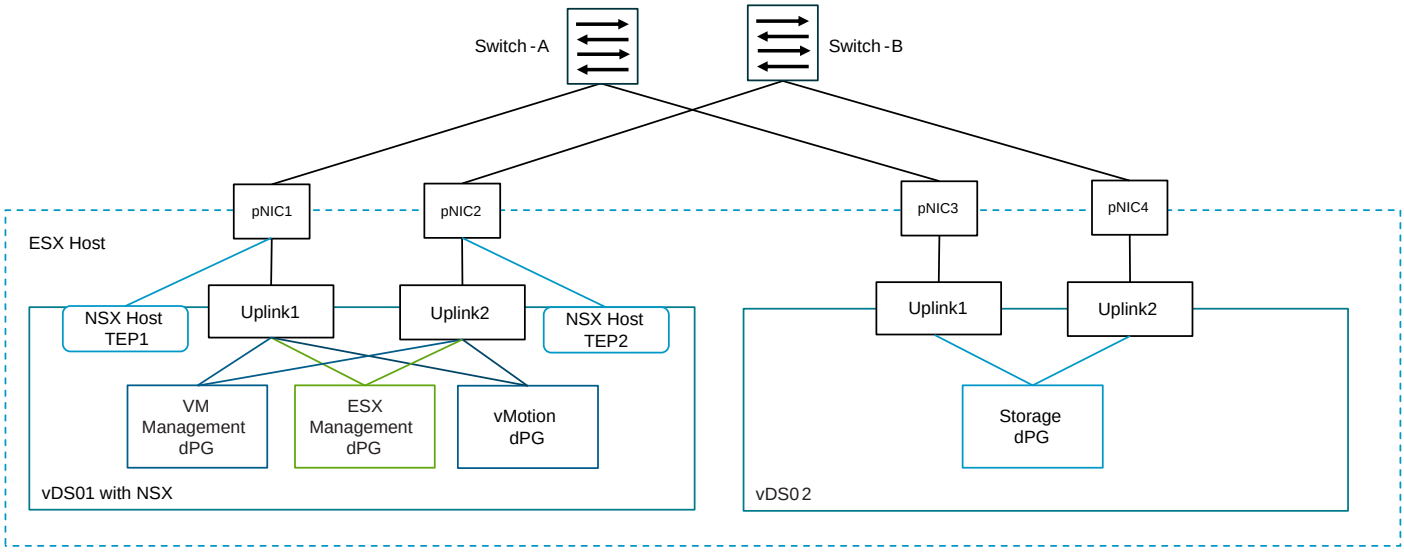
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-SC-RCMD-CFG-001	Create a host TEP IP pool for each availability zone with a subnet that is allocated per availability zone and a gateway that is configured for the network.	• Provides the host TEP IP addressing for each availability zone. • Enables a Layer 3 routed TEP traffic between availability zones. • Using an external DHCP server for the host overlay VLANs in both availability zones is not required.	Requires additional subnet per availability zone.
VCF-VNI-SC-RCMD-CFG-002	Configure an NSX sub-transport node profile for hosts in availability zone 2.	• The NSX transport node profile can remain attached when using two separate VLANs for host TEPs at each availability zone. • You can use static IP pools for the host TEPs in each availability zone.	Changes to the host transport node configuration are done at the vSphere cluster level.

Storage Separation Distributed Switch Model

The *Storage Separation Distributed Switch Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Storage Separation Distributed Switch Model	A *Storage Separation Distributed Switch Model* in VMware Cloud Foundation provides separation of storage traffic from all other network services running on the vSphere cluster.

Figure 147: Storage Separation Distributed Switch Model With Single Network Fabric in VMware Cloud Foundation



Storage Separation Distributed Switch Model Attributes

A *Storage Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Shared for Management, vSphere vMotion, NSX workloads. • Dedicated for IP storage.

Storage Separation Distributed Switch Model Options

A *Storage Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (storage traffic isolated to dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two or more network cards. • Each traffic type is spread across separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Common Distributed Switch Design Requirements

Storage Separation Distributed Switch Model Design Requirements

Table 375: Storage Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-SS-REQD-CFG-001	Use two vSphere Distributed Switch per vSphere cluster: • vDS01 — Management — vMotion — NSX • vDS02 — Storage	• Separates storage traffic onto a dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for storage traffic (NFS or vSAN) with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switch required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than the simple model.
VCF-VDS-SS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion	Provides the required network services for the vSphere cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-SS-REQD-CFG-003	Create the following distributed port groups on vDS02: • Storage	Provides connectivity for the hosts in the cluster to the storage network.	vSphere Distributed Switch storage traffic separation profile provides the required configuration.
VCF-VDS-SS-REQD-CFG-004	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switches for each additional rack: vDS01: • ESX management • vMotion vDS02: • Storage	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack
VCF-VDS-SS-REQD-CFG-005	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switches for availability zone 2 vDS01: • ESX management • vMotion vDS02: • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2

Common Distributed Switch Design Recommendations

Table 376: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Storage Separation Distributed Switch Model Design Recommendations

Table 377: Storage Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-DPG-001	Use the **Failover Order** teaming algorithm for the storage port group.	Provides a consistent traffic flow through a single physical ToR switch without needing to traverse Inter Switch Link during normal operations.	It must be manually configured during deployment using vSphere Distributed Switche custom profile option.
VCF-VDS-SS-RCMD-NIO-001	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than Principal storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-SS-RCMD-NIO-002	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as virtual machine traffic.	During times of network contention, vMotion takes longer than usual to complete.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-NIO-003	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in the SDDC. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-SS-RCMD-NIO-004	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

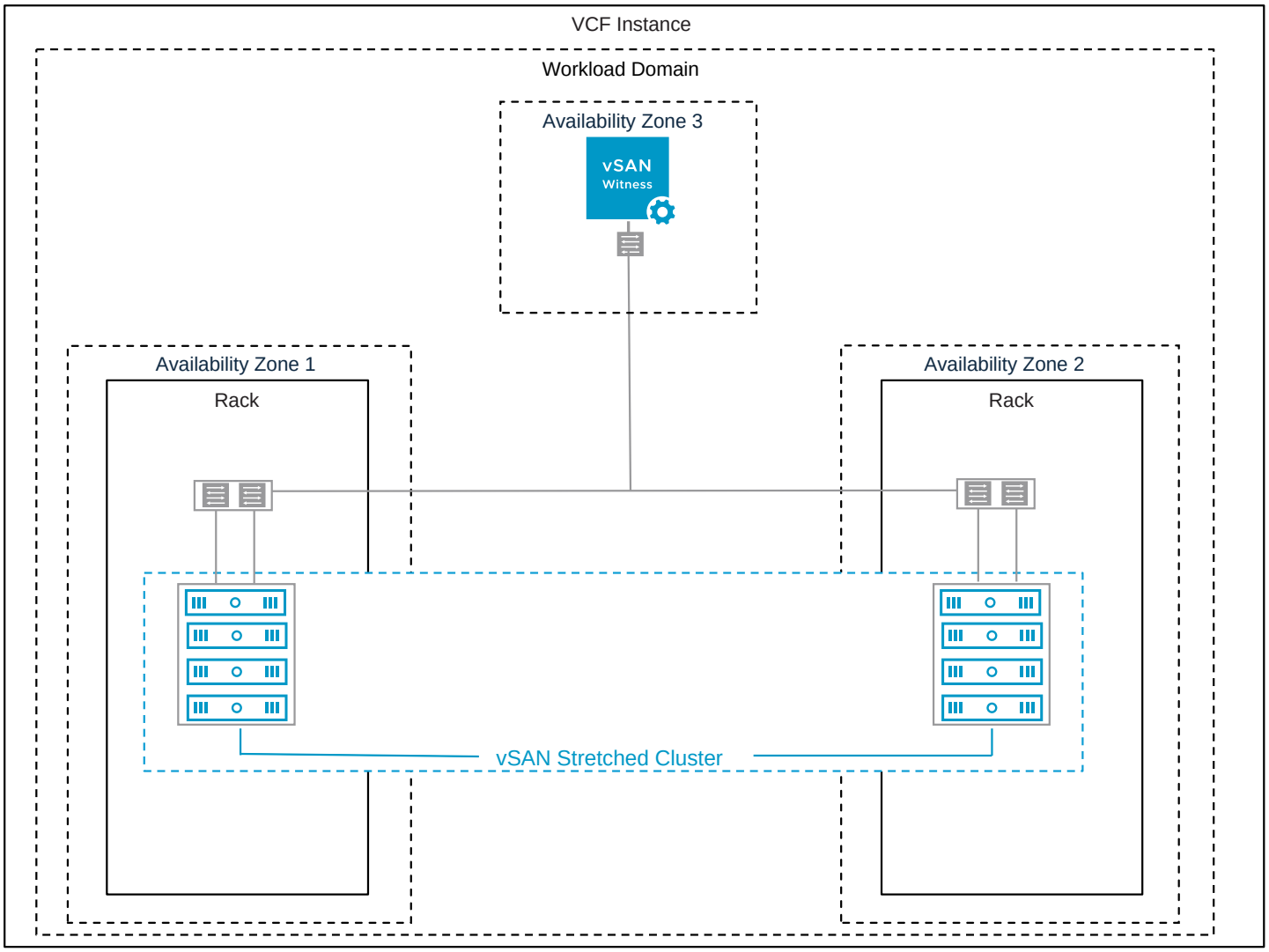
Stretched vSAN Storage Model

The *Stretched vSAN Storage Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrasturcture blueprint.

Selected Model	Additional Information
Stretched vSAN Storage Model	A *Stretched vSAN Storage Model* is a cluster configuration where the ESX hosts, that comprise a single vSAN cluster, reside in two availability zones.

Figure 148: Stretched vSAN Storage Model in VMware Cloud Foundation

Stretched vSAN Storage Model depicting three availability zones for vSAN. Two availability zones are designated for vSAN data nodes and a third availability zone is designated for a vSAN witness node.


VCF Instance
Workload Domain
Availability Zone 3
vSAN
Witness
Availability Zone 1
Rack
Availability Zone 2
Rack
vSAN Stretched Cluster

Stretched vSAN Storage Model Attributes

A *Stretched vSAN Storage Model* has the following attributes.

Attribute	Detail
Fault Domain / Avilability zone	• Stretched vSAN clusters comprise of three (3) Fault domains. A fault domain comprises of one or more vSAN nodes. • A fault domain or availability zone can represent a physical location. Fault domains / availability zones can either be two distinct data centers in a metro distance, or two safety or fire sectors (data halls) in the same large-scale datacenter.
vSAN Traffic Type	• Stretched vSAN clusters utilize vSAN Traffic type for data traffic. • Stretched vSAN clusters utilize vSAN Witness Traffic type for vSAN meta-data traffic. **Note:** Stretched vSAN Storage clusters (ESA) can be configured to use vSAN Storage Cluster Client traffic for vSAN compute client clusters. See Stretched vSAN Storage Cluster Storage Model.
vSAN Witness Appliance	A witness node is a pre-configured appliance that runs ESX and is distributed as an OVA file.

Common vSAN Storage Model Design Requirements

Table 378: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Common vSAN Host Network Model Design Requirements

Table 379: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

vSAN ESA Storage Model Design Requirements

Table 380: vSAN ESA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-ESA-REQD-CFG-001	Ensure vSAN certified Drives used for vSAN ESA clusters.	• When using VMware Cloud Foundation automated workflows, vSAN ESA cluster creation maybe blocked unless vSAN certified drives are detected. • vSAN Auto-RAID will be enabled when vSAN certified components are detected. This allows vSAN to generate an optimal default storage policy, based on the cluster topology and the number of ESX hosts.	None.
VCF-VSAN-ESA-REQD-CFG-002	1.6 TB drive size or higher for vSAN ESA devices for storage pools.	Minimum supported vSAN ESA drive size is 1.6TB.	vSAN ESA supports up to 24 drives per node. Smaller capacity drives may constrain raw capacity of a vSAN node.
VCF-VSAN-ESA-REQD-CFG-003	16 Cores or higher per vSAN node. vSAN ESA has a minimum value of 16 cores per vSAN Node	Minimum vSAN ESA core count per node is 16 for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	Smaller number of cores per node may require more vSAN nodes to support targeted workloads.
VCF-VSAN-ESA-REQD-CFG-004	Memory size per vSAN Ready node 128GB or higher	Minimum vSAN ESA memory is 128GB for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	• Smaller memory footprint per node will limit number of vSAN drives per node. • Smaller memory footprint per node for vSAN ESA will limit VM workloads being placed on vSAN nodes.

Stretched vSAN Storage Model Design Requirements

Table 381: Stretched vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-001	Two (2) availability zones and a third site for the witness appliance location.	Fault domains are mapped to availability zones to provide logical and physical ESX host separation and ensure a copy of vSAN data is always available even when an availability zone goes offline.	• This requires both availability zones to have an equal number of ESX hosts residing in each site, and are connected using a high bandwidth/low latency connection known as an inter-site link, or ISL. • The third site should host the witness appliance and must not have a logical or physical dependency on either availability zone.
VCF-VSAN-SC-REQD-CFG-002	The minimum latency required between the data sites hosting virtual machine objects should not exceed 5 milliseconds (ms) in round trip time (RTT), or 2.5ms one way.	The latency minimums exist largely to ensure that applications can write data resiliently across two sites in a timely manner.	Stretched cluster network latency requirements will dictate the realistic geographic distance between sites.
VCF-VSAN-SC-REQD-CFG-003	The minimum latency required between witness site and the data sites should not exceed 200ms RTT, or 100ms one way.	The witness site only stores small amounts of metadata, and is not a part of the vSAN data path.	None.
VCF-VSAN-SC-REQD-CFG-004	vSAN Stretched Cluster Data Site to Data Site Bandwidth minimum is no less than 10GbE. • The vSAN Stretched Cluster Bandwidth Sizing document can be used for general guidance in sizing for vSAN OSA and ESA architectures.	The network requirements for the data-site to data site in a stretch cluster remain the same for both vSAN ESA and OSA.	• VM workloads will ultimately determine the network bandwidth demand between data-sites. • Please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing
VCF-VSAN-SC-REQD-CFG-005	vSAN Stretched Cluster Data Site to Witness Site Bandwidth is based on number of VMs that is protected by a stretch cluster storage policy.	• Network communication from the data sites to the witness host appliance is comprised entirely of metadata and this is wholly dependent on the number of VM and components protected by vSAN Stretch Cluster polices. • There is no minimum bandwidth defined as estimating the bandwidth required for the witness site is based on the number of VM components.	For accurate sizing please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-006	Management Domain initial cluster Minimum of eight (8) nodes for vSAN stretched cluster for high availability (HA) install	Management domain initial clusters deployed in high availability (HA) mode have a requirement for 4 nodes in a non-stretched deployment. Stretching a cluster is a day two operation and will require 4 additional hosts on the second availability zone.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure
VCF-VSAN-SC-REQD-CFG-007	Workload domain clusters (except for management Domain initial cluster) Minimum of six (6) nodes for vSAN stretched clusters	This will provide minimum requirements for site availability and site local availability, tolerating a site failure and one failure on each availability zone. This will give a balance between a host failure on a local availability zone and a full availability zone failure.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure with minimum three (3) nodes per site. However with three nodes per site, vSAN will not automatically rebuild objects that are impacted by a local failure.
VCF-VSAN-SC-REQD-CFG-008	For vSAN OSA architecture Modify the default Storage Policy as follows • **Site disaster tolerance:** Site mirroring • **Failure to tolerate :** One (1) failure RAID-1 (Mirroring)	Provides the minimum necessary protection for VMs in each availability zone, with the ability to recover from an availability zone outage. **Remember:** • For vSAN ESA , Auto Policy configuration will automatically alter the default storage policy to reflect the number of hosts and underlying vSAN topology, when the cluster is converted to a stretch cluster. • With 9.1 vSAN Auto-RAID will be configured by default to all new vSAN ESA clusters and automticlly configure the optimal policy based on the cluster topology	• Additional Storage is necessary to satisfy the storage policy requirements in both availability zones. • Additional policies can be configured for specific workloads depending on their individual performance or availability requirements which may differ from what the default vSAN stretched cluster policy.
VCF-VSAN-SC-REQD-CFG-009	Deploy a vSAN witness appliance in a location that is not co-located to the ESX hosts in any of the availability zones.	Ensures availability of vSAN witness appliance in the event of a failure of one of the availability zones.	You must provide a third availability zone that is logically and physically independent of the data sites The third site must be able to provide a environment capable to host the witness appliance.
VCF-VSAN-SC-REQD-CFG-010	Deploy a witness appliance that corresponds to vSAN Architecture, (ESA or OSA).	Ensures the witness appliance deployed is compatible to the correct vSAN Storage architecture	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-011	Connect the first VMkernel (vmk0) adapter of the vSAN witness appliance to the management network in the witness site.	The ESX host management networks in both availability zones must be routed to the vSAN witness host management network in the third witness site.	When using witness traffic separation, the second interface on vSAN witness host (vmk1) can be removed.
VCF-VSAN-SC-REQD-CFG-012	Configure vSAN Witness host identity as per standard ESX host recommendations.	• Allows vSAN Witness host to be managed identically as a ESX host. • Ensures host identity, DNS and NTP is consistently configured	Apply the requirements for External Services Design
VCF-VSAN-SC-REQD-CFG-013	Enable vSphere Lifecycle Manager images on the vSAN witness host.	• VMware Cloud Foundation 9.0 requires vSphere Lifecycle Manager images for ESX hosts. • This will prevent warnings or vSAN health check alarms.	This requires either deploying or converting a vSAN witness host as a vSphere Lifecycle Manager image managed entity.

Common vSAN Storage Model Design Recommendations

Table 382: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Common vSAN Host Network Model Design Recommendations

Table 383: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

vSAN ESA Storage Model Design Recommendations

Table 384: vSAN ESA Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-001	• Select a specific size of drives used according to the requirements of target workloads. • Use multiple storage devices per vSAN node • NVMe TLC devices with a performance Class F (100000 - 349999) or higher with endurance class 3 DWPD or higher.	• The number of storage devices has an impact on the performance of vSAN ESA. • NVMe drives with mixed-use endurance and performance class typically classified on the vSAN HCL as "3DWPD" (3 Drive Writes per Day) will give a balance between performance and durability.	• Smaller drive sizes will impact overall capacity of vSAN ESA. Large drive sizes will introduce larger capacity fault domain in case of failure. • Sizing vSAN Nodes requires determining the workload capacity requirements prior to deployment. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-002	Use vSAN ready nodes with high core count.	vSAN Express Storage Architecture (ESA) is designed to be highly parallel and leverages multiple CPU cores for improved performance, making it multithreaded, and optimized for NVMe- TLC based flash devices.	There will be a correlation between target workload, number of cores and NVMe devices per vSAN Node. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-003	Use vSAN ready nodes with high memory capacity per vSAN node.	When enabled vSAN has required memory overheads and must have adequate resources account for VM workloads.	Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsanizer/home.
VCF-VSAN-ESA-RCMD-CFG-004	At least four (4) nodes for vSAN cluster size.	This ensures in the event of a fault domain failure, vSAN objects can be re-protected to maintain policy compliance. In conjunction with auto policy management, four (4) node vSAN ESA clusters will allow vSAN to automatically create a default storage policy that is capacity optimized.	None. **Note:** New vSAN clusters in VCF 9.1 will use clusters will automatically be configured to use Auto-RAID.
VCF-VSAN-ESA-RCMD-CFG-005	• Select the same minimum requirements for vSAN Storage cluster nodes as vSAN ESA HCI nodes. • Review vSAN Storage minimum node requirements	• vSAN Storage Cluster Nodes use the same architecture as regular vSAN ESA nodes. • vSAN ESA nodes can be re-purposed for vSAN Storage Clusters or vSAN HCI Clusters	None.

Stretched vSAN Storage Model Design Recommendations

Table 385: Stretched vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-RCMD-001	Use Layer 3 (routed) networking between the two data sites, and the witness site. • The data sites will only be able to communicate with each other using the ISL • The witness host should only be able to communicate with hosts in each data site directly, and not through another data site.	Using Layer 3 helps avoid spanning tree protocol (STP) from redirecting traffic across an undesirable link, such as the more bandwidth constrained link to the witness site	Ensure Layer 3 gateways are highly available such that it tolerates the failure of an entire availability zone.
VCF-VSAN-SC-RCMD-002	Use witness traffic separation when configuring a stretched vSAN cluster.	• Network Flexibility. You can configure different VLANs or subnets for vSAN data and witness traffic, allowing for better network segmentation and resource allocation. • Mixed MTU. This will allow vSAN witness traffic type to use a different MTU size than the vSAN data traffic. This avoids configuring static routes between vSAN networks. • Simplified Troubleshooting. Identifying and resolving network issues related to witness traffic becomes easier when it is isolated from the main vSAN data traffic.	This requires the vSAN witness network to be routable to both the ESX host management networks on either availability zone as well to VM management network if configured.
VCF-VSAN-SC-RCMD-CFG-003	Configure the vSAN witness host to use the first VMkernel adapter (vmk0), that is the management interface, for vSAN witness traffic.	• Removes the requirement to have two IP addresses allocated to the vSAN witness host. • Removes the requirement to have static routes in the witness as the witness traffic is routed over the management networks	• The management networks in both availability zones must be routed to the management witness site. • Ensure to use the same MTU size as the host management interfaces. **Note:** Default MTU size is 1500 for management interfaces

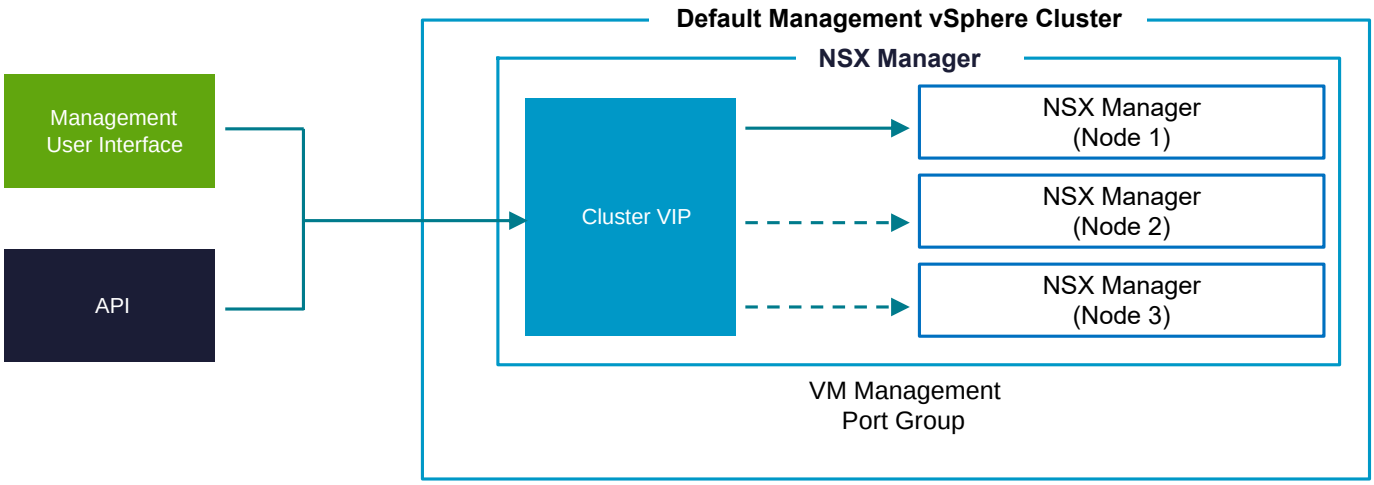
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-RCMD-CFG-004	For vSAN OSA clusters, configure vSAN storage policies with **Force Provisioning**.	vSAN objects are created if an availability zone becomes unavailable because of a site maintenance operation or a failure.	This would require modifying vSAN OSA policies used as part of the stretched cluster. **Tip:** **Force Provisioning** is no longer an applicable storage rule in a stretched cluster using Auto-RAID. Objects will continue to be created in the event of site mainteance.
VCF-VSAN-SC-RCMD-CFG-005	Ensure DRS Host Groups, VM Groups, and VM/Host Rules are in conjunction with vSAN storage policies.	• VM and vSAN data resilience desired outcomes are satisfied. • Suggests or forces a behavior of VM placement in a vSAN stretched cluster while in steady state. • vSphere HA will trigger the desired behavior in response to a failure.	• Requires creation of additional DRS Host groups, VM Groups, and VM/Host DRS Rules. • The placement rules should match the desired failure response.
VCF-VSAN-SC-RCMD-CFG-006	vSphere HA Admission control fail-over capacity is configured to reserve 50% of the CPU and memory capacity of the entire stretched cluster.	This ensures that VMs can fail between AZs and have adequate resources.	Requires adequate capacity guarantee successful fail-over of protected workloads in the event of failure of an entire availability zone.
VCF-VSAN-SC-RCMD-CFG-007	Do not enable reserved Capacity in a stretched vSAN cluster.	vSAN capacity reserve is not supported in conjunction with the following vSAN cluster topologies: • Three (3) node clusters. • Stretched vSAN clusters. • vSAN clusters using fault domains.	This will require manually calculating vSAN slack space. Refer to the vSAN sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

High Availability NSX Manager Model

The *High Availability NSX Manager Model* was chosen for the V*CF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.

Figure 149: High Availability NSX Manager Model in VMware Cloud Foundation



High Availability NSX Manager Model Attributes

The *High Availability NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Three-node cluster of NSX Manager appliances with internal virtual IP
Availability	• Highly available cluster of virtual IP NSX Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.
API Request Scale	Internal virtual IP directs API requests to a single NSX Manager appliance.

Common NSX Manager Design Requirements for All Models

Table 386: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

High Availability NSX Manager Model Design Requirements

Table 387: High Availability NSX Manager Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-005	Deploy three (3) NSX Manager nodes in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports high availability of the NSX Manager cluster.	You must have sufficient resources in the default cluster of the management domain to run three NSX Manager nodes.

Common NSX Manager Model Design Recommendations for All Models

Table 388: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

High Availability NSX Manager Model Design Recommendations

Table 389: High Availability NSX Manager Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-003	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Manager appliances.	Keeps the NSX Manager appliances running on different ESX hosts for high availability.	You must allocate at least four physical hosts so that the three NSX Manager appliances continue running if an ESX host failure occurs.

Common NSX Manager Design Recommendations for Stretched Clusters for All Models

Table 390: Common NSX Manager Design Recommendations for Stretched Clusters for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-002	Add the NSX Manager appliances to the virtual machine group for the first availability zone.	Ensures that the NSX Manager appliances are powered on an ESX host in the primary availability zone.	None.

Physical Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
The physical infrastructure layer constitutes the tangible hardware and facilities layer that provides the foundational computing, storage, and networking resources upon which all digital services depend. This layer encompasses the actual

physical assets—servers, storage arrays, network equipment, and supporting facilities—that form the bedrock of modern IT operations.

Multi Availability Zone Network Fabric Model

The *Multi Availability Zone Network Fabric Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Multi Availability Zone Network Fabric Model	This section describes the VLAN and gateway requirements to support a vSphere cluster within multiple availability zones.

This design requires stretching a minimum number of VLAN segments between each availability zone. The physical network fabric stretches these segments to provide a single Layer 2 domain across zones, enabling virtual machine mobility during an availability zone outage. For these fabric-stretched VLAN segments, the first-hop gateway must remain highly available across both availability zones. You must give special consideration to VLAN segments for NSX Edge uplinks when using the Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model, as BGP peering from the NSX Edge node to the fabric occurs over these networks.

Note:

VLAN IDs are switch-level numerical labels identifying a Layer 2 domain. In some fabric configurations, the same VLAN ID may identify different domains, or different VLAN IDs may identify the same domain. A distributed port group can be associated with a single VLAN ID. Any distributed port groups supporting virtual machines must have the same VLAN ID and Layer 2 domain mapping across the entire vSphere cluster. You may use different VLAN IDs for separate vSphere clusters that do not share a common vSphere Distributed Switch.

Table 391: Infrastructure VLAN Scope and First Hop Gateway Availability for Multiple AZ

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management component VMs*		Global across AZs	Highly Available - Across AZs **	1500	9000
ESX host	ESX management VMK	AZ Local***	Highly Available - Within AZ**		1500
	IP Storage VMKs (VSAN, NFS, iSCSI)		Highly Available - Within AZ **		9000
	vMotion VMK		Highly Available - Within AZ **		
	Host TEP		Highly Available - Within AZ **	1700	
NSX Edge node	Refer to Dual Availability Zone Based on NSX Edge Node HA Model				
	NSX Edge Node Management	AZ Local	Highly Available - Within AZ **	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available - Within AZ** BGP - None	1700	
	Refer to Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model				
	NSX Edge Node Management	Global across AZs	Highly Available - Across AZs *	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
	NSX Edge Uplink		Static Routing - Highly Available - Across AZs ** BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Dual Availability Zones				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
	BGP-EVPN Peering	Across AZs	Single-Hop BGP - None Multi-Hop BGP - Highly Available - Across AZs **	1500	1500

* When using separate VLAN segments for management component virtual machines and ESX VMkernels.
** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.
**In this table, ESX host infrastructure VLANs are limited to an individual availability zone. Stretching of these VLANs between availability zones is supported, however the first hop gateway must also be highly available between availability zones.

Table 392: Workload VLAN Scope and First Hop Gateway Availability for Multiple Availability Zones

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Global between AZs	Highly Available - Across AZs**	1500	9000
Distributed VLAN Connectivity Model	Global between AZs*	Highly Available - Across AZs**	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model	Refer to Edges			
NSX Segment Connectivity Model	Refer to Edges			

* All Node of a VNA cluster must share a common L2 network. Optional stretching of this VLAN between racks will enable seamless maintenance and recovery across racks.
** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.

IP Connectivity and Symmetry

Traffic must be routed between availability zones for all VLAN segments that require cross-site communication. For example, vSAN networks in each availability zone must have a route to each other to support synchronous data replication.

For VLAN segments that stretch between availability zones, the physical network fabric determines the ingress and egress paths. In most cases, to simplify troubleshooting, symmetrical routing is recommended.

For workloads, the ingress and egress path depends upon the Network Consumption Models chosen. For the Centralized Connectivity Modelingress and egress paths depend on the specific NSX Edge Cluster Models for Dual Availability Zonesand routing model:

- **Static Routing**: The physical network fabric determines the ingress and egress paths.
- **eBGP**: A combination of the physical fabric and the NSX Edge gateway determines the ingress and egress paths.

For all stretched VLAN network segments (infrastructure networks, VLAN Networking Connectivity Model or Distributed VLAN Connectivity Model), or Distributed VXLAN Connectivity Model for Version 9.1.0, the physical network fabric must provide a highly available Layer 3 gateway at the first hop. This ensures that virtual machines maintain connectivity on the same IP address when moving between availability zones.

Witness Location

For vSAN stretched cluster, an independent witness site is required. This requires independent network paths from each availability zone to the witness site, such that the failure of an individual availability zone does not impact the traffic path from the other availability zones to the witness site.

When configured with vSAN Witness Traffic Separation (WTS), the vSAN witness traffic flows between the ESX management VMkernels and the vSAN Witness node. This traffic uses the MTU of the ESX management VMkernels.

Availability Zone Data Center Interconnect

The data center inter-connect between the availability zones must be appropriately sized to handle all infrastructure, workload, and storage replication traffic. The interconnect is important to the operations of the vSAN Cluster. It should have the appropriate level of resilience to achieve business objectives.

Shared Design Selection between Management and Workload Domain Region B

Shared Design Selections between Management and Workload Domain are the models that are the same across the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Virtual Infrastructure Layer for VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
Virtual Infrastructure Software represents the foundational technology stack that underpins all applications, services, and workloads within an organization's computing environment. This essential layer provides the fundamental capabilities required to run, scale, and manage modern enterprise systems, serving as the building blocks upon which higher-level services and applications are constructed.

vCenter Model
The *vCenter Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
vCenter Model	The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

vCenter Model Attributes

A *vCenter Model* has the following attributes.

Design Area	Attributes
vCenter Deployment Type	• One vCenter instance for the management domain that manages the management components of the VCF Instance. • Optionally, additional vCenter instances for workload domains to support workloads. • vSphere HA protecting all vCenter appliances.
vCenter Single Sign-On	• Each vCenter has a distinct vCenter Single Sign-On domain.

Methods for Protecting vCenter

VMware Cloud Foundation supports multi-rack management vSphere clusters, vSAN stretched clusters, or 3rd party synchronous replication solutions such as metro clusters that are designed to provide disaster recovery and/or high availability for the entire management stack (vCenter, NSX Manager, etc.) as opposed to a single vCenter.

Table 393: Methods for Protecting vCenter in VMware Cloud Foundation

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere High Availability	Yes	You must ensure sufficient resources in the cluster to allow virtual machines to restart on another ESX host in the event of an ESX host failure.
vCenter High Availability (vCenter HA)	No	• vCenter services must fail over to the passive node so there is no continuous availability. • Recovery time can be up to 15 mins. • You must meet additional networking requirements for the private network. • vCenter HA requires additional resources for the passive and witness nodes. • Lifecycle management is complicated because you must manually delete and recreate the standby virtual machines during a lifecycle management operation.
vSphere Fault Tolerance (vSphere FT)	No	• The vCPU limit of vSphere FT vCPU would limit vCenter appliance size to medium. • You must provide a dedicated network.

vCenter Model Design Requirements and Recommendations

Table 394: vCenter Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-CFG-001	Deploy a dedicated vCenter appliance for the management domain of the VCF Instance.	• Isolates vCenter failures to management or customer workloads. • Isolates vCenter operations between management and business workloads. • Supports a scalable vSphere cluster design where you can reuse the management components as more customer workloads are added to the VCF Instance. • Simplifies capacity planning for customer workloads because you do not consider management workloads for the workload domain vCenter. • Improves the ability to upgrade the vSphere environment and related components by enabling for explicit separation of maintenance windows: — Management workloads remain available while you are upgrading the tenant workloads. — Workloads remain available while you are upgrading the management nodes. • Supports clear separation of roles and responsibilities to ensure that only administrators with granted authorization can control the management workloads. • Facilitates quicker troubleshooting and problem resolution. • Simplifies disaster recovery operations by supporting a clear separation between recovery of the management components and tenant workloads. • Provides isolation of potential network issues by introducing network separation of the vSphere clusters in the VCF Instance.	The vSphere cluster must have sufficient resources.
VCF-VCS-REQD-NET-001	Place all workload domain vCenter appliances on the VM management network in the management domain.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-SSO-001	Create all vCenter instances within a VCF Instance in their own unique vCenter Single Sign-On domain.	• Sharing vCenter Single Sign-On domains is not supported. • Enables isolation at the vCenter Single Sign-On domain layer for increased security separation.	• Each vCenter instance is managed through its own pane of glass using a different set of administrative credentials. • You must manage password rotation for each vCenter Single Sign-On domain separately.

Table 395: vCenter Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VCS-RCMD-CFG-001	Deploy an appropriately sized vCenter appliance for each workload domain.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate compute resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-002	Deploy a vCenter appliance with the appropriate storage size.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate storage resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-003	Protect workload domain vCenter appliances by using vSphere HA.	vSphere HA is the only supported method to protect vCenter availability in VCF.	vCenter becomes unavailable during a vSphere HA failover.
VCF-VCS-RCMD-CFG-004	In vSphere HA, set the restart priority policy for the vCenter appliance to high.	vCenter is the management and control plane for physical and virtual infrastructure. In a vSphere HA event, to ensure that the rest of the SDDC management stack comes up faultlessly, the workload domain vCenter must be available first, before the other management components come online.	If the restart priority for another virtual machine is set to highest, the connectivity delays for the management components are longer.

ESX Model

The *ESX Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
ESX Model	In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCF domain vSphere cluster.

ESX Hardware and Sizing Considerations

For more information on host sizing, refer to the VCF Planning and Preparation Workbook.

Table 396: ESX Hardware and Sizing Considerations

Hardware Element	Considerations
BIOS/Firmware	Ensure the ESX host hardware is on the Hardware Compatability List (HCL) and that you are running the latest supported BIOS and firmware.
CPU	• Total CPU requirements for the workloads that are running in the cluster. • Host failure and maintenance scenarios. • CPU over-commitment ratio guidelines – Management Domain • To ensure optimal performance and maintain resilience during host failures or maintenance, keep the management domain vCPU-to-pCPU over-commitment ratio at or below 2:1. **Note:** While higher ratios such as 3:1 or 4:1 are supported, you must closely monitor CPU ready times to ensure that the management cluster has enough capacity to run the workloads. – Workload domains • CPU over-commitment ratios for workload domains should be calculated on a case by case basis depending on the CPU intensity of the workloads. You must closely monitor CPU ready times to ensure optimal performance. • Number of physical cores, not logical cores. – Simultaneous multi-threading (SMT) in CPUs improve CPU performance however the performance enhancement will not be equivalent to 100% more CPU power.
Memory	• Total memory requirements for the workloads that are running in the cluster. • When sizing memory for the ESX hosts in a cluster, to reserve the resources of one host for failover or maintenance, set the admission control setting to N+1, which reserves the resources of one host for failover or maintenance. • *vSAN* OSA (Original Storage Architecture). – To support the maximum number of disk groups, you must provide 32 GB of RAM. • *vSAN* ESA (Express Storage Architecture). – You must provide a minimum of 128GB of RAM.
Storage	• Use high-endurance devices such as a hard drive or SSD for boot device. • Use a 128GB or greater boot device to maximize the space available for ESX-OS Data. • Use hosts with homogeneous configuration.
Network	• Use multiple physical NICs connected to multiple ToR switches to ensure redundancy. • 25GbE is recommended • Use physical NICs of the same speed, vendor, and model across all host within a cluster.
Power	• Use redundant power supplies, preferably connected to separate circuits to ensure availability. • Consider the impact of enabling Power Management on your ESX hosts. In order to allow ESX to control CPU power-saving features you need to set power management in the BIOS to *OS Controlled Mode*.

ESX Model Design Requirements and Recommendations

In your ESX design, you apply certain best practices.

Table 397: ESX Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-ESX-REQD-CFG-001	Install no less than the minimum number of ESX hosts required for the vSphere cluster model being deployed.	• Ensures that availability requirements are met. • If one of the ESX hosts is not available because of a failure or maintenance event, the CPU over-commitment ratio becomes 2:1.	None.
VCF-ESX-REQD-CFG-002	Ensure each ESX host matches the required CPU, memory, and storage specification.	Ensures that workloads will run without contention even during failure and maintenance conditions.	Assemble the server specification and number according to the sizing in the VCF Planning and Preparation Workbook which is based on projected deployment size.
VCF-ESX-REQD-SEC-001	Regenerate the certificate of each ESX host after assigning the ESX host an FQDN.	Establishes a secure connection with the VCF Installer during the initial deployment and prevents man-in-the-middle (MiTM) attacks.	You must manually regenerate the certificates of the ESX hosts before the deployment.

Table 398: ESX Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-001	When using vSAN as primary storage in the management domain, use vSAN ReadyNodes for each ESX host.	Your management domain is fully compatible with vSAN at deployment. **Note:** For information about the models of physical servers that are vSAN-ready, see vSAN Compatibility Guide for vSAN ReadyNodes.	• Hardware choices might be limited. • If you plan to use a server configuration that is not a vSAN ReadyNode, your CPU, disks, and I/O modules must be listed on the VMware Compatibility Guide under *CPU Series* and *vSAN Compatibility List* aligned to the ESX version.
VCF-ESX-RCMD-CFG-002	Allocate ESX hosts with uniform configuration across the default management vSphere cluster.	A balanced vSphere cluster has these advantages: • Predictable performance even during hardware failures. • Minimal impact of resynchronization or rebuild operations on performance.	You must apply vendor sourcing, budgeting, and procurement considerations for uniform server nodes on a per vSphere cluster basis.
VCF-ESX-RCMD-CFG-003	When sizing CPU, do not consider multi-threading technology and associated performance gains.	Although multi-threading technologies increase CPU performance, the performance gain depends on running workloads and differs from one case to another.	Because you must provide more physical CPU cores, costs increase and hardware choices become limited.
VCF-ESX-RCMD-CFG-004	Install and configure all ESX hosts in the default management vSphere cluster to boot using a 128-GB device or larger.	Provides ESX hosts that have large memory, that is, greater than 512 GB, with enough space for the scratch partition when using vSAN.	None.

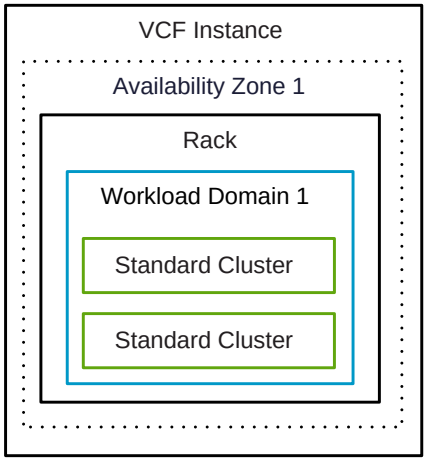
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-005	Use the default configuration for the scratch partition on all ESX hosts in the default management vSphere cluster.	• If a failure in the vSAN cluster occurs, the ESX hosts remain responsive and log information is still accessible. • If using vSAN storage its not possible to use a vSAN datastore for the scratch partition.	None.
VCF-ESX-RCMD-CFG-006	For workloads running in the default management vSphere cluster, save the virtual machine swap file at the default location.	Simplifies the configuration process.	Increases the amount of replication traffic for management workloads that are recovered as part of the disaster recovery process.
VCF-ESX-RCMD-NET-001	Place the ESX hosts in each management domain vSphere cluster on the ESX management network that is separate from the VM management network.	• Enables the separation of the physical VLAN between ESX hosts and the other management components for security reasons. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Increases the number of VLANs required.
VCF-ESX-RCMD-NET-002	Place the ESX hosts in each workload domain on a separate ESX host management VLAN-backed network.	Enables the separation of the physical VLAN between the ESX hosts in different workload domains for security reasons.	Increases the number of VLANs required. For each workload domain, you must allocate a separate management subnet.
VCF-ESX-RCMD-NET-003	Use dedicated VLANs for each traffic type.	Enables the separation of the physical VLAN for vSAN, vMotion, NFS & backup traffic.	Increases the number of VLANs required. For each traffic type you must allocate a separate VLAN and subnet.
VCF-ESX-RCMD-SEC-001	Set the advanced setting `UserVars.SuppressShellWarning` to `0` across all ESX hosts in the management domain.	• Ensures compliance with the *vSphere Security Configuration Guide* and with security best practices • Enables the warning message that appears in the vSphere client every time SSH access is activated on an ESX host.	You must turn off SSH enablement warning messages manually when performing troubleshooting or support activities.

Single-Rack vSphere Cluster Model

The *Single-Rack vSphere Cluster Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack vSphere Cluster	The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.

Figure 150: Single-Rack vSphere Cluster Model in VMware Cloud Foundation



Single-Rack vSphere Cluster Model Attributes

A *Single-Rack vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter.
Physical Rack Mapping	All cluster hosts exist in one physical rack.
Resilience	vSphere HA protects against host failures.

Single-Rack vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere which supports your storage type and VCF domain considerations.

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
Minimum number of ESX hosts	vSAN	Simple	Three (3)	• 33% CPU and memory • Tolerates one host failure	Three (3)	• 33% CPU and memory • Tolerates one host failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
	NFS or FC	Simple	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Important:

vSphere clusters are configured allowing for the failure of a single ESX host while still having enough resources to run all vSphere cluster workloads.

Attention:

* Designs should be completed in consultation with your storage vendor.

Common vSphere Cluster Design Requirements

Table 399: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Table 400: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Common vSphere Cluster Design Recommendations

Table 401: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Table 402: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

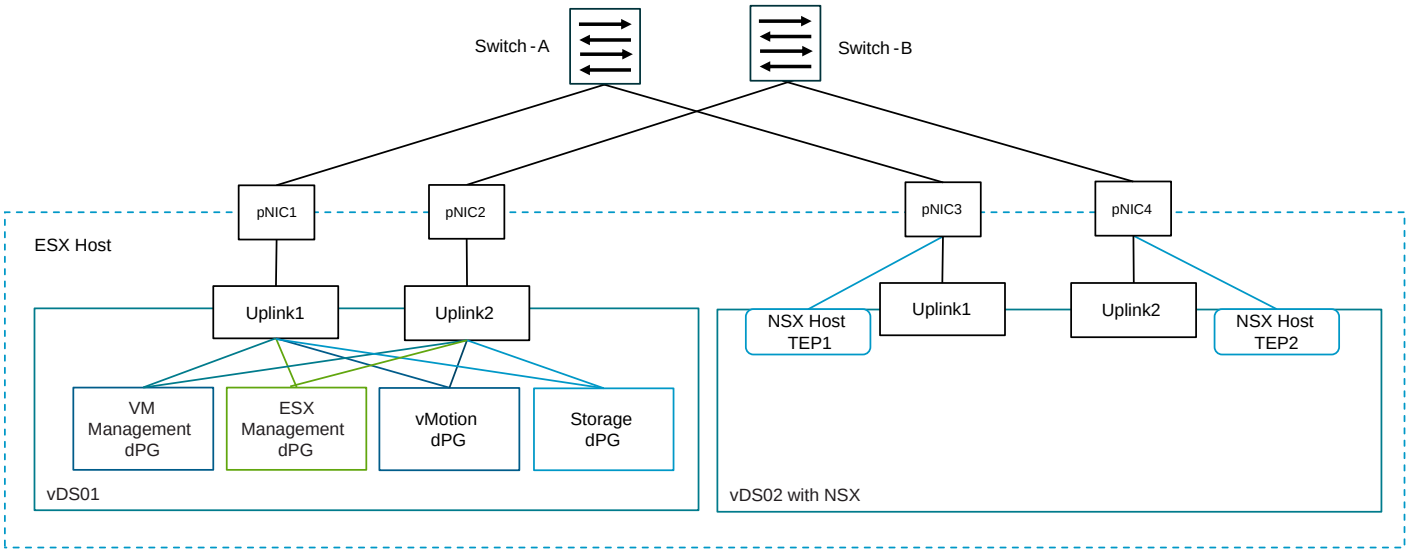
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Workload Separation Distributed Switch Model

The *Workload Separation Distributed Switch Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Workload Separation Distributed Switch Model	A *Workload Separation Distributed Switch Model* in VMware Cloud Foundation provides a separation of workload traffic running on NSX segments from all other network services running on the vSphere cluster.

Figure 151: Workload Separation Distributed Switch Model in VMware Cloud Foundation



Workload Separation Distributed Switch Model Attributes

A *Workload Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Shared for Management, vSphere vMotion, vSAN. • Dedicated for workloads on NSX.

Workload Separation Distributed Switch Model Options

A *Workload Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (workload traffic using NSX and VPC networks isolated to a dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two separate 2-port network cards. • Each traffic type is spread across two separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Common Distributed Switch Design Requirements

Workload Separation Distributed Switch Model Design Requirements

Table 403: Workload Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-WS-REQD-CFG-001	Use two (2) vSphere Distributed Switches per vSphere cluster: • vDS01 — ESX Management — vMotion — Storage • vDS02 — NSX	• Separates NSX workload traffic onto dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for workload traffic with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switches required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than the simple model.
VCF-VDS-WS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion • Storage	Provides the required network services for the cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain clusters.
VCF-VDS-WS-REQD-CFG-003	Configure vDS02 for NSX.	Provides NSX networking capabilities for the workloads on the cluster.	None.
VCF-VDS-WS-REQD-CFG-004	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switch for each additional rack: • ESX management • vMotion • Storage (vSAN/NFS)	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack.
VCF-VDS-WS-REQD-CFG-005	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switch for availability zone 2: • ESX management • vMotion • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2.

Common Distributed Switch Design Recommendations

Table 404: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Workload Separation Distributed Switch Model Design Recommendations

Table 405: Workload Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-WS-RCMD-DPG-001	Use the **Route based on physical NIC load** teaming algorithm for the storage distributed port group on vDS01 (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-WS-RCMD-NIO-001	Enable Network I/O Control on vSphere Distributed Switch vDS01 (default setting).	Increases resiliency and performance when network traffic shares uplinks.	Network I/O Control might impact network performance for critical traffic types if misconfigured.
VCF-VDS-WS-RCMD-NIO-002	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than principal storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.

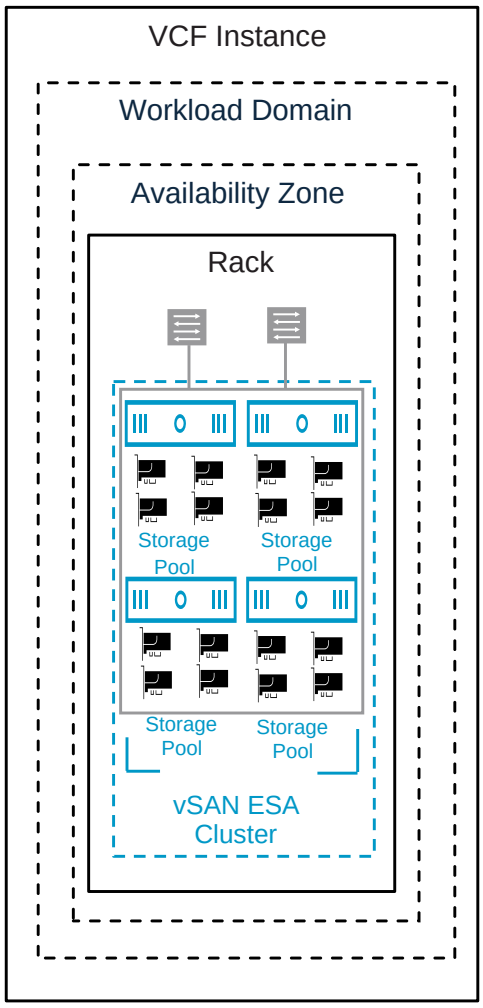
Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-WS-RCMD-NIO-003	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as storage traffic.	During times of network contention, vMotion takes longer than usual to complete.
VCF-VDS-WS-RCMD-NIO-004	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in an VCF Instance. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-WS-RCMD-NIO-005	Set the share value for principal IP-based Storage traffic to High for vDS01 (default setting).	During times of network contention, storage traffic needs guaranteed bandwidth to support virtual machine performance.	None.
VCF-VDS-WS-RCMD-NIO-006	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

Single-Rack vSAN ESA Storage Model

The *Single-Rack vSAN ESA Storage Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack vSAN ESA Storage Model	vSAN ESA uses a single-tier architecture that uses high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.

Figure 152: Single-Rack vSAN ESA Storage Model in VMware Cloud Foundation



Single-Rack vSAN ESA Storage Model Attributes

A *Single-Rack vSAN ESA Storage Model* has the following attributes.

Attribute	Detail
Single Tier	All data is stored on the same type of storage device, regardless of access frequency or importance.
Storage Pool	All NVMe Flash Storage.
Data Services	• Compression (enabled by default). • Encryption. (in transit and at rest) • vSAN Global Deduplication **Restriction:** vSAN Global Deduplication is not currently compatible with vSAN Stretched Clusters and Two Node vSAN Clusters. • Integrated vSAN File Services. • vSAN iSCSI Target Service. • vSAN Storage Clusters. • Stretched Cluster. • Fault domains. • Kubernetes persistent storage services (block and file). • Native scalable snapshots

Attribute	Detail
Virtual Machine Storage policies	vSAN storage policies (also known as Virtual Machine Storage Policies or VM Storage Policies) are rules defining virtual machine (VM) storage requirements that vSAN automatically enforces across a cluster. vSAN 9.1 automatically configures vSAN Auto-RAID policy on new vSAN ESA clusters. vSAN Auto-RAID policy manages the resilience for data residing in a cluster, regardless of the size or topology of the vSAN cluster. vSAN Auto-RAID creates a single policy within the vCenter server,

Single-Rack vSAN ESA Storage Model Options

A *Single-Rack vSAN ESA Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Deduplication	vSAN Global Deduplication is supported with VMware Cloud Foundation 9.1 and can be enabled as a day 2 operation.
Data In Transit Encryption	Data In Transit is supported to be enabled as day 0 or day 2 operation.
Data at Rest encryption	Data at Rest Encryption is supported as a day 2 operation.
vSAN Networking	vSAN ESA can utilize 10GbE , 25GbE or higher. See Table 764: vSAN Host Network Model Options.

Common vSAN Storage Model Design Requirements

Table 406: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Common vSAN Host Network Model Design Requirements

Table 407: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

vSAN ESA Storage Model Design Requirements

Table 408: vSAN ESA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-ESA-REQD-CFG-001	Ensure vSAN certified Drives used for vSAN ESA clusters.	• When using VMware Cloud Foundation automated workflows, vSAN ESA cluster creation maybe blocked unless vSAN certified drives are detected. • vSAN Auto-RAID will be enabled when vSAN certified components are detected. This allows vSAN to generate an optimal default storage policy, based on the cluster topology and the number of ESX hosts.	None.
VCF-VSAN-ESA-REQD-CFG-002	1.6 TB drive size or higher for vSAN ESA devices for storage pools.	Minimum supported vSAN ESA drive size is 1.6TB.	vSAN ESA supports up to 24 drives per node. Smaller capacity drives may constrain raw capacity of a vSAN node.
VCF-VSAN-ESA-REQD-CFG-003	16 Cores or higher per vSAN node. vSAN ESA has a minimum value of 16 cores per vSAN Node	Minimum vSAN ESA core count per node is 16 for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	Smaller number of cores per node may require more vSAN nodes to support targeted workloads.
VCF-VSAN-ESA-REQD-CFG-004	Memory size per vSAN Ready node 128GB or higher	Minimum vSAN ESA memory is 128GB for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	• Smaller memory footprint per node will limit number of vSAN drives per node. • Smaller memory footprint per node for vSAN ESA will limit VM workloads being placed on vSAN nodes.

Common vSAN Storage Model Design Recommendations

Table 409: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Common vSAN Host Network Model Design Recommendations

Table 410: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

vSAN ESA Storage Model Design Recommendations

Table 411: vSAN ESA Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-001	• Select a specific size of drives used according to the requirements of target workloads. • Use multiple storage devices per vSAN node • NVMe TLC devices with a performance Class F (100000 - 349999) or higher with endurance class 3 DWPD or higher.	• The number of storage devices has an impact on the performance of vSAN ESA. • NVMe drives with mixed-use endurance and performance class typically classified on the vSAN HCL as "3DWPD" (3 Drive Writes per Day) will give a balance between performance and durability.	• Smaller drive sizes will impact overall capacity of vSAN ESA. Large drive sizes will introduce larger capacity fault domain in case of failure. • Sizing vSAN Nodes requires determining the workload capacity requirements prior to deployment. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-002	Use vSAN ready nodes with high core count.	vSAN Express Storage Architecture (ESA) is designed to be highly parallel and leverages multiple CPU cores for improved performance, making it multithreaded, and optimized for NVMe- TLC based flash devices.	There will be a correlation between target workload, number of cores and NVMe devices per vSAN Node. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

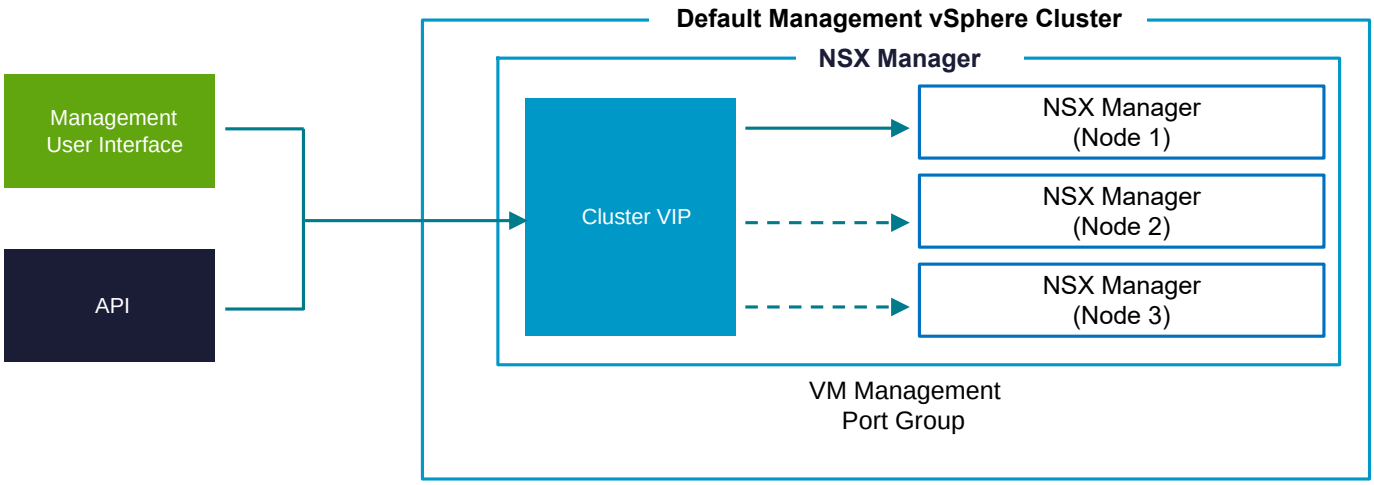
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-003	Use vSAN ready nodes with high memory capacity per vSAN node.	When enabled vSAN has required memory overheads and must have adequate resources account for VM workloads.	Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsanizer/home.
VCF-VSAN-ESA-RCMD-CFG-004	At least four (4) nodes for vSAN cluster size.	This ensures in the event of a fault domain failure, vSAN objects can be re-protected to maintain policy compliance. In conjunction with auto policy management, four (4) node vSAN ESA clusters will allow vSAN to automatically create a default storage policy that is capacity optimized.	None. **Note:** New vSAN clusters in VCF 9.1 will use clusters will automatically be configured to use Auto-RAID.
VCF-VSAN-ESA-RCMD-CFG-005	• Select the same minimum requirements for vSAN Storage cluster nodes as vSAN ESA HCI nodes. • Review vSAN Storage minimum node requirements	• vSAN Storage Cluster Nodes use the same architecture as regular vSAN ESA nodes. • vSAN ESA nodes can be re-purposed for vSAN Storage Clusters or vSAN HCI Clusters	None.

High Availability NSX Manager Model

The *High Availability NSX Manager Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
High Availability NSX Manager Model	The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.

Figure 153: High Availability NSX Manager Model in VMware Cloud Foundation



High Availability NSX Manager Model Attributes

The *High Availability NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Three-node cluster of NSX Manager appliances with internal virtual IP
Availability	• Highly available cluster of virtual IP NSX Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.
API Request Scale	Internal virtual IP directs API requests to a single NSX Manager appliance.

Common NSX Manager Design Requirements for All Models

Table 412: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

High Availability NSX Manager Model Design Requirements

Table 413: High Availability NSX Manager Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-005	Deploy three (3) NSX Manager nodes in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports high availability of the NSX Manager cluster.	You must have sufficient resources in the default cluster of the management domain to run three NSX Manager nodes.

Common NSX Manager Model Design Recommendations for All Models

Table 414: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

High Availability NSX Manager Model Design Recommendations

Table 415: High Availability NSX Manager Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-003	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Manager appliances.	Keeps the NSX Manager appliances running on different ESX hosts for high availability.	You must allocate at least four physical hosts so that the three NSX Manager appliances continue running if an ESX host failure occurs.

Physical Infrastructure Layer

The physical infrastructure layer constitutes the tangible hardware and facilities layer that provides the foundational computing, storage, and networking resources upon which all digital services depend. This layer encompasses the actual physical assets—servers, storage arrays, network equipment, and supporting facilities—that form the bedrock of modern IT operations.

Single-Rack Network Fabric Model

The *Single-Rack Network Fabric Model* was chosen for the *VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)* infrastructure blueprint.

Selected Model	Additional Information
Single-Rack Network Fabric Model	The *Single-Rack Network Fabric Model* describes the VLAN and gateway requirements to support the Single-Rack vSphere Cluster Model.

The data center network fabric provides connectivity to the vSphere clusters. The vSphere Cluster Models and Storage Models chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a Single-Rack vSphere Cluster Model. Depending upon the storage type chosen, additional VLAN requirements may apply.

Note:

If this rack will contain NSX Edge nodes, additional VLAN Requirements will apply. NSX Edge Cluster Models for Single Availability Zone

Table 416: Infrastructure VLAN Scope and First Hop Gateway Availability for Single Rack

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management VM Appliances*		Rack Local	Highly Available	1500	1500
ESX host	ESX management VMK *				1500
	IP Storage VMKs (vSAN, NFS, iSCSI)				9000
	vMotion VMK				
	Host TEP			1700	
NSX Edge node	Refer to Host Fault Tolerant NSX Edge Cluster Model or Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters				
	NSX Edge Node management	Rack Local	Highly Available	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available Static Routing with BFD - Optional BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
	BGP-EVPN Peering	Across Racks	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* When using separate VLAN segments for management component VMs and ESX VMKs.					

Table 417: Workload VLAN Scope and First Hop Gateway Availability for Single Rack

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VLAN Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model Attributes	Refer to Edges			

VCF Fleet Management Blueprint

Fleet management in VCF Operations helps build, manage, and scale your private cloud infrastructure while providing that consistency across each of its components.

Introduction

This document defines the operational guidelines for fleet management functions in VCF Operations , serving as the authoritative benchmark for achieving security and operational excellence at scale. By moving beyond siloed administration toward a global governance model, this blueprint provides prescriptive guidance on the following core capabilities:

1. Identity and Access Management
2. Certificate Management
3. Account and Password Management
4. Fleet Configuration Management
5. VCF Tag Management for vSphere

These guidelines are recommended for new or transforming environments, ensuring infrastructure is aligned for standardization, automation, and predictability.

Deviation from guidelines and best practices in this blueprint is permissible to meet unique business needs, but any such deviation:

- Requires evaluation of the impact on long-term supportability.
- May diverge from operational efficiency goals and could impair subsequent automation, lifecycle management, and platform consistency.

Audience

Platform engineers: Implement and maintain fleet-wide infrastructure components such as SSO, certificate, password, and configuration management. They operationalize automation frameworks, APIs, and Infrastructure as Code (IaC) pipelines.

Cloud Operations Engineers: Execute daily operations across VCF instances and fleets, perform routine compliance validations, and ensure audit readiness.

Cloud and Infra Architects: Define the overarching VCF fleet architecture, ensuring that designs align with enterprise governance, scalability, and automation principles. They make key design decisions on topology, identity, certificate, and configuration models.

Fleet Management Objectives

Maximize Operational Efficiency: Reduce administrative overhead through standardized fleet-wide operations, automated lifecycle management, and centralized governance.

Ensure Unified Governance & Consistency: Enforce consistent policy application and lifecycle management across all fleet instances.

Scale with Automation: Standardize architectural design to support robust, API-driven automation and predictable scalability.

Fleet Management Foundational Principles

Centralization Strategy: VCF Operations acts as the single management interface for fleet-wide services, providing a unified view for operations and compliance.

Automation-First: Day 2 operations should be executed via API or workflow automation through VCF Operations or integrated automation platforms to reduce human error and dependency on individual operator accounts.

Governance & Consistency: Unified configuration baselines should be enforced across the Fleet. Administrators should utilize VCF continuous configuration drift detection to verify the infrastructure configuration remains synchronized with the central validated baseline. Inconsistent configurations should trigger timely remediation alerts or automated correction.

Identity and Access Management

VCF Identity and Access Management (IAM) defines the framework for centralized identity governance and access control across a VCF Fleet. It enables operational teams to design and manage access to VCF Operations, VCF Automation, vCenter, NSX, and so on, at fleet scale.

Prerequisites

The guidance in this section assumes the following environmental state:

- A VCF Fleet where all management components are deployed and operating normally, and VCF SSO is deployed and enabled.
- An authoritative external Identity Provider (IdP) is available to manage user and group identities and to perform authentication for access to the VCF Fleet.
- Network connectivity is established to allow secure communication between the Identity Broker and the external IdP. When System for Cross-domain Identity Management (SCIM) is used, the inbound connectivity from the external Identity Provider to the identity broker should be allowed to support real-time provisioning events.
- [**Optional**] Log Management is recommended for centralized collection, retention, and auditability of identity-related events across the VCF Fleet.

Scope

This section provides operational guidance for the following use cases:

Use Case Number	Description
UC 1.1	How to choose between specific Modern Identity Providers (such as Okta, Ping Identity, or Microsoft Entra ID) and generic SAML 2.0 or OpenID Connect (OIDC) IdP when configuring IdP.
UC 1.2	How to select an appropriate user and group provisioning approach for VCF SSO, based on environment constraints and identity lifecycle requirements.
UC 1.3	How to design roles and manage role assignments to achieve consistent access control across the VCF Fleet, enabling standardized operations while supporting intentional scoping aligned with organizational responsibilities.
UC 1.4	How to structure programmatic access using API clients and tokens to support automation and system integrations, while minimizing operational dependency and access risk.

To implement the recommendations and best practices discussed in this section, refer to Managing Identity and Access With VCF Single Sign-On and the relevant portions of the fleet management guide for procedural guidance.

Configuring VCF SSO

VCF SSO enables unified identity management across the entire VCF fleet. The SSO configuration is a multi-steps process.

Choosing an Identity Provider Type

VCF SSO supports several types of Identity Providers (IdP), ranging from directory-based services like AD or LDAP to modern IdPs such as Okta and Microsoft Entra ID, as well as generic SAML 2.0 or OIDC options. It is recommended to select the IdP type that most closely matches your specific provider. For instance, although Okta supports both SAML 2.0 and OIDC protocols, you should choose the Okta IdP option rather than generic SAML 2.0 or OIDC IdP option. Use the generic IdP options only if your specific provider is not listed. This approach ensures that the environment uses a more deeply validated integration.

VCF SSO User and Group Provisioning Options

VCF SSO supports several user and group provisioning options: AD/LDAP On-demand Provisioning, AD/LDAP pre-provisioning, Just-in-Time (JIT) users and groups on-demand provisioning, group pre-provisioning, and SCIM 2.0.

Table 418: VCF SSO User and Group Provisioning Options

Provisioning Option	Implication	Considerations
AD/LDAP On-demand Provisioning	Users and groups are provisioned as a Day2 operation. Group memberships are discovered in real time from Active Directory upon user authentication. Provisioned users and groups are removed from the Identity Broker only when they are deleted from Active Directory.	On-demand provisioning suits environments with high turnover or frequent changes to Active Directory users and groups. Real-time AD lookups during authentication introduce higher latency and can impact performance, For a given AD, this option cannot coexist with other provisioning options, such as AD/LDAP Pre-provisioning.
AD/LDAP Pre-provisioning	Users and groups are periodically synchronized from the external directory into VCF SSO. Role assignments can be defined in advance and can remain independent of user login activity.	Access changes and offboarding are applied based on synchronization frequency, which may introduce delay at fleet scale. Where the default weekly synchronization fails to meet access grant and revocation requirements, update the interval to comply with operational standards.
JIT Users and Groups Provisioning	Users and groups are created in VCF SSO only after a successful first login. Role assignments cannot be preconfigured and must be applied post-login.	Role assignments are applied after the first login, which limits pre-defined access control. User deactivation and cleanup require operational processes to prevent a buildup of inactive user accounts.
JIT with Group Pre-Provisioning	Groups can be pre-provisioned in VCF SSO, enabling group-based role assignments before first access. User objects are created only upon first login.	Improves role consistency compared to pure JIT by allowing pre-defined group-based access. User deactivation and cleanup require operational processes to prevent a buildup of inactive user accounts.

Provisioning Option	Implication	Considerations
SCIM	Users and groups are provisioned into VCF SSO independently of user login. Role assignments can be fully defined and applied before the first login.	Introduces dependency on inbound connectivity from the external IdP to the identity broker for provisioning events. User deactivation is driven by the external IdP. Users are often disabled, not deleted, so past access and activity can still be reviewed. Disabled accounts can accumulate over time, requiring periodic cleanup.

Important:

No single user and group provisioning method is universally applicable across all VCF deployments. Environmental constraints such as regulatory requirements, network security policies, and identity lifecycle expectations influence which provisioning approach is feasible and operationally sustainable. The considerations that are outlined are intended to help organizations evaluate trade-offs and select a provisioning strategy that aligns with their fleet architecture, governance model, and compliance requirements.

Roles and Assignments for Consistent Access Control

Role-Based Access Control (RBAC) is the access control model that is used in VCF. VCF supports defining and managing roles at fleet scope, enabling consistent role usage across systems. Predefined VCF-level roles are provided for common administrative functions, and custom roles can be created to meet specific operational requirements.

VCF Custom Role Definition

Custom roles enable operational teams to align access permissions with their organizational structure and support model.

Important:

You can only choose built-in roles from NSX, VCF Operations HCX, VCF Operations orchestrator when defining a VCF custom role. Custom roles from VCF Operations, VCF Automation, and vCenter are available for VCF custom role definitions.

Figure 154: VCF Custom Role Definition Example

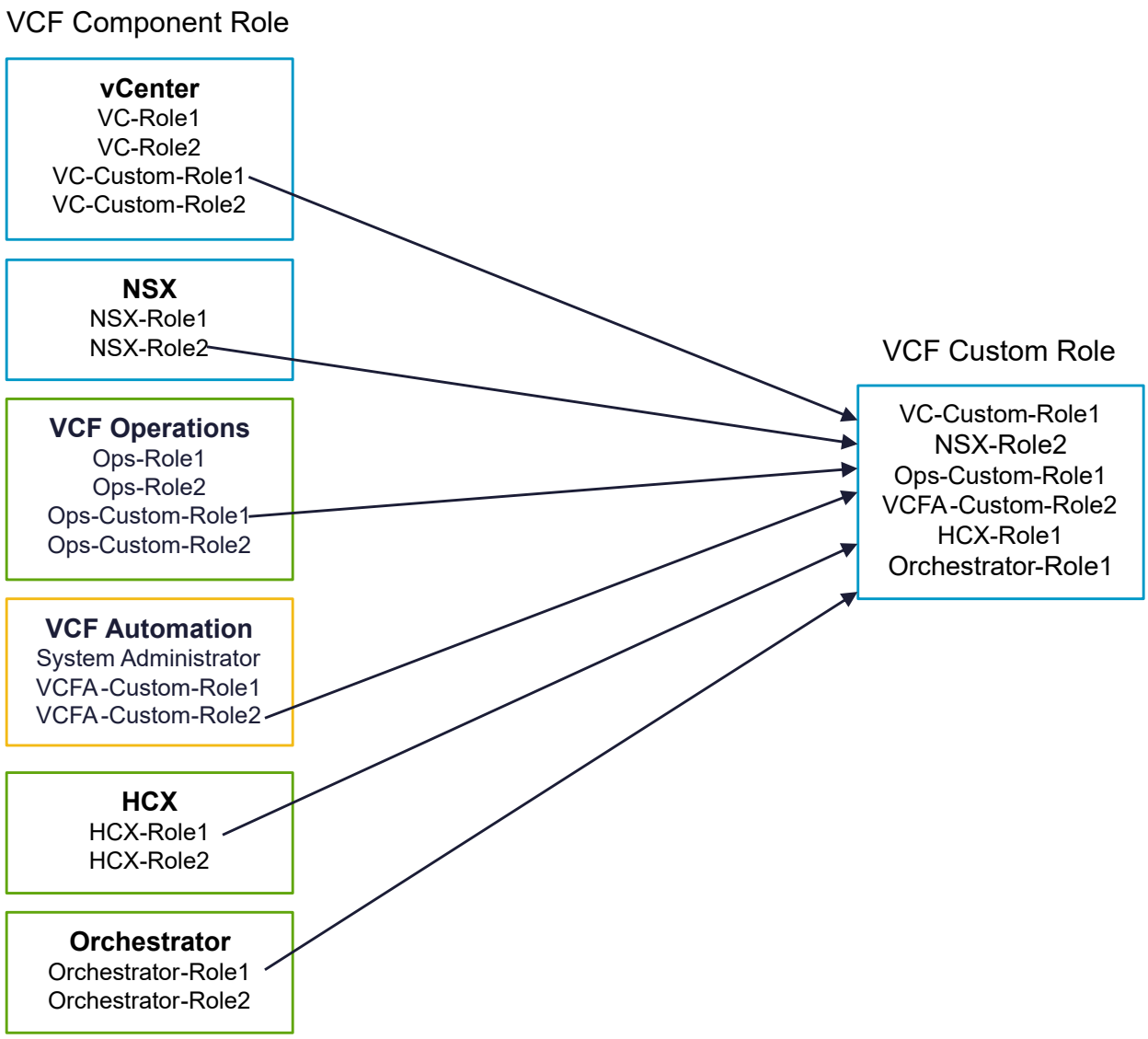


Table 419: Recommendations for Custom Role Definition

Category	Recommendation	Justification	Implication
VCF Custom Role	Administrators should design Custom roles to map directly to defined job functions or team responsibilities.	Ensure that each role represents a clear operational purpose rather than a collection of ad hoc permissions.	None.
VCF Custom Role	The principle of least privilege should be applied when aggregating component permissions into a custom role.	Reduce the impact of role misconfiguration and avoid unintended access.	None.
VCF Custom Role	Consistent role naming conventions should be established and applied across the fleet.	Improve configuration consistency and traceability.	None.

VCF Role Assignment

Role assignment in a VCF Fleet is governed through the combined use of scope, role selection, and validity controls. These guardrails are used to ensure that access is appropriately constrained, auditable, and aligned with operational responsibilities.

Table 420: Recommendations for VCF Role Assignment

Category	Recommendation	Justification	Implication
VCF Role Assignment	Role assignments should be scoped intentionally to align with operational responsibilities and separation-of-duties requirements.	Ensure that granted access reflects actual support boundaries.	None.
VCF Role Assignment	Roles assigned to users or API clients should reflect the minimum access required for their function.	Enforces the Principle of Least Privilege to minimize the attack surface and simplify security auditing.	None.
VCF Role Assignment	Use time-bound access whenever possible.	Mandates periodic reviews and reduce the risk of lingering privileges	None.

Important:

A user effective permissions are determined by the aggregation of their VCF-level roles and their specific component-level roles.

Managing vCenter Role Consistency

VCF Operations Identity and Access Management provide a centralized mechanism to define vCenter custom roles and propagate them across a VCF Fleet. Roles are defined once and can be propagated to all vCenters in a single VCF instance or across multiple VCF instances, based on the selected provisioning scope.

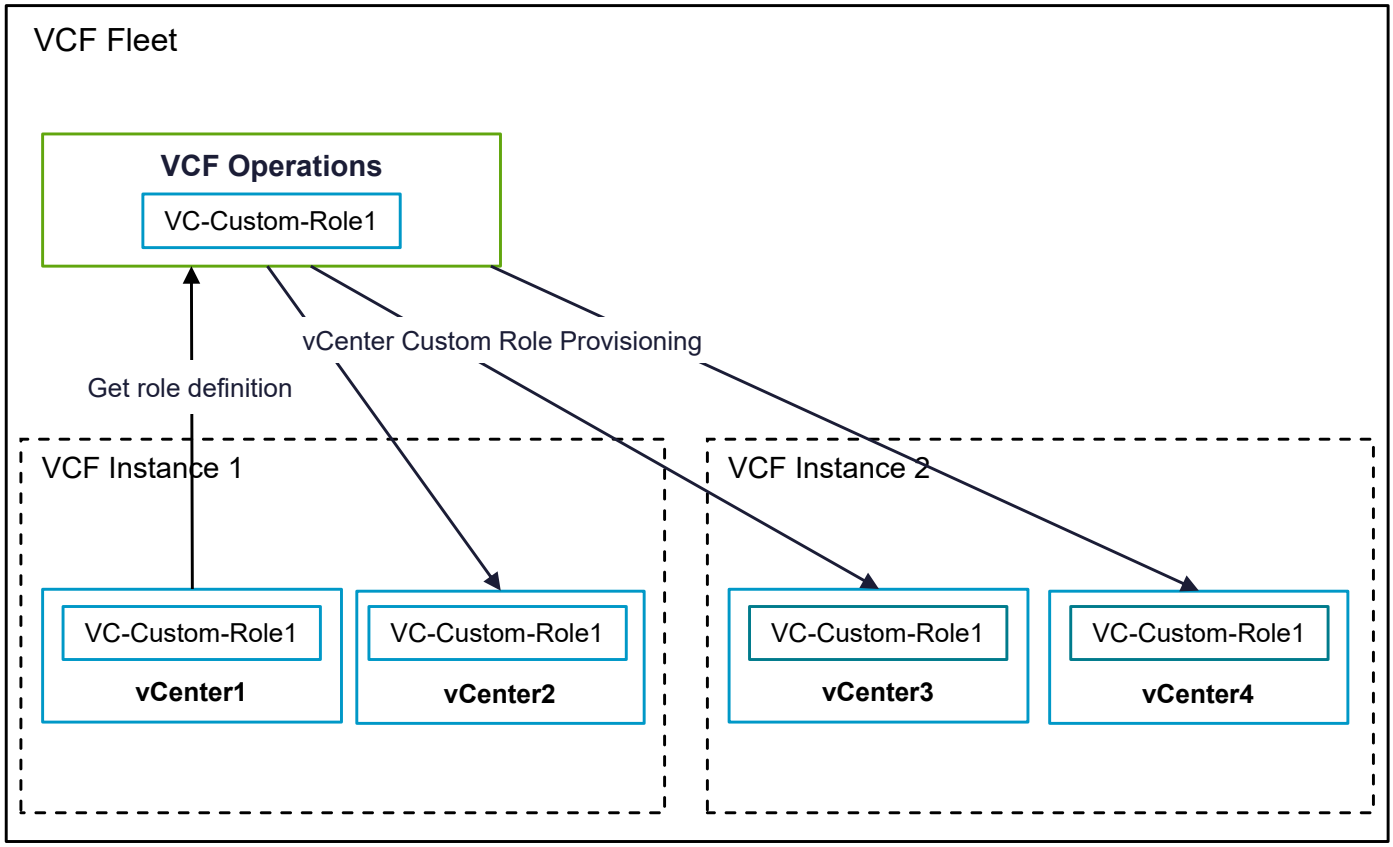
Considerations for Role Provisioning Scope

When defining the propagation scope, the operational model and support boundaries should be considered:

Table 421: vCenter Custom Role Provisioning Scope

Scope	Implication	Considerations
Fleet-Wide	The vCenter custom role is provisioned in all vCenters in a VCF fleet.	Reduces administrative overhead for standardized environments.
Instance	The vCenter custom role is provisioned in all vCenters in specific instances in a VCF fleet.	Supports separation between different contexts, such as Production and Dev/Test, or between tenants with distinct responsibility models.

Figure 155: vCenter Custom Role Provisioning



vCenter Custom Role Configuration Drift Management

VCF Operations detects configuration drift for centrally managed vCenter roles within the defined propagation scope. Any manual change made directly on a vCenter to a propagated role is treated as configuration drift.

Category	Recommendation	Justification	Implication
Drift Management	Establish a Standard Operating Procedure (SOP) for handling detected role definition changes. This protocol should distinguish between tracked updates and configuration drift, such as changes directly on an individual vCenter, and should define remediation actions to restore approved role definitions.	Provides the team with a clear playbook to quickly identify out-of-band changes and remediate promptly.	None.
Drift Management	When the default daily drift detection interval does not align with operational and compliance requirements, schedule extra drift checks using the fleet management API in VCF Operations.	Ensures the unapproved changes fast enough to fix them before they cause a security or compliance issue.	Requires custom automation to schedule additional drift checks through the fleet management API in VCF Operations.

Programmatic Access to the VCF Fleet

This section defines how operational teams should structure and manage programmatic access in a VCF Fleet to support automation and reduce dependency on individual user accounts.

VCF API Client Management

VCF Operations API clients are managed within VCF Operations and define the effective permissions for all API tokens that are issued under them.

Figure 156: VCF SSO API Client API Token Flow

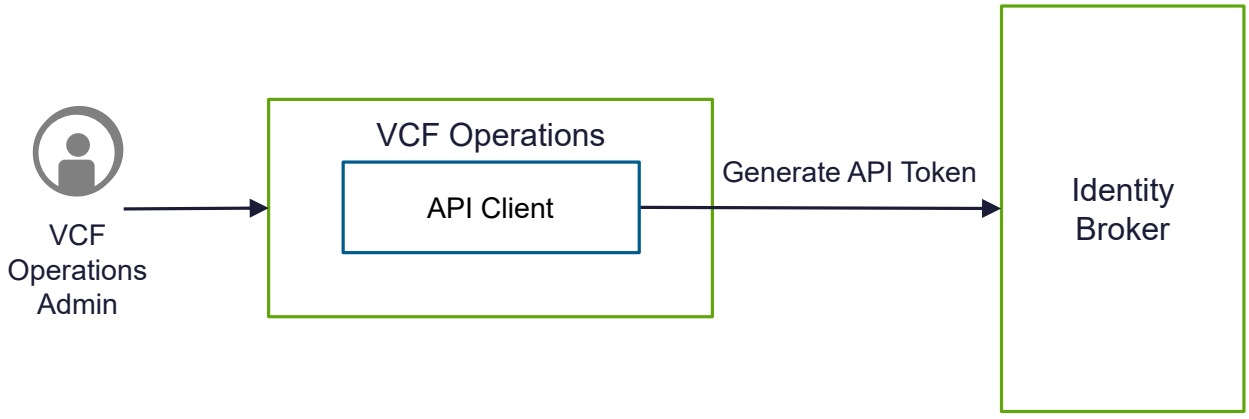


Table 422: Recommendations for API Client Management

Category	Recommendation	Justification	Implication
API Client Management	Manage API clients as first-class operational objects with clear ownership.	Ensures accountability by linking every API client to a specific owner, making it easier to track lifecycle changes and security responsibilities.	None.
API Client Management	Separate API clients should be created for distinct operational functions, such as monitoring or backup.	Ensures granular access control by matching permissions to specific functions while providing clear traceability for auditing and troubleshooting.	Requires the establishment and enforcement of a standardized naming convention to ensure each API client is easily identifiable during audits and troubleshooting.
API Client Management	API client scope and role assignment should reflect the minimum access that is required for the intended function.	Enforces the Principle of Least Privilege to reduce the attack surface and limit potential damage if there is a compromise.	None.

VCF API Token Management

VCF API access is granted through two identity types—SSO users (for humans) and API clients (for systems). Both generate API tokens that are exchanged for short-lived bearer access tokens that are used across the VCF Fleet. Operational teams should manage these tokens to ensure automation reliability and contain security risks.

Important:

API token permissions are inherited exclusively from the access levels that are assigned to SSO users or API clients at the VCF level. Tokens do not inherit SSO user permissions that are configured directly at the component level.

For example, an SSO user might be assigned the **VCF Viewer** role at the fleet level while holding a **vCenter Administrator** role directly on a specific vCenter. In this case, any access tokens that are generated by the user only reflect the VCF **Viewer** permissions.

Figure 157: VCF SSO User API Token Flow

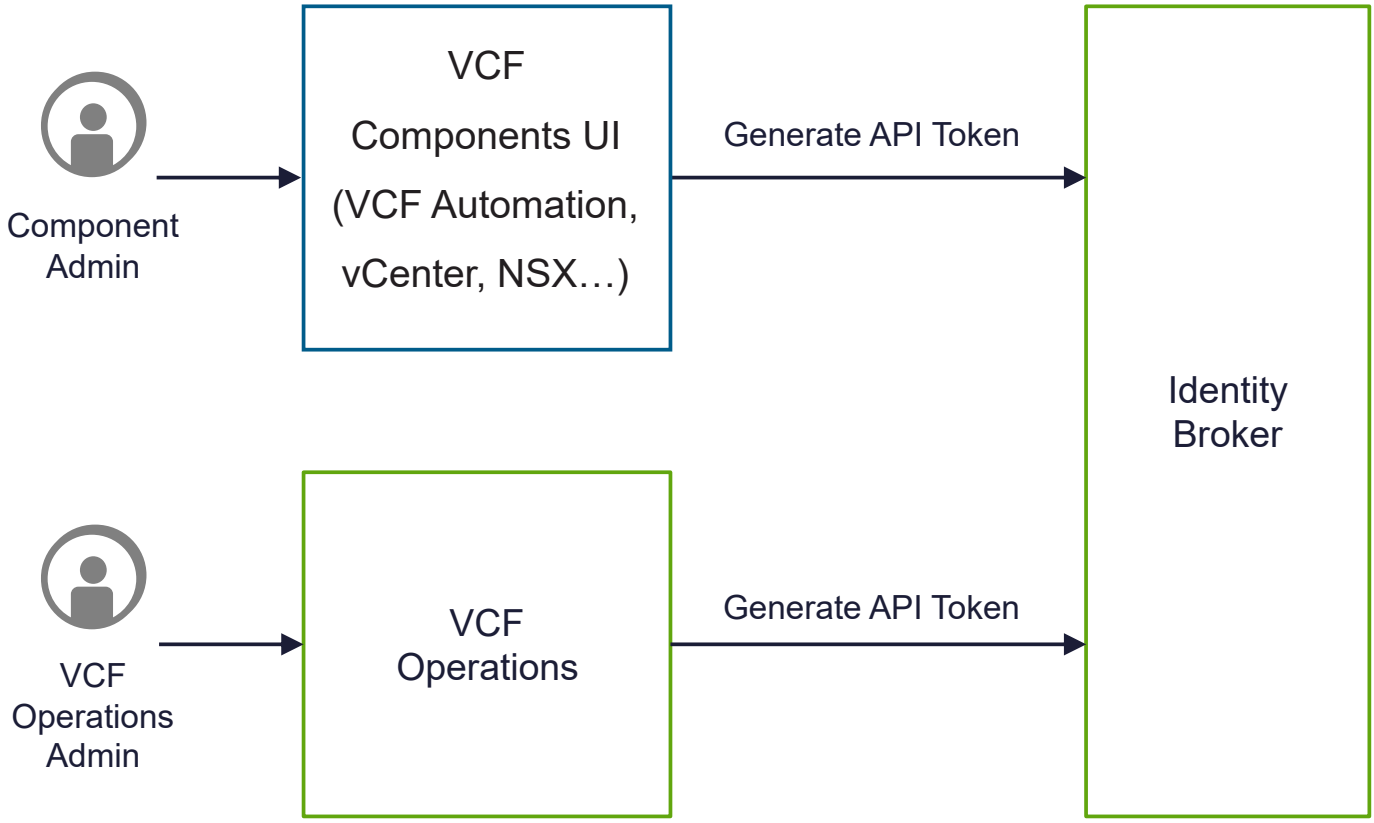


Table 423: Recommendations for API Token Management

Category	Recommendation	Justification	Implication
API Token Management	Use API Client–based tokens for automation or system-to-system integrations.	API client tokens remain active even if the account that generated them is disabled or deleted, ensuring that critical integrations are not disrupted.	None.
API Token Management	Restrict user-based API tokens for manually triggered activities on-demand activities.	User-based API tokens are tied to personal accounts and would fail when the account is disabled.	Requires the establishment and enforcement of a standardized naming convention to ensure each API token is easily identifiable during audits and troubleshooting.
API Token Management	Token lifetimes should be defined intentionally to balance operational usability and timely invalidation of access.	Reduces the risk of unintended lingering access, as bearer tokens remain valid until expiry even if the owner is deleted.	None.
API Token Management	Separate tokens should be used for separate operational functions.	Limits impact during API token revocation or rotation.	None.
API Token Management	Review and update Token Lifecycle and Security settings within VCF Operations to align with organizational governance requirements for both SSO users and API clients.	The default settings may not meet organizational security requirements.	None.

VCF API Emergency Access

Emergency access is intended to support break-glass scenarios where standard access mechanisms are unavailable. The emergency API access token provides high-privilege access and is scoped per identity broker instance.

Table 424: Recommendations for Emergency Access Token Management

Category	Recommendation	Justification	Implication
Emergency Access	Emergency client access tokens should be stored securely using an enterprise-grade secret management solution.	Prevents high-privilege credentials from being exposed, ensuring they are only accessible through a secure, audited vault.	Requires an enterprise-grade vault system.
Emergency Access	Access to manage or retrieve emergency client credentials should be restricted to a small, well-defined set of operational roles with VCF Identity and Access “Manage” Access.	Ensures that high-privilege access is strictly controlled.	None.
Emergency Access	Emergency client access tokens should be regenerated regularly.	Limits the window of risk for leaked emergency access tokens and ensures that emergency access is always verified and up to date.	None.

Category	Recommendation	Justification	Implication
Emergency Access	Ensure that the emergency API client is not used for daily operations. Configure event filters in Log Management to explicitly track, audit, and alert on any authentication or activity events associated with the Emergency Client ID.	Mandates tracking and alerting for all emergency access token activity to proactively prevent and investigate misuse.	Requires Log Management is deployed.

Certificate Management

Certificate management ensures secure communication and authentication integrity within a VCF Fleet. This section provides guidance on certificate management for VCF management components such as VCF Operations, VCF Automation, vCenter, NSX, and SDDC Manager, and infrastructure components (ESX hosts).

Prerequisites

The guidance in this section assumes the following environmental state:

- A VCF Fleet where all management components are deployed and operating normally
- An enterprise Certificate Authority (CA) is available for integration.
- [**Optional**] A defined governance model for certificate lifecycle operations, including approval policies, certificate standards, and CA issuance controls.
- [**Optional**] Log Management is recommended for the audit trail required for certificate-related compliance validation across the fleet.

Scope

This section provides operational guidance for the following use cases:

Use Case Number	Description
UC 2.1	How to choose the right approach to manage the certificates in a VCF Fleet.
UC 2.2	How to manage ESX host certificates while balancing security requirements and operational efficiency.
UC 2.3	How to manage vCenter machine SSL certificates in a VCF fleet.

VCF Certificate Lifecycle Management

This section details the certificate management and lifecycle automation strategies for VCF certificates.

VCF Certificate Management

VCF supports multiple options for certificate management. Using VCF Operations for VCF certificate management whenever possible.

Table 425: VCF Certificate Management Options

Management Option	Scope and Coverage	Recommended Use
VCF Operations	Fleet-wide	It is the recommended method since it aligns with the fleet management model and provides a unified console and APIs.
SDDC Manager	Instance-level	SDDC Manager Certificate Management APIs are only for backward compatibility during the transition to VCF Operations.

Management Option	Scope and Coverage	Recommended Use
vCenter	vSphere Certificates	Avoid using vCenter for VCF certificate management unless the use case is not supported by VCF Operations or SDDC Manager.

VCF Certificate Lifecycle Automation

VCF fleet administrators should select a certificate management automation approach that balances efficiency with other factors.

Evaluate the following decision drivers:

- Operational Scale: As fleet size increases, manual intervention becomes a reliability risk. For large-scale fleets, Auto-Renewal or API-Based Automation is preferred to standardize execution.
- Automation Maturity: API-Based Automation assumes that the organization is able to design, operate, and govern custom certificate lifecycle workflows. This includes ownership of workflow logic, failure handling, validation, and auditability across the fleet. Organizations should assess whether they can sustainably maintain this operational responsibility before adopting this approach.
- Regulatory and Compliance: Regulated environments often mandate strict issuance policies or manual approval gates. Since native auto-renewal relies on *post-event* auditing, organizations requiring *pre-event* approval (change control) should use API-Based Automation to enforce these checks before issuance.
- Key Custody: If security policies require private keys to be generated externally (for example, on a Hardware Security Module), API-Based Automation is the only viable path. Native Auto-Renewal relies on VCF generating the keys.

Table 426: VCF Certificate Lifecycle Automation Options

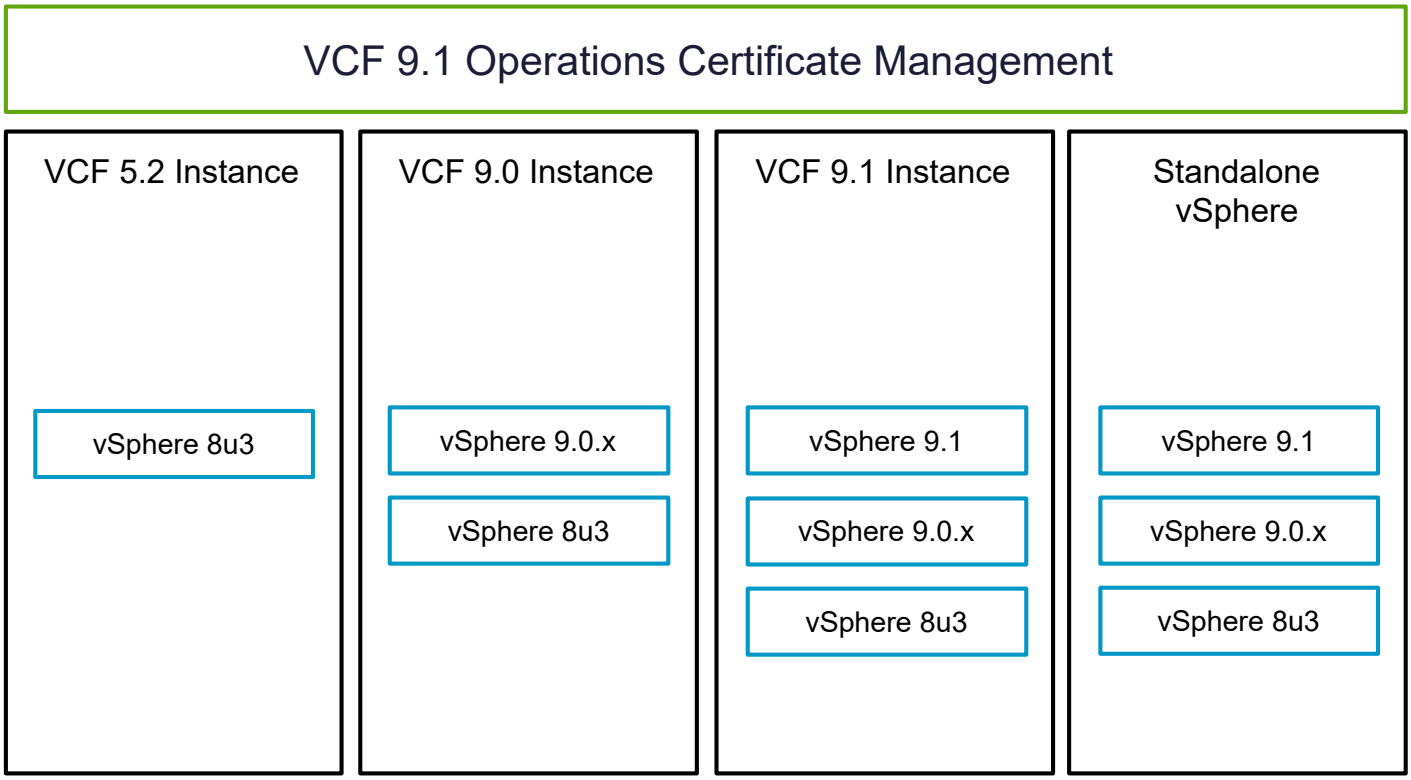
Automation Option	Description	When to Use	Considerations
Auto-Renewal	Certificates are automatically renewed according to a system-defined schedule through VCF Operations built-in integration with a supported external CA.	Fleets prioritizing operational efficiency and consistent certificate lifecycle management at scale.	Simplifies certificate lifecycle operations across many endpoints. Relies on continuous connectivity to external CA services. The built-in renewal mechanism offers no administrative control over task scheduling. Since the system dictates all operation timing, it is unsuitable for environments requiring short-lived certificates (less than 60 days) or manual approval workflows.
API-Based Automation	Certificate lifecycle operations are performed via APIs and integrated with enterprise automation platforms and workflows.	Large enterprises with established automation ownership and governance. Environments requiring lifecycle behavior not fully addressed by native auto-renewal, such as customized renewal schedules.	Enables greater control over certificate lifecycle policies and governance integration. Shifts lifecycle responsibility to organization-managed automation workflows. Requires sustained operational ownership for validation, error handling, and auditability of certificate operations.

Automation Option	Description	When to Use	Considerations
Manual	Certificate lifecycle operations are performed manually, including CSR generation, submission to a CA, and certificate installation.	Small or isolated environments with limited automation requirements. Temporary approach during initial deployment or transition toward automated models.	Introduces high operational overhead at scale. Increases reliance on human processes, which may impact consistency and reliability. Does not align well with long-term fleet management efficiency objectives.

vSphere Certificate Management

VCF 9.1 extends certificate management support to include VCF 5.2 and standalone vSphere 8.0 U3 and above environments to ensure consistent infrastructure operations during VCF adoption. This section provides guidance on vSphere certificate management in VCF.

Figure 158: VCF Operations vSphere Certificate Management Support Matrix



ESX Certificate Management

Managing ESX host identities requires balancing operational efficiency with organizational security requirements for trust boundaries.

While ESX hosts typically generate their own private keys, some highly regulated environments require keys to be created and kept externally. In these cases, the keys remain outside the visibility of VCF Operations, so VCF Operations cannot be used to manage those ESX host certificates.

Table 427: ESX Host Certificate Management Options

ESX Certificate Issuer	Description	When to Use	Considerations
Default VMCA	ESX host certificates are issued by the VMware Certificate Authority (VMCA) and fully managed by VCF Operations.	The default and recommended option for VCF fleets. Environments prioritizing standardized lifecycle management and minimal operational overhead at scale.	Fully aligns ESX certificate lifecycle with VCF fleet management and automation. Simplifies renewal and replacement operations across large fleets. ESX host certificates do not chain to the enterprise Certificate Authority, which may not align with corporate Public Key Infrastructure (PKI) trust standards.
Enterprise CA (VCF Operations)	ESX host certificates must chain to an enterprise CA, with all lifecycle operations that are centrally managed by VCF Operations.	Fleets requiring direct enterprise CA–signed ESX certificates while keeping ESX certificate lifecycle operations centrally managed and orchestrated by VCF Operations.	Introduces a one-time certificate replacement activity when transitioning from the default VMCA-issued model. Introduces operational dependency on enterprise CA availability, policies, and issuance processes. Native auto-renewal can be used when supported by the enterprise CA; otherwise, organization-managed automation is required for certificate auto-renewal.
Enterprise CA (vSphere / Custom Automation)	ESX host certificates are signed directly by an enterprise Certificate Authority and managed via vSphere or custom automation	Environments where ESX host private keys are generated and retained outside the ESX hosts due to security, regulatory, or compliance requirements.	Falls outside native VCF certificate lifecycle automation and visibility. Requires organization-managed processes for renewal, replacement, and governance. Increases operational overhead, coordination effort, and long-term lifecycle complexity at fleet scale.

Important:

Whenever ESX certificate operations are performed through VCF Operations (UI or API), the ESX certificate mode in vCenter is changed to Custom. Therefore, all subsequent renewals must be managed centrally by using VCF Operations.

VI administrators can automate ESX host certificate renewals by using either the native auto-renewal feature or custom automation using the VCF Operations fleet management APIs.

Important:

VCF does not support configuring the VMCA as a customer intermediate CA for both greenfield and brownfield deployments that are converged or upgraded to VCF 9.1.

vCenter Machine SSL Certificates Management

This use case focuses on the vCenter Machine SSL certificate, which serves as the primary endpoint identity. The selection determines how vCenter aligns with enterprise PKI standards.

Table 428: vCenter Machine SSL Certificate Management Options

vCenter Certificate Issuer	Description	When to Use	Considerations
Enterprise CA	The vCenter Machine SSL certificate is signed directly by the trusted Enterprise Certificate Authority and managed by VCF Operations.	Recommended for VCF environments where vCenter must meet enterprise PKI standards to avoid browser and client trust warnings.	Introduces a one-time certificate replacement activity and requires ongoing certificate lifecycle operations.
Default VMCA	The vCenter Machine SSL certificate is issued and signed by the default VMCA.	Suitable for lab or small, isolated environments where integrating with corporate PKI is not feasible or necessary.	Requires minimal operational effort to manage. vCenter Machine SSL certificate does not chain to the enterprise PKI, resulting in client and browser trust warnings.

Recommendations for VCF Certificate Management

Follow these guidelines to ensure consistent certificate lifecycle management and maintain a secure trust model across the VCF fleet.

Table 429: Certificate Management Operational Guidelines

Category	Recommendation	Justification	Implication
Certificate Management	Perform certificate management for all supported components exclusively through VCF Operations.	The practice aligns with the fleet management model and provides a unified console and APIs.	None.
Certificate Management	Avoid using the SDDC Manager for certificate management; use SDDC Manager APIs only for backward compatibility during the transition to VCF Operations.	The SDDC Manager UI is deprecated in VCF 9.0. Restricting its API use to legacy automation ensures a structured migration to the modern VCF Operations fleet management API.	None.
Certificate Management	Do not use vCenter (API, UI, CLI) to manage ESX host or vCenter certificates once those certificates are being managed by VCF Operations.	Direct management at the component level creates orphaned CSRs in vCenter and results in a "Source of Truth" conflict.	None.
Certificate Management	Default CA-signed certificates on VCF management components should be replaced with certificates that are issued by a trusted enterprise Certificate Authority before going to production.	Using trusted certificates establishes a consistent trust baseline and reduces operational friction during the lifecycle management and compliance activities.	None.

Category	Recommendation	Justification	Implication
Certificate Management	Use VCF Operations built-in alerts and notifications mechanism to actively monitor certificate expiration.	Automated monitoring ensures that administrators receive advanced warning of pending certificate expirations, allowing for timely rotation and preventing unexpected downtime.	Requires integrations with the notification plugin in VCF Operations for outbound alerting.
Certificate Management	Configure Log Management to retain all certificate-related events in accordance with regulatory or organizational requirements.	These logs are critical for historical audit validation and compliance reviews.	Requires Log Management is deployed.
Certificate Management	Use Audit Records in VCF Operations to centrally trace the certificate lifecycle operations.	Provides a forensic audit trail for compliance verification and incident response across all certificate lifecycle events.	None.
Certificate Management	Assign Certificate "Manage" and "View" permissions to specific user roles within VCF Operations based on their job requirements.	Ensures users only have the access they need, reducing the risk of unauthorized changes or accidental errors.	None.

To implement the recommendations and best practices discussed in this section, refer to Managing Certificates in VMware Cloud Foundation and the relevant portions of the fleet management guide for procedural guidance.

Account and Password Management

VCF Password Management ensures security hardening, compliance with organizational policies, and operational integrity within a VCF Fleet. It covers the generation and update of credentials used by VCF Fleet management components, such as VCF Operations and VCF Automation, as well as core infrastructure components like vCenter, NSX, SDDC Manager, and ESX hosts in a VCF instance. Also, this section provides guidance on the lifecycle management of service accounts that are used for cross-component integration.

Prerequisites

The guidance in this section assumes the following environmental state:

- A VCF Fleet where VCF Operations are deployed and operating normally.
- [**Optional**] Log Management is recommended for the audit trail required for password-related compliance validation across the fleet.

Scope

This section focuses on the following core operational use cases:

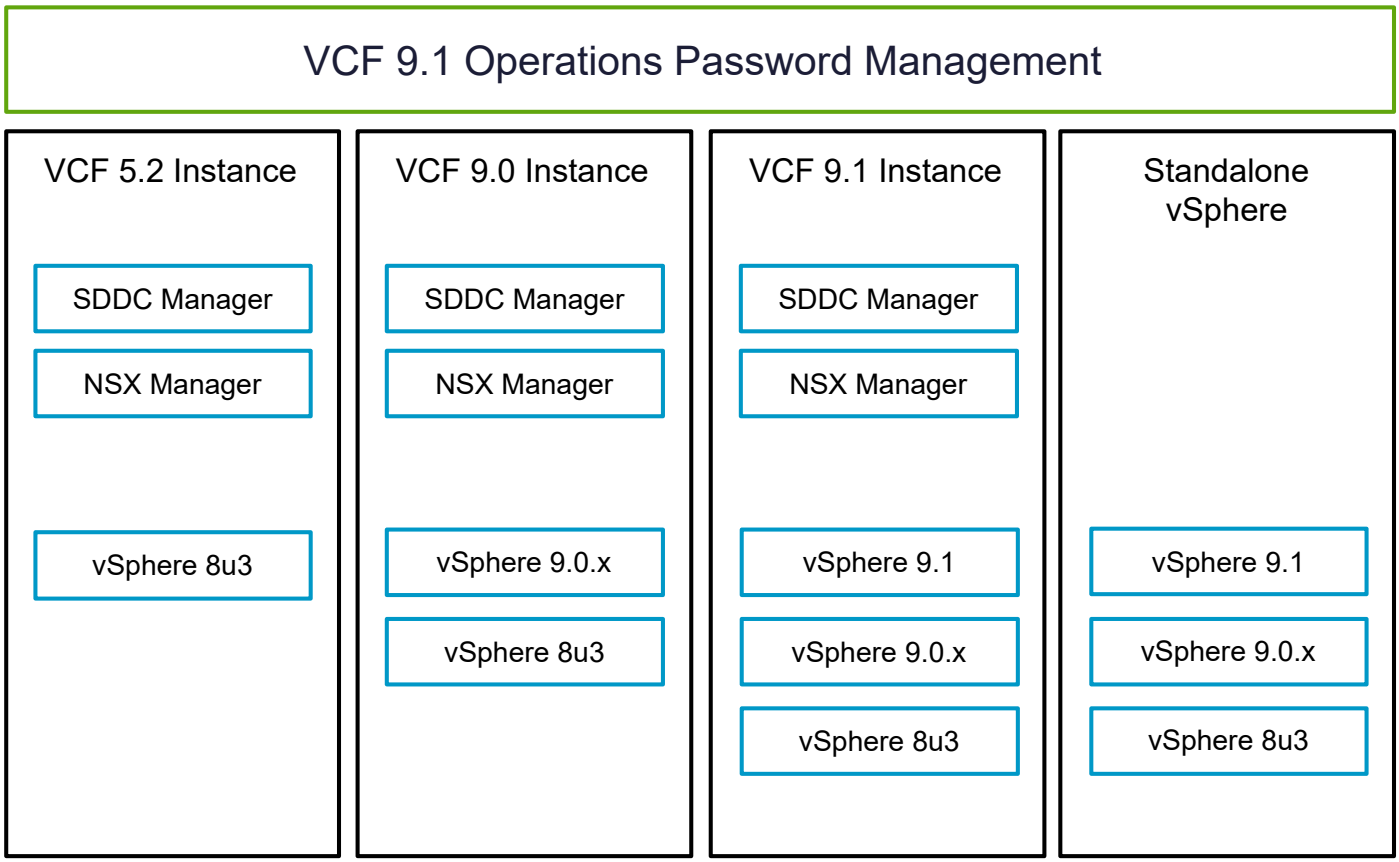
Use Case Number	Description
UC 3.1	How do I choose an appropriate approach to manage VCF break-glass accounts' passwords
UC 3.2	How to centrally manage and enforce a uniform password policy across the VCF Fleet
UC 3.3	How to design service account usage for VCF component integration

VCF Password Lifecycle Management

This section outlines the different options for managing the lifecycle of VCF credentials across the fleet.

VCF 9.1 extends password management support to include VCF 5.2 instances and standalone vSphere 8.0 U3 and above environments to ensure consistent infrastructure operations during VCF adoption.

Figure 159: VCF 9.1 Password Management



Note:

This figure is a representative view and does not include all supported components by VCF Password Management, such as VCF Operations, VCF Automation, VCF services runtime, VCF Operations HCX, NSX Edge, and Avi Load Balancer.

VCF Password Management Options

VCF supports multiple methods for password management. Using VCF Operations for VCF password management whenever possible.

Management Option	Scope and Coverage	Recommended Use
VCF Operations	Fleet-wide	It is the recommended method since it aligns with the fleet management model and provides a unified console and APIs.
SDDC Manager	Instance-level	This method is maintained for backward compatibility and is useful for customers transitioning from older versions to VCF 9.
Component	Component-specific	This approach should be used for emergency or break-glass scenarios only, such as password reset. **Note**: Password reconciliation in SDDC Manager is required when updating NSX and vSphere passwords on the systems directly to prevent integration failures.

VCF Password Management Automation

Depending on an organization's operation automation maturity and specific Privileged Access Management (PAM) requirements, administrators can choose between integrated automation, API-based orchestration, or manual management to ensure operational integrity and security hardening.

Table 430: VCF Password Lifecycle Automation Options

Automation Option	Description	When to Use	Considerations
Integrated Automation (CyberArk)	Credentials are managed via the CyberArk VCF plugin provided by CyberArk. This approach is used to eliminate manual secret handling and provide enterprise-grade auditing to justify compliance.	Customers who have adopted and plan to adopt CyberArk as their enterprise PAM.	Offers a robust audit trail but introduces a hard dependency on plugin availability and version compatibility.
API-Based Automation	Lifecycle operations are orchestrated via VCF Operations APIs integrated into external automation. This improves operational efficiency by embedding rotation into existing CI/CD pipelines.	Large enterprises with mature automation pipelines but no native PAM-VCF integration.	Requires custom development and ongoing maintenance of scripts and secure secret handling methods.
Manual	Administrators manually rotate and manage passwords using the VCF Operations UI.	Small-scale environments or as a temporary, transitional approach before automation adoption.	Labor-intensive and difficult to scale; highly susceptible to human error and potential lockout scenarios.

VCF Password Policy Management

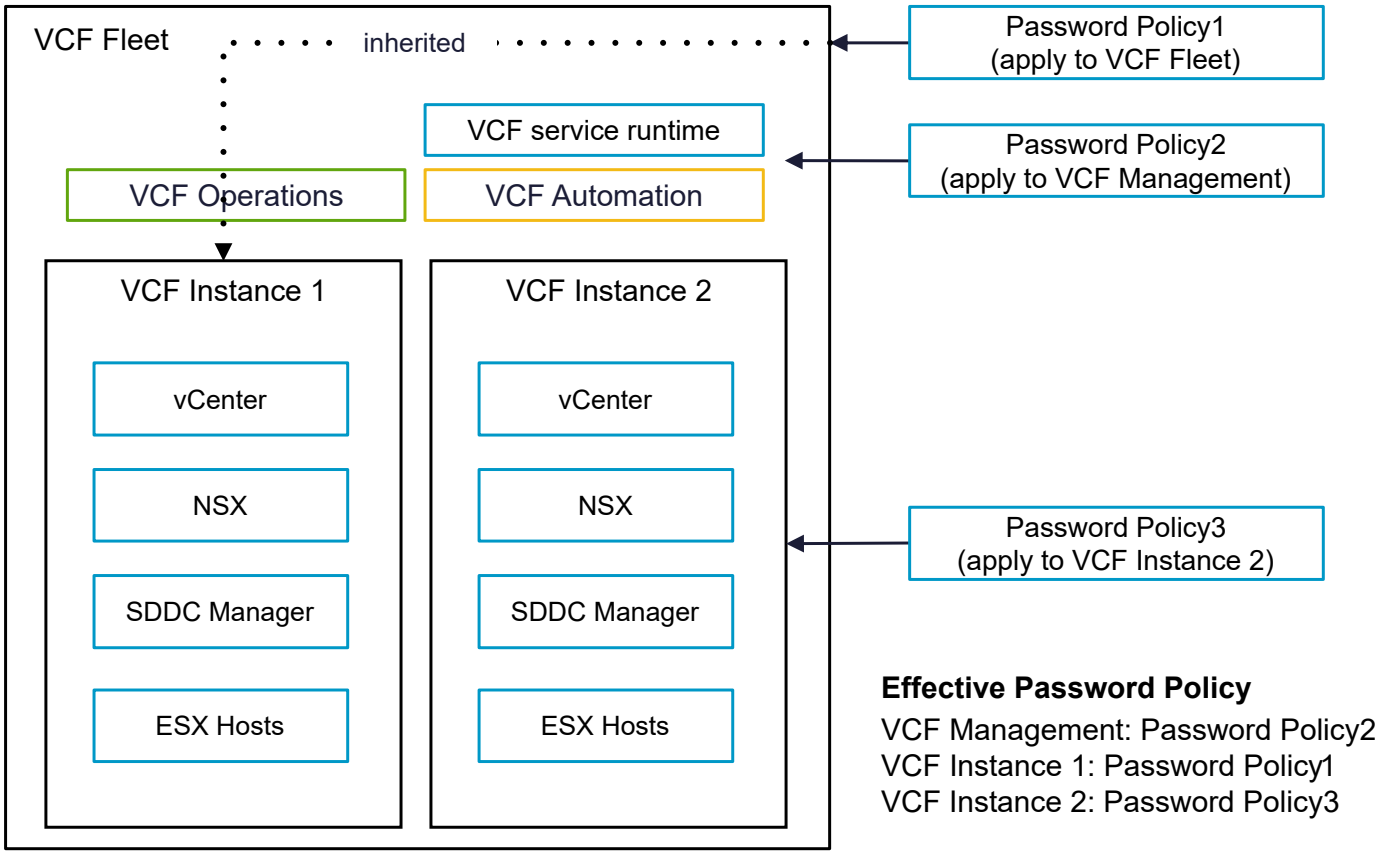
Managing password policies within a VCF Fleet requires balancing uniform compliance with the need for granular control. This section defines the scope of policy application, allowing administrators to enforce either a single fleet-wide password policy for centralized governance or instance-specific policies to accommodate unique tenant or regulatory requirements.

Policy Scope	Description	When to Use	Considerations
VCF Fleet-wide	Enforces a single password policy across the entire VCF Fleet. This ensures uniform governance and significantly reduces administrative overhead.	Enterprise customers require a consistent security posture across the whole VCF Fleet.	Uniform compliance is guaranteed, but it does not allow for customization to meet specific tenant or regulatory requirements
VCF Instance	Uses different password policies for different VCF instances. This provides the operational flexibility that is needed to satisfy unique regulatory or tenant-specific mandates.	Multi-tenant environments or instances where tenants have different complexity, expiration, and lock out requirements.	Offers granular control for instance-specific requirements for VCF credentials.

Important:

Password policies that are assigned to specific VCF Instances or VCF Management take precedence over and override the policy that is assigned at the VCF Fleet level.

Figure 160: Effective Password Policy



VCF Service Accounts

In VCF, accounts are categorized into three distinct types:

- OS Account: These accounts are provisioned directly within the operating system of a virtual appliance and are primarily used for Secure Shell (SSH) access. A common example is the vCenter Server root account.
- Application Account: These accounts are designed for interactive logins, granting users access to the VCF component UI and API. A typical example of a user account is the vCenter administrator@vsphere.local account.
- Inter-Component Account: These accounts that are used for one VCF component to communicate with another VCF component. An example of an inter-component account is *svc-ops-vc-vcf-ops-cluster-uid@vsphere.local* that is used for communication from VCF Operations to vCenter.

Starting with VCF 9.0, the VCF platform offers the capability to manage inter-component accounts for component integration, natively handling their automated creation and credential lifecycle.

For more information on active service accounts, see Account Management Design

Recommendations for VCF Account and Password Management

Follow these guidelines to ensure consistent password management operations across the fleet.

Table 431: Password Management Operational Guideline

Category	Recommendation	Justification	Implication
Password Management	Regularly update credentials as per customer password standard.	Reduces the risk of credential compromise and limits the window of opportunity for attackers.	None.
Password Management	Store all VCF credentials in an Enterprise Vault system when using API or manual processes.	Ensure that the vault remains the authoritative source of truth for all secrets.	Requires an enterprise vault system.
Password Management	Monitor and alert on credential retrieval attempts from SDDC Manager via API or CLI access using VCF Operations for Logs	Direct retrieval of break-glass credentials is commonly considered a security and compliance risk in highly regulated environments.	Requires Log Management is deployed.
Password Management	Use VCF Operations built-in alerts and notifications mechanism to actively monitor password expiration.	Automated monitoring ensures that administrators receive advanced warning of pending password expirations, allowing for timely rotation and preventing unexpected downtime.	Requires integrations with the notification plugin in VCF Operations for outbound alerting.
Password Management	Use Audit Records in VCF Operations to centrally trace the password lifecycle (generation, rotation, retrieval) and correlate operations with associated actors and timestamps.	Ensures a verifiable audit trail for compliance and accountability for all credential lifecycle events.	Requires Log Management is deployed.
Password Management	Configure Log Management to retain all credential-related events in accordance with regulatory or organizational requirements.	These logs are critical for historical audit validation and compliance reviews.	Requires Log Management is deployed.
Password Management	Assign Password "Manage" and "View" permissions to specific user roles within VCF Operations based on their job requirements.	Ensures users only have the access they need, reducing the risk of unauthorized changes or accidental errors.	None.

Category	Recommendation	Justification	Implication
Password Policy	Once the policy is active, update your current password immediately to ensure compliance.	VCF Password Policy complexity settings (password length and character limits) do not apply to the existing passwords.	None.
Password Policy	Periodically review the password policy settings (complexity, lockout, expiration, history).	Ensure that password policies remain aligned with evolving internal security standards and external regulatory requirements.	None.
Password Policy	Establish a Standard Operating Procedure (SOP) to handle password policy configuration drift detected by VCF Operations.	Provides the team with a clear playbook to quickly identify an unauthorized change and remediate to a known good state.	None.
Service Accounts	Use VCF-managed inter-component accounts for integration between components whenever possible.	Reduces operational overhead and minimizes human error as the platform natively handles account creation and ongoing password management.	None.
Service Accounts	Do not change inter-component account passwords or permissions directly on VCF components.	Manual changes may break the integration between VCF components and may cause unnecessary service outages.	None.
Service Accounts	Enable SDDC Manager password auto-rotation for inter-component accounts that can only be managed by SDDC Manager.	Reduces operational overhead and minimizes human error by ensuring security compliance without manual intervention.	The rotation frequency is limited to 30 days, 60 days, or 90 days.

To implement the recommendations and best practices discussed in this section, refer to Managing Passwords for VMware Cloud Foundation Components and the relevant portions of the fleet management guide for procedural guidance.

Fleet Configuration Management

VCF Configuration Management can provide configuration consistency and operational integrity across a VCF Fleet. By using vCenter configuration templates and vSphere cluster templates, administrators can establish standardized baselines to monitor vSphere infrastructure configuration at scale. While current capabilities are focused on drift detection, this approach significantly reduces reliability risks by providing clear visibility into manual or unintended changes that occur as the fleet size increases.

Prerequisites

The guidance in this section assumes the following environmental state:

- A VCF Fleet where VCF Operations is deployed and operating normally.
- One or more vSphere Clusters are managed with vSphere Configuration Profiles in a vCenter instance that serves as the authoritative source for the cluster template.
- All target vCenter Server instances and vSphere clusters are under the VCF Operations management.
- [**Optional**] Log Management is configured as the central logging platform to provide the audit trail for configuration changes.

Scope

This section provides operational guidance for the following use cases:

Use Case Number	Description
UC 4.1	How to choose the appropriate scope for template application: Fleet-wide or Selected Objects
UC 4.2	How to choose a management approach for template storage and versioning: VCF Operations (Local) vs. Git (External).

Defining Template Application Scope

Managing configuration templates requires balancing uniform governance with the need for granular control.

The template application scope is determined by the Group and Objects within a VCF Operations Policy that is linked to a Configuration Management Template. Administrators should structure their Operations policies and corresponding object groups to align with their VCF fleet deployment and operational requirements.

Template Application Scope	Description	When to Use	Considerations
VCF Fleet-wide	Establish a single configuration baseline across the entire VCF Fleet. This is typically achieved by selecting the Default Operations Policy with a Configuration Template.	Enterprise customers require a consistent vSphere configuration across the whole VCF Fleet.	Monitors for configuration consistency but does not allow for customization to meet specific tenant requirements.
Selected Objects	Associate templates to specific vCenters or clusters via VCF Operations custom policy.	Multi-tenant environments where tenants have varying settings. Heterogeneous Environments where Production and Non-production assets (vCenter instances or vSphere clusters) are mixed within the same VCF Fleet, VCF Instance, or vCenter.	Provides operational flexibility to accomodate varied business logic but increases overhead as the number of templates grows. Criteria-based Custom Groups enable dynamic membership updates; however, precise criteria are essential to avoid unintended policy application.

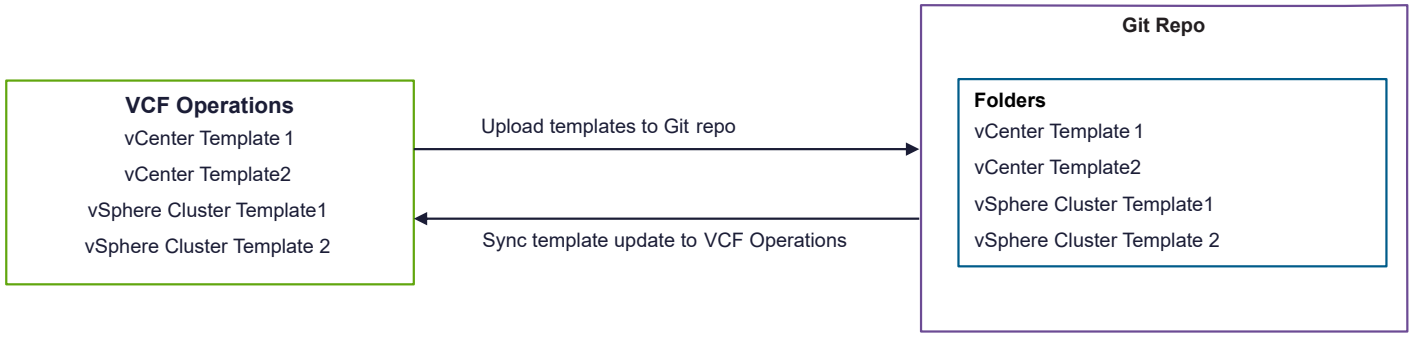
Template Management and Versioning

Organizations must decide where their primary records for configurations reside based on their automation maturity. This decision impacts how templates are versioned, reviewed, and applied across the VCF Fleet.

Table 432: Template Management Options

Management Option	Description	When to Use	Considerations
VCF Operations	Templates are created, stored, and managed directly within the VCF Operations.	Small-scale environments or as a temporary approach before Git Repository adoption.	Labor-intensive and difficult to scale; highly susceptible to human error during manual updates.
Git Repository	Templates are treated as code stored in an external Git repository.	Large enterprises with mature automation pipelines and established governance.	Since VCF Operations Git Integration supports a single branch and repo, follow these best practices to minimize misconfiguration: • Code review is enabled by default during VCF Operations integration and should not be disabled without a valid reason. • Because each template is stored in a folder that is named after the template in the Git repo, always use unique and meaningful names for your templates.

Figure 161: VCF Configuration Management Git Integration



Recommendations for VCF Configuration Management

Follow these guidelines to consistently manage configuration across the fleet.

Table 433: vSphere Configuration Management Operational Guideline

Category	Recommendation	Justification	Implication
Template Management	Establish a standardized VCF Operations template and policy naming convention.	Standardized naming prevents administrative confusion in VCF Operations, especially as the number of templates and custom policies grows to satisfy unique tenant or regulatory mandates.	None.
Template Management	Periodically review vCenter and cluster configuration templates to ensure configuration consistency and minimize misconfiguration that is caused by human error.	These reviews help ensure that templates remain aligned with evolving corporate VCF configuration standards.	None.
Template Management	Defines a single template per object (e.g., a vCenter or a cluster) as its baseline config.	Prevents baseline conflicts where multiple templates might provide contradictory settings.	None.
Drift Detection	Leverage VCF Operations Automation Central to schedule regular drift detection tasks for the desired state.	These automated jobs verify that effective configurations on target instances have not deviated from defined fleet-wide or custom baselines.	Requires a brief administrative task to create the configuration drift check job.
Drift Detection	Enable email notifications when creating recurring drift detection jobs in VCF Operations Automation Central.	Ensures the operations team is notified of unapproved changes so drift can be remediated before causing security or compliance issues.	Requires integration with an email server using the Standard Email Plugin in VCF Operations for outbound email notifications.
Drift Detection	Establish a Standard Operating Procedure (SOP) to handle "Configuration Drift" detected by scheduled jobs.	This protocol should dictate how to validate if a deviation is an approved exception or an unauthorized change requiring remediation.	None.
Drift Detection	Update VCF Operations default Configuration management drift history Data retention to meet operational requirements.	Provides a deeper historical lookback to align with operational audit cycles and ensures that configuration drift remains visible for forensic analysis beyond the 7 days default.	Requires a brief administrative task to update the retention setting, which is technically capped at a maximum of 30 days.

To implement the recommendations and best practices discussed in this section, refer to Managing the Configuration of your Environment and the relevant portions of the fleet management guide for procedural guidance.

VCF Tag Management for vSphere

VCF Tag Management supports vSphere tag management across a VCF Fleet. This capability extends beyond traditional tagging that is confined to individual components like vCenter Server, enabling VCF Fleet-wide tag visibility and orchestration. This ensures that tags that are used for policy enforcement are consistent across all managed vCenter instances.

Prerequisites

The guidance in this section assumes the following environmental state:

- A VCF Fleet where VCF Operations is deployed and operating normally.
- [**Optional**] Log Management is configured as the central logging platform to provide the audit trail for configuration changes.

Scope

This section provides operational guidance for the following use cases:

Use Case Number	Description
UC 5.1	How to choose a management approach for vSphere tag management in a VCF fleet.
UC 5.2	How to define the tag provisioning scope in a VCF fleet.

VCF Tag Lifecycle and Ownership for vSphere

VCF fleet administrators should select a tag management approach which is based on their organizational structure and how they intend to balance global standardization with local operational flexibility.

Table 434: Tag Management Options

Management Option	Description	When to Use	Considerations
Centralized Tag Management	VCF Operations is used for all tag lifecycle stages, including creation, update, deletion, synchronization, and tag assignment to objects in vCenter.	Organizations with a "Platform Engineering" team that manages the entire global estate.	Simplifies auditing; requires the central team to have visibility into the local resource context.
Centralized Tag Definition and Delegated Assignment	VCF Operations is used for tag creation, update, deletion, and synchronization, while tag assignment is performed at each vCenter.	Geographically distributed teams (e.g., APJ vs. US) where local context and regional autonomy matter.	Guarantees standardized tag definition while allowing local teams to control resource mapping.

Important:

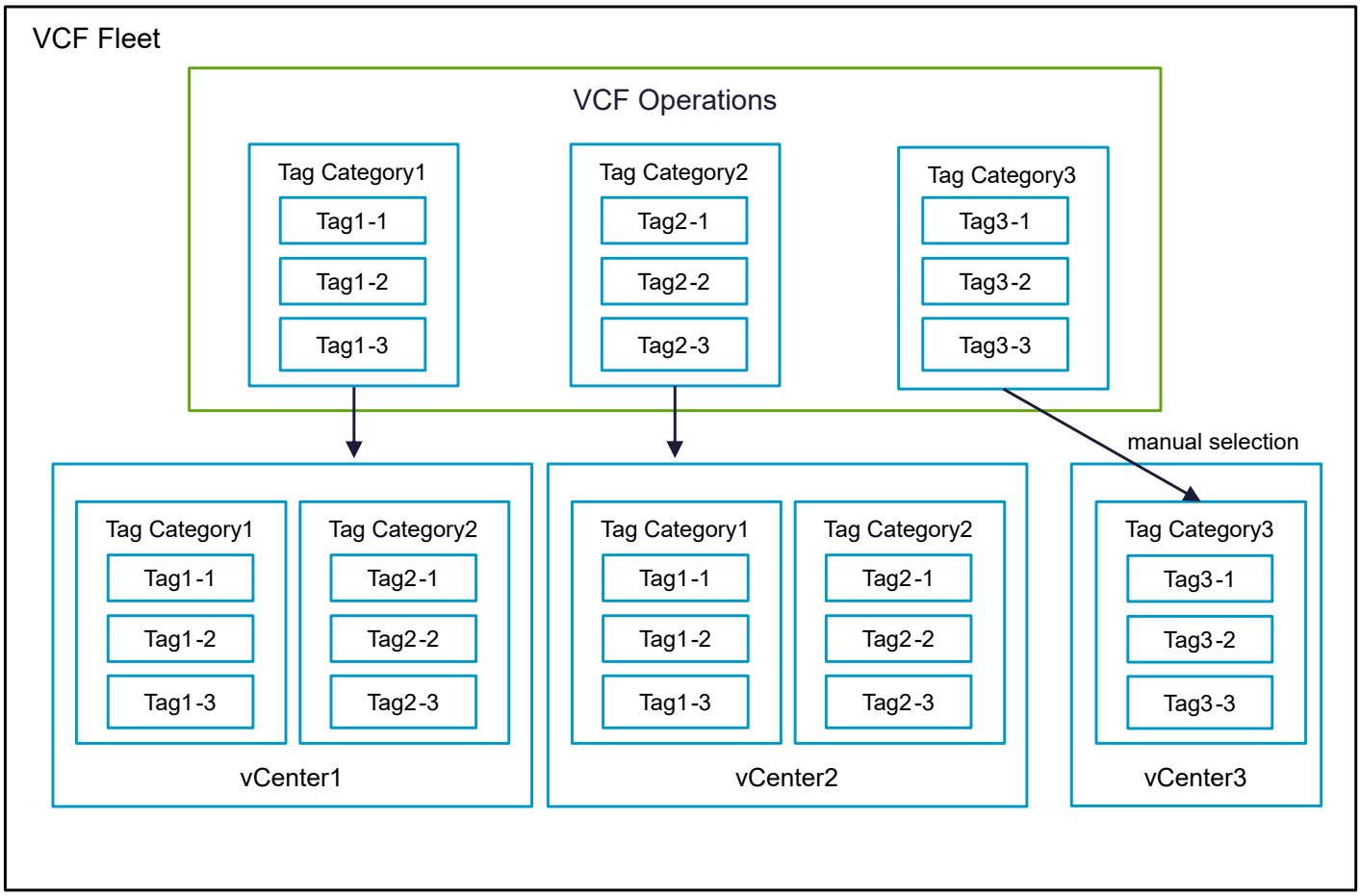
Once a tag is created in VCF Operations, only the description can be changed in VCF Operations but not the tag name.

Define VCF Tag Provisioning Scope for vSphere

Once a management approach is selected, administrators must define the scope of where tags are published. Defining the provisioning scope ensures that tag categories are only synchronized to relevant environments.

Provisioning Scope	Description	When to Use	Considerations
Manual Selection	Limits tag synchronization to manually selected vCenter instances.	Small-scale fleets or for staging/testing new tag schemas on a single instance before a global rollout.	High administrative overhead at scale; requires manual updates whenever a new vCenter is added to the fleet.
Custom Group	Custom Groups (Group Type: Tags Group) defined in VCF Operations.	Large enterprises requiring automated, policy-based tag delivery across specific environment types (e.g., all "Production" sites).	Minimizes human error by ensuring new vCenters automatically join the custom group and inherit the correct tags.

Figure 162: vSphere Tag Provisioning



Recommendations for VCF Tag Management for vSphere
Follow these guidelines to maintain consistent vSphere tag management operations across the fleet.

Category	Recommendation	Justification	Implication
Tag Management	Implement a VCF Private Cloud or fleet-wide naming and casing convention for all tag categories and tags.	Uniform naming improves the config consistency and reduces errors of tag-based automation.	None.
Tag Management	Use VCF Operations Custom Groups to define the provisioning scope for tag categories instead of manual vCenter selection.	Ensures that a new vCenter automatically inherit the required tag schema, reducing the overhead of manual configuration and human errors.	None.
Tag Management	Establish a Standard Operating Procedure (SOP) to remediate configuration drift and resolve conflicts when moving to centralized tag management.	Ensures a clear remediation path for tag conflicts and unauthorized configuration changes.	None.
Tag Management	Configure Log Management with a specific filter—such as logger contains `com.vmware.cis.core.tagging.vapi.TagProviderImpl` - to monitor and alert on tag creation and update events at the vCenter level.	Ensures the VCF Operations remains the authoritative source of truth.	None.

To implement the recommendations and best practices discussed in this section, refer to Tags and Categories Overview and the relevant portions of the fleet management guide for procedural guidance on creating, managing, and assigning tags at scale within your VCF Operations instance.

VCF Monitoring and Alerting Blueprint

This document provides guidance for monitoring a VMware Cloud Foundation platform using the monitoring components Real-time metrics, Log management, and VCF Operations for networks. Effective monitoring is crucial for maintaining the stability, performance, and health of the SDDC infrastructure.

Extending Monitoring Capabilities with Components

During the initial installation or converge of a VMware Cloud Foundation platform, VCF Operations is installed as the default monitoring solution. It provides monitoring, alerting, capacity planning, and charge-back capabilities for the platform. VCF Operations can be used as is or extended with extra capabilities, services, and also consumed through API, or integrated with 3rd-party monitoring systems. The components in this section are part of the VMware Cloud Foundation license entitlement and are highly recommended additions to the initial installation.

Log Management

Log Management collects, analyzes, and visualizes logs across your VMware Cloud Foundation platform. In VMware Cloud Foundation 9.1 this component runs on top of VCF management services and is deployed using VCF Operations.

VCF Operations for Networks

VCF Operations for Networks delivers intelligent operations for software-defined networking and security. It helps customers build an optimized, highly-available, and secure network infrastructure across multi-cloud environments. VCF Operations for networks accelerates micro-segmentation planning and deployment, enables visibility across virtual and physical networks, and provides operational views to manage and scale the NSX component.

Real-Time Metrics

Real-time metrics allows you to collect the real-time, granular data from the underlying core infrastructure, such as vCenter, ESX, NSX, and vSAN. To collect detailed performance data for effective troubleshooting, you can configure different tiers of metric collection within your monitoring policies. Essentials metrics are always enabled by default and cannot be turned off.

Adding Components to VCF Operations

Note:

If you navigate to a UI section that requires Log management, Real-time metrics, or VCF Operations for Networks you will see a button "Install <Component_Name>" that will land you in the component management page, where you can install the required component.

Prerequisites:

The Software depot service should be populated and the desired install binaries should be downloaded.

Workflow:

1. Login into VCF Operations as a user with administrator privileges
2. Navigate to: **Build** -> **Lifecycle** -> **VCF Management** -> **Components**
3. Click on “**ADD COMPONENT**” and select the component that you want to install
4. Follow the wizard to install the component.
5. Enable components for the desired VCF Instances/resources if necessary.

More details could be found in the "Completing the Deployment of Your Platform".

Dashboards

VCF Operations provides effective monitoring through a collection of pre-built, curated dashboards. These dashboards are organized into distinct sections. They are designed to align with key operational areas, offering comprehensive visibility into both the SDDC infrastructure and the associated workloads, making them the essential starting point for your monitoring efforts.

Default Dashboards

Dashboard Group	Purpose	Example Dashboards
Availability	Availability covers uptime of the object now and the uptime trend over time. The availability of hybrid clouds should be tracked at both the provider and consumer layers to understand the availability of the environment. These dashboards show the current uptime and the uptime percentage over the past month.	VM Availability vSphere Availability
Configuration	Configuration management ensures a consistent configuration that meets the desired standards and settings, which can help understand configuration drifts and change requests. These dashboards show the configuration of consumer and provider objects.	VM Configuration Cluster Configuration ESXi Configuration Network Configuration vSAN Configuration
Inventory	Inventory shows the number of resources in the environment. These dashboards help quantify the inventory based on various perspectives and relationships within the environment.	vSphere Inventory Summary vSphere Compute Inventory vSphere Network Inventory vSphere Storage Inventory
Performance	Performance is about ensuring workloads get the necessary resources. Key Performance Indicators (KPI) can be used to identify when workloads may have performance problems, and these KPIs can be used to define SLAs associated with tiers of service. These dashboards use KPIs to show the performance of workloads at the consumer layer and the aggregate performance of workloads at the provider layer.	Cluster Performance Datastore Performance VM Performance vSAN Performance

Dashboard Group	Purpose	Example Dashboards
Capacity	Capacity quantifies the resources used, resources remaining, and opportunities to reclaim unused resources. Projections of the demand provide a proactive view of capacity. These dashboards show capacity in terms of time remaining before capacity is projected to run out, the amount of capacity remaining, the number of VMs that may fit in the remaining capacity, and reclaimable resources that can increase the available capacity.	VM Capacity Reclamation VM Rightsize vSAN Capacity Cluster Capacity Datastore Capacity Capacity Overview
Cost	Running an efficient cloud requires insight into the cost of the infrastructure. The cost of the infrastructure drives the cost of the workloads. Understanding the cost of workloads helps influence consumer behaviors to become more cost-efficient. These dashboards show the cost of the infrastructure, cost breakdown, cost efficiency, and cost reporting for showback purposes.	Total Cost of Ownership Cost Optimization Overview Showback Chargeback

Tip:

In addition to **Dashboards**, you could also use **VCF Health** overview located in **Operate** tab in VCF Operations**.** It is designed to reduce downtime, improve visibility, and increase operational resilience by offering actionable insights into VCF software stack, affecting compute, networking, storage, and the management layer. The structure of **VCF Health** allows a drill-down to the hierarchy from a VCF instance to its management and workload domains, vCenter instances, NSX instances, vSAN, and ESX hosts.

Tip:

The dashboards that are frequently used could be added to the "Favorites" section by clicking the star in the tile or in the dashboard itself.

Tip:

Dashboards could be added to "Home" as Tabs.

Workflow: Open a dashboard, click actions, and select "Add to Product Home"

Tip:

A dashboard that is most important for you could be set as "Dashboards Home". This Dashboard is loaded by default once you click the Dashboards in the left navigation panel.

Workflow: Open a dashboard, click actions, and select "Set as Dashboards Home"

Alerts

Alerts are a way to be notified of anything that is new or an issue that could be potentially dangerous to your environment. Whenever there is a problem in the environment, alerts are generated. You can also create alert definitions so that the generated alerts inform you about the problems in the monitored environment.

Alerts in VCF Operations

The alerts in VCF Operations have tree main components:

- **Symptoms:** A symptom might be "CPU Usage > 90%" or "vCenter Heartbeat Lost". Also an alert might be triggered when a specific combination of symptoms occurs simultaneously.
- **Alert Definitions:** Defines the "criticality" of the alert (e.g. Info, Warning, Immediate, or Critical) and determines which objects (e.g. aVM, vCenter, or all production clusters) the alert applies to.
- **Recommendations:** Alerts in VCF Operations could include a list of human-readable steps or automated **Actions** to resolve the issue.

Checking Open Alerts

You could see the active alerts in the VCF Operations UI under: **Operate** -> **Alerts** -> **Alerts.** This view by default is filtered to show only the Active alerts, you could filter further, expand groups, and go through the other tabs in the Alerts section.

Configure Event Notifications

Notifications are alert notifications that meet the filter criteria in the notification rules before they are sent outside VCF Operations. You can configure notification rules for the supported outbound alerts so that you can filter the alerts that are sent to the selected external system.

Why I may need an event notification

Since each environment may have different critical components, the administrator of the environment should map and analyze it. This analysis should be used to create a plan for proactive notifications. The recommended way is to start with fewer notifications and gradually increases them based on learning and analysis. If you decide to send all alerts as notifications, your notification channel is overloaded and you are risking missing the important messages.

Some common areas that could provide a starting point for defining the critical points in your notification plan:

1. Availability & "Hard" Failures ("Is it Up?")
   These are the most common high-priority notifications. They trigger immediately because they represent a service outage.
   a. **Host Not Responding / PSOD:** An ESXi host has crashed or lost network connectivity.
   b. **VM Unexpected Reboot:** Useful for identifying hardware issues or underlying storage drops that caused a Guest OS to restart.
   c. **Management Component Down:** Critical VCF-specific alerts for when **SDDC Manager**, a **vCenter Service**, or an **NSX Edge Node** becomes unreachable.
2. Performance & Contention ("Is it Slow?")
   These are often "Symptom-based" alerts. Instead of triggering the moment that a spike happens, they usually wait for a **Wait Cycle** (e.g., 3 consecutive collections) to ensure that the problem is persistent.
   a. **CPU/Memory Contention:** Alerts when a VM is "starving" for resources because the physical host is over-committed.
   b. **Disk Latency:** Triggers when storage response times exceed a threshold (e.g., >20ms), often used to identify "Noisy Neighbors" on a shared Datastore.
   c. **vSAN Health Alarms:** High-level notifications when vSAN disk groups are degraded or resyncing.
3. Capacity & Planning ("Will it fit?")
   These are **Intelligent Alerts**. VCF Operations uses AI to "look ahead" and notify you before the failure actually happens.
   a. **Days Remaining Alert:** A notification that says, *"Based on your current growth, Datastore X will be full in 14 days".* This is far more useful than a simple "Disk Full" alert.
   b. **Snapshot Accumulation:** Notifies you if a VM has a snapshot older than a week, or larger than a specific size, preventing storage exhaustion.
4. VCF Infrastructure ("Management")
   Unique to VCF environments, these alerts monitor the health of the management connections between SDDC components.
   a. **Password or Certificate Expiration:** Notifies you 30–60 days before an SDDC Manager or vCenter certificate expires (a common cause of VCF upgrade failures).
   b. **NSX Controller Stability:** Alerts if the NSX management cluster loses its "stable" status or if a Controller node goes down.
   c. **BGP/Routing Peering Down:** Critical for networking teams to know if the physical-to-virtual network link has been severed.

How to Create an Event Notification

In this section, we go over a scenario for creating a notification that will be send to the team that is responsible for the support of vSphere infrastructure. The example creates a single notification for any critical alarm for an ESX host and it could be used as a template to create more notifications.

Configure the Desired Protocol

As part of the notification strategy you should decide what will be the best way to present the information the the interested parties. You could choose between several options including Email; SNMP Trap; Webhook; Slack ServiceNow. Choose the one that the recipient of the notification pays attention to. If different teams prefer different channels, you could configure multiple protocols and could choose them later based on the recipient.

In this example, we decided to send an email to the infrastructure team mailbox, so we must configure the respective plugin (if it is not configured).

Procedure:

1. Login into VCF Operations as a user with administrator privileges
2. Navigate to: **Operate** -> **Administration** -> **Configurations** -> **Alerts -> Outbound Settings -> Outbound Instances**
3. Click “**ADD**” and in “**Plugin Type**” select “Standard Email Plugin”
4. Fill in the details and click “**Test**” - if the configuration is correct a test email is sent to “**Receiver Email Address**”
5. Confirm that the email is received and click “**Save**”

Create a Notification Rule

The notification rule is used as a filter for alerts. Only the alerts that match this filter are sent as a notification with the selected protocol.

Procedure:

1. Login into VCF Operations as an user with administrator privileges
2. Navigate to: **Operate** -> **Administration** -> **Configurations** -> **Alerts -> Notifications**
3. Click “**ADD**”
4. In “**Notification**” tab fill-in **Name** (e.g. “*CRITICAL - ESX Host*”), fill in Description to help you identify the notification later and click **Next**
5. In “**Define Criteria**” tab fill-in:
   a. In **Criteria**: Select “**Object Type**”; in “**Select an Object Type**” dropdown go to “**vCenter->Host System**” or directly type “**Host System**”.
      Selected the ESX host as the object for the filter.
   b. In Alert Scope we will choose:
      Category: <Do not change>
      By default all the categories will be matched. If this generates too many alerts we will reconsider this in the future and choose a subset of categories.
      Criticality: Critical
      We want only the Critical alarms to trigger notification. The other states are triaged as a dayly routine, since they are with less priority.
      Control State: Open
      Filter only alarms that are in an open state.
   c. In Notify On select:
      Status: New
      The message is sent only for new events.
   d. *We will not enable* **"Notification Heartbeat"** *since this will send a new notification for each collection cycle leading to multiple notifications for the same object.*
   e. Click **Next**
6. In **Set Outbound Method** choose**:**
   a. **Outbound Method -> ‘‘Standard Email Plugin"** -> Choose the name of the instance and click **Next**
7. In the “**Select Payload Template**"
   a. **Recepient(s)** - the mailbox for the infrastructure team.

   b. **Notify again** - *sent another notification in 2h if the issue is not resolved - ensures that the team gets fresh notification if the first one has been missed - so enter 120 here.*
   c. **Max Notifications -** *we will send no more than 4notifications in total*
   d. **Delay to notify** - *we will delay the notification for 10 min to avoid notificaitons for alarms that are temporaly.* And click **Next**
8. In the “Test Notification” tab click on **Initiate Process**
   a. Enable “Filter the alert definitions and objects based on criteria outlined in the Define Criteria section”, select an **Alert Definition** and click **Next**
   b. In **Objects** select one hosts and click “**Validate Configuration**”
9. If the notification is sent successfully, click “**Create**”. In case of errors fix them and click again on **Initiate Process**

With this notification in place the infrastructure team will get a notification once the alert criteria is matched. We could consider adding one more notification that will be used as an "escalation notification" if we have alert open for a longer period of time. The easiest way to do this is to clone an existing notificaiton and change few parameters in it.

Procedure:

1. Login into VCF Operations as an user with administrator privileges
2. Navigate to: **Operate** -> **Administration** -> **Configurations** -> **Alerts -> Notifications**
3. Choose an existing notification that you want to add an escalation option, click on the three dots before the name and select **clone.**
4. Add "**-escalation**" suffix to the name and description in order to identify those notificaitons in the future.
5. Navigate to "Select Payload Template" and change:
   a. **Recipient(s)** - Enter the escalation manager email
   b. **Delay to notify** - We want this issue to be escalated if it is open for more than 48h, so we need to enter 2880 (48h*60min).
6. Click **Update**

Pause Alerts and Notifications
Alerts and notifications could be paused during expected downtime - e.g. hardware upgrade that requires servers downtime, power maintenance, servers relocation. Doing so will not create an "alert storm" that could mask some real issues in the environment. When an object is identified as being in maintenance mode, VCF Operations does not collect metrics from the object or generate anomalies or alerts for it. In addition, VCF Operations cancels any active symptoms and alerts for the object.

Manual Maintenance

Manual maintenance mode is used during single events like hardware or software maintennace. In the below example we need to add aditional hardware in a single host.

Procedure:

1. Login into VCF Operations
2. Navigate to: **Operate > Inventory** and select the VCF instance that contains the ESX host which we want to manage.
3. Click the **Environment tab.**
4. Use the filter on the right side of Environment tab and type the ESX hostname or part of it.
5. Select the ESX host name (you could select several hosts if needed) and click **Start Maintenance** from the tool bar in the **Objects** tab.

   Tip:

   In this example we will choose to end maintenance on a specific date - tomorrow. In this way even if we forget to end maintenance after the host is up and running VCF Operations will do it for us. Choose the right option for your case.
6. Select **End Maintenance on** and choose the date.
7. Confirm with OK. No
8. Use your procedure to enter host in Maintenance Mode through vCenter and to Shutdown the ESX host.
9. Complete hardware maintenance, run hardware tests if needed, and put the host back in service through vCenter.

10. Repeat steps 1 to 4, select the hosts that are back in operation and click **End Maintenance** from the tool bar in the **Objects** tab.

Scheduled Maintenance

You could also create a scheduled maintenance window for periodical processes, avoiding false alarms during a known maintenance windows.

In the example below we create a scheduled maintenance for one of our remote clusters in a production plant. Every quarter the power of the facility is stopped for a day to perform an equipment maintenance. During this period, the connection to the main datacenter is lost and we have alarms in VCF Operations, that we want to prevent.

We must perform three steps in order to attach a scheduled maintenance to a resource:

1. Create the Scheduled Maintenance.
2. Create a custom profile with the Scheduled Maintenance.
3. Attach the Custom Profile to a resource.

Procedure (create Scheduled Maintenance):

1. Login into VCF Operations.
2. Navigate to: **Operate > Administration** > **Configurations** and click on **Maintenance Schedules** tile under **Others.**
3. Click **ADD.**
4. Populate a **Schedule Name** that will help you to identify the schedule (e.g. "Production plant maintenance").
5. Select the right time zone for the Schedule.
6. Maintenance happens at the last day of each quarter and finishes for a working day, so we enter the following values:
   Recurrence: Monthly
   Start on: 3/31/26 **at:** 7:30AM
   Stop after: 12 hours
   Repeat every: Mar; Jun; Sep; Dec
   Days: Last
   Expires on: 12/31/30
7. Click **SAVE** and check the values in the table.

Scheduled Maintenance shold be attached to a Custom Profile in order to be used.

Procedure (create a Custom Profile):

1. Login into VCF Operations.
2. Navigate to: **Operate > Administration** > **Configurations** and click on **Policy Definition** tile under **Policies.**
3. Click **ADD.**
4. In Create **New Policy** enter a name and description that will help you to identify this policy on a later stage. For this example we will use:
   Name: "Production Facility Maintenance"
   Description: "Power outage in production plan at the end of every quarter"
5. Select the policy that you want to be used as a temlate and click **Create Policy**. For this example "Default Policy" has the corect atributes so we select it in the **Inherit From** dropdown.
6. Click on the **Maintenance Schedule** tile
7. We want the maintenance to be applied on the cluster, ESX hosts, datastores, and VMs.
   a. In the **"Select Object Type"** dropdown, type "**Cluster Compute Resource**" and select: **vCenter** > **Cluster Compute Resource**, click on the padlock in front of "**Maintenance Schedule**" and select the Maintenance Schedule that we created in the previous step (e.g. "Production plant maintenance").
   b. In the **"Select Object Type"** dropdown, type "**Datastore**" and select: **vCenter** > **Datastore**, click on the padlock in front of "**Maintenance Schedule**" and select the Maintenance Schedule that we created in the previous step (e.g. "Production plant maintenance").
   c. In the **"Select Object Type"** dropdown, type "**Host System**" and select: **vCenter** > **Host System**, click on the padlock in front of "**Maintenance Schedule**" and select the Maintenance Schedule that we created in the previous step (e.g. "Production plant maintenance").

   d. In the **"Select Object Type"** dropdown, type "**Virtual Machine**" and select: **vCenter** > **Virtual Machine**, click on the padlock in front of "**Maintenance Schedule**" and select the Maintenance Schedule that we created in the previous step (e.g. "Production plant maintenance").
8. (Optionally) Click on "**View Unsaved Changes**" and check that the correct objects and attributes are set.
9. Click on **Save.**

The custom profile should be attached to a resource.

Procedure (attach Custom Profile to the desired resource):

1. Login into VCF Operations.
2. Navigate to: **Operate > Inventory**
3. Navigate to the object that you want to assing policy to. In our example it is a cluster so **VCF instances** > "The name of VCF instance" > "The name of Workload Domain" > "vCenter Server FQDN" > "Datacenter name" > "Cluster name"
4. In the top right corner click on "**Policy**" dropdown and select "Assign Policy"
5. Chose the policy that we created in the previous step (Production Facility Maintenance). Note that it will be a child object to the policy that we selected in "**Inherit From**", so it under "Default Policy".
6. Select "**Include child objects**", leave the default **Deph 10** and click OK.

With those steps the cluster and all its child objects have our new policy asigned and they will go in maintenance during the specified intervals.

Create a Custom Dashboard

Custom dashboards allow administrators to consolidate relevant information from various VCF components into a single, focused view that is tailored to specific roles or operational needs.

Why I may need a custom dashboard

VCF Operations comes with dozens of powerful out-of-the-box dashboards, they are designed to be "one size fits most". Users could move toward custom dashboards to transition from generic monitoring to targeted operations.

Why I may want a custom dashboard:

1. Unified Visibility
   Standard dashboards often focus on a single layer (e.g., vSAN or just NSX). A custom dashboard can bridge these gaps:
   a. **The Full Stack View:** You can place a widget for Physical Switch health right next to a widget for VM Latency. This allows an admin to see if a networking issue is the root cause of an application performance drop without switching tabs.
   b. **Multi-vCenter Aggregation:** While OOTB dashboards often drill into one vCenter at a time, a custom dashboard can show "Top 10 most stressed VMs" across your entire global VCF footprint on one screen.
2. Role-Specific Personalization
   Different teams care about different metrics. Custom dashboards allow you to strip away the "noise":
   a. **Executive/Manager View:** High-level "Green/Yellow/Red" health scores and "Money Saved" via rightsizing, without the technical weeds of IOPS or CPU cycles.
   b. **App Owners:** A view filtered strictly to the VMs and databases that belong to *their* specific application, rather than the entire infrastructure.
   c. **Help Desk:** A simplified "search-and-status" dashboard where they can quickly type in a VM name and see its current health and recent alerts.
3. Resource & Cost Optimization
   Create custom views to justify spending or prove efficiency:
   a. **Rightsizing Dashboards:** Specifically highlighting VMs that are over-provisioned (wasted money) or under-provisioned (performance risk).
   b. **Showback/Chargeback:** Visualizing resource consumption by department so IT can show business units exactly what they are using in terms of VCF resources.
4. Focused Troubleshooting

a. **Incident Dashboards:** Build a "War Room" dashboard that displays only the specific components that are involved in a recurring issue (e.g., specific Edge Nodes, Load Balancers, and Backend Web Servers) with high-frequency data collection.
b. **Additional troubleshooting options and ideas** in the Trobleshooting Guide.

5. Branding and Reporting
   a. **NOC Displays:** Creating "Wallboard" dashboards with large, high-contrast widgets that are designed to be displayed on large monitors in a Network Operations Center.
   b. **Custom Labels:** Using "Views" within dashboards to translate technical metric names into business-friendly language (e.g., instead of **sys|uptime_latest**, labeling it as **Service Availability**).

Tip:

- If you want to create a custom dashboard you could use an existing dashboard as a template. To do so, clone the existing dashboard and edit the clone. Editing an existing dashboard is not a good practice, since it could be used by other users or teams.
- You could export Dashboards and could import them in other VCF Operations. This approach provides a uniform configuration across environments.
- Consider using a source control for your exported configurations for a better version management and reverting to a previous working verstion if needed.

VCF Troubleshooting Blueprint

The troubleshooting guide is intended to help guide troubleshooting efforts for typical scenarios experienced in an enterprise environment. The focus of the guide is on incident management - identifying the cause of an actively occurring problem, and determining potential solutions.

Overview

This guide assumes typical enterprise workloads running on a VMware Cloud Foundation platform, with VCF Operations 9.1. A subset of the described capabilities may be available in prior versions.

It will leverage the data and analysis capabilities from:

- VCF Operations
- Log Management
- VCF Operations for Networks
- Real-Time Metrics

Having all of these components installed and configured will provide the best capabilities for troubleshooting effectively, however, it is still possible to troubleshoot with a subset of the capabilities. At a minimum VCF Operations is required.

Example scenarios:

- An application is performing slower than typical
  - What process should be followed to find the cause
  - What data points and which areas should be used to help
- A workload is not able to connect to a database
  - Where should the investigation start
  - What are some of the tools that can help pinpoint the issue
- Sometimes a file transfer operation fails
  - How to investigate when the symptoms are not occurring
  - What additional measure to take to capture details during a recurrence
- Unplanned reboot/crashes
  - How to investigate a crash/reboot scenario

Components for VCF Troubleshooting

Before experiencing an incident that requires investigation, it is important to have the appropriate data collection and analysis tools deployed and configured to capture metrics and data that will be important during an investigation. For the purposes of this guide, the following components should be installed:

- VCF Operations
  - This component is core to VCF 9, and provides the central User Interface for the other components referenced in this guide. It also provides capabilities for environment configuration and management, which are covered in a separate guide.
  - All VCF 9 deployments will have one instance of VCF Operations per VCF Fleet. If you have multiple Fleets in your environment, each will have its own VCF Operations instance, managed through its own UI. As of VCF 9.1, the core inventory and metrics from multiple VCF Fleets' Operations can be connected together and viewed in a single VCF Operations instance, providing easier visibility. However, for more advanced troubleshooting including logs, network and real-time data, each VCF Operations instance will need to be accessed directly. Therefore the steps in this guide should be considered incompatible with the Federation feature of VCF Operations.
  - VCF Operations collects inventory data for the entire fleet, as well as a number of metrics for each inventory object pertaining to its performance, health and activity, which are stored to various resolutions - higher for shorter periods of time and lower resolution data retained over extended periods.
- Log Management
  - This component collects the contents of log files for components in the VCF Fleet, as well as any external components that have been configured to send their log (syslog) data to the logs collector.
  - Log messages contain valuable information on what is occurring in a system, and sometimes includes performance information as well (either through data embedded in messages or by analyzing the time stamps for logs identifying specific states or actions). Logs are also a key component for root cause analysis through event re-creation.
  - Log data needs to be retained for the period of time where troubleshooting may need to refer back to. Longer retention provides more history, but it also consumes more storage capacity and can make searching through logs slower and more difficult. Older logs rarely provide significant value, but can help determine when an issue first started occurring, or confirm if it happened previously.
- VCF Operations for Networks
  - This component tracks network flows - communication sessions between 2 endpoints over the network. It is useful for understanding what workloads are connecting to and how much data they are sending or receiving to those endpoints.
  - Traffic is categorized into "flows" which is traffic part of the same session, or generally the same "conversation". It includes the source, destination, protocols used and the type of traffic. It does not contain the actual data being exchanged in most cases, unless that data is encoded in the network packet header. It also tracks the number and size of packets, which can provide the total amount of data sent and received.
- Real-time data
  - This component is new for VCF 9.1, and provides high-resolution metrics from ESX hosts, vSAN, and NSX Edges for detailed troubleshooting. It allows tracking the different counters on the hypervisor (ESX) that describe a VM's activities and how its performance is affected by available resources, and how the host's resources are being allocated.

Troubleshooting Concepts

Methodologies and Glossary of concepts used in this guide and for Incident Management in general.

Workbench

Workbench is a module in VCF Operations that provides a framework to guide the troubleshooting process as a session. Based on a selected context (such as a VM or other vSphere object), you can refine details like the time range, or expand to related objects. You are shown potential evidence - metrics, events, or changes that stand out, and can expand to additional metrics, events, flows, and logs within the defined context. You can add text notes or save charts or graphs to the notes, and save the investigation to refer to later on and compare to other data points that are identified as the investigation proceeds. It also enables sharing the session with others so that they can assist or take over the troubleshooting process, making collaboration on an incident much easier.

While it is optional, in this guide we recommend using Workbench for all incidents of significance. It provides a solution to track and share what was captured and reviewed during the troubleshooting process, which can help resolve the incident more quickly and make hand-offs between team members easier. It also retains this history for post-incident review and for future recall comparing against new similar incidents that can also help with problem management and root-cause identification.

Best Practices when using Workbench

- **Define Precise Time Windows:** Start by narrowing the scope of the workbench analysis to the exact period when the issue was detected. This focus filters out "noise" from unrelated system events, allowing the tool to surface the most relevant evidence and configuration changes.
- **Start with top-level impacted object for scope**: Workbench can automatically identify all related objects and expand or collapse using relationship mapping. For example, selecting an impacted VM will identify the dependent host, network, datastores, etc. to include in the analysis.
- **Prioritize Root Cause Evidence:** Rather than manually scanning individual logs, focus on the "Potential Evidence" tab. Use the workbench to automatically correlate metric spikes with recent configuration changes, which frequently point directly to the source of the problem.
- **Leverage Metric Correlation:** Use the workbench to compare multiple metrics simultaneously. If a virtual machine is underperforming, correlate its CPU and memory usage with the underlying host's resource availability to determine if the issue is contained within the VM or stems from the physical hardware.
- **Review Recent Configuration Changes:** Infrastructure instability often follows a change in the environment. Check the workbench for "Events" that indicate new patches, setting updates, or hardware reconfigurations.
- **Compare with Regular Performance:** Review metrics for periods when the issue was not occurring to understand the baseline and use that to identify variances to "normal" behavior.
- **Incorporate Log Insight Integration:** While the workbench excels at high-level correlation, use its integrated view of log data once the issue has been narrowed down to a specific component or timestamp.

Isolation

Isolation can have two distinct meanings in troubleshooting:

Isolation (segregated environment): When the environment or workload with the issue is prevented from communicating or interacting with external components. This method is typically used on a clone of the production environment that can then be tested in ways that cannot be done on a live environment without fear of impact. This is often done as part of an issue reproduction (see below) or when the problem cannot be identified sufficiently on the live environment. In cases of malware infection, a live environment may need to be isolated to prevent the spread of the malware to other environments while it is investigated and cleaned.

Isolation (Narrow the scope): This is a process where the source or cause of the issue is focused on a more specific area - ruling out various services and components so the investigation can concentrate on a smaller area. It’s important to isolate an issue to specific components at the start of the troubleshooting process to know where to look more closely. A number of methods may be used to rule out components (for example other services sharing the same components but are not affected by the same problem) or where the issue can be triggered across different workloads by using a common infrastructure component.

Problem Theory Development

Once the issue has been isolated to a specific component or area, the observed symptoms and configuration are matched to possible chains of actions that could fully represent the problem. A theory may start out being relatively vague, and then through observing additional configurations, logs or behaviours from the impacted systems, can be refined to a full chain of events scenario that encompasses all the symptoms and observations of the problem scenario within the affected environment’s configuration.

Problem Theory Validation

Once a theory for the cause of the problem has been identified, it needs to be validated through testing to confirm that the theory is actually the source of the issue. A full validation test requires confirming both making a change causes the issue, and reversing the change causes the issue to stop. The detail of the theory as to the full chain of the problem and aligning all symptoms to it will help prepare more accurate validation tests. This step is critical to ensure that the theory is properly validated. If the theory is too vague when validation is performed, it may lead to a scenario where it is validated correctly but does not accurately confirm the full problem chain.

Issue Reproduction

It may be helpful in some scenarios to create a different environment in a controlled fashion that mimics the configuration of the impacted environment and attempt to reproduce the issue in this test environment. This is necessary when the production environment is not suitable for testing - either in high security, high availability, or extremely critical environments, or when more destructive/impactful configurations need to be put in place to help trace difficult to identify issues (for example debug logging on a busy production environment may generate too much data to capture, or cause a performance impact itself). A Reproduction environment is generally kept fully isolated from the live production environment to avoid cross-contamination of data used for testing.

Incident Management

Incident management addresses the process of identifying and resolving an impactful problem to the environment. It aims to restore functionality to eliminate or reduce the impact to customers, end users and applications.

Problem Management

Problem management delves deeper into issues at a wider, more systemic level and works to reduce the likelihood of a recurrence of issues and/or prevent future similar or related issues in the original or other systems. It is generally done after an incident has been resolved, and includes root-cause analysis of prior incidents.

Troubleshooting Scenarios

Sample scenarios that walk through a typical problem situation and use the available VCF components to scope, analyze, and resolve an issue.

Workload Performance Issue

The primary challenge with identifying the cause of a performance issue is to sift through the layers of data to identify the root cause. Often the overloaded components will be easily identifiable in metrics, but the reason for the increased demand can be many layers deep, and each layer needs to be peeled back one at a time until the underlying cause is identified. In some cases this will just be a scale limitation, where the design's limits have finally been reached. It can also be a misconfiguration that results in a high-load operation being performed too frequently.

Another challenge with performance issues is that they can also be intermittent issues. That scenario is covered under its own section.

Resources

VCF Operations (required), VCF Operations Networks (optional), Real-time Data (optional)

[other]:

Sample scenario:

Customers report a 2-tier app called WorkDataApp is very slow to respond after clicking a button in its UI. Reports of this slowness are coming in from all locations. They said that it was working fine yesterday, but this morning it is unusably slow.

Step 1: Initial Triage and Isolation

1. **Set clear performance expectations:** This should identify the scope of the problem (what is affected), and the definition of what a "good" state is. The goal for incident management is to address a specific performance issue, not to engage in performance optimization of an application or system. The target should be based on what the performance was prior to the incident (or as close as can be determined when not known).
2. **Isolate the Impact:** Determine the scope of the performance degradation. Is it affecting a single VM, a specific ESXi host, an entire cluster/fleet, or a particular application/service?
3. **Determine the timeline:** Did the issue start on a particular date/time? Is it consistently better or worse at specific times of day or days of the week?
4. **Establish a Baseline:** If possible, compare current performance against historical metrics (before the issue started) to quantify the degradation and identify the resource under stress (CPU, Memory, Disk, Network).

5. **Use VCF Operations Workbench:** Create a new Workbench entry for the incident. Attach initial charts showing the degraded metrics, relevant logs, and a description of the symptoms for collaboration and historical tracking. If you don’t know the exact scope of the issue, enter the most specific object you know is impacted (or choose one if multiple are impacted). For example, select a VM that is part of the impacted application. Workbench will provide all the related objects to help you further narrow the scope.

   [other]:

   Sample scenario:

   Identify the VMs involved in this application:

   Web/App Server: WDA-App1

   DB Server: WDA-DB1

   Start a workbench session and enter the App VM as the scope.

   Enter the Time Range for a confirmed period when the issue was occurring, or enter one for a time a bit earlier than when the issue was known to have started to capture the transition.

Step 2: Data Analysis

Host & VM Metrics (Historical Aggregate)

*Requires VCF Operations.

VCF Operations core metrics provide the long-term view necessary to identify trends and overloaded components.

Metric Type	Location	Key Indicators to Check	Correlation with Root Cause	Possible Causes
CPU	VCF Operations - Host or VM object: Metrics->CPU	CPU Usage (Host/VM), CPU Ready Time	High Ready Time often points to host contention (over-commitment). High VM usage suggests the workload is demanding more resources.	Imbalanced workloads, oversized or undersized VMs, insufficient resources in cluster, guest OS or application-level configuration causing high utilization
Memory	VCF Operations - Host or VM object: Metrics->Memory	Active Memory, Memory Ballooning, Swapping, Consumed Memory	Ballooning or Swapping indicates host memory pressure. High Active Memory suggests the VM needs more resources.	Imbalanced workloads, oversized or undersized VMs, insufficient resources in cluster, guest OS or application-level configuration causing high utilization
Storage	VCF Operations - Host or VM object: Metrics->Storage	Latency (ms), IOPS, Throughput	High latency is a primary indicator of storage bottleneck. Check read/write ratios and IOPS compared to baseline.	Follow the latency to identify the scope of the bottleneck and address on that object (Host, switch, array)
Network	VCF Operations - Host or VM object: Metrics->Network	Throughput (Host/VM), Packet Loss, Dropped Packets	High throughput or packet drops can indicate a network bottleneck either at the virtual switch or physical layer. Consistently increasing drops can also be caused by FW rules, reverse-path forwarding (RPF) checks, or other configuration issues.	Follow the high traffic to identify the object(s) contributing the most to the total, and address at the app/guest OS level, or rebalance workloads to minimize network path or separate from other guests with high throughput.

Actionable Steps:

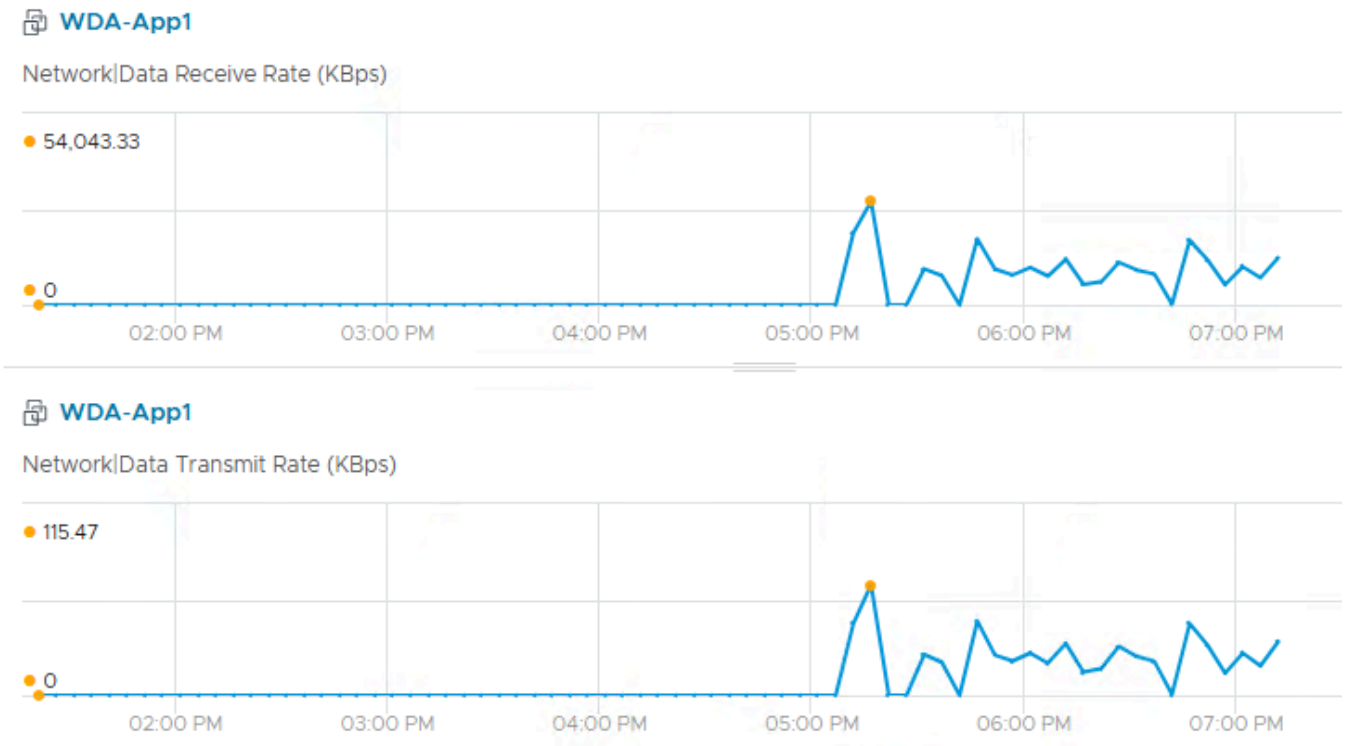
- Review the last 7 days of performance graphs for the impacted object(s) (VM or Host) based on what was identified in Step 1. Use the Troubleshoot an Application view to see the typical metrics that apply to an application across multiple infrastructure objects.
- Pinpoint the time the degradation started and check for any correlated configuration changes or increases in workload by comparing against other metrics.
- Compare against related components: for example high load on a database server correlating to high load on application servers, or different load on different servers performing similar tasks. To accomplish this, leverage the Correlation tool on the metrics graphs for the affected workloads to compare across other metrics on the same workload, or the same metric on other workloads for the same time period.
- Trace high load metrics to the lowest level object(s) showing similar utilization patterns - spikes or peaks at the same times. This can help to define the scope of the issue, and trace back to a root cause - for example, an application misconfiguration creating a large number of network connections impacting an entire network segment or host.
- The goal of the analysis is to develop a problem theory, which can then be tested in the next step. Continue to look for matching data points to support or disprove the interim theories.

[other]:

Sample scenario:

On the workbench session for the WDA-App1, select the VM as scope, then under Metrics, expand Metrics->Network and add Data Receive Rate and Data Transmit Rate. Add Virtual Disk -> Read Throughput and Virtual Disk -> Write Throughput. Add CPU-> Run (ms).

Note that all graphs have a spike starting at the same time, around 5:00PM last night.



This indicates that data is being read from disk and sent over the network. It shows a new level of activity that started just after 5PM, but by itself is insufficient to determine a problem theory, other than to identify that there is new traffic coming in over the network

Host & VM High-Resolution Metrics

*Requires VCF Real-time Data component

Data Point	Purpose	How to Use
ESX Performance Counters	Provides granular, second-by-second data on resource usage on the host.	Use to confirm if a spike in a metric (e.g., CPU Ready) is sustained or short-lived, which is often missed by lower-resolution metrics.
VM Kernel Resource Allocation	Details how the ESX scheduler is allocating physical resources (CPU/Memory/Disk) to the struggling VM.	Helps determine if the hypervisor is throttling the VM (e.g., due to resource limits, shares, or reservations) or if the host is genuinely saturated.
Network Microbursts	High-frequency network traffic spikes that can overwhelm network components.	Used to identify brief, intense network demands that cause intermittent latency but don't show up in 5-minute averages.

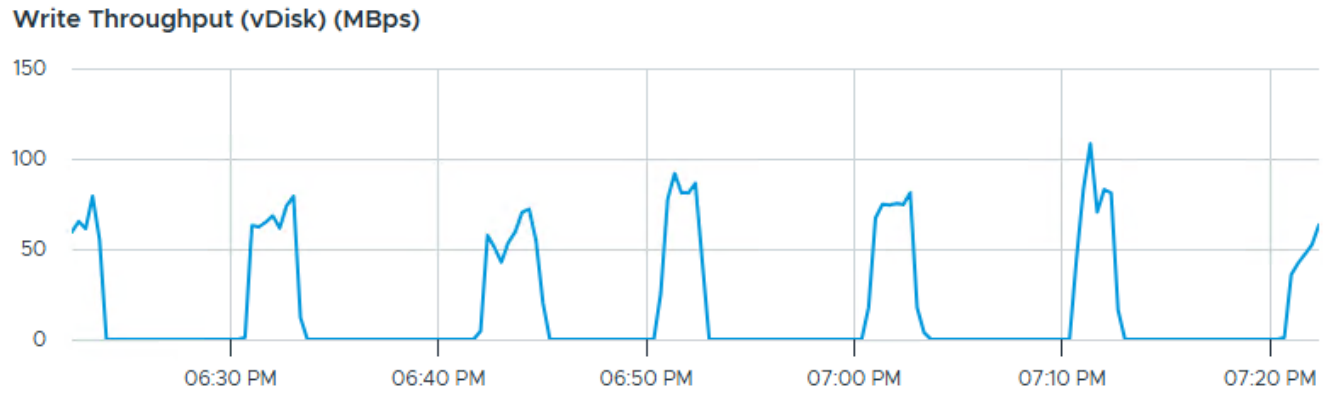
Actionable Steps:

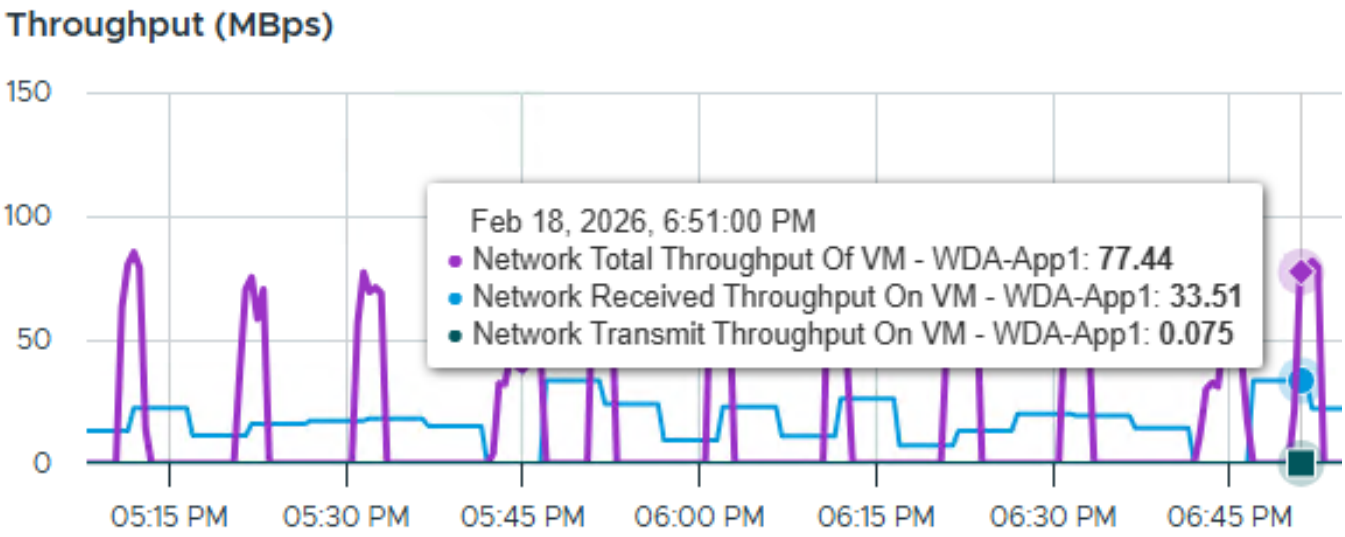
- When the issue is actively occurring, launch Real-time Data for the affected host/VM.
- Monitor counters like **%USED** (consumption) and **%READY** (contention) simultaneously to quickly distinguish between high demand and resource contention.

[other]:

Sample scenario:

On the Real-Time Analysis tab, select the TOP N section and then look at the Network Total Throughput of VM, the Write Throughput (vDisk), we see that there is a periodic spike in both that occurs every 10 minutes and lasts about 3 minutes. The data indicates disk writes. If we look at the Overview page, VM Network Summary shows more packets being received than sent.





This graph suggests that some kind of scheduled or automatic activity is generating the spikes. We can also infer that the traffic is originating from outside the VM, being sent over the network to the WDA-App1 VM where it is written to disk, as we see much higher traffic on Network Received Throughput than Transmit Throughput, and the times of the peaks align with the disk writes.

Traffic and Flow Analysis

*Requires VCF Operations Networks component

Key Area	Focus	Performance Implication
Traffic Volume	Analyze inbound/outbound throughput for the VM or host.	Unusually high traffic to/from a specific endpoint (internal or external) suggests a potential service overload or misconfigured application.
Flow Counts	Check the total number of simultaneous network connections (flows).	Extremely high flow counts can saturate network control plane resources (e.g., firewall session tables), leading to connection failures or latency.
Endpoint Identification	Identify the top "talkers" and "listeners" for the impacted VM.	If performance slows during communication with a known problematic external service, the bottleneck may be external to VCF.

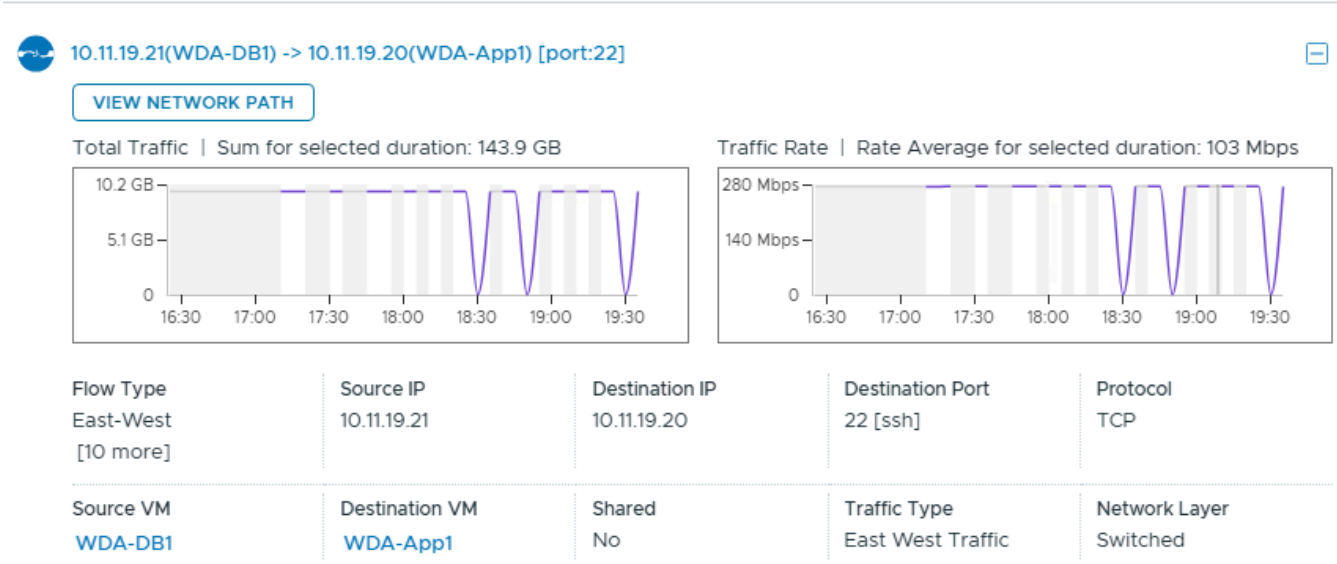
Actionable Steps:

- Filter Network Flows based on the IP address of the impacted VM during the time of the incident.
- Look for a dramatic change in flow volume or a large increase in traffic to an unexpected internal service (e.g., communication storms).

[other]:

Sample scenario:

Since we observed that the traffic is originating from outside the VM in the previous section, we can look at the network flows that occur at the same time to determine where the traffic is occurring and investigate that source. Select the Flows tab. Then view by Top Talkers. Look at the list by Flow Volume, and we can click on the top flow, which is most likely our flow in question. Narrow down the time range to start just before 5PM which we identified as the start time of the issue. The flow data shows a similar periodic pattern as our other graphs. Although the flow data is broken down into 5-minute intervals on the :05 and :10 minute marks, the alignment is still apparent. If we expand the details we can see that this traffic is originating from a VM called WDA-DB1 on port 22.



Log Analysis

*Requires Log Management.

Logs are the least likely source of information for performance issues, as most applications don't self-monitor or report on performance issues, however there are some logs that might provide indications that are observed as performance issues:

- Repeated service or VM restarts. If a VM or an application service fails and restarts, that can produce performance impacts as the affected node will be unavailable to process data while it is restarting. Look for messages containing the words restart, reboot, failure, crash as examples.
- Connectivity issues - if there are intermittent connectivity losses between application components, this can also be observed as slow performance, as the application will be stalled while the connection re-establishes. High packet loss on a connection can also lead to performance issues as the data will need to be re-transmitted resulting in delays. Look for connections being terminated and/or re-established, for example to databases. Terms such as disconnect, reconnect, timeout, drop can be indicators of such an issue.
- Some systems can log messages at performance thresholds - for example the vDefend IDS/IPS system can log when different utilization thresholds are hit, and if any packets are dropped due to oversubscription of inspection capacity.

[other]:

Sample scenario:

To see if there is an underlying infrastructure issue related to the periodic spikes, we can enter the VM's name into the Logs search panel, and adjust the timeframe to the 5PM start time. While the search turns up some standard housekeeping log entries from services like fdm (vSphere HA), and some messages from Hostd about monitoring and metrics tracking, nothing in the logs appears to indicate any abnormalities in this case.

Step 3: Problem Theory and Validation

Once the data points converge on a suspected bottleneck (e.g., high storage latency, high CPU ready time, or high network flow counts):

1. **Formulate a Problem Theory:** *Example Theory: "The performance issue is caused by high CPU Ready Time on Host X, which is driven by a resource-intensive nightly backup job running on VM Y, confirmed by a spike in historical CPU metrics at 1:00 AM."*
2. **Validate the Theory:** Test the theory. In the example above, try disabling or rescheduling VM Y's backup job. If performance immediately improves, the theory is validated. If the issue persists, the investigation must pivot to the next most likely cause.

[other]:

Sample scenario:

We can theorize that there is a task running every 10 minutes on the VM WDA-DB1 that is transferring data over the network on port 22. If we start a workbench for WDA-DB1 and look at the same metrics, we can see that there is both Disk Read and Write traffic, as well as Network Transmit in a pattern that matches what we saw on WDA-App1. To further validate our theory and identify the task, we will need to look at the guest OS itself, and observe what processes are generating the disk and network I/O.

In this case, we observe that there is a cron job scheduled every 10 minutes under the root account that is generating data to disk, transferring that data to WDA-App1 and then deleting it from disk. We'll need to identify the administrator who created this job to understand what the goal was, and find an alternative solution that does not impact the performance of the application.

```
root@photon-machine [ ~ ]# crontab -l
*/10 * * * * /root/diskio >/dev/null 2>&1
```

Connectivity Problem

This is when an endpoint is not able to communicate with an external entity. Sometimes this can appear intermittent, or may only affect certain communications. All these details can be valuable clues to isolating where the problem is.

The most important thing to keep in mind when dealing with a connectivity issue is that in order for a connection to be established, all points in the path must be operating. This includes the application service at the destination. If a database service isn't running, then a remote client will not be able to connect to it, and that failure to connect is most often returned as a "connection error". This does not imply a network problem. This is why isolation of the issue should be the first step with a connection issue - start at the destination and validate each component, validating one step towards the source until the entire end-to-end chain is validated. Many application services have a proxy service that establishes the connection to the external clients, so ensure to validate if this is the case - it can be possible to establish a connection to this proxy service, but the proxy cannot then reach its back end application because of an issue with the application. This is often seen as an error 502 - bad gateway, because the proxy cannot reach its gateway (the application).

When troubleshooting a connectivity problem, the objective is to methodically isolate the failure point along the path from the source to the destination, which involves checking application, operating system, hypervisor, virtual network, and physical network layers.

Resources

VCF Operations (required), VCF Operations Networks (optional but recommended), Log Management (optional)

[other]:

Sample scenario:

An application implemented a new workflow to provide a report to users, and it is not working. They checked their app and the data is being generated, but the application is not able to show it.

Data is generated on VM: WDA-DB1

Application presents the report to users from VM: WDA-App1.

The report was implemented on Friday at 5PM.

Step 1: Initial Triage and Isolation

1. **Start with a clear problem definition:** Define exactly when the problem started (did this work previously?), what is affected (certain types of communications, traffic between specific endpoints, etc.), what the type of desired communication is/are, and any other defining characteristics of the problem.
2. **Start at the Destination (Application Layer):** Before investigating network components, confirm the target service is running and accessible locally.
   – *Actionable Step:* Log into the destination VM/server and confirm the target application service (e.g., database, web server) is active and listening on the expected port (e.g., using netstat or service status checks).
   – *Problem Theory:* If the service is down, the connection failure is an application issue, not a network problem.
3. **Test Local Connectivity (OS Layer):** Verify the source VM's OS can reach its default gateway and itself.
   – *Actionable Step:* Use simple connectivity tests (e.g., ping or traceroute or nc) from the source VM to its own virtual IP and its gateway.
4. **Work from the destination back to the source**: Test at each accessible point until the spot where the communication fails is identified, and then focus on that component.

[other]:

Sample scenario:

Application data from WDA-DB1 to WDA-App1 fails to transfer

This worked in DEV + TEST, but when implemented in PRODUCTION it did not work.

Confirmed that the service on WDA-App1 is listening, and data can be transferred from other hosts.

Ping between the 2 hosts is successful.

The data transfer is being done with SCP, which uses the same port/service as SSH: TCP port 22.

Step 2: Data Analysis

Flow Verification

*Requires VCF Operations for Networks

VCF Networks is crucial for verifying if traffic is even *leaving* the source and *reaching* the destination's virtual switch. It's also important to use NSX as a data source to ensure incomplete flows are tracked (e.g. when communication is initiated but the session is not established).

VCF Operations for networks provides a guided network troubleshooting workflow that can assist with identifying relevant flow data based on the initial triage. Follow the steps on this page to start a troubleshooting incident, and view data related to the selected objects: relationships, alerts, metrics available for analysis. Any anomalies detected in the metrics will be highlighted for easy identification. Another benefit of creating a troubleshooting incident is that the investigation and any notes you add will be recorded and available for any future investigations on related components which can help accelerate future incident resolution.

Key Area	Focus	Connectivity Implication
Flow Existence	Check for active or recently terminated flows between the source and destination IP/Port.	The absence of a flow confirms traffic is blocked or dropped *before* reaching the VCF Network monitoring point (likely source firewall or host networking).
Flow Details	Analyze the protocol, ports, and state of the flow (e.g., SYN_SENT, ESTABLISHED).	If the flow shows **SYN_SENT** but never establishes, the connection is reaching a point (likely a firewall or the destination OS) that is dropping the reply packet (a classic asymmetric or firewall issue).
Endpoint Details	Confirm the path traversal by checking the source and destination logical ports and segments.	Helps trace if the traffic is being misdirected by NSX rules, T0/T1 gateways, or security policies.

[other]:

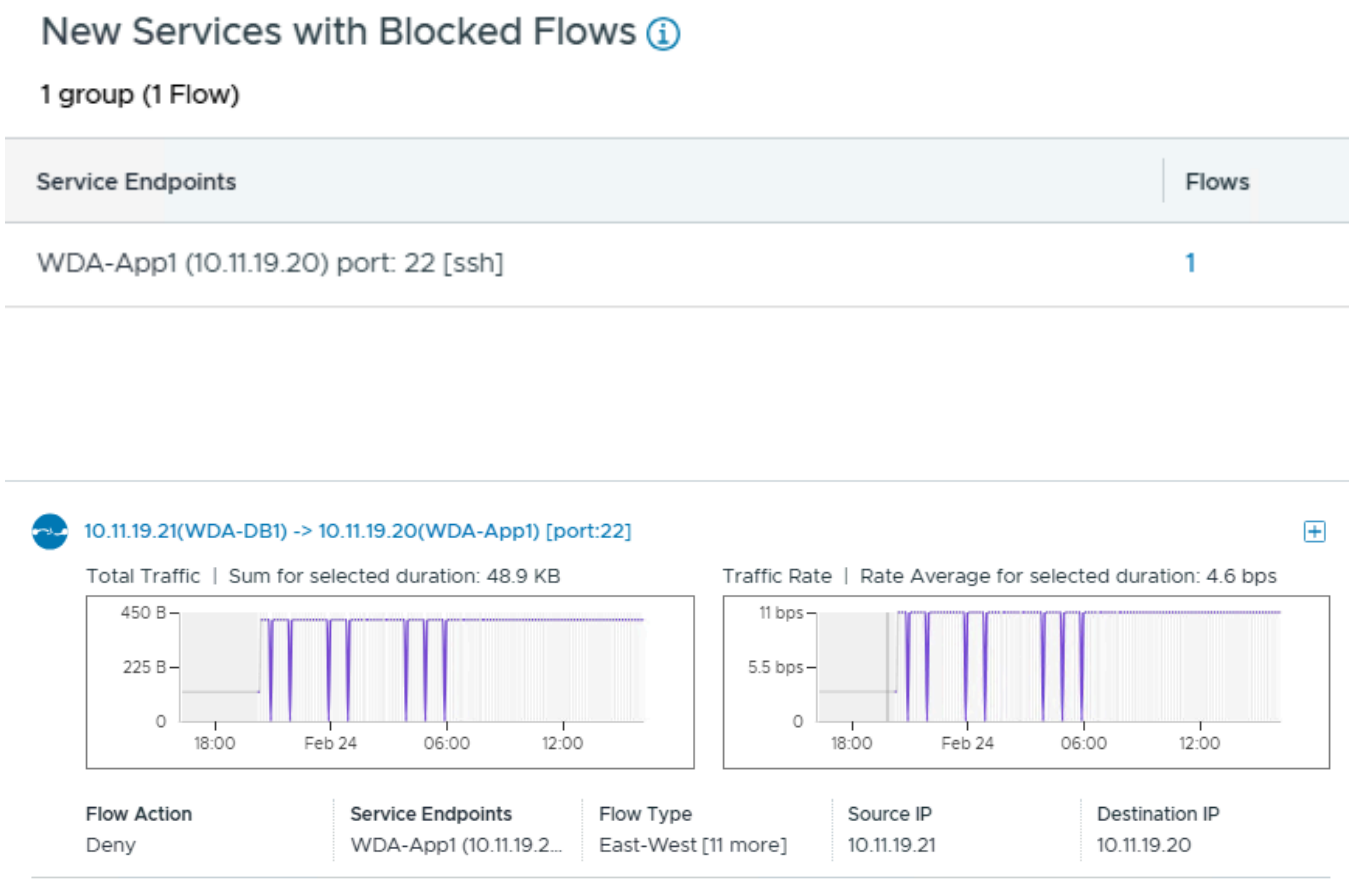
Sample scenario:

Start a workbench session with the context of VM WDA-App1.

Go to the Flows tab, and under Top Talkers, select Destination VM, since the traffic is coming from an external VM (VM-WDA-DB1) to WDA-App1. If you click on one of the bars, it will show the detailed list. Notice there are no flows from WDA-DB1 to WDA-App1 on port 22. This means the flow isn't getting created.

Tracing the network path backwards, we land on the source VM, WDA-DB1. We can create a workbench session for this VM and examine its flows and logs as well, although we can expect to find no flows there.

If NSX vDefend is licensed and enabled on cluster, then ipfix collection from NSX can be enabled, and this would provide a direct view of flows blocked by the firewall under Recent Flows.



We can see that the flow from WDA-DB1 -> WDA-App1 on port 22 has a Deny from the vDefend Firewall.

Log Analysis (Security and Network Components)

*Requires Log Management

Component	Log Focus	Connectivity Implication
NSX Edge/Firewall	Check firewall logs for **DENY** or **DROP** or **REJECT** messages corresponding to the source/destination IP and port.	The presence of a DENY rule confirms a security policy is blocking the traffic. This is a configuration problem, not a physical network failure.
ESXi Host	Check hostd/vmkernel logs for reports of port errors, vSwitch issues, or physical NIC failures.	Error messages here can indicate an underlying hypervisor or physical network interface card (pNIC) problem causing intermittent or permanent drops.

[other]:

Sample scenario:

On the Logs tab of the Workbench, there are a few terms we can search for to find entries from firewall rules that might be dropping the traffic. “DROP” will find any logs of rules that are dropping traffic. “REJECT” is another possible way the firewall can block traffic.

Timestamp	Log
2026-02-20 18:31:03 +00:00	2026-02-20T18:31:03.272Z sfo01-m01-r01-esx04.sfo.rainpole.io FIREWALL-PKTLOG 2113318 - [esx@4413] 2026-02-20T18:31:03.185Z 11360879 INET match DROP 2024 OUT 60 TCP 10.11.19.21/50600->10.11.19.20/22 S
2026-02-20 18:30:30 +00:00	2026-02-20T18:30:30.715Z sfo01-m01-r01-esx04.sfo.rainpole.io FIREWALL-PKTLOG 2113318 - [esx@4413] 2026-02-20T18:30:30.673Z 11360879 INET match DROP 2024 OUT 60 TCP 10.11.19.21/50600->10.11.19.20/22 S
2026-02-20 18:30:14 +00:00	2026-02-20T18:30:14.563Z sfo01-m01-r01-esx04.sfo.rainpole.io FIREWALL-PKTLOG 2113318 - [esx@4413] 2026-02-20T18:30:14.545Z 11360879 INET match DROP 2024 OUT 60 TCP 10.11.19.21/50600->10.11.19.20/22 S

In the logs, we can see a firewall rule dropping traffic from 10.11.19.21, which is our source VM’s IP, destined to 10.11.19.20, our destination VM, on port 22 - which is what SCP and SSH use. This is a match - so now we know there is a firewall rule that is blocking the intended traffic, and we can raise that with the security team to determine what the reason for that rule is and whether it can be removed.

Metrics (Utilization/Congestion Check)

*Requires VCF Operations.

Review Network Metrics to rule out congestion masquerading as a connectivity issue.

- *Actionable Step:* Check **Network Throughput** and **Packet Loss/Drops** for the affected VM's virtual NIC and the corresponding ESXi host's pNIC.
- *Problem Theory:* High packet loss or drops often indicate a severe bottleneck (saturation) either at the physical switch or the host's pNIC, making it *appear* as if the connection is failing intermittently.

Step 3: Problem Theory and Validation

1. **Formulate a Problem Theory:** Based on the flow analysis, metrics, and logs, pinpoint the single most likely point of failure.

- *Example Theory:* "Connectivity is failing because an explicit **NSX Distributed Firewall DENY** rule is blocking TCP port 1433 between the application and database segments, as confirmed by firewall logs and the absence of established flows in VCF Networks."

1. **Validate the Theory:** Take action on the suspected component and immediately retest the connection.

- *Example Validation:* Disable the suspected NSX rule (or create an explicit ALLOW rule) and retest the connection. If the connection immediately succeeds, the theory is validated, and the resolution is a policy fix. If it fails, pivot the investigation.

[other]:

Sample scenario:

Our sample scenario had a vDefend Distributed Firewall (DFW) rule blocking the desired traffic. When we had the NSX ipfix data being collected, this gave an explicit report of what the issue was. In cases where the traffic was being blocked by either an external firewall or a Guest OS-based firewall, we would have seen the connection being initiated by the source, but no flow established. This would require some additional steps to test the traffic from the source and isolate where it was being blocked, but the troubleshooting process would follow the same process.

Intermittent Problems

This type of issue refers to a problem that only appears on occasion, and at other times presents no symptoms. The challenge with this type of issue is that much or all of the isolation and reproduction efforts must be done by looking at historical information - logs, metrics, network flows rather than observing live systems.

The process for troubleshooting intermittent issues should follow the same general process based on the incident type (e.g. intermittent connectivity should follow the connectivity issue process, intermittent performance should follow the performance issue process). On top of that some additional considerations are required:

1. Create a workbench session. While using workbench is a good practice for any type of issue, it’s especially important for intermittent issues so that when they do occur, all the prior information gathered can be immediately at hand.
2. Create a detailed timeline and history. Since the incident is not constant, it’s important to understand when the issue does occur to address 2 main situations:
   a. Is there a pattern to when it is triggered?
   b. Identifying the times when it has occurred to review other datasets at those times to look for similarities that can point to the source of the problem.

Reboot or Crashing problems

This type of issue is where a system hits an unrecoverable error that causes the system to halt operations. It is referred to a BSOD (Blue Screen of Death) or PSOD (Purple Screen of Death) as the systems generate a core dump and display it on the console with a background of Blue for windows OS and Purple for ESX. The core dump contains an image of the memory and system registers at the time of the crash, which can help to determine what occurred just prior to the failure to help isolate the issue to a specific module and/or scenario.

Resources

VCF Operations (required), Log Management (optional)

[other]:

Sample scenario:

Periodically, an ESX host in one cluster will crash with a Purple Screen of Death (PSOD).

Step 1: Initial Triage and Isolation

1. Review the data on the screen or console when the crash occurs. Save an image or photo of the screen so that the data can be analyzed later.
2. Record the time and host/resources that experienced the crash to identify any patterns.
3. Track the resolution - does a reboot resolve the issue? A full power disconnect/reconnect?

[other]:

The PSOD always starts with the line "#PF Exception 14 in world 2098030:qfle3_stats_ IP". The issue occurs about once a week. It happens on a different host each time, but always within the same cluster. The times vary throughout the day/week.

Step 2: Data Analysis

Knowledgebase search

Visit https://support.broadcom.com, and search for the key terms from the PSOD message. Look for the Exception (i.e. "#PF Exception 14" and the details from the world: "qfle3_stats_ IP". This will indicate if there is a known match that could have a fix identified.

Log Analysis

*Requires Log Management

Component	Log Focus	Implication
Affected ESXi Host	Look for indications of a boot: bootstate=0 in vmkernel.log	Provides specific timestamps for when the crash event occurred. Look immediately prior to the crash time to determine what events may have led up to the crash.
other ESXi Hosts in the cluster, vCenter	vSphere HA detected a possible host failure of host The vSphere HA availability state of the host[...]has changed to Host Failed Host experienced a crash. More details will be available in the generated vmkernel-zdump. Cannot synchronize host [...]. An error occurred while communicating with the remote host	Provides specific timestamps for when the crash event occurred. Look for messages of issues communicating with the affected host for example vSphere HA (v

[other]:

Sample scenario:

We find:

2026-03-26 15:38:04.732 sfo01-m01-r01-esx01.sfo.rainpole.io vcenter-server: The vSphere HA availability state of the host sfo01-m01-r01-esx01.sfo.rainpole.io in cluster in sfo-vcf91-cl01 in sfo-vcf91-dc01 has changed to Unreachable

2026-03-26 15:38:09.567 sfo01-m01-r01-esx01.sfo.rainpole.io vcenter-server: vSphere HA detected a possible host failure of host sfo01-m01-r01-esx01.sfo.rainpole.io in cluster sfo-vcf91-cl01 in datacenter sfo-vcf91-dc01

This gives us a timestamp for the crash, which we can use to search logs on the host and cluster for the period just prior to identify any unusual activity that could indicate what scenario led to the crash.

Step 3: Problem Theory and Validation

1. **Formulate a Problem Theory:** Based on the flow analysis, metrics, and logs, pinpoint the single most likely point of failure.

- *Example Theory:* "A driver for hardware used in this cluster is causing a crash."

1. **Validate the Theory:** Take action on the suspected component and immediately retest the connection.

- *Example Validation:* Contact the vendor of the hardware, and/or look for updated versions of the driver(s) to see if a similar reported issue was fixed in a more recent version. If an update is available, or if the vendor has another fix, test that version or fix to see if it resolves the issue.

VCF Edge: Single Node with Argo CD Design Blueprint

VMware Cloud Foundation Edge (VCFE) extends the centralized management capabilities of the platform to geographically dispersed edge locations. Common edge use cases that are supported by VCF Edge are remote office / branches, manufacturing factories, retail stores, or mission-critical battle vehicles for the Tactical edge. Design Library for VMware Cloud Foundation outlines the most commonly adopted and recommended infrastructure design options to accommodate those use cases.

In a traditional fleet deployment, vCenter "pushes" configurations to the edge. For Edge sites that only have intermittent connectivity, if the WAN link is down, the configuration update fails as Edge ESX hosts are not reachable. A GitOps-based approach uses a "Pull" model that is inherently resilient to the intermittent connectivity typical of edge sites can be a good alternative. Configuration updates for workloads are staged in a central Git repository, and Edge sites retrieves the workload update only when there is connectivity.

To enable easier adoption of GitOps, VCF bundles the Broadcom-built Argo CD Operator. Unlike the upstream Argo CD Operator, this operator is purpose-built by Broadcom to integrate directly into the VCF platform. This Operator can be consumed in the datacenter and edge, enabling seamless lifecycle management, namespace-level access control, and native support for hybrid workloads. Argo CD Operator can be used across all VCF edge designs patterns. The following patterns are the most applicable to one-node integration:

VCF Edge Pattern 8: Single Host vSphere Supervisor at the Edge in a VCF Instance

VCF Edge Pattern 9: Leveraging Existing VCF Operations for Far Edge Expansion

VCF Edge Option 1: VCF Operations with vCenter for Distributed Edge

Those designs unlock a wide range of GitOps-driven workflows across infrastructure and application layers on a single edge node:

- Create, update, and configure VKS clusters.
- Declarative Virtual Machine provisioning, updating or scaling.
- Deployment of containers as vSphere Pods to simplify the deployment abstraction.
- Application deployment and configuration into VKS clusters using version-controlled manifests.

Single Host Design Criteria

A single-node VCF Edge deployment consolidates the ESX hypervisor, vSphere Supervisor, and workload plane onto one physical host. Every hardware and software decision must account for the absence of HA failover between hosts. The following criteria govern a production-grade single-host edge design.

Design Requirement	Details	Note
Management Domain	One	Dedicated Cluster for all management components of VCF Fleet, such as VCF Operations, vCenter, SDDC Manager, and NSX. Customers have the flexibility to host the management domain either locally at the edge location or at a central data center
vSphere Cluster	One or Multiple	Use the vCenter to create a one-node vSphere Cluster based on your requirement.
VCF Networking	vSphere Distributed Switch (vDS)	Configure vSphere Distributed Switch for vSphere Supervisor
vSphere Supervisor	One	Supervisor must be enabled on a single node cluster by logging into vCenter.
Supervisor Zone	One	Assign zone during the Supervisor activation process.

Design Requirement	Details	Note
VCF Storage	Local Storage or External Storage	Local or external storage are both supported for one node cluster
Licensing	VCF or VCF Edge License	Deploy VCF Fleet and apply either VCF or VCF Edge license.
Latency and bandwidth (vCenter to ESX)	300 ms and 10mbps	If your Edge sites are managed remotely from a central location, latency and bandwidth must be within the supported range. Refer to Latency Details for each component.
WAN link	Dedicated dual-link connection	A primary and secondary WAN link is recommended to maintain constant connectivity between vCenter and vSphere Supervisor-enabled hosts.

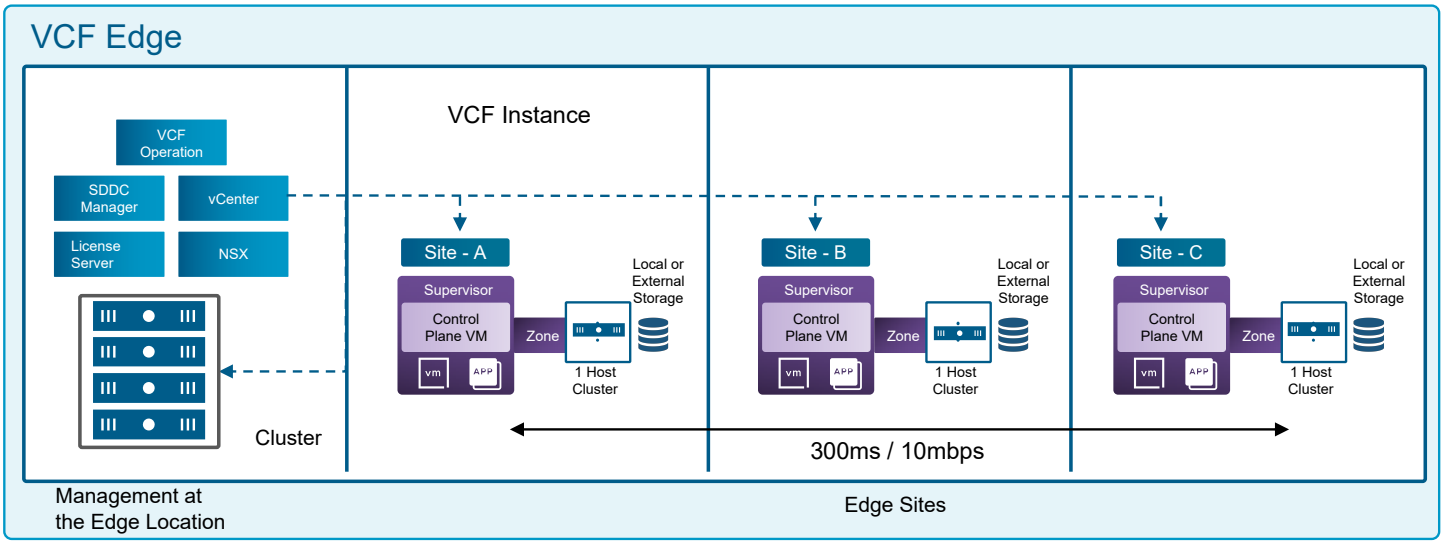
VMware VCF distribution of the Argo CD operator offers flexibility in design and deployment strategies. This section reviews the design considerations of the operator and how it impacts the architecture of other GitOps tooling components.

The location of components required to operate GitOps influences the amount of tuning and customization that are required to meet the availability, monitoring, and performance of the overall Architecture.

In most edge deployments, this decision is often based on the reliability of the wide area network and required operations that when the WAN isn't available. The availability of Argo CD determines when configuration updates can be applied. Similarly the location of Git Repository, Content Library, and Container Registry location impacts the quality of the WAN required to operate the Edge consistently.

Single Node vSphere Infrastructure Design

Figure 163: VCF Edge Pattern 8: Single Host vSphere Supervisor at the Edge in a VCF Instance



Central Management with Remote Edge hosts consists of the following major components:

- Centralized Management Domain
  - A Central management domain is used to host a dedicated set of management components for the Edge sites. Customers have the flexibility to host the management domain either locally at the edge location or at a central data center. Management components are tasked with managing the inventory and lifecycle of the remote edge clusters across the WAN. For the one node design, Required management components consist of:
    - vCenter: Resource and LCM management of vSphere Cluster
    - VCF Ops: Monitoring and licensing of Edge

    SDDC Manager and NSX manager are not required for the one node architecture, but can be used to manage thick Edge locations.
  - The WAN link must satisfy the following latency and bandwidth requirements:
    - Up to 300ms RTT between vCenter and ESX (Please refer official Latency details for each component.)
    - Bandwidth: Minimum 10-Mbps throughput
- Remote Edge Clusters: Each edge site is represented by a single vSphere Cluster. Characteristics of the edge cluster include:
  - vSphere Cluster: Single-Node with vSphere DRS and HA enabled.
  - vSphere Supervisor: vSphere Supervisor enables the deployment and management of Kubernetes workloads directly on the ESX host.
  - Workloads: Supports a mix of traditional Virtual Machines (VMs), native Containers, and vSphere Pods.
  - Edge Services to support GitOps Integration: ArgoCD and Harbor that is deployed as part of the Supervisor Services.

Single Node Management Design

The Management Domain is a foundational component of the VCF architecture. It is highly flexible and can be tailored to accommodate the environmental constraints of an Edge deployment.

The design of a Management Domain factors in edge security requirements, WAN bandwidth, latency, and the overall scale of the edge locations. Hosting Edge management components begins with placement. If the latency between the central data center and the edge is below 300ms, centralizing the Management Domain at the central DC is the most efficient and simplest approach. For locations where latency exceeds 300ms, deploying the Management Domain closer to the edge locations is a viable alternative. Regardless of the placement strategy, the following design recommendations apply to all Management Domain architectures.

Table 435: Single Node Management Domain Design Requirements

Design Decision	Design Details
Size the number of Management domain hosts based on the type of External Storage	Management Domain starts with 2 ESXi hosts when used with external storage such as NFS or FC. 3 Nodes for vSAN. Single-Rack Cluster Model Sizing Considerations
Design the Management domain storage based on available storage on Management domain hosts	Management Storage Design based on type of the type of storage • vSAN – vSAN Single-Rack HCI ESA Storage Model – vSAN Single-Rack HCI OSA Storage Model • Fibre Channel Storage Model • NFS Storage Model
Design the Management Distributed Switch based on uplink NIC configuration on Management domain hosts	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Use VCF Installer to deploy management components.	Deploy VCF Fleet and apply either VCF or VCF Edge license. The Dedicated Edge vCenter instance design must align with VCF9 vCenter design recommendations.

Single Node Cluster Design

A core tenet of edge design is the ability to support VM and Kubernetes workloads at the edge that offers operational predictability and simplified lifecycle management.

The fundamental unit of resources for each edge site is a vSphere cluster. Even with a single node edge, each host must be a member of a single node vSphere cluster. This cluster must have both vSphere HA (no admission control) and DRS must be enabled as these features are required by VKS. Storage on the one node can be VMFS based on local disk or external storage using FC or NFS.

A single VDS with a minimum of two NICs is required for link level redundancy on a single node. NIC speed can be GE or higher.

Table 436: Single Node Cluster Design Requirements

Design Decision	Design Consideration	Design Implication
Use the Edge Router or switch as the source of NTP and set the service policy to *"Start and stop with host."* on the one node ESX host.	NTP configuration can be set on the host using VMware Host/vSPhere Client, API, or vSphere Configuration Profile.	The router must be configured to authenticate against trusted time sources
Use vLCM image to manage the desired ESX version and OEM vendor add-on	Reference Managing the Lifecycle of Clusters withvSphere Lifecycle ManagerImages at Cluster Level for associating an Image with a vSphere Cluster. The vSphere Lifecycle Manager Depot to store ESX base images, vendor add-ons, third-party components, and legacy patches and updates vSphere Lifecycle Manager Remediation Settings to configure the behavior of vSphere Lifecycle Manager during remediation. Lifecycle Management of Clusters and Standalone Hosts at Scale	
Single vSphere Cluster for each Edge Site	vSphere cluster settings for DRS and HA: • DRS: Fully Automated • vSphere HA: – Admission control disabled – Host Failure Monitoring - Restart VM – VM Monitoring – Disabled	Settings are required to deploy Supervisor and VKS. Refer to the Multi-host edge design for sites with multiple hosts.
vDS Design	VDS Design must be based on the number of uplinks that are required to satisfy security compliance requirements for isolation. • Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model The number of uplinks impacts which model is applicable. A host with 2 uplinks can only support a single vDS with redundancy. Refer to the VCF detailed distributed switch design guide for design recommendations	When consolidating multiple types of traffic on a single VDS, classifying and tagging traffic is essential for maintaining both cluster stability and deterministic performance. If ESX host with four uplinks can dedicate two uplinks for either Storage traffic or VM traffic. **Note:** Multiple VDS is not typical for a single node configuration. Consider a dedicated storage VDS if: • VM with low network throughput, but high write IO Consider a dedicated VDS for workload if: • High network bandwidth application, low storage IO

Design Decision	Design Consideration	Design Implication
Local Storage Design	Local VMFS storage volume using local disks Use a Hardware RAID Controller: HW raid Controller must be VMware-compatible (HCL). Ensure that the RAID controller has BBWC or FBWC cache for write IO performance. Populate the server with the number of disks that are required to support the I/O and resiliency requirements of the workload When creating the datastore in the vSphere Client, always select VMFS6. VMFS6 natively supports 4K sector drives and features automatic, asynchronous space reclamation (UNMAP), which tells the underlying storage array to recover deleted blocks automatically.	A host with a bare minimum number of disks is forced to sacrifice space efficiency for redundancy. When there are few disks (e.g., 2 disks), the option for disk redundancy is Mirroring (RAID 1). With 1 extra disk, RAID5 configuration can yield 66% space efficiency
Create a dedicated VM Kernel NIC for each type of management traffics	STIG compliance design mandates strict network isolation for different types of management traffic. In a single node: • Management VMK • Storage (example: NFS) VMK	Extra IP addresses required for separate VMK interfaces.
All data at rest must be encrypted using vSphere Virtual Machine Encryption (VM Encryption)	Reference the Virtual Machine Encryption documentation for implementation details	Encryption is performed in ESX software and can incur a minor performance overhead. Overhead can be mitigated by enableAES-NI hardware acceleration.
External Storage	Refer to the VCF Storage Design documentation on • NFS Storage Model • Fibre Channel Storage Model	Install storage hardware acceleration in vSphere (VAAI) for external storage deployments. VAAI capability must be verified with the storage vendor

Important:

Adding a single host to an existing vCenter environment is a manual step. For larger-scale edge onboarding or automation workflows, customers can use the VCF Edge at Scale PowerShell Script. Currently, automated single-host Workload Domain creation through the VCF operations workflow is not available.

Single Node vSphere Supervisor Design

Single node vSphere Supervisor is crucial for edge use cases due to several factors that align with the unique demands and constraints of edge deployments:

- Get the most out of a single physical server by efficiently running both virtual machines and containerized applications, managed by a unified control plane
- Process data and run applications directly at the source, crucial for real-time analytics, AI/ML inference, and critical operational control at the edge.
- Native integration with vSphere Content Library, Argo CD, and Harbor to deliver pull and desired state-based orchestration for Containers and Virtual Machines workloads.

Table 437: Edge vSphere Supervisor Design Requirements

Design Decision	Design Consideration	Design Implication
Single Zone supervisor per edge location	Each edge site has only one vSphere cluster, one zone. Reference the Simple vSphere Supervisor Control Plane Model	Since each edge site is an independent, single host. There is no redundancy at the physical infrastructure layer. Refer to the Multi-host edge design for sites with multiple hosts if redundancy is required.
One Supervisor Control VM instance per vSphere Cluster	One Supervisor VM per vSphere cluster Small Size is sufficient for a single node vSphere Cluster. Refer to Deploy a Supervisor withFoundation Load Balancer and VDS Networking for implementation steps	Supervisor is enabled on a single node cluster by logging into vCenter Supervisor Scale up is supported if the workload exceeds small Supervisor Size capacity.
Use FLB based supervisor Load Balancer integration	FLB is a L4 load-balancing solution for environments using Virtual Distributed Switch. FLB handles the load balancing and VIP allocations for the Kubernetes API and external applications tight integration with supervisor activation flow. The FLB deployment is handled entirely by the standard Supervisor enablement workflow. FLB Small with 2vCPU / 4-GB memory can be used for small edge sites with limited workloads Refer to the Foundation Load Balancer documentation for all FLB Requirements	FLB is not a L7 load balancer.
Latency and bandwidth	300 ms and 10mbps between vCenter and ESXi host and Supervisor. Refer to Latency Details for each component.	Management domain must be deployed closer to the edge if latency exceeds 300 ms.
Supervisor with FLB PortGroups	A minimum of three distinct Distributed Port Groups (DVPGs). To ensure proper traffic isolation, security, and routing, each of these logical networks should ideally map to its own dedicated VLAN and subnet. • DVPG forManagement plane • DVPG for workload • DVPG for Load Balancer VIP Refer to Supervisor Architecture with Foundation Load Balancer Supported Topologies for deployment options.	IP address ranges are expected to not overlap when networks are overloaded for multiple functions.
Define Tag-based Storage Policies to map Local datastore to Supervisor namespaces.	Datastore that is reserved for the Supervisor workload must be tagged. VM Storage policy reflects: • VMFS Datastore specific rules • VMFS Rules – Volume Allocation Type: Fully Initialized – Tag based datastore filter Reference the Storage Policy Design documentation for implementation details	The design assumes that tags are managed at vCenter level. Tag management with VCF Ops is a new capability in VCF 9 and will be included in future updates to the Design guide.

GitOps Components Placement

The following components are the minimum components that are needed for GitOps:

- Argo CD - Argo CD is used to implement a GitOps-based continuous delivery model by synchronizing application configurations stored in Git repositories with edge clusters. Two placement choices for Argo CD Operator and instance across edge sites: A per edge site design versus A centralized Argo CD instance. The appropriate approach depends on factors such as network reliability, scale of edge sites, operational overhead, and organizational governance requirements.
- Git Repository - Git repositories serve as the source of truth for infrastructure and application configurations in a GitOps workflow. A decision must be made whether to host these repositories using a third-party git platform as a service or through an internally managed solution.
- Content Caching - Two types of Content Caching: Content Library stores VM OVAs and Container registries store. In distributed or edge environments, organizations often evaluate whether to rely on a third-party cloud-based container registry or deploy an internally managed private registry with caching or replication capabilities. Organizations also must determine if VM OVAs must be centrally managed or must reside locally at the edge.

Argo CD Placement

Two placement choices for Argo CD across multiple edge sites: a per edge site design versus a centralized ArgoCD instance.

Argo CD communication involves periodically checking Git repositories for configuration updates and communicating with each cluster to detect configuration drift. Depending on the placement of the Argo CD server, if deployed at the edge, Git repository access requires WAN bandwidth, and configuration drift checks require LAN bandwidth. If deployed centrally, configuration drift check requires WAN bandwidth and Git repository access does not require bandwidth from the edge.

The total amount of network traffic depends on the size of the applications and the number of clusters at the edge site. In general, configuration drift check generates a high volume of traffic compared to repository access. The configuration check is low bandwidth because only the Git commit hash id is used to determine if the branch has been updated. If the commit SHA remains unchanged, ArgoCD does not fetch the repository content again.

Local Argo CD Instance Per Edge - Distributed Argo CD

In the per Edge Site design, each edge site hosts its own dedicated Argo CD instance managing VKS clusters or Supervisor namespaces local to the edge location. This design works well in environments with limited WAN. Credentials required to manage the edge site can be scoped locally (see section on Authentication design). If VM images and application binaries are also hosted locally using the Content Library and Harbor container registry, workload bootstrapping does not need to leave the site. Since each site is independent, local Argo CD also helps reduce the blast radius by isolating failures to individual sites.

For thin edges that support only a handful of edge applications, capacity planning of the Argo CD instance can also be simplified. There are typically no sharding requirements of Argo CD Application controllers, lots of the parallelism and concurrency settings can be left at default or even reduced for thin edge sites to minimize consumption and configuration management overhead (see section on sizing).

While this approach simplifies Argo CD sizing and tuning, especially as the number of sites increases to thousands. Since each site requires its own Argo CD instance, complexity in monitoring and number of deployment steps increase as site counts gets larger. Although both can be managed effectively through deployment automation and monitoring. VCF Edge Automation is developed to reduce deployment complexity. The automation framework addresses the ease of on-ramp by simplifying deployment of Supervisor and services with idempotent deployment. VCF Edge Automation is available on the Powershell Gallery and on Github.com (Open Source).

Central Argo CD for all Edge Sites - One Argo CD managing all remote Namespaces and Clusters

The centralized Argo CD setup involves a single Argo CD instance managing remote VKS clusters or Supervisor namespaces across edge sites. The centralized approach allows for simpler control, since there is only a single Argo CD endpoint, one place to add repository credentials, API keys, Argo CD CLI/API and etc.

The steps to add remote VKS clusters to Argo CD are similar to the distributed setup. However it's important to make sure the Argo CD has direct access to the Kubernetes API server at each location and there is sufficient WAN bandwidth. Argo CD relies heavily on communication with each Kubernetes cluster to capture data to ensure that the desire state is always maintained. In a centralized setup, this communication must traverse the WAN. The larger the edge location (more

applications), the more bandwidth is consumed by Argo CD. Careful bandwidth monitoring is critical in the centralized deployment as any issues with the WAN at the central instance could impact configuration updates for all sites.

Attention:

Managing remote namespaces requires creating an additional service account and role binding in each namespace before it can be added to ArgoCD. Currently this step is manual and must be centrally maintained.

Performance tuning is also a key consideration in this architecture. Cluster states, can grow significantly over time. Sharding Application Controller replicas across clusters to improve performance can be complex and often requires manual intervention at scale. As the number of sites and applications increases, careful resource planning and ongoing performance optimization become essential.

Key Considerations:

- **Local Argo CD** design ensures better fault isolation and easier scalability at a high number of locations. Simple Local Network design. Works well with limited WAN. Overhead of monitoring and patching are high as each site must be managed separately.
- **Centralized Argo CD** carries the risk of failure impacting all sites, and it needs careful attention to performance and resource management. Complex Network design. Does not operate well with limited WAN, edge connectivity. Day-2 activities such as monitoring and patching are centralized.

The summary table comparing two approaches to deploy Argo CD in a multi-site environment:

Table 438: Local vs. Central Argo CD Design Characteristics

Design Characteristics	Local Argo CD	Central Argo CD
Blast Radius	Small Argo CD failure only impacts Desire state updates for a single site.	Large Any issues with at the central datacenter could impact configuration updates for all sites
Deployment Steps	Simple. Use VCFE PowerShell automation to deploy the Argo CD operator and instance.	Simple. Deploy once and shared across many sites, but requires extensive monitoring for performance
ArgoCD Performance Design	Simple. Edge locations have a limited number of apps. multiple approaches to scale to increase availability and performance	Complex. Performance tuning involves load distribution across shards and concurrency tuning
Global View of applications deployment through ArgoCD UI	No. Each edge instance has a local Argo CD UI	Yes. Single Argo CD UI for all applications.
Network Connectivity Design	Simple. Only connectivity to edge K8s cluster endpoints are required within an edge site	Complex. Argo CD must have full connectivity to all K8s endpoints across all edge sites
Edge cluster Access Control	High. Argo CD RBAC settings can be scoped to a single edge site, but increases the overhead of auditing.	High. Argo CD RBAC is centrally maintained. Service accounts that are required for each site and namespace.
WAN Design	Simple. Each site runs independently. Each site can use different tuning options to reduce WAN dependency	Complex. Desired state check requires a reliable and consistent WAN. Must design for the lowest common denominator, sites with the most unreliable connection.
Argo CD Scalability Design	Simple. Each edge site has a limited number of K8s endpoints and applications. Enterprise can come up with different T-Shirt can replicate the design to many locations.	Complex. Careful resource planning and ongoing performance optimization are necessary as more sites and applications are added.

Argo CD monitoring and patching	Complex. Each location must be monitored and updated separately.	Simple. Single set of control planes to monitor and update.

Overall, the per-edge-site local ArgoCD approach is favored for limited WAN. Unlike the central ArgoCD design, distributed approach does not require global reachability and easier performance tuning. The increase in deployment complexity, can be handled through VCFE automation scripts.

Git Repository and Container Registry Placement

This section outlines the architecture and operational considerations for implementing GitOps for Edge sites. It focuses on placement impact of key GitOps components and configurations that enable scalable, reliable, and secure GitOps workflows.

GitOps Infrastructure Design

The Supervisor enables the deployment and management of workloads in a declarative manner through Supervisor services, supporting modern, containerized applications and must be enabled at each edge site. This section reviews three possible design patterns:

Infrastructure Design 1: Local Argo CD at Edge with Public Git Repository and Container Registry

Git repositories and container registries are consumed as services from GitHub, GitLab, Docker, Artifactory, or other third-party hosted platforms. ArgoCD is deployed at each Edge site for simplified performance, RBAC, and networking design.

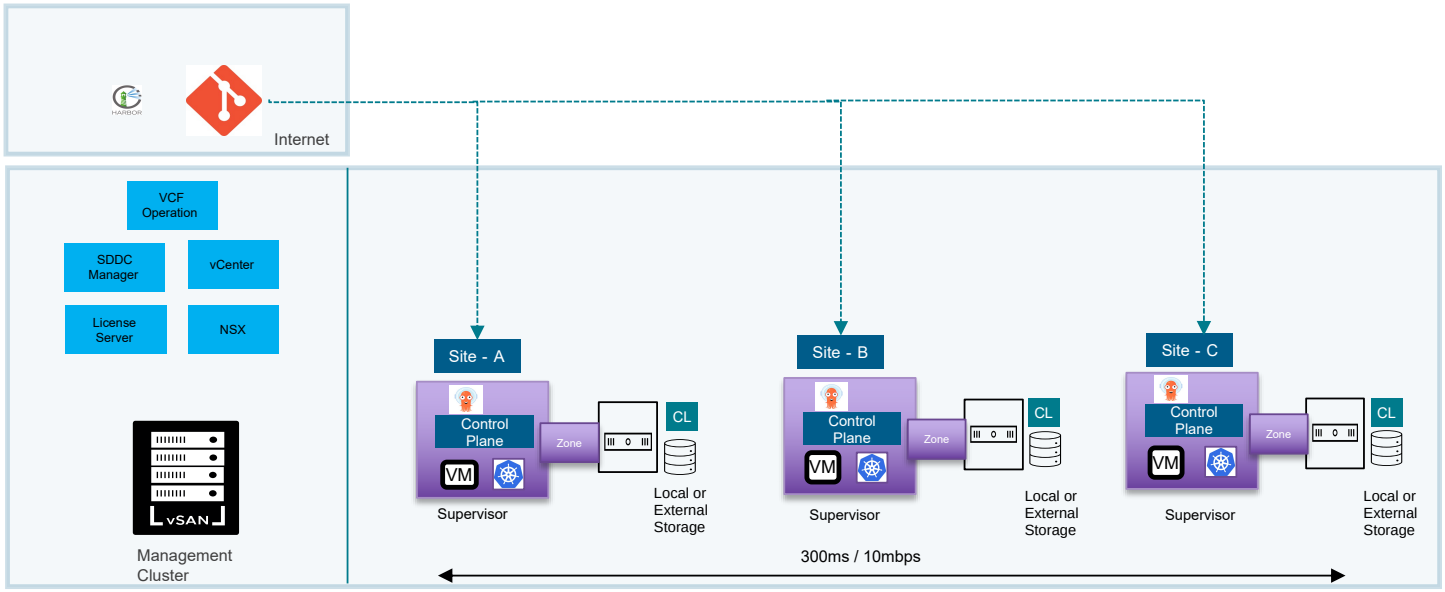
Infrastructure Design 2: Local Argo CD at Edge with Private Git Repository with Local Harbor Registry

Private Git repositories are deployed in the central VCF instance or a mirror copy of the central Git repository are deployed in the management domain of the VCF instance. ArgoCD is deployed at each Edge site for simplified performance, RBAC, and networking design. Harbor registry and Content library are also deployed at each edge site to service virtual machine or VKS cluster deployment.

Git Repository and Container Registry Placement Design 1

Git repositories and container registries are consumed as services from GitHub, GitLab, Docker, Artifactory, or other third-party hosted platforms. ArgoCD is deployed at each Edge site for simplified performance, RBAC, and networking design.

Design 1 : Local ArgoCD at Edge with Public Git Repository and Container Registry


Internet
VCF Operation
SDDC Manager
vCenter
License Server
NSX
vSAN
Management Cluster
Site - A
Control Plane
VM
Zone
CL
Supervisor
Local or External Storage
Site - B
Control Plane
VM
Zone
CL
Supervisor
Local or External Storage
Site - C
Control Plane
VM
Zone
CL
Supervisor
Local or External Storage
300ms / 10mbps
Management at the Edge Location
Edge Sites

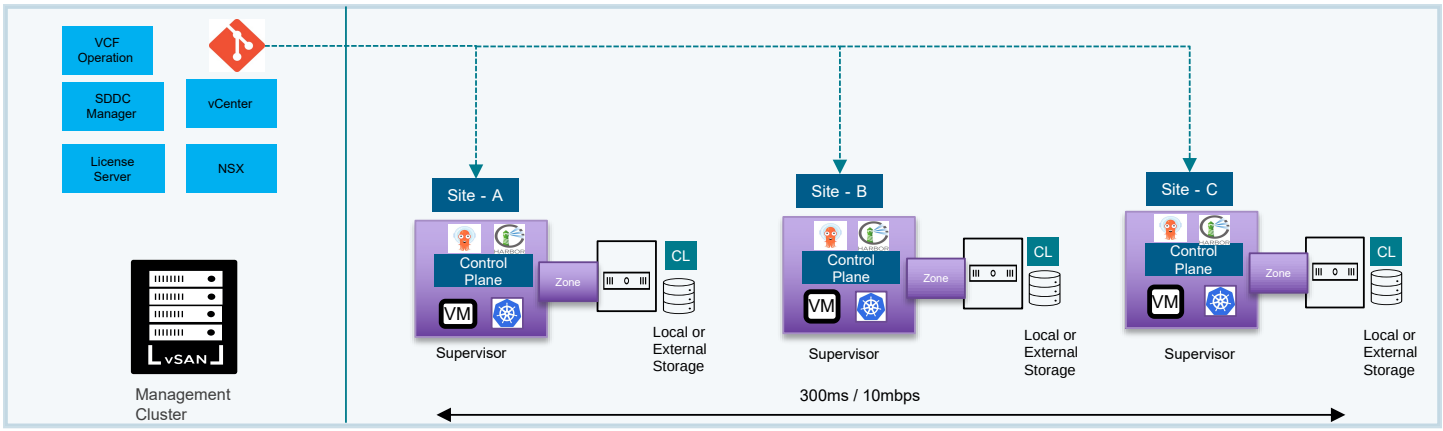
Container Registry and Git repository that is consumed as a service from public Git and artifact providers	Public Git and Artifact providers are typically designed for global reach. They use third-party globally distributed Content Delivery Networks (CDNs) to cache Git repository contents and container images closer to the edge network, enabling faster downloads and reduced latency.	Public Git and Artifact manage all infrastructure, patching, upgrades, and maintenance (including high availability/ redundancy). Edge Admins do not have to worry about uptime or capacity.	Since public Git and Artifact services are multi-tenant and accessed over the Internet, more security policies are required for access control.
Edge Content Library	Content libraries are Local to the edge and can be created on vSAN, Shared, or Local datastore. Configured to sync up from the Central Content Library, used to pre-stage VM OVAs to reduce dependency to WAN.	Design the content library to enable publishing of golden VM OVAs and subscription at each edge location. With local CL, VM templates are transferred **only once** when the content is synchronized instead of every time a new workload is created. For organizations paying for bandwidth, especially metered connections, minimizing WAN traffic leads to substantial operational cost savings.	Edge Admins must allocate and maintain the storage capacity necessary to host the entire replicated Content Library at each edge site. Ensuring that the correct versions of templates are pushed to the correct sites requires careful policy management. A mistake can lead to deploying an incorrect or outdated template at a remote edge.
WAN Optimization: Increase *reposerver.git.request.timeout* to extend time to wait for git commands to return from the Git provider.	The default timeout is 60s. Increase the timeout if there are large network delays between edge sites and the Git Repository server.	If the Git operation takes longer than the **reposerver.git.request.timeout** limit, the repo-server abruptly kills the process. ArgoCD UI as a generic **"context deadline exceeded"** or **"git fetch timed out"** error, and applications fail to sync or refresh	
WAN Optimization: Increase *server.repo.server.timeout.seconds* to mitigate against edge locations with high network latency to the Internet.	*server.repo.server.timeout.seconds* is the maximum amount of time the Argo CD Application Controller waits for a response from the Repo Server before giving up and failing the application reconciliation/refresh	A higher timeout prevents premature failures during the transfer of large manifest files or low transfer speed due to high WAN latency.	Not efficient use of Argo CD resources with high timeout value.

WAN Optimization: Increase *timeout.reconciliation* to reduce the frequency of automatic sync attempts	The controller polls Git every 3 minutes by default. If configuration updates are not frequent, you can increase to several hours to reduce the frequency.	Reduced frequency reduces the load on the Git server as well Argo CD application controller. **Reduced API Server Load:** The reconciliation process involves multiple Kubernetes API calls to fetch the live state of resources in the cluster. Increasing the interval reduces the frequency of these API calls. Lowers the load on both the ArgoCD Application Controller and the Kubernetes API Server under management. **Reduced Git Polling:** Fewer polls mean less frequent CPU cycles that are dedicated to Git operations and reduced traffic/ load on the Git server itself. **Fewer Reconciliation Queue Entries:** By lengthening the interval, you give the controller more room to finish the previous cycle before the next one starts, helping the reconciliation queue clear out.	Slower drift detection Slower propagation of controlled changes in Git.

Git Repository and Container Registry Placement Design 2

This section reviews Design consideration of Local Argo CD at Edge with Private Git Repository and Local Harbor Registry.

Private Git repositories are deployed in the central VCF instance or a mirror copy of the central Git repository are deployed in the management domain of the VCF instance. Argo CD is deployed at each Edge site for simplified performance, RBAC, and networking design. Harbor registry and Content library are also deployed at each edge site to service virtual machine or VKS cluster deployment.


VCF Operation
SDDC Manager
License Server
vCenter
NSX
vSAN
Management Cluster
Site - A
Site - B
Site - C
Control Plane
VM
Zone
CL
Supervisor
Local or External Storage
300ms / 10mbps
Edge Sites

Deploy a Private Git repository at the Central VCF instance or Edge VCF instance management domain	Git Repository can be deployed in the management domain if there are sufficient capacities. Otherwise, deploy the Git Repository as a workload in the workload domain. Multiple Git repository instances can be set up to mirror each other.	Private Git repositories are behind the corporate firewall, offering better access control, private issues tracking, and security compliance. Git Repository Server TLS certificate must be loaded as part of the Argo CD instance private repository configuration.	Edge Admins have to manage all infrastructure, patching, upgrades, and maintenance as well as uptime and capacity for the Git Repository server.
Deploy Content Library and Harbor Container at the edge	Content libraries are Local to the edge and can be created on vSAN, shared or local datastore. Configured to sync up from the Central Content Library, used to pre-stage VM OVAs to reduce dependency to WAN. Edge Harbor container registry Configured to sync from central Harbor to reduce WAN usage.	Design the content library to enable publishing of golden VM OVAs and subscription at each edge location. Use the Check-in/Check-out feature for tracking template changes. When updating, convert to a VM, patch, snapshot if needed, then update the library item. Design Harbor replication policies to control image distribution between instances, or used as a pull-through proxy for local caching depending on the use case. With local CL and Harbor, VM templates and container images are transferred **only once** when the content is synchronized instead of every time a new workload is created. For organizations paying for bandwidth, especially metered connections, minimizing WAN traffic leads to substantial operational cost savings.	Edge Admins must allocate and maintain the storage capacity necessary to host the entire replicated Content Library and Harbor Registry at each edge site. Ensuring that the correct versions of templates are pushed to the correct sites requires careful policy management. A mistake can lead to Edge storage exhaustion and WAN bandwidth contention.
WAN Optimization: Increase *reposerver.git.request.timeout* to extend time to wait for git commands to return from the Git provider.	The default timeout is 60s. Increase the timeout if there are large network delays between edge sites and the Git Repository server.	If the Git operation takes longer than the *reposerver.git.request.timeout* limit, the repo-server abruptly kills the process. Argo CD UI as a generic "context deadline exceeded" or "git fetch timed out" error, and applications fail to sync or refresh	
WAN Optimization: Increase *server.repo.server.timeout.seconds* to mitigate against edge locations with high network latency to the Internet.	*server.repo.server.timeout.seconds* is the maximum amount of time the Argo CD Application Controller waits for a response from the Repo Server before giving up and failing the application reconciliation/refresh	A higher timeout prevents premature failures during the transfer of large manifest files or low transfer speed due to high latency.	Not efficient use of Argo CD resources with high timeout value.

WAN Optimization: Enable controller.sync.timeout.seconds	The timeout value can be updated in the *argocd-cmd-params-cm, i*t is a global timeout value, not per application. Set the timeout value based on the number of edge locations and latencies across each location. The default is to never time out. This value should be set to several hours if there are many edge sites.	Addresses issues where the sync process gets stuck indefinitely, often due to resources failing to start properly and continuously retrying. Subsequent deployments are blocked until sync completes.	The timeout value must be adjusted so it catches the worst case, but not timeout too quickly.
WAN Optimization: Increase *timeout.reconciliation* to reduce the frequency of automatic sync attempts	The controller polls Git every 3 minutes by default. If configuration updates are not frequent, you can increase to several hours to reduce the frequency.	Reduced frequency reduces the load on the Git server as well Argo CD application controller. **Reduced API Server Load:** The reconciliation process involves multiple Kubernetes API calls to fetch the live state of resources in the cluster. Increasing the interval reduces the frequency of these API calls. Lowers the load on both the Argo CD Application Controller and the Kubernetes API Server under management. **Reduced Git Polling:** Fewer polls mean less frequent CPU cycles that are dedicated to Git operations and reduced traffic/ load on the Git server itself. **Fewer Reconciliation Queue Entries:** By lengthening the interval, you give the controller more room to finish the previous cycle before the next one starts, helping the reconciliation queue clear out.	Slower drift detection Slower propagation of controlled changes in Git.

Git Repository Design

Repository design stands out as one of the most critical elements for operational scalability, maintainability, and team efficiency. A well-structured repository enforces a clear separation between source code and deployment code, which is essential for managing environments and automating deployments effectively. For smaller projects, a monorepo approach can be a simple starting point. However, as systems grow and multiple teams manage different applications, adopting a multi-repo strategy can be more effective.

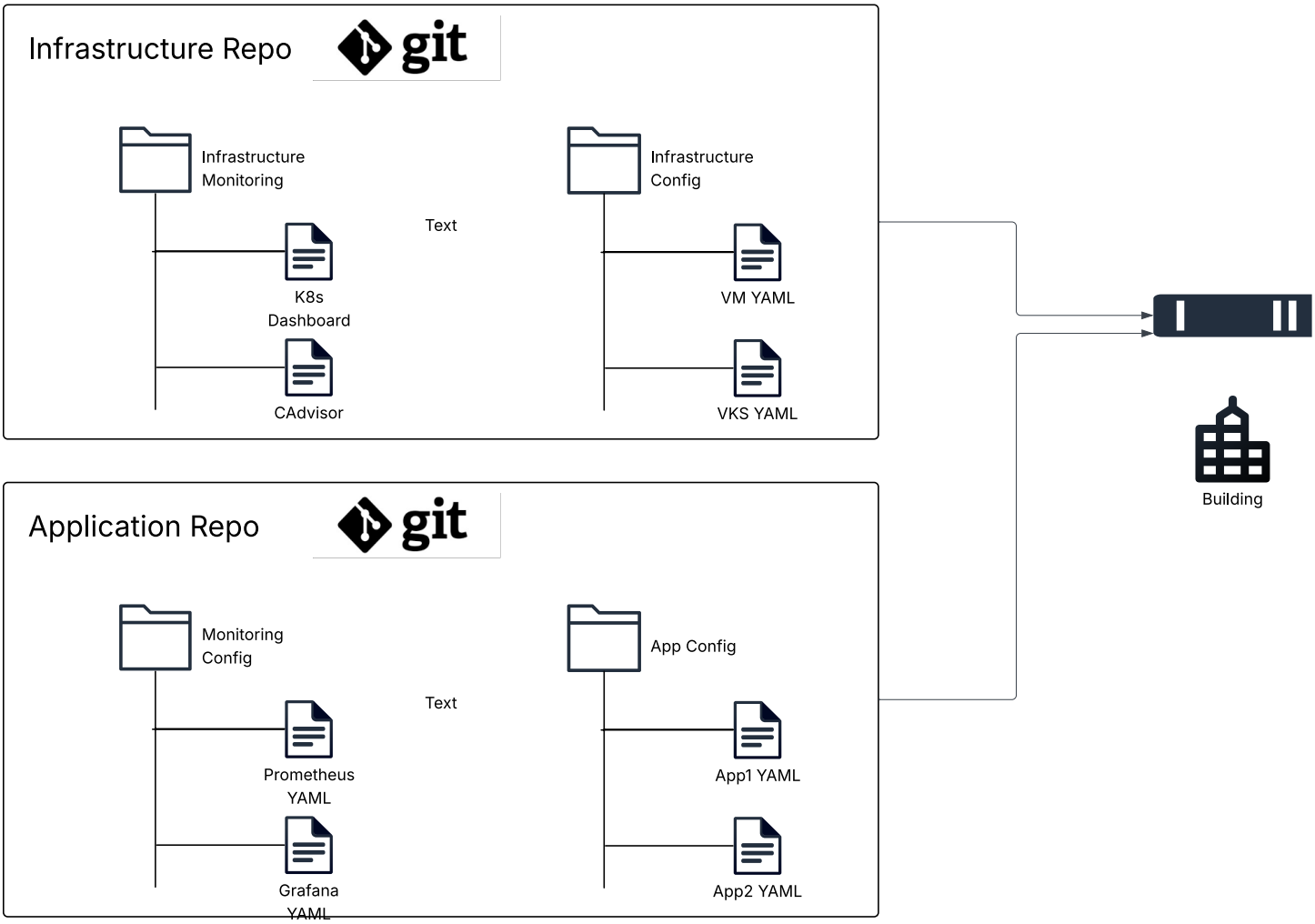
One of the most important best practices is to maintain a clear separation between source code repositories and deployment configurations (defined using YAML). Combining source code and deployment configurations in a single repository introduces unnecessary coordination across teams and makes auditing changes difficult. Keeping them separate helps prevent accidental deployment and simplify access control.

Mono Repository

Use a single Git repository for all Argo CD deployments, infrastructure, and applications. This repository does not contain any source code. Instead, it stores the deployment definitions for all applications, even when those applications are unrelated to each other. Monorepo is a logical choice when starting out with GitOps. Using one repository simplifies maintenance and improves overall visibility. However as the edge environment grows and more teams and applications are involved in the GitOps process, this approach introduces coordination challenges and deployment conflicts.

Git Repository Per Team

A single Repository works great as a starting point, another approach is to group applications into groups, based on team or projects. Each team would have a dedicated repository. Members of each team would be given write access to the repository while read access or no access are provided to members of other teams.


Infrastructure Repo
git
Infrastructure Monitoring
K8s Dashboard
CAdvisor
Text
Infrastructure Config
VM YAML
VKS YAML
Application Repo
git
Monitoring Config
Prometheus YAML
Grafana YAML
Text
App Config
App1 YAML
App2 YAML
Building

A minimum of two repositories are created when starting out: one for application and another for Infrastructure. The infrastructure repository contains workload configurations that run directly on top of the virtualization layer, such as VM and VKS. The application layer includes both App Configuration and App monitoring deployment definitions. Over time as edge applications grow, more application repositories can be created to align with each application development team, and a common repository to share dependencies required by all teams. With a dedicated repository, devOps engineers can check out only the Git repository that contains the relevant components. This approach helps avoid Git conflicts and makes it easier to apply extra security constraints without running into cross-organization challenges.

Key Considerations:

- **Mono Repository** design works well when it's desirable to view all changes across all environments and services in one place. But it's harder to scale to large projects with multiple teams.
- **Multi Repository** provides autonomy. Each team manages its own pipeline, approvals, and dependency. It provides scoped access control, but much harder to apply updates across services. Global changes must be repeated across repositories. Tracking deployment and changing history across repositories can be cumbersome.

Table below is a summary comparison of the two approaches:

Table 439: Repository Approaches

	Mono Repository	Multi Repository
Change Isolation	Change in one directory can impact others	Each repo is isolated. Misconfiguration is isolated to each repository.
Change Visibility	All changes are tracked at one place, including history.	Changes are spread across repos, harder to get full visibility
Scalability	Harder for large teams and projects	Scales better across many teams.
Team Autonomy	Must coordinate changes across multiple teams	Changes can be coordinated within a team.
Access control RBAC	Hard to limit permission per team.	Simple. Team controls who gets access to the repository

Repository structure to Model Edge Sites

Software Development organizations are familiar with using Git branches to represent a release and promote a branch only after it passes all automated tests. When adopting GitOps, it may seem natural on paper to extend the same branching strategy that is used for application release management to edge configuration management. However in practice managing each environment as a separate branch is highly complex and should be avoided.

The best practice in GitOps is to map all environments to a single branch and use dedicated folders to represent each environment. This "environment-per-folder" approach is similar to how Helm operates - a set of base configuration that can be shared, and a site-specific folder that contains parameters unique to each deployment. Configuration updates are done by creating a short-lived branch, modify configuration in the site-specific folder, and immediately commit back to main for deployment. Long lived branches should not be used to model edge environments.

A summary comparison of the two approaches:

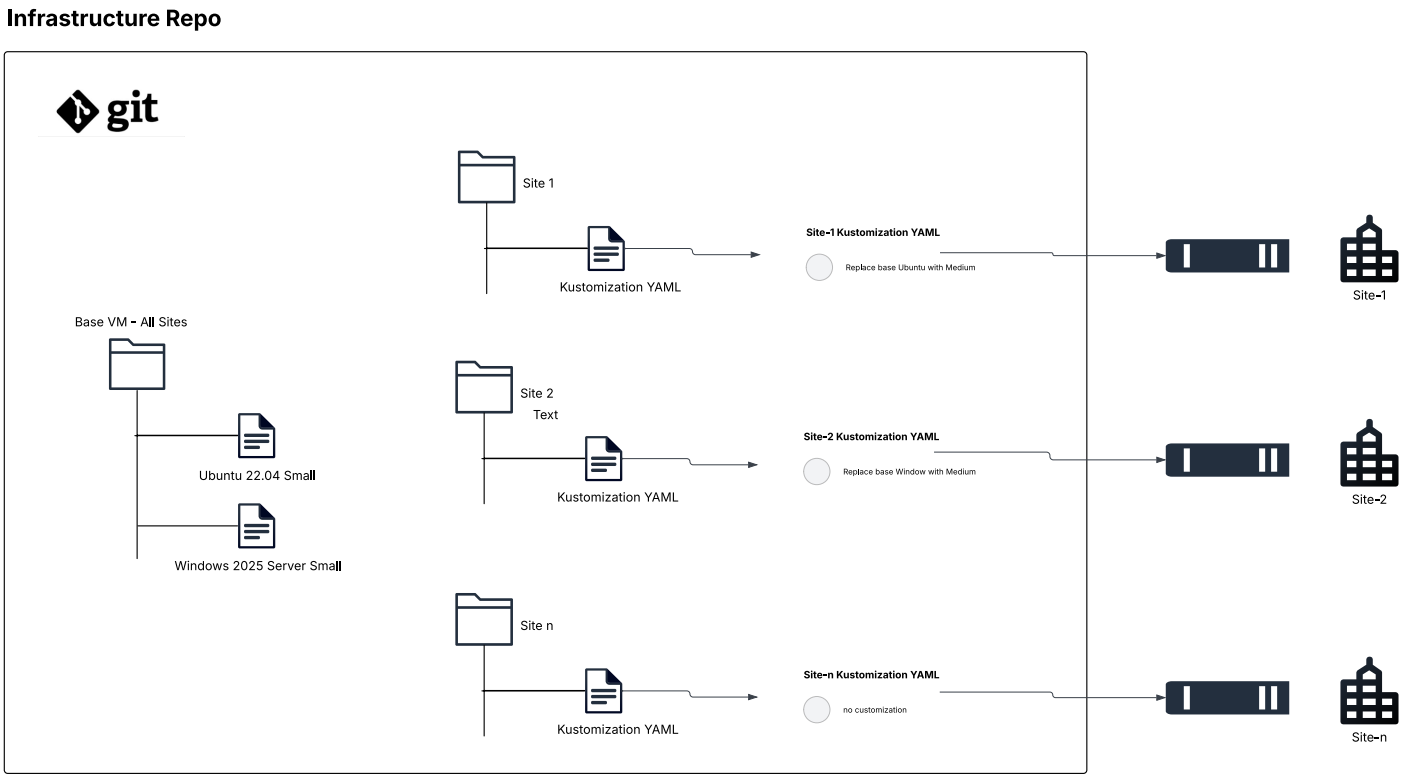
	Environment Per Folder	Environment Per Branch
Branch Management	Simple - trunk based. Short lived branches and changes always merge back to the main branch.	Complex - Applying changes to multiple long lived environments across multiple branches are difficult and hard to maintain.
View Site specific configuration	Simple. If implemented correctly, only configuration deltas are maintained in each folder.	Complex - Complexity in associating branch to environment. Lacks consistency in addressing environment-specific configuration delta.
Change Auditing	Simple. All change histories are maintained in a single place	Complex - Change histories are specific to a branch & environment

Base and Overlay

In GitOps, configuration readability and maintainability are critical. Less configuration, reduces the likelihood to introduce configuration errors and is easier to understand the configuration delta between sites.

Configuration optimization is often accomplished with the Kustomization operator using base and overlays. Base consists of defining a set of base configuration that applies to all edge locations, and overlays are site-specific modifications of the base configuration. The Kustomize operator traverses base configuration manifests to add, remove, or update configuration in line, without requiring templates. More details can be found here. The following figure is an example using base and overlay to define the configuration for a VM:

Figure 164: Base and Overlay Structure



In this example, each location has two VMs deployed: one for Windows and one for Linux. The default VM size is *Small* for Windows and *Large* for Linux. Kustomization YAML is used to override the base configuration. In this scenario, Site 1 deploys Small flavors for both Linux and Windows VMs, while Site 2 deploys a Large Linux VM and a Small Windows VM.

Base YAML

Windows:

```
apiVersion: vmoperator.vmware.com/v1alpha4
kind: VirtualMachine
metadata:
  name: vm-xv8z
  namespace: argocd
  labels:
    vm-selector: vm-xv8z
spec:
  className: best-effort-small
  imageName: windows2025-v0.2.55
  storageClass: vsan-default-storage-policy
  powerState: PoweredOn
```

Linux:

```
apiVersion: vmoperator.vmware.com/v1alpha4
kind: VirtualMachine
metadata:
  name: vm-xv8z
  namespace: argocd
  labels:
    vm-selector: vm-xv8z
spec:
  className: best-effort-large
  imageName: ubuntu2k04-v1.2.55
  storageClass: vsan-default-storage-policy
  powerState: PoweredOn
```

Sample Site1 customization to change the VM name and VM T-Shirt class:

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
bases:
- ../../base
patches:
  - target:
      group: vmoperator.vmware.com
      version: v1alpha4
      kind: VirtualMachine
      name: vm-xv8z
    patch: |-
      - op: replace
        path: /spec/className
        value: best-effort-small
  - target:
      group: vmoperator.vmware.com
      version: v1alpha4
      kind: VirtualMachine
```

```
    name: vm-xv8z
  patch: |-
    - op: replace
      path: /metadata/name
      value: site1-vm-xv8z
```

If there are multiple applications, there should be a set of bases and overlays for each application.

Table 440: Repository Design Decisions

Design Attribute	Detail	Justification	Implication
Separate application config from source code	The CI pipeline is responsible for the source code repo. The pipeline pulls the source repo builds the image The CD pipeline uses a separate process to update the configuration repo and deploys to each environment.	Source code reviews and Infrastructure code reviews involve different teams and people Major differences in branching strategy due to development methodology and code change frequently. Change audit processes are different.	None
Create a repository for each type of edge application	Start with two repositories: A Git repository for infrastructure A Git repository for edge application	DevOps teams can check out only configuration relevant to its project. This approach helps avoid Git conflicts and makes it easier to apply extra security constraints without running into cross-organization challenges	More repositories to manage Cross-service changes require multiple PRs
Environment-per-directory with Kustomize	Define a set of base configuration that applies to all edge locations Create a folder for each environment/edge site. Dev, staging, production edge sites are each represented by a folder. Each folder contains site-specific overlays that modify the base configuration for that location	Environment differences are captured as overlay. Easy to spot changes at each site.	Test escape is high if overlay configuration manifests are drastically different across environments. Base Configuration may not have defined correctly.

Bootstrap and Operational Flow

Edge site activation involves configuration ESX host into a single node vSphere edge cluster, activate the Supervisor service, and install Argo CD and register the Git Repository with Harbor.

The Day 0 Bootstrap phase establishes the foundation by provisioning ESX hosts using Zero Touch Provisioning or manual ISO installation, followed by enabling the vSphere cluster and Supervisor through vCenter. Once the infrastructure is ready, the Broadcom Argo CD Operator is deployed via the Supervisor catalog, and the Git repository is securely registered to trigger the initial synchronization.

Day 1 Ongoing Operations, the environment shifts to a continuous delivery model where platform operators push manifest updates directly to Git, which Argo CD continuously monitoring for state drift. Argo CD automatically detects and applies

configuration updates. This GitOps approach streamlines lifecycle maintenance, allowing workloads to self-recover after central VCF-driven events.

Day 0 - Bootstrap	Description	Day 1 - Ongoing Operations	Description
ESX Host Provisioning	Use Zero Touch Provisioning with a host profile that is applied from a central or locally managed vCenter Or perform a manual ISO installation.	**Manifest Update (GitOps)**	Platform Operator pushes changes to Git; Argo CD detects and applies automatically.
Enable vSphere Cluster and Supervisor	vSphere cluster and Supervisor activation through vCenter	**Drift Detection**	Argo CD continuously reconciles desired vs actual state; alerts on out-of-sync
Argo CD Operator Install	Broadcom Argo CD Operator deployed using Supervisor catalog through vCenter	**ESX Patching**	VCF lifecycle patch applied from Central or Regional DC; workloads self-recover post-reboot using Argo CD
Git Repo Registration	Git Repository token secret applied; initial sync triggered	**VKS Version Upgrade**	Update VKS Kubernetes Release reference in Git; Argo CD triggers rolling upgrade

Argo CD Application Components Design

In Argo CD, the journey from deploying a single application to managing a global fleet of edge sites is bridged by three progressive concepts: the Application, the App-of-Apps pattern, and the ApplicationSet.

At its foundation, an Application is the core building block that connects one specific Git directory to one target cluster and namespace. However manually creating these individual resources becomes a tedious bottleneck. The App-of-Apps pattern was created to bootstrap an entire edge cluster's application from a single root manifest. For multiple cluster environments, the ApplicationSet uses "generators" (example: scan Git repo for application folders) to dynamically template and spawn dozens or hundreds of individual Applications on the fly.

Table 441: Argo CD Scaling Hierarchy

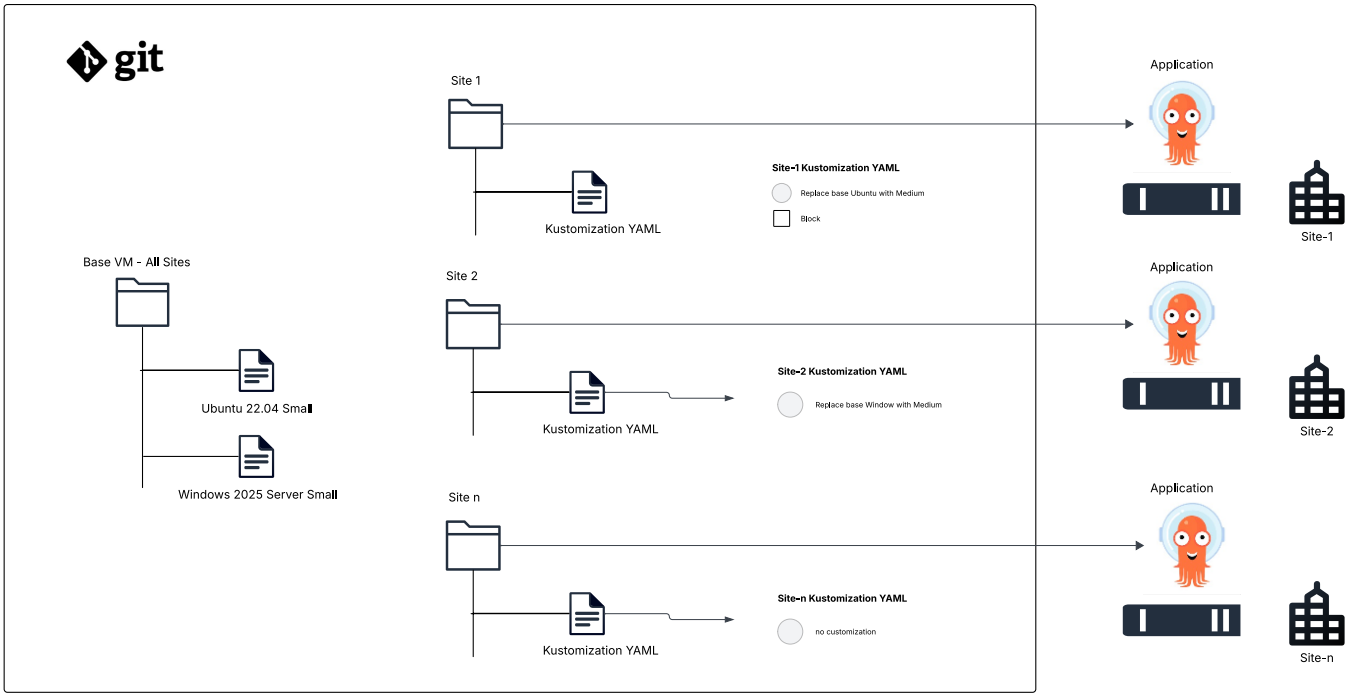
	What	What it does	Best Used for
Application	A Custom Resource linking one Git path to one destination.	Replaces manual *kubectl apply* with automated GitOps syncs.	Single deployments, simple microservices
App-of-App	A pattern where one Application deploys other Applications.	Bootstrapping multiple apps without clicking through the UI many times.	Managing a static set of apps for a single cluster.
ApplicationSet	A controller that automatically generates multiple Argo CD Applications using templates and rules.	Managing multi-cluster scaling, multi-tenant environments, and avoiding YAML duplication.	Deploying a single app across 20 clusters, or auto-deploying a new app whenever a new Git folder with new Apps is introduced

App-of-AppSets is another design pattern for more advanced use cases and we cover it in a later update to the design guide if necessary.

Argo CD Component Design Pattern #1

Each Application specifies a Git repository, a path within that repository where the Kubernetes manifests reside, and the destination cluster and namespace. This approach works for a few applications, but can be overwhelming to deploy and manage if the number of applications grows or many sites. A typical structure looks similar to the following diagram:

Infrastructure Repo



Base VM Application:

```
apiVersion: vmoperator.vmware.com/v1alpha4
kind: VirtualMachine
metadata:
  name: vm-xv8z
  namespace: argocd
  labels:
    vm-selector: vm-xv8z
spec:
  className: best-effort-xsmall
  imageName: ubuntu-24-04
  storageClass: local-default-storage-policy
  powerState: PoweredOn
  bootstrap:
    cloudInit:
      cloudConfig:
        users:
        - name: ubuntu
          lock_passwd: false
          passwd:
            name: my-vm-boot-data
            key:  jdoe-passwd
          shell: /bin/bash
          sudo: ALL=(ALL) NOPASSWD:ALL
          ssh_authorized_keys:
          - ssh-rsa <removed>
---
apiVersion: v1
kind: Secret
metadata:
  name:      my-vm-boot-data
  namespace: argocd
stringData:
  jdoe-passwd: <removed>
```

Site1 VM Customization:

```
kind: Kustomization
namespace: demo1
bases:
- ../../base
patches:
  - target:
      group: vmoperator.vmware.com
      version: v1alpha4
      kind: VirtualMachine
    patch: |-
      - op: replace
        path: /spec/cLassName
        value: guaranteed-small
```

And VM application for site1:

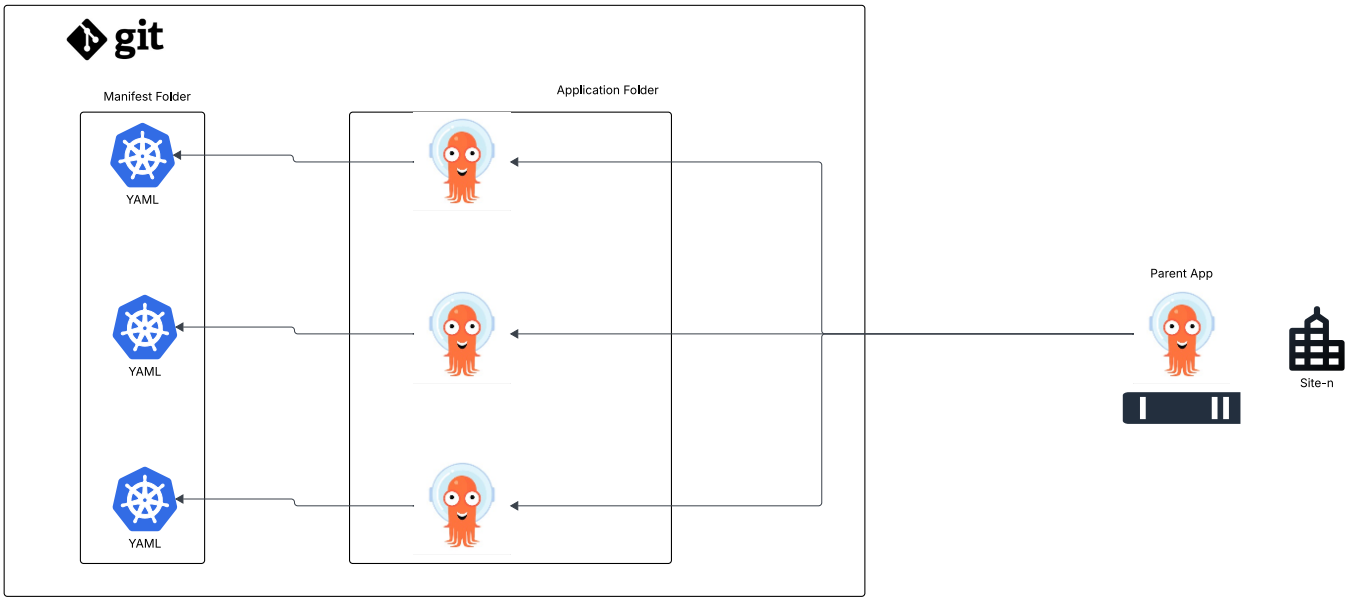
```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: site1-vm-app
  namespace: demo1
spec:
  project: default
  source:
    repoURL: https://github.com/xiaoGaoBroadCom/argocd-example.git
    path: apps/vms/overlays/site1/
    targetRevision: HEAD
  destination:
    server: https://192.168.30.161:443
    namespace: demo1
```

- **spec.source**: Defines the location of the Application manifests.
- **spec.destination**: Defines the cluster and namespace where the manifests should be deployed

Argo CD Component Design Pattern #2

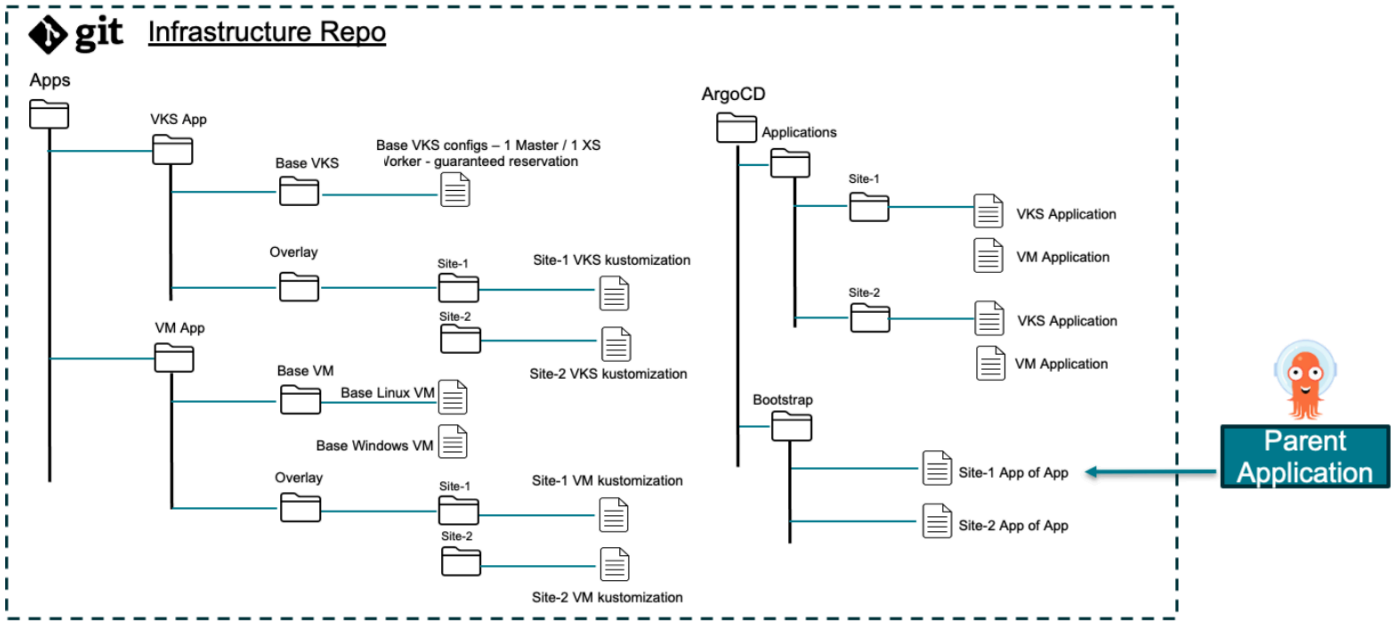
The App-of-App design involves defining a parent application referencing a directory in a Git repository that holds the definitions of other child applications. Instead of manually deploying each application, the root Argo CD application automatically detects and deploys all the child applications. The parent application tracks changes and propagates updates, configurations, and policies to its child applications, without manually accessing the Argo CD UI or CLI. This design pattern uses the Application controller, and is installed by the Argo CD operator by default. App of App is typically used for Argo CD bootstrapping, and a programmatic way to control application startup dependencies. A typical directory folder structure looks similar to the following diagram:

Infrastructure Repo



The following diagram is a sample mapping of App-of-Apps.

Argo CD Example Repository layout for App-of-App



When a cluster is created, a parent Application is created in argo CD pointing to the bootstrap YAML. The bootstrap YAML references applications defined in the Application folder. Each site has a separate bootstrap App of App YAML referencing applications that are required for each location.

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: site1-bootstrap
  namespace: demo1
spec:
  project: default
  source:
    repoURL: https://github.com/xiaoGaoBroadCom/argocd-example.git
    path: argocd/application/site1
    targetRevision: HEAD
  destination:
    server: https://192.168.30.161:443
    namespace: demo1
```

VKS applications as Define in the site1 folder:

```
apiVersion: argoproj.io/v1alpha1
```

```
kind: Application
metadata:
  name: site1-vks-app
  namespace: demo1
spec:
  project: default
  source:
    repoURL: https://github.com/xiaoGaoBroadCom/argocd-example.git
    path: apps/vkscluster/overlays/site1/
    targetRevision: HEAD
  destination:
    server: https://192.168.30.161:443
    namespace: demo1
```

And VM application in the same site1 folder:

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: site1-vm-app
  namespace: demo1
spec:
  project: default
  source:
    repoURL: https://github.com/xiaoGaoBroadCom/argocd-example.git
    path: apps/vms/overlays/site1/
    targetRevision: HEAD
  destination:
    server: https://192.168.30.161:443
    namespace: demo1
```

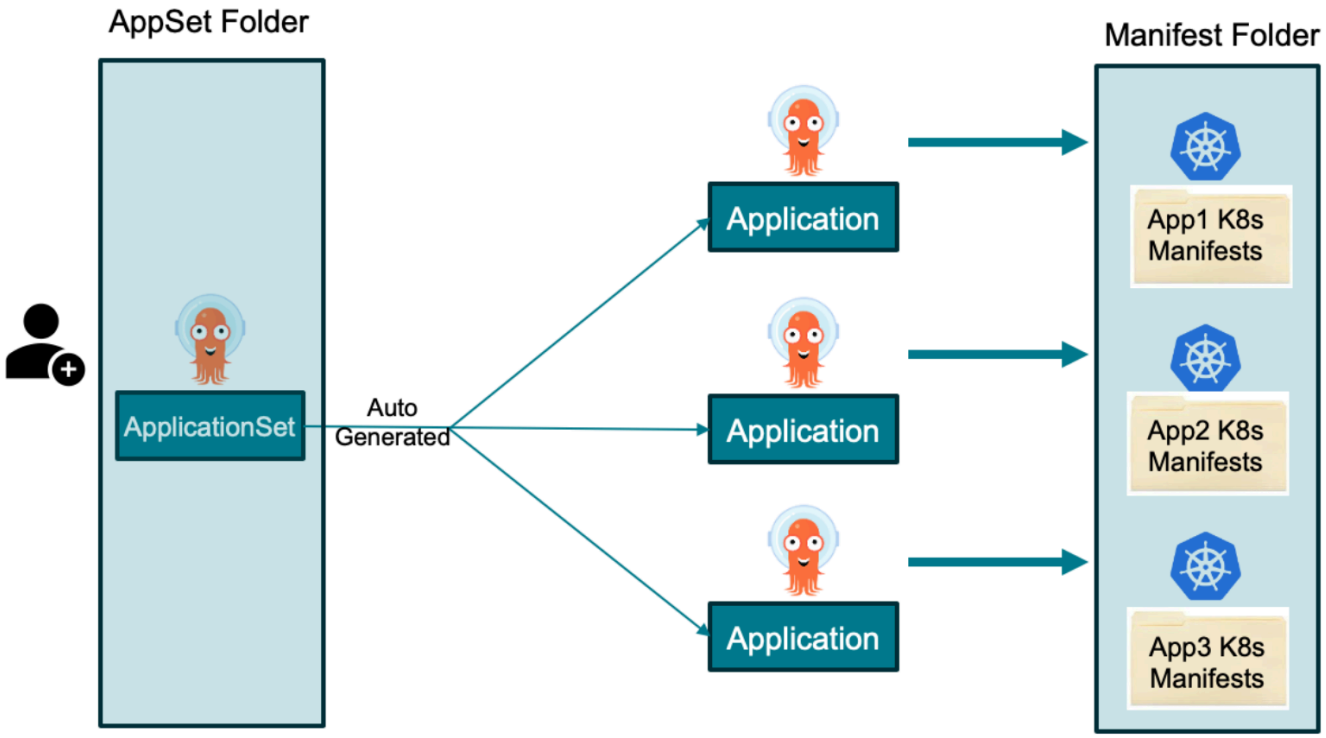
Argo CD Component Design Pattern 3:

When managing many edge sites spread across many geographically diverse sites, creating and mapping Argo CD application YAMLs can be repetitive. ApplicationSet significantly reduces the manual effort that is involved in building Argo CD Application YAML files. It uses a template to define the base configuration for applications, which is then combined with parameters from a generator to produce multiple Argo CD Applications.

The ApplicationSet controller currently supports several types of generators, each offering different ways to define and generate application parameters. The Cluster generator can be powerful in a centralized Argo CD design with lots of edge clusters deploying identical sets of Apps. If Argo CD is deployed at an edge site, the Git generator could be more appropriate to deploy multiple application workloads to a site. To add dependency or order of boot-up sequence, ProgressiveSync feature was introduced in Argo CD v.2.6.0, but is in an early stage of adoption, not ready for production deployments.

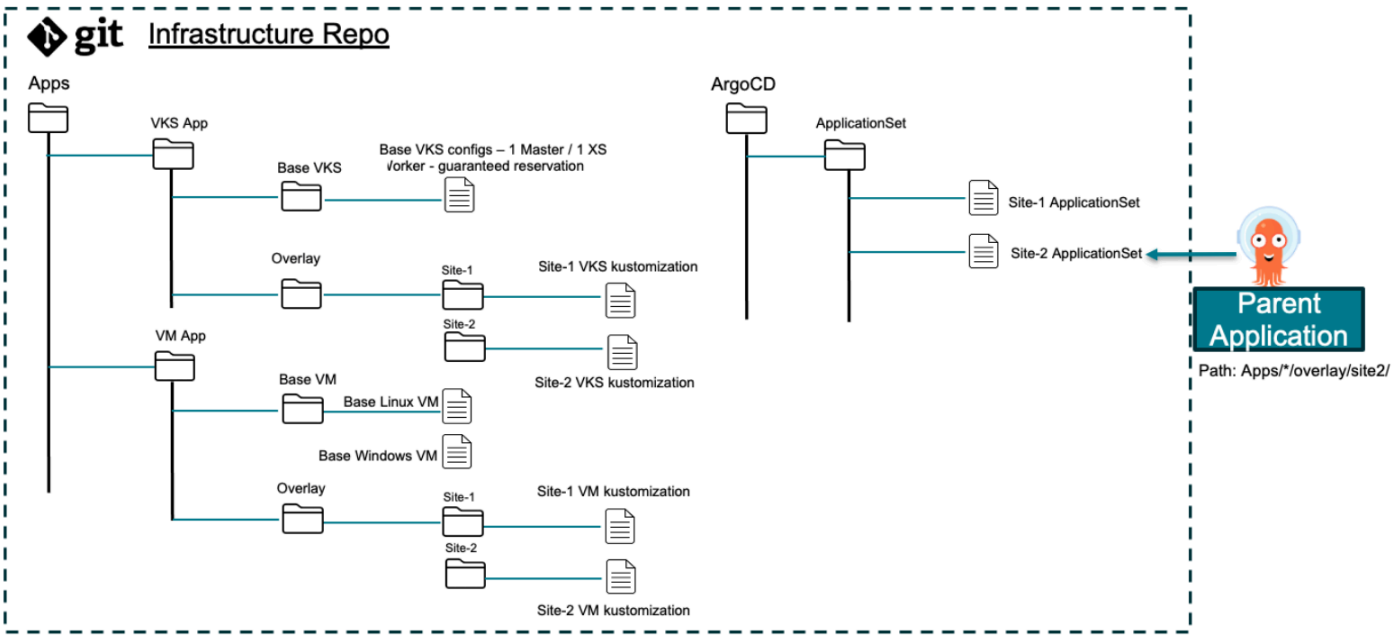
The ApplicationSet controller is bundled with the VMware Argo CD Operator, but must be explicitly enabled.

ApplicationSet Design Pattern



Below YAML is an example design. This design assumes that the Git generator is used. Instead of creating Application YAML, ApplicationSet YAML is used to auto generate all applications, in this case VKS clusters, and virtual machines.

ApplicationSet Design Pattern Example



The Git Directory generator generates parameters using the directory structure of the Argo CD repo. The application metadata name is derived from indexing array elements making up the directory path (3 and 1 in this example). Source is set to the directory path that matches path “apps/*/overlays/site1” deploying to the destination cluster specified in the destination server endpoint.

```
apiVersion: argoproj.io/v1alpha1
kind: ApplicationSet
metadata:
 name: site1-appset
 namespace: argocd
spec:
 goTemplate: true
 goTemplateOptions: ["missingkey=error"]
 generators:
 - git:
     repoURL: https://github.com/xiaoGaoBroadCom/argocd-example.git
     revision: HEAD
     directories:
     - path: apps/*/overlays/site1
 template:
   metadata:
     name: '{{index .path.segments 3}}-{{index .path.segments 1}}'
   spec:
     #Referto https://github.com/argoproj/argo-cd/issues/18643 when adding repo to project.
     project: default
```

```
  source:
    repoURL: https://github.com/xiaoGaoBroadCom/argocd-example.git
    targetRevision: HEAD
    path: '{{.path.path}}'

  destination:
    server: https://10.50.12.202:443
    namespace: argocd
```

This ApplicationSet automatically discovers every directory under repoURL that matches the pattern apps/*/overlays/site1 and creates a separate Argo CD Application for each one and dynamically names the generated application. Some of the key parameters:

- **metadata.name**: '{{index .path.segments 3}}-{{index .path.segments 1}}':

The path.segments variable breaks the matched directory path into parts that are separated by "/".

For a path like apps/vmapp/overlays/site1:

- Index 1 is vmapp (since the path starts with apps).
- Index 3 is site1.

The final name for the VM application is *vmapp-site1*

Argo CD Resources Design

Scalability of Argo CD deployment is a function of configuration options for its three core components: the application server, repository server, and API server.

For the Application controller, factors that influence the sizing includes:

- Number of applications
- Number of clusters under management
- Rate of APIs requests (Client QPS)
- Tuning configurations regarding parallelism, operational processors, etc
- Number of shards (If applicable)

Argo CD Repo server handles Git repository operations. Operations such as cloning, syncing, and manifest generation. Variables to consider include:

- Size of the manifest
- Type of tooling (Helm vs. Kustomization)
- Parallelism / Concurrent processing

The primary function of the Argo CD API server is to expose platform functionalities to users via UI, API, and CLI.

Each Service is configured with a default resource Request / Limits based on the K8s resource specification definition. The following table outlines the default resource limits associated with each service.

	CPU Limit	Memory Limit	Ephemeral Storage
Repo Server	1	0.5G	5G
Argocd-Server	1	0.25G	1G
Redis	1	0.25G	1G
Notification	1	1G	1G
application-controller	2	2G	1G
applicationSet-controller	2	1G	1G

The above settings are a good starting point for PoC with a limited number of applications and clusters. Platform administrators are expected to monitor actual usage based on their deployment and configure appropriate resource

requests and limits for their production edge deployments. Options to reduce CPU and memory utilization involve reducing the number of parallel operations:

- Reposerver.parallelism.limit - Default value is 0, which means no limit. In environments with many applications, introducing a limit can prevent performance degradation and OOM kills.
- Controller.kubectl.parallelism.limit - Default value is 20, which means Argo CD can use 20 concurrent kubectl sessions to interact with the K8s cluster. Decreasing this value reduces resource utilization, but might slow down sync operations.
- controller .operation.processors and controller.status.processors - Both settings control parallel processors for reconciliation and sync operations. Decreasing this value reduces cpu and memory utilization, but slows down sync performance.
- ArgoCD_K8S_Client_QPS and ARGOCD_K8S_CLIENT_BURST - Both parameters can indirectly impact Argo CD utilization

Networking Resource Design

In environments with infrequent changes, and where self-healing is not enabled, WAN bandwidth usage is minimal when Argo CD is deployed at the edge. However, in scenarios with frequent application updates, Argo CD tuning may be required to reduce the frequency of configuration manifest downloads.. Argo CD design options that can help reduce WAN usage include:

Timer	Default	Recommendation
Application reconciliation timeout *timeout.reconciliation*	3 minutes	Increase *timeout.reconciliation* to reduce the frequency of automatic sync attempts
ARGOCD_EXEC_TIMEOUT *reposerver.exec.timeout*	90 seconds	Increase timeout is one way to deal with slow WAN to edge cluster *reposerver.exec.timeout* should be tuned according to the characteristics of the available WAN link
Server.repo.server.timeout.seconds	60 seconds	Increase *server.repo.server.timeout.seconds* to mitigate against edge locations with high network latency to the Internet.
controller.sync.timeout.seconds	The default is to never time out.	The timeout value must be adjusted so it catches the worst case, but not timeout too quickly.
reposerver.git.request.timeout	60 seconds	Increase the timeout if there are large network delays between edge sites and the Git Repository server.

In all cases, a side effect of increasing the timeout could reduce the CPU load on the Argo CD controllers.

Network capacity should match the number of clusters and users involved. It is important to monitor the right set of Argo CD Application metrics reflecting the deployment architecture.

Argo CD Authentication Design

The VMware distribution of the Argo CD Operator supports authentication using either local accounts or Single Sign-On (SSO) using the OpenID Connect (OIDC) protocol. Protocols such as SAML and LDAP requires external delegation capabilities and can be accomplished using VCF Automation.

Design Option 1 (Central OIDC Provider)

This design assumes reliable external access from the edge to either the Internet or intranet. In Central OIDC Provider setup, Argo CD is configured to redirect all UI and CLI access requests to a centralized Identity Provider (IdP). IdP can be hosted either on-premises or in the cloud. List of certified OIDC-compliant providers are listed on the ArgoCD documentation page. A sample integration with Okta is listed in the Appendix.

Design Option 2 (Local Accounts)

In scenarios where external connectivity is unavailable and only a small group of users, local Argo CD accounts are the best alternative that can be used to ensure uninterrupted access. Local accounts are typically created as part of the Argo

CD instance creation, and can be updated via kubectl patching and Argo CD CLI. JWT tokens are still used even for local accounts. When a local user is authenticated, a non-expiring JWT token is signed and issued by the Argo CD API server. This token can be revoked by updating the user password.

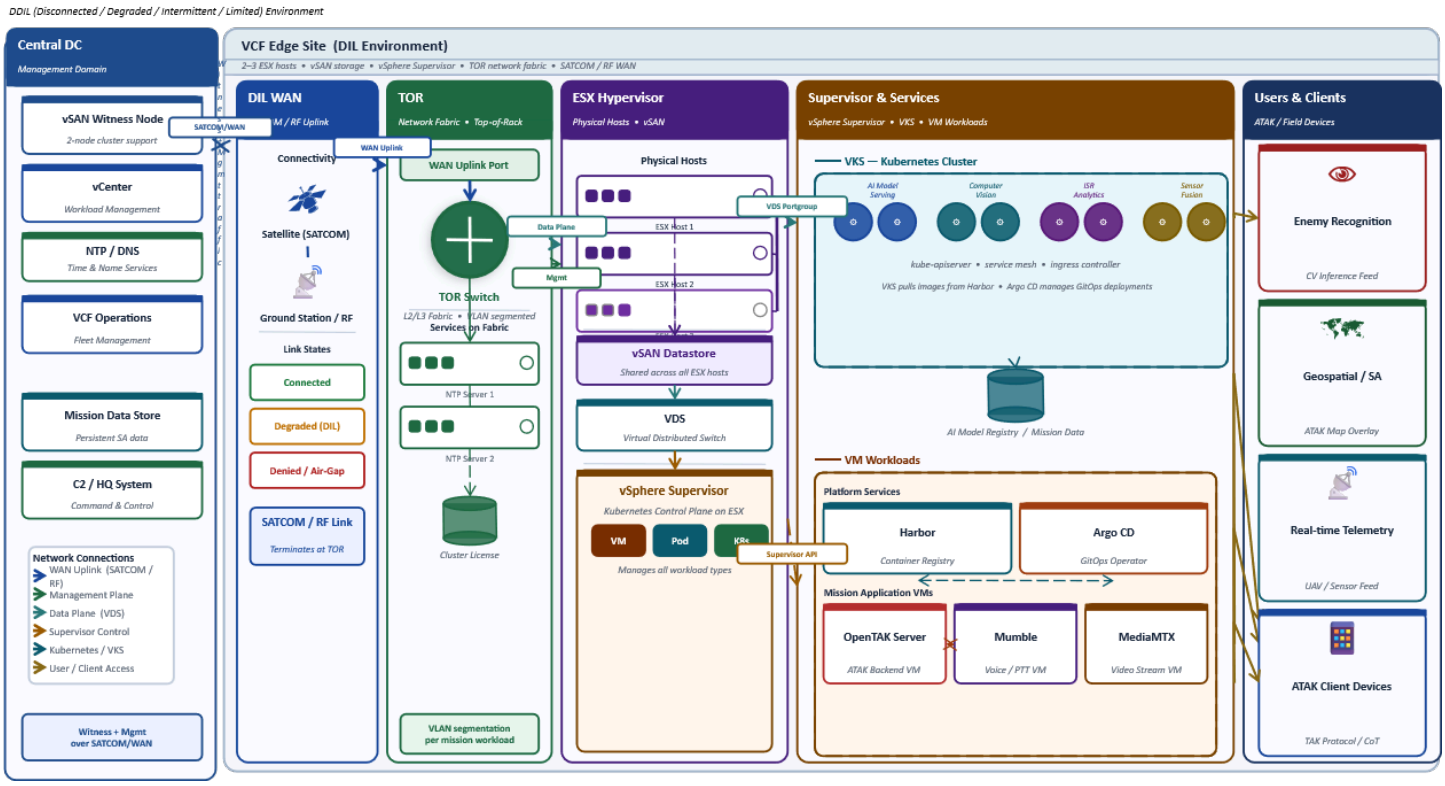
Important:

Based on this blueprint design, to deploy and onboard single-host edge sites, including Argo CD integration, customers can use the **VCF Edge at Scale automation script** to simplify and streamline edge deployments.

VCF Edge Design Blueprint for Government and Defense Use Case

This design focuses on the Defense Use Case for VCF Edge to address the unique operational, security, and compliance needs of federal agencies on and off the battlefield.

Figure 165: VCF Edge Reference Design for Defense Use Case



This design blueprint focuses on government, federal, and defense use cases for VMware Cloud Foundation Edge. The Blueprint addresses the unique operational, resiliency, and compliance requirements of mission environments both within centralized facilities and at remote or tactical edge locations.

The architecture is designed to support operational environments where connectivity may be intermittent, degraded, or fully disconnected. Type of deployments include operating in remote field locations, mobile operational environments, forward operating sites, and other mission-critical edge scenarios.

This reference design demonstrates the integration of centralized infrastructure services, localized edge operations, Kubernetes and VM-based workloads, and security controls into a unified operational model. This model supports both connected and disconnected states. Key design objectives of this architecture include:

- enabling localized operational autonomy at the tactical edge

- maintaining centralized governance and policy control where connectivity permits
- supporting mission continuity during WAN or SATCOM disruption
- securing sensitive workloads through segmented networking and layered security controls
- supporting sovereign and compliance-sensitive operational environments
- enabling scalable deployment of mission applications and AI-enabled workloads at the edge
- supporting both VM and Kubernetes-based application platforms in constrained edge environments

This architecture also supports multiple operational deployment models depending on mission and site requirements, including:

- centrally managed edge operations
- regionally autonomous edge hubs
- fully disconnected or degraded edge operations
- lightweight edge deployments with constrained infrastructure footprints
- locally managed edge environments requiring operational independence

The design blueprint progressively decomposes the architecture into the specific domains, including the infrastructure, networking, security, lifecycle management, and workload.

Operational Context

VMware Cloud Foundation Edge (VCFE) extends the centralized management capabilities of the platform to geographically dispersed edge locations supporting a wide range of use cases and verticals. VCFE edge locations can be as small as one node, two node cluster with vSAN witness or external storage, or multi-node ESX clusters up to 256 Cores per edge location. A typical VCFE customer can start with as little as 10 physical locations and can scale to thousands of sites. There’s no upper adoption limit on the total number of edge sites that can be deployed. The number of edge locations are driven by the use case.

This guide focuses on various Military use cases and focuses on the thin edge design, which supports vSphere cluster deployments ranging from a single host up to four ESX hosts. The document addresses these deployment scenarios by detailing three common architectures.

Operational Aspects of Federal Design for Air-Gapped Edge Environments

All design options are based on the assumption of **air-gapped deployment** with either **intermittent or no WAN connectivity**.

Design Criteria	Design Options	Key Components and Deployment	Connectivity Assumption	Applicable Host Count
Central Management, Remote Edge with Intermittent Connectivity	VCF Edge manages compute resources centrally for simplified operations and consistent infrastructure across distributed locations. • Based on VCF Edge Pattern 1 for edge locations needing advanced multi-layer LAN routing and switching. • Based on VCF Edge Pattern 10 for edge locations with existing vCenter and NSX manager • Based on VCF Edge Pattern 9 for edge sites with basic LAN connectivity.	Fleet management components in central DC including a shared vCenter using override license feature(VCF9.1) or a dedicated edge vCenter. A subset of control plane services are deployed at each remote site for local services and content caching.	Intermittent	Single ESX Host to up to 256 Cores per edge across multiple hosts.
Remote Management with Multiple hosts	Each edge site runs a full set of control plane services. • Based on VCF Edge Option 1, use one vCenter and one VCF Operations for unified management and monitoring. • Based on Edge Pattern #5 for deployments needing advanced network services and multi-layer routing/ switching	Option 1: One vCenter and one VCF Operations and configured using VCF Edge Powershell automation to each edge site. Edge Pattern 5: VCF Fleet components deployed using VCF Installer.	N/A (local management)	Multiple Hosts (up to 256 cores per edge)
Consolidated Remote Management with Single host	Based on VCF Edge Option 1, use one vCenter for compute resource management	One vCenter and configured using VCF Edge Powershell automation script.	N/A (local management)	Single Host

Government or Federal Use Cases

Military platforms, from heavily armored tanks and high-performance fighter jets to the individual soldier's wearable gear, rely on highly specialized, mission-critical applications to achieve operational success. These software systems, collectively known as C4ISR (Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance), must function with absolute reliability in the harshest environments. Application stacks that are hosted on this platform provide real-time situational awareness and precise targeting, whether the asset is traversing rugged terrain, flying at supersonic speeds, or operating in a **Disconnected, Intermittent, and Limited (DIL)** bandwidth environment. Use cases that are applicable to this guide:

Table 442: Government or Federal Design Use Cases

Use Case	Core Edge Compute Applications	Key Functions	Deployment Type
Dismounted Soldier (Backpack)	Tactical Communication and Networking, Mission Processing, Integrated Power, and Data Systems	Secure data and voice transmission, Blue Force Tracking (BFT), navigation support, sensor fusion, and power management.	Single Node or Multiple Nodes
Moving Ground Vehicles (e.g., Tanks)	Fire Control Systems (FCS), Battlefield Situational Awareness (BMS), Integrated Health Management	Real-time ballistic calculation, primary weapon stabilization, sensor fusion, Blue Force Tracking (BFT), and predictive maintenance capabilities.	Single Node or Multiple Nodes
Aircraft (e.g., Fighters, Cargo, Bombers, Surveillance)	Avionics Systems, Flight Control, Weapons Management, Load Management, Targeting Systems, Data Fusion	Enhanced agility for combat operations, logistical support and navigation, stand-off weapons deployment, and intelligence collection and analysis.	Single Node or Multiple Nodes
Naval Vessels (e.g., Destroyers, Carriers, Submarines, Logistics Ships)	Command and Control Systems, Integrated Bridge Systems, Damage Control Systems, Combat Systems, Air Traffic Control, Acoustic Processing	Maintenance of a Unified Common Operational Picture (COP), navigation, structural integrity monitoring, missile defense, covert detection, and troop/cargo logistical management.	Single Node or Multiple Nodes

In all cases, ruggedized, small form factor edge hardware supporting VM or container applications are mandatory. All use cases designed to operate in limited WAN or disconnected modes and support autonomous operations via a desired state and mission readiness dashboard. Refer to the VMware Compatibility guide for partner hardware compatibility.

Feature	Central Management, Remote Edge Host(s) with Intermittent Connectivity	Remote Management with Single Host	Remote Management with Multiple Hosts
Primary Use Cases	- Edge compute nodes in high-performance fighter jets (short missions)	- Edge compute nodes in soldier backpacks	- Edge compute nodes in a backpack kit
	- Edge compute nodes on navy aircraft carriers or low-altitude helicopters	- Edge compute nodes in long-range bombers/surveillance planes	- Edge compute nodes in military cargo/aircraft
	- Edge compute nodes on aircraft carrier type of ships	- Edge compute nodes in navy submarines	- Edge compute nodes in a fleet of tanks
		- Edge compute nodes in moving vehicles (e.g., tanks)	- Edge compute nodes on aircraft carriers or logistics cargo ship
Central Management Components	License Server, VCF Ops, VCF Management Services, Edge vCenter	NA	NA
Remote Management / Control Plane Components	Supervisor, Foundational Load Balancer (FLB), Argo CD, Harbor, Content Library	VCF Ops, vCenter, Supervisor, FLB, Argo CD, Harbor, Content Library	VCF Ops, vCenter, Supervisor, FLB, NSX, Argo CD, Harbor, Content Library
Edge Compute and Storage Design	- One or multiple nodes in a cluster	- Single node in a cluster	- Multiple nodes in a cluster

Feature	Central Management, Remote Edge Host(s) with Intermittent Connectivity	Remote Management with Single Host	Remote Management with Multiple Hosts
	- Single or multiple NICs (10G+ recommended, lower acceptable for low throughput when deployed directly from the Edge vCenter)	- Single or multiple NICs (10G+ recommended, lower acceptable for low throughput when deployed directly from the Edge vCenter)	- Single or multiple NICs (10G+ recommended, lower acceptable for low throughput when deployed directly from the Edge vCenter)
	- Local VMFS, vSAN, or External shared storage	- Local VMFS or External shared storage	- Local VMFS, vSAN, or External shared storage
	- vSphere Cluster with full DRS and HA	- vSphere cluster with full DRS and HA (no admission control)	- vSphere Cluster with full DRS and HA
Edge Networking Design	- VCF Networking with NSX VLAN segments or VDS	- VCF Networking with VDS	- VCF Networking with NSX VLAN segments or VDS
Local Edge Services	- DNS/NTP	- DNS/NTP	- DNS/NTP
	- Content Library	- Content Library	- Content Library
	- Supervisor Services (VM Service operator, VKS, Argo CD, Harbor)	- Supervisor Services (VM Service operator, VKS, Argo CD, Harbor)	- Supervisor Services (VM Service operator, VKS, Argo CD, Harbor)
	- Monitoring	- Monitoring	- Monitoring
Edge Activation	- ESX UEFI Network boot for self-activation	- Manual Installation	- Manual Installation of the first node, remaining nodes can use ESX UEFI Network boot for self-activation
	- vSphere Config Profile for infrastructure desired state enforcement.	- Partial Automation (staging for installation, then shipped for application deployment using Argo CD)	- Partial Automation (staging for installation, then shipped for application deployment using Argo CD)
Edge Access Security	- ESX Security (Secure boot, TPM, Entropy, lockdown mode)	- ESX Security (Secure boot, TPM, Entropy, vTPM, lockdown mode)	- ESX Security (Secure boot, TPM, Entropy, vTPM, lockdown mode)
	- Fleet SSO or local auth with smartCards	- Local auth	- SSO, AD, or Local auth
	- Fleet Password rotation or local	- Manual password rotation	- Manual password rotation
	- Fleet Certification rotation or local	- Manual certification rotation	- Manual certification rotation
Edge Compliance	VCF Ops FISMA and DISA compliance reporting	VCF Ops FISMA and DISA compliance reporting	VCF Ops FISMA and DISA compliance reporting
Edge Monitoring	**Central Only:** Key VCF Metrics (usage - Node, storage, Supervisor; Health) and External Services Metrics Management Pack (MP) for third-party components, Custom Dashboards)	Key VCF Metrics (usage - Node, storage, Supervisor; Health) and External Services Metrics (MP for third-party components, Custom Dashboards)	Key VCF Metrics (usage - Node, storage, Supervisor; Health) and External Services Metrics (MP for third-party components, Custom Dashboards)
Disconnected Mode Support	- Short WAN outage (minutes or days)	- Extended WAN outage (several months)	- Extended WAN outage (several months)
	- Extended WAN outage (several months)	- No WAN connectivity after initial deployment	- No WAN connectivity after initial deployment
Day 2 Autonomous Operations	- vSphere Configuration Profile	- vSphere Configuration Profile	- vSphere Configuration Profile
	- Workload desired state with Argo CD	- Workload desired state with Argo CD	- Workload desired state with Argo CD

Design Options

This section expands on the three options that are outlined in the overview section.

Design Options:

- Central Management with Remote Edge Hosts and Intermittent Connectivity
- Remote Management with Multiple Hosts

Central Management with Remote Edge Hosts and Intermittent Connectivity

Central Management with Remote Edge hosts consists of the following major components:

- Centralized Management Domain
  - A dedicated set of management components hosted near the edge locations. Management Domain components are tasked with managing the inventory and lifecycle of the remote edge clusters across the WAN. The WAN link must satisfy the following latency and bandwidth requirements:
    - Latency: Up to 300ms RTT between vCenter and ESX
    - Bandwidth: Minimum 10-Mbps throughput
- Remote Edge Clusters: Each edge site is represented by a single vSphere Cluster. Characteristics of the edge cluster include:
  - vSphere Cluster: Single-Node or Multi-Node ESX hosts with DRS and HA enabled.
  - vSphere Supervisor: vSphere Supervisor enables the deployment and management of Kubernetes workloads directly on the ESX host. A "Control Plane VM" and Load Balancer are deployed for the Supervisor functionality.
  - Workloads: Supports a mix of traditional Virtual Machines (VMs), native Containers, and vSphere Pods. Native Containers running within the vSphere Kubernetes Service (VKS) cluster. Virtual Machine and vSphere Pods run directly on top of the hypervisor.
  - GitOps Integration: Argo CD deployed as part of the Supervisor Services. Argo CD enables declarative application deployment directly to the edge cluster. Deployments can be infrastructure or applications running inside the infrastructure.

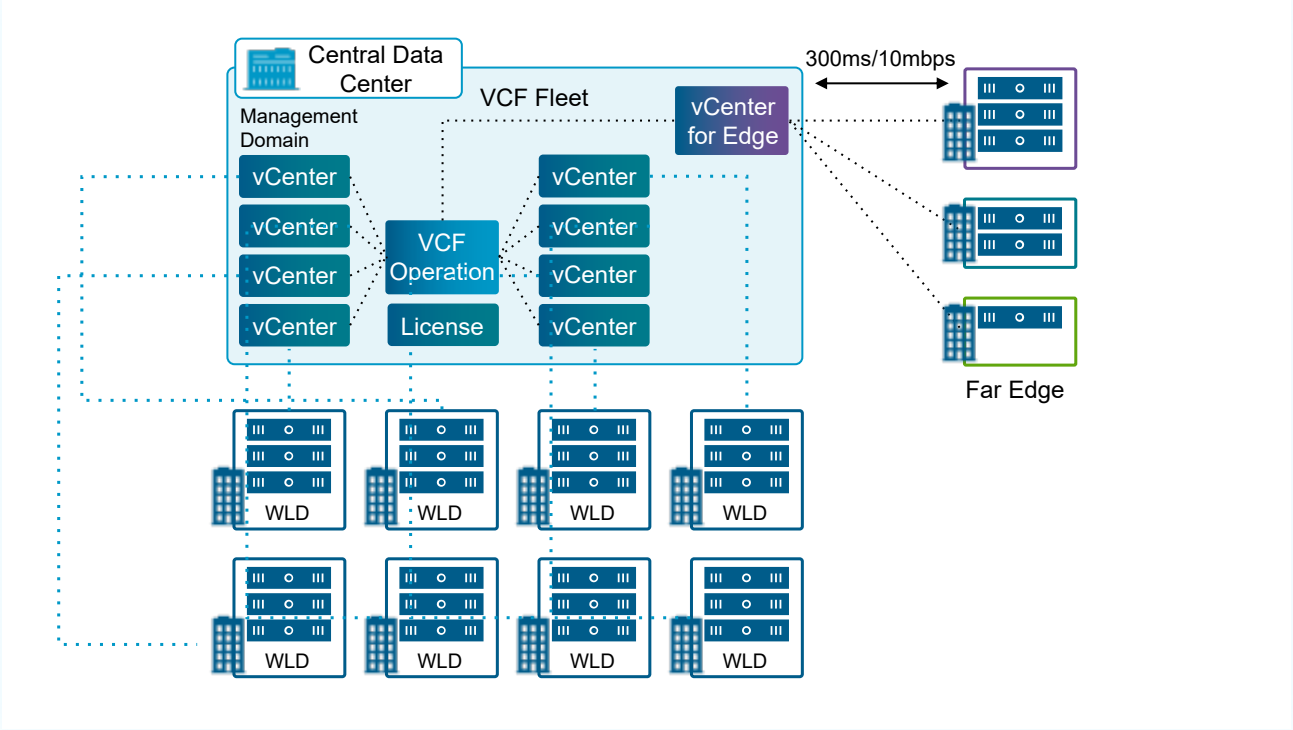
A core tenet of this design is the standardization of the edge deployment. By maintaining a uniform host configuration, network topology, and storage configuration across all remote sites ensures operational predictability and simplified lifecycle management. This standardized approach is a foundational best practice for distributed edge infrastructure and is assumed throughout this document.

Management Component Design

The management domain is based on the design options that are published in the VCF9.1 Design Library. Management component design to support the edge must be aligned with the specific functional requirements of the edge. Sites that do not require multi-layer routing and security addon with vDefend can extend existing Fleet with a new edge vCenter and can share VCF Operations for licensing and monitoring. This approach is documented as Edge Pattern #9.

Figure 166: Scenario 1 - Using Existing VCF Operations for Far Edge Expansion - Option 1

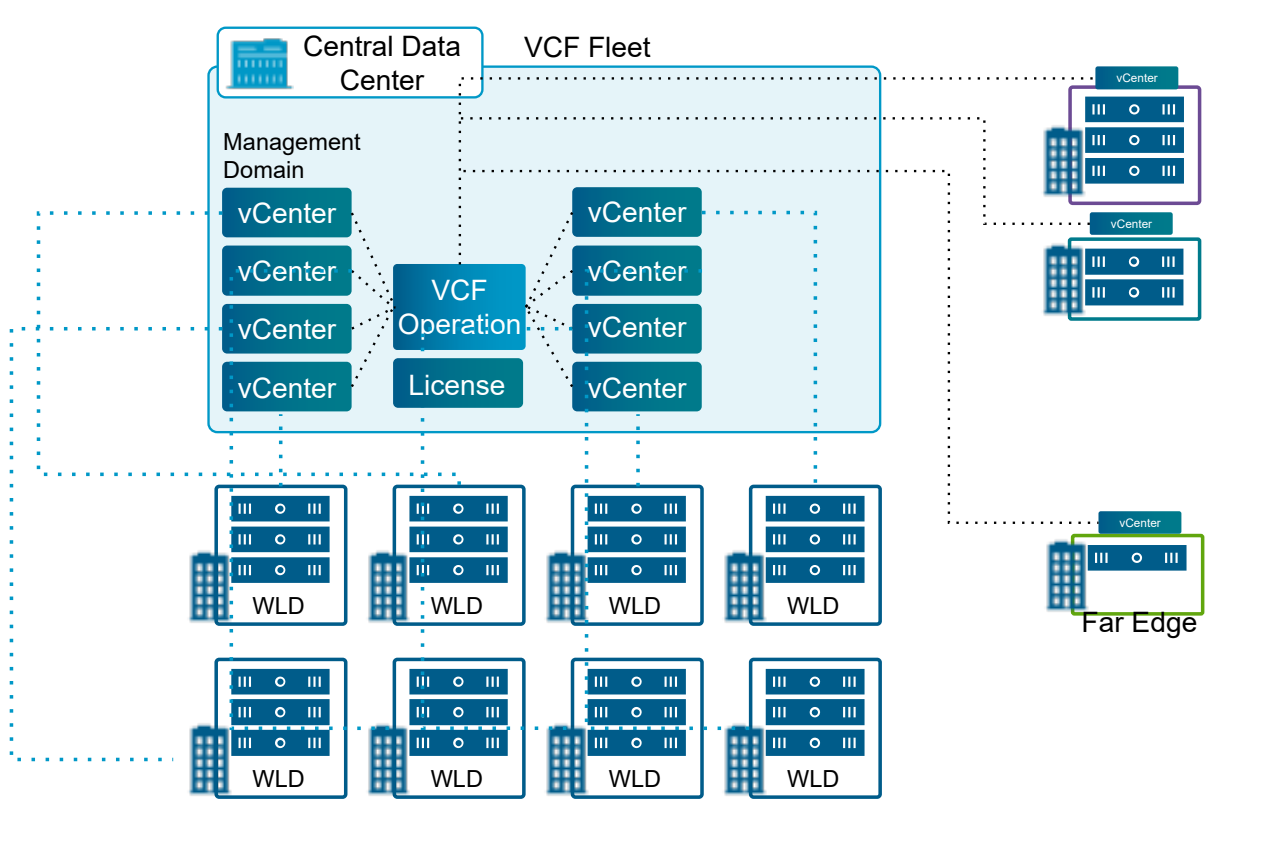
Centrally Managed Edge Locations


Central Data Center
VCF Fleet
Management Domain
vCenter
vCenter
vCenter
vCenter
VCF Operation
License
vCenter
vCenter
vCenter
vCenter
vCenter for Edge
300ms/10mbps
Far Edge
WLD
WLD
WLD
WLD
WLD
WLD
WLD
WLD

If the latency between edge sites and the Central Data Center exceeds 300ms or WAN bandwidth is less than 10mbps, a distributed vCenter topology can be used. Instead of a centralized vCenter, Edge Pattern #9 can easily be extended to support local vCenters at each edge location. Central VCF Operations is used for license management and monitoring.

Figure 167: Scenario 1 - Using Existing VCF Operations for Far Edge Expansion with Local VC - Option 2

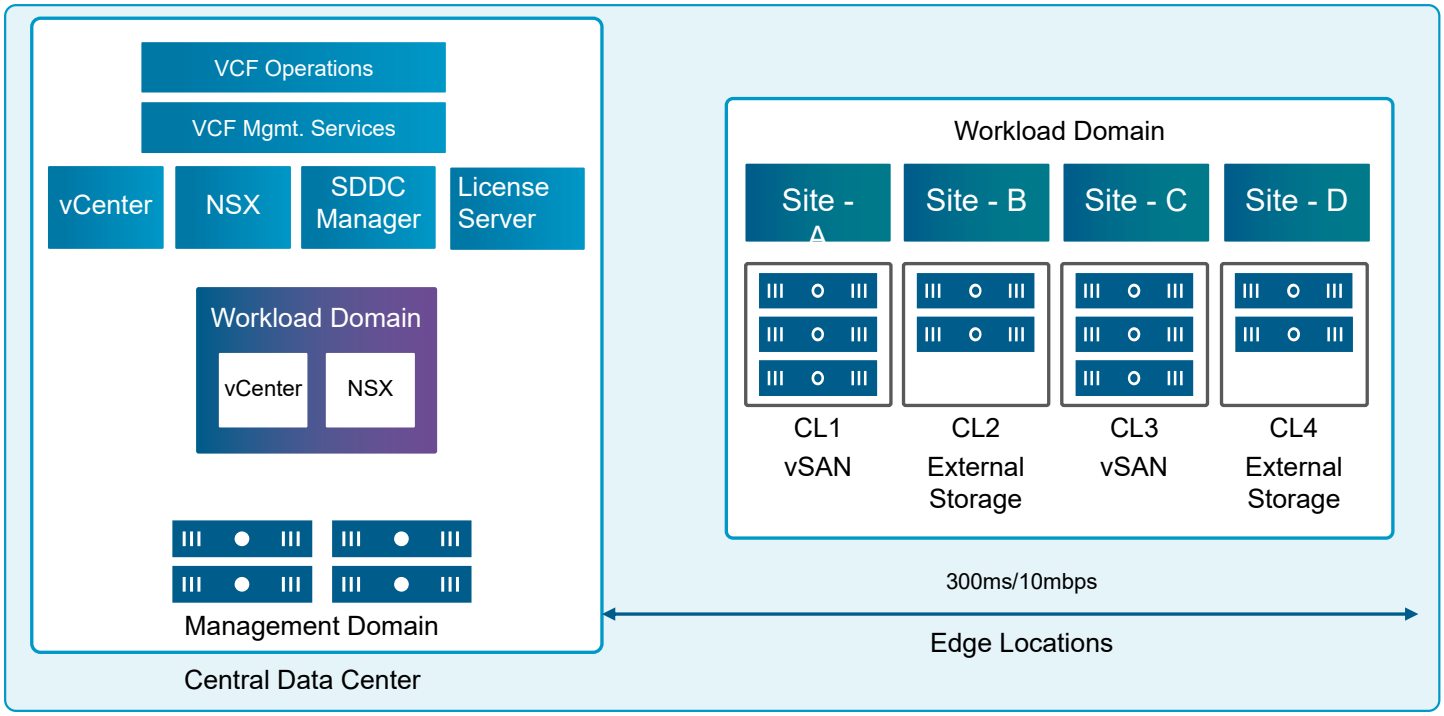
Edge locations with local VC


Central Data Center
VCF Fleet
Management Domain
vCenter
vCenter
vCenter
vCenter
VCF Operation
License
vCenter
vCenter
vCenter
vCenter
vCenter
vCenter
vCenter
Far Edge
WLD
WLD
WLD
WLD
WLD
WLD
WLD
WLD

Conversely edge environments requiring comprehensive, multi-layer routing, switching, and enhanced security addon with vDefend, should use NSX to provide advanced stateful services directly at the workload level. This approach is documented in VCF Edge Pattern #1. Edge Pattern 1 is ideally suited for central management of compute resources for simplified operations and consistent infrastructure across distributed locations.

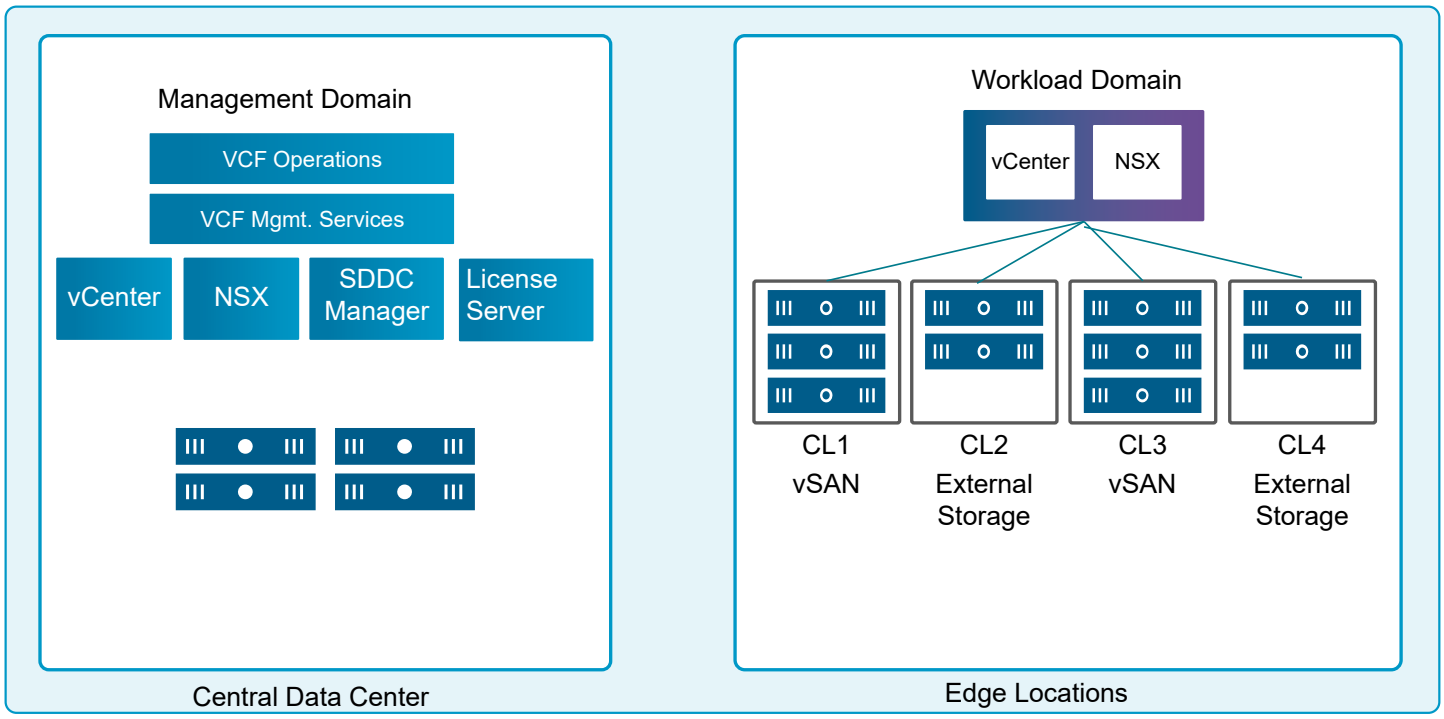
Figure 168: Scenario 2 - Centrally Managed Edge Compute Clusters in a Workload Domain - Option 1

Multi-layer routing, switching, and security using NSX at the Edge.


VCF Operations
VCF Mgmt. Services
vCenter
NSX
SDDC Manager
License Server
Workload Domain
vCenter
NSX
Management Domain
Central Data Center
Workload Domain
Site - A
Site - B
Site - C
Site - D
CL1
vSAN
CL2
External Storage
CL3
vSAN
CL4
External Storage
300ms/10mbps
Edge Locations

A distributed topology can be used if the edge site already has a vCenter and/or NSX manager. Instead of a centralized Workload Domain, Edge Pattern #10 can easily be extended to import local vCenters and NSX Manager at each edge location into a Fleet and consolidate LCM centrally with VCF Operation. There is a limit of 24 workload domain per VCF instance.

Figure 169: Scenario 2 - Centrally Management Domain with Edge Compute Clusters in a Workload Domain - Option 2



Design Decision	Design Details	Design Implication
Deploy a central VCF workload domain (WLD) to manage Edge sites needing • Advanced multi-layer routing/switching. • Integration with vDefend • Latency between Central DC and Edge locations less than 300ms. • Requires Central Monitoring, licensing, and fleet level management from VCF Ops	Documented in VCF Edge Pattern #1.. The vSphere management stack in the central management domain consists of: • vCenter. • NSX Manager • VCF Ops (shared) • VCF Management Services (shared) • License Server (shared) • SDDC Manager • Collector The vSphere Management stack at each edge location consists of: • None Edge vCenter design should align with VCF9 vCenter design recommendations. NSX manager design should follow the scale and failure tolerance that is outlined in the VCF9 NSX design library. Latency between Edge vCenter and ESX host must be less than 300 ms. Latency between NSX manager and Edge ESX hosts must be less than 150 ms. Please refer Latency Details for each component.	VCF management domain capacity planning must be updated to reflect the new Edge workload domain management components. Max 24 workload domains per VCF instance are supported. Use the VCF Ops sizing tool to check if existing VCF Ops have sufficient capacity for expansion. Refer to Edge Cluster Activationfor IP Addressing requirements
Use brownfield import to add edge environments with an existing vCenter to VCF9.1 fleet to manage Edge sites needing • Advanced multi-layer routing/switching. • Integration with vDefend • Central Monitoring, licensing, and fleet level management from VCF Ops	Documented in VCF Edge Pattern #10. The vSphere management stack in the central management domain consists of: • VCF Ops (shared) • VCF Management Service (shared) • License Server (shared) • SDDC Manager (shared) The vSphere Management stack at each edge location consists of: • vCenter • NSX Manager • VCF Ops Collector Key Design Considerations: • The Edge vCenter design should align with VCF9 vCenter design recommend ations. • Latency between Edge vCenter and SDDC manager must be less than 100 ms. Please refer Latency Details for each components. • Deploy VCF Ops Collector and activate data persistence to avoid data gaps during temporary connectivity issues. • Latency between VCF Ops collector and VCF Operations must be less than 300 ms. • Size the Ops Collector disk based on one-hour data persistence.	Max 24 workload domains per VCF instance are supported. VCF Ops Collector can store data for a maximum duration of one hour. Future host addition must be done from VC for brownfield imported VC Refer to Edge Cluster Activationfor IP Addressing and brownfield import sequences.

Edge Core Service Design

The availability of foundational services such as DNS, NTP, and workload templates are critical for autonomous operations. Provided the services themselves do not consume excessive CPU or memory capacity that are intended for workloads, deploying close to the edge is recommended. In addition, deploying foundational services at the edge reduces WAN bandwidth contention between workloads and the control plane.

To support the disconnected mode of operation required for autonomous operations, infrastructure services that are required to support the workload runtime dependencies must be available.

- **Network and Foundational Services:** STIG-compliant infrastructure relies on FIPS-validated encryption and a robust Public Key Infrastructure (PKI) to ensure session integrity and data-in-transit security. Modern cloud-native workloads have deep, systemic dependencies on these foundational services:
  - **DNS:** Service discovery in both traditional and modern applications relies heavily on DNS. If DNS is unavailable, edge applications may fail to operate.
  - **NTP:** Encryption is highly sensitive to temporal accuracy. If the system clock drifts even slightly, certificate validation may fail. STIG compliant infrasturcture relies on encryption and certificates to ensure session integrity.
  - **Availability Requirements:** Because both DNS and NTP services are pre-requisites for the rest of the stack, they must be architected for high availability to prevent a single point of failure from cascading through the entire environment.
- **Artifact Repository:** Local Artifact repository enables fast provisioning without WAN dependency and offline operations during WAN disruptions.
  - **Local Content Library:** Content libraries are vSphere container objects for storing and sharing OVF and OVA packages, VM templates, vApp templates, and other types of files.
  - **Container Registry:** If a Kubernetes pod crashes or a node reboots while the WAN link is down, the host (depending on imagePullPolicy) pulls an image from its configured registry. If the registry is central and can't be reached, the workload stays down (ImagePullBackOff). With a local Harbor registry, the Images can be stored at the edge. Hosts pull images over the high-speed local LAN rather than the restricted WAN link. Harbor can be consumed as a supervisor service or deployed as a VM appliance. Harbor replication policies can be used to control image distribution between instances, or used as a pull-through proxy for local caching. In both scenarios, an image is retrieved from the central registry once and served locally to all endpoints within the edge cluster. This architectural approach prevents bandwidth saturation, and significantly accelerates application startup times by using high-speed local LAN connectivity.
- **Pull based Configuration Updates:** In a traditional fleet deployment, vCenter "pushes" configurations to the edge. For Edge sites that only have intermittent connectivity, if the WAN link is down, the configuration update fails as Edge ESX hosts are not reachable. A GitOps-based approach uses a "Pull" model that is inherently resilient to the intermittent connectivity typical of edge sites. Configuration updates for workloads are staged in a central Git repository, and Edge sites retrieves the workload update only when there is connectivity.

Table 443: Edge Enablement Services

Design Decision (Disconnected Mode Support)	Design Details	Design Implication
Deploy a redundant DNS service at the Edge to support infrastructure and application requirements.	Deploy DNS as VMs with strict "Anti-Affinity" DRS rules. Anti-Affinity rules ensure that a single physical host failure cannot take down all DNS instances simultaneously. Configure Affinity rules so that if Edge Cluster fails, DNS service can be quickly located and brought back online. • The Edge application & node should point to a local DNS resolver first. This local resolver should contain a cached copy of all local infrastructure records (ESX hosts, vCenter, vSAN nodes, and local gateways). • The local resolver should be configured to forward queries it can't answer to the central data center DNS over the WAN. • DNS servers must be configured for both forward and reverse DNS records. Every Fully Qualified Domain Name (FQDN) used must resolve to a unique IP address.	DNS VMs consume CPU/Memory resources from the Edge Cluster. Increased management cost with: • **Syncing Records:** Must ensure that a change that is made to a domain record propagates to every edge server simultaneously. • **Security Patching:** Every edge DNS server is a new entry point that must be secured, patched, and monitored. Redundant DNS service for a single node isn't necessary.
Use the Edge Router or switch as the source of NTP and set the ESX service policy to *"Start and stop with host."*	NTP configuration can be set on the host using VMware Host/vSPhere Client, API, or vSphere Configuration Profile. Reference VMware documentation for time synchronization best practices specific to the VMware Infrastructure.	Edge routers must be configured to prevent "Time Spoofing" attacks.
Create a Local Content Library at each Edge Location to store virtual machine templates and serve as the "Image Repository" for the VMware Supervisor services.	Content libraries are local to the edge and on the local datastore. Local Content library can be used to enable and upgrade Supervisor or pre-stage VM OVAs to reduce dependency to WAN. Use VCF documentation for Content Library usage details	Content Library consumes local storage.
Deploy VCF Supervisor at the Edge to reduce WAN dependency	Supervisor at the edge site enables edge services to simplify edge workload deployment and management. • Deploy a Tiny or Small Supervisor with a single FLB load balancer to support [VM Service,VKS, Argo CD, Harbor]. VM Operator and VKS are core services that are activated as part of the supervisor deployment. Harbor is optional, needed if edge workloads are container-based. ArgoCD's GitOps principles are used for edge workloads (refer to the Desired State Management section for details). Refer Edge Cluster Activation for activation details.	A sufficient set of IP addresses must be set aside for all edge Supervisor Services: • Management VLAN: 7 IPs (Supervisor / FLB same network) Budget sufficient Core and Memory: • 2 cores and 12G for the supervisor/FLB. • 3 cores and 4G for ArgoCD. • 4 cores and 5G for Harbor • Content Library consumes local storage

Design Decision (Disconnected Mode Support)	Design Details	Design Implication
Deploying Harbor as a Supervisor Service or Standalone OVA	If Harbor is used to service container images for edge workloads, deploy Harbor as a supervisor service. On the other hand, deploy Harbor as a VM appliance if used to bootstrap the Supervisor services. Refer to the VCF documentation on "*Deploying Supervisor Services from a Private Container Image Registry*"	4 cores and 5G RAM for Harbor running as vSphere Pods.
Enable Harbor proxy cache to populate STIG-compliant container images at the Edge	Proxy cache simplifies image distribution. Instead of synchronization using policy, proxy cache allows local edge Harbor to cache only images that are required for the edge workload from an upstream registry. For Harbor proxy cache, configure pod manifests to pull images from the proxy cache project instead of the target registry. Example <harbor_servername>/ <proxy_project_name>/ as a prefix to the image tag	upstream registry must serve only images that are STIG hardened and scanned against the DISA General-Purpose Operating System SRG with reports available.
Extend Harbor Cache retention policy to tolerate extended WAN outage	Harbor proxy cache has a default 7-day retention policy for each new proxy cache project. Refer to the Harbor documentation to update Tag Retention Rules.	Excess storage consumption and out dated container images when the retention policy is set for long retention.
Deploy ArgoCD as a Supervisor Service	Argo CD agents run locally on the edge cluster. They continuously monitor a Git repository for changes. • If the connection to the central data center drops, the edge site continues to run in its "Last Known Good" state. • The moment connectivity is restored, the local Argo CD agent "pulls" any updates that were committed to Git while the site was offline and applies them immediately. Refer to the Argo CD Documentation for implementation details	3 cores and 4G RAM for ArgoCD vSphere Pods. The Git Repository that is required for ArgoCD sync is deployed outside of VCF.

With services running locally, even without vCenter connectivity, the following actions remain operational at the host level.

- ESX Host Resilience: A physical host can be rebooted. Once initialized, the hypervisor remains in its last known good configuration and resumes managing local resources.
- VM Workload Continuity: Existing Virtual Machines can be restarted and resumed. Because the VM configuration files (.vmx) reside on shared or local storage, the ESX host can power them on without vCenter intervention.
- VKS / Containerized Workloads: vSphere Kubernetes Service (VKS) workloads can be restarted provided vCenter assisted operation is not required. An example of such operation is Persistent Volume (PV) attachment to containers. This is contingent on the host having access to the container image repository to pull images if necessary.

The following limitation applies when operating in disconnected mode:

- Administrative and maintenance tasks that rely on the vCenter API must be deferred and scheduled once connectivity is restored.
- Any workload requiring the creation, attachment, or modification of Persistent Volumes.
- Any operation specifically requiring vCenter, Distributed Resource Scheduler (DRS) balancing, creating VMs, or modifying virtual networks.

Edge Compute Hardware and Network Design

Military edge compute hardware must be ruggedized to tolerate extreme temperatures and environmental fluctuations. Depending on the specific use case, systems may be required to adhere to MIL-STD-810H for environmental survivability and MIL-STD-461G for electromagnetic compatibility, ensuring signal stealth in contested environments. Beyond physical durability, the hardware must be optimized to support VMware ESX in alignment with the DISA STIG framework. This includes a hardened BIOS and hardware-assisted virtualization (Intel VT-x or AMD-V), TPM 2.0 modules to satisfy CCI-000366, which mandates hardware-rooted Secure Boot and host attestation. If shared storage is required, the hardware should ideally be vSAN ReadyNode certified. If vSAN ReadyNode is not available, at a minimum, edge hardware should be listed on the VMware Compatibility Guide (VCG) for VCF 9.

Depending on the edge requirements, edge hardware must be equipped with sufficient disk count and capacity to support ESX host imaging and persistent data storage. To accommodate use cases requiring strict network isolation, servers must also feature enough of physical NICs, each providing sufficient throughput to meet application performance requirements. Edge workloads using external storage such as NFSv3 and FC can also be used as primary storage. Refer to VCF9 documentation for external storage support.

Table 444: Edge Hardware Design Decisions

Design Decision	Design Details	Design Implication
Deploy MIL-STD-810H and MIL-STD-461G compliant hardware that is validated on the VMware Compatibility Guide (VCG/HCL)	Edge node with minimum 8 Cores with TPM module. Edge node with UEFI HTTPS network boot support in BIOS ECC memory	TPM module activation requires a secure boot and an SHA256 Hash in the BIOS configuration. TPM recovery key must be available locally before attempting to perform hardware maintenance vCenter certificate must be trusted by the edge node for HTTPS boot.
Deploy systems with minimum two physical uplink NICs	Two uplinks provide redundancy against link failure. 10G or higher NIC, but lower speed is acceptable for low throughput edge deployment	More physical resources required • Physical switchport • Cable and SFP
Standardize on Copper SFP+ based interfaces for use cases involving rugged terrains.	Copper is highly durable and less likely to be impacted by extreme environmental shifts such as dust and bend.	Copper generates more heat for fanless systems.
Single Node Deployment: Two local SSD disks at a minimum.	Local disks reduce external dependency and remove external storage as a failure domain. One ESX bootdisk, another disk for capacity. Both disks are formatted for VMFS. Uplink speed must be sized based on the workload.	Migrating to a redundant setup requires vSphere re-provisioning Minimal configuration does not provide HW redundancy. Any loss of HW (Disk, Memory, CPU) results in complete outage. Edge locations that do not have heavy storage read/write requirements but desire disk redundancy, Boot Optimized Storage Solutions such as the Dell BOSS or HPE NS204i often include hardware RAID. These boot optimized solutions can be used to extend storage to accommodate both the ESX boot partition and a VMFS datastore for workloads. Sufficient WAN bandwidth is required as UEFI boot downloads ESX and workload binaries over the WAN links.

Design Decision	Design Details	Design Implication
Two+ Node Compute with vSAN: minimum 3 disks per node	**OS/Boot Device:** 1 OS Boot disk is the minimum. Boot disk is where ESX is installed. **vSAN Data Disks: 2 Minimum:** • vSAN OSA Minimum: – One Cache Disk – One Capacity Disk • vSAN ESA Minimum: – NVME disk per pool Networks must be 10G or above. Only vSAN OSA can be activated with link speed below 10G.	Each host must have a minimum of 32G of memory vSAN OSA or 128G for vSAN ESA A minimal disk setup forces the least efficient storage policies. Workloads are restricted to basic RAID-1 (Mirroring), which requires 200% capacity overhead. There may not be enough hosts or disks to use space-efficient RAID-5 or RAID-6 Erasure Coding, which can provide the same protection for less raw capacity.
vSAN Nodes: Enable AES-NI (Advanced Encryption Standard New Instructions) in the BIOS to reduce CPU utilization and improve application performance	AES-NI is often disabled in the BIOS of some ruggedized motherboards to save power. You must check two places: The BIOS/UEFI: Look under Security or CPU Configuration for "Intel AES-NI" or "AES Instructions." must be set to enabled. From Within ESX: esxcli hardware cpu cpuid raw list -c 0 Ensure the "vSAN Health" status for "AES-NI Enabled" is Green	Encryption is CPU-intensive. **AES-NI** enables hardware acceleration for data encryption.

Edge Compute Node Design

The fundamental unit of resources for each edge site is a vSphere cluster. Since edge sites are geographically separated and follow a traditional hub-and-spoke topology, resources are localized to each site, making a single-site cluster design the most appropriate. Depending on the host count, number of uplinks, and storage profile, each cluster type follows a specific design pattern.

- **Single-Node Edge:** vSphere cluster settings even in a single node requires vSphere HA (no admission control) and DRS, as these features are required by VKS. Local VMFS storage volume using local disks is sufficient for a Single Node configuration. A single vDS with a minimum of two NICs is required for link level redundancy. NIC speed can be GE or higher.
- **Two-Node Edge:** A two-node edge cluster can use both vSAN or third-party external storage such as NFS or Fibre Channel. If a two-node vSAN design is used for storage, then a witness node is required to maintain quorum and store metadata. VMware Cloud Foundation (VCF) at the edge fully supports both dedicated witness and shared witness models. Witness appliance can be local or remote, in the cloud or at the Edge. Both vSAN OSA and ESA are supported. A single vDS with a minimum of two NICs is required as a starting point. Multiple vDS with redundant NICs is required to support use cases that require physical isolation between workloads.
- **Three+ Node Edge:** Similar to the two-node design, three-node edge vSphere cluster can use both vSAN or third-party external storage such as NFS or Fibre Channel. Cluster using vSAN as storage can use both OSA and ESA depending on the type of disks and network connectivity. A single vDS with a minimum of two NICs is required as a starting point. Multiple vDS with redundant NICs is required to support use cases that require physical isolation between workloads.

In all cases, data must be encrypted both in transit and at rest, adhering to FIPS 140 requirements. Traffic isolation is achieved through dedicated physical interfaces (where hardware permits) or VLAN tagging when constrained to a single interface.

Design Decision	Edge Node Count	Design Consideration	Design Implication
Use the Edge Router or switch as the source of NTP and set the service policy to *"Start and stop with host."*	All	NTP configuration can be set on the host using VMware Host/ vSPhere Client, API, or vSphere Configuration Profile.	The router must be configured to authenticate to official DoD-only servers using authentication that is cryptographically based to prevent "Time Spoofing" attacks.
Use vLCM image to manage the desired ESX version and OEM vendor add-on	All	Reference Managing the Lifecycle of Clusters withvSphere Lifecycle ManagerImages at Cluster Level for associating an Image with a vSphere Cluster. The vSphere Lifecycle Manager Depot to store ESX base images, vendor add-ons, third-party components, and legacy patches and updates vSphere Lifecycle Manager Remediation Settings to configure the behavior of vSphere Lifecycle Manager during remediation. Lifecycle Management of Clusters and Standalone Hosts at Scale	If a cluster contains different models of server, a composite image is required to manage each generation. vLCM handles the remediation of the vSAN hosts sequentially
Single vSphere Cluster for each Edge Site	All	Deploy each edge as symmetrical clusters, of which the hosts contribute equal amounts of resources (CPU, Memory, Networking, and Disk) across the cluster	Asymmetrical configurations impose design are harder to operate and may expose bottlenecks that are only exposed during failure conditions
	All	Single Rack Cluster design is applicable for use cases that have a single availability zone.	The single Rack Cluster design protects against compute failure, but does not protect against failure of the chassis/rack.
	Single Node	Enable vSphere cluster: • DRS: Fully Automated • vSphere HA: Admission control disabled	Settings are required to deploy Supervisor and VKS. No impact to traditional VM workloads
	Three Nodes+	Layer 2 Multi-Rack cluster Model is applicable for use cases that have multiple fault domains within a single availability zone.	L2 VLAN Network must be stretched across all computes in the cluster. Single subnet for each network Bandwidth and Latency Requirements between different rack of a Navy vessel or moving vehicles must be met

Design Decision	Edge Node Count	Design Consideration	Design Implication
VDS Design	All	vDS Design must be based on the number of uplinks that are required to satisfy security compliance requirements for isolation. Refer to the VCF detailed distributed switch design guide for design recommendations • Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model • Dedicated vSphere Cluster for NSX Edges Distributed Switch Model • vSAN Storage Cluster Separation Distributed Switch Model	When consolidating multiple types of traffic on a single vDS, classifying and tagging traffic is essential for maintaining both cluster stability and deterministic performance. Refer to the table on "*VDS VLAN Tagging and QoS Marking*" for best practice recommendations. The number of uplinks impacts which model is applicable. A host with 2 uplinks can only support a single VDS with redundancy. vSAN host with four uplinks can dedicate two uplinks for either Storage traffic or VM traffic. Consider a dedicated VDS for vSAN if: • vSAN ESA/OSA is used, but there is no dedicated 10 GE + Switchports on the TOR • VM with low network throughput, but high write IO Consider a dedicated VDS for workload if: • High network bandwidth application, low storage IO
vSAN Storage Design	Singe Node	NA. Local VMFS storage volume using local disks is sufficient	NA
	Two Nodes	Use direct NIC connections between nodes. Connect the vSAN ports of Node 1 directly to Node 2 using a ruggedized crossover cable Removing the network switch also reduces the edge compute kits weight and power draw	Communication with the remote witness appliance must be kept physically separate from the high-bandwidth "Direct Connect" data traffic.
		Use vSAN Witness Traffic Separation(WTS) to separate traffic for the witness host appliance from the traffic for data Refer to the Two Node vSAN Deployments documentation for implementation details.	Configuration applies to the two Data Nodes in a 2-Node cluster, not the witness appliance.
		Configure a different MTU for the intersite vSAN traffic from the vSAN Witness Separation Traffic.	Improves vSAN performance for host-to-host connections that support larger MTU sizes that are possible on the WAN link to the witness site

Design Decision	Edge Node Count	Design Consideration	Design Implication
	Three Nodes	Refer to the VCF documentation on single site vSAN Design • Single-Rack Cluster Storage Models • Multi-Rack Cluster Storage Models vSAN ESA requires a network throughput of 10GbE and higher.	When implementing vSAN ESA on 10GE networks, use **esxcfg-advcfg -s 1 /VSAN/DOMNetworkSchedulerThrottleCompon** To control vSAN resynchronization traffic.
	All	Define vSAN storage policies that are based on cluster hardware capabilities to guarantee availability and performance. Refer to the vSAN storage Profile documentation for implementation details	Workloads cannot deploy to the vSAN cluster unless the underlying hardware can support the availability and performance requirements of the virtual machine.
Create a dedicated VM Kernel NIC for each type of management traffic	All	STIG compliance design mandates strict network isolation for different types of management traffic. Create • Management VMK • vSAN VMK • vMotion VMK	Increased provisioning time for each edge location. Extra routable IP addresses required for separate VMK interfaces.
	Two-Node vSAN	Create logically separate VMK interfaces for vSAN Witness Separation Traffic (WTS) • vSAN Data Port: Tagged for "vSAN" traffic only. (Direct Connect path). • vSAN Witness Port: Tagged for "Witness" traffic. (WAN path).	WTS requires ~2 Mbps depending on vSAN I/O. It is better to use a stable "rugged" port with less link speed than a link with higher link speed but sensitive to the dust and vibration. Select a link with the highest availability over the highest bandwidth.
	Three-Node with NSX	VMK interface for NSX overlay. Each ESX host typically requires two TEP IP addresses for redundancy and load balancing	Ensure the VCF Network Pool for NSX TEPs has enough IPs to cover (Number of Hosts) x 2
Enable vSAN Data-in-Transit Encryption at the cluster level.	All	vSAN Data-in-Transit encryption is a STIG requirement. Any sensitive metadata leaving the physical enclosure must be encrypted. This ensures the handshake between the data nodes and the remote witness is AES-256 encrypted.	CPU utilization increases. The CPU encrypts every packet before it hits the wire and decrypts it upon arrival.
All vSAN data at rest must be encrypted using AES-256	All	Data at rest encryption using AES-256 is a STIG requirement. Reference the vSAN Encryption Design guide for design considerations.	The TPM 2.0 module on each host should be used to store encryption keys locally. Backing up the NKP: vSphere prevents enabling encryption until the primary key is exported and backed up.

Design Decision	Edge Node Count	Design Consideration	Design Implication
External Storage	All	Refer to the VCF Storage Design documentation on • NFS Storage Model • Fibre Channel Storage Model	Install storage hardware acceleration in vSphere (VAAI) for external storage deployments. VAAI capability must be verified with the storage vendor

When sharing a common uplink for all traffic, classifying and tagging traffic is essential for maintaining both cluster stability and deterministic performance. By assigning vSAN traffic to a dedicated VLAN with High NIOC priority, you protect the storage backplane from latency spikes that could cause host desynchronization. vMotion and Management traffic are isolated on their own VLANs to prevent bulk data transfers from saturating the link, ensuring that administrative functions to the hosts remain accessible. Finally, categorizing VM traffic with a lower relative priority ensures that a surge in application demand or a guest-level network issue cannot starve critical infrastructure services of bandwidth. The following table outlines the recommended VLAN and QOS marking based on traffic type:

Table 445: VDS VLAN Tagging and QoS Marking

Traffic Type	VLAN Recommendation	Priority (NIOC)
vSAN	Dedicated	High
vMotion	Dedicated	Standard
MGMT	Dedicated	Standard
VM Traffic	App Specific	Low/Standard

Edge Access Security Design

Security for virtual infrastructure is a defense-in-depth model where each layer must be individually hardened to prevent lateral movement and unauthorized access. This layered approach starts at the hypervisor and moves up through the management and network control planes.

While the vSphere stack is secure by default, mission-critical environments, especially at the Edge require more configurations to meet DISA STIG standards.

Edge environments lack the protection of a traditional data center. This section focuses on hardening security at the ESXi layer and management components that resides at the edge. Ensuring that even when unauthorized physical access is gained, the integrity of the host and its data remains intact is critical. Key security decisions include:

- **Secure Boot & TPM 2.0**: In the DISA STIG for vSphere, Secure Boot and TPM 2.0 form the foundation of the Hardware Root of Trust. Secure Boot ensures that only digital signatures that it trusts are loaded. If an attacker attempts to replace the ESX bootloader with a malicious version, Secure Boot detects the absence of a valid VMware signature and refuses to start the server. The TPM 2.0 chip enables Remote Attestation, allowing vCenter to cryptographically verify that the hypervisor has not been compromised by unauthorized kernel-level modifications. Together, these technologies ensure that the boot process and the operating state of the host remain untampered and compliant with the federal security mandates.
- **VIB Acceptance Levels:** For DISA STIGs, VIB Acceptance Levels are critical because they establish the "Chain of Trust" for every piece of code running at the kernel level. If an attacker can inject a malicious VIB, they gain total control over the hypervisor and all hosted Virtual Machines. To prevent the execution of unauthorized code set to a minimum of PartnerSupported. In high-security or tactical environments, acceptance level is often restricted further to VMwareCertified.
- **Encryption in Transit:** vSphere now runs in FIPS-compliant mode by default. Both TLS1.2 and TLS1.3 are available. TLS1.3 protocol is on by default and falls back to TLS1.2 as well to support legacy product integration.
- **Access control and Account Security:** Lockdown restricts access to ESX host. At the Edge, where physical security may be lower than in a central data center. Limiting ESX host access with lockdown mode is vital. Normal Lockdown Mode allows specific users to access the DCUI, while Strict Lockdown Mode disables the DCUI entirely, forcing all management through vCenter.. Implementing Active Directory for centralized identity management. Enable ESX Shell and SSH services only when required for troubleshooting

- **Traffic & Network Isolation:** infrastructure stability and security depend on the strict logical and physical separation of critical system traffic. Administrators must classify Management, vSAN, and vMotion traffic into dedicated VLANs to prevent unauthorized interception and "noisy neighbor" contention. To mitigate the risk of MAC spoofing and packet sniffing, vSwitch security policies must be set to Reject for Promiscuous Mode, MAC Address Changes, and Forged Transmits. The DISA STIG for VMware ESX mandates a "Least Privilege" approach to network security. For every enabled service (like SSH, vMotion, or Web Access), you must disable the "Allow connections from any IP address" setting. Limiting firewall ports on ESX hosts to a specific set of hosts prevents attack surface

Table 446: Security Design for ESX at Edge

Design Decision	Design Details	Design Implication
Enable secure boot and TPM to secure ESX configuration	The ESX STIG mandates Secure Boot enforcement because it moves security from a software-only check that can be configured to bypass to a hardware-enforced check. For tactical kits it ensures if a kit is captured or tampered with in transit, the data remains cryptographically protected.	"Community Supported" or unsigned drivers can no longer be loaded. Without explicit validation. Secure Boot block the host from booting with unauthorized VIB. Use /usr/lib/vmware/secureboot/bin/secureBoot.py -c command to validate current drivers before enabling the feature Host attestation failure if TPM report does not match vCenter known good values.
Use TLS Profiles to administer TLS parameters on ESX and vCenter.	Both TLS 1.3 and 1.2 are supported. Enable NIST_2024_TLS_13_ONLY if TLS 1.2 is not required. A more restrictive TLS profile that meets NIST 2024 standards by allowing only AES-GCM ciphers (TLS_AES_256_GCM_SHA384 and TLS_AES_128_GCM_SHA256) Refer to the vSphere TLS Configuration guide for detailed steps	Non-VMware and partner vSphere Client Remote plug-ins may need upgrade to use conformant encryption libraries. Refer to Developing Remote Plug-ins with the vSphere Client SDK
Enable ESX in normal lockdown mode	Lockdown mode is a critical configuration for tactical kits or remote edge sites where vCenter might become unreachable due to a satellite link failure. Normal lockdown mode allows a user with administrator role physical access to log in locally and troubleshoot the host. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	If the ESX Shell or SSH is activated and the host is in lockdown mode, accounts on the Exception Users list who have administrator privileges can use these services
Provide an allowlist of users for break glass ESX access.	The following rules should be followed for the allowlist: • Use Named Accounts: Never use the root account as your primary exception user if possible. Create a dedicated "Emergency Admin" local account. This ensures that the audit log shows who used the break-glass access. • Limit the List: The STIG requires that the Exception Users list be "verified and legitimate." Adding unnecessary users defeats the purpose of lockdown. Usually, this list contains only one emergency admin and perhaps a backup service account. • Permissions: A user added to the exception list must have the Administrator role assigned. A user with a non administrative role won't have enough permission to do anything once they log in. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	The Exception Users list is meant for service accounts that perform specific tasks, and not for administrators.
Enable ESX Entropy and set entropySource to Zero STIGS compliance	entropySources = 0. allows an ESX host to use a combination of the local CPU (RDRAND), interrupts, and any available hardware. If RDSEED is supported, the default is FIPS compliance. Otherwise	Must set entrySource to external (8) for Strict EAL4. requires a reliable external HSM.

Design Decision	Design Details	Design Implication
Enable secure boot and TPM to secure ESX configuration	The ESX STIG mandates Secure Boot enforcement because it moves security from a software-only check that can be configured to bypass to a hardware-enforced check. For tactical kits it ensures if a kit is captured or tampered with in transit, the data remains cryptographically protected.	"Community Supported" or unsigned drivers can no longer be loaded. Without explicit validation. Secure Boot block the host from booting with unauthorized VIB. Use /usr/lib/vmware/secureboot/bin/secureBoot.py -c command to validate current drivers before enabling the feature Host attestation failure if TPM report does not match vCenter known good values.
Use TLS Profiles to administer TLS parameters on ESX and vCenter.	Both TLS 1.3 and 1.2 are supported. Enable NIST_2024_TLS_13_ONLY if TLS 1.2 is not required. A more restrictive TLS profile that meets NIST 2024 standards by allowing only AES-GCM ciphers (TLS_AES_256_GCM_SHA384 and TLS_AES_128_GCM_SHA256) Refer to the vSphere TLS Configuration guide for detailed steps	Non-VMware and partner vSphere Client Remote plug-ins may need upgrade to use conformant encryption libraries. Refer to Developing Remote Plug-ins with the vSphere Client SDK
Enable ESX in normal lockdown mode	Lockdown mode is a critical configuration for tactical kits or remote edge sites where vCenter might become unreachable due to a satellite link failure. Normal lockdown mode allows a user with administrator role physical access to log in locally and troubleshoot the host. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	If the ESX Shell or SSH is activated and the host is in lockdown mode, accounts on the Exception Users list who have administrator privileges can use these services
Provide an allowlist of users for break glass ESX access.	The following rules should be followed for the allowlist: • Use Named Accounts: Never use the root account as your primary exception user if possible. Create a dedicated "Emergency Admin" local account. This ensures that the audit log shows who used the break-glass access. • Limit the List: The STIG requires that the Exception Users list be "verified and legitimate." Adding unnecessary users defeats the purpose of lockdown. Usually, this list contains only one emergency admin and perhaps a backup service account. • Permissions: A user added to the exception list must have the Administrator role assigned. A user with a non administrative role won't have enough permission to do anything once they log in. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	The Exception Users list is meant for service accounts that perform specific tasks, and not for administrators.
Enable ESX Entropy and set entrySource to Zero STIGS compliance	entropySources = 0. allows an ESX host to use a combination of the local CPU (RDRAND), interrupts, and any available hardware. If RDSEED is supported, the default is FIPS compliance. Otherwise the default is all entropy sources except entropyd.	Must set entrySource to external (8) for Strict EAL4. requires a reliable external HSM.

Design Decision	Design Details	Design Implication
Enable secure boot and TPM to secure ESX configuration	The ESX STIG mandates Secure Boot enforcement because it moves security from a software-only check that can be configured to bypass to a hardware-enforced check. For tactical kits it ensures if a kit is captured or tampered with in transit, the data remains cryptographically protected.	"Community Supported" or unsigned drivers can no longer be loaded. Without explicit validation. Secure Boot block the host from booting with unauthorized VIB. Use /usr/lib/vmware/secureboot/bin/ secureBoot.py -c command to validate current drivers before enabling the feature Host attestation failure if TPM report does not match vCenter known good values.
Use TLS Profiles to administer TLS parameters on ESX and vCenter.	Both TLS 1.3 and 1.2 are supported. Enable NIST_2024_TLS_13_ONLY if TLS 1.2 is not required. A more restrictive TLS profile that meets NIST 2024 standards by allowing only AES-GCM ciphers (TLS_AES_256_GCM_SHA384 and TLS_AES_128_GCM_SHA256) Refer to the vSphere TLS Configuration guide for detailed steps	Non-VMware and partner vSphere Client Remote plug-ins may need upgrade to use conformant encryption libraries. Refer to Developing Remote Plug-ins with the vSphere Client SDK
Enable ESX in normal lockdown mode	Lockdown mode is a critical configuration for tactical kits or remote edge sites where vCenter might become unreachable due to a satellite link failure. Normal lockdown mode allows a user with administrator role physical access to log in locally and troubleshoot the host. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	If the ESX Shell or SSH is activated and the host is in lockdown mode, accounts on the Exception Users list who have administrator privileges can use these services
Provide an allowlist of users for break glass ESX access.	The following rules should be followed for the allowlist: • Use Named Accounts: Never use the root account as your primary exception user if possible. Create a dedicated "Emergency Admin" local account. This ensures that the audit log shows who used the break-glass access. • Limit the List: The STIG requires that the Exception Users list be "verified and legitimate." Adding unnecessary users defeats the purpose of lockdown. Usually, this list contains only one emergency admin and perhaps a backup service account. • Permissions: A user added to the exception list must have the Administrator role assigned. A user with a non administrative role won't have enough permission to do anything once they log in. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	The Exception Users list is meant for service accounts that perform specific tasks, and not for administrators.
Enable ESX Entropy and set entrySource to Zero STIGS compliance	entropySources = 0. allows an ESX host to use a combination of the local CPU (RDRAND), interrupts, and any available hardware. If RDSEED is supported, the default is FIPS compliance. Otherwise the default is all entropy sources except entropyd.	Must set entrySource to external (8) for Strict EAL4. requires a reliable external HSM.

Design Decision	Design Details	Design Implication
Enable secure boot and TPM to secure ESX configuration	The ESX STIG mandates Secure Boot enforcement because it moves security from a software-only check that can be configured to bypass to a hardware-enforced check. For tactical kits it ensures if a kit is captured or tampered with in transit, the data remains cryptographically protected.	"Community Supported" or unsigned drivers can no longer be loaded. Without explicit validation. Secure Boot block the host from booting with unauthorized VIB. Use /usr/lib/vmware/secureboot/bin/ secureBoot.py -c command to validate current drivers before enabling the feature Host attestation failure if TPM report does not match vCenter known good values.
Use TLS Profiles to administer TLS parameters on ESX and vCenter.	Both TLS 1.3 and 1.2 are supported. Enable NIST_2024_TLS_13_ONLY if TLS 1.2 is not required. A more restrictive TLS profile that meets NIST 2024 standards by allowing only AES-GCM ciphers (TLS_AES_256_GCM_SHA384 and TLS_AES_128_GCM_SHA256) Refer to the vSphere TLS Configuration guide for detailed steps	Non-VMware and partner vSphere Client Remote plug-ins may need upgrade to use conformant encryption libraries. Refer to Developing Remote Plug-ins with the vSphere Client SDK
Enable ESX in normal lockdown mode	Lockdown mode is a critical configuration for tactical kits or remote edge sites where vCenter might become unreachable due to a satellite link failure. Normal lockdown mode allows a user with administrator role physical access to log in locally and troubleshoot the host. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	If the ESX Shell or SSH is activated and the host is in lockdown mode, accounts on the Exception Users list who have administrator privileges can use these services
Provide an allowlist of users for break glass ESX access.	The following rules should be followed for the allowlist: • Use Named Accounts: Never use the root account as your primary exception user if possible. Create a dedicated "Emergency Admin" local account. This ensures that the audit log shows who used the break-glass access. • Limit the List: The STIG requires that the Exception Users list be "verified and legitimate." Adding unnecessary users defeats the purpose of lockdown. Usually, this list contains only one emergency admin and perhaps a backup service account. • Permissions: A user added to the exception list must have the Administrator role assigned. A user with a non administrative role won't have enough permission to do anything once they log in. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	The Exception Users list is meant for service accounts that perform specific tasks, and not for administrators.
Enable ESX Entropy and set entrySource to Zero STIGS compliance	entropySources = 0. allows an ESX host to use a combination of the local CPU (RDRAND), interrupts, and any available hardware. If RDSEED is supported, the default is FIPS compliance. Otherwise the default is all entropy sources except entropyd.	Must set entrySource to external (8) for Strict EAL4. requires a reliable external HSM.

Design Decision	Design Details	Design Implication
Enable secure boot and TPM to secure ESX configuration	The ESX STIG mandates Secure Boot enforcement because it moves security from a software-only check that can be configured to bypass to a hardware-enforced check. For tactical kits it ensures if a kit is captured or tampered with in transit, the data remains cryptographically protected.	"Community Supported" or unsigned drivers can no longer be loaded. Without explicit validation. Secure Boot block the host from booting with unauthorized VIB. Use /usr/lib/vmware/secureboot/bin/ secureBoot.py -c command to validate current drivers before enabling the feature Host attestation failure if TPM report does not match vCenter known good values.
Use TLS Profiles to administer TLS parameters on ESX and vCenter.	Both TLS 1.3 and 1.2 are supported. Enable NIST_2024_TLS_13_ONLY if TLS 1.2 is not required. A more restrictive TLS profile that meets NIST 2024 standards by allowing only AES-GCM ciphers (TLS_AES_256_GCM_SHA384 and TLS_AES_128_GCM_SHA256) Refer to the vSphere TLS Configuration guide for detailed steps	Non-VMware and partner vSphere Client Remote plug-ins may need upgrade to use conformant encryption libraries. Refer to Developing Remote Plug-ins with the vSphere Client SDK
Enable ESX in normal lockdown mode	Lockdown mode is a critical configuration for tactical kits or remote edge sites where vCenter might become unreachable due to a satellite link failure. Normal lockdown mode allows a user with administrator role physical access to log in locally and troubleshoot the host. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	If the ESX Shell or SSH is activated and the host is in lockdown mode, accounts on the Exception Users list who have administrator privileges can use these services
Provide an allowlist of users for break glass ESX access.	The following rules should be followed for the allowlist: • Use Named Accounts: Never use the root account as your primary exception user if possible. Create a dedicated "Emergency Admin" local account. This ensures that the audit log shows who used the break-glass access. • Limit the List: The STIG requires that the Exception Users list be "verified and legitimate." Adding unnecessary users defeats the purpose of lockdown. Usually, this list contains only one emergency admin and perhaps a backup service account. • Permissions: A user added to the exception list must have the Administrator role assigned. A user with a non administrative role won't have enough permission to do anything once they log in. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	The Exception Users list is meant for service accounts that perform specific tasks, and not for administrators.
Enable ESX Entropy and set entrySource to Zero STIGS compliance	entropySources = 0. allows an ESX host to use a combination of the local CPU (RDRAND), interrupts, and any available hardware. If RDSEED is supported, the default is FIPS compliance. Otherwise the default is all entropy sources except entropyd.	Must set entrySource to external (8) for Strict EAL4. requires a reliable external HSM.

Design Decision	Design Details	Design Implication
Enable secure boot and TPM to secure ESX configuration	The ESX STIG mandates Secure Boot enforcement because it moves security from a software-only check that can be configured to bypass to a hardware-enforced check. For tactical kits it ensures if a kit is captured or tampered with in transit, the data remains cryptographically protected.	"Community Supported" or unsigned drivers can no longer be loaded. Without explicit validation. Secure Boot block the host from booting with unauthorized VIB. Use /usr/lib/vmware/secureboot/bin/secureBoot.py -c command to validate current drivers before enabling the feature Host attestation failure if TPM report does not match vCenter known good values.
Use TLS Profiles to administer TLS parameters on ESX and vCenter.	Both TLS 1.3 and 1.2 are supported. Enable NIST_2024_TLS_13_ONLY if TLS 1.2 is not required. A more restrictive TLS profile that meets NIST 2024 standards by allowing only AES-GCM ciphers (TLS_AES_256_GCM_SHA384 and TLS_AES_128_GCM_SHA256) Refer to the vSphere TLS Configuration guide for detailed steps	Non-VMware and partner vSphere Client Remote plug-ins may need upgrade to use conformant encryption libraries. Refer to Developing Remote Plug-ins with the vSphere Client SDK
Enable ESX in normal lockdown mode	Lockdown mode is a critical configuration for tactical kits or remote edge sites where vCenter might become unreachable due to a satellite link failure. Normal lockdown mode allows a user with administrator role physical access to log in locally and troubleshoot the host. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	If the ESX Shell or SSH is activated and the host is in lockdown mode, accounts on the Exception Users list who have administrator privileges can use these services
Provide an allowlist of users for break glass ESX access.	The following rules should be followed for the allowlist: • Use Named Accounts: Never use the root account as your primary exception user if possible. Create a dedicated "Emergency Admin" local account. This ensures that the audit log shows who used the break-glass access. • Limit the List: The STIG requires that the Exception Users list be "verified and legitimate." Adding unnecessary users defeats the purpose of lockdown. Usually, this list contains only one emergency admin and perhaps a backup service account. • Permissions: A user added to the exception list must have the Administrator role assigned. A user with a non administrative role won't have enough permission to do anything once they log in. Refer to the vSphere 9.0 documentation to Configuring and Managing Lockdown Mode on ESX Hosts	The Exception Users list is meant for service accounts that perform specific tasks, and not for administrators.
Enable ESX Entropy and set entrySource to Zero STIGS compliance	entropySources = 0. allows an ESX host to use a combination of the local CPU (RDRAND), interrupts, and any available hardware. If RDSEED is supported, the default is FIPS compliance. Otherwise the default is all entropy sources except entropyd.	Must set entrySource to external (8) for Strict EAL4. requires a reliable external HSM.

Edge Cluster Activation
Edge activation starts at the edge location. With the introduction of Zero Touch Provisioning (ZTP) in VCF 9, new edge hosts can be imaged, provisioned, and brought into service using VCFE PowerShell automation without manual intervention.

The transition of a new host from power-on to an active member of the remote cluster follows a highly orchestrated automated workflow. The process initiates at the edge when a device powers on and requests an IP address with DHCP. The DHCP server provides an IP along with a global vCenter boot URL. The edge ESX server uses this boot URL to securely retrieve its boot configuration. vCSA uses values from SMBIOS data from the Edge server to decide which cluster and image to assign. If DHCP is not used, this boot URL can be statically programmed in the BIOS. The activation process varies between deployment scenarios.

Scenario 1: vCenter only and basic L2 connectivity at edge location.

- Design 1: Central vCenter with remote edge vSphere cluster, no NSX
- Design 2: Edge vSphere cluster with vCenter local to the cluster, no NSX

Scenario 2: vCenter and NSX at the edge with multi-layer routing/switching.

- Design 1: Central vCenter and NSX manager, remote edge cluster with 3+ hosts
- Design 2: Remote edge cluster with 3+ hosts, vCenter, and NSX local to the cluster

Activation Scenario 1:

Activation for scenario 1 involves an vSphere edge cluster of 1 - 2 nodes.

Table 447: Design #1: Central vCenter with remote vSphere edge cluster, no NSX

Design Decision	Design Details	Design Implication
Use ZTP to Boot and Install ESX Image	Ship hosts to Edge site Boot and install ESX image on host using ZTP Enable vSwitch HealthCheck for VLAN, Teaming, and failover and MTU (disable after all checks pass)	vCenter certificate must be trusted by the Edge Host. vSwitch HealthCheck consumes network bandwidth and CPU cycles. Check should be disabled after initial configuration. Sufficient WAN bandwidth to download the ESX image over the link.
Use the Edge Automation script to complete the infrastructure setup for edge locations requiring basic L2 connectivity and standard vSphere security at edge locations	Automation Script configures: • Edge vSphere cluster (DRS/HA, VDS, Storage) • Enable vSphere Desired State • Deploy Supervisor and Foundation Load Balancer • Deploy Edge services (ArgoCD and Harbor)	Infrastructure parameters supplied to the automation script must align with DISA STIG compliance requirements specified earlier sections. Edge hosts and vCenter must remain connected during the duration of the provisioning phase.

Design Decision	Design Details	Design Implication
Activate vSphere Configuration Profile	• **Select a Reference Host:** single ESX host with desired configuration in networking, Storage, NTP, syslog, security settings, etc. • **Extract the Schema:** vSphere extracts these settings and generates a JSON configuration document representing the cluster's desired state. • **Handle Overrides:** Remaining hosts need specific static IP addresses for vSAN & vMotion, these are handled in the "host-specific" section of the JSON document. • **Apply and Enforce:** The profile is applied to the cluster. vSphere handles the orchestration, ensuring all hosts conform to the blueprint.	Desired State Enforcement Remediation is manually triggered
Configure Argo CD for Pull-based Orchestration	• **Connect Git Repository (The Source of Truth)** - If the repository is private, authentication credentials, such as an SSH private key or an HTTPS personal access token must be provided to Argo CD • **Create an Argo CD Application** - **Source:** The Git repository URL, the specific branch or revision to track, and the directory path containing Kubernetes manifests (Helm charts, Kustomize files, or raw YAML). **Destination:** The target Supervisor Namespace or VKS cluster and the specific namespace where the resources should be deployed. • Enable Automated Syncing: Once enabled, Argo CD continuously pull Git repository (by default, every 3 minutes). On a new Git commit, Argo CD automatically pulls the new manifests and reconciles the cluster's live state to match your desired state in Git	Git Repository structure is prepared outside of the Edge Activation.

Table 448: Design #2: Edge vSphere Cluster with vCenter local to the Cluster, no NSX

Design Decision	Design Details	Design Implication
Use ZTP to Boot and Install ESX Image	Ship hosts to Edge site Boot and install ESX image on host using ZTP Enable vSwitch HealthCheck for VLAN, Teaming, and failover and MTU (disable after all checks pass)	vCenter certificate must be trusted by the Edge Host. vSwitch HealthCheck consumes network bandwidth and CPU cycles. Check should be disabled after initial configuration. Sufficient WAN bandwidth to download the ESX image over the link.
Deploy vCenter by loading VCSA Appliance on one of the edge ESX hosts	An unattended deployment can be performed from the CLI or traditional method using the UI	Environment specific JSON file must be prepared before CLI-based deployment.

Design Decision	Design Details	Design Implication
Use the Edge Automation script to complete the infrastructure setup pointing to the New VCSA appliance	Automation Script configures: • Edge vSphere cluster (DRS/HA, VDS, Storage) • Deploy Supervisor and Foundation Load Balancer • Deploy Edge services (ArgoCD and Harbor)	Infrastructure parameters supplied to the automation script must align with DISA STIG compliance requirements specified earlier sections. Edge hosts and vCenter must remain connected during the duration of the provisioning phase.
Activate vSphere Configuration Profile	• **Select a Reference Host:** single ESX host with desired configuration in networking, Storage, NTP, syslog, security settings, etc. • **Extract the Schema:** vSphere extracts these settings and generates a JSON configuration document representing the cluster's desired state. • **Handle Overrides:** Remaining hosts need specific static IP addresses for vSAN & vMotion, these are handled in the "host-specific" section of the JSON document. • **Apply and Enforce:** The profile is applied to the cluster. vSphere handles the orchestration, ensuring all hosts conform to the blueprint.	Desired State Enforcement Remediation is manually triggered
Configure Argo CD for Pull based Orchestration	• **Connect Git Repository (The Source of Truth)** - If the repository is private, authentication credentials, such as an SSH private key or an HTTPS personal access token must be provided to Argo CD • **Create an Argo CD Application** - **Source:** The Git repository URL, the specific branch or revision to track, and the directory path containing Kubernetes manifests (Helm charts, Kustomize files, or raw YAML). **Destination:** The target Supervisor Namespace or VKS cluster and the specific namespace where the resources should be deployed. • Enable Automated Syncing: Once enabled, Argo CD continuously pull Git repository (by default, every 3 minutes). On a new Git commit, Argo CD automatically pulls the new manifests and reconciles the cluster's live state to match your desired state in Git	Git Repository structure is prepared outside of the Edge Activation.

Activation Scenario 2:

Activation Scenario 2 covers vSphere edge locations with more than three hosts per edge location.

Table 449: Design 1 - Central vCenter and NSX Manager and remote vSphere edge cluster with 3+ hosts

Design Decision	Design Details	Design Implication
Use ZTP to Boot and Install ESX Image	Ship hosts to Edge site Boot and install ESX image on host using ZTP Enable vSwitch HealthCheck for VLAN, Teaming, and failover and MTU (disable after all checks pass)	vCenter certificate must be trusted by the Edge Host. vSwitch HealthCheck consumes network bandwidth and CPU cycles. Check should be disabled after initial configuration. Sufficient WAN bandwidth to download the ESX image over the link.
Use VCF Ops to create a WLD	A new WLD domain consists of • one vCenter • NSX manager (simple mode or with HA)	Depending on the WLD topology. Minimum number of required IP with FQDN: • 1 IP for vCenter with existing NSX Manager • 3 IP address for one vCenter and one NSX Simple Mode • 5 IP adddress for one vCenter and NSX manager HA 1 VLAN and subnet for Distributed VPC connectivity
Use the Edge Automation script to complete the infrastructure setup pointing to the New VCSA appliance	Automation Script configures: • Deploy Edge services (ArgoCD and Harbor)	Edge hosts and vCenter must remain connected during the duration of the provisioning phase.
Activate vSphere Configuration Profile	• **Select a Reference Host:** single ESXi host with desired configuration in networking, Storage, NTP, syslog, security settings, etc. • **Extract the Schema:** vSphere extracts these settings and generates a JSON configuration document representing the cluster's desired state. • **Handle Overrides:** Remaining hosts need specific static IP addresses for vSAN & vMotion, these are handled in the "host-specific" section of the JSON document. • **Apply and Enforce:** The profile is applied to the cluster. vSphere handles the orchestration, ensuring all hosts conform to the blueprint.	Desired State Enforcement Remediation is manually triggered

Design Decision	Design Details	Design Implication
Configure Argo CD for Pull based Orchestration	• **Connect Git Repository (The Source of Truth)** - If the repository is private, authentication credentials, such as an SSH private key or an HTTPS personal access token must be provided to Argo CD • **Create an Argo CD Application** - **Source:** The Git repository URL, the specific branch or revision to track, and the directory path containing Kubernetes manifests (Helm charts, Kustomize files, or raw YAML). **Destination:** The target Supervisor Namespace or VKS cluster and the specific namespace where the resources should be deployed. • Enable Automated Syncing: Once enabled, Argo CD continuously pull Git repository (by default, every 3 minutes). On a new Git commit, Argo CD automatically pulls the new manifests and reconciles the cluster's live state to match your desired state in Git	Git Repository structure is prepared outside of the Edge Activation.

Table 450: Design 2: Edge vSphere Cluster with vCenter and NSX Manager local to the Cluster

Design Decision	Design Details	Design Implication
Use ZTP to Boot and Install ESX Image	Ship hosts to Edge site Boot and install ESX image on host using ZTP Enable vSwitch HealthCheck for VLAN, Teaming, and failover and MTU (disable after all checks pass)	vCenter certificate must be trusted by the Edge Host. vSwitch HealthCheck consumes network bandwidth and CPU cycles. Check should be disabled after initial configuration. Sufficient WAN bandwidth to download the ESX image over the link.
Deploy vCenter by loading VCSA Appliance on one of the edge ESX hosts	An unattended deployment can be performed from the CLI or traditional method using the UI	Environment specific JSON file must be prepared before CLI-based deployment.
Use the Edge Automation script to complete the infrastructure setup pointing to the New VCSA appliance	Automation Script configures: • Edge vSphere cluster (DRS/HA, VDS, Storage) • Enable vSphere Desired State • Deploy Supervisor and Foundation Load Balancer • Deploy Edge services (ArgoCD and Harbor)	Edge hosts and vCenter must remain connected during the duration of the provisioning phase.
Use the Brownfield import workflow to deploy an NSX manager co-located at the edge location.	Reference the VCF documentation for supported brownfield import scenarios	

Design Decision	Design Details	Design Implication
Activate vSphere Configuration Profile	• **Select a Reference Host:** single ESX host with desired configuration in networking, Storage, NTP, syslog, security settings, etc. • **Extract the Schema:** vSphere extracts these settings and generates a JSON configuration document representing the cluster's desired state. • **Handle Overrides:** Remaining hosts need specific static IP addresses for vSAN & vMotion, these are handled in the "host-specific" section of the JSON document. • **Apply and Enforce:** The profile is applied to the cluster. vSphere handles the orchestration, ensuring all hosts conform to the blueprint.	Desired State Enforcement Remediation is manually triggered
Configure Argo CD for Pull based Orchestration	• **Connect Git Repository (The Source of Truth)** - If the repository is private, authentication credentials, such as an SSH private key or an HTTPS personal access token must be provided to Argo CD • **Create an Argo CD Application** - **Source:** The Git repository URL, the specific branch or revision to track, and the directory path containing Kubernetes manifests (Helm charts, Kustomize files, or raw YAML). **Destination:** The target Supervisor Namespace or VKS cluster and the specific namespace where the resources should be deployed. • Enable Automated Syncing: Once enabled, Argo CD continuously pull Git repository (by default, every 3 minutes). On a new Git commit, Argo CD automatically pulls the new manifests and reconciles the cluster's live state to match your desired state in Git	Git Repository structure is prepared outside of the Edge Activation.

Once Edge activation and vSphere infrastructure configuration are complete, having critical services available locally even when there's no external connectivity is a critical part of DoD autonomous operations requirement.

Day 2 Operations

Maintaining consistent ESX host configurations is critical. Even a minor drift in key settings such as firewall rules, password policies, DNS configuration, or NTP can have far-reaching consequences. Once the edge stack is activated, **Desired State Configuration (DSC)** enables version control of the intended configuration of edge locations. In doing so, it is possible to continuously enforce consistency and ensure that all environments remain aligned with the desired state configurations. This desired state enforcement happens at two layers:

- **vSphere Layer:** The vSphere Configuration Profile (VCP) delivers desired state configuration across vSphere clusters. Administrators can:
  - Set the desired configuration at the cluster in the form of a JSON document.
  - Check that hosts are compliant with the desired configuration.
  - If non-compliant, remediate hosts to bring them into compliance.

- **Workload Layer:** ArgoCD synchronizes the runtime state of the workload with the desired state configuration that is defined in Git. Configuration changes are made in Git and Argo CD updates the runtime state to reflect the Git. With Argo CD:
  - Set the source of workload desired state by pointing to a git Repo
  - Monitor workload runtime compliance by using the Argo CD UI
  - Update the Git Repo to change the desired state

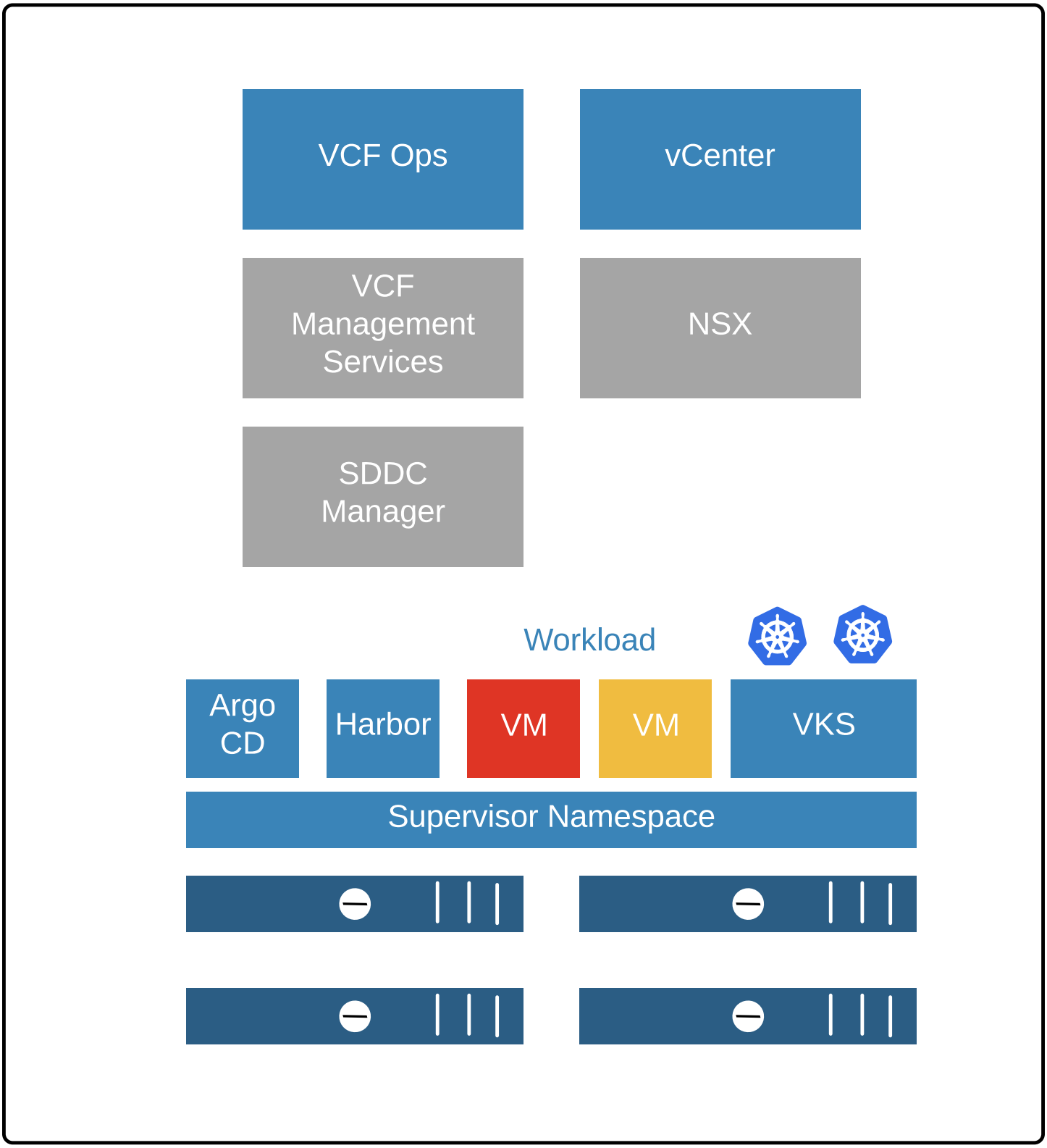
Table 451: Day 2 Configuration Management

Design Decision	Design Details	Design Implication
Enable VCP across all edge vSphere clusters	Use the Edge Automation script to complete the edge cluster setup followed by importing a configuration from a reference host. Reference Configuration Management using vSphere Configuration Profiles to activate VCP and enforce desired state configuration.	Updates to the vSphere Desired state configuration must be done in the form of a Draft and must Apply the Draft to the cluster, not directly from the VC.
Use Argo CD to manage workload configuration for VMs and containers.	Virtual Machines must be defined in YAML format using the VirtualMachine CRD. vSphere Kubernetes Service uses Cluster API(CAPI) and the Cluster API Provider for vSphere (CAPV) to provision and manage the lifecycle of Kubernetes clusters. Reference the VM Service Operator documentation and VKS Cluster documentation for implementation details.	Changes must be made in Git, not directly against the virtual infrastructure.
Use vSphere Lifecycle Manager to manage ESXi image updates and patches	Create a single vLCM image that contains the ESX version updates and OEM vendor add-on	In an air gapped environment with offline depot or no depot, the VCF Download Tool is used for creating vSphere Lifecycle Manager images in the vSphere Client. Reference VCF documentation: • Download ESX Component Data to an Offline Depot • Synchronize ESX Component Data for a VCF Instance
Use Staging as part of ESX image lifecycle management to reduce WAN dependency	Staging enables pushing ESXi images to a remote edge cluster without applying updates until the maintenance window begins. Refer to the vSphere 9.0 documentation on staging for implementation details.	Staging requires sufficient capacity in / scratch partition and might fail due to insufficient space.

Local Management with Multiple Hosts

Local Management with Multiple hosts is an air-gapped architecture that is designed for total self-sufficiency. Edge cluster consists of three or more hosts and the management control plane is runs directly on the edge cluster. This "Local Management" design ensures that the edge cluster remains fully operational and secure without any external connectivity.

Figure 170: Local Management Design



The Local Management Design is built around a vSphere cluster that provides foundational infrastructure, networking, storage, and lifecycle management in a unified platform. Networking is delivered through one or more vSphere Distributed Switches (VDS), sized according to uplink capacity and service isolation needs. Storage is powered by vSAN (OSA or ESA), with architecture determined by disk types and networking. Lifecycle management is handled through vLCM using a single cluster image.

The vSphere management plane runs as virtual appliances within the same cluster as workloads. Management components include vCenter for ESX resource management, and NSX for deployments needing advanced network services and multi-layer routing/switching. Depending on the type and duration of the mission, VCF Ops and fleet management components may run local to the cluster or remotely.

Modern container workloads and traditional VM applications are both fully supported at the edge by enabling the VCF Supervisor. Modern applications can run as vSphere Pods directly on ESXi or within VMware Kubernetes Service (VKS) clusters for standard Kubernetes cluster deployments. Traditional virtual machines operate alongside containers within the same namespace boundaries. Workload management is based on the operational model. Either directly through vSphere or GitOps-based using Argo CD.

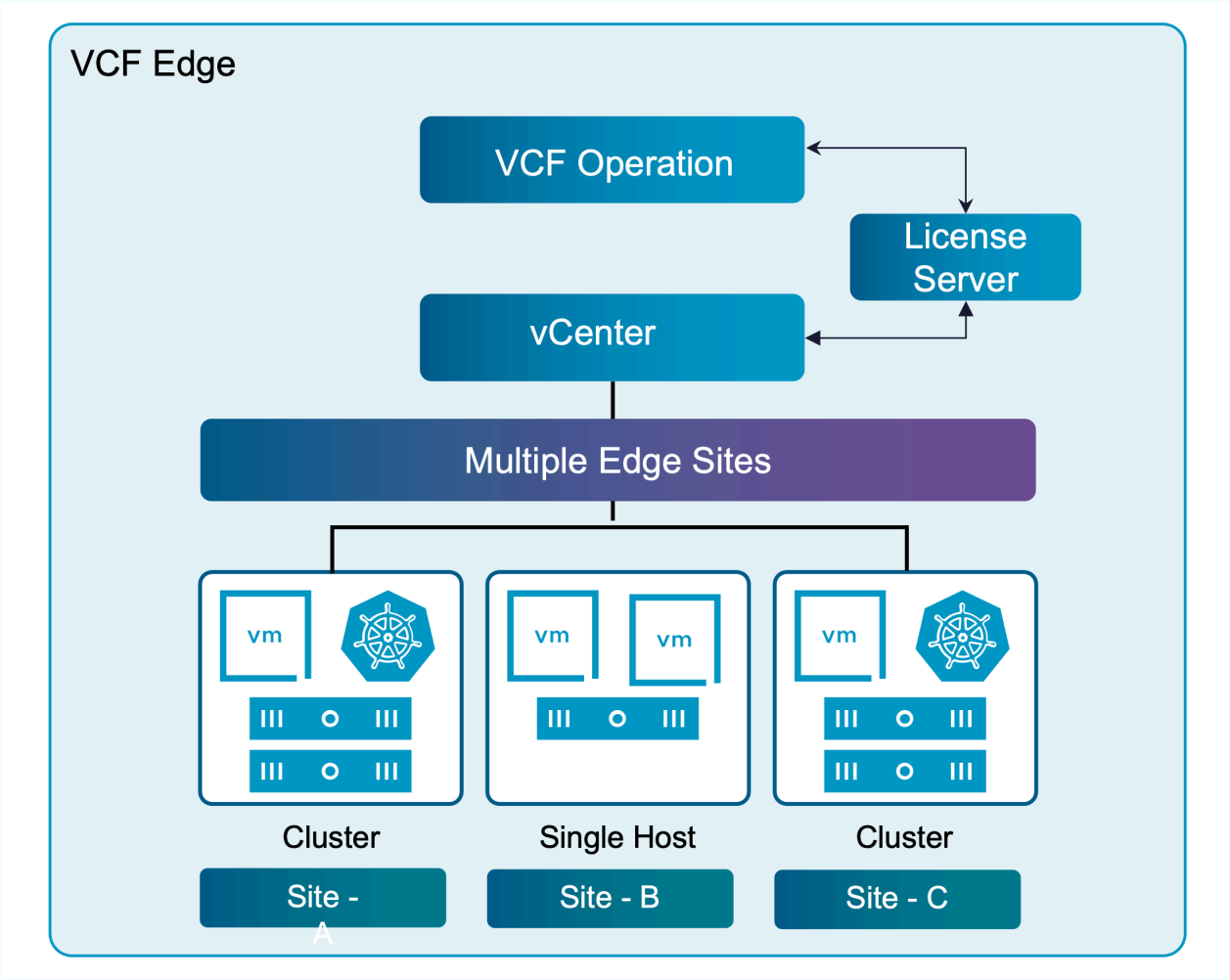
Management Components

Management components to support the "*local Management with Multiple Hosts"*design must be aligned with the functional requirements of the Edge.

Edge sites that do not require hypervisor-based security only needs a new edge vCenter, license server, and VCF Operations for license management and monitoring. The management component layout is similar to Edge Option #1, but for a single location. Management domain is also used for workloads.

Figure 171: VCF Edge Basic

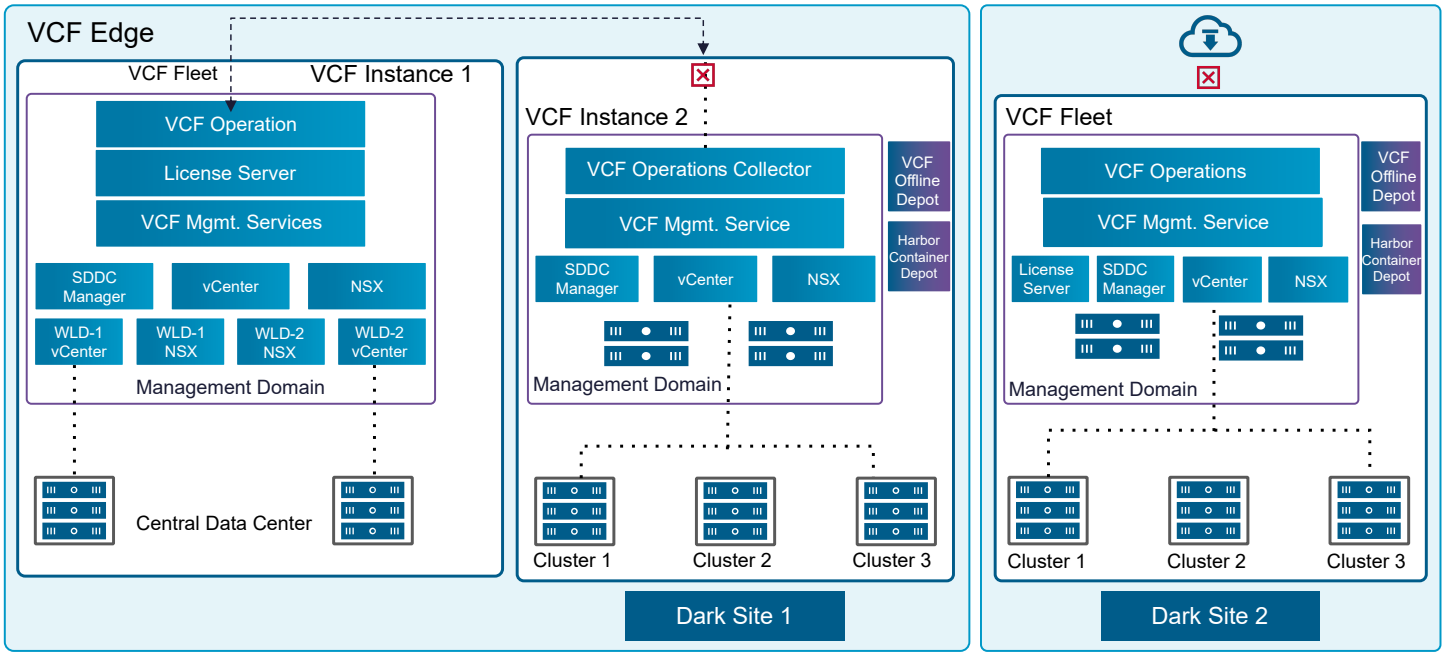
VCenter only with VCF Operations and VMSP for monitoring and day-2 services



Edge sites that require multi-tenant networking, advanced network services such as DHCP and stateless FW needs NSX in addition to vCenter. The management component layout is similar to Edge Pattern #5.

Figure 172: VCF Edge Advanced

VCF Edge with advanced networking


VCF Edge
VCF Fleet
VCF Instance 1
VCF Operation
License Server
VCF Mgmt. Services
SDDC Manager
vCenter
NSX
WLD-1 vCenter
WLD-1 NSX
WLD-2 NSX
WLD-2 vCenter
Management Domain
Central Data Center
VCF Instance 2
VCF Operations Collector
VCF Mgmt. Service
SDDC Manager
vCenter
NSX
Management Domain
VCF Offline Depot
Harbor Container Depot
Cluster 1
Cluster 2
Cluster 3
Dark Site 1
VCF Fleet
VCF Operations
VCF Mgmt. Service
License Server
SDDC Manager
vCenter
NSX
Management Domain
VCF Offline Depot
Harbor Container Depot
Cluster 1
Cluster 2
Cluster 3
Dark Site 2

Standard VCF instance can be used for edge locations that do not require local monitoring. A VCF instance comes with vCenter for resource management, multi-layer hypervisor networking with NSX, and a full set of components to support lifecycle management. Only mission-critical components must be continuously running. If infrastructure updates such as patching are only permitted at mission completion, SDDC-manager, VCF management services, and cloud proxy can be powered off to reduce CPU/memory consumption.

For long-duration misions where continuous infrastructure monitoring, patching and ESX LCM are mandatory, a single VCF fleet can be deployed. With the VCF fleet, administrators can benefit from a full set of day-2 operational benefits using VCF Operations, including centralized monitoring, logging, certificate, password, identity management.and LCM.

Management stack activation processes are documented in the Edge Activation section.

Design Decision	Design Details	Design Implication
Deploy Edge Option #1 to manage edge deployments requiring only basic L2 connectivity and standard vSphere security at edge locations	Edge vCenter: Edge vCenter sizing should align with VCF9 vCenter design recommendations.	No advanced multi-layer routing and switching No network services such as DHCP and stateless firewall Up to 16 IP address required for Activation DNS and NTP servers are required for the deployment. 6 FQDN requiring both forward and reverse lookup
Deploy a VCF instance Edge Pattern #5 for Edge deployments needing advanced network services and multi-layer routing/switching without the mandatory requirement for local monitoring	VCF Instance Simple deployment mode is sufficient. Simple model comes with • VCF management services - Small • vCenter - Small • NSX Manager - Medium • Cloud Proxy - Small the sizing and scale for simple mode is outlined in the VCF9 Design document, vCenter and NSX. Sizing guide	Simple Mode uses vSphere HA for redundancy and a single instance of the management component is deployed for efficient sizing Up to 20 IP address required for Activation DNS and NTP servers are required for the deployment. 8 FQDN requiring both forward and reverse lookup
Deploy a VCF fleet Edge Pattern #5 for Edge deployments needing advanced multi-layer routing/switching, central VCF ops for monitoring, patching, and ESX LCM	VCF Fleet Simple deployment mode is sufficient. Simple model comes with • VCF Operations - Small • VCF management services - Small • vCenter - Small • NSX Manager - Medium • Cloud Proxy - Small • License Server - Small the sizing and scale for simple mode is outlined in the VCF9 Design document, vCenter and NSX. Sizing guide	Simple Mode uses vSphere HA for redundancy and a single instance of the management component is deployed for efficient sizing 23 IP address required for Activation without VCFA - 7 more routable IP with VCFA DNS and NTP servers are required for the deployment. 10 FQDN requiring both forward and reverse lookup

Edge Service Design

The availability of foundational services such as DNS, NTP, and workload templates are critical for autonomous operations. Deploying those services at the edge is required.

To support the disconnected mode of operation required for autonomous operations, infrastructure services that are required to support the workload runtime must be deployed locally.

- **Network and Foundational Services:** STIG-compliant infrastructure relies on FIPS-validated encryption and a robust Public Key Infrastructure (PKI) to ensure session integrity and data-in-transit security. Modern cloud-native workloads have deep, systemic dependencies on these foundational services:
  - **DNS:** Service discovery in both traditional and modern applications relies heavily on DNS.If DNS is unavailable, edge application fails to operate.

  - **NTP:** Encryption is highly sensitive to temporal accuracy. If the system clock drifts even slightly certificate validation may fail. STIGs compliant infrasturcture relies on encryption and certificates to ensure session integrity.
  - **Availability Requirements:** Because both DNS and NTP services are "pre-requisites" for the rest of the stack, they must be architected for high availability to prevent a single point of failure from cascading through the entire environment.
- **Artifact Repository:** Local Artifact repository enables fast provisioning without WAN.
  - **Local Content Library:** Content libraries are vSphere container objects for storing and sharing OVF and OVA packages, VM templates, vApp templates, and other types of files.
  - **Container Registry:** If a Kubernetes pod crashes or a node reboots in a disconnected environment, the host (depending on imagePullPolicy) will try to pull an image from its configured registry. With a local registry, required images are always available at the Edge. VMware Harbor is a CNCF compliant container registry and can be consumed as a supervisor service or deployed as a VM appliance.
- **Pull based Configuration:**
  - In a traditional VCF Edge deployment, vCenter "pushes" configurations to the infrastructure. A GitOps-based approach uses a "Pull" model that allows configuration updates to be staged in a Git repository, and applied as infrastructure code update instead of manual "pushes". ArgoCD is a CNCF Continuous Delivery (CD) tool that follows the GitOps methodology and can be consumed as a supervisor service.

Design Decision (Disconnected Mode Support)	Design Details	Design Implication
Deploy a redundant DNS service at the Edge to support infrastructure and application requirements.	Deploy DNS as VMs with strict "Anti-Affinity" DRS rules. Anti-Affinity rules ensure that a single physical host failure cannot take down all DNS instances simultaneously. Configure Affinity rules so that if Edge Cluster fails, DNS service can be quickly located and brought back online. DNS servers must be configured for both forward and reverse DNS records for all edge resources. Every Fully Qualified Domain Name (FQDN) used must resolve to a unique IP address.	DNS VMs consume CPU/Memory resources from the Edge Cluster. Increased management cost with: • Updating DNS server as edge resources change • **Security Patching:** Every edge DNS server is a new entry point that must be secured, patched, and monitored.
Use the Edge Router or switch as the source of NTP and set the service policy to *"Start and stop with host."*	NTP configuration can be set on the host using VMware Host/vSPhere Client, API, or vSphere Configuration Profile. Reference VMware documentation for time synchronization best practices specific to the VMware Infrastructure.	Edge routers must be configured to prevent "Time Spoofing" attacks.
Create a Local Content Library at each Edge Location to store virtual machine templates and serve as the "Image Repository" for the VMware Supervisor VM Operator services.	Content libraries are Local to the edge and on the Local datastore. Local Content library can be used to enable and upgrade Supervisor or pre-stage VM OVAs to reduce dependency to WAN. In a GitOps and Edge context, this integration is the mechanism that allows your declarative YAML manifests to reference a specific OS version. Use VCF documentation for Content Library usage details	Content Library consumes local storage. The Content Library is the source for the VM Operator. When you submit a YAML file for a VirtualMachine, the spec.imageName field must match the name of a template that is stored in the associated Content Library.

Design Decision (Disconnected Mode Support)	Design Details	Design Implication
Deploy VCF Supervisor at the Edge to enable Supervisor services.	Supervisor at the edge site enables edge services to simplify edge workload deployment and management. • Deploy small Supervisor with a single FLB load balancer to support [VMoperator,VKS, Argo, Harbor]. VM Operator and VKS are core services that are activated as part of the supervisor deployment. • FLB deployed in one arm mode to reduce number of NICs and Networks Sharing Management VLAN is possible between FLB and Supervisor to reduce network VLAN dependency. Workload VLAN must be different from Management VLAN Refer to the Deploying vSphere Supervisor with Foundation Load Balancer	FLB deployment is supported with vSphere Distributed Switch A sufficient IP address must be set aside for all edge Supervisor Services: • Management VLAN: – 5 IP Supervisor – 2 IP FLB Budget sufficient Core and Memory: • 2 cores and 16 GB for the supervisor • 1 Core and 4 GB for FLB. • Content Library consumes local storage NSX VPC-based deployment requires Edge VM to be deployed. Refer to Supervisor Architecture with VCF Networking with VPC
Deploying Harbor as a Supervisor Service	Deploy Harbor as a supervisor service to store container images for Application workloads and Edge Supervisor Services.	4 cores and 5G RAM for Harbor running as vSphere Pods.
Load container images for workload and supervision services into the edge Harbor Registry so workloads can restart in an air-gapped site	Refer to the VCF documentation on uploading Supervisor services: "*Deploying Supervisor Services from a Private Container Image Registry*" Refer to the VCF documentation on loading application container images into Harbor	Deploying images from an edge specific private registry requires updates to the image path in the service YAML manifest.
Deploy ArgoCD as a Supervisor Service	Argo CD implements a "Pull" configuration model that is inherently resilient to the intermittent connectivity typical of edge sites. Argo CD agents run locally on the edge cluster and continuously monitor a Git repository for changes. • If the Git repository is lost, the edge site continues to run in its "Last Known Good" state. • The moment Git is restored, the local Argo CD agent "pulls" any updates that were committed to Git and applies them immediately. ArgoCD is necessary only if edge workload deployments use GitOps principles (refer to the Desired State Management section for details). Refer to the ArgoCD Documentation for implementation details	3 cores and 4G RAM for ArgoCD vSphere Pods. 6 IPs on the workload network The Git Repository for ArgoCD sync is deployed outside of VCF.

Edge Cluster Design

Edge cluster design consists of the following aspects:

- Edge Hardware:
  - **Environmental and Electromagnetic Compliance:** Select hardware that meets rigorous standards for environmental survivability and electromagnetic compatibility.
  - **Hardware-Rooted Security:** Implement security foundations that are based on TPM 2.0 and Secure Boot to ensure platform integrity.
- vSphere Cluster:

  - **Compute Design:** Define the CPU core and memory requirements necessary for both the management plane and edge workloads.
  - **Storage Architecture:** Determine the optimal shared storage strategy that is based on the disk configuration of the compute
  - **Network Configuration:** Develop a VDS design based on the available number of physical NICs and the required traffic isolation for management and workload data.
  - **Availability and Optimization:** DRS (Distributed Resource Scheduler) automation levels and establish affinity rules to support the most optimal resource utilization. vSphere HA settings to ensure management stack failover under a single-node failure.
- **Security and Compliance Alignment**
  - **STIG Compliance:** Ensure full alignment with the DISA STIG implementation guidelines for all cluster components.
  - **Traffic Isolation:** Implement robust traffic isolation based on service type to prevent lateral movement.
  - **Identity and Access Management:** Restrict cluster access using Strong Identity, Credential, and Access Management (ICAM)
  - **FIPS-Validated Cryptography:** Use FIPS validated cryptographic modules for both vSAN data-at-rest encryption and data-in-transit protection.

Design Decision	Considerations	Design Implication
Edge Hardware	Refer to Remote-Management Hardware Design	Edge hardware must be validated and certified by the OEM vendor against VCF 9
Compute Node Design	Refer to Edge Compute Node Design	
Cluster	Multiple computes deployed in a single rugged chassis	Single Rack Cluster design is applicable for use cases that have a single availability zone.
	Multiple fault domains within a single availability zone. The networks are stretched across racks using a single broadcast domain for each network.	Layer 2 Multi-Rack cluster Model is applicable for use cases that have multiple fault domains within a single availability zone. L2 VLAN Network must be stretched across all computes in the cluster Single subnet for each network
Edge Access Security	Refer to Edge Access Security Design	

Management At Edge Activation

Edge activation starts in a staging environment and shipped to the field at the start of each mission. With the introduction of Zero Touch Provisioning (ZTP), the transition of a new host from power-on to an active member of the remote cluster follows a highly orchestrated automated workflow. Along with PowerShell automation scripts, new edge hosts can be imaged, provisioned, and brought into service without manual intervention. The process is initiated at the central staging data center. Once a mission starts, vSphere cluster LCM is frozen until mission completion and equipment is returned back to the staging environment. Based on the desired topology, there are three activation approaches:

Scenario 1: vCenter only to manage edge deployments requiring only basic L2 connectivity and standard vSphere security at edge locations.

Scenario 2: Deploy a new VCF instance and power-off components that are not required for a mission. This approach requires a pre-existing VCF Ops instance and assumes that the cluster may rejoin the fleet for LCM after each mission.

Scenario 3: Deploy a new VCF fleet where central VCF ops for continuous infrastructure monitoring, patching, and ESX LCM are mandatory. This approach allows administrators to have a full set of day-2 operational capabilities including centralized monitoring, logging, LCM of cluster image, management certificate, password, and identity.

Table 452: Scenario 1: vCenter Only

vCenter only to manage edge deployments requiring only basic L2 connectivity and standard vSphere security at edge locations.

Design Decision	Design Details	Design Implication
Use ZTP to Boot and Install ESX Image	Ship hosts to a staging data center Boot and install ESX image on host using ZTP	The staging datacenter has an existing VCF management domain with Staging VC preinstalled. DHCP, DNS, and NTP servers are required for the deployment. ESX Hosts FQDN must be forward and must reverse resolvable for cluster formation vCenter certificate must be trusted by the Edge Host.
Deploy vCenter, VCF Ops, VMSP, License Server	Use VCF installer with VVF option and then entitle VCF Edge license with VCF Operation.	16 IP address required for Activation 6 FQDN requiring both forward and reverse lookup
Use the Edge Automation script to complete the infrastructure setup	Automation Script configures: • Edge vSphere cluster (DRS/HA, VDS, Storage) • Enable vSphere Desired State • Deploy Supervisor and Foundation Load Balancer • Deploy Edge services (ArgoCD and Harbor)	A sufficient IP address must be set aside for the Supervisor and FLB: • Management VLAN: 7 IP (Supervisor / FLB same network) Edge hosts and vCenter must remain connected during the duration of the provisioning phase.
Activate vSphere Configuration Profile	• **Select a Reference Host:** single ESXi host with desired configuration in networking, Storage, NTP, syslog, security settings, etc. • **Extract the Schema:** vSphere extracts these settings and generates a JSON configuration document representing the cluster desired state. • **Host Overrides:** Remaining hosts need specific static IP addresses for vSAN & vMotion, overrides are handled in the "host-specific" section of the JSON document. • **Apply and Enforce:** The profile is applied to the cluster. vSphere handles the orchestration, ensuring all hosts conform to the blueprint.	Desired State Enforcement Remediation is manually triggered

Design Decision	Design Details	Design Implication
Configure Argo CD for Pull-based Orchestration	• **Connect Git Repository (The Source of Truth)** - If the repository is private, authentication credentials, such as an SSH private key or an HTTPS personal access token must be provided to Argo CD • **Create an Argo CD Application** - **Source:** The Git repository URL, the specific branch or revision to track, and the directory path containing Kubernetes manifests (Helm charts, Kustomize files, or raw YAML). **Destination:** The target Supervisor Namespace or VKS cluster and the specific namespace where the resources should be deployed. • Enable Automated Sync: Once enabled, Argo CD continuously poll Git repository. On a new Git commit, Argo CD automatically pulls the new manifests and reconciles the cluster live state to match your desired state in Git	Git Repository structure must be implemented based on GitOps best practice. See Git Repository Design.
Power-off VMSP cluster, VCF Operations, License Server if mission does not allow LCM updates or configuration changes	Update the password expiration to extend beyond the duration of the mission Refer to Shut Down the Management Domain documentation for the sequence of steps of shutdown.	Monitoring, add /remove hosts, and Image updates possible without critical services online.
Power-on VMSP cluster, VCF Operations, License Server once mission completes for LCM or configuration updates	Refer to Start the Management Domain documentation for the sequence of steps for startup.	None

Table 453: Scenario 2: New VCF Instance

Deploy a new VCF Instance and power off non mission-critical components

Design Decision	Design Details	Design Implication
Use ZTP to Boot and Install ESX Image	Ship hosts to a staging data center with an existing VCF management domain Boot and install ESX image on host using ZTP	The staging datacenter has an existing VCF management domain with Staging VC preinstalled. DHCP, DNS, and NTP servers are required for the deployment. ESX Hosts FQDN must be forward and must reverse resolvable for cluster formation vCenter certificate must be trusted by the Edge Host BIOS.
Use the VCF installer to deploy a new VCF instance. Management Components are deployed on the new ESX hosts.	Using the VCF installer UI, select • Existing VCF Ops • New vCenter and NSX Manager instances • Simple Deployment Number of Management IPs with FQDN • 20 IP Addresses for the Management stack • 8 FQDN requiring both forward and reverse lookup	The smallest size possible may be larger than required for the edge vSphere Cluster: • VCenter -Small • NSX Manager - Medium Monitoring not available with standard VCF instance deployment.

Design Decision	Design Details	Design Implication
Use the Edge Automation script to complete the infrastructure setup	Automation Script configures: • Deploy Edge services (ArgoCD and Harbor)	Infrastructure parameters supplied to the automation script must be generated prior before deployment Edge hosts and vCenter must remain connected during the duration of the provisioning phase.
Activate vSphere Configuration Profile	• **Select a Reference Host:** single ESXi host with desired configuration in networking, Storage, NTP, syslog, security settings, etc. • **Extract the Schema:** vSphere extracts these settings and generates a JSON configuration document representing the cluster desired state. • **Host Overrides:** Remaining hosts need specific static IP addresses for vSAN & vMotion, overrides are handled in the "host-specific" section of the JSON document. • **Apply and Enforce:** The profile is applied to the cluster. vSphere handles the orchestration, ensuring all hosts conform to the blueprint.	Desired State Enforcement Remediation is manually triggered
Configure Argo CD for Pull-based Orchestration	• **Connect Git Repository (The Source of Truth)** - If the repository is private, authentication credentials, such as an SSH private key or an HTTPS personal access token must be provided to Argo CD • **Create an Argo CD Application** - **Source:** The Git repository URL, the specific branch or revision to track, and the directory path containing Kubernetes manifests (Helm charts, Kustomize files, or raw YAML). **Destination:** The target Supervisor Namespace or VKS cluster and the specific namespace where the resources should be deployed. • Enable Automated Syncing: Once enabled, Argo CD continuously poll Git repository (by default, every 3 minutes). On a new Git commit, Argo CD automatically pulls the new manifests and reconciles the cluster's live state to match your desired state in Git	Git Repository structure must be implemented based on GitOps best practice. See Git Repository Design.
Power-off VMSP cluster, VCF Operations, SDDC Manager, Cloud Proxy if mission does not allow LCM updates or configuration changes	Update the password expiration to extend beyond the duration of the mission Refer to Shut Down the Management Domain documentation for the sequence of steps for shutdown.	Monitoring, add /remove hosts, and Image updates are not available with the VMSP cluster powered off.
Power-on VMSP cluster, SDDC Manager, Cloud Proxy once mission completes for LCM or configuration updates	Refer to Start the Management Domain documentation for the sequence of steps for startup.	

Table 454: Option 3: New VCF Fleet

Design Decision	Design Details	Design Implication
Use ZTP to Boot and Install ESX Image	Ship hosts to a staging data center. Boot and install ESX image on host using ZTP	The staging datacenter has an existing VCF management domain with Staging VC preinstalled. DHCP, DNS, and NTP servers are required for the deployment. ESX Hosts FQDN must be forward and must reverse resolvable for cluster formation vCenter certificate must be trusted by the Edge Host.
Use the VCF Installer to create a VCF fleet using new ESX hosts. Management Components are deployed on the new ESX hosts.	Using the VCF installer UI, select • Deploy VCF Automation Later • Simple Mode Deployment Number of Management IPs with FQDN • 23 IP address required for Activation without VCFA • 10 FQDN requiring both forward and reverse lookup	Simple Mode uses vSphere HA for redundancy and a single instance of the management component is deployed for efficient sizing
Use the Edge Automation script to complete the infrastructure setup	Automation Script configures and deploys: • Enable vSphere Desired State • Edge services (ArgoCD and Harbor)	Infrastructure parameters supplied to the automation script must be generated prior before deployment Edge hosts and vCenter must remain connected during the duration of the provisioning phase.
Activate vSphere Configuration Profile	• **Select a Reference Host:** single ESXi host with desired configuration in networking, Storage, NTP, syslog, security settings, etc. • **Extract the Schema:** vSphere extracts these settings and generates a JSON configuration document representing the cluster desired state. • **Host Overrides:** Remaining hosts need specific static IP addresses for vSAN & vMotion, overrides are handled in the "host-specific" section of the JSON document. • **Apply and Enforce:** The profile is applied to the cluster. vSphere handles the orchestration, ensuring all hosts conform to the blueprint.	Desired State Enforcement Remediation is manually triggered

Design Decision	Design Details	Design Implication
Configure Argo CD for Pull-based Orchestration	• **Connect Git Repository (The Source of Truth)** - If the repository is private, authentication credentials, such as an SSH private key or an HTTPS personal access token must be provided to Argo CD • **Create an Argo CD Application** - **Source:** The Git repository URL, the specific branch or revision to track, and the directory path containing Kubernetes manifests (Helm charts, Kustomize files, or raw YAML). **Destination:** The target Supervisor Namespace or VKS cluster and the specific namespace where the resources should be deployed. • Enable Automated Syncing: Once enabled, Argo CD continuously poll Git repository (by default, every 3 minutes). On a new Git commit, Argo CD automatically pulls the new manifests and reconciles the cluster's live state to match your desired state in Git	Git Repository structure must be implemented based on GitOps best practice. See Git Repository Design.

Example Day 2 Deployment

Modern naval vessels are an example of a multi-node air gap use case. Modern navy ships rely on highly resilient edge-computer architectures to support mission-critical operations. Unlike conventional terrestrial data centers, shipboard IT infrastructure must maintain continuous availability under stringent operational constraints, including hardware degradation, power transients, and the potential for kinetic impact. The onboard compute hosts a diverse set of workloads essential to the operation from combat and navigation systems to engineering telemetry and logistical support.

To optimize resource allocation without sacrificing operational integrity, applications are classified into three distinct tiers based on mission criticality:

Table 455: Application Classifications & Recovery Patterns

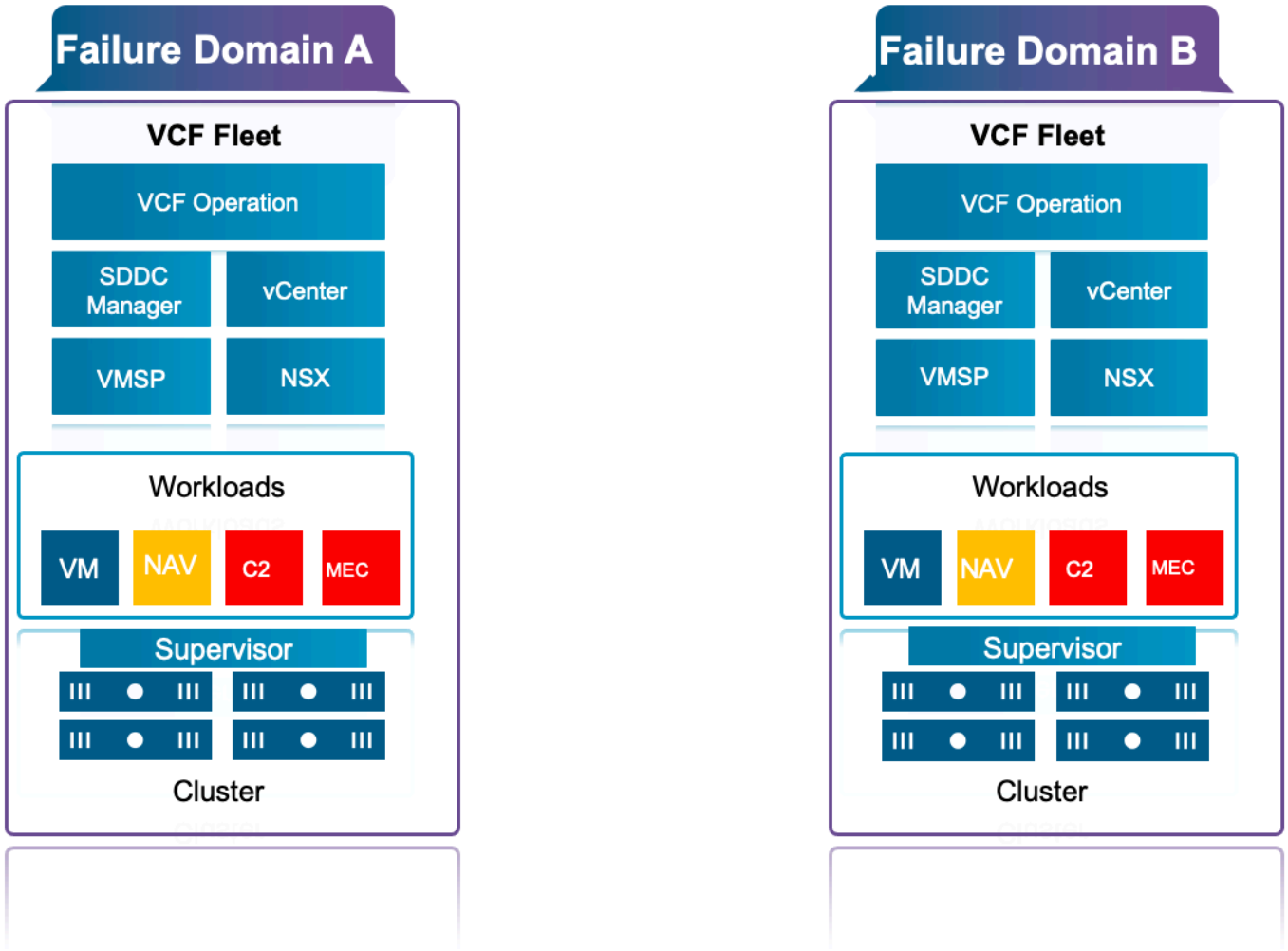
	Classification	Pattern	Recovery
Combat & Command and Control (C2)	Tier 0: Ultra Critical	Active-active or hot-standby across 2 isolated domains with continuous synchronous state replication	Instantaneous assumption of operational load by the surviving node/domain. Compute capacity may shrink, but execution remains uninterrupted.
Engineering Control (Eng)	Tier 0: Ultra Critical	Localized Programmable Logic Controllers (PLCs) acting as independent "control islands" adjacent to physical machinery, with centralized supervisory applications	Local PLCs continue autonomous execution even if the central compute cluster or internal network fails.
Tier 1: Navigation & Ship Control (Nav)	Tier 1: Mission-Critical	Primary and backup architecture that is separated across domains, with explicit manual fallback	The secondary controller assumes authority upon primary failure. If all compute domains fail, manual bridge operations bypass the IT infrastructure entirely.

	Classification	Pattern	Recovery
Tier 2: Business & Administrative Operations	Tier2: Routine / Admin	Standard Enterprise Application. Does not require	Application Restart is acceptable and can be powered off to reduce resource consumption during hardware degradation.

Infrastructure Platform Orchestration

To operationalize the high-availability and failure recovery patterns that are previously defined, the underlying hardware must be abstracted and managed by a robust VCF fleet architecture. In this design, a locally managed edge deployment model based on four nodes is used. This model is explicitly engineered for environments that anticipate severe network degradation or complete physical isolation.

Figure 173: Independent Fleet



To provide the highest availability, architecture is based on two sets of fully autonomous management planes. As illustrated in the deployment diagram, two physical failure domains (Domain A and Domain B) are provisioned. Each physical Failure Domain is provisioned with an independent, self-contained VCF Fleet management stack and operates independent of each other.

- **Independent Infrastructure:** Each domain operates its own localized instance of SDDC Manager, vCenter Server, and NSX Manager
- **Localized Operations:** VCF Operations (monitoring, logging, and performance analytics) are isolated to their respective domains.

By deploying an independent locally managed control plane, the architecture ensures that if one failure domain suffers catastrophic physical destruction, the resources in the failover domain experiences zero interruption.

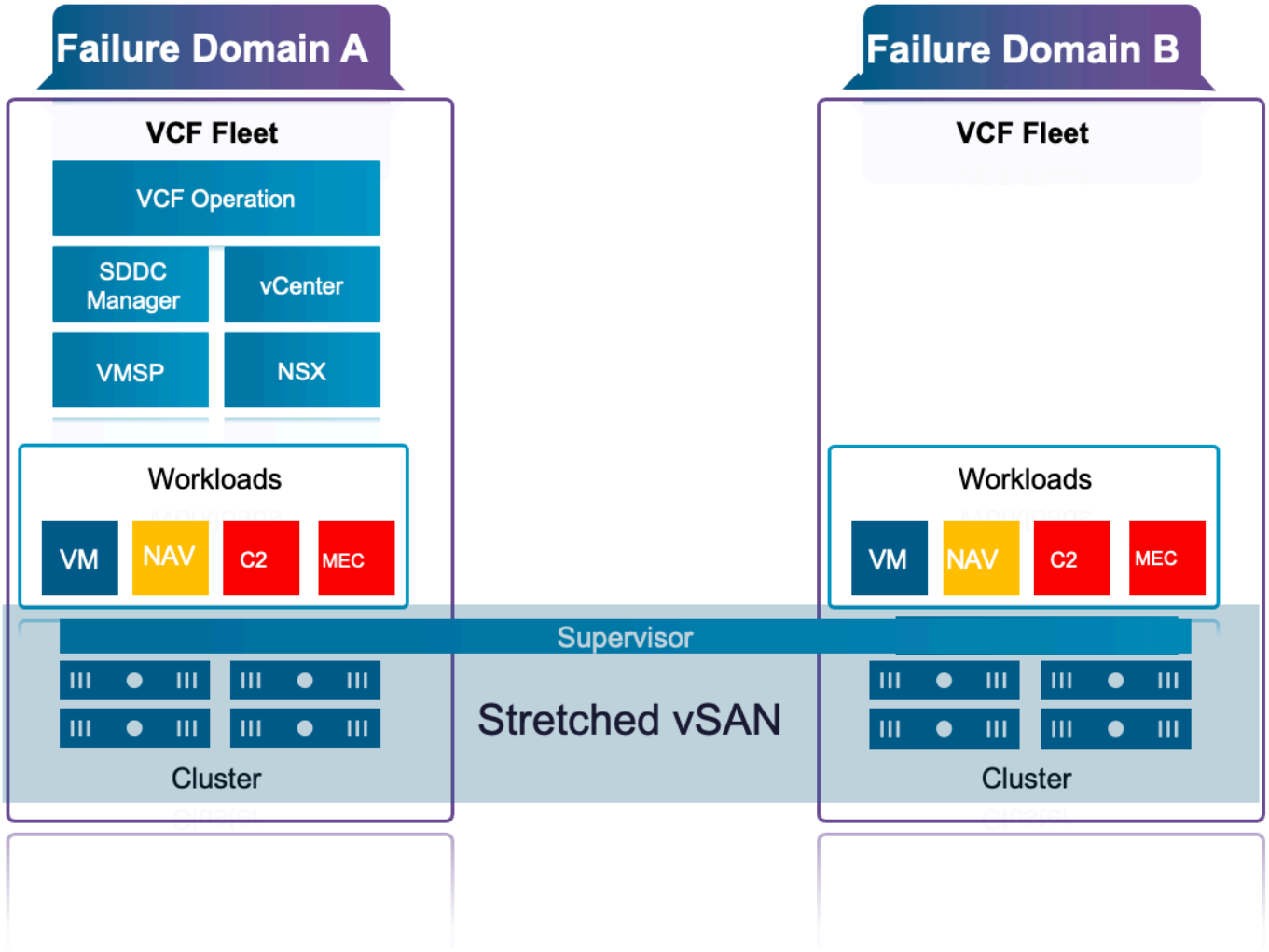
The distribution of workloads across these autonomous VCF instances directly correlates to the application availability tiers defined in the Application Classification table.

- **Mission-Critical Instantiation (Tier 0 & 1):** The highest-priority workloads, example Combat/C2 (Tier-0), Navigation (Tier-1), and Engineering Control(Tier-0), are symmetrically instantiated across both Failure Domain A and B. Symmetrical instantiation ensures an active-active or hot-standby placement that are required by Tier 0/1 applications execute continuous state replication or autonomous local control.
- **Routine Instantiation (Tier 2):** Conversely, lower-priority business and administrative workloads do not require redundant instantiation across all domains. Email server application, for example, does not determine the mission outcome and can reside exclusively in Domain A or Domain B. In a failure event, these Tier 2 applications can be manually restarted based on the availability of compute and network.

Because the infrastructure layers are independent, data must be replicated at the vSphere layer or natively within the application. An alternative deployment model uses a synchronously stretched storage fabric to reduce the need for vSphere replication.

Stretched Fleet Design

Figure 174: Stretched Fleet



As depicted in the diagram, this configuration consolidates the VCF Fleet management stack entirely within Failure Domain A. Failure Domain B acts as a compute and storage extension. The foundational element of this design is the Stretched vSAN cluster, which aggregates the local storage of both Domain A and Domain B into a single, logical

datastore. Deploying a Stretched vSAN configuration offers specific advantages, primarily by shifting the burden of high availability from the application layer to the underlying infrastructure:

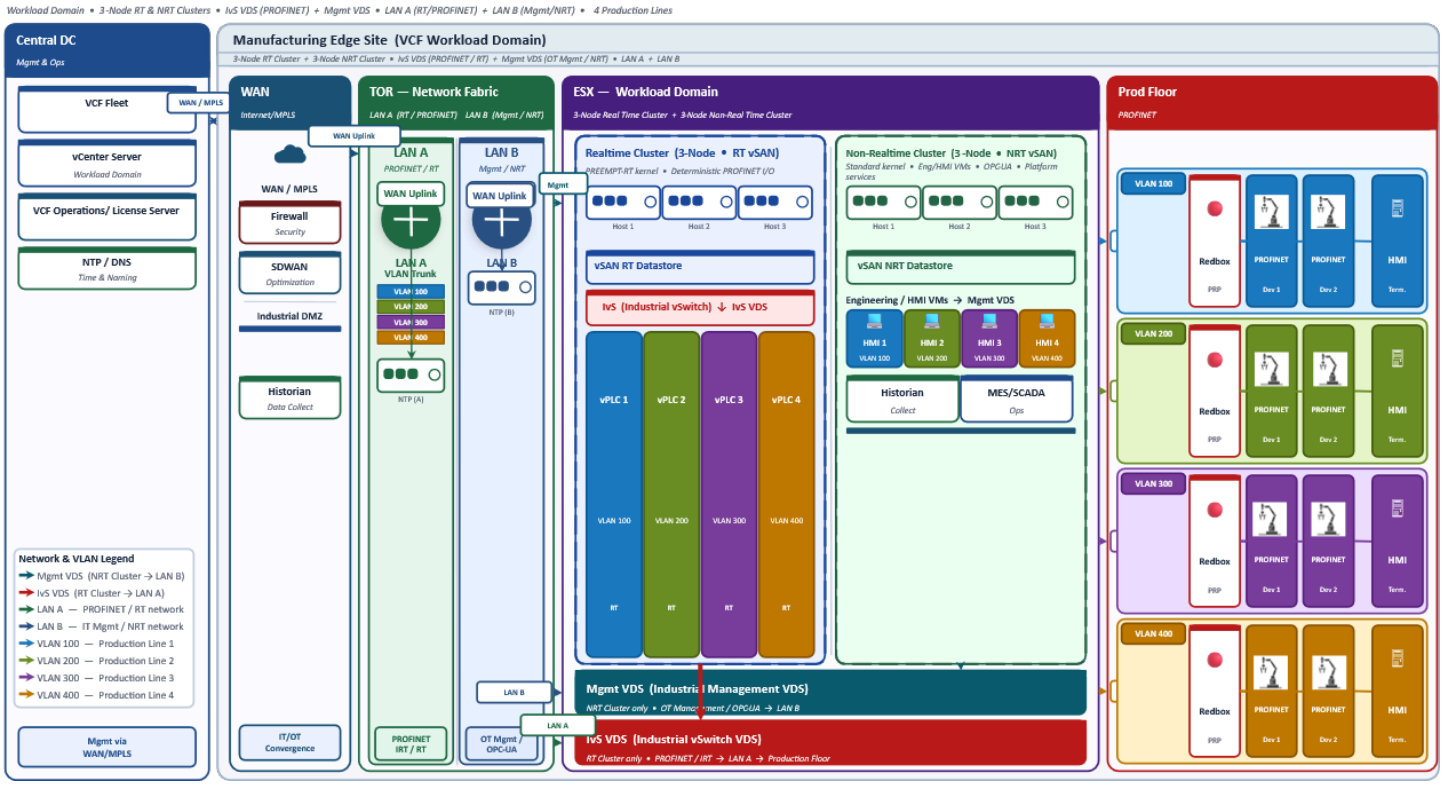
- **Zero Recovery Point Objective (RPO):** Stretched vSAN implements synchronous, block-level I/O replication across the two failure domains. Every write operation is committed to both physical domains simultaneously, ensuring zero data loss if an entire compartment is destroyed.
- **Infrastructure-Level High Availability:** This model facilitates seamless vSphere High Availability (HA). If Failure Domain A experiences a failure, virtual machines can be automatically restarted on the surviving hardware in Failure Domain B without requiring complex, application-specific failover implementation.
- **Legacy Application Support:** Applications that lack native clustering or state replication capabilities can achieve high availability purely through the infrastructure layer, simplifying the deployment of legacy or off-the-shelf (COTS) software.
- **Single Pane of Operations:** VCF Operations (monitoring, logging, LCM, and performance analytics) for the entire vessel, across Failure Domains A and B.

VCF Edge Design Blueprint for Manufacturing Use Case

This design blueprint provides architectural guidance for deploying VMware Cloud Foundation Edge in manufacturing environments where IT and OT workloads must operate together within resilient, secure, and low-latency edge infrastructure. The blueprint focuses on workload isolation, industrial network segmentation, operational resiliency, and supporting both traditional industrial applications and modern Kubernetes-based workloads at the manufacturing edge.

Figure 175: Manufacturing VCF Edge reference design

Manufacturing VCF Edge reference design supporting converged IT/OT operations, real-time industrial workloads, and resilient edge infrastructure for factory and industrial environments.



This design focuses on manufacturing and industrial edge use cases for **VMware Cloud Foundation Edge** and addresses the operational, resiliency, segmentation, and low-latency requirements of converged IT and OT (Operational Technology) environments.

Manufacturing environments often require infrastructure platforms capable of supporting both traditional enterprise workloads and sensitive industrial control systems operating under strict performance, availability, and network isolation requirements. These environments may include real-time industrial communication protocols, segmented production networks, deterministic workloads, manufacturing execution systems, historian platforms, SCADA systems, and edge analytics platforms operating close to production systems.

This reference architecture demonstrates how centralized operational governance can be combined with localized industrial edge infrastructure to support resilient and scalable manufacturing operations across distributed factory environments.

The architecture illustrates how:

- centralized VCF operations integrate with manufacturing edge sites
- IT and OT networks remain logically segmented while operating on shared infrastructure
- real-time and non-real-time workloads can coexist within the same VCF workload domain
- industrial protocols and deterministic traffic patterns can be isolated through dedicated networking constructs
- operational continuity can be maintained during WAN degradation or disruption
- modern application platforms and Kubernetes workloads can be integrated into manufacturing edge environments

Key architectural themes that are represented in this design include:

- convergence of IT and OT operational environments
- segmentation between industrial and enterprise workloads
- support for real-time and deterministic industrial workloads
- resilient edge infrastructure for manufacturing operations
- industrial network isolation using VLAN and distributed switch segmentation
- operational continuity during a network disruption or degraded connectivity conditions
- support for industrial telemetry, analytics, and manufacturing application platforms

This architecture also demonstrates how manufacturing edge environments can support multiple workload classes within the same operational framework, including:

- real-time industrial workloads
- non-real-time operational workloads
- manufacturing execution systems
- SCADA and historian platforms
- HMI workloads
- industrial telemetry and analytics platforms

The design patterns and implementation models presented throughout this document progressively decompose this architecture into infrastructure, networking, workload isolation, resiliency, lifecycle management, and operational design considerations required for manufacturing and industrial edge deployments.

Industrial Operational Context

For OT and IT Infrastructure administrators and architects, selecting the right design pattern for a stable and resilient VCF Workload Domain capable of running sensitive real-time workloads can be challenging due to numerous options. The optimal choice depends on factors such as the specific use case, budget, and workload sensitivity.

Begin by familiarizing yourself with the general VCF architecture and terminology. Refer to the VCF Product Overview and review the VCF Release Notes. For information on deploying, converging, or upgrading VCF, consult the Overview of VCF Deploy, Converge, and Upgrade Guide.

These Design Options cover the most common use cases, though not every configuration option is detailed. Architectures that align with official documentation, such as the maximum settings within the Hardware Compatibility List (HCL), are supported.

Design Options

Edge Design Options for Manufacturing

This section explores design options, focusing primarily on replacing traditional Physical Logic Controllers (PLCs) with virtual PLCs (vPLCs) hosted in data centers. While PLCs are most commonly found on the shopfloor, this approach is also applicable to other environments like building control or logistics centers. Establishing a scalable, future-proof Edge Infrastructure is essential for global manufacturing organizations. VCF 9 supports diverse Edge Deployment scenarios, managed through a comprehensive Management Plane.

Virtual Desktop Infrastructure (VDI) for Shopfloor IPC Replacement

VDI offers an alternative to traditional Shopfloor Industrial PCs (IPCs), which are used for tasks such as worker guidance, device control (e.g., documenting and applying torque), and as Human-Machine Interfaces (HMIs). This involves centralizing the IPC functionalities on virtual machines (VDI Desktops) within a data center.

This transition yields significant Capital Expenditure (CAPEX) savings, as Thin Clients are substantially cheaper than IPCs. Furthermore, Thin Clients prove more cost-effective over time because IPCs frequently require replacement due to the demanding production environment.

(Note: VDI use cases are mentioned for context only, as they are not the primary focus of this specific design.)

Core vPLC Design Leveraging VCF 9

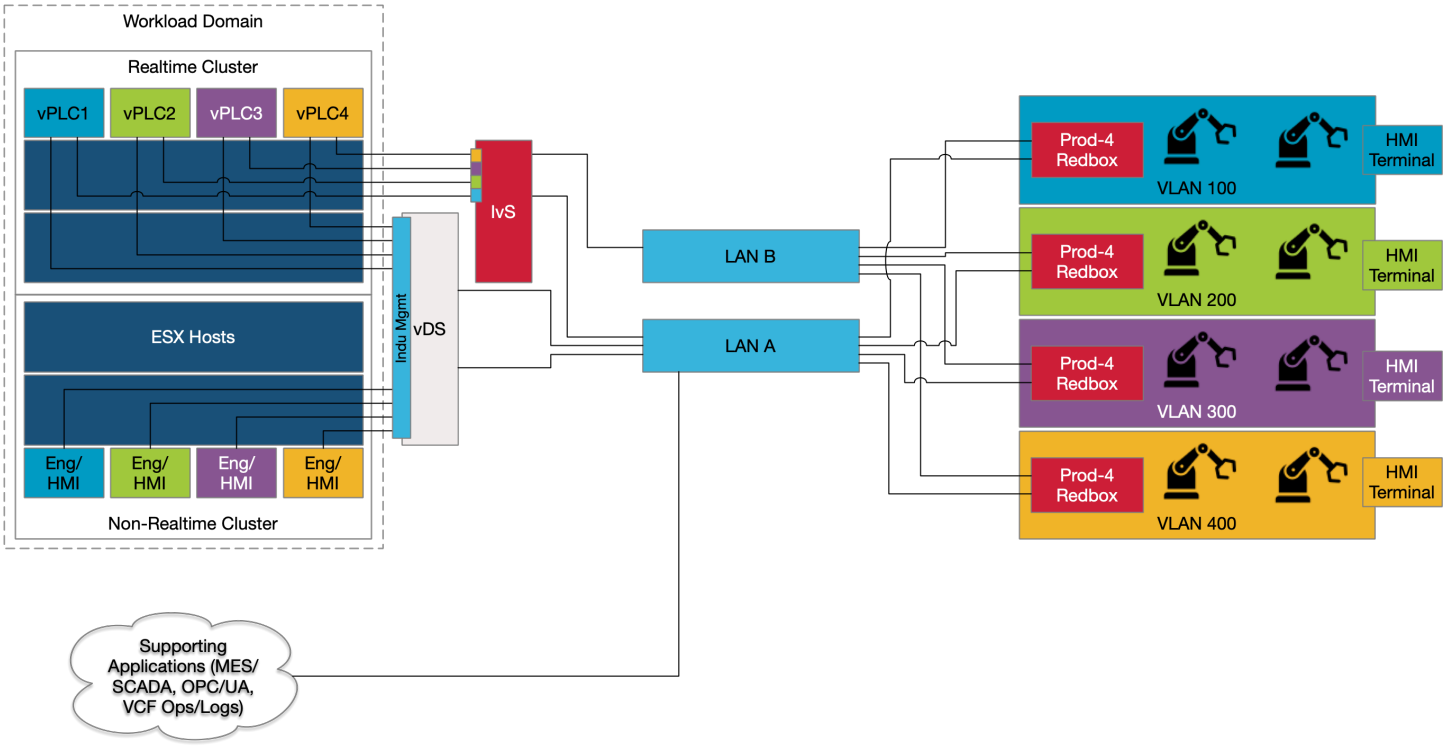
This design capitalizes on VCF 9's enhancements to support real-time workloads, such as vPLCs, on a VCF infrastructure. The typical architectural approach involves creating a separate Workload Domain for each location to ensure operational and physical separation and clearly defined responsibilities. Each domain generally comprises two clusters: one dedicated to real-time workloads (vPLCs) and another for non-real-time workloads (e.g., Engineering Solutions, HMIs).

Networking utilizes the Parallel Redundancy Protocol (PRP), which needs two distinct LAN Fabrics, referred to as LAN A and LAN B. LAN A is typically the LAN which connects to other Services on the network.

The overall general architecture is structured as follows:

Figure 176: Real-time OT Overview

The example illustrates four virtual PLCs (vPLCs), each managing a separate production line. These lines are segmented into distinct VLANs, represented by different colors. Each production line is composed of a vPLC, its corresponding Engineering Workstation (which also hosts the Human-Machine Interface or HMI), and PROFINET devices. These devices operate behind an industrial switch configured as a Redundancy Box (Redbox).


Workload Domain
Realtime Cluster
vPLC1
vPLC2
vPLC3
vPLC4
ESX Hosts
Eng/ HMI
Eng/ HMI
Eng/ HMI
Eng/ HMI
Non-Realtime Cluster
Indu Mgmt
vDS
IvS
LAN B
LAN A
Prod-4 Redbox
HMI Terminal
VLAN 100
Prod-4 Redbox
HMI Terminal
VLAN 200
Prod-4 Redbox
HMI Terminal
VLAN 300
Prod-4 Redbox
HMI Terminal
VLAN 400

Supporting Applications (MES/ SCADA, OPC/UA, VCF Ops/Logs)

Real-Time Platform Design Considerations
To effectively execute real-time workloads, it is crucial to establish a cluster that guarantees deterministic performance. The platform design is composed of several layers, each requiring distinct considerations during the planning phase of the target platform.

Server Design and Settings
Reliable Realtime Workload Platform performance hinges on properly configured ESXi Hosts. Incorrect configurations can lead to timing issues in Realtime VMs, causing PLC and production stops. To prevent this, a valid server configuration is crucial.

For easy integration, a single-socket server with at least 32 cores is recommended. This setup avoids NUMA architecture complexities and provides enough cores to run up to 12 PRP vDANs, which is the current maximum supported.

Design Attribute	Details
Server CPU Make and Model	Today, only Intel XEON Gold and higher CPUs are supported. Only P-Cores are supported. Consult VMware HCL for detailed BOM
Server Performance Settings	To ensure maximum performance, deactivate all power-saving mechanisms in the server's BIOS and firmware. Set the workload profile to "manual" or "custom," depending on your server vendor, and configure all relevant settings to "High Performance" or similar. Specifically, disable features such as C-State and frequency shifting capabilities like Turbo Speed. Do not use a pre-defined High Performance or Turbo Profile. Example Settings
Realtime Networking Adapter Support	Choose a Network Adapter with Support for ENS. Consult VMware HCL
Network Adapter count	At least 5 NICs: • Two for VCF management and Non-RT VM Traffic and • Two for Real-time PRP-Profinet Traffic. • One for TSA passthrough More Adapters are possible if target design characteristics demand this (max. 4 per IvS). See Maximum Configuration Tool Implement workload separation distributed switch mode. Detailed Network configuration Designs can be found here

Parallel Redundancy Protocol Design
Industrial networks require low latency and minimal jitter, standard Active/Passive redundancy methods are inadequate for maintaining production continuity during component failures. To address industrial networks requirements, the IEC 62439-3 Parallel Redundancy Protocol (PRP) was developed to ensure fault tolerance for network traffic.

Parallel Redundancy Protocol Overview

Industrial networks require low latency and minimal jitter. Because standard Active/Passive redundancy methods are inadequate for maintaining production continuity during component failures, the IEC 62439-3 Parallel Redundancy Protocol (PRP) was developed to ensure fault tolerance for network traffic.

Parallel Redundancy Protocol (PRP) is a Layer 2 redundancy protocol designed to eliminate service interruptions, packet loss, and delays. Unlike traditional protocols like spanning tree that take time to reconverge after a failure, PRP achieves instant failover by transmitting identical data simultaneously across two separate active networks. One network is designated as LAN A and another as LAN B. To manage this dual-traffic, PRP uses sequence numbers to identify and drop duplicate frames. Latency between the two networks must be matched. If one network is slower than the other, the sequence tracking fails and the connection can drop.

Key PRP Components:

- **DAN (Dual Attached Node):** A PRP node that connects to both LAN A and LAN B simultaneously.
- **SAN (Single Attached Node):** Standard devices (like PCs or servers) that do not natively support PRP.
- **RedBox (Redundancy Box):** Often an industrial switch with PRP support (or a specialized NIC adapter), this allows SANs to join a PRP network.

VMware Industrial vSwitch (IvS) Architecture

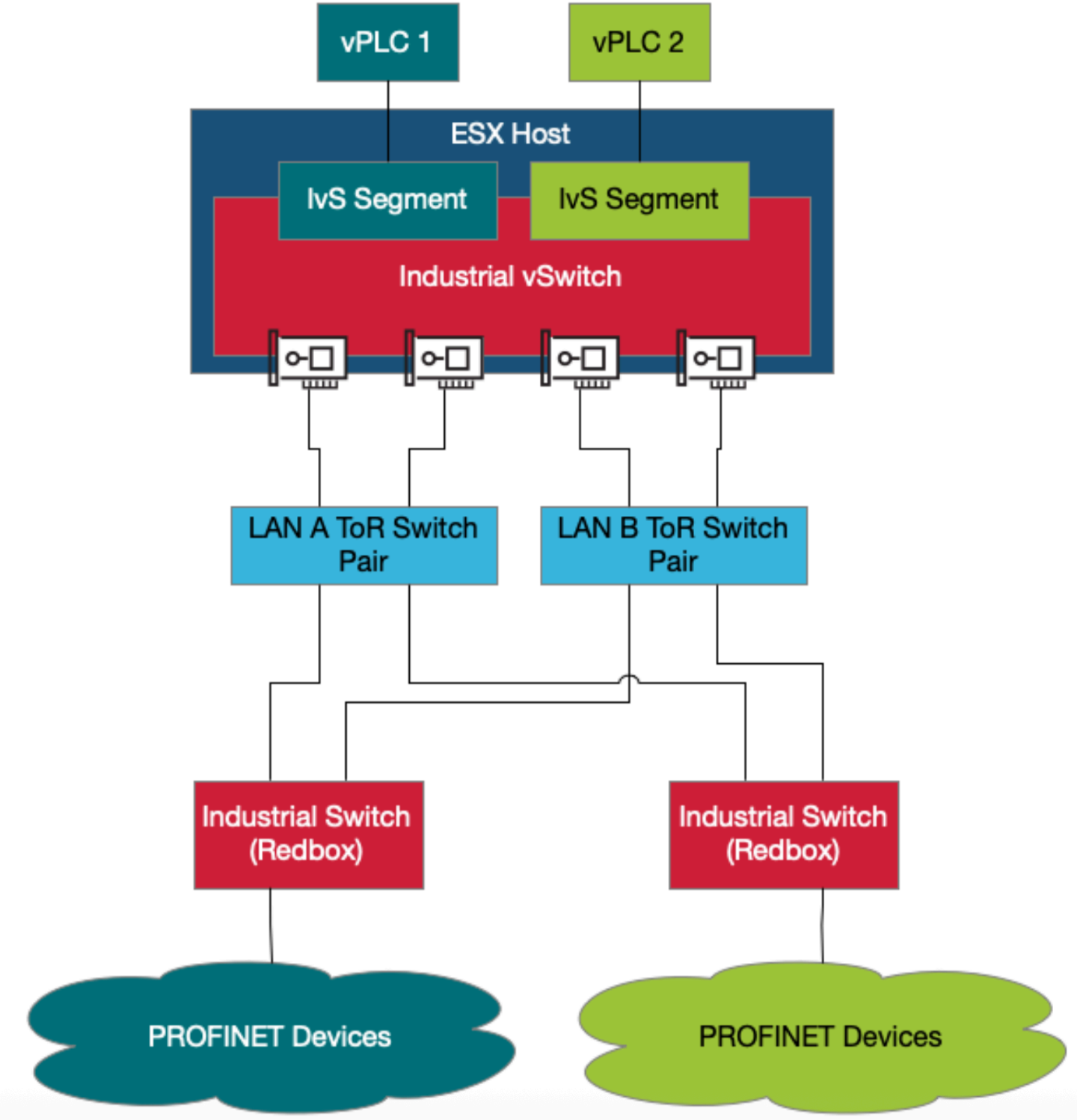
Available since VCF 9, the Industrial vSwitch (IvS) can function as a RedBox, enabling virtual machines to establish redundant PRP connections.

The IvS comprises two primary components:

1. The IvS incorporates a VMware Distributed Switch, which is configured at the vCenter level. Each ESXi Host within the IvS contains a Data Plane, which is optimized for stringent latency demands and offers support for Profinet and PRP.
2. NSX Manager facilitates the creation and management of the IvS configuration, including all network-related settings. This encompasses NIC failover settings, establishing security and functional boundaries through Transport Zones, and segmenting each production cell with isolated VLANs. activate or deactivate PRP is based on the NSX Segment Profile using the NSX Manger.

Both the physical and virtual layers must strictly control latency and jitter. The IvS, designed for the virtual layer, aims to maintain latency within **250 microseconds** with a jitter of **±10%**. These exact same rules must be applied to the underlying physical layer to ensure stability.

Figure 177: Parallel Redundancy Protocol Topology Overview



The general architecture, illustrated in the preceding diagram, routes each ESXi host uplink belonging to the IvS to either LAN A or LAN B.

The following guidelines are important for PRP design:

- **Minimum Requirement:** PRP requires at least one uplink that is connected to LAN A and one connected to LAN B.
- **Best Practice:** For enhanced redundancy and to eliminate single points of failure, two uplinks per LAN are recommended.
- **Uplink Limitations:** You can use a maximum of two uplinks per LAN. Because the IvS only supports failover mode for redundant NICs per LAN, these dual uplinks operate in an active/standby failover sequence.
- **PRP Management:** PRP can be dynamically activated or deactivated using NSX Segment Profile. More about Segment Profiles here

Parallel Redundancy Protocol Use Cases

Not all industrial network connections need PRP. The specific design choice depends on the needs of your particular use case. Furthermore, each production cell should be isolated with its own dedicated VLAN to create distinct security and responsibility boundaries.

Table 456: Parallel Redundancy Protocol Design Consideration

Consideration	No PRP Needed	PRP Support Needed
Topology	• IO devices and vPLC that are located in close proximity (example: an IPC mounted in a production cell shelf). • End-to-end redundancy is not feasible (example: only one WAN connection is available).	vPLC and IO devices are located across locations with multiple network hops.
Failover Tolerance	Failover latency of approximately 100 ms is acceptable.	Zero packet loss or delay is required.
App-Level Alternatives	Application-level service redundancy is available (example: time services for Safety PLCs).	NA
VM Compatibility	Windows VMs / standard Linux VM without RT Kernel	Linux VMs with RT Kernel and vmxnet3 Adapter driver 1.9.0 or later.

Traffic Routing and Configuration Details

- **Non-PRP Traffic Handling:** Typically, LAN A serves as the primary connection, handling all traffic, including non-PRP traffic originating from SANs located behind a RedBox. Although LAN B is technically identical, using both LAN A and LAN B to connect to standard services adds unnecessary complexity for operation, maintenance, and troubleshooting.
- **Virtual Machine SANs:** Virtual machines without PRP support (such as Windows VMs used for HMI) only communicate on LAN A. In the event of a persistent failure on LAN A, this non-vDAN traffic can be manually changed to LAN B (see command syntax in Troubleshooting).

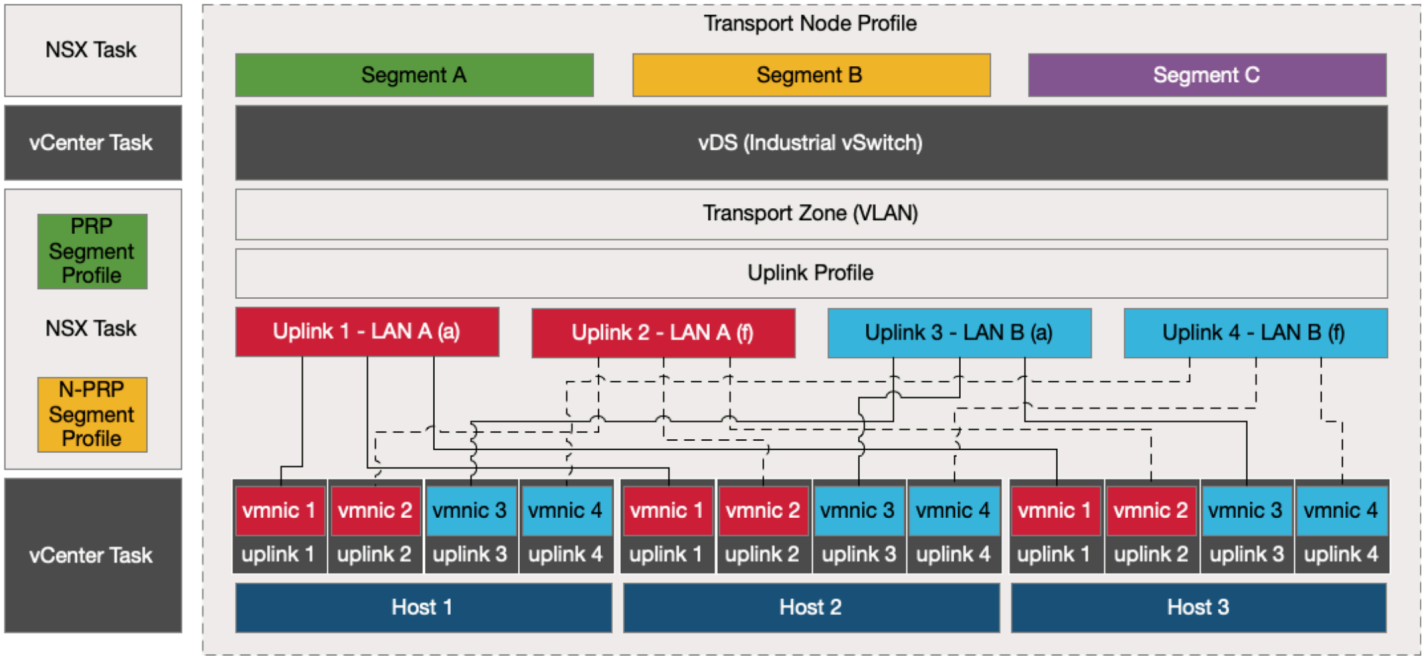
Industry vSwitch Integration with vSphere Details

The IvS is composed of the following key components:

- **vDS (vCenter Level):** The foundational distributed switch. Must be version 9 or later.
- **VLAN Transport Zone (NSX):** Defines the scope for VLAN-backed segments.
- **Uplink Profile (NSX):** Decouples the vDS uplink from the NSX uplink and specifies the active and failover uplinks.
- **Segment Profiles (NSX):** One or more profiles that are created, which may include Parallel Redundancy Protocol (PRP) settings.
- **Transport Node Profiles:** Defines, which Hosts shall inherit the IvS Configuration
- **NSX Segments:** Created and assigned a specific IvS Segment Profile, segregated via VLANs

Refer to the official VCF Documentation to create an Industrial vSwitch. Refer to the Appendix for Host and NSX config details.

Figure 178: Example Three Node IvS enabled Workload cluster



Feature	Setting	Remarks/Considerations
vMotion	Partly	vMotion for a vPLC (virtual Programmable Logic Controller) is only permissible during planned maintenance, such as when no vPLC project is active. This restriction is due to the short STUN generated by vMotion, which exceeds the maximum network interruption tolerance for vPLCs. non-Realtime VMs, including for SCADA monitoring, Engineering, or HMI can use vMotion.
Cold Migration	Supported	

Fully automated DRS	Not Supported	non-Realtime VMs, including those for SCADA, Engineering, or HMI, can safely utilize vMotion. Running a vPLC project prevents the use of vMotion. As a result, DRS-initiated vMotion, which is employed by Fully Automated DRS, can only be used if vMotion operation can be managed.
Partially Automated DRS	Supported	Supported. VMs should be operationally grouped to align host maintenance cycles or outages with manufacturing cells.
Manual DRS	Supported	Supported. Should be considered over partially DRS.

vPLC Storage Design

Virtual PLCs (vPLCs) must offer the same resilient features as their physical counterparts to ensure continuous production. Physical PLCs utilize a dedicated hardware capacitor to detect power loss and locally store transient data. Because vPLCs lack this physical capacitor, vendors have developed a software-based method to achieve similar availability.

The most intensive storage requirement involves saving "retain and remanent" data to disk. Remanent data captures the state and position of downstream I/O devices before a vPLC failure. This data is critical because it allows downstream devices to seamlessly resume operations without requiring a manual reset. To ensure this availability, vPLCs continuously save this information directly to disk after each program cycle, based on parameters defined by the PLC software developer.

The constant saving of remanent data results in a continuous, 100% I/O write pattern per vPLC. Current data indicates a write rate of approximately **1 Gb/s per host**.

In scaled-up environments, this constant data stream can significantly load the storage backend. To manage storage cluster fragmentation, mitigate contention, and reserve capacity for supplementary workloads (like non-real-time VMs), you must adhere to the following constraint:

- **Scale Limit:** Access to the Storage Cluster is limited to a maximum of **20 hosts**.

Disk latency thresholds are typically dictated by the vPLC vendor, as they depend entirely on how their specific Retain Data mechanism is implemented.

Every storage technology presents unique advantages and disadvantages. Your selection depends on performance requirements, staff skill sets, site technology, and organizational processes.

> **Note:**
>
> *While VCF 9 is compatible with any supported storage solution, VMware typically recommends vSAN due to its operational advantages and because all platform testing was conducted using it. Consolidating both platform and storage operations under a single team significantly accelerates time-to-resolution for troubleshooting.*

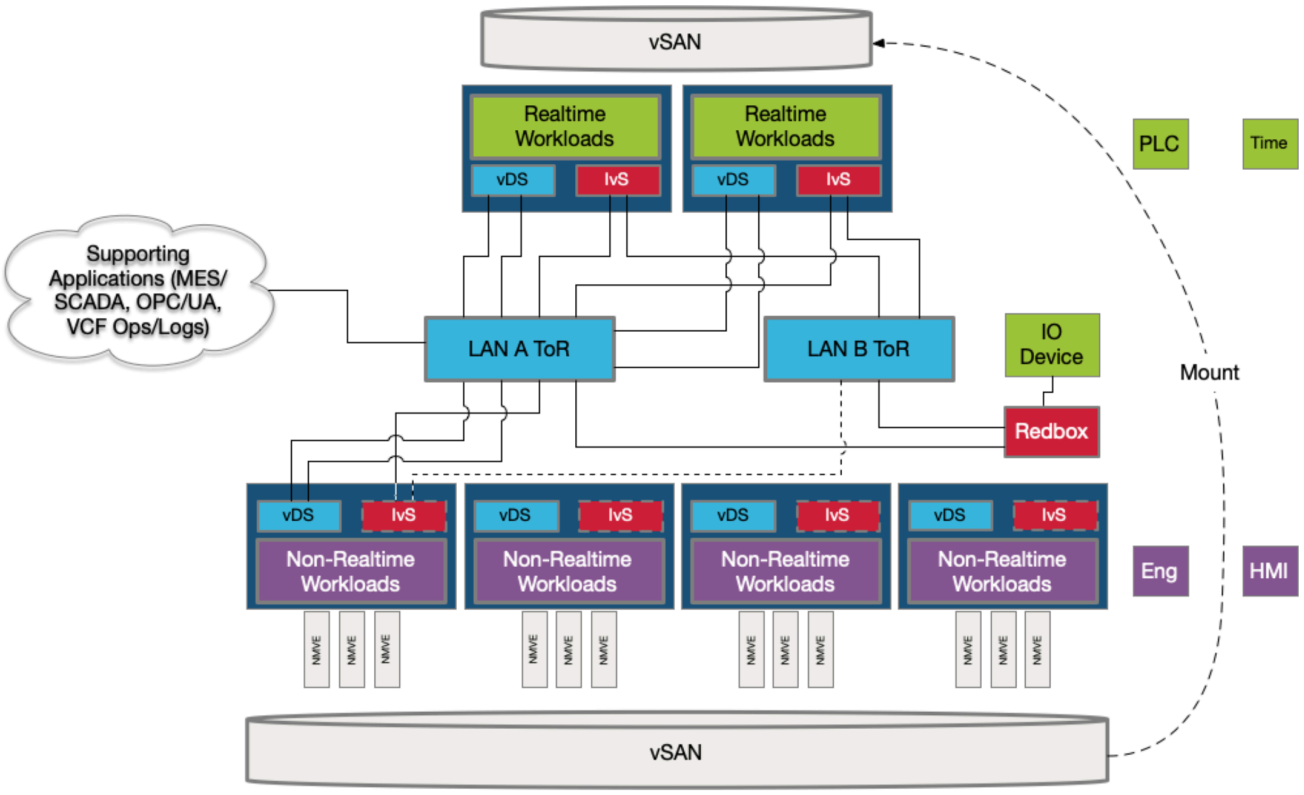
For discussing different Storage models, also consult our documentation.

Storage Solution	Supported	Consideration
Fibre Channel	Yes	A dedicated Storage Area Network (SAN) is necessary, requiring skilled labor and specialized network components. Asynchronous scaling, which involves adding more workloads without accounting for increased storage requirements, can lead to high disk latency and significantly impact production.

iSCSI	Yes	To optimize performance and availability, consider using dedicated networks for storage traffic, depending on bandwidth and availability requirements. While iSCSI is a SAN protocol that shares performance and workload dependencies with Fibre Channel, its reliance on standard L3 Ethernet traffic means that network issues can lead to performance bottlenecks and availability issues.
NFS	No	Due to the nature of NFS as a shared FIlesystem, we cannot support this technology at the moment.
vSAN	vSAN MAX	For real-time workloads, which are currently restricted to Compute-only Clusters, implement vSAN Max. A robust networking infrastructure is crucial for vSAN. Always follow Broadcom's vSAN Best Practices and refer to the official vSAN Design Guide.
	HCI	non-real-time workloads such as Engineering, HMI, or SCADA systems, they are not suitable for vPLCs or Time Services in 9.1.

Recommended vSAN Architecture.

Figure 179: vSAN MAX supporting vPLC



The manufacturing edge design features two distinct clusters:

1. **vSAN / HMI Cluster:** Dedicated to providing vSAN storage and running non-real-time workloads, such as engineering workstations, HMIs, and other production-dependent applications.
2. **Real-Time (Compute-Only) Cluster:** Designed for deterministic RT VMs, including vPLCs and supporting applications like Time Services

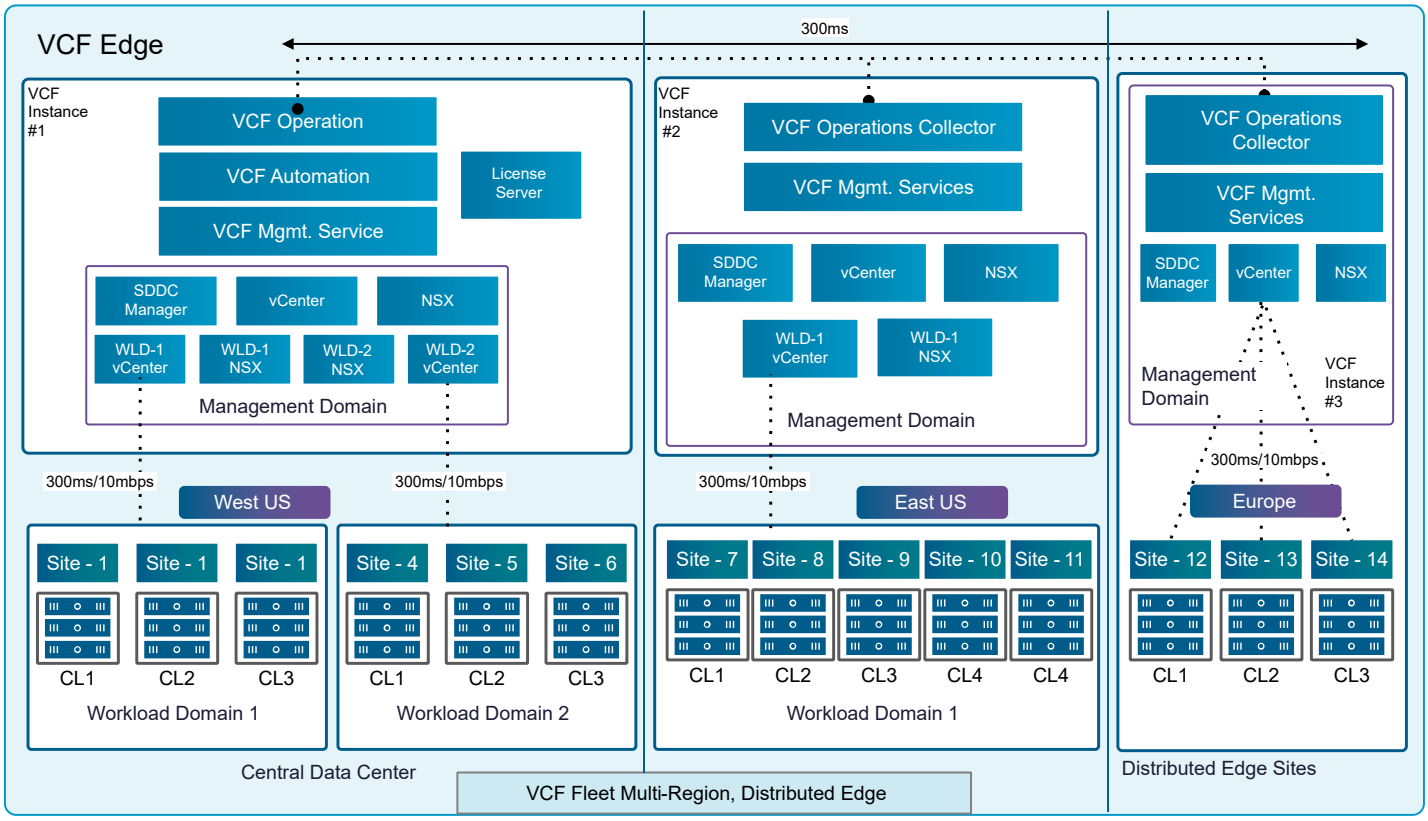
The Real-time Cluster accesses the vSAN Datastore that is provided by the vSAN Non Real-time Cluster,. While architecturally L3 routed vSAN is a valid design, it is not advisable to use a Layer 3 routed vSAN due to potential unpredictability in L3 network performance.

Disk latency thresholds are typically defined by the vPLC vendor, as these depend on the specific implementation of the Retain Data mechanism. Stretch cluster can be used for HA/DR design. For a stretched Cluster configuration, refer to the vSAN Design Guide.

Edge Deployments
Edge Deployments use standard VCF Edge Deployment procedures. For comprehensive deployment instructions, consult our Edge Design Guides. A key consideration is determining the necessity of a dual-site deployment versus the sufficiency of a single-site deployment.

Modern multinational manufacturers operate on a massive scale, relying on numerous distributed edge locations that span across multiple geographical regions to power their global operations. To navigate the operational complexities, manufacturers employ a sophisticated dual-level management approach that seamlessly combines high-level centralized control with dedicated regional administration to optimize both enterprise-wide efficiency and scalability. This strategy features localized control within each specific region, ensures compliance with strict regional data regulations but also enhances operational resilience against potential inter-region network disruptions. Ultimately, this robust architectural framework guarantees the reliable, ultra-low latency performance that is required directly on each distributed factory floor to support advanced, real-time manufacturing use cases. Edge Pattern 4 is most suited for this design.

Figure 180: VCF Fleet with Multi Region, Distributed Edge



Within each VCF instance, each factory is represented as a VCF workload domain (WLD). A workload domain can consist of a single or multiple availability zone.

Use a VCF 9 WLD with Industrial vSwitch with or without PRP in one Availability Zone

For smaller or dependent deployments, consolidating all workloads within a single datacenter can be an effective strategy. This approach, likely the most common architecture, simplifies security by confining all necessary components, networks, and VLANs to one site. Details about Single Site deployments can be found here.

Considerations:

Design Component	Consideration	Detailed Design Options
Networking	• VLANs, and Networking for PRP can be limited to a single site. • Network latency and jitter are only relevant within a single site. • Inter-site links can be disregarded. • A long downtime can be expected in the event of a disaster.	Network Design
Storage	• Storage is not required to be stretched or replicated. • Storage latency is only a concern on a single site. • Different Storage technologies can be used per site • No extra license fees for replication are necessary if external storage is used. • Workload placement is simplified.	Storage Design
Cluster configuration	• Facilitates simplified workload distribution across a site. • Eliminates dependencies between sites.	Cluster Design
Operations	Each site functions autonomously, preventing conflicts related to essential downtimes or lifecycle management. However, this independence necessitates separate maintenance for each site's vCenter, NSX and inventory, including individual configuration and documentation of monitoring and alarming systems. Consequently, implementing potential failover scenarios becomes more complex.	N/A

Use a VCF 9 WLD with Industrial vSwitch with or without PRP in multiple Availability Zones

Deploying industrial workloads across multiple availability sites is crucial because it significantly enhances service availability. This multi-site deployment provides robust protection against various disasters, ensuring continuous operation and minimizing downtime.

When utilizing multiple locations, it is not always necessary to establish a single management plane, workload domain, or clusters across these sites. However, this approach is worth considering. A disaster could bring down a centralized, non-stretched alarming system, leading to alarming outages and impacting downtime. Details about multiple Site Deployments can be found here.

Another valid approach is to operate workloads for different production lines in separate locations, while still allocating resources for a manual failover in the event of a disaster. Consider those Architectural decisions.

For a detailed deployment guide, focused on vSAN Stretched Cluster, see here.

The following table should help to decide the right deployment model.

Component	Consideration
vPLC	• **Production Stoppages:** Any power down or halt, including those caused by vMotion stuns, results in a production stop. • **Disk I/O Load:** vPLCs can generate significant disk I/O load due to writing remanent data after each cycle. Replicating this data across an Inter-Site Link could negatively impact other workloads. • **Stateless Operation:** vPLCs can be treated as stateless machines. Upon power-on, the project is loaded into the vPLC. Having cold-standby vPLCs per site allows a fast restart of the production.
Networking	• **Network Consistency:** All sites must have access to all networks and VLANs, including storage and industrial networks. • **Inter-Site Bandwidth:** Sufficient bandwidth is required to handle all replication traffic within specified timeframes (synchronous vs. asynchronous). • **Firewall Rules:** Firewall rules are necessary to allow management and monitoring systems to access workloads across all sites.
Storage	• **High-Performance Inter-site Link:** The necessity of a high-performance link between sites is contingent on the specific storage technology employed. • **vSAN Implementation:** – **Storage Profiles:** When utilizing vSAN, it is recommended to create multiple storage profiles tailored to diverse workload requirements. For instance, synchronous mirroring for vPLCs is only beneficial if the remanence data is deemed critical also for disasters. – **Availability:** To meet availability demands with vSAN, a minimum of three locations is required. Further details can be found here. • **Synchronous Mirroring:** Designing for synchronous mirroring of remanence data needs careful attention to the potential I/O load on both the disks and the inter-storage link. Each virtual PLC (vPLC) contributes significantly to this load, with approximately 1000 IOPs and typically 6-7MB of throughput, all of which are write operations. These figures are subject to variation based on the vPLC vendor, technology, and data volume. This combined activity results in an I/O load of nearly 1 Gb per host. Therefore, deploying a high number of vPLCs across multiple hosts risks generating enough traffic to potentially saturate the storage link that connects the sites. A saturated Storage Link leads to higher Disk Latency and therefore potential PLC Stops or inconsistent data.

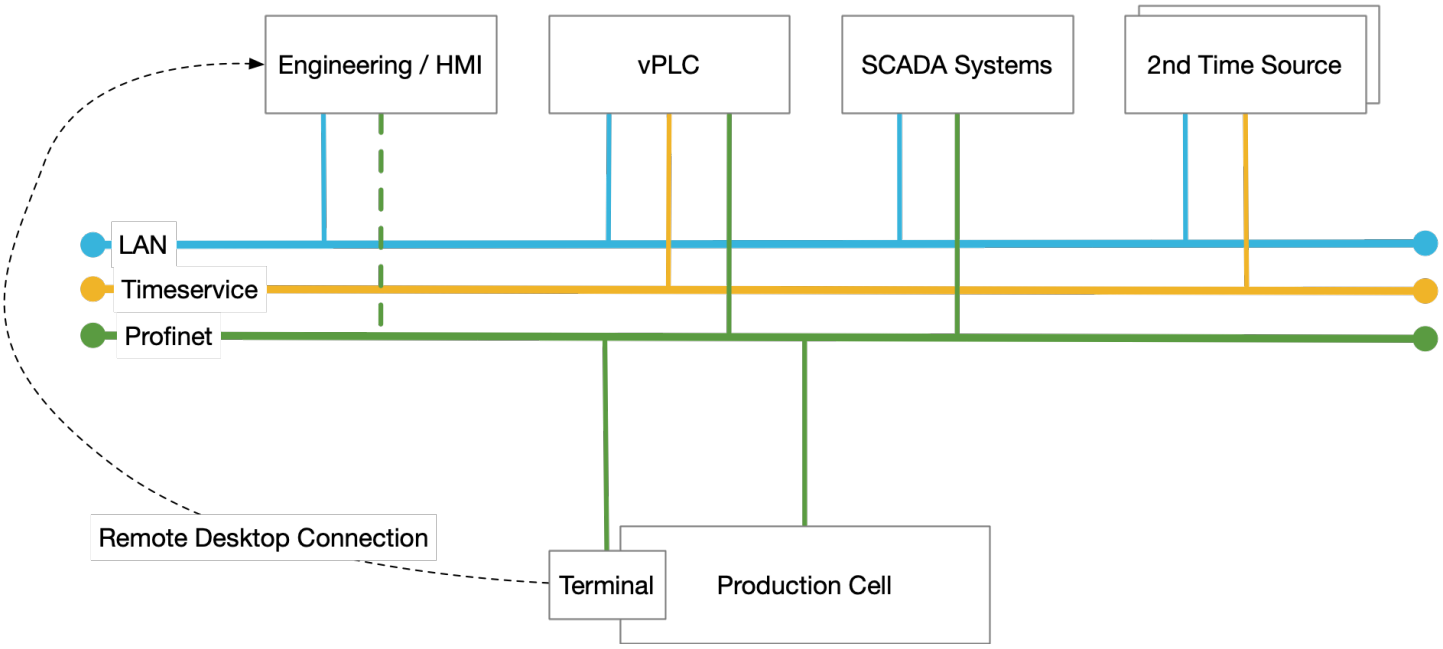
Network Design Overview

To prevent production outages, a reliable network is crucial when separating PLCs from their PROFINET IO Devices.

Reliability is achieved by using the PRP Protocol, enabling full active/active networking. Also, if a Safety vPLC is used, a separate IVS-backed network handles the Time Services.

For management, monitoring, and connecting to SCADA systems for production planning and data collection, a regular Non-realtime Network is employed.

Figure 181:



In a typical manufacturing setting, the logical network connectivity features an industrial network connection that links the production cell (or line) to an industrial vSwitch. This vSwitch, which is not considered a PROFINET Device, supports PROFINET as a bus protocol and includes PRP for redundancy. A vPLC, located elsewhere, is also connected to this industrial vSwitch.

A second industrial vSwitch facilitates the transmission of a time signal between the vPLC and a dedicated Time Server. Both components are supplied by the same PLC vendor and are essential for Safety vPLCs. The vendor also provides the detailed implementation for the time service.

Network Feature and Traffic	Details
Industrial vSwitch	The Industrial vSwitch facilitates deterministic PROFINET communication and end-to-end monitoring between a virtual PLC and PROFINET Devices on dedicated industrial networks. A maximum of two Industrial vSwitches are permitted per Host. See Configuration Maximum Tool Within the production cell, an industrial Switch with PRP Support, functioning as a RedBox (Redundancy Box), is deployed. SCADA Systems, which are used for monitoring and production planning, are also connected via Profinet, depending on the specific solution implemented. That may include OPC/UA Traffic, like Pub/Sub traffic.
PRP Support	PRP (Parallel Redundancy Protocol) requires two independent network fabrics, which are designated as LAN A and LAN B. Network traffic is transmitted simultaneously over both paths. Each LAN can accommodate multiple Network Interface Cards (NICs) configured for failover. Currently, only one Industrial vSwitch per Host supports PRP.
Regular Network Traffic	Non-Realtime network traffic (VM Traffic, vSAN, vMotion, and etc). This includes OPC/UA Traffic, like pub/sub. See Network Fabric Models

Setup Real-time VMs

Achieving Deterministic Behavior in Manufacturing Edge Designs

Programmable Logic Controllers (PLCs) demand deterministic behavior to maintain consistent speed and latency for their operations. Failure to meet this requirement can lead to PLC program halts, production downtime, and significant financial losses.

The entire technology stack must support the deterministic execution of PLC programs. To guarantee deterministic behavior, virtual machine resources must be configured with guaranteed allocation. Real-time capabilities are built upon guarantee the following three fundamental pillars:

- **Networking**
- **CPU**
- **Memory**

Real-time and PRP Networking Prerequisites:

For real-time networking, the prerequisites include a VMXNet3 Adapter and the latest driver (version 1.9.0 or higher). This driver is integrated into the upstream Linux Kernel and should be included with all current Linux distributions (from Kernel 6.14 onward). Deterministic networking becomes available for a specific VM when its network adapter is connected to the Industrial vSwitch.

It is important to note that only Linux VMs equipped with this updated VMXNET3 driver are fully compatible with a PRP network. Systems such as Windows VM or linux VM using older VMXNET3 drivers or different virtual NIC types are not PRP capable. Non PRP capable systems can only be connected to a single LAN fabric. In most OT designs, only LAN A of the PRP network as LAN-B is typically isolated.

For virtual machines using industrial bus protocols not supported by IvS (e.g., EtherCAT, EtherNet/IP), DirectPath I/O (PCI passthrough) of a dedicated physical NIC (VMNIC) is required. However, PCI passthrough introduces architectural constraints. Standard vSphere capabilities such as VM snapshot, HA, and vMotion are not available for VM with passthrough NICs. In addition, large number of passthrough NICs imposed a large constraint on the number of available physical NICs on a single host.

Operating System Considerations:

The Guest Operating System requires a real-time Kernel to achieve fully deterministic performance. As of now, only Linux RT-Kernel has been validated. Windows-based vPLC isn't available from PLC partners and has not been validated against the iVS Switch.

When tuning a linux based VM for real-time workload, The following settings are the most important for optimizing latency:

- **Real-Time (RT) Kernel:** Install the Linux RT Preemptive Kernel to ensure predictable behavior for time-sensitive applications.
- **Tickless (NO_HZ):** Implement the tickless feature in the Linux Kernel to reduce kernel overhead. This can significantly improve both performance and application latency. Refer to your specific Linux Distribution's Manual for detailed instructions.
- **Tuned Profiles:** Use the tuned utility, which offers pre-defined profiles such as "latency-performance" or "real time." This tool can also help isolate workloads to specific virtual CPUs (vCPUs) and prevent Linux Kernel scheduling that adds latency. For maximum optimization, a customized tuned profile may be necessary.
- **Large Pages:** Using Large Pages (up to 2 MB) at guest OS Level and Application helps eliminate latency that is associated with memory allocation operations.

For more comprehensive information, consult Broadcom's Performance Whitepaper.

Table 457: VM Real-time Settings

Setting	Value
Virtual Hardware Version	Real-time Networking requires Virtual Hardware Version 22 or higher. Ensure that the virtual hardware version is correct before reconfiguring the VM, as earlier versions may not provide all necessary settings. vPLC vendors might deliver their vPLC images with an earlier virtual hardware version.
Guest Operating System	Linux-based real-time capabilities across compute, memory, and network contexts.
CPU	To achieve 100% reservation, determine the GHz of a single CPU core on your server. Multiply this value by the number of vCPUs allocated to the VM, and then add the resulting figure to the Reservation field.
Memory	Set the checkmark at “Reserve all guest memory (All locked)”
Network Adapter	For the machine network (e.g., Profinet), add a vmxnet3 adapter. Select an IvS Segment for this adapter, as connecting to an IvS Segment configures the VM settings accordingly. Keep IvS latency settings at their default values
Latency Sensitivity	On the "VM Options" page, navigate to "Advanced." From there, under "Latency Sensitivity," choose either "High" or "High with Hyperthreading."
Optional Settings	• Under "VM Options," expand "Advanced" and deselect "Enable Logging." This is an optional parameter that should only be disabled if the VM operates flawlessly. While it may help free up CPU cycles, disabling it prevents VMware Support from analyzing potential issues with the VM. • To disable latency measurement, deselect the "Latency measurement" setting on the IvS-connected network adapter.

Safety Time Source Redundancy Design Options

vPLC Design Option for Safety vPLCs

Human safety PLCs require a redundant time source to prevent time drift from compromising program reliability. In physical installations, this often involves using specialized PLCs equipped with a secondary, built-in PTP clock to serve as a fail-safe.

In a virtualized setting, a virtual PLC (vPLC) initially relies on the underlying ESXi host for its primary time reference. Therefore, a secondary time source must be introduced through an external application. The approach to implementing this redundancy differs based on the vPLC vendor, generally falling into one of two design and implementation models.

- Design 1: dedicated Time Service IvS for time service and dedicated IvS for Profinet-PRP
- Design 2: Single IvS for time service and Profinet-PRP

Both approaches have their pros and cons. Design 1 requires more CPUs to support a second IVS switch, but it offers better protection against latency spikes. This is because time-synchronization processing is mapped to a dedicated CPU on the second IvS, minimizing the risk of slowing down the PROFINET processing. Conversely, Design 2 carries a higher risk of latency spikes but is more resources and operationally efficient, as the same resource pool supports multiple application flows. Refer to the following sections for details on both design options.

Design 1 - Dedicated Time IVS with Time Service VM with PTP passthrough

This design uses a dedicated physical network adapter to obtain a PTP time signal. The adapter, which is attached to a PTP grandmaster clock, is passed through directly to the Time Service VM (TSA). The TSA VM then uses the NIC hardware oscillator to acquire a qualified PTP time signal,. Once PTP time signal is acquired, the TSA distributes it across a VMXNET3 vNIC attached to a dedicated Time Service IvS instance. The primary difference between the Time Service IvS and the Profinet-PRP IvS is their NIC composition: the Time Service IvS can be deployed with zero NICs if TSA redundancy is not required, or with two NICs for host-level TSA redundancy.

Also attached to the Time Service IvS switch are vPLCs local to the host. Each vPLC configured to receive PTP traffic from the local TSA, and for redundancy, a remote TSA server running on a different Real-time host. PTP signal from the local TSA is preferred over redundant TSA running on another host. Synchronization between the TSA and vPLC are based on PTP, but the actual PTP implementation can be vendor-specific.

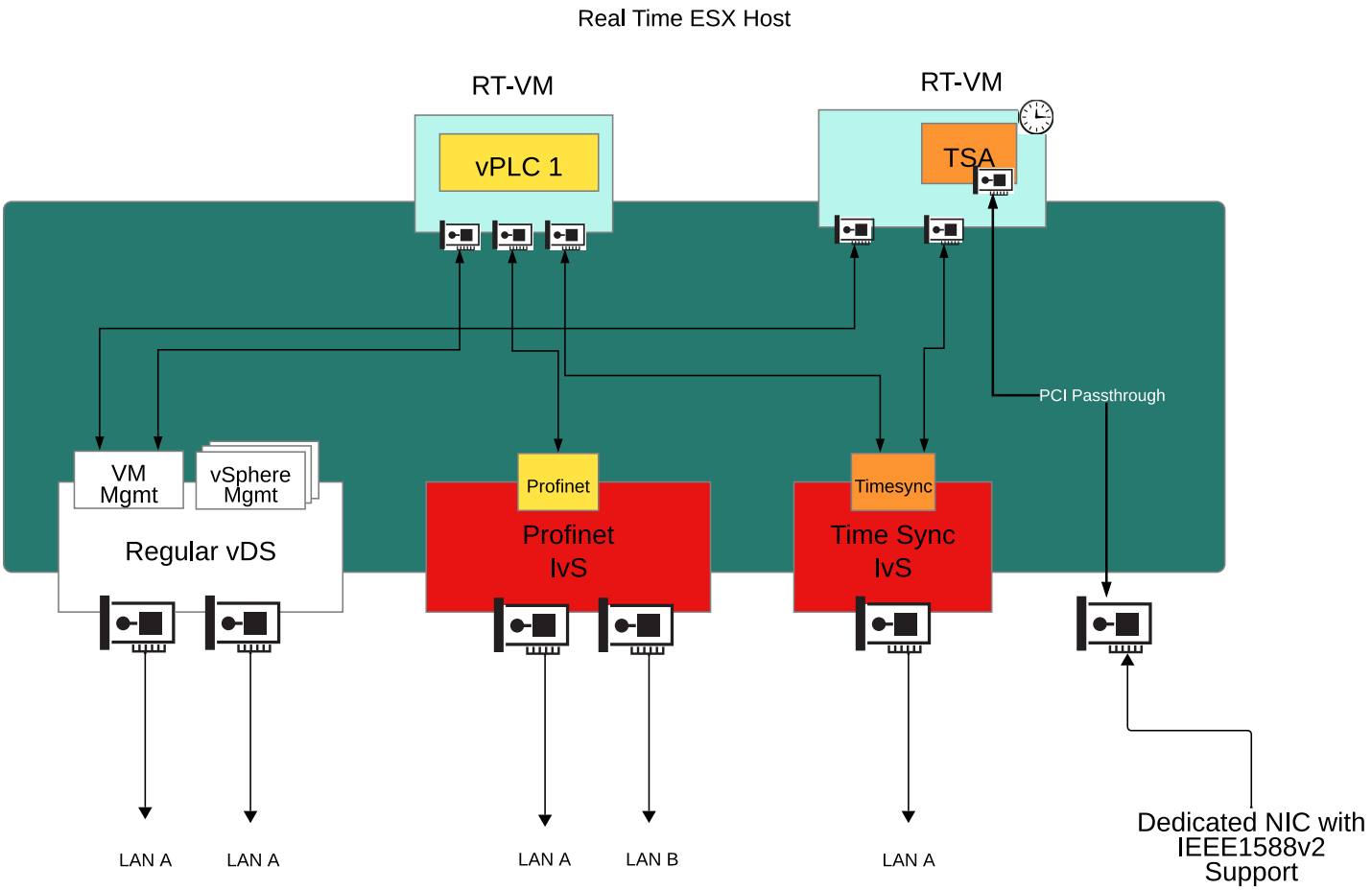
From a resource perspective:

- Two IvS switches per Host
  - One Time-Service IvS
  - One Profinet PRP IvS
- 1 Time Service VM
- 12 vPLCs VMs, each connects to a unique Profinet VLAN.
- All vPLC VMs share the Time Service VLAN on the second Time-Service IvS
- A total of 5 ENS LCores required to support 12 vPLC

Below is a high-level diagram of Option 1:

Figure 182: Time Service Design Option 1

Time Service VM with PTP passthrough


Real Time ESX Host
RT-VM
RT-VM
vPLC 1
TSA
PCI Passthrough
VM Mgmt
vSphere Mgmt
Regular vDS
Profinet
Profinet IvS
Timesync
Time Sync IvS
LAN A
LAN A
LAN A
LAN B
LAN A
Dedicated NIC with IEEE1588v2 Support

Design 2 - Single IVS, different Port group for Time Service

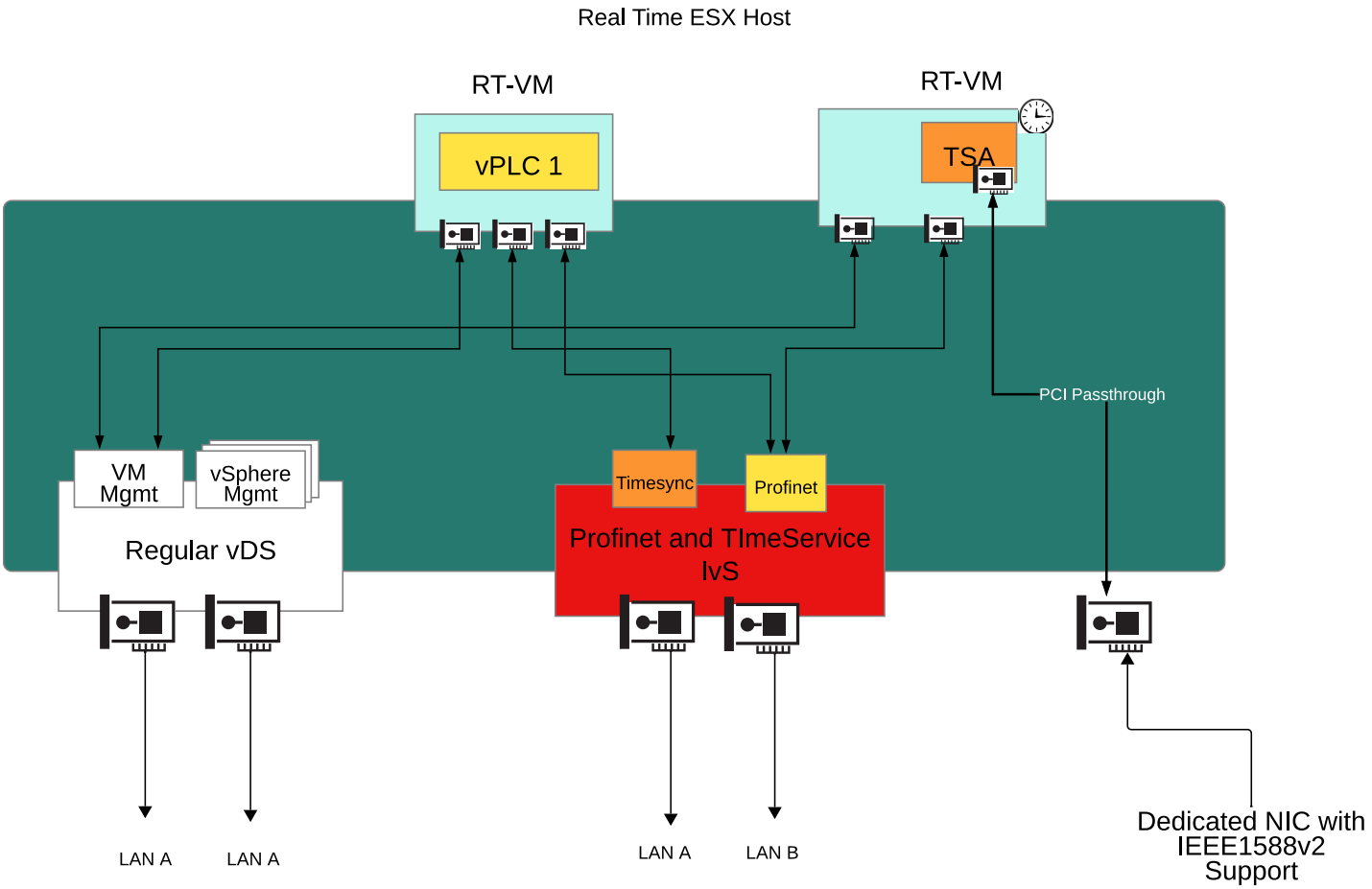
Instead of a dedicated IVS switch, this approach uses the same IVS switch to obtain the PTP time signal. Similar to the previous design, a dedicated network adapter that is attached to a PTP grandmaster clock is passed through directly to the Time Service VM (TSA). The TSA VM then uses the NIC hardware oscillator to acquire a qualified time signal. The acquired time signal is then distributed by the TSA across a VMXNET3 vNIC attached to the IvS.

In this design, the same Industrial Virtual Switch (IvS) used for Profinet-PRP is used, vLAN, and port group is used to segregate the time service and Profinet-PRP traffic. Each vPLC has two VMXNET3 vNICs attached to the IvS switch, one vNIC for PRP and a second vNIC for PTP Time service. For redundancy, vPLC is configured to receive PTP traffic from the local TSA and from a redundant TSA server running on a different Real-time host. PTP signal from the local TSA is preferred over redundant TSA running on another host. Synchronization between the TSA and vPLC are based on PTP, but the actual PTP implementation can be vendor-specific.

From a resource perspective:

- One IvS Switch per Host
  - 12 vPLC, each connects to a unique Profinet VLAN
  - All vPLC VMs share Time Service VLAN, Port group
- 1 Time Service VM per Host
- A total of 4 ENS LCores required to support 12 vPLC

Figure 183: Single IVS, different Port group for Time Service



Industrial vSwitch Monitoring

Monitoring metrics and Alarms are crucial for Day-2 operation. VCF delivers multiple Tools, which are tailored for Real-time Monitoring.

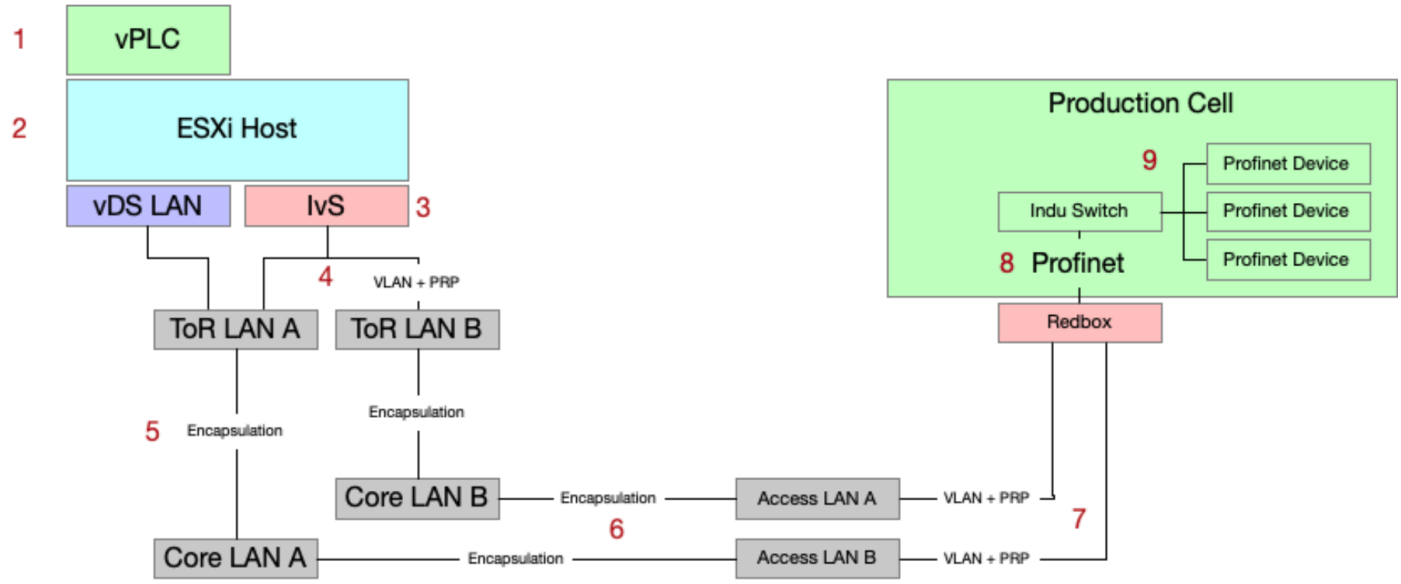
Monitoring and Alarming responsibilities are split across IT and OT, including maintenance staff on the actual shopfloor. It is crucial to define a notification chain to make sure that all necessary parties are involved.

Assuming the VCF Operations team is responsible for the compute and virtualization infrastructure, the OT maintenance team manages the Profinet devices, and the Network Operations team oversees the physical PRP LAN, an example of this division of responsibilities is as follows:

Issue	Alarming System	Responsibility
vPLC Shutdown	VCF Operations Alarm	VCF Operations Team
Link loss vPLC Network	VCF Operations Alarm	VCF Operations Team
IvS Latency Threshold breach	VCF Logs/Ops Alarm	VCF and Network Operations Team
Link loss uplinks	vSphere/VCF Operations Alarm	VCF Operations and Network Operations Team
Encapsulation loss	Network Management Tool/SNMP Trap	Network Operations Team
Profinet Traffic (BUS Failure)	SCADA Tools	VCF Operations Team, Network Operations Team, OT Maintenance Team
Profinet Devices (loss, error, etc.)	SCADA Tools	OT Maintenance Team

The image below illustrates the logical structure of a manufacturing platform incorporating vPLCs. The entities requiring monitoring and troubleshooting are identified by the red numbers. The accompanying table details the objects to be monitored or troubleshot, along with the responsible team. In cases where multiple teams are listed, each team manages its respective components in close coordination.

Figure 184: Component Level Monitoring for the IvS solution



Object to monitor	Level	Tool	Leading Team
1	VM	VCF Ops, vCenter	VCF Ops Team
	Application	Vendor-specific Engineering Tool, VCF Ops	Shopfloor Automation Team
2	Host	VCF Ops, vCenter	VCF Ops Team
3	Industrial vSwitch	VCF Ops, VCF Logs	VCF Ops Team
4	Network Links	VCF Ops, vCenter, VCF Logs	VCP Ops Team, Network Ops Team
5	ISL	VCF Ops (Network MP), VCF Ops for Networks, Network vendor-specific Toolset	Network Ops Team
	Encapsulation	VCF Ops - Networks, Network vendor-specific Toolset	Network Ops Team
6	ISL	VCF Ops (Network MP), VCF Ops for Networks, Network vendor-specific Toolset	Network Ops Team

	Encapsulation	VCF Ops for Networks, Network vendor-specific Toolset	Network Ops Team
7	ISL	VCF Ops for Networks, Network vendor-specific Toolset	Network Ops Team
	PRP	VCF Ops for Networks, Network vendor-specific Toolset	Network Ops Team
8	ISL	Network vendor-specific Toolset	Network Ops Team
	Profinet	SCADA Toolset	Shopfloor Maintenance Team
9	Network Link	Network vendor-specific Toolset	Network Ops Team
	Profinet	SCADA Toolset	Shopfloor Maintenance Team, Shopfloor Automation Team

Appendix: Manufacturing

Troubleshooting Commands and Real Time Network Adapter Settings

Host Setup for Network Adapter

Realtime Networking Adapter Setting	Deactivate DRSS in the driver: $vmkload_mod -u nmlx5_core **#unload NIC Driver** $esxcli system module parameters set -m nmlx5_core -p "DRSS=0" **#deactivate DRSS** $vmkload_mod nmlx5_core **#reload NIC Driver** $pkill -HUP vmkdevmgr **#Allow ESXi Kernel rescan Adapter** Replace “nmlx5_core” with the driver name of your Adapter. Check the actual Parameter name and current setting with: $esxcli network nic list **#find the name of the driver module** $esxcli system module parameters list -m <module_name> Find more information here.

Troubleshooting Commands

Set the sampling rate in the IvS Segment Profile to "1" when taking latency measurements.

The following commands should be executed on the ESXi Host, and the outputs are specific to that host.

Command	Comment
nsxdp-cli vswitch instance list	All distributed switches are displayed. vPLCs, equipped with the correct NIC, are linked to a hidden vmnic (vmnic128). This specific vmnic handles PRP packet duplication and discard.
nsxdp-cli ens prp vdan list | awk 'NR==1 {print} $3 == *VLANID*'	Shows vDANs in a specific VLAN and which uplink is used for LAN A and LAN B.
nsxdp-cli ens prp stats vdan list | awk 'NR==1 {print} $3 == *VLANID* {print; getline; print}'	Shows detailed information about traffic on LAN A and LAN B for all vDANs in the specified VLAN.
nsxdp-cli ens prp node list | awk 'NR==1 {print} $3 == *VLANID*'	Shows all PRP Nodes on a specific VLAN. Shows, whether they are Dual- or single attached
nsxdp-cli ens switch list	Displays a list of all ENS Switches that are associated with a Host. The number in the "swID" column is required to identify the specific Switch for certain commands, typically indicated by the -s parameter.
nsxdp-cli ens port list -sX	shows the number of tx and rx queues per vnic (must be 1)
nsxdp-cli ens latency system clear -s 0	Before running Latency checks, clear the statistic counters.
nsxdp-cli ens latency system dump -s 0	Shows a latency histogram for all VMs, connected to the IvS. Latency has to be below 250us.

pktcap-uw --uplink|switchport vmnicX|PortID -o filename.pcap	Collect Packet Captures for in- and outgoing packets, collected at the IvS Switchport, or at the Uplink Port. See KB 341568 for details.
nsxdp-cli ens prp tunable get	Shows actual settings for PRP Redbox
nsxdp-cli ens prp tunable set -dl LANB	Switch active LAN for non vDAN Traffic to LAN B

To monitor latency on an ESXi Host, configure a threshold warning. This warning sends messages to logging facilities, which can then trigger alarms on a syslog collector, such as VCF Logs. To create these alarms, set a filter for "LatencyType" or "LatencyValue".

Setting	Remarks
nsxdp-cli ens latency lcore logchannel set -s <sw_id> --enable/--disable	Activates or deactivates Latency threshold on IvS. Get the correct Switch ID with “nsxcli -c get ens switch list” on an ESXi Host
nsxdp-cli ens latency lcore logchannel tunable get	Retrieve current settings
nsxdp-cli ens latency lcore logchannel tunable set --help	Detailed information about settings
nsxdp-cli ens latency lcore logchannel tunable set -ut 65	Set threshold to 65us

Setup Tasks for an Industrial vSwitch.

vCenter Tasks to create an Industrial vSwitch.

Task	Setting	Remarks
Name and location	Select a meaningful name and select the Datacenter	
Add new Distributed Switch	Version 9.0.0 and later	IvS is introduced with VCF 9
Configure Settings	Define the Number of Uplinks and Select “Enable Industrial vSwitch”	Up to 4 uplinks are supported
Configure vSwitch	Tag each uplink to be part of either LAN A or LAN B	

NSX Configuration for configuring an Industrial vSwitch

Task	Sub task	Setting	Remark
Check vCenter connection		System->Fabric->Compute Managers	“Registration Status” and “Connections Status” need to be green
Create a Transport Zone		System->Fabric->Transport Zones	Only VLAN Transport Zones are supported for IvS.
Create a Uplink Profile		System->Fabric->Profiles->Uplink Profiles	Select the Industrial vSwitch, created earlier.
	Create Teaming Policy in Uplink Profile	Uplink Profiles->Teamings	Configure uplinks for failover only. Under "Active Uplinks" and "Standby Uplinks," select the vDS uplinks, using the LAN A/ LAN B tags as guidance. Active and Standby configurations require one LAN A and one LAN B uplink. Leave the Field for Standby Uplinks empty, if there are only two uplinks available.

Create Transport Node Profile		System->Fabric->Hosts->Transport Node Profile	To ensure the correct Host Switch (IvS) is selected, configure the appropriate vCenter. Then, add the previously created Transport Zone, the correct Uplinks Profile, and the vDS.
	Set Teaming Policy Mapping	Transport Node Profile->Teaming Policy Mapping	Assign the NSX Uplinks to the vDS Uplinks.
	Resource Allocation	Transport Node Profile->Advanced configuration	Go to CPU config and choose the NUMA Node, where the RT PLCs are running on (typically 0) and assign 4 LCores (add 4 to the field)
	Optional Settings	Transport Node Profile->Advanced configuration	Latency Measurements and PRP (Parallel Redundancy Protocol) can be activated or deactivated as required, along with other configurable values. Note that PRP activation is limited to a single Industrial virtual Switch (IvS). In larger environments featuring multiple NSX Managers and PRP networks, the Supervision Multicast Address is modifiable. However, it is advised to use the default settings unless a specific necessity dictates a change.
Create a Segment Profile		Networking->Segments->Profiles->Add Segment Profile->Industrial vSwitch	You can add one or more Segment Profiles. The settings, already made at the TNP Level are optional and can be changed. It is best practice to create both a PRP and a Non-PRP Segment Profile.
Create a Segment		Networking->Segments->NSX-Add Segment	Choose the appropriate IvS Transport Zone and enter the VLAN ID. Save these settings. Proceed to "Segment Profiles" and select the previously created Segment Profile under "Industrial vSwitch". No further modifications are required.

Application Modernization

Organizations are evolving application platforms to support Kubernetes-based workloads, self-service infrastructure consumption, AI-enabled services, and modern private cloud operating models. This section provides architectural blueprints and design guidance for designing and operating modern application platforms on **VMware Cloud Foundation** environments. The following blueprints provide deployment patterns, operational models, and design considerations for different application modernization scenarios and platform requirements.

Enable Self-Service Infrastructure and Cloud Services for App Teams	
Self-Service Multi-Tenant Private Cloud Consumption Design Blueprint	The *Self-Service Multi-Tenant Private Cloud Consumption* blueprint provides guidance for implementing a self-service, multi-tenant private cloud. The blueprint delivers a robust, scalable, and automated platform that offers agility and efficiency in resource consumption, while adhering to commonly adopted design patterns.
Build, Run, and Manage Kubernetes and Other Modern Workloads	

VMware vSphere Kubernetes Service Consumption Blueprint	The *VMware vSphere Kubernetes Service Consumption Blueprint* provides architectural guidance for selecting and implementing the appropriate VMware vSphere Kubernetes Service (VKS) consumption model within VMware Cloud Foundation platforms.
Build, Run, and Manage Private AI Apps	
Private AI Services Consumption Blueprint	The *Private AI Services Consumption Blueprint* describes the design selections and architectural components required to deploy and operate a GenAI platform on VMware Cloud Foundation. It is intended to be used in conjunction with an underlying infrastructure blueprint.

Self-Service Multi-Tenant Private Cloud Consumption Design Blueprint

This blueprint aims to provide a structured design guide for implementing a self-service multi-tenant private cloud solution using VMware Cloud Foundation. The primary objective is to empower customers with a robust, scalable, and automated platform that offers agility and efficiency in resource consumption, while adhering to commonly adopted design patterns within the VCF private cloud enviornments.

Use Case and Objective

This guide provides the most widely used design patterns for deploying and operating VMware Cloud Foundation (VCF) with VCF Automation across Enterprise and Cloud Service Provider (CSP) environments. It is intended for cloud architects, platform engineers, and operations teams responsible for designing scalable, multi-tenant cloud platforms.

- Define and differentiate the primary tenancy models that are supported by VCF with VCF Automation
- Establish reference network topologies that are aligned to tenancy and consumption patterns.
- Provide prescriptive IdP (Identity Provider) design guidance for tenant onboarding scenarios.
- Detail chargeback and billing model for single-tenant and multi-tenant environments.
- Design patterns for VKS (VMware Kubernetes Service) with VCF Automation, covering single and multi-region deployments.
- PAIS (Private AI Service) integration patterns with VCF Automation
- DSM integration pattern with VCF Automation

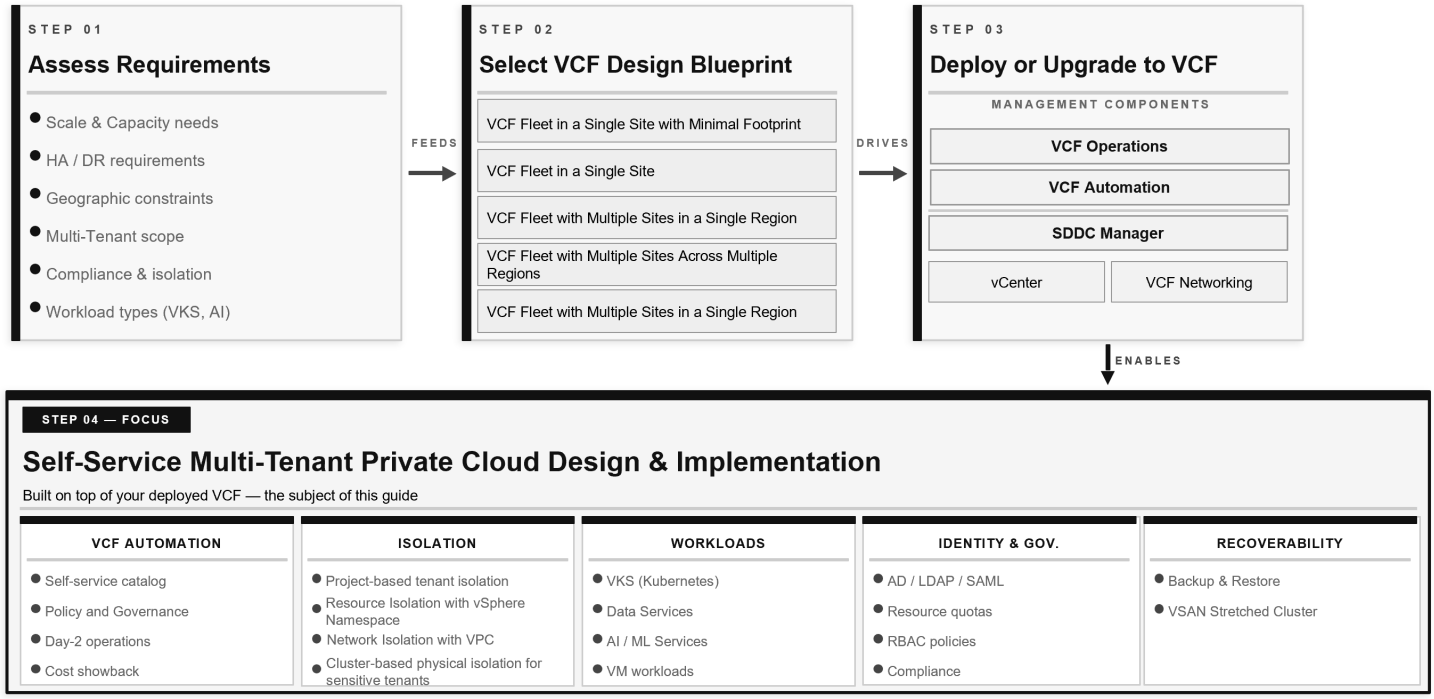
Design Prerequisite and Profile Assessment

Before initiating the design for a self-service multi-tenant private cloud, potential tenants or cloud administrators must assess their deployment needs against existing, validated VCF infrastructure blueprints. This ensures the foundational infrastructure meets the necessary scale, availability, and geographic requirements.

The following conceptual design flow outlines key design profile considerations and essential prerequisites that must be confirmed *prior* to starting the multi-tenant design work.

Conceptual Design Flow

Required sequence: VCF deployment must precede Self-Service Multi-Tenant Private Cloud Design.



Deployment Design Decisions

Designing a **Self-Service Multi-Tenant Private Cloud** requires a layered decision framework that starts with the core VCF foundation and progressively defines how the platform is deployed, separated, and consumed.

Each decision influences the next layer of the design, including fleet architecture, VCF Automation topology, tenant separation, and the way users consume resources through projects, namespaces, and VPCs.

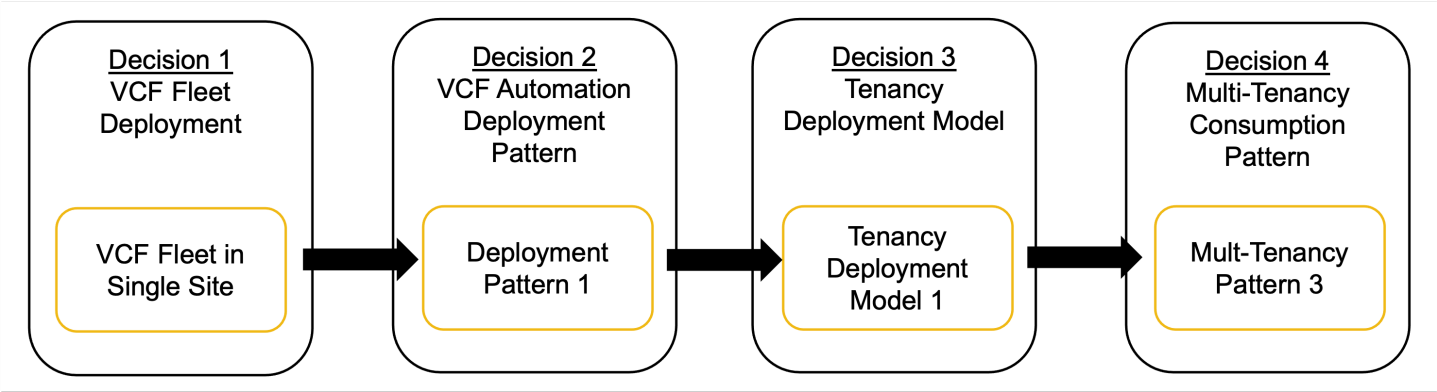
The following table summarizes the primary deployment design decisions in sequence, helping architects move from **VCF fleet design → automation deployment → tenancy model → consumption model** in a structured way.

Deployment Design Decisions (in order)	Objective	Design Choices
VCF Fleet Deployment	Deploy Core VCF Fleet Architecture	Single vs multiple VCF fleet models
VCF Automation Deployment	Deploy base VCF Automaton Topology	Single vs multiple domain models Internal only vs Externally network exposure
Tenancy Deployment Design	Define physical and operational separation between tenants	Single vs Multiple Tenants Tenant mapping to underlying domain topology

Deployment Design Decisions (in order)	Objective	Design Choices
Multi-Tenancy Consumption Design	Define consumption model within tenant	Project design Namespace design within Project(s) VPC design within Namespace(s)

Example Self-Service Multi-Tenant Deployment Design Decisions

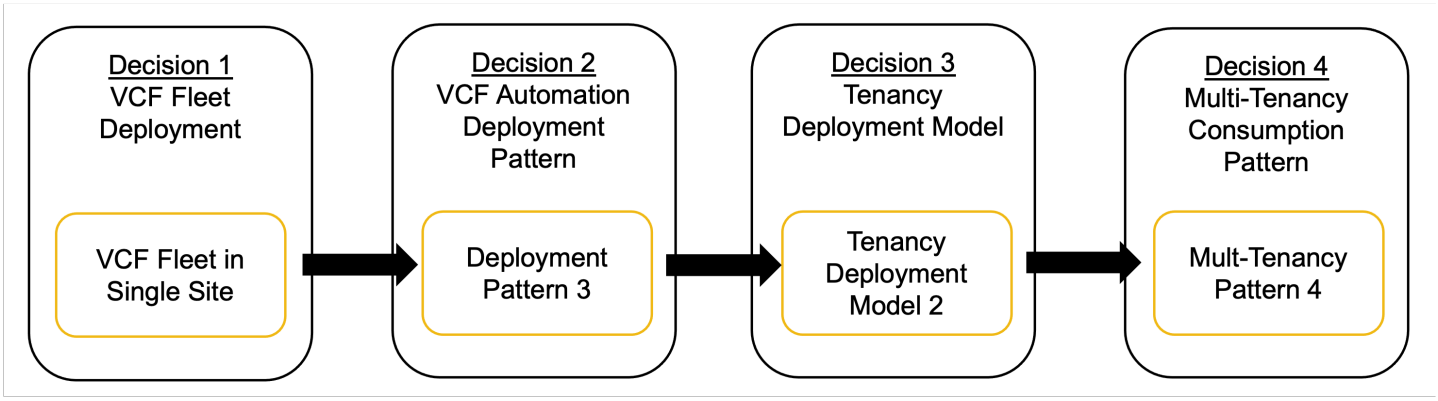
Figure 185: Example Single-Tenant (Enterprise) Self-Service Blueprint Design Decisions



The above shows the flow to design VCF Automation with the following parameters:

- A single VCF Fleet.
- Management and organization workloads are placed in a different vSphere cluster of the same management domain.
- One domain that hosts the management components and organization workloads.
- Resource isolation for multiple lines of business (LOBs) by consolidating them within a single Virtual Private Cloud (VPC) that spans three vSphere Supervisor Zones.

Figure 186: Example Multiple Organizations - Multi-Tenant or Cloud Service Provider Self-Service Blueprint Design Decisions



The above shows the flow to design VCF Automation with the following parameters:

- A single VCF Fleet.
- External-facing deployment of VCF Automation.
- Uses an external load-balancer.
- Separation of management and workloads into different domains.
- Multiple organizations share a single workload domain where isolation is achieved through vSphere Namespaces and network segmentation.
- Resource isolation for multiple organizations by providing each with a dedicated Virtual Routing and Forwarding (VRF) instance, ensuring completely separate network routing and forwarding domains.

Foundational VCF Infrastructure Design Blueprints

There are multiple VCF fleet design blueprints available, and the appropriate choice depends on factors such as geographic footprint, availability requirements, and organizational scale (for example, single site, multi-site, or multi-region deployments).

Selecting the right fleet model is the first foundational decision, as it establishes the underlying infrastructure architecture on which all higher-level platform services are built.

Once the fleet architecture is defined, organizations can then determine how to deploy and expose VCF Automation to enable private cloud consumption. This includes decisions around domain topology, external access, and how tenants and application teams will interact with the platform.

The table below outlines the available foundational VCF infrastructure design options and helps guide the selection of the most appropriate starting point for building a self-service, multi-tenant private cloud.

Design Profile Consideration	Description	Choices
Target Infrastructure Blueprint	Which of the VCF design blueprints serves as the foundation? (e.g., Single Region, Multi-Site).	1. VCF Fleet in a Single Site with Minimal Footprint 2. VCF Fleet in a Single Site 3. VCF Fleet with Multiple Sites in a Single Region 4. VCF Fleet with Multiple Sites Across Multiple Regions 5. VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)

Pre-Requisite Design Choices for VCF Automation Deployment

Before deploying VMware Cloud Foundation (VCF) Automation, architects and platform engineers must evaluate and document a defined set of infrastructure design choices across several interdependent components. These pre-requisite decisions are not optional post-deployment configurations - they are foundational commitments that shape the topology, scalability, security posture, and operational model of the entire VCF platform for consumption. Getting these decisions right before Day 1 significantly reduces rework, avoids architectural debt, and ensures the platform is fit-for-purpose from the outset.

The scope of these design choices varies depending on the deployment model being adopted. A single-enterprise deployment - where VCF Automation manages a single vSphere environment and a single organizational boundary - requires fewer integration touchpoints and simpler identity and network models. A multi-enterprise deployment, where multiple business units or tenants share the platform, introduces additional considerations around federated identity, cross-domain networking, namespace isolation, and centralized governance. Deployments operating within regulated industries or subject to compliance frameworks such as PCI-DSS, HIPAA, or SOC 2 must also incorporate security hardening, audit logging, encryption, and DFW policy decisions from the start.

Each design area includes a summary of the key decision points, a structured design-choice table with recommended options and justification, and a reference to the detailed section of this guide where full configuration guidance can be found. For environments with unique constraints -such as air-gapped deployments, regulated data residency requirements, or existing brownfield network topologies - require additional design considration.

Note:

The design choices that are documented in this guide represent recommended practices for production VCF Automation deployments. Some decisions may be adjusted based on organizational policy, existing infrastructure constraints, or licensing availability.

Pre-Requisite Design Choice for VCF Automation Deployment

Pre-Requisite Design Area	Key Design Considerations	VCF Automation Supported Choices
vSphere Supervisor Design	• vSphere Supervisor cluster topology • vSphere Supervisor Zone Design • Control plane VM Placement and sizing (Small / Medium / Large)	• vSphere Supervisor Zone Models • vSphere Supervisor Control Plane Availability Models
Network Connectivity Design Choices	Choose the connectivity design model supported from the VCF Automation All Apps Orgs Consumption Model. VPC required for all VCF Automation and workload consumption.	• Centralized Connectivity Model • Distributed VLAN Connectivity Model
Load Balancer Choices	• Default - NSX Edge Tier-1 or Virtual Network Appliance Load Balancer • Avi Load Balancer	• Avi Load Balancer Model • NSX Edge VPC Load Balancer Model • Virtual Network Appliance Load Balancer Model

Note:

Any of the design options described in the table above can be implemented in combination with the deployment models outlined below, depending on the customer's operational, connectivity, governance, and workload requirements.

VCF Automation Deployment Topologies

The placement and network exposure of the VCF Automation platform are critical design decisions that directly impact security, accessibility, and tenancy model support. While VCF Automation can be deployed in various configurations, the choice is usually driven by whether the platform is intended for internal consumption only or for external, multi-tenant public access.

This guide details three primary deployment patterns, focusing particularly on the **External-Facing Deployment** as it is essential for the self-service multi-tenant cloud blueprint.

Available VCF Automation Deployment Patterns:

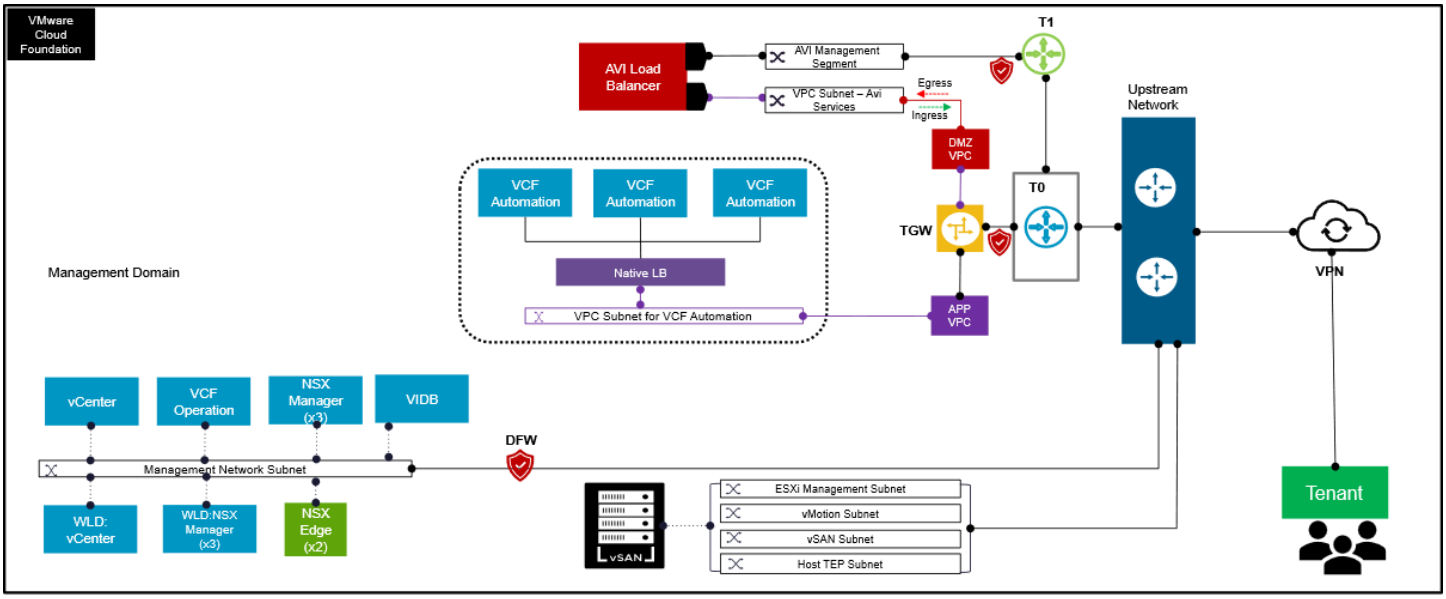
Deployment Pattern Choices	Description	Network Exposure	Design Library Reference Link
Deployment Pattern 1: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters	VCF Automation is deployed in a management domain and organization workloads are placed in a different vSphere cluster of the same management domain.	Internal (Management Network)	Deployment Pattern 1: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters
Deployment Pattern 2: VCF Automation in a VCF Instance with multiple domains	VCF Automation is deployed in a management domain and organization workloads are placed in a workload domain.	Internal (Management Network)	Deployment Pattern 2: VCF Automation in a VCF Instance with multiple domains
Deployment Pattern 3: External-Facing deployment of VCF Automation	External-Facing Deployment of VCF Automation refers to implementing VCF Automation to seamlessly interact with external networks or services in a secure manner.	External (Public Internet / VPN)	Deployment Pattern 3: External-Facing deployment of VCF Automation

As outlined in the table above, there are multiple deployment patterns available for customers to choose from based on their requirements. However, for the context of this design guide, we focus on a comprehensive deployment model where customers require segregation of the VCF Automation deployment for security and compliance reasons, while also exposing VCF Automation for external (internet-facing) access to end users.

This model is referred to as:

External-Facing Deployment (Deployment Pattern 3)

Figure 187: External Facing Deployment of VCF Automation



For a **Self-Service Multi Tenant Private Cloud** environment (both Enterprise with strict governance, and especially the **CSP Multi-Organization Model**), the VCF Automation platform must be externally accessible to tenants while maintaining robust isolation from the VCF Management Network.

Key Design Elements:

1. **Dedicated Network Segment:** VCF Automation components are placed on a dedicated VPC overlay subnet that is isolated from the VCF Management Network. Any required connectivity to management services is controlled through explicit routing, firewall policies, and least-privilege access rules rather than direct flat network adjacency. AVI Service Engines are placed in a separate DMZ VPC subnet to isolate north-south application traffic from management and automation components.
2. **Public Access Exposure:**
3. **Security Gateways:** Strict DFW and Gateway Firewall policies are enforced to ensure that:
   - Only necessary inbound ports (443, 80) are permitted to the VCF Automation public VIPs.
   - No inbound traffic from the tenant public segment is allowed to the VCF Management Network.
   - Outbound communication from VCF Automation to VCF components (vCenter, NSX, SDDC Manager) is restricted to required API calls only.

This pattern is critical for Cloud Service Providers (CSPs) as it allows external customers to access the self-service catalog without compromising the core infrastructure management plane.

Important:

Current Platform Constraint: At the time of this design, deployment of the VMware Avi Load Balancer Control Plane that is integrated with NSX requires the Avi Management segment to be connected through a Tier-1 (T1) segment. This is currently the supported deployment model for Avi Control Plane communication and integration with NSX networking services.

However, as illustrated in the architecture diagram, Avi Service Engine traffic handling ingress, and egress application flows can still be deployed within a dedicated DMZ VPC subnet. This enables separation between the Avi management/control plane traffic and externally exposed application traffic paths, improving network segmentation and security isolation for tenant-facing services.

Tenancy Deployment Models

This section details the various VMware Cloud Foundation (VCF) tenancy deployment models available, providing a foundational understanding for architects and implementers. Although the VCF design library contains a broad range of patterns to meet various organizational requirements, we have picked the two most widely used deployed tenancy models in this blueprint: the **Single Organization Model** and the **Multi-Organization Cloud Service Provider (CSP) Model**. These two primary models are the central focus of the subsequent detailed design guidance.

The choice of a VCF tenancy model is a critical decision dictating:

- Resource allocation.
- How tenants are separated.
- The operational model for management and consumption of the cloud environment.

The following table provides a complete reference of the documented VCF Tenancy Deployment Models, serving as a comprehensive directory for all potential tenancy deployment architectures.

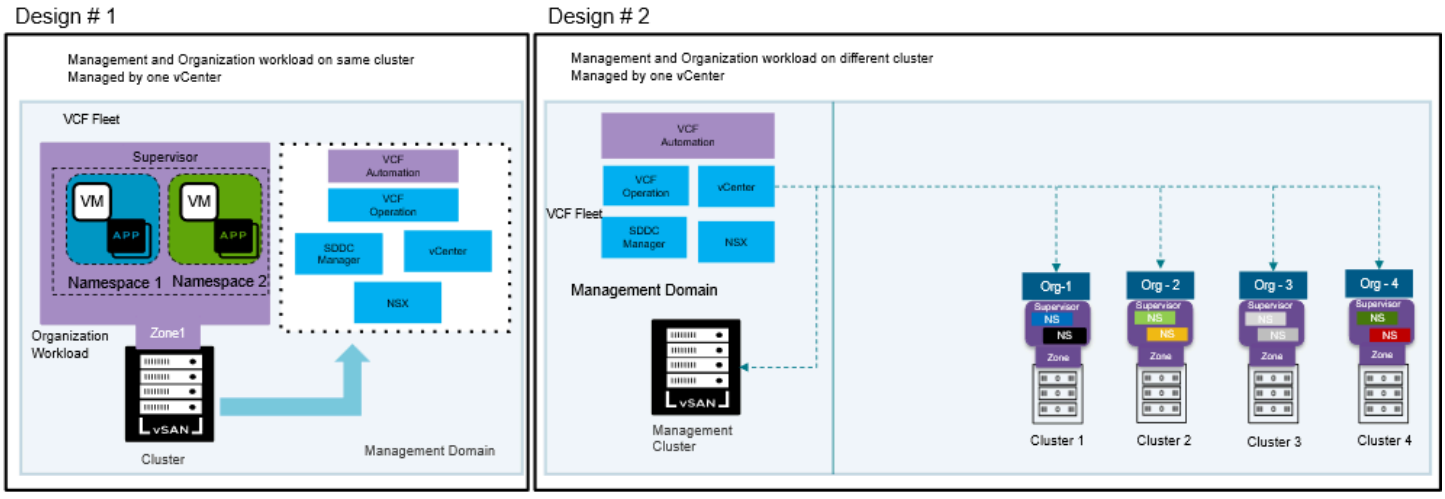
Complete Set of VCF Tenancy Deployment Models:

Tenancy Deployment Models	Description	Primary Use Cases
Tenancy Deployment Model 1: Consolidated Management and Organization Workloads in a VCF Instance	This tenancy deployment model consists of one VCF workload domain that hosts the management components and organization workloads.	For organizations with smaller environments and limited resource requirements, consolidating workloads can provide a cost-effective and manageable solution.
Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance	This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.	• Cloud Service Providers (CSPs) Offering Multi-Tenant Cloud Services • Large Enterprises with Multiple Business Units or Departments • Organizations Offering Internal PaaS or Developer Platforms • Educational Institutions or Research Organizations with Multiple Projects or Teams
Tenancy Deployment Model 3: Centralized Management with Dedicated Workload Domains per Organization in a VCF Instance	This model provides a robust and secure multi-tenant environment by leveraging VCF Automation and dedicated workload domains. Each organization gets a dedicated environment, ensuring security, performance, and independent management. The central and shared management domain provides centralized control, while organization-specific components offer flexibility and autonomy.	• Performance-Sensitive Workloads • Specialized Hardware or Dedicated Hardware • Data Sovereignty and Geographic Restrictions • Mergers and Acquisitions

Tenancy Deployment Models	Description	Primary Use Cases
Tenancy Deployment Model 4: Centralized VCF Automation Managing Dedicated VCF Instances per Organization	This model offers a dedicated VCF Instance (consisting of a management domain and one or more workload domains) for every tenant. Each tenant receives dedicated resources, in addition to common management components such as VCF Operations and VCF Automation managing both instances.	• Large enterprises with autonomous business units • Service Providers offering highly isolated environments • Government agencies with separate departments • Global organizations with regional data sovereignty needs • Organizations Undergoing Mergers and Acquisitions
Tenancy Deployment Model 5: Dedicated VCF Fleet with Independent VCF Automation Supporting Single or Multiple Domains per Tenant	This model illustrates a dedicated tenant approach where each tenant has its own isolated environment within a VCF Instance single domain or multiple domain construct. This means each tenant has their own VCF Automation as well as dedicated compute and storage infrastructure supported by dedicated vSphere Supervisor and NSX deployments.	• Organizations with strict isolation requirements but also wanting to optimize resource usage. • Multi Geo - Complying with data sovereignty regulations • Service Provider - Offering hosted private clouds with dedicated infrastructure. • Supporting regulated industries with specific compliance needs. • Providing high-value services with strict SLAs. Facilitating M&A scenarios with isolated environments.

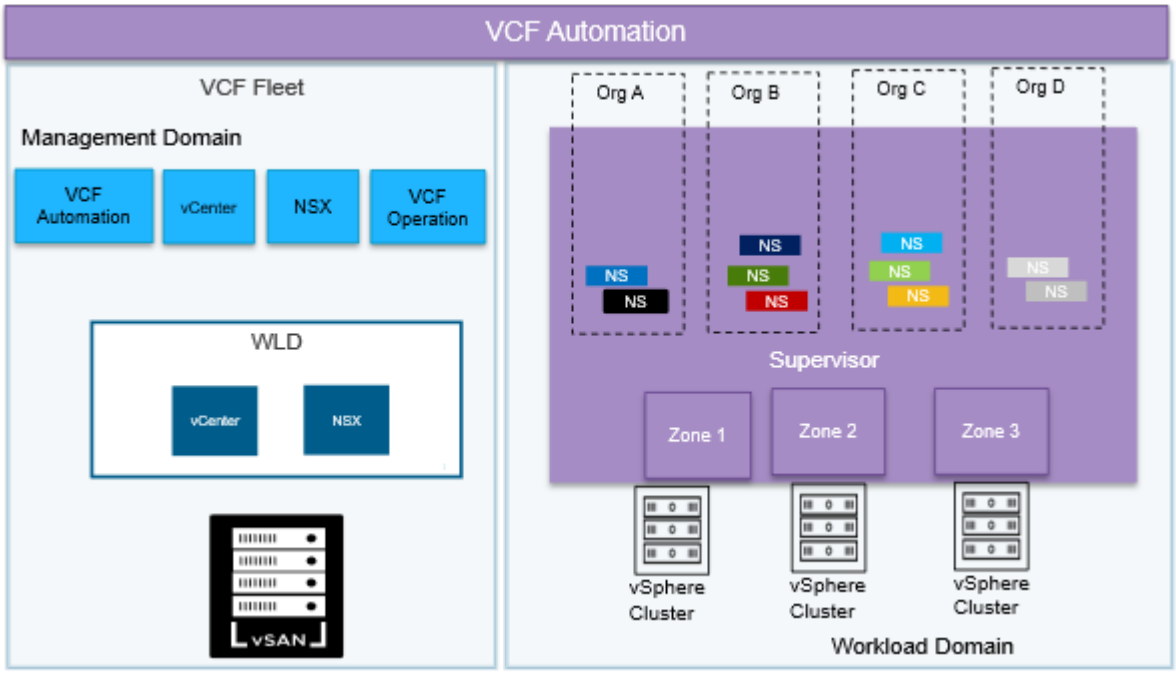
While the VCF supports multiple deployment models to address complex enterprise and sovereign cloud requirements, this design blueprint focuses on the two most widely adopted patterns. These following two patterns represent the fundamental architectural divergence between internally consumed IT services and externally provided cloud services:

- **Single Organization - Single Tenant (Tenancy Deployment Model 1):** This model prioritizes resource sharing and a unified operational plane for different internal business units. The focus is on *logical isolation* (VPC Subnets, vSphere namespaces) within a single trust domain. This pattern is elaborated in the **Enterprise – Single Organization** section below**.**



- **Multiple Organizations - Multi-Tenant or Cloud Service Provider (Tenancy Deployment Model 2):** This model prioritizes *operational and identity isolation* between distinct external customers. The focus is on mapping each

customer to a dedicated Aria Organization and ensuring strict separation via network segmentation (Tier-1 per tenant) and per-tenant IDP federation. This pattern is elaborated in the **Cloud Service Provider – Multiple Organizations** section below.



Single Organization - Single Tenant

In the Enterprise model, a single organization owns and operates the VCF stack. All workloads, management components, and platform services belong to one administrative domain. While multiple business units or projects may be isolated using virtual constructs (VPC Subnets, vSphere namespaces, or VCF Automation Projects), there is a single root of trust and a centralized operations model.

Key Characteristics

- Centralized identity federation via a corporate IdP (Active Directory / LDAP / SAML).
- Shared physical infrastructure with logical isolation per team or department.
- Chargeback reports or internal cost center allocation using VCF Operations.
- Platform services (VKS, PAIF) deployed in a shared or dedicated namespace per BU.

When to Use

- Single-enterprise IT operating model with a central cloud team.
- Regulatory environments where a single audit scope is required.
- Environments where shared services (DNS, NTP, monitoring) span all workloads.

Multiple Organizations - Multi-Tenant or Cloud Service Provider

In this model, a single infrastructure stack is partitioned to serve multiple distinct organizations (tenants). Each tenant is operationally isolated and may have independent IdP integration, dedicated network segments, private compute domains, and separate billing. The CSP operator maintains a shared infrastructure plane while tenants interact only with their own resource pools.

Key Characteristics

- Multiple Organizations, each mapped to a distinct company.
- Tenant-specific IDP federation (per-tenant SAML / OIDC providers).
- Resource and network isolation is enforced at per tenant.
- Independent chargeback with metered billing per tenant.
- Shared or dedicated physical clusters depending on compliance/isolation requirements.

Multi-Tenancy Consumption Patterns

While the **Tenancy Deployment Model** defines the physical and operational separation between organizations, the **Consumption Design** defines how tenants, business units, application teams, and end users consume infrastructure, networking, and platform services within the private cloud environment.

These consumption patterns describe different approaches for implementing logical isolation, network segmentation, organizational boundaries, and shared versus dedicated resource models across VPCs, Supervisor Zones, gateways, and workload domains.

The appropriate pattern depends on factors such as:

- organizational structure and tenant boundaries
- required level of network and resource isolation
- scalability and operational management requirements
- regional deployment strategy
- shared versus dedicated infrastructure preferences
- governance and compliance considerations

The following patterns illustrate common multi-tenancy consumption models that can be implemented using VCF Automation for a modern private cloud architectures.

Complete Set of VCF Multi-Tenancy Consumption Patterns:

Multi-Tenancy Consumption Pattern	Description/Use Cases
Multi-Tenancy Consumption Pattern 1: Resource Isolation with Single VPC for Multiple Businesses in One Supervisor Zone	This design follows a multi-tenancy consumption pattern where resources are shared among multiple tenants, but logical isolation is provided through multiple levels within the private cloud stack.
Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone	This approach provides robust resource isolation for different lines of business (LOBs) within a single vSphere Supervisor Zone.
Multi-Tenancy Consumption Pattern 3: Resource Isolation with Single VPC for Multiple Lines of Business in Three Supervisor Zones	This architecture provides resource isolation for multiple lines of business (LOBs) by consolidating them within a single Virtual Private Cloud (VPC) that spans three vSphere Supervisor Zones.
Multi-Tenancy Consumption Pattern 4: Resource Isolation for Multiple Organizations with dedicated VRFs	This design achieves strong resource isolation for multiple organizations by providing each with a dedicated Virtual Routing and Forwarding (VRF) instance, ensuring completely separate network routing and forwarding domains.
Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways	This design provides multi-tenant resource isolation across multiple regions by combining dedicated gateways for strict tenant separation with shared gateways for efficient common services.
Multi-Tenancy Consumption Pattern 6: Multi-Organization resource isolation with shared NSX across workload domains	This design provides multi-organization resource isolation across distinct workload domains while using a shared NSX platform to centrally manage and logically segment network and security services for each tenant.

Network Consumption Model:

However, the ability to implement these patterns effectively depends heavily on the underlying software-defined networking architecture.

Networking is not just an implementation detail - it is the control plane for isolation, connectivity, and policy enforcement across tenants. The choice of networking model directly impacts how well you can achieve:

- Tenant isolation (organization vs line of business)
- Scalability across Supervisor Zones and regions
- Operational simplicity vs flexibility
- Shared vs dedicated services

VMware Cloud Foundation provides multiple networking constructs - such as VPCs, VRFs, shared and dedicated gateways, and VPC - based subnets that enable these consumption patterns.

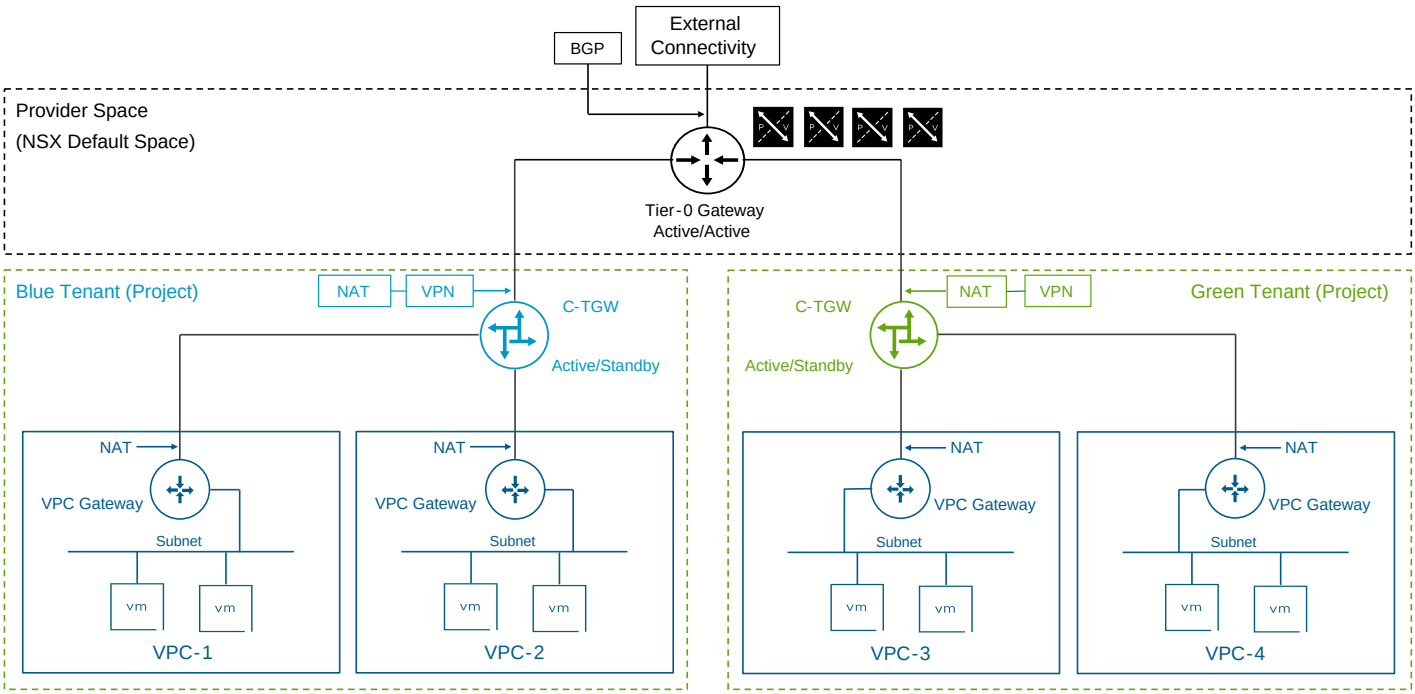
Building on these networking constructs, there are multiple ways to implement multi-tenant connectivity in VMware Cloud Foundation. In practice, these approaches can be broadly categorized into two representative deployment models:

- **Centralized Connectivity Model**
- **Distributed VLAN Connectivity Model**

Centralized Connectivity Model

The model uses an NSX Edge cluster with Tier-0 gateway to provide VPC-compatible workload construct with network connectivity established via a Centralized Transit Gateway.

Figure 188: Centralized Connectivity Model



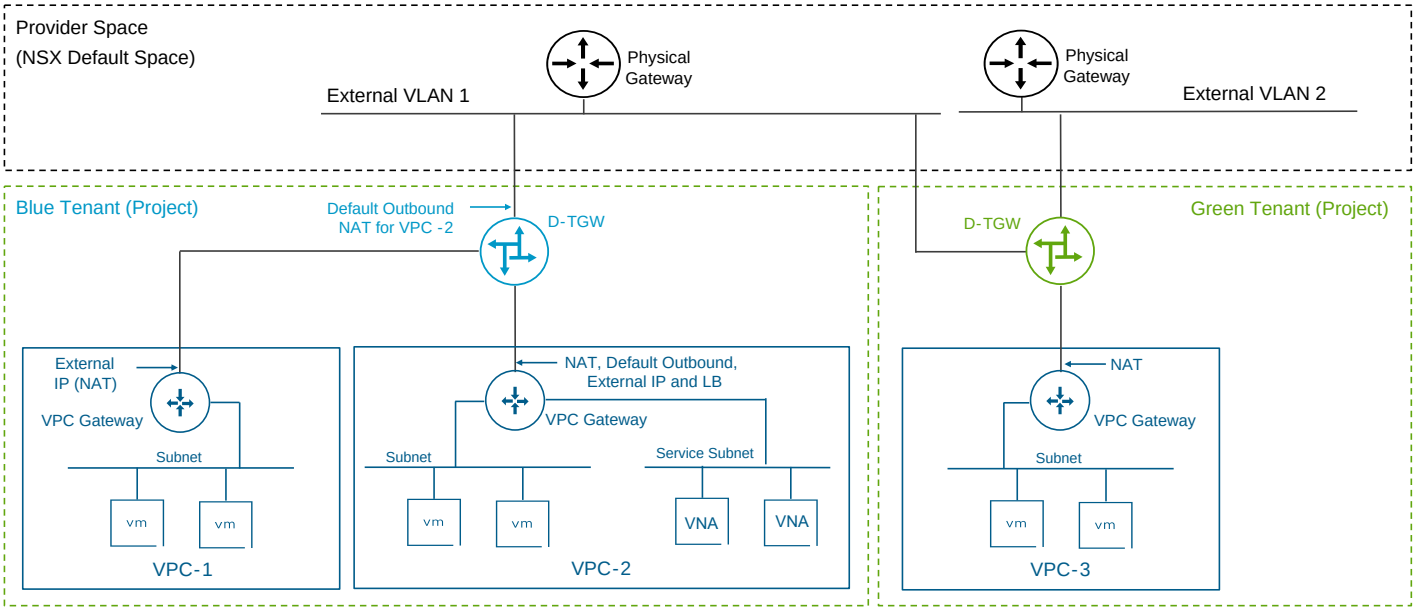
Note:

While the above design uses a Active/Active T0, Active/Standby design is also supported.

Distributed VLAN Connectivity Model

The Distributed VLAN Connectivity Model enables rapid adoption of VPC network virtualization without requiring NSX Edge VM deployment. It uses a Distributed Transit Gateway (DTGW) that connects via external VLANs in the physical network fabric, utilizing a distributed forwarding model at the ESX host level, achieving close to host pNIC line rate. Stateful network services such as Default Outbound NAT (Auto-SNAT) , custom NAT, and Load Balancing for vSphere Supervisor and VKS workloads and VCFA All Apps Orgs are supported when using Virtual Network Appliance

Figure 189: Distributed VLAN Connectivity Model



These models illustrate different design choices in how network services - such as routing, NAT, and external connectivity - are placed and consumed across tenants. Each approach represents a trade-off between centralized control and operational simplicity versus distributed isolation and flexibility.

Workload Connectivity Model	Attributes	Benefits	Implications
Centralized Connectivity Model	• Requires NSX Edge nodes. • Active/active or active/ standby Tier-0 gateway.	• Support for VPC network • Support for stateful services (i.e., Default Outbound NAT (Auto-SNAT), NAT, Gateway Firewall and load balancing) • Support for VPN.	• North-south throughput is determined by the number and type of NSX Edge nodes deployed.
Distributed VLAN Connectivity Model	• Requires VNA nodes for stateful services. • Support for VPC network and stateful services (i.e., Default Outbound NAT, NAT and load balancing)	• Can achieve close to host pNIC line rate for traffic not being redirected to the VNA cluster • Does not require NSX Edge nodes.	• No support for VPN. • No support for Multi-Rack Layer 3 vSphere Cluster Model

Supported Network Services on TGW

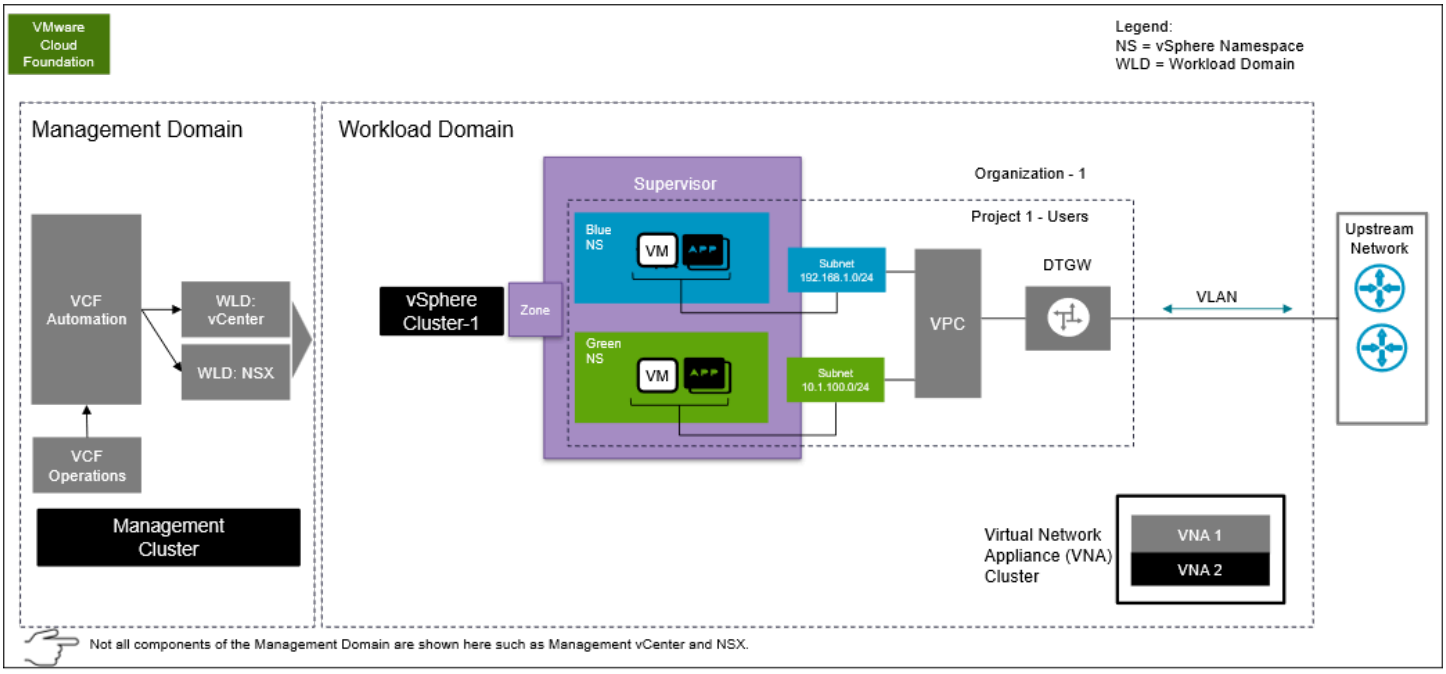
Network Services	Centralized TGW Model (T0 - A/A and TGW -A/S)	Distributed VLAN TGW with VNA Model	Notes
External IP (1:1 NAT)	✓	✓	Required for inbound connectivity to workloads in both models
NAT (SNAT/DNAT)	✓	✓	Core north-south translation supported in both architectures
VPC Outbound NAT (N:1 NAT also called Auto-SNAT)	✓	✓	Enables shared outbound internet access
DHCP	✓	✓ (Only DHCP Server)	Required for workload IP management
NSX LB for Supervisor / VKS	✓	✓	Supports Kubernetes control plane and services
Avi VPC LB Plugin	✓	✓	Provides advanced load balancing for applications
VPN	✓	✖	Typically centralized in CTGW model for shared external connectivity

The following sections present reference design for each model and demonstrate how they map to the multi-tenancy consumption patterns discussed earlier, enabling end-to-end tenant consumption of VCF resources.

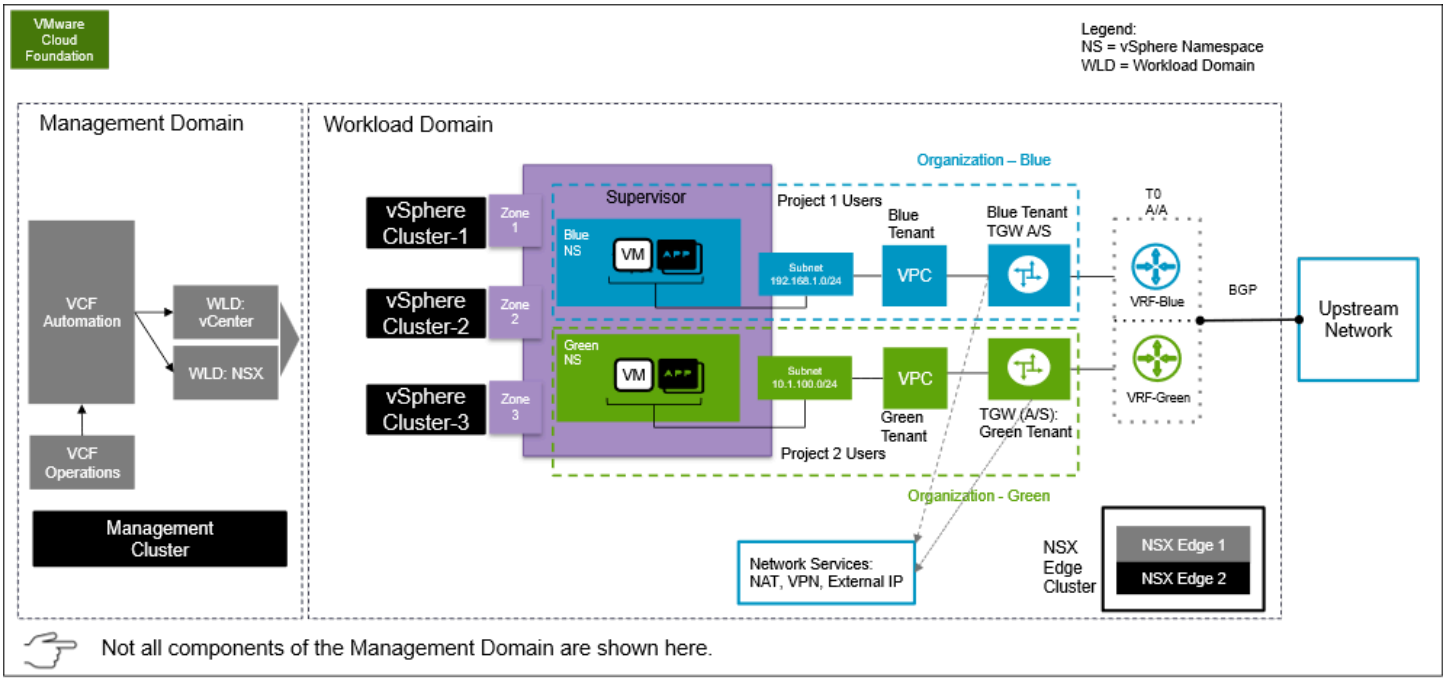
The multi-tenancy consumption patterns that are outlined above describe how organizations and tenants logically consume resources within VCF Automation.

For clarity, the following sections highlight two representative consumption models aligned to common use cases: one for a **single enterprise environment**, and another for a **multi-enterprise (multi-tenant) environment**. These examples demonstrate how architectural and operational design choices vary based on requirements for isolation, governance, scalability, and operational independence. They also illustrate when to adopt a **Centralized Connectivity Model** versus a **Distributed Connectivity Model** as part of the overall private cloud operating model within VMware Cloud Foundation.

Example : Single Enterprise Environment



Example: Multiple Enterprise Environment:



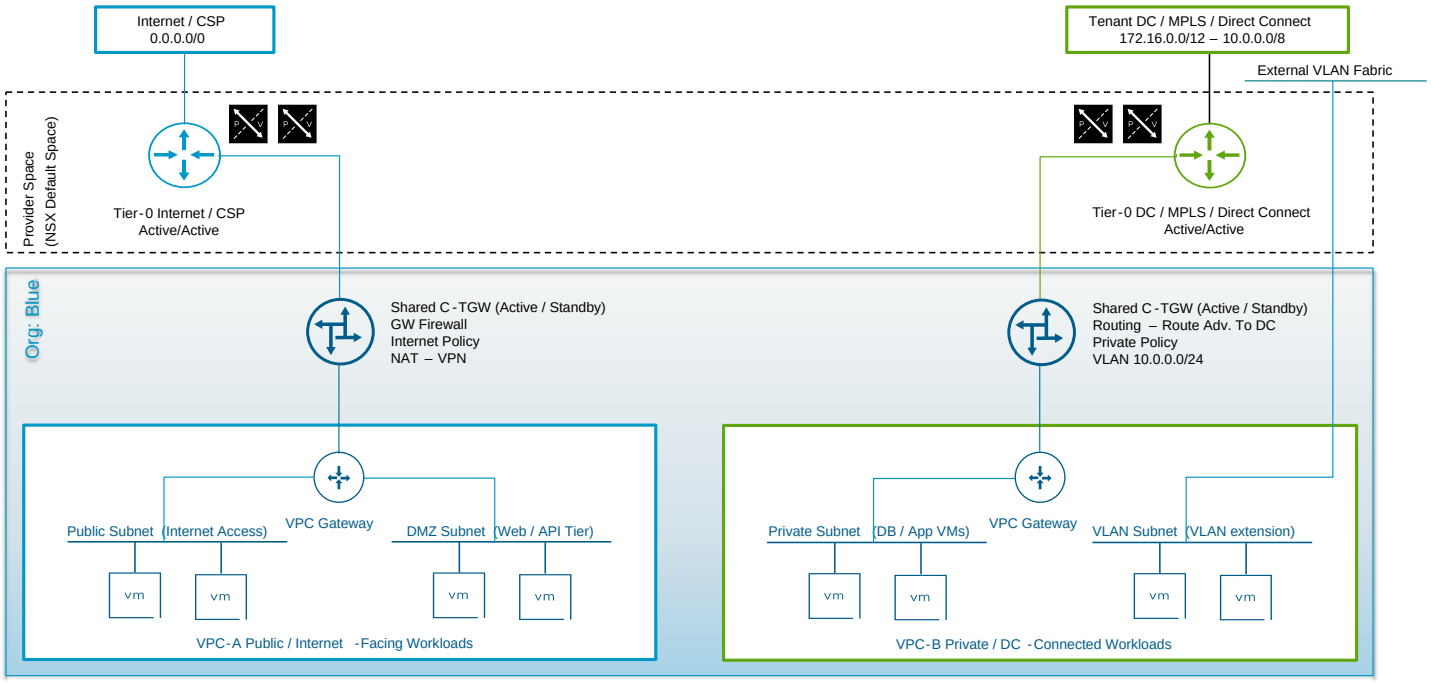
As outlined in the previous networking design choce tables for single-enterprise and multi-enterprise environments, the selection between a centralized connectivity model and a distributed connectivity model should be based on the customer's operational, scalability, isolation, and governance requirements.

Below are additional example scenarios demonstrating how customers can design their networking models for both single-organization and multi-organization environments based on operational, connectivity, security, and tenant isolation requirements.

Additional Network Consumption Models

Figure 190: Multiple Transit Gateway Architecture for Segregated Connectivity in VCF Automation

VCF Automation Organization: Single Org, Dual Dedicated T0s (or VRFs)


Internet / CSP
0.0.0.0/0
Tenant DC / MPLS / Direct Connect
172.16.0.0/12 – 10.0.0.0/8
External VLAN Fabric
Provider Space
(NSX Default Space)
Tier-0 Internet / CSP
Active/Active
Tier-0 DC / MPLS / Direct Connect
Active/Active
Org: Blue
Shared C-TGW (Active / Standby)
GW Firewall
Internet Policy
NAT – VPN
Shared C-TGW (Active / Standby)
Routing – Route Adv. To DC
Private Policy
VLAN 10.0.0.0/24
Public Subnet (Internet Access)
VPC Gateway
DMZ Subnet (Web / API Tier)
vm
vm
vm
vm
VPC-A Public / Internet -Facing Workloads
Private Subnet (DB / App VMs)
VPC Gateway
VLAN Subnet (VLAN extension)
vm
vm
vm
vm
VPC-B Private / DC -Connected Workloads

This architecture illustrates a single-organization or tenant deployment model in VCF Automation where a tenant organization within a Cloud Service Provider (CSP) environment requires strict separation between:

- internet-facing external connectivity
- private enterprise or on-premises data center connectivity

To achieve this separation, the design uses:

- two dedicated Centralized Transit Gateways (CTGWs)
- separate connectivity domains for Internet and private DC traffic
- dedicated Tier-0 (T0) gateways or isolated VRFs
- separate VPCs aligned to workload communication requirements

This architecture enables the tenant organization to securely consume public Internet services while simultaneously maintaining private MPLS, Direct Connect, or enterprise WAN connectivity back to their own data center environment.

Design Justification

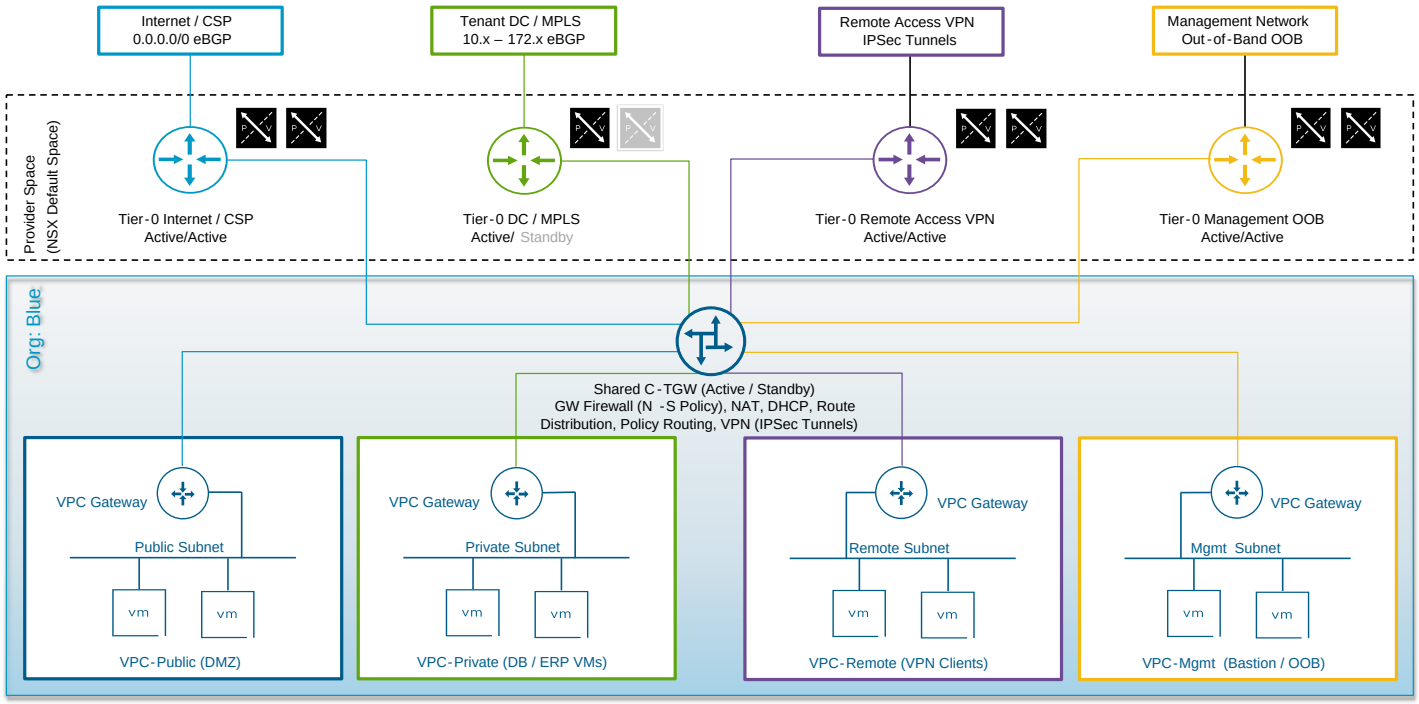
This design is intended for tenant organizations operating within a CSP hosted VCF Automation environment that require different connectivity and security boundaries for internet-facing workloads versus enterprise-connected workloads.

By separating these connectivity domains through independent Transit Gateways and dedicated routing constructs, organizations can:

- enforce different security policies for public and private traffic
- isolate routing domains between external and internal workloads
- reduce security exposure between Internet-facing and enterprise-connected applications
- maintain private enterprise WAN connectivity while still supporting external-facing services
- support hybrid application architectures spanning cloud and enterprise data center environments

> **Note:**
>
> Following is an alternate design levegarding a single CTGW but multiple paths into the tenant.

Single Transit Gateway (TGW) Architecture for Unified Connectivity in VCF Automation:


Internet / CSP
0.0.0.0/0 eBGP
Tenant DC / MPLS
10.x – 172.x eBGP
Remote Access VPN
IPSec Tunnels
Management Network
Out-of-Band OOB
Provider Space
(NSX Default Space)
Tier-0 Internet / CSP
Active/Active
Tier-0 DC / MPLS
Active/ Standby
Tier-0 Remote Access VPN
Active/Active
Tier-0 Management OOB
Active/Active
Org: Blue
Shared C-TGW (Active / Standby)
GW Firewall (N -S Policy), NAT, DHCP, Route
Distribution, Policy Routing, VPN (IPSec Tunnels)
VPC Gateway
Public Subnet
vm
vm
VPC-Public (DMZ)
VPC Gateway
Private Subnet
vm
vm
VPC-Private (DB / ERP VMs)
VPC Gateway
Remote Subnet
vm
vm
VPC-Remote (VPN Clients)
VPC Gateway
Mgmt Subnet
vm
vm
VPC-Mgmt (Bastion / OOB)

This architecture illustrates a single-organization or tenant deployment model in VCF Automation where multiple connectivity types are consolidated through a **shared Centralized Transit Gateway (C-TGW)**. Instead of deploying separate transit gateways for each connectivity domain, this design uses one shared C-TGW to provide unified routing and network services across multiple VPCs.

The tenant organization may require connectivity to:

- Internet or CSP networks
- Private data center or MPLS networks
- Remote access VPN users
- Out-of-band management networks

To support these requirements, the design uses:

- Multiple Tier-0 gateways for upstream connectivity domains
- A shared C-TGW for centralized routing and network services
- Separate VPCs for public, private, remote access, and management workloads
- Segmented subnets aligned to workload and access requirements

The shared C-TGW provides common services such as gateway firewall, NAT, DHCP, route distribution, policy routing, and VPN services. Each VPC remains logically separated, but connectivity is centralized through a common transit layer.

Design Justification

This design is intended for customers who want a simplified connectivity model while still supporting multiple types of north-south access for a tenant organization.

By using a single shared Transit Gateway, organizations can:

- reduce the number of gateway constructs to deploy and manage
- centralize routing and policy enforcement
- simplify connectivity between VPCs and upstream networks
- provide a common services layer for NAT, DHCP, firewall, route distribution, and VPN
- support multiple workload zones such as public, private, remote access, and management VPCs

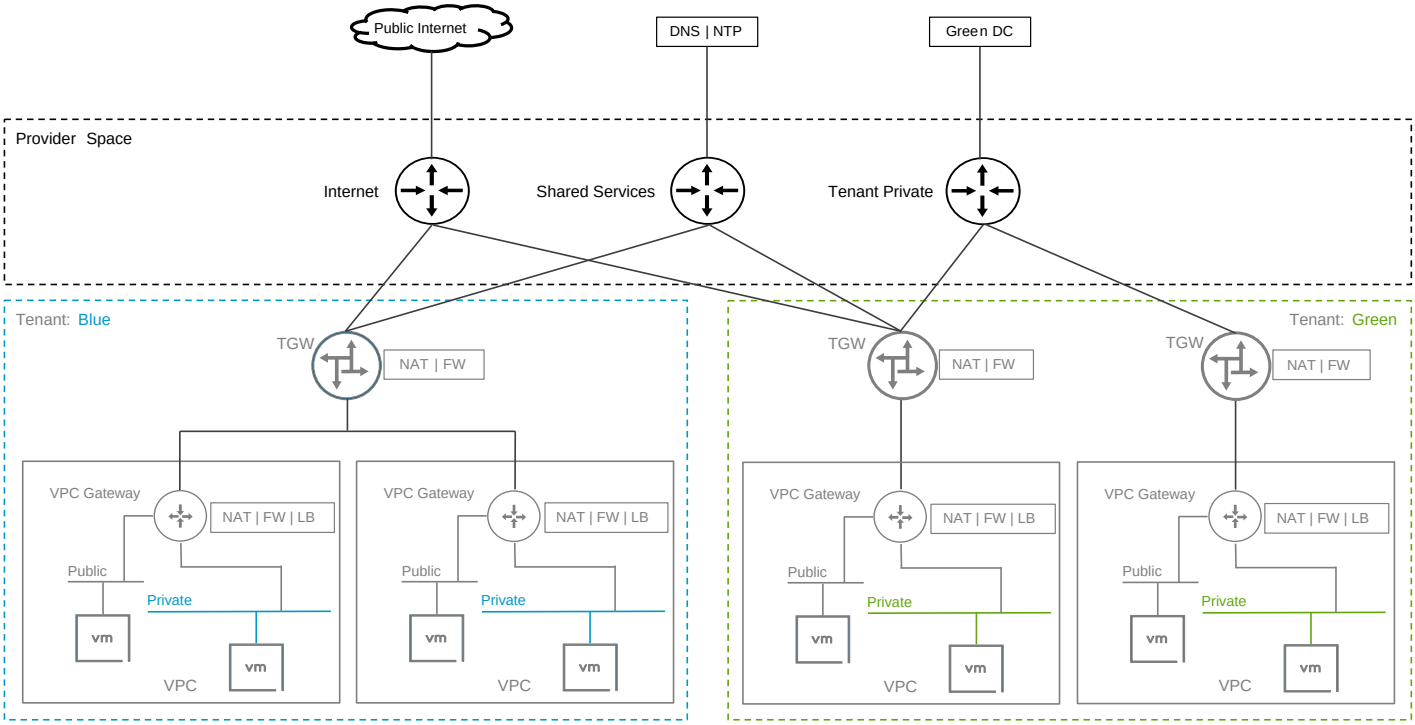
This model is well suited when the tenant requires multiple connectivity paths, but does not require strict separation of routing domains through dedicated transit gateways.

Design Attributes

Design Attribute	Description
Single Shared C-TGW	Provides centralized routing and shared network services for multiple VPCs within the organization.
Multiple Upstream T0 Gateways	Supports Internet/CSP, DC/MPLS, Remote Access VPN, and Management/OOB connectivity.
VPC Segmentation	Uses separate VPCs for public, private, remote access, and management workloads.
Centralized Network Services	Provides NAT, DHCP, gateway firewall, route distribution, policy routing, and VPN services from a common transit layer.
Simplified Operations	Reduces operational overhead compared to maintaining multiple dedicated transit gateways.
Unified Connectivity Model	Allows different workload VPCs to consume shared upstream connectivity services through one transit architecture.

> **Note:**
>
> Following example is an advanced topology design leveraging a transit gateway (TGW) with multiple external connections

Transit Gateways with multiple external connections


Public Internet
DNS | NTP
Green DC
Provider Space
Internet
Shared Services
Tenant Private
Tenant: Blue
TGW
NAT | FW
VPC Gateway
NAT | FW | LB
Public
Private
vm
VPC
Tenant: Green

This architecture illustrates a multi-tenant VCF Automation deployment model where each tenant is provided with one or multiple dedicated Transit Gateway (TGW) that connects to multiple external connectivity domains through shared provider infrastructure.

The design enables tenant environments to simultaneously access:

- Public Internet connectivity
- Shared enterprise services (e.g., DNS, NTP)
- Private data center (DC) connectivity

This is achieved by integrating tenant-specific TGWs with multiple upstream Tier-0 gateways located in the provider space.

Architecture Overview

In this model:

- Each tenant (e.g., Blue and Green) is assigned dedicated TGWs
- Each TGW provides:
  - NAT services
  - Gateway firewall (N-S policy enforcement)
  - Routing and connectivity control
- TGWs connect upstream to:
  - Internet Tier-0 gateway for public access
  - Shared Services Tier-0 gateway for platform services (DNS/NTP)
  - Tenant Private Tier-0 gateway for DC/MPLS connectivity

Within each tenant:

- Multiple **VPCs** are deployed
- Each VPC contains:
  - Public and private subnets
  - Integrated NAT, firewall, and load balancing capabilities
- Workloads are isolated per tenant and per VPC

Design Justification

This architecture is designed for multi-tenant environments where each tenant requires:

- Independent network control and policy enforcement
- Access to multiple external connectivity domains
- Secure and isolated communication paths

By using dedicated TGWs per tenant, the design enables:

- Tenant-level isolation of routing and security policies
- Controlled and secure access to shared and external resources
- Separation of tenant traffic while still leveraging shared provider infrastructure

Key Design Attributes

Design Attribute	Description
Per-Tenant Transit Gateways	Each tenant has its own TGW, ensuring isolation of routing, NAT, and firewall policies.
Multiple External Connectivity Domains	Supports Internet, shared services, and private DC connectivity simultaneously.
Provider-Managed Tier-0 Gateways	Centralized Tier-0 gateways provide upstream connectivity across tenants.
VPC-Based Segmentation	Workloads are segmented into VPCs with public/private subnet separation.

Design Attribute	Description
Centralized Security with Tenant Control	Firewall and NAT policies are enforced at the TGW level per tenant.
Scalable Multi-Tenant Design	New tenants can be onboarded by provisioning new TGWs and VPCs without impacting existing tenants.

VPC VLAN Extension with VCF Automation

VPC VLAN Extension is a VMware Cloud Foundation networking capability that enables Layer 2 (L2) and Layer 3 (L3) connectivity between traditional physical network VLANs and VCF-managed Virtual Private Cloud (VPC) subnets. This capability allows workloads running inside VPC-based environments to integrate directly with existing enterprise VLAN-backed network architectures without requiring immediate network redesign or large-scale IP renumbering efforts.

By extending VLAN-backed network connectivity into VPC-managed environments, organizations can support coexistence between traditional VM-based applications and modern VPC or cloud-native workload constructs within the same operational framework. This enables a gradual transition toward VPC-based consumption models while preserving compatibility with existing enterprise networking and operational dependencies.

Migration Use Case: Transitioning Traditional VM Workloads into VCF Automation Constructs

One of the primary use cases for VPC VLAN Extension is enabling migration of traditional virtual machine workloads into VCF Automation consumption constructs without requiring immediate rearchitecture of the existing network environment.

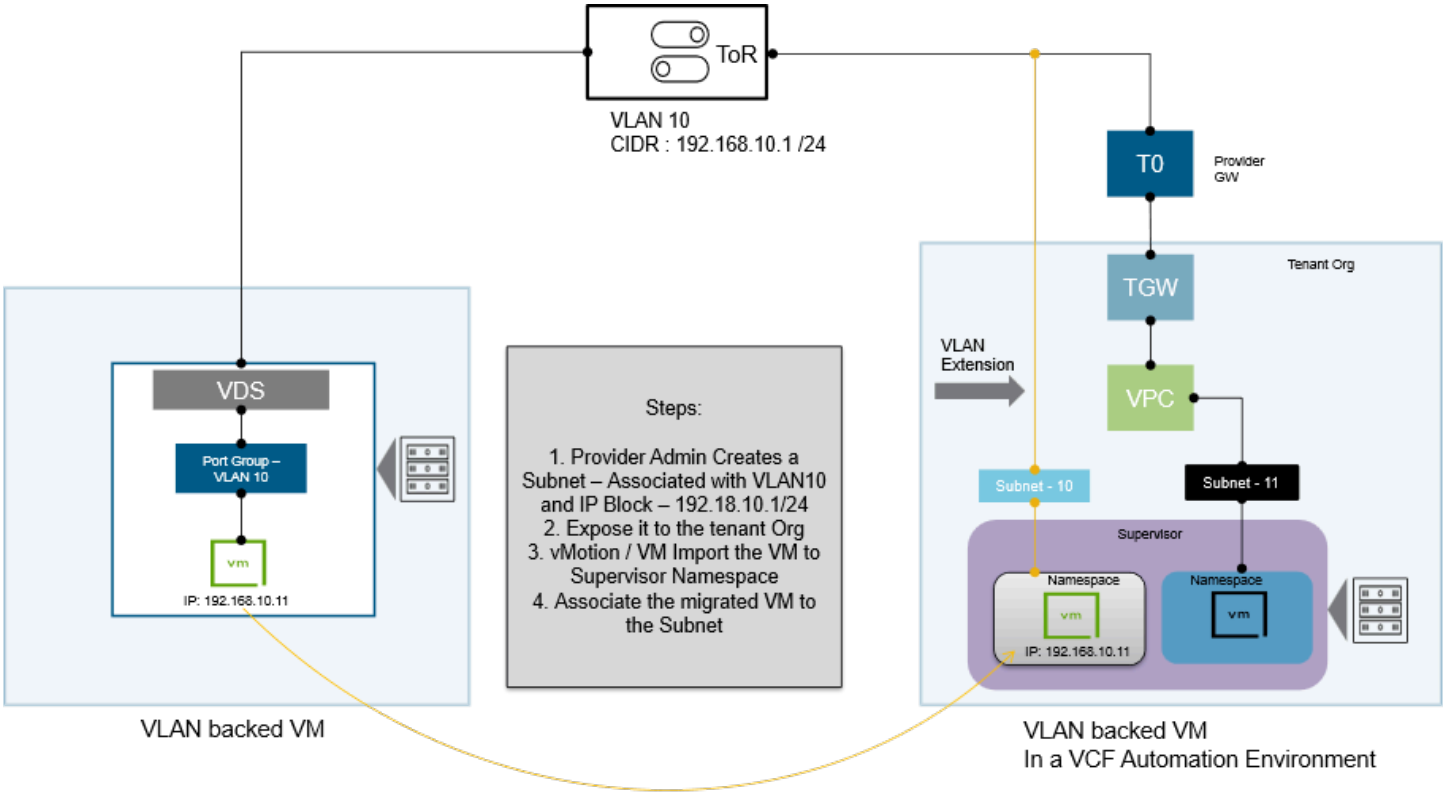
In many enterprise environments, traditional workloads are tightly coupled to existing VLAN-backed network designs, static IP addressing schemes, firewall policies, routing dependencies, and operational processes. Replatforming these workloads into VPC-based automation environments often introduces operational risk if applications must be immediately re-IPed or migrated into entirely new network segments.

VPC VLAN Extension helps address this challenge by allowing existing VLAN networks to be extended into VPC-managed environments so that workloads can retain their existing network context during migration.

Example Workflow

1. A provider administrator creates a VPC subnet associated with an existing enterprise VLAN and IP range.
2. The subnet is exposed to a tenant organization within **VMware Aria Automation**.
3. Existing virtual machines are migrated or imported into the tenant namespace or VPC construct.
4. The migrated workloads continue operating within the same IP addressing and network context as the original VLAN-backed environment.
5. Over time, workloads can be progressively modernized or transitioned into fully cloud-native VPC networking models if desired.

Figure 191: VPC VLAN Extension with VCF Automation - Migration Use Case



Best Fit Scenarios:
• VPC VLAN Extension bridges the gap between brownfield VLAN-based infrastructure and greenfield VPC-based deployments. • It enables customers to modernize incrementally without forklift migrations or re-IP exercises. • Direct L2 adjacency between physical VLANs and VPC subnets eliminates the need for NAT, firewall hairpins, or complex routing overlays in hybrid workload scenarios. • VM Migration: Lift-and-shift VMs from physical hosts to VCF without re-IP. VMs retain their physical VLAN X IPs and default gateway. • Legacy Application Compatibility: Apps with hardcoded IP/GW configurations that cannot tolerate re-IP (ERP systems, license servers, etc.).

Identity and Access Management Design

Identity and Access Management (IAM) is a foundational pre-requisite for VCF Automation deployment. Configuring authentication correctly determines who can access the platform, at what level of privilege, and through which identity source. VCF Automation enforces a two-tier access model built around two distinct personas - the Provider and the Tenant / Line-of-Business (LOB) - each with its own authentication requirements, supported identity providers (IdP), and portal scope. These personas do not share authentication configuration; each must be independently integrated with an enterprise identity provider before operational use begins.

VCF Automation Personas

VCF Automation defines two persona levels that represent distinct administrative boundaries. Understanding these personas - their responsibilities, their authentication options, and their portal access scope is essential for designing a correct IAM model before deployment begins.

Area	Provider Persona	Line of Business / Tenant Persona
Responsibilities	• Installs and manages core VCF Automation infrastructure • Provider / Enterprise level administrative functions • Manages shared content libraries published to organizations Creates organizations, assigns resource quotas, regional networking, and first tenant user	• Manages organizations • Creates and manages projects • Creates and manages vSphere Namespaces • Configures Content Hub • Creates blueprints Provides organization-level user management
Supported Identity Providers	• VCF SSO (via VCF Identity Broker - embedded or appliance) • LDAP (AD or OpenLDAP connection) • SAML (upstream SAML identity provider) Default Local Admin account (created at installation, full rights, one account only)	• OpenID Connect (OIDC) • LDAP (single AD or OpenLDAP connection) • SAML (upstream SAML identity provider) First User account (created during org deployment, Org Administrator role, local account)
Portal Access Scope	• Access the Provider Administrator Portal • Manage platform-wide resource quotas and regional networking • Publish content libraries to all organizations Bootstrap organizations and assign the first tenant user	• Access the Organization Portal • Manage projects, blueprints, and namespaces within their organization • Consume cloud resources via VCF Automation self-service catalog Manage organization-level user assignments and roles

IAM Recommendations

Provider Persona Requirements:

The following items must be confirmed before and/or completed after VCF Automation deployment.

- VCF Identity Broker deployed (embedded or appliance) and integrated with the target identity provider.
- VCF SSO configured and validated across VCF platform components (vCenter, NSX, SDDC Manager).
- Provider-level administrator accounts that are created in a directory and assigned provider admin roles in VCF Automation.
- Local admin password stored securely in a privileged access management (PAM) vault — not shared.

Tenant / LOB Persona Requirements:

The following items must be confirmed before and/or completed as part of organizational onboarding (tenant creation).

- Target identity provider that is configured and accessible from the VCF Automation management network.
- Tenant organization that is created by provider admin with first user account credentials stored securely.
- OIDC/LDAP/SAML IdP configured within the tenant organization before additional users are onboarded.
- Organization-level roles (Organization Administrator, Organization Auditor, and Organization User) mapped to directory groups - not individual users.
- Project-level roles (Project Administrator, Project Auditor, Project Advanced User, and Project User) mapped to directory groups - not individual users.

IAM Design Criteria and Recommendations

Persona	Design Criteria	Design Recommendation
Provider	Single sign-on (SSO) across VCF Fleet management components for administrator-level functions.	Integrate with VCF SSO for administrator-level account single sign-on across VCF Fleet management components for administrator-level functions.
Tenant	The First User account is the only local account exists within an Organization. All other account access must be maintained via integration with Identity Provider. Organizations do not require SSO access to other VCF Fleet management components.	Direct integration with an Identity Provider via OIDC, LDAP or SAML.

Figure 192: VCF Automation Enterprise Authentication Model in VMware Cloud Foundation

This example shows both the Provider and Tenants sharing the same IdP, where the Provider is integrated with the the Identity Broker Service to enable VCF SSO and the Tenants are directly integrated with the IdP.

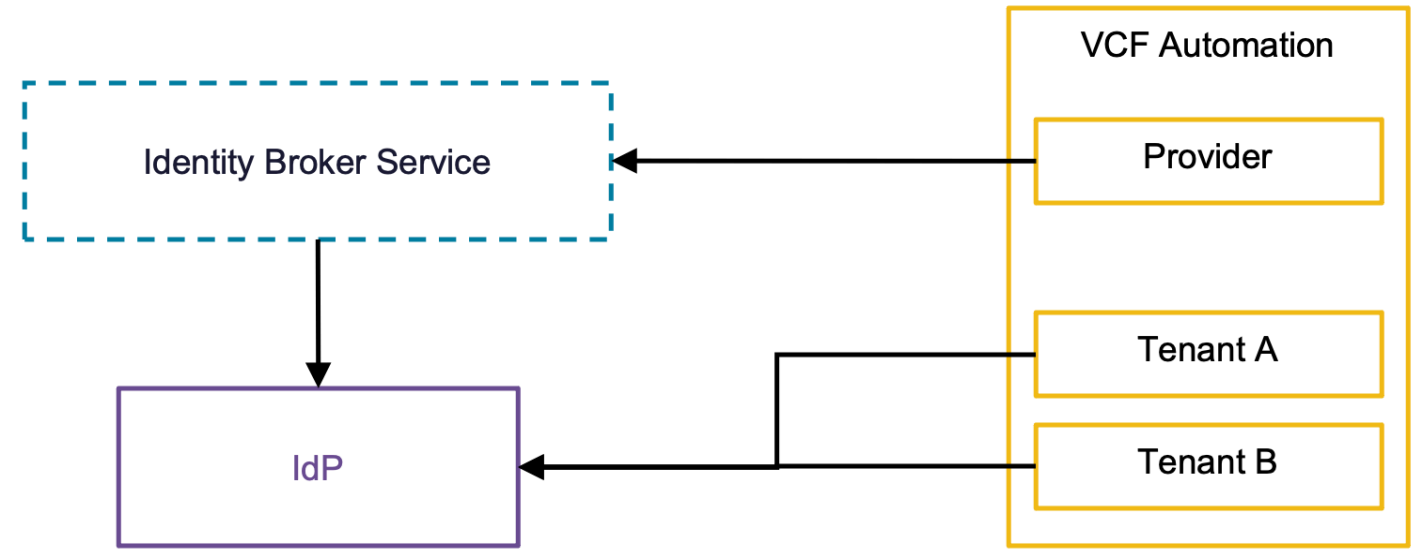
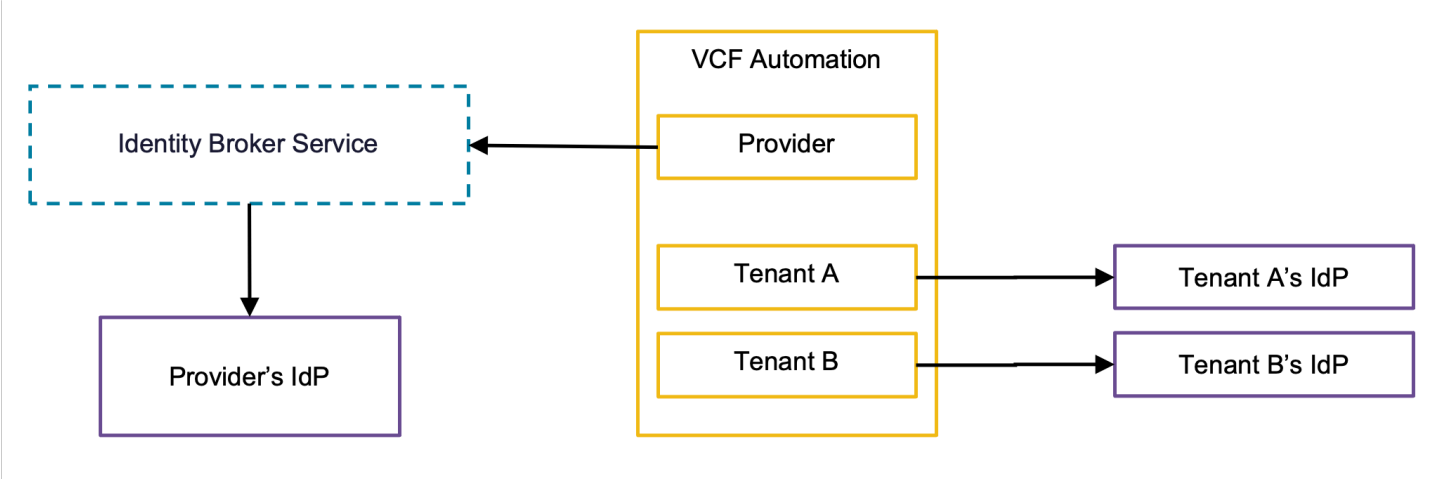

VCF Automation
Identity Broker Service
Provider
Tenant A
Tenant B
IdP

Figure 193: VCF Automation Service Provider Authentication Model in VMware Cloud Foundation

This example showt he Provider and Tenants using different IdPs. The Provider is integrated with the the Identity Broker Service to enanle VCF SSO. The Tenants are directly integrated with the IdP with their individual IdP.


VCF Automation
Identity Broker Service
Provider
Tenant A
Tenant A's IdP
Tenant B
Tenant B's IdP
Provider's IdP

For detailed information on IAM for VCF Automation review the VCF Automation Design Library section on VCF Automation Identity and Access Management Design.

VCF Operations Orchestrator Design

VCF Operations orchestrator is a workflow automation solution that is designed to simplify the automation of complex IT tasks. There are two deployment models as shown in the table below.

Orchestrator Deployment Options	Remarks	Recommended Scenario
Embedded	Installed, by default, as part of VCF Automation deployment.	• Used by the Provider Consumption Organization • Used by a Provider Admin (with system administrator rights) to create workflows that can be published to Organizations
External	Is installed manually.	• External VCF Operations orchestrator should be used by Organizations that want to create and publish their own workflows. • Each Organization can have only one VCF Operations orchestrator instance. • The Provider Admin (with System Administrator rights) deploys, manages, and provides access to the external VCF Operations orchestrator used by organizations.

VCF Operations Orchestrator Deployments for the VCF Automation Enterprise Model

The VCF Operations Orchestrator deployment depends on whether the target design includes one or multiple VCF Automation Organizations.

1. Single VCF Automation Organization - Deploy a single VCF Operations Orchestrator
2. Multiple VCF Automation Organizations - Deploy a VCF Operations Orchestrator per Organization on a 1:1 basis.

VCF Operations Orchestrator Deployments for VCF Automation Cloud Service Provider Model

Since the Cloud Service Provider model assumes a multiple VCF Automation Organization design, deploy a VCF Operations Orchestrator per Organization on a 1:1 basis.

For detailed design information, refer to the VCF Automation Design Library section on VCF Operations Orchestrator.

Chargeback and Billing

Chargeback is a cost management solution that is designed to help service providers accurately track and distribute expenses within a private cloud environment. Enabling both providers and tenants to gain insights into resource usage and ensures fair cost allocation-based on consumption. As a plug-in for tenants, Chargeback provides a metering and billing interface that simplifies usage tracking, measurement, and invoicing for services delivered through VCF Automation and VCF Operation.

For more design information, refer to the VCF Automation Design Library section on Chargeback and Billing.

Platform Protection and Multi Tenant Workload Protection Design

Backup/ Restore and Disaster Recovery (DR) for a VCF private cloud must address two distinct protection domains: the platform infrastructure itself, and the tenant workloads running on that platform. These are not the same problem. Platform DR protects the management and control plane components such as - VCF Fleet, vCenter, NSX Manager, VCF Automation, and the Identity Broker - that the platform cannot function without. Tenant workload DR protects the application VMs running in tenant Supervisor Namespaces and VPCs, governed by per-tenant SLA commitments that may have very different RPO and RTO requirements from one tenant to the next.

Both protection domains must be designed before production go-live. A platform that can recover tenant workloads but not its own management plane cannot execute that recovery. Equally, a platform that can recover its management plane

but has no tenant workload replication cannot meet the SLA obligations made to tenant organizations. This section covers both domains and documents the design choices, recommended technologies, and pre-requisite steps for each.

Platform Protection Blueprints

Protection Blueprints assist with choosing a protection and recovery model for your specific business and technology requirements. Providing a set of pre-defined architecture topologies based on specific design profiles. Each protection blueprint provides a prescriptive set of requirements and implementation steps.

Protection Blueprint	Overview	Link
VCF Component Backup and Restore	Regularly backing up the management components of a VMware Cloud Foundation platform is essential to ensure the resilience and continuity of your VCF private cloud. In the event of data loss or system failure, having up-to-date backups allows for efficient restoration of critical services, minimizing downtime and maintaining operational integrity. Implementing a consistent backup strategy safeguards your infrastructure against unforeseen disruptions.	VCF Backup and Restore - Blueprint
VCF Instance Backup and Restore	VCF Instance Recovery involves the recovery of an entire VCF Instance that has failed, and must be recovered in the same site.	VCF Instance Recovery - Blueprint
VCF Fleet Disaster Recovery	A VCF Fleet has failed completely, or the location it resides on as failed, and components must be recovered in another location, where that location is typically outside metro radius.	VCF Disaster Recovery - Blueprint
Cyber Recovery for VMware Cloud Foundation	This *Cyber Recovery for VMware Cloud Foundation* protection blueprint provides a clean room environment on a vCenter instance that allows you to inspect, analyze, and recover infected virtual machines before restoring them to a production environment.	Cyber Recovery for VMware Cloud Foundation - Blueprint

Multi Tenant DR Design Choices

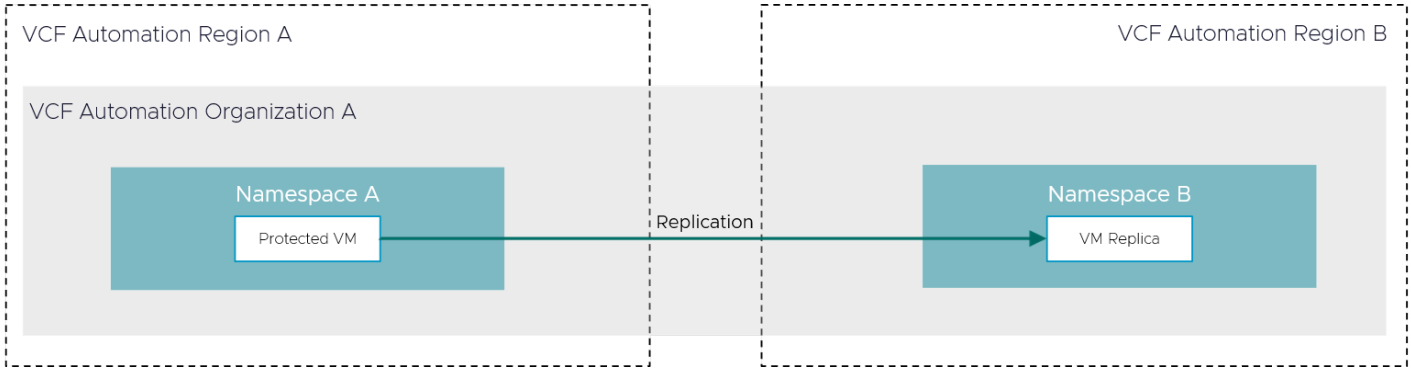
Multi-tenant workload Disaster Recovery (DR) in VCF Automation is the capability to protect, replicate, and orchestrate the recovery of tenant application VM workloads across two independent VCF Instances within a single VCF Fleet - a primary instance where workloads run day-to-day, and a recovery instance at a geographically separate site that is ready to receive and run those VM workloads in the event of a primary site failure. This section describes how that capability is designed, what technologies deliver it, what design decisions must be made per tenant, and what the pre-requisite configuration looks like within the VCF Automation architecture.

In VCF Automation, tenant workloads are isolated by design - each tenant has their own vSphere Namespace, their own Project, their own VPC, and their own Transit Gateway. This isolation model is not abandoned at the DR layer; it is carried through it.

Protection and Recovery for VCF Automation Topologies

Protection and recovery for VCF Automation allows authorized Organization users to replicate VM Service VMs across regions that each have a region quota within a vSphere Supervisor namespace.The most basicprotection and recovery topology is where VMs residing in a vSphere namespace from a single organization are replicated between two regions (for example, region A and region B) of a VCF fleet.

Figure 194:



In the diagram above, the contents of protected VMs registered in namespace A are replicated over the network to storage capacity in namespace B. Replicas of protected virtual machines can be used to recover VMs in Namespace B.

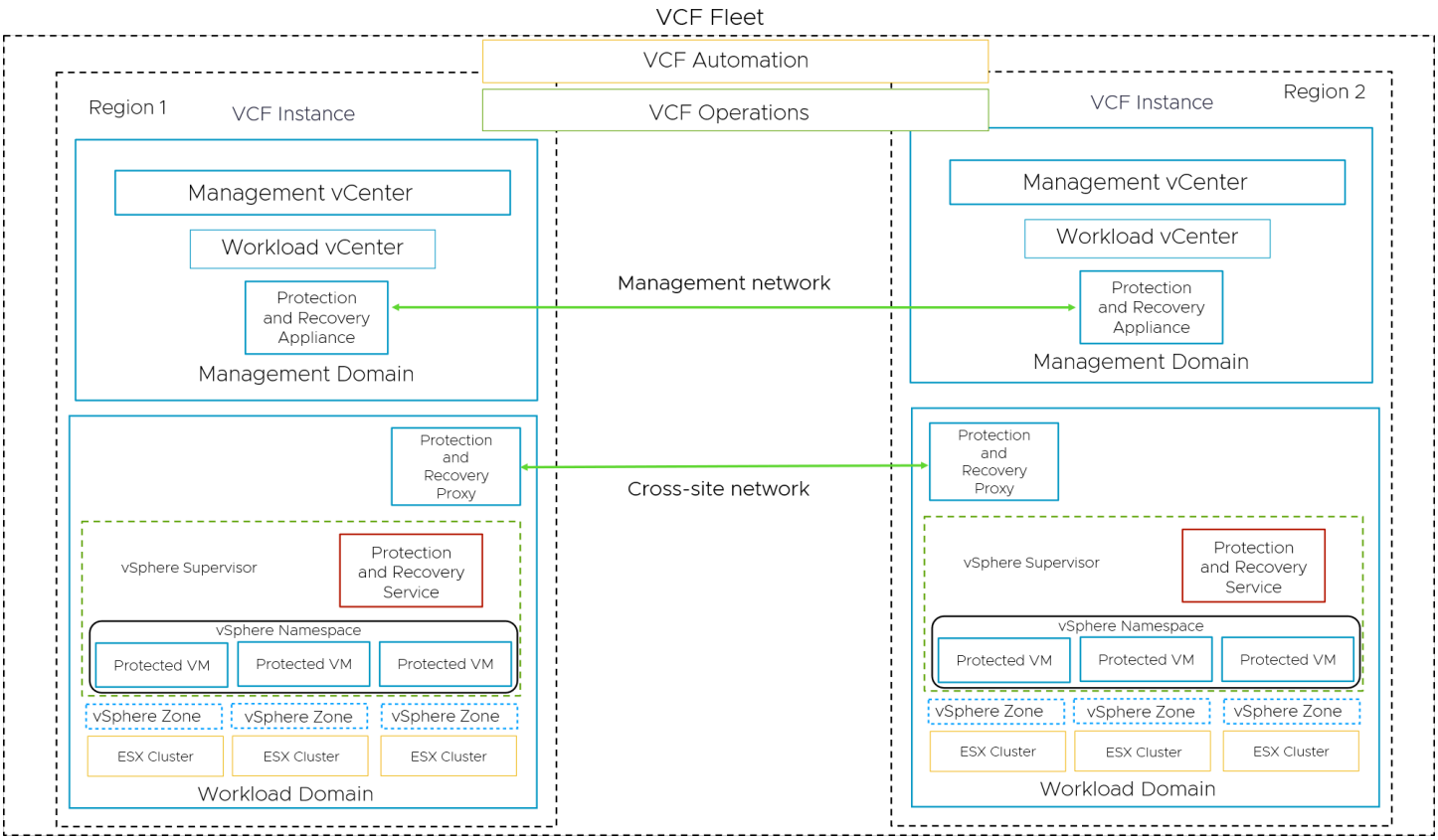
Workload Protection Design Scenario#1 :One VCF Fleet with Two VCF Instances Across Multiple Regions

A commonly supported and recommended topology for protection and recovery in VCF Automation is a single VCF fleet spanning multiple regions, where each region is backed by its own independent VCF instance. Within each region, both a management domain and one or more workload domains are deployed.

While this approach requires deploying a dedicated management domain per region, it is intentionally designed to isolate failures and limit the blast radius of any disaster event to a single region.

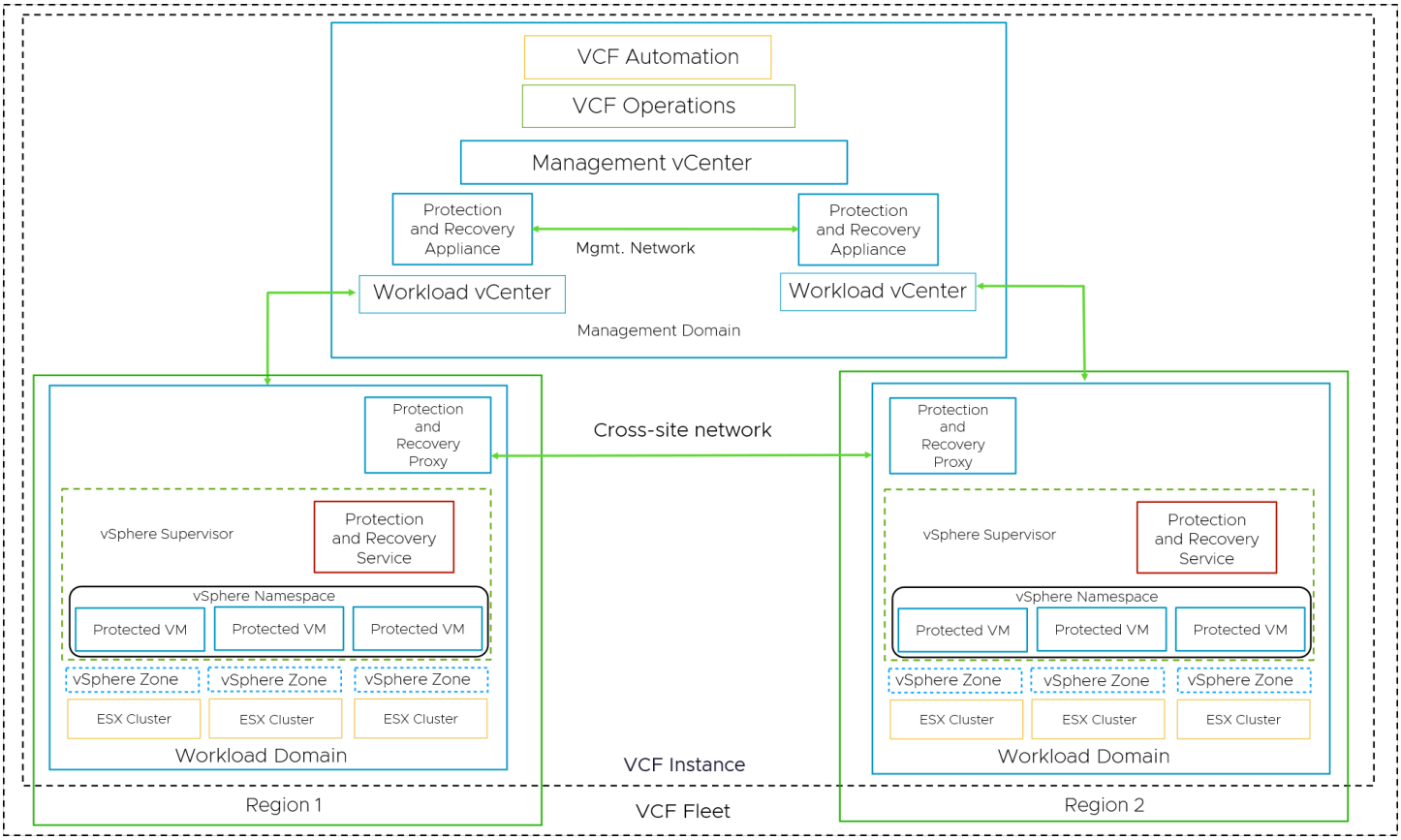
In the event that a management domain hosting VCF Automation becomes unavailable, the service can be restored in another region’s management domain. This enables continued access for organization users and allows for the recovery of workloads without dependency on the affected region.

Figure 195: One VCF Fleet with Two VCF Instances Across Multiple Regions



Workload Protection Design Scenario#1 :One VCF Fleet with Two VCF Instances Across Multiple Regions

Figure 196:


VCF Automation
VCF Operations
Management vCenter
Protection and Recovery Appliance
Mgmt. Network
Protection and Recovery Appliance
Workload vCenter
Workload vCenter
Management Domain
Protection and Recovery Proxy
Cross-site network
Protection and Recovery Proxy
vSphere Supervisor
Protection and Recovery Service
vSphere Namespace
Protected VM
Protected VM
Protected VM
vSphere Zone
vSphere Zone
vSphere Zone
ESX Cluster
ESX Cluster
ESX Cluster
Workload Domain
Region 1
VCF Instance
VCF Fleet
vSphere Supervisor
Protection and Recovery Service
vSphere Namespace
Protected VM
Protected VM
Protected VM
vSphere Zone
vSphere Zone
vSphere Zone
ESX Cluster
ESX Cluster
ESX Cluster
Workload Domain
Region 2

In this topology, a VCF fleet is composed of one VCF instance, with workload domains deployed to separate geographic regions. Each workload domain is managed by a separate workload vCenter Server Appliance (VCSA).You should only use this topology for to test and evaluation use cases where the reduced hardware overhead is desirable, but where less resilience to disaster events is acceptable.

Please refer to the following product documentation for guidance on implementing a multi-tenant data protection strategy described in the document.

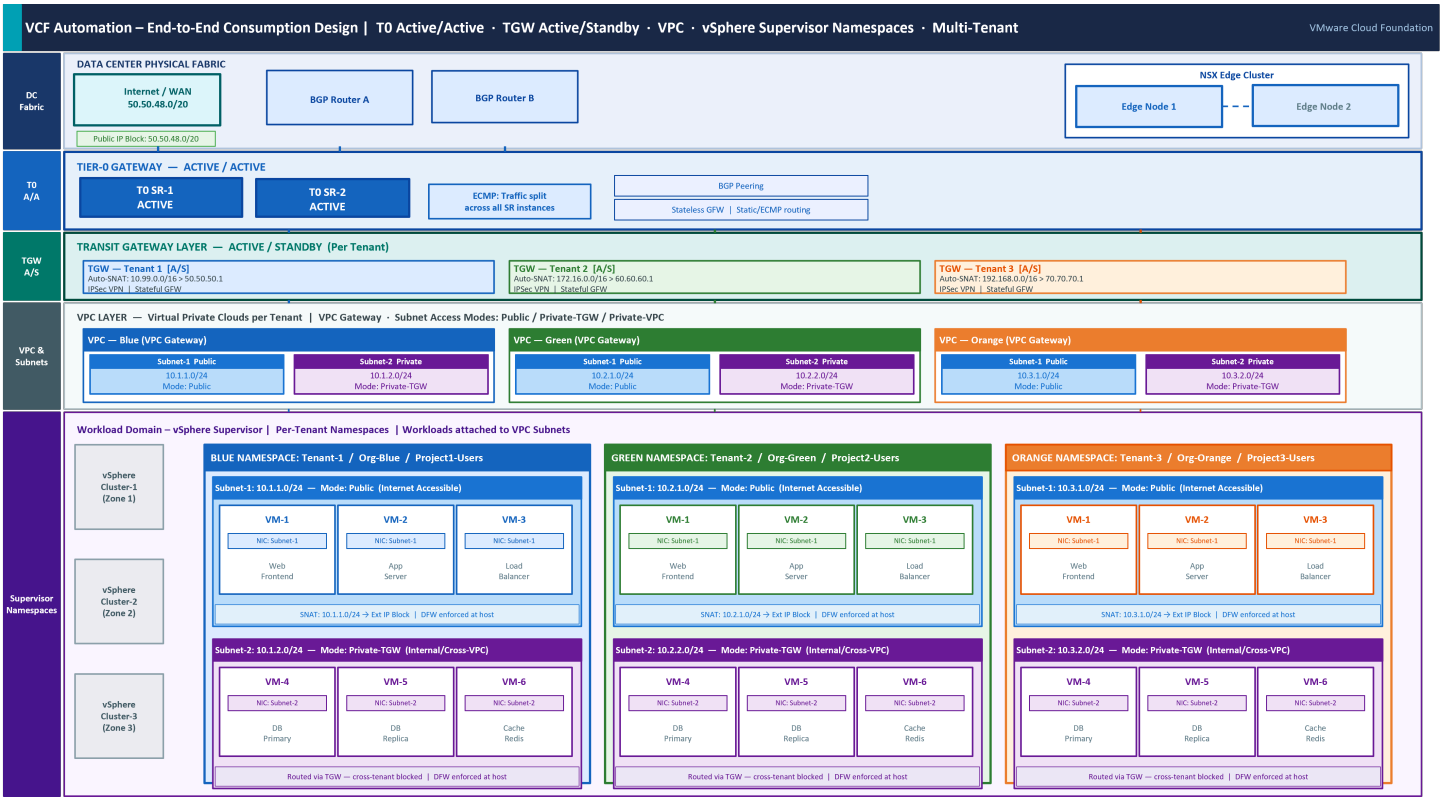
Using Protection and Recovery for VCF Automation

Deploy and Configure Protection and Recovery for VCF Automation.

> **Important:**
>
> Tenant workload protection does not eliminate dependency on the platform control plane. In a disaster scenario where VCF Automation or the management domain is lost, recovery must occur in two stages: first restore the platform management services in another VCF instance, then initiate tenant workload recovery. This makes platform recovery a prerequisite for tenant recovery.

Outcome - End to End Multi Tenant Reference Design

Figure 197: Reference Design - Self-Service Multi-Tenant Private Cloud



The blueprint reference design and its layer-by-layer explanation serve two purposes: first, to validate that all individual design decisions made earlier in this guide are coherent and consistent when assembled together; and second, to provide a concrete, deployable reference architecture that cloud architects can use as the starting point for production VCF Automation environments.

The diagram above illustrates the complete end-to-end architecture across all layers of the blueprint design: the physical DC fabric, the T0 Provider Gateway, the per-tenant Transit Gateway layer, the VPC and subnet layer, and the vSphere Supervisor Namespace workload layer.

The blueprint models a three-tenant private cloud deployment - representing three independent lines of business or organizational units (Blue, Green, and Orange) - each consuming dedicated networking, compute, and workloads through VCF Automation self-service workflows. Despite sharing the same physical infrastructure, each tenant is completely

isolated at the networking, security, identity, and resource governance layers. This is the target state that the VCF Automation platform is designed to deliver.

Attention:

This blueprint is a reference design that demonstrates how the design choices from each section of this guide work together in a production multi-tenant deployment. The IP addressing, tenant names, and workload types are illustrative. Adapt them to match your organizational IP management plan and tenant structure before deployment.

VCF Service Consumption: Extending the Foundation

The design and architecture documented each section of this blueprint design establish a complete, production-grade private cloud with VMware Cloud Foundation Automation. That foundation - the Supervisor cluster, VPC networking, Transit Gateway, VCF Automation multi-tenant model, IAM, load balancer, and disaster recovery design -is not the end state. It is the starting point.

Once your foundation is in place, VMware Cloud Foundation exposes a rich catalog of platform services that your tenant users and application developers can consume directly through the same self-service model established in this design guide. Each of these services - VMware vSphere Kubernetes Service (VKS), Data Services Manager (DSM), and Private AI infrastructure - runs on the vSphere Supervisor you have already designed and deployed. There is no separate infrastructure to build. The network, security, identity, and governance model you have already defined automatically extends to every service you add.

This section explains how the foundational architecture supports service consumption, describes the primary VCF services available, and provides the reference table of Design Libraries and Implementation Blueprints that guide the deployment of each service.

Whether you have designed VCF Automation for a single enterprise with internal teams, or for multiple independent tenant organizations sharing the same infrastructure, the consumption architecture is the same. The difference is the scope of isolation - not the capability. Both deployment models have access to the full VCF service catalog.

The following table catalogs the dedicated Design Libraries and Implementation Blueprints available for each VCF service. Each Design Library is the authoritative architectural reference for the service - it defines the design decisions, prerequisites, and configuration guidance that is required to deploy the service on VMware Cloud Foundation.

VCF Services	Description	Reference Link
VMware vSphere Kubernetes Service	The guide focuses on the architectural, operational, networking, governance, and multi-tenancy considerations that influence how organizations should consume Kubernetes services using VKS. Rather than presenting deployment steps alone, this document provides a design decision framework intended to help architects align VKS consumption patterns with organizational operating models, governance requirements, scalability objectives, and private cloud strategies.VMware vSphere Kubernetes Service Consumption Blueprint.	VMware vSphere Kubernetes Service
Private AI Services Consumption Blueprint	The *Private AI Services Consumption Blueprint* describes the design selections and architectural components required to deploy and operate a GenAI platform on VMware Cloud Foundation. It is intended to be used in conjunction with an underlying VCF infrastructure blueprint.	Private AI Service Consumption Blueprint

VCF Services	Description	Reference Link
Data Service Manager (DSM)	VMware Data Services Manager (DSM) is designed to provide a "Database-as-a-Service" (DBaaS) solution for private cloud. DSM is delivered as an Advanced Service for VMware Cloud Foundation. DSM simplifies and automates database lifecycle management for database engines that are deployed in VCF environments: PostgreSQL, MySQL, and Microsoft SQL Server.	VMware Data Service Manager

Transition Scenario – Aria Automation to VCF Automation (All Apps Model)

This section addresses customers who are currently using **Aria Automation with a VM-based consumption model** and are looking to transition to **VCF Automation** to enable a more modern private cloud experience with **self-service, multi-tenancy, and Kubernetes-native consumption.**

In these environments, Aria Automation is typically used for VM provisioning with limited or externally implemented tenancy and governance constructs. As organizations evolve toward platform-centric operating models, there is a need to move beyond VM-only consumption and adopt a unified model that supports both **virtual machines and modern application workloads**.

VCF Automation introduces the **All Apps consumption model**, which provides:

- Built-in multi-tenancy using organizations and projects
- Policy-driven governance with quotas and RBAC
- Self-service catalog and API-driven provisioning
- Native support for Kubernetes through VKS
- Integrated networking using NSX VPC-based constructs

For customers transitioning from Aria Automation, this shift represents not just a tooling change, but an operational model transformation-from infrastructure-centric provisioning to platform-based consumption.

The following guidance outlines recommended approaches for:

- Transitioning existing VM-based workloads and automation workflows
- Introducing multi-tenant constructs using organizations and projects
- Enabling self-service consumption for both VM and Kubernetes workloads
- Designing VPC-based networking aligned with modern application architectures

Customers can adopt this model incrementally by continuing to operate existing VM workloads while enabling **new application deployments through the VCF Automation All Apps model**, allowing a phased transition toward a fully integrated private cloud platform.

> **Note:**
>
> For detailed design information, refer to the Bimodal Consumption Design for VM Apps and All Apps Workloads

Figure 198: Transitional Architecture: VLAN Backed workload with Aria Automation to VCF Automation Consumption

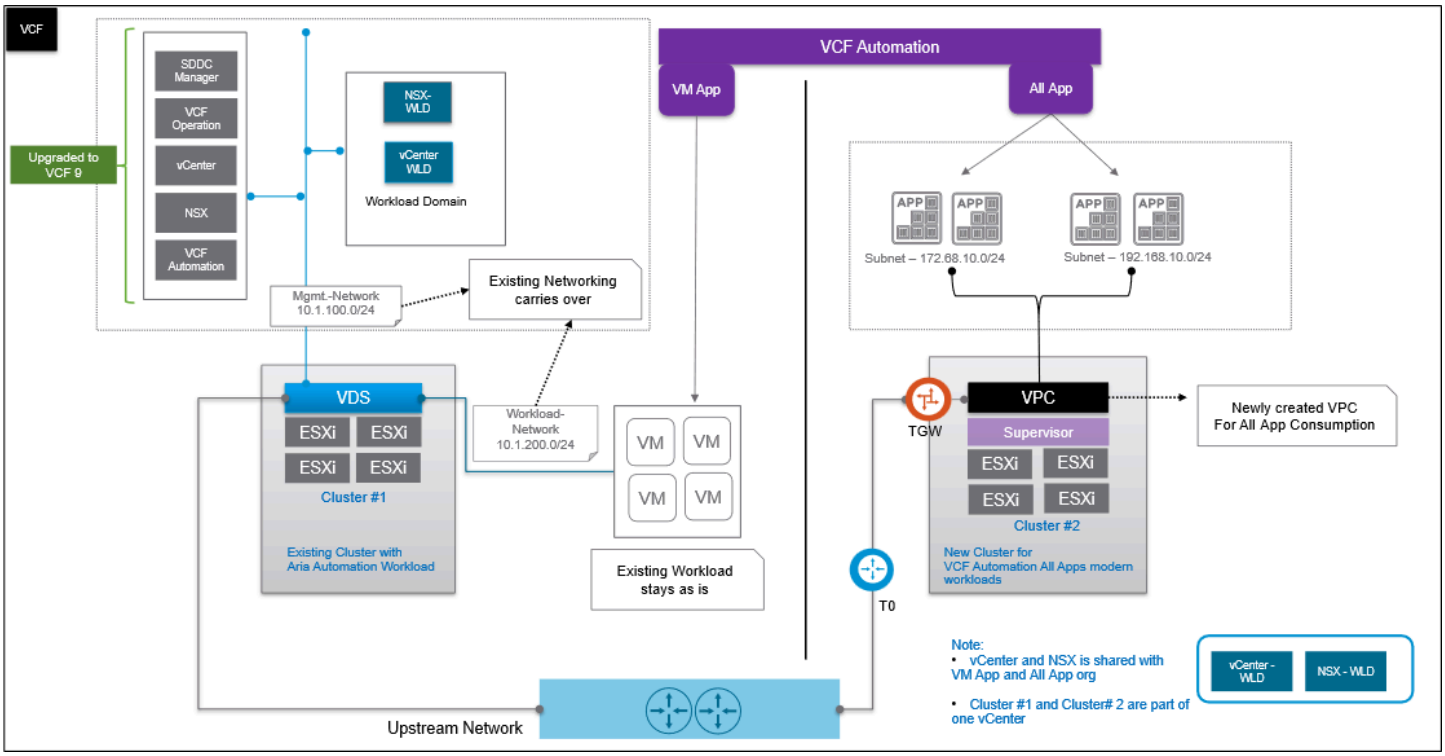
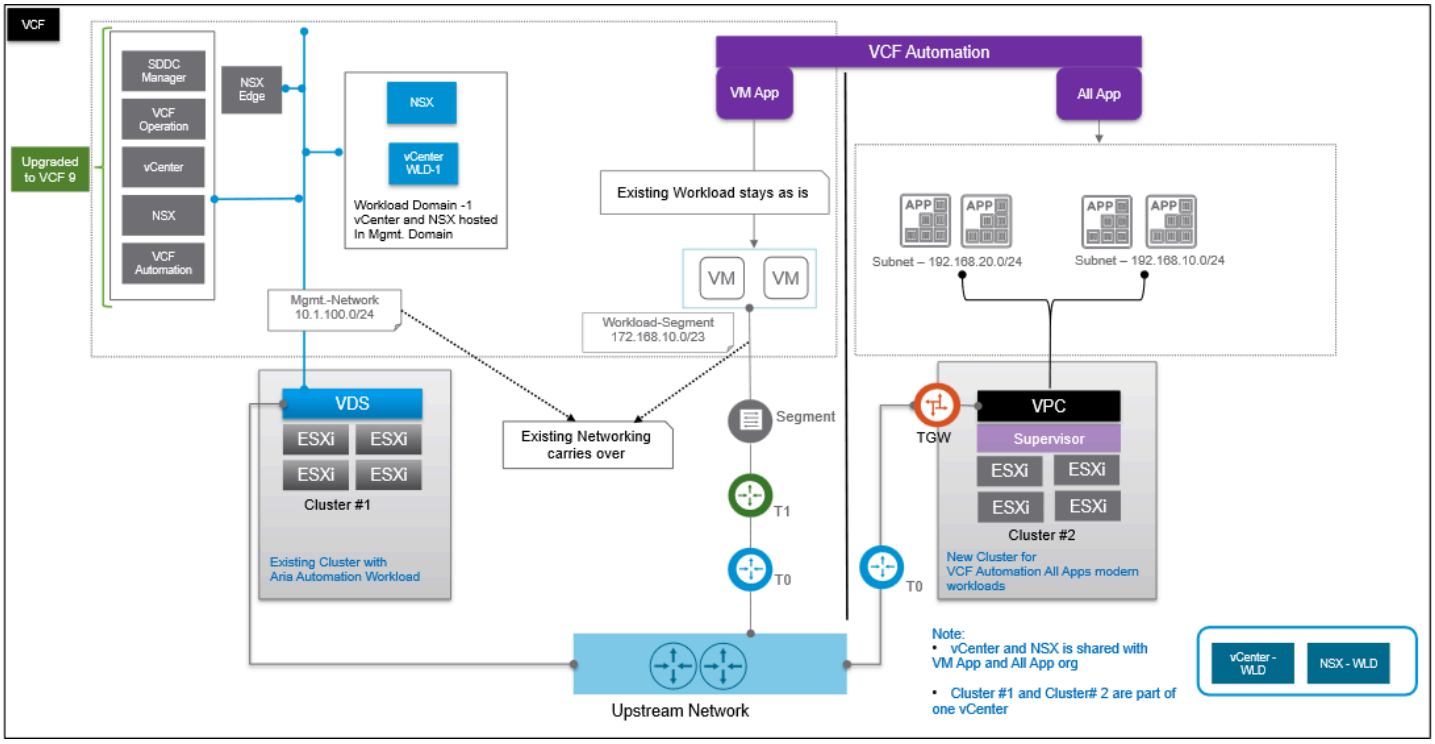


Figure 199: Transitional Architecture: NSX- backed workload with Aria Automation to VCF Automation



Design Context

This architecture illustrates a transitional adoption model for organizations migrating from existing VMware Aria Automation environments to VMware Cloud Foundation with VCF Automation while preserving existing workload operations and enabling gradual adoption of modern application consumption models.

In these scenarios, organizations may already operate existing Aria Automation environments using either:

- traditional VLAN-backed networking through a VDS-based architecture
- or existing NSX-backed networking and segment-based workload connectivity models

Following the transition to VCF Automation:

- existing workloads continue operating on their current networking and infrastructure constructs
- existing VLAN-backed or NSX-backed networking configurations remain unchanged
- existing workloads continue to be consumed through the **VM App organization model**
- operational disruption to existing workloads, routing models, firewall policies, and application connectivity is minimized

This design enables organizations to preserve existing operational and networking models while incrementally introducing modern platform consumption capabilities through the VCF Automation All App model.

Note:

For detailed design information, refer to the Bimodal Consumption Design for VM Apps and All Apps Workloads

Implementation of Self-Service Multi-Tenant Private Cloud Consumption Blueprint

The example phases for implementing the *Self-Service Multi-Tenant Private Cloud Consumption Blueprint*.

This is an example implementation for deploying a **Multiple Organizations - Multi-Tenant or Cloud Service Provider Self-Service Blueprint Design**.

Deployment Criteria for the example Multiple Organizations - Multi-Tenant or Cloud Service Provider Self-Service Blueprint Implementation
• A single VCF Fleet. • External-facing deployment of VCF Automation. • Uses an external load-balancer. • Separation of management and workloads into different domains. • Multiple organizations share a single workload domain where isolation is achieved through vSphere Namespaces and network segmentation. • Resource isolation for multiple organizations by providing each with a dedicated Virtual Routing and Forwarding (VRF) instance, ensuring completely separate network routing and forwarding domains.

Table 458: Multiple Organizations - Multi-Tenant or Cloud Service Provider Self-Service Blueprint Design Implementation Procedure.

Procedure Grouping	Procedure	Description
Deploy VCF Fleet in Single Site	Implementation of VCF Fleet in a Single Site	Follow these steps to deploy a VCF Fleet in a Single Site.
Important: The Implementation of VCF Fleet in a Single Site procedure assumes the VCF Automation: • Deployed as part of the base VCF Fleet • Deployed on the same network as the other VCF Fleet management components • You are deploying VCF Automation Using Deployment Patterns 1 or 2. If you intend to deploy VCF Automation using a third-party load balancer as in Deployment Pattern 3: External-Facing deployment of VCF Automation, **DO NOT DEPLOY VCF Automation by default as part of the VCF Fleet deployment.** VCF Automation will be deployed Day-2 as separate step below.		
Deploy VCF Automation	Deploying VCF Automation as a Day-N Operation	This will deploy VCF Automation on a dedicated network in the DMZ.
Configure VCF Automation for self-service consumption - Provider Portal Level	Managing VCF Automation Provider Users and Groups	Configure Access Control (users and groups) Note: This assumes that VCF SSO was configured as your Identity Provider as part of the VCF Fleet in Single Site Blueprint.
	Create a Region in Your VCF Automation	Configure a region in VCF Automation to make vSphere compute, and storage resources available.
	Managing IP Blocks in the VCF Automation Provider Management UI	This section provides the steps necessary to configure External IP blocks.
	Managing External Connections in the VCF Automation Provider Management UI	This section provides the steps necessary to configuring an External Connection using either the Centralized or Distributed VLAN model

Procedure Grouping	Procedure	Description
Configure Organization for All Apps - Provider Portal Level	Create a VCF Automation Organization for All Applications	Create an organization for All Apps in VCF Automation. Note: This includes: • Creating and assigning an initial Region Quota to the organization • Creating and assigning an initial Networking (external connection) • Creating and assigning the Organization Administrator to the First User Account
	Configure the Regional Networking of a VCF Automation Organization	As necessary, configure additional regional networking for your organization in VCF Automation.
	Create a Region Quota for a VCF Automation Organization	As necessary, configure region quota for your organization in VCF Automation.
	Create a Content Library	Create an initial Provider Content Library.
	Upload a Content Library Item	Upload initial content library items (ISO, OVA or OVF).
Hand off to Organization Admin	N/A	Provide the First User account and default to organization URL to the Organization Admins so that they can configure the Organization through the organization portal
Configure Organization for All Apps - Organization Portal Level Note: The organization admin initially uses the First User account to configure the Identity Provider and subsequent users and groups. Once this is complete, the organization admin can switch to an appropriate Identity Provider integrated account as necessary.	Managing Identity Providers in VCF Automation Organizations	Configure Identity Provider
	Managing Predefined User Roles in VCF Automation	Configure Access Control (users and groups)
	Deploy VCF Operations Orchestrator	Deploy stand-alone VCF Operations orchestrator (if intending to integrate the Organization with VCF Operations orchestrator). Note: This step is performed outside of VCF Automation by an administrator with direct access to the target vCenter.
	Integrate VCF Automation with VCF Operations orchestrator	Integrate stand-alone VCF Operations orchestrator (if intending to integrate the Organization with VCF Operations orchestrator). Note: This step is performed by initially loggin into the Provider Portal with a Provider Administrator level account.
Deploy and Configure Protection and Recovery for VCF Automation	Deploy and ConfigureProtection and Recoveryfor VCF Automation	Protection and recovery for VMware Cloud Foundation Automation is a multi-organization replication and disaster recovery solution for VM Service VMs.
Configure Replication and Protect VMs created within VCF Automation Organizations for All Apps	Using Protection and Recovery for VCF Automation	Protection and recovery for VCF Automation 9.1 provides protection and recovery for regular and multi-tenant providers that want to replicate workload VMs between a pair of supervisors, within two region quotas, in the same VCF Automation instance and organization. After you have deployed and configured Protection and Recovery for VCF Automation, you can start creating VMs, configuring replication, and protecting / reprotecting VMs.

Remember:

For each new Organization for All Apps you will:

- Create new External IP Blocks (as necessary)
- Create new External Connections (as necessary)
- Repeat the the following sections
  - Configure Organization for All Apps - Provider Portal Level
  - Configure Organization for All Apps - Organization Portal Level
  - Deploy and Configure Protection and Recovery for VCF Automation
  - Configure Replication and Protect VMs created within VCF Automation Organizations for All Apps

Securing VCF Automation Deployment Pattern 3 with vDefend and Avi
This runbook, describes how to deploy VCF Automation on an external network using Avi as the external load balancer.

Overview

The VMware Cloud Foundation design library has a section that covers different VCF Automation Deployment Patterns for VCF private clouds. For the first 2 deployment patterns listed:

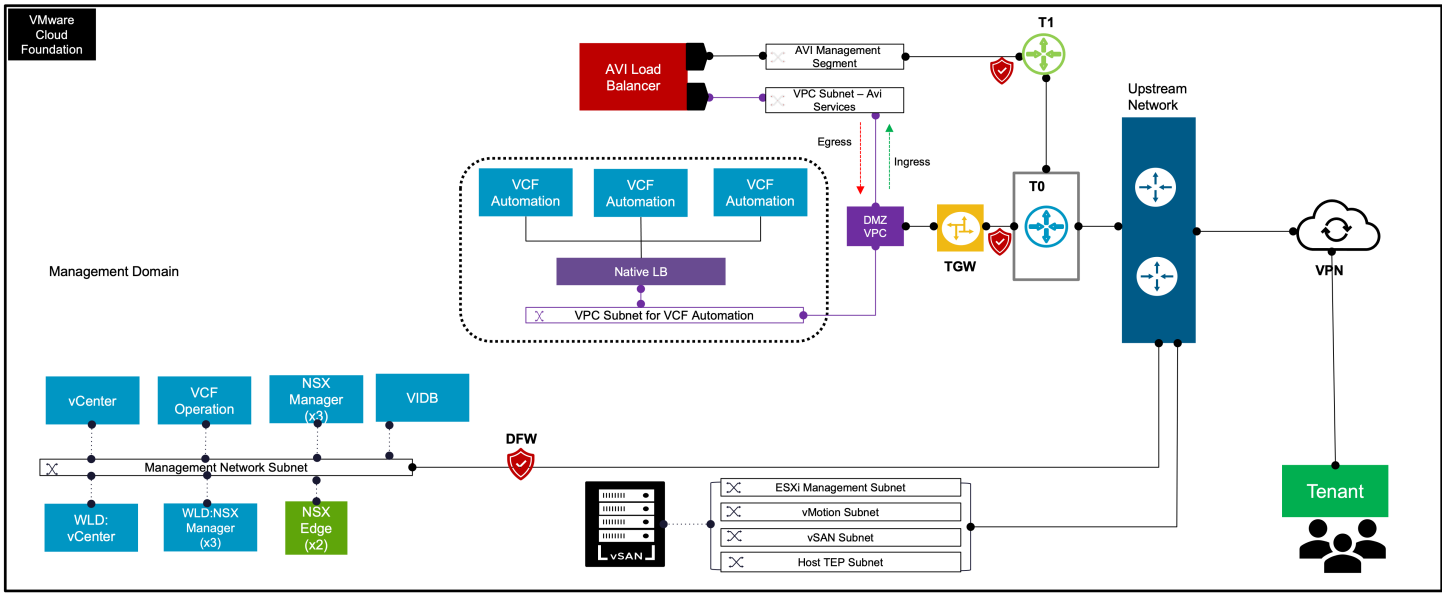
- Deployment Pattern 1: VCF A in a VCF Instance with Single Domain and Multiple Clusters.
- Deployment Pattern 2: VCF Automation in a VCF Instance with multiple domains.

VCF Automation is deployed in the same network as the other management components reside(to include: vCenter, NSX Manager, SDDC Manager, VCF Operations Manager, etc.). Securing such architecture is covered in detail in the published Detailed Design for Lateral Security for VMware Cloud Foundation with VMware vDefend.

The final design model, Deployment Pattern #3: External-Facing Deployment of VCF Automation, details implementing VCF Automation to seamlessly interact with external networks or services in a secure manner while minimizing the attack surface, containing potential security breaches, and providing the necessary controls for meeting various security and compliance standards for VCFA. In this document, we will be covering how to secure this deployment pattern.

Deployment Overview

Before proceeding with the technical details of securing the VCF Automation, we need to ensure that the environment is configured per VMware recommendation. The following diagram illustrates the architecture covered in this run book:



The following sections will go into detail but here are the highlights In this model:

1. **VCF Automation** is deployed as an appliance - in a single node (Simple) or 3 node (High Availability) model within a VPC specifically created as the DMZ for VCF Automation (DMZ VPC). The VPC includes:
   a. An external IP block defining the publicly facing IP range available for this deployment.
   b. Subnets.

    a. A manually configured VPC public subnet for VCF Automation components.
        a. IP addressing is pulled from the external IP block.
        b. Must be a /28 or larger CIDR range.
    b. An automatically created VPC services subnet (private) where the Avi service engines providing load balancing services for VCF Automaton are deployed.
2. The **Avi controllers**, the management and control plane components of the Avi Load Balancers - are connected to a VLAN-backed management subnet (VCF Management). It must be fully routed to an overlay-backed NSX segment that is connected to the NSX T1 gateway where the Service Engine management NICs reside. This T1 gateway will be used for securing Avi controllers to Avi Service Engine communications with gateway firewall (GFW) functionality.
3. The **Avi Service Engines**, which runs the advanced load balancer that is used for load balancing traffic for VCF Automation, is connected to the VPC services subnet in DMZ VPC. The one-arm load balancing function for VCF Automation will be hosted here. This is the same VPC that has the subnet for VCF Automation. Please note that:
    a. The VPC services subnet is a private subnet.
    b. The VPC subnet where the VCF Automation components are located e is a VPC public subnet.
4. The **DMZ VPC** has one public VPC subnet:
    a. The public VPC subnet will provide all IPs used to deploy VCF Automation and configure the AVI-based virtual service.
    b. The public VPC subnet is part of the external IP block and is connected to the transit gateway (TGW) for onramp/offramp function. This TGW will also provide gateway firewall (GFW) services.
5. **VCF Management** components - such as vCenter, NSX, VCF Operations etc. - are connected to a VLAN-backed management subnet. vDefend DFW will be used here for security services.

The next section, we will cover summary order of Operations, assumptions and pre-requisites related to installation and configuration.

Order of Operations Summary

The following list summarizes the order of operations for integrating VCF Automation and Avi:

1. Create a NSX project with a VCF Automation Subnet for the DMZ VPC for this deployment
    a. Recommend that this is a dedicated NSX Project for the VCF Automation DMZ VPC.
    b. Recommend assigning an external IP block to the NSX Project that is at least a /27 (32 IPs).
    c. From that external IP block, you will need a public subnet that is at least a /28 (16 IPs) as all the IPs used will be public IPs.
2. Deploy Avi and integrate with the target management domain
3. In Avi, pre-VCF Automaton installation
    a. In Avi
        a. Create a virtual service for the target VCF Automation FQDN.
        b. Configure the virtual service to allow L4 passthrough on port 443 to allow for VCF Automation installation.

        Important:

        Post install it is recommended to reconfigure the virtual service to support HTTP/HTTPS with SSL offloading leveraging either the self-signed certificate created at VCF Automation install or a custom certificate (also applied to VCF Automation).
    b. In the Management Domain NSX Manager, ensure Avi SEs are added to the distributed firewall exclusion list in the target NSX Manager

        Important:

        VCF Management Components are automatically in an exclusion list.You will remove them from the exclusion list in a later step.
4. Deploy VCF Automation via VCF Operations fleet management API (see section 1.2 VCF Automation Deployment Procedure for more information)
5. Once VCF Automation is deployed following these documents for configuration
    a. For provider-level configuration, refer to the Provider Management guide.
    b. For organization-level configuration, refer to the Organization Management guide.

6. Secure the VCF Automation deployment
   a. In Avi
      a. Reconfigure virtual service to support HTTP/HTTPS with SSL offloading.
      b. Configure WAF rules for the virtual service.
   b. In NSX Manager
      a. Remove VCF Management Components from the distributed firewall exclusion list.
      b. Configure gateway and distributed firewall rules to secure communication with VCF Automation.

Assumptions

for the VCF Automation application cluster. For more information on VCF Automation design refer to the VCF Automation Design Library section of the VCF Design Guide.

The following is a list of assumptions for how the target environment should be configured:

- VCF 9.1 Fleet infrastructure.
  - Deployed and configured.
  - VCF Automation is NOT deployed with the base VCF Fleet deployment.
- Avi
  - Installed in the target VCF management domain through VCF Operations fleet management.
  - Integrated with the VCF management domain instance.
- VCF Automation
  - Will be deployed as a Day-2 activity.
  - Will be deployed in a network separate (i.e. DMZ network) from the other VCF Fleet components.
  - The target separate network will be a VPC subnet.
- The VCF Fleet, Avi, and vDefend Security component licenses can be entitled.

VCF Automation Deployment Procedure

Deploying VCF Automation with external load balancer integration requires:

- VCF Automation is installed as a day-2 activity after initial VCF Fleet deployment.
- VCF Automation is deployed through VCF Operations fleet lifecycle API.

This section reviews the applicable:

- API commands.
- JSON deployment template.
- How to derive the bearer token for authentication.
- Example curl command including the API and example JSON.
- Explanation of how to determine the data for fields in the JSON.

VCF Automation Prerequisites

- VCF Automation install binaries are downloaded.
- Fully Qualified Domain Names (FQDN) registered and resolvable.
- IP addresses are defined.

API Command

```
curl --request POST \
  --url https://<Fleet Lifecycle FQDN>/fleet-lcm/v1/components \
  --header 'authorization: Bearer <Token>' \
  --header 'content-type: application/json' \
  --data '{
  "componentSpecs": [
    {
     <JSON Payload>
    }
```

```
  ]
}'
```

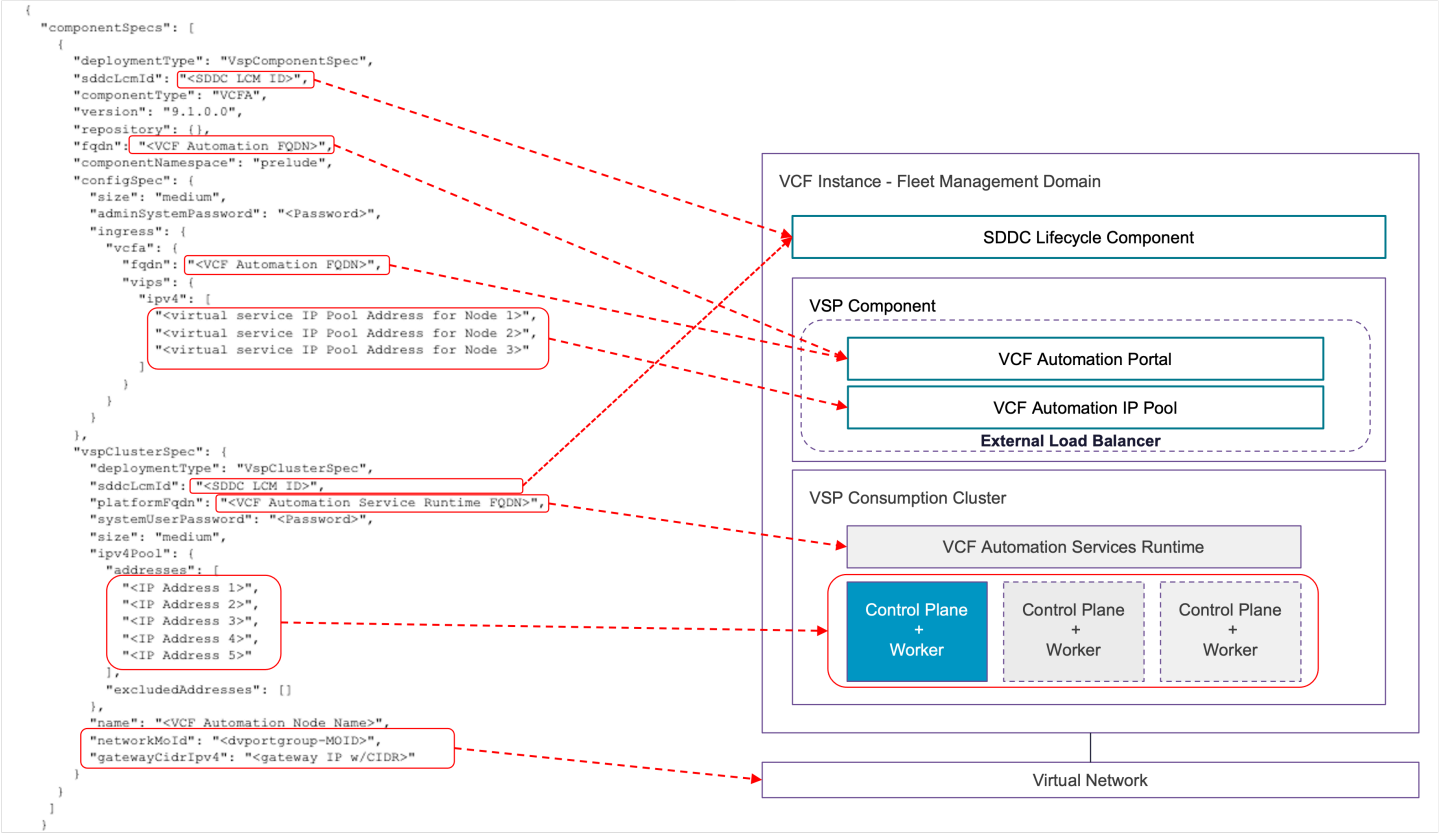
Deployment JSON Template

```
{
  "componentSpecs": [
    {
      "deploymentType": "VspComponentSpec",
      "sddcLcmId": "<SDDC LCM ID>",
      "componentType": "VCFA",
      "version": "9.1.0.0",
      "repository": {},
      "fqdn": "<VCF Automation FQDN>",
      "componentNamespace": "prelude",
      "configSpec": {
        "size": "medium",
        "adminSystemPassword": "<Password>",
        "ingress": {
          "vcfa": {
            "fqdn": "<VCF Automation FQDN>",
            "vips": {
              "ipv4": [
                "<virtual service IP Pool Address for Node 1>",
                "<virtual service IP Pool Address for Node 2>",
                "<virtual service IP Pool Address for Node 3>"
              ]
            }
          }
        }
      },
      "vspClusterSpec": {
        "deploymentType": "VspClusterSpec",
        "sddcLcmId": "<SDDC LCM ID>",
        "platformFqdn": "<VCF Automation Service Runtime FQDN>",
        "systemUserPassword": "<Password>",
        "size": "medium",
        "ipv4Pool": {
          "addresses": [
            "<IP Address 1>",
            "<IP Address 2>",
            "<IP Address 3>",
            "<IP Address 4>",
            "<IP Address 5>"
          ],
          "excludedAddresses": []
        },
        "name": "<VCF Automation Node Name>",
        "networkMoId": "<dvportgroup-MOID>",
        "gatewayCidrIpv4": "<gateway IP w/CIDR>"
      }
    }
  ]
}
```

Deployment JSON Example and Considerations

Deployment JSON Example and Explanation

The following image shows the mapping of the JSON variables to the relevant VCF Components.



Considerations for the JSON Example

IP Addressing and FQDN

- Total number of required IP Addresses
  - Total IPs = 10
  - IP breakdown by function
    - Platform FQDN (aka: VCF Automation Services Runtime FQDN)
      a. The address assigned to the FQDN of the “vspClusterSpec > platformFqdn” section of the example
      b. IPs required = 1
    - VCF Automation FQDN (aka: Avi Virtual Service VIP)
      a. Assigned to the virtual services VIP
      b. “Fqdn” fields in the example
      c. IPs required = 1
    - Ingress IPs (aka: Avi IP Pool)
      a. “virtual service IP Pool” IP addresses in the “ingress > vcfa > vips > ipv4” section of the example
      b. IPs required = 3
    - Pool IPs
      a. The addresses in the “vspClusterSpec > ipv4Pool > addresses” section of the example
      b. IPs required = 5
- IP Addressing for the VCF Automation components
  - IP Pool and Ingress IP Addresses - Assigned from the VPC public subnet created within the DMZ VPC for this deployment.
  - Platform FQDN IP Address - Assigned from the VPC public subnet created within the DMZ VPC for this deployment.

  VCF Automation FQDN IP Address
  - Statically assigned within Avi (see section 2.3.2.1 Virtual service configuration for the process).

  - IP address is pulled from the external IP block assigned to the DMZ VPC.
  - Not an address within the VPC public subnet.

  For example:

  The external IP Block is 172.16.49.0/24.

  The VPC public subnet is 172.16.49.32/28.

  The VCF Automation FQDN IP Address is any other IP in the external IP block such as 172.16.49.200.

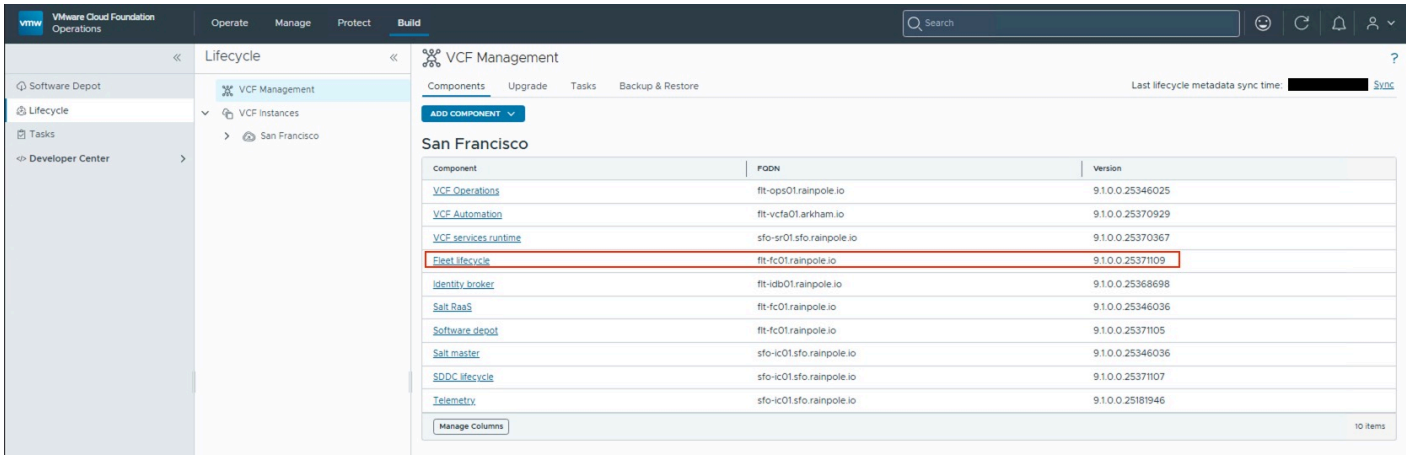
- FQDNs
  - VCF Automation FQDN for the VIP assigned to the virtual service configured in Avi - The “fqdn” fields in this example.
  - Platform FQDN (aka: VCF Automation Services Runtime FQDN)- The “platformFqdn” field in this example.

Node Sizing and High Availability

- The json example assumes a three-node, highly available deployment.
- The json example sets the “medium” node size. If you require large nodes, replace medium in the “size” fields with “large.”

Identifying the Fleet Lifecycle FQDN

- It is found in VCF Operations.
- Navigate to Build > Lifecycle > VCF Management > Components.



Identifying the SDDC LCM ID

- Run the following command
- The SDDC LCM ID is found in the “id” field in the response payload

```
curl --request GET \
  --url https://<Fleet Lifecycle FQDN>/fleet-lcm/v1/sddc-lcms \
  --header 'authorization: Bearer <Token>' \
  --header 'content-type: application/json'
```

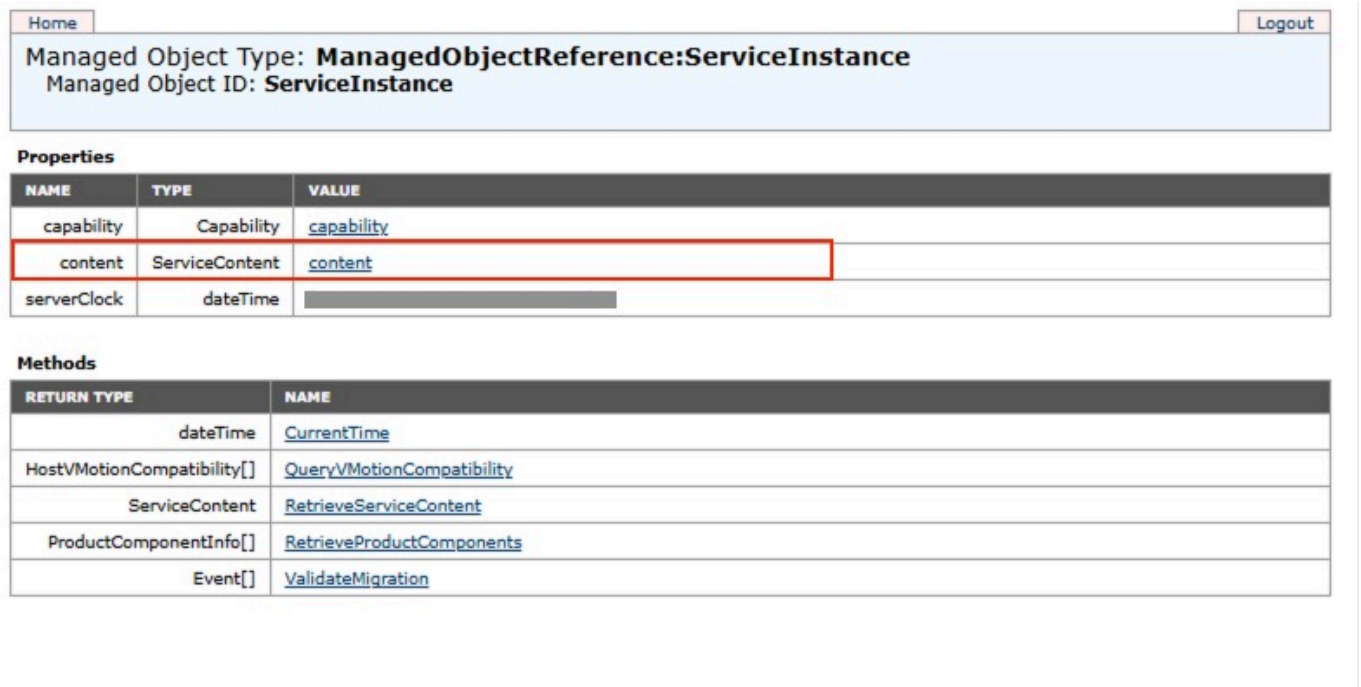
Identifying the Network MOID

- MOID - Network Management Object ID.
- Associated with the VPC Public Subnet (dvportgroup) where VCF Automation will be deployed and connected.
- To find the MOID, there are two methods:
  - Manual Method.
  - Scripted Method.

Manual Method

- Uses the vCenter Managed Object Browser (MOB).
- In vCenter 9,the MOB is disabled by default. To enable the MOB, refer to KB:https://knowledge.broadcom.com/external/article/401669/mob/.
- This is the network management object id of the VPC Subnet in which VCFA will be deployed and connected to.

1. Login to the vCenter managed object browser.
   a. https://<target vcenter>/mob/
   b. Use SSO credentials such as adminstrator@vsphere.local
2. Navigate to “content” in the left-hand column, then in the right-hand column of the same line, click the content.



Home | Logout

Managed Object Type: **ManagedObjectReference:ServiceInstance**
Managed Object ID: **ServiceInstance**

Properties

NAME	TYPE	VALUE
capability	Capability	capability
content	ServiceContent	content
serverClock	dateTime	

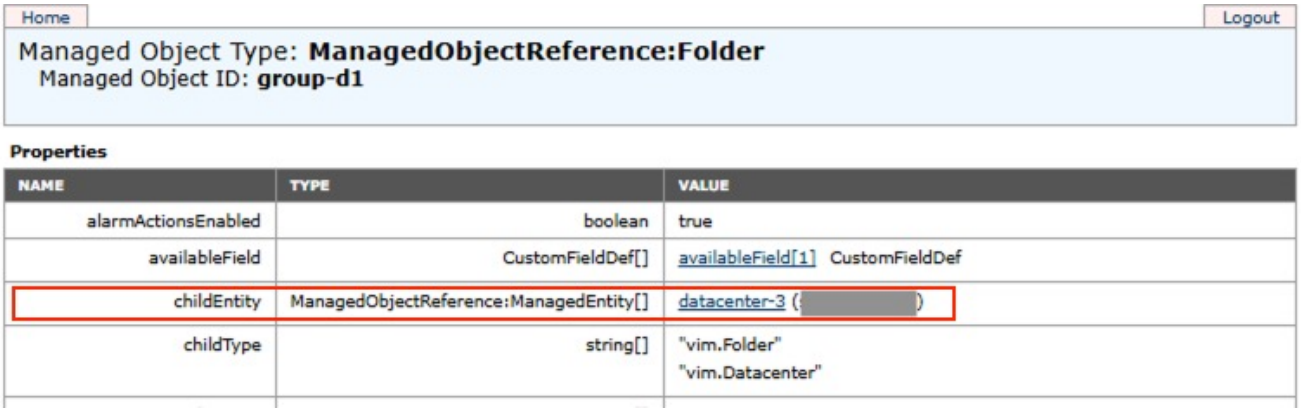
Methods

RETURN TYPE	NAME
dateTime	CurrentTime
HostVMotionCompatibility[]	QueryVMotionCompatibility
ServiceContent	RetrieveServiceContent
ProductComponentInfo[]	RetrieveProductComponents
Event[]	ValidateMigration

3. Navigate to “roofolder” in the left-hand column, then in the right-hand column of the same line, click the link. It should be “group-d1 (Datacenters).”

customizationSpecManager	ManagedObjectReference:CustomizationSpecManager	CustomizationSpecManager
datastoreNamespaceManager	ManagedObjectReference:DatastoreNamespaceManager	DatastoreNamespaceManager
diagnosticManager	ManagedObjectReference:DiagnosticManager	DiagMgr
directPathProfileManager	ManagedObjectReference:DirectPathProfileManager	DirectPathProfileMgr
dvSwitchManager	ManagedObjectReference:DistributedVirtualSwitchManager	DVSManager
eventManager	ManagedObjectReference:EventManager	EventManager
extensionManager	ManagedObjectReference:ExtensionManager	ExtensionManager
failoverClusterConfigurator	ManagedObjectReference:FailoverClusterConfigurator	FailoverClusterConfigurator
failoverClusterManager	ManagedObjectReference:FailoverClusterManager	FailoverClusterManager
fileManager	ManagedObjectReference:FileManager	FileManager
guestCustomizationManager	ManagedObjectReference:VirtualMachineGuestCustomizationManager	GuestCustomizationManager
guestOperationsManager	ManagedObjectReference:GuestOperationsManager	guestOperationsManager
healthUpdateManager	ManagedObjectReference:HealthUpdateManager	HealthUpdateManager
hostProfileManager	ManagedObjectReference:HostProfileManager	HostProfileManager
hostSpecManager	ManagedObjectReference:HostSpecificationManager	HostSpecificationManager
ioFilterManager	ManagedObjectReference:IoFilterManager	IoFilterManager
ipPoolManager	ManagedObjectReference:IpPoolManager	IpPoolManager
licenseManager	ManagedObjectReference:LicenseManager	LicenseManager
localizationManager	ManagedObjectReference:LocalizationManager	LocalizationManager
overheadMemoryManager	ManagedObjectReference:OverheadMemoryManager	OverheadMemoryManager
ovfManager	ManagedObjectReference:OvfManager	OvfManager
perfManager	ManagedObjectReference:PerformanceManager	PerfMgr
propertyCollector	ManagedObjectReference:PropertyCollector	propertyCollector
rootFolder	ManagedObjectReference:Folder	group-d1 (Datacenters)
scheduledTaskManager	ManagedObjectReference:ScheduledTaskManager	ScheduledTaskManager

4. Navigate to “childEntity” in the in the left-hand column, then in the right-hand column of the same line, click the link for the target datacenter that includes the cluster with the target dvportgroup (likely your management domain datacenter).


Home

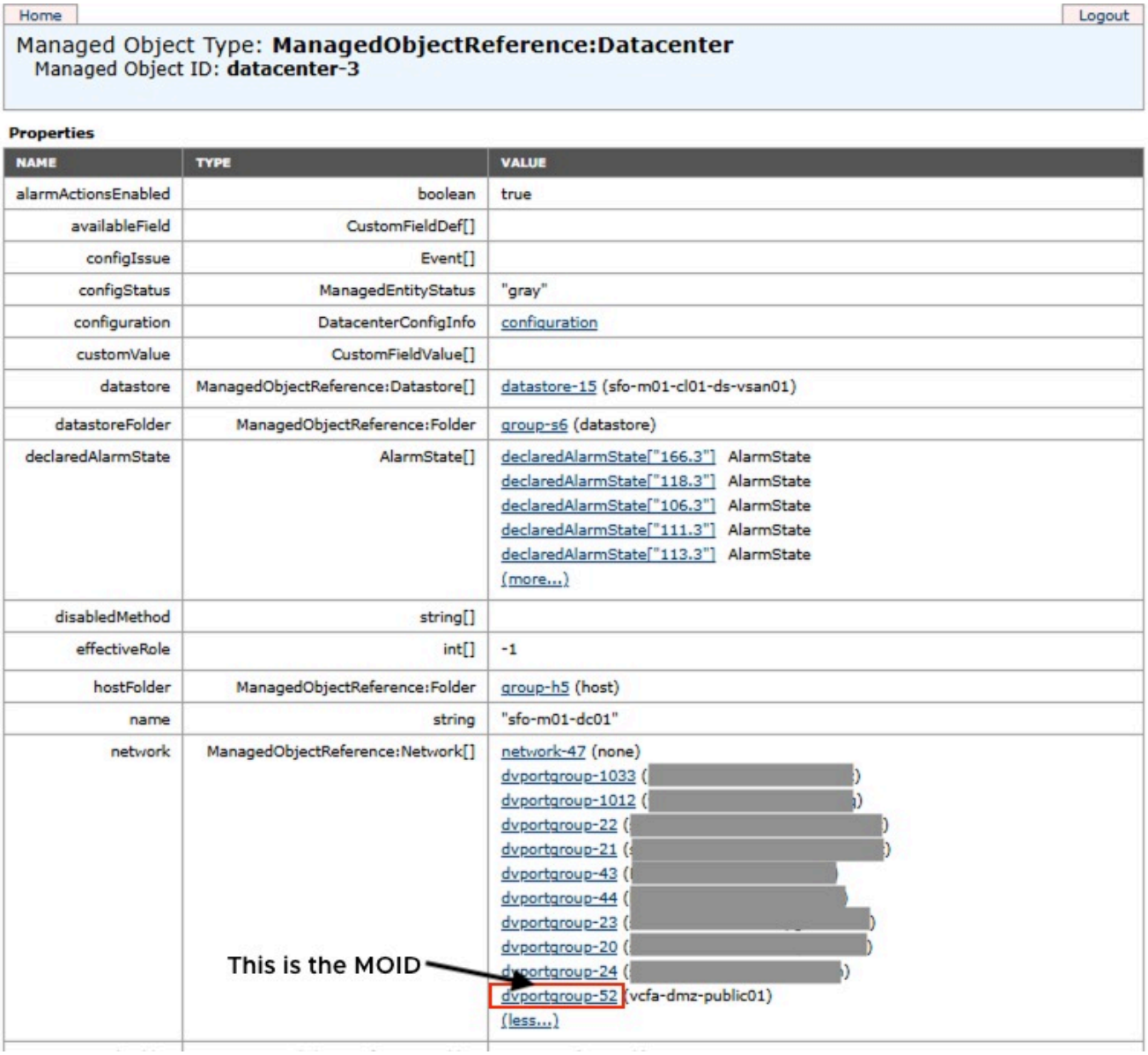
Logout

Managed Object Type: **ManagedObjectReference:Folder**
Managed Object ID: **group-d1**

Properties

NAME	TYPE	VALUE
alarmActionsEnabled	boolean	true
availableField	CustomFieldDef[]	availableField[1] CustomFieldDef
childEntity	ManagedObjectReference:ManagedEntity[]	datacenter-3 ( )
childType	string[]	"vim.Folder" "vim.Datacenter"

5. Navigate to “network” in the left-hand column, then in the right-hand column of the same line find your target network.
   a. The target network name will be in parentheses.
   b. The MOID is the dvportgroup-# next to the name in parentheses.
   c. In example the network is named “vcfa-dmz-public01” and the MOID is “dvportgroup-52.”



Scripted Method

1. Run the following powershell commands on a computer with connectivity to the target vCenter server.

```
# Connect to your vCenter 9 server
Connect-VIServer -Server "<target vCenter Server>"

# Retrieve all networks and format them into a clean table
Get-VirtualNetwork | Select-Object Name, @{Name="MOID";Expression={$_.ExtensionData.MoRef.Value}},
 NetworkType | Format-Table -AutoSize
```

2. The command will display all MOIDs within your target vCenter server.
3. Example Output - Find your target network within the list and the associated MOID for that network.

```
Name                           Port  User
----                           ----  ----
sfo-m01-vc01.sfo.rainpole.io   443   VSPHERE.LOCAL\Administrator


Name                               MOID               NetworkType
----                               ----               -----------
none                              Network-40          Network
```

```
AviMgmt                              dvportgroup-49  Distributed
Edge-Interface-DCHaSUY0Pd            dvportgroup-37  Distributed
Edge-Interface-yzF6JAXmhp            dvportgroup-36  Distributed
sfo-m01-cl01-vds01-pg-vmotion        dvportgroup-24  Distributed
sfo-m01-cl01-vds01-pg-mgmt           dvportgroup-23  Distributed
sfo-m01-cl01-vds01-pg-vsan           dvportgroup-25  Distributed
sfo-m01-cl01-vds01-pg-vm-mgmt        dvportgroup-22  Distributed
sfo-m01-cl01-vds-DVUplinks-20        dvportgroup-21  Distributed
VCFA-DMZ-PublicSubnet                dvportgroup-56  Distributed
Services Subnet                      dvportgroup-54  Distributed
```

Optional - Split DNS and Network Configuration

The VCF Automation default design calls for a single FQDN/VIP combination that is then exposed both internally and externally. Additionally, the default example used in this guide assumes the FQDN/VIP combination is on the same overall IP range as the internally facing VCF Automation components (Node IPs, VCF Automation service runtime, and the load balancer virtual service IP Pool).

Depending on your organization's security posture, the default deployment could potentially expose the internally facing VCF Automation components, previously mentioned, unnecessarily. This issue can be mitigated by deploying VCF Automation using a split DNS and network configuration.

What is Split DNS and Networking Configuration?

The overall concept is to split the communication between internal users and external users so that the internally facing VCF Automation components are not directly exposed to external access.

The overall process:

1. Assigning VCF Automation components to separate network ranges within the external IP block
2. Separate internal and external DNS configurations for the target VCF Automation FQDN.
3. Internally the VCF Automation FQDN will resolve to the virtual service IP Pool addresses.
4. Externally the VCF Automation FQDN will resolve to the virtual service VIP address.

Creating a Split DNS & Network Configuration

> **Important:**
>
> ***This uses the IP structure from the JSON example in Example CURL Command with Example JSON Payload section of this guide.***

Process

1. **Configure Split Networking** - Configure two separate ranges in the external IP block for the DMZ VPC.
   For example:
   a. External IP block range #1
      a. 172.16.49.0/24.
      b. Used for internally facing VCF Automation components (Node IPs, VCF Automation service runtime, and the load balancer virtual service IP Pool).
   b. External IP block range #2
      a. 10.200.0.0/24.
      b. Used for external IP block for the VCF Automation FQDN/VIP.
2. **Configure Split DNS**
   a. Internally facing DNS - Configure the internally facing DNS to resolve the VCF Automation FQDN to the IP Pool IP addresses.
      For example: vcfa01.rainpole.io resolves to 172.16.49.43,172.16.49.44, and 172.16.49.45 for internal usage.

      > **Important:**
      >
      > ***The VCF Automation service runtime FQDN will only be advertised by the internally facing DNS.***

b. Externally facing DNS - Configure the externally facing DNS to resolve the VCF Automation FQDN to the IP Pool IP addresses
   For example: vcfa01.rainpole.io resolves to 10.200.0.100
3. **Secure Deployment** - Follow guidance in this guide to secure VCF Automation network traffic through WAF and firewall configuration.

Result

1. All users access VCF Automation through the same FQDN (in this example vcfa01.rainpole.io).
2. Internal users will resolve the FQDN to the “internal” IP Pool IP addresses associated with the load balancer virtual service.
3. External tenants will resolve the FQDN to the “external” virtual services VIP address associated with the load balancer virtual service.

Important:

The JSON file for the split DNS configurations is the same as the standard JSON (non-split DNS). During the deployment of the split DNS configuration, the system will assign the IP Pool IPs as the VCF Automation FQDN within VCF Operations. This will ensure that the VCF Fleet components communicate with VCF Automation using the “internal” path. External users will still access VCF Automation through the externally facing IP address.

Example CURL Commands with Example JSON Payload

Example CURL Command to Derive the Bearer Token for Authentication

```
curl --request POST \
  --url https://<Fleet Lifecycle FQDN>/api/v1/identity/token \
  --header 'content-type: application/x-www-form-urlencoded' \
  --data grant_type=password \
  --data username=admin@vsp.local \
  --data 'password=<Password>'
```

Example Deployment CURL Command with JSON Payload

```
curl --request POST \
  --url https://flt-fc01.rainpole.io/fleet-lcm/v1/components \
  --header 'authorization: Bearer
 eyJhbGciOiJFZERTQSIsImtpZCI6InBOeTB6bnVkZXJ3WHUzTXgwNmY3cWRGNWhBd253ZmV5WnRGYnQyYThpeE0iLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJ
 \
  --header 'content-type: application/json' \
  --data '{
  "componentSpecs": [
    {
      "deploymentType": "VspComponentSpec",
      "sddcLcmId": "3828ce42-270d-4453-ac9c-c9caef51a115",
      "componentType": "VCFA",
      "version": "9.1.0.0",
      "repository": {},
      "fqdn": "flt-vcfa01.rainpole.io",
      "componentNamespace": "prelude",
      "configSpec": {
        "size": "medium",
        "adminSystemPassword": "<Password>",
        "ingress": {
          "vcfa": {
            "fqdn": "flt-vcfa01.rainpole.io",
            "vips": {
              "ipv4": [
                "172.16.49.43",
                "172.16.49.44",
```

```
                "172.16.49.45"
              ]
            }
          }
        }
      },
      "vspClusterSpec": {
        "deploymentType": "VspClusterSpec",
        "sddcLcmId": "3828ce42-270d-4453-ac9c-c9caef51a115",
        "platformFqdn": "flt-vcfa-sr.rainpole.io",
        "systemUserPassword": "<Password>",
        "size": "medium",
        "ipv4Pool": {
          "addresses": [
            "172.16.49.36",
            "172.16.49.37",
            "172.16.49.38",
            "172.16.49.39",
            "172.16.49.40"
          ],
          "excludedAddresses": []
        },
        "name": "vcfa-node",
        "networkMoId": "dvportgroup-57",
        "gatewayCidrIpv4": "172.16.49.33/28"
      }
    }
  ]
}'
```

Network Traffic Segregation with vDefend and Avi Load Balancer

In the following section, we will go into detailed design and methodologies to secure the deployment pattern #3 , including the scope, conventions as well as detailed implementation steps.

- Network Traffic
- Configuration Inventory

Network Traffic

External Traffic

An external connection defines how a transit gateway (TGW) in Virtual Private Cloud (VPC) connects to networks outside of NSX and into the outside environment as well as services it supports. This runbook is based on using centralized TGW (CTGW) connection, where the transit Gateway uses a Tier-0 gateway provisioned on NSX for external connection. Distributed TGW (DTGW) connection, where the transit gateway connects directly to VLANs provisioned in the fabric is not in the scope.

Tenant Traffic

External traffic, originating from the Tenant Organizations into VCF Automation. This traffic traverses through NSX T0 Gateway, NSX Transit Gateway (TGW) and DMZ VPC Gateway into the VPC subnet that hosts the load balancer services.

Management Traffic

Internal traffic includes bi-directional communication channels between VCF Automation and other required entities. This includes VCF management components, e.g. vCenter, NSX Manager etc for access to the underlying resources. as well as VCF Automation user interface for the VCF Automation provider to perform configuration and administrative activities. The traffic in this case traverses through the NSX T1 gateway for Avi management and TGW gateway for the

rest. vDefend distributed firewall (DFW) controls traffic to the management network segment that hosts VCF management components mentioned, such as vCenter, NSX manager, VCF Operations etc.

Please see the diagram in the Deployment Overview Section at the beginning of this blueprint.

Configuration Inventory

Before implementing firewall policies and rules, we will utilize various inventory management options available in NSX to organize and give uniformity to the various elements used.

Groups and Service

In order to secure VCF Automation, NSX objects are created to provide structure and streamline the process by organizing various components into logical groups. This includes creating dynamic groups using attributes or properties, such as IP, VM name etc as well as static groups that aggregates similar objects, e.g. VCF Management Components. This will ensure that the constructs are defined once and used in various areas, such as the NSX gateway firewall as well as NSX distributed firewall. The following section covers the convention and logic used to create those components:

- **Groups**: NSX allows logically grouping various objects into groups. The groups used in this document start with **grp-** and include name to indicate what it includes, e.g.
  - Groups, grp-vcenters and grp-nsx-managers include vCenter servers and NSX Managers respectively.
  - Group, grp-vcf-a includes all of the VMPS management IPs used by VCF Automation
  - Groups, grp-vcf-fleet includes every VCF Operations component, such as VCF Ops, VCF Fleet Management, SDDC manager etc.
  - Groups, ext-* for any component used for external traffic, e.g. ext-tenant-one for traffic sourced from external tenant, tenant-one
- **Services**: NSX includes pre-defined services that cover most of the common protocol+port combination. Any new services created to formally organize and re-use new services are created with **svc-** followed by the services e.g.
  - Service, svc-vsphere-supervisor to identify TCP:6443 used by vSphere Supervisor
  - Service, svc-vc-webconsole to identify TCP:902 used for web console to the VM form VCF Automation

Naming Convention

Firewall Policies & Rules: vDefend firewall policies and rules follow the guidelines below where applicable:

- Policy names use the convention **policy-** to identify what the policy is created for, e.g. Policy-Avi is related to Avi vs Policy-VCFA for VCFA Automation
- Rule names use the convention [**allow | deny | reject]-<serviceName>** to identify what the rule action (allow, deny or reject) will be for a given service, e.g. allow-web to allow HTTP/HTTP traffic with TCP 89/443 and allow-ssh to allow SSH service

If the same security policy needs to be in multiple places - TGW FW, T1 GFW and DFW - names for groups, services and FW policies and rules are reused for consistency.

Configuring Security Policies and Rules

In the following sections, we will cover how to use vDefend firewall as well as Avi WAF rules to provide secure channels for relevant traffic.

- Protecting Tenant Traffic with vDefend Gateway Firewall on TGW
- Protecting Tenant Traffic with vDefend and Avi
- Avi Virtual Service Configuration
- Avi HTTP Configuration
- Avi Web Application Firewall (WAF) Configuration
- Protecting Management Traffic

Protecting Tenant Traffic with vDefend Gateway Firewall on TGW

In this design, along with the ingress and egress to the VPC, the Transit Gateway (TGW) also provides security with its built-in gateway firewall (GFW) functions to control and isolate the internal, management traffic from that of the external, tenant traffic. This TGW has a VPC for DMZ where a public VPC subnet, **Avi Services** hosts the Avi Load Balancers. The

services that provide external tenant network traffic into VCF Automation flows through this subnet in DMZ VPC and into TGW. The firewall security policies on this TGW allow only the required services access into the VCF Automation user interface. The following table shows sample firewall policies required in the transit gateway's gateway firewall:

Name	Source	Destination	Protocol & Port (Service)		Description
Allow Rules					
Policy-Infra-Services					
DNS	Any	DNS-Servers	DNS		DNS Traffic
LDAP	Any	LDAP-Servers	LDAP & LDAP-UDP		LDPA Traffic
Policy-Avi					
allow-web	Any	Avi SE	TCP	80 & 443	Web (HTTP/HTTPS) access to Avi SE
Policy-VCFA					
VCFA-to-Orchestrator	VCFA	Orchestrator	HTTPS		Web (HTTP/HTTPS) access to VCFA VIP
VCFA-to-Supervisor	VCFA	vSphere Supervisor	TCP	6443	vSphere Supervisor Traffic
gw-health-check	Any	VCF Automation VIP	TCP	8008	
allow-web	Any	VCF Automation VIPVCF Automation Management IPs	HTTP/HTTPS		SSH from Bastion/ Jump hosts
api-server	Any	vSphere Supervisor	TCP	6443	
SSH	Bastion Jumphosts	VCF Automation Management IPs	SSH		Management Access
ICMP Pings	- Bastion - Jumphosts - Tenant IP	VCFA VIP VCFA Management IPs	ICMP		Ping tests from Bastion/Jump hosts
VCF Management	VCFA VIPVCFA Management IPs	VCF Management Components	HTTPS		To all VCF management components - Ops, SDDC, etc
VC Webconsole	VCFA Management IPs	ESX Hosts	HTTPS		VM webconsole access
Node Health Check - GW Config	Any	VCFA VIP	TCP	22462	
Synthetic Checkers	Any	VCFA VIP	TCP	30005-30006	
Node Health Check - VMSP GW	Any	VCFA VIP	TCP	30136	
Deny Rules					
Default-Deny	Any	Any	Any		Deny the rest

Protecting Tenant Traffic with vDefend and Avi

The native load balancer deployed with VCFA provides only layer 4 services. In this deployment pattern, an external VMware Avi Load Balancer will be used with VCFA deployment for advanced layer 7 LB features. This Avi Load Balancer will:

- Provide a Virtual Service that will SNAT for the external traffic to the VCF Automation appliances.
- Provide SSL termination.
- Provide certificates in addition to those assigned at deployment for the VCF Automation tenant users to access the service portal.
- Front-end traffic to the internal load balancer configured at VCF Automation deployment.

- Provide security filtering using the Avi WAF engine.
- Provide analytics on the traffic destined for the VCF Automation portal.
- Prevent tenant users from accessing the provider portal via redirection.
- (Optional) Prevent tenant users from accessing other tenant's portals.
  - This requires knowing the client source IPs of the tenant users.

WAF protection of the VCF-A VIP

- Prevent client subnets from getting to provider portal.
- Prevent Tenant 1 subnets from getting to Tenant 2 portal when we know the source IP (no possibility if we don't know client subnet).
- Configure VIP and pool with HMs appropriate to VCF-A, use the new 9.1 deployment model to have 3 internal VIPs and configure those to be the pool members.

Best Practices:

- Static IP must be configured via Avi CLI.

High level requirements:

- A Certificate on vip with matching FQDN SAN.
- Persistence, using HTTP cookie.
- WAF configurations.
- SEs in NSX exclusion list.
- HTTP policies to prevent access based on source IP to provider portal.
  - Also other tenant portals as an option, would be cumbersome in a large deployment and does require known source IP ranges.

Important:

All objects should be created in the tenant corresponding to the VPC that VCF Automation is deployed in.

Avi Virtual Service Configuration

- Configure a VSVip, using the Avi CLI.

```
[admin:10-11-10-92]: > switchto tenant <tenant>
# the tenant that corresponds to the NSX project where the VCF-A deployment VPC is>

[VCFA-DMZ-VPC-Project:10-11-10-92]: > show vrfcontext
# gather the VRF context name that corresponds to the VCF-A DMZ VPC

[VCFA-DMZ-VPC-Project:10-11-10-92]: > show cloud
#gather cloud name for the NSX cloud

[VCFA-DMZ-VPC-Project:10-11-10-92]: > configure vsvip <vsvip name>
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip> vrf_context_ref <vrf name>
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip> cloud_ref <cloud name>
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip> tier1_lr /orgs/default/projects/<NSX project name>/vpcs/<VCF-A VPC
 name>
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip> vip
New object being created
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip:vip> ip_address <VIP static ip>
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip:vip> prefix_length 32
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip:vip> index 1
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip:vip> subnet 0.0.0.0/32
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip:vip> subnet_uuid <name of VCF-A NSX Project>_AVISEPARATOR_<name of
 the VCF-A VPC>_AVISEPARATOR_PUBLIC
## example value  VCFA-DMZ-VPC-Project_AVISEPARATOR_VCFA-Internal-Public-VPC_AVISEPARATOR_PUBLIC
```

```
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip:vip> save
[VCFA-DMZ-VPC-Project:10-11-10-92]: vsvip> save
```

- Configure a custom health monitor according to the health monitoring guide.
  - Type should be HTTPS
  - For user input , use **GET /api/server_status HTTP/1.1**
  - Choose **2xx** and **3xx** for response codes
  - Select **Enable SSL Attributes**
    - For **TLS SNI server name**, use the FQDN you're using for the VCF-A vip
    - Use **System-Standard** for the SSL Profile
- Configure a pool according to the following the create server pool guide.
  - Attach the health monitor you created earlier
  - Use **System-Persistence-Http-Cookie** as the persistence profile
  - We're re-encrypting on the server side, so use **System-Standard** SSL profile and check the box for **Enable TLS SNI**
  - **Remember to set the pool default port to 443**
- Configure a virtual service according to the create virtual service guide.
  - Cloud will be the automatically configured NSX cloud that was created via the VCF Operations deployment process
  - VRF will correspond to the name of the VPC that VCF Automation was deployed in to.
  - Application type should be **HTTP/HTTPS**
  - Application Profile should be **System-HTTP**

Avi HTTP Configuration

IP/String Group Configuration for HTTP Policies

We will utilize IP groups for configuring HTTP Policies, as well as string groups for WAF positive security model rules.

Reference for String Group

Reference for IP Group

- IP group for Provider Portal users, this should be the subnet where management stations will be sourced.
- (Optional) IP groups for each tenant where you have known tenant client IP addresses.
- String group for "safe" characters that will be utilized by the Positive Security Model rules.
  - This string group should be created with a single key (VCFA_BLUEPRINTS as an example) and value, and the value should contain the following regex.

```
^[0-9A-Za-z._ \t:,!?+*=@#\-\$\(\)\&\'\/\[\]]*$
```

HTTP Policy Configuration

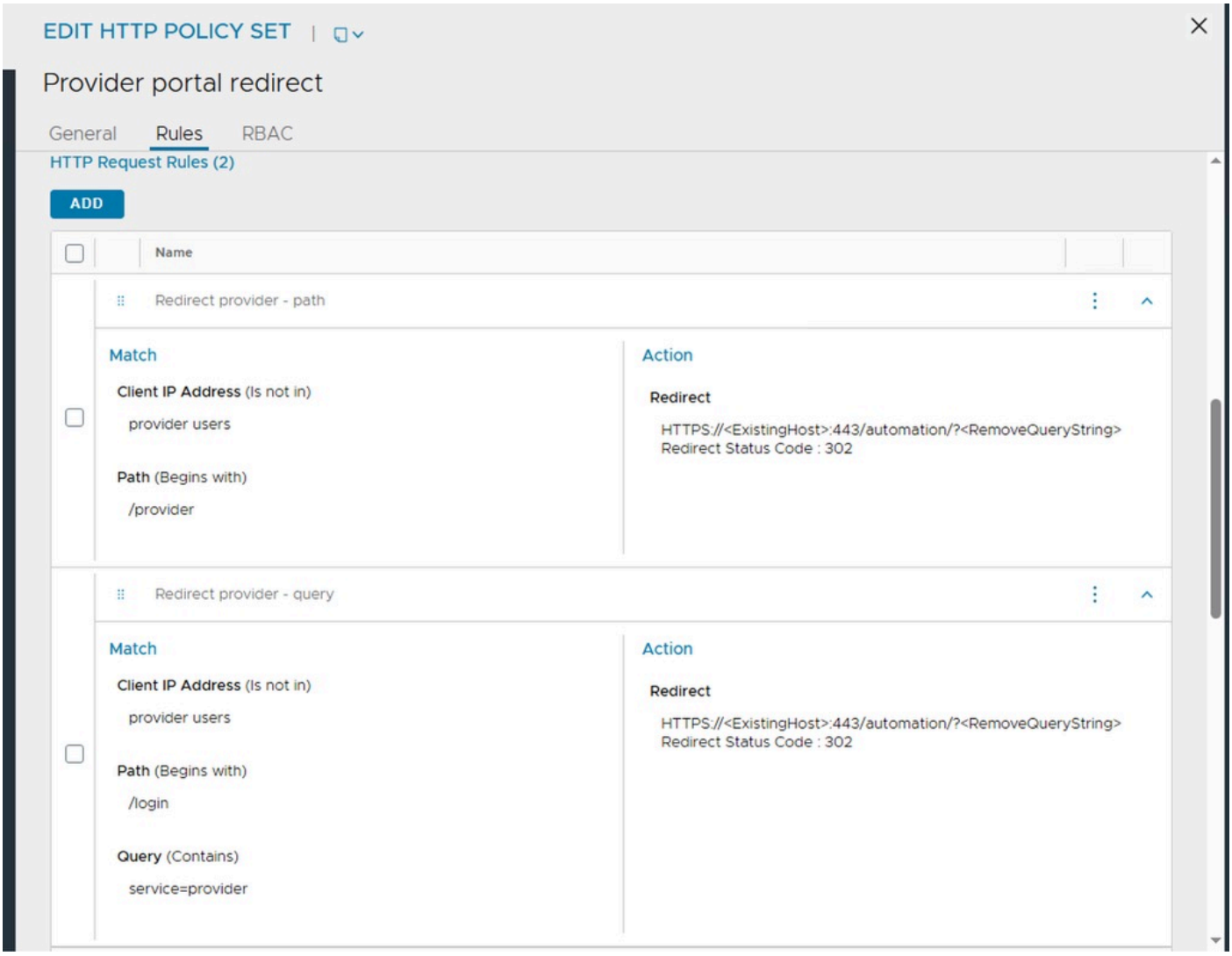
For preventing end user access to the VCF Automation provider portal, we use HTTP policies to redirect this traffic based on the client IP address. For flexibility, this is broken up into two HTTP Policy Sets.

Reference for HTTP Policy Sets

Provider Portal Policy

- Create a new **HTTP Policy Set** at **Templates/Policies/HTTP Policy Set.**
- We will create two **HTTP Request Rules.**
  - One rule for redirection based on path (example name: Redirect Provider - Path).
  - One rule for redirection based on query parameters (example name: Redirect Provider - Query).

Configuration summary



Provider request rule - Path

Match criteria

- Client IP
  - Condition **Is Not In**
  - String group **Provider Users**
- Path
  - Criteria **Begins With**
  - String **/provider**

Action - **Redirect**

- Path **automation/**
- Uncheck **Keep Query**

Provider request rule - Query

Match criteria

- Client IP
  - Condition **Is Not In**
  - String group **Provider Users**
- Path
  - Criteria **Begins With**

  - String **/login**
- Query
  - Enter custom value **service=provider**

Action - **Redirect**

- Path **automation/**
- Uncheck **Keep Query**

Tenant Portal Policy

The configuration for these policies are largely the same as the provider portal. Using the example of tenant1 and tenant 2:

Tenant1 policy

Provider request rule - Path

Match criteria

- Client IP
  - Condition **Is Not In**
  - String group **Tenant1 Users**
- Path
  - Criteria **Begins With**
  - String **/tenant/tenant1**

Action - **Redirect**

- Path **automation/**
- Uncheck **Keep Query**

Provider request rule - Query

Match criteria

- Client IP
  - Condition **Is Not In**
  - String group **Tenant1 Users**
- Path
  - Criteria **Begins With**
  - String **/login**
- Query
  - Enter custom value **service=tenant:tenant1**

Action - **Redirect**

- Path **automation/**
- Uncheck **Keep Query**

Avi Web Application Firewall (WAF) Configuration

There are two parts to the WAF configuration, creating a WAF profile and then creating a PSM group for preventing false positives.

WAF Profile

- Under **Templates/WAF/WAF profile**, click create.
- For **Allowed Methods** add the following to the existing entries.
  - PUT
  - PATCH
  - DELETE
- Scroll to **Content Type mapping** and add.

  - Content Type - **application/*+json** Request Parser - **JSON.**
  - Content Type - **application/*+xml** Request Parser - **XML.**

Positive Security Group

- Under **templates/WAF/Positive Security,** click create.
- Provide a **Name.**
- Ensure **Enable Group** is checked.
- Change **Miss Action** to **Flag or Reject.**

Here we will split the rules up into two sets, one set for All-Apps organizations, and one for VM-Apps organizations. You do not need to use the VM-Apps organization rules if you have no plans to implement a VM-Apps organization.

All-Apps rules

- Still within the positive security group edit screen, under **Location** click Add.
- Provide a name for the rule, ex. **AllApps org rules.**
- For match criteria add a **Path** rule:
  - Criteria - **Begins With**
  - Custom Value string - **/blueprint/api/blueprints**
- Under **Argument Rules** click Add.
  - Ensure **Enable Rule** is checked
  - Provide a unique rule ID, ex 10001
  - Provide a **Name**
  - Under **Match Elements / Value:**
    - Choose **String Group / Key**
    - Choose the string group you created earlier
    - Provide the key for the regex in the string group, ex. **VCFA_BLUEPRINTS**
  - Scroll to the **Match Elements** / **Match Elements** subsection and click Add (we will create two match elements rules)
    - Name - **ARGS ,** Criteria - **Equals** , Sub Element - **content**
    - Name - **ARGS_NAMES ,** Criteria - **Equals** , Sub Element - **$select**
- Save twice to return to the Positive Security Group. If using VM apps orgs, continue with the following steps, else click Save.
- For VM apps:
- Provide a name for the rule, ex. **VMApps org rules**
- For match criteria add a **Path** rule:
  - Criteria - **Begins With**
  - Custom Value strings:
    - **/project-service/api/projects**
    - **/provisioning/uerp**
    - **/form-service/api/forms**
    - **/provisioning/mgmt**
    - **/provisioning/resources**
- Under **Argument Rules** click Add
  - Ensure **Enable Rule** is checked
  - Provide a unique rule ID, ex 10002
  - Provide a **Name**
  - Under **Match Elements / Value:**
    - Choose **String Group / Key**
    - Choose the string group you created earlier
    - Provide the key for the regex in the string group, ex. **VCFA_BLUEPRINTS**
  - Scroll to the **Match Elements** / **Match Elements** subsection and click Add (we will create six match elements rules)
    - Name - **ARGS ,** Criteria - **Equals** , Sub Element - **$filter**

    - Name - **ARGS_NAMES ,** Criteria - **Equals** , Sub Element - **$select**
    - Name - **ARGS ,** Criteria - **Equals** , Sub Element - **odataQuery**
    - Name - **ARGS ,** Criteria - **Equals** , Sub Element - **network.instanceAdapterReference**
    - Name - **ARGS ,** Criteria - **Contains** , Sub Element - **constraints**
    - Name - **ARGS ,** Criteria - **Equals** , Sub Element - **size**
- Click save 3 times to complete configuration of the **Positive Security Group**

Lastly, we need to create a WAF Policy

WAF Policy

- Under **Templates/WAF/WAF policy**, click create
- Provide a name for the policy
- For **WAF profile** choose the profile we created earlier
- Change **Policy mode** to **Enforcement**
- Scroll to **Positive Security** and add the PSM group we created earlier
- Scroll to **Signatures / Pre-CRS Groups** and click add
- Provide a name for this group
- Under **Rules/ Pre-CRS Rules** click add
  – Provide a name for the rule
  – In the **Rule** box, paste the following string
  – **SecRule REQUEST_HEADERS:Accept "@beginsWith application/json" "phase:1,id:5000020,nolog,pass,t:lowercase,ctl:ruleRemoveById=920600"**
- In **CRS Groups,** click the vertical ellipsis to edit the **CRS_920_Protocol_Validation** group
  – Look for rule **921130 - Check for HTTP Response Splitting (2/2)** and edit that rule
  – Add an exception with the following attributes
    - Subnet - **Any**
    - Path - **Any**
    - Match Element - **ARGS:body**
  – Click save twice to return you to the base WAF policy screen
- Scroll through the **CRS Groups** list until you find **CRS_941_Application_Attack_XSS** , which could be on a second or third page depending on your CRS rule set and the number of items per page
  – Click **Add** to add an exception that will apply to the entire group using the following attributes:
    - Subnet - **Any**
    - Path - **Any**
    - Match Element - **ARGS:body**
- Click save until you have exited all edit screens and are back at the **WAF Policy** listing screen

Remember:

Ensure that the newly created WAF policies and HTTP policies are applied to the Virtual service!

This can be done by editing the **virtual service** and scrolling to **Policies**, add the **provider redirect policy set** as well as any **tenant specific policy sets** you may be using. Then scroll to Security and choose the **WAF Policy** we created. Click **save** to apply the changes.

Protecting Management Traffic

The VCF Automation nodes are deployed in a private VPC subnet - **VCF Automation Subnet** in the same VPC as the Service Subnet where Avi Load Balancer resides. The VCF Automation nodes provide ingress via a built-in native load balancer to the VCF Automation services.This design with public VPC subnet for Avi Load Balancer data traffic and private VPC subnet for management traffic maintains a separation between the tenant organization traffic from that of the internal application network for VCF Automation. There are two main types of management traffic that needs to be secured in this design::

- Avi management traffic
- VCF Automation Management Traffic

The following sections discuss securing both types of management traffic

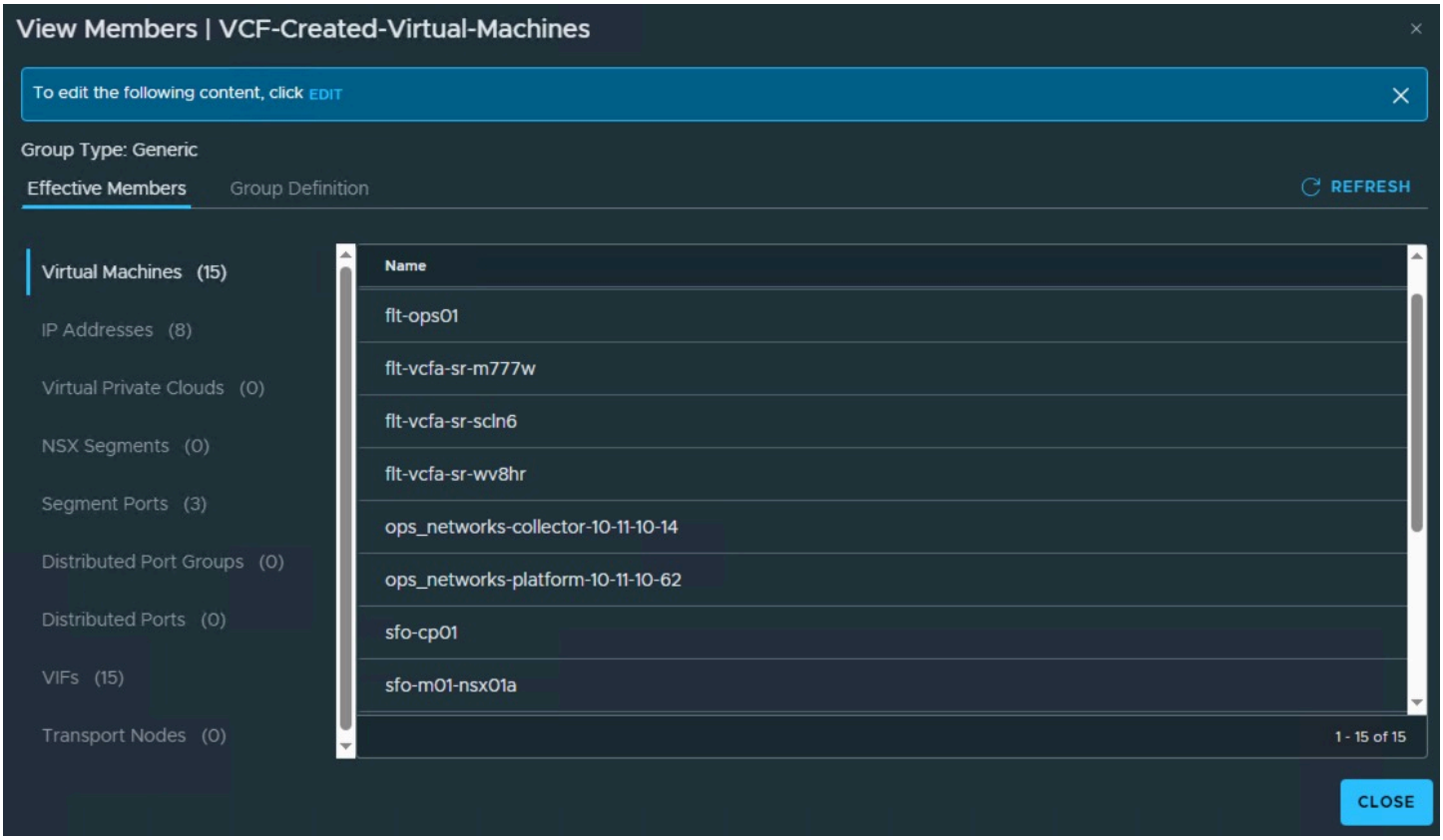
Protecting Avi Management Traffic in T1 GW

Avi management network is on a NSX segment that is backed by a NSX T1 Gateway which controls traffic to the Avi management plane. The gateway firewall GFW functions in this gateway is used for security control to the Avi management flows. The following table shows sample firewall policies required in the T1 gateway GFW:

Name	Source	Destination	Protocol & Port (Service)		Description
Allow Rules					
plcy-Avi-UX					
allow-keyx-channel	Avi SE	AVI Controllers	TCP	8443	Secure key exchange
allow-ssh	Avi SE	AVI Controllers	SSH		Secure communication among Avi SE and controllers
allow-NTP	Avi SE	AVI Controllers	NTP		NTP sync for Avi SE
SE-object-store	Avi SE	AVI Controllers	TCP	9001	SE image for deployment(?)
Avi-backend-pool	Avi SE	VCF Automation VIPVCF Automation VMs	TCP	8443	
VCFA & VIP	Avi SE	VCF Automation VIPVCF Automation VMs	HTTPS		SE to VCFA web traffic
Deny Rules					
Default Policy					
Default Rule	Any	Any	Any		[Deny]

Protecting VCF Automation Management Traffic

VCF Installer creates an User Excluded Groups called “VCF-Created-Virtual-Machines” in NSX by default. This group’s members include all VCF Management applications including VCF Automation. From NSX web interface, navigating to Security > Distributed Firewall : Settings > User Excluded Groups shows the members of this group as seen here:



Being a member of this group means that the following security posture is applicable:

1. All of the VCF management components are automatically excluded from any of the firewall rules created in DFW
2. Inter-communication between all of those components are unrestricted

However, VCF Automation management network in this pattern is different from the rest of the management components; it's hosted in a VPC subnet in DMZ VPC whereas the rest are in a management network segment, VCF Automation traffic needs a communication channel to the VCF management components. TGW FW configuration covered in section 2.3.1 explicitly allows those communication channels.

Lateral Security for VCF management with vDefend for granular security:

For an organization seeking to have a more granular control with centralized enterprise-grade security using the VMware vDefend security, the lateral Security with vDefend, in the security modernization section of the design blueprint for VMware Cloud Foundation covers in great depth. The design blueprint goes into details of related planning, design, and security policy and rules implementation as seen in the following sample of infrastructure services protection policies:

Infrastructure Services Policy (DFW Infrastructure Category)

SOURCE	DESTINATION	SERVICE	CONTEXT PROFILE	APPLIED TO	ACTION / LOGGING
VCF Fleet Infrastructure Policy			Applied To: VCF_FLEET		
VCF_FLEET	DNS_SVC	DNS-TCP DNS-UDP	DNS	DFW	Allow
VCF_FLEET	NTP_SVC	NTP	None	DFW	Allow
VCF_FLEET	DHCP_SVC	DHCP-Server	None	DFW	Allow
VCF_FLEET	AD_SVC	Microsoft Active Directory V1	None	DFW	Allow
VCF_FLEET	VCF_OPS_LOGS SYSLOG_SVC	Syslog (TCP) Syslog (UDP) TCP/9543	None	DFW	Allow
VCF_FLEET	Any	DNS-TCP DNS-UDP NTP DHCP-Server Syslog(TCP, UDP) TCP/9543	None	DFW	Drop / Log

Please refer to the following link for more information:

https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-1/design/design-blueprints-for/security-modernization/vdefend-lateral-security.html

Configuration Validation

The following sections cover some assumptions that were met before validating the overall solution recommendations in this document.

Assumptions

Active/Standby Centralized Connection

In 9.1,TGW HA is decoupled from the T-0 (A/S TGW with A/A T-0).

vDefend Firewall Licenses

Before configuring vDefend firewall features, ensure that the required licenses are applied. Once it is activated, DFW must be activated for the vSphere cluster from NSX Manager > Security > Disturbed Firewall > Settings.

Avi Licensing

Avi requires a valid license before Service Engines can come online. See the following guide.

Validate Tenant Traffic

The validation steps includes the following:

1. Access VCFA tenant portal using the external DNS entry
2. Validate the Integrated Access Control for the Tenant Organization
3. Additionally, the NSX enterprise or the security administrator can also check:

a. The firewall rule hit counters for DFW, TGW FW and GFW
b. Change the rule to reject and try to re-login to VCFA tenant portal

Additional References

Lateral Security for VMware Cloud Foundation with VMware vDefend:

https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vvs/9-X/lateral-security-for-vmware-cloud-foundation-with-vmware-vdefend.html

Manage a Firewall Exclusion List:

https://techdocs.broadcom.com/us/en/vmware-security-load-balancing/vdefend/vdefend-firewall/9-0/vdefend-distributed-firewall/configuring-distributed-firewall/about-firewall-rules/manage-a-firewall-exclusion-list.html

Guidance to Write Efficient vDefend Firewall Rules:

https://techdocs.broadcom.com/us/en/vmware-security-load-balancing/vdefend/vdefend-firewall/9-0/vdefend-distributed-firewall/configuring-distributed-firewall/about-firewall-rules/guidance-to-write-efficient-and-secure-firewall-rules.html

VMware vSphere Kubernetes Service Consumption Blueprint

The *VMware vSphere Kubernetes Service Consumption Blueprint* provides architectural guidance for selecting and implementing the appropriate VMware vSphere Kubernetes Service (VKS) consumption model within VMware Cloud Foundation platforms.

The guide focuses on the architectural, operational, networking, governance, and multi-tenancy considerations that influence how organizations should consume Kubernetes services using VKS. Rather than presenting deployment steps alone, this document provides a design decision framework intended to help architects align VKS consumption patterns with organizational operating models, governance requirements, scalability objectives, and private cloud strategies.VMware vSphere Kubernetes Service Consumption Blueprint.

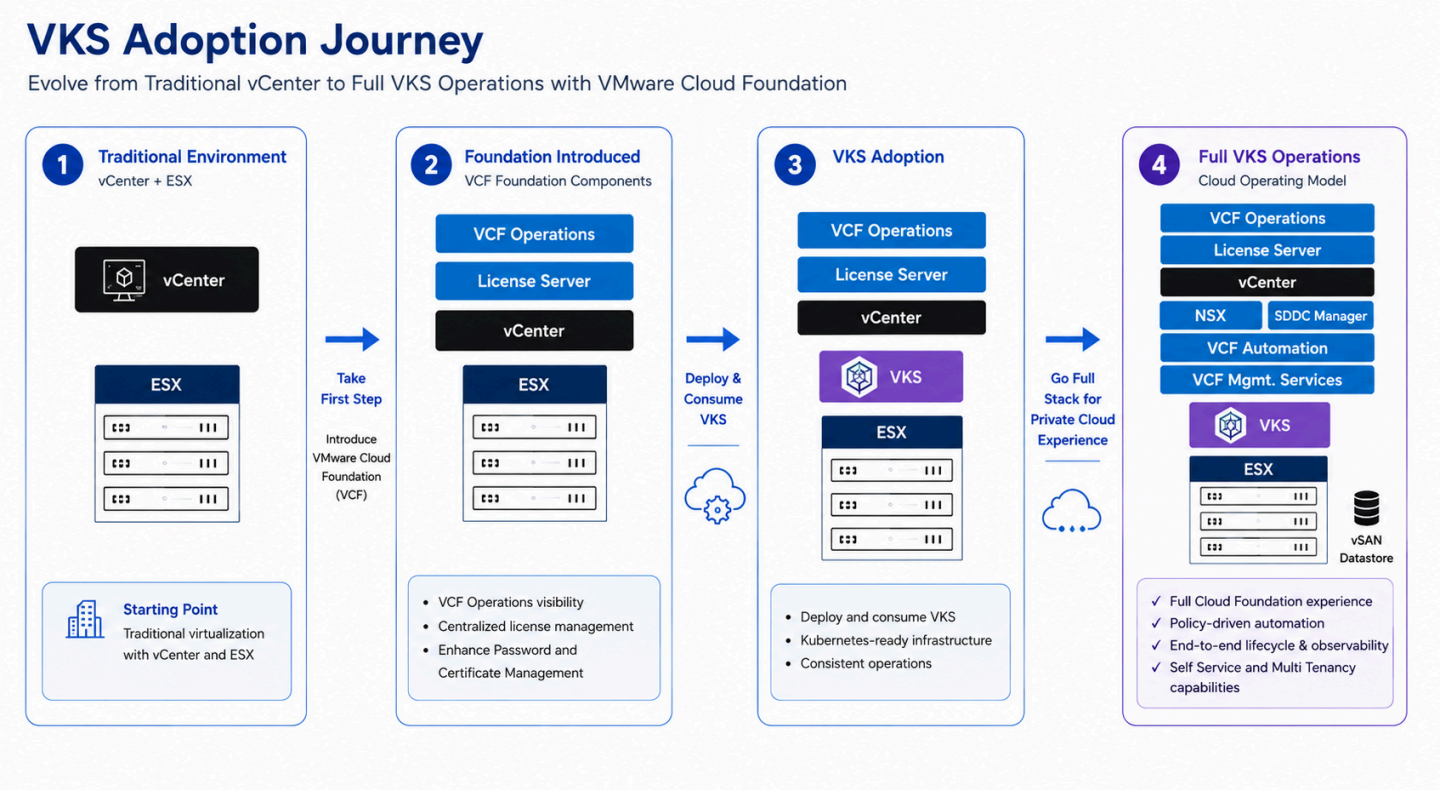
VKS Consumption Adoption Journey

Before selecting a specific **VMware vSphere Kubernetes Service** consumption model or deployment architecture, it is important to recognize that Kubernetes platform adoption is typically an incremental operational journey rather than a single deployment event.

Organizations often evolve their platform strategy over time based on operational maturity, existing infrastructure investments, governance requirements, modernization goals, and long-term private cloud strategy. As a result, VKS adoption may occur at different stages of operational and architectural maturity depending on the organization.

The following adoption journey illustrates a common progression path organization may follow as they transition from traditional virtual infrastructure operations toward modern Kubernetes-enabled and private cloud operating models.

Figure 200: VKS Consumption Adoption Journey



This figure illustrates a progressive adoption journey for organizations transitioning from traditional **VMware vSphere** environments toward a modern Kubernetes-enabled private cloud platform using **VMware vSphere Kubernetes Service** and **VMware Cloud Foundation**.

The journey demonstrates how organizations can incrementally modernize their infrastructure and operational model without requiring an immediate transition to a full private cloud architecture on day one.

Stage 1 – Existing vSphere Environment

The journey begins with an existing vSphere 8.x environment operating traditional virtualized workloads that are managed through **VMware vCenter**.

At this stage:

- infrastructure operations remain VM-centric
- Kubernetes services are not yet enabled
- operational management is focused on traditional virtualization administration

This phase represents the baseline operational state for many existing enterprise environments.

Stage 2 – Platform Foundation Modernization

Organizations begin modernizing the platform by upgrading to newer platform services and introducing foundational operational components.

Typical activities include:

- upgrading to vSphere 9.1
- deploying VCF Operations capabilities
- integrating centralized licensing services

This phase establishes the operational foundation required for future Kubernetes and private cloud adoption.

Stage 3 – VMware vSphere Kubernetes Service (VKS) Enablement

After the foundational platform services are established, organizations enable vSphere Supervisor for VKS to begin consuming Kubernetes services directly on the existing infrastructure platform.

At this stage:

- Kubernetes workloads can coexist with traditional VM workloads
- organizations begin operational adoption of Kubernetes platforms
- platform teams gain operational experience with Kubernetes lifecycle management

This phase represents the transition from traditional virtualization operations toward modern application platform operations.

Stage 4 – Full Private Cloud Operating Model

As platform maturity increases, organizations can expand toward a more complete private cloud operating model by introducing additional VMware Cloud Foundation services and operational capabilities.

This may include:

- VCF Automation
- NSX
- SDDC Manager
- Fleet management services
- platform governance and automation capabilities

At this stage, organizations evolve toward a fully integrated private cloud platform capable of supporting both traditional and modern application workloads under a unified operational model.

Key Architectural Message

The primary architectural message of this journey is that organizations do not need to adopt the full private cloud stack immediately in order to begin consuming Kubernetes services.

Instead, organizations can progressively evolve:

- from traditional virtualization operations
- toward Kubernetes-enabled infrastructure platforms
- and eventually toward a fully integrated private cloud operating model based on their operational maturity, organizational readiness, governance requirements, and modernization strategy.

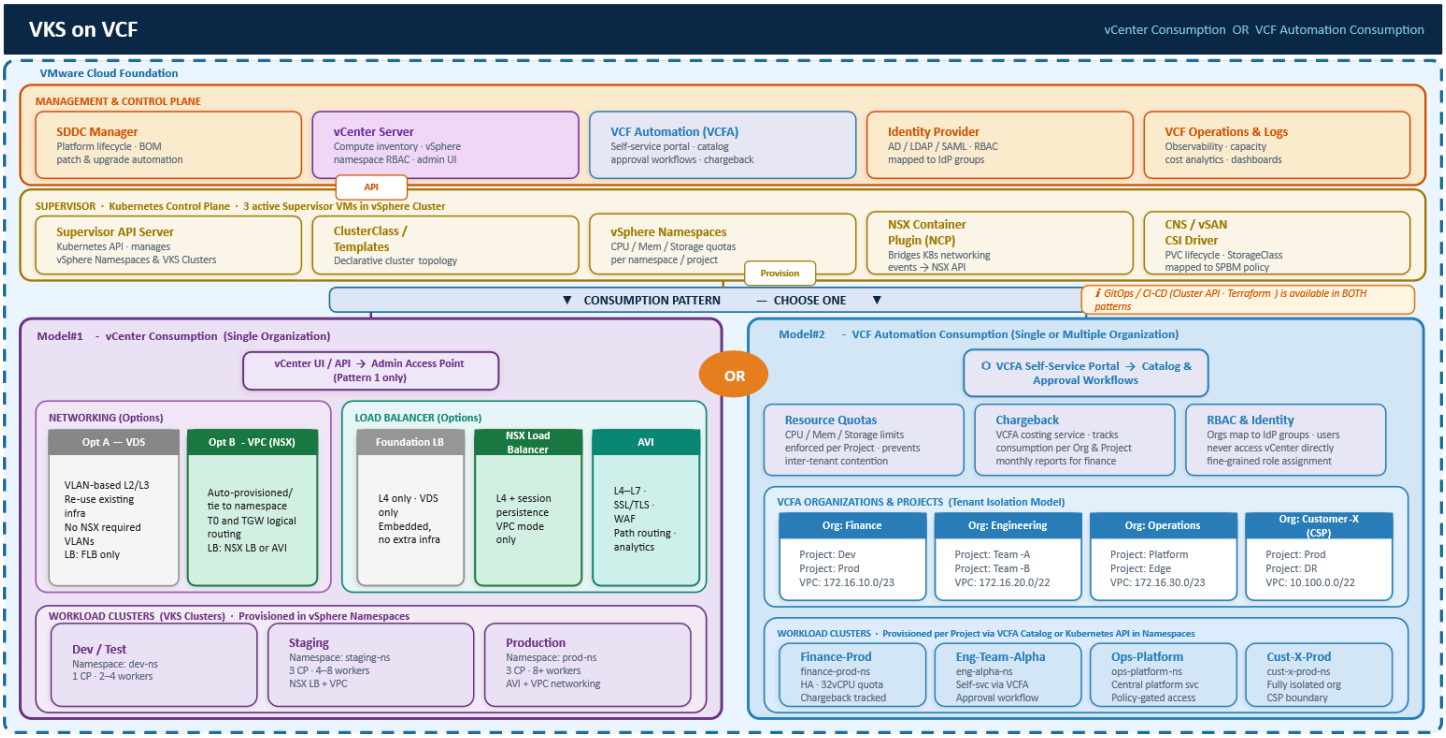
Consumption Patterns

The adoption journey that is outlined above demonstrates that organizations may adopt VMware vSphere Kubernetes Service at different stages of operational and platform maturity.

As organizations progress through this journey, selecting the appropriate VKS consumption model becomes a critical architectural decision that influences networking architecture, governance models, operational workflows, and long-term platform strategy.

The following sections provide a design decision framework for evaluating the available VKS consumption patterns within VMware Cloud Foundation.

Figure 201: Consumption Patterns



Design Context

This design blueprint focuses on the architectural decision framework for consuming and operating VMware vSphere Kubernetes Service within VMware Cloud Foundation 9 environments.

The reference design shown above represents the operational and architectural models available for deploying and consuming VKS across different organizational structures, operational models, governance approaches, and infrastructure consumption patterns.

Within VCF 9 and later, organizations can consume VKS through two primary consumption models:

- **Model 1 – vCenter Consumption Pattern**
- **Model 2 – VCF Automation Consumption Pattern**

These patterns represent fundamentally different operational consumption approaches and platform operating models for Kubernetes services. The architecture also illustrates how supporting platform services including networking, load balancing, identity integration, governance controls, quota management, and Kubernetes lifecycle operations -integrate into the overall VKS consumption architecture.

Design Philosophy

The core design principle of this guide is that VKS architecture decisions should be driven primarily by:

- operational ownership and consumption model requirements
- organizational operating structure and governance boundaries
- desired level of infrastructure abstraction
- lifecycle and operational management approach
- standardization and automation requirements
- scalability of operational processes and platform consumption models
- long-term platform operating strategy and organizational growth

The decision between vCenter-based VKS consumption and VCF Automation-based VKS consumption is fundamentally an operational architecture decision rather than purely a technical implementation choice. Both approaches can support varying levels of governance, isolation, and automation depending on how the environment is designed and operated. The appropriate model should therefore be selected based on the desired operational experience, organizational structure, lifecycle management approach, and long-term platform strategy rather than on individual technical features alone.

As a result, selecting the appropriate consumption model early in the design process is critical, since networking architecture, governance models, operational workflows, lifecycle boundaries, and consumption experiences are directly influenced by this decision.

Model Comparison

Characteristic	Model 1: vCenter	Model 2: VCF Automation
Organization Support	Typically aligned to a single operational organization or centralized infrastructure team	Designed for single or multiple organizational consumption models
Multi-Tenancy Model	Can be implemented operationally through infrastructure segmentation, namespaces, RBAC, or external governance models	Natively organization, project, quota, and policy-driven organizational abstraction built into the platform
Self-Service Consumption	Can be implemented through custom automation, APIs, or external self-service platforms integrated with vCenter	Built-in catalog-driven consumption experience with integrated workflows and API-driven provisioning
Network Options	VDS, NSX Classic or NSX (VPC)	NSX VPC-based networking model
Load Balancer	Foundation Load Balancer or Avi Load Balancer	NSX default Load Balancer or Avi Load Balancer
Consumption Interface	vCenter UI/API	VCF Automation Portal/API
Resource Governance	Operationally managed through infrastructure administration processes and controls	Integrated policy-based governance with quotas and organizational controls

Greenfield and Brownfield Deployment Considerations

This document distinguishes between two deployment scenarios that carry different constraints and recommendations:

- **Greenfield:** Net-new VCF deployments with no existing VKS infrastructure. Architects have maximum freedom to choose the optimal pattern and networking model from Day 1.
- **Brownfield:** Brownfield deployments include existing VMware vSphere 8.x environments that may already be running VMware vSphere Kubernetes Service, as well as environments planning to adopt VKS after upgrading to VMware Cloud Foundation 9 or later. The upgrade and adoption journey may vary depending on the existing networking, load balancer architecture, and target operational model.

The selected consumption pattern determines how administrators, platform teams, and developers interact with the vSphere Supervisor platform and its downstream Kubernetes workload clusters. This consumption model is distinct from how Kubernetes workloads themselves are operated after deployment.

Operational models such as:

- GitOps workflows
- Helm-based deployments
- Argo CD operations
- CI/CD integration pipelines remain consumption-pattern agnostic and can operate across both vCenter-based and VCF Automation-based VKS environments.

Greenfield Scenarios

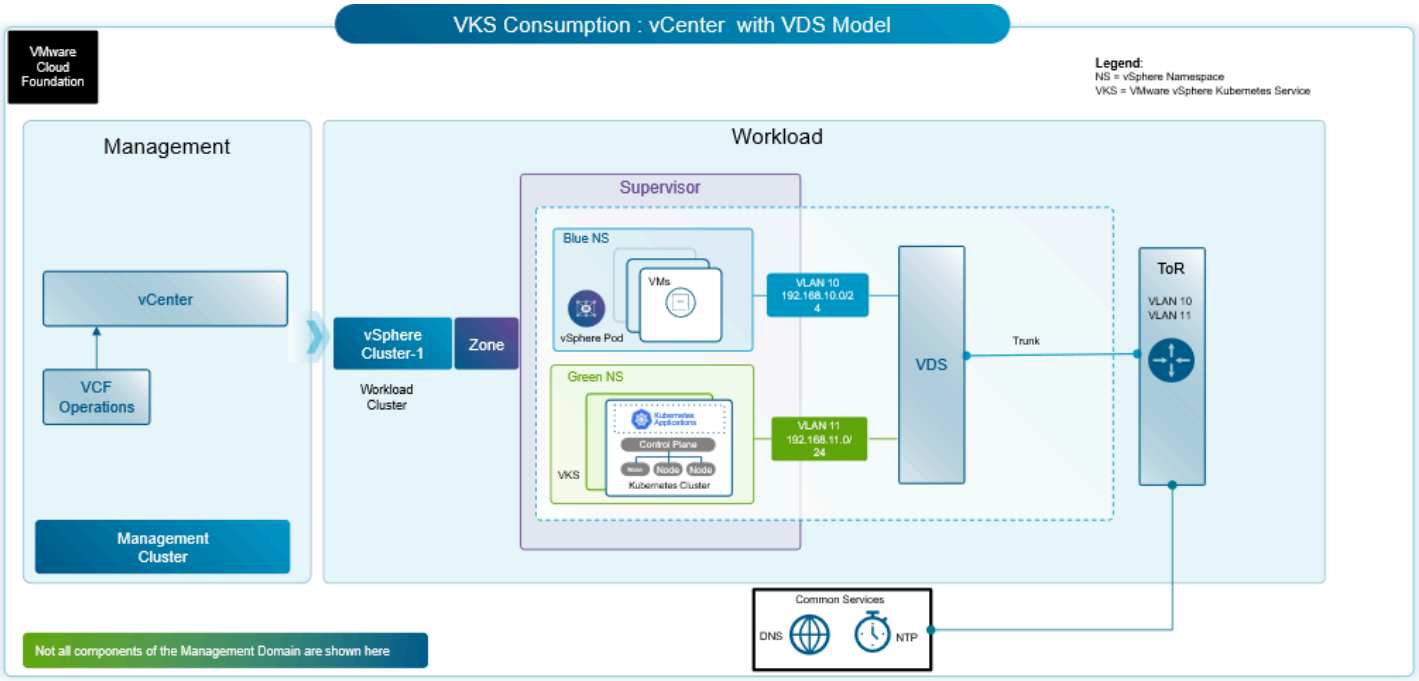
Following this section, the design guide presents:

- Example consumption patterns for both vCenter-based and VCF Automation-based VKS consumption
- Deployment models and architectural patterns that map these consumption approaches to real-world design scenarios

Pattern 1: Single Organization -VKS Consumption with vCenter and VLAN based network

Pattern 1 enables direct consumption of VKS Clusters through the VMware vCenter interface and is designed for organizations operating under a single administrative and operational model.

Figure 202:



This architecture illustrates the logical design for consuming VMware vSphere Kubernetes Service directly through a VMware vCenter-based operational model.

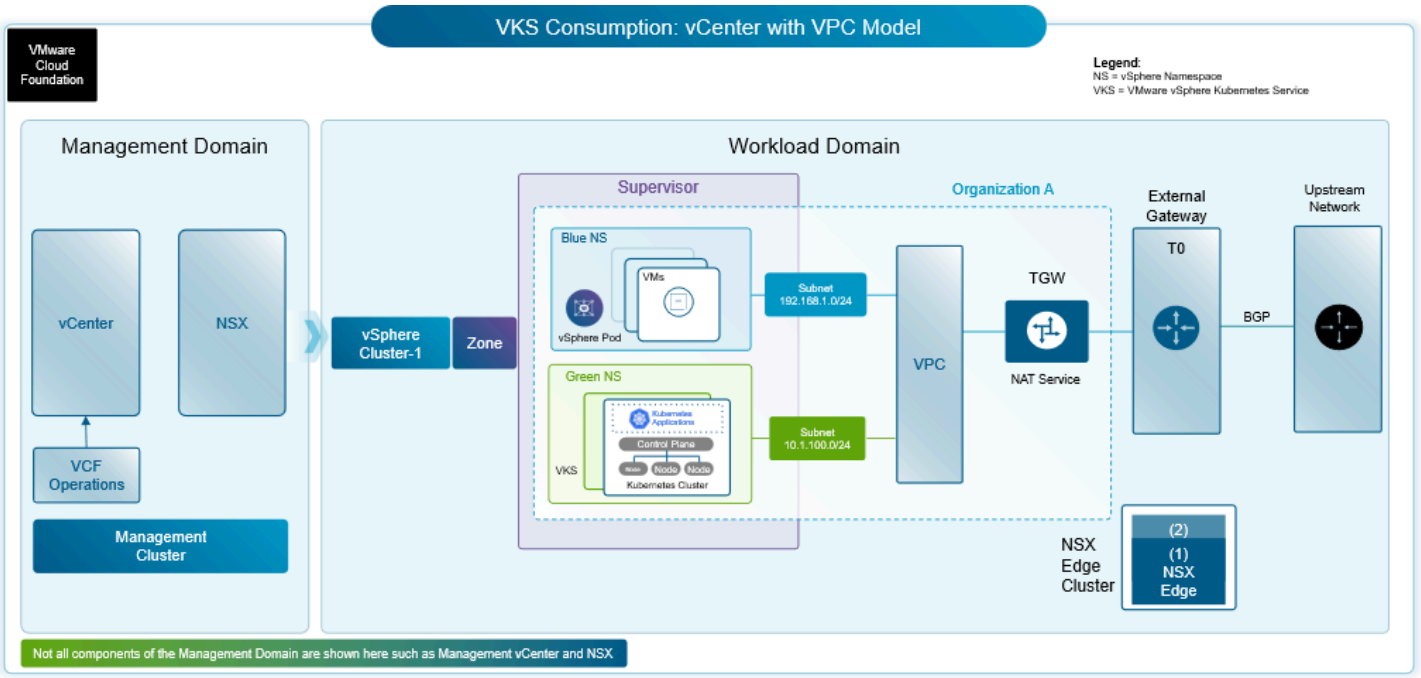
In this consumption pattern, Kubernetes services are provisioned and managed directly through vCenter and vSphere Supervisor constructs without introducing additional platform abstraction layers such as VCF Automation.

This model aligns closely with traditional infrastructure-centric operational practices where virtualization administrators manage Kubernetes infrastructure alongside existing virtual machine operations.

The architecture uses VLAN-backed networking through a VMware vSphere Distributed Switch (VDS) and integrates Kubernetes workloads directly into existing enterprise Layer 2 and Layer 3 networking environments.

Pattern 2: Single Organization - VKS Consumption with vCenter and VPC Networking model

Figure 203:



This architecture illustrates the logical design for consuming VMware vSphere Kubernetes Service directly through a VMware vCenter-based operational model while leveraging NSX VPC-based networking.

In this consumption pattern, Kubernetes services are still provisioned and managed directly through vCenter and vSphere Supervisor constructs, without introducing additional platform abstraction layers such as VCF Automation. However, unlike the VDS-based model, networking is implemented using NSX VPC constructs, enabling a more modern and flexible networking architecture.

This model maintains an infrastructure-centric operational approach - where virtualization administrators retain direct control - while introducing enhanced networking capabilities such as logical isolation, scalable segmentation, and policy-driven connectivity for Kubernetes workloads.

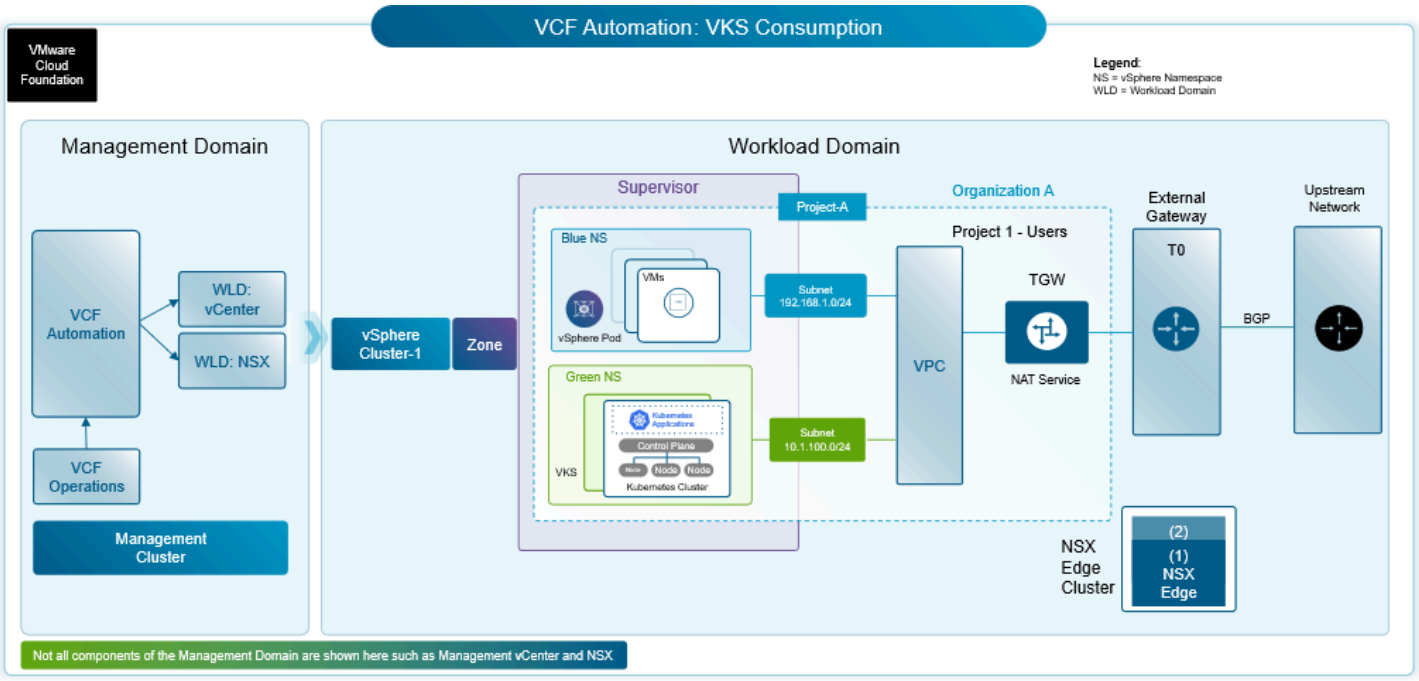
The architecture uses NSX VPC-based networking to provide:

- logical network isolation for workloads within a single organization
- abstraction from traditional VLAN-based networking constructs
- integrated routing and connectivity through NSX gateways
- support for modern application networking patterns within Kubernetes environments

This approach is well suited for organizations that want to retain vCenter-based operational control while adopting more advanced, cloud-like networking capabilities for Kubernetes workloads.

Pattern 3: Single Organization– VKS Consumption with VCF Automation

Figure 204:



This architecture illustrates the logical design for consuming VMware vSphere Kubernetes Service using VMware Cloud Foundation Automation within a single-organization model, where VPC-based networking is the default and required construct.

In this consumption pattern, Kubernetes services are provisioned and managed through the VCF Automation platform rather than directly via VMware vCenter. The vSphere Supervisor still provides the underlying Kubernetes control plane, but all consumption, governance, and lifecycle operations are abstracted through VCF Automation constructs such as organizations, projects, and policies.

This model introduces a platform-centric operational approach, where infrastructure is abstracted and consumed through standardized workflows, APIs, and catalog-driven provisioning, enabling consistent governance and operational control even within a single organization.

The architecture uses NSX VPC-based networking as the default model, providing:

- logical network isolation through VPC constructs
- integrated routing and connectivity via TGW and NSX gateways
- built-in NAT capabilities for workload egress
- policy-driven networking aligned with project and organizational boundaries
- abstraction from underlying VLAN or traditional network constructs

Kubernetes workloads are deployed within VPC-backed subnets and are logically grouped under organizational and project constructs, enabling structured consumption and governance even in a single-organization environment.

This approach is best suited for organizations that want to:

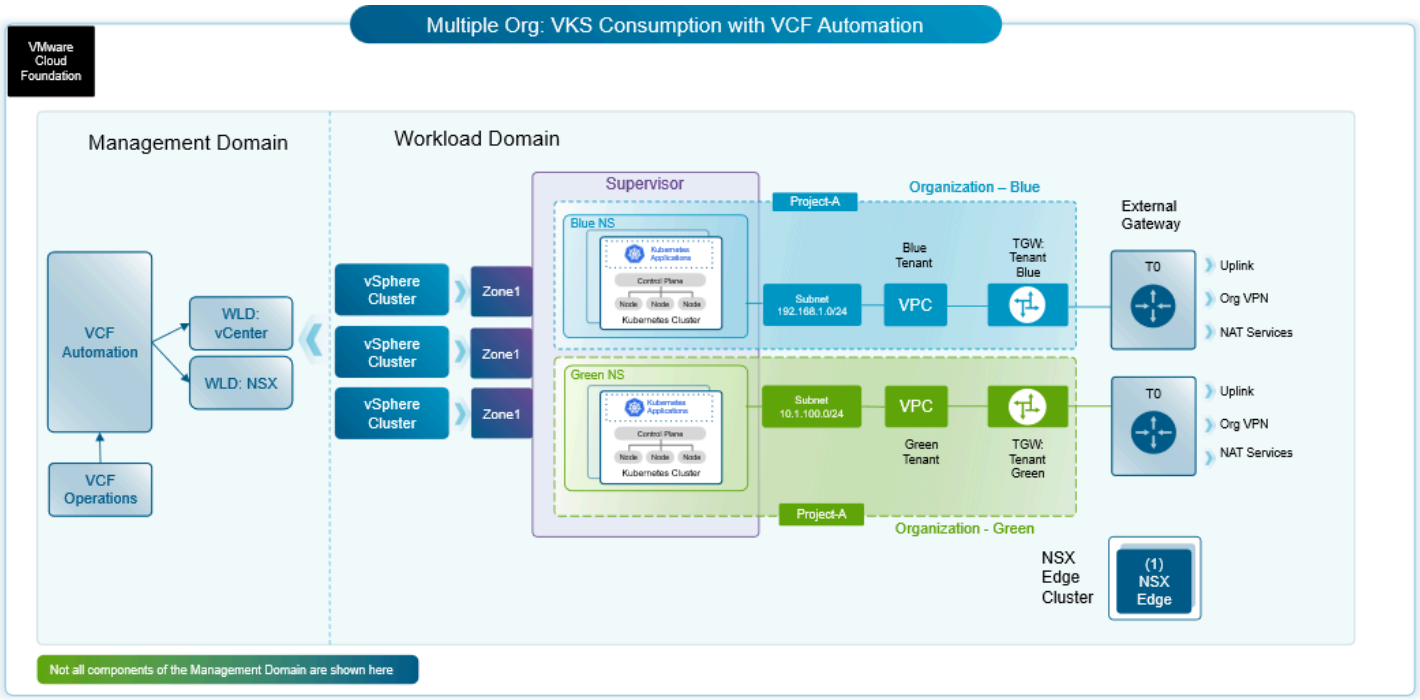
- standardize Kubernetes consumption through a platform layer

- introduce governance, quotas, and policy-driven operations
- adopt cloud-like networking and application delivery models
- prepare for future multi-tenancy expansion without redesigning the architecture

Even in a single-organization setup, this model establishes a foundation for scalable, governed, and automated Kubernetes platform operations, aligning closely with modern private cloud operating models.

Pattern 4: Multiple Organization– VKS Consumption with VCF Automation

Figure 205:



This architecture illustrates the logical design for consuming VMware vSphere Kubernetes Service using VMware Cloud Foundation Automation in a multi-organization (multi-tenant) model.

In this consumption pattern, Kubernetes services are provisioned and governed through VCF Automation constructs such as organizations, projects, and policies, rather than direct interaction with VMware vCenter. The vSphere Supervisor provides the shared control plane, while VCF Automation enforces logical separation and operational boundaries across multiple tenants.

This model introduces a true multi-tenant platform architecture, where each organization is isolated at the networking, policy, and consumption layers while sharing underlying infrastructure resources.

The architecture uses NSX VPC-based networking as the default, enabling:

- **Tenant-level isolation** through dedicated VPCs per organization
- **Independent networking domains** with separate subnets, routing, and NAT policies
- **Dedicated connectivity paths** via TGW and Tier-0 gateways per tenant

- **Policy-driven segmentation** aligned with organizational and project boundaries
- **Secure multi-tenant access** to upstream networks through controlled egress

Kubernetes workloads are deployed within organization-specific namespaces and mapped to VPC-backed subnets, ensuring that each tenant operates within a logically isolated environment while leveraging a shared Supervisor and infrastructure platform.

This approach is best suited for environments that require:

- **Strong multi-tenancy and tenant isolation** (e.g., enterprise business units, service providers)
- **Centralized platform governance with decentralized consumption**
- **Chargeback / showback models** aligned to organizations and projects
- **Scalable onboarding of multiple tenants without infrastructure redesign**

Key Takeaway

VCF Automation enables multiple organizations to safely share the same underlying VKS platform while maintaining strict isolation, independent networking, and controlled access - making it the preferred model for scalable, multi-tenant private cloud environments.

Pre-Requisite Design Choices for VMware vSphere Kubernetes Service

Before deploying and consuming **VMware vSphere Kubernetes Service (VKS)**, customers must prepare the supporting platform services required by the selected consumption pattern. These prerequisites apply regardless of whether VKS is consumed through **vCenter** or **VCF Automation**, but the available networking and load balancer choices differ by pattern.

Pre-Requisite Design Area	Pattern 1 and Pattern 2: vCenter Consumption	Pattern 3 and Pattern 4: VCF Automation Consumption	Design Guidance	Key Design Considerations	Design Choices
vSphere Supervisor Design	Required	Required	A vSphere Supervisor must be designed and enabled before VKS clusters can be consumed. Supervisor design should account for cluster placement, namespace strategy, storage policies, capacity, availability, and lifecycle requirements.	• vSphere Supervisor cluster topology • vSphere Supervisor Zone Design • Control plane VM Placement and sizing (Small / Medium / Large)	• vSphere Supervisor Zone Models • vSphere Supervisor Control Plane Availability Models
Networking Model	• VLAN-backed/VDS • NSX VPC	NSX VPC only	• vCenter consumption supports traditional VLAN-backed networking or NSX- VPC designs.	If considering to use NSX -VPC model with vCenter consumption then choose connectivity design model which best fit to your scenarios.	• Centralized Connectivity Model • Distributed VLAN Connectivity Model

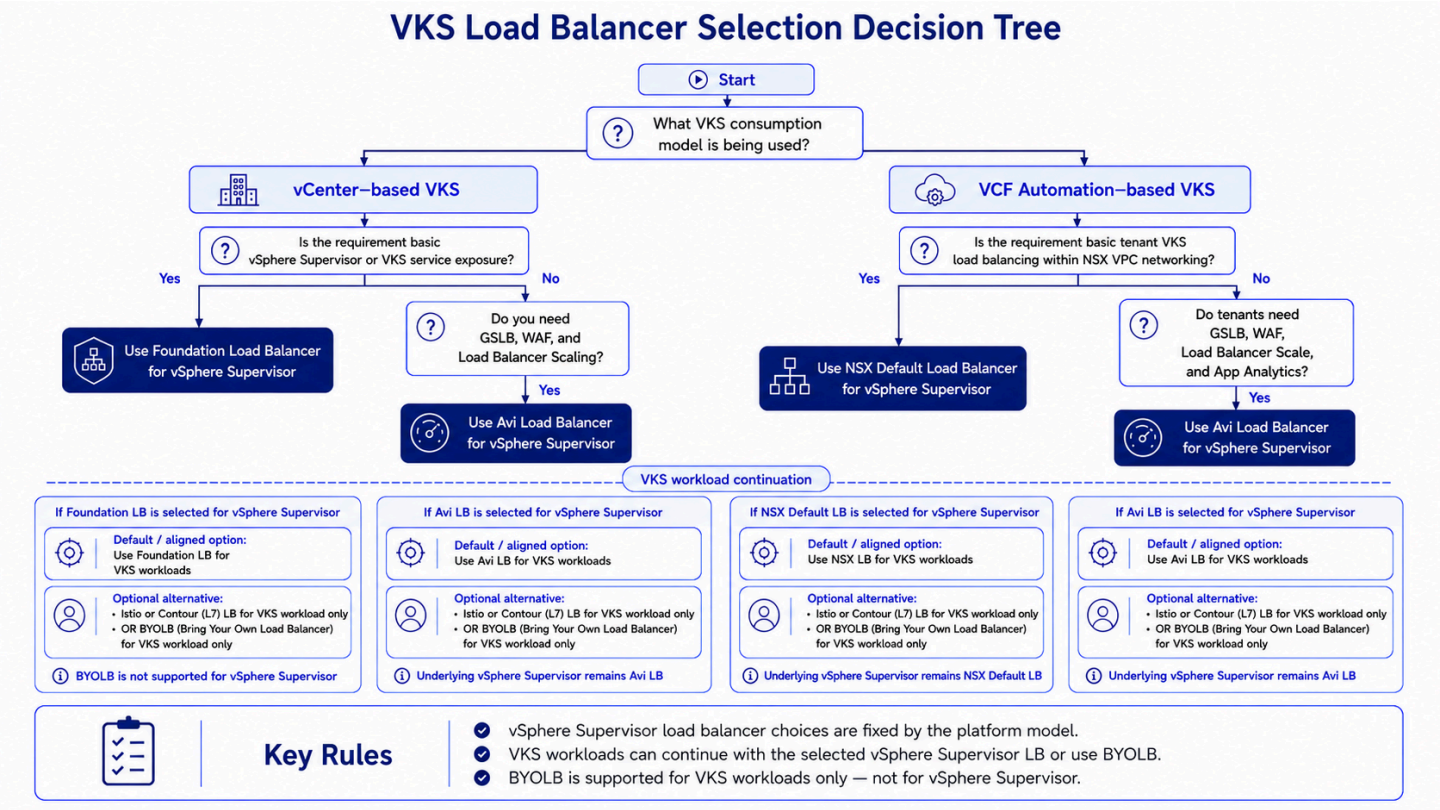
Pre-Requisite Design Area	Pattern 1 and Pattern 2: vCenter Consumption	Pattern 3 and Pattern 4: VCF Automation Consumption	Design Guidance	Key Design Considerations	Design Choices
			• VCF Automation consumption requires NSX VPC to support the platform consumption model and automated networking workflows.	Choose the connectivity design model supported from the VCF Automation All Apps Orgs Consumption Model. VPC required for all VCF Automation and workload consumption.	
Load Balancer	• Foundation Load Balancer • Avi Load Balancer	• NSX default load Balancer • or Avi Load Balancer	Load balancer selection should align with the networking model, scale, application delivery needs, and operational requirements.	• Default - NSX Load Balancer • Or Avi Load Balancer	• Avi Load Balancer Model • NSX Edge VPC Load Balancer Model • Virtual Network Appliance Load Balancer Model • Foundation Load Balancer Model

Attention:

Design Note:

The selected VKS consumption pattern should be decided before enabling the environment for VKS. The choice between **vCenter-based consumption** and **VCF Automation-based consumption** directly influences the required networking model, load balancer architecture, governance constructs, and operational workflows.

Load Balancer Decision Tree



Load Balancer Selection

The load balancer selection depends on the VKS consumption model, the vSphere Supervisor networking model, and the level of application delivery capability required. The design decision should be viewed in two layers:

1. **vSphere Supervisor load balancer selection**
2. **VKS workload load balancer continuation**

The vSphere Supervisor load balancer is selected as part of the platform design. Once selected, it becomes the default aligned load balancer option for VKS workloads. However, customers may choose to bring their own load balancer for Kubernetes workloads using the BYOLB model. This BYOLB option applies only to VKS workloads and does not replace or change the load balancer used by the vSphere Supervisor.

vCenter-Based VKS Consumption:

For **vCenter-based VKS**, the primary design decision is whether the environment requires basic vSphere Supervisor and Kubernetes service exposure or advanced application delivery.

The **Foundation Load Balancer** is appropriate when the primary requirement is basic vSphere Supervisor and VKS service exposure with minimal operational complexity. This model is best suited for foundational VKS enablement, small dev environments, or simpler Kubernetes deployments where advanced application delivery services are not required.

When Foundation Load Balancer is selected for the vSphere Supervisor, it also becomes the default aligned option for VKS workloads. Customers can continue to use Foundation Load Balancer for VKS workload exposure. Alternatively, they may use Bring Your Own Load Balancer (BYOLB) for VKS workloads only, if they have a separate Kubernetes workload load balancing requirement.

Avi Load Balancer should be selected for vCenter-based VKS when the customer requires production-grade Kubernetes ingress, advanced L7 services, application analytics, autoscaling, GSLB, WAF, or stronger application delivery control.

When Avi Load Balancer is selected for the vSphere Supervisor, Avi also becomes the default aligned option for VKS workloads. Customers can continue with Avi for both vSphere Supervisor and VKS workload load balancing, or they can use BYOLB for VKS workloads only. In this case, the underlying vSphere Supervisor load balancer remains Avi.

VCF Automation-Based VKS Consumption:

For **VCF Automation-based VKS**, the load balancer decision is tied to NSX VPC-based tenant networking and the level of application delivery required by tenants or application teams.

The **NSX Default Load Balancer** is appropriate when the customer needs basic L4 load balancing integrated with NSX VPC-based tenant networking and VCF Automation consumption. This model aligns well with standard tenant Kubernetes clusters where platform-provided networking services are sufficient.

When NSX Default Load Balancer is selected for the vSphere Supervisor, it also becomes the default aligned option for VKS workloads. Customers can use NSX load balancing for VKS workload exposure. Alternatively, they may use BYOLB for VKS workloads only. In this case, the underlying vSphere Supervisor load balancer remains NSX Default Load Balancer.

Avi Load Balancer should be selected for VCF Automation-based VKS when tenants or application teams require advanced self-service application publishing, Kubernetes ingress, L7 routing, app visibility, WAF, GSLB, or enterprise-grade application delivery.

When Avi Load Balancer is selected for the vSphere Supervisor, Avi becomes the default aligned option for VKS workloads. Customers can continue with Avi for workload load balancing or use BYOLB for VKS workloads only. The underlying vSphere Supervisor load balancer remains Avi.

VKS Workload Load Balancer Alternatives:

For VKS workloads only, customers may use workload-specific alternatives instead of continuing with the selected Supervisor load balancer.

Supported workload-level options include:

- **Istio**
- **Contour**
- **OR Bring Your Own Load Balancer (BYOLB)**

These options are useful when application teams have existing Kubernetes ingress standards, service mesh requirements, third-party load balancer preferences, or specialized application delivery patterns.

However, BYOLB does not replace the load balancer required by the vSphere Supervisor. The vSphere Supervisor must still use the platform-supported load balancer choice based on the selected consumption and networking model.

A key design consideration is that **BYOLB is a workload-level option, not a vSphere Supervisor load balancer option**.

VKS Workload Load Balancer Design Choices

VKS Workload Option	When to Use	Design Considerations	Notes
Selected vSphere Supervisor Load Balancer	Use when the customer wants a platform-aligned model and does not need a separate Kubernetes workload LB pattern	Keeps the design simple and consistent with the Supervisor LB choice	If Foundation LB, NSX LB, or Avi is selected for the vSphere Supervisor, customers can continue with the same LB option for VKS workloads where supported
Istio L7	Use when the customer needs service mesh capabilities or advanced traffic management for Kubernetes workloads	Useful for mTLS, traffic splitting, retries, observability, and service-to-service control	Best suited when the customer has a service mesh strategy
Contour L7	Use when the customer needs Kubernetes ingress using Envoy-based L7 routing	Good fit for HTTP/S ingress, path-based routing, and Kubernetes-native app exposure	Simpler than a full service mesh; commonly used for ingress-focused use cases
Avi with AKO (L7)	Use when the customer has Avi entitlement and wants enterprise-grade L7 load balancing for Kubernetes workloads	Provides Kubernetes integration through Avi Kubernetes Operator	Best fit for production-grade ingress, GSLB and WAF
Bring Your Own Load Balancer / BYOLB	Use when the customer has an existing third-party or standardized Kubernetes workload LB solution	Useful when app teams already use a preferred ingress controller, cloud-native LB, or enterprise LB platform	BYOLB applies to VKS workloads only and does not replace the vSphere Supervisor load balancer

vSphere Supervisor Design for VKS Consumption

The vSphere Supervisor is the foundational control plane that enables VMware vSphere Kubernetes Service consumption within a vSphere environment. It provides the Kubernetes API endpoint, namespace abstraction, and integration with underlying vSphere infrastructure required to provision and manage Kubernetes workload clusters.

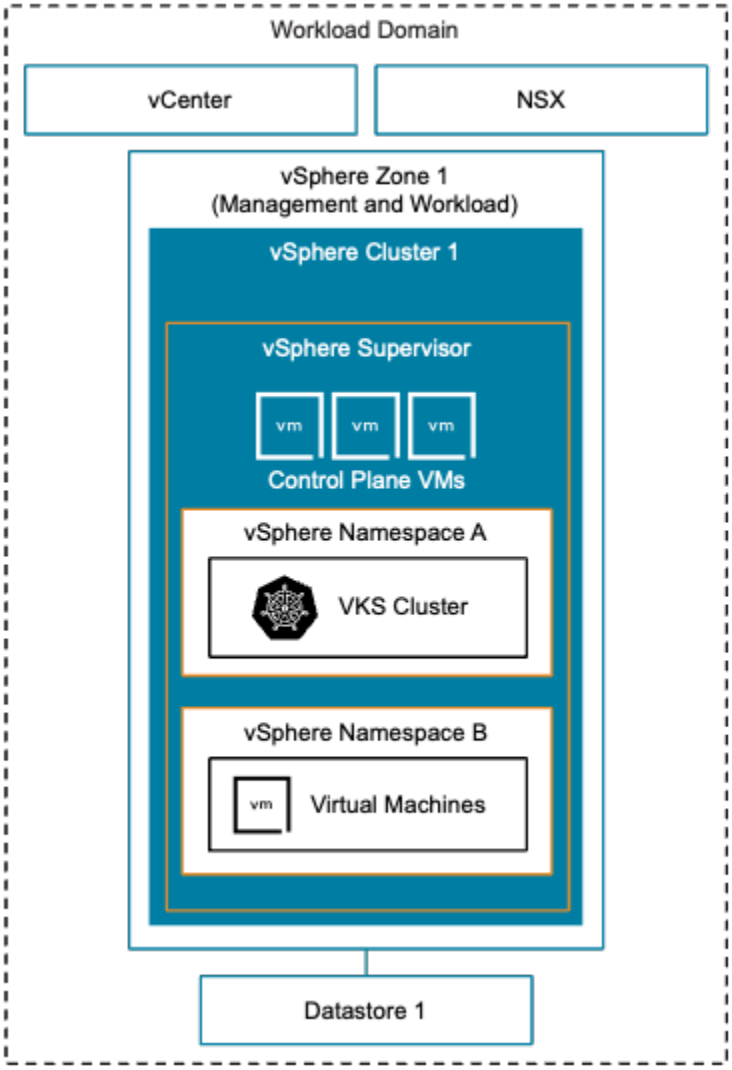
Designing the Supervisor correctly is critical, as it directly influences networking architecture (VDS vs NSX VPC), load balancing, resource isolation, and operational scalability. The Supervisor design also determines how Kubernetes workloads are exposed, secured, and integrated with enterprise networking and platform services.

The following scenarios explore different Supervisor deployment models and design patterns based on networking choices, operational requirements, and consumption models.

Single Management Zone with Combined Workload Zones Model Design

Figure 206: Single Management Zone with Combined Workload Zones Model Diagram

This diagram illustrates an example of a *Single Management Zone with Combined Workload Zones Model* deployment.



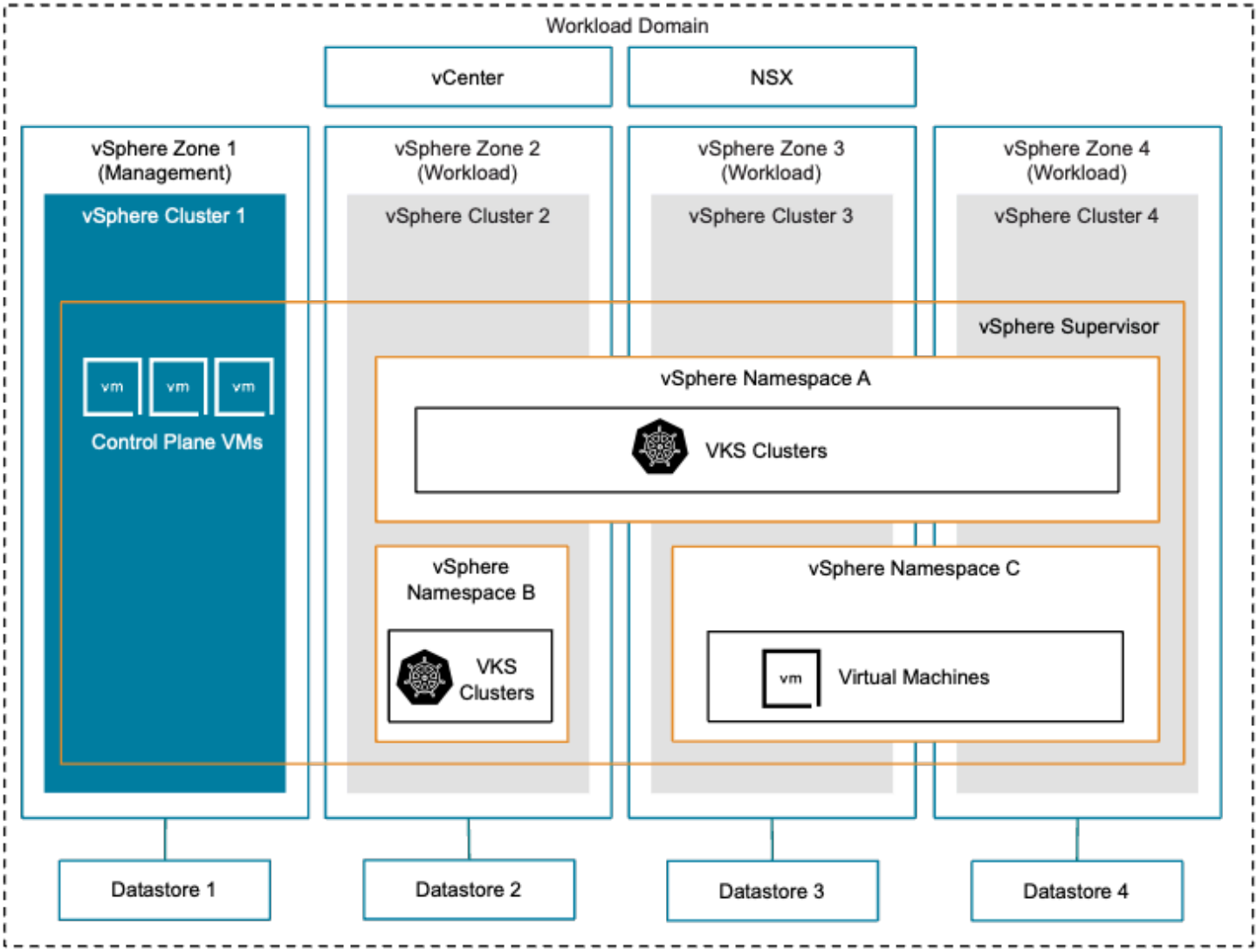
Design Attributes:

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single - Combined with management
vSphere Zone Datastore Type	Zonal datastore
Activation	• At workload domain creation. OR • From vCenter.

Single Management Zone with Isolated Workload Zones Model Design

Figure 207: Single Management Zone with Isolated Workload Zones Model

This diagram illustrates an example of a *Single Management Zone with Isolated Workload Zones Model* deployment.



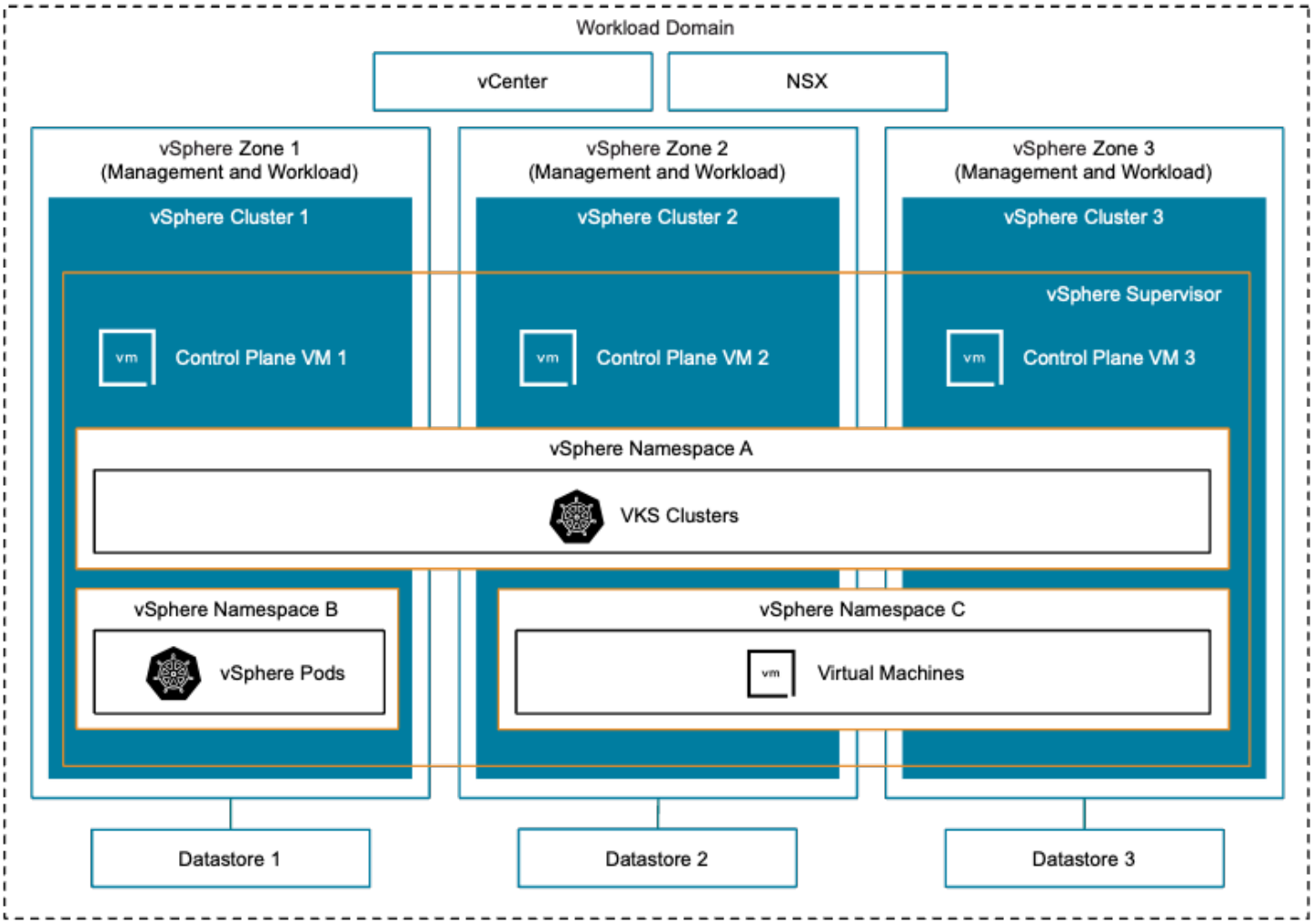
Design Attributes:

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single or multiple - Isolated from management
vSphere Zone Datastore Type	**Management Zone** • Zonal datastore **Single Workload Zone** • Zonal datastore **Multiple Workload Zones** • Cross-zone datastore • Multi-zone datastore
Activation	• At workload domain creation. OR • From vCenter.

Three Management Zone with Combined Workload Zones Model Design

Figure 208: Three Management Zones with Combined Workload Zones Model

This diagram illustrates an example of a *Three Management Zones with Combined Workload Zones Model* deployment.



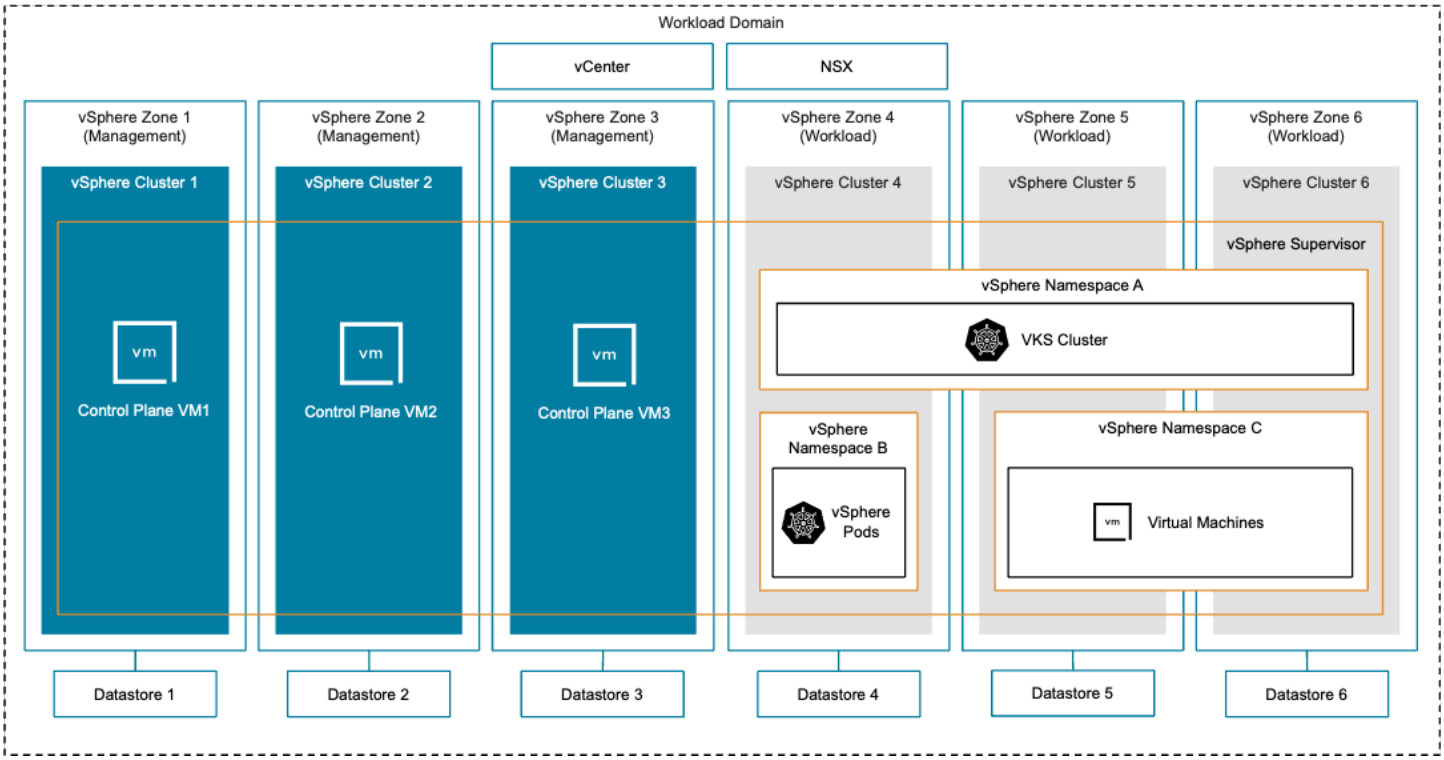
Design Attributes:

Attribute	Detail
vSphere Supervisor Management Zone	Three
vSphere Supervisor Workload Zones	Three - Combined with management
vSphere Zone Datastore	• Cross-zone datastore • Multi-zone datastore
Activation	Activated from vCenter API after workload domain creation.

Three Management Zone with Isolated Workload Zones Model Design

Figure 209: Three Management Zones with Isolated Workload Zones Model

This diagram illustrates an example of a *Three Management Zones with Isolated Workload Zones Model* deployment.



Design Attributes:

Attribute	Detail
vSphere Supervisor Management Zone	Three
vSphere Supervisor Workload Zones	Single or multiple - Isolated from management
vSphere Zone Datastore	**Management Zone** • Cross Zone Datastore • Multi Zone Datastore **Single Workload Zone** • Single Zone Datastore **Multiple Workload Zones** • Cross Zone Datastore • Multi Zone Datastore
Activation	Activated from vCenter API after workload domain creation.

Deployment Models

After selecting the appropriate consumption pattern for VMware vSphere Kubernetes Service, the next critical step is defining the deployment model of the vSphere Supervisor.

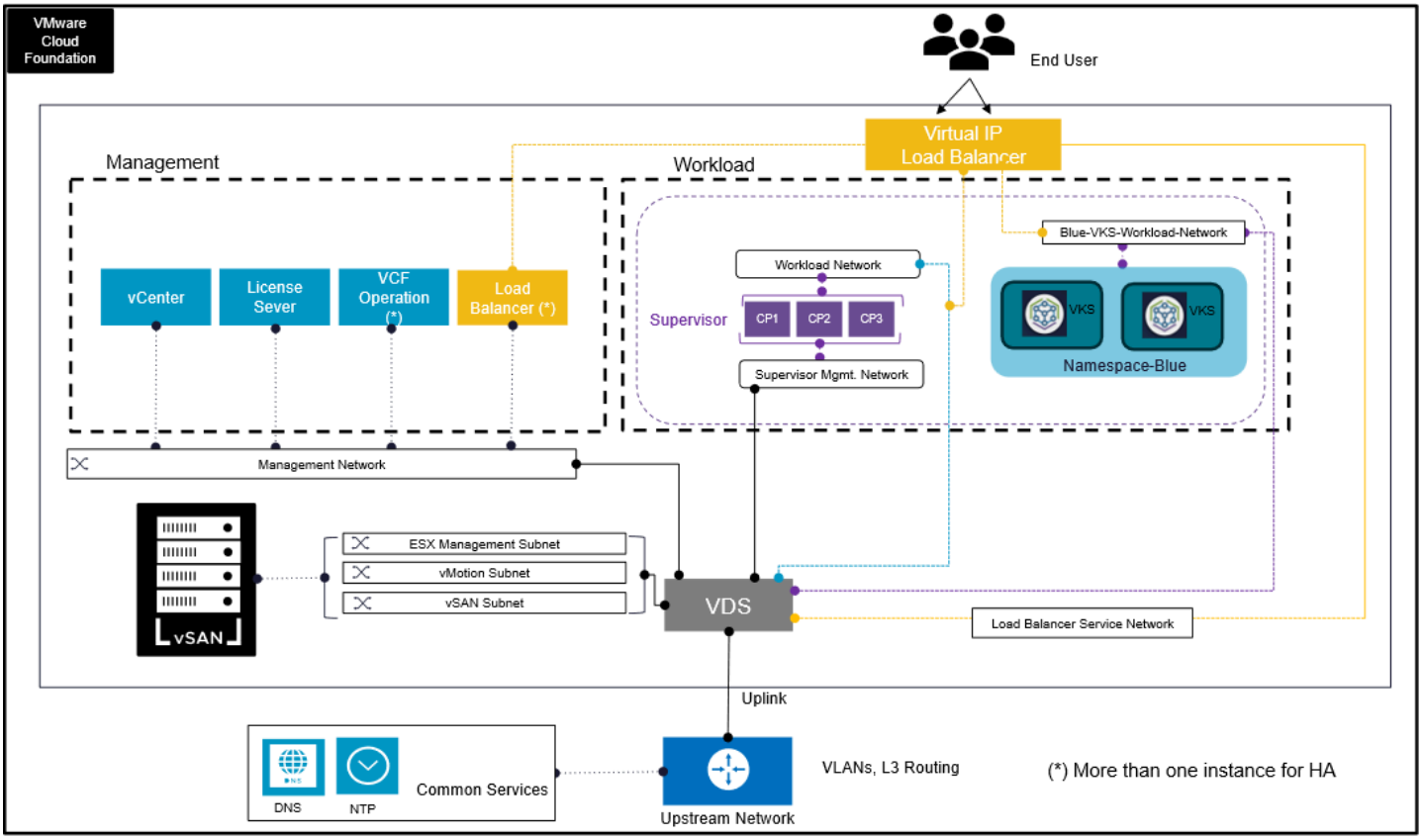
The Supervisor deployment model determines how Kubernetes control plane services are integrated with the underlying networking architecture, how workloads are connected and exposed, and how operational responsibilities are structured across infrastructure and platform teams. It also directly impacts key design areas such as network segmentation, load balancing, scalability, and workload isolation.

In VCF environments, two primary Supervisor deployment models are commonly used:

- **VDS-based networking model**, which leverages traditional VLAN-backed networking using vSphere Distributed Switch and aligns with existing enterprise network architectures
- **VPC-based networking model**, which uses NSX VPC constructs to provide cloud-like network abstraction, enhanced isolation, and platform-oriented networking capabilities

The following sections describe these deployment models in detail and provide guidance on when to use each approach based on operational requirements, networking strategy, and long-term platform design objectives.

Figure 210: vSphere Supervisor Deployment Model for VKS Consumption Using VDS-Based Networking



Architecture Details

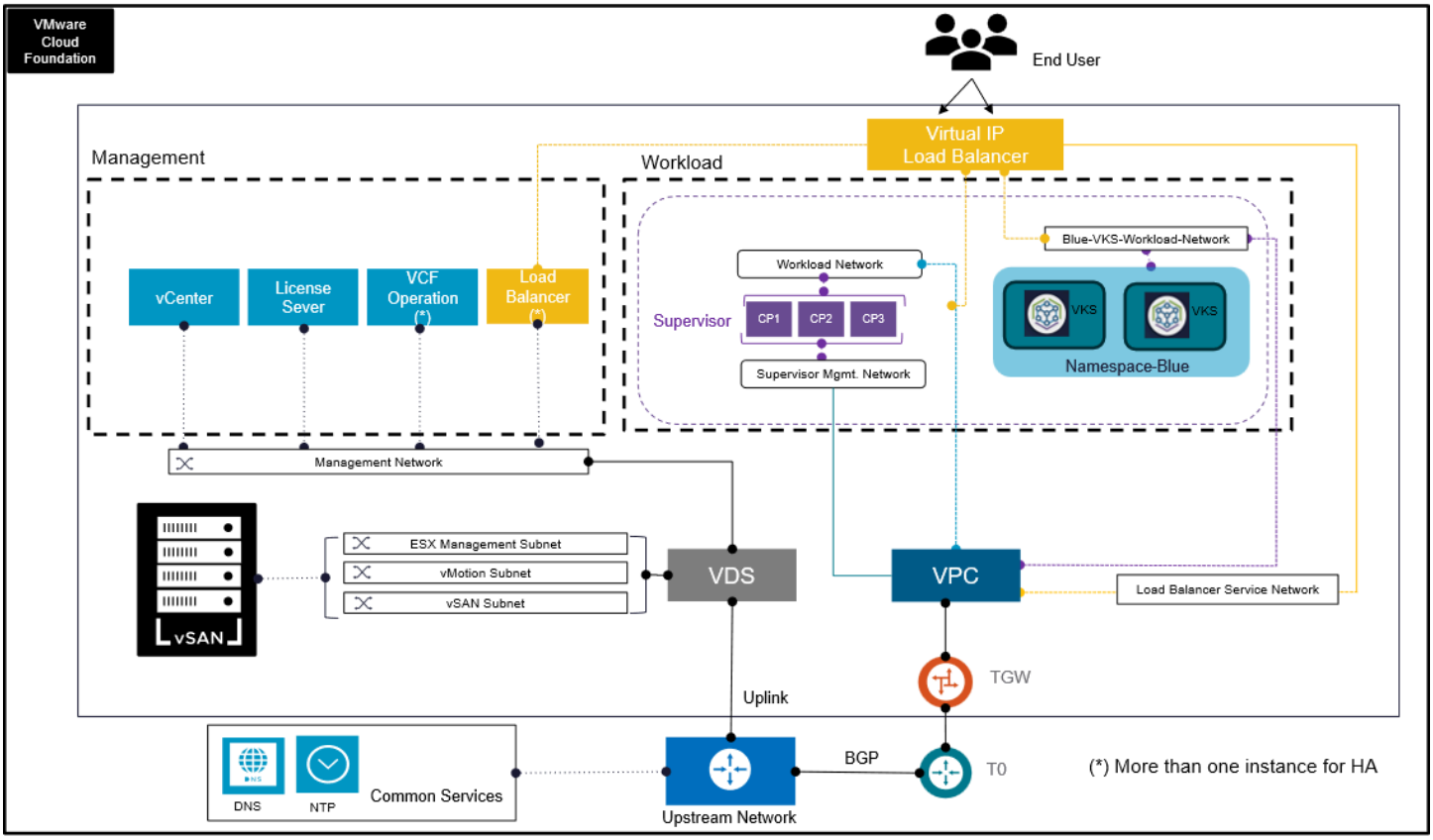
This architecture illustrates the deployment model for vSphere Supervisor and VMware vSphere Kubernetes Service consumption using a traditional VDS-backed networking architecture within a VMware Cloud Foundation enviornment.

In this model, Kubernetes services are deployed on top of existing VLAN-backed networking infrastructure using a vSphere Distributed Switch (VDS) architecture rather than NSX VPC-based networking. The design enables organizations to consume VKS while preserving existing Layer 2 and Layer 3 networking constructs commonly used in traditional enterprise virtualization environments.

This deployment model is commonly used in environments where organizations want to adopt Kubernetes services while maintaining operational alignment with existing VLAN-based networking architectures and operational processes.

Architecture Component	Description
Management Cluster	Hosts the core infrastructure and operational platform services that are required to operate the VKS environment
vSphere Supervisor	Provides the Kubernetes control plane required for **VMware vSphere Kubernetes Service** consumption
VKS Workload Clusters	Kubernetes workload clusters provisioned inside vSphere namespaces
VDS-Based Networking	Uses VMware vSphere Distributed Switch as the primary networking construct for Supervisor and workload connectivity
Load Balancer	Provides ingress access and service exposure for Supervisor and Kubernetes workloads

Figure 211: vSphere Supervisor Deployment Model for VKS Consumption Using VPC-Based Networking



Architecture Details

This architecture illustrates the deployment model for vSphere Supervisor and VMware vSphere Kubernetes Service consumption using an NSX VPC-based networking architecture within a VMware Cloud Foundation environment.

Unlike traditional VLAN-backed networking models, this architecture uses NSX VPC networking constructs to provide workload isolation, logical network abstraction, scalable routing, and platform-oriented networking capabilities for Kubernetes workloads and services.

This model is commonly aligned with modern platform operating models where Kubernetes workloads require stronger workload isolation, scalable tenant segmentation, automated networking workflows, and cloud-like application consumption patterns.

Architecture Components

Architecture Component	Description
Management Domain Services	Hosts centralized platform management services required for operating the VCF environment
Workload Domain	Dedicated workload infrastructure domain hosting Supervisor and Kubernetes workloads
NSX Edge	Provides north-south routing and connectivity services for VPC-based workloads
vSphere Supervisor	Provides the Kubernetes control plane for VKS consumption
VPC Networking	Provides logical application and workload networking abstraction for Kubernetes workloads
VKS Workload Clusters	Kubernetes workload clusters provisioned within namespace boundaries using VPC networking
CTGW (Centralized Transit Gateway)	Provides centralized north-south routing and connectivity services
T0 Gateway	Provides upstream routing integration
Load Balancer Integration	Provides ingress and service exposure for Kubernetes workloads

Brownfield Scenarios

Many organizations adopting VMware vSphere Kubernetes Service within VMware Cloud Foundation environments are not starting from a net-new infrastructure deployment. Instead, they are evolving existing environments that may already contain established VKS deployments, networking architectures, operational processes, workload clusters, automation platforms, and governance models.

These brownfield environments introduce unique architectural considerations because existing operational models, networking dependencies, load balancer integrations, and workload placement strategies often need to be preserved during the modernization journey.

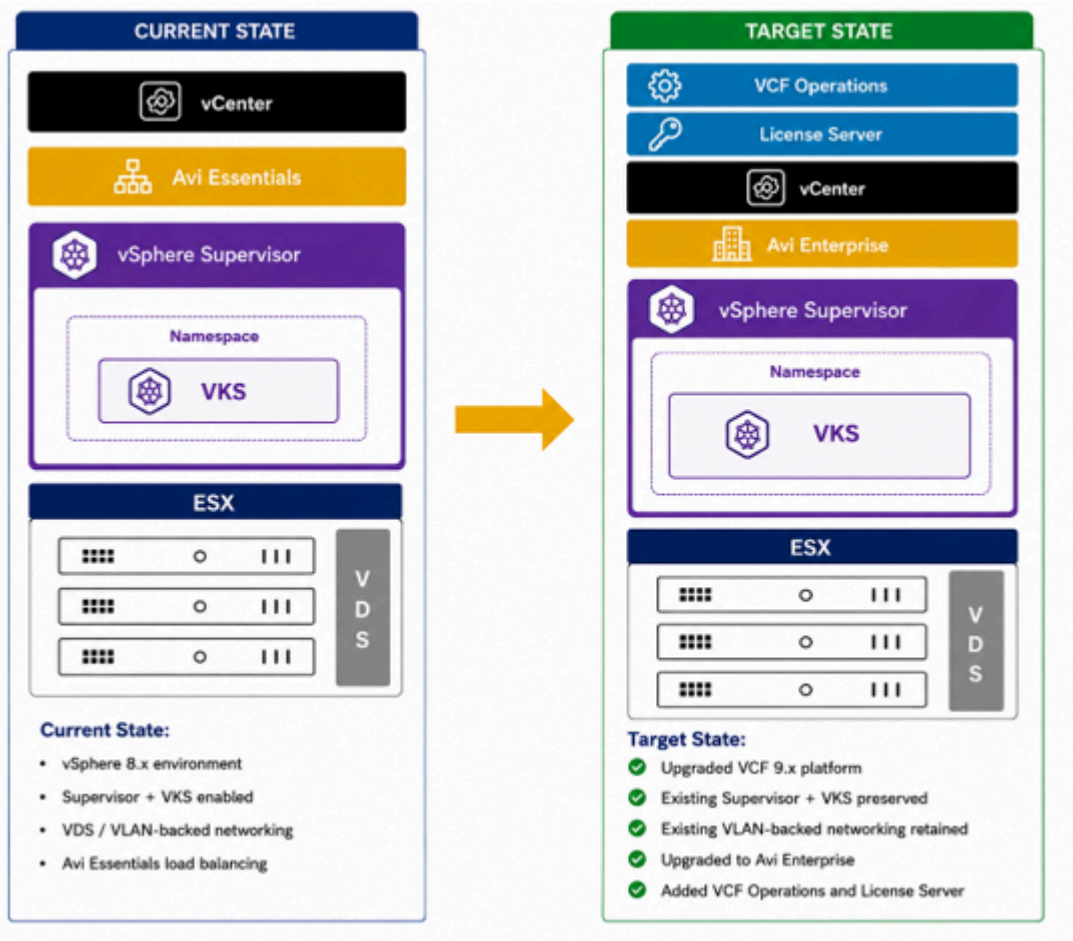
Brownfield adoption scenarios may include:

- existing vSphere environments with no Kubernetes deployments
- existing VKS deployments operating on VLAN-backed networking
- existing NSX-backed Kubernetes environments
- environments requiring coexistence between existing workloads and new modern workloads with VCF Automation

The following sections outline common brownfield deployment scenarios and describe architectural considerations, transition models, networking implications, and operational design choices associated with each approach.

> **Attention:**
>
> **Note:** The brownfield scenarios described in this document provide high-level design and transition guidance for moving to **VCF 9 or later**. They are intended to help architects understand possible upgrade paths, design considerations, and target-state options.
>
> This section is not a step-by-step upgrade guide and does not replace the official product upgrade documentation. When planning an actual upgrade, refer to the official VCF upgrade documentation and follow the prescribed upgrade procedures, compatibility checks, and sequencing guidance.

Scenario 1: vSphere 8.x + VKS + VDS Networking + Avi Essentials



Current State:

This brownfield scenario represents an existing **VMware vSphere** 8.x environment operating:

- vSphere 8.x
- vSphere Supervisor
- VMware vSphere Kubernetes Service (VKS)
- VDS-based VLAN-backed networking
- Avi Essentials load balancing

The existing environment consumes VKS using a VLAN-backed networking architecture integrated with Avi Essentials for Kubernetes services.

Transition Path to Target State:

The target architecture modernizes the environment to a vSphere 9.x and VCF 9.x-aligned deployment model while preserving the existing VDS-based networking architecture and Supervisor/VKS operational model.

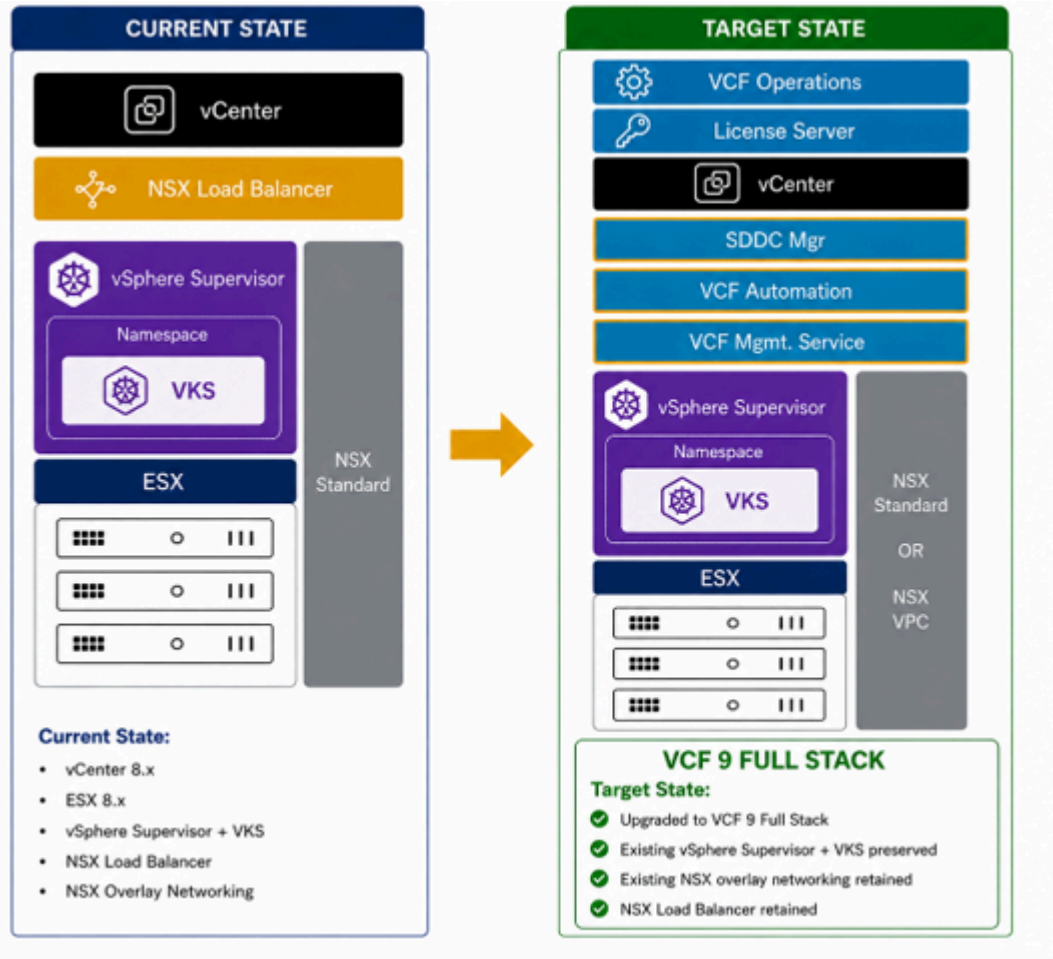
In this scenario, Avi Essentials must be upgraded to **VMware Avi Load Balancer Enterprise** to align with VCF 9.x capabilities and support production-grade load balancing features.

High Level Upgrade Workflow:

- Plan the platform and Avi upgrade strategy prior to upgrading to vSphere 9.x.
- Upgrade Avi Essentials to Avi Enterprise, ensuring licensing and feature compatibility.
- Upgrade vCenter from 8.x to 9.x.
- Upgrade ESX hosts to 9.x.
- Existing Supervisor and VKS workloads are preserved but it is still recommended to keep a workload backup before you plan for the transition.

- Existing VLAN Backed networking retained
- Deploy VCF Operations.
- Deploy License Server for centralized license management.

Scenario 2: vSphere 8.x + VKS + NSX Overlay Networking + NSX Load Balancer



Current State:

This brownfield scenario represents an existing **VMware vSphere** 8.x environment operating:

- vSphere 8.x
- vSphere Supervisor
- VMware vSphere Kubernetes Service (VKS)
- NSX overlay networking
- NSX Load Balancer

The existing environment consumes VKS using NSX-based overlay networking, where Kubernetes workloads are integrated with NSX overlay segments and leverage the NSX Load Balancer for ingress and service exposure.

Transition Path to Target State:

The target architecture upgrades the existing environment to a **VMware Cloud Foundation 9 or later full-stack deployment model**.

In this scenario, because the environment is already based on NSX overlay networking, the supported and recommended path is to transition to a full VCF stack rather than a phased or incremental modernization approach as described above in the VKS consumption adoption journey section.

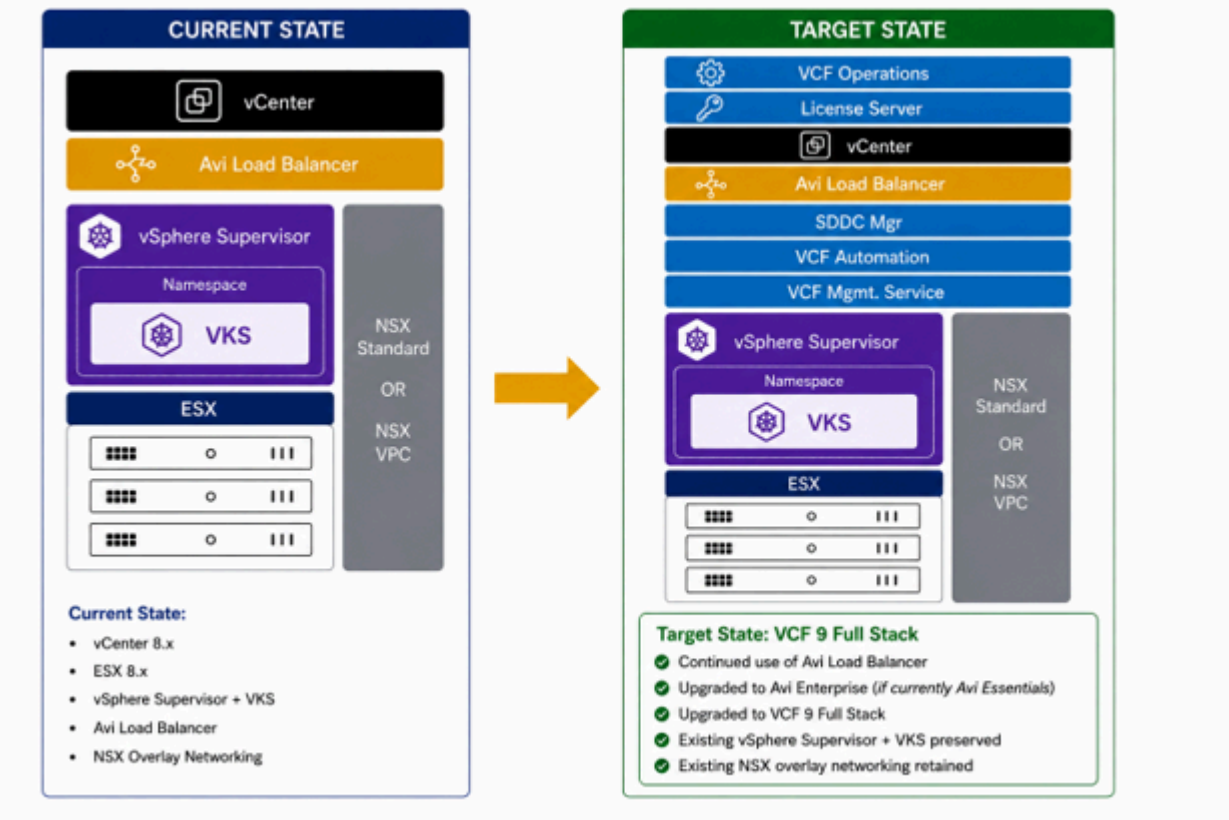
This requires converging or upgrading the existing environment using the VCF Installer and following the official VCF upgrade process.

High Level Upgrade Workflow:

- Follow the official product upgrade guide and plan the transition to VCF 9.x full-stack and validate compatibility of all components, including NSX, vCenter, ESX, Supervisor, and VKS.
- Use the VCF Installer to **converge or upgrade** the existing environment into a VCF 9.x deployment.
- Upgrade vCenter and ESX as part of of the VCF Upgrade workflow.
- Upgrade NSX components (NSX Manager and Edge nodes and Load Balancer Services) as part of the VCF upgrade workflow.
- Supervisor upgrades are done as a separate step and need to be done after the brownfield import.
- Existing Supervisor and VKS workloads are preserved but it is still recommended to keep a workload backup before you plan for the transition.
- Existing NSX Overlay network and NSX Load Balancer retained.

> **Important:**
>
> After upgrading to VMware Cloud Foundation 9.0 or later, you can choose to consume **VMware vSphere Kubernetes Service** using the VCF Automation (All Apps) model.
>
> However, this requires deploying a **new vSphere cluster** with **vSphere Supervisor** enabled using **VPC-based networking**.
>
> This can be achieved by either:
>
> - adding a new cluster within the existing vCenter, or
> - deploying a new workload domain with a dedicated vCenter and NSX instance
>
> Existing clusters and workloads remain unchanged and cannot be directly converted to the VPC-based All Apps consumption model. Please refer **Scenario 5** for more information.

Scenario 3: vSphere 8.x + VKS + NSX Networking + Avi Load Balancer



Current State

This brownfield scenario represents an existing **VMware vSphere** 8.x environment operating:

- vSphere 8.x
- vSphere Supervisor
- VMware vSphere Kubernetes Service
- NSX overlay networking
- **VMware Avi Load Balancer** (Avi Essentials or Enterprise)

The environment is already tightly integrated with NSX for networking and uses Avi Load Balancer for Kubernetes ingress and service exposure. Kubernetes workloads are deployed on NSX overlay segments and rely on NSX and Avi for networking, routing, and load balancing services.

This architecture reflects a modern NSX-based deployment model that is aligned with cloud-like networking constructs and is well-positioned for transition into a full VCF stack.

Transition Path to Target State

The target architecture upgrades the existing environment to a **VMware Cloud Foundation 9 or later** full-stack deployment model.

In this scenario, because the environment is already based on NSX overlay networking, the supported and recommended path is to transition to a **full VCF stack**, rather than a phased or incremental modernization approach as described earlier in the VKS consumption adoption journey.

This requires converging or upgrading the existing environment using the VCF Installer and following the official VCF upgrade process.

During this transition:

- Existing vSphere Supervisor and VKS workloads are **preserved**
- NSX overlay networking is **retained**
- Avi Load Balancer continues to provide Kubernetes ingress and service exposure
- Platform services such as VCF Operations and License Server are introduced as part of the VCF stack

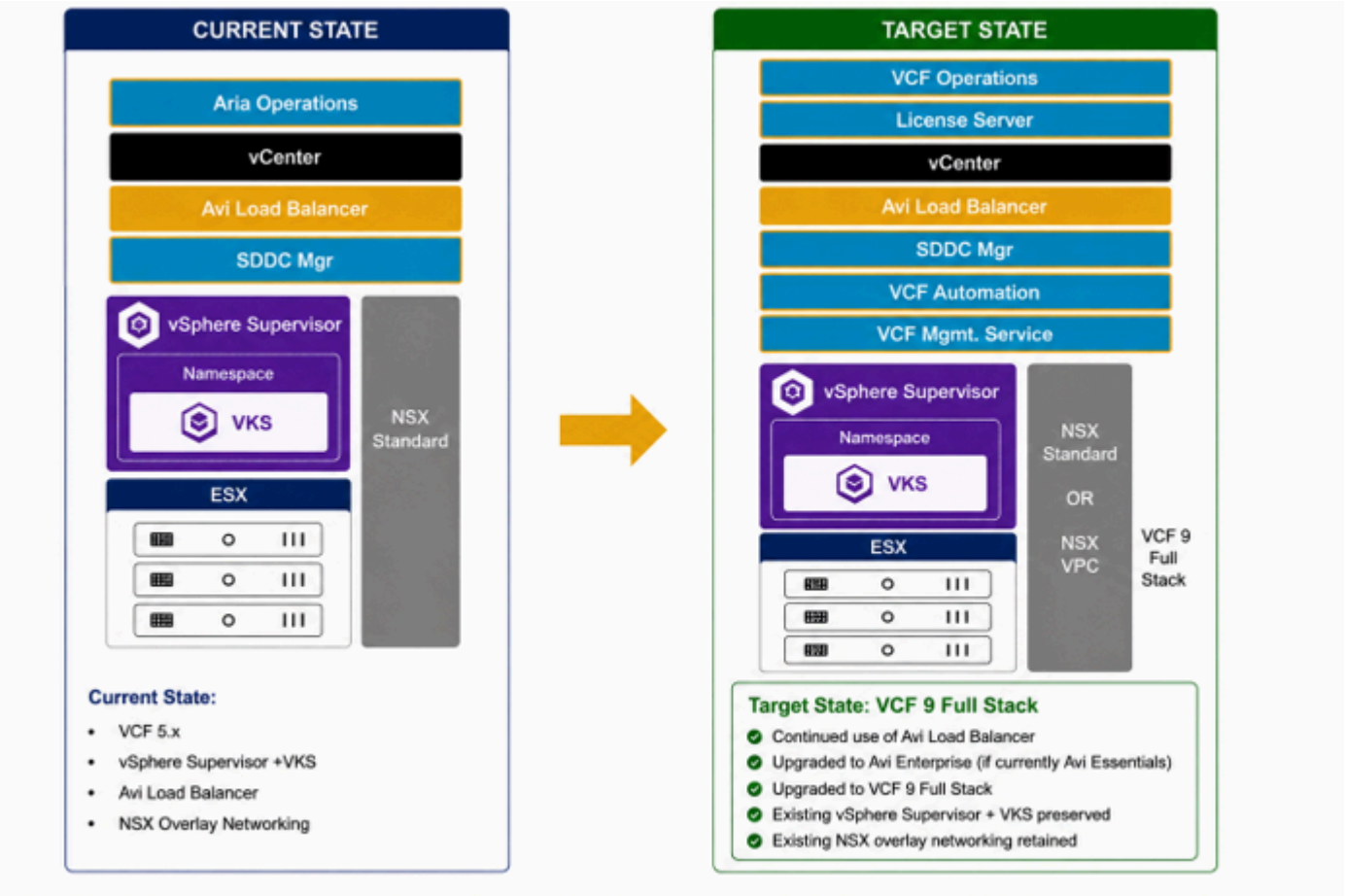
High-Level Upgrade Flow

Plan and validate compatibility of all components, including vCenter, ESX, NSX, **VMware Avi Load Balancer**, vSphere Supervisor, and **VMware vSphere Kubernetes Service**.

> **Note:**
>
> **If Avi Essentials is deployed, upgrade to Avi Enterprise prior to any platform upgrade.**

- Upgrade the Avi Load Balancer (Controller and Service Engines) to a version compatible with the target VCF release.
- Validate Avi integration with vSphere Supervisor and ensure Kubernetes ingress and service exposure are functioning correctly.
- Prepare the environment for **VMware Cloud Foundation** conversion, including networking, DNS, NTP, and resource requirements.
- Use the VCF Installer to **converge or upgrade** the existing vSphere environment into a VCF 9.x deployment.
- Upgrade NSX components (NSX Manager and Edge nodes) as part of the VCF upgrade workflow.
- Upgrade vCenter and ESX hosts to versions aligned with the VCF release.
- Validate NSX overlay networking, transport nodes, and connectivity.
- Validate vSphere Supervisor and VKS cluster health post-upgrade.
- Validate Kubernetes workloads, ingress connectivity, and application functionality.

Scenarios 4: VCF 5.x + VKS + NSX Networking + Avi Enterprise



Current State

This brownfield scenario represents an existing **VMware Cloud Foundation** 5.x environment operating:

- VCF 5.x platform (SDDC Manager, vCenter, NSX)
- vSphere Supervisor
- VMware vSphere Kubernetes Service
- NSX overlay networking
- **VMware Avi Load Balancer** (Essentials or Enterprise)

The environment is already deployed as a full VCF stack with integrated lifecycle management through SDDC Manager. Kubernetes workloads run on NSX-backed networking and leverage Avi Load Balancer for ingress and service exposure.

Transition Path to Target State

The target architecture upgrades the environment from VCF 5.x to **VCF 9.x (or later)** using the standard VCF Upgrade process.

Since the environment is already operating on a full VCF stack:

- Existing VKS clusters and workloads are **preserved**
- NSX overlay networking is **retained**
- Avi Load Balancer continues to provide Kubernetes ingress and service exposure

However, to align with VCF 9.x lifecycle and platform requirements:

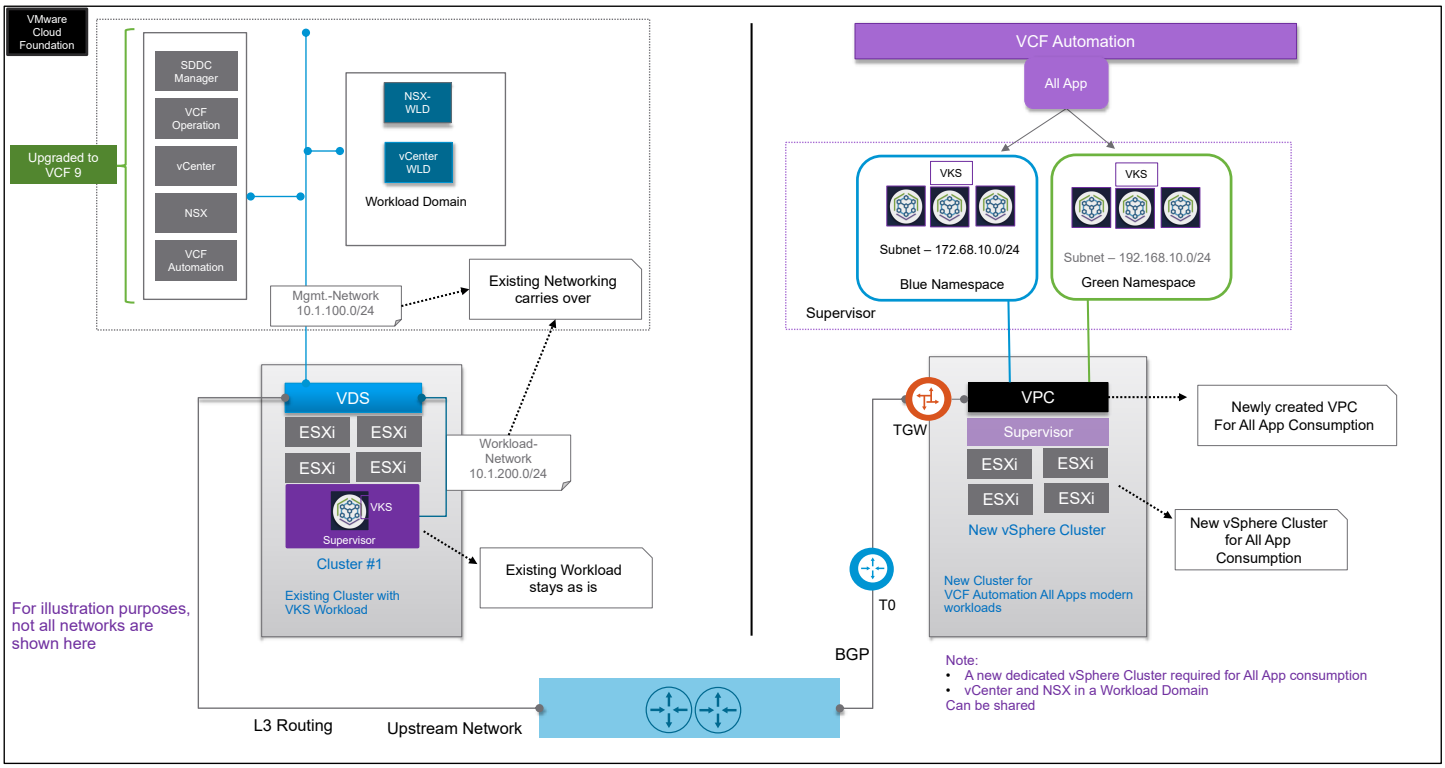
- **Avi Essentials must be upgraded to Avi Enterprise** (if applicable)

High-Level Upgrade Flow:

- Plan and validate compatibility of all VCF components against the target VCF 9.x Bill of Materials (BOM).

- **If Avi Essentials is deployed, upgrade to Avi Enterprise as the first step.**
- Upgrade the **VMware Avi Load Balancer** Controller and Service Engines to a version compatible with the target VCF release.
- Validate Avi integration with vSphere Supervisor and ensure Kubernetes ingress and service exposure are functioning correctly.
- Follow the VCF upgrade sequencing defined in the upgrade guide
- Validate NSX overlay networking, transport nodes, and connectivity.
- Validate vSphere Supervisor and VKS cluster health post-upgrade.
- Validate Kubernetes workloads, ingress connectivity, and application functionality.

Scenario 5: Brownfield VKS Transition to VCF 9 with VCF Automation (All Apps)



The target architecture upgrades the environment to **VMware Cloud Foundation 9** and introduces **multi-tenant VKS consumption using VCF Automation (All Apps)** while preserving existing workloads.

In this scenario:

- Existing VKS clusters and workloads **remain unchanged**
- Existing networking architecture **carries forward**
- No replatforming or migration of existing workloads is required

Multi-tenancy is introduced by deploying **new infrastructure alongside the existing environment**, rather than modifying the existing cluster.

Recommended Approach:

After upgrading or converging the environment into VCF 9:

Step 1 – Preserve Existing Environment

- Existing vSphere cluster continues to run current VKS workloads
- No changes to Supervisor or networking are required
- Existing workloads remain operational (“stay as-is”)

Step 2 – Introduce Multi-Tenant VKS (All Apps)

Customers can choose one of the following deployment models:

Option A – Same Workload Domain (Side-by-Side Cluster)

- Deploy a **new vSphere cluster** in the existing vCenter
- Enable **vSphere Supervisor with VPC networking**
- Integrate with **VCF Automation (All Apps)**

Use this cluster for:

- Multi-tenant VKS consumption
- New application workloads

Option B – New Workload Domain (Dedicated vCenter + NSX)

Deploy a **new workload domain** with:

- A dedicated vCenter
- A dedicated NSX instance
- Deploy a **new vSphere cluster** in this domain
- Enable **Supervisor with VPC networking**
- Integrate with **VCF Automation (All Apps)**

> **Important:**
>
> Scenario 5 represents a side-by-side consumption pattern for brownfield VKS environments and VCF Automation All App model. This approach enables customers to deploy and consume new VKS clusters through the VCF Automation All Apps consumption model while existing brownfield VKS environments continue to operate separately. It does not provide an in-place migration path for existing VKS clusters into the VCF Automation All Apps model. Migration support for brownfield VKS environments into VCF Automation All Apps is planned for a future release.

Post-Deployment Operational Enablement

After selecting a VKS consumption pattern and successfully deploying the VMware vSphere Kubernetes Service platform, organizations can enable additional application delivery and platform operations capabilities.

These capabilities are independent of the chosen consumption model (vCenter-based or VCF Automation-based) and can be layered on top of the deployed Supervisor and downstream Kubernetes workload clusters.

The following table outlines common Day 2 operational services and integration runbooks.

Capability	Description	Reference
GitOps / Continuous Delivery (Argo CD)	Declarative application deployment and lifecycle management using Git as the source of truth, integrated with Supervisor	Continuous Integration and Delivery with the Argo CD Supervisor Service
CI/CD Integration (Harness)	Enterprise CI/CD platform integration for automated build, test, and deployment workflows integrated with Kubernetes workloads	Continuous Integration and Delivery with Harness
VCF Edge: Single Node with ArgoCD Design Blueprint	VMware Cloud Foundation Edge (VCFE) extends the centralized management capabilities of the platform to geographically dispersed edge locations. Common edge use cases that are supported by VCF Edge are remote office / branches, manufacturing factories, retail stores, or mission-critical battle vehicles for the Tactical edge.	VCF Edge: Single Node with ArgoCD Design Blueprint
Other vSphere Supervisor Services	Additional Supervisor-integrated services that extend Kubernetes platform capabilities for application delivery, operations, and ecosystem integrations	Using vSphere Supervisor Services

Private AI Services Consumption Blueprint

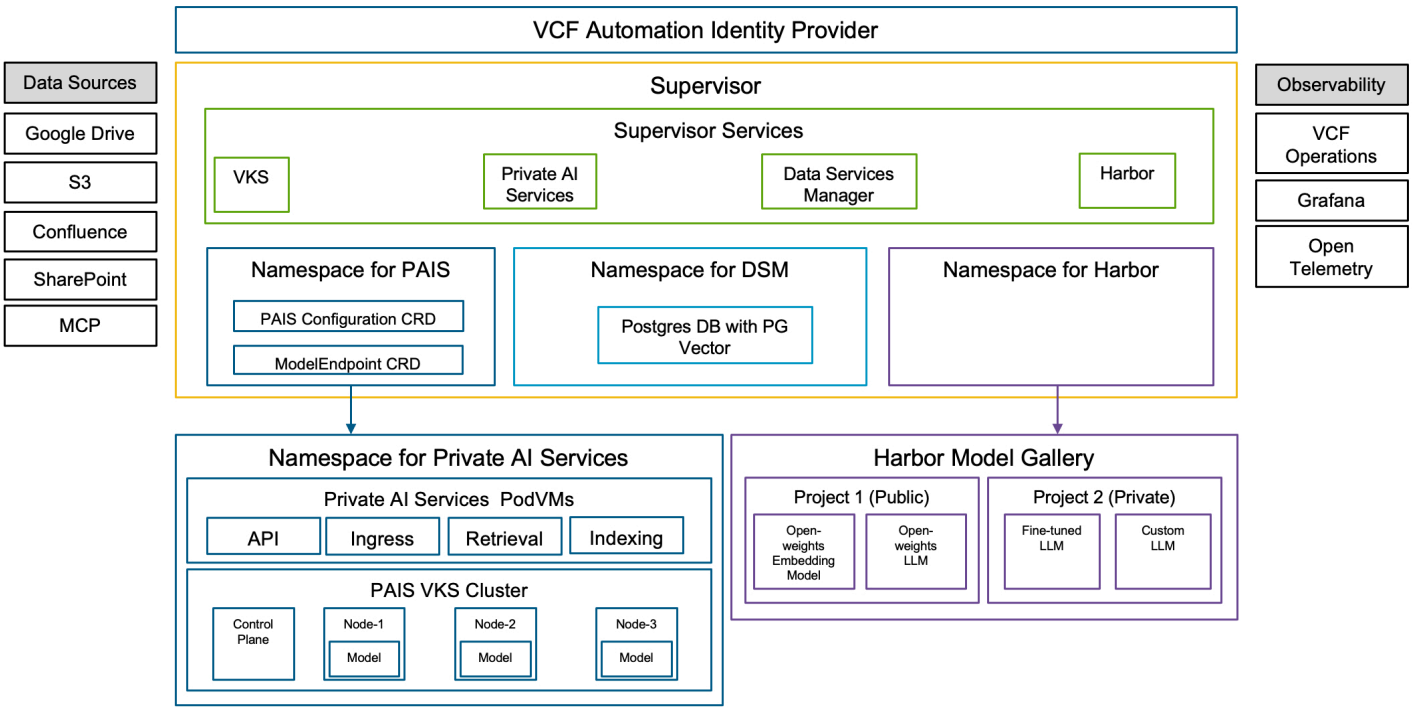
The *Private AI Services Consumption Blueprint* describes the design selections and architectural components required to deploy and operate a Generative AI platform on VMware Cloud Foundation. It is intended to be used in conjunction with an underlying infrastructure blueprint.

Blueprint Objectives

This blueprint is designed to address a specific set of business outcomes enabled by the Private AI platform running on VMware Cloud Foundation. Each objective maps a concrete business driver to the platform capability that satisfies it, giving stakeholders a clear view of the value delivered. Use this table to validate that the blueprint scope aligns with your organization's AI strategy before proceeding with this design selection.

Business Objective	Capability Enabled
Accelerate Generative AI application development	Self-service deployment of model endpoints and agents via GUI/API, minimal Kubernetes, or VM expertise required
Enable private, sovereign AI inference	On-premises LLM inference via OpenAI-compatible APIs, data never leaves the datacenter boundary
Ground AI responses in enterprise knowledge	Automated RAG pipeline: ingest, chunk, embed, and retrieve from corporate data sources (Google Drive, Confluence, SharePoint, S3, MCP servers)
Deliver enterprise conversational AI	Stateful agents with session management, automated RAG, and MCP tool integration for acting on live systems
Reduce MLOps operational complexity	Platform abstracts VM, Kubernetes, and microservice management. Engineers operate at the knowledge base and model endpoint level

Figure 212: Private AI Services Consumption Blueprint



Private AI Services Consumption Blueprint Components

The table below presents the function of the main components part of the *Private AI Services Consumption Blueprint*.

Component	Function
VCF Automation Identity Provider	Authenticates the Organization Admin and ML/AI Platform Admin personas to the VCF Automation portal.
Private AI Services Identity Provider	Authenticates the GenAI Developer persona to the Private AI Services UI and API.
Harbor Model Gallery	Central registry for AI model artifacts, organized into public projects (open-weights models) and private projects (fine-tuned and custom models).
Open-weights LLM	Completion model served by the Model Runtime; generates text responses via an OpenAI-compatible API endpoint.
Open-weights Embedding Model	Embedding model served by the Model Runtime; produces vector representations used for semantic search operations.
Fine-tuned LLM	Completion model variant stored in a private Harbor project; consumed by the Model Runtime.
Custom LLM	Customer-provided completion model stored in a private Harbor project; consumed by the Model Runtime.
Postgres DB with PG Vector	External PostgreSQL database with the pgvector extension; stores and retrieves vector embeddings produced by the embedding model.
Data Sources (Google Drive, S3, Confluence, SharePoint, MCP)	External content repositories indexed by the Data Indexing and Retrieval module; documents are split into chunks and embedded as vectors.
Private AI Services PodVMs (API, Ingress, Retrieval, Indexing)	Runtime components of the Private AI Services application; collectively implement model serving, data retrieval, and agent orchestration.
VCF Operations	Provides infrastructure-level visibility for VI admins; correlates Private AI Services workload demands with cluster capacity and GPU resource utilization across namespaces.

Component	Function
Grafana	Visualizes Prometheus metrics through pre-built dashboards that display GPU hardware metrics alongside vLLM inference performance and pod health.
OpenTelemetry	Collects OpenTelemetry-compliant LLM trace data,including request flows, prompt/completion pairs, and token-level timing, for consumption by an external observability platform such as Phoenix or Langfuse.

Personas

Successful deployment and operation of the Private AI platform spans multiple organizational roles, each with distinct responsibilities across the infrastructure and application layers. This table defines the personas involved, from the Cloud Admin who provisions the underlying VCF fleet to the GenAI Developer who consumes model endpoints and builds AI-powered applications. Understanding these role boundaries is essential for planning access control, onboarding sequences, and operational handoffs. In some organizations, the role of multiple personas is fulfilled by the same individual.

Persona	Responsibilities
VI Admin	• Deploys and operates the VCF fleet, VCF instances, and workload domains • Set up virtual infrastructure and accelerators • Create the VCF Automation Organization and assign resource quotas to it (It should include GPU VM Classes)
Organization Admin	• Create a VCFA Project and associate a Namespace with GPU and non-GPU VM Classes with it • Assign the ML/AI Platform Admin user to the VCF Automation Project
ML/AI Platform Admin	• Deploy PostgreSQL Database via VCF Automation Data Services • Collect OIDC Identity Provider Details for Private AI Services • Enable Private AI Services • Integrate Private AI Services with observability tools • Deploy Completion and Embeddings Models
GenAI Developer	• Configures Private AI Services agents with session management and MCP tool connections. • Builds LLM-powered applications consuming Private AI Services endpoints and agents

Infrastructure Dependencies

The Private AI Services blueprint is not a standalone design; it builds on an existing VCF infrastructure blueprint and inherits its design attributes and its storage, compute, and connectivity models. This table identifies the underlying infrastructure components and the specific models that are compatible with the Private AI Services blueprint. These requirements ensure that prerequisite design selections are in place before Private AI Services are enabled.

Underlying Infrastructure	Compatible Models
Infrastructure Modernization Blueprint	• VCF Fleet in a Single Site with Minimal Footprint • VCF Fleet in a Single Site
Network Consumption	• VCF Automation All Apps Orgs Consumption Model
Workload Connectivity	• Centralized Connectivity Model • Distributed VLAN Connectivity Model
vSphere Supervisor	• Single Management Zone with Combined Workload Zone Model
vSphere Cluster	• Single-Rack vSphere Cluster Model

Blueprint layering

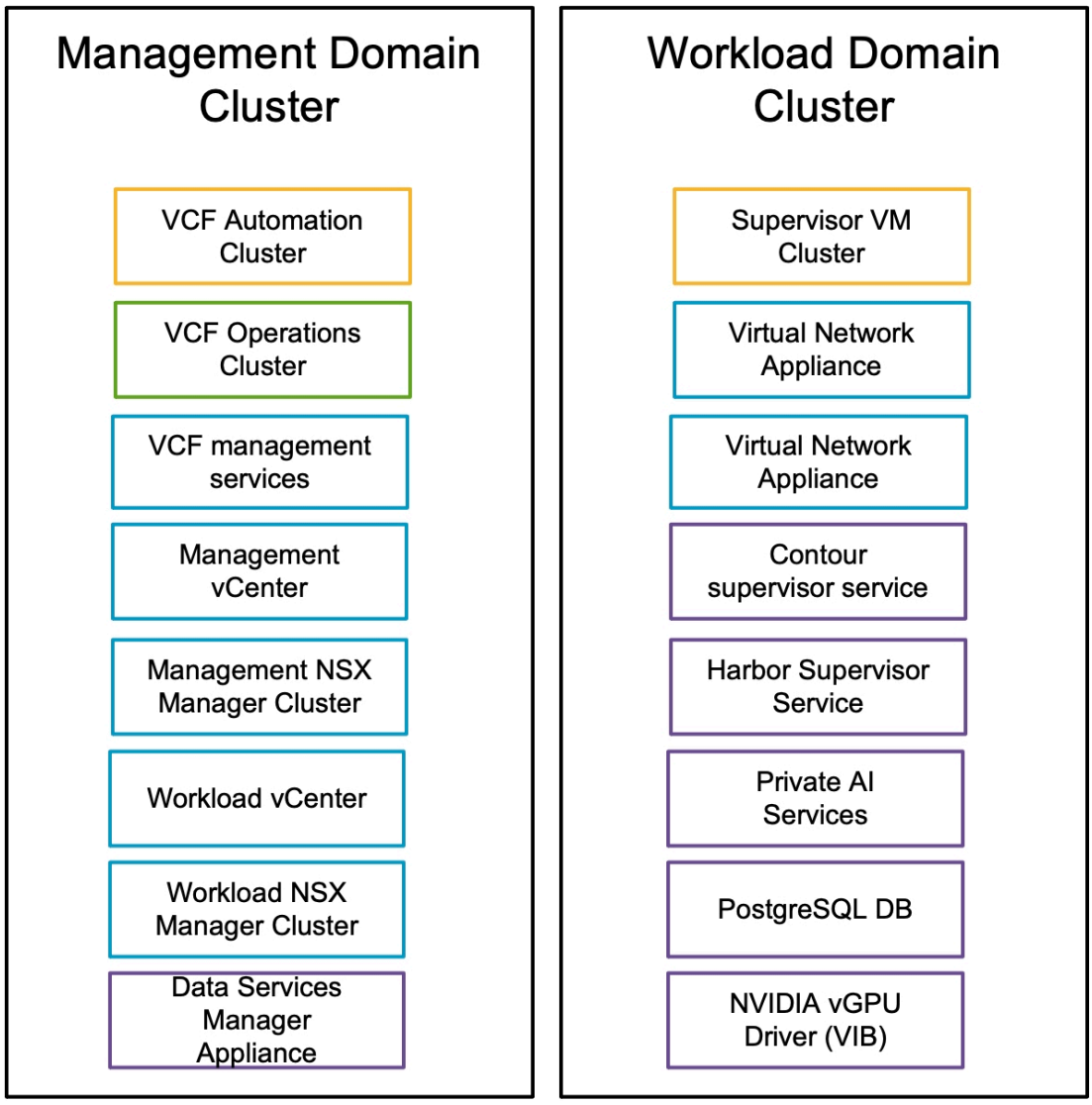
The following diagram illustrates how the Private AI Services blueprint layers additional components on top of an existing infrastructure blueprint. In this case, the VCF Fleet in a Single Site blueprint is presented. In the management domain, the Cloud Admin deploys the Data Services Manager appliance alongside the existing Fleet and VCF Instance management components.

The workload domain cluster already hosts Supervisor cluster VMs. Depending on the selected workload connectivity model, it also hosts either NSX Virtual Network Appliances (VNAs) for the Distributed VLAN model, or NSX Edge nodes for the Centralized Connectivity model.

Within the workload domain cluster, the Cloud Admin adds the NVIDIA vGPU driver (VIBs) to the cluster vLCM image. The Cloud Admin then enables the following Supervisor Services: Harbor, Contour, Data Services Manager, and Private AI Services.

The Private AI Services application and its PostgreSQL database are then deployed trough VCF Automation by the ML/AI Platform Admin. The components added by the Private AI Services blueprint are highlighted in purple.

Figure 213: Private AI Services Blueprint on top of VCF Fleet in a Single Site infrastructure blueprint



Private AI Services Design Selection

The following design selections define the architectural decisions made at each layer of the Private AI platform to satisfy the objectives and design profile described above. Each selection represents the recommended model for this blueprint, along with the rationale and any alternative models that remain compatible with the overall design. Deviating from these selections may require revisiting the infrastructure dependencies and operational considerations described in the implementation steps and design elements sections of the document.

Layer	Selected Model	Additional Information
Private AI Platform	Private AI Services Model	Fastest path to operational GenAI endpoints. Additional Platform models (Deep Learning VMs or AI VKS Clusters) can run in the same VCF Automation Namespace if GPU resources are available.
Vector Store	Data Services Manager	DSM is a native VCF service providing DBaaS lifecycle management. Tenant self-service provisioning is available directly via VCF Automation without additional tooling.
Private AI Compute	vGPU Compute Model and CPU-Only Model (Optional)	The vGPU compute model provides the highest flexibility, from fractional GPUs to the ability to pool the resources from multiple GPUs to run a large LLM. As part of this design, the CPU-only model can be used to run Embedding Models when GPU resources are unavailable. Supplemental guidance is provided in Accelerator Design Based on AI/ML Workloads Requirements

Implementation of the Private AI Services Consumption Blueprint

To implement the *Private AI Services Consumption Blueprint* follow the procedures in the table below.

Implementation procedure

Procedure Grouping	Procedure	Persona
Virtual Infrastructure Preparation	Add NVIDIA VIBs to the cluster vLCM image and remediate the cluster	VI Admin
	Configure NVIDIA vGPU on the ESX hosts	VI Admin
	Configure VM Classes with vGPUs	VI Admin
	Configure a License Server instance on the NVIDIA Licensing Portal and generate a client configuration token.	VI Admin
	Deploy VMware Data Services Manager	VI Admin
Supervisor Services	Install the Data Services Manager Consumption Operator as a supervisor service on the cluster where Private AI Services will be enabled	VI Admin
	Install Harbor as a supervisor service on the cluster where Private AI Services will be enabled (Use NGINX Option)	VI Admin
	Install Private AI Services as a supervisor service on the cluster where Private AI Services will be enabled	VI Admin
VCF Automation	Configure Data Services in VCF Automation	VI Admin
	Set up VCF Automation for Private AI Services	VI Admin and Organization Admin

Procedure Grouping	Procedure	Persona
	Create a Data Service Policy and assign it to the organization where Private AI Services will be deployed	VI Admin
	Create a PostgreSQL Database Using Data Services	ML/AI Platform Admin
	ActivatePrivate AI Services in an Organization Namespace by Using the VCF Automation UI	ML/AI Platform Admin
Private AI Services	Set up Observability for Private AI Services	ML/AI Platform Admin
	Delivering Generative AI Applications by using Private AI Services (Operational procedures)	ML/AI Platform Admin and GenAI Developer

Design Elements for the Private AI Services Consumption Blueprint

Table 459: AI/ML Platform based on Private AI Services Design Recommendations

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-001	Enable the Private AI Services in a namespace with access to a single vSphere zone	While it is supported to deploy the Private AI Services in a multi-zone namespace, critical components including the VKS cluster control plane are deployed as single instances rather than three instances across supervisor zones, preventing cross-zone fault tolerance. Replicas of the same model endpoint must be located in the same zone.	Private AI Services infrastructure lacks zone-level high availability. A zone failure causes a complete PAIS service outage, including loss of model inference endpoints and control plane functionality. Organizations must implement alternative resilience strategies such as multiple independent single zone deployments.
PAIF-PL-RCMD-PAIS-002	Deploy Private AI Services in a namespace with at least 5 GHz of CPU reservation	Default Private AI Services components are defined with minimal CPU reservations. They will not start in a namespace without enough reserved resources.	Reservation may need to be increased depending on the additional workloads running in the namespace
PAIF-PL-RCMD-PAIS-003	Deploy Private AI Services in a namespace with memory reservation that accounts for GPU memory allocation	vSphere reserves the full system memory of VMs that have GPU devices that are attached, ensuring guaranteed memory availability for GPU operations. When configuring VM Classes that include GPUs, the VM's system memory should be set to around 2X the GPU memory to account for the GPU memory allocation plus operating system, vLLM runtime, model loading buffers, and KV cache overhead. For example, a VM with an A100 80GB GPU should be configured with at least 160 GB of system RAM.	This requires advanced capacity planning around the number of model endpoints and their replicas to be hosted in the namespace. Calculate the total namespace memory reservation as: (number of GPU VMs) × (2 × GPU memory per VM). For instance, hosting 3 replicas of models on A100 80-GB GPUs requires planning for at least 480 GB (3 × 160 GB) of namespace memory reservation for GPU workloads alone.
PAIF-PL-RCMD-PAIS-004	Deploy the Harbor Model Registry as a supervisor service.	Harbor provides an OCI compatible registry where an approved model can be stored and consumed. Enabling Harbor as a supervisor service provides the most integrated experience.	The Harbor Supervisor Service is not enabled by default and must be manually enabled.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-005	Expose the Harbor Model Registry through the NGINX load balancer service by setting **enableNginxLoadBalancer: true** and **enableContourHttpProxy: false** in the Harbor data values file.	Removes the Contour Supervisor Service from the Private AI Services stack entirely, eliminating one deployment step, and one service to lifecycle-manage. The design already requires manual edits to the same data values file to size the registry volume, so no additional operational step is introduced.	Harbor consumes one Virtual IP from the load balancer IP pool for the lifetime of the service. Harbor cannot share a VIP with other Ingress-based services on the Supervisor. Harbor Supervisor Service **2.11.2 or later** must be used, as earlier versions do not support the NGINX load balancer exposure method
PAIF-PL-RCMD-PAIS-006	Increase the size of the Harbor registry volume before enabling the service.	The registry volume stores all OCI artifacts including model weight files, which for large LLMs can be 50-150GB per model, with multiple versions and formats multiplying storage needs. A collection of 10-20 LLMs with their configurations and multiple versions can easily consume 2-5TB of storage**.**	Manual modification of the data YAML file is required when enabling the harbor supervisor service. The same edit applies the registry volume sizing and the NGINX load balancer flags so both changes are made in a single operation.
PAIF-PL-RCMD-PAIS-007	Replicate the Harbor registry content to a secondary Harbor registry in separate infrastructure.	Provides recoverability for the stored models.	Requires deploying an additional Harbor registry instance with sufficient storage to mirror all OCI artifacts. Harbor replication policies must be configured and monitored.
PAIF-PL-RCMD-PAIS-008	Configure VCF Automation as the authorization provider for Private AI Services when VCF Automation is part of the deployment.	VCF Automation federates each tenant's upstream identity provider and issues the tokens Private AI Services trusts, so the deployment integrates with VCF Automation rather than separately with whatever identity solution each customer or organization operates	Requires VCF Automatin to be be part of the design The permitted organizations must be declared on the Private AI Services instance and the access role assigned within VCFA, so entitlement is managed in two places.
PAIF-PL-RCMD-PAIS-009	Use VCF Automation API tokens for non-interactive access to the Private AI Services API, rather than implementing an OIDC flow in client tooling.	Long-lived VCFA API tokens work directly as bearer credentials against the Private AI Services API, allowing scripts, CI/CD pipelines, OpenAI SDK clients, and cross-instance model consumption to authenticate without an interactive login or a token refresh implementation.	Long-lived credentials must be stored securely by the consuming client and rotated according to the organization's credential policy.
PAIF-PL-RCMD-PAIS-010	Do not use the local authentication provider for production deployments.	The provider is explicitly scoped to lab, demonstration, and proof-of-concept use. It has no tenancy model and no audit logging for issued tokens.	Deployments that begin with local authentication require a migration to VCF Automation or a customer identity provider before production use, including reissuing any personal access tokens held by non-interactive clients.
PAIF-PL-RCMD-PAIS-011	Integrate the Private AI Services deployment with a Postgres Database deployed via Data Service Manager (DSM).	Data Service Manager provides a simplified way to deploy a compatible vector database.	The DSM appliance must be deployed and integrated with vCenter.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-012	Deploy the Postgres database via VCF Automation rather than via the Data Services Manager console.	• The Private AI Services enablement workflow in the VCF Automation UI will automatically discover databases that are deployed via VCF Automation. • Connection details for other compatible databases will have to be provided manually.	• VCF Automation must be part of the design • The Data Services Manager supervisor service must be enabled the supervisor
PAIF-PL-RCMD-PAIS-013	Size the database disk to 5-10X the size of the source documents that will be indexed into knowledge bases.	PostgreSQL with pgvector stores not only vector embeddings but also the original text chunks, metadata, full-text search indexes, and database overhead.	The namespace class storage limit must be sized accordingly to accommodate the storage requirements of the database instance. For example, if planning to index 500 GB of source documents, provision 2.5-5TB of database storage. DSM database is deployed in a different namespace than Private AI Services.
PAIF-PL-RCMD-PAIS-014	Enable Prometheus metric collection in the PAISConfiguration custom resource. The DCGM exporter and vLLM metric endpoints are automatically deployed and maintained.	Prometheus collects critical metrics across the Private AI Services infrastructure. They provides complete visibility into GPU health, inference performance, and pod status. These metrics include GPU hardware telemetry (temperature, utilization, power, memory, clock speeds, and interconnect bandwidth), vLLM inference performance data (request latencies, token throughput, queue depths, and cache utilization), and Kubernetes pod health indicator.	Enabling Prometheus adds minimal computational overhead from metric scraping and requires storage for time-series data.
PAIF-PL-RCMD-PAIS-015	Use the pre-built Private AI Services Grafana Dashboards to visualize Prometheus metrics.	Grafana transforms raw Prometheus metrics into interactive dashboards that make infrastructure state immediately visible and actionable. The pre-configured dashboards display GPU hardware metrics alongside vLLM inference performance and pod health, enabling operators to correlate issues across layers (such as identifying that high GPU memory usage is causing increased inference latency).	Grafana requires a separate deployment (containerized or standalone), ongoing version management, and user access control configuration. Teams need basic training on dashboard navigation and interpretation.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-016	Enable LLM traces in the PAISConfiguration custom resource.	Enabling LLM traces in the PAISConfiguration custom resource sends OpenTelemetry-compliant trace data to observability tools like Phoenix or Langfuse, providing application-level visibility that complements infrastructure metrics. While Prometheus/Grafana show system-level performance, traces capture the application developer's view: individual request flows, prompt/completion pairs, token-level timing, model parameters used, and semantic context of each inference call. This enables developers to debug specific user requests, and provides insights that infrastructure metrics alone cannot provide.	Traces contain sensitive application data, requiring careful consideration of where trace data is stored. Organizations must choose between SaaS observability platforms (easier setup, managed infrastructure, but data leaves the premises) or self-hosted deployments (full data control and confidentiality, but requires deploying and managing Phoenix/Langfuse infrastructure on-prem). Trace sampling rates must be configured to balance observability coverage with storage and processing costs, and access controls must restrict trace visibility to authorized developers only.
PAIF-PL-RCMD-PAIS-017	Visualize LLM traces in an observability platform that is dedicated to the Private AI Services instance.	Deploying a self-hosted observability platform dedicated exclusively to the Private AI Services ensures that sensitive trace data containing prompts, completions, and model interactions never leaves the on-premises environment, maintaining complete data sovereignty and compliance with privacy requirements. A dedicated deployment simplifies role-based access control (RBAC) by eliminating the complexity of multi-tenant isolation.	A dedicated observability platform requires provisioning separate resources exclusively for tracing data, increasing infrastructure footprint and operational overhead.
PAIF-PL-RCMD-PAIS-018	Configure the Telegraf supervisor service to scrape Private AI Services controller metrics, enabling visualization in VMware Cloud Foundation (VCF) Operations.	While the Private AI Services Grafana dashboards serve application developers focused on inference performance and model behavior, VCF Operations serves VI admins who must correlate Private AI Services workload demands with cluster capacity, identify GPU resource contention across namespaces, and plan infrastructure scaling	Configuring Telegraf to scrape Private AI Services controller metrics requires manual setup.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-019	Deploy MCP (Model Context Protocol) servers as POD VMs in the same namespace where the PAIS service is enabled.	Deploying MCP servers as Pod VMs within the same vSphere namespace where PAIS is enabled creates a self-contained system where all components share network locality and administrative boundaries. MCP servers can be exposed via ClusterIP services that remain local to the namespace without consuming routable IP addresses from the broader network infrastructure. Co-locating MCP servers with Private AI Services in the same namespace consolidates the entire LLM application stack into a single namespace, enabling users with namespace-level RBAC permissions to manage both PAIS model endpoints and their associated MCP servers through a unified access control model.	Pod VMs consume compute and memory resources from the same vSphere namespace quota as Private AI Services workloads, requiring careful capacity planning to avoid resource contention between model inference and MCP server operations.
PAIF-PL-RCMD-PAIS-020	Instantiate the Private AI Services VKS control plane as a VMClass XLarge.	It ensures that all Private AI services that are part of this design run smoothly with sufficient compute and memory resources.	VMClass XLarge consumes 4 vCPUs and 32 GB of memory, requiring capacity planning at the namespace level to accommodate both the control plane and workload resources.
PAIF-PL-RCMD-PAIS-021	Deploy Model endpoints with replica = 2 or higher.	Configuring model endpoints with at least 2 replicas ensures high availability for inference workloads by distributing pods across separate VKS worker nodes. This architecture provides redundancy that is critical for those GPU-accelerated models that cannot leverage vSphere's vMotion capabilities. Models running GPUs in passthrough mode or vGPU requiring Unified Virtual Memory (UVM) must be shut down for host maintenance or failure scenarios, causing service interruptions for single-replica deployments. With 2+ replicas, the Private AI Services inference gateway automatically routes requests to healthy replicas while a worker node undergoes maintenance or recovers from failure, maintaining continuous inference service availability.	Deploying multiple replicas multiplies the resource consumption per model endpoint. This significantly impacts namespace capacity planning and increases infrastructure costs proportionally to the replica count.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-022	Deploy a dedicated Private AI Services instance for each team or tenant that requires a distinct set of models or a distinct data boundary.	Access to a Private AI Services instance grants access to everything configured in it: model endpoints, agents, knowledge bases, and MCP tools Access cannot be scoped to a subset of those resources. The enforcement boundary is therefore the instance itself. Consuming models from a shared serving instance makes this practical, as each team's instance can draw on a common pool of GPU-backed models rather than requiring its own GPU capacity.	Each instance requires its own namespace, authentication configuration, agents, knowledge bases, indexes, and vector database. Content and tools needed by more than one team must be configured in each instance.
PAIF-PL-RCMD-PAIS-023	When deploying a Private AI Services instance dedicated to shared model serving, do not co-locate tenant application workloads in the same namespace.	The serving instance determines inference performance for every consumer, and application workloads compete for the same namespace resources as inference. Separation keeps maintenance windows independent.	Requires an additional namespace and instance operated by the provider or platform team.
PAIF-PL-RCMD-PAIS-024	Size model endpoint replicas on a shared serving instance for the aggregate demand of all consuming instances.	A shared model receives the combined load of every consuming team, so replica count must be driven by aggregate concurrency rather than by availability alone.	Capacity planning requires demand estimates from each consuming team before the model is deployed, and must be revisited as teams are onboarded. Replica count multiplies GPU consumption and interacts with the memory reservation calculation
PAIF-PL-RCMD-PAIS-025	Issue a dedicated service account and API token to each user consuming Private AI Services instance.	A credential shared between consumers cannot be revoked or rotated for one of them without interrupting all of them. Distinct credentials also allow requests arriving at the serving instance to be attributed to their originating instance.	The number of credentials grows with the number of consuming instances.
PAIF-PL-RCMD-PAIS-026	Track the expiry of the ingress CA certificate of the shared serving Private AI Services instance and rotate it before it expires.	Every consuming instance trusts the same certificate, so its expiry causes remote model routes to fail simultaneously across all of them.	The certificate must be recorded in a certificate inventory and monitored. Rotation requires a coordinated trust bundle update on every consuming instance, involving each tenant administrator in a multi-tenant environment.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-027	Use strict TLS verification for inference gateway routes that point to model endpoints outside the organization.	Requests to a model endpoint carry prompt content and return completions, both of which routinely contain sensitive data. Both **strict** and **caOnly** validate the endpoint certificate chain against a CA registered in the Private AI Services trust bundle; the difference is the identity check. **strict** additionally verifies that the FQDN or IP address used in the route matches an entry in the certificate Subject Alternative Name, confirming that the endpoint presenting the certificate is the one the route intended to reach. **caOnly** omits that check and should be used only where the endpoint is legitimately reached through an address absent from the SAN. Disabling verification permits plaintext transport and should not be used outside testing.	The CA certificate of each external endpoint must be added to the trust bundle, and the certificate must include the FQDN or IP address used in the route in its SAN. Endpoints reached by an address not present in the SAN require either a reissued certificate or a documented exception using **caOnly**.
PAIF-PL-RCMD-PAIS-028	Deploy Private AI Services instances that consume only remote models in namespaces without GPU capacity	An instance with no local model endpoints does not require GPU-backed VM classes reducing the namespace reservation to what the control plane, gateway, and application components require.	The instance depends entirely on the serving instance and the network path to it for all inference. The CPU reservation floor in PAIF-PL-RCMD-PAIS-003 continues to apply.

Table 460: Data Services Manager Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-PC-DSM-REQD-001	Deploy Data Services Manager in the management domain and register the plugin with a separate workload domain vCenter instance.	• Provides clear separation between management and workload domains. • The Data Services Manager appliance itself deploys in the management domain which typically hosts core VCF components: vCenter, NSX Managers, and VCF Automation. • The Data Services Manager plugin connects to the vCenter instance managing a separate workload domain. This is where applications and DSM-provisioned databases are deployed. • Data Services Manager appliance maintains a 1:1 relationship to a vCenter instance.	Requires management domain and separate workload domain.
VCF-PC-DSM-REQD-002	Protect the Data Services Manager appliance using vSphere HA.	Ensures availability of Data Services Manager appliance without requiring manual intervention during an ESX host failure event.	None.
VCF-PC-DSM-REQD-003	Use shared storage on management domain cluster to place Data Services Manager provider VM.	Ensure that the Data Services Manager appliance can support vSphere vMotion, vSphere DRS, and vSphere HA.	Requires shared storage and enough storage capacity to satisfy the minimum requirements of the appliance.
VCF-PC-DSM-REQD-004	Allocate a single IPv4 address for the Data Services Manager management IP on the same network as core VCF management components.	Simplifies deployment insuring the same management vlan is used for all management components on VCF.	Requires the IP address to be registered with the DNS server for both forward and reverse lookup.
VCF-PC-DSM-REQD-005	Enable vSphere Supervisor on workload clusters for VCF Automation.	vSphere Supervisor enables a Kubernetes-based IaaS control plane on vSphere. Data Services Manager uses a Consumption Operator (installed as a Supervisor Service) to listen for requests from VCF Automation.	Requires vSphere Supervisor be enabled on the targeted workload cluster. For more information, see vSphere Supervisor Models.
VCF-PC-DSM-REQD-006	Use VPC networking for databases (centralized or distributed) provisioned by Data Services Manager.	With NSX VPC, each project or tenant has its own networking policies and security controls, which allows for tenants to avail of • Simplified Management: Database owners can now manage their VPC subnets. • Automate VPC deployment through VMware Cloud Foundation Automation.	• Requires enablement of vSphere Supervisor on the target workload domain. • Requires deployment of NSX Edge cluster for centralized VPC or Virtual Network Appliance cluster for distributed VPC.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-PC-DSM-REQD-007	Size VPC subnet for the workload databases for desired database topology and maintenance operations.	• While a database cluster operates within a topology, it permanently requires one IP address for each node, and two extra cluster-level IP addresses for the database and control plane access. • If a VPC subnet segment is exhausted, you can no longer create database clusters. Scaling or upgrading existing clusters might lead to database unavailability.	Requires configuring a VPC subnet for network resources within a supervisor deployment. For more information, see Workload Connectivity Models
VCF-PC-DSM-REQD-008	Configure a local Data Services Manager administrator user.	If the primary Identity Provider system is down, allows for local access and a break glass account.	• The local user must be in an email address format and stored locally on the appliance. • Dedicate a unique persona that has a valid email address that is associated with the user ID.
VCF-PC-DSM-REQD-009	Configure Active Directory or OpenLDAPS as a Directory Service for Data Services Manager.	• Data Services Manager provides role-based access for two roles. – DSM Admin - A super user that manages the platform. – DSM User creates and maintains individual databases. • This centralizes user management, enhances security through standardized, group-based permissions. **Note:** VCF Automation provisioned data services are managed by VCF Automation independently.	• Data Services Manager requires Lightweight Directory Access Protocol over TLS (LDAPS) as an external identity source. • The servers certificate must be self-signed, or signed by a Certificate Authority (CA).

Design Requirement ID	Design Requirement	Justification	Implication
VCF-PC-DSM-REQD-010	Configure a S3-compatible object store with TLS enabled.	• Data Services Manager's backup architecture requires S3-compatible object storage—external to vSphere datastores. • S3 provider repositories are used for core VMware Data Services Manager storage, backup, logs, and database backups • Create multiple database backup locations to segregate data. • Data Services Manager requires at a minimum — A bucket for Provider Logs. — A bucket for Provider Backups. — Database backup storage. • Use policies to control which bucket each database uses. This provides granular control over data protection and compliance requirements.	• The S3 compatible storage should be external to VCF infrastructure where Data Services Manager appliances and workload are deployed to. • S3 endpoints must be resolvable by the Data Services Manager management IP • For the Provider VM to be able to access S3-compatible storage, you must configure Identity and Access Management (IAM) policies at the S3 storage provider end.
VCF-PC-DSM-REQD-011	Configure an SMTP server for system emails.	• Email notifications for Database operations and alerts are not possible without a configured SMTP server. • Data Services Manager facilitates sending an email to a system user when the user initiates a Forgot Password password reset. To enable this functionality, you must configure an SMTP server for the installation.	• Requires connectivity to SMTP server. • To receive email notifications, directory service users must have a valid email address attribute (`mail` or `rfc822mailbox` in the LDAPS server).
VCF-PC-DSM-REQD-012	Deploy the Data Services Manager Consumption Operator service.	Development teams can leverage VCF Automation's integration to automate database creation and modification to define database resources. The Consumption Operator processes these requests and provisions databases through Data Services Manager.	You must download and install the Data Services Manager Consumption Operator service on vSphere Supervisor.

Table 461: Data Services Manager Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-PC-DSM-RCMD-001	Configure VCF SSO and vCenter Groups on VCF Operations to deploy DSM appliance.	• This allows a VCF SSO role assigned multiple vCenters to deploy DSM appliance. Appropriate permissions can be assigned to SSO role for DSM Deployment and administration. • This allows DSM installer access to both management domain vCenter for appliance placement and Workload vCenter for DSM plugin registration.	• Requires VCF SSO and vCenter Groups configured within the VCF Operations console. • This allows a VCF SSO role that is assigned to multiple vCenters to deploy DSM appliance. • For more information please review. • Identity and Access Management
VCF-PC-DSM-RCMD-002	Configure Metrics forwarding to VCF Operations.	• With metrics forwarding enabled, platform teams gain a single operational view that correlates VM-level resource metrics that are already collected by VCF Operations with database-specific metrics published by Data Services Manager. • Data Services Manager forwards metrics for all three supported database engines: PostgreSQL, MySQL, and Microsoft SQL Server.	Requires the cloud proxy component certificate to establish connectivity.
VCF-PC-DSM-RCMD-003	Configure log forwarding to Log Management.	• Allows you to configure Log Forwarding for VMware Data Services Manager provider VM and provisioned databases. • Logs can be analyzed to improve productivity of the system, deduce trends, or to troubleshoot, as required.	Requires Log managementdeployed within a VCF Fleet.
VCF-PC-DSM-RCMD-004	For air-gapped deployments, provide access to a private container registry.	• For air-gap deployments, a private container registry must be available to host DSM Consumption Operator images. • Data Services Manager does not ship with Microsoft SQL Server images. • MS SQL Server container images must be uploaded to a private container image registry	Requires deployment of a private container image registry that supports the OCI (Open Container Initiative).

Table 462: Accelerator Design Recommendations Based on AI/ML Workload Requirements

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-001	Size GPU memory to accommodate model weights plus KV cache for expected concurrent requests at target context length. Sizing GPU memory at 2.5× model weight for production workloads is a general practical guideline.	Model weights require 2 bytes per parameter (FP16) as a fixed baseline. KV cache grows dynamically based on the number of simultaneous requests and their context lengths. Each model has a specific KV cache size per token (calculated from model architecture: 2 × 2 bytes × n_layers × n_kv_heads × d_head). Example: 8B model with 32 layers and 8-KV heads has 0.000122 GiB/token cache size. For 10 simultaneous requests each using 4,000 tokens: need 16 GB (weights) + 4.88 GB (10 × 4,000 × 0.000122) = 21 GB minimum. Formula: Total_Memory ≥ Model_Weights + (concurrent_requests × tokens_per_request × KV_cache_per_token).	Underestimating concurrent load or context length causes OOM failures during production traffic. Overestimating wastes expensive GPU memory that sits idle. Different model families have vastly different KV cache sizes even at the same parameter count.
PAIF-ACC-RCMD-002	Procure GPUs with adequate memory bandwidth to meet token generation throughput requirements	Throughput (tokens per second) measures how fast the model generates output tokens during the streaming response phase. Token generation is memory-bound because the GPU must load all model weights from High Bandwidth Memory (HBM) to the cache for each token, spending the majority of time on memory operations rather than computation. Generation time = model_bytes ÷ memory_bandwidth. Example: 16-GB model on 864-GB/s GPU takes 18.5 ms/token; on 3,350-GB/s GPU takes 4.8 ms/token - nearly 4× faster despite similar memory capacity.	Memory bandwidth cannot be improved through software, as it's hardware-limited. Maximizing throughput encourages premium GPU purchases even when compute capabilities are underutilized.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-003	Procure GPUs with adequate compute (TFLOPS) to meet Time-to-First-Token (TTFT) requirements	Time-to-First-Token (TTFT) measures the latency from submitting a prompt until the first output token appears, which is critical for interactive responsiveness. The prefill phase is compute-bound because it processes many prompt tokens in parallel, reusing loaded weights across tokens. Prefill time = model_params × 2 ÷ GPU_TFLOPS. Example: 8B model on 181 TFLOPS GPU takes 0.088 ms/token prefill; on 989 TFLOPS takes 0.016 ms/token, 5.5× faster.	Some applications tolerate higher TTFT latency, making this requirement unnecessary. Must size based on the expected prompt length distribution in the actual workload. Must validate that the selected GPU meets TTFT targets for expected prompt lengths in the actual workload.
PAIF-ACC-RCMD-004	Target sub-200ms Time-to-First-Token for interactive applications	200 ms represents the human visual reaction time threshold. Delays beyond this value create perceptible lag in a conversational interface. Time-to-first-token equals prompt length multiplied by per-token prefill time: TTFT = prompt_tokens × (model_params × 2 ÷ GPU_TFLOPS). Example: 8B model on mid-range GPU (181 TFLOPS) processes tokens at 0.088 ms each, supporting a maximum of ~2,300 token prompts within a 200-ms TTFT. High-end GPU (989 TFLOPS) at 0.016 ms/token supports ~12,500 tokens. The prefill phase is compute-bound and benefits from higher TFLOPS GPUs.	Severely constrains GPU selection for long-context applications like document analysis or code review. Some applications tolerate higher latency (batch processing, background tasks, non-interactive workflows), making this target unnecessary.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-005	Target minimum tokens-per-second generation rate based on application responsiveness requirements. Plan GPU allocation using a 30 tokens/sec minimum throughput for chat applications.	Generation throughput determines perceived responsiveness during streaming output. Human reading speed averages several words per second; generating significantly faster ensures the user never waits. Memory-bound generation time = model_bytes ÷ memory_bandwidth. Example: 16-GB model on 864-GB/s GPU generates at 54 tokens/sec (18.5 ms/token); on 3,350 GB/s generates at 208 tokens/sec (4.8 ms/token). Real-time chat applications typically require higher throughput than asynchronous workflows. The average reading speed is estimated to be between 200-300 words per minute. In comparison, a model generating 30 tokens per second (approximately 90 words per second) can produce around 1,350 words per minute.	Throughput requirements may eliminate entire GPU classes from consideration. The formula calculates per-request generation speed, which remains relatively constant regardless of batching. However, batching multiple requests allows the GPU to amortize weight loading across many users, dramatically increasing total system throughput (tokens/sec across all users) while maintaining individual user experience.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-006	Size GPU memory for the average context window, not the maximum model capacity	Context window refers to the total tokens in the KV cache per user (prompt + generated response). Models advertise maximum context lengths (8K, 128K, or more), but real workloads use much shorter contexts. Sizing for maximum dramatically reduces concurrent capacity since KV cache memory scales linearly with context length. Formula: Max_concurrent_users = (Total_GPU_Memory - Model_Weights) ÷ (context_window × KV_cache_per_token). Example: 8B model on 48GB GPU sized for 4K average context supports 64 concurrent users; sizing for 8K maximum halves capacity to 32 users; sizing for 128K reduces to 2 users. Typical applications: RAG workflows use prompt (1-2K) + retrieved docs (2-3K) + response (1K) = 4-6K total; chat uses 1-3K context. TTFT also scales with context length: Total_TTFT = context_window × prefill_time_per_token. Longer contexts increase prefill time linearly - an 8K context takes 2× longer to process than 4K. Optimizing for realistic average usage maximizes both concurrent capacity and TTFT performance.	Must monitor actual context length distribution in production to validate sizing assumptions and identify when undersizing causes user impact.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-007	When scaling the GPU infrastructure for a model endpoint, use data parallelism (independent model replicas) over tensor parallelism (model sharding across GPUs) when possible	Data parallelism deploys complete, independent copies of the model on separate GPUs with load balancing, distributing requests across replicas. Each GPU holds full model weights and processes requests independently. Tensor parallelism splits a single model across multiple GPUs - each GPU holds a fraction of the model's layers or weights and must communicate with other GPUs during every forward pass to complete inference. Example: 8B model (16 GB) can run as 4 independent replicas on 4×48GB GPUs (data parallelism) or as 1 model split across 4 GPUs with 4 GB weights per GPU (tensor parallelism). Tensor parallelism is theoretically more memory efficient, pooling memory across GPUs for shared KV cache enables higher concurrent capacity. Example: Single model across 2× 48-GB GPUs has 80 GB for KV cache (after 16-GB weights); 2 independent replicas have only 64-GB total for KV cache. However, data parallelism achieves near-linear throughput scaling without inter-GPU communication overhead, offers simpler deployment and better fault isolation. Single replica failure reduces capacity but does not crash the entire system (vs tensor parallelism, where one GPU failure brings down the model). Tensor parallelism requires high-speed interconnects (NVLink/ NVSwitch) and introduces latency from GPU-to-GPU communication every forward pass.	Data parallelism multiplies total memory requirements. 5 replicas need 5× the model weight storage (80-GB total for 8B model) vs 1× with tensor parallelism (16 GB). Prevents KV cache memory pooling across GPUs, reducing effective concurrent capacity per GPU compared to shared memory in tensor parallelism.
PAIF-ACC-RCMD-008	Provision VM/container RAM at 1-2× total GPU memory for inference workloads	Modern LLM inference frameworks load models directly into GPU memory and should not swap to system RAM during normal operation. VM RAM supports the operating system, inference framework, monitoring tools, and request handling overhead. System RAM below this ratio risks framework instability or slow startup times.	Over-provisioning wastes resources since inference primarily uses GPU memory.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-009	Size physical server RAM at 2-3× total GPU memory to accommodate host overhead and ancillary workloads	Physical servers require additional RAM beyond VM allocations for hypervisor overhead and potential non-GPU workloads.	Over-provisioning wastes resources since inference primarily uses GPU memory.
PAIF-ACC-RCMD-010	Provision 5-10× storage capacity relative to source document size for RAG embedding databases	RAG systems multiply your source document storage due to vector embeddings, indexes, and chunk overlap. This multiplier increases with higher-dimensional embeddings, greater chunk overlap, or storing multiple embedding models.	Difficult to estimate precisely without testing actual documents and parameters. The database grows continuously as documents are added.
PAIF-ACC-RCMD-011	Start with 4-8 vCPUs per GPU and monitor CPU utilization. Scale based on actual bottlenecks, not a fixed ratio.	CPU cores handle request preprocessing, tokenization, output post-processing, batching logic, and framework orchestration while the GPU performs model inference. CPU requirements scale with the number of GPUs and expected request concurrency. Insufficient CPU cores bottleneck request processing, even when the GPU is underutilized.	CPU is typically not the bottleneck. Over-provisioning wastes resources and increases costs.

Table 463: Common Design Requirements for vGPUs and DirectPath (Passthrough) GPUs Compute Models

ID	Design Recommendations	Justification	Implication
PAIF-COM-REQD-COM-001	Enable Intel VT-d or AMD I/O VT in the ESX server BIOS	Required for IOMMU functionality that enables Direct Memory Access (DMA) isolation and allows the VMkernel to assign GPU devices to VMs directly. Provides a direct path to hardware while maintaining complete isolation of resources. Without VT-d or AMD-V enabled, passthrough GPUs and vGPUs cannot function	Verify CPU support and BIOS enablement before deployment.
PAIF-COM-REQD-COM-002	Enable Memory Mapping Above 4 GB in the ESX server BIOS	Modern GPUs require larger contiguous MMIO address spaces (beyond the 256-MB limit in the sub-4GB region). Enables the system to map entire GPU BARs above the 4-GB threshold, reducing fragmentation and improving stability. Without this setting, GPUs with BAR sizes exceeding 256 MB may fail to initialize or be prevented from use entirely. The device may have to incorporate multiple BARs during operations, increasing complexity and overhead, negatively impacting performance.	Verify BIOS enablement before deployment.
PAIF-COM-REQD-COM-003	Workloads must use a 64-bit operating system	32-bit operating systems can only address up to 4 GB of memory, which is insufficient for modern GPU memory capacities. A 64-bit OS can address up to 16 million TB, enabling full utilization of GPU memory and large MMIO spaces.	Private AI Services model endpoints VMs, DLVMs, and VKS clusters deployed using VCF Automation catalog use 64-bit OSs. Verify the OS of custom workloads.
PAIF-COM-REQD-COM-004	Workloads using GPUs must be configured with the EFI firmware boot option	BIOS firmware only supports BARs up to 256 MB in the MMIO-low region. GPUs with BAR1 sizes exceeding 256 MB require EFI firmware to map their memory spaces above 4 GB properly. GPUs with BAR1 sizes larger than 256 MB cannot be properly initialized with BIOS firmware.	Firmware cannot be changed after OS installation without reinstalling the guest OS.

ID	Design Recommendations	Justification	Implication
PAIF-COM-REQD-COM-005	Workloads must have all their memory reserved	Memory reservation "pins" VM memory pages to fixed physical addresses, preventing ESXi from swapping or ballooning them during memory pressure. Memory reservation is required because GPU DMA operations use host physical addresses and cannot handle page faults mid-transfer. Without memory reservation, ESX may swap out or may balloon memory pages during DMA operations, causing page faults that cannot be serviced.	Any VM configured with a PCI devices has their full memory that is automatically reserved, no user configuration is required. Memory overcommitment is not possible for GPU-enabled workloads.
PAIF-COM-REQD-COM-006	Set pciPassthru.set.use64bitMMIO = true in the workload VM advanced configuration parameters	Enables 64-bit MMIO addressing for the VM, allowing access to MMIO regions above 4 GB. The default 32GB MMIO space is often insufficient for modern GPUs with large BAR sizes. Without this setting, the VM is limited to the default 32GB MMIO space, which is insufficient for modern GPUs requiring larger MMIO regions. GPU initialization will fail if BAR requirements exceed available MMIO space. https://knowledge.broadcom.com/external/article/312208/vsphere-vmdirectpath-io-and-dynamic-dire.html	This setting is automatically applied to Private AI Services model endpoints VMs, VKS, and supervisor managed DLVM workloads configured with GPUs. Manual configuration is required for DLVM deployed from OVA.
PAIF-COM-REQD-COM-007	Specify a value for the pciPassthru.64bitMMIOSizeGB setting in the workload VM advanced configuration parameters. The value is based on the total memory of the GPUs that the VM has access to. Calculated as: sum of GPU memory capacities, round up to power of 2, then use the next power of 2 value (e.g., 40GB GPU → 64 GB → 128GB MMIO).	Defines the total MMIO address space for all GPUs assigned to the VM. MMIO space must accommodate all GPU BARs with natural alignment, meaning memory addresses must be allocated on boundaries that match their size (e.g., a 64GB BAR must start on a 64-GB boundary). GPUs expose multiple BARs (BAR0 for control, BAR1 for frame buffer, BAR3 for device-specific functions) that must be properly aligned within the MMIO space. Without a properly sized MMIO space, the system cannot naturally align GPU BARs, preventing GPU initialization. Insufficient MMIO allocation will cause the GPU device to fail to start or function incorrectly.	This setting is set to 512 GB to Private AI Services model endpoints VMs, VKS, and supervisor managed DLVM workloads configured with GPUs. Manual configuration is required for DLVM deployed from OVA.

ID	Design Recommendations	Justification	Implication
PAIF-COM-REQD-COM-008	Use non-reserved (best-effort) GPU VM Classes. Avoid reserved GPU VM Classes unless tenant isolation requirements explicitly demand them.	GPU resources remain in a shared pool accessible to all namespaces associated with the vSphere cluster, preventing artificially stranded capacity. When GPU hosts are added to the cluster, their capacity becomes immediately available to existing workloads without requiring the administrator to update zone reservations, VCFA Automation organization quotas, and namespaces.	In environments where organizations or teams require a guaranteed and pre-allocated number of GPU slots reserved, GPU VM classes are the appropriate mechanism. Without reservation, there is no guarantee that GPU capacity will be available when a workload attempts to start.

Table 464: Design Recommendations for vGPU Compute Model

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-001	Configure a high-bandwidth vMotion network (100 GigE recommended for vGPU workloads)	GPU frame buffer memory must be copied during vMotion. VMs can have multiple GPUs assigned (up to 8 in NVSwitch configurations). vMotion stun time is directly proportional to the data volume and inversely proportional to network bandwidth. Insufficient bandwidth causes extended stun times that may exceed the 100-second default timeout, resulting in failed vMotion operations. 10-GigE networks may be inadequate for VMs with large or multiple vGPU profiles.	Higher cost
PAIF-COM-RCMD-VGPU-002	Deploy VCF 9.0+ with supported processors (Intel Sapphire Rapids+, AMD Milan+) for better vMotion performance	ESX 9.0 or later enables vMotion to identify static versus actively changing GPU memory pages. Static data (~70% for LLM inference - model weights/ parameters) is copied during the pre-copy phase while the VM runs. Only changing data (KV cache, activations) is copied during stun time, reducing stun times by 3-4x. Without this capability, all GPU memory is copied during stun time regardless of whether it's actively changing.	VCF 9.0 or later is required

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-003	Increase vMotion default Stun Time Limit (for large vGPU profiles)	The default vMotion stun time limit is 100 seconds. For VMs with large vGPU profiles (multiple GPUs or GPUs with large frame buffer memory), the time that is required to copy GPU memory during vMotion may exceed this default. vSphere allows administrators to set a custom stun time limit that their workload can tolerate. If the stun time limit is set too low for the VM's vGPU configuration, vMotion operations will time out and fail. The VM remains on the source host, preventing maintenance operations or DRS load balancing. vSphere calculates its own estimate and warns if the allocated time is insufficient. https://knowledge.broadcom.com/external/article?legacyId=88271	ESX evacuation operations during maintenance activities may take more time.

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-004	Enable DRS automation for vGPU VMs by setting PassthroughDrsAutomation=1 and LBMaxVmotionPerHost=1 in the DRS cluster advanced settings	By default, DRS sets Load Balancer recommendations to Manual Remediation for vGPU VMs due to extended vMotion downtimes. Setting PassthroughDrsAutomation=1 enables automatic DRS load balancing and Maintenance Mode evacuations for vGPU VMs, provided the stun time is acceptable according to DRS calculations. Setting LBMaxVmotionPerHost=1 limits DRS to one vMotion per host per scan cycle, reducing the number of concurrent vMotions during consolidation and preventing network saturation from multiple large GPU memory transfers. Without PassthroughDrsAutomation enabled, DRS will not automatically migrate vGPU-enabled VMs for load balancing or maintenance mode evacuation. Administrators must manually approve each recommendation. Without LBMaxVmotionPerHost, DRS may initiate multiple vMotions simultaneously, potentially overwhelming the vMotion network when multiple VMs with large vGPU profiles are migrated concurrently. https://knowledge.broadcom.com/external/article?articleNumber=313479	ESX evacuation operations during maintenance activities may take more time.

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-005	Enable VgpuVmConsolidation for GPU Capacity Optimization	When set to 1, this DRS advanced setting causes DRS to tightly pack VMs with smaller vGPU profiles (e.g., single vGPU) onto hosts with extra compatible GPU capacity. This consolidation frees up larger contiguous GPU capacity on other hosts, allowing VMs with multiple full-memory vGPU devices to power on. Without this setting, GPU resources may become fragmented across the cluster. VMs requiring multiple GPUs may fail to power on even when sufficient total GPU capacity exists, because the capacity is scattered across multiple hosts rather than consolidated on fewer hosts. https://knowledge.broadcom.com/external/article/312021/vgpu-vm-placement-may-result-in-suboptim.html	Manual configuration is required. This optional setting may not be beneficial in deployments leveraging homogenous vGPU profiles.
PAIF-COM-RCMD-VGPU-006	Enable vMotion stun time validation at VM power-on by setting PassthroughRequireDrs=1 in the DRS cluster advanced settings	When set to 1, this setting performs checks at VM power-on time to ensure that the VM can be moved using vMotion and that its stun time fits within the estimate calculated by DRS based on vGPU profile sizes and configured vMotion network bandwidth. Without this setting, VMs may be powered on without validation of their vMotion feasibility, resulting in VMs that cannot be migrated during maintenance or load balancing and only discovered when vMotion is attempted. https://knowledge.broadcom.com/external/article/313178/insufficient-resources-fault-when-using.html	VMs may be prevented from being powered on even if the required resources are available on the cluster. It may not be appropriate in deployments leveraging multi-GPU profiles or UVM, where VMs cannot be live-migrated via vMotion.

Table 465: Design Recommendations for CPU-only Compute Model

ID	Design Recommendation	Justification	Implication
PAIF-COM-RCMD-CPU-001	Configure full memory reservation for CPU-only inference VMs running llama.cpp	llama.cpp loads the entire quantized model into system RAM and relies on fast, consistent memory access for token generation. Memory reservation prevents ESX from swapping or ballooning VM memory during memory pressure, which would cause severe performance degradation.	Memory reservation reduces cluster memory overcommitment flexibility. Reserved memory cannot be reclaimed by ESX for other VMs, potentially reducing overall cluster density. For development or non-production workloads where performance consistency is less critical, this trade-off may not be justified.

Security Modernization

To assist with choosing a protection and recovery model for your specific business and technology requirements, the protection blueprints section provides a set of pre-defined architecture topologies, based on specific design profiles. To deliver a ready-to-consume solution, providing the full set of capabilities (end-to-end design) offered by the VMware Cloud Foundation platform, each protection blueprints provides a prescriptive set of requirements and implementation steps.

Bolster Lateral Security and Intrusion Detection and Prevention	
Lateral Security with vDefend	The *Lateral Security with vDefend* blueprint describes the architectural components and design selections required to provide centralized, enterprise-grade security services for a VMware Cloud Foundation platform using VMware vDefend security solutions.
Accelerate Recovery from Ransomware and Other Disasters	
VCF Component Backup and Restore	The *VCF Component Backup and Restore* blueprint provide guidance on ensuring the resilience and continuity of your VCF private cloud. In the event of data loss or system failure, having up-to-date backups allows for efficient restoration of critical services, minimizing downtime and maintaining operational integrity. Implementing a consistent backup strategy protects your infrastructure against unforeseen disruptions.
VCF Instance Backup and Restore	The *VCF Instance Backup and Restore* blueprint provides guidance on the recovery of an entire VCF Instance that has failed, and must be recovered in the same site.
VCF Fleet Disaster Recovery	This *VCF Fleet Disaster Recovery* protection blueprint provides the ability to handle a VCF Instance, or site failure, and recover components in another location, where that location is typically outside metro radius.
Cyber Recovery for VMware Cloud Foundation	This *Cyber Recovery for VMware Cloud Foundation* protection blueprint provides a clean room environment on a vCenter instance that allows you to inspect, analyze, and recover infected virtual machines before restoring them to a production environment.

VCF Component Backup and Restore

Regularly backing up the management components of a VMware Cloud Foundation platform is essential to ensure the resilience and continuity of your VCF private cloud. In the event of data loss or system failure, having up-to-date backups allows for efficient restoration of critical services, minimizing downtime and maintaining operational integrity. Implementing a consistent backup strategy safeguards your infrastructure against unforeseen disruptions.

Overview

This *Protection Blueprint* applies to both the management domain and workload domains where you must recover one or more management components.

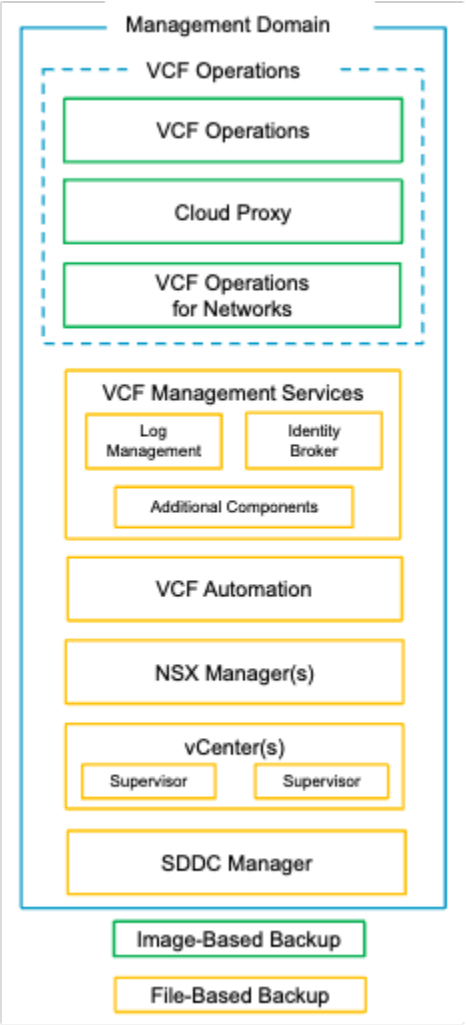
After you deploy a VCF fleet, backing up management components ensures that you can keep your environment operational if a data loss or failure occurs. You implement backups to prepare for:

- A critical failure of any management component.
- An upgrade of any management component.
- A certificate update of any management component.

You take the following backup types:

- Scheduled backups, which ensure that at any given point in time, you can restore from a recent backup.
- Manual backups before a system update, which ensure that if the operation is unsuccessful, you can restore to a point in time immediately before the operation.
- Manual backups after the recovery of a failed part of the system.

Figure 214: Backup Methods for VMware Cloud Foundation



Category	Component	Supported Backup Method(s)	Additional Information
Management	VCF Operations	Image-Based	• You must back up all cluster nodes at the same time. • It is recommend to take the cluster offline during backup. If the cluster cannot be taken offline, do not quiesce the file system. • Do not perform a backup when dynamic threshold (DT) calculations are running because the backup might lead to performance problems or loss of nodes. By default, DT calculations run at 2 a.m.
	VCF management services	File-based	SFTP only
	Log management	Included as part of VCF management services backup	
	Identity broker		
	VCF Operations for networks	Image-Based	

Category	Component	Supported Backup Method(s)	Additional Information
	SDDC Manager	File-based or Image-Based*	SFTP only*
	License server	None	You must deploy a new license server and reconfigure licensing.
Consumption	VCF Automation	File-based	SFTP only
Core	vCenter	File-based or Image-Based*	SFTP recommended
	vSphere Supervisor	Included in vCenter backup	
	ESX	N/A	
	NSX Manager	File-based	SFTP only
	NSX Edge	NSX Edge configuration is included in NSX Manager backup	

* This design recommends the file-based backup method for this component.

Planning for VCF Component Backup and Restore

Aspect	Requirement
File-based backups	Provision an SFTP server
Image-based backups	Use a third-party backup tool

Implementation of VCF Component Backup and Restore

Detailed implementation steps for backup and restore are available in Component Backup and Restore of VMware Cloud Foundation.

Design Elements for VCF Component Backup and Restore

The following section contains a list of design elements for this protection blueprint.

VCF Component Backup Design Requirements

Table 466: VCF Core Component Backup Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-BCK-REQD-SFTP-001	Provide an SFTP server for file-based backups and ensure that sufficient storage is available to store backups (including required retention) for the following components: • vCenter • NSX Manager • SDDC Manager • VCF Automation • VCF management services	The SFTP server handles the backup of the core management stack components. You require sufficient storage to store all backups within your retention period.	You must provide more secondary storage capacity to accommodate increased disk requirements.
VCF-BCK-REQD-SFTP-002	Create a dedicated sub-directory on the SFTP server for each VCF Instance in the fleet, if more than one VCF Instance is sharing the SFTP server.	Isolates the backups from each VCF Instance to a dedicated folder avoiding any confusion when selecting a backup to restore from.	You must manually maintain the directory structure and permissions.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-BCK-REQD-SFTP-003	Create a dedicated service account **(e.g. svc-vcf-bck)** on the SFTP server for file-based backups. Limit the permissions on the SFTP target folder to this user.	Using a dedicated account for backup and restore protects the backup data from being accidentally or maliciously accessed by other users.	You must create a password rotation policy on the dedicated account, and must put procedures in place to manage the rotation.
VCF-BCK-REQD-SFTP-004	Replicate backups to another site if they are to be used as part of VCF fleet disaster recovery.	Ensures backup data is available to recover the fleet management components on another site.	You must manage the replication of backup data.
VCF-BCK-REQD-VC-001	Use the vCenter management interface or API to configure vCenter file-based backups.	Supported way to configure vCenter file-based backups.	You must manually configure file-based backups for each vCenter.
VCF-BCK-REQD-VC-002	Configure a daily schedule for vCenter file-based backups.	Ensures a Recovery Point Objective (RPO) of 24 hours or less.	You lose data that changed after the last backup up to 24 hours ago.
VCF-BCK-REQD-VC-003	Configure retention as per your requirements for vCenter file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.
VCF-BCK-REQD-VC-004	Encrypt vCenter file-based backups using an encryption password.	Protects sensitive backup data.	You must create a password rotation policy and must put procedures in place to manage the rotation.
VCF-BCK-REQD-VC-005	Select vSphere SupervisorSupervisor Control Plane to be included in the backup if Supervisor is activated on the associated vCenter.	Captures the vSphere Supervisor Control Plane state as part of the vCenter backup.	vCenter file-based backup size increases.
VCF-BCK-REQD-SDDC-001	Configure an external SFTP backup target for each VCF Instance.	Ensures SDDC Manager and NSX Manager backups are stored on an external SFTP server.	You must manually configure the SFTP server using the VCF Operations UI.
VCF-BCK-REQD-SDDC-002	Configure a daily schedule for SDDC Manager and NSX Manager file-based backups.	Ensures a Recovery Point Objective (RPO) of 24 hours or less.	You may lose data that changed after the last backup up to 24 hours ago.
VCF-BCK-REQD-SDDC-003	Configure retention as per your requirements for SDDC Manager and NSX Manager file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.

Table 467: VCF Management Component Backup Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-BCK-REQD-VCFMS-001	Configure an external SFTP backup target for each VCF Instance, using a dedicated sub-directory on the SFTP server.	Ensures VCF management services and VCF Automation backups are stored on an external SFTP server.	You must manually configure the SFTP server using the VCF Operations UI.
VCF-BCK-REQD-VCFMS-001	Configure a backup schedule for VCF management services per your requirements.	Provides the ability to restore to a desired point in time.	You must manually configure the backup schedule using the VCF Operations UI.
VCF-BCK-REQD-VCFMS-002	Configure retention as per your requirements for VCF management services file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-BCK-REQD-VCFA-003	Configure backup schedules and retention for VCF Automation per your requirements.	Provides the ability to restore to a desired point in time.	You must manually configure the backup schedules and retention using the VCF Operations UI.
VCF-BCK-REQD-VCFMS-004	Configure retention as per your requirements for VCF Automation file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.

VCF Component Backup Design Recommendations

Table 468: VCF Backup Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-BCK-RCMD-NET-001	Use a dedicated VLAN for backup traffic.	Isolates backup traffic from other management traffic.	Requires manual configuration of the vSphere Distributed Port Group.
VCF-BCK-RCMD-NET-002	Create a VMkernel on each ESX host with vSphere Backup NFC enabled and assign to the backup VLAN.	Ensures that traffic isEnsures that traffic is correctly tagged as backup traffic.	Requires manual configuration of the VMkernel.
VCF-BCK-RCMD-SFTP-001	Use a highly available SFTP server that is hosted is a different availability zone from the components that are using it as a backup target.	Ensures availability of the SFTP server in the event of a failure of the primary availability zone.	Requires an additional availability zone.
VCF-BCK-RCMD-SFTP-002	Use an immutable storage system for the SFTP server.	Prevents files from being altered or deleted after they are written to SFTP.	Requires an immutable storage system.
VCF-BCK-RCMD-SFTP-003	Use a dedicated service account for each connection to the SFTP server.	Using a dedicated account for each connection to the SFTP server protects the backup data from being accidentally or maliciously accessed by other users.	You must create a password rotation policy on the dedicated account, and must put procedures in place to manage the rotation.
VCF-BCK-RCMD-VC-001	Back up the vSphere Distributed Switch configurations from each vCenter instance.	Enables restore of a vSphere Distributed Switch independently of a vCenter restore.	You must manually take the backups after a configuration change.

VCF Instance Backup and Restore

VCF Instance Backup and Restore involves the recovery of an entire VCF Instance that has failed, and must be recovered in the same site.

This *Protection Blueprint* applies to both the management domain and workload domains where you must recover all management components.

Example Failure Scenarios

- Complete VCF Instance failure.
- Datastore corruption.

Supported Topologies

Several topologies of VCF exist according to the number of availability zones and VCF Instances. See Design.

The following topologies are supported for recovery using this guidance.

Supported Topology
• Single VCF fleet. • Workload domains with a single or multiple vSphere clusters. • vSphere clusters with a single or multiple vSphere Distributed Switches. • Workload domains deployed in a single availability zone. • vSphere Supervisor with a single or multiple zones. • vSphere content libraries running on NFS storage.

VCF Components to Back Up

All important data in a VMware Cloud Foundation platform should be backed up to a remote backup site, or local backups should be duplicated to an offsite location. Include all virtual machines in the management domain and virtual machines in workload domains that require data protection.

After initial deployment, the management domain contains a core set of virtual machines to manage the VCF fleet. When you deploy a workload domain, VCF deploys additional virtual machines to manage the workload domain.

Note:

If you have other virtual machines deployed in the management domain that require backup, for example, Microsoft SQL servers, Microsoft Active Directory servers, backup software virtual mahines, and so on, plan to back them up too. These compoents will not be restored as part of the Instance Recovery process and need to be carried out manually

Follow the VCF Component Backup and Restore protection blueprint to ensure all required management compoents are backed up.

Documenting the System Configuration of VCF

Keeping detailed as-built documentation on the system configuration eases the recovery process if a failure in your VMware Cloud Foundation platform occurs.

While the processes in this guidance retrieve much of the information below, keep a record of the following items. Save this information on a secure secondary storage.

- Topology diagrams of the VMware Cloud Foundation platform
  - Physical networking
  - vSphere Distributed Switch networking
  - NSX networking
  - Workload domain configuration
  - Cluster configuration for each cluster in a workload domain
  - ESX hosts assigned to each cluster
  - vSphere Supervisor configuration
- Networking information
  - ESX host vmnic-to-switch port mappings
  - VM virtual NIC to distributed port group mappings
  - IP address information of the VMkernel interfaces on the ESX hosts
  - IP address information of virtual machines
  - DNS, NTP, AD, and other well-known services used by the VMware Cloud Foundation platform

    Note:

    The Planning & Preparation workbook used to deploy the platform is a good source for capturing this detail.

Planning for VCF Instance Backup and Restore

Before you start the implementation of the *VCF Instance Backup and Restore* protection blueprint, you should follow the procedures in the following table.

Table 469: Preparation Tasks VCF Instance Backup and Restore

Task	Task Name	Additional Information
1	Locate the most recent known good file-based backups for the following components: • SDDC Manager • All vCenter instances • All NSX Manager instances • VCF Automation • VCF management services	
2	Locate the most recent known good image-based backups for the following components: • VCF Operations • VCF Operations for networks	
3	Gather Information from the SDDC Manager Backup	
6	Locate version and build number for vCenter, NSX Manager, and SDDC Manager.	• You can see the vCenter build number in the `backup-metadata.json` in the vCenter file-based backup. • You can see the version and can build number for NSX Manager from the file name of the backup file.
7	Download the OVA files of vCenter, NSX Manager, and SDDC Manager for the required version and build number.	
8	Download the Refresh SSH Key Script from Broadcom Knowledge Base article 79004	
9	Re-install ESX on the hosts.	See Deploying a New VCF or vSphere Foundation Platform.

Implementation of VCF Instance Backup and Restore

Detailed implementation steps for the *VCF Instance Backup and Restore* protection blueprint are available in VCF Instance Backup and Restore.

Design Elements for VCF Instance Backup and Restore

The following section contains a list of design elements for this protection blueprints.

Important:

The foundation of *VCF Instance Backup and Restore* is having backups for each component in the VCF Instance. The *VCF Instance Backup and Restore* protection blueprint uses the design elements from the *VCF Component Backup and Restore* protection blueprint.

VCF Instance Backup and Restore Design Requirements

Table 470: VCF Core Component Backup Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-BCK-REQD-SFTP-001	Provide an SFTP server for file-based backups and ensure that sufficient storage is available to store backups (including required retention) for the following components: • vCenter • NSX Manager • SDDC Manager • VCF Automation • VCF management services	The SFTP server handles the backup of the core management stack components. You require sufficient storage to store all backups within your retention period.	You must provide more secondary storage capacity to accommodate increased disk requirements.
VCF-BCK-REQD-SFTP-002	Create a dedicated sub-directory on the SFTP server for each VCF Instance in the fleet, if more than one VCF Instance is sharing the SFTP server.	Isolates the backups from each VCF Instance to a dedicated folder avoiding any confusion when selecting a backup to restore from.	You must manually maintain the directory structure and permissions.
VCF-BCK-REQD-SFTP-003	Create a dedicated service account **(e.g. svc-vcf-bck)** on the SFTP server for file-based backups. Limit the permissions on the SFTP target folder to this user.	Using a dedicated account for backup and restore protects the backup data from being accidentally or maliciously accessed by other users.	You must create a password rotation policy on the dedicated account, and must put procedures in place to manage the rotation.
VCF-BCK-REQD-SFTP-004	Replicate backups to another site if they are to be used as part of VCF fleet disaster recovery.	Ensures backup data is available to recover the fleet management components on another site.	You must manage the replication of backup data.
VCF-BCK-REQD-VC-001	Use the vCenter management interface or API to configure vCenter file-based backups.	Supported way to configure vCenter file-based backups.	You must manually configure file-based backups for each vCenter.
VCF-BCK-REQD-VC-002	Configure a daily schedule for vCenter file-based backups.	Ensures a Recovery Point Objective (RPO) of 24 hours or less.	You lose data that changed after the last backup up to 24 hours ago.
VCF-BCK-REQD-VC-003	Configure retention as per your requirements for vCenter file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.
VCF-BCK-REQD-VC-004	Encrypt vCenter file-based backups using an encryption password.	Protects sensitive backup data.	You must create a password rotation policy and must put procedures in place to manage the rotation.
VCF-BCK-REQD-VC-005	Select vSphere SupervisorSupervisor Control Plane to be included in the backup if Supervisor is activated on the associated vCenter.	Captures the vSphere Supervisor Control Plane state as part of the vCenter backup.	vCenter file-based backup size increases.
VCF-BCK-REQD-SDDC-001	Configure an external SFTP backup target for each VCF Instance.	Ensures SDDC Manager and NSX Manager backups are stored on an external SFTP server.	You must manually configure the SFTP server using the VCF Operations UI.
VCF-BCK-REQD-SDDC-002	Configure a daily schedule for SDDC Manager and NSX Manager file-based backups.	Ensures a Recovery Point Objective (RPO) of 24 hours or less.	You may lose data that changed after the last backup up to 24 hours ago.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-BCK-REQD-SDDC-003	Configure retention as per your requirements for SDDC Manager and NSX Manager file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.

Table 471: VCF Management Component Backup Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-BCK-REQD-VCFMS-001	Configure an external SFTP backup target for each VCF Instance, using a dedicated sub-directory on the SFTP server.	Ensures VCF management services and VCF Automation backups are stored on an external SFTP server.	You must manually configure the SFTP server using the VCF Operations UI.
VCF-BCK-REQD-VCFMS-001	Configure a backup schedule for VCF management services per your requirements.	Provides the ability to restore to a desired point in time.	You must manually configure the backup schedule using the VCF Operations UI.
VCF-BCK-REQD-VCFMS-002	Configure retention as per your requirements for VCF management services file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.
VCF-BCK-REQD-VCFA-003	Configure backup schedules and retention for VCF Automation per your requirements.	Provides the ability to restore to a desired point in time.	You must manually configure the backup schedules and retention using the VCF Operations UI.
VCF-BCK-REQD-VCFMS-004	Configure retention as per your requirements for VCF Automation file-based backups.	Provides the ability to restore to a desired point in time.	Depending on your retention requirements, you may require additional storage.

VCF Instance Backup and Restore Design Recommendations

Table 472: VCF Backup Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-BCK-RCMD-NET-001	Use a dedicated VLAN for backup traffic.	Isolates backup traffic from other management traffic.	Requires manual configuration of the vSphere Distributed Port Group.
VCF-BCK-RCMD-NET-002	Create a VMkernel on each ESX host with vSphere Backup NFC enabled and assign to the backup VLAN.	Ensures that traffic isEnsures that traffic is correctly tagged as backup traffic.	Requires manual configuration of the VMkernel.
VCF-BCK-RCMD-SFTP-001	Use a highly available SFTP server that is hosted is a different availability zone from the components that are using it as a backup target.	Ensures availability of the SFTP server in the event of a failure of the primary availability zone.	Requires an additional availability zone.
VCF-BCK-RCMD-SFTP-002	Use an immutable storage system for the SFTP server.	Prevents files from being altered or deleted after they are written to SFTP.	Requires an immutable storage system.
VCF-BCK-RCMD-SFTP-003	Use a dedicated service account for each connection to the SFTP server.	Using a dedicated account for each connection to the SFTP server protects the backup data from being accidentally or maliciously accessed by other users.	You must create a password rotation policy on the dedicated account, and must put procedures in place to manage the rotation.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-BCK-RCMD-VC-001	Back up the vSphere Distributed Switch configurations from each vCenter instance.	Enables restore of a vSphere Distributed Switch independently of a vCenter restore.	You must manually take the backups after a configuration change.

VCF Fleet Disaster Recovery

This *VCF Fleet Disaster Recovery* blueprint, provides the ability to handle a VCF Instance, or site failure, and recover fleet components in another location, where that location is typically outside metro radius.

Architecture

You use a combination of Protection and Recovery and restore from backup to implement disaster recovery between two VCF Instances for the management domain and workload domains in the VCF fleet.

Figure 215: VCF Fleet Disaster Recovery Architecture

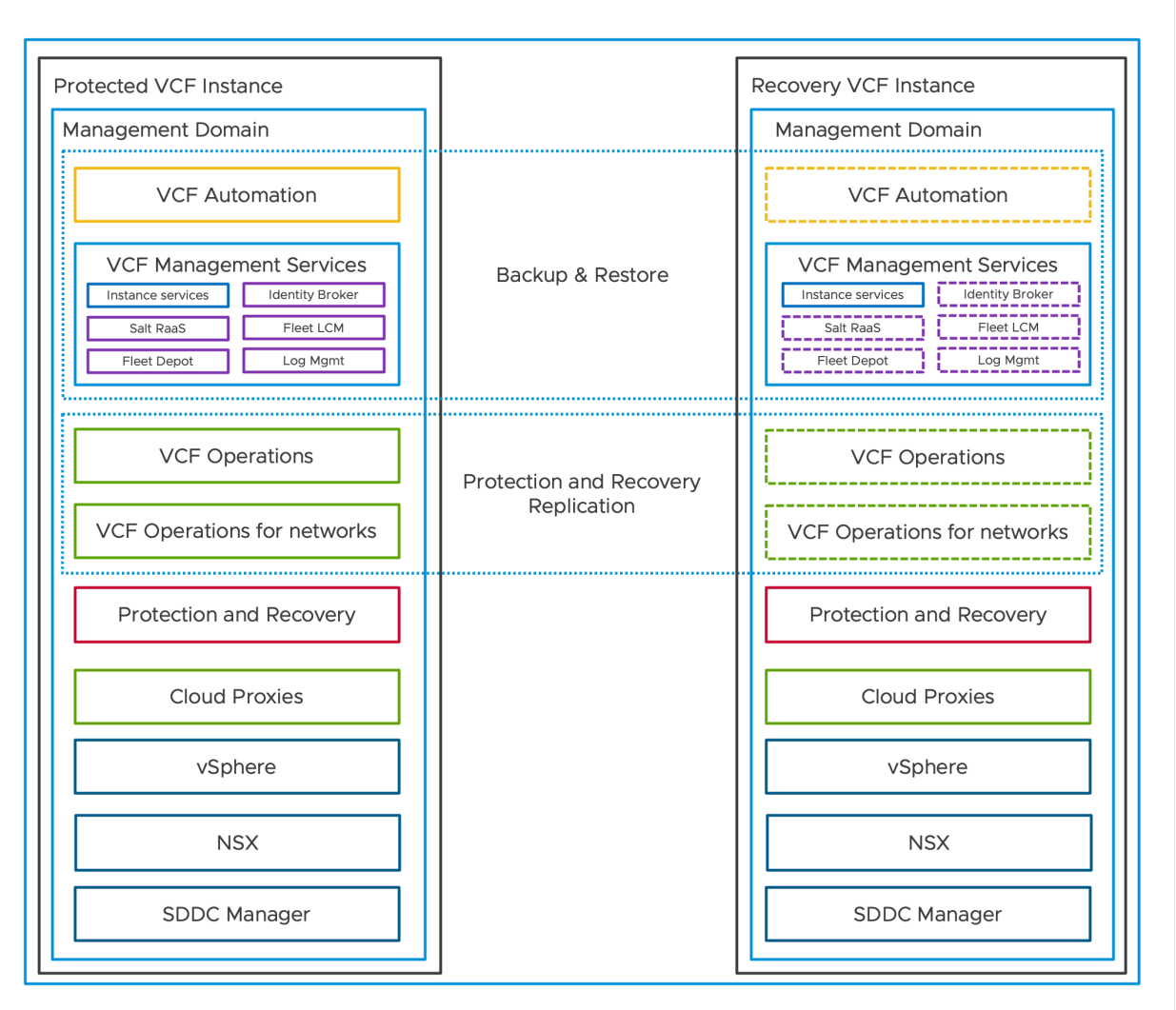


Table 473: Design Objectives of VCF Fleet Disaster Recovery

Objective	Description
Main objective	Provide disaster recovery capabilities for VMware Cloud Foundation fleet components in a multi-instance environment.
VMware Cloud Foundation architecture support	Multiple VCF Instances with IP mobility
VCF domain type support	• Management domain • Workload domain
Scope of guidance	• Deployment and initial configuration of site protection and disaster recovery components for the management domain. • Operational guidance for the site protection and disaster recovery components such as failover and failback of management applications.
Scope of implementation	Deployment and configuration of solution components: • Protection and Recovery Configuration for failover and failback for VMware Cloud Foundation components: • VCF Operations • VCF Operations for networks Backup & restore is used for the following components • VCF Automation • VCF management services

Planning for VCF Fleet Disaster Recovery

Before implementing the *VCF Fleet Disaster Recovery* blueprint, verify that the environment meets all compute, storage, and networking prerequisites, and that all required external services are accessible to the solution components.

Use the Planning and Preparation Workbook to capture environment-specific input values that are required during the implementation.

Implementation of VCF Fleet Disaster Recovery

To provide disaster recovery of VCF fleet components, you use the following recovery methods.

Fleet Component	Recovery Method
• VCF Operations • VCF Operations for networks	• Protection & Recovery Replication & Recovery Plan * • Restore from image-based backup * • VCF Operations Continuous Availability between sites can be leveraged if the latency requirements are met. For more see Continuous Availability VCF Operations Model **Note:** * Requires IP mobility
VCF Automation	Deploy a new VCF Automation instance on the recovery site and restore from backup **Note:** Does not require IP mobility
VCF management services fleet components • Identity broker** • Fleet lifecycle • Software depot** • Salt RaaS • Log management	Restore each component onto the existing VCF management services instance on the recovery site **Note:** Does not require IP mobility

** To enable recovery of this component, another instance of the component should not have been deployed on the recovery site VCF management services instance. Only one instance of each component can exist on a given VCF management services instance.

Detailed implementation steps for the *VCF Fleet Disaster Recovery* protection blueprint are available in VCF Fleet Disaster Recovery.

Design Elements for VCF Fleet Disaster Recovery

The following section contains a list of design elements for this blueprint.

Table 474: Protection and Recovery Deployment Design Requirements

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-DR-REQD-001	Deploy a Protection and Recovery instance in the protected and recovery VCF Instances.	Allows you to orchestrate the recovery of the VMware Cloud Foundation management components in another VCF Instance.	Requires sufficient resources in the management domain.
VCF-PB-DR-REQD-002	Deploy each Protection and Recovery instance in the management domain.	Provides a consistent deployment model for all management applications.	Requires sufficient resources in the management domain.
VCF-PB-DR-REQD-003	Deploy the Protection and Recovery virtual appliance using the Light deployment type.	Provides the highest level of availability by protecting the management components. This size further accommodates the following setup: • The number of protected management virtual machines as defined in Management workloads with failover support. • Two protection groups.Two recovery plans.	None.
VCF-PB-DR-REQD-004	Place the Protection and Recovery instances on the vm management network.	Places the Protection and Recovery instance on the same network as the VMware Cloud Foundation components that the appliance must communicate with.	None.
VCF-PB-DR-REQD-005	Allocate and assign a static IP address to the Protection and Recovery instances.	Using assigned IP addresses removes the constraints and risks that are associated with providing and managing DHCP on your management networks.	The use of static IP addresses requires precise IP address management.
VCF-PB-DR-REQD-006	Configure both forward (A) and reverse (PTR) DNS records for each Protection and Recovery instance.	Protection and Recovery is accessible using a fully qualified domain name.	DNS infrastructure services must be available in the environment.
VCF-PB-DR-REQD-007	Configure the Protection and Recovery instances to use NTP servers rather than using VMTools to synchronize with the ESX hosts on which it is running.	Ensures that Protection and Recovery has accurate time synchronization.	NTP services must be available in the environment.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-DR-REQD-008	Use a dedicated vSphere Distributed Switch portgroup for Protection and Recovery traffic.	Isolates Protection and Recovery traffic from other system traffic.	You must manually create the portgroup.
VCF-PB-DR-REQD-009	Use a dedicated VLAN for Protection and Recovery traffic.	Isolates Protection and Recovery traffic from other system traffic.	You must manually create the VLAN.
VCF-PB-DR-REQD-010	Use a dedicated VMkernel interface on each ESX host for Protection and Recovery traffic.	Isolates Protection and Recovery traffic from other system traffic.	You must manually create the VMkernel interface.
VCF-PB-DR-REQD-011	Assign a static IP address to the dedicated VMkernel interface on each ESX host for Protection and Recovery traffic.	Using assigned IP addresses removes the constraints and risks that are associated with providing and managing DHCP on your management networks.	The use of static IP addresses requires precise IP address management.
VCF-PB-DR-REQD-012	Lifecycle management of Protection and Recovery is provided using the native tools in the appliance.	Protection and Recovery is not managed by VCF Operations.	Deployment, patching, updates, and upgrades of Protection and Recovery are performed without native automation.
VCF-PB-DR-REQD-013	Use vSphere Replication in Protection and Recovery as the protection method for virtual machine replication.	• Allows for flexibility in storage usage and vendor selection between the two disaster recovery VCF Instances. • Minimizes administrative overhead that is required to maintain Storage Replication Adapter compatibility between two VCF Instances of disaster recovery.	All management components must be in the same cluster.
VCF-PB-DR-REQD-014	Use Protection and Recovery to automate the recovery of the following management components: • VCF Operations • VCF Operations for networks	• Provides an automated run book for the recovery of the management components in the event of a disaster. • Ensures that the recovery of management applications can be delivered in a recovery time objective (RTO) of 4 hours or less.	None.
VCF-PB-DR-REQD-015	Do not activate guest OS quiescing in the policies for the management virtual machines in Protection and Recovery.	Not all management virtual machines support the use of guest operating system quiescing. Using the quiescing operation might result in an outage.	The replicas of the management virtual machines that are stored in the target VCF Instance are crash-consistent rather than application-consistent.
VCF-PB-DR-REQD-016	Activate network compression on the management virtual machine policies in Protection and Recovery.	• Ensures the Protection and Recovery traffic over the network has a reduced footprint. • Reduces the amount of buffer memory that is used on the Protection and Recovery virtual machines.	To perform compression and decompression of data, Protection and Recovery VM might require more CPU resources on the source site as more virtual machines are protected.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-DR-REQD-017	Configure a recovery point objective (RPO) of 15 minutes on the management virtual machine policies in Protection and Recovery.	Ensures that the management application that is failing over after a disaster recovery event contains all data except any changes before 15 minutes of the event.	Any changes that are made up to 15 minutes before a disaster recovery event are lost.
VCF-PB-DR-REQD-018	Configure point-in-time (PIT) instances, keeping 3 copies over a 24-hour period on the management virtual machine policies in Protection and Recovery.	Ensures application integrity for the management application that is failing over after a disaster recovery event occurs.	Increasing the number of retained recovery point instances increases the disk usage on the vSAN datastore.
VCF-PB-DR-REQD-019	Use a prioritized startup order for VCF Operations nodes.	Ensures that the individual VCF Operations nodes are started in such an order that the operational monitoring services are restored after a disaster.	You must maintain the customized recovery plan if you increase the number of nodes in the VCF Operations cluster.
VCF-PB-DR-REQD-020	Use a prioritized startup order for VCF Operations for networksnodes.	Ensures that the individual nodes within VCF Operations for networks are started in such an order that cloud automation services are restored after a disaster.	You must maintain the customized recovery plan if you increase the number of nodes in the VCF Operations for networkscluster.
VCF-PB-DR-REQD-021	In an environment with multiple VCF Instances, configure the DNS settings for each protected component to use the DNS servers across all VCF Instances.	Each protected component can resolve DNS from DNS servers during a planned migration or disaster recovery between VCF Instances.	As you scale from a single VCF Instance to multiple VCF Instances, you must update the DNS settings on each protected component.
VCF-PB-DR-REQD-022	In an environment with multiple VCF Instances, configure the NTP settings for each protected component to use the NTP servers across all VCF Instances.	Each protected component can resolve NTP from NTP servers during a planned migration or disaster recovery between VCF Instances.	As you scale from a single VCF Instance to multiple VCF Instances, you must update the NTP settings on each protected component.
VCF-PB-DR-REQD-023	Ensure a backup location is configured for the protect and recovery site VCF Instances in VCF Operations.	Required to enable the restore of the fleet components on the recovery site.	You must add the backup location manually.
VCF-PB-DR-REQD-024	Ensure the backups from the primary site are available on the recvovery site SFTP server under the path configured on the backup location for that site.	Required to enable the restore of the fleet components on the recovery site.	You must either use the same SFTP server or configure replication of the backups from the primary site.

Table 475: Protection and Recovery Deployment Design Recommendations

Decision ID	Design Decision	Justification	Implication
VCF-PB-DR-RCMD-001	Configure a service account in each vCenter for application-to-application communication from Protection and Recovery to vSphere. This user account must be a member of the vCenter Single Sign-On administrators group.	Provides the following access control features: • Protection and Recovery accesses vSphere with the required set of permissions to perform disaster recovery failover orchestration and site pairing. • In the event of a compromised account, the accessibility in the destination application remains restricted. • You can introduce improved accountability in tracking request-response interactions between the components of the VCF Instance.	You must maintain the service account's life cycle outside of VMware Cloud Foundation to ensure its availability.
VCF-PB-DR-RCMD-002	Configure the password expiration policy for the Protection and Recovery appliance.	• You configure the password expiration policy for theProtection and Recovery appliance to align with the requirements of your organization which might be based on industry compliance standards. • The policy is applicable only to the local users for Protection and Recovery.	You can manage the password expiration policy on the Protection and Recovery appliance by using the virtual appliance console or a Secure Shell (SSH) client.
VCF-PB-DR-RCMD-003	Configure the password complexity policy for the Protection and Recovery appliance.	• You configure the password complexity policy for the Protection and Recovery appliance to align with the requirements of your organization which might be based on industry compliance standards. • The policy is applicable only to the local users for Protection and Recovery.	You can manage the password complexity policy on the Protection and Recovery appliance by using the virtual appliance console or a Secure Shell (SSH) client.
VCF-PB-DR-RCMD-004	Configure the account lockout policy for the Protection and Recovery appliance.	• You configure the account lockout policy for the Protection and Recovery appliance to align with the requirements of your organization which might be based on industry compliance standards. • The policy is applicable only to the local users for Protection and Recovery.	You can manage the account lockout policy on the Protection and Recovery appliance by using the virtual appliance console or a Secure Shell (SSH) client.

Decision ID	Design Decision	Justification	Implication
VCF-PB-DR-RCMD-005	Rotate the Protection and Recovery appliance **root** and **admin** passwords on a recurring or event-initiated schedule by using the virtual appliance console or a Secure Shell (SSH) client.	By default, the password for the Protection and Recovery **root** and **admin** accounts never expire.	You must routinely perform the password rotation for the **root** and **admin** accounts by using the virtual appliance console or a Secure Shell (SSH) client.
VCF-PB-DR-RCMD-007	Replace the default self-signed certificate in each Protection and Recovery instance with a CA-signed certificate.	Ensures that all communication to the externally facing web UI of Protection and Recovery and cross-product communication are encrypted.	You must have access to a Public Key Infrastructure (PKI) to acquire certificates.
VCF-PB-DR-RCMD-008	Install the Protection and Recovery management pack for VCF Operations.	Establishes the communication between VCF Operations and Protection and Recovery endpoints.	You must install the management pack manually.
VCF-PB-DR-RCMD-009	Configure the following endpoints to use the cloud proxy or collector group for the VCF Instance: • VMware Live Site Recovery • vSphere Replication	Local-instance components are configured to use the cloud proxy or collector group for the VCF Instance. This offloads data collection for local management components from the analytics cluster.	None.
VCF-PB-DR-RCMD-010	Configure a service account in vCenter with global permissions, for application-to-application communication from the Protection and Recovery adapters in VCF Operations to vSphere and Protection and Recovery, and assign the **Read Only** role.	Provides the following access control features: • The adapters in VCF Operations access vSphere and Protection and Recovery with the minimum set of permissions that are required to collect metrics. • In the event of a compromised account, the accessibility in the destination application remains restricted. • You can introduce improved accountability in tracking request-response interactions between the components of the SDDC.	You must maintain the life cycle and availability of the service account outside of the SDDC stack.

Decision ID	Design Decision	Justification	Implication
VCF-PB-DR-RCMD-011	Configure a service account in each vCenter with global permission, for application-to-application communication from the Protection and Recovery adapters in VCF Operations to vSphere and Protection and Recovery, and assign the **VRM replication viewer** role.	Provides the following access control features: • The adapters in VCF Operations access vSphere and Protection and Recovery with the minimum set of permissions that are required to collect metrics. • In the event of a compromised account, the accessibility in the destination application remains restricted. • You can introduce improved accountability in tracking request-response interactions between the components of the SDDC.	You must maintain the life cycle and availability of the service account outside of the SDDC stack.
VCF-PB-DR-RCMD-012	When using Protection and Recovery, install and configure the log management agent on the Protection and Recovery appliance.	Simplifies configuration of log sources in the SDDC that are packaged with the log management agent.	You must configure the log management agent to forward logs to the log management VIP.
VCF-PB-DR-RCMD-013	Configure the log management agent to forward logs from the Protection and Recovery instance to the log management system using the default `cfapi` protocol and port `9543`, authenticated with a secret created in VCF Operations.	Ensures the transmission of logs from the Protection and Recovery instance to log management are encrypted.	You must generate the secret in VCF Operations and update the configuration on the Protection and Recovery instance to enable log transmission to the log management system. Edit the agent configuration file (/etc/liagent.ini) and include the secret along with the ssl_ca_path parameter.

Cyber Recovery for VMware Cloud Foundation

This *Cyber Recovery for VMware Cloud Foundation* protection blueprint provides a clean room environment on a vCenter instance that allows you to inspect, analyze, and recover infected virtual machines before restoring them to a production environment.

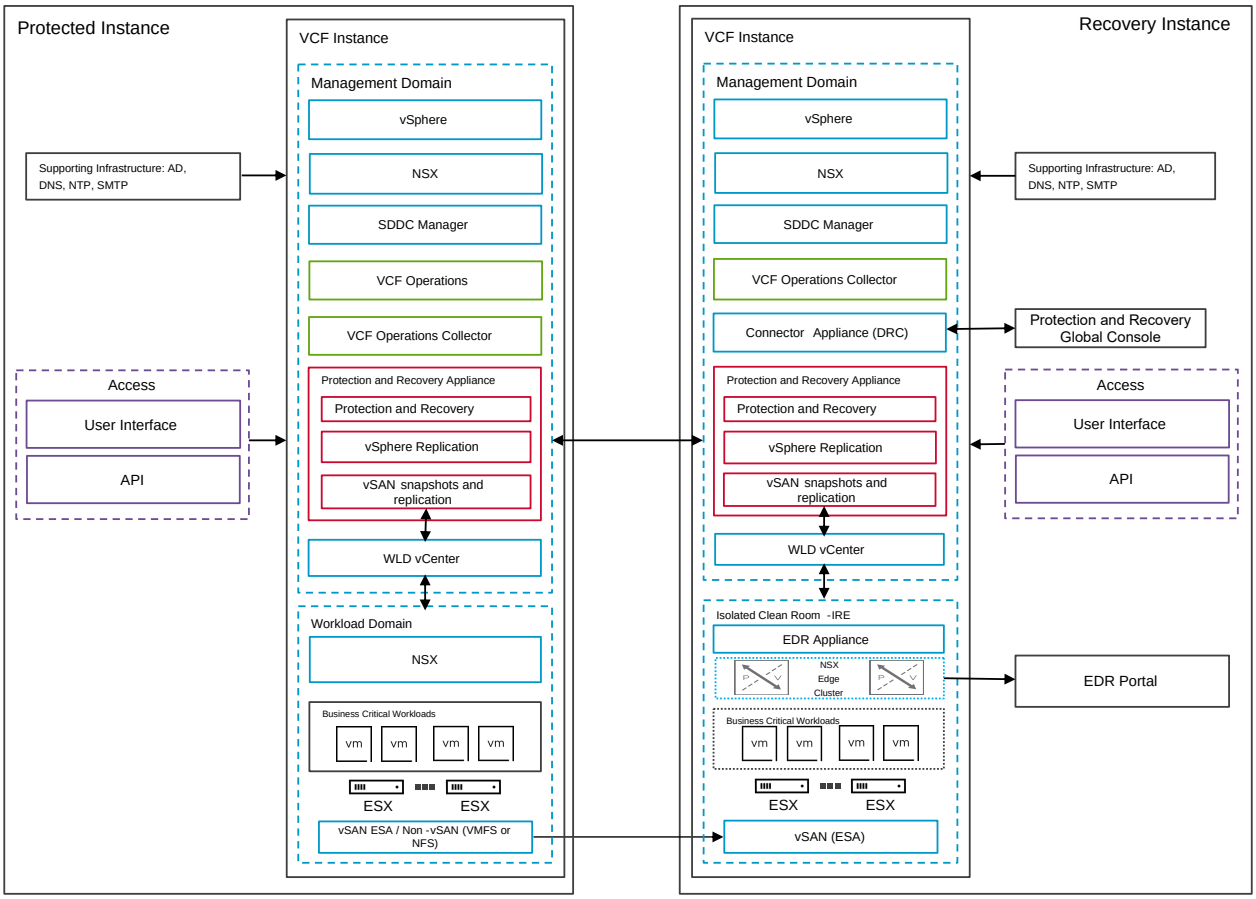
Architecture

You use Protection and Recovery to protect business critical workloads with remote vSAN snapshots against cyber threats.

After a cyber attack or ransomware attack, virtual machine snapshots might also be infected. An isolated network in a clean room is used to perform analysis, remediation, and validation of compromised virtual machines without risking infection of other production workloads.

Once the virtual machines are validated as clean, they can be recovered and reprotected to production.

Figure 216: Cyber Recovery for VMware Cloud Foundation Architecture



Cyber Recovery for VMware Cloud Foundation 9.1 has the following main elements:

Protected instance
The protected VMware Cloud Foundation or VMware vSphere Foundation platform provides business-critical data center services.

Recovery instance
The recovery VMware Cloud Foundation platform is an alternative infrastructure where business critical workloads are replicated.

It consists of the following

- Management domain
- Isolated clean room (Isolated Recovery Environment)

Clean room orchestrator
A cloud component used to create isolated clean rooms in the recovery instance.

Isolated clean room
- An isolated workload domain in the recovery instance.
- In the event of a ransomware attack, the business critical workloads are recovered to the isolated clean room in the recovery instance for forensic analysis and validation.
- Contains integrated Endpoint Detection and Response (EDR) used by the cyber response team to verify, cleanse, and recover business-critical workloads.
- Provides push-button network isolation to apply different isolation rules to one or more virtual machines under validation.
  - In VCF 9.1.0, push-button network isolation is available by using VMware vDefend Firewall.
  - In VCF 9.1.1, push-button network isolation is available by using VMware vDefend Firewall or NSX VPC networking.
- Workload domain consists of the following clusters.
  - 1x Single-Rack vSAN Compute Cluster Storage Model
  - 1x Single-Rack vSAN Storage Cluster Storage Model
  - 1x Host Fault Tolerant NSX Edge Cluster Model

    Or
  - 1x Single-Rack vSAN ESA Storage Model
  - 1x Host Fault Tolerant NSX Edge Cluster Model
- vSphere Compute cluster
  - Provides a dedicated, secure network environment that enables you to safely power-on ransomware-infected virtual machines.
- vSAN Storage cluster
  - Backed by vSAN ESA, it provides a high-performance enterprise storage system.
  - The replicated business-critical workloads are stored on the storage cluster's datastore, which is mounted remotely to the vSphere cluster.

Replication of virtual machine data
You configure remote replication of business-critical workloads using the integration of Protection and Recovery.

You configure Protection and Recovery to use the multiple-point-in-time snapshot feature providing more flexibility for data recovery of protected business critical workloads on the recovery instance.

vSAN Protection groups
Place virtual machines in a protection group to schedule and manage snapshots for all VM members of the group and to replicate virtual machines to the recovery VCF Instance.

Recovery plans
A recovery plan specifies how Protection and Recovery recovers virtual machines in the protection groups.

Endpoint Detection and Response (EDR)
Access to the EDR portal to perform security analysis on ransomware-infected virtual machines.

Design Objectives of Cyber Recovery for VMware Cloud Foundation

Objective	Description
Main Objective	Recovery of the on-premises business critical workloads within an Isolated clean room in a VMware Cloud Foundation platform, in the event of a ransomware attack in the protected site.
VMware Cloud Foundation Architecture Support	• VCF Fleet with Multiple Sites Across Multiple Regions • VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
Number of VCF Instances	2 (two)

Objective	Description
VCF Domain(s)	• 2 x management domains • 1 x workload domain (protected instance) • 1 x workload domain (recovery instance)

Planning for Cyber Recovery for VMware Cloud Foundation

Before implementing the *Cyber Recovery for VMware Cloud Foundation* protection blueprint, verify that the environment meets all compute, storage, and networking prerequisites, and that all required external services are accessible to the solution components.

Use the VMware Cloud Foundation Planning and Preparation Workbook to capture environment-specific input values that are required during the implementation.

Implementation of Cyber Recovery for VMware Cloud Foundation

To provide ransomware recovery for business critical workloads within VMware Cloud Foundation, you deploy the protection and recovery appliances across the protected and recovery instances. Then, you configure replication and create recovery plans.

To implement the *Cyber Recovery for VMware Cloud Foundation* protection blueprint, follow the procedures in the following table.

Before implementing *Cyber Recovery for VMware Cloud Foundation*, onboard to the service and verify that the clean room vCenter instance meets the NSX requirements. See the Onboard to Cyber Recovery and NSX Requirements.

Procedure Grouping	Procedure	Description
Prepare the Planning and Preparation workbook	Download and populate the Planning and Preparation Workbook	Download the Planning and Preparation Workbook from the Broadcom Technical Documentation site and populate with details needed to complete the implementation.
Deploy Protection and Recovery appliance	Deploy the Protection and Recovery Appliance	Deploy the Protection and Recovery appliance both at the protected and at the recovery site.
Configure Protection and Recovery appliance	Configure the Protection and Recovery Appliance	Configure Protection and Recovery to connect to the vCenter instance on both the protected and recovery sites using a user account that is a member of the vCenter Single Sign-On administrator group.
Isolate Network Traffic	Isolating the Network Traffic of vSphere Replication	Isolate the network traffic of vSphere Replication from all other traffic in a data center's network.
Create Site and Cluster Pairing	Create Site and Cluster Pairing	Connect the instances on the protected and recovery sites.
Apply Protection and Recovery License	Online: • Connect Protection and Recovery to the Protection and Recovery Global Console • Activate Protection and Recovery for the Site Pair Offline: • Set Up Protection and Recovery to Work in Offline Mode	Connect the protection and recovery pair to the Protection and Recovery Global Console, or create and apply an offline key.
Set Up the Clean Room Orchestrator	Set Up the Clean Room Orchestrator	Set up the clean room Orchestrator, which deploys all the necessary components to enable ransomware recovery in a clean room environment.
Enabling Ransomware Recovery Services	Enabling Ransomware Recovery Services	To use ransomware recovery with integrated security and vulnerability analysis, you must first enable the service.

Procedure Grouping	Procedure	Description
Set up Clean Room	Create and Name the Clean Room	The first step to creating a clean room requires creating and naming the clean room.
Deploy Connector appliance	Cyber Recovery Connector System Requirements Download the Connector OVA Deploy the Connector from the vCenter UI	To create a clean room, you must download the Cyber Recovery Connector OVA and deploy it to the recovery site of the site pair.
Configure the Connector appliance	Configure the Connector VM from VM Console	Configure the cyber recovery connector appliance from the VM console inside the vSphere client on the recovery site.
Register vCenter with Clean Room	Register vCenter with Clean Room	After you deploy the cyber recovery connector to an on-premises site and configure it, you must register the clean room with vCenter,
Select an Isolated Recovery VPC	Select an Isolated Recovery VPC	Configure the NSX VPC that allows virtual machines that are deployed on the network to connect to the Internet, to enable the setup of a clean room for ransomware recovery with Carbon Black, or CrowdStrike.
Cyber Recovery Sensor Installer Appliance	Cyber Recovery Sensor Installer Appliances	Deploy Cyber Recovery Sensor Installer Appliance. Choose between two Endpoint Detection and Response (EDR) technologies: Carbon Black, which is fully managed by Broadcom, or CrowdStrike, which you must manage and provide your own license.
Configure Mappings between the Protected and the Recovery Instances	Configure Inventory Mappings	Inventory mappings provide default objects in the inventory on the recovery site for the recovered virtual machines to use when you run recovery.
Create a vSAN Protection Group	Create a vSAN Protection Group	Place business critical workload virtual machines in a protection group to schedule and manage snapshots for all group members and to replicate the virtual machines to a peer site.
Create a Ransomware Recovery Plan	Create a Ransomware Recovery Plan	Create a ransomware recovery plan to define how protection and recovery processes restore infected virtual machines.
Perform a Ransomware Recovery	Perform a Ransomware Recovery	Workflow to perform a ransomware recovery.

Design Elements for Cyber Recovery for VMware Cloud Foundation

The following section contains a list of design elements for the *Cyber Recovery for VMware Cloud Foundation* protection blueprint.

Cyber Recovery for VMware Cloud Foundation Deployment Design Requirements

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-001	Deploy a Protection and Recovery appliance in both the protected and recovery VCF Instances.	Allows you to orchestrate the ransomware recovery of the VMware Cloud Foundation business workloads.	Requires sufficient resources in the management domain.
VCF-PB-CR-REQD-002	Deploy each Protection and Recovery instance in the management domain.	Provides a consistent deployment model for all management applications.	Requires sufficient resources in the management domain.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-003	Place the Protection and Recovery instances on the VM management network.	Places the Protection and Recovery instance on the same network as the VMware Cloud Foundation components that the appliance must communicate with.	None.
VCF-PB-CR-REQD-004	Allocate and assign a static IP address to the Protection and Recovery instances.	Using assigned IP addresses removes the constraints and risks that are associated with providing and managing DHCP on your management networks.	The use of static IP addresses requires precise IP address management.
VCF-PB-CR-REQD-005	Configure both forward (A) and reverse (PTR) DNS records for each Protection and Recovery instance.	Protection and Recovery instance is accessible using a fully qualified domain name.	DNS infrastructure services must be available in the environment.
VCF-PB-CR-REQD-006	Configure the Protection and Recovery instances to use NTP servers rather than using VM Tools to synchronize with the ESX hosts on which it is running.	• Ensures that the Protection and Recovery instance has accurate time synchronization. • Assists in the prevention of time mismatch between the management components.	NTP services must be available in the environment.
VCF-PB-CR-REQD-007	Isolated clean room: Configure a workload domain with the following cluster deployed and configured in the recovery VCF Instance. • 1x Single-Rack vSAN Compute Cluster Storage Model • 1x Single-Rack vSAN Storage Cluster Storage Model • 1x Host Fault Tolerant NSX Edge Cluster Model Or • 1x Single-Rack vSAN ESA Storage Model • 1x Host Fault Tolerant NSX Edge Cluster Model	• vSAN ESA is mandatory, providing native snapshots and replication, and improving performance, efficiency, and scalability by using NVMe-based TLC flash devices. • Provides a network-restricted environment that is disconnected from the production data center used as an isolated place to safely and securely power-on, inspect, and recover ransomware-infected workloads.	Additional infrastructure and operational overhead involved.
VCF-PB-CR-REQD-008	Isolated clean room: Identify or create a VPC and a subnet with public access mode for isolated recovery, and ensure that the subnet has outbound Internet access.	The recovered virtual machine must access Endpoint Detection and Response (EDR) for sensor installation and security analysis.	Configure the organization's firewall to allow outbound traffic from the isolated clean room to the EDR.
VCF-PB-CR-REQD-009	Isolated clean room: Configure the VPC subnet with a DHCP service for the network segment in the isolated clean room.	Ensures that the recovered test virtual machines get their IP addresses and DNS settings from the DHCP server.	None
VCF-PB-CR-REQD-010	Isolated clean room: Configure a separate DNS server on the recovery instance.	Isolated clean room must utilize a DNS server that is distinct from the DNS infrastructure that is used in the protected instance.	Additional infrastructure and operational overhead involved.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-011	Isolated clean room: Configure a separate NTP server that is different from the protected instance.	The path to the Internet from the isolated clean room should not follow the same path as production workloads.	Open access to an Internet-based NTP server.
VCF-PB-CR-REQD-012	Isolate the network traffic of vSphere Replication from all other traffic in both the Protection and Recovery instances. In the recovery instance, perform network isolation on the ESXi hosts of the vSAN storage cluster or vSAN ESA HCI cluster where the replicas of the business-critical workloads are stored.	Provides security isolation with a dedicated vSphere Replication VLAN for replication traffic.	Configuration of network requirements involved.
VCF-PB-CR-REQD-013	Use a dedicated vSphere Distributed Switch port group and VLAN for vSphere Replication traffic.	Isolates vSphere Replication traffic from other system traffic.	You must manually create the distributed portgroup and VLAN.
VCF-PB-CR-REQD-014	Use a dedicated VMkernel interface on each ESX host for vSphere Replication traffic.	Isolates vSphere Replication traffic from other system traffic.	You must manually create the VMkernel interface.
VCF-PB-CR-REQD-015	Assign a static IP address to the dedicated VMkernel interface on each ESX host for vSphere Replication traffic.	Using assigned IP addresses removes the constraints and risks that are associated with providing and managing DHCP on your management networks.	The use of static IP addresses requires precise IP address management.
VCF-PB-CR-REQD-016	Configure a service account in each vCenter for application-to-application communication from Protection and Recovery to vSphere. This user account must be a member of the vCenter Single Sign-On administrators group.	Provides the following access control features: • Protection and Recovery accesses vSphere with the required set of permissions to perform recovery failover orchestration and site pairing. • Accesses vSphere with the required set of permissions to perform site to site replication of virtual machines. • In the event of a compromised account, the accessibility in the destination application remains restricted. • You can introduce improved accountability in tracking request-response interactions between the components of the VCF Instance.	You must maintain the service account's life cycle outside of VMware Cloud Foundation to ensure its availability.
VCF-PB-CR-REQD-017	Set up a Clean Room Orchestrator in the VMware cloud on AWS using the Protection and Recovery global console.	To create one or more clean rooms and deploy all necessary components to enable ransomware recovery within isolated clean room environments.	Requires additional configuration and operational management.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-018	Clean Room Orchestrator: Activate integrated security and vulnerability analysis to perform security analysis on the recovered virtual machine. **Attention:** Decide whether to use Carbon Black or CrowdStrike for ransomware services integration based on your organization's business and security requirements.	To use ransomware recovery with integrated security and vulnerability analysis, you must enable the service. **Note:** Carbon Black integrated analysis is fully integrated into protection and recovery software and is managed by Broadcom.	• If CrowdStrike is selected, the organization must have valid licenses and active instances of CrowdStrike Falcon Prevent and CrowdStrike Spotlight. • Your organization must allow the outbound traffic to the EDR.
VCF-PB-CR-REQD-019	Clean room: Deploy a connector appliance in the recovery VCF Instance using the Isolated Clean Rooms option in the global console.	The component that executes commands on behalf of the cloud orchestration engine.	• The connector appliance must have network connectivity to the Clean Room Orchestrator, the isolated clean room vCenter (TCP 80/443), and the Protection and Recovery appliance. • Appropriate firewall rules, routing, and security policies must be configured to permit this communication.
VCF-PB-CR-REQD-020	Clean room: Register the isolated clean room vCenter instance with the connector in the recovery VCF Instance using the Isolated Clean Rooms option in the global console.	Connects the isolated clean room vCenter instance with the connector to enable recovery of business critical workloads.	• The connector appliance must be deployed and configured in the recovery VCF Instances. • User must have vCenter Administrator privileges.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-021	Clean room: Select one VPC from the isolated clean room vCenter instance. The recovered virtual machines that are connected to the VPC, remain isolated from production and failover networks, and have controlled Internet access.	• Using a dedicated VPC within the isolated clean room ensures that recovered virtual machines operate in a secure, contained environment. • Once selected, the Clean Room Orchestrator creates firewall rules in the VPC that function as network isolation controls during validation. • Network isolation prevents the lateral spread of malware between virtual machines on the segment and allows for granular behavioral analysis at different phases of the recovery process. • Internet connectivity is required to install the Carbon Black or CrowdStrike sensor on the recovered virtual machines and to enable integrated behavior and security analysis during validation.	• The isolated clean room vCenter instance must be registered. • In VCF 9.1.0, VMware vDefend Firewall (separate subscription required) must be installed and configured in the isolated clean room vCenter instance to enable network isolation. • In VCF 9.1.1, you can choose either vDefend Firewall or VPC NSX for per-VM isolation during your cyber recovery operations. • Network isolation in the recovery VCF Instance requires that the Distributed Firewall on the relevant cluster, the Gateway Firewall on the relevant Tier-0 Gateway, and the N-S Gateway Firewall on the VPC Security Profile be enabled.
VCF-PB-CR-REQD-022	Clean room: Deploy a dedicated Carbon Black or CrowdStrike sensor installer appliance in the recovery VCF Instance using the Isolated Clean Rooms option in the global console.	Auto-installing the sensor when starting virtual machine validation requires a sensor installer appliance.	• Requires sufficient resources in the isolated clean room vCenter instance. • The appliance must be deployed in a network segment that can reach the virtual machines being validated.
VCF-PB-CR-REQD-023	Configure site and cluster pairing using the vCenter UI.	You must pair a vCenter site with a peer vCenter site to enable replication.	• Verify that you are a member of the Administrators group. • Ensure that your non-vSAN cluster meets the following requirement in the protected instance: • Must have a minimum of one NFS or VMFS datastore. • Paired with a peer vSAN ESA cluster or vSAN 9.0 or higher compute cluster.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-024	Configure mappings between the Protection and Recovery instances, and use the VPC network segment for ransomware recovery network.	• Mappings allow you to specify how the protection and recovery server maps virtual machine resources on the protected site to resources on the recovery site. • During a recovery, when virtual machines start on the recovery site, the virtual machines use the resources on the recovery site that you specify in the mappings.	None.
VCF-PB-CR-REQD-025	Create a vSAN Protection Group through the vCenter UI to protect your virtual machines with remote vSAN snapshots against cyber threats. Use vSAN snapshots and replication in Protection and Recovery as the protection method for virtual machine replication.	• Place virtual machines in a protection group to schedule and manage snapshots for all virtual machine members of the group and to replicate virtual machines to a peer site. • Allows quick recovery of virtual machines from operational failures or ransomware attacks using snapshots that are stored on the cluster. • With Enhanced vSphere Replication, replication traffic goes directly to the host at the target site.	• You must purchase an Advanced Cyber Compliance. • The target site must be paired and configured with a vSAN ESA storage cluster. • Enhanced replications require TCP network connectivity on port 32032 from the ESX hosts on which the replicated virtual machines are running to the ESX hosts of the vSAN storage cluster and Protection and Recovery appliance in the recovery instance.
VCF-PB-CR-REQD-026	Enable immutability mode on the vSAN protection group to enhance security.	An immutable snapshot is a read-only copy of data that cannot be modified or deleted, even by an attacker with administrative privileges.	The protection group and its associated snapshots cannot be edited or deleted.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-027	Configure the Recovery Point Objective (RPO) and retention policy based on organizational requirements. The RPO must be configurable between one minute and 24 hours. Specify the time and duration of the snapshot retention. Optionally select the Ransomware Recovery template to quickly apply predefined retention settings.	• Protection against and recovery from a ransomware attack poses distinct challenges compared to traditional data loss scenarios caused by disasters. • The period between the initial intrusion and the detection of the breach can last for weeks or even months. It is likely that the most recent replicas have also been infected and partially encrypted. • The retention specifies the number of replication snapshots to maintain on the peer site. • Ransomware recovery requires a comprehensive point-in-time history to identify the moment that an application was infected with ransomware for effective removal. You can keep hourly, daily, weekly, and monthly snapshots, as required.	• The maximum snapshots that you can retain are 200. • The maximum duration of snapshot retention is three years. • Increasing the number of retained recovery point instances increases the disk usage on the vSAN ESA datastore. • Lower RPO values increase replication frequency, bandwidth usage, and system load.
VCF-PB-CR-REQD-028	Configure ransomware recovery with Integrated Analysis when creating a recovery plan using the Protection and Recovery UI.	Automatically install sensors when virtual machines are restored in the isolated clean room to detect security threats and vulnerabilities.	You can use only vSAN protection groups for ransomware recovery.

Decision ID	Design Decision	Design Justification	Design Implication
VCF-PB-CR-REQD-029	Use Protection and Recovery to automate recovery of business-critical workloads.	• In a cyber recovery plan, virtual machines progress through four states: Backup, Validation, Staged, and Recovered. • Virtual machines in the In validation state have been powered-on in the clean room and are managed by the clean room Orchestrator. While in the In validation state, security sensors are installed on virtual machines so vulnerability and behavioral analysis and malware signature scanning begins. During this state, you can log into virtual machines and can patch discovered vulnerabilities, change network isolation levels, try a different snapshot. • Staged virtual machines can be recovered to a recovery site, or returned to validation to find a better recovery candidate. • In the Recovered state, virtual machines have been recovered to the recovery site.	• A separate license is required for the EDR. • Your organization must allow the outbound traffic to the EDR. • To connect to the clean room, you must log in to the Protection and Recovery global console as a user who has the Recovery Administrator role. • You must manually patch vulnerabilities and remove any malware.
VCF-PB-CR-REQD-030	Configure the DNS settings for each protected business critical workloads to use the DNS servers across Protection and Recovery instances.	Each protected business critical workload can resolve DNS using DNS servers when powered-on recovered test virtual machines are in the IRE or during a planned migration or disaster recovery.	None
VCF-PB-CR-REQD-031	Configure the NTP settings for each protected business critical workload to use the NTP servers across Protection and Recovery instances.	Each protected business critical workload can receive NTP from the NTP servers when powered-on recovered test virtual machines are in the IRE or during a planned migration or disaster recovery.	None
VCF-PB-CR-REQD-032	Life-cycle management of Protection and Recovery instance is provided using the native tools in the appliance.	Protection and Recovery instance is not managed by VCF Operations.	Deployment, patching, updates, and upgrades of Protection and Recovery are performed without native automation.

Cyber Recovery for VMware Cloud Foundation Deployment Design Recommendations

Decision ID	Design Decision	Justification	Implication
VCF-PB-CR-RCMD-001	Configure to use Cyber Recovery ReadyNodes for vSAN Storage Cluster.	Pretested for optimal storage performance.	Additional infrastructure and operational overhead involved.
VCF-PB-CR-RCMD-002	Configure the Cyber Recovery option on the vSAN ESA storage cluster.	A vSAN ESA storage cluster that is configured for cyber recovery is recommended for performing ransomware recovery operations.	• You should only use the cyber recovery cluster for ransomware recovery workflows. • Additional infrastructure and operational overhead involved.
VCF-PB-CR-RCMD-003	Configure the password expiration policy for the Protection and Recovery appliance.	• You configure the password expiration policy for the Protection and Recovery appliance to align with the requirements of your organization which might be based on industry compliance standards. • The policy is applicable only to the local users for Protection and Recovery	You can manage the password expiration policy on the Protection and Recovery appliance by using the virtual appliance console or a Secure Shell (SSH) client.
VCF-PB-CR-RCMD-004	Configure the password complexity policy for the Protection and Recovery appliance.	• You configure the password complexity policy for the Protection and Recovery appliance to align with the requirements of your organization which might be based on industry compliance standards. • The policy is applicable only to the local users for Protection and Recovery.	You can manage the password complexity policy on the Protection and Recovery appliance by using the virtual appliance console or a Secure Shell (SSH) client.
VCF-PB-CR-RCMD-005	Configure the account lockout policy for the Protection and Recovery appliance.	• You configure the account lockout policy for the Protection and Recovery appliance to align with the requirements of your organization which might be based on industry compliance standards. • The policy is applicable only to the local users for Protection and Recovery.	You can manage the account lockout policy on the Protection and Recovery appliance by using the virtual appliance console or a Secure Shell (SSH) client.
VCF-PB-CR-RCMD-006	Change the Protection and Recovery appliance root password on a recurring or event-initiated schedule by using the appliance console or a Secure Shell (SSH) client.	By default, the root account password for theProtection and Recovery appliance expires after 90 days.	You must routinely perform the password change for the root account by using the appliance console or a Secure Shell (SSH) client.
VCF-PB-CR-RCMD-007	Change the Protection and Recovery appliance admin account password on a recurring or event-initiated schedule by using the appliance console or a Secure Shell (SSH) client.	By default, the password for the Protection and Recovery appliance admin account never expires.	You must routinely perform the password change for the admin account by using the appliance console or a Secure Shell (SSH) client.

Decision ID	Design Decision	Justification	Implication
VCF-PB-CR-RCMD-008	Replace the default self-signed certificate in each Protection and Recovery instance with a CA-signed certificate.	Ensures that all communication to the externally facing web UI of Protection and Recovery and cross-product communication are encrypted.	• You must have access to a Public Key Infrastructure (PKI) to acquire certificates. • Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered.
VCF-PB-CR-RCMD-009	Install the Protection and Recovery Adapter management pack for VCF Operations.	Establishes the communication between VCF Operations and Protection and Recovery endpoints.	You must install the management pack manually.
VCF-PB-CR-RCMD-010	Configure a service account in each vCenter with global permissions, for application-to-application communication from the Protection and Recovery adapters in VCF Operations to vSphere and Protection and Recovery, and assign the **Read Only** role.	Provides the following access control features: • The adapters in VCF Operations access vSphere and Protection and Recovery with the minimum set of permissions that are required to collect metrics. • In the event of a compromised account, the accessibility in the destination application remains restricted. • You can introduce improved accountability in tracking request-response interactions between the components of the SDDC.	You must maintain the life cycle and availability of the service account outside of the SDDC stack.
VCF-PB-CR-RCMD-011	Configure a service account in each vCenter with global permission, for application-to-application communication from the vSphere Replication adapters in VCF Operations to vSphere and vSphere Replication, and assign the **VRM replication viewer** role.	Provides the following access control features: • The adapters in VCF Operations access vSphere and vSphere Replication with the minimum set of permissions that are required to collect metrics. • In the event of a compromised account, the accessibility in the destination application remains restricted. • You can introduce improved accountability in tracking request-response interactions between the components of the SDDC.	You must maintain the life cycle and availability of the service account outside of the SDDC stack.
VCF-PB-CR-RCMD-012	Configure the following adapters to use the cloud proxy or collector group for the VCF Instance: • Protection and Recovery Adapter • vSphere Replication Adapter	Local-instance components are configured to use the cloud proxy or collector group for the VCF Instance. This offloads data collection for local management components from the analytics cluster.	None.

Decision ID	Design Decision	Justification	Implication
VCF-PB-CR-RCMD-013	Forward events from the Protection and Recovery clean room to log management.	• Log management is recommended as it is natively integrated with VMware Cloud Foundation and simplifies SDDC log configuration. • Ensures the transmission of logs to log management are encrypted.	• The log management component must be deployed, and the Protection and Recovery connector must have network connectivity to the log management endpoint. • log management must be configured with SSL using a trusted certificate.
VCF-PB-CR-RCMD-014	When using Protection and Recovery, install and configure the log management agent on the Protection and Recovery appliance.	Simplifies configuration of log sources in the SDDC that are packaged with the log management agent.	You must configure the log management agent to forward logs to the log management VIP.
VCF-PB-CR-RCMD-015	Configure the log management agent to transmit logs from the Protection and Recovery instance to log management in the VMware Cloud Foundation instance using the log management ingestion API, `cfapi`, on port `9543`.	Ensures the transmission of logs from the Protection and Recovery instance to log management are encrypted.	You must update the configuration on the Protection and Recovery instance to send logs to log management by using the ingestion API, `cfapi`, on port `9543` by editing the agent configuration (`/etc/liagent.ini`).

Lateral Security with vDefend

The *Lateral Security with vDefend* blueprints provides centralized enterprise-grade security services for a VMware Cloud Foundation platform using the VMware vDefend security solutions.

Understanding the Lateral Security with vDefend Blueprints

A blueprint is:

- A prescriptive end-to-end security design created to accelerate time-to-value when protecting the VCF private cloud platform.
- Has a prescriptive selection of deployment and configurations models that are chosen from the design library to conform to the design profile of the blueprints.
- Provides a set of planning, implementation, and design elements (requirements and recommendations) based on the chosen deployment models.

Choosing Lateral Security with vDefend Blueprint

This design guide covers the following pre-defined blueprints:

- License Hub for vDefend and Avi Load Balancer
- Security Services Platform for VMware Cloud Foundation
- Management Domain Security
- Workload Domain Security
- Distributed Virtual Patching

vDefend Distributed Firewall for VMware Cloud Foundation

VMware vDefend is built into the VCF private cloud, which is activated by providing a valid license key. vDefend Firewall configurations for your workloads are done through the respective NSX Manager instances for the workload domains. An NSX instance can manage a single or multiple workload domains. To provide security services, vDefend addresses the configurations for the respective workload domains.

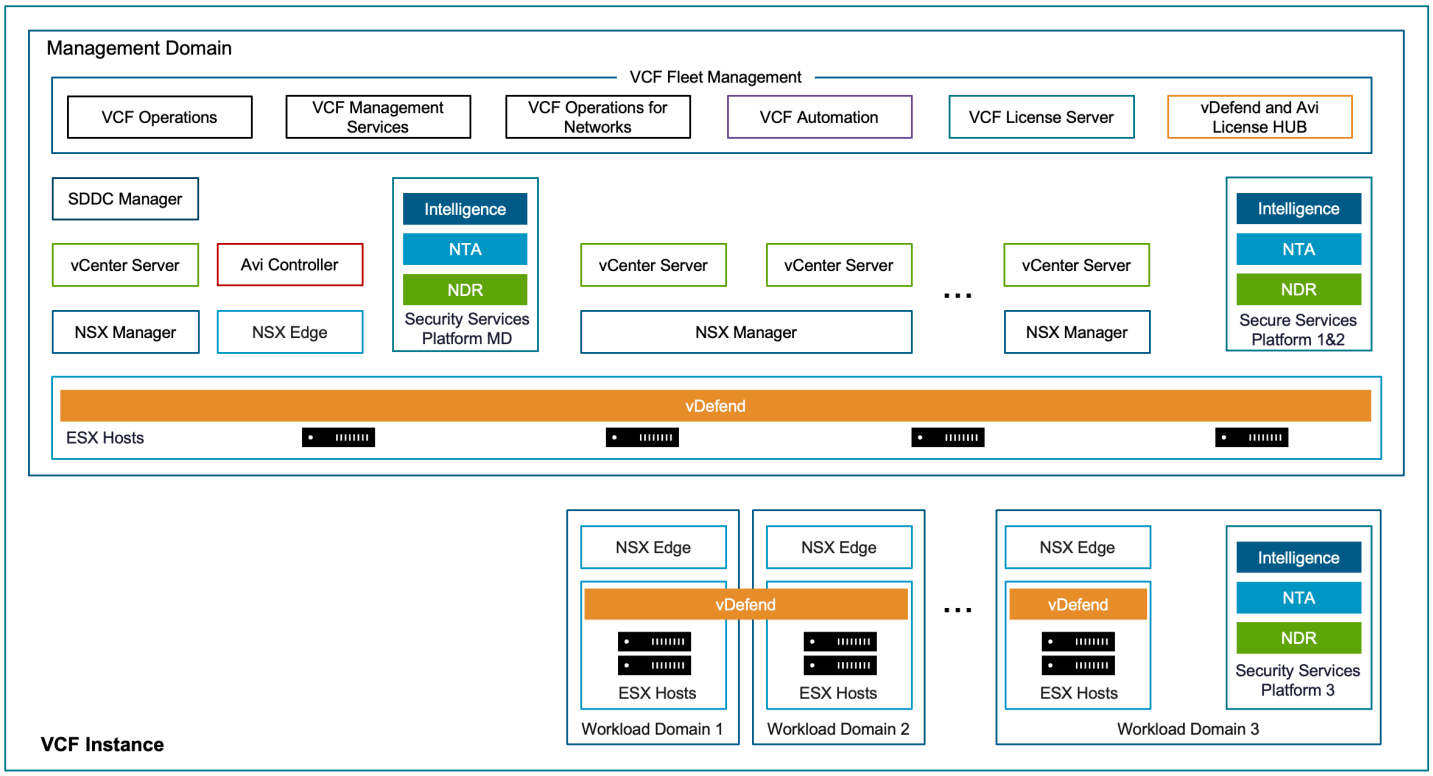
Note:

Starting VCF 9.1, the management domain NSX Manager can be shared with workload domains.

Architecture Overview

vDefend Distributed Firewall (DFW) provides complete Layer 2 to Layer 7 visibility and enforcement with automated policy delivery for VMware Cloud Foundation private cloud workloads and management components. The DFW is implemented as a software-defined architecture, with a centralized management plane, hybrid control plane, and distributed data plane. This software-defined architecture enables firewalling at the scale and speed that is required for dynamic private cloud environments. The DFW delivers lateral security for different workload types: virtual machines, containers, and bare metal workloads.

Figure 217: Lateral Security with vDefend for VMware Cloud Foundation



Note:

The DFW operates on every virtual NIC (vNIC) of each VM, monitoring every packet entering or exiting without hairpinning or network re-architecture. Attached to the VM's vNIC, the DFW policy, and state seamlessly follow the VM during vMotion events, maintaining its security posture. Kubernetes clusters running the Antrea Container Network Interface (CNI) plugin can be easily integrated with NSX, thus providing an efficient centralized security management policy for container workloads using VMware vDefend. Applications running on a Bare Metal server can also be protected with the vDefend DFW. The latter is facilitated by the Security Services Platform, which implements a new, horizontally scalable architecture designed to support, alongside advanced security services, micro-segmentation for Bare Metal servers.

Lateral Security with vDefend Design Objectives

Objective	Description
Main objective	Provide centralized enterprise-grade security services for VMware Cloud Foundation using the VMware vDefend security solutions.
VMware Cloud Foundation architecture support	• VCF Fleet in a Single Site with Minimal Footprint • VCF Fleet in a Single Site • VCF Fleet with Multiple Sites in a Single Region **Note:** Federated NSX environments across workload domains in single or multiple VMware Cloud Foundation are not in the scope of the Lateral security with vDefend Blueprints..
Scope of implementation	• Deployment of solution components: – License Hub for vDefend and Avi Load Balancer • Deployment of solution components: – Security Services Platform • Configuration for VMware Cloud Foundation management domain: – vDefend Distributed Firewall • Configuration for VMware Cloud Foundation workload domain: – vDefend Firewall Security Journey

Support Matrix

The Lateral Security with vDefend blueprints are compatible with specific versions of the VMware products that are used for implementing the solution. For more information on product version interoperability and lifecycle phase, see VMware Product Interoperability Matrix.

Product Group	Version	Details
License Hub for vDefend and Avi Load Balancer	2.0	License Hub 2.0 for vDefend and Avi Load Balancer Release Notes
Security Services Platform	5.2	Security Services Platform 5.2 Release Notes

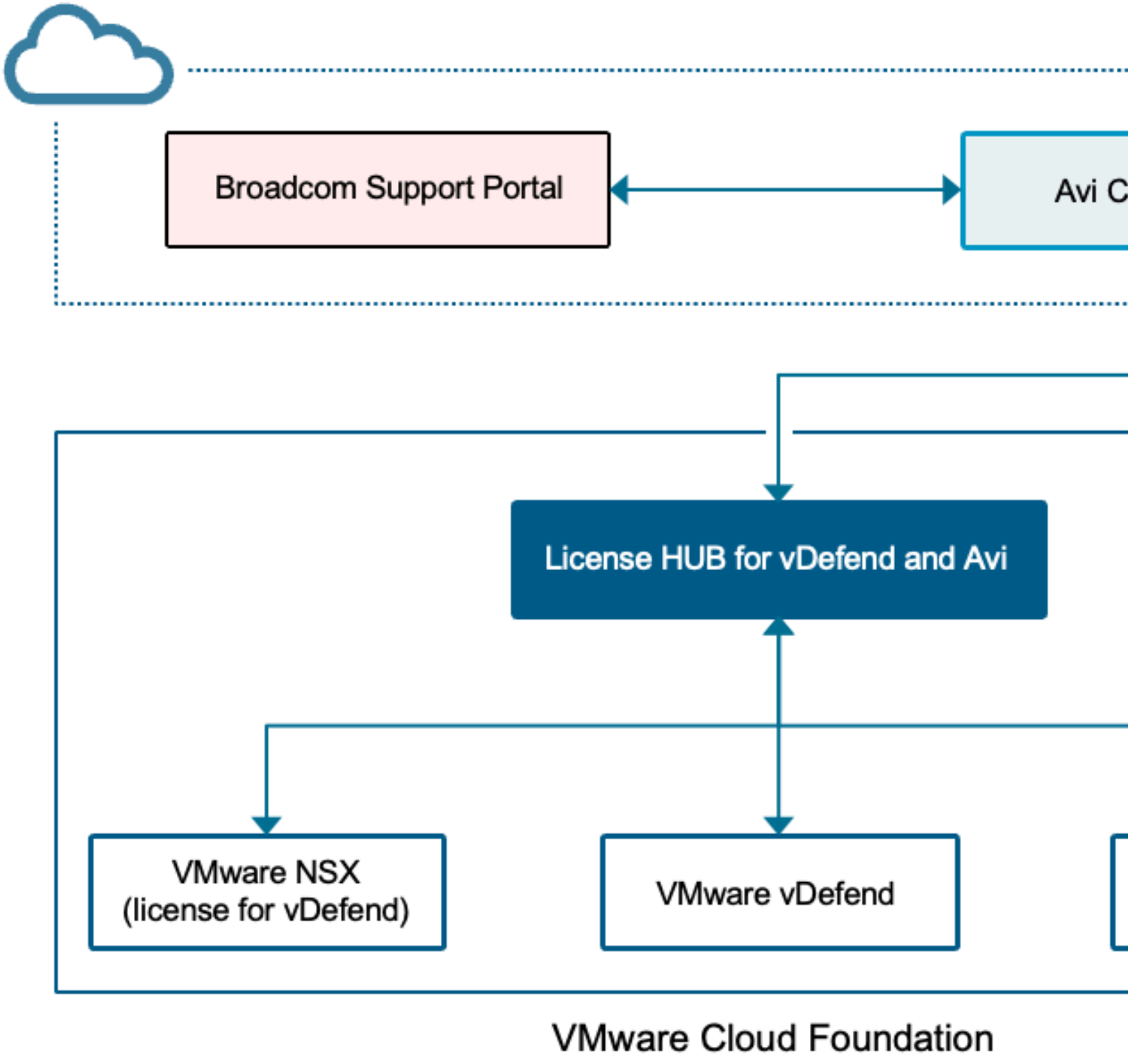
License Hub for vDefend and Avi Load Balancer

License Hub for VMware vDefend and VMware Avi Load Balancer is a centralized licensing platform that manages and distributes subscription-based licenses for VMware vDefend and VMware Avi Load Balancer. The License Hub is a mandatory component when using the VMware vDefend and Avi Load Balancer add-ons. This section provides the design details and decisions for deploying License Hub on VCF.

Architecture Overview

License management for VMware vDefend and VMware Avi Load Balancer involves using digitally signed subscription license keys to activate features corresponding to the purchased license editions. License utilization reports are instrumental in monitoring product usage and consumption across diverse deployments.

Figure 218: License Hub Architecture



License Hub Core Components

Component	Component Details
Avi Cloud Console	A cloud-based platform that serves as the source for VMware vDefend and VMware Avi Load Balancers licenses.
License Hub appliance	This appliance manages the License Hub service. You can perform tasks such as upgrading the License Hub service or viewing diagnostic information.
License Hub service	Also known as License Hub. This service is started by the License Hub appliance. You can manage licenses and perform related tasks through this service.
License Edition	See the vDefend Feature and Edition Guide for feature entitlement details for VMware vDefend products.
Endpoints	Endpoints are the downstream components that consume license from the License Hub: • NSX Manager Cluster • Security Services Platform • Avi Controller Cluster

Operating Modes

Mode	Details
Connected Mode	• The License Hub has Outbound HTTPS Internet access to the Avi Cloud Console portal. The License Hub supports Proxy configuration for secure Internet communications. • Registration is automatic, and licenses are polled from the Avi Cloud Console to the License Hub every 15 minutes without file exchange. • License usage report is consolidated and provided to the Avi Cloud Console every 24 hours.
Disconnected Mode	• No direct connectivity to Avi Cloud Console. • Suitable for isolated or air-gapped environments. • Registration is file-based: the file is manually exported from the License Hub and uploaded to the Avi Cloud Console. • The activation file is downloaded from Avi Cloud Console and imported to the License Hub. • The license file is downloaded from the Avi Cloud Console and manually imported to License Hub. • In order to avoid license expiration, the license must be manually uploaded at least once every six months.
Private Mode	• For highly secure, classified, or operationally sensitive environments where even manual Internet access for licensing is restricted. This mode is only for Broadcom-certified private customers.

System Requirements

The License Hub appliance has the following system requirements.

vCPU	Memory, GB	Storage, GB
6	12	256

Networking Requirements

Any supported network type, including DVPG, NSX Segment (VLAN or Overlay), and VPC subnet, can be used to deploy the License Hub as long as it has Layer 3 connectivity to the endpoints that needs to be licensed.

- Three IP addresses:
  - Single IP address for appliance management.
  - One address pool with two contiguous IP addresses for the License Hub service. It is not required that the management IP address and the IP address pool be on the same subnet as long as they are reachable.
- Two DNS records:

  - One FQDN for the appliance. This FQDN must map to the management IP address of the appliance (for example, licensehub20.example.com maps to 172.16.111.39).
  - One FQDN for Kafka. This FQDN must map to the second IP address of the IP address pool (for example, licensehub-kafka.example.com maps to 172.16.111.41).
- Latency:
  - Network latency between License Hub and the endpoints is 150 ms or less.
  - Storage latency is 10 ms or less.
- Ports and Protocols:
  - License Hub management port - TCP/5480.
  - License Hub Services - TCP/443 and TCP/9092
  - Refer to the Ports and Protocols web page for the full list of the ports and protocols that are required by the License Hub Service.
- License Hub endpoint connectivity types:
  - Dynamic - connection type for the endpoints that are not behind a load balancer or NAT.
  - Static - connection type for the endpoints that are behind a load balancer or NAT.
- Avi Cloud Console connectivity:
  - When in Connected mode, License Hub connects to Avi Cloud Console, which needs Internet access. If the management network can't connect directly to the Internet, a proxy can be set up.

Scale

- A single License Hub supports up to 120 endpoints to be connected, where endpoints could be a mix of NSX Manager Clusters, vDefend Security Services Platform, Avi Controller Clusters.
- Multiple License Hub can be deployed when the number of endpoints to license is above 120.

License Hub Design Requirements for All Models

Table 476: License Hub Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
LH-REQD-LIC-001	Maintain a 1:1 mapping between License Hub and VCF Fleet.	Consistency in architecture and operations is standardized.	If the number of endpoints within a VCF Fleet exceeds the limit of 120, then deploy another License Hub.
LH-REQD-LIC-002	Deploy the License Hub appliance in the management domain of the first VCF Instance of the Fleet.	• License Hub provides centralized licensing management for VMware vDefend and VMware Avi Load Balancer add-ons for VCF • Facilitates frequent communication with endpoints.	The management domain needs proper resource sizing.
LH-REQD-LIC-003	Connect the License Hub appliance to the VCF VM Management network.	Allows for ease of management.	Network latency between License Hub and the endpoints is 150 ms or less.
LH-REQD-LIC-004	Provision a single IP address from the VCF VM management network CIDR.	The License Hub appliance required a single IP address for management.	The VCF VM management network must be configured with a sufficient CIDR block.
LH-REQD-LIC-005	Provision an address pool with two contiguous IP addresses from the VCF VM management network.	The License Hub service requires two contiguous IP addresses.	The VCF VM management network must be configured with a sufficient CIDR block.

Design Requirement ID	Design Requirement	Justification	Implication
LH-REQD-LIC-006	Configure a DNS (A) record for the License Hub appliance.	The License Hub appliance is accessible by an easy-to-remember FQDN for management.	Assumes that DNS infrastructure is available.
LH-REQD-LIC-007	Configure a DNS (A) record for the License Hub service (Kafka).	Used by the License Hub service for: • Registration & Reporting • License Management • Endpoint Management	Assumes that DNS infrastructure is available.
LH-REQD-LIC-008	Use storage policy that includes one eligible shared datastore, with all cluster ESX hosts having read and write access.	License Hub requires shared storage for deployment.	Shared storage latency must be 10 ms or less.

License Hub Design Recommendations for All Models

Table 477: License Hub Design Recommendations

Design Requirement ID	Design Requirement	Justification	Implication
LH-RECD-LIC-001	Change the self-signed Ingress interface certificate with a validated CA-Signed Certificate via an Ingress CSR.	Secures client connections against browser trust alerts and mitigates internal corporate security scan violations.	The certificate chain must follow strict cryptographic ordering: Ingress Cert -> Intermediate Cert -> Root CA Cert.
LH-RECD-LIC-002	Configure local account password rotation policies targeting a threshold of every 150 days.	Local admin/audit account tokens expire after 180 days. Aligning to a 150-day window ensures corporate rotation compliance before platform flags trip.	Failure to rotate will lock admin access.
LH-RECD-LIC-003	Integrate the License Hub instance with an external identity platform using Active Directory over LDAP or OpenLDAP.	Promotes Role-Based Access Control compliance by mapping existing corporate user stores directly to system operational rights.	Only one active identity provider configuration can run concurrently on the platform engine.
LH-RECD-LIC-004	Configure License Hub backups.	Mitigates data loss and ensures a fast recovery path.	Backups should target secure SFTP locations.
LH-RECD-LIC-005	Configure external logging with a central Syslog Server.	Simplifies security auditability and log preservation.	None.
LH-RECD-LIC-006	Perform lifecycle management of License Hub instances via the the License Hub appliance (port 5480) interface.	License Hub appliance management interface is used to upgrade.	License Hub upgrade packages need to be downloaded manually from the Broadcom support portal and uploaded to the License Hub appliance.

License Hub for vDefend and Avi Model

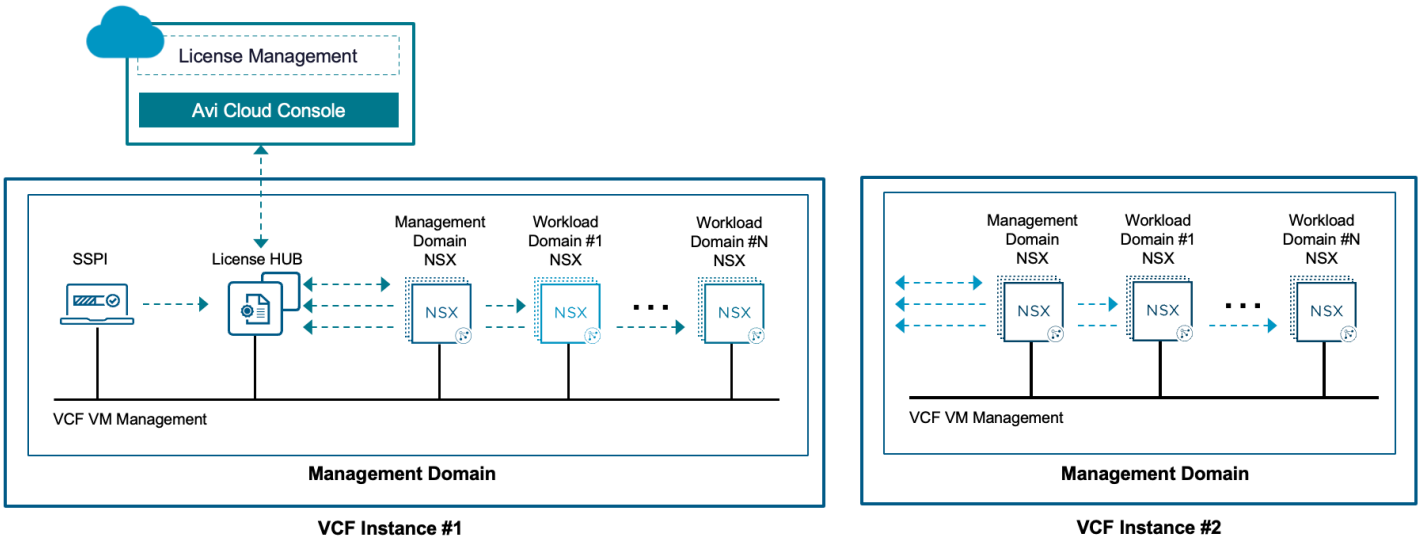
This section covers the design details for deploying the License Hub for vDefend and Avi for VMware Cloud Foundation.

Deployment Model

Deploy the License Hub for VMware vDefend and VMware Avi Load Balancer instance in the first VCF instance management domain of the VCF Fleet.

License Hub Scale

- Total Endpoints Connected to a single License Hub instance = 120, where Endpoints could be a mix of NSX Manager clusters, vDefend Security Services Platform, Avi Controller.
- Multiple License Hub instances can be deployed when the number of endpoints to license is above 120


License Management
Avi Cloud Console
SSPI
License HUB
Management Domain NSX
Workload Domain #1 NSX
Workload Domain #N NSX
NSX
VCF VM Management
Management Domain
VCF Instance #1
Management Domain NSX
Workload Domain #1 NSX
Workload Domain #N NSX
NSX
VCF VM Management
Management Domain
VCF Instance #2

License Hub Components

Component	Component Details
Avi Cloud Console	A cloud-based platform that serves as the source for VMware vDefend and VMware Avi Load Balancers licenses.
License Edition	Two license editions are available to use for the VMware vDefend product line: • VMware vDefend • VMware vDefend Advanced Threat Prevention Add-On See the vDefend Feature and Edition Guide for feature entitlement details for VMware vDefend products.
License Hub Instance	• The License Hub is deployed as a Security Service Platform instance comprising a single worker node and a single controller node. • License Hub acts as the local license broker between the Avi Cloud Console and the downstream endpoints. • The License Hub instance is identified by a unique asset ID that is established during registration with Avi Cloud Console.
Endpoint Type	Endpoints are the downstream components that consume license from the License Hub: • NSX Manager cluster • Security Services Platform • Avi Controller

License Hub Operating Modes

Mode	Details
Connected Mode	• The License Hub has live outbound connectivity to the Avi Cloud Console portal. • Registration is automatic, and licenses are polled from the Avi Cloud Console to the License Hub every 15 minutes without file exchange. • License usage report is consolidated and provided to the Avi Cloud Console every 24 hours.
Disconnected Mode	• No direct connectivity to Avi Cloud Console. • Suitable for isolated or air-gapped environments. • Registration is file-based: the file is manually exported from the License Hub and uploaded to the Avi Cloud Console. • The activation file is downloaded from Avi Cloud Console and imported to the License Hub. • The license file is downloaded from the Avi Cloud Console and manually imported to License Hub. • Manually uploading the license at least once every six months is required to prevent license expiration.

License Hub System Requirements

Component	vCPU	Memory, GB	Storage, GB
Installer	4	6	400
License Hub Controller	2	8	75
License Hub Worker	4	16	75

- Content library storage: 50 GB
- Persistent Volume Claim (PVC) storage: 50 GB

Note:

As per License Hub 5.1.2, the Installer is an individual appliance, separate from the standard Security Services Platform Installer.

If a resource reservation is activated, you must add one more worker node when you calculate the required amount of resources.

License HUB instance IP address requirements

A License Hub instance requires a single IP subnet with two distinct IP pools: one for the controller and worker nodes, and another for License Hub services. The Installer requires a single IP from a subnet that has L2/L3 connectivity to the License Hub pool subnet.

Installer	Node IP Pool	Service IP Pool
1	4	4

- Refer to the VMware Ports and Protocols web page for detailed information on the ports and protocols that are required by the License Hub.
- Refer to the License Hub System Requirements for detailed information on the system specifications that are required for deploying and managing the License Hub.

Design Elements for Security Services Platform for VMware Cloud Foundation

Deployment Specification

Table 478: Design Decisions on Deployment of Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-001	Deploy the Security Services Platform Installer virtual machines and Security Services Platform instances in the VMware Cloud Foundation management domain primary vSphere cluster or dedicate a vSphere cluster. **Note:** In the case of insufficient CPU/ Memory/Storage resources in the management domain deploy the Security Services Platform instance in the respective workload domain.	• TheSecurity Services Platform Installer and Security Services Platformperform management and operational functions for VMware Cloud Foundation private cloud. • Network latency between Security Services Platform Installer, Security Services Platform, and the respective NSX Manager cluster that is used must be 10 ms or less.	• Each SSP instance reserves resources (CPU, memory, storage) and may affect resource availability for the VMware Cloud Foundation management domain. • You must ensure that the VMware Cloud Foundationmanagement domain is properly sized.
SSP-REQD-CFG-002	Deploy Security Services Platform Installer in the VMware Cloud Foundation management domain	• Security Services Platform Installer automates the deployment, configuration and lifecycle management of the Security Services Platform. • Each Security Services Platform Installer can manage up to five(5) Security Services Platform instances.	If there are more than five(5) instances of Security Services Platform, an additional Security Services Platform Installer is required, which can manage an additional 5 Security Services Platform instances.

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-003	Deploy individualSecurity Services Platforminstances for each NSX Manager domain.	A singleSecurity Services Platform can onboard one NSX Manager cluster.	• You must individually deploy and manage a Security Services Platformper NSX Manager cluster. • You must maintain Security Services Platformwithin its scalability and configuration limit maximums.
SSP-REQD-CFG-004	Protect the Security Services Platform Installer and Security Services Platform nodes with vSphere High Availability.	Supports the availability objectives for Security Services Platform without requiring manual intervention during an ESX host failure event.	None.
SSP-REQD-CFG-005	Do not apply vSphere Distributed Resource Scheduler (DRS) anti-affinity rules for the Security Services Platform instance nodes. **Note:** vSphere DRS is mandatory on the cluster used to deploy the Security Services Platform instance.	The Security Services Platform instance runs a component called "Machine Health Check" (MHC) that monitors all its nodes(controller & worker nodes)health and automatically replaces unhealthy nodes.	Since the MCH can dynamically redeploy failed Security Services Platform node(s), manual anti-affinity rule could be ineffective.

Table 479: Design Decisions on Deployment of Security Services Platform in Multiple Availability Zones

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-006	When using two availability zones, add the Security Services Platform Installer and the Security Services Platform nodes to the VM group of the first availability zone.	Ensures that the Security Services Platform Installer and the Security Services Platform nodes are powered on by default in the primary availability zone hosts group.	After the implementation of the second availability zone for the management domain, you must update the VM group for the primary availability zone virtual machines to include the Security Services Platform Installer and the Security Services Platform nodes.
SSP-REQD-CFG-007	Disable "Machine Health Check" (MHC) on the Security Services Platform instance.	"Machine Health Check" might dynamically redeploySecurity Services Platform VMs on the second availability zone if still enabled.	The Security Services Platform instance high availability relies only on vSphere HA (host health) and not on the Security Services Platform VMs health.

Table 480: Design Decisions on Sizing of Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-008	Deploy the Security Services Platform with a minimum of 2 worker nodes, known as Medium form factor which supports the following core security features in a production-grade deployment: Security Intelligence, Rule Analysis, and Metrics. **Note:** To activate addtional features, the Security Services Platform first needs to be scaled-out to a Large or Extra Large (XL) form factor as covered in the section below.	The core security features supportd by a medium form factor can be a starter.	Scaling-out the worker nodes to support additional features is not allowed unless the form factor is changed from Medium to Large or XL. **Note:** In a brownfield deployment utilize the Host Capacity Dashboad, which gives the recommendation related to optimal scaling.
SSP-REQD-CFG-009	Change the Security Services Platform form factor to Large, which can support activating all of the available features.	• A Large form factor Security Services Platform can support all of the available features. • Number of worker nodes may need to be scaled out to support activating additonal features.	Both the form factor and the number of deployed worker nodes of the Security Services Platform instance impacts the compute and storage resource requirements of the target vSphere cluster.
SSP-REQD-CFG-010	Change the Security Services Platform form factor to Extra Large if the NSX Manager inventory exceeds the configuration limits supported by the Security Services Platform large form factor.	• An XL form factor Security Services Platform can support all of the available features. • Number of worker nodes may need to be scaled out to support activating additonal features.	The XL form factor is required if the configuration inventory in the NSX manager exceeds the limits supported by a large form factor Security Services Platform.
SSP-REQD-CFG-011	Utilize the vSphere resource pool that is created by the Security Services Platform instance deployment.	vSphere resource pool is essential to manage computing resources efficiently, ensuring fair distribution, improved performance, and flexible resource allocation.	None.
SSP-REQD-CFG-012	Deploy the Security Services Platform instance with resource reservation enabled.	Enables the allocation and proper usage of CPU and Memory resources and optimal performance for the Security Services Platform instances.	• With Resource Reservation activated, resource for an additional one worker node is required, increasing the total resources requirement to 80 vCPU, 286GB Mem, 3496GB Storage. • If the resource demand exceeds available resources, the virtual machine may not be able to get enough resources, resulting in performance degradation.

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-013	For vSAN datastore use the vSAN Default Storage Policy.	• Security Services Platform instance deployment requires the use of vSphere Storage Based Policy Management (SPBM). • Storage usage is optimized when thin provisioning is used; there is no need to create a custom storage policy.	None.
SSP-REQD-CFG-014	For datastores different than vSAN, create a vSphere Storage Policy Based Management (SPBM) using vSphere tag.	Tag based SPBM policy is supported by Security Services Platform.	This must be done manually or via automation.

Network Design

Table 481: Design Decisions on the Network Segment for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-001	Deploy theSecurity Services Platform Installer on the VMware Cloud Foundation VM management network.	• Allows for ease of management for the Security Services Platform Installer • The Security Services Platform Installer requires a single IP address.	The VMware Cloud Foundation VM management network must be configured with a sufficient CIDR block.
SSP-REQD-NET-002	Deploy the Security Services Platform instances on a dedicated network (DVPG, NSX Segment ot VPC Public subnet) **Note:** This network should have IP connectivity to the Security Services Platform Installer, NSX Manager cluster and the ESX Transport Nodes.	Allows for the design of the Security Services Platform instances' IP space/ s independently from the VMware Cloud Foundation VM management network.	• The physical network fabric must be configured to route the traffic from the Security Services Platform instances to the NSX Manager cluster, ESX Transport Nodes management interface (vmk0) and the vCenter Server. • The network must be created manually or via automation.

Table 482: Design Decisions on IP Addressing for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-003	Use static IP assignment for the Security Services Platform Installer from the VMware Cloud Foundation VM management network.	The Security Services Platform Installer uses management IPs to communicate with vCenter Server, NSX Manager cluster and the Security Services Platform instances.	The VMware Cloud Foundation VM management network must have sufficient IP space available.
SSP-REQD-NET-004	Dedicate two IP ranges from the Security Services Platform network subnet: • Node IP Pool - 22 IPs. • Service IP Pool - 11IPs.	Designing the Security Services Platform IP addressing for max scale deployment ensures straightforward future platform scalability.	• The IP addresses must be reserved. • The Service IP pool range must be contiguous.

Table 483: Design Decisions on Name Resolution for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-005	Configure a DNS (A) record for the Security Services Platform Installer.	• The Security Services Platform Installer is accessible by an easy-to-remember FQDN or directly with an IP address. • Security Services Platform uses the Security Services Platform Installer FQDN to access the registry.	Assumes DNS infrastructure is available.
SSP-REQD-NET-006	Configure a DNS (A) record for the Security Services Platform ingress.	• The Security Services Platform UI is accessible by an easy-to-remember FQDN or directly with an IP address. • Used by NSX Managers and NSX Transport Nodes to send Metrics to Security Services Platform.	Assumes DNS infrastructure is available.
SSP-REQD-NET-007	Configure a DNS (A) record for the Security Services Platform messaging service.	Used by NSX Transport Nodes to send flows to Security Services Platform instance.	Assumes DNS infrastructure is available.

Table 484: Design Decisions on Time Synchronization for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-008	Configure time synchronization by using an NTP time for the Security Services Platform Installer and the Security Services Platform instances.	Prevents from time synchronization issues.	An operational NTP service must be available in the environment.

Security Services Platform for VMware Cloud Foundation

The Security Services Platform for VMware Cloud Foundation blueprint covers the design details and decisions, planning and preparation, and implementation steps for deploying the vDefend Security Services Platform on VCF.

Architecture Overview

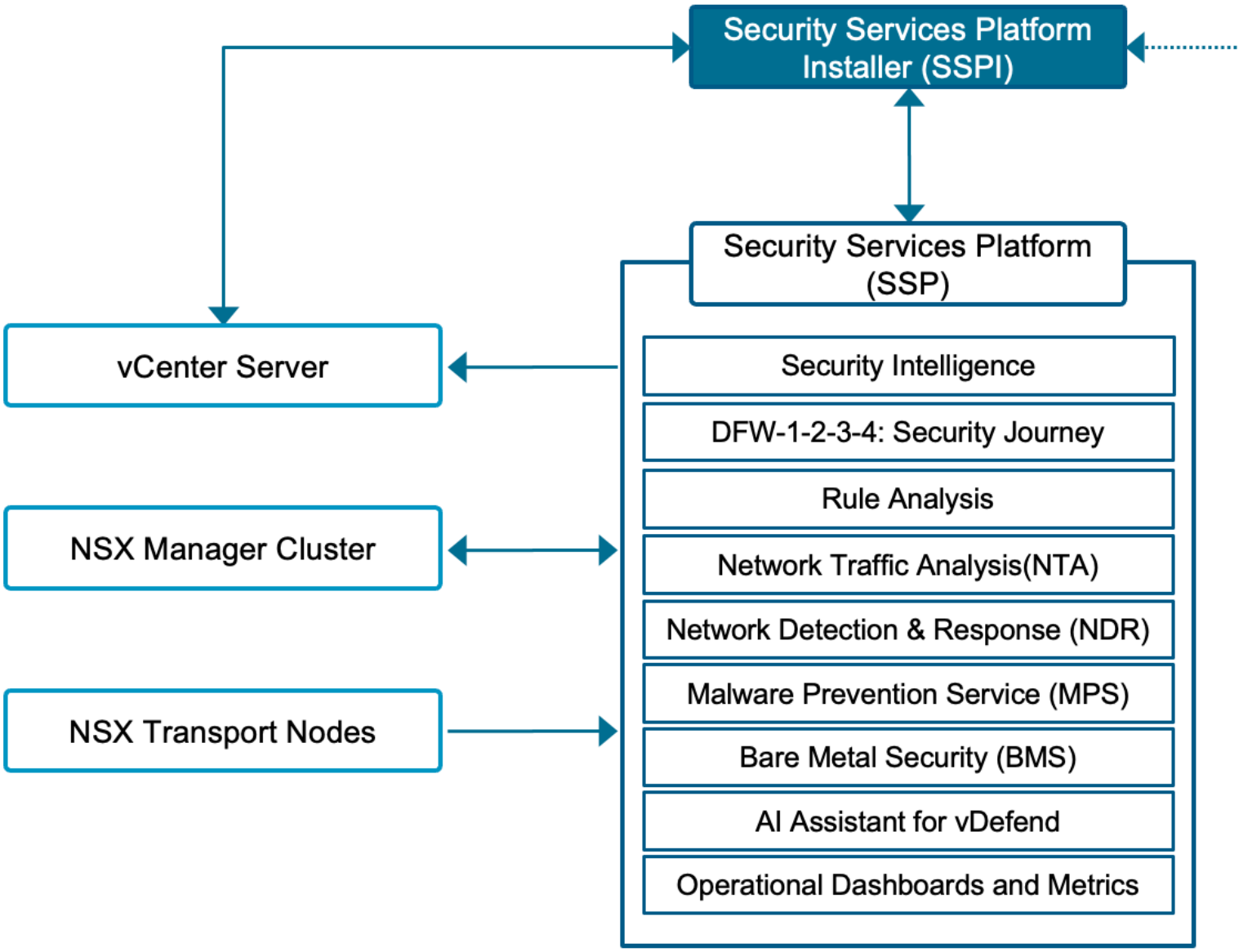
The Security Services Platform (SSP) is VMware vDefend next-generation solution that provides visibility and advanced security capabilities in VMware Cloud Foundation. SSP is a high-performance, scalable platform that runs vDefend security services, including Security Intelligence, Network Detection and Response, Malware Prevention, and more.

Security services provided by the Security Services Platform:

- Segmentation and Advanced Threat Prevention Score along with Security Assessment Report to help users assess their security posture.
- vDefend Security Journey: Built-in automated workflows to speed-up DFW and Advanced Threat Prevention(ATP) deployments.
- Firewall Rule Analysis: Identify potential security gaps in the DFW configuration and manage the DFW rules more efficiently.
- Security Intelligence: Visibility about the security posture and the applications network behavior, and streamlines Distributed Firewall Security Policies configuration thanks to a recommendation engine.
- Network Traffic Analysis (NTA): Detects abnormal network behaviors, such as DNS tunneling, Data Exfiltration, or port scanning

- Network Detection and Response (NDR): Collects, scores, and correlates atomic security events from IDPS, NTA, and Malware Prevention into campaign blueprints. This feature enables security teams to reduce the Mean Time To Detect/ Respond and focus on the incidents that really matter.
- Malware Prevention Service (MPS): Detects malware being transferred over the network and prevents them from writing to the disk. An optional cloud sandboxing can also be used to analyze Day-0 malware activity.
- Security Metrics: Collects and stores metrics coming from various platform components - used for multiple purposes, such as sizing, monitoring, or troubleshooting. Provides data for Top N analysis, lifetime counters, and historical trends.
- Distributed Firewall for Bare Metal Workloads
- Out-of-band NDR sensor for threat visibility beyond VCF workloads
- AI Assistant for VMware® vDefend™ covering both Distributed Firewall and Threat Defense.

Figure 219: Security Services Platform Architecture



Security Services Platform Detailed Design

The following models for the Security Services Platform aligns with the design profile for this blueprint.

Layer	Selected Model	Additional Information
Virtual Infrastructure	Security Services Platform Deployment Model	• Management Domain Deployment Model - Security Services Platform Installer and Security Services Platform instances are all deployed in the management domain. • Workload Domain Deployment Model - Security Services Platform Installer is deployed in the management domain and Security Services Platform instances are deployed in their respective workload domains.
	Security Services Platform Sizing	Security Services Platform compute, memory, storage, and reservations requirements details.
Networking	Security Services Platform Networking	Security Services Platform networking requirements details.

Security Services Platform for VCF Model

This section covers the design details and decisions for implementing the Security Services Platform for VMware Cloud Foundation.

vDefend Security Services Platform

The Security Services Platform for VMware vDefend employs an appliance-based deployment model. It is delivered through the Security Services Platform Installer (SSPI). The Security Services Platform Installer also serves various critical functions in managing the operational integrity of the Security Services Platform , providing upgrades, scale-out, backup and restore capabilities, password management, and platform diagnostic functionalities. Furthermore, the Security Services Platform Installer hosts the packages that are needed to deploy, operate, and upgrade the Security Services Platform.

A single Security Services Platform Installer can deploy and manage up to five independent Security Services Platform instances

- Security Services Platform Deployment Model
- Security Services Platform Sizing
- Security Services Platform Networking

Security Services Platform Deployment Model

Security Services Platform is deployed to support the design objectives for solution component availability and the number of workloads that the solution must support.

Deployment Models

Even though a single Security Services Platform Installer can deploy and manage up to five Security Services Platform instance, each of thoseSecurity Services Platform instances can only onboard a single NSX Manager cluster. This results in a M:1:1 mapping for deployment model alignment.

In VMware Cloud Foundation, you can have one or more NSX Manager clusters (single NSX Managers in a Simple NSX Manager Model deployment):

- One for the management domain.
- One for the first workload domain.
- Additional workload domains may share an existing workload domain NSX Manager cluster or have dedicated NSX Manager clusters.

Depending on the number of NSX Manager clusters, one or multiple Security Services Platform instances may be required. The Security Services Platform instance is not bound to a workload domain, but to an NSX Manager cluster. Workload domains, which share a single NSX Manager also share the Security Services Platform instance.

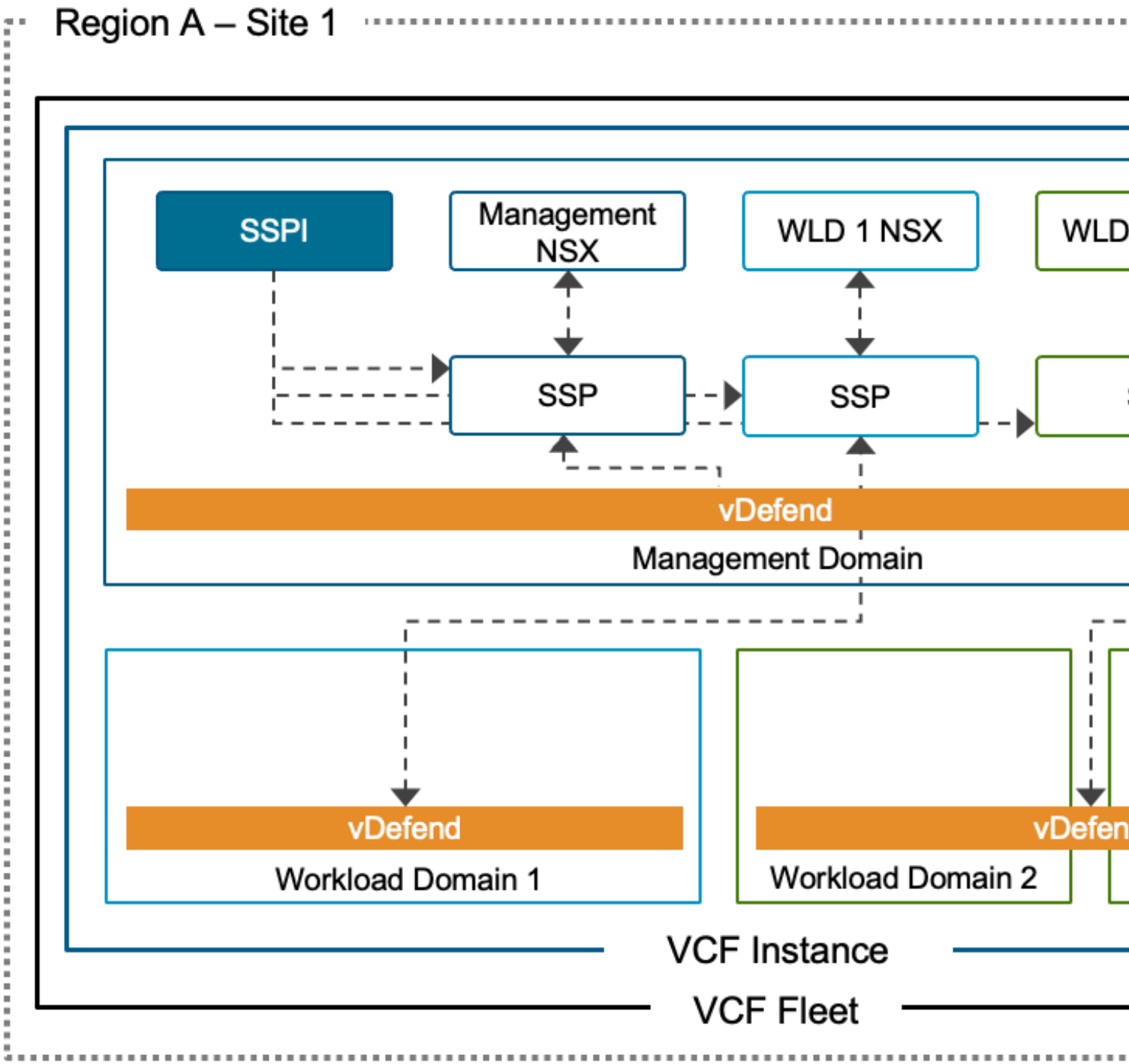
vSphere High Availability protects the platform by restarting the Security Services Platform Installer and Security Services Platform instance virtual machines on an alternate ESX host in the vSphere cluster, if affected by an ESX host failure.

The Security Services Platform, Kubernetes-based architecture, and declarative operational model ensure high availability for the services running upon it.

Management Domain Deployment Model

Attributes	Benefits	Implications
The Security Services Platform Installer as well as the Security Services Platform instances are deployed in the management domain.	• Consistent deployment with other VCF management components. • Enables centralized deployment and management for all SSP instances. • Enables a unified deployment model for any VCF architecture. • Enables separate life cycle management of SSP and workload domains. • Predictable performance with appropriately sized management domain. • Centralized security for all SSP instances with vDefend DFW. • Minor dependency on existing physical Firewalls.	• You may need to expand the resources in the management domain to accommodate the deployment of the required SSP instances. • Proper resource reservation is required to limit the potential impact on the performance of VCF management components. Using a separate vSphere cluster is recommended. • Accurate SSP instance sizing to address future workload domain growth.

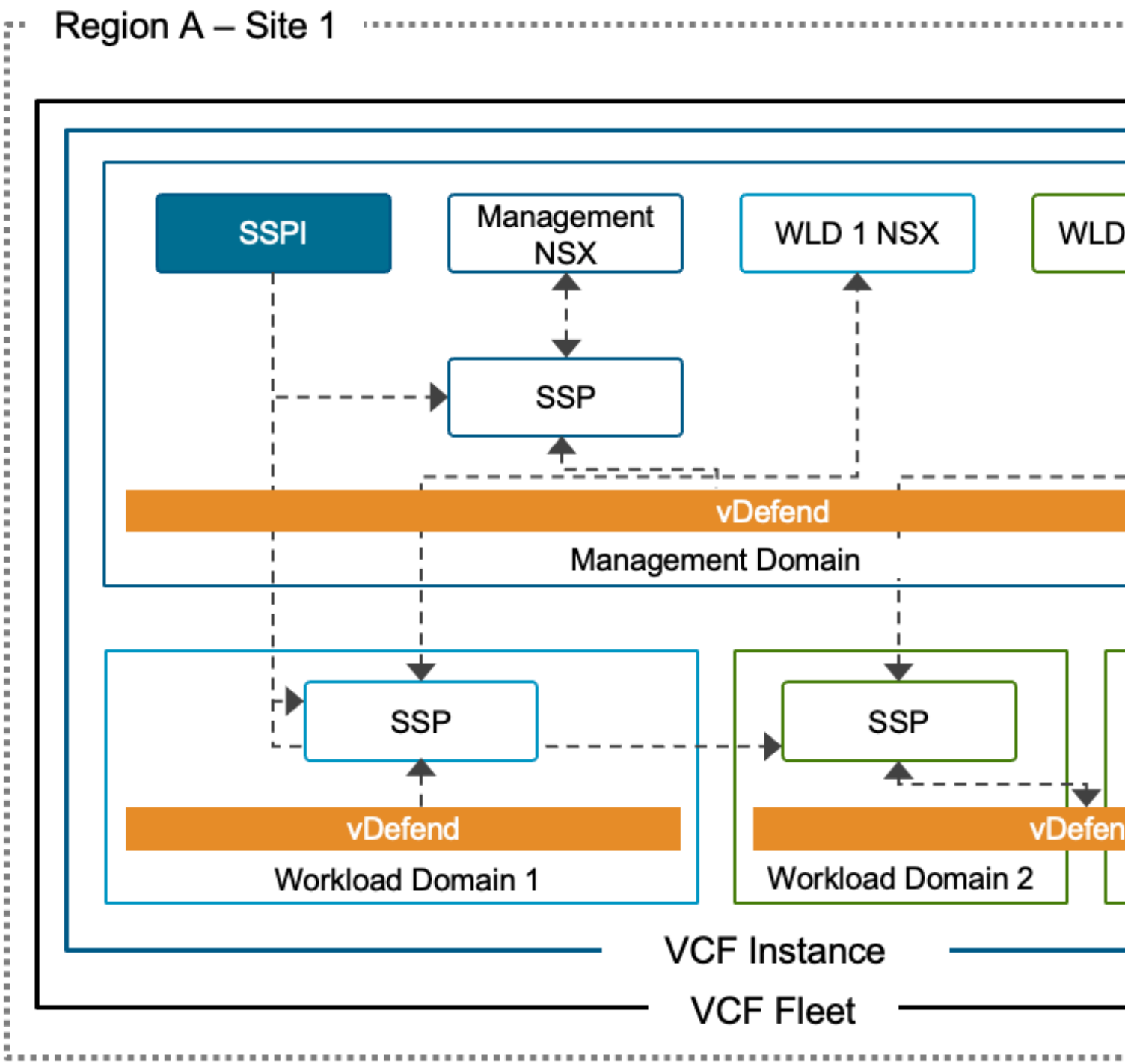
Figure 220: Security Services Platform Management Domain Deployment



Workload Domain Deployment Model

Attributes	Benefits	Implications
Security Services Platform Installer deployed in the management domain and Security Services Platform instances deployed in their respective workload domain.	• Resources availability (CPU/Memory/ Storage) in the workload domain/s is more common than in the management domain. • Enables SSP to scale easily with the workload domain growth. • Enables SSP placement proximate to the ingestion components (ESX, NSX Edge Nodes, SVMs).	• Dependency on the workload domain life cycle management. • Proper resource reservation is required to limit the potential impact on SSP. • There may be an increased dependence on physical firewall changes. • Potential network latency dependency.

Figure 221: Security Services Platform Workload Domain Deployment



Security Services Platform Deployment Model Design Requirements

Table 485: Design Decisions on Deployment of Security Services Platform

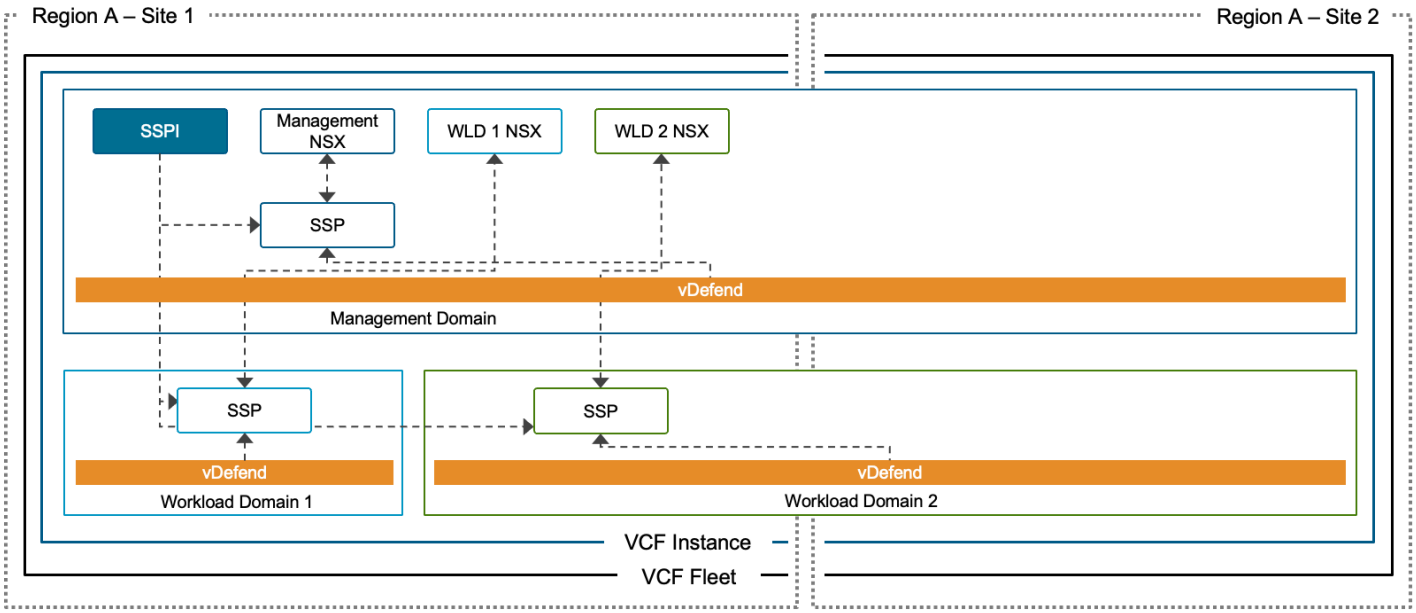
Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-001	Deploy the Security Services Platform Installer virtual machines and Security Services Platform instances in the VMware Cloud Foundation management domain primary vSphere cluster or dedicate a vSphere cluster. **Note:** In the case of insufficient CPU/ Memory/Storage resources in the management domain deploy the Security Services Platform instance in the respective workload domain.	• TheSecurity Services Platform Installer and Security Services Platformperform management and operational functions for VMware Cloud Foundation private cloud. • Network latency between Security Services Platform Installer, Security Services Platform, and the respective NSX Manager cluster that is used must be 10 ms or less.	• Each SSP instance reserves resources (CPU, memory, storage) and may affect resource availability for the VMware Cloud Foundation management domain. • You must ensure that the VMware Cloud Foundationmanagement domain is properly sized.
SSP-REQD-CFG-002	Deploy Security Services Platform Installer in the VMware Cloud Foundation management domain	• Security Services Platform Installer automates the deployment, configuration and lifecycle management of the Security Services Platform. • Each Security Services Platform Installer can manage up to five(5) Security Services Platform instances.	If there are more than five(5) instances of Security Services Platform, an additional Security Services Platform Installer is required, which can manage an additional 5 Security Services Platform instances.
SSP-REQD-CFG-003	Deploy individualSecurity Services Platforminstances for each NSX Manager domain.	A singleSecurity Services Platform can onboard one NSX Manager cluster.	• You must individually deploy and manage a Security Services Platformper NSX Manager cluster. • You must maintain Security Services Platformwithin its scalability and configuration limit maximums.
SSP-REQD-CFG-004	Protect the Security Services Platform Installer and Security Services Platform nodes with vSphere High Availability.	Supports the availability objectives for Security Services Platform without requiring manual intervention during an ESX host failure event.	None.

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-005	Do not apply vSphere Distributed Resource Scheduler (DRS) anti-affinity rules for the Security Services Platform instance nodes. **Note:** vSphere DRS is mandatory on the cluster used to deploy the Security Services Platform instance.	The Security Services Platform instance runs a component called "Machine Health Check" (MHC) that monitors all its nodes(controller & worker nodes)health and automatically replaces unhealthy nodes.	Since the MCH can dynamically redeploy failed Security Services Platform node(s), manual anti-affinity rule could be ineffective.

Deployment for Multiple Availability Zones

In an environment with multiple availability zones, the Security Services Platform instance/s runs in the first availability zone. If a failure occurs in the first availability zone, the Security Services Platform nodes are failed over to the second availability zone. The "Machine Health Check" feature must be disabled on the Security Services Platform instance.

Figure 222: Security Services Platform Deployment for Multiple Availability Zones



Multiple Availability Zones Design Requirements

Table 486: Design Decisions on Deployment of Security Services Platform in Multiple Availability Zones

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-006	When using two availability zones, add the Security Services Platform Installer and the Security Services Platform nodes to the VM group of the first availability zone.	Ensures that the Security Services Platform Installer and the Security Services Platform nodes are powered on by default in the primary availability zone hosts group.	After the implementation of the second availability zone for the management domain, you must update the VM group for the primary availability zone virtual machines to include the Security Services Platform Installer and the Security Services Platform nodes.
SSP-REQD-CFG-007	Disable "Machine Health Check" (MHC) on the Security Services Platform instance.	"Machine Health Check" might dynamically redeploySecurity Services Platform VMs on the second availability zone if still enabled.	The Security Services Platform instance high availability relies only on vSphere HA (host health) and not on the Security Services Platform VMs health.

Security Services Platform Sizing

Platform Sizing & Scaling for the Security Services Platform requires some careful planning since this requirement involve features activated, configuration inventory and the flows collected by the platform. For example, lowering the number of features frees up resources for additional flows and configuration inventory limits, and vice-versa. Utilizing the collected data for trending, the platform also makes recommendations based on various metrics and published configuration maximums.

Sizing Compute and Storage Resources

Summary steps for sizing can be summarized into the following 3 broad recommendations:

1- Choose Form Factor(FF) based on NSX configuration inventory limits: Select from medium, large, or extra-large(XL) Security Services Platform FF based on the VM counts, Groups, & Security Policies + Rules. Note that during the onboarding pre-checks, Security Services Platform will give a warning if NSX inventory scale exceeds 80% of limits and error if NSX inventory scale exceeds 90% for Medium/Large FF or 99% for XL FF. After Onboarding, Security Services Platform generates an alarm if NSX inventory exceeds the supported limits of the deployed instance.

2 - Choose the initial number of Worker Nodes based on the required features: Note that advanced security capabilities demand higher node counts and that the order of feature activation could impact resource utilization since some features use more resources than the others

3 - Scale the number of Worker Nodes based on Flows: The platform administrator can utilize both the flow capacity dashboard (requires a minimum of 3-days of Metrics collection) along with system alarms to aid in scaling recommendations.

For a production deployment with the following core security features: Security Intelligence, Rule Analysis, and Metrics, the Security Services Platform has the following minimum CPU, memory, and storage resources requirements.

Table 487: Components

Component Type	Function	Number of VMs
Security Services Platform Installer	SSP Installer VM	1
Controller nodes	Security Services Platform controller node VMs	3
Worker nodes	Deploy security features	2

Table 488: Total Resource Requirements

Component Type	vCPUs	Memory	Storage
SSP Installer VMs	4 vCPUs	6 GB	400 GB
Three controller node VMs	3 * 4 = 12 vCPUs	3 * 8 = 24 GB	3 * 155 = 465 GB
Two worker node VMs	2 * 16 = 32 vCPUs	2 * 64 = 128 GB	2 * 155 = 310 GB
Content Library	N/A	N/A	50 GB
PVC storage	N/A	N/A	2201 GB
Reserved for upgrade	16 vCPUs	64 GB	155 GB
Total resources	**64 vCPUs (32 cores)**	**222 GB**	**3.26 TB**

Note:

Each core has 2 vCPUs and for each vCPU 1 GHz is reserved.

To activate additional vDefend Advanced Threat Prevention capabilities, such as Network Traffic Analysis, Network Detection and Response, NDR Sensor, and Malware Prevention Service etc., you will need to add additional worker node VMs, each with the following resources:

Table 489: Extra Worker Node VM Resource Requirements

Resource Type	Extra Node VM Resource
vCPUs	16 vCPUs
Memory	64 GB
Storage	155 GB

Refer to the following Platform Details and Available Features section, Security Features Form Factor, for detailed information on the system requirement specifications that are required for deploying and managing the Security Services Platform based on required features. In the following design decision table, we will focus on the minimum system requirements supported for a production environment.

Table 490: Design Decisions on Sizing of Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-008	Deploy the Security Services Platform with a minimum of 2 worker nodes, known as Medium form factor which supports the following core security features in a production-grade deployment: Security Intelligence, Rule Analysis, and Metrics. **Note:** To activate addtional features, the Security Services Platform first needs to be scaled-out to a Large or Extra Large (XL) form factor as covered in the section below.	The core security features supportd by a medium form factor can be a starter.	Scaling-out the worker nodes to support additional features is not allowed unless the form factor is changed from Medium to Large or XL. **Note:** In a brownfield deployment utilize the Host Capacity Dashboad, which gives the recommendation related to optimal scaling.
SSP-REQD-CFG-009	Change the Security Services Platform form factor to Large, which can support activating all of the available features.	• A Large form factor Security Services Platform can support all of the available features. • Number of worker nodes may need to be scaled out to support activating additonal features.	Both the form factor and the number of deployed worker nodes of the Security Services Platform instance impacts the compute and storage resource requirements of the target vSphere cluster.
SSP-REQD-CFG-010	Change the Security Services Platform form factor to Extra Large if the NSX Manager inventory exceeds the configuration limits supported by the Security Services Platform large form factor.	• An XL form factor Security Services Platform can support all of the available features. • Number of worker nodes may need to be scaled out to support activating additonal features.	The XL form factor is required if the configuration inventory in the NSX manager exceeds the limits supported by a large form factor Security Services Platform.
SSP-REQD-CFG-011	Utilize the vSphere resource pool that is created by the Security Services Platform instance deployment.	vSphere resource pool is essential to manage computing resources efficiently, ensuring fair distribution, improved performance, and flexible resource allocation.	None.
SSP-REQD-CFG-012	Deploy the Security Services Platform instance with resource reservation enabled.	Enables the allocation and proper usage of CPU and Memory resources and optimal performance for the Security Services Platform instances.	• With Resource Reservation activated, resource for an additional one worker node is required, increasing the total resources requirement to 80 vCPU, 286GB Mem, 3496GB Storage. • If the resource demand exceeds available resources, the virtual machine may not be able to get enough resources, resulting in performance degradation.

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-CFG-013	For vSAN datastore use the vSAN Default Storage Policy.	• Security Services Platform instance deployment requires the use of vSphere Storage Based Policy Management (SPBM). • Storage usage is optimized when thin provisioning is used; there is no need to create a custom storage policy.	None.
SSP-REQD-CFG-014	For datastores different than vSAN, create a vSphere Storage Policy Based Management (SPBM) using vSphere tag.	Tag based SPBM policy is supported by Security Services Platform.	This must be done manually or via automation.

Security Services Platform Networking

Network Connectivity

IP reachability is required among the Security Services Platform Installer, Security Services Platform, the NSX Manager cluster, and NSX Transport Nodes (ESX hosts and NSX Edge Nodes). Furthermore, the Security Services Platform requires Layer 2 (L2) connection for their inter-node (controller and worker) communications.

This validated solution outlines a deployment model for SSP using vSphere Distributed Port Group (DVPG). However, deployment model using either types of NSX segments - Overlay or VLAN, as well as the NSX Virtual Private Cloud (VPC) subnets are supported. Public VPC subnets are recommended since it does not require NAT when allowing SSP to ingest data from components such as ESX and NSX Edge transport nodes, and NSX Managers.

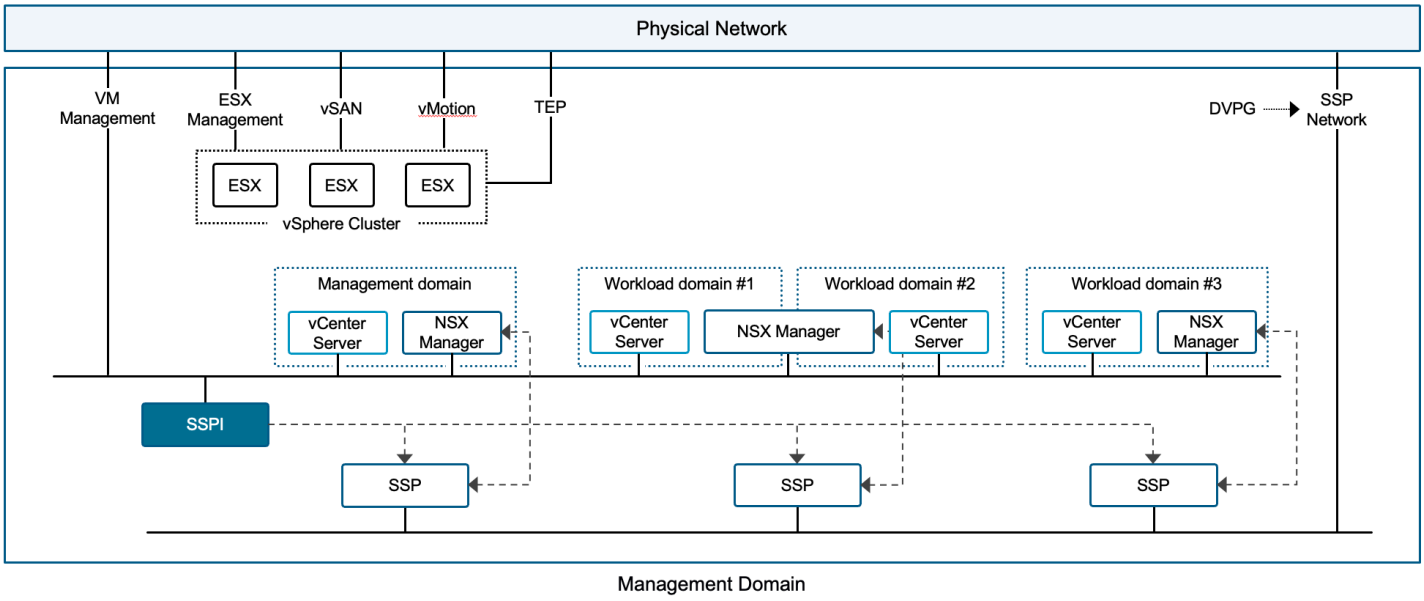
Requirement	Description
Management network	Used to connect the Security Services Platform Installer. The existing VMware Cloud Foundation VM Management DVPG can be used.
Security Services Platform instance network	Used to connect the SSP instance nodes.

Network Design for deployment in the Management domain

This design option places the appliances within the management domain of the VMware Cloud Foundation instance. It ensures connectivity and close proximity to vCenter and NSX Manager. The Security Services Platform Installer appliance is connected to the VCF VM management network. The Security Services Platform nodes are placed on a dedicated network.

> **Note:**
>
> The Security Services Platform can be deployed on the VM Management network if sufficient IP address space is available.

Figure 223: Security Services Platform deployment in the Management domain



Note:

NSX segments (Overlay or VLAN) or VPC public subnets are also supported. Bidirectional communication with the VCF physical networks (ESX Management and VM Management) has to be allowed.

Network Design for deployment in the Workload domain

This design option solution places the Security Services Platform Installer appliance in the management domain and the SSP instances in their respective workload domain of the VMware Cloud Foundation instance. This design solves potential management domain resource constraints. The SSPI appliance is connected to the VCF VM management network; whereas the SSP nodes are placed on a dedicated network (DVPG, NSX segment, or VPC subnet) in the workload domain. Note that when using a VPC, the subnet type of the VPC must be public to avoid complex DNAT configuration for SSP ingestion traffic.

Note:

1. The Security Services Platform network needs routed connectivity with the VCF VM Management network.
2. If traffic between the Security Services Platform network and the VCF VM management network passes through a firewall, make sure that the required ports are opened as posted here: https://ports.broadcom.com/.

Figure 224: Security Services Platform deployment in the Workload domain

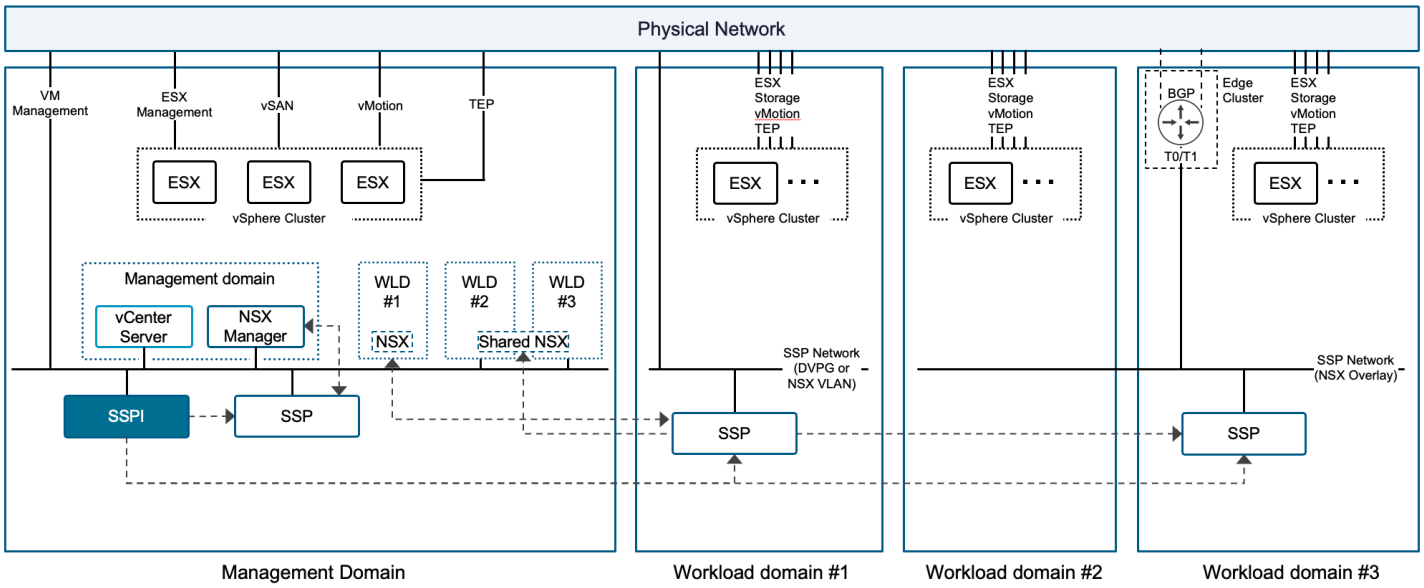


Table 491: Design Decisions on the Network Segment for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-001	Deploy theSecurity Services Platform Installer on the VMware Cloud Foundation VM management network.	• Allows for ease of management for the Security Services Platform Installer • The Security Services Platform Installer requires a single IP address.	The VMware Cloud Foundation VM management network must be configured with a sufficient CIDR block.
SSP-REQD-NET-002	Deploy the Security Services Platform instances on a dedicated network (DVPG, NSX Segment ot VPC Public subnet) **Note:** This network should have IP connectivity to the Security Services Platform Installer, NSX Manager cluster and the ESX Transport Nodes.	Allows for the design of the Security Services Platform instances' IP space/ s independently from the VMware Cloud Foundation VM management network.	• The physical network fabric must be configured to route the traffic from the Security Services Platform instances to the NSX Manager cluster, ESX Transport Nodes management interface (vmk0) and the vCenter Server. • The network must be created manually or via automation.

IP Addressing

The Security Services Platform IP address requirements are:

1. Management network:
   a. Connects the Security Services Platform Installer appliance;
   b. Requires 1 IP address.
2. Security Services Platform instance network:
   a. Connects the Security Services Platform controllers and worker nodes.
   b. Single subnet is required with two separate IP ranges:
      a. Controller and worker node VMs;
      b. Security Services Platform services (messaging).
   c. With the Security Services Platform v5.1 onwards:
      a. You can append a new IP pool for extra resources but you can't edit or remove an existing IP pool
      b. The addresses within each IP pools must be contiguous..

The following table lists the Security Services Platform instance IP addresses requirements.

Table 492: Security Services Platform instance IP address requirements

Form Factor	Controller Nodes	Worker Nodes	Node IP Pool	Service IP Pool
Medium	3	2	7	5
Large	3	4	9	5
Large	3	5	10	5
Large	3	6	11	6
Large	3	7	12	6
Large or Extra Large	3	8	13	7
Large or Extra Large	3	9	14	7
Large or Extra Large	3	10	15	8
Large or Extra Large	3	11	16	8

Form Factor	Controller Nodes	Worker Nodes	Node IP Pool	Service IP Pool
Large or Extra Large	3	12	17	9
Large or Extra Large	3	13	18	9
Extra Large	3	14	19	10
Extra Large	3	15	20	10
Extra Large	3	16	21	11
Extra Large	3	17	22	11

Note:

Node IP Pool requirements provision one reserved IP for scale-out and replacement of failed node scenarios.

Table 493: Design Decisions on IP Addressing for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-003	Use static IP assignment for the Security Services Platform Installer from the VMware Cloud Foundation VM management network.	The Security Services Platform Installer uses management IPs to communicate with vCenter Server, NSX Manager cluster and the Security Services Platform instances.	The VMware Cloud Foundation VM management network must have sufficient IP space available.
SSP-REQD-NET-004	Dedicate two IP ranges from the Security Services Platform network subnet: • Node IP Pool - 22 IPs. • Service IP Pool - 11IPs.	Designing the Security Services Platform IP addressing for max scale deployment ensures straightforward future platform scalability.	• The IP addresses must be reserved. • The Service IP pool range must be contiguous.

Note:

When activating Security Services Platform features - Bare Metal Security or NDR Sensor - additional Service IP Pool addresses are required.

Name Resolution

Name resolution provides the translation between an IP address and a fully qualified domain name (FQDN). One FQDNs is required for theSecurity Services Platform Installer and two FQDNs are required for each of the Security Services Platform instances with a valid internal DNS forward (A) record.

Requirement	Description
Security Services Platform Installer FQDN	• Access to the Security Services Platform Installer UI/CLI. • Security Services Platform access the Security Services Platform Installer registry. • Mapped to the Security Services Platform Installer VM IP address.

Requirement	Description
For each instance of the Security Services Platform:	
Security Services Platform Instance FQDN	• Access to the Security Services Platform UI/API. • NSX Managers and NSX Transport Nodes to send Metrics to Security Services Platform. • Mapped to the 1st IP address of the Service IP Pool.
Security Services Platform Messaging FQDN	• Used by NSX Transport Nodes to send Flows to Security Services Platform. • Mapped to the 2nd IP address of the Service IP Pool.

Table 494: Design Decisions on Name Resolution for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-005	Configure a DNS (A) record for the Security Services Platform Installer.	• The Security Services Platform Installer is accessible by an easy-to-remember FQDN or directly with an IP address. • Security Services Platform uses the Security Services Platform Installer FQDN to access the registry.	Assumes DNS infrastructure is available.
SSP-REQD-NET-006	Configure a DNS (A) record for the Security Services Platform ingress.	• The Security Services Platform UI is accessible by an easy-to-remember FQDN or directly with an IP address. • Used by NSX Managers and NSX Transport Nodes to send Metrics to Security Services Platform.	Assumes DNS infrastructure is available.
SSP-REQD-NET-007	Configure a DNS (A) record for the Security Services Platform messaging service.	Used by NSX Transport Nodes to send flows to Security Services Platform instance.	Assumes DNS infrastructure is available.

Time Synchronization

Time synchronization provided by the Network Time Protocol (NTP) is important to ensure that all components within the VMware Cloud Foundation instance are synchronized to the same time source.

Table 495: Design Decisions on Time Synchronization for Security Services Platform

Design Requirement ID	Design Requirement	Justification	Implication
SSP-REQD-NET-008	Configure time synchronization by using an NTP time for the Security Services Platform Installer and the Security Services Platform instances.	Prevents from time synchronization issues.	An operational NTP service must be available in the environment.

Load Balancing

Ingress Load Balancing is included and configured as part of Security Services Platform installation. The platform does not require an external load-balancing solution.

Planning and Preparation for Security Services Platform for VMware Cloud Foundation

Before the Security Services Platform deployment, the environment must meet the required compute, storage, and network configuration, and provides external services to the components of the solution.

Planning and Preparation

Security Services Platform

Security Services Platform Installer is deployed manually, using an OVA template from the VMware Cloud Foundation management domain vCenter Server. Make sure to download the latest version of the OVA from the support website ahead of the deployment time.

Package Management

The Security Services Platform Installer package management allows the latest packages to be uploaded to install and upgrade the Security Services Platform Installer and Security Services Platform. Make sure to download the latest bundle TAR file from the support website ahead of deployment time.

External Services

You use services that are external to VMware Cloud Foundation infrastructure when implementing the *Lateral Security with vDefend* protection blueprint.

External Service	Description
Domain Name Services (DNS)	Domain Name Services is used to ensure that components are resolvable by FQDN and by IP address.
Network Time Protocol (NTP)	Network Time Protocol is used to synchronize time consistently across components.
SSH File Transport Protocol Server (SFTP)	File Transport Protocol server is used as a target for backup storage using the SFTP protocol.

Implementation of Security Services Platform for VMware Cloud Foundation

To implement the Security Services Platform for VMware Cloud Foundation Blueprint follow the procedures in the following table.

Implementation

To implement the Security Services Platform for VCF blueprint, follow the procedures in the following table.

Procedure Grouping	Procedure	Description
Prepare the Planning and Preparation workbook	Download and populate the Planning and Preparation Workbook	Download the Planning and Preparation Workbook from the Broadcom Technical Documentation site and populate with details that are needed to complete the implementation.
Interoperability Requirements	Interoperability Matrix.	Check the interoperability requirements to verify that the version of VCF 9.X is compatible with the Security Services Platform.
Ports and Protocols	Ports and Protocols	Verify that the necessary network ports are open to allow the Security Services Platform to effectively communicate and perform its functions within your environment.
Configuration Limits	Configuration Limits	The Configuration Maximums tool provides the officially supported configuration limits for Security Services Platform. These limits define the maximum values that are tested and validated to ensure reliable performance and support.
Valid vDefend License	vDefend Feature and Edition Guide	Make sure that you have a valid license to activate the security features that operate on the Security Services Platform , for uninterrupted access and functionality of the platform.
Prepare Software Bundles	Download the Security Services Platform Installerand the Security Services Platform package	Download the Security Services Platform Installer OVA from the Broadcom Support portal and make sure it is available on the machine that you use to access the vSphere Client. Download the Security Services Platform package (TAR) file from the Broadcom Support portal.
Deploy Security Services Platform Installer	Deploying Security Services Platform Installer	Configure and deploy the Security Services Platform Installer.

Procedure Grouping	Procedure	Description
Deploy Security Services Platform Instance	Deploy Security Services Platform	Configure and deploy the Security Services Platform Instances.
Onboarding and Managing theSecurity Services Platform	Onboarding and Managing theSecurity Services Platform	Onboard the Security Services Platform using VCF Management or Workload domain NSX Manager and activate services.

Management Domain Security

The Management Domain Security blueprint covers the design details and decisions, planning and preparation, and implementation steps for securing the VCF management plane.

Architecture Overview

The management domain of the first VMware Cloud Foundation instance of a VCF fleet hosts the management components of the VCF Instance and the management components of the Fleet itself.

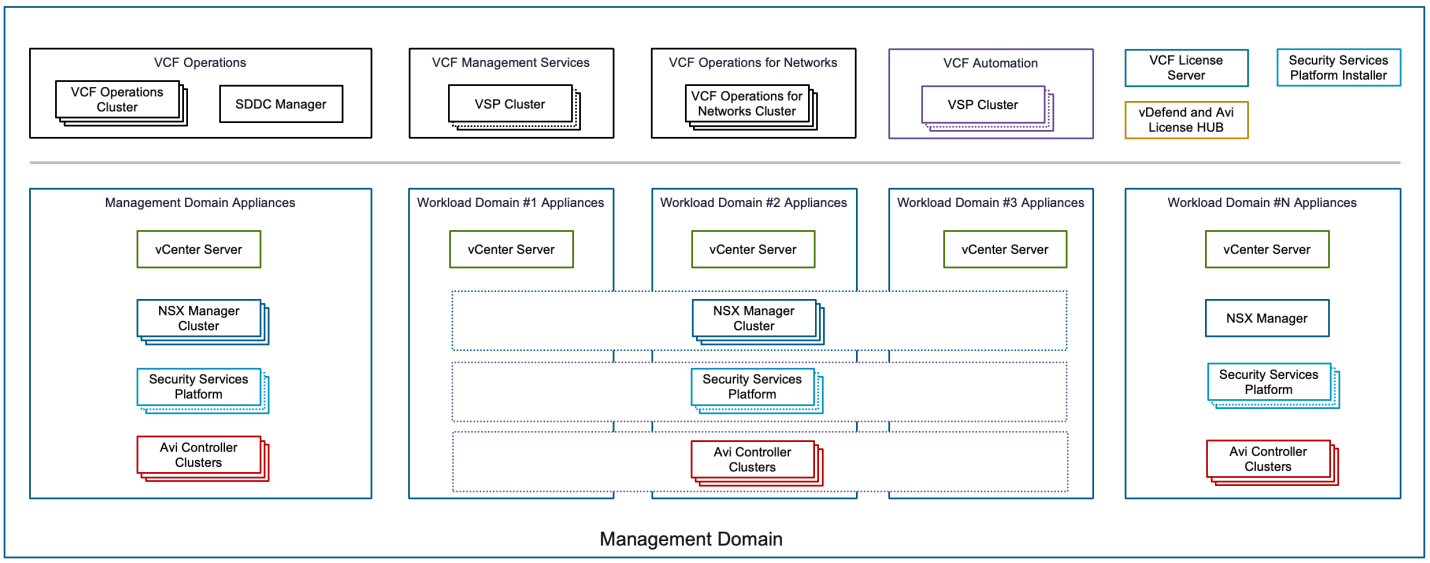
The VCF instance management components: VCF Operations, VCF Automation, VCF Management Services, vCenter Servers, NSX Managers, vDefend Security Services Platform(s), and other appliances that are deployed in the management domain are secured via the vDefend DFW, configured from the management domain NSX Manager. The Management Domain Model also supports running customer business workloads, with management components and workloads secured by a single vDefend Firewall instance.

Table 496: VCF Fleet, VCF Instance Management Components

Component	Description	Operation Model
VCF fleet management components:		
• VCF Operations	VCF Fleet central operating management	• Simple VCF Operations Model • High Availability VCF Operations Model • Continuous Availability VCF Operations Model
• VCF Operations collector		
• VCF Management Services	VCF Fleet key platform services into a common framework	• VCF Management Services in the First VCF Instance Model • VCF Management Services in an Additional VCF Instance Model
• VCF Operations for Logs (optional)	VCF Fleet centralized log management framework	• Simple Log Management Model • High Availability Log Management Model • Multi-Instance Log Management Model
• VCF Operations for Networks (optional)	VCF Fleet an intelligent networking operations console	• Simple VCF Operations for Networks Model • High Availability VCF Operations for Networks Model
• VCF Automation (optional)	VCF Fleet self-service cloud management platform	• Simple VCF Automation Model • High Availability VCF Automation Model • VCF Automation Deployment Patterns
VCF Instance management components - contains the following management appliances for all workload domains:		
• SDDC Manager	VCF Instance life-cycle management	• Single appliance
• vCenter	VCF workload domains' compute, storage, and networking management platform.	• Single appliance
• NSX Manager	VCF workload domain' software-defined networking platform	• Simple NSX Manager Model • High Availability NSX Manager Model • Shared or Dedicated NSX Manager

Component	Description	Operation Model
Optional components deployed in the management domain:		
• License Hub for vDefend and Avi Load Balancer	Centralized licensing platform that manages and distributes subscription-based licenses for VMware vDefend and VMware Avi Load Balancer	Single appliance
• Security Services Platform Installer • Security Services Platform	Optional components of this validated solution, which can be deployed in the VMware Cloud Foundation management domain or in a workload domain.	Single Security Services Platform Installer appliance Security Services Platform delivered as a Kubernetes cluster.
• Avi Load Balancer Controller	Centralized enterprise-grade load-balancing solution for VMware Cloud Foundation	Three nodes cluster

Figure 225: First VCF Instance Management Components



Note:

The number of appliances (VCF Operations, VCF Management Services, VCF Operations for Networks, VCF Automation, NSX Manager) depends on the chosen Operational Model (single or high availability).

Management Domain Security Detailed Design

The following models for the management domain align with the design profile for this blueprint.

Layer	Selected Model	Additional Information
Security	Segmentation Planning	Definitions of VCF management components within diverse security environments (zones).
	Management Components Grouping	Specification details regarding effective mechanisms for the management of dynamic and/or static inventory grouping.
	Infrastructure Services Protection	vDefend Distributed Firewall policies to protect communication between the VCF management components and common infrastructure services.
	Environment Level Protection	vDefend Distributed Firewall policies to protect communication between VCF management domain environments (zones).

Management Domain Security Model

This section covers the security design details and decisions for securing the management domain components of the first VMware Cloud Foundation instance of a VCF Fleet.

- Segmentation Planning
- Management Components Grouping
- Infrastructure Services Protection
- Environment Level Protection

Segmentation Planning

VCF Management segmentation planning design details, covering VMware Cloud Foundation management component definitions in various security environments (zones).

The primary strategy for developing a segmentation model is alignment with the inherent boundaries of the data center's virtual infrastructure. Segmentation by environment (or zone) is an efficient way to implement traffic controls, mitigate lateral movement between zones, and restrict access to critical data. Using this security architectural technique, you partition the VMware Cloud Foundation management domain components into distinct, logical groups with similar security requirements.

- Management components with VCF Fleet access needs:
  - VCF Operations
  - VCF Management Services
    - Fleet lifecycle
    - Salt RaaS
    - Log management
  - VCF Operations for Networks
  - VCF Automation
  - VCF License Server
  - vDefend and Avi License HUB
- Management components with VCF Instance access needs:
  - VCF Management Services
    - SDDC lifecycle
    - Telemetry
    - Salt master

    - Identity broker
    - Real-time metrics
    - Software depot
  - SDDC Manager
  - VCF Operations collectors
  - VCF Operations for Networks collectors
- Management components with VCF workload domain access needs:
  - vCenter
  - NSX Manager
  - Security Services Platform
- External to the VCF components: - jump hosts, automation, and any additional operation and management tools.
- Standard infrastructure services - common shared services like DNS Servers, AD Servers, NTP Servers, Backup Servers.

Important:

The VCF 9.1.0 the VCF Fleet components, SDDC Manager, vCenter Servers, NSX Managers are placed by default in the management domain NSX, DFW User Excluded Groups list.

Figure 226: Management Domain Segmentation Planning

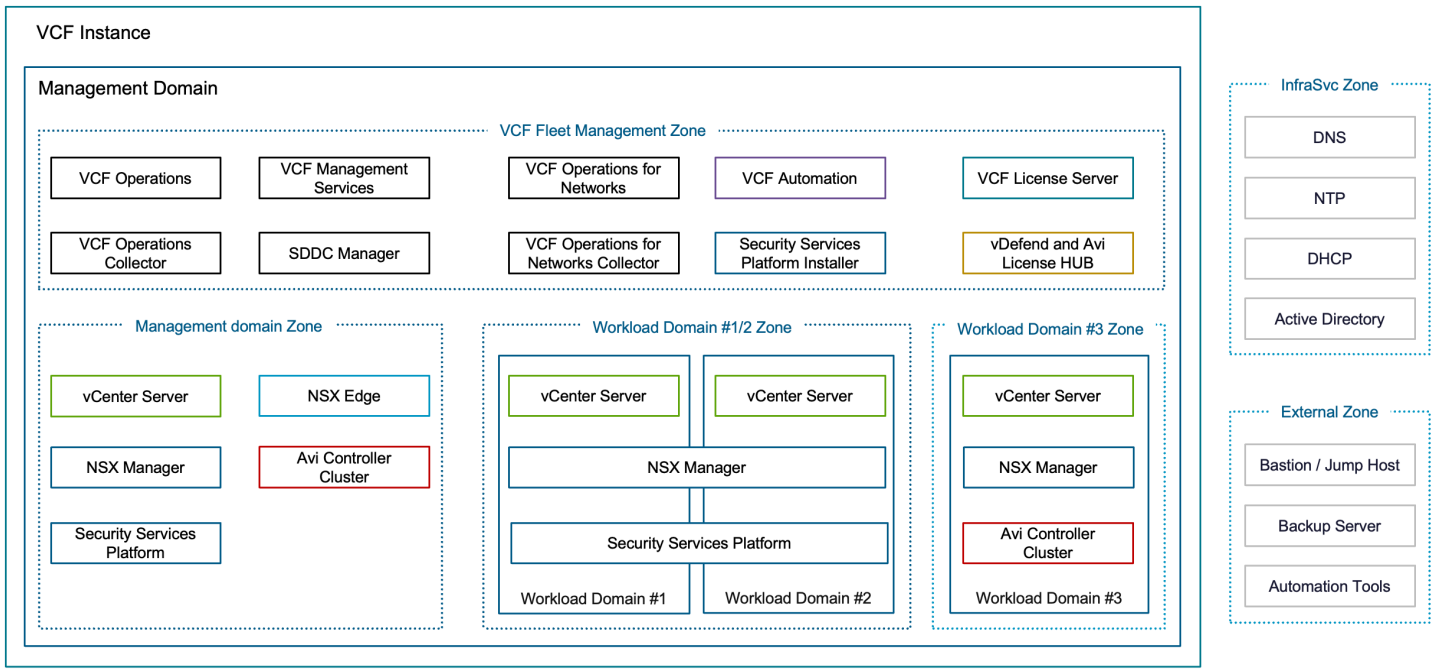


Table 497: Management Domain Segmentation Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-001	• Segment the VMware Cloud FoundationManagement domain componenets into security zone. • Align with the natural boundaries of the VMware Cloud Foundation infrastructure.	• Attack Surface Reduction with vDefend DFW Environment-level protection. • Isolation without need for deep application knowledge.	• You must create the NSX security groups manually or using automation in advance. • For tag-based grouping, the respective components (VMs) must be manually or via automation.

Management Components Grouping
The management components grouping specification details the design decisions for proper dynamic and/or static management domain NSX inventory grouping mechanisms.

VMware Cloud Foundation Management and Workload Domains Grouping

Grouping for management and workload domains for VMware Cloud Foundation:

Componenet	Group Type	Membership Criteria
Management and/or workload domains vCenter Servers	Generic	Virtual Machine name or name subset (i.e. Equals, Contains, Starts With, Ends With)
Management and/or workload domains NSX Managers	Generic	Virtual Machine name or name subset (i.e. Equals, Contains, Starts With, Ends With)
Management and/or workload domains ESX hosts	IP Addresses Only	IP Addresses or CIDR
Management and/or workload domains NSX Edge Nodes	IP Addresses Only	IP Addresses or CIDR
Management and/or workload domains Security Services Platform	Generic	Virtual Machine name or name subset (i.e. Equals, Contains, Starts With, Ends With)
Management and/or workload domains Security Services Platform Service Pool	IP Addresses Only	IP Addresses or CIDR
Management and/or workload domains Avi Controllers	Generic	Virtual Machine Tag
Management and/or workload domains Avi Service Engines	IP Addresses Only	IP Addresses or CIDR
Workload domains Supervisor management	IP Addresses Only	IP Addresses or CIDR
Management and/or workload domains IP-based Storage	IP Addresses Only	IP Addresses or CIDR
All management and/or workload domains components	The groups above	Nested

Note:

1. NSX Edge Nodes are included in the DFW System Exclusion List, and as a result, the configured DFW policies do not apply to them. However, creating groups for these nodes is beneficial for configuring DFW rules that designate them as either sources or destinations.
2. Object-based grouping criteria (like VM Name, NSX tags, VIFs), do not work for the management domain NSX Edge Nodes. They must be grouped using their management IPs.

Figure 227: VCF Management Domain Grouping Model

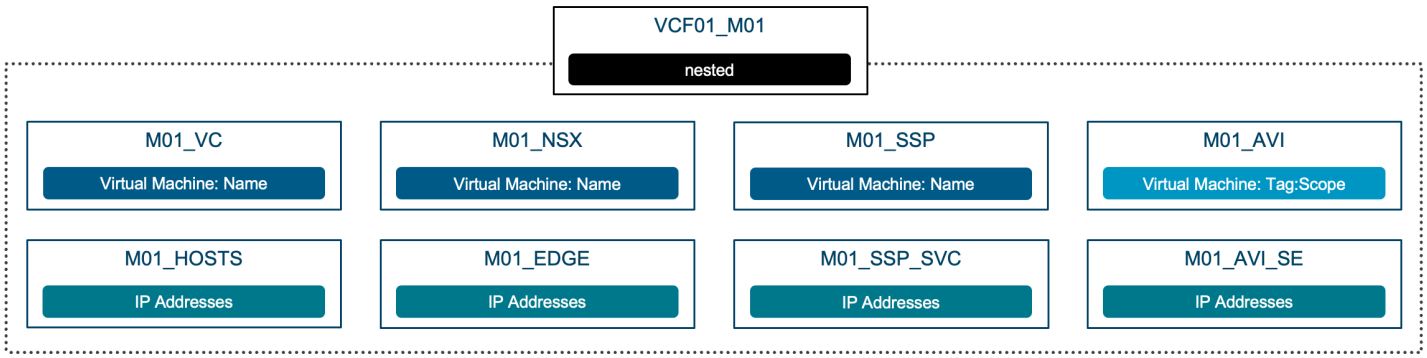
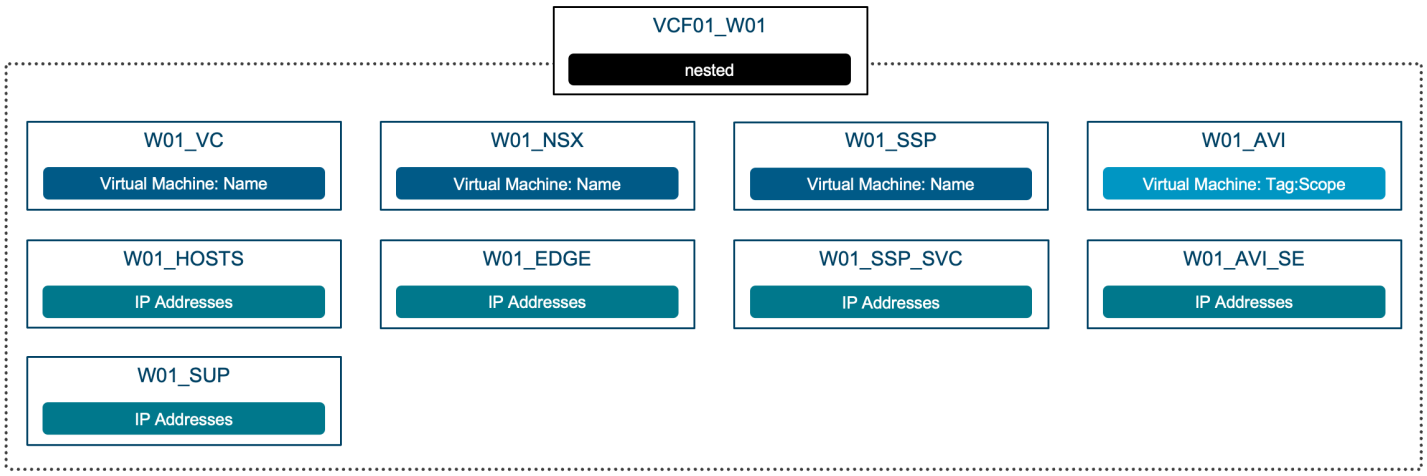


Figure 228: VCF Workload Domain Grouping Model



Important:

The grouping methodology employed in this validated solution primarily relies on VM names, which enables automatic grouping of components at the exact moment of creation. Nevertheless, tags may also be applied as alternative dynamic membership criteria to accommodate the specific characteristics of your environment.

VMware Cloud Foundation Fleet Management Grouping

VMware Cloud Foundation provided flexibility for the deployment of the fleet level management components using a different networking model. By default, the fleet-level management components are connected to the VCF VM management network, which is used for deploying the management domain core components, such as vCenter and NSX Manager.

Table 498: Fleet Level Components Networking Models

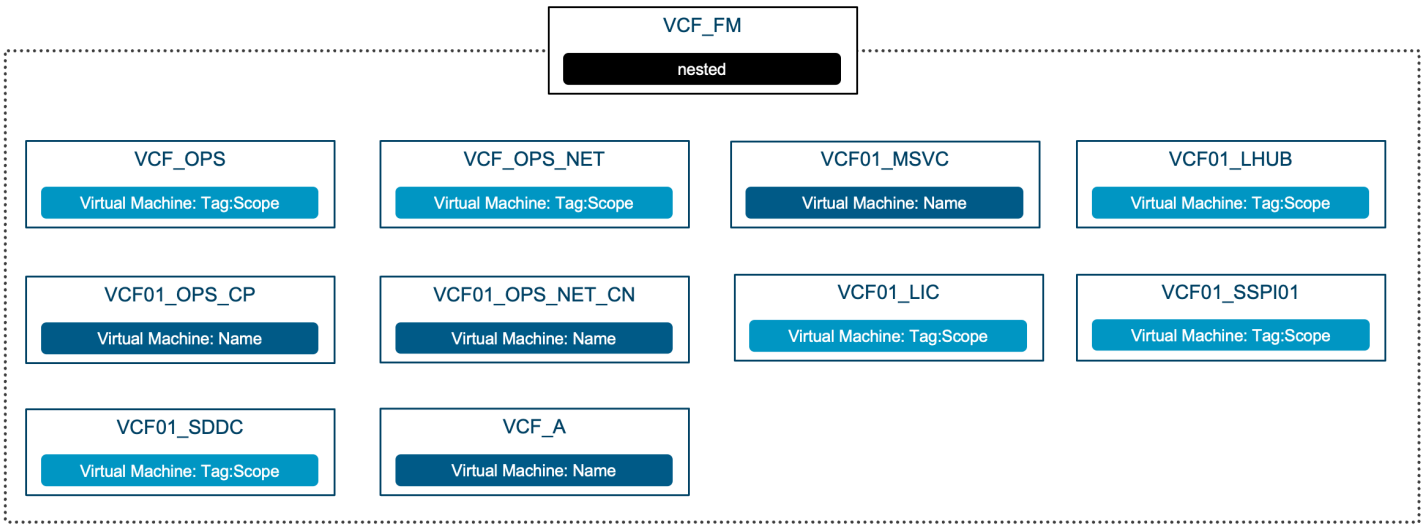
Networking Model	Description
VCF Management Shared VLAN Network Model	Fleet Level components that are connected on the VCF VM management distributed port group
VCF Management Dedicated VLAN Network Model	Fleet Level components that are connected on a dedicated distributed port group
VCF Management Dedicated VLAN and NSX Overlay Segment Network Model	Fleet Level components that are connected on NSX VLAN and Overlay Segment
VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model	Fleet Level components connected on an NSX stretched global segment (NSX Federation deployment)

VCF Fleet Management component groups use dynamic membership criteria based on Virtual Machine Name, remaining agnostic to the networking model.

Grouping for VMware Cloud Foundation Fleet Management:

Componenet	Group Type	Membership Criteria
VCF Operations	Generic	Virtual Machine Tag
VCF Operations Collector Node	Generic	Virtual Machine name or name subset (i.e. Equals, Contains, Starts With, Ends With)
VCF Operations for Networks	Generic	Virtual Machine Tag
VCF Operations for Networks Collector Node	Generic	Virtual Machine name or name subset (i.e. Equals, Contains, Starts With, Ends With)
SDDC Manager	Generic	Virtual Machine Tag
VCF Management Services	Generic	Virtual Machine name or name subset (i.e. Equals, Contains, Starts With, Ends With)
VCF Automation	Generic	Virtual Machine name or name subset (i.e. Equals, Contains, Starts With, Ends With)
VCF License Server	Generic	Virtual Machine Tag
vDefend and Avi License Hub	Generic	Virtual Machine Tag
Security Services Platform Installer	Generic	Virtual Machine Tag
All VCF Fleet Management components	The groups above	Nested

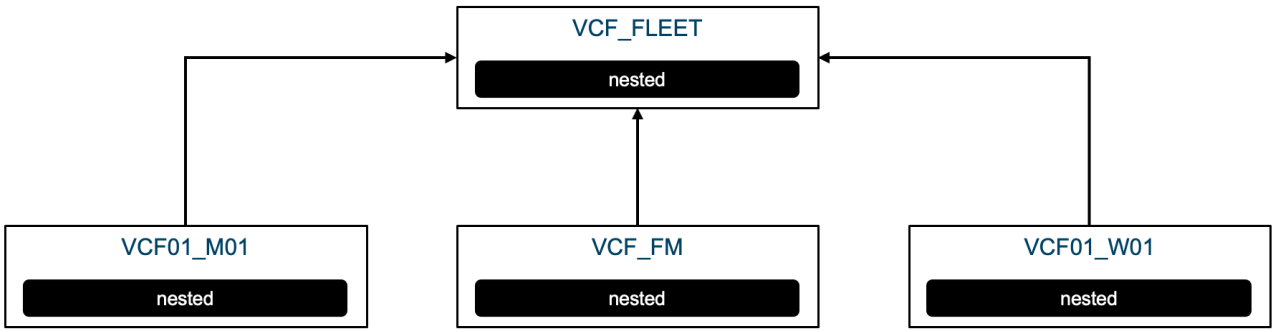
Figure 229: VCF Fleet Level Components Grouping Model



VMware Cloud Foundation Fleet Grouping

- All VCF Fleet components - nested group, using static membership criteria.

Figure 230: VCF Fleet Grouping Model



Infrastructure Services and External Components Grouping

- The Infrastructure Services and External Components Grouping is static - IP based.

Figure 231: Infrastructure Services and External Components Grouping Model

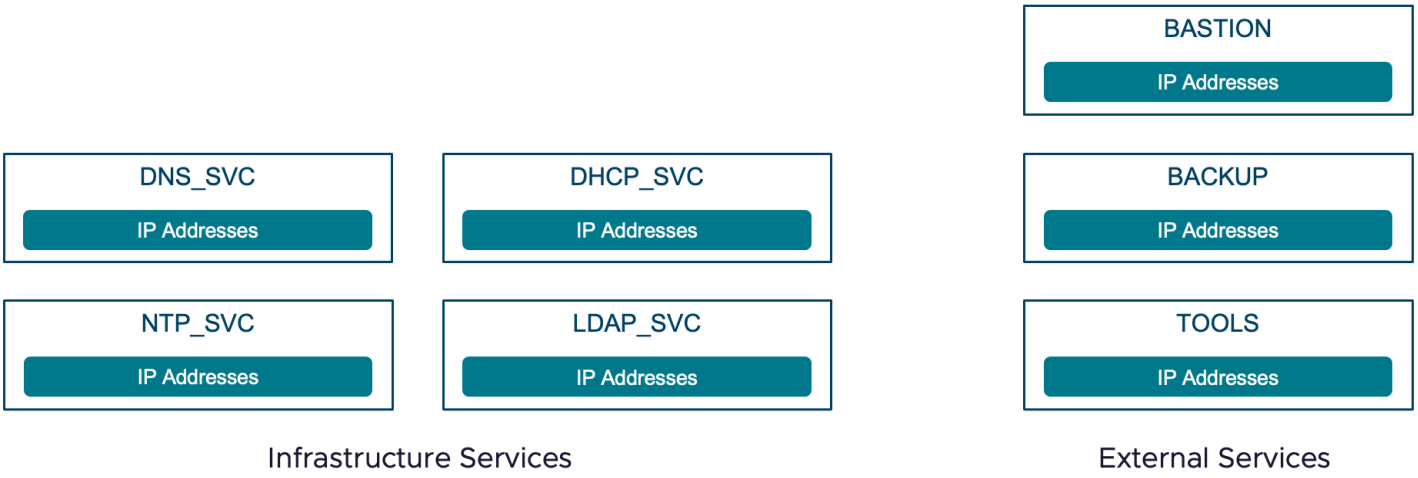


Table 499: Management Domain Grouping Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-002	For the management domain and each workload domain vCenter Server create an individual group with dynamic membership based on VM name.	• Enables connectivity between a new workload domain's vCenter Server and VCF Fleet management components upon deployment. • VCF Operations will perform a Reduced Downtime Upgrade (RDU) by default. The RDU workflow instantiates and configures a vCenter VM, migrates the database and services, stops and deletes the original vCenter. • An NSX tag will be lost during a vCenter reduced downtime upgrade (RDU), because the existing vCenter VM will be deleted.	Groping based on VM names requires a strict naming convention.
MDS-REQD-CFG-003	For the management domain and each workload domain NSX Manager cluster create an individual group with dynamic membership based on VM name.	Enables connectivity between a new workload domain's NSX Manager cluster and VCF Fleet management components upon deployment.	Groping based on VM names requires a strict naming convention.
MDS-REQD-CFG-004	• For the management domain and each workload domain, Avi Controller clsuters, create groups with dynamic membership based on NSX tags. • Utilize the tag to uniquely identify the management component. • Utilize the scope of the tag to identify workload domain.	• Provides best flexibility for the dynamic security posture of the of the core management components. • Allow abstraction of workload from the underlying infrastructure topology. • Facilitates dynamic and flexible Distributed Firewall policy configuration.	• You must define the NSX tags and apply them to the respective VM manually or using automation in advance. • The manual process of creating the security groups could be a cumbersome task. • Newly created workload domain management components must be manually tagged afterward.
MDS-REQD-CFG-005	• For the Security Security Services Platform Installercreate group with dynamic membership based on tags. • Utilize the tag to uniquely identify the management component. • Utilize the scope of the tag to identify workload domain.	• Provides best flexibility for the dynamic security posture of the of the core management components. • Allow abstraction of workload from the underlying infrastructure topology. • Facilitates dynamic and flexible Distributed Firewall policy configuration.	• You must define the NSX tags and apply them to the respective VM manually or using automation in advance. • The manual process of creating the security groups could be a cumbersome task. • Newly created workload domain management components must be manually tagged afterward.

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-006	• For the Security Services PlatformInstance create groups using dynamic membership on the VM name. • Use matching criteria: Starts With, including the required (existing and future) Security Services Platformnodes.	• The Security Services PlatformInstance name is added as a prefix to each node. • Allow dynamically to add the respective Security Services Platformnodes into the group during scale out operation. • Distributed Firewall rules using the group will automatically apply to any new VM matching the criteria.	More NSX groups being created and managed.
MDS-REQD-CFG-007	Using static membership based on IPs, create individual groups for each VMware Cloud Foundation workload domain respective: • ESX hosts management VMkernel. • NSX Edge Nodes. • IP-based Storage (for instance Network File System (NFS) if exists.	These objects fall outside the management domain's Distributed Firewall policy boundary, yet they maintain active communication with VCF management components.	The manual process of creating the security groups could be a cumbersome task.
MDS-REQD-CFG-008	For IP-based groups, utilize the entire (including future use) dedicated CIDR block instead of individual IPs.	Using the entire dedicated CIDR block instead of individual IPs makes the Distributed Firewall policy LCM easier because it automatically allows the required traffic in the case of host commissioning, new workload domains of cluster creation, and NSX Edge nodes deployment.	The VMware Cloud Foundation IP schema has to be precisely architected in advance.
MDS-REQD-CFG-009	• For each workload domain, create a nested group that includes the respective workload domain management components (vCenter, NSX Managers, and Security Services PlatformIf deployed) and its respective ESX hosts, NSX Edge Nodes, and IP-based storage (if used). • Use matching criteria based on: — VMs tag scope — Child group tag — Static inclusion	• Optimize the computation during the publishing of the Distributed Firewall policy.	More NSX groups being created and managed.

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-010	• For VCF Automation instance create groups using dynamic membership on the VM name. • Use matching criteria: "Starts With", matching the VCF Automation nodes' name prefix.	• The VCF Automation instance name is added as a prefix to each node. • Allow dynamically adding the respective nodes to the group in case of recreation. • Distributed Firewall rules using the group will automatically apply to any new VM matching the criteria.	More NSX groups being created and managed.
MDS-REQD-CFG-011	• For VCF Management Services instance create groups using dynamic membership on the VM name. • Use matching criteria: "Starts With", matching the VCF Management Services nodes name prefix.	• The VCF Management Services instance name is added as a prefix to each node. • Allow dynamically adding the respective nodes to the group in case of recreation. • Distributed Firewall rules using the group will automatically apply to any new VM matching the criteria.	More NSX groups being created and managed.
MDS-REQD-CFG-012	• For VCF Operations (platform and cloud proxy, VCF Operations fleet management, VCF Operations for Networks (platform and collector nodes), SDDC Manager, License Server, License Hub and Security Services Platform Installer create an individual group with dynamic membership based on Tags.	• Provides best flexibility for the dynamic security posture of the workload. • Allow abstraction of workload from the underlying infrastructure topology. • Facilitates dynamic and flexible Distributed Firewall policy configuration.	• You must define the NSX tags and apply them to the respective VM manually or using automation in advance. • The manual process of creating the security groups could be a cumbersome task. • Newly created management components (VCF Automation nodes) must be manually tagged afterward.
MDS-REQD-CFG-013	Create a nested group, that includes all VCF Fleet Management components:	• Use a single group, which includes all VMware Cloud Foundation Fleet management componenets to simplify the Distributed Firewall management. • Use a single group at the policy or rule *Applied To* filed.	• The manual process of creating the security groups could be a cumbersome task. • More NSX groups being created and managed.
MDS-REQD-CFG-014	Create a nested group, that includes all VCF Fleet components:	• Use a single group, which includes all VMware Cloud Foundation Fleet to simplify the Distributed Firewall management. • Use a single group at the policy or rule *Applied To* filed.	• The manual process of creating the security groups could be a cumbersome task. • More NSX groups being created and managed.

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-015	For each Infrastructure service and External service create individual group matching on IP address or CIDRs.	The VMs which provide shared Infrastructure services are not part of the VMware Cloud Foundation instance.	• The manual process of creating the security groups could be a cumbersome task. • More NSX groups being created and managed.

Distributed Firewall for vSphere VLAN Distributed Port Groups

By using the vDefend Distributed Firewall, you can implement granular security policies to protect the VMware Cloud Foundation management components. This approach ensures that VMware Cloud Foundation infrastructure is effectively monitored and controlled based on predefined security criteria.

Table 500: Distributed Firewall for vSphere Distributed Port Groups Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-016	Verify existing Distributed Firewall policy before enabling "NSX on DVPGs".	The existing Distributed Firewall policy immediately takes effect on VMs connected to DVPGs, when applicable to them, for instance, rules with "Apply To" set to Distributed Firewall.	Manual verification of the Distributed Firewall policy is required.
MDS-REQD-CFG-017	Activate *NSX on DVPGs* on the management domain vSphere cluster and vSphere Distributed Switch, where the VCF Fleet and core management componenets resides.	Enable vDefend Distributed Firewall on vSphere VLAN distributed port groups on the VMware Cloud Foundation management domain.	Any existing Distributed Firewall policy immediately affects the vSphere VLAN distributed port groups connected workload after activation.
MDS-REQD-CFG-018	Remove the VCF management components from the management domain NSX Distributed Firewall User Excluded Group: **VCF-Created-Virtual-Machines**. **Note:** In VCF 9.1.0, the User Excluded Group VCF-Created-Virtual-Machines along with the static membership criteria, also uses a dynamic based on VM names. The dynamic membership criteria apply for the VCF Management Services, VCF Operations for Networks, and VCF Automation. To remove these management components, you have to delete the dynamic membership criteria.	Enforce Distributed Firewall on the VCF Fleet management components.	• You must manually remove these VMs from the Distributed Firewall User Excluded Group.

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-019	Retain the management domain NSX Manager appliances in the Distributed Firewall Excluded Group.	The management domain NSX Manager appliances must remain on the Distributed Firewall User Excluded list to prevent possible Distributed Firewall misconfiguration that could block access to them.	• Management domain NSX Manager appliances cannot be protected with Distributed Firewall. • Applies for Workload Domains NSX Manager that is shared with the Management domain.
MDS-REQD-CFG-020	Use the default Security, IP Discovery segment profiles for DVPGs.	• After activating *NSX on DVPGs*, default segment profiles are automatically applied to the discovered vSphere VLAN distributed port groups. • The default IP Discovery and Security profiles for DVPGs are tuned for vSphere distributed port groups.	You must manually implement the change

Infrastructure Services Protection
In this stage, you design a vDefend DFW policy to protect communication between the VCF Management components and common (shared) services, such as: DNS, NTP, DHCP, etc.

Infrastructure Services Distributed Firewall Policy Design

Protecting shared infrastructure services, such as DNS, NTP, DHCP, Active Directory (AD), and centralized logging systems is critically important because they are the foundational trust boundary on which VMware Cloud Foundation private cloud operations, security, and visibility depend. Compromising just one of these services can lead to total network control loss, widespread service outages, or complete security blind spots.

The Infrastructure service protection is structured in a single DFW policy that grants access strictly based on a destination server and a service port requirement. A unique allow rule is configured for each service, assuming each service runs on an individual server.

The last policy rule is a lockdown that prevents potential traffic from reaching destinations other than enterprise-allowed ones. Enable logging on the lockdown rule to monitor for potential policy misconfiguration and avoid service disruption.

The following table provides an example of the VMware Cloud Foundation management domain vDefend DFW policy and rules for infrastructure services protection - based on the designed segmentation model.

Infrastructure Services Policy (DFW Infrastructure Category)

Table 501: Infrastructure

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
VCF Fleet Infrastructure Policy			**Applied To: VCF_FLEET**			
Any	DNS_SVC	DNS-TCP DNS-UDP	DNS	DFW	Allow	Out
Any	NTP_SVC	NTP	None	DFW	Allow	Out
Any	DHCP_SVC	DHCP-Server	None	DFW	Allow	Out
Any	AD_SVC	Microsoft Active Directory V1	None	DFW	Allow	Out
Any	SYSLOG_SVC	Syslog (TCP) Syslog (UDP) TCP/9543	None	DFW	Allow	Out

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
Any	Any	DNS-TCP DNS-UDP NTP DHCP-Server Syslog(TCP, UDP) TCP/9543	None	DFW	Drop / Log	Out

Note:

1. The preceding example presumes that the Infrastructure services are outside the VCF instance. If these services run as VMs in the VCF management domain, the policy must be updated to include the respective workload domains ESX hosts and Edge Nodes.
2. Other ports may need to be added for extra features. For a complete list of communication port list protocols, review https://ports.broadcom.com/.
3. The Applied To field is populated at the policy level. The policy level Applied To take precedence over the individual rules Applied To scope.
4. The Malicious IP Feeds policy exists in the DFW Infrastructure category and is enabled by default.
5. For automated upgrade through SDDC Manager In-Place upgrade is integrated with the live patch process in VMware Cloud Foundation lifecycle management. When upgrading to an NSX hot patch or express patch, VMware Cloud Foundation initiates a live patch when possible and triggers an In-Place upgrade. An In-Place upgrade is not supported when Layer 7 context profile or Identity Firewall rules are enabled. Maintenance mode must be used.

Table 502: Infrastructure Level Protection Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-021	Configure Distributed Firewall Infrastructure category policy for the VMware Cloud Foundation Fleet.	Ensures your security policies are systematically arranged and managed, clearly defined, and easily accessible.	None.
MDS-REQD-CFG-022	Configure policy level *Applied To* for group: • VCF_FLEET	The policy will be automatically applied to new VCF management components, like new workload domain vCenter and NSX Managers, when properly tagged.	The policy level *Applies To* takes precedence over the *Applied To* scope of the individual Distributed Firewall rule.
MDS-REQD-CFG-023	Configure an individual rule for each Infrastructure service: DNS, NTP, DHCP, LDAP, SYSLOG etc.	• VMware Cloud Foundation management components require access to shared infrastructure services. • Infrastructure services are typically provided on dedicated servers for each individual service.	• DNS L7 inspection is required if FQDN filtering rules are utilized. • In-Place upgrade via the SDDC Manager is not supported when Layer 7 or Identity Firewall rules are enabled. Manual NSX Manager upgrade needs to be utilized.
MDS-REQD-CFG-024	Configure infrastructure services Lockdown rule. Enable rule Logging	• Deny any other access fromVMware Cloud Foundationmanagement components to infrastructure services. • Log potential traffic drops to avoid misconfigurations.	Misconfigured Infrastructure services policy and rules can result in excessive traffic, impacting system performance.

Environment Level Protection
In this stage, you design a vDefend DFW policy to protect communication between VCF Management domain Environments (zones).

Environments (Zones) Distributed Firewall Policy Design

Securing traffic between different Environments (often referred to as zone-to-zone security or macro-segmentation) is one of the most critical and foundational elements of a robust VCF private cloud security strategy.

With respect to the VCF management domain, this control point is important because it enforces a trust boundary between components with different management and operational functions, and security requirements, e.g., VCF Fleet-level access needs, VCF Instance-level access needs, VCF management, and/or workload domain-level access needs.

The VCF management domain Environments DFW design is based on configuring a separate policy for each zone (see Management Domain Segmentation Planning for the zones definition). The policies are organized by zone-level access permissions, with the VCF Fleet external communications policy processed before the VCF Fleet management policy, and the VCF management and workload domain policies processed after.

Each policy, aside from the VCF Fleet and external components communication that does not have a "lockdown" rule, follows a unified structure:

- The policy is applied to the group representing the environment (policy-level Applied To).
- First rule for controlling intra-zone traffic. This rule has a specific action: "Jump to Application", which creates a dynamic policy chain, allowing for the delegation of traffic inspection and enforcement. So the traffic that matches this rule in the Environment category policy will continue to be inspected by the Application category policies and rules.
- Rules for required inter-zone communications with application-level granularity.
- Inter-zone communication rules are duplicated in individual policies to ensure policies remain order-independent.
- Environment rule(s) with unknown source and/or destination (Any) have an action: Jump to Application, for more granular inspection.
- The last rule of each policy is a "lockdown" rule designed to prevent any other inbound/outbound zone communication.

This approach offers a flexible DFW design that begins with segmentation at the Environment level and allows for the configuration of more specific Application category rules at a later stage, without necessitating modifications to the Environment policies. Also, it enhances troubleshooting capabilities by pinpointing which specific policy may be obstructing traffic that should otherwise be permitted.

Note:

1. Although the VCF management domain design is prescriptive and the management component communication requirements are known, variations may exist across customer environments.
2. An initial action of "Allow and Log" within a lockdown rule should be followed by a transition to "Drop and Log" or "Drop" based on the level of confidence in the policy.
3. This policy design offers flexibility and is effective in prescriptive environments such as the VCF management domain. However, it may not be appropriate for large workload domain environments, where the outbound and inbound communication patterns of various applications can differ significantly. Therefore, safeguarding this traffic at the environment level may prove to be impractical.

The following table provides an example of the VMware Cloud Foundation management domain vDefend DFW policy and rules for environments protection - based on the designed segmentation model.

Environments Polices - (DFW Environment Category)

Table 503: VCF Fleet Environment (Zone)

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
VCF Fleet Policy			**Applied To: VCF_FLEET**			
BASTION	VCF_FLEET	HTTPS SSH ICMP RDP	None	DFW	Allow	In
TOOLS	VCF_FLEET	HTTPS SSH	None	DFW	Allow	In
Any	VCF_A	HTTPS	None	DFW	Jump to Application	In
VCF_FLEET	BACKUP_SVC	ICMP HTTPS FTP SSH	None	DFW	Allow	Out
VCF_FLEET	SMTP_SVC	SMTP SMTP_SSL	None	DFW	Allow	Out
VCF_FLEET	Any	HTTPS	INTERNET_FQDNS	DFW	Allow	Out

Table 504: VCF Fleet Management Environment (Zone)

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
VCF Fleet Management Policy			**Applied To: VCF_FM**			
VCF_FM	VCF_FM	Any	None	DFW	Jump to Application	In-Out
VCF_FM	VCF01_M01 VCF01_W(...)	HTTPS	None	DFW	Allow	Out
VCF01_SDDC	VCF01_M01 VCF01_W(...)	SSH TCP/5480 ICMP Echo Request	None	DFW	Allow	Out
M01_NSX W(...)_NSX	VCF01_SDDC	SSH	None	DFW	Allow	In
VCF01_OPS_CP	VCF_FM VCF01_M01 VCF01_W(...)	ICMP Echo Request	None	DFW	Allow	Out
VCF01_M01 VCF01_W(...)	VCF01_MSVC	HTTPS TCP/4505-4506 Syslog (TCP) Syslog (UDP) TCP/9543	None	DFW	Allow	In
VCF01_M01 VCF01_W(...)	VCF01_OPS_NET_CN	HTTPS TCP/1991 UDP/2055	None	DFW	Allow	In
W(...)_SUP	VCF_A VCF01_OPS_CP	HTTPS	None	DFW	Allow	In
VCF_A	W(...)_SUP	TCP/6443	None	DFW	Allow	Out

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
M01_VC W(...)_VC	VCF01_LIC	HTTPS	None	DFW	Allow	In
VCF01_LHUB	VCF01_M01 VCF01_W(...)	HTTPS TCP/2049 TCP/3260	None	DFW	Allow	Out
VCF01_M01 VCF01_W(...)	VCF01_LHUB	HTTPS TCP/9092	None	DFW	Allow	In
VCF01_SSPI01	VCF01_M01 VCF01_W(...)	HTTPS TCP/6443	None	DFW	Allow	Out
VCF01_M01 VCF01_W(...)	VCF01_SSPI01	HTTPS	None	DFW	Allow	In
Any	Any	Any	None	DFW	Allow (Drop) / Log	In-Out

Note:

- For the policy catch-all (lockdown) rule, start with an action **Allow and Log**, and change to **Drop and Log** after ensuring policy accuracy.
- Configure a log label for easy troubleshooting.

Table 505: VCF01 Management Domain Environment (Zone)

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
VCF01 Management Domain Policy			**Applied To: VCF01_M01**			
VCF01_M01	VCF01_M01	Any	None	DFW	Jump to Application	In-Out
VCF_FM	VCF01_M01	HTTPS	None	DFW	Allow	In
VCF01_SDDC	VCF01_M01	SSH TCP/5480 ICMP Echo Request	None	DFW	Allow	In
M01_NSX	VCF01_SDDC	SSH	None	DFW	Allow	Out
VCF01_OPS_CP	VCF01_M01	ICMP Echo Request	None	DFW	Allow	In
VCF01_M01	VCF01_MSVC	HTTPS TCP/4505-4506 Syslog (TCP) Syslog (UDP) TCP/9543	None	DFW	Allow	Out
VCF01_M01	VCF01_OPS_NET_CN	HTTPS TCP/1991 UDP/2055	None	DFW	Allow	Out
M01_VC	VCF01_LIC	HTTPS	None	DFW	Allow	Out
VCF01_LHUB	VCF01_M01	HTTPS TCP/2049 TCP/3260	None	DFW	Allow	In
VCF01_M01	VCF01_LHUB	HTTPS TCP/9092	None	DFW	Allow	Out
VCF01_SSPI01	VCF01_M01	HTTPS TCP/6443	None	DFW	Allow	In
VCF01_M01	VCF01_SSPI01	HTTPS	None	DFW	Allow	Out

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
Any	Any	Any	None	DFW	Allow (Drop) / Log	In-Out

Note:

- For the policy catch-all (lockdown) rule, start with an action **Allow and Log**, and change to **Drop and Log** after ensuring policy accuracy.
- Configure a log label for easy troubleshooting.

Table 506: VCF01 Workload Domain (...) Environment (Zone)

Source	Destination	Service	Context Profile	Applied To	Action / Logging	Direction
VCF01 Workload Domain (...) Policy			**Applied To: VCF01_W(...)**			
VCF01_W(...)	VCF01_W(...)	Any	None	DFW	Jump to Application	In-Out
VCF_FM	VCF01_W(...)	HTTPS	None	DFW	Allow	In
VCF01_SDDC	VCF01_W(...)	SSH TCP/5480 ICMP Echo Request	None	DFW	Allow	In
W(...)_NSX	VCF01_SDDC	SSH	None	DFW	Allow	Out
VCF01_OPS_CP	VCF01_W(...)	ICMP Echo Request	None	DFW	Allow	In
VCF01_W(...)	VCF01_MSVC	HTTPS TCP/4505-4506 Syslog (TCP) Syslog (UDP) TCP/9543	None	DFW	Allow	Out
VCF01_W(...)	VCF01_OPS_NET_CN	HTTPS TCP/1991 UDP/2055	None	DFW	Allow	Out
W(...)_VC	VCF01_LIC	HTTPS	None	DFW	Allow	Out
VCF01_LHUB	VCF01_W(...)	HTTPS TCP/2049 TCP/3260	None	DFW	Allow	In
VCF01_W(...)	VCF01_LHUB	HTTPS TCP/9092	None	DFW	Allow	Out
VCF01_SSPI01	VCF01_W(...)	HTTPS TCP/6443	None	DFW	Allow	In
VCF01_W(...)	VCF01_SSPI01	HTTPS	None	DFW	Allow	Out
Any	Any	Any	None	DFW	Allow (Drop) / Log	In-Out

Note:

- WLD(...) = All associated workload domains (WLD01, shared WLD02/03, etc.).
- For the policy catch-all (lockdown) rule, start with an action **Allow and Log**, and change to **Drop and Log** after ensuring policy accuracy.
- Configure a log label for easy troubleshooting.

Table 507: Environment (Zone) Level Protection Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-025	Configure Distributed Firewall Environment category policy for VCF Fleet external communications.	Required for VCF fleet traffic management with external components.	None.
MDS-REQD-CFG-026	Configure policy level *Applied To* for group: • VCF_FLEET	The policy will be automatically applied to new VCF fleet components.	The policy level *Applies To* takes precedence over the *Applied To* scope of the individual Distributed Firewall rule.
MDS-REQD-CFG-027	Configure a Distributed Firewall rule for the external components to the VCF fleet communication: • Bastion / JumpHost • Automation Tools	Allow required external components to communicate with VCF fleet on respective services: • HTTPS • SSH • ICMP • RDP (if required)	None.
MDS-REQD-CFG-028	Configure a Distributed Firewall rule for VCF Automation Portal access with action: Jump to Application.	• Create a policy chain for VCF Automation cloud portal access on port HTTPS. • Allows granular Application category policy inspection.	Requires an Application category policy to protect the VCF Automation cloud portal access: • Native load balancer IP • External load balancer IP to VCFA (in the case external load balancer is used)
MDS-REQD-CFG-029	Configure a Distributed Firewall rule for the VCF fleet to external components communication: • Backup Server • SMTP Server • SIEM Server	Allow required VCF fleet communication with external resources on respective services: • ICMP • HTTPS • FTP • SSH • SMTP • SMTP_SSL	None.

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-030	Configure a Distributed Firewall rule for the VCF fleet management components HTTPS traffic to Internet. • depot.vmware.com • depot.broadcom.com • hostupdate.vmware.com • vapp-updates.vmware.com • vcsa.vmware.com • vcgw-updates.vmware.com • partnerweb.vmware.com • *.esp.vmware.com • *.prod.nsxti.vmware.com **Note:** To verify the most recent list of FQDNs to be accessed over the Internet, see https://ports.broadcom.com	• Allow download of VMware Cloud Foundation upgrade and patch binaries. • Allow update downloads for vSphere Lifecycle Manager and Update Manager. • Allow CEIP and Skyline Health for vSphere Health Check. • Allow in-Product Feedback Form in the vSphere Client. • Allow vSAN Health Service Hardware. Compatibility Downloads. • Allow NSX Manager to NTICS service deployed in the cloud.	• A custom context profile configuration is requited for FQDN filtering. • FQDN filtering requires DNS L7 inspection.
MDS-REQD-CFG-031	Configure a Distributed Firewall Environment section policy for each Environment (Zone) based on the segmentation model outlined in Design Decision SEC-MD-CFG-001.	Ensures your security policies are systematically arranged, managed, clearly defined, and easily accessible.	None.
MDS-REQD-CFG-032	Configure policy level *Applied To* for respective group: • VCF_FM • VCF01_M01 • VCF01_W(...)	The policy will be automatically applied to new VCF fleet and VCF instance management components.	The policy level *Applies To* takes precedence over the *Applied To* scope of the individual Distributed Firewall rule.
MDS-REQD-CFG-033	For each policy, configure Distributed Firewall rules to permit all traffic within the Environment (Zone) with the Action set to Jump to Application.	• Permit all traffic within the Environment (Zone), while allowing for more detailed protection through policies in the Distributed Firewall Application category section. • Provides flexibility to begin with Environment (Zone) protection and improve it later without altering the Distributed Firewall Environment policies.	Requires more manual rules creation and management.
MDS-REQD-CFG-034	• For each Environment (Zone) policy, configure Distributed Firewall rules for the required inter-zone communications with application-level granularity.	Permit the required traffic to VCF management components that require fleet and instance-level communication.	None.

Design Requirement ID	Design Requirement	Justification	Implication
MDS-REQD-CFG-035	• At the end of each Environment (Zone) policy, configure a Distributed Firewall "lockdown" rule. • Start with action **Allow and Log**, and change to **Drop and Log** after ensuring policy accuracy. • Configure rule log label.	• Prevent any additional inbound or outbound zone communications. • Identify the policy of interest in the Distributed Firewall syslog easily.	• Requires ESX servers to send logs to a Syslog solution, like VCF Operations for Logs. • Requires Distributed Firewall log monitoring to prevent policy misconfiguration.

Planning and Preparation for Management Domain Security

Planning and Preparation

Before you implement the Management Domain Security blueprint, follow these procedures to ensure a smooth deployment.

Procedure Grouping	Procedure	Description
Valid vDefend License	vDefend Feature and Edition Guide	Make sure that you have a valid license to use vDefend Distributed Firewall.
Activate the vDefend Distributed Firewall (DFW) on vSphere Distributed Port Groups	Activate NSX on Distributed Virtual Port Groups (DVPGs)	The vDefend Distributed Firewall is inactive for VCF greenfield deployments on vSphere Distributed Port Groups. You enable "**NSX on DVPGs"** on the respective VCF management domain vSphere cluster to activate the DFW for VMs connected to DVPGs

Implementation of Management Domain Security

Implementation

Use the Terraform provider for VMware NSX to secure the management domain for VMware Cloud Foundation.

Procedure Grouping	Procedure	Description
Terraform Provider for VMware NSX	Prepare a management host and install Terraform	The Terraform Provider for VMware NSX is a Terraform plug-in that allows you to interact with VMware NSX and it is officially supported by Broadcom.
Management Domain DFW configuration with Terraform	Example Terraform configuration files for securing the VMware Cloud Foundation management domain components with VMware vDefend - GitHub repository.	To deliver vDefend DFW policies and protect the VMware Cloud Foundation management domain, you create the required Terraform resources (groups, services, DFW policies, and rules). You use variables and data sources to fetch dynamic information from the NSX manager that is required for the resource definition.
Remove VCF management VM from User Excluded Group	Manage a Firewall Exclusion List	To activate the DFW on the respective VMware Cloud Foundation management domain VMs, you remove them from the DFW User Excluded Group.

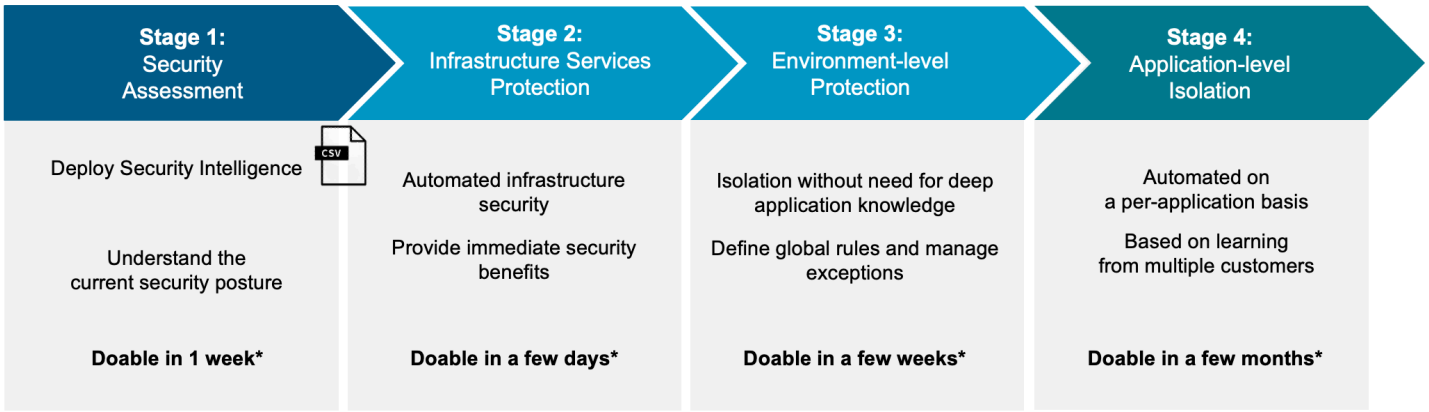
Workload Domain Security

The Workload Domain Security blueprint covers the design details and decisions, planning and preparation, and implementation steps for securing customer workloads using the vDefend DFW-1-2-3-4 staged prescriptive workflow.

Architecture Overview

Attribute	Value
Security Assessment	Before implementing VCF workload domain security, it is recommended to conduct a security segmentation assessment to understand the current workload risk posture. Security Intelligence evaluates the current ruleset and traffic flows to assign points to various security areas. These areas include the status of the Distributed Firewall, the status of Malicious IP protection, the percentage of infrastructure traffic protected, the percentage of protected environment pairs, the application protection status, and the percentage of workloads using obsolete operating systems or risky protocols. All these factors are used to calculate a vDefend Security Segmentation Score. A Security Segmentation Report can also be generated, including detailed information about the risk posture and actionable recommendations to improve it.
Secure Infrastructure	Involves creating policy to allow only necessary network connectivity from servers and endpoints to critical common services in the VI workload domain. These shared services will most likely be services like Active Directory, DNS, patching, logging, and more. The idea is to limit connectivity to known enterprise workloads and prevent unauthorized connectivity from external IPs or untrusted endpoints. Older operating systems are at greater risk and should be protected with more fine-grained security policy instead of broad-based. Insecure protocols like Telnet, SMBv1, LLMNR, FTP, and any other unencrypted ports and protocols should also be denied with security policy.
Secure Environment	Applications in the network are separated into multiple environments or zones such as production, user acceptance testing, quality assurance, or development. These zones can be utilized to create zone-based policy. Production servers should be protected from the other zones, and only necessary network connectivity should be allowed between them. If a workload in one of these zones is compromised, the inter-zone policy will protect lateral movement and potential data exfiltration.
Secure Application	Applications can require both inter-app policy and micro-segmentation policy. • Inter-app Policy: An application can be made up of one or more workloads. Security policy is created to limit what ports and protocols other applications or workloads are allowed to access within the application. Application policy can be created with no dependency on the network architecture. The workloads within or between applications can be separated at a Layer 3 boundary or even live within the same Layer 2 broadcast domain. Policy between applications should be created based on the documentation provided by the vendor or application owners. Only necessary services should be allowed while all others are denied. Inter-app policy can be created at a macro-level (App 1 to App 2 on port 80) or to a specific tier (Web, App, DB, etc) within the application depending on your security policy requirements. • Micro-segmentation Policy: Network connectivity, external or internal, in the datacenter should be inherently untrusted. Security policy is created to limit connectivity even between workloads grouped within a single application. Traditionally, application communication was limited between logical tiers like web, application, and database. Workloads within a tier would have free access between each other and are susceptible to potential malicious lateral movement. In a micro-segmentation framework, even workloads within the same tier are subjected to security policy. Workload policy within an application should only allow necessary ports and protocols based on the documentation and block all other services to prevent potential lateral movement.

Figure 232: Workload Domain Security Prescriptive Workflow



Workload Domain Security Design Selections

The following models for the workload domain align with the design profile for this blueprint.

Layer	Selected Model	Additional Information
Security	vDefend DFW Stage Prescriptive Segmentation Design	vDefend DFW 1-2-3-4 prescriptive deployment process that follows lateral traffic patterns to secure VCF workloads quickly.
	vDefend Grouping and Tagging Design	Identify and organize the VM inventory within a workload domain, using various object-grouping techniques to simplify the design of the security policy.
	vDefend DFW Policy Design	Strategic framework for architecting a Zero Trust security posture for VCF workloads by applying hierarchical use of DFW Policies.

Workload Domain Security Model

This section covers the security design details and decisions for securing the workload domain for VMware Cloud Foundation.

- vDefend DFW Stage Prescriptive Segmentation Design
- vDefend Grouping and Tagging Design
- vDefend DFW Policy Design

vDefend DFW Stage Prescriptive Segmentation Design
vDefend DFW 1-2-3-4 provides a 4-stage prescriptive deployment process that follows lateral traffic patterns to quickly secure VCF workloads, with built-in guidance that mirrors lateral traffic components and policy categories within vDefend DFW.

Stage 1: Security Segmentation Assessment

The initial phase involves assessing the current state of the environment to determine existing protection levels. This foundational step helps establish a baseline and a measurable security score. The assessment tracks progress as new rules are defined and enforced, providing a starting point for the entire segmentation initiative.

The score is structured using multiple factors, with a maximum of 95 points, all based on the latest 30 days of traffic.

Score Category	Points	Description
Distributed Firewall in Operation	5	Points are awarded for having the Distributed Firewall activated along with Malicious IP Protection
Infrastructure Protection	5	Percentage of critical infrastructure traffic (14 predefined services) that is protected by an explicit rule in the Infrastructure category.
Environment Protection	5	Percentage of environment pairs (e.g., Prod to Dev) that have a default rule action set to Drop or Reject, and the coverage of workloads by at least one environment rule.
Application Protection	70	Largest component, weighted by the percentage of application workload and traffic coverage, as well as the presence of application or datacenter lockdown rules.
Workloads without Obsolete OS & Risky Protocols	10	Percentage of workloads that are not running an obsolete operating system or allowing connections using risky protocols.

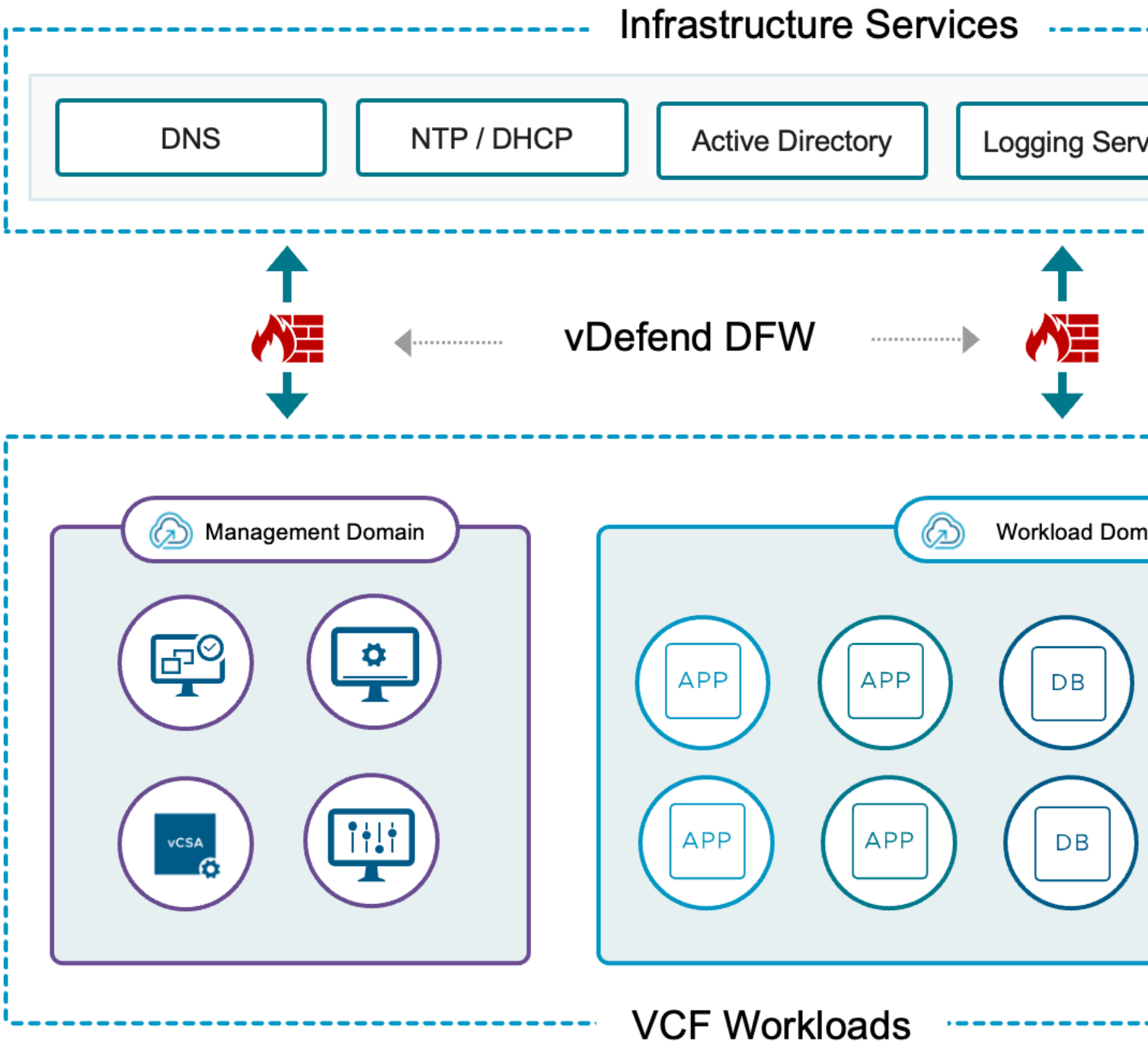
The assessment serves as a benchmark and a continuous progress tracker. By following the prescriptive workflows in **Stage 2 (Secure Infrastructure)**, **Stage 3 (Secure Environments)**, and **Stage 4 (Secure Applications)**, you will be implementing new policies and lockdown rules. As you complete these stages and your security coverage increases, you can periodically rerun the score calculation to observe your progress. The score will steadily increase, validating your progress toward a mature Zero Trust segmentation architecture.

The detailed report will also provide specific projects and recommendations to help you further improve your score.

Stage 2: Infrastructure Services Protection (Secure Infrastructure)

This stage focuses on identifying and protecting critical, shared infrastructure services within the workload domain, such as Active Directory, DNS, patching, and logging.

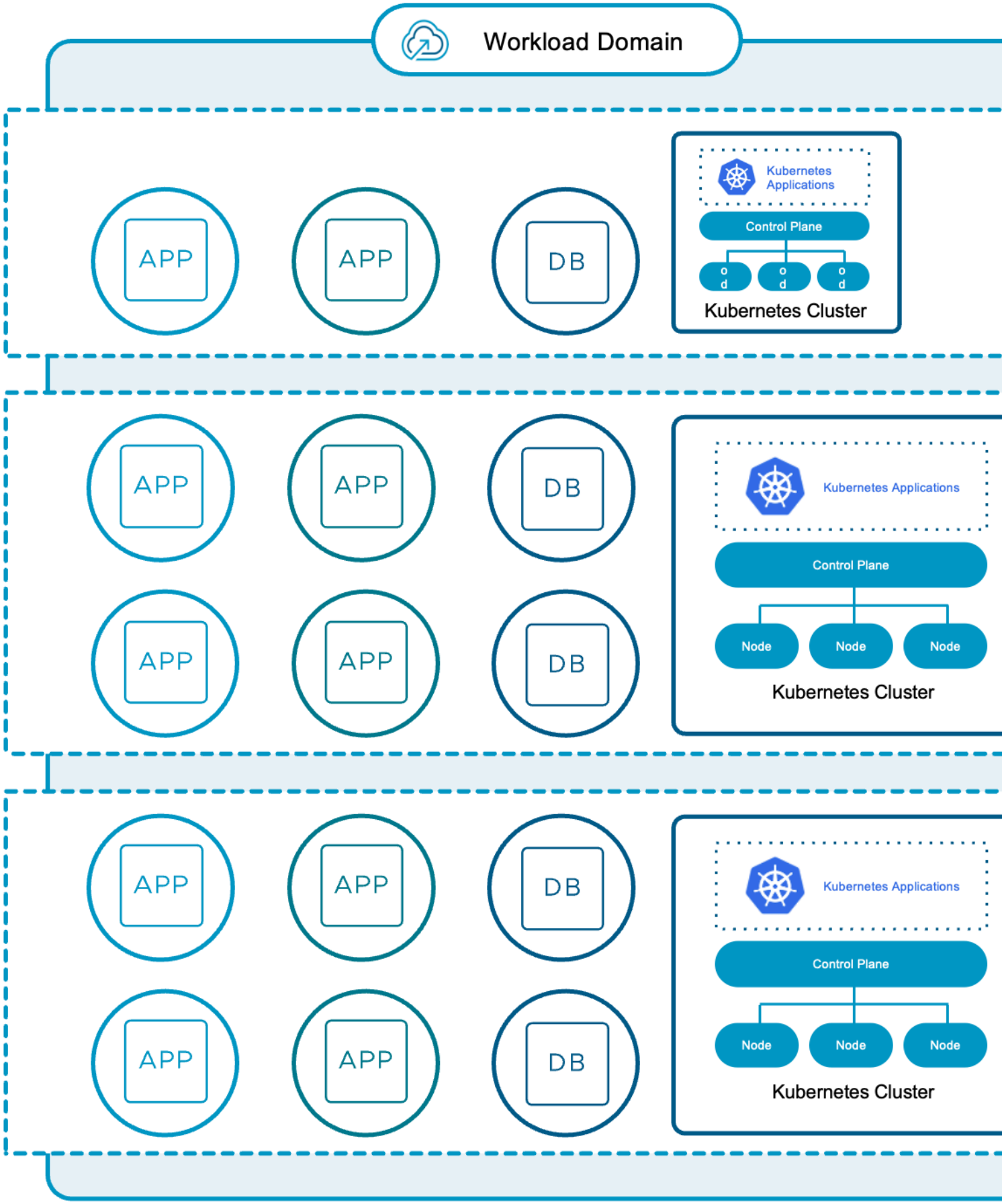
- **Policy Creation:** The primary goal is creating a policy to allow only necessary network connectivity from servers and endpoints to these core services.
- **Boundary Control:** Connectivity must be limited to known enterprise workloads, preventing unauthorized access from external IPs or untrusted endpoints.
- **Risk Mitigation:** Older operating systems, which are at greater risk, should be protected with more fine-grained security policies.
- **Insecure Protocol Denial:** Insecure protocols like Telnet, SMBv1, LLMNR, FTP, and any other unencrypted ports and protocols should be explicitly denied with security policy.



Stage 3: Environment Segmentation (Secure Environment)

This stage involves dividing the data center or network into broader security zones or environments.

- **Zone Definition:** Applications are separated into logical environments such as production, user acceptance testing (UAT), quality assurance (QA), or development.
- **Zone-Based Policy:** These zones are utilized to create zone-based policy, ensuring production servers are protected from the other zones.
- **Inter-Zone Control:** Only necessary network connectivity should be allowed between environments. If a workload in one zone is compromised, the inter-zone policy will prevent lateral movement and potential data exfiltration to more sensitive areas.


Workload Domain
APP
APP
DB
Kubernetes Applications
Control Plane
Kubernetes Cluster
APP
APP
DB
APP
APP
DB
Kubernetes Applications
Control Plane
Node
Node
Node
Kubernetes Cluster
APP
APP
DB
APP
APP
DB
Kubernetes Applications
Control Plane
Node
Node
Node
Kubernetes Cluster

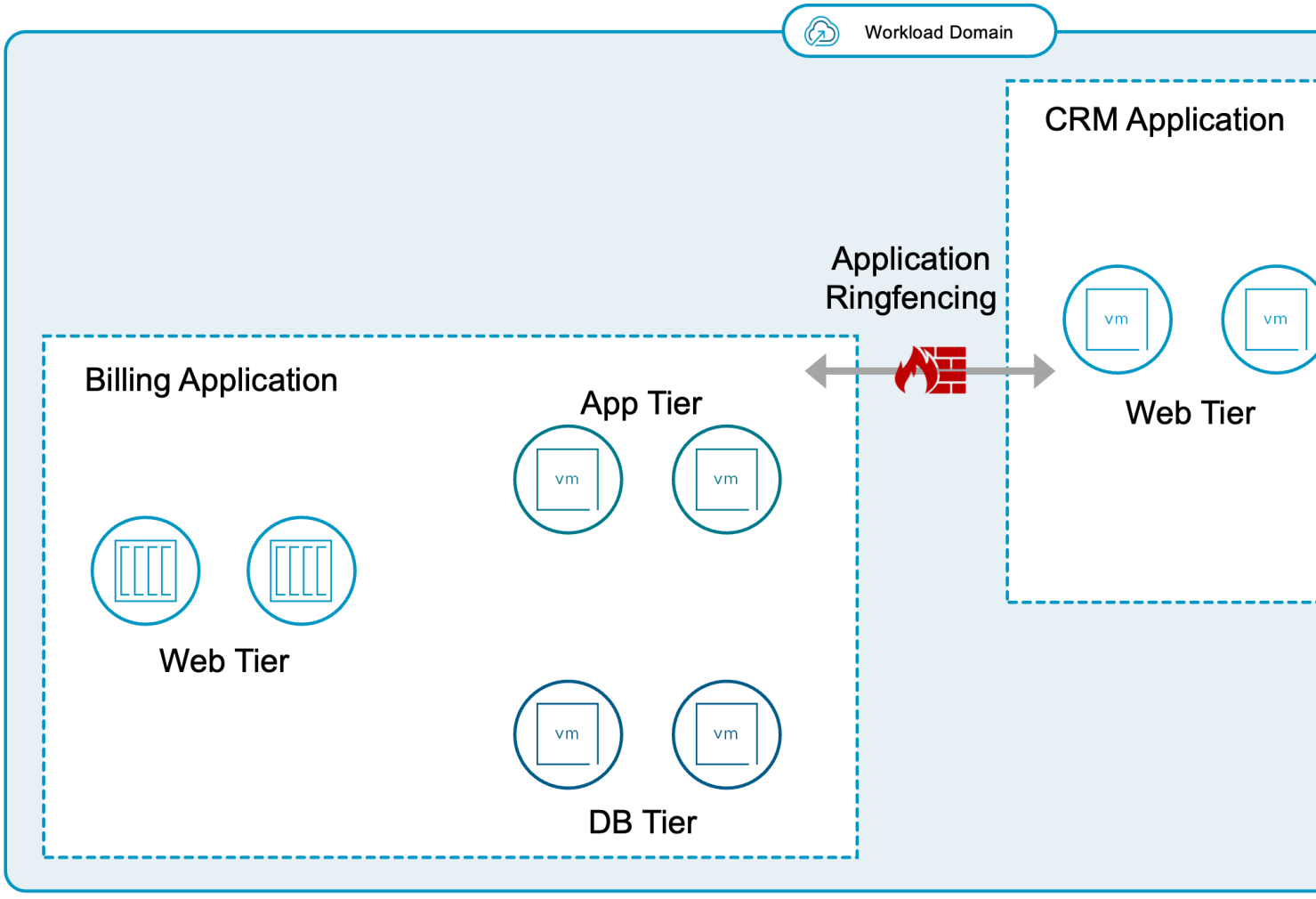
Stage 4: Application Segmentation (Secure Application)

This final stage implements the most granular security policies, which can be further refined into inter-application and intra-application controls.

1. **Inter-application Policy (App-to-App or Ring Fencing)**: An application may be composed of one or more workloads. This policy focuses on controlling access between different applications.
   - **Access Limitation:** Security policy is created to limit which ports and protocols other applications or workloads are allowed to access the application.
   - **Network Agnostic:** Application policy can be created with no dependency on the underlying network architecture, even if workloads are separated at a Layer 3 boundary or live within the same Layer 2 domain.
   - **Documentation-Based:** Policy between applications should be based on documentation provided by the vendor, application owners, or Security Intelligence, allowing only necessary services and denying all others. Policy can be created at a macro-level (e.g., App 1 to App 2 on port 80).

   Figure 233: Policy Example 1

   Example of using Application Ring Fencing policy

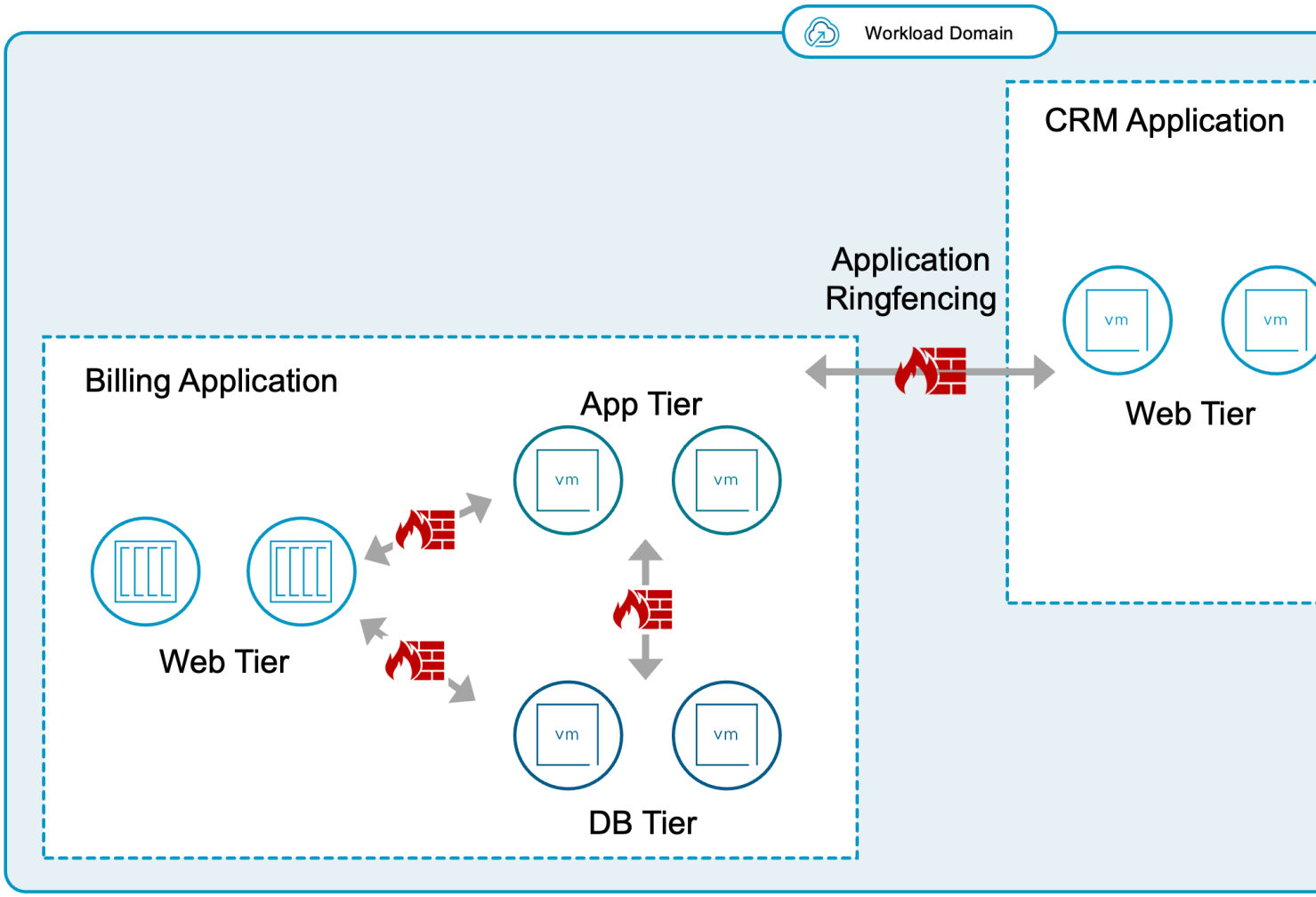


2. **Intra-application policy (Micro-Segmentation or Zero Trust)** : This is the most fine-grained level, operating on the principle that all network connectivity, internal or external, should be inherently untrusted.
   - **Intra-Tier Control:** Security policy is created to limit connectivity *even between* workloads grouped within a single application tier (e.g., between two web servers).

- **Inter-Tier Control:** Security policy is created to limit connectivity between workload tiers within a single application tier (e.g., between Web, App, DB tiers).
- **Lateral Movement Prevention:** Traditionally, workloads within a tier had free access to each other, making them susceptible to malicious lateral movement. In a micro-segmentation framework, all workloads are subjected to security policy.
- **Rule Set:** Workload policy within an application should only allow necessary ports and protocols based on documentation or Security Intelligence and block all other services to prevent potential lateral movement.

Figure 234: Policy Example 2

Example of using both Inter-App and Application Ring Fencing policy



vDefend Grouping and Tagging Design
Before designing a security policy to secure workloads within a workload domain, it is important to identify and organize the VM inventory within the workload domains of a VCF Fleet. This is achieved by leveraging various object grouping techniques to simplify the design of the security policy finally implemented.

Groups

A group is a logical construct that allows grouping of workloads into a common collection based on static (IP Address/NSX objects like segments) and/or dynamic (VM names/VM tags) membership criteria.

Table 508: Groups with static members can be created from the following NSX Objects

NSX Object	Description
IP Addresses or Subnets	Grouping of IP addresses and subnets.
Segments	All VMs/vNICs connected to this segment/ logical switch segment will be selected.
Segment Ports	This particular vNIC instance will be selected.
Distributed Port Groups	All VMs/vNICs connected to this distributed port group will be selected.
Distributed Ports	This particular vNIC instance connected to the respective distributed port group will be selected.
Virtual Interfaces (VIFs)	This particular VIF instance will be selected.
Virtual Machines	The particular VM will be selected.
AD Group	Grouping based on Active Directory groups for Identity Firewall (VDI/RDSH) use case.
MAC Addresses	Selected MAC sets that in turn contain a list of MAC addresses will be used.

Table 509: Groups with dynamic members can be created from the following NSX objects and criteria

VM Properties	Description
VM Name	All VMs that contain/equal/starts/not-equals with the string as part of their name.
VM Tag	All VMs that are applied (equals/contains/start with/end with) with specified NSX security tags
VM OS Name	All VM with specific operating system type and version
VM Computer Name	All VMs that contain/equal/starts/not-equals with the string as part of their hostname.

NSX Segment Properties	Description
NSX Segment Tag	All VMs connected to the NSX Segment applied with the tag (equal/not-equals).
NSX Segment Port Tag	The VM connected to the NSX Segment Ports applied with the tag (equal/not-equals/not in).

Distributed Port Group Properties	Description
Distributed Port Group Tag	All VMs connected to the Distributed Port Group applied with the tag (equal/not-equals).
Distributed Port Tag	The VM connected to the Distributed Port applied with the tag (equal/not-equals/not in).

IP-based Properties	Description
IP Set Tag	All IPs from the IP Set applied with the tag
Group (IP Address Only) Tag	IP Address Only based Groups applied with the tag

Nested Security Groups

Groups can be nested. A Group may contain multiple groups or a combination of groups and other grouping objects. A security rule applied to the parent Group is automatically applied to the child Groups.

Table 510: Design Decisions on Workload Grouping in the Workload Domain

Design Requirement ID	Design Requirement	Justification	Implication
WDS-REQD-CFG-001	For environments (zones) with distinct IP schemas, establish IP-based groups with static membership aligned with the CIDR block assigned to each environment.	Improves efficiency and reduces resource consumption.	• Not resilient to network architecture changes. • The static-membership group must be maintained periodically to ensure the security posture remains up to date. • Cannot be used in the Distributed Firewall "Applied To" field.
WDS-REQD-CFG-002	Create a group with dynamic membership for more granular zones, application boundary, and different tiers of an application.	More flexible and allows firewall rules to be independent of the underlying network architecture.	When new workloads are commissioned, automation workflows must ensure that the workloads are appropriately tagged by a user with correct permissions.
WDS-REQD-CFG-003	Use grouping criteria based on the DVPG, NSX segment, or VPC subnet (static assignment or dynamic based on tags) when VMs from different application tiers are connected to specific segments (e.g., web-tier, db-tier).	Streamlines the grouping methodology, aligning with the application architecture while ensuring robust application inter-tier protection.	Additional groups must be defined if application intra-tier protection and/or zero trust are required.
WDS-REQD-CFG-004	Use VM-based grouping criteria based with static assignment or dynamic assignment based on tags or names when all application VMs are connected to a single network (DVPS, NSX Segment, VPC subnet.	Aligning grouping with the application architecture when deployed on a flat network.	Tags had to be manually created and assigned to the application VMs.
WDS-REQD-CFG-005	Create IP based groups for external inventory within the datacenter either in the form of physical servers or non-NSX managed virtual workloads and for external IPs.	Simplifies security policies between workloads in a Workload Domain and external inventory.	An IP-based group cannot be used in the Distributed Firewall "Applied To" field.
WDS-REQD-CFG-006	Limit the number of levels of nesting in nested groups to 3 or less.	Reduces the complexity of troubleshooting, minimizes unintentional policy results and optimizes the computational burden of your publishing policy.	None.

Tags

A tag is a user-defined label that can be associated with a virtual machine or group of virtual machines or NSX objects like segments, segment ports etc.

Each tag has the following two attributes:

- Tag (refers to the tag name. It is required and must be unique and case-sensitive)
- Scope (optional)

Scope is analogous to a key and tag name is analogous to a value. For example, if all virtual machines are to be tagged based on their operating system (Windows, Mac, Linux). You can create three tags, such as Windows, Linux, and Mac, and set the scope of each tag to OS. Tags are used to group workloads logically, which can be helpful when defining

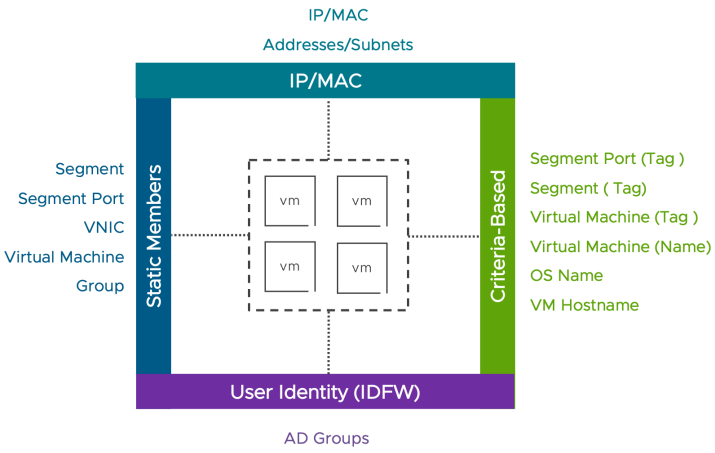
security policies. They can be used to simplify and automate firewall rules and apply different security settings specific to the workloads they are being applied to.

Naming Conventions and Tags

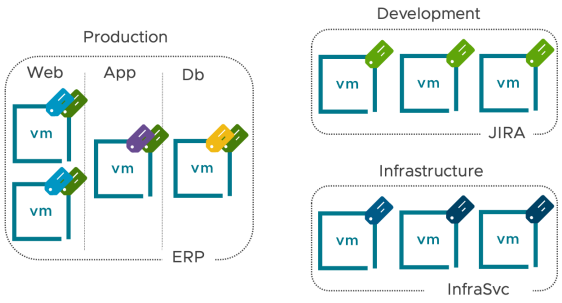
1. Use a descriptive name: Give your group a name that describes what it protects. Example: All workloads belonging to "PROD" can be a part of group "SG_PROD".
2. If the number of tag and group requirements are within the supported scale limit then keep it simple, have multiple individual tags, do not have a combined tag.
3. If the number of tag and group requirements are beyond the supported scale limit (ref. VMware Configuration Maximums):
4. Combine scope and tag names to reduce the number of tags required.
5. Modify the name of the tag to accommodate for the description of multiple layers it can help protect (combined tag)

Example: For a workload that belongs to the zone: PROD, Application: APP1 and Tier: Web create a single tag: PROD_APP1_Web instead of associating three separate tags with the workload.

Groups



Tags



Application	Tag:Scope
Engineering JIRA	jira:dev
Finance ERP System	erp:prod
Active Directory	mad:infra
DNS Servers	dns:infra

Application Tier	Tag:Scope
Web Tier	web:
App Tier	app:
Db Tier	db:

Table 511: Design Decisions on Workload Tagging in the Workload Domain

Design Requirement ID	Design Requirement	Justification	Implication
WDS-REQD-CFG-007	Use Tags to manage VMs and NSX objects (Groups, DVPGs, NSX segments, VPC subnets, ports, etc.) metadata.	• Enables dynamic, attribute-based application grouping and security policy. • Decouple security policy membership from network topology, allowing for granular micro-segmentation. • Ensures that automated workflows' firewall rule placement based on the presence of specific user-defined metadata.	Tags had to be manually created and assigned to the application VMs or NSX objects.
WDS-REQD-CFG-008	Deliver a standardized tagging schema by utilizing both Tags and Scope values (i.e., tag = app; scope=environment)	Provides a flexible, automated posture for the workload within a hierarchical grouping structure.	Both the tag and scope have to be defined.
WDS-REQD-CFG-009	Use a descriptive naming convention for tags.	Provides instant context and ensures that the workload automatically lands in the correct group.	None.

vDefend DFW Policy Design

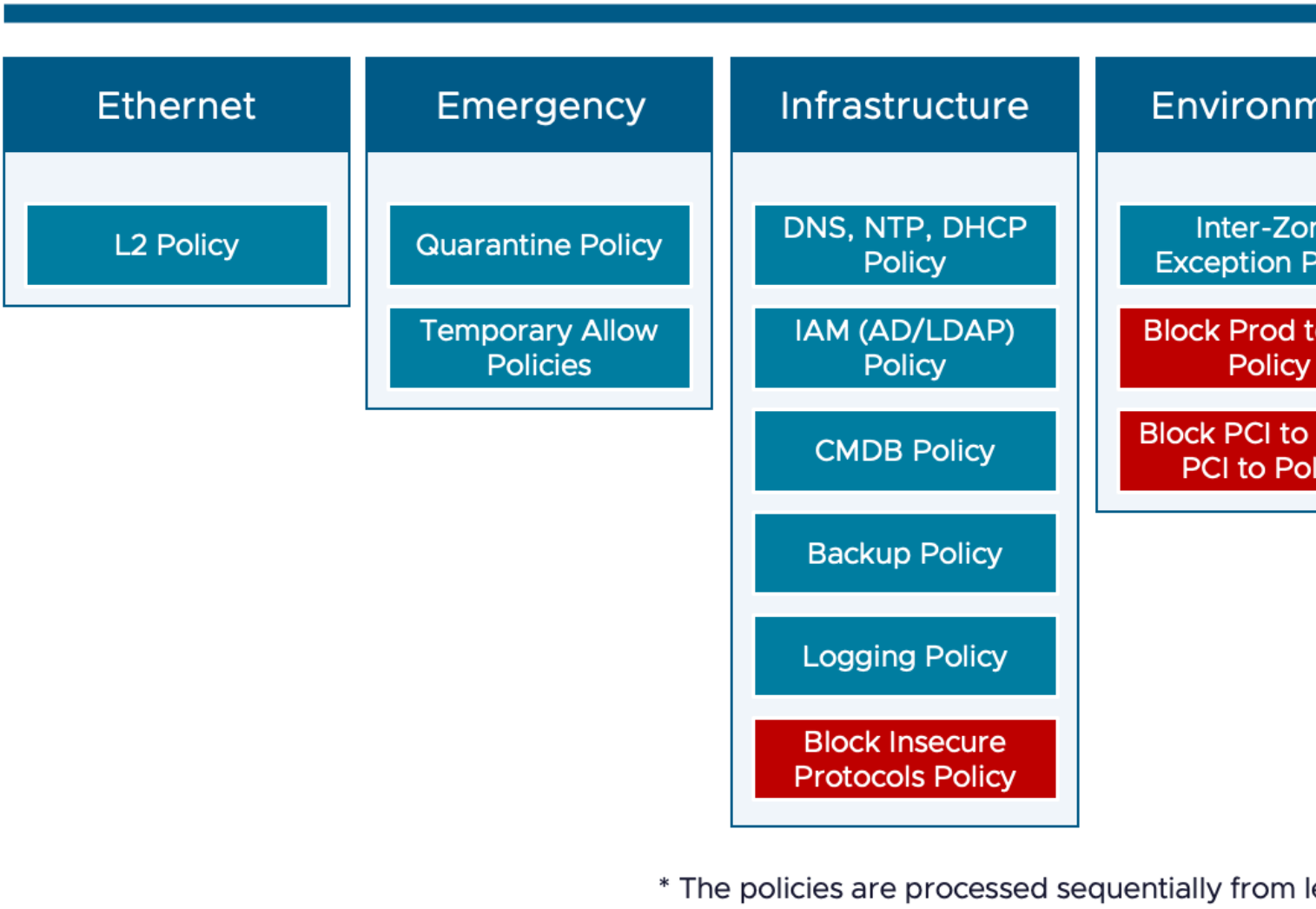
This section outlines the strategic framework for architecting a Zero Trust security posture for VCF workloads by leveraging hierarchical use of DFW Policy Categories (Infrastructure, Environment, and Application) and the critical importance of scoping rules in the "Applied To" field to minimize resource overhead.

Policy Design

Policy is a collection of one or more rules. Rules are a set of instructions that determine whether a packet should be allowed or blocked. Rules in a policy are evaluated from top to bottom, and a match stops further processing when the action is set to "Allow", "Drop", or "Reject". In the Environment category, a fourth action called "Jump to Application" will stop rule evaluation in the Environment category and proceed to the Application category for further rule evaluation.

The vDefend Distributed Firewall has several categories where you can place rules. Policies and rules are assessed and enforced starting with the Ethernet category. After rules are assessed from top to bottom, the flow is compared to policies and rules in the Emergency category, followed by Infrastructure, Environment, and finally Application. Once a flow has matched a rule with a allow/drop or reject action, it's not matched against any further rules.

Figure 235: vDefend Distributed Firewall Policy Design for VMware Cloud Foundation



Ethernet

The ethernet category is for Layer 2 policy only. These policies have source and destination groups that only have MAC addresses as members.

Emergency

The emergency category should be used for temporary rules only. These rules are created to block connectivity during a security incident or allow connectivity to and from an application if there are connectivity issues. These rules are eventually moved to the infrastructure, environment, or application sections that are explained below

Infrastructure

The infrastructure category is where all common services rules are placed. These rules should be created to protect critical infrastructure services like Active Directory, DNS, patching, logging, and more. Broad rules to protect common services from insecure protocols like Telnet, SMBv1, LLMNR, FTP, and any other unencrypted ports and protocols should be placed here as well.

Table 512: Design Decisions on Infrastructure Category Distributed Firewall Rules for the Workload Domain

Design Requirement ID	Design Requirement	Justification	Implication
WDS-REQD-CFG-010	Create a common services policy to secure workloads in the infrastructure category.	Each common service like Active Directory, DNS, patching, SIEM, and others, should have their own sections that secures communication to and from the workloads.	Groups might include RFC1918 as a source group if specific endpoints are not know.
WDS-REQD-CFG-011	Block insecure protocols to each common service in the Infrastructure Category.	Prevents unencrypted, insecure protocols and ports from being used to exploit workloads with potentially older or unpatched operating systems.	You must configure exceptions in the common services policy.

Environment

The environment category is where inter-zone security policies are placed. These rules protect workloads that serve different functions like production, user acceptance testing, quality assurance, and development. The broad-based policy to deny inter-zone communication should be placed near the bottom and exception to the policy be placed at the top.

Table 513: Design Decisions on Environment Category Distributed Firewall Rules for the Workload Domain

Design Requirement ID	Design Requirement	Justification	Implication
WDS-REQD-CFG-012	Create inter-zone exception rules.	At times, traffic must be allowed between zone segments. These exceptions rules must be created here and documented for compliance.	Creates potential for lateral movement.
WDS-REQD-CFG-013	Block each environment's inter-zone based traffic to and from all other zones.	Create policy to block traffic to and from each zone. This is to prevent unauthorized lateral movement if a workload is compromised.	Might block legitimate traffic. Exceptions should be placed above if application traffic patterns are not well understood.

Application

The application category is where inter-application and micro-segmentation policy are placed. Each application should have it’s own policy section with rules placed inside them. More broad-based inter-app policies that don’t require micro-segmentation should be placed near the top while fine grained micro-segmentation policy are placed below. These rules will be created based on the application owner or vendor documentation.

Rules to protect applications from insecure protocols like Telnet, SMBv1, LLMNR, FTP, and anything other unencrypted port and protocol should be placed at the bottom to allow exceptions to be included in application policy rules.

The default Layer 3 rule should be set to ANY/ANY/ANY/ALLOW while migrating applications into the workload domain. Once all applications have been migrated and have policy implemented, a principal of least privilege model can be used where the default rule is switched to deny. This also allows you to delete the “block insecure protocols policy” at the bottom of the application category since the default rule will now drop that traffic.

Table 514: Design Decisions on Application Category Distributed Firewall Rules for the Workload Domain

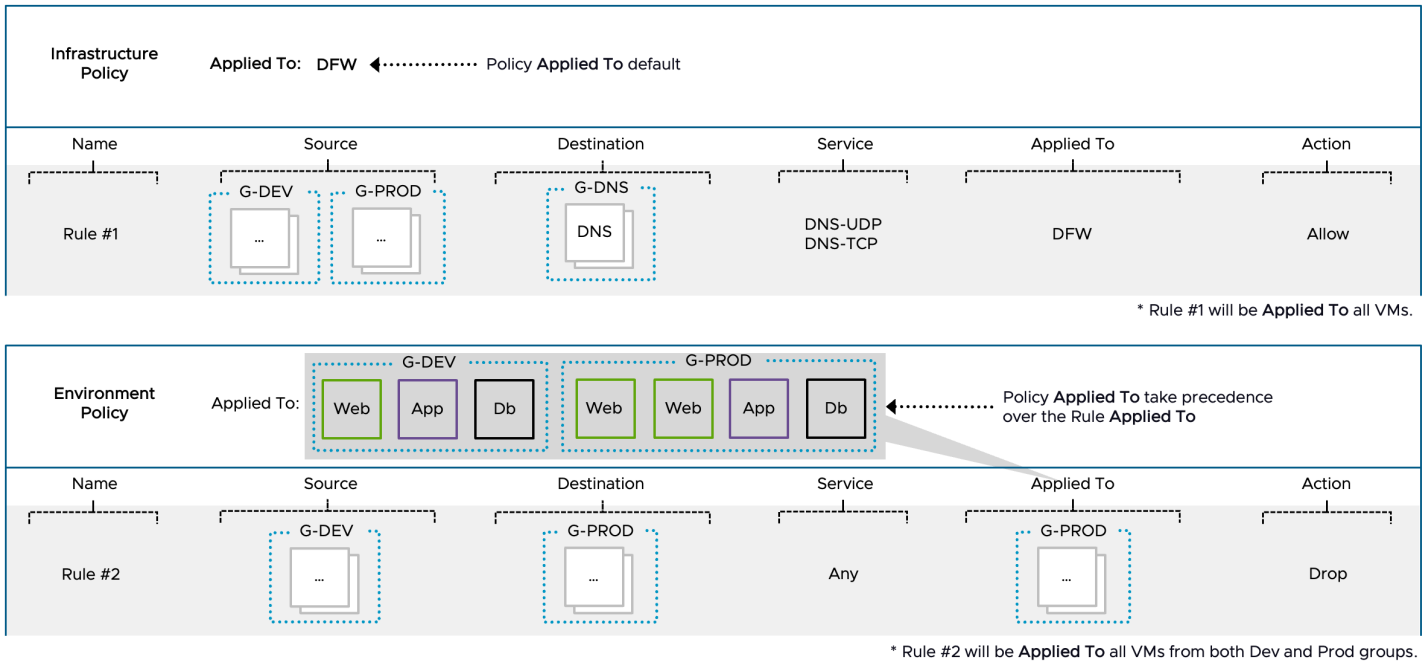
Design Requirement ID	Design Requirement	Justification	Implication
WDS-REQD-CFG-014	Create application-ring fencing (inter-app) policy.	The policy secures the communication between applications. These applications may not need or are not prepared for a full micro-segmentation policy.	Does not prevent lateral movement within applications. Needs to be well understood for full micro-segmentation and zero-trust.
WDS-REQD-CFG-015	Create micro-segmentation policies per application.	The policy secures traffic between applications and workloads within the application.	The policy is more complex to configure but enables the zero-trust model.
WDS-REQD-CFG-016	Block insecure protocols policy.	The policy blocks all other traffic using insecure protocols like Telnet, FTP, SMBv1, LLMNR, etc.	Can block legitimate traffic if exceptions are not configured in the specific application sections.

vDefend Distributed Firewall "Applied To"

Policy can be assigned groups in the "Applied To" field. This determines whether rules will be applied to all DFW protected vNICs or to specific vNICs defined in one or many groups. Broad-based policies for infrastructure and environment categories should set the "Applied To" field to groups identifying different workloads environments or left to the default value - DFW.

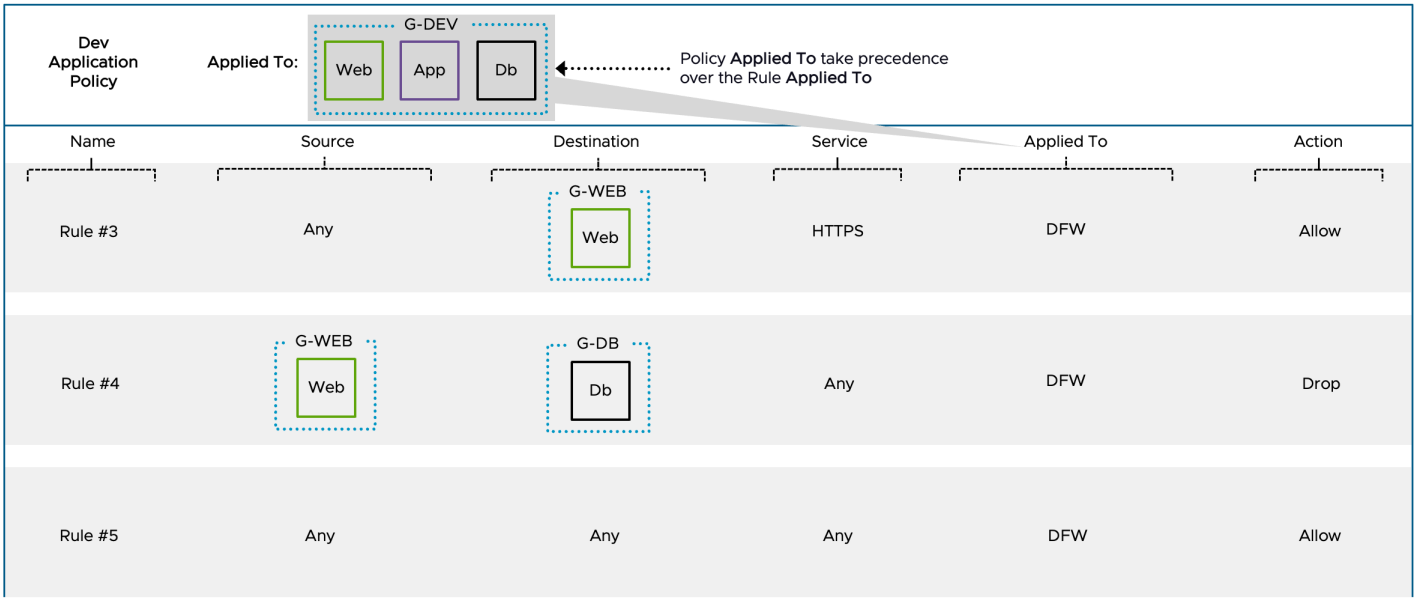
Rules should utilize the "Applied To" field to ensure efficient rule enforcement on the VMs that the rules pertain to. This also enables the ability to use a scoped deny ANY/ANY/ANY rule that enforces principle of least priviledge for targeted application workloads defined using the "Applied To" field without affecting any other application.

Figure 236: Distributed Firewall "Applied To" at Policy Level - Example 1



More fine-grained policies in the application category should set the "Applied To" to the group(s) of VMs or vNICs that are part of the application or specify the "Applied To" field for each rule individually.

Figure 237: Distributed Firewall "Applied To" at Policy Level - Example 2

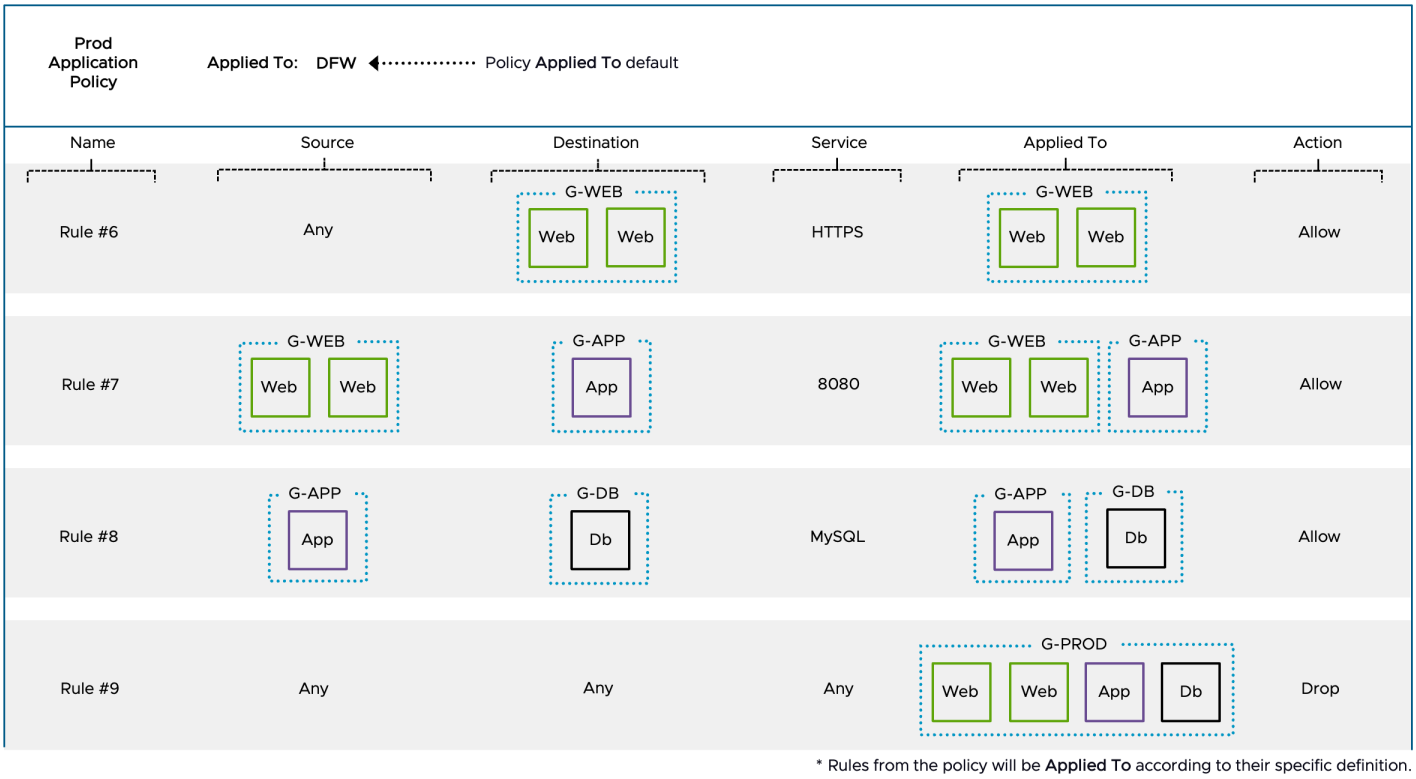


* All rules from the policy will be **Applied To** all VMs from the Dev group.

Rules

Rules can have an "Applied To" field set to DFW or one or multiple groups. If a policy has the "Applied To" field set to a group or multiple groups, it will override the "Applied To" field set in the rule. The source and destination of a rule is set using groups.

Figure 238: Distributed Firewall "Applied To" at Rule Level



* Rules from the policy will be **Applied To** according to their specific definition.

Note:

When implementing policies with a rule-level "Applied To" (policy-level is not configured), it is crucial to ensure that all rules in the policy have an "Applied To" definition (i. e. DFW is not used). Failure to do so will result in the rule being universally applied to all workloads.

Table 515: Design Decisions on Distributed Firewall "Applied To" Field for the Workload Domain

Design Requirement ID	Design Requirement	Justification	Implication
WDS-REQD-CFG-017	Leverage “Applied To” field in firewall policies or rules to restrict which vNICs will receive the firewall rule in question.	To prevent very large firewall tables from being loaded on vNICs, which will in turn negatively affect performance.	IP-Based groups, and AD groups cannot be used in the "Applied To" field
WDS-REQD-CFG-018	Use policy-level "Applied To" on infrastructure and environment categories policies.	Distributed Firewall policies in the infrastructure and environment categories define broad-based control.	• All Distributed Firewall rules in a particular policy will be applied to any workload parf og the "Applied To" group. Certain workloads may receive rules that are not relevant to them. • Rule-level "Applead To" is ignored when a policy-level "Applead To" is defined.
WDS-REQD-CFG-019	Use rule-level "Applied To" on application category policies.	Distributed Firewall policies in the application category define application-specific control.	• Could lead to more complex rule definition and manageability overhead. • Policy-level "Applied To" should not be used in conjunction with rule-level.

Table 516: Design Decisions for DFW "Applied To" field

Important:

For the most optimized rule enforcement, the "Applied To" field on the rule should be set to one or more groups. This ensures only the required rules are enforced on the VM vNIC and rules that do not pertain to the workload will not be installed or evaluated. The "Applied To" field at the policy level will need to include all groups that all the rules use to ensure policy is installed correctly for all workloads. The result of using "Applied To" at the policy level is rules that might not pertain to a workload will be installed on the VM’s vNIC resulting is evaluation of rules that will never match and suboptimal rule installation.

Layer 7 Policy Considerations

A Layer 7 rule context profile should be coupled with the correlating service (port/protocol). An example being if you configure a rule to inspect traffic to confirm the payload is actually HTTP traffic, the service should also be set to HTTP (TCP/80) or whatever port you expect the HTTP traffic to utilize in the workload. L7 rules with multiple protocols (ie DNS, SSH, etc) should be avoided. Create separate rules for each protocol and correlating context profile. While setting a service to any and defining a context profile is allowed, it should be avoided to prevent large amounts of unnecessary packets being sent to the deep packet inspection engine. Some examples are below:

Source	Destination	Service	Context-Profile	Action
RFC1918	Web-Server	HTTP (TCP/80)	HTTP	Allow
App1	DB-Server	MYSQL (TCP/3306)	MYSQL	Allow

Source	Destination	Service	Context-Profile	Action
Web-Server	App-Server	TCP/8080	HTTP	Allow

Table 517: Design Decisions on Distributed Firewall L7 Rules for the Workload Domain

Design Requirement ID	Design Requirement	Justification	Implication
WDS-REQD-CFG-020	Layer 7 rule services should be coupled with the correlating context profile.	Ensures optimized Distributed Firewall Deep Packet Inspection (DPI) flow inspection.	More rules need to be created in the Distributed Firewall

Planning and Preparation for Workload Domain Security

Security Intelligence

Security intelligence provides a graphical user interface to visualize the security posture and network traffic flows that occur in your environment.

A Security Services Platform, and therefore Security Intelligence is linked to an NSX manager cluster. Depending on your private cloud needs, you may have multiple Security Services Platform instances and multiple Security Intelligence within a single VMware Cloud Foundation instance.

Enabling Security Intelligence is a mandatory step for the Security Segmentation Journey.

Segmentation Planning

The metadata in CSV files is used to tag VMs and create groups, which are foundational for defining security policies in the subsequent stages. The CSV file doesn't have to contain ALL assets, and multiple iterative imports are possible. The CSV can be exported from a CMDB that maintains datacenter assets metadata.

The CSV file must contain at least 1 column that will be used by the Security Services Platform to find the corresponding asset from the NSX inventory. Commonly used fields are VM names, IP addresses or UUIDs. Multiple columns can also be used as identifiers – this is useful when some identifiers might be duplicated: for example, if multiple VMs have the same IP address, the VM name or the UUID can be used as a differentiator.

The fields in the CSV file define a hierarchy for your datacenter assets, which directly correlates to the segmentation levels used in the Security Journey.

Table 518: CSV File Schema

CSV Field	Description	Correlation to Security Journey Stages
asset type	Defines the type of asset, such as VIRTUAL_MACHINE, DVPG, SEGMENT, or GROUP. If this field is not specified, the asset type is assumed to be VIRTUAL_MACHINE.	Used in **Stage 2: Secure Infrastructure** and **Stage 4: Secure Applications** to identify the asset receiving the security policy.
asset id	A unique identifier for the asset (e.g., UUID).	Used by SSP to find the corresponding asset in the NSX inventory.
asset ip	The IP address or IP range of the asset.	Used by SSP as an identifier, particularly as a differentiator if asset name is duplicated.
asset name	The name of the asset (e.g., a VM name).	A commonly used field by SSP to find the corresponding asset in the NSX inventory.
region	The top-level geographic location of the asset.	Part of the hierarchy used for segmentation and group creation.
zone	A more granular area within a region (e.g., a specific datacenter or availability zone).	Part of the hierarchy used for segmentation and group creation.

CSV Field	Description	Correlation to Security Journey Stages
environment	The classification of the workload (e.g., Production, Test, Dev, Staging).	Directly correlates to **Stage 3: Secure Environments**, where policies are defined to control traffic between these security zones (e.g., preventing Dev from communicating with Prod).
application	The business application the asset belongs to (e.g., Oracle-Finance, InHouse-Logistics).	Directly correlates to **Stage 4: Secure Applications**, enabling the creation of application-level segmentation policies (Inter-application control).
application tier	A layer within the application (e.g., DB, App, Web, nginx, mysql).	Directly correlates to the most granular level of **Stage 4: Secure Applications**, allowing for fine-grained **Micro-segmentation Policy** (Intra-application/Intra-tier control).
infrastructure service	Specifies a core service the asset is related to (e.g., dns:udp/53, ntp:udp/123).	Used in **Stage 2: Secure shared services** (Infrastructure Services Protection) to identify and protect critical shared services like DNS or NTP.

Example CSV File

Below is an example dataset in a CSV file that will be imported into the Security Segmentation Planning in SSP using the Segmentation Planning page.

asset type	asset id	asset ip	asset name	region	zone	environment	application	application tier	infrastructure service
VIRTUAL_MACHINE		192.168.150.100	rvejda0048	AMER	US-East	prod	jira	Web	
VIRTUAL_MACHINE		192.168.150.101	gukiqw9818	AMER	US-East	prod	jira	App	
VIRTUAL_MACHINE		192.168.150.102	qovsrx6257	AMER	US-East	prod	jira	DB	
VIRTUAL_MACHINE		192.168.150.103	ymosfn5317	AMER	US-East	dev	servicenow	Web	
VIRTUAL_MACHINE		192.168.150.104	xfqter0292	AMER	US-East	dev	servicenow	App	
VIRTUAL_MACHINE		192.168.150.105	oaprnh4333	AMER	US-East	dev	servicenow	DB	
VIRTUAL_MACHINE		192.168.150.111	wdnywk5352	AMER	US-East				ntp:udp/123
VIRTUAL_MACHINE		192.168.150.112	jkuwvh3796	AMER	US-East				ntp:udp/123
VIRTUAL_MACHINE		192.168.150.113	wqafba9935	AMER	US-East				LDAP:udp/389
VIRTUAL_MACHINE		192.168.150.114	oiuagx0503	AMER	US-East				LDAP:udp/389

Implementation of Workload Domain Security

Implementation

Security Segmentation Journey is used to help secure your datacenter using the vDefend DFW 1-2-3-4 stages to help protect workloads from malicious lateral movement.

Procedure Grouping	Procedure	Description
Valid vDefend License	vDefend Feature and Edition Guide	Make sure that you have a valid license to use vDefend Distributed Firewall.
Activate Security Intelligence	Activate Security Intelligence	Security Intelligence activation occurs through the Security Services Platform user interface.
Configure Security Intelligence	Configure Security Intelligence Settings	By default, Security Intelligence collects network traffic data on all prepared standalone hosts and clusters. If necessary, data collection from a standalone host or cluster can be deactivated.
Stage1: Assess Segmentation Posture	Assessing Security Segmentation	The purpose is to provide a comprehensive understanding of your current security posture and identify areas that require immediate attention as you move towards a Zero Trust deployment. The score is computed across several categories, indicating the health and completeness of your segmentation policy.
Stage 2: Secure Infrastructure Services	Monitoring Infrastructure Services	Quickly establish security rules for infrastructure, ensuring that approved servers are protected by specific rules while blocking communication with unauthorized servers.
Stage 3: Secure Environments	Monitoring Environment	Secure communication between environments (i.e. DEV, TEST, PROD) by allowing only required traffic as exceptions and blocking all other traffic. This isolation between security zones immediately improves security posture by reducing the threat blast radius without requiring deep knowledge of application traffic.
Stage 4: Secure Applications	Monitoring Applications	Secure each application by defining both inter-application (traffic between applications) and intra-application (traffic within a single application) policies.

Distributed Virtual Patching

The Distributed Virtual Patching blueprint covers design details and decisions, planning and preparation, and implementation details to deploy vDefend Intrusion Detection and Prevention System (IDPS) to protect unpatched workloads in VMware Cloud Foundation (VCF).

Architecture Overview

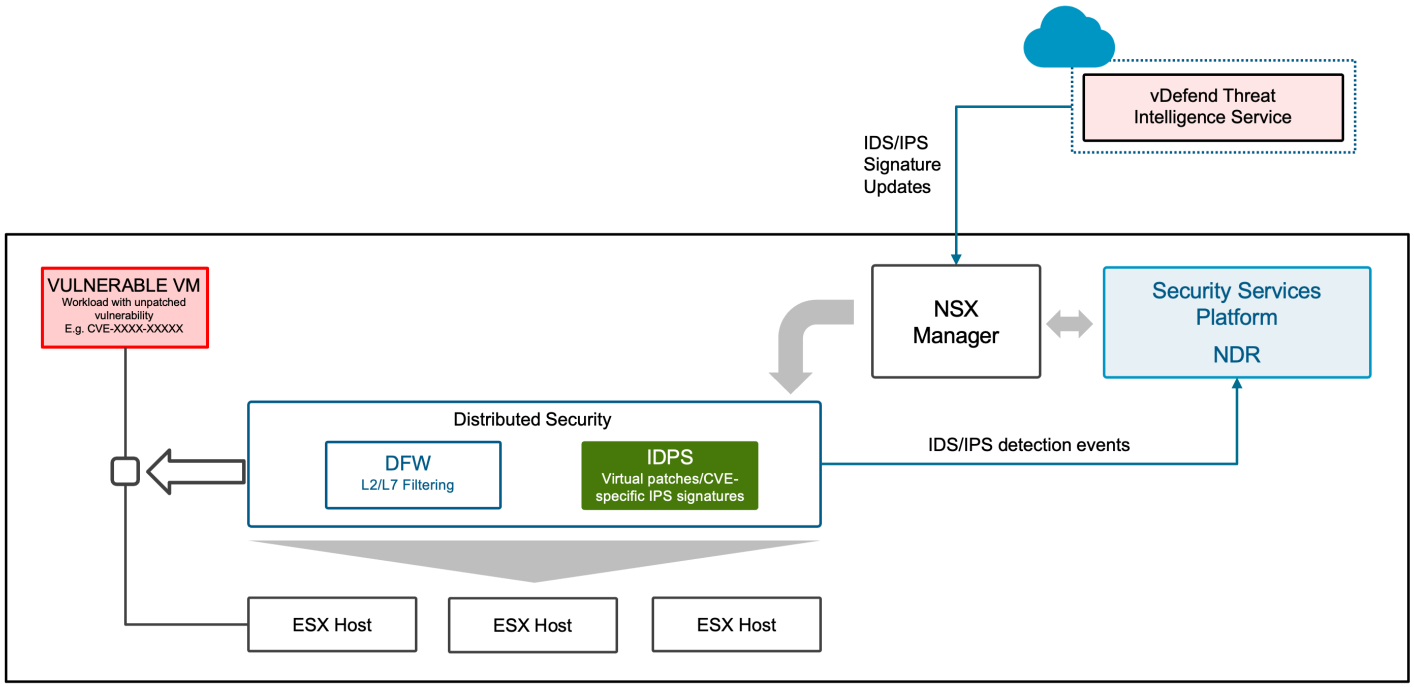
The mandate for the use of a compensating control, such as Distributed Virtual Patching, is intensified by the rapid identification of novel vulnerabilities via Frontier AI Models. Consequently, these controls must be deployed in a highly efficient and targeted manner. Incorporating these Intrusion Detection and Prevention System (IDPS) rules into the Emergency policy section enables the architecture to evaluate highly specific traffic at the inception of the ruleset. This ensures that subsequent standard threat rules are processed with minimal redundancy. This strategic placement is paramount; the virtual patch control must be configured strictly to detect and prevent potential vulnerability exploitation effectively while simultaneously minimizing the incidence of false positives.

To achieve this objective, the meticulous selection and targeted evaluation of IDPS signatures remain imperative. Identifying only the most pertinent signatures is critical to preventing the inadvertent blocking of legitimate network traffic. Furthermore, optimization of the Source and Destination fields within the IDPS rule should be pursued whenever feasible.

Additionally, utilization of the Services field is recommended to isolate the specific ports and protocols associated with the exploit mechanism.

By deploying these rules within the hypervisor-embedded layer at the virtual network interface card (vNIC) of each respective workload, vDefend effectively isolates threats prior to their propagation across the physical infrastructure or penetration into the unpatched operating system kernel. This network-layer defense mitigates high-risk vulnerability exposure, granting enterprise operations teams the requisite latency to systematically validate, schedule, and execute permanent software remedies without inducing disruptive, emergency application downtime.

Figure 239: Distributed Virtual Patching Architecture



Design Profile

The VMware vDefend Distributed IDS/IPS represents a fundamental shift in how network intrusion detection and prevention are architected within virtualized and private cloud environments. Instead of traditional models that treat security as a perimeter-based or discrete hardware function, vDefend builds intelligence directly into the fabric of the virtual infrastructure.

At its core, the vDefend Distributed IDS architecture eliminates the concepts of centralized inspection chokepoints and network "hairpinning." It achieves this by moving the threat inspection engine directly into the **ESXi hypervisor kernel**, attaching a purpose-built, software-defined IDS/IPS sensor to the virtual network interface card (**vNIC**) of every single workload. Whether a workload is a virtual machine or a container running on VKS, it receives an individualized security perimeter that travels with it dynamically.

Attribute	Value
Hypervisor-Embedded Native Architecture	Traditional network security forces internal "East-West" traffic out of the virtual layer and onto physical appliances to be inspected, halving link capacity and demanding complex VLAN/network re-architecting. vDefend re-engineers this paradigm via kernel-level sensors: the inspection engine runs natively within the ESXi hypervisor layer, turning every host into an independent, high-performance security sensor. The result is **Zero Network Disruption** — the control sits directly inside the data path of the hypervisor, inspecting 100% of packets passing through the vNIC without requiring network taps, span ports, or modifications to physical switches and routing topologies.
Context-Aware Curated Signature Distribution	One of the most critical flaws of legacy IDS/IPS systems is performance degradation caused by evaluating thousands of irrelevant signatures against every packet. vDefend allows you to target the most potent and relevant signatures against the most appropriate workloads, reducing the active signature checklist from tens of thousands down to a highly optimized, single-digit fraction while maintaining high efficiency and security.
	Security Services Platform compute, memory, storage, and reservations requirements details.
Linear Elastic Scalability	Centralized hardware appliances scale vertically — organizations must over-provision expensive hardware to handle peak data center volumes, creating inescapable bottlenecks. vDefend utilizes the distributed, idle CPU capacity across all existing ESXi hosts in the cluster. As the private cloud environment grows and more compute nodes are added, threat inspection capacity scales out linearly alongside it.
Complete East-West Visibility	While perimeter firewalls successfully defend the boundaries ("North-South" traffic), they remain completely blind to malicious lateral movements once an attacker establishes initial access. vDefend provides absolute visibility inside the data center corridor, capturing and neutralizing tactics and techniques like lateral discovery, Remote Desktop Protocol (RDP) abuse, and internal data exfiltration.

Distributed Virtual Patching Design Selections

The following models align with the design profile for this blueprint.

Layer	Selected Model	Additional Information
Security	Virtual Patching Prescriptive Workflow	
Security	Virtual Patching Policy Design	Virtual patching rules are placed in the emergency policy section so they are evaluated before standard Threat Rules.

Layer	Selected Model	Additional Information
Security	Virtual Patching Profile Design	New in VCF 9.1: IDPS Profiles can be created on a per-signature basis rather than relying on broader categories. Select relevant signatures based on CVE or specific threat.
Operations	Virtual Patching Events Correlation	Centralized event collection, proactive alerting, and optional context enrichment through vDefend Network Detection and Response (NDR). Configure Logging
Operations	Virtual Patching Lifecycle Management	

Distributed Virtual Patching Model

- Virtual Patching Prescriptive Workflow
- Virtual Patching Policy Design
- Virtual Patching Profile Design
- Virtual Patching Events Correlation
- Virtual Patching Lifecycle Management

Virtual Patching Prescriptive Workflow

The blueprint follows a 4-stage prescriptive workflow that takes an organization from risk identification through to control retirement.

Stage 1: Identify Vulnerabilities. Use vulnerability scanners or Vulnerability Assessment Solutions to flag missing patches. You will utilize this information to pinpoint the exact IDS signature to apply as the compensating control. These scanning solutions should be run against the entirety of the workloads in the data center to ensure that as little time passes between the discovery or disclosure of a vulnerability and the application of the Distributed Virtual Patch.

Stage 2: Identify the Compensating Control. Use the vTIS portal for IDPS Signature research. The output from the scanners is likely one or more CVE numbers. A CVE number is a unique ID for a computer security flaw. It stands for **Common Vulnerabilities and Exposures**. It acts like a standard dictionary name so people worldwide know they are talking about the exact same software vulnerability.

Ensure that all relevant threats details are considered for inclusion into the overall IDPS rule. This may include one or more IDPS signatures and will most often focus on one network port. Once the relevant IDPS Signatures are identified, they will become the integral part of an IDPS Profile. This profile is simply a collection of one or more IDPS Signatures that will be used to prevent any exploitation of the unpatched vulnerability against the workload.

While there are multiple methods that can be used to create this IDPS Profile, the recommended method is to use the CVE itself. This will allow for a dynamic Profile of all of the relevant IDPS signatures that may be released or updated for the protection against the exploitation of vulnerability. Choosing the signatures manually within this IDPS profile is possible but does not allow for any automatic updates to be provided automatically as threat research evolves over time.

Stage 3: Create and Enforce the Virtual Patching Ruleset. Within the NSX Manager, build a granular IDPS profile and emergency-section rule with the action set to Detect and Prevent. Ensure the use of NSX Groups made with Tags (Dynamic Groups) and utilize the ApplyTo Field to tightly control where this control is applied.

Service or ports and protocols are critical here to ensure that the Ruleset is as tightly focused as possible. This will allow for as little as possible overlap between the Distributed Virtual Patching Rule and all other IDPS Rules. This is due to the way in which the IDPS Ruleset is evaluated. The processing of this rulebase is done Emergency first and then the Threat Rules immediately after. Like all Firewall Rules, the processing of the traffic will stop on the first match. This means that the more focused the Distributed Virtual Patching Rule is the less chance of overlapping a more broadly created rule. It is important to set the Distributed Virtual Patching Rule to Detect and Prevent (IPS) to ensure the potential exploit of the vulnerability is stopped automatically. Most often, the IDPS rules within the Threat Rules Section are set to Detect (IDS) only.

Stage 4: Monitor, Integrate, and Decommission. Forward events to a SIEM (Security Information and Event Management) program that are enriched by the NDR. It is critical to be alerted to the Distributed Virtual Patching Rule

firing as it can mean that someone is actively trying to exploit the vulnerability. Optionally, there may be a process to remove the virtual patch once the actual OS or Application patch is aplies. This will become part of the organization's change control process and remove the virtual patch once the real software patch is applied.

Risk Assessment and Vulnerability Identification

Table 519: Risk Assessment and Vulnerability Identification Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
DVP-RECD-VUL-001	Conduct a risk assessment to identify and quantify unmanaged risk before deploying virtual patches, using the foundational scoping questions and information-gathering tools.	Ensures the compensating control is targeted at genuine, prioritized risk and that stakeholders and event consumers are identified up front.	Requires cross-team input (NetSec, SOC, application owners) and access to assessment tooling such as Security Intelligence or BAS tools.
DVP-RECD-VUL-002	Use a vulnerability scanner and assessment tools to identify missing patches and the exact IDS signature to apply.	Scanner output highlights missing patches and pinpoints the corresponding IDPS signature, enabling rapid and accurate creation of the compensating control.	Requires commercial scanning tooling.

Virtual Patching Policy Design

The following section lays out the design requirements to apply a fine-grained vDefend IDPS policy.

Virtual Patching Ruleset

When creating the ruleset for Distributed Virtual Patches, ensure the control is applied as granularly as possible. Utilizing the emergency policy section is the best practice: entries in the emergency section are evaluated before standard policies in the Threat Rules section, and rules are evaluated from top to bottom, stopping after the first match.

> **Note:**
>
> The Distributed IDPS is not Layer 7 (AppID) aware.

Table 520: Virtual Patching Ruleset Design Requierements

Design Requirement ID	Design Requirement	Justification	Implication
DVP-REQD-CFG-001	Place Distributed Virtual Patching rules in the emergency policy section.	Emergency-section entries are evaluated before standard Threat Rules, allowing rapid reaction to active exploit vectors.	Emergency rules must be tracked as temporary controls and removed once the underlying vulnerability is patched.
DVP-REQD-CFG-002	Narrow the scope of each rule using the four available variables: Source, Destination, Port/Protocol, and Applied To (e.g., incoming TCP/80 traffic to the NSX Group TargetVMs).	Granular scoping minimizes performance impact and limits the rule to precisely the vulnerable workloads.	The DIDPS is not Layer 7 (AppID) aware, so scoping is limited to L3/L4 constructs; NSX Groups for the vulnerable workloads must exist.
DVP-REQD-CFG-003	Set the rule action to **Detect and Prevent**.	Enforcement is crucial to successfully blocking the exploit of an unpatched vulnerability, rather than only alerting on it.	False positives will drop legitimate traffic; event monitoring is required to identify and troubleshoot them.

Virtual Patching Profile Design

The following section lays out the design requirements to define CVE-based IDPS profiles.

Intrusion Detection and Prevention System Profiles

New in VCF 9.1 is the ability to create an IDPS Profile on a CVE basis rather than relying on broader categories.

Table 521: IDPS Profiles Design Requierements

Design Requirement ID	Design Requirement	Justification	Implication
DVP-REQD-CFG-004	Create the IDPS Profile on a per CVE basis rather than relying on broader signature categories.	Granular management allows you to cherry-pick specific signatures, ensuring tailored protection within the compensating control.	Administrators must specifically identify and manually select individual signature IDs rather than applying broad category-based protections. Requires VCF 9.1.

Virtual Patching Events Correlation

This chapter details the requirement of IDPS logging and alerting and Security Information and Event Management (SIEM) integration.

Logging, Alerting, and SIEM Integration

The solution must generate an event every time a rule fires and blocks an attack. Keeping these events isolated within the NSX Manager is insufficient, as it requires manual log-ins to discover them. These events are vital for troubleshooting false positives and hold significant value outside the vDefend system.

A SIEM is a centralized software platform that aggregates, normalizes, and analyzes log data and security events from across an entire IT ecosystem. In a typical enterprise, thousands of independent systems — firewalls, servers, endpoints, cloud services, and applications — generate massive volumes of telemetry daily. A SIEM acts as a single pane of glass, ingesting this disparate data to look for patterns, detect potential cyberattacks, and help security teams manage incidents.

Table 522: Logging, Alerting, and SIEM Integration Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
DVP-RECD-INT-001	Forward all Distributed IDPS events to a centralized collection site (Syslog or SIEM).	Events isolated in NSX Manager require manual log-ins to discover; centralization supports false-positive troubleshooting and enterprise-wide correlation.	Requires Syslog/SIEM infrastructure and log-forwarding configuration.
DVP-RECD-INT-002	Configure proactive alerting to notify the appropriate engineering groups the moment a Distributed Virtual Patching event occurs.	Ensures immediate awareness of active exploit attempts against unpatched workloads.	Alert routing and on-call ownership must be defined in the SIEM or event-management platform.
DVP-RECD-INT-003	Where possible, use vDefend NDR to ingest and enrich Distributed IDPS events with VCF contextual data (such as the actual VM name) before export to the SIEM.	Enrichment saves analysts extensive manual correlation to understand the scope and impact of an event.	Not strictly mandatory; introduces an additional vDefend component to deploy and operate.

Virtual Patching Lifecycle Management

Effective lifecycle management of virtual patches requires systematically decommissioning rules via change control once official software patches are applied.

Lifecycle Management and Operations

Once the compensating control is live and events are being monitored, you must manage the lifecycle of these rules. After the actual software patch is tested and successfully applied to the host OS or application, plan for the removal of the corresponding Distributed Virtual Patch.

Maintaining overlapping controls serves no long-term structural purpose. Ensure that your change control process includes a post-patch task to decommission the IDPS rule when it is no longer needed.

As these Distributed Virtual Patching rules are very targeted at removing the risk from new and unpatched vulnerabilities in the environment, it is crucial not to omit the remainder of the potential IDPS Controls. Having IDPS inspect traffic across the associated Risk Profile is crucial to reducing overall risk in the environment. The Distributed Virtual Patching rules are written in a manner to ensure that as little as possible overlap in rules or traffic occurs so as to not reduce the overall effectiveness.

Table 523: Lifecycle Management and Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
DVP-RECD-LCM-001	Decommission the corresponding Distributed Virtual Patch once the actual software patch is tested and successfully applied to the host OS or application.	Maintaining overlapping controls serves no long-term structural purpose.	The change control system must include a post-patch task to remove the IDPS rule when it is no longer needed.

Planning and Preparation for Distributed Virtual Patching

The Mechanics of Virtual Patching

Because vDefend operates at the vNIC layer, it enables a highly effective defensive strategy called **Distributed Virtual Patching**. When a high-severity vulnerability is disclosed (e.g., Log4j or an enterprise-wide OS flaw), the window between discovery and manual patching is historically when organizations are most vulnerable. vDefend mitigates this risk immediately.

By applying a network-level signature tailored to detect and block the specific exploit mechanism at the vNIC boundary, the threat is neutralized at the earliest possible stage. Even if the application running inside the VM remains unpatched and vulnerable, the exploit packet is detected and dropped before it can ever reach the OS kernel or application. This allows the infrastructure teams the required weeks or months to schedule safe, orderly application maintenance windows.

The goal of a Distributed Virtual Patching rule is to reduce risk within the organization. The reduction of Risk can be measured in terms or Exposure time as well as the risk of having unprotected and vulnerable workloads within the data center.

IDPS Performance

Before implementing Distributed Virtual Patching, it is important to understand the overall health and remaining capacity of the IDPS Engine. There are several methods that can be used to determine the overall performance and health of the IDPS. Starting in SSP 5.2, there are built in dashboards that offer at a glance health and capacity graphs that will help ensure that the engine is operating normally and has available capacity. However, if these dashboards are showing that the engine is running at or near capacity, it may be prudent to review the existing rules to ensure that they are all configured correctly according to the best practices.

Intrusion Detection and Prevention System (IDPS) resource utilization refers to the consumption of system assets - specifically CPU and memory - required to perform deep packet inspection and signature matching.

Security Services Platform (SSP) monitors resource utilization through an asynchronous collection framework to help identify whether performance bottlenecks originate from general infrastructure overhead or high volumes of inspected traffic. Metrics are collected from the host transport nodes and are categorized to provide visibility into the security service (L7/IDPS) footprint. The IDPS service operates as a specialized pipeline within the host capacity and may impact overall host health even if other background processes remain low.

Security Services Platform evaluates historical utilization against defined thresholds to assign health status based on two statistical percentiles over a 7-day window: P95 (representing worst-case peak hours) and P80 (representing typical high-load periods).

Status	Green	Yellow	Red
Notes	The service has significant headroom and can handle traffic spikes.	The service is approaching its allocated resource threshold and monitoring is recommended.	The service is oversubscribed, which may lead to data path disruptions that affect the overall performance.

The status transitions as follows:

- Normal (green): The service has significant headroom and can handle traffic spikes. The P95 metric is well below the maximum threshold limit.
- High (yellow): The service is approaching its allocated resource threshold and monitoring is recommended. The P95 metric exceeds or equals the threshold, but the P80 metric remains safely below the threshold (occasional traffic spikes).
- Critical (red): The service is oversubscribed, which may lead to data path disruptions that affect the overall performance. The P95 metric exceeds or equals the threshold.

Host Capacity Usage represents how much host capacity is actively in use, allowing administrators to assess resource pressure and plan capacity accordingly.

Key Components

- Host CPU/Memory Utilization: The total resource consumption of the physical transport node, including hypervisor overhead and all running virtual machines.
- Security Service (L7/IDPS) CPU Utilization (Turbo mode): The specific percentage of CPU cycles dedicated to the security pipeline for inspecting Layer 7 traffic and executing IDPS signature matching.
- Security Service (L7/IDPS) Memory Utilization (Turbo mode): The dedicated memory footprint used by the security engine to maintain session states and signature databases.

If you see hosts that have Critical (Red) or High (Yellow) Alerts, then it is recommended not to enable additional Security controls without considering a rebalance of the Virtual machines across the cluster. Enabling the Detailed Host Metrics for those hosts will show where each individual Virtual Machine is running at from a performance perspective. You can choose to vMotion any VM that is towards the higher end of the performance to a host that is less used. This should free up the resources needed for implementation of the IDPS Control.

NSX Manager also has a number of Alarms that will signal if the IDPS engine is having any issues at or near capacity in terms of resources. It is important to acknowledge open alerts within the Distributed IDPS area. Use these alarms to consider the Security Dashboard within the SSP to evaluate the root cause of the capacity issue. Further troubleshooting steps can also be taken from these alarms such as in the case of an IDPS engine failure or similar.

Many customers may also use VCF Operations for the active monitoring of the IDPS solution. There is also a community provided IDPS Dashboard that will highlight any potential issues with the overall health and capacity of the engine.

Before adding any additional rules, it is important to review the current operational state of the IDPS engine using any of these monitoring solutions.

Assessing and Defining IT Security Risk

A fundamental task in IT security is identifying and defining the areas within an organization's infrastructure that carry some level of unmanaged risk. The primary goal of this step is to comprehensively understand and quantify this risk in relation to the specific assets within the customer's environment, so that the organization can properly identify which compensating controls are necessary to implement.

To accurately define the scope of a risk, security teams should begin by asking several foundational questions:

- What is the specific problem that needs to be addressed?
- What is the scope of the risk?
- What existing architecture and security controls are currently in place?

- How would the implementation of a new control effectively reduce this risk?
- What is the prioritization level of the issue?
- Who are the key stakeholders, and who will be consuming the resulting data or security events?

To fully qualify the risk, teams must also evaluate the broader context of the environment: the potential impact of unauthorized access, existing vulnerabilities (OS or application flaws) alongside current mitigation strategies such as patching procedures, user roles and access permissions, network restrictions already in place to prevent lateral movement, and existing outbound access control and filtering policies.

Information Gathering

Organizations can utilize a variety of specialized tools and reports to help identify vulnerabilities and risks across their infrastructure:

- SSP with Security Intelligence
- Security Segmentation Journey Reports
- Vulnerability Scanners
- Breach and Attack Simulation (BAS) Tools

Reducing Risk by Mapping to Security Controls

Once risks are clearly defined and quantified, the next step is applying appropriate controls to reduce the risk surface by directly mapping specific threat vectors to the controls designed to neutralize them.

Threat Vector	Mitigating Control
Lateral Movement	Distributed Firewall (DFW) configurations (1-2-3-4)
Unauthenticated Access	Identity Firewall (IDFW)
Network-Based Exploits	Intrusion Detection and Prevention System (IDPS)
Malicious Files or Scripts	Malware Prevention and Detection
LOLbins (Living Off the Land Binaries)	Network Traffic Analysis (NTA) alongside Behavioral IDPS

Aligning Controls with Security Personas

Risk reduction strategies must align with the specific operational goals of different security personas within the organization.

Persona	Primary Focus	Relevant Controls
NetSec (Network Security) Teams	High-fidelity detection and prevention; blocking malicious activity and keeping bad actors out of the network.	DFW, IDPS, NTA, Malware Prevention
SOC (Security Operations Center) Teams	Investigating end-to-end attacks and maintaining deep visibility through security signals; forensic investigations.	NDR for enhanced logging, event enrichment, and correlation

Ultimately, establishing risk definitions and deploying controls is an ongoing journey rather than a final destination. Organizations should start by assessing the risk associated with a specific service or area of interest, and then decide whether broad or deep coverage is required based on their unique, customer-dependent needs.

Finding Vulnerabilities

One way to identify missing patches is through the use of a vulnerability scanner, such as those commercially available from Tenable and Qualys. These tools reach out over the network and scan workloads for missing patches, flagging when a specific VM is vulnerable to a known exploit. The detailed output of a scan will highlight missing patches and pinpoint the exact IDS signature to apply as the compensating control.

Users with a Broadcom support account can log into the vTIS Portal to search for specific vulnerabilities and view available IDPS signatures for protection.

Parallel Operation with Traditional Patching

Typically, identification of a missing patch kicks off a lengthy process of testing and negotiating for an outage window, which historically takes several days or even weeks. As a parallel effort, a compensating control like Distributed Virtual Patching allows the Distributed IDS to protect against the exploit immediately — without touching or modifying the running virtual machine or its operating system.

Implementation of Distributed Virtual Patching

Pre-requisites

Before implementing Distributed Virtual Patching, the following prerequisites must be met:

Pre-requisite	Details
Licensing	A valid vDefend license (add-on for VCF) that includes the Distributed IDPS capability.
VCF version	VCF 9.1 or later is required to create IDPS Profiles on a CVE basis.
Vulnerability identification source	A vulnerability scanner (e.g., Tenable, Qualys), or mature vulnerability management program to flag missing patches, plus a Broadcom support account for the vTIS Portal to look up available IDPS signatures.
Traffic visibility	The control must be able to inspect network traffic in clear text (non-encrypted); signature availability also depends on access to exploit code.
Grouping constructs	NSX Groups represent the vulnerable workloads (e.g., TargetVMs) so rules can be scoped tightly.
Centralized logging	A Syslog or SIEM endpoint to receive IDPS events; events isolated within NSX Manager are insufficient.
Change control	A change control system capable of tracking the virtual patch and carrying a post-patch decommissioning task.
Vulnerbilites Identified	Perform risk assessment and scoping by scanning workloads for unpatched vulnerabilities.

Implementation

Procedure Grouping	Procedure	Description
Valid vDefend License	vDefend Feature and Edition Guide	Make sure a valid license is in place to use the vDefend Distributed IDPS.
Stage 1: Create the Virtual Patch	VDefend Advanced Threat Prevention	• Configure the Distributed IDPS, IDPS profile, and emergency rule • Create a per-signature IDPS Profile (VCF 9.1), then create a tightly scoped rule in the emergency policy section with the action set to Detect and Prevent. **:** Insert step-by-step setup and configuration details for the Distributed IDPS here.
Stage 2: Monitor and Integrate	Manage Logging and Alerts	• Configure logging, alerting, SIEM, and NDR. • Forward IDPS events to Syslog/SIEM, configure proactive alerting, and optionally enrich events with vDefend NDR.
Stage 3: (Optional) Lifecycle Management and Operations	Consider the removal of the Virtual Patch if other appropriate controls have been introduced	• Decommission the virtual patch. • After the real software patch is applied, execute the post-patch change-control task to remove the corresponding IDPS rule.

Securing VCF Against Frontier AI Era Threats

The *Securing VCF Against Frontier AI Era Threats* blueprint outlines how to secure your VCF private cloud across different risk areas to defend against AI-driven threats.

Frontier AI Threat Overview

Frontier AI models accelerate innovation, software development, and operational efficiency, but they also allow attackers to take advantage of vulnerabilities almost instantly.

Modern frontier AI models have a deep understanding of software systems, networking protocols, and cryptographic constructs. They can identify known vulnerabilities across vast codebases, chaining disparate flaws together and discovering complex Day-0 vulnerabilities in legacy systems without human instruction.

Securing VCF requires a secure-by-design, resilient platform baseline that systematically minimizes the exposed attack surface.

Consider the implementing the following end-to-end security measures:

- Enforcing robust user and identity security controls, such as centralized single sign-on, federated multi-factor authentication, and dynamic least-privilege role management
- Restricting lateral threat movement
- Ensuring rapid, uncorrupted recovery when incidents occur

Frontier AI Impact on the Threat Landscape

The integration of frontier AI into the threat landscape has fundamentally changed the economics and dynamics of cyberattacks in terms of pace, ease, and the overall scale of cyber incidents.

Pace

Reduces vulnerability-to-exploitation timelines from weeks to minutes. AI automates code audits, vulnerability discovery, and payload creation, making manual patch cycles obsolete.

Human security teams no longer have enough time to manually review security advisories, test patches, and schedule maintenance windows. Defense mechanisms must now operate continuously and autonomously to keep pace.

Ease

Lowers technical barriers. Entry-level attackers can launch high-sophistication cyberattacks which previously required elite nation-state offensive teams.

AI transforms the entire kill chain into a unified, automated workflow with the following advantages:

- Reconnaissance and social engineering: Frontier AI models perform automated reconnaissance in seconds by ingesting leaked databases and public profiles to build detailed target maps, while crafting context-aware, hyper-personalized spear-phishing lures that perfectly mimic internal executive communications.
- Chaining vulnerabilities: Frontier AI models combine vulnerabilities in the following way:
  - Chain together vulnerabilities in creative ways to run complex attacks.
  - Dynamically rewrite malware payloads on the fly, rendering traditional, signature-based antivirus scanners, and perimeter defenses obsolete.
- Autonomous: Frontier AI agents can work with other malicious agents to carry out automated attacks without human control.
- Lateral movement: Once inside a boundary, AI agents map internal infrastructure locally, moving horizontally across virtual networks and escalating privileges. These agents do not need to rely on slow command-and-control communication channels back to a human operator.

Cyber Incidents

AI-assisted cyber threats represent systemic risks to organizations and financial infrastructure. Threat actors who use automated AI can find vulnerabilities, bypass legacy controls, and access sensitive enterprise data before detection teams can analyze the incoming alert.

Risk-Based Security Posture Assessment

The desired security posture of a VCF environment depends on the risk profile of your organization, workload placement, and the workload itself. Consider the following items:

Likelihood of Initial Exploit Execution	Also known as the "front door" risk. Determining the right security baseline begins with analyzing the probability of initial exploitation. This dimension assesses exposure to threat actors, attack surface, and the accessibility of management and workload boundaries. Environments that host public-facing endpoints, demilitarized zones (DMZs), or Internet-accessible container services have a higher risk of initial compromise compared to isolated corporate networks. Mitigating exploitation risk requires preventative controls across the user security and platform security pillars. Organizations must reduce the exposed attack surface through the following measures: • Strict identity federation • Mandatory multi-factor authentication (MFA) • Timely patching cycles • Continuous configuration drift monitoring to ensure that vulnerabilities are fixed before threat actors can exploit them
The Blast Radius of an Exploit	Also known as the "impact" risk. You must determine the potential extent of operational, financial, and reputational damage if an exploit breaches the perimeter. If an attacker can move laterally across the infrastructure or access sensitive data, a single compromised entry point can result in catastrophic failure even in low-risk environments. Addressing blast radius risk requires defense-in-depth mechanisms that consider lateral security and application security. Organizations must also consider the effort that is required to recover from a potential incident. Organizations can contain threat actors to a single isolated segment and prevent lateral movement by implementing micro-segmentation, zero-trust network boundaries, and strict multi-tenant isolation policies. Organizations can recover quickly from cyber incidents by implementing immutable recovery cleanrooms.

Securing VCF Against Frontier AI Era Exploits

You can transition from reactive patch management to an automated, continuous security posture that handles the speed and scale of frontier AI-era threats. Organizations must consider a risk-based security posture assessment across the following five critical areas:

- People and Process for Securing VCF Against Frontier AI Era Threats
- User Security for Securing VCF Against Frontier AI Era Threats
- Platform Security for Securing VCF Against Frontier AI Era Threats
- Lateral Security for Securing VCF Against Frontier AI Era Threats
- Application Security and Recovery for Securing VCF Against Frontier AI Era Threats

People and Process for Securing VCF Against Frontier AI Era Threats

Establish operational rigor, rapid incident response, and predictable lifecycle governance in your VCF private cloud.

The strength of any technical architecture is related to the operational practices of the team managing it. This pillar evaluates organizational readiness, patch cadence, and lifecycle strategies. A high-maturity posture requires moving away from reactive "don't fix unless broken" mentalities toward standardized Service Level Agreements (SLAs), particularly for VMware Security Advisories (VMSAs). Establishing dedicated rapid response teams, formal incident protocols, and continuous vulnerability assessment schedules ensures that modern security baselines are maintained and operational drift is caught early.

Table 524: Design Recommendations for People and Process for Defending Against Frontier AI Era Exploits

Design Recommendation ID	Design Requirement	Justification	Implications
VCF-SEC-RCMD-PPR-001	Standardize infrastructure upgrade cycles on the latest available VCF software release.	• Running the latest software release proactively closes vulnerability exposure windows before automated exploit chains can proliferate. • Fixes might originate from public security advisories or internally identified code improvements. • Running older releases (N-1 or older) leaves known vulnerabilities that are exposed to automated scanning engines.	Requires shortened qualification cycles, automated testing pipelines, and tight operational alignment with release cadences.
VCF-SEC-RCMD-PPR-002	Perform general private cloud platform upgrades within a strict SLA that is measured in weeks.	A multi-week operational upgrade window significantly shrinks the window of exposure, minimizing the time threat actors have to scan, identify, and exploit infrastructure vulnerabilities before updates are applied.	Demands image-based lifecycle management and pre-authorized maintenance windows across workload domains to avoid application downtime.
VCF-SEC-RCMD-PPR-003	Apply patches for VMSAs across the private cloud platform within a strict SLA that is measured in days.	Threat actors quickly analyze public VMSA disclosures to create exploit payloads. Enforcing a multi-day patching SLA significantly compresses the exposure window and reduces the time VCF components remain vulnerable to targeted exploit attempts.	Requires pre-staged patch pipelines, emergency change authorization pathways, and non-disruptive live patching capabilities.
VCF-SEC-RCMD-PPR-004	Before deployment in production, complete qualification testing for general software upgrades within an SLA that is measured in days.	• Accelerating qualification testing to a multi-day SLA minimizes operational delay between patch availability and deployment. • Reduces the duration of exposure during which known vulnerabilities could be exploited.	Demands automated synthetic validation test suites and dedicated staging environments to replace manual regression testing bottlenecks.
VCF-SEC-RCMD-PPR-005	Before deployment in productin, complete qualification testing for high-severity VMSAs within an emergency SLA that is measured in days.	• High-severity VMSAs represent prime targets for rapid exploitation. • Establishes an expedited multi-day qualification window. • Reduces the time critical vulnerabilities remain unverified for production. • Allowing teams to authorize and roll out mitigations before threat actors can exploit the gap.	Requires establishing dedicated, expedited qualification pathways specifically for VMSAs, independent of standard feature releases.

Design Recommendation ID	Design Requirement	Justification	Implications
VCF-SEC-RCMD-PPR-006	Adopt new platform security and operational capabilities upon General Availability (GA).	• Adopting new security releases promptly minimizes the duration during which the platform operates under legacy defense baselines. • Reduces the time frame during which the infrastructure remains exposed to evolving threat vectors and security capability gaps.	Operational teams must maintain continuous competency updates and agile architectural review boards to rapidly evaluate and validate new capabilities.
VCF-SEC-RCMD-PPR-007	Standardize private cloud infrastructure on the VCF 9.1.x architecture.	• Upgrading to VCF 9.1.x provides native fleet-wide management, automated compliance drift remediation, and integrated security operations • Reduces the window of time configuration gaps and legacy management silos remain exposed to potential threats.	Might require architectural migration planning, hardware compatibility validation, and operational runbook updates.
VCF-SEC-RCMD-PPR-008	Enforce guest operating system and application patch management protocols that can apply security updates within an SLA that are measured in days.	• Rapid guest OS and application patching significantly reduces the duration that known vulnerabilities remain accessible • Reduces the window of opportunity for an initial workload compromise to be weaponized or escalated across the platform.	Requires close operational alignment between infrastructure platform teams and application DevOps teams to automate guest OS patch delivery and testing pipelines.
VCF-SEC-RCMD-PPR-009	Establish and maintain a dedicated, permanent internal Rapid Response Team for security incident management.	• Maintaining a permanent rapid response team ensures pre-coordinated incident protocols and active threat-hunting capabilities. • Reduces time-to-detect and time-to-contain during security incidents. • Reduces the duration that an active threat can operate within the environment.	Requires dedicated headcount allocation, ongoing threat-hunting duties, and pre-authorized executive authority to execute emergency containment actions across the cloud fabric.
VCF-SEC-RCMD-PPR-010	Enforce strict use of predefined protocol tasklists and incident runbooks during security events.	• Execution using standardized protocols minimizes decision latency and operational ambiguity during security events. • Reduces the total duration that is required to contain active threats and limiting the potential window of disruption.	Involves drafting detailed incident runbooks, performing regular dry-run simulation drills, and establishing automated containment triggers.

Design Recommendation ID	Design Requirement	Justification	Implications
VCF-SEC-RCMD-PPR-011	Align post-incident procedures with established compliance and security governance frameworks, such as ISO, NIST, SOC 2, or NERC.	• Standardized post-incident frameworks enforce structured root-cause remediation • Reduces the time systemic vulnerabilities or architectural gaps remain in the environment to reduce the duration of exposure to recurring exploit attempts.	Demands formal audit logging, rigorous forensic root-cause documentation, and executive and regulatory reporting mechanisms.
VCF-SEC-RCMD-PPR-012	Perform regular third-party vulnerability assessments and penetration tests, sharing findings with security leadership and relevant stakeholders.	• Independent security assessments and penetration testing validate posture integrity. • Minimizes the duration that undetected configuration drift, unpatched attack surface gaps, or legacy exposures remain present in the environment.	Requires allocating recurring budget for third-party security services and establishing formal remediation tracking for identified findings.

User Security for Securing VCF Against Frontier AI Era Threats

Enforce zero-trust identity controls, strict perimeter isolation, and immutable auditability in your VCF private cloud.

User security protects the VCF management plane by restricting administrative privileges across the fabric. It removes identity fragmentation through centralized authentication with federated enterprise identity providers. Access is strictly controlled by using mandatory multi-factor authentication (MFA) and granular role-based access controls (RBAC). Also, it manages SSL/TLS certificate lifecycles and uses dynamic privileged access management (PAM) vaulting for service accounts. Centralized, encrypted audit logs ensure complete traceability for all administrative actions across the cloud environment.

Table 525: Design Recommendations for User Security for Defending Against Frontier AI Era Exploits

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-USR-001	Aggregate administrative and platform audit logs across the private cloud stack.	• Real-time log correlation connects API tokens directly to identity sessions, significantly reducing investigation latency. • Minimizes the duration during which unauthorized API activity or compromised sessions remain undetected.	Demands central log forwarding configuration with token-level session tracking and encrypted transport.
VCF-SEC-RCMD-USR-002	Federate administrative identities by using a central identity plane with mandatory MFA and granular, job-specific role-based access control.	• Centralized federation with MFA and strict role-based access control reduces reliance on static credentials and restricts privilege boundaries. • Minimizes the window of opportunity for credential misuse or unauthorized privilege escalation.	Requires integrating VCF Single Sign-On with an enterprise identity provider, such as Okta, or Entra ID, and mapping granular VCF roles.

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-USR-003	Centrally automate and continuously monitor SSL/TLS certificate lifecycles across the fleet using enterprise certificate authority (CA) integration.	• Centralized, automated certificate lifecycle management maintains consistent cryptographic baselines. • Reduces the duration that expired, misconfigured, or weak certificates create management plane exposure or unexpected outages.	Shifts management from manual per-appliance scripts to centralized, automated CA renewal workflows that persist across platform updates.
VCF-SEC-RCMD-USR-004	Centrally establish, continuously audit, and automatically enforce global password complexity, aging, and account lockout policies across all infrastructure systems.	• Centralized fleet-wide password policy enforcement eliminates configuration drift and local credential variations • Minimizes the duration that non-compliant or weak authentication baselines remain exposed.	Requires establishing automated password complexity, rotation, and lockout profiles across all infrastructure management components.
VCF-SEC-RCMD-USR-005	Vault and automatically rotate high-privilege operating system, application, and service account credentials using native Privileged Access Management (PAM) integration.	• Dynamic PAM vaulting and continuous credential rotation compress the usable lifespan of secrets. • Minimizes the duration that harvested or exposed service account credentials can be exploited for lateral movement.	Requires integrating high-privilege service accounts, API clients, and OAuth tokens with an enterprise PAM solution.

Platform Security for Securing VCF Against Frontier AI Era Threats

Harden core infrastructure, enforce cryptographic standards, and eliminate configuration drift in your VCF private cloud.

Platform security forms the baseline defense for physical hosts, hypervisors, and storage fabrics. Automated alignment with VMware Security Configuration Guides (SCG) and compliance frameworks, such as PCI-DSS, FIPS, and DISA STIG, provides secure operations. Data security is provided both in transit (mandatory vMotion and cluster replication encryption) and at rest (AES storage encryption with automated key rotation). Hardware-rooted trust, such as Trusted Platform Module (TPM) 2.0 remote attestation and UEFI Secure Boot and using non-disruptive ESX live patching enables continuously hardened infrastructure without sacrificing availability.

Table 526: Design Recommendations for Platform Security for Defending Against Frontier AI Era Exploits

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-PLT-001	Standardize, continuously audit, and strictly enforce TLS 1.3/1.2 protocols and FIPS-validated cryptographic profiles across all management appliances.	• Enforcing modern TLS protocols and FIPS-validated cryptographic profiles eliminates legacy cipher support. • Minimizes the window of opportunity for traffic interception, decryption, or man-in-the-middle attacks.	Requires configuring fleet-wide TLS and FIPS enforcement profiles across all management appliances.
VCF-SEC-RCMD-PLT-002	Automate the deployment, continuous auditing, and remediation of security baselines aligned with VMware Security Configuration Guides (VCF SCG).	• Automated compliance drift detection and remediation significantly compress the duration configuration gaps exist • Minimizes the timeframe during which VCF infrastructure components remain vulnerable to exploitation.	Demands automated benchmark deployment and drift monitoring using central compliance toolchains.
VCF-SEC-RCMD-PLT-003	Require fleet-wide data-in-transit encryption across all live migrations (vMotion), fault tolerance streams, inter-host storage communication and storage replication.	• Enforcing mandatory transport encryption neutralizes unencrypted data exposure during live operations. • Minimizes the duration active workload RAM and replication streams are susceptible to packet sniffing or memory extraction.	Requires setting vMotion encryption to 'Required' globally and enabling host hardware acceleration.
VCF-SEC-RCMD-PLT-004	Mandate AES data-at-rest encryption across all storage datastores, integrated with enterprise Key Management Servers (KMS) and automated Key Encryption Key (KEK) rotation.	• Automated KEK rotation that is paired with datastore encryption reduces static cryptographic exposure. • Minimizes the time window during which extracted storage snapshots or compromised physical media could expose underlying data.	Requires integrating native vSAN / vSphere Encryption with an enterprise Key Management Server (KMS).
VCF-SEC-RCMD-PLT-005	Centralize proactive telemetry, security event correlation, and observability across the SDDC fabric using VCF Operations fleet management.	• Unified observability connects multi-layer security telemetry in real time. • Reduces mean time to detect (MTTD) and minimizing the duration anomalous cross-stack behavior goes unnoticed.	Requires centralizing telemetry and security event correlation across all SDDC management layers.
VCF-SEC-RCMD-PLT-006	Execute platform lifecycle updates and emergency security maintenance using Quick Patch, Reduced Downtime, and rebootless Live Patching capabilities.	• Rebootless patching and non-disruptive lifecycle workflows eliminate the operational friction of scheduled downtime. • Reduces the exposure window between patch release and active host remediation.	• Requires a VCF release that supports Live Patching. • Live Patching is not supported in conjunctions with some features.

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-PLT-007	Enforce hardware-rooted trust across host infrastructure using UEFI Secure Boot integrated with automated TPM 2.0 remote attestation.	• Hardware attestation and Secure Boot protect hypervisor firmware integrity. • Minimizes the time 'boot kit' modifications or unauthorized low-level kernel changes can persist without detection.	Requires enabling UEFI Secure Boot on host hardware, installing TPM 2.0, and configuring vCenter remote attestation.
VCF-SEC-RCMD-PLT-008	Enforce multi-tenant compute, storage, and network isolation across tenant environments using VPCs and Supervisor Zones.	• Structural VPC isolation enforces strict hypervisor and network boundaries. • Contains threat blast radius and minimizing the opportunity for a compromised tenant workload to pivot into adjacent tenant spaces.	Requires implementing VPCs for VCF Automation and NSX multi-tenancy constructs to enforce strict logical boundaries.
VCF-SEC-RCMD-PLT-009	Automate compute isolation and workload placement policies using vSphere tags, VCF Automation Placement Policies, and tenant resource quotas.	• Automated placement policies and quotas ensure that high-security workloads execute on dedicated physical hosts. • Minimizes the likelihood and duration of noisy-neighbor resource exhaustion or targeted Denial of Service (DoS) conditions.	Demands configuring VCF Automation placement policies, vSphere tags, and explicit organization resource quotas.
VCF-SEC-RCMD-PLT-010	Automate continuous compliance enforcement and drift remediation using built-in security benchmarks that are paired with fleet-wide CVE visibility.	• Continuous benchmark tracking and integrated CVE insights replace periodic auditing cycles. • Reduces the duration non-compliant states or unpatched vulnerabilities remain hidden across the fleet.	Demands enabling VCF Security Posture Management with automated benchmark mapping and integrated CVE tracking across the fleet.

Lateral Security for Securing VCF Against Frontier AI Era Threats

Contain lateral threat movement using deep East-West network visibility and micro-segmentation in your VCF private cloud.

Traditional perimeter firewalls are insufficient once a breach occurs within internal network boundaries. Lateral Security mitigates the blast radius by enforcing distributed, zero-trust network policies down to the individual Virtual Machine or container pod. Using deep East-West traffic inspection, Intrusion Detection/Prevention Systems (IDS/IPS), and Network Detection and Response (NDR), this pillar restricts unauthorized communication between workloads and sensitive shared

services (DNS, AD, Syslog). Micro-segmenting management domains and deploying Web Application Firewalls (WAF) ensures that threat actors cannot navigate horizontally across tenant boundaries.

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-LAT-001	Establish complete East-West traffic observability and flow monitoring across all VCF workload domains.	• Deep East-West traffic visibility uncovers hidden lateral movement patterns and unauthorized API probing. • Minimizes the duration malicious reconnaissance or intra-VLAN threat propagation can occur undetected.	Requires deploying distributed flow monitoring across all virtual switches and workload domains.
VCF-SEC-RCMD-LAT-002	Enforce zero-trust micro-segmentation policies at the individual workload level (virtual machines, bare-metal hosts, and container pods).	• Workload-level ring-fencing restricts lateral connectivity. • Reduces the blast radius of a compromised asset • Minimizes the time and pathways that an attacker has to move horizontally across internal segments.	Requires defining stateful micro-segmentation policies using distributed firewalling directly on host hypervisors.
VCF-SEC-RCMD-LAT-003	Restrict and enforce stateful security controls on all communication between workloads and shared infrastructure services, such as DNS, NTP, Syslog, and Active Directory.	• Enforcing strict default-deny rules between workloads and shared services minimizes the exposure window for Active Directory abuse or service poisoning. • Helps prevent compromised workloads from exploiting core infrastructure.	Requires creating strict default-deny rules allowing only verified ports and protocols between workloads and core management services.
VCF-SEC-RCMD-LAT-004	Isolate access to VCF management domains and infrastructure management interfaces by using dedicated micro-segmentation policies.	• Micro-segmenting management planes prevents compromised tenant workloads from probing or attacking SDDC infrastructure APIs. • Minimizes the risk and duration of management plane exposure.	Requires applying distributed firewall rules around management domain hosts, vCenter, and other VCF components.
VCF-SEC-RCMD-LAT-005	Enforce simultaneous container security controls across both Kubernetes Pod and Node layers.	• Dual pod-and-node security limits the blast radius of container breakouts. • Minimizes the opportunity and timeframe for an exploited pod to compromise underlying worker nodes or adjacent microservices.	Requires integrating CNI-level micro-segmentation for Kubernetes container networks.
VCF-SEC-RCMD-LAT-006	Aggregate and correlate network and host security alerts in real time using Network Detection & Response (NDR).	• Automated real-time event correlation stitches disparate network signals into threat campaigns. • Reduces mean time to detect (MTTD) and minimizing alert fatigue for SOC teams.	Demands deploying threat correlation engines capable of aggregating network traffic analysis and host alerts.

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-LAT-007	Deploy an Advanced Threat Prevention (ATP) suite—including Distributed IDS/IPS, Network Traffic Analysis (NTA), malware Prevention, and NDR—across all VCF workload clusters.	• Full-stack threat inspection detects polymorphic malware and zero-day exploit attempts in real time • Minimizes the duration fileless execution or AI-driven command-and-control behavior can operate undetected.	Requires activating distributed intrusion prevention, malware inspection, and traffic analysis across all cluster hosts.
VCF-SEC-RCMD-LAT-008	Enforce and continuously audit regulatory compliance controls (PCI-DSS, HIPAA, FIPS, NERC) across East-West traffic flows.	• Continuous compliance enforcement ensures that sensitive data zones maintain audit-ready isolation. • Reduces the duration non-compliant security postures or unmonitored traffic paths can persist.	Requires mapping compliance benchmarks directly to distributed firewall and threat inspection rules.
VCF-SEC-RCMD-LAT-009	Automate the creation, monitoring, and optimization of security policies via programmatic pipelines and Infrastructure-as-Code (IaC).	• Programmatic policy automation ensures Zero Trust controls are applied instantly as workloads scale. • Minimizes manual intervention delays and preventing security policy drift.	Requires integrating firewall rule management into IaC pipelines, for example, Terraform or Ansible, and security automation platforms.
VCF-SEC-RCMD-LAT-010	Continuously review, audit, and optimize firewall rule sets to eliminate redundant, shadow, or stale rules.	• Continuous rule optimization prevents policy bloat. • Minimizes the timeframe over which overly permissive or obsolete firewall rules remain active in the environment.	Requires implementing automated rule analysis tools to identify shadow rules and recommend cleanup.
VCF-SEC-RCMD-LAT-011	Secure, load balance, and manage containerized application services using enterprise-grade Kubernetes API gateways.	• Software-defined ingress gateways deliver elastic traffic routing and TLS termination. • Minimizes exposure to container ingress vulnerabilities and traffic overload attacks.	Demands deploying software-defined load balancers tightly integrated with Kubernetes cluster control planes.
VCF-SEC-RCMD-LAT-012	Protect external-facing applications using an active, analytics-driven WAF with zero-day virtual patching.	• Active WAF inspection and virtual patching compress the exposure window between the discovery of a vulnerability and the application code fixes. • Blocks OWASP Top 10 threats in real time.	Requires deploying central WAF policies with positive security models and signature-based threat detection.
VCF-SEC-RCMD-LAT-013	Automate SSL/TLS certificate lifecycles for applications to meet shortened certificate expiration requirements without manual intervention.	• Automated certificate renewal eliminates manual handling bottlenecks. • Minimizes the risk and window of application outages that are caused by unexpected certificate expiration.	Requires integrating load balancers and ingress gateways directly with enterprise CAs by using automated APIs.

Application Security and Recovery for Securing VCF Against Frontier AI Era Threats

Safeguard containerized and virtualized workloads while ensuring automated ransomware resilience in your VCF private cloud.

Application security and recovery is the final line of defense focuses on workload-level hardening and rapid operational restoration following a cyber incident. This pillar handles continuous CVE vulnerability management across guest OS and container runtimes, which is coupled with desired-state compliance enforcement. Automation provides an isolated cleanroom recovery orchestration in case of a destructive attack or ransomware incident. Immutable snapshots, Next-Gen antivirus (NGAV) fileless malware validation in isolated environments, and unified recovery workflows, enable organizations to minimize Recovery Time Objectives (RTO) and restore as trusted state with confidence.

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-REC-001	Orchestrate cyber and ransomware recovery processes through an end-to-end automated workflow across infrastructure, network, and security tooling.	• End-to-end automated recovery workflows eliminate human error and manual handoffs during incident response. • Significantly compresses operational downtime and minimizes the window during which business operations remain disrupted.	Requires deploying unified recovery orchestration platforms capable of automating network quarantine and workload restoration.
VCF-SEC-RCMD-REC-002	Restore workloads into an Isolated Recovery Environment (IRE) or Clean Room before re-introducing them into production.	• Restoring into an isolated recovery environment prevents compromised restore points from re-contaminating production networks. • Restricts malware payload execution to a quarantined space. • Minimizes the duration of operational risk during validation.	Requires building an isolated recovery network boundary (cleanroom) with dedicated compute and network quarantine rules.
VCF-SEC-RCMD-REC-003	Perform automated validation of restore point candidates in isolated cleanrooms using Next-Gen antivirus (NGAV) to detect fileless and file-based malware.	• Automated NGAV inspection on powered-on candidates identifies dormant malware and vulnerabilities before production failback. • Reduces the risk and timeframe of re-introducing threats back into production.	Requires integrating automated sandbox analysis and behavioral antivirus inspection into the cyber recovery pipeline.
VCF-SEC-RCMD-REC-004	Standardize on a unified management platform for cyber recovery across infrastructure, networking, orchestration, and restore point validation.	• Consolidates recovery operations into a single management plane eliminates tool fragmentation and cross-team friction. • Reduces RTO and minimizing the duration of business interruption during an incident.	Requires consolidating legacy backup, replication, and disaster recovery tools into an integrated platform.

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-SEC-RCMD-REC-005	Automate backup schedules to generate immutable, tamper-proof snapshots that resist deletion or modification by compromised administrative credentials.	• Cryptographically locked, immutable snapshots guarantee restore point integrity. • Prevents ransomware or malicious administrators from wiping recovery data and minimizing the window of irrecoverable data loss.	Requires enabling immutable cloud or air-gapped storage repositories with locked retention policies.
VCF-SEC-RCMD-REC-006	Test and validate cyber recovery plans at least quarterly using automated execution workflows in isolated sandboxes.	• Regular non-disruptive testing ensures that operational playbooks stay aligned with application changes. • Minimizes the time that is required to initiate recovery and preventing plan failures during live incidents.	• Requires adequate operational process to ensure that testing is prioritized to occur every 90 days. • Demands scheduling non-disruptive, automated DR dry-runs in isolated sandboxes every 90 days.
VCF-SEC-RCMD-REC-007	Automate workload compliance and vulnerability management across virtual machines and container workloads using desired-state enforcement and continuous CVE tracking.	• Guest-level vulnerability tracking and desired-state enforcement eliminate manual patching bottlenecks. • Reduces the duration application-layer vulnerabilities and configuration drift remain exposed to exploitation.	Requires deploying continuous workload compliance scanning and guest OS desired-state remediation tools.
VCF-SEC-RCMD-REC-008	Establish and validate a recovery architecture capable of meeting an RTO measured in hours for critical systems following a cyber incident.	• Achieving an RTO of hours minimizes the total timeframe of business disruption, financial loss, and operational outage. • A cyber attack or ransomware encryption event may result in destruction of data.	Demands automated, live-mount instant recovery capabilities from immutable cloud storage to local hypervisor clusters.

Design Library for VMware Cloud Foundation

The design library section contains detailed designs for each component of the VCF private cloud.

Functional Area	Detailed Designs
Consumption	• VCF Automation Detailed Design • vSphere Supervisor Detailed Design • VMware Data Services Manager Detailed Design • Workload Connectivity Detailed Design • Load Balancer Detailed Design • VCF Edge Detailed Design • Private AI Platform Detailed Design • Private AI Compute Detailed Design
Operations	• VCF Management Services Component Detailed Design • VCF Management Services Detailed Design • VCF Management Network Detailed Design • VCF Operations Detailed Design • License Server Detailed Design • Log Management Detailed Design • VCF Operations for Networks Detailed Design
Security Governance and Compliance	• Identity Broker Detailed Design • VCF Single Sign-On Detailed Design • Information Security Detailed Design
Virtual Infrastructure	• VCF Domain Detailed Design • vSphere Detailed Design • vSphere Cluster Detailed Design • Distributed Switch Detailed Design • Storage Detailed Design • NSX Manager and Control Plane Detailed Design • NSX Edge Cluster Detailed Design • Virtual Network Appliance Cluster Detailed Design
Physical Infrastructure	• External Services Detailed Design • Physical Data Center Network Design • Network Fabric Detailed Design • Infrastructure IP Addressing Detailed Design • Workload IP Addressing Detailed Design

VCF Automation Detailed Design

This section describes the options, requirements, and recommendations for each VCF Automation deployment model.

VCF Automation can be deployed during initial deployment of your VMware Cloud Foundation platform by using VCF Installer, or at a later stage by using VCF Operations to use a custom network configuration.

Table 527: VCF Automation Deployment Paths

Deployment Path	Implications
Deployment by using VCF Installer see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard	• Part of an initial net new VCF deployment along with other core management domain components (for example, vCenter, NSX, VCF Operations, SDDC Manager, etc.). • Uses the VM management distributed port group by default.
Deployment by using VCF Operations fleet management for custom networking configuration.	• Post VCF core management components deployment. • Allows for custom network configurations.

VCF Automation enables the modern private cloud. Playing a crucial role in both single-tenancy and multi-tenancy scenarios by providing a unified platform for designing a modern private cloud platform with VMware Cloud Foundation.

Modern private cloud platform from the service provider perspective:

- Single tenancy: service providers offer dedicated private cloud environments to customers who require strong isolation, compliance, or specific performance guarantees.
- Multi-tenancy: service providers use multi-tenancy to deliver cost-effective, shared cloud services to a broader customer base.

Modern private cloud platform from the Enterprise perspective:

- Single tenancy: an enterprise using a single organization to isolate sensitive workloads, critical applications, or different business units with specific requirements.
- Multi-tenancy: an enterprise using multiple organizations to completely segregate internal IT infrastructure to improve efficiency, maintain strict security, and simplify management of less critical applications.

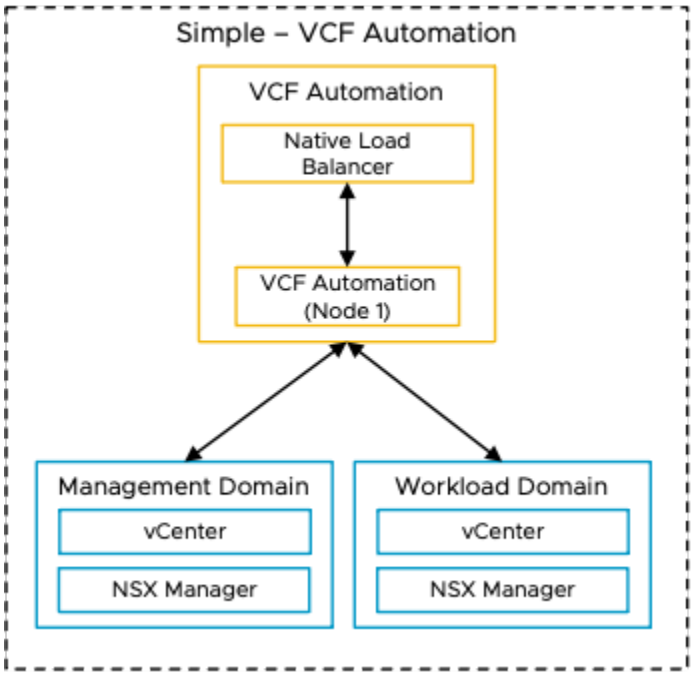
This design section is intended to guide both enterprise and service providers in designing and deploying a modern private cloud with VCF Automation tailored to their organizational needs.

- Simple VCF Automation Model
- High Availability VCF Automation Model

Simple VCF Automation Model

The *Simple VCF Automation Model* consists of a single node that is deployed in a vSphere cluster. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Figure 240: Simple VCF Automation Model in VMware Cloud Foundation



Simple VCF Automation Model Attributes

A *Simple VCF Automation Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out to high the availability model. **Note:** Scale-up to a larger node size and scale-out to three nodes at the same time.
Recoverability	Restore through a backup.

Simple VCF Automation Model Options

A *Simple VCF Automation Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Networking	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model • VCF Management Dedicated VLAN and NSX Overlay Segment Network Model • VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model

Simple VCF Automation Model Design Requirements

Table 528: Simple VCF Automation Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AUTO-REQD-SIM-001	Deploy VCF Automation as a single node in the default management domain vSphere cluster.	• Supports the simple deployment objective. • VCF Automation supports scale-out to a high availability model.	• The simple deployment model defaults to the small appliance size. • Administrators must be aware of the scale and performance implications of a single small-size node to their design.
VCF-AUTO-REQD-SIM-002	Protect the VCF Automation node using vSphere HA.	Ensures availability of VCF Automation without requiring manual intervention during an ESX host failure event.	None.

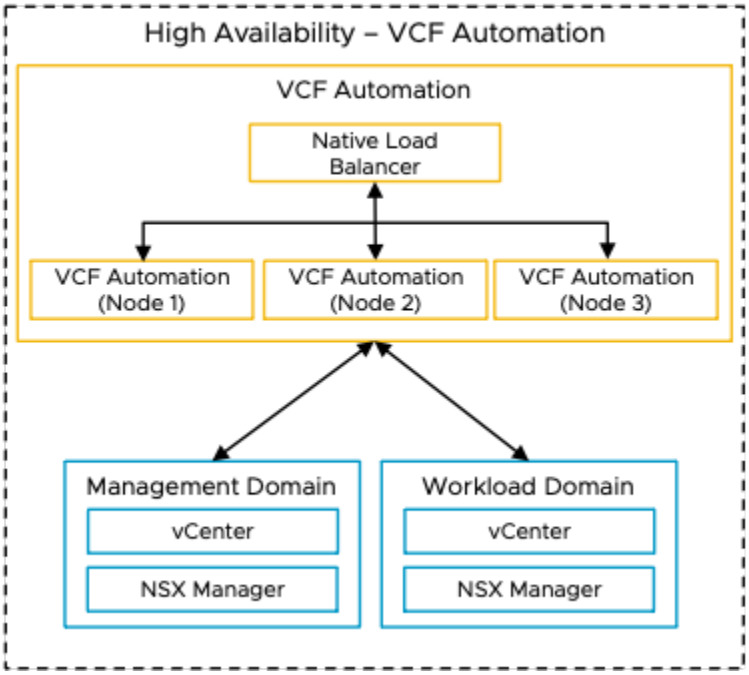
High Availability VCF Automation Model

The *High Availability VCF Automation Model* consists of a three-node deployment. This model provides resiliency at both the application level through the clustered deployment and at the vSphere cluster level through vSphere HA and vSphere DRS functionality.

The high availability discussion in this section focuses on the VCF Automation appliance level. For discussions on high availability design for components supporting VCF Automation, see the following:

- vSphere Clusters - See vSphere Cluster Detailed Design for more details.
- vSphere Supervisor deployments - See vSphere Supervisor Detailed Design for more details.
- NSX Edges - See NSX Edge Cluster Detailed Design for more details.

Figure 241: High Availability VCF Automation Model in VMware Cloud Foundation



High Availability VCF Automation Model Attributes

A *High Availability VCF Automation Model* has the following attributes.

Attribute	Detail
Availability	• Application level availability provided by the three node cluster • Appliance level level availability provided by vSphere HA
Scalability	Supports scale-up and scale-down of node sizes.
Recoverability	Restore through a backup.

High Availability VCF Automation Model Options

A *High Availability VCF Automation Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Option
Networking	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model • VCF Management Dedicated VLAN and NSX Overlay Segment Network Model • VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model
External Load Balancing	• External load balancer configuration is a post deployment configuration. • The native Kubernetes load balancer is configured automatically during deployment of VCF Automation and cannot be removed. • The external load balancer must be configured to send traffic to the native Kubernetes load balancer configured at deployment. • The external load balancer requires VIP addressing, FQDN, as well as a certificate in addition to those assigned at deployment.

High Availability Automation Model Design Requirements

Table 529: High Availability VCF Automation Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AUTO-REQD-HA-001	Deploy VCF Automation as a three node cluster.	Supports high availability objectives.	• Analysis is required to determine which node size is appropriate for your deployment. • Node can be resized after deployment as necessary.
VCF-AUTO-REQD-HA-003	Protect all VCF Automation nodes using vSphere HA.	Ensures availability of VCF Automation without requiring manual intervention during an ESX host failure event.	None

VCF Automation Deployment Patterns

This section explains different deployment patterns of VCF Automation within VMware Cloud Foundation single and multi-domain architectures.

- Deployment Pattern 1: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters
- Deployment Pattern 2: VCF Automation in a VCF Instance with Multiple Domains
- Deployment Pattern 3: External-Facing Deployment of VCF Automation

Deployment Pattern 1: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters

VCF Automation is deployed in a management domain and organization workloads are placed in a different vSphere cluster of the same management domain.

This deployment pattern separates the management and compute resources into two vSphere clusters in a single domain. The management domain hosts the core management components of VCF. In this deployment pattern, different subnets are used for various traffic types, ensuring security and performance. A transit gateway is used to connect upstream network and it serves as a high-level network routing hub, linking multiple subnets, and networks across the architecture.

The vSphere cluster that hosts the organization workloads has NSX VPC configured and vSphere Supervisor enabled to consume a modern private cloud with VCF Automation.

This deployment pattern:

- Enables high availability through the deployment of multiple instances of components to provide redundancy and ensure reliability.
- Supports the use of either centralized gateway (CTGW) or decentralized gateway (DTGW) architectures.

Benefits	Drawbacks
• Fast deployment. • Simplified integrated management. • Dedicated VCF Instance for a tenant or a line of business. • Near-edge deployments smaller environments and no plan of growth. • Smaller hardware footprint. • VCF Automation HA is an option.	• Limited scalability compared to a multiple domain architecture. • Management Overhead to manage each VCF Instance for a dedicated tenant. • VCF Automation HA deployment consumes more infrastructure resources.

Figure 242: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters - Centralized Gateway (CTGW) Design

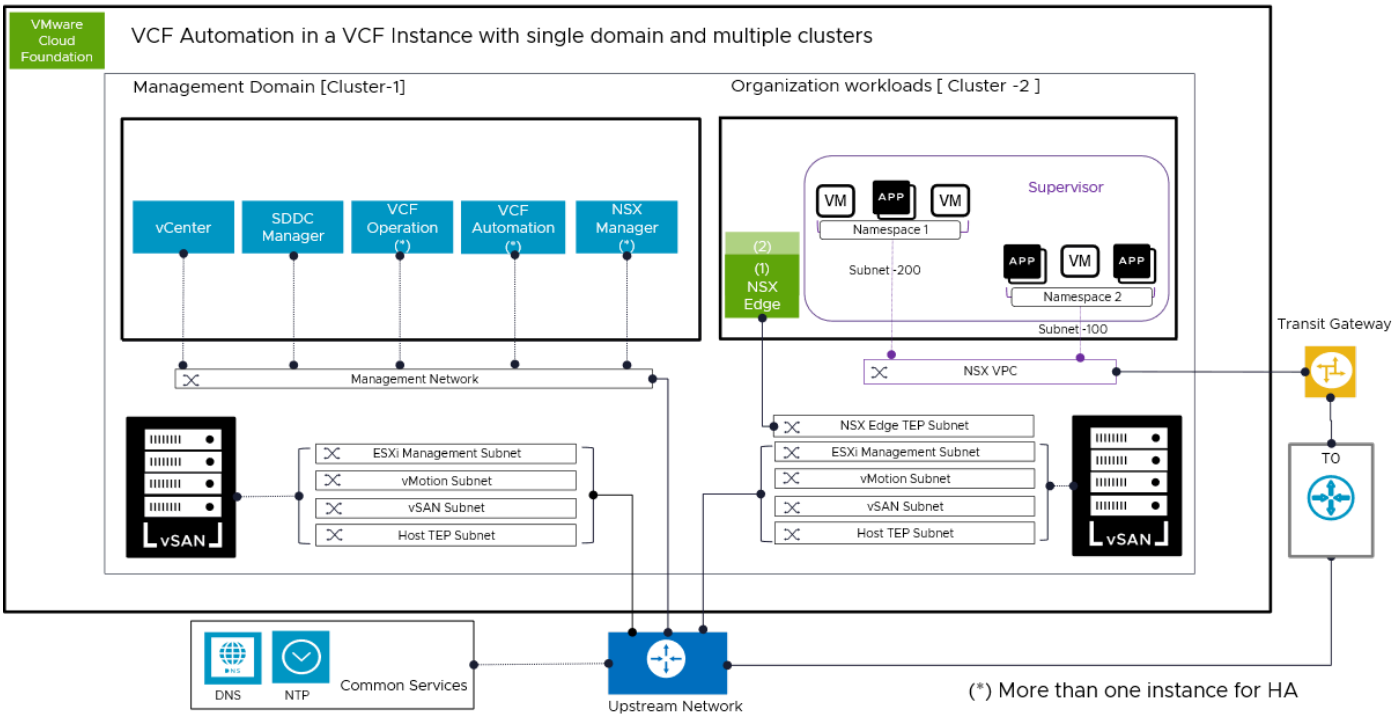
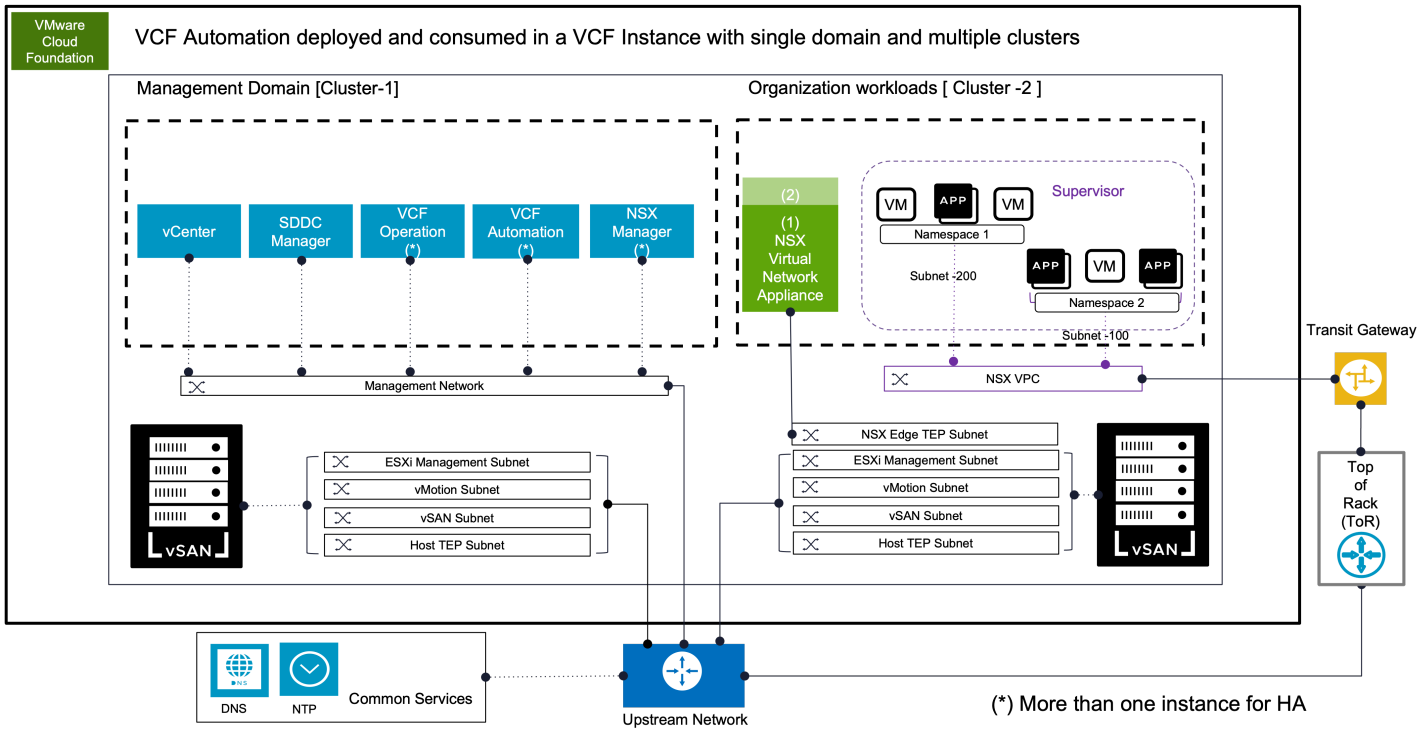


Figure 243: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters - Distributed Gateway (DTGW) Design



Deployment Pattern 2: VCF Automation in a VCF Instance with Multiple Domains

VCF Automation is deployed in a management domain and organization workloads are placed in a workload domain.

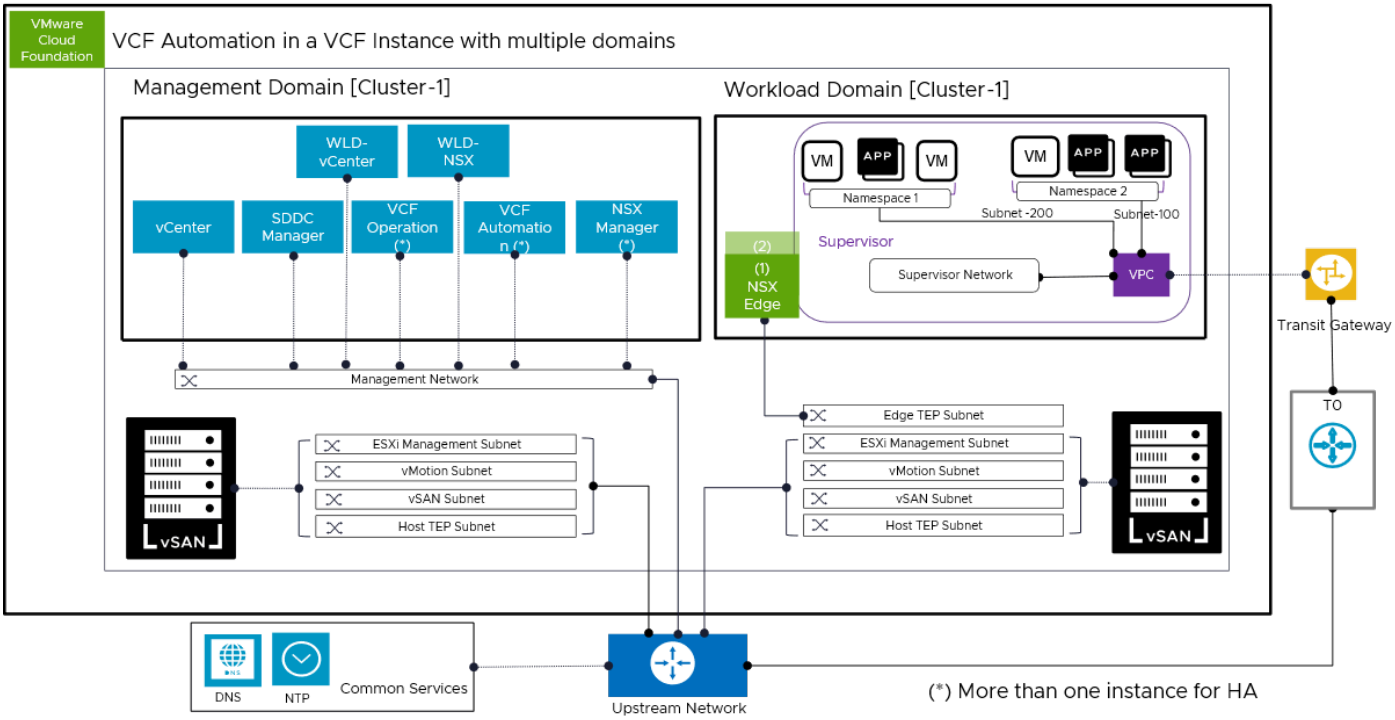
The management domain hosts the core management components of VCF. The workload domain hosts the tenant virtual machines. Different subnets are used for various traffic types, ensuring security and performance. In this deployment pattern, different subnets are used for various traffic types, ensuring security and performance. A Transit Gateway is used to connect upstream network and it serves as a high-level network routing hub, linking multiple subnets, and networks across the architecture.

The vSphere cluster that hosts the organization workloads has NSX VPC configured and vSphere Supervisor enabled to consume a modern private cloud with VCF Automation.

This deployment pattern enables High Availability through the deployment of multiple instances of components to provide redundancy and ensure reliability.

Benefits	Drawbacks
• Simplified integrated management • Highly scalable through independent scaling of compute, storage, and networking components • Distributed architecture • Greater long-term flexibility to adapt to changing needs or growth • Highly Available	• Larger hardware footprint • Larger cost footprint

Figure 244: VCF Automation in a VCF Instance with Multiple Domains - Centralized Gateway (CTGW) Design



Important:

Although the diagram shows the centralized gateway (CTGW) architecture the decentralized gateway (DTGW) architecture is also supported.

Deployment Pattern 3: External-Facing Deployment of VCF Automation

External-Facing Deployment of VCF Automation refers to implementing VCF Automation to seamlessly interact with external networks or services in a secure manner.

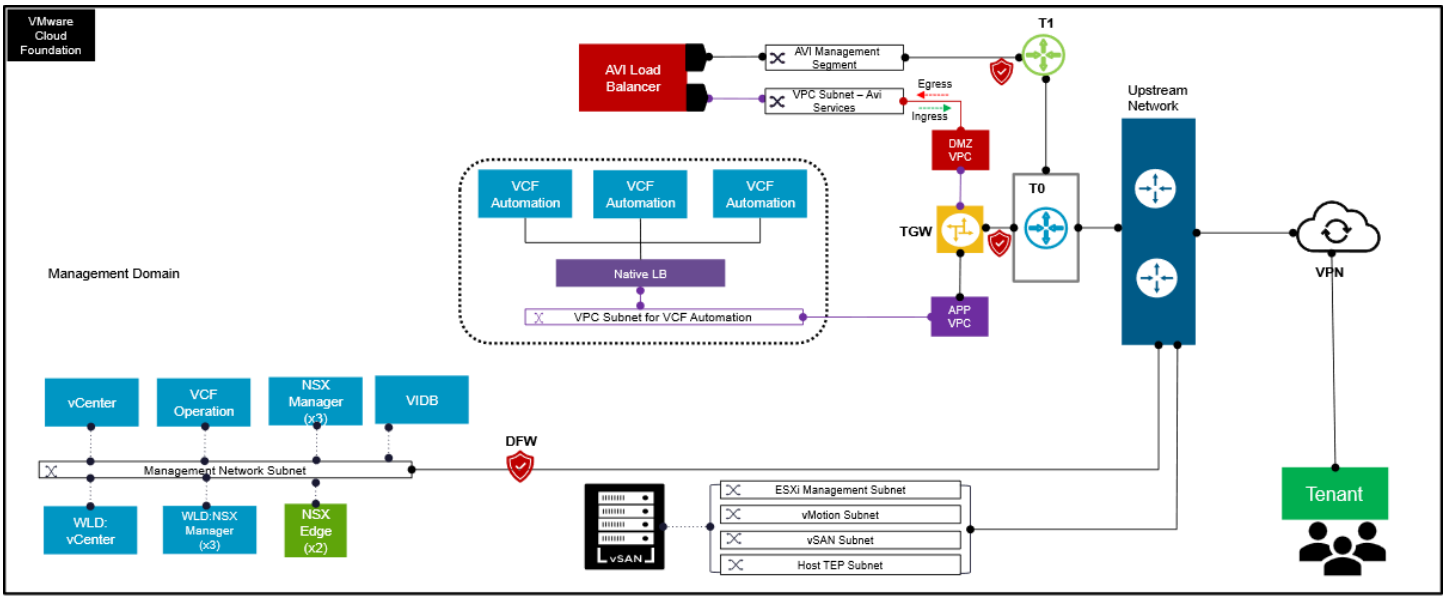
This deployment pattern helps to minimize the attack surface, contain potential security breaches, and provides the necessary controls for meeting various security and compliance standards for VCF Automation.

The DMZ VPC accommodates external facing services (like load balancers) and serves as a security barrier between the Internet and the internal application network. To provide an additional layer of network isolation, this deployment pattern also uses a second VPC for the VCF Automation instances. The Transit Gateway routes traffic between the two VPCs and to the upstream network, enabling safe channels of communication.

The Tier-0 Gateway provides north and south routing as well as stateful firewall features. This offers a reliable and secure design for hosting VCF Automation that is accessible over the Internet. Provider Admin can apply east/west firewall on the Mgmt App VPC to protect access to VCF Automation appliances and limit lateral movement.

Benefits	Drawbacks
• Easier to manage and update one set of VCF Automation appliances. Reduces operational overhead. • Limits lateral movement and isolates external-facing services from core management and workloads. • Micro-segmentation approach securely exposes services through DMZ network and Load Balancer for strong protection • Distributes traffic efficiently across multiple automation appliance instances for scale and availability.	• Needs robust security (e.g., firewalls, WAF, and API security in place). • Load balancer misconfigurations could cause downtime or abnormal traffic routing. • Adds network complexity; managing routing, firewall rules, and subnet allocations needs careful planning for each organization or tenants.

Figure 245: External-Facing Deployment of VCF Automation - Centralized Gateway (CTGW) Design



Important:

Although the diagram shows the centralized gateway (CTGW) architecture the decentralized gateway (DTGW) architecture is also supported.

Network Design Options for VCF Automation Instance Deployment

VMware Cloud Foundation provides flexibility in network segment design for a VCF private cloud. VCF Automation supports two basic network designs.

Network Design Options for VCF Automation

VCF Automation supports two network design models:

1. *Shared Subnet Model* - VCF Automation resides on the same subnet as the other VCF fleet components.
2. *Dedicated Subnet Model* - VCF Automation resides on a dedicated subnet separate from the other VCF fleet components.

Shared Subnet Design

The *Shared Subnet Model* focuses on VCF Automation residing on the same subnet as the other VCF fleet components. For this model, refer to the VCF Management Network Detailed Design section of this guide.

This model applies to:

- Deployment Pattern 1: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters
- Deployment Pattern 2: VCF Automation in a VCF Instance with Multiple Domains

Dedicated Subnet Design

The *Dedicated Subnet Model* is used where there is a requirement to place VCF Automation on a dedicated subnet in parallel to the subnet for the other VCF fleet components. Most commonly this design is used where VCF Automation must be placed in a DMZ (demilitarized zone).

This model applies to Deployment Pattern 3: External-Facing Deployment of VCF Automation.

Network Design Options	Justification	Implication
Shared Subnet Model	• Default deployment through VCF Installer. • Quick and easy deployment as it uses the default management network.	• Management network must have enough available IP addresses to support the VCF Automation deployment. • Moving to a separate network from the default management network may require a redeployment of VCF Automation.
Dedicated Subnet Model	• Would like to separate user-facing networks from management networks to meet regulatory and/or security requirements. • Where VCF Automation must be deployed in a DMZ.	• If this option is chosen, VCF Automation deployment cannot happen through VCF Installer. • The subnet must be created before VCF Automation deployment. • VCF Automation installation must be deployed through VCF Operationsfleet management API.

IP Addressing and FQDN Requirements

VCF Automation runs on a dedicated VCF management services deployment (referred to as the VCF services runtime) this runs in parallel to the primary VCF management services which services the VCF fleet components. The following table defines the IP and FQDN requirements for VCF Automation.

VCF Automation Component	FQDN Requirement	IP Address Requirement
VCF Automation VIP	1 FQDN	1 IP address associated with FQDN
Dedicated VCF services runtime	1 FQDN	1 IP address associated with FQDN
Nodes for dedicated VCF services runtime	N/A	5 IP addresses

Tenancy Deployment Models

This section describes how different tenancy models relate to the VCF fleet and VCF domain concepts, which impacts how tenants are provisioned and managed within a private cloud environment. Businesses can select a deployment model (or models) based on operational flexibility, compliance needs, and scalability requirements.

- Tenancy Deployment Model 1: Consolidated Management and Organization Workloads in a VCF Instance
- Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance
- Tenancy Deployment Model 3: Centralized Management with Dedicated Workload Domains per Organization in a VCF Instance
- Tenancy Deployment Model 4: Centralized VCF Automation Managing Dedicated VCF Instances per Organization
- Tenancy Deployment Model 5: Dedicated VCF Fleet with Independent VCF Automation Supporting Single or Multiple Domains per Tenant

Tenancy Deployment Model 1: Consolidated Management and Organization Workloads in a VCF Instance

This tenancy deployment model consists of one VCF domain that hosts the management components and organization workloads.

Management components and workloads can share one vSphere cluster, or can be deployed on different vSphere clusters.

When adopting this model, carefully evaluate the potential implications, including resource contention, performance impact, security risks, and reduced availability that may arise from hosting organization workloads within the same cluster as management components.

Figure 246: Management and Organization Workload Sharing One vSphere Cluster within a vCenter

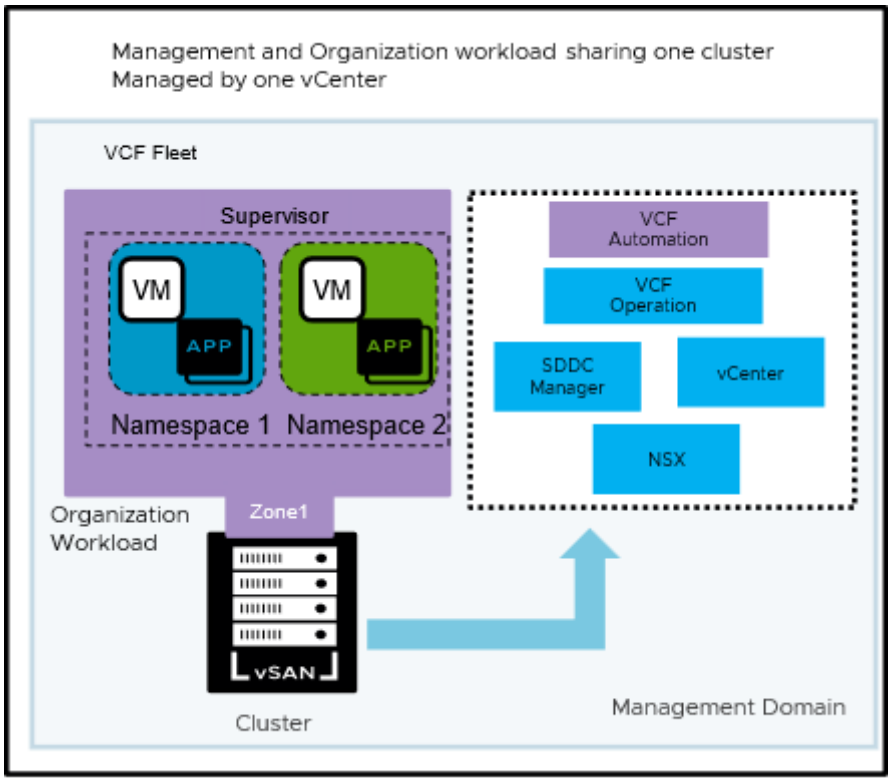
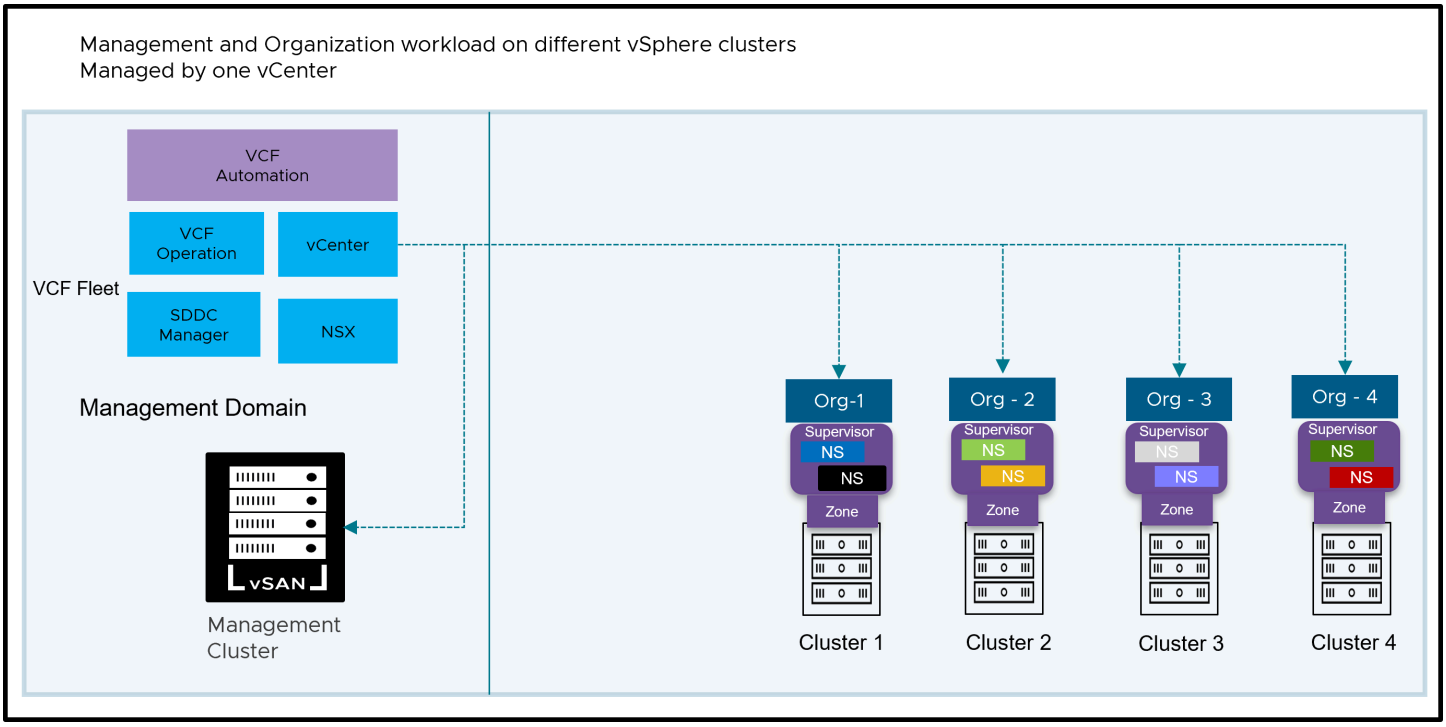


Figure 247: Management and Organization Workload on Different vSphere Clusters within in a vCenter

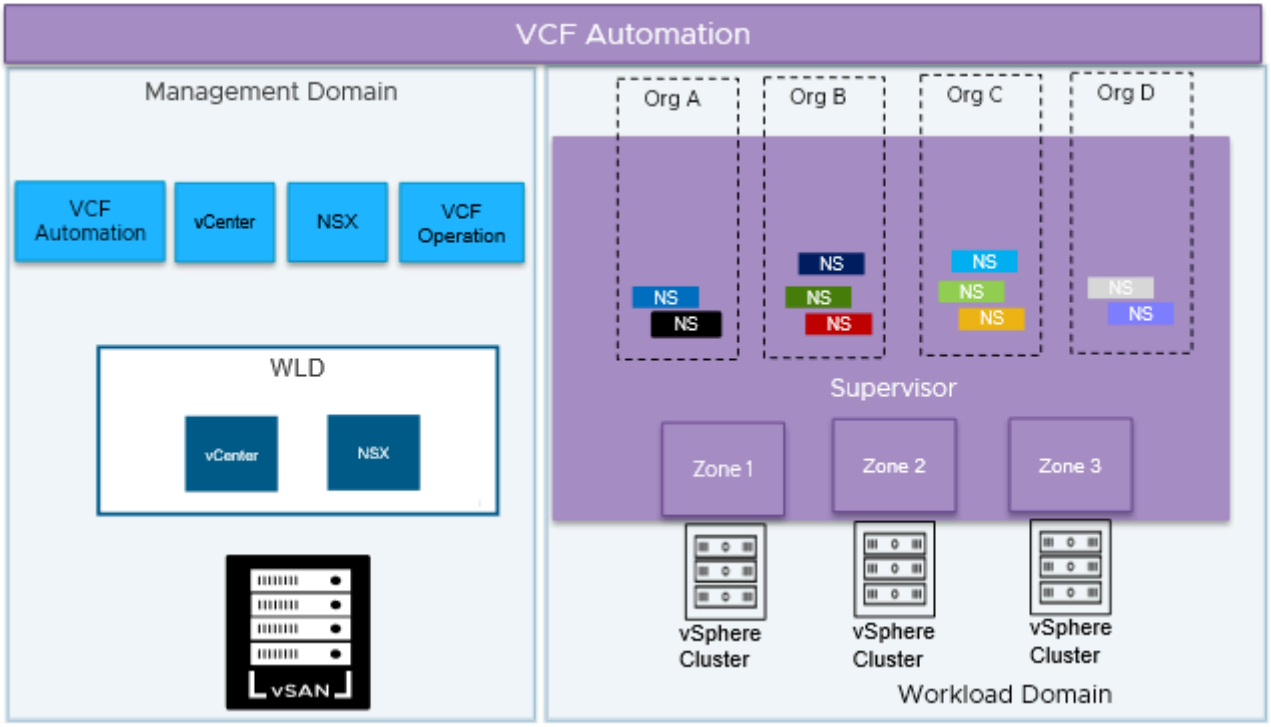


Benefits	Best Fit Scenarios
• Consolidating workloads can streamline the initial setup of a VCF Instance, which results in fewer physical hosts and clusters to configure, which speeds up the deployment process. • Using a single cluster lowers the overall hardware requirements that lead to cost savings in terms of server hardware, networking equipment, and data center space.	For organizations with smaller environments and limited resource requirements, consolidating workloads can provide a cost-effective and manageable solution.

Tenancy Deployment Model 2: Shared Workload Domain for Multiple Organizations in a VCF Instance

This model offers a multi-tenant VMware Cloud Foundation platform where multiple organizations share a single workload domain. Isolation is achieved through vSphere Namespaces and network segmentation.

Figure 248: Shared Workload Domain for Multiple Organizations in a VCF Instance



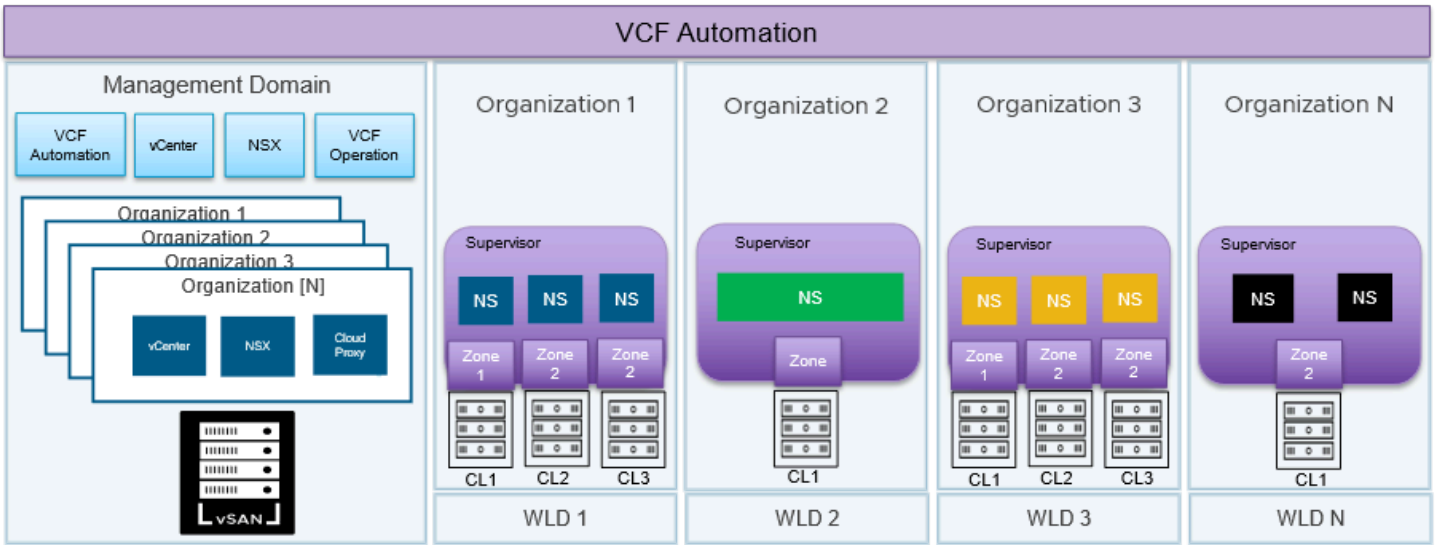
Benefits	Best fit scenarios
• Resource Sharing: Tenants share the underlying infrastructure (workload domain, vCenter, NSX), leading to cost savings and efficient resource utilization. • Simplified Management and Operational Consistency: While providing tenant isolation, the centralized management of the VCF Management Domain simplifies overall infrastructure operations, including patching, upgrades, monitoring, and capacity planning. Consistent policies can be applied across all tenants. • Scalability and Flexibility: The model is highly scalable. As the number of tenants or their resource demands grow, the underlying workload domain can be scaled up by adding more hosts. Resource allocations within Namespaces can also be adjusted dynamically. If the requirement comes, the service provider can add more workload domains for multiple other organizations to share or can dedicate a workload domain per tenant like explained in Model 2. • vSphere Namespace Isolation: vSphere Namespaces provide logical resource isolation for tenant workloads, preventing interference. • Network Segmentation: Network policies and security groups can be used to further segment tenant traffic and can enhance security.	• Cloud Service Providers (CSPs) Offering Multi-Tenant Cloud Services: This model allows the CSP to host multiple customers within a single VCF instance, maximizing resource utilization across their infrastructure. The Supervisor Cluster ensures strong isolation between each customer's workloads and data. Each customer gets their own vSphere Namespace, providing them with a secure and logically separated environment for deploying VMs and containers. Network segmentation (via NSX) further guarantees that customer traffic remains isolated. • Large Enterprises with Multiple Business Units or Departments: A large corporation has several independent business units (e.g., Finance, Marketing, R&D) that require their own isolated IT environments but can share underlying infrastructure. • Organizations Offering Internal PaaS or Developer Platforms: An organization wants to provide a self-service platform for its internal development teams to deploy and manage their applications (including containerized microservices) in an isolated and governed manner. • Educational Institutions or Research Organizations with Multiple Projects or Teams: A university hosts various research projects or has different departments with distinct IT requirements and security needs.

Tenancy Deployment Model 3: Centralized Management with Dedicated Workload Domains per Organization in a VCF Instance

This model provides a robust and secure multi-tenant environment by leveraging VCF Automation and dedicated workload domains. Each organization gets a dedicated environment, ensuring security, performance, and independent

management. The central and shared management domain provides centralized control, while organization-specific components offer flexibility and autonomy.

Figure 249: Centralized Management with Dedicated Workload Domains per Organization in a VCF Instance



Benefits	Best fit scenarios
• Centralized Management: Offers one control point for overseeing the entire VCF environment for service providers handling several tenants. • Simplified Life-Cycle Management: Optimizes the updating and patching of essential VCF elements. • Providers of multi-tenant services: Provide safe and separate environments for various clients. • Major corporations with separate business divisions: Grant independence and seclusion to various departments. • Industries with extensive regulations: Adhere to rigorous compliance standards for data protection and confidentiality (such as healthcare, finance, public sector). • Organizations handling sensitive information: Provide increased security against unauthorized access and data leaks.	• Performance-Sensitive Workloads: Applications with demanding performance needs (e.g., high-throughput databases, real-time processing) require consistent and predictable resource availability. Dedicated hardware eliminates the "noisy neighbor" effect, where the activity of other tenants can negatively impact performance. • Specialized Hardware or Dedicated Hardware: Some organizations may require specific hardware configurations (e.g., GPUs, specialized processors, high-performance networking) that differ from the standard infrastructure. In certain cases different organization may have a different licensing requirement depending on their hardware configurations such as different CPU cores. Dedicated workload domains allow them to use their required hardware without imposing it on other tenants. • Data Sovereignty and Geographic Restrictions: Some countries have data sovereignty laws that require data to be stored and processed within their borders. Dedicated workload domain can be used to create isolated environments within specific geographic locations to comply with these regulations. • Mergers and Acquisitions: Maintaining separation, When two organizations merge or one acquires another, they may require separate IT environments for a period (or even permanently) due to: Security policies, Compliance requirements, different IT processes, phased integration plans. Dedicated workload domains allow them to keep their infrastructure separate while still benefiting from a centralized private cloud platform.

Tenancy Deployment Model 4: Centralized VCF Automation Managing Dedicated VCF Instances per Organization

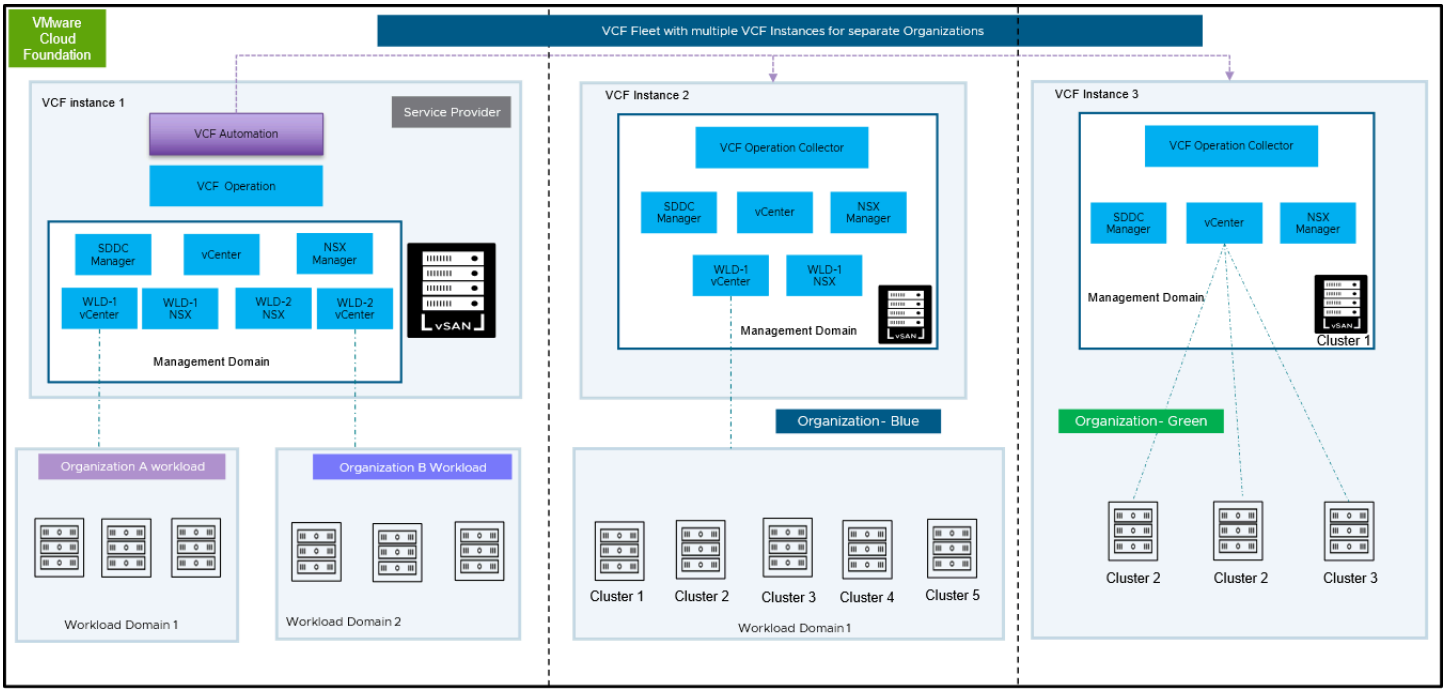
This model offers a dedicated VCF Instance (consisting of a management domain and one or more workload domains) for every tenant. Each tenant receives dedicated resources, in addition to common management components such as VCF Operations and VCF Automation managing both instances.

The bandwidth and latency requirements for day-to-day administrative user operations in VCF Automation can vary significantly. These requirements depend on several factors, including:

- The number of users accessing VCF Automation.
- The geographical distribution of users and resources.
- The volume and nature of user transactions.

> **Attention:**
>
> Refer to the Broadcom Ports and Protocols website for all latency information.

Figure 250: Centralized VCF Automation managing dedicated VCF Instances per Organization



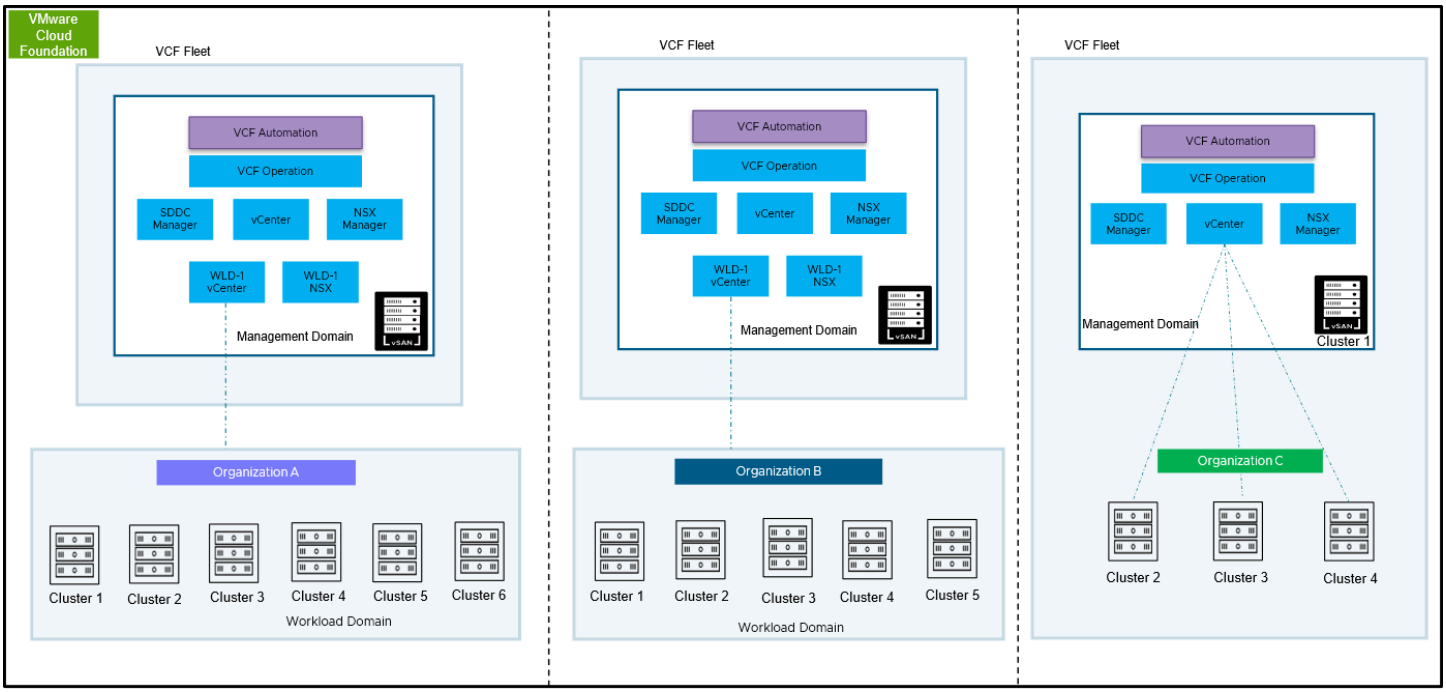
Benefits	Best fit scenarios
• This model offers a dedicated VCF Instance (comprising a Management Domain and Workload Domains) for every tenant. However, every VCF instances are consumed through one shared VCF Automation. This signifies that every tenant receives fully separate resources, encompassing compute, storage, and networking, in addition to their specific management components such as vCenter and NSX. • Improved Security: Robust isolation stops the actions of one tenant from impacting others, reducing security threats, and supporting adherence to data privacy and compliance. • Committed Resources: Every tenant has assured access to their designated resources, guaranteeing stable performance and preventing resource conflict problems. • Autonomous Management: Tenants possess complete authority over their environment, enabling them to oversee their resources and implement applications based on their individual requirements and regulations. Streamlined Operations: Dedicated instances reduce complexity and simplify management tasks, as each tenant environment is self-contained.	• Large enterprises with autonomous business units: Corporations with distinct business units that have their own IT teams, budgets, and operational processes are ideal candidates for a VCF Fleet. Each business unit can have its own VCF instance that is tailored to its specific needs. • Service Providers offering highly isolated environments: Cloud Services Platforms that must provide isolated and secure environments for different enterprise customers benefit from a VCF Fleet. Each customer gets a dedicated VCF instance, ensuring maximum security and control. • Government agencies with separate departments: Government entities with different departments that have strict security and operational autonomy requirements can deploy a VCF Fleet to ensure the necessary isolation and control for each department's workloads. • Global organizations with regional data sovereignty needs: Multinational corporations with requirements for data to reside within specific geographic regions can deploy VCF instances in different locations, adhering to data sovereignty laws while maintaining a consistent VCF architecture.

Tenancy Deployment Model 5: Dedicated VCF Fleet with Independent VCF Automation Supporting Single or Multiple Domains per Tenant

This model illustrates a dedicated tenant approach where each tenant has its own isolated environment within a VCF Instance single domain or multiple domain constructs. This means each tenant has their own VCF Automation as well as dedicated compute and storage infrastructure supported by dedicated vSphere Supervisor and NSX deployments.

You can use the same vCenter for managing both management and compute workloads for Tenants.

Figure 251: Dedicated VCF Fleet with independent VCF Automation supporting single or multiple domains per tenant



Benefits	Best fit scenarios
• Provides a dedicated VCF Fleet to each tenant. Each fleet operates with its own VCF Automation and VCF Operations instances, ensuring complete operational isolation. Within each VCF Fleet, service providers or enterprises have the flexibility to either deploy a single workload domain with multiple clusters or multiple workload domains, each containing multiple clusters. This model offers full design freedom to allocate and dedicate resources based on specific tenant or organizational needs. • Resource Isolation and Efficiency: This model strikes a balance between providing dedicated resources for isolation and consolidating infrastructure for efficiency.	• Organizations with strict isolation requirements but also wanting to optimize resource usage. • Multi Geo - Complying with data sovereignty regulations • Service Provider - Offering hosted private clouds with dedicated infrastructure. • Supporting regulated industries with specific compliance needs. • Providing high-value services with strict SLAs. Facilitating M&A scenarios with isolated environments.

Organization Consumption Model

This section focuses the organizational (tenant) layer within VCF Automation. How resources are consumed by administrators and end users through an organization.

- Understanding VCF Automation All Apps Organization Multi Tenancy Model
- Provider Consumption Organization
- Provider Admin vs Organization Admin
- Understanding Supervisors, Regions, vSphere Zones, vSphere Namespaces, & Classes
- Understanding Content Libraries
- Understanding Organizations and Projects
- Understanding Workload Networking within VCF Automation
- Understanding VCF Automation Blueprints and Catalog

Understanding VCF Automation All Apps Organization Multi Tenancy Model

The All Apps Organizations are the primary multi-tenancy model within VCF Automation. All Apps Organizations uses vSphere Namespaces within vSphere Supervisor (Supervisors) to abstract tenant users and groups from the underlying vSphere and NSX infrastructure.

This Supervisor-based model:

- Completely isolates resources between All Apps Organizations.
- Within an All Apps Organization, allows for further isolation at the Project and vSphere Namespace levels.
- Controls user access to the resources.

The resulting Supervisor-based command and control model

- Completely removes the need for direct infrastructure access via individual tenant or service accounts.
- Removes potential security concerns that are created by the service account access models.

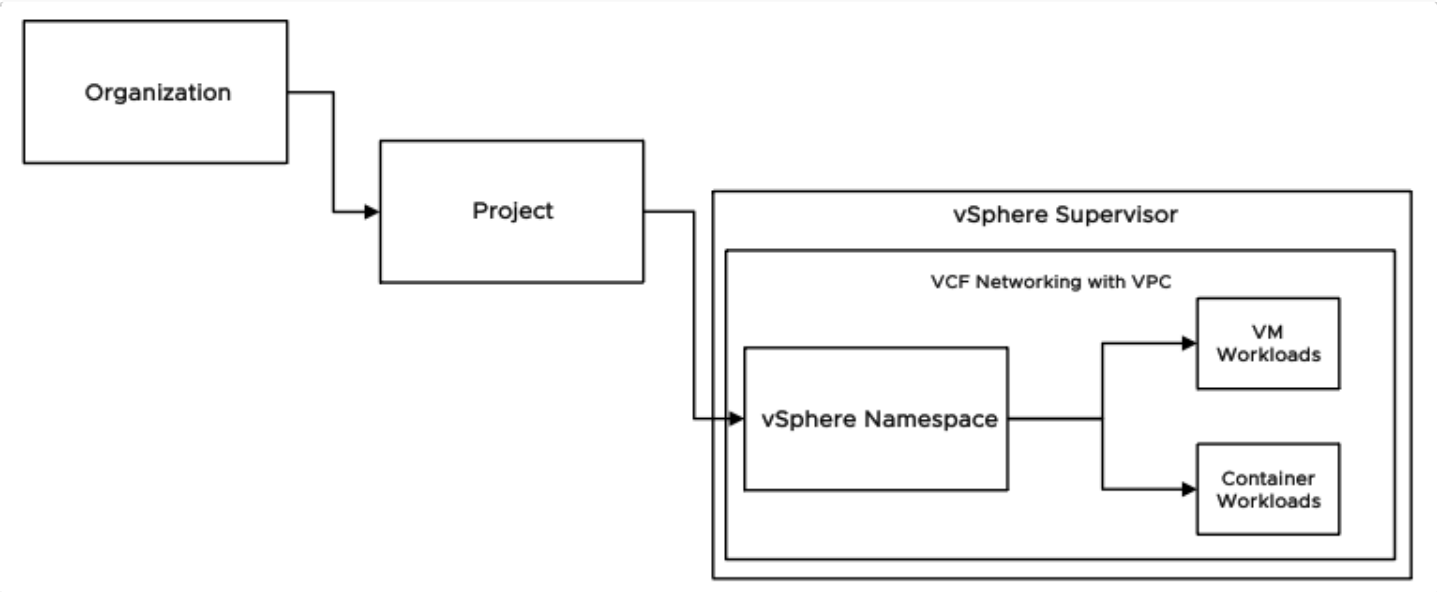
> **Remember:**
>
> The resources defined by a vSphere Namespace include: compute, memory, storage, and network resources (network defined through VPCs).

The new structure guarantees tenant visibility to assigned infrastructure (vSphere Namespace) while protecting tenant A from impacting tenant B. Total isolation is assured by the vSphere Namespace boundary between tenants working in concert with the NSX Project boundary (for networking). Thus providing a modern, true cloud experience with secure multi-tenancy for VCF Automation customers.

Implementing the All Apps Organization model within VCF Automation requires a few underlying components:

VCF Automation Structure	Details
Organization	• A top-level entity that is used to group and manage resources, users, policies, IaaS services, and catalog entities, while maintaining a secure boundary from other organizations. • Can be defined as a department or a line of business within an enterprise or a tenant or customer for a service provider • Offers a centralized structure for billing, access control, and resource sharing, facilitating collaboration and governance across teams, departments. • Maps to one or more Supervisor Clusters through Region Quota to provide the resources (compute, memory, storage, and networking) available for assignment within vSphere Namespaces.
vSphere Supervisor	• A Supervisor is a capability that turns vSphere into a platform that can host multiple IaaS services. • Supervisor defines the scope of vSphere IaaS control plane on vCenter. • A supervisor is a collection of one or more clusters on a vCenter instance and can span across multiple zones for fault tolerance.vSphere Supervisor is a management plane for Infrastructure-as-a-Service. • Provides compute, memory, and storage resources to an Organization. • Provides the structure on which vSphere Namespaces are deployed. • A single Supervisor Cluster can be mapped to multiple organizations through a Region Quota
Region	• A region consists of one or more Supervisor-enabled Clusters. • A region is a collection of compute, memory, storage, and networking resources that can span across vCenter instances.
Region Quota	• A Region Quota is a subset, or allocation, of resources from a region. • Defines the amount of capacity and the number of availability zones that are made available from a region to an organization. • On a per vSphere Zone basis, defines — CPU limit/reservation — Memory limit/reservation — Available VM classes — Available Storage Classes (also defines storage limits
Project	• Resides within an organization • There can be many projects within a single organization • Manage users & resources at scale using projects — Add users or groups to projects to give them access to IaaS services, — Add catalog items to be shared with the project. • Provide approvals and automate resource reclamation using project policies. • Govern total allocated resources and access to VM and storage classes using vSphere Namespaces in projects.
vSphere Namespace	• Resides within a Project. • There can be many vSphere Namespaces within a single Project. • Defines resource limits for CPU, memory, and storage for workloads. • Assigns available networks to workloads. • vSphere Namespaces are bound to the scope of the project with which they are associated.
vSphere Namespace Class	• Reusable templates for defining guardrails on parameters that are used during vSphere Namespace creation. • Defines at the vSphere Namespace level — CPU limit/reservation — Memory limit/reservation — Available Storage Classes (also defines storage limits) — VM classes available — Content Libraries
VPC	• A set of networking resources that provide a public-cloud experience to VCF customers. • Can be assigned to one or many vSphere Namespaces within an Organization.

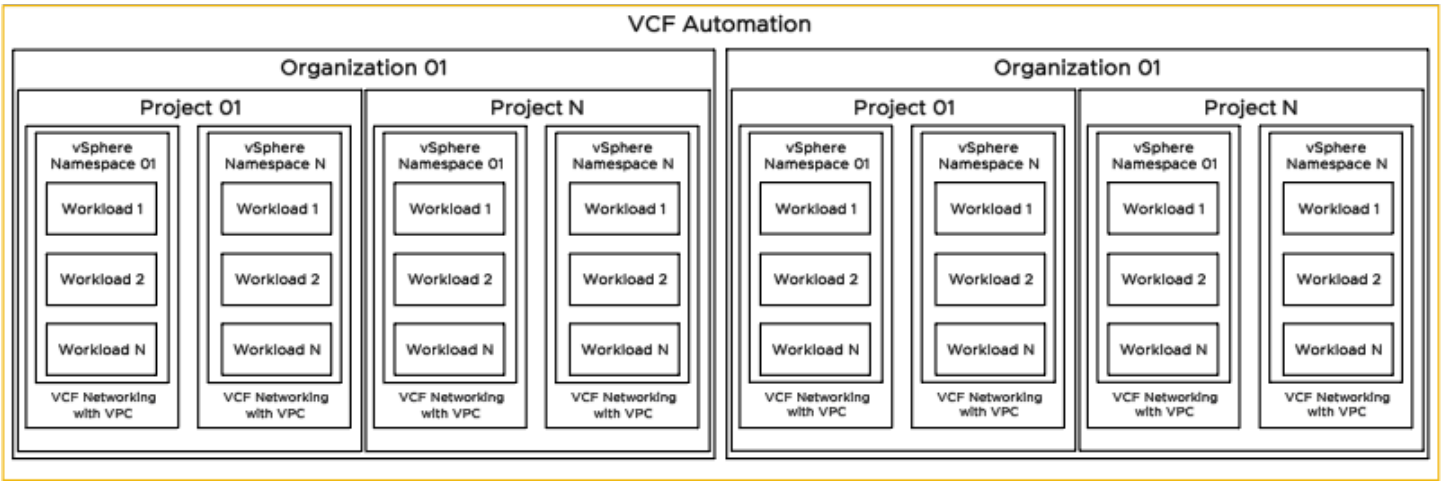
VCF Automation Structure	Details
Workload	• Virtual machine or containers that are deployed within a vSphere Namespace. • The number of workloads that are deployed in an organization is constrained by – The overall amount of resources assigned via Region Quota – The amount of resources that are then allocated to a particular vSphere Namespace



During All Apps Organization creation,

- All Apps Organizations are mapped to one or more vSphere Supervisors through Regions.
- Each organization then holds one or more projects.
- Each project then holds one or more vSphere Namespaces.
- vSphere Namespaces are instantiated within vSphere Supervisors that are mapped to the Organization.

Traditional virtual machine or VMware Kubernetes Service-based container workloads are then deployed utilizing the Supervisor Cluster resources defined within vSphere Namespaces.



Provider Consumption Organization

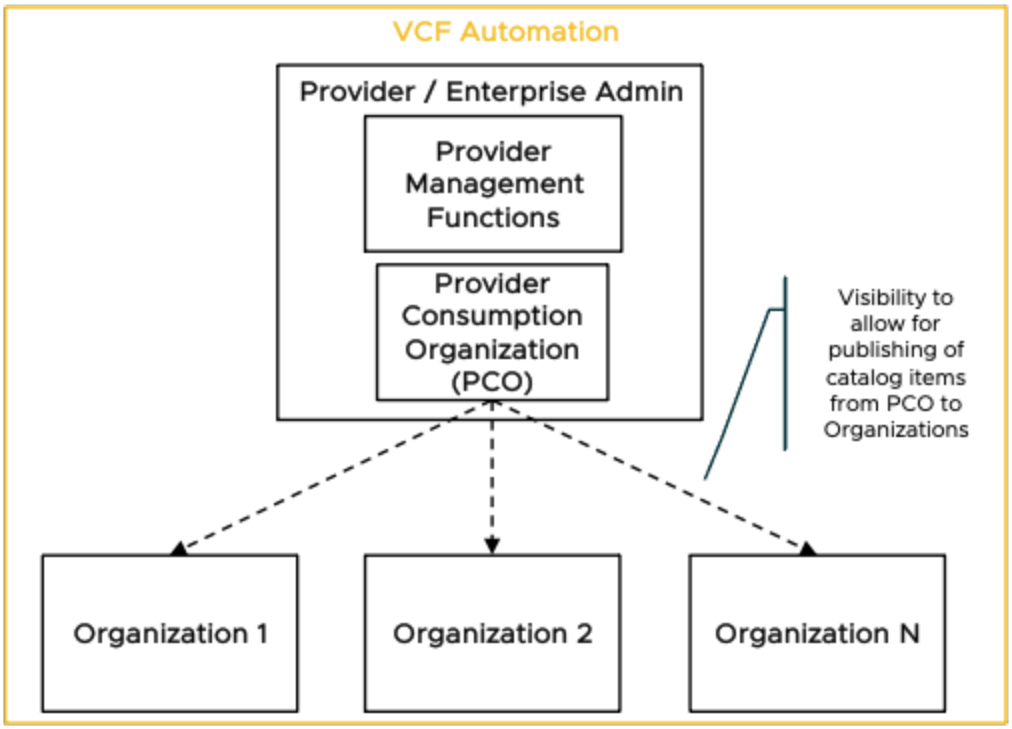
The Provider Consumption Organization (PCO) is a special purpose All Apps Organization that is used by the provider persona (the Provider or Enterprise-level admin)

Provider Consumption Organization allows you to:

- Consume their own resources
- Publish blueprints and catalog items to other Organizations
- Run published workflows from the embedded VCF Operations Orchestrator

Important:

The PCO is largely identical to all other All Apps Organizations. The primary difference is a special privilege allowing the PCO to publish catalog items to other Organizations.



The publishing permissions do not change the management structure of the other Organization. This means that the PCO cannot perform any resource management for other Organizations.

Table 530: PCO Functionality Breakdown

Functions the PCO can perform	Functions the PCO cannot perform
See all other Organizations in the system in order to publish catalog items	Context switch into any other tenants
Publish catalog items to those Organizations	Use an identity provider different from the provider org
Login with the same credentials as they log in to the system level	Manage other Organizations (AKA: be a sub-provider)
Manage user/group importing and role assignment within the PCO separately from the system level	Perform resource management for other Organizations
Provide isolated resources for use by the Provider Organization	

Provider Admin vs Organization Admin

Providers vs Organization

VCF Automation introduces a multi-tenancy model that is based on a vSphere Supervisor running as a management plane for all IaaS services including both VMs and containers.

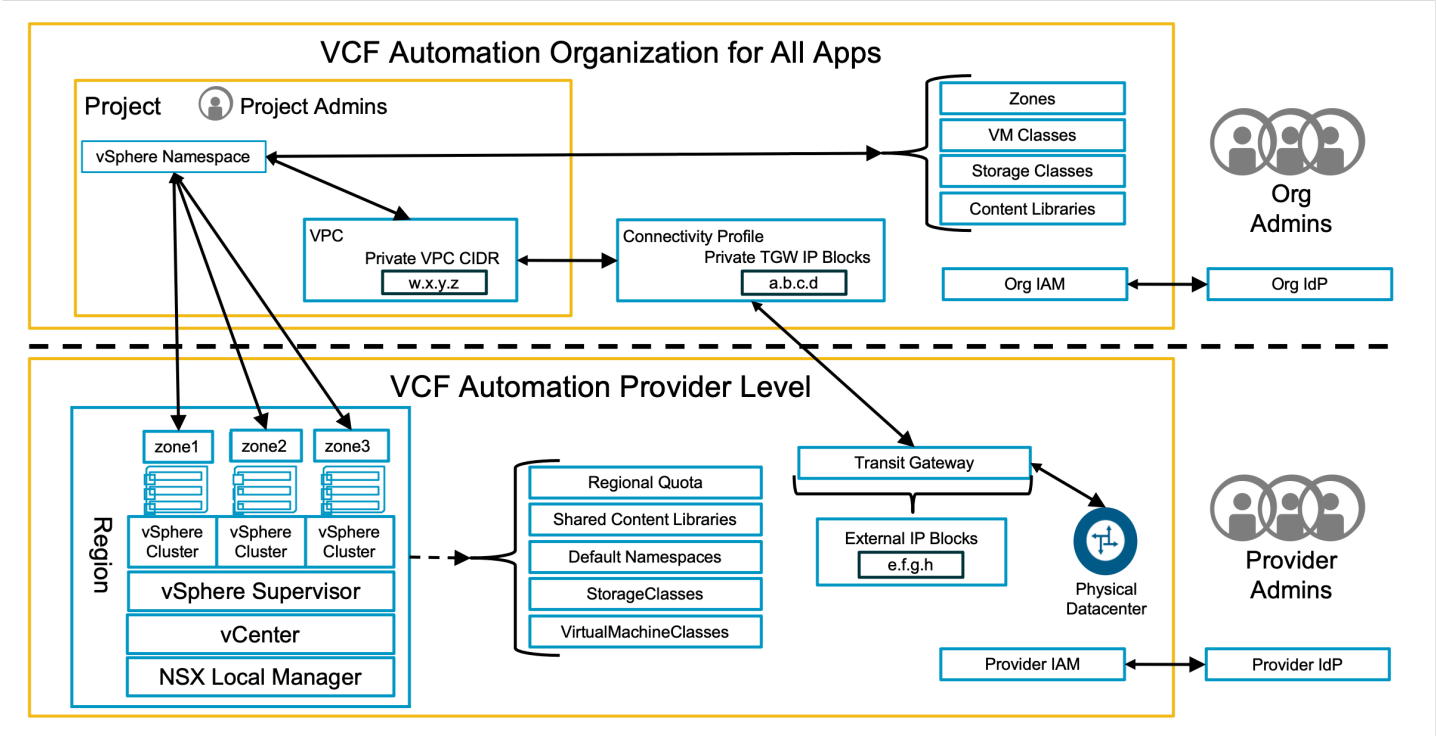
- Controls access via vSphere Namespaces within the Supervisor Cluster
- Completely removes the need for direct infrastructure access by tenants
- Provides complete isolation between tenants and limits each tenant to only view their own environment

The organizational model is separated into two primary levels - Provider and Organization or Tenant. An additional layer of separation exists within a tenant: Projects (which are covered in the Understanding VCF Automation All Apps Organization Multi Tenancy Model section).

- Provider Administrators are responsible for
  - The underlying infrastructure
  - Allocating and assigning infrastructure resources to tenants.
- Organization Administrators
  - Consume from their assigned resources and allocate them to different projects within their organization through vSphere Namespaces
  - Control over cross-project networking by managing Transit Gateways that can interconnect multiple VPCs, and their upstream connectivity
  - Also manages identity and access within the tenant

Figure 252: VCF Automation Organizational model

A graphical representation of the Provider and Tenant capabilities


VCF Automation Organization for All Apps
Project
Project Admins
vSphere Namespace
VPC
Private VPC CIDR
w.x.y.z
Connectivity Profile
Private TGW IP Blocks
a.b.c.d
Zones
VM Classes
Storage Classes
Content Libraries
Org IAM
Org IdP
Org Admins
VCF Automation Provider Level
Region
zone1
zone2
zone3
vSphere Cluster
vSphere Cluster
vSphere Cluster
vSphere Supervisor
vCenter
NSX Local Manager
Regional Quota
Shared Content Libraries
Default Namespaces
StorageClasses
VirtualMachineClasses
Transit Gateway
External IP Blocks
e.f.g.h
Physical Datacenter
Provider IAM
Provider IdP
Provider Admins

Understanding Supervisors, Regions, vSphere Zones, vSphere Namespaces, & Classes

The Supervisor is the component that owns the physical infrastructure and runs the organization workloads on those resources. A region is a collection of one or more Supervisors that share a common NSX Local Manager. A vSphere Zone is how the Supervisor maps to resources within a vSphere cluster. A supervisor can have multiple vSphere Zones. Supervisors and vSphere Zones are defined in vCenter.

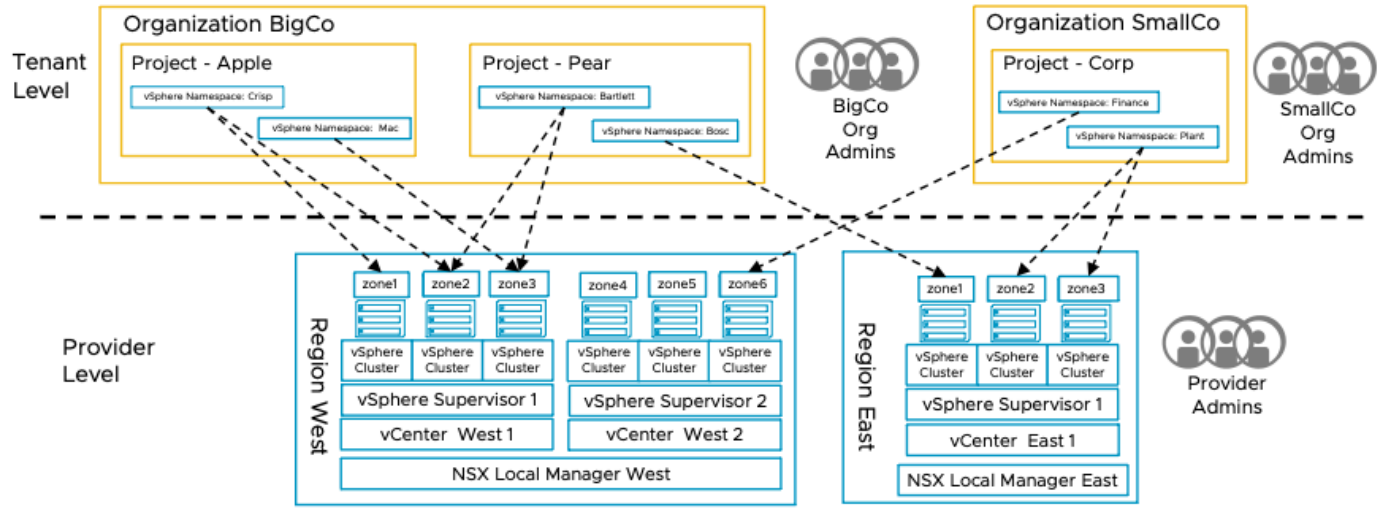
Supervisors and their vSphere Zones are assigned into Regions by the Provider. vSphere Namespaces are created within a Project by the organization administrator using the resources that are allocated to the organization through Region Quotas. vSphere Namespaces are defined using a vSphere Namespace class. vSphere Namespace classes define the CPU, memory, and storage resource limits and reservations and Content Library assignments for a vSphere Namespace. Each organization includes three default vSphere Namespace classes (small, medium, and large). Organization administrators can customize the default vSphere Namespace classes or can define new ones as necessary.

Table 531: Provider and Organization Administrator Roles for Supervisor and Namespace objects

Object	Provider	Organization
Supervisor	Uses Supervisors to create Regions. Exposes the underlying vSphere Zones, vm classes, and storage classes available for allocation via Region Quota.	Consumes indirectly (through vSphere Namespaces)
Region Quota	Creates Region Quotas	Assigns resources to vSphere Namespaces from the assigned Region Quota.
vSphere Namespace	Does not manage	Creates vSphere Namespaces.
vSphere Namespace Class	Does not manage	Creates vSphere Namespace Classes.

Figure 253: VCF Automation Supervisor relationships

How Organizations, Projects, Supervisors, and vSphere Namespaces interrelate



Remember:

If a VCF Automation Region is required to span multiple workload domains, share the single NSX Local Manager deployment across all the target workload domains.

If greater scale or physical separation is required, using a separate VCF Automation Regions allows control over where the workloads run through the vSphere Namespace configuration. vSphere Namespaces are scoped to a single region.

Note:

Please refer to vSphere Supervisor Detailed Design to understand supervisor availability models.

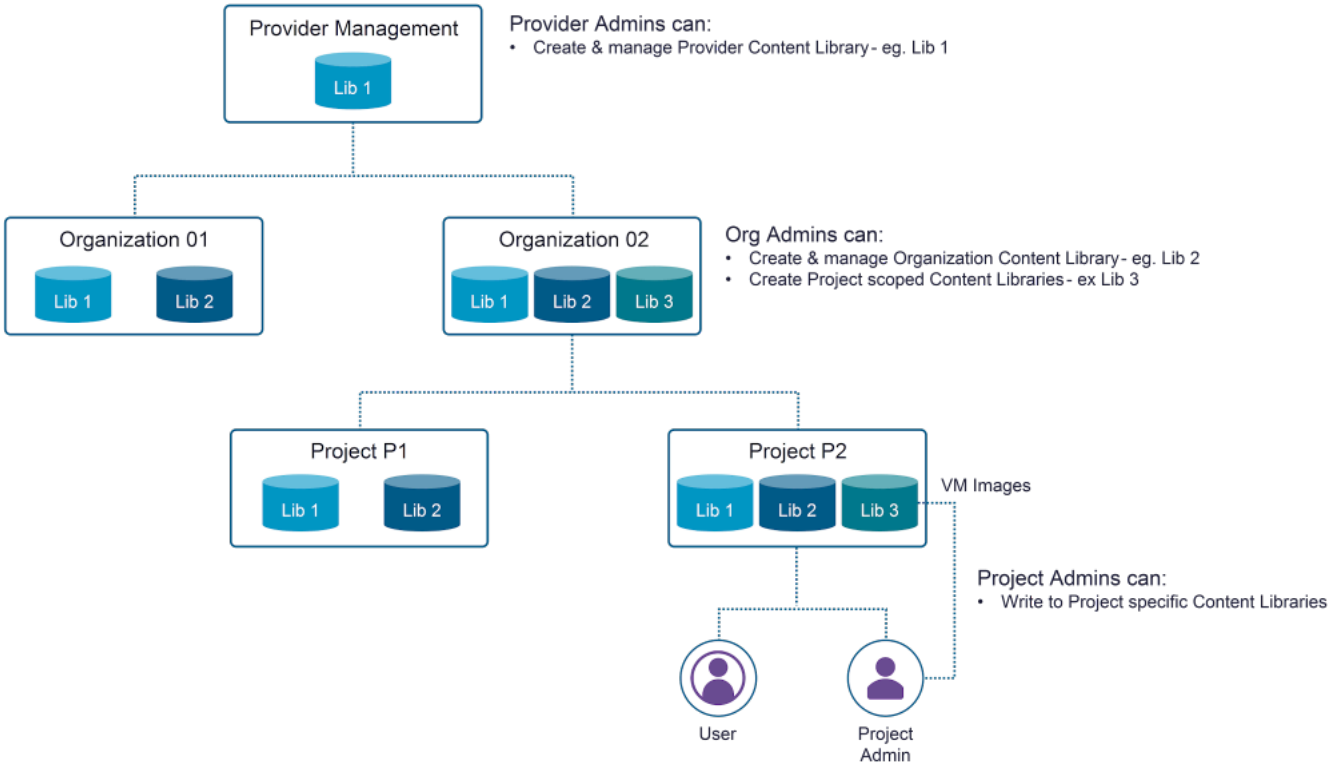
Design Objectives	Design Decisions
Assignment of clusters/Zones to Supervisors, and Supervisors to Regions	Define geographical alignment and boundaries for allocating infrastructure to organizations. All Clusters/Zones in a supervisor must be managed by the same vCenter. All Supervisors in a Region should have their networking provided by the same NSX Local Manager.
Define the parameters for which resources (compute, memory, and storage) are able to be consumed by end users for their workloads.	**vSphere Namespace Class:** vSphere Namespace classes are templates for the resources that are assigned to a vSphere Namespace. During vSphere Namespace creation a vSphere Namespace class is assigned, which determines the CPU, memory, and storage available to that vSphere Namespace. Default vSphere Namespace classes (small, medium, and large) are part of the base organization deployment and can be used as is. Organization administrators can also define their own classes or edit the default ones as needed. **VM Class:** VM classes define the CPU & Memory properties for individual workloads that can be created. The class includes t • The number of CPUs as well as how much of the CPU is reserved or allocated through best effort • How much memory is allocated, along with how much of that memory is reserved or allocated through best effort. The provider administrator assigns which VM classes are available to the organization during Region Quota creation. The organization administrator can then determine which available VM classes are assigned to a particular vSphere Namespace Class. **Storage Class** Storage Classes define the storage policy (e.g., RAID5 on cluster 1) and the capacity of that policy. The provider administrator assigns which storage classes are available to the organization during Region Quota creation. The organization administrator can then determine which available storage classes are assigned to a particular vSphere Namespace Class.

Understanding Content Libraries

Content libraries are centralized repositories of VM images, like OVA, OVF, and ISO images, and scripts that are used to deploy workloads. VCF Automation supports content libraries at the provider level and at the organization level. Within the organization, administrators can create content libraries for use in specific projects for better resource isolation and management.

VCF Automation provides two levels of content libraries:

- Provider-level
  - Provider Content Library
- Organization-level
  - Organization Content Library
  - Project Content Library



Content Library Type	Details	Use Cases/Caveats
Provider Content Library	• Created in the Provider Portal. • Automatically shared with read-only access to all organizations. • Supports OVA, OVF, and ISO images. • Stored as a vCenter Content Library.	Intended to share OVAs, OVFs, and ISOs created and/or supported by the Provider (e.g., generic Linux image).
Organization Content Library	• Created in the Organization Portal. • Supports OVA, OVF, and ISO images. • Specific to the organization and can only be shared within an organization. • Stored as a vCenter Content Library.	When an organization wants to use OVAs, OVFs, and ISOs that are directly managed/maintained by the organization administrator, for example: • A customized Linux image. • An ISO that is not available within the Provider Content Library.

Content Library Type	Details	Use Cases/Caveats
Project Content Library	• Created in the Organization Portal. • Supports OVA and OVF images • Does not support ISO images • Specific to one or more projects within an Organization • Project content libraries cannot be shared with other organizations. • Images are stored in the vCenter inventory within a new "content libraries" folder structure as powered-off VM templates instead of the native vCenter content library feature.	• Used as the location to store the blueprints that are derived from the capture blueprints as a namespace function. • Makes VM images available to specific users and resources within a project, compared to organization or provider content libraries that are always available to all users within the organization that the library is shared with.

Important:

Content Libraries must be assigned to one or more Regions. When multiple regions are selected, the content libraries are replicated with local copies that are stored in each region.

Design Objectives	Design Decisions
Define Content Library strategy/design - Provider-Level	Define OVAs, OVFs, and ISOs that: • Must be managed/maintained at the Provider-Level. • Must be shared across organizations.
Define Content Library strategy/design - Organization-Level	• Organization Content Library — Define if there are requirements to manage/maintain specific or custom OVAs, OVFs, and/or ISOs within an organization. — If the above is required, determine whether the Organization Content Library should be assigned to all current and future namespaces. • Project Content Library — Determine if the project requires specific or custom OVAs or OVFs outside of what is available from the Provider or Organization Content Libraries

Understanding Organizations and Projects

Organization and projects are the top-level objects that are used to define isolated sets of users and resources. An organization is fully isolated from other organizations, with separation at the resource, network, and identity layers. One organization has no knowledge of any other organizations. Typically used when multiple separate entities share a common infrastructure either with:

- A services provider model.
  or
- A centralized corporate IT department providing resources with self-service workload deployment and billing to different business units within the broader organization.

Within an organization, there is further grouping possible.

- Projects are the structures wherein vSphere Namespaces are created and access control to the resources within the vSphere Namespaces is defined.
- Compute, storage, and network resources are defined independently for each vSphere Namespace.
- VPCs can be shared or dedicated as needed between any projects and/or vSphere Namespace within the organization.

These separations help reduce the need for requests to central infrastructure teams (Provider and Infrastructure Managers) and moving the operations closer to the consumers of the service.

Each Organization has a dedicated NSX project that is created when the provider creates the Region Network Configuration. Providing each Organization their own Transit Gateway and IP block allocation which they can carve up into VPCs and subnets to deploy different workloads.

Design Objectives	Design Decisions
Define organizations based on the business structure to maintain the required isolation and access management. Assign administrators for each of these groups, allocating them appropriate resources for their areas, which they can re-allocate as needed. Provide each development or application team with compute and networking resources.	For Enterprise use cases without the need for full isolation between departments, such as when a shared IT team manages all infrastructure, a single Organization can be used. A degree of isolation and self-service between departments, using the shared organization, can be maintained using Projects. This can simplify the networking as the same External and Private TGW networks can be used. Each department will have its own Private TGW network space, used when workloads require communication between projects. Private VPC network space is defined at the project level and cannot communicate with other projects without using NAT. The External network space is generally used for defining NAT rules to allow inbound access while the default SNAT enables outbound connectivity. Organization each has a separate directory to define their users. When multiple tenants or departments use the same directory, it may be preferable to use a single organization with project-level separation.

Note:

Refer to the Role Base Access Control (RBAC) section for more information.

Understanding Workload Networking within VCF Automation

VCF Automation uses the Virtual Private Cloud (VPC) Network Virtualization Model for workload networking. Specifically, either the:

- Centralized Connectivity Model
  or
- Distributed VLAN Connectivity Model

For detailed information on both, refer to the Workload Networking Detailed Designs section of this document.

Important:

When using VCF Automation Organizations for All Apps, administrators perform all workload network configuration with the VCF Automation Provider or Organization Portals.

Do not configure any workload networks (VPCs) targeted for VCF Automation consumption outside of VCF Automation.

VCF Automation defines workload networks at two levels:

Provider-level Networking Components

- Pertains to the functions managed within the Provider Portal
- Defines external connectivity - External IP Blocks, External Connections, VLAN-backed subnets
- Assigns external connectivity to organizations

- Defined default private TGW and private VPC ip blocks assigned to organizations

Provider Level Networking Component	Definition
External IP Blocks	• Previously called IP Spaces, used to centrally manage IP addressing for organizations. • External IP blocks are associated with external connections. Subject to quotas, organizations can allocate individual IPs (for NAT and load-balancing use-cases). • Organizations can also create subnets from these blocks for direct workload connectivity without NAT
External Connections	• External Connections define how organizations' NSX Transit Gateways (TGW) can be connected to outside networks. • Either centralized or distributed VLAN connections. • An External Connection also associates provider-managed External IP Block with organizations.
Centralized Connections	• A type of External Connection. • Define external network connectivity via NSX Tier-0 or Tier-0 VRF. • Provides connectivity to designated Remote Networks and facilitates the advertisement of Subnets and IPs originating from IP Blocks associated with this connection. • Can connect to the default gateway if 0.0.0.0/0 is used rather than a specific Remote Network. • Enables a full set of stateful networking services within organizations, such as NAT, Load-balancing, IPsec VPN and Gateway Firewall.
Distributed VLAN Connections	• A type of External Connection. • Define external network connectivity through a VLAN on the physical networking infrastructure. • Offer a simplified way to connect organizations' Transit Gateways (TGW) to outside networks without using an NSX Tier-0 gateway.
Edge Clusters	• A logical grouping of transport nodes providing stateful services for VPCs. • Used with Centralized Connections.
VNA Clusters	• A logical grouping of transport nodes providing stateful services for VPCs. • Used with Distributed VLAN Connections.
Subnets	• Create and manage subnets that are extended to a VLAN on a physical networking infrastructure. Once subnets are shared with an organization, workloads there can be directly connected to these subnets. • Ensure the VLAN is available on all network switch ports that connect to hosts within the region quotas shared with the organization.
Default Private IP Blocks	• When an organization is created a default VPC is instantiated to include – Default Private TGW IP Block – Default Private VPC IP Block • Upon installation of VCF Automation, there are initial settings for these IP blocks. • Are inherited by every organization • Can be customized for each organization during Regional Network Configuration for an organization

Organization-level Networking Components

- Pertains to the functions managed within the Organization Portal
- Inherits default network configuration set during organization creation in the Provider Portal
- Define new VPC that is created and managed at the organization level
  - Any VPC created at the organzation-level will use the external IP block (or blocks) assigned to the organization
  - Will allow the use of an existing connectivity profle and transit gatewat or the create new ones

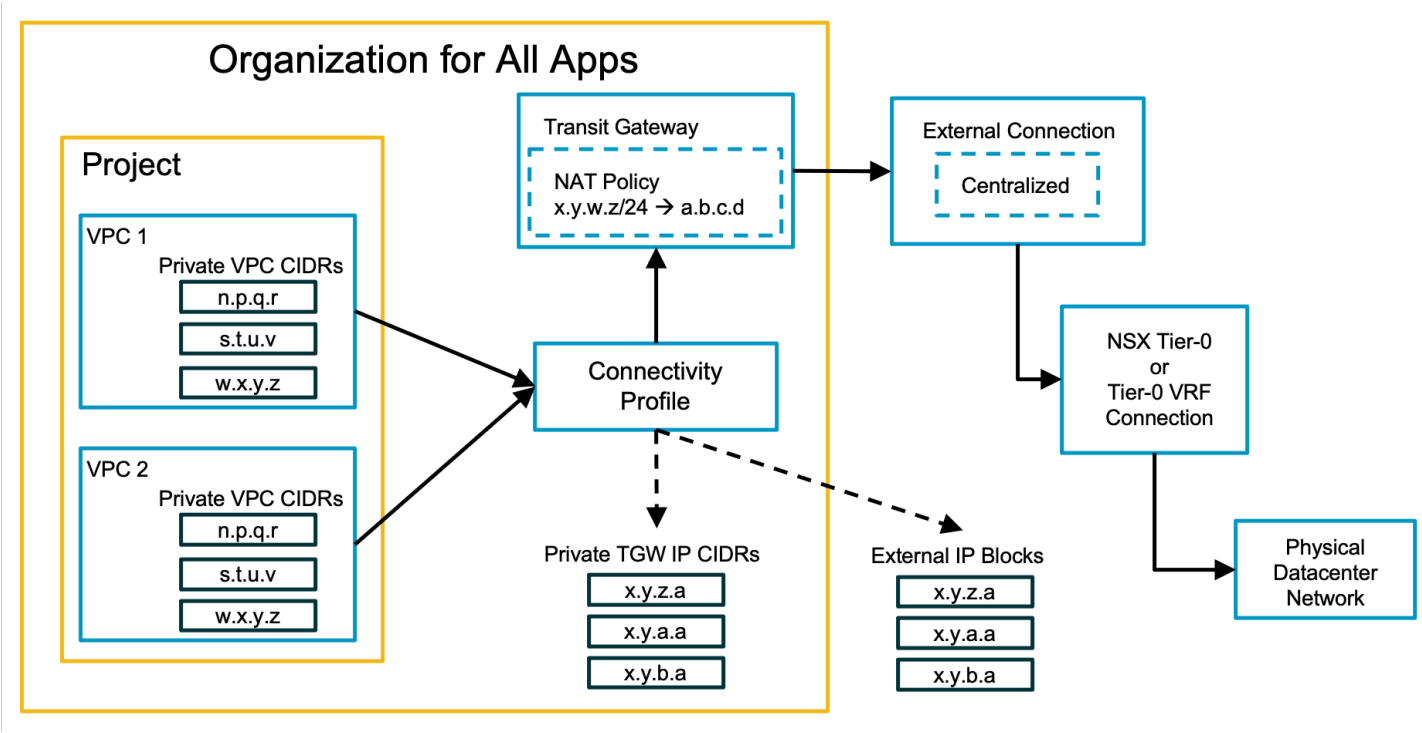
- Organization-level subnets to be shard with one or more namespaces.

Organization Level Networking Component	Definition
Virtual Private Clouds	• VPC provides an isolated domain of networking resources that can be associated with one or more isolated domains of compute resources (vSphere namespace). • Create, view, and manage VPC configuration here. Use a Networking service under a specific namespace in Build & Deploy to create, update, and age resources within the VPC for specific namespaces. • You can also create subnets Subnets in VPCs that are usable by multiple namespaces in this organization.
Connectivity Profiles	• A connectivity profile functions as a template intended to streamline the configuration of new VPCs. • Permits the specification of one or more external and private TGW IP address blocks, which are then used by VPCs configured with that specific connectivity profile. • A default profile is provisioned by the infrastructure administrator.
Transit Gateways	• Link your Virtual Private Clouds together and provide external access through External Connections shared to this Organization. • If connecting to a Centralized Connection they also enable perimeter services, such as NAT, VPN, and Gateway Firewalls.
External Connections	• Defined at the Provider level and shared with the Organization. • One or more of these connections may be attached to a Transit Gateway to enable external connectivity.
Subnets	• Created from existing VPCs within the Organization • Can be shared with one or more Namespaces. • Allows workloads from a given Namespace to connect to Subnets in different VPCs and/or spanning multiple Namespaces

Figure 254: Workload Networking Logical Overview

The diagrams provide a high-level visual representation of the workload networking.

- The first diagram shows the flow for the centralized connectivity model
- The second diagram shows the flow for the distributed VLAN model


Organization for All Apps
Project
VPC 1
Private VPC CIDRs
n.p.q.r
s.t.u.v
w.x.y.z
VPC 2
Private VPC CIDRs
n.p.q.r
s.t.u.v
w.x.y.z
Transit Gateway
NAT Policy
x.y.w.z/24 → a.b.c.d
Connectivity Profile
Private TGW IP CIDRs
x.y.z.a
x.y.a.a
x.y.b.a
External Connection
Centralized
NSX Tier-0 or Tier-0 VRF Connection
External IP Blocks
x.y.z.a
x.y.a.a
x.y.b.a
Physical Datacenter Network

Design Objectives	Design Decisions
Define a networking model providing the appropriate level of isolation, scale, and connectivity for each organization's workloads. Every organization should have sufficient external IP address allocations and guidance on reachability for external services to self-manage connectivity for their workloads.	• Determine connectivity model (centralized vs distributed VLAN) • Determine the external connection model – What are the external IP blocks targetted for use – How are the external IP blocks mapped to target Organizations • Determine the model for Organization networking – Whether to use the default VPC assigned to the Organization, if yes • Whether to customize the default private TGW IP block • Whether to customize private VPC IP block – Creation of custom VPCs (in parallel with or instead o the default VPC) – Mapping of VPCs to target namespaces – Use provider or organization defined Subnets • VPC Mapping: – vSphere Namespaces are managed by tenants, which includes mapping a vSphere Namespace to a VPC. Tenants can create a new VPC for each vSphere Namespace, or re-use the same VPC for multiple vSphere Namespaces. – Different VPCs should be used for workloads that require some level of network segregation (when on private VPC subnets) but also support communication between other workloads (on private TGW subnets). – When no network separation is required, the same VPC can be used across the region. – VPCs only span a single region, so Namespaces in different regions will always be mapped to different VPCs.

Understanding VCF Automation Blueprints and Catalog

Blueprints

Blueprints offer a declarative, YAML-based Infrastructure-as-Code (IaC) methodology to facilitate the automated provisioning of cloud services. The IaC methodology provides significant flexibility in both blueprint definition and how one creates a blueprint.

Blueprints define the configuration of networks, virtual machines, container architectures, applications, and services deployed through VCF Automation into a vSphere Namespace. They are used to enforce standards and governance by using constructs such as vm and storage classes to define cpu, memory, and storage configurations. Blueprint can vary in complexity. They can be as simple as a single virtual machine or as complex as a design with multiple VMs, networks, services, and other components.

> **Attention:**
>
> You can capture the contents of an existing VCF Automation deployed vSphere Namespace as a reusable blueprint. The resulting blueprint captures the entire configuration of the vSphere Namespace.

Organization administrators or project administrators have a few methods of blueprint creation:

Blueprint Creation Method	Use Cases/Caveats
Organization Portal Blueprint Design Page	• Manual blueprint creation. • Editing existing blueprints.
Capture a vSphere Namespace as a blueprint	• Without the need to manually, define or export infrastructure details, can be used to enable – Rapid environment cloning – Migration – Recovery • An alternative to traditional, manually authored VCF Automation blueprints and provide a way to create reusable deployments directly from existing workloads. • Used to deploy identical or differentiated environments from the same blueprint. • Suitable for use in: – Hands-on labs – Training environment – Similar scenarios where reusable, preconfigured environments are required. • After creation, can be edited using the Blueprint Design page.

Important:

To use a blueprint, it must be associated with one or more projects and published to the VCF Automation Catalog.

VCF Automation Catalog

The VCF Automation catalog is the streamlined user interface that you use to request and deploy infrastructure on a self-service basis. Catalog items include VCF Automation blueprints, VM images, storage, networks, and so on, that you can request for deployment. The items that are available to you depend on your project membership. At request time, the information that you must provide depends on how the item was configured by your administrator. When you deploy an item, it is provisioned based on the infrastructure available to the selected project. Projects also determine your options at deployment time.

Multi-Tenancy Consumption Patterns

This section describes the *Multi-Tenancy Consumption Patterns* aligned with the *Tenancy Deployment Model* described in the *Tenancy Deployment Model* section. It explains the different ways multiple organizations or tenants can consume shared compute, memory, storage, and networking resources, depending on the chosen *Tenancy Deployment Model*.

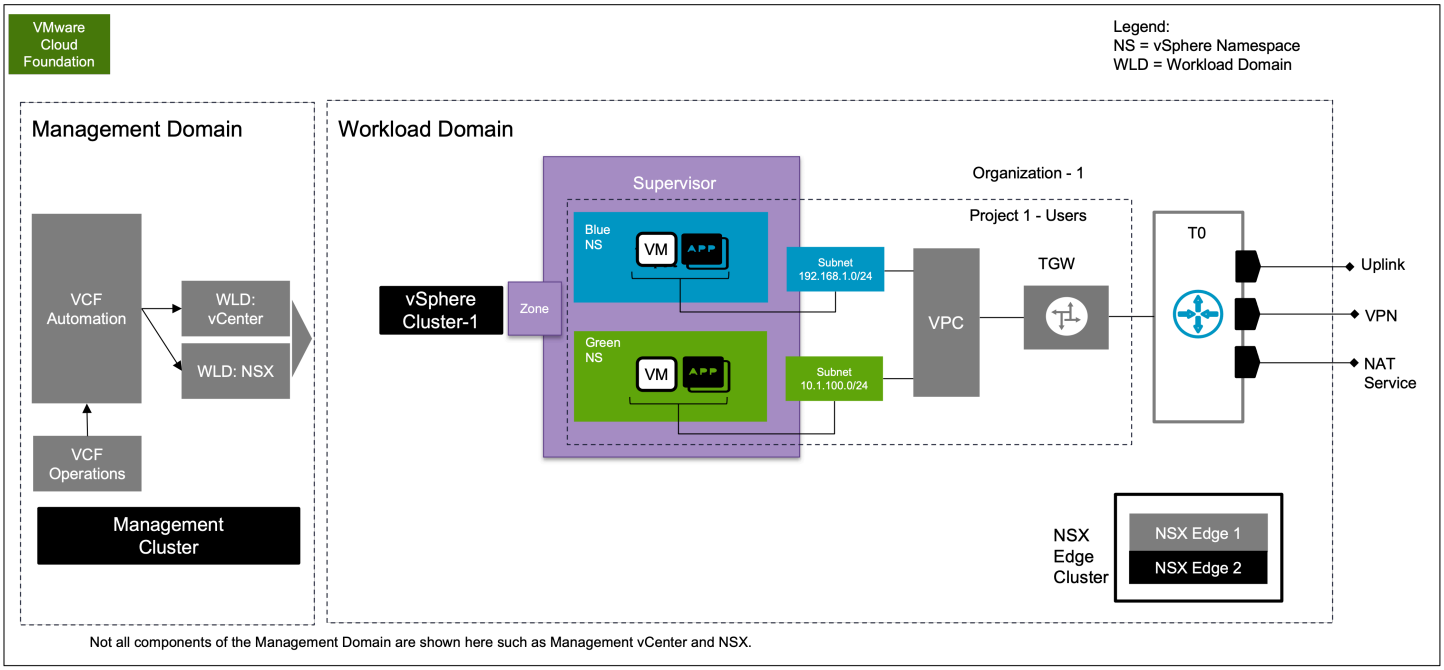
- Multi-Tenancy Consumption Pattern 1: Resource Isolation with Single VPC for Multiple Businesses in One Supervisor Zone
- Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone
- Multi-Tenancy Consumption Pattern 3: Resource Isolation with Single VPC for Multiple Lines of Business in Three Supervisor Zones
- Multi-Tenancy Consumption Pattern 4: Resource Isolation for Multiple Organizations with dedicated VRFs or Transit Gateways (TGW)
- Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways
- Multi-Tenancy Consumption Pattern 6: Multi-Organization resource isolation with shared NSX across workload domains
- Bimodal Consumption Design for VM Apps and All Apps Workloads

Multi-Tenancy Consumption Pattern 1: Resource Isolation with Single VPC for Multiple Businesses in One Supervisor Zone

This design follows a multi-tenancy consumption pattern where resources are shared among multiple tenants, but logical isolation is provided through multiple levels within the private cloud stack.

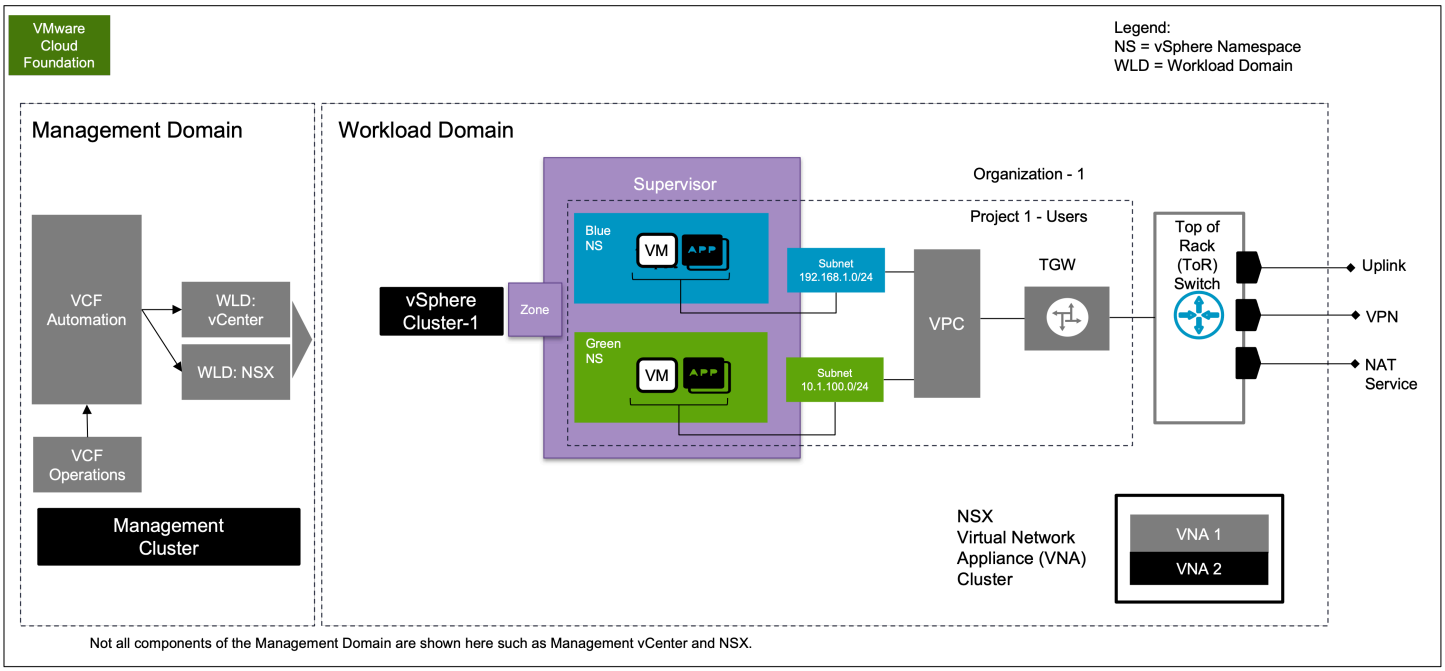
This consumption pattern prioritizes cost optimization and simplified management over strict tenant isolation.

Figure 255: Multi-Tenancy Consumption Pattern 1a: Resource Isolation with Single VPC for Multiple Businesses in One Supervisor Zone - Centralized Gateway (CTGW) Design



Design Attributes	Details	Diagram Representation
Number of vSphere Clusters	One	A compute and vSAN storage cluster used in a workload domain.
Number of vSphere Supervisors	One	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	One	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Project	One	Represented by "Project 1 - Users". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One	The Tier-0 Gateway provides external connectivity. This is where NAT and VPN services are configured.
vSphere Namespaces	Two	Blue and Green line of businesses have dedicated vSphere Namespaces for workload isolation.
VPC	One	All lines of businesses / tenants share a single VPC.
Transit Gateway	One	A single Transit Gateway connects all the VPCs to the Provider Gateway.
Tenant Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Figure 256: Multi Tenancy Consumption Pattern 1b: Resource Isolation with Single VPC for Multiple Businesses in One Supervisor Zone - Distributed Gateway (DTGW) Design

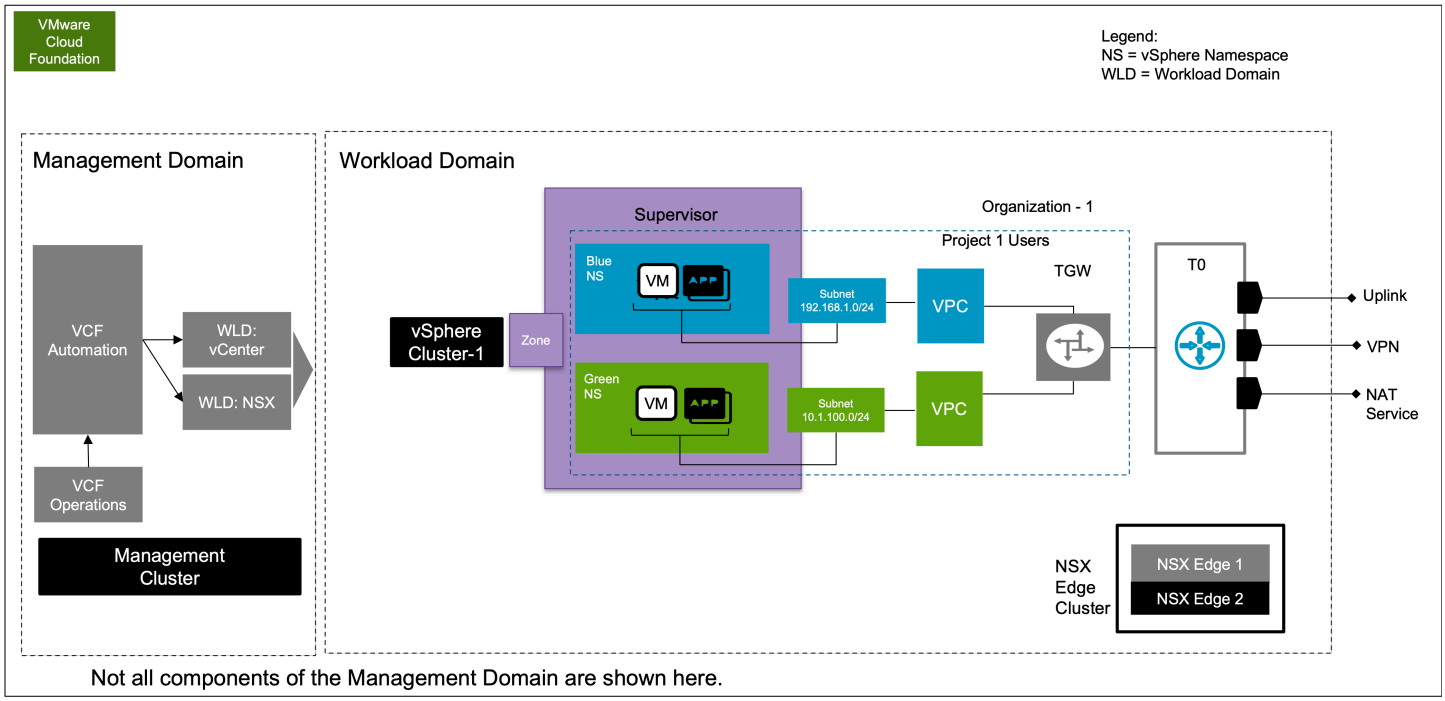


Design Attributes	Details	Diagram Representation
Number of vSphere Clusters	One	A compute and vSAN storage cluster used in a workload domain.
Number of vSphere Supervisors	One	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	One	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Project	One	Represented by "Project 1 - Users". This represents a logical project or user ownership within the environment.
Top of Rack (ToR)	One	Transit Gateways connect directly to the ToR via mapping to a VLAN-backed external IP block mapped to an external connection.
vSphere Namespaces	Two	Blue and Green line of businesses have dedicated vSphere Namespaces for workload isolation.
VPC	One	All lines of businesses / tenants share a single VPC.
Transit Gateway	One	A single Transit Gateway connects all the VPCs to the Provider Gateway.
Tenant Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone

This approach provides robust resource isolation for different lines of business (LOBs) within a single vSphere Supervisor Zone.

Figure 257: Pattern 2: Resource Isolation with multiple VPCs for multiple lines of business in one Supervisor zone - Centralized Gateway (CTGW) Design



Important:

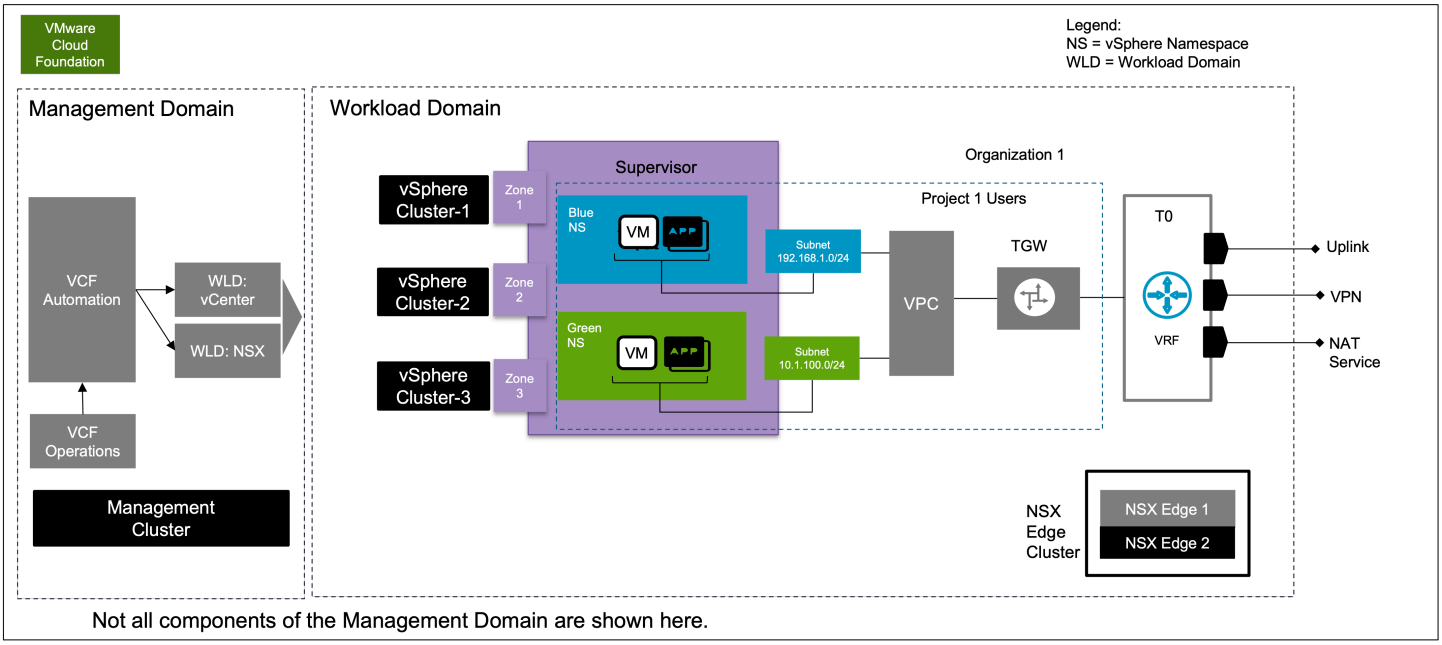
Although the diagram shows the centralized gateway (CTGW) architecture the decentralized gateway (DTGW) architecture is also supported.

Design Attributes	Details	Diagram Representation
Organization	One	Represents as one Organization
Line of Business	Two	Describes Blue and Green as separate LOB
Number of vSphere Clusters	One	A compute and vSAN storage cluster used in a workload domain construct
Number of vSphere Supervisors	One	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	One	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Projects	One	Represented by "Project 1 - Users". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One	The T0 gateway provides external connectivity. This is where NAT and VPN services are configured.
vSphere Namespaces	Two	Blue and Green LOB has dedicated vSphere Namespace for workload isolation.
VPCs	Two	Each LOB/ tenant has their own dedicated VPC for network isolation.
TGWs (Transit Gateway)	One	A single TGW connects all the VPCs to the Provider Gateway.
Tenant Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Multi-Tenancy Consumption Pattern 3: Resource Isolation with Single VPC for Multiple Lines of Business in Three Supervisor Zones

This architecture provides resource isolation for multiple lines of business (LOBs) by consolidating them within a single Virtual Private Cloud (VPC) that spans three vSphere Supervisor Zones.

Figure 258: Pattern 3: Resource Isolation with single VPC for multiple lines of business in three Supervisor zone - Centralized Gateway (CTGW) Design



Important:

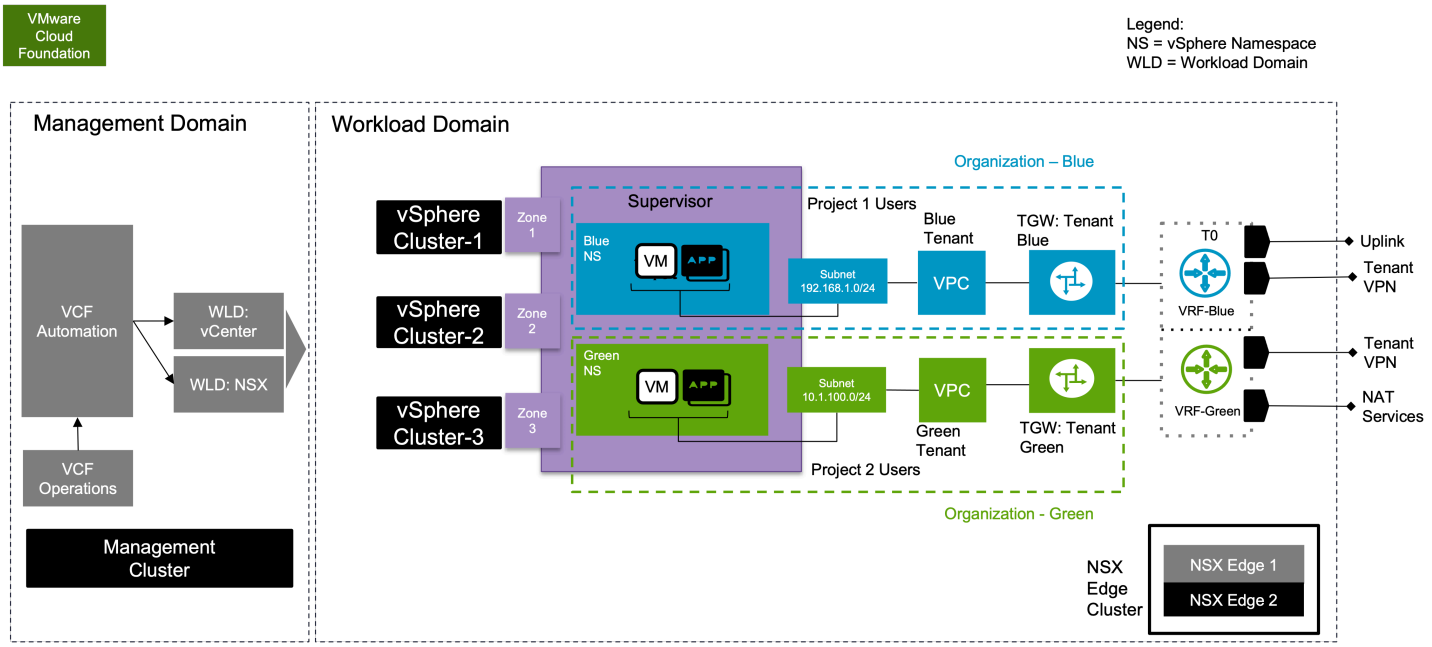
Although the diagram shows the centralized gateway (CTGW) architecture the decentralized gateway (DTGW) architecture is also supported.

Design Attributes	Details	Diagram Representation
Organization	One	Represents as one Organization
Line of Business	Two	Describes Blue and Green as separate LOB
Number of vSphere Clusters	Three	Three compute and vSAN storage cluster used in a workload domain construct
Number of vSphere Supervisors	One	vSphere Supervisor defines the scope of vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	Three	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Projects	One	Represented by "Project 1 - User". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One	The T0 gateway provides external connectivity. This is where NAT and VPN services are configured.
vSphere Namespaces	Two	Blue and Green LOB has dedicated vSphere Namespace for workload isolation.
VPCs	One	All LOB/ tenant sharing a single VPC
TGWs (Transit Gateway)	One	A single TGW connects all the VPCs to the Provider Gateway.
Organization Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Multi-Tenancy Consumption Pattern 4: Resource Isolation for Multiple Organizations with dedicated VRFs or Transit Gateways (TGW)

This design achieves strong resource isolation for multiple organizations by providing each with a dedicated Virtual Routing and Forwarding (VRF) instance, ensuring completely separate network routing and forwarding domains.

Figure 259: Pattern 4a: Resource Isolation for multiple organizations with dedicated per organizations VRFs - Centralized Gateway (CTGW) Design

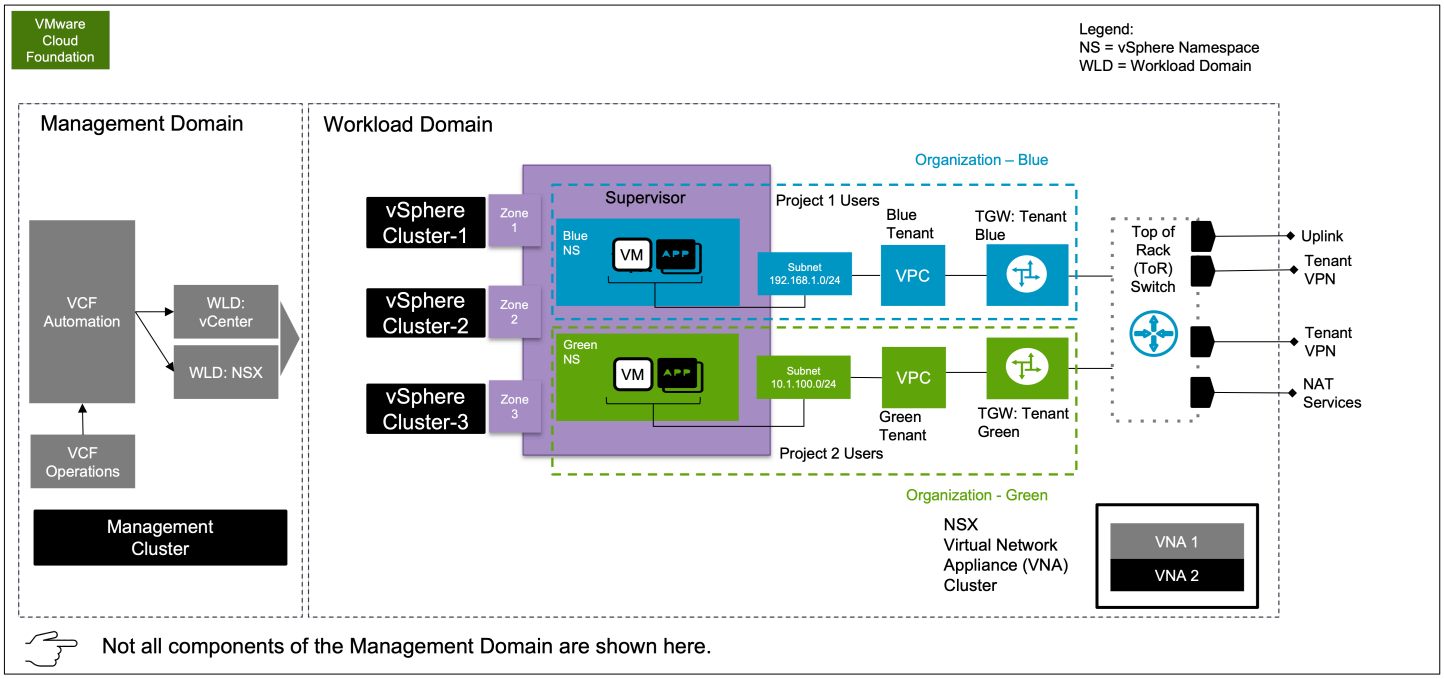


Not all components of the Management Domain are shown here.

Design Attributes	Details	Diagram Representation
NSX Local Manager	One	Dedicated one NSX Local Manager per region
Organization	Two	Represents two separate Organizations. Such as Blue and Green
Number of vSphere Clusters	Three	Three compute and vSAN storage cluster used in a workload domain construct. Shared by both the organizations.
Number of vSphere Supervisors	Three	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	Three	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Project	Two	Represented by "Project 1 - Users" and "Project 2- Users". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One	A single T0 gateway provides external connectivity, including Uplink, VPN, VRF (Virtual Routing and Forwarding), and NAT services.
VRF	Two	Each tenant has a dedicated VRF instance ("VRF: Tenant Blue" and "VRF: Tenant Green") for enhanced network segmentation and isolation. VRF resides in the T0 construct.
vSphere Namespace	Two	Blue and Green tenant has dedicated vSphere Namespace for workload isolation.
VPC	One	All LOB/ tenant sharing one VPC
TGW (Transit Gateway)	Two	Each tenant has a dedicated TGW, "TGW: Tenant Blue" and "TGW: Tenant Green", for connecting their respective VPCs.
Organization Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Figure 260: Pattern 4b: Resource Isolation for multiple organizations with dedicated TGW per tenant - Distributed Gateway (DTGW) Design

Note: In the case of the DTGW architecture, VRFs are not used. The TGW is connected directly to the ToR via mapping to a VLAN-backed external IP block mapped to an external connection.

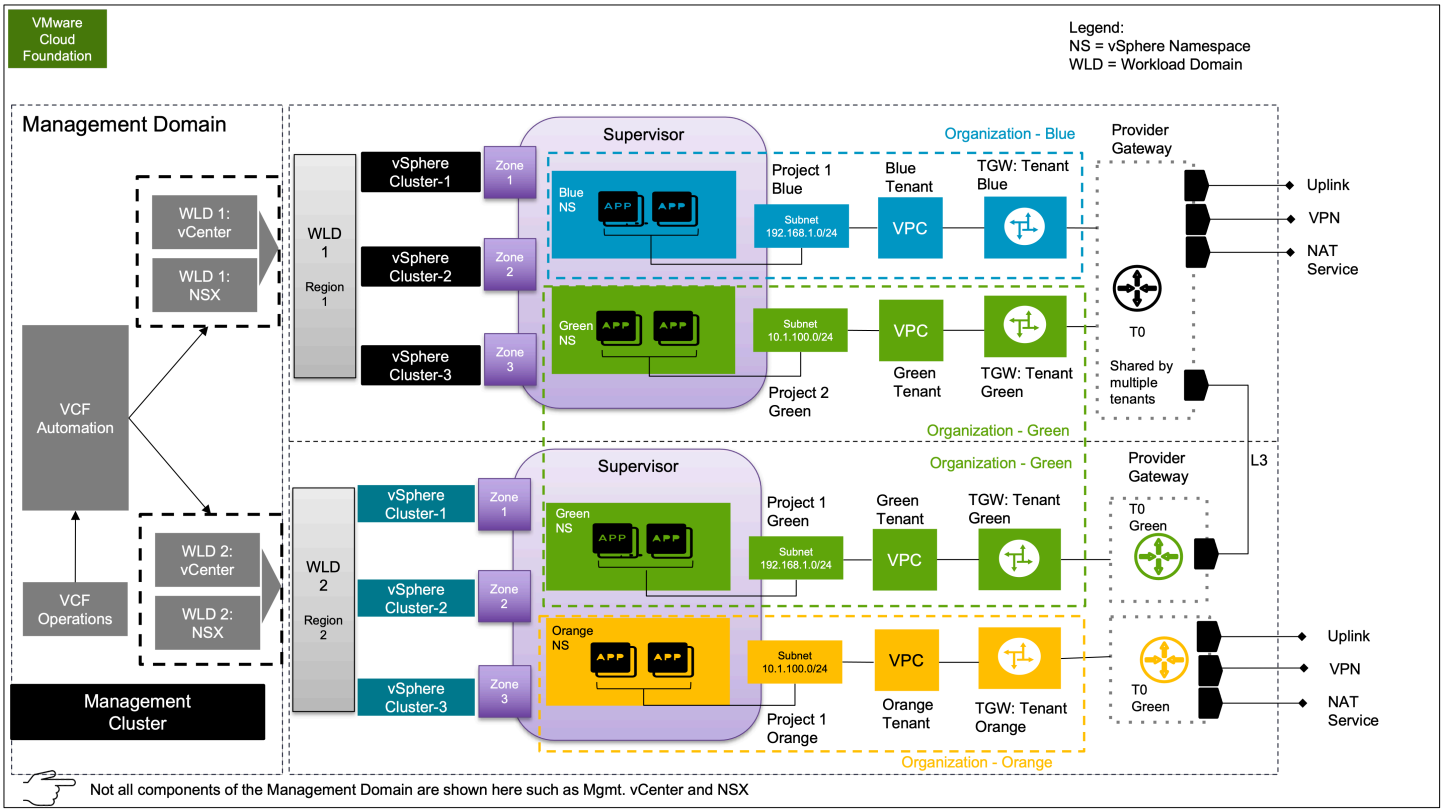

VMware Cloud Foundation
Legend:
NS = vSphere Namespace
WLD = Workload Domain
Management Domain
Workload Domain
VCF Automation
WLD: vCenter
WLD: NSX
VCF Operations
Management Cluster
Organization – Blue
vSphere Cluster-1
vSphere Cluster-2
vSphere Cluster-3
Zone 1
Zone 2
Zone 3
Supervisor
Blue NS
Green NS
VM
APP
Project 1 Users
Project 2 Users
Subnet 192.168.1.0/24
Subnet 10.1.100.0/24
Blue Tenant
Green Tenant
VPC
TGW: Tenant Blue
TGW: Tenant Green
Top of Rack (ToR) Switch
Uplink
Tenant VPN
Tenant VPN
NAT Services
Organization - Green
NSX Virtual Network Appliance (VNA) Cluster
VNA 1
VNA 2
Not all components of the Management Domain are shown here.

Design Attributes	Details	Diagram Representation
NSX Local Manager	One	Dedicated one NSX Local Manager per region
Organization	Two	Represents two separate Organizations. Such as Blue and Green
Number of vSphere Clusters	Three	Three compute and vSAN storage cluster used in a workload domain construct. Shared by both the organizations.
Number of vSphere Supervisors	Three	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	Three	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Project	Two	Represented by "Project 1 - Users" and "Project 2- Users". This represents a logical project or user ownership within the environment.
Top of Rack (ToR)	One	Transit Gateways connect directly to the ToR via mapping to a VLAN-backed external IP block mapped to an external connection.
vSphere Namespace	Two	Blue and Green tenant has dedicated vSphere Namespace for workload isolation.
VPC	One	All LOB/ tenant sharing one VPC
TGW (Transit Gateway)	Two	Each tenant has a dedicated TGW, "TGW: Tenant Blue" and "TGW: Tenant Green", for connecting their respective VPCs.
Organization Networks	Two	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Multi-Tenancy Consumption Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways

This design provides multi-tenant resource isolation across multiple regions by combining dedicated gateways for strict tenant separation with shared gateways for efficient common services.

Figure 261: Pattern 5: Multi-Region, Multi-Tenant Resource Isolation with Dedicated and Shared Gateways - Centralized Gateway (CTGW) Design



Important:

In the context of VCF Automation Region

- A region consists of one or more vSphere Supervisor.
- A region is a collection of compute, memory, storage, and networking resources that can span across vCenter instances.

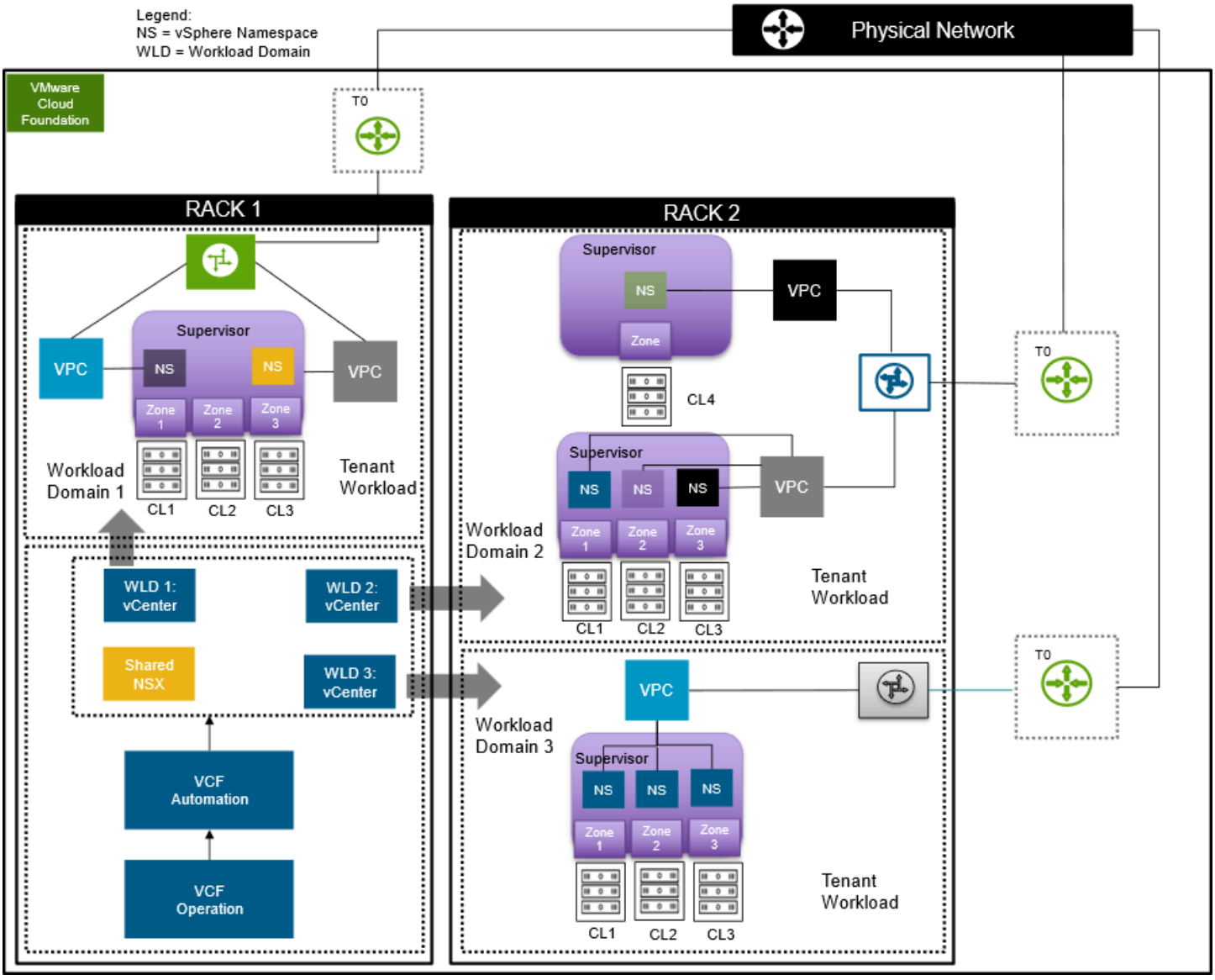
Although the diagram shows the centralized gateway (CTGW) architecture the decentralized gateway (DTGW) architecture is also supported.

Design Attributes	Details	Diagram Representation
Management Domain	One	Centralized Management: Includes VCF Automation, workload domain vCenter, NSX, and VCF Operations. Consist of a vSphere Cluster -the underlying hardware for the management domain.
Workload Domains	Two	A workload domain with three compute vSphere clusters for tenant workloads.
NSX Local Managers	Two	Each workload domain has one NSX local manager
Organizations	Three	Represents three separate Organizations. Such as Blue, Green and Orange
Number of vSphere Clusters	Six	Six compute and vSAN storage cluster used in two workload domain construct.
Number of vSphere Supervisors	Six	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter.
Number of vSphere Supervisor Zones	Six	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability.
VCF Automation Projects	Three	Represented by "Project 1 - User" and "Project 2- User". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One in each region	A single T0 gateway provides external connectivity, including Uplink, VPN, VRF (Virtual Routing and Forwarding), and NAT services.
VRFs	Two	Tenant "Green" and "Orange" has a dedicated VRF instance in a T0. Both tenant sharing a T0 but still they have network isolation.
vSphere Namespaces	Two	Blue and Green tenant has dedicated vSphere Namespace for workload isolation.
VPCs	One	Each LOB/ tenant sharing a single VPC
TGWs (Transit Gateway)	Two	Each tenant has a dedicated TGW, "TGW: Tenant Blue" and "TGW: Tenant Green", for connecting their respective VPCs.
Tenant Networks	Four	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Multi-Tenancy Consumption Pattern 6: Multi-Organization resource isolation with shared NSX across workload domains

This design provides multi-organization resource isolation across distinct workload domains while using a shared NSX platform to centrally manage and logically segment network and security services for each tenant.

Figure 262: Pattern 6: Multi-Organization resource isolation with shared NSX across workload domains - Centralized Gateway (CTGW) Design



Important:

For more information around workload networking, see Workload Connectivity Detailed Design.

Although the diagram shows the centralized gateway (CTGW) architecture the decentralized gateway (DTGW) architecture is also supported.

Design Attributes	Details	Diagram Representation
Management Domain	One	Centralized Management: Includes VCF Automation, workload domain vCenter, NSX, and VCF Operations. Consist of a vSphere Cluster -the underlying hardware for the management domain.
Workload Domains	Three	A workload domain with three compute vSphere clusters for tenant workloads.
Shared NSX Local Managers	One	Workload domains share an NSX Manager deployment.
Organizations	Three	Each workload domain represents as one organization workload.
Number of vSphere Clusters	Ten	Ten compute and vSAN storage clusters used in a three workload domain construct.
Number of vSphere Supervisors	Ten	vSphere Supervisor defines the scope of the vSphere IaaS control plane on vCenter. In this design Workload Domains 1 and 3 each have a 3 zone vSphere Supervisor. Workload Domain has a 3 zone vSphere Supervisor as well as a separate 1 zone vSphere Supervisor for a total of 10 zones.
Number of vSphere Supervisor Zones	Ten	Zones are logical constructs within a vSphere Supervisor that represent failure domains or availability zones. Each zone is mapped to a specific vSphere cluster, ensuring that workloads deployed in a particular zone are physically located on separate hardware for high availability. In this design Workload Domains 1 and 3 each have a 3 zone vSphere Supervisor. Workload Domain has a 3 zone vSphere Supervisor as well as a separate 1 zone vSphere Supervisor for a total of 10 zones.
VCF Automation Projects	Three	Represented by "Project 1 - User" and "Project 2- User". This represents a logical project or user ownership within the environment.
Tier-0 (T0)	One in each region	A single T0 gateway provides external connectivity, including Uplink, VPN, VRF (Virtual Routing and Forwarding), and NAT services.
vSphere Namespaces	Nine	Various vSphere Namespaces across the vSphere Supervisors.
VPCs	Five	Various VPCs assigned across the vSphere Namespaces
TGWs (Transit Gateway)	Three	Each Workload Domain has a dedicated TGW.
Tenant Networks	Multiple	Distinct subnets are used for each vSphere Namespace, providing network segmentation.

Bimodal Consumption Design for VM Apps and All Apps Workloads

VCF Automation with VMware Cloud Foundation 9.1 introduces a key capability that enables a **bimodal consumption model** for VM App and All App organizations. This enhancement allows both organizations to **share a common vCenter Server and NSX instance**, improving resource utilization and reducing operational overhead for selected deployment topologies.

> **Important:**
>
> **Design Recommendation:** Although this capability enables shared consumption between VM App and All App organizations within a common VMware vCenter and NSX environment, the primary intent of this enhancement is to provide an easier transition path for existing VMware Aria Automation customers moving to VCF Automation. This allows organizations to continue supporting traditional VM-based consumption models while progressively adopting the modern All App consumption model side by side. For net-new greenfield environments without existing Aria Automation operational dependencies, the recommended approach is to adopt the VCF Automation All App consumption model from the start to align with the long-term cloud operating model and platform architecture direction.

In this model, an **Organization for All Apps and an Organization for VM Apps operate on shared infrastructure** while remaining logically isolated through policy, networking, and access controls. This enables traditional VM-based workloads and modern application platforms to coexist efficiently within the same environment.

Prior to VCF 9.1, each organization typically required a **dedicated workload domain**, including its own vCenter and NSX instances, leading to higher infrastructure footprint. VCF Automation 9.1 addresses this by enabling a shared infrastructure approach, which is activated through the feature flag configuration setting: **“Mixed Tenancy Mode”**

This capability provides greater flexibility in how infrastructure is consumed, supporting both **consolidated (shared) and isolated (dedicated)** deployment models based on workload and organizational requirements.

1. **Shared Cluster Deployment:** Both All App and VM App organizations can deploy their respective workloads and services onto the *same vSphere Cluster* that is managed by the single vCenter.
2. **Isolated Cluster Deployment:** Alternatively, the organizations can deploy to *different vSphere clusters* (e.g., Cluster A for VM App, Cluster B for All App) that are still managed by the same shared vCenter.
3. **VM App workload deployment across VCF Fleets -** VM App-based workloads can be deployed and managed across multiple VCF Fleet or workload domains, enabling centralized governance with distributed execution for scale and locality.

The unified management plane allows the organizations to effectively utilize shared physical and virtual resources (e.g., compute, storage, networking) while streamlining operational tasks such as monitoring and patching that need only be performed on the single shared vCenter and NSX Local Manager pair.

A shared deployment serves as an ideal bridge for organizations looking to strategically transition from an existing Aria Automation environment to the modern, expanded capabilities of VCF Automation Organization for All Apps. Rather than executing immediate migration, the organization can:

- **Co-exist:** Run existing workloads and consumption models in the VCF VM Apps organization.
- **Pilot:** Gradually introduce new services, modern application development teams, or specific workloads into the VCF Automation All Apps organization.
- **Migrate:** Systematically move eligible workloads and users from the Organization for VM Apps to the Organization for All Apps over time, ensuring business continuity and allowing teams to adjust to the new platform capabilities at a manageable pace.

This phased approach minimizes risk, maximizes return on existing investments, and ensures a smoother, more controlled adoption of the full VCF Automation with VMware Cloud Foundation.

> **Attention:**
>
> Whenever the Organization types share a vCenter and NSX Local Manager pair, the Organization for VM Apps must be created before the Organization for All Apps. The Organization for VM Apps must be configured (including mapping the vCenter and NSX Manager cloud accounts) before a VCF Automation Region is configured using the same vCenter and NSX Manager).

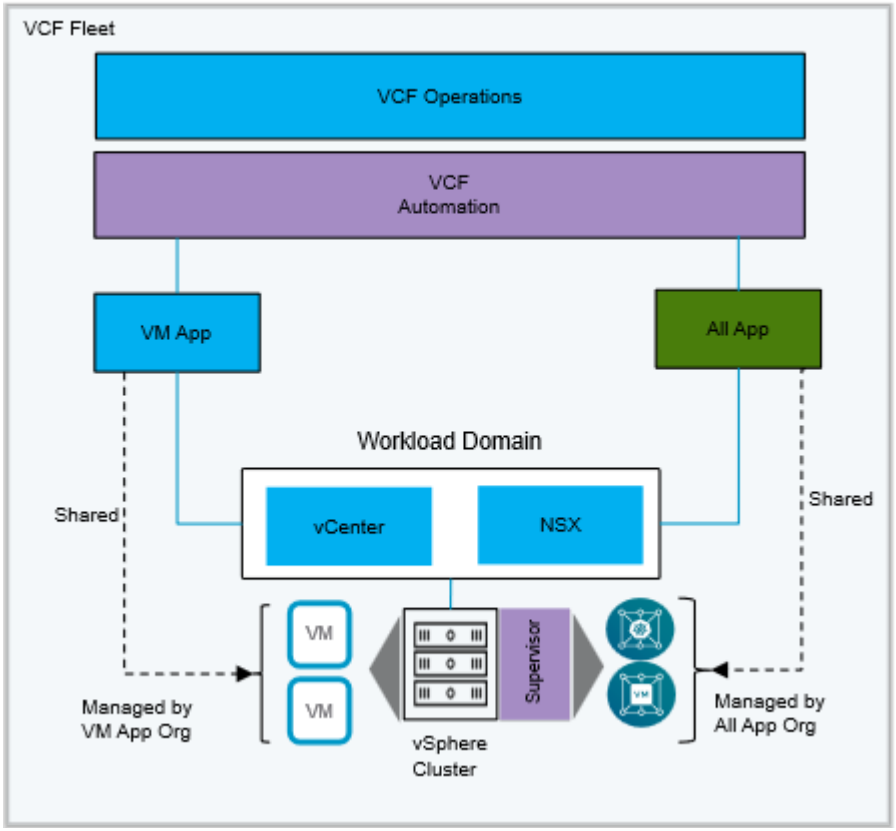
During the upgrade from a legacy Aria Automation to VCF Automation, the legacy environment is converted to an Organization for VM Apps. The converted Organization for VM Apps will be the first Organization deployed in the new VCF Automation instance.

Design Choices and Justification for Single Cluster vs Multiple Cluster within the same Workload Domain

Design Choice	Description	Pros	Cons	When to Consider
Sharing workload in a Single vSphere Cluster (Design #1)	A unified cluster houses all functional groups (e.g., VM App, All App org) and workloads.	**Lower Initial Hardware Cost**	**Resource Contention:** High potential for competition between diverse functional groups, leading to performance instability and "noisy neighbor" issues.	**Small/Pilot Environments:** Best suited for small deployments with homogenous workloads, low SLA requirements, and minimal risk of resource competition.
Workloads in separate vSphere Clusters (Design #2)	Creating distinct, dedicated clusters for VM App Org and All App Org workloads.	**Performance Isolation:** Provides true resource isolation, mitigating the "noisy neighbor" problem and ensuring stable performance for critical SLAs.	**Higher Initial Hardware Cost**	**Enterprise Environments:** Mandatory for large deployments with diverse, mission-critical workloads, stringent SLA requirements, and a need for strong performance isolation between functional groups.

Figure 263: Design #1 VM App and All App Deployments on a Single Cluster Within a Shared vCenter

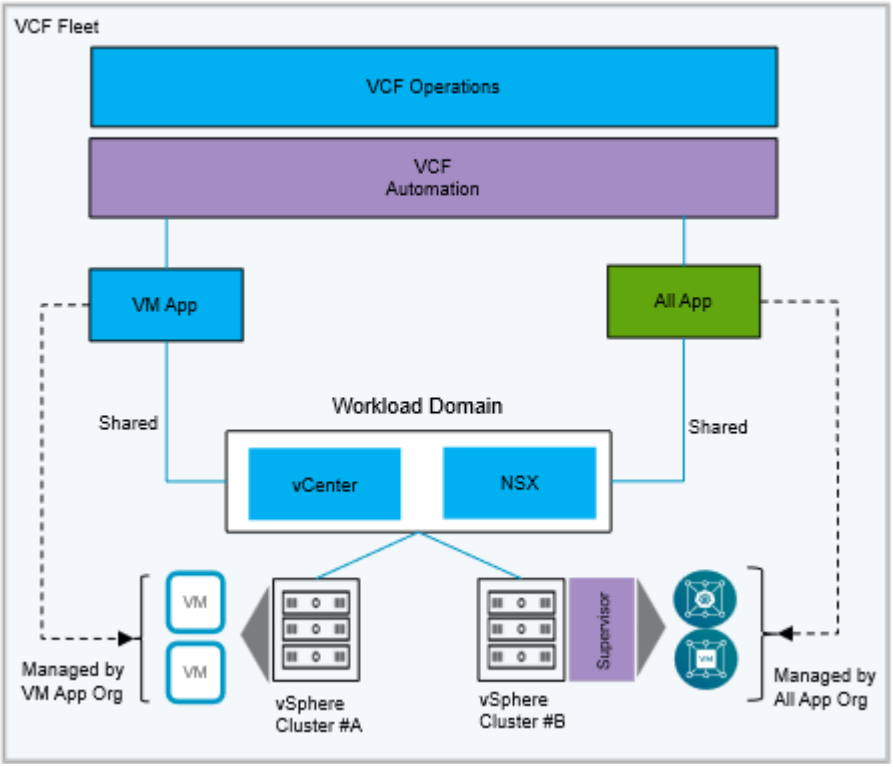
VM App and All App Deployments on a Single Cluster Within a Shared vCenter



This design enables both VM-based workloads (VM App) and modern platform services (All App) to be deployed on a shared vSphere cluster managed by a single vCenter, optimizing resource utilization and simplifying infrastructure operations. By consolidating workloads onto a common cluster, organizations benefit from reduced management overhead, consistent policy enforcement, and improved capacity efficiency. This model is well suited for environments prioritizing operational simplicity and cost efficiency, while still supporting a diverse mix of traditional and cloud-native workloads.

Figure 264: Design #2 VM App and All App Deployments on Separate Clusters Within a Shared vCenter

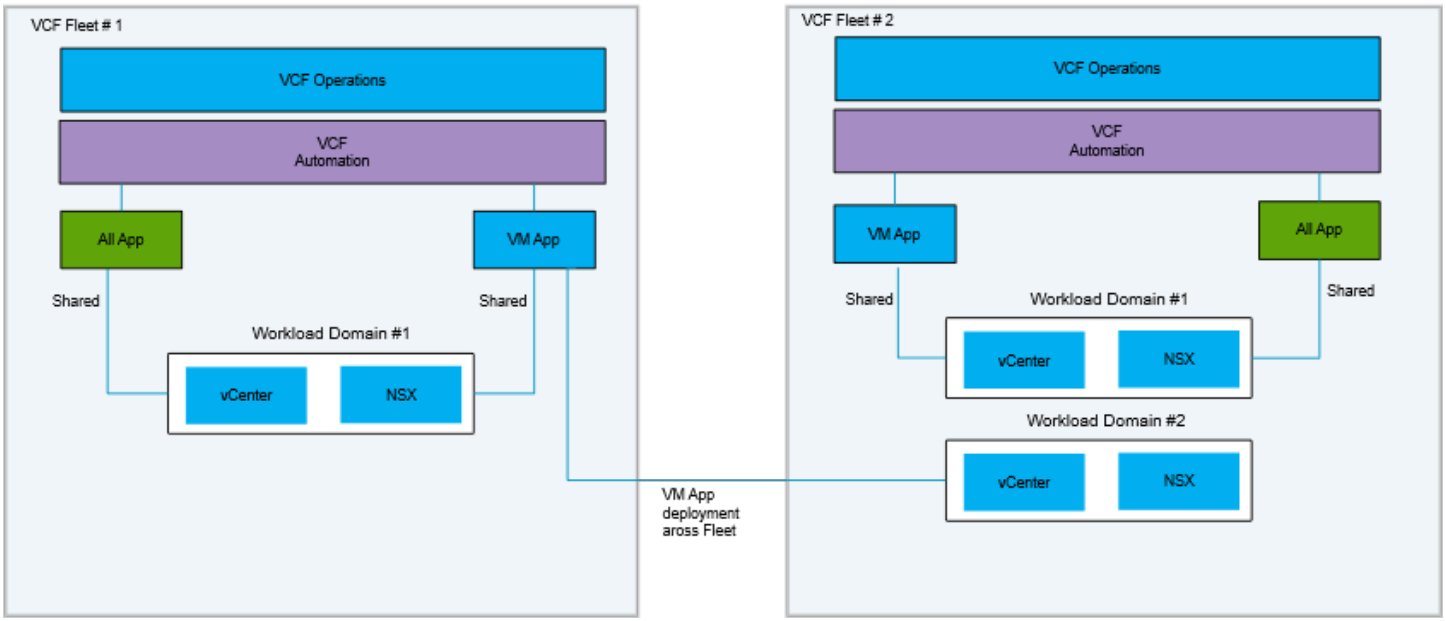
VM App and All App Deployments on Separate Clusters Within a Shared vCenter.



This design separates VM-based applications (VM App) and modern platform services (All App) into dedicated vSphere clusters while still operating under a single shared vCenter. By isolating workloads at the cluster level, organizations gain stronger performance boundaries, improved resource predictability, and clearer operational ownership for different workload types. This approach is particularly valuable for environments with differing SLAs, security requirements, or lifecycle needs between traditional and cloud-native applications. At the same time, the shared vCenter maintains centralized management, consistent governance, and simplified operations across clusters, enabling a balance between isolation and operational efficiency.

Figure 265: Design #3 VM App Workload Deployment Across Multi-Fleet VCF Environments

VM App Workload Deployment Across Multi-Fleet VCF Environments



This design enables VM-based applications (VM Apps) to be deployed and managed across multiple VCF Fleet or workload domains (in a multi fleet environment), providing flexibility to scale workloads beyond a single environment. It supports centralized governance and consistent policy enforcement while allowing workloads to run closer to users, data sources, or specific regions. This approach is well suited for organizations operating in multi-region or distributed environments, where workload portability, resiliency, and geographic alignment are critical.

Important:

The multi-fleet workload design **only supports** the Organization for VM Apps consumption model. It **does not work** with the Organization for All Apps model.

Brownfield Scenarios for Aria Automation

Figure 266: Transitional Architecture: VLAN Backed workload with Aria Automation to VCF Automation Consumption

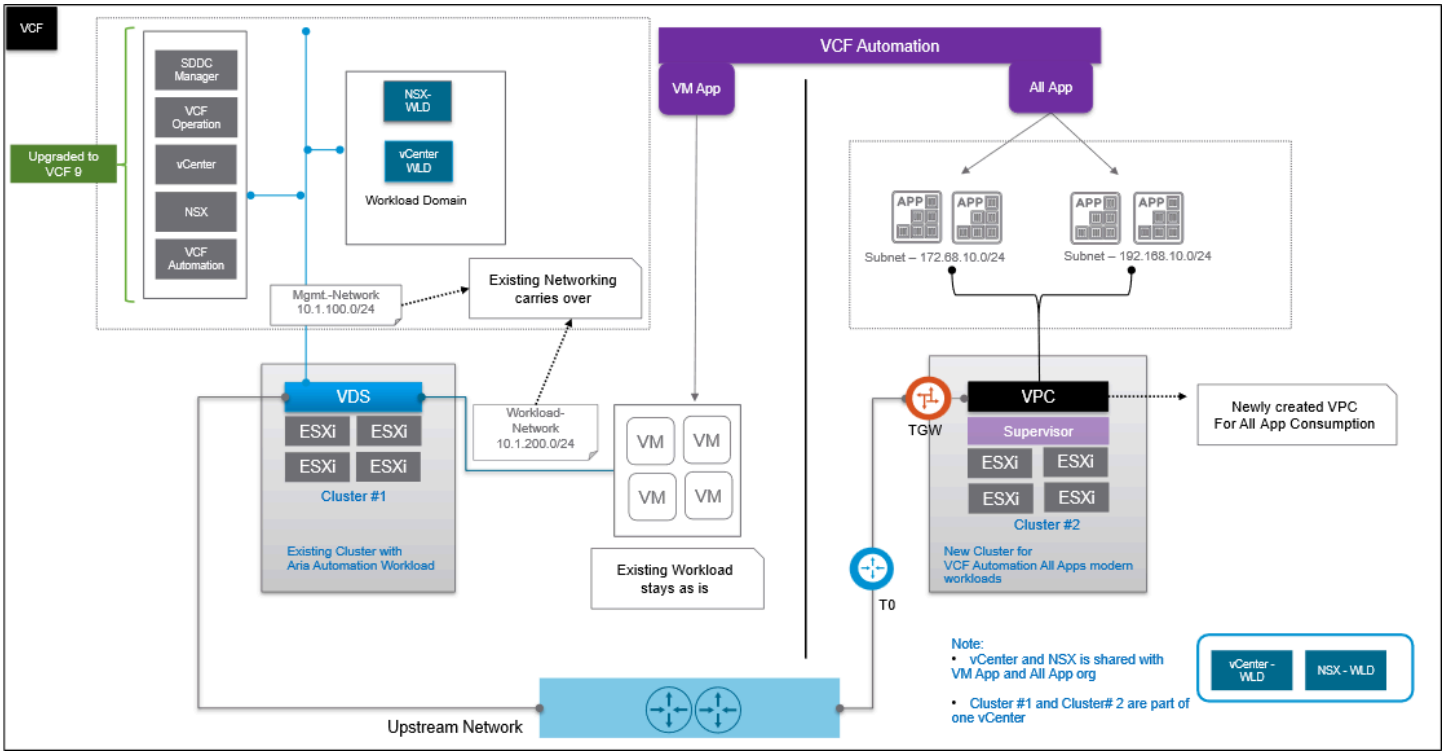
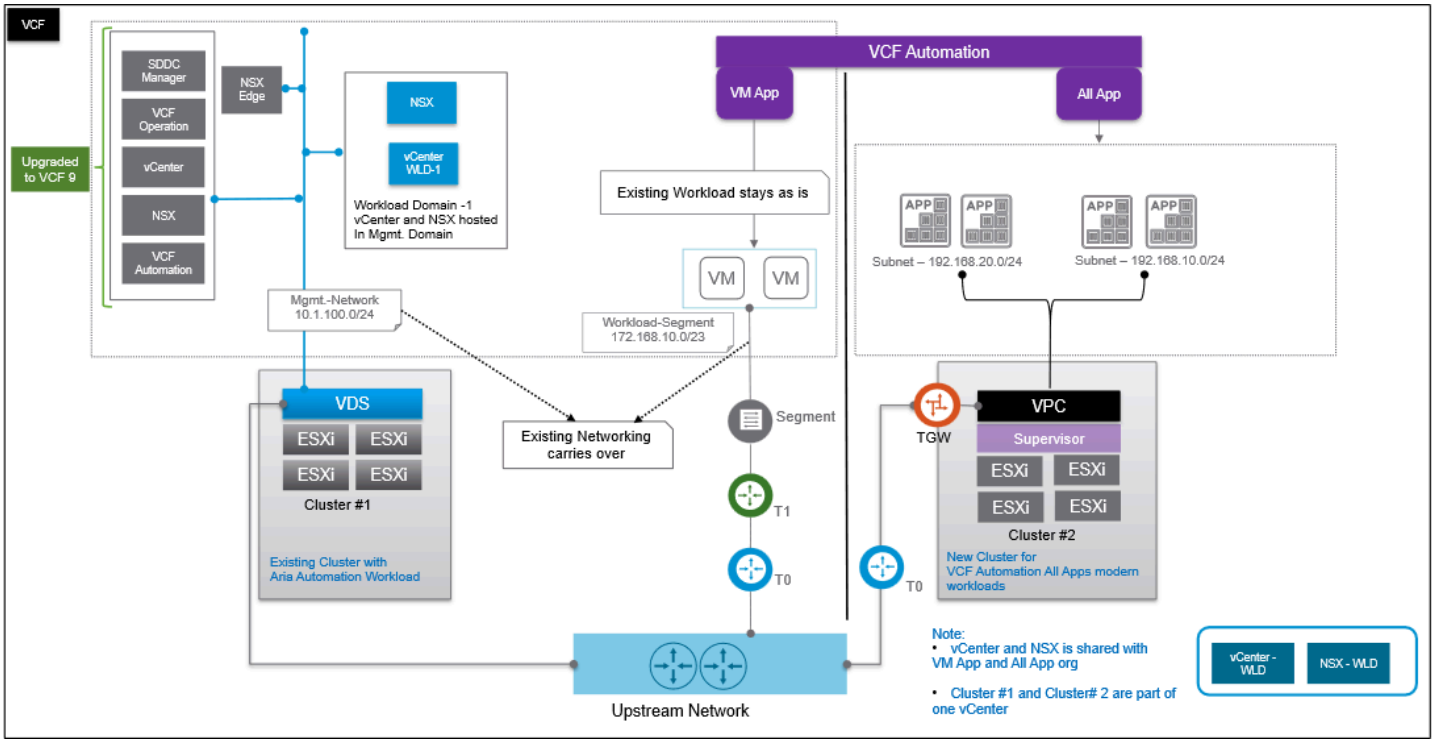


Figure 267: Transitional Architecture: NSX- backed workload with Aria Automation to VCF Automation



Design Context

This architecture illustrates a transitional adoption model for organizations migrating from existing VMware Aria Automation environments to VMware Cloud Foundation with VCF Automation while preserving existing workload operations and enabling gradual adoption of modern application consumption models.

In these scenarios, organizations may already operate existing Aria Automation environments using either:

- traditional VLAN-backed networking through a VDS-based architecture
- or existing NSX-backed networking and segment-based workload connectivity models

Following the transition to VCF Automation:

- existing workloads continue operating on their current networking and infrastructure constructs
- existing VLAN-backed or NSX-backed networking configurations remain unchanged
- existing workloads continue to be consumed through the **VM App organization model**
- operational disruption to existing workloads, routing models, firewall policies, and application connectivity is minimized

This design enables organizations to preserve existing operational and networking models while incrementally introducing modern platform consumption capabilities through the VCF Automation All App model.

Transitional Bimodal Consumption Model

This architecture demonstrates how organizations can operate both:

- traditional VM-centric consumption models
- modern application-centric consumption models

Using the VCF Automation bimodal consumption model, organizations can simultaneously support:

Consumption Model	Purpose
VM App Organization	Supports traditional VM-based workloads operating on existing VLAN-backed or NSX-backed networking models
All App Organization	Supports modern platform consumption models for services such as VM Service, **VMware vSphere Kubernetes Service**, Database Services Manager (DSM), and modern application workloads using VPC-based networking

This allows organizations to modernize incrementally without requiring immediate migration of existing workloads into new networking or operational constructs.

Network Architecture Transition Model

A key focus of this architecture is demonstrating how networking evolves during the transition from traditional infrastructure consumption models toward modern VPC-based application consumption models.

Existing VM App Workloads

Existing workloads continue operating on:

- existing VDS-backed VLAN networking architectures
- or existing NSX segment-based networking architectures
- existing Layer 2 and Layer 3 routing constructs
- existing workload IP addressing and operational models

This allows organizations to preserve:

- existing firewall and security policies
- routing dependencies
- VLAN or NSX segment operational models
- existing operational tooling and processes

New All App Workloads

Modern workloads introduced through the All App consumption model operate using:

- VPC-based networking
- Supervisor-enabled infrastructure
- modern application platform constructs
- tenant-aware networking and workload isolation models

This requires enabling a **vSphere Supervisor** cluster either:

- on the existing cluster (if sufficient capacity exists)
- or on a newly deployed cluster operating alongside existing workloads within the same vCenter environment

The architecture also demonstrates how modern VPC-based networking environments can coexist alongside existing VLAN-backed or NSX-backed networking environments during the transition period.

Shared Infrastructure Model

Using the VCF Automation bimodal architecture introduced in **VCF 9.1**, both VM App and All App consumption models can share:

- the same **VMware vCenter**
- the same **NSX** environment

This enables organizations to modernize application consumption models without requiring completely separate workload domains for traditional and modern workloads.

Key Design Benefits

Benefit	Description
Incremental Modernization	Allows gradual adoption of modern VCF Automation consumption models
Existing Network Preservation	Existing VLAN-backed or NSX based networking remains operational
Reduced Migration Risk	Existing workloads remain unchanged during the transition
Bimodal Operations	Traditional and modern application models coexist simultaneously
Shared Infrastructure Efficiency	Shared vCenter and NSX infrastructure reduces operational overhead
Flexible Adoption Path	Organizations can modernize workloads progressively over time
Operational Continuity	Existing VM workloads and operational tooling remain functional

Best-Fit Use Cases

Use Case	Why This Model Fits
Existing Aria Automation customers upgrading to VCF 9	Preserves existing operational and networking models
Organizations with large workload estates with VLAN-backed or NSX-backed network	Avoids immediate network rearchitecture
Enterprises adopting modern platform services gradually	Supports phased modernization
Mixed traditional and modern workload environments	Enables coexistence of VM and cloud-native platforms
Organizations reducing migration disruption risk	Allows controlled transition toward All App consumption models

Design Recommendation

This architecture is best suited for organizations seeking a phased modernization strategy where existing VM-centric operational models must remain operational while modern application platforms and VPC-based consumption models are introduced progressively within the same VCF Automation environment.

VCF Automation Identity and Access Management Design

As part of the VCF Automation, you configure authentication to grant user access and provide granular control of resources.

VCF Automation contains two persona levels. The provider persona manages the core infrastructure and the tenant / line-of-business (LOB) persona manages the individual tenants within that infrastructure. Each of these personas has different authentication options.

VCF Automation Personas	Capabilities
Provider	• Installs and manages core VCF Automation infrastructure. • Provider / Enterprise-level administrative functions. • Manages shared content libraries published to organizations. • Creates organizations, assigns resource quotas, regional networking, and first tenant user.
Tenant / Line of Business	• Manages organizations. • Creates and manages projects. • Creates and manages vSphere Namespaces. • Configures Content Hub. • Creates blueprints. • Provides organization-level user management.

Provider Persona Attributes

The provider level supports three primary identity providers along with the default local **admin** account.

Attribute	Details
VCF Single Sign-On	• Embedded or instance identity broker. • Provides consistent Single Sign-On across the VMware Cloud Foundation platform. • Identity broker supports multiple different simultaneous LDAP and / or Active Directory connections.
LDAP	Allows for the configuration of a single Active Directory or OpenLDAP connection.
SAML	Assumes an upstream SAML identity provider.
Default Local Admin Account	• Local admin account is created during the initial installation of VCF Automation. • Full **admin** rights. • Only one local **admin** account allowed.

Table 532: VCF Automation Provider Authentication Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-PROV-001	Configure an identity provider for VCF Automation Provider Management.	• Allows a cloud provider to use an enterprise directory. • Enables named access control to Infrastructure level constructs. • Required to provision administrative access to the Provider Administrator Portal for any account other than the default local admin account.	If not configured, the only access to the VCF Automation Provider Managementportal will be by using the default local admin account created during installation.
VCF-AUTO-REQD-PROV-002	Use the same identity provider across all VCF fleet components, such as identity broker.	• Provides Single Sign-On with the rest of the VMware Cloud Foundation components. • Applies the authentication direction for VMware allowing support for future enhancements.	None.

Tenant Persona Attributes

The tenant level accepts three primary identity providers along with the default local admin account.

Attribute	Details
OpenID Connect (OIDC)	Allows for configuration of a OIDC identity provider.
LDAP	Allows for the configuration of a single Active Directory or OpenLDAP connection.
SAML	Assumes an upstream SAML identity provider.
First user	• Created during initial deployment of a tenant. • Organizational **administrator** role. • Is a local account.

Table 533: VCF Automation Tenant Authentication Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-TEN-001	Configure an identity provider for VCF Automation organizations.	• Allows tenants to use enterprise directory. • Enables named access control to tenant level constructs. • Required to provision administrative or user access to the Organization portal for any account other than the first user account established during Organization creation.	If not configured, the only access to the Organization Portal will be through the first user account established during Organization creation.

VCF Automation Authentication Models

Since VCF Automation deployments consist of a Provider and a Tenant persona. This section describes the design requirements, recommendations, and choices for each VCF Automation authentication model.

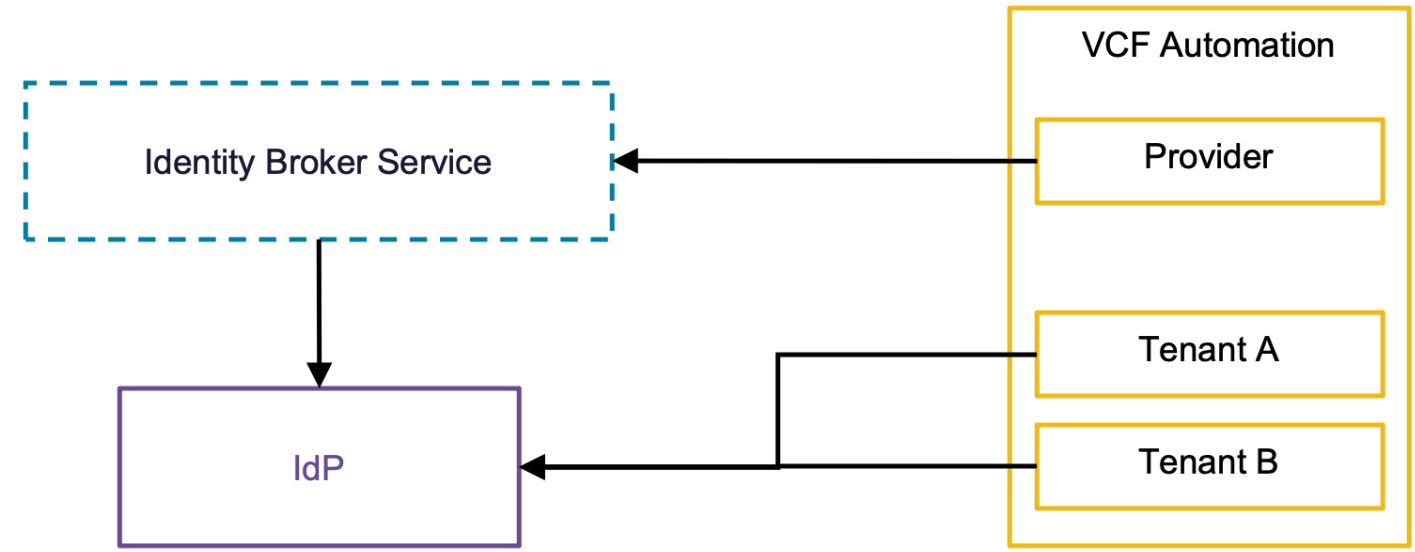
Table 534: Provider / Tenant Models

Attributes	Details
Enterprise Model	In this model, both the provider and tenant authenticate to the same identity provider.
Service Provider Model	In this model, the provider and tenant authenticate to different identity providers.

VCF Automation Enterprise Authentication Design

The *VCF Automation Enterprise Authentication Model* configures both the provider and tenant to authenticate to the same identity provider.

Figure 268: VCF Automation Enterprise Authentication Model in VMware Cloud Foundation



VCF Automation Enterprise Authentication Design Attributes

A *VCF Automation Enterprise Authentication Model* has the following attributes.

Attribute	Detail
Provider	• The provider organization connects to the identity broker service. • Identity broker sources the users and groups from the identity provider. • The provider uses the same identity provider as the tenant.
Tenant	• The tenant organization connects directly to the identity provider through either OIDC, LDAP, or SAML. • The tenant uses the same identity provider as the provider.
Single Sign-On scope	• The provider log-ins have Single Sign-On with the rest of VCF components. • The tenant log-ins do not have Single Sign-On with the rest of VCF components.

VCF Automation Enterprise Authentication Model Options

A *VCF Automation Enterprise Authentication Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Identity Broker Models	• Embedded Identity Broker Model • Instance Identity Broker Model

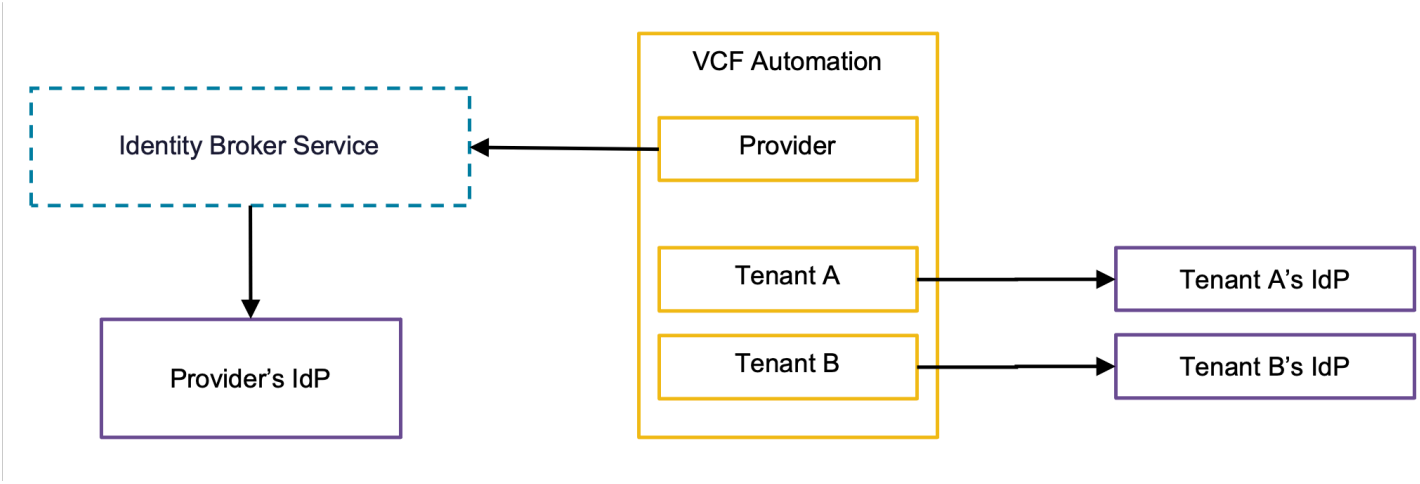
Table 535: VCF Automation Enterprise Authentication Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-IDP-001	The provider uses the same IdP as the tenant.	• Allows consistent authentication across provider and tenant	None

VCF Automation Service Provider Authentication Design

The *VCF Automation Service Provider Authentication Model* configures the provider and tenant to authenticate to different identity providers.

Figure 269: VCF Automation Service Provider Authentication Model in VMware Cloud Foundation



VCF Automation Service Provider Authentication Design Attributes

Attribute	Detail
Provider	• The provider organization connects to the identity broker service. • The identity broker sources the users and group from the identity provider. • The provider uses a different identity provider than the tenant.
Tenant	• The tenant organization connects directly to the identity provider through OIDC, LDAP, or SAML. • The tenant uses a different identity provider than the provider.
Single Sign-On scope	• The provider log-ins have Single Sign-On with the rest of VCF components. • The tenant log-ins do not have Single Sign-On with the rest of VCF components.

VCF Automation Service Provider Authentication Model Options

A *VCF Automation Service Provider Authentication Model*can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Identity Broker Models	• Embedded Identity Broker Model • Instance Identity Broker Model

VCF Automation Service Provider Authentication Design Requirements

Table 536: VCF Automation Service Provider Authentication Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-AUTO-REQD-IDP-002	The provider uses a different IdP than the tenant.	• Supports separate authentication mechanisms for provider and tenant.	None

VCF Automation Role Base Access Control

Managing multi-tenant cloud environments demands a clearly established role-based access control (RBAC) strategy to guarantee restricted access, tenant separation, and secure operations.

Table 537: Best Practices for Efficient Role-Based Access Control

Best Practice	Recommendation
Principle of Least Privilege	Ensure users are granted only the permissions absolutely necessary for their specific roles, minimizing security risks.
Regular Review & Updates	Continuously review and update RBAC assignments to ensure they align with any changes in user roles and responsibilities.
Logging & Auditing	Implement logging and auditing to track user access and actions, providing transparency and accountability.
User Education	Provide ongoing training on the importance of role-based access and secure credential handling to prevent security breaches.
Integrate Third-Party Solutions for RBAC Optimization	Consider using external tools for more streamlined RBAC management, detailed reporting, and enhanced control.

VCF provides a platform to create and manage secure and multi-tenant environments. Implementing role-based access control for the management components offers the following benefits.

Table 538: Role-Based Access Control Design for Service Providers

Attribute	Detail
Separate Administrator roles	Establish distinct roles for service provider admins, such as managing infrastructure, provisioning resources, and troubleshooting tenant issues.
Provider Admin levels	To control access to sensitive areas, differentiate roles for basic resource management, advanced configuration, and network administration.
Multi-Factor Authentication for all privileged accounts	Implement Multi-Factor Authentication (MFA) for all privileged accounts, including both tenant and service provider administrators.

Table 539: Role-Based Access Control Design for Tenants

Attribute	Detail
Predefined Tenant Roles	Create predefined roles such as **tenant user**, **tenant operator**, and **tenant administrator** with permissions based on specific needs.
Custom Roles for Granularity	Implement custom roles for specific workflows, such as a **deployment engineer**, with permissions to deploy templates without managing infrastructure.
Least Privilege	Apply the principle of least privilege to grant only the minimum permissions needed for each role to perform their tasks effectively.

VCF Automation Default Provider Level Global Roles

These are the default system-level roles found in the Provider Admin Portal.

- They are automatically published to all Organizations
- Custom roles can be created at Provider Admin level and then published to one, many, or all Organizations as necessary

Table 540: VCF Automation Default Provider Level Global Roles

Role	Description	Published to other Organizations
Defer to Identity Provider	Rights will be determined based on information received from IdP	Yes, Default - All Organizations
Organization Administrator	Owns an Organization and can add users/ groups to the organization	Yes, Default - All Organizations
Organization Auditor	Audit an Organization	Yes, Default - All Organizations
Organization User	Rights given to an organization user	Yes, Default - All Organizations
Custom role	Use "new" role button in the Access Control screen to create a custom role as necessary	Can be granularly published to one, many or all Organizations as necessary

VCF Automation Default Organization Level Roles

These are the default Organization level roles.

- Organization level roles govern who can log into the Organizational level portal and with what rights
- Organizations can also inherit custom roles created and subsequently published from the Provider Admin level
- Organizations Administrators can also create custom roles at the Organization level

Table 541: VCF Automation Default Organization Level Roles

Role	Description	Considerations
Defer to Identity Provider	Rights will be determined based on information received from IdP	• Inherited from Provider Level Global Roles • Cannot be edited at Organization level
Organization Administrator	Owns an Organization and can add users/ groups to the organization	• Inherited from Provider Level Global Roles • Cannot be edited at Organization level
Organization Auditor	Audit an Organization	• Inherited from Provider Level Global Roles • Cannot be edited at Organization level
Organization User	Rights given to an organization user	• Inherited from Provider Level Global Roles • Cannot be edited at Organization level
Custom role	Use "new" role button in the Access Control screen to create a custom role as necessary	• Created specifically for the Organization • Cannot be published to other Organizations

VCF Automation Default Organization Project Level Roles

Even if a user has access to the Organization portal, they still need to be assigned specific rights at the Project level to access and manage resources.

- Once a Project is created, the administrator must assign users to a specific role
- The role assigned to the user applies to all Namespaces created within the Project

Table 542: VCF Automation Default Organization Project Level Roles

Role	Description
Project Administrator	Can manage members and catalog content
Project Advanced User	Can request catalog items and use IaaS services
Project User	Can request catalog items
Project Auditor	Can view all project content but cannot request

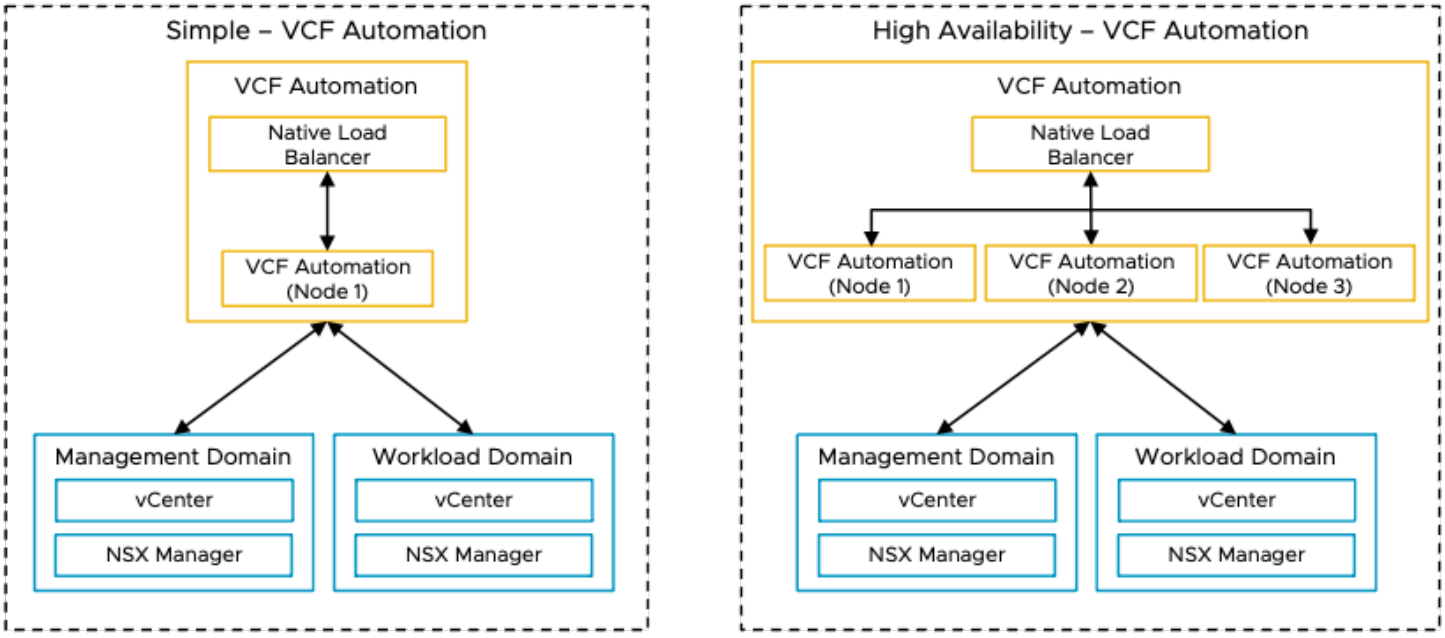
VCF Automation Load-Balancing Design

VCF Automation runs within a Kubernetes-based container environment. The native load balancer is the default automatically configured load balancer for both the simple and highly available deployment models.

Important:

Note: This section does not cover load balancing for subsidiary functions, such as:

- Workload load balancing
- Load balancing of single sign-on solutions such as VMware Identity Broker
- vSphere Supervisors



If VCF Automation must be deployed in a manner requiring additional security and/or the use of an external load balancer, refer to:

- Deployment Pattern 3: External-Facing Deployment of VCF Automation section of the VCF Automation Detailed Design.
- Securing VCF Automation Deployment Pattern 3 with vDefend and Avi design blueprint.

Default Load Balancer	Additional Load Balancer Options
Native Load Balancer	Deploy VCF Automation with external, 3rd-party load balancer integration (such as, VMware Avi Load Balancer)

VCF Operations Orchestrator Design

VCF Operations orchestrator is a workflow automation solution that is designed to simplify the automation of complex IT tasks.

VCF Operations orchestrator helps improve service delivery efficiency, operational management, and IT agility. VCF Operations orchestrator is built with an open and flexible architecture that system administrators and IT operations staff can use to streamline tasks and integrate functions with third-party software through workflows.

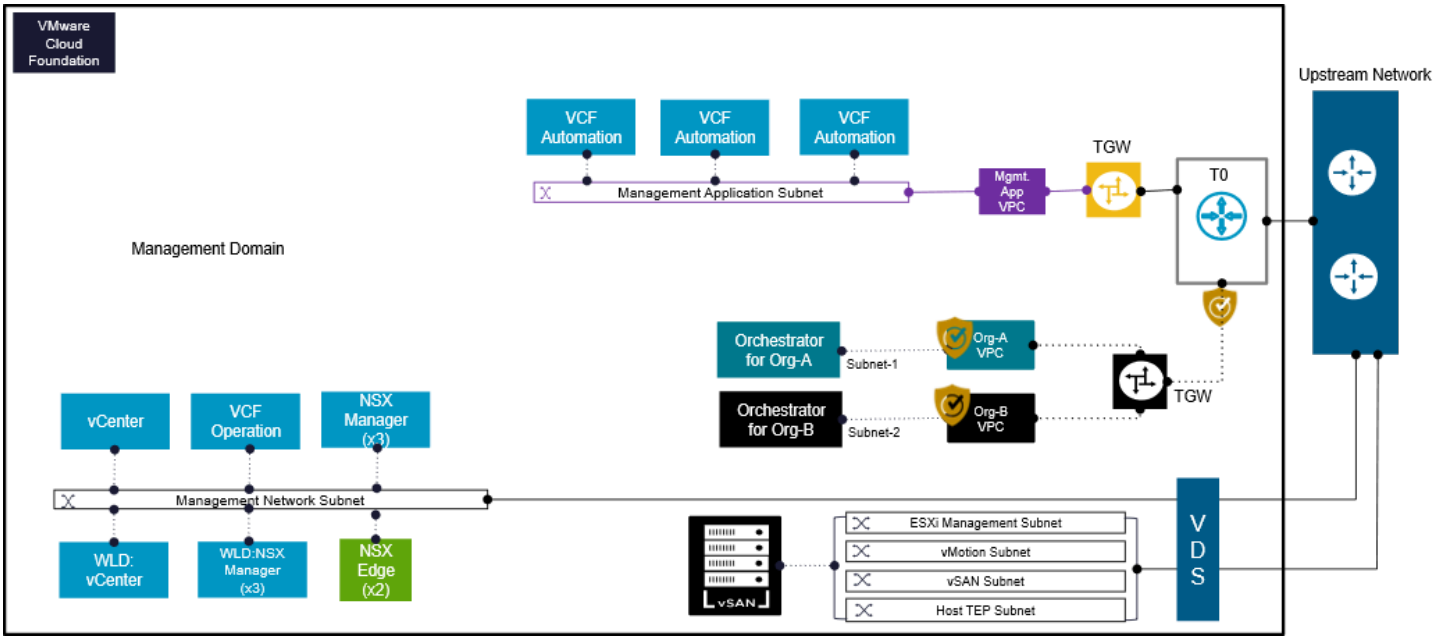
VCF Operations Orchestrator Deployment Options

Orchestrator Deployment Options	Remarks	Recommended Scenario
Embedded	Installed, by default, as part of VCF Automation deployment.	• Used by the Provider Consumption Organization • Used by a Provider Admin (with system administrator rights) to create workflows that can be published to Organizations
External	Is installed manually.	• External VCF Operations orchestrator should be used by Organizations that want to create and publish their own workflows. • Each Organization can have only one VCF Operations orchestrator instance. • The Provider Admin (with System Administrator rights) deploys, manages, and provides access to the external VCF Operations orchestrator used by organizations.

This design outlines a robust and scalable architecture for deploying VCF Operation orchestrator in a multi-tenant environment, catering to the distinct needs of multiple organizations in a Cloud Service Provider environment.

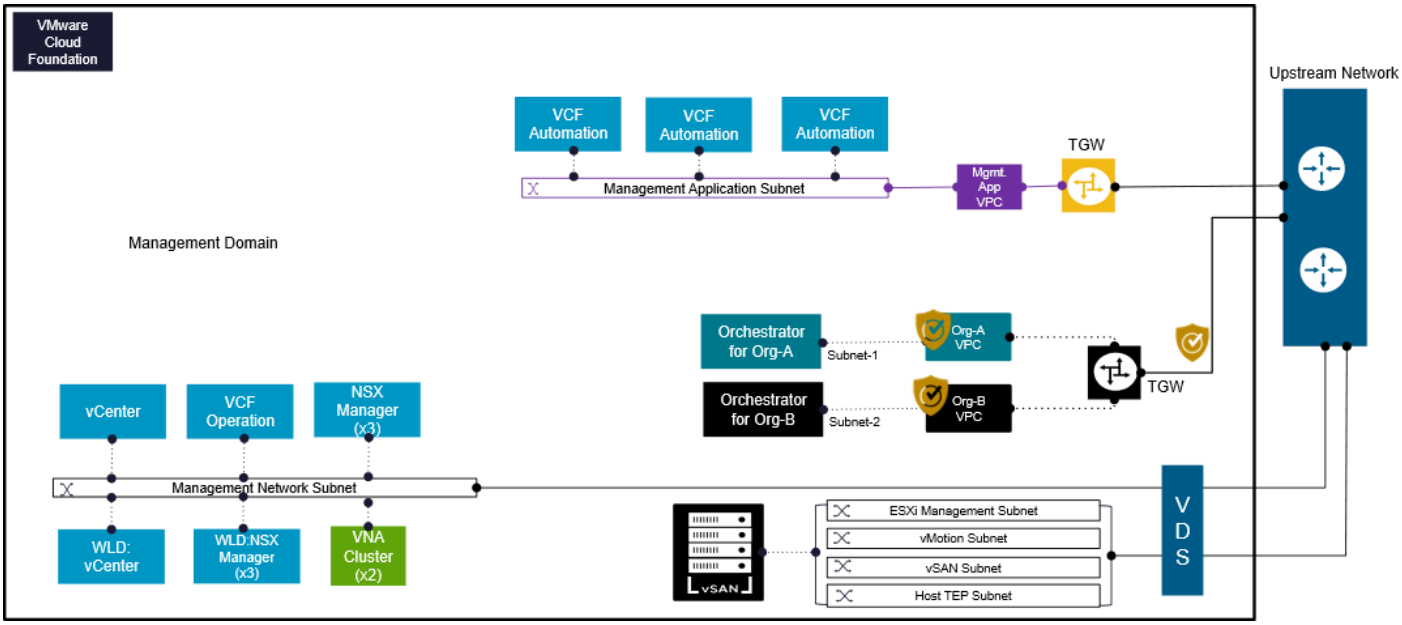
VCF Operation Orchestrator Deployment for Multiple Organizations with CTGW Model

Figure 270: VCF Operation Orchestrator Deployment for Multiple Organizations with CTGW Model



VCF Operation Orchestrator Deployment for Multiple Organizations with DTGW Model

Figure 271: VCF Operation Orchestrator Deployment for Multiple Organizations with DTGW Model



The architecture introduces separate, isolated VCF Operations orchestrator instances for every organization in a CSP environment (e.g., "Orchestrator for Org-A", "Orchestrator for Org-B"). These organizational VCF Operations orchestrator instances are each hosted in a logically distinct network segment or a Virtual Private Cloud (VPC). These dedicated VCF Operations orchestrator instances ensure controlled and safe communication by establishing a secure connection to the centralized VCF management components. Thus enabling any organization to take advantage of the economies of scale provided by the shared underlying VCF infrastructure. Simultaneously the pattern allows for the use of unique automation workflows and operational tasks with logical separation and autonomy.

Design Attributes	Details
VCF Operations orchestrator Model	A dedicated VCF Operations orchestrator instance is deployed for each organization (e.g., "Orchestrator for Org-A," "Orchestrator for Org-B").
Network Isolation and Segmentation	• Each organization's VCF Operations orchestrator instance resides within its own distinct Virtual Private Cloud (VPC), providing inherent network isolation (e.g., "Org-A VPC," "Org-B VPC"). • Specific subnets are allocated within each organizational VPC for the VCF Operations orchestrator instances (e.g., "Subnet-1" for Org-A, "Subnet-2" for Org-B).
Firewall and Security Enforcement	Add-on services such as vDefend can be used to create east-west and north-south firewall policies to further protect traffic patterns. These policies will control and secure traffic flow to and from each organization's Orchestrator and its associated workloads, ensuring proper access control and segmentation. Provider Admin can apply DFW policies on the NSX Project level in the case of individual NSX Projects.

VCF Operations Orchestrator Rights Bundle within VCF Automation

VCF Automation maintains a specific rights bundle for VCF Operations orchestrator that are known as the "Orchestrator Rights Bundle." After the VCF Operations orchestrator is deployed and integrated with the target VCF Automation Organization, the rights needs configuration:

1. From the Provider Portal, publish the Orchestrator Rights Bundle to the target Organization
2. From the Organization Portal, assign the rights to target users and groups

For detailed instructions, refer to the Managing VMware Cloud Foundation Operations Orchestratordocumentation.

VCF Operations Orchestrator for the Provider Consumption Organization

The VCF Automation deployment includes the embedded VCF Operations orchestrator by default.

After the provider admin creates a workflow in the embedded VCF Operations orchestrator, the workflow can be published to the catalog, either:

- As a catalog item for sole use by the provider admin
- As as catalog item that is available to all Organizations

Any workflow that is built in the PCO then published to all Organizations runs on the embedded VCF Operations orchestrator, even when run by a user in a non-PCO Organization.

If an Organization wishes to create and publish their own workflows within their Organization, they cannot use of the PCO embedded VCF Operations orchestrator. The Organization must use an external, dedicated VCF Operations orchestrator instance.

Chargeback and Billing with VCF Automation

Chargeback is a cost management solution that is designed to help service providers accurately track and distribute expenses within a private cloud environment. Enabling both providers and tenants to gain insights into resource usage and ensures fair cost allocation-based on consumption. As a plug-in for tenants, Chargeback provides a metering and billing interface that simplifies usage tracking, measurement, and invoicing for services delivered through VCF Automation and VCF Operation.

Chargeback is an essential component of cloud services, particularly for service providers using VMware Cloud Foundation. Involving assigning the costs of cloud resources and services to the users or departments that actually use them. This practice is critical for both service providers and tenants, as it creates a transparent and clear method for billing and payment.

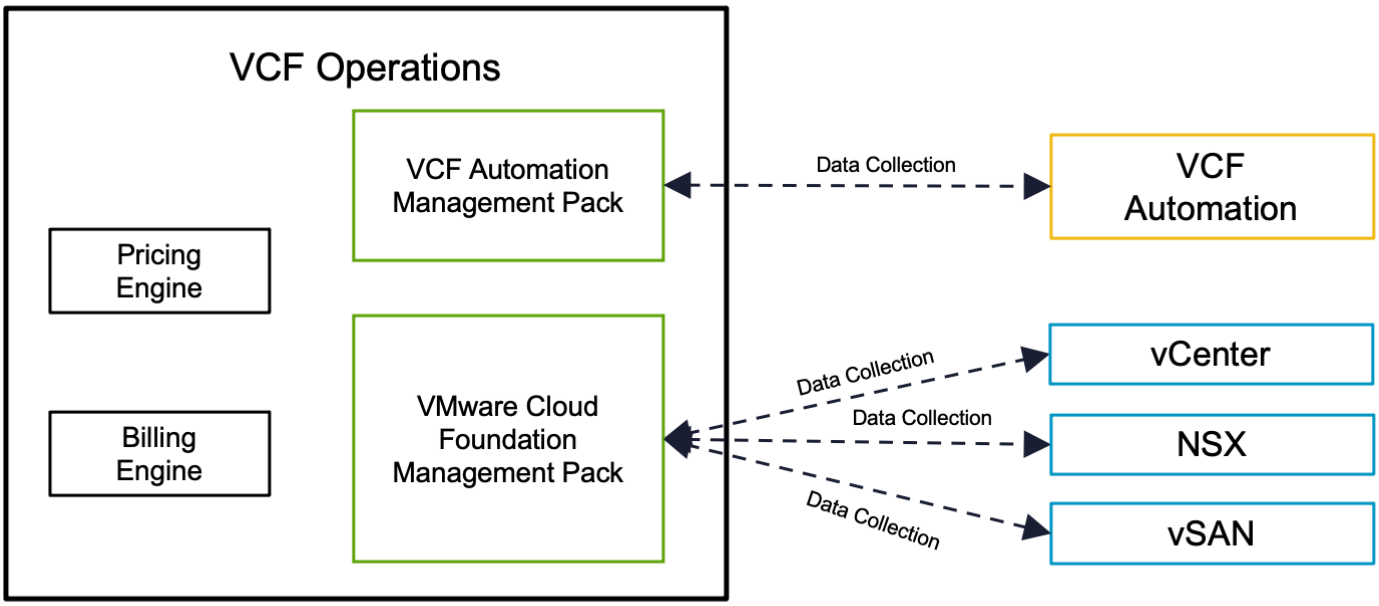
In older versions of VCF, service providers used VMware Cloud Director (VCD) to handle things like pricing, billing, and dashboards. Starting with VCF 9.x, however, these tasks are managed directly through VCF Automation. This change likely aims to streamline operations, reduce dependencies on external systems, and provide a more integrated experience for providers within the VCF platform.

The following figure describes a high-level overview of how Chargeback and Billing are integrated part of VCF Operation. VCF Operations acts as the central management hub and the starting point for data collection and integration.

Management Packs (MPs):

VCF Operations utilizes Management Packs (MPs) to collect data from specific components. These MPs are like plugins or modules that enable integration. Management packs shown are the VCF Automation MP and VMware Cloud Foundation MP (collects vCenter, NSX, and vSAN data). This suggests a direct integration between each MP and its respective data sources.

Figure 272: Chargeback Integration with VCF Operations



Key Use Cases:
Flexible Pricing (Rate Cards): • Service providers need the ability to set their own prices for different resources (like storage, compute, network) based on regions or quotas. Allowing them to offer tailored pricing plans to their customers. **Accurate Billing (Bills):** • Providers must be able to generate and provide detailed bills to their customers, showing usage and charges for organizations or specific resource allocations (quotas) in different regions. **Usage Monitoring (Dashboards):** • Providers need ready-to-use dashboards that give them a clear view of how their resources are being used by customers. **Customized Provider Interface (Control Panel and Launchpad):** • Providers need a customized interface (control panel and launchpad) that allows them to manage and configure their services and resources in a way that suits their specific needs.

High-level Steps to Initiate Billing

vSphere Supervisor Detailed Design

A VCF domain can include one or more of the supported *vSphere Supervisor Models*. The following sections details the design of each deployment model types for vSphere Supervisor.

- Simplified vSphere Supervisor Model
- vSphere Supervisor Zone Models
- vSphere Supervisor Control Plane Availability Models

Simplified vSphere Supervisor Model

VMware Cloud Foundation allows you to create a *Simplified vSphere Supervisor Model* requiring fewer resources and configuration steps than a typical deployment. This simplified deployment allows access to a limited subset of vSphere Supervisor capabilities. You can then expand the deployment over time to build out to full vSphere Supervisor functionality.

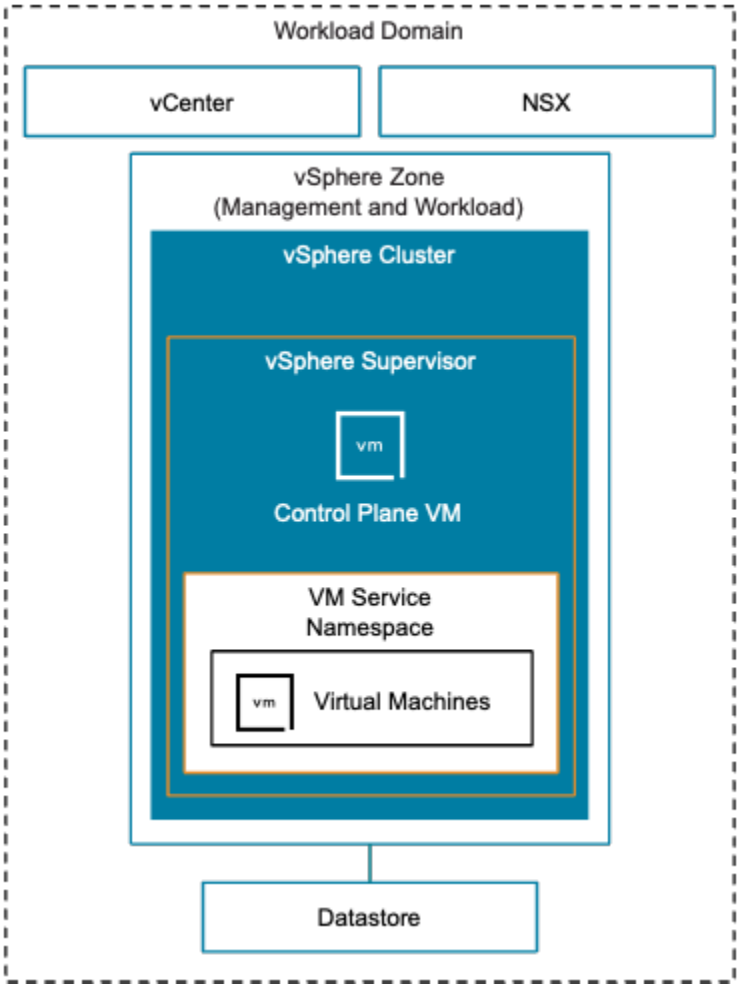
For information on how to deploy or scale the *Simplified vSphere Supervisor Model*, refer to Deploying Supervisor with a Simplified Deployment Flow.

Important:

The *Simplified vSphere Supervisor Model* is not supported for use by VCF Automation.

Figure 273: Simplified vSphere Supervisor Model in VMware Cloud Foundation

This diagram illustrates an example of a *Simplified vSphere Supervisor Model* deployment.



Simplified vSphere Supervisor Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single
vSphere Supervisor Workload Zone	Single - Combined with management
Control Plane VMs	Single
Supported Service	VM Service
Scalability	Scalable to full feature Single Management Zone with Combined Workload Zones Model or Single Management Zone with Isolated Workload Zones Model.
Activation	From vCenter

vSphere Supervisor Zone Models

vSphere Supervisor management workloads can be deployed to one or three management zones. Workloads in vSphere namespaces can be deployed to the same or separate dedicated zones to isolate them from management workloads. This section details the different deployment models to choose from.

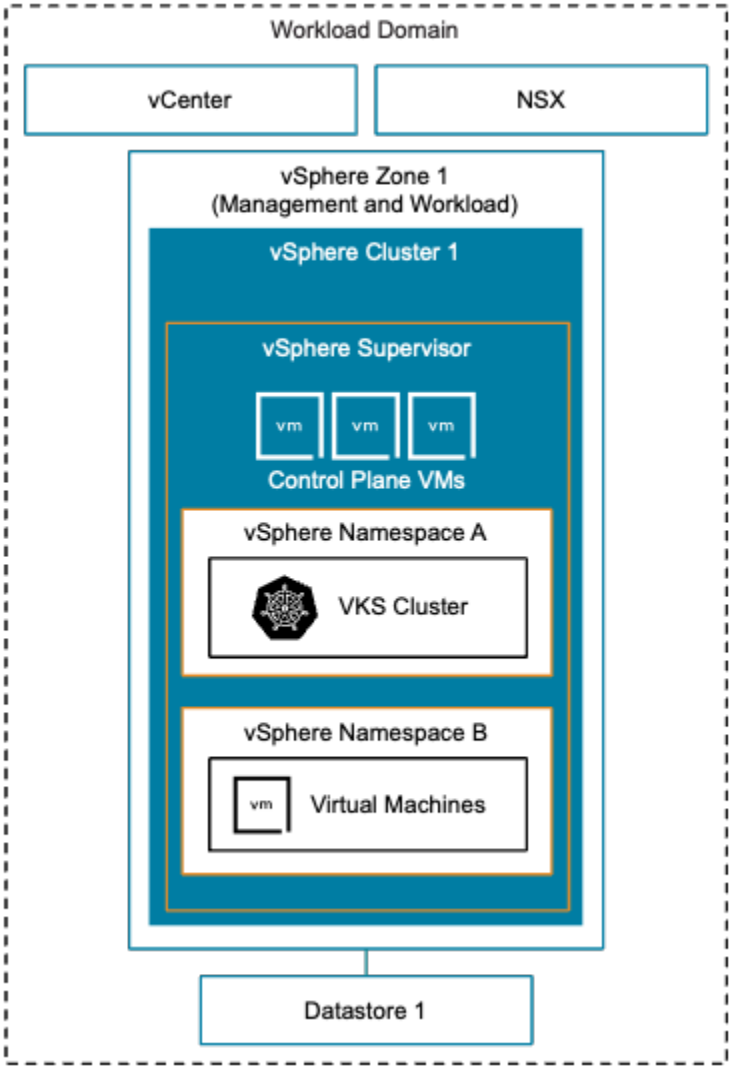
- Single Management Zone with Combined Workload Zones Model
- Single Management Zone with Isolated Workload Zones Model
- Three Management Zones with Combined Workload Zones Model
- Three Management Zones with Isolated Workload Zones Model

Single Management Zone with Combined Workload Zones Model

A *Single Management Zone with Combined Workload Zones Model* utilizes resources from a single vSphere Zone for management and workloads. Availability protection is provided using vSphere HA.

Figure 274: Single Management Zone with Combined Workload Zones Model Diagram

This diagram illustrates an example of a *Single Management Zone with Combined Workload Zones Model* deployment.



Single Management Zone with Combined Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single - Combined with management
vSphere Zone Datastore Type	Zonal datastore
Activation	• At workload domain creation. OR • From vCenter.

Single Management Zone with Combined Workload Zones Model Options

Design Area	Choice
vSphere Cluster Models	• Single-Rack vSphere Cluster Model • Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model • Stretched vSphere Cluster Model
Control Plane Availability Models	• Simple vSphere Supervisor Control Plane Model • High Availability vSphere Supervisor Control Plane Model
Workload Connectivity Models	• Centralized Connectivity Model • Distributed VLAN Connectivity Model • NSX Segment Connectivity Model (API Only) • VLAN Networking Connectivity Model
Load Balancer Models	• Avi Load Balancer Model • Virtual Network Appliance Load Balancer Model • NSX Edge VPC Load Balancer Model • NSX Edge Tier-1 Load Balancer Model • Foundation Load Balancer Model
Storage Models	**Single-Rack vSAN Storage Models** • Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model • Single-Rack vSAN Storage Cluster Storage Model **Multi-Rack vSAN Storage Models** • Multi-Rack vSAN Storage Model **Stretched vSAN Storage Models** • Stretched vSAN Storage Model • Stretched vSAN Storage Cluster Storage Model

Note:

Workload Connectivity Model choice will limit *Load Balancer Model* choice. Refer to the Load Balancer Models and Network Connectivity Model Support Matrix for vSphere Supervisor in Architectural Options.

Single Management Zone with Combined Workload Zones Model Design Requirements and Recommendations

Table 543: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

Table 544: vSphere Supervisor Single Management Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-SMZ-REQD-CFG-001	One available vSphere Zone that is backed by a supported vSphere cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	vSphere Zone for Supervisor management workload use.	The selected vSphere Zone is dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.

Table 545: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

Table 546: vSphere Supervisor Design Recommendations

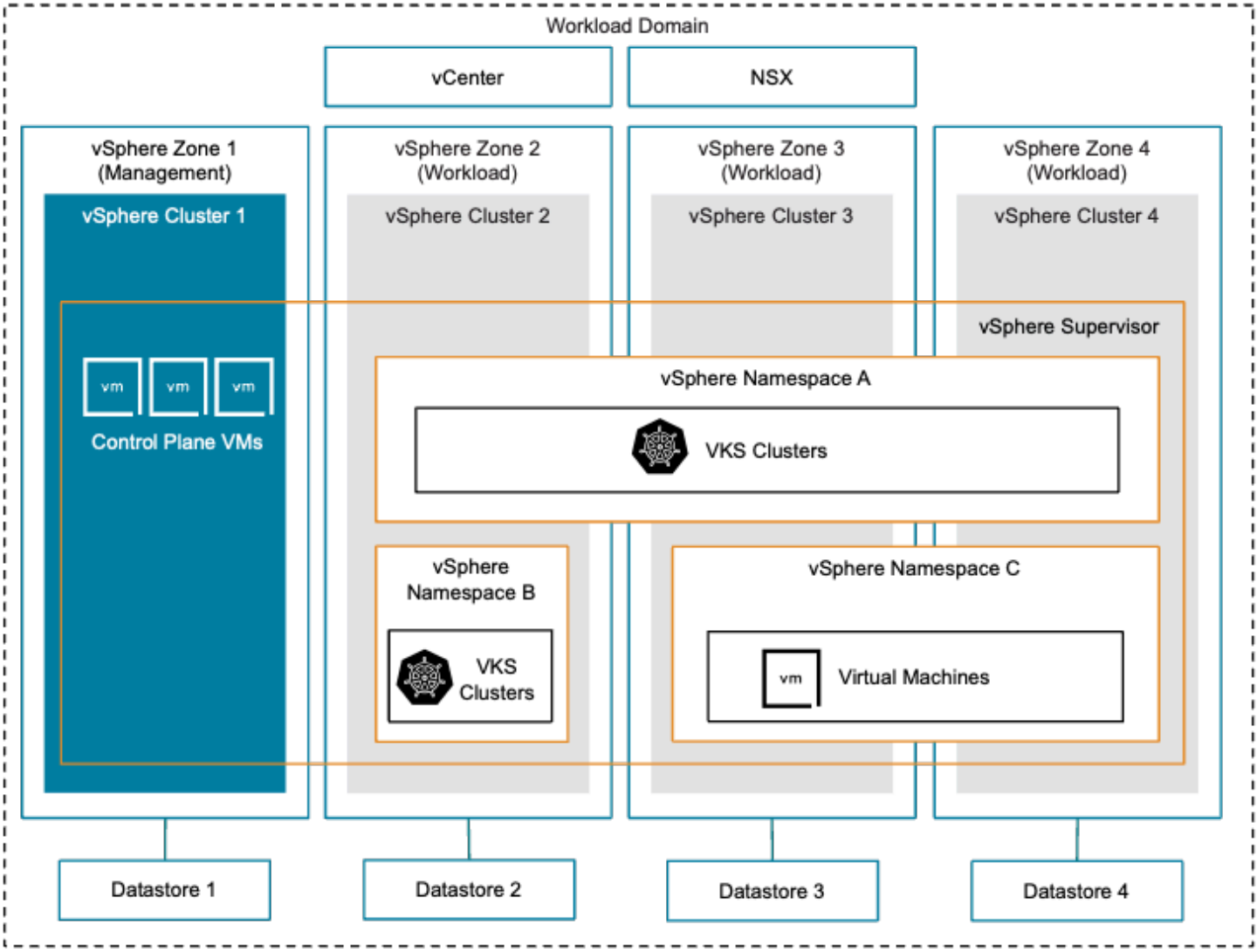
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

Single Management Zone with Isolated Workload Zones Model

A *Single Management Zone with Isolated Workload Zones Model* utilizes resources from a single management zone for control plane VMs. Workloads are isolated from the Management Zone by deploying into additional vSphere Zones. Management workload availability protection is provided using vSphere HA. Workloads can be distributed across multiple zones for increased availability.

Figure 275: Single Management Zone with Isolated Workload Zones Model

This diagram illustrates an example of a *Single Management Zone with Isolated Workload Zones Model* deployment.



Single Management Zone with Isolated Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Single (Default)
vSphere Supervisor Workload Zones	Single or multiple - Isolated from management
vSphere Zone Datastore Type	**Management Zone** • Zonal datastore **Single Workload Zone** • Zonal datastore **Multiple Workload Zones** • Cross-zone datastore • Multi-zone datastore
Activation	• At workload domain creation. OR • From vCenter.

Single Management Zone with Isolated Workload Zones Model Options

Design Area	Choice
vSphere Cluster Models	• Single-Rack vSphere Cluster Model • Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model • Stretched vSphere Cluster Model
Control Plane Availability Models	• Simple vSphere Supervisor Control Plane Model • High Availability vSphere Supervisor Control Plane Model
Workload Connectivity Models	• Centralized Connectivity Model • Distributed VLAN Connectivity Model • NSX Segment Connectivity Model (API Only) • VLAN Networking Connectivity Model
Load Balancer Models	• Avi Load Balancer Model • Virtual Network Appliance Load Balancer Model • NSX Edge VPC Load Balancer Model • NSX Edge Tier-1 Load Balancer Model • Foundation Load Balancer Model
Storage Models	**Single-Rack vSAN Storage Models** • Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model • Single-Rack vSAN Storage Cluster Storage Model **Multi-Rack vSAN Storage Models** • Multi-Rack vSAN Storage Model **Stretched vSAN Storage Models** • Stretched vSAN Storage Model • Stretched vSAN Storage Cluster Storage Model

Note:

Workload Connectivity Model choice will limit *Load Balancer Model* choice. Refer to the Load Balancer Models and Network Connectivity Model Support Matrix for vSphere Supervisor in Architectural Options.

Single Management Zone with Isolated Workload Zones Model Design Requirements and Recommendations

Table 547: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

Table 548: vSphere Supervisor Single Management Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-SMZ-REQD-CFG-001	One available vSphere Zone that is backed by a supported vSphere cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	vSphere Zone for Supervisor management workload use.	The selected vSphere Zone is dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.

Table 549: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

Table 550: vSphere Supervisor Design Recommendations

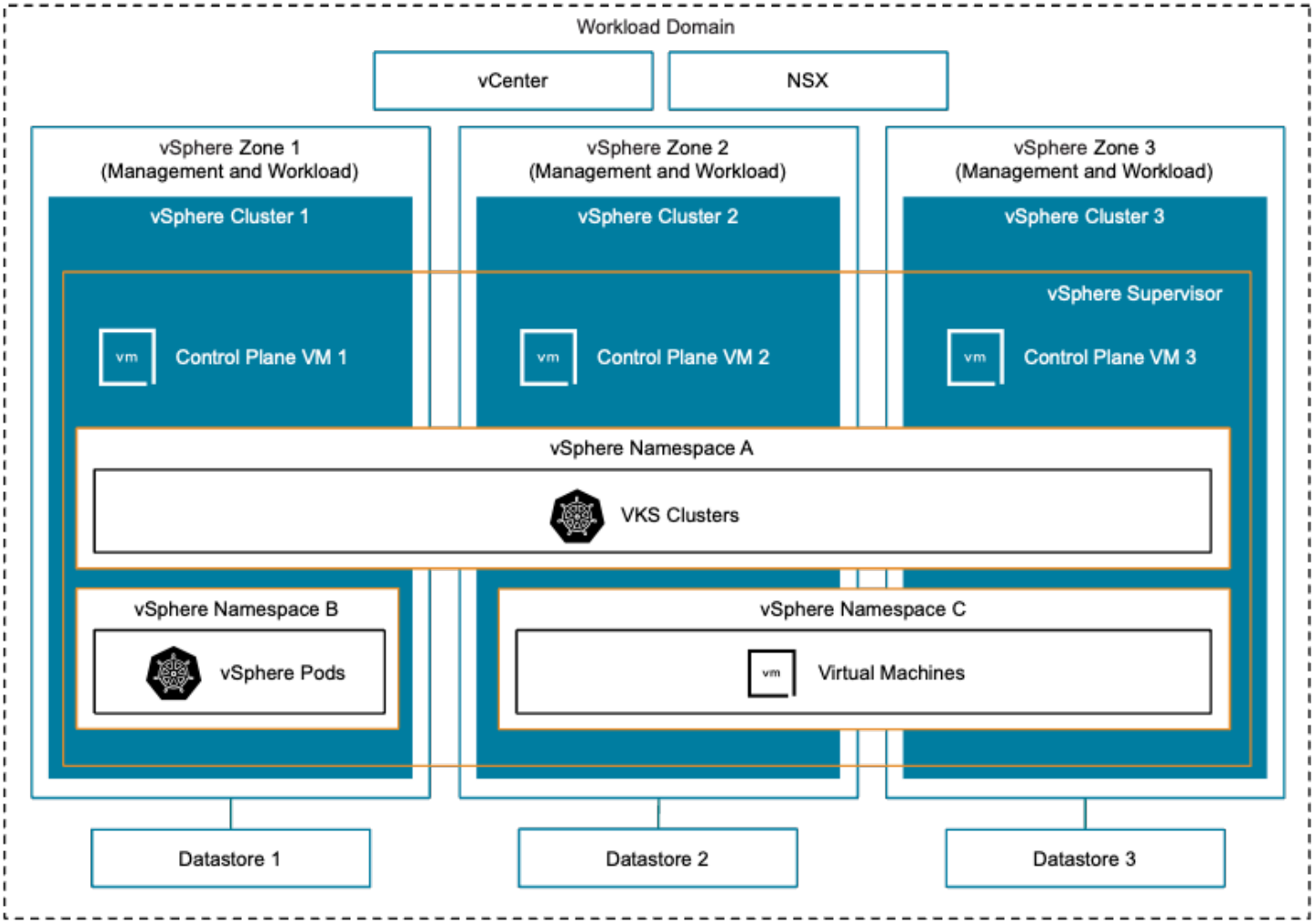
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

Three Management Zones with Combined Workload Zones Model

A *Three Management Zones Supervisor Model with Combined Workload Zones Model* utilizes resources from three vSphere Zones for control plane VMs. Workloads share resources from the same vSphere Zones. Management workload availability protection is provided using vSphere HA and can tolerate a single cluster failure. Workload availability protection is provided using vSphere HA and can be distributed across multiple zones for increased availability.

Figure 276: Three Management Zones with Combined Workload Zones Model

This diagram illustrates an example of a *Three Management Zones with Combined Workload Zones Model* deployment.



Three Management Zones with Combined Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Three
vSphere Supervisor Workload Zones	Three - Combined with management
vSphere Zone Datastore	• Cross-zone datastore • Multi-zone datastore
Activation	Activated from vCenter API after workload domain creation.

Three Management Zones with Combined Workload Zones Model Options

Design Area	Choice
vSphere Cluster Models	• Single-Rack vSphere Cluster Model • Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model
Control Plane Availability Models	High Availability vSphere Supervisor Control Plane Model
Workload Connectivity Models	• Centralized Connectivity Model • Distributed VLAN Connectivity Model • NSX Segment Connectivity Model (API Only) • VLAN Networking Connectivity Model

Design Area	Choice
Load Balancer Models	• Avi Load Balancer Model • Virtual Network Appliance Load Balancer Model • NSX Edge VPC Load Balancer Model • NSX Edge Tier-1 Load Balancer Model • Foundation Load Balancer Model
Storage Models	**Single-Rack vSAN Storage Models** • Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model **Multi-Rack vSAN Storage Models** • Multi-Rack vSAN Storage Model **Stretched vSAN Storage Models** • Stretched vSAN Storage Model • Stretched vSAN Storage Cluster Storage Model

Note:

Networking Model choice will limit Load Balancer Model Choice. Refer to the Load Balancer Models and Network Connectivity Model Support Matrix for vSphere Supervisor in Architectural Options.

Three Management Zones with Combined Workload Zones Model Design Requirements and Recommendations

Table 551: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

Table 552: vSphere Supervisor Three Management Zones Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-TMZ-REQD-CFG-001	Three available vSphere Zones of a supported vSphere Cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	• Dedicated vSphere Zones for vSphere Supervisor management workload use. • Individual zones provide single cluster level failure scenarios.	The selected vSphere Zones are dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.
VCF-SUP-TMZ-REQD-CFG-002	High-availability control plane model option selected.	Ensures even distribution and a quorum is available across three management zones.	Additional resources are required to host additional control plane nodes.
VCF-SUP-TMZ-REQD-CFG-003	Opt-out of vSphere Supervisor activation when deploying a workload domain.	Activating vSphere Supervisor as part of the workload domain creation defaults to a Single Management Zone model.	vSphere Supervisor activation must be performed from vCenter as a Day 2 operation.

Table 553: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

Table 554: vSphere Supervisor Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

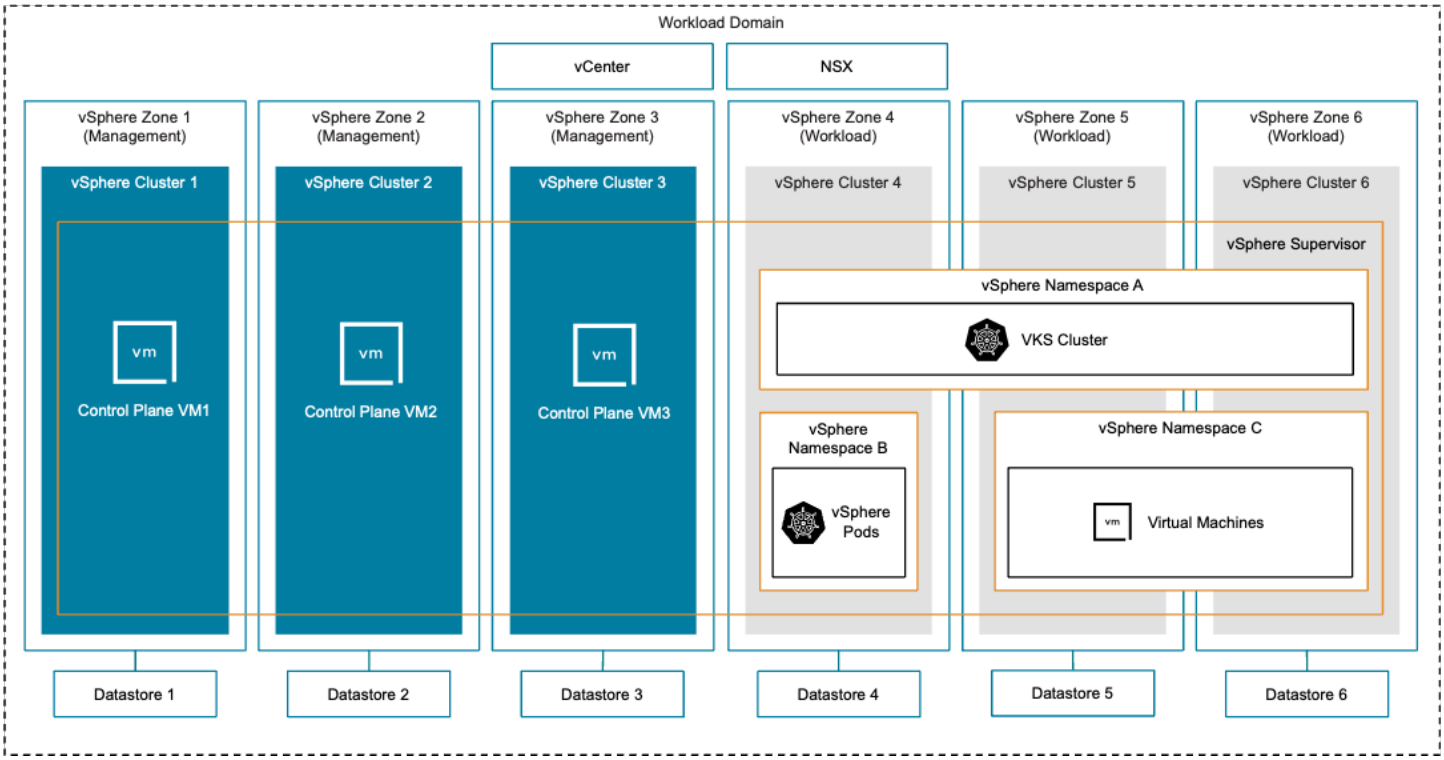
Three Management Zones with Isolated Workload Zones Model

A *Three Management Zones Supervisor Model with Isolated Workload Zones Model* utilizes resources from three dedicated vSphere Zones for control plane VMs. Workloads are isolated from the Management Zone by deploying into

additional vSphere Zones. Management workload availability protection is provided using vSphere HA and can tolerate a single cluster failure. Workload availability protection is provided using vSphere HA and can be distributed across multiple zones for increased availability.

Figure 277: Three Management Zones with Isolated Workload Zones Model

This diagram illustrates an example of a *Three Management Zones with Isolated Workload Zones Model* deployment.



Three Management Zones with Isolated Workload Zones Model Attributes

Attribute	Detail
vSphere Supervisor Management Zone	Three
vSphere Supervisor Workload Zones	Single or multiple - Isolated from management

Attribute	Detail
vSphere Zone Datastore	**Management Zone** • Cross Zone Datastore • Multi Zone Datastore **Single Workload Zone** • Single Zone Datastore **Multiple Workload Zones** • Cross Zone Datastore • Multi Zone Datastore
Activation	Activated from vCenter API after workload domain creation.

Three Management Zones with Isolated Workload Zones Model Options

Design Area	Choice
vSphere Cluster Models	• Single-Rack vSphere Cluster Model • Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model
Control Plane Availability Models	High Availability vSphere Supervisor Control Plane Model
Workload Connectivity Models	• Centralized Connectivity Model • Distributed VLAN Connectivity Model • NSX Segment Connectivity Model (API Only) • VLAN Networking Connectivity Model
Load Balancer Models	• Avi Load Balancer Model • Virtual Network Appliance Load Balancer Model • NSX Edge VPC Load Balancer Model • NSX Edge Tier-1 Load Balancer Model • Foundation Load Balancer Model
Storage Models	**Single-Rack vSAN Storage Models** • Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model • Single-Rack vSAN Storage Cluster Storage Model **Multi-Rack vSAN Storage Models** • Multi-Rack vSAN Storage Model **Stretched vSAN Storage Models** • Stretched vSAN Storage Model • Stretched vSAN Storage Cluster Storage Model

Note:

Workload Connectivity Model choice will limit *Load Balancer Model* choice. Refer to the Load Balancer Models and Network Connectivity Model Support Matrix for vSphere Supervisor in Architectural Options.

Three Management Zones with Isolated Workload Zones Model Design Requirements and Recommendations

Table 555: vSphere Supervisor Design Requirements for all Zone Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-AMZ-REQD-CFG-001	vSphere clusters must have vSphere HA enabled.	Required feature to automatically recover from host failure.	Sufficient spare ESX host resources required for vSphere HA functionality.
VCF-SUP-AMZ-REQD-CFG-002	vSphere clusters must have vSphere DRS enabled in Fully Automated or Partially Automated mode.	Required feature to automatically balance resource utilization within the cluster.	Sufficient host resources are required for vSphere DRS functionality.
VCF-SUP-AMZ-REQD-CFG-003	vSphere Zones name must be configured and it must be compliance with RFC 1123.	Kubernetes objects and resource types must follow DNS label standards for their names.	• A vSphere Zone name is generated for a Single Management Zone Model if not configured and cannot be changed. • vSphere Zones must start with a lowercase letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-004	A vSphere Supervisor name must be configured and RFC 1123 compliant.	Kubernetes objects and resource types must follow DNS label standards for their names.	vSphere Supervisor name must start with a lower case letter or number and contain only alphanumeric characters or hyphens.
VCF-SUP-AMZ-REQD-CFG-005	A compatible vSphere Zone must be available.	• A vSphere Zone must be configured and available during vSphere Supervisor activation. • For multi-zone the required number of zones must exist depending on the deployment model.	None.
VCF-SUP-AMZ-REQD-CFG-006	Suitable storage policies must be assigned for datastore placement for Supervisor control plane VMs, Ephemeral Disks, and Image Cache.	• Storage policies should have placement and protection considerations that are aligned with the use case. • Policy should align to the datastore topology (zonal, multi-zone, or cross-zone) depending on the intended supervisor model and workload placement behavior.	Additional storage policies are required.

Table 556: vSphere Supervisor Three Management Zones Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-TMZ-REQD-CFG-001	Three available vSphere Zones of a supported vSphere Cluster model must be available and not already assigned to another vSphere Supervisor instance at activation.	• Dedicated vSphere Zones for vSphere Supervisor management workload use. • Individual zones provide single cluster level failure scenarios.	The selected vSphere Zones are dedicated to that vSphere Supervisor instance and cannot be reused by another supervisor within the same vCenter.
VCF-SUP-TMZ-REQD-CFG-002	High-availability control plane model option selected.	Ensures even distribution and a quorum is available across three management zones.	Additional resources are required to host additional control plane nodes.
VCF-SUP-TMZ-REQD-CFG-003	Opt-out of vSphere Supervisor activation when deploying a workload domain.	Activating vSphere Supervisor as part of the workload domain creation defaults to a Single Management Zone model.	vSphere Supervisor activation must be performed from vCenter as a Day 2 operation.

Table 557: vSphere Supervisor Common Networking Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-SUP-NET-RCMD-CFG-001	The IP Assignment Mode for the control plane VMs should be Static.	Removes the need for an external DHCP server for the control plane VMs.	Requires precise IP address management.
VCF-SUP-NET-RCMD-CFG-002	When deploying Supervisor on a vSAN stretched cluster, place the control plane VMs on a management network that is extended across both availability zones, such as a stretched Layer 2 VM Management Network.	Provides the required network reachability for Supervisor control plane VMs across the stretched vSAN cluster topology.	• Requires precise IP address management. • vSAN cluster must be stretched before deploying Supervisor.

Table 558: vSphere Supervisor Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SUP-RCMD-CFG-002	Deploy the control plane in high-availability mode.	High availability for vSphere Supervisor control plane.	Additional hosting resources are required.

vSphere Supervisor Control Plane Availability Models

vSphere Supervisor can be deployed with one or three control plane VMs. This section details the deployment models.

- Simple vSphere Supervisor Control Plane Model
- High Availability vSphere Supervisor Control Plane Model

Simple vSphere Supervisor Control Plane Model

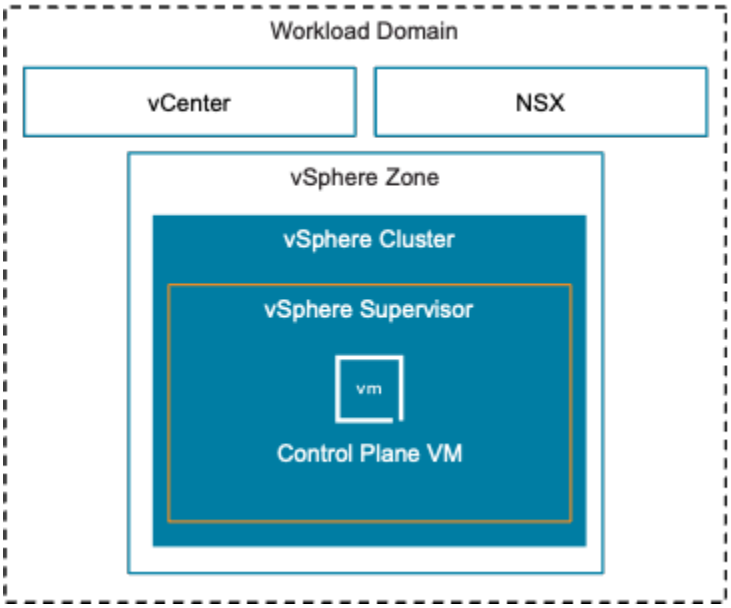
The *Simple vSphere Supervisor Control Plane Model* deploys a single vSphere Supervisor control plane VM into the selected vSphere Supervisor management zone. This deployment model can be scaled to three control plane VMs to convert the model to the *High Availability vSphere Supervisor Control Plane Model*.

Note:

The *Simple vSphere Supervisor Control Plane Model* is not an option when deploying a vSphere Supervisor three management zones model.

Figure 278: Simple vSphere Supervisor Control Plane Model

This diagram illustrates an example of a *Simple vSphere Supervisor Control Plane Model* deployment.



Simple vSphere Supervisor Control Plane Model Attributes

Attribute	Detail
Control Planes Nodes	One
Availability	Relies on vSphere HA
Scalability	Scale up and scale out
Placement	Hosted in a single vSphere Supervisor management zone.
Applicable Zone Models	• Simplified vSphere Supervisor Model • Single Management Zone with Combined Workload Zones Model • Single Management Zone with Isolated Workload Zones Model

Simple vSphere Supervisor Control Plane Model Design Requirements and Recommendations

Table 559: vSphere Supervisor Control Plane Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-CP-REQD-CFG-001	Five consecutive management IP addresses routable to vCenter NSX, and optionally Avi management networks if Avi Load Balancer are used.	Required for integration and management communications.	Requires precise IP address management.
VCF-SUP-CP-REQD-CFG-002	Three vSphere Zones are required for the High-Availability Model in a Three Management Zone Model.	Control pane nodes are evenly placed with one per zone to protect against a single vSphere cluster failure.	Additional vSphere zones each back by separate supported vSphere clusters are required.

High Availability vSphere Supervisor Control Plane Model

The *High Availability vSphere Supervisor Control Plane Model* deploys three Supervisor control plane VMs into the selected Supervisor management zones. This deployment model can be scaled from the *Simple vSphere Supervisor Control Plane Model*.

Figure 279: *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Single Management Zone Model deployment.

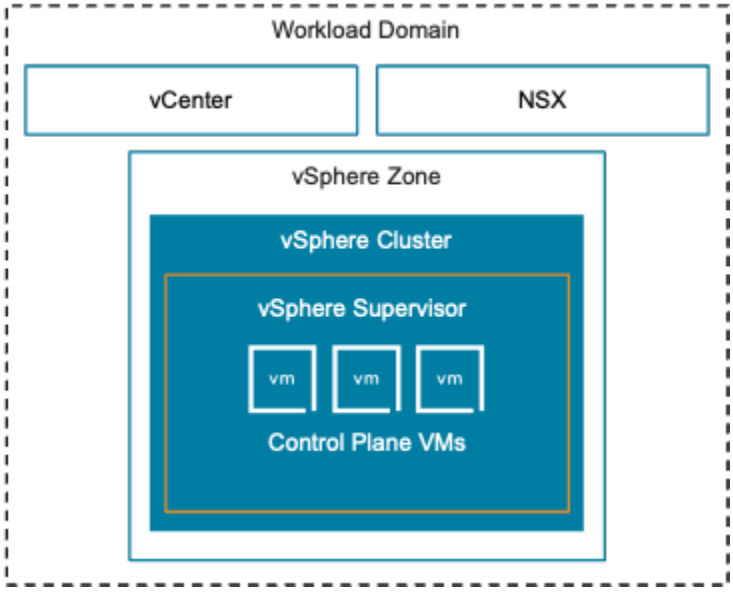
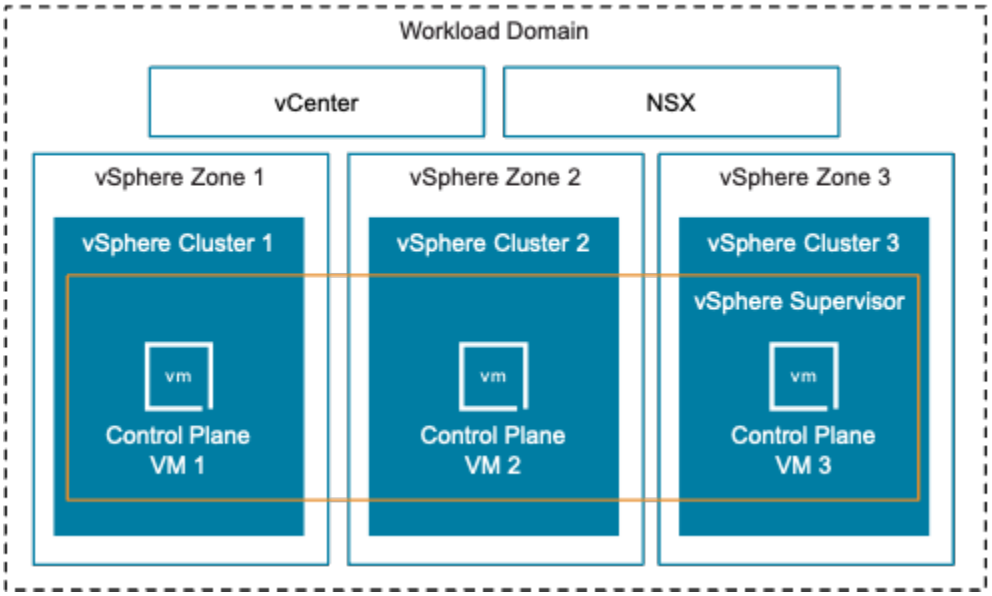


Figure 280: *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model Diagram

This diagram illustrates an example of a *High Availability vSphere Supervisor Control Plane Model* in a Three Management Zones Model deployment.



High Availability vSphere Supervisor Control Plane Model Attributes

Attribute	Detail
Control Planes Nodes	Three
Availability	• Relies on vSphere HA. • Three management zones, which are protected from single cluster failure.
Scalability	Scale up only

Attribute	Detail
Placement	• Single management zone, all nodes in a single management zone. • Three management zones, one node per management zone.
Applicable Zone Models	• Single Management Zone with Combined Workload Zones Model • Single Management Zone with Isolated Workload Zones Model • Three Management Zones with Combined Workload Zones Model • Three Management Zones with Isolated Workload Zones Model

High Availability vSphere Supervisor Control Plane Model Design Requirements and Recommendations

Table 560: vSphere Supervisor Control Plane Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SUP-CP-REQD-CFG-001	Five consecutive management IP addresses routable to vCenter NSX, and optionally Avi management networks if Avi Load Balancer are used.	Required for integration and management communications.	Requires precise IP address management.
VCF-SUP-CP-REQD-CFG-002	Three vSphere Zones are required for the High-Availability Model in a Three Management Zone Model.	Control pane nodes are evenly placed with one per zone to protect against a single vSphere cluster failure.	Additional vSphere zones each back by separate supported vSphere clusters are required.

Table 561: vSphere Supervisor Control Plane Design Recommendations

Design Requirement ID	Design Recommendation	Justification	Implication
VCF-SUP-CP-RCMD-CFG-001	Deploy the supervisor control plane in high availability mode with three control plane VMs. *Applicable to Non-Simplified Supervisor models only.	Three control plane VMs allow for continuous availability of the Supervisor control plane during rolling upgrades and single node outages.	Additional resources are required to support the extra control plane VMs.

VMware Data Services Manager Detailed Design

This section describes the options, requirements, and recommendations to deploy and configure VMware Data Services Manager.

Architecture

VMware Data Services Manager (DSM) is designed to provide a "Database-as-a-Service" (DBaaS) solution for private cloud. DSM is delivered as an Advanced Service for VMware Cloud Foundation. DSM simplifies and automates database lifecycle management for database engines that are deployed in VCF environments: PostgreSQL, MySQL, and Microsoft SQL Server.

The architecture is built on a central control plane that manages the lifecycle of these data services. DSM provides centralized provisioning, backup, cloning, patching, scaling, and clustering for database environments. The DSM Virtual Appliance (or DSM Provider) is deployed as a virtual appliance (OVA). DSM serves as the management control plane providing an API Gateway & UI interfaces for self service.

VMware Data Services Manager integrates with VMware Cloud Foundation Automation to include databases as part of a larger application blueprint. VMware Data Services Manager also supports GenAI and RAG applications by applying PostgreSQL with pgvector for advanced vector search capabilities.

Figure 281: VMware Data Services Manager Architecture

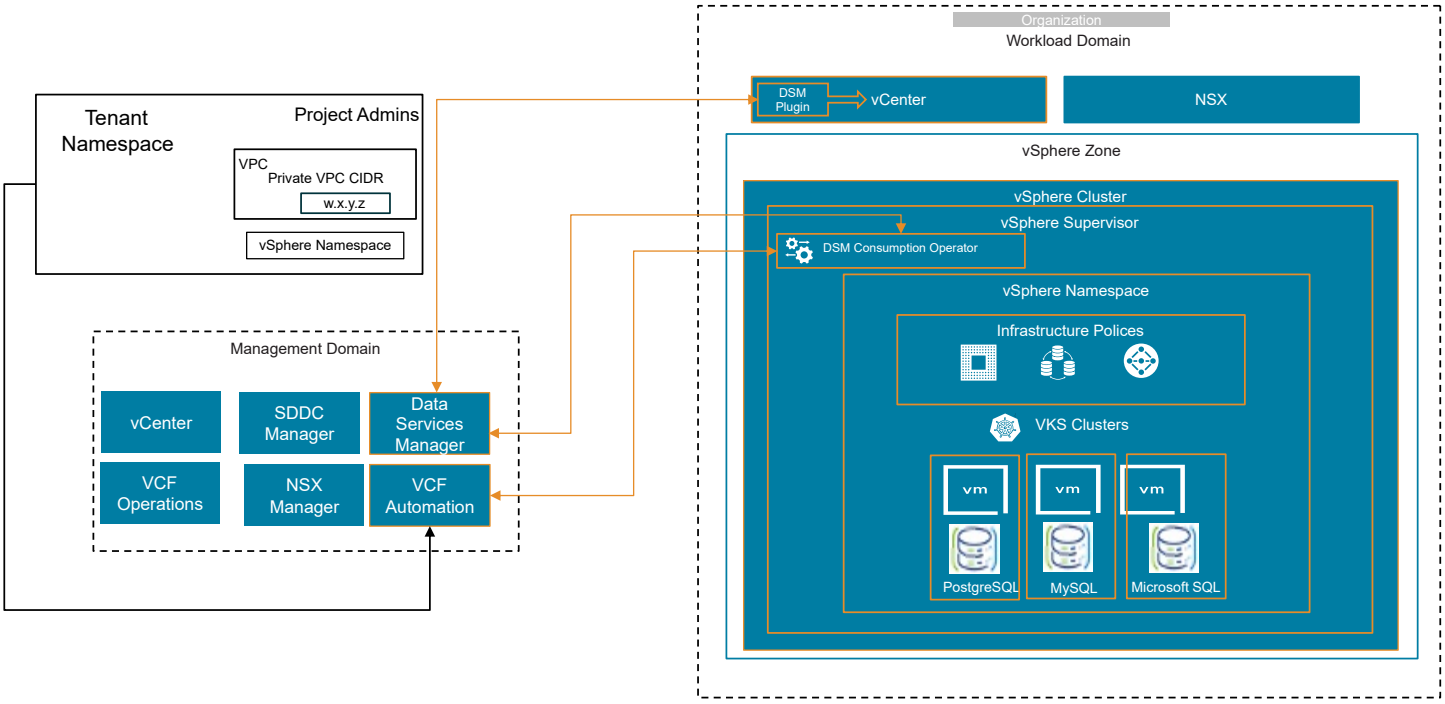
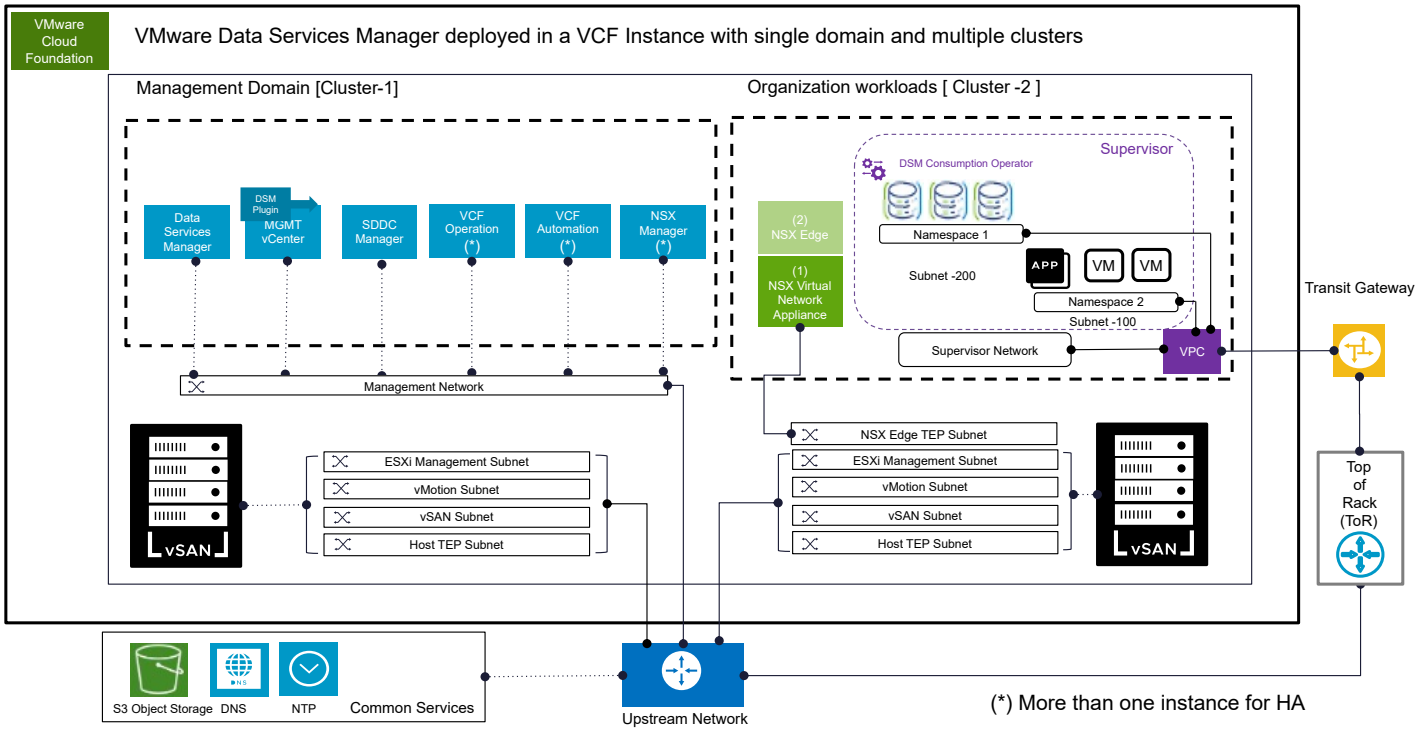


Table 562: VMware Data Services Manager Deployment Options

Deployment Path	Implications
VMware Data Services Manager OVA is deployed to the management domain and the Data Services Management Plugin is registered to the vCenter instance. This aligns to VCF Automation deployment Pattern 1 Deployment Pattern 1: VCF Automation in a VCF Instance with Single Domain and Multiple Clusters This architecture consolidates the management and compute resources into two primary clusters in a single domain.	• The management domain hosts the core management components of VCF such as vCenter, NSX, SDDC Manager, VCF Operations, and VCF Automation • The management domain is where the actual Tenant virtual machines (VMs) and applications run. • The plugin must be registered to the targeted management domain vCenter instance during deployment. • NSX VPC and Supervisor Cluster are enabled in the management domain to consume a modern private cloud with VCF Automation and Data Services.
VMware Data services manager OVA is deployed to the VCF management domain and the Data Services Management Plugin is registered to a vCenter workload domain vCenter instance. This aligns to VCF Automation deployment Pattern 2 Deployment Pattern 2: VCF Automation in a VCF Instance with Multiple Domains The management domain hosts the core management components of VCF and the workload domain hosts the tenant virtual machines.	• The Management Domain hosts the core management components of VCF such as vCenter, NSX, SDDC Manager, VCF Operation, and VCF Automation • The workload domain is where the actual Tenant virtual machines (VMs) and applications run. • The plugin must be registered to the targeted VCF Workload Domain vCenter instance during deployment • NSX VPC and Supervisor Cluster are enabled in the Workload Domain to consume a modern private cloud with VCF Automation and Data Services

Figure 282: Data Services Manager Deployment

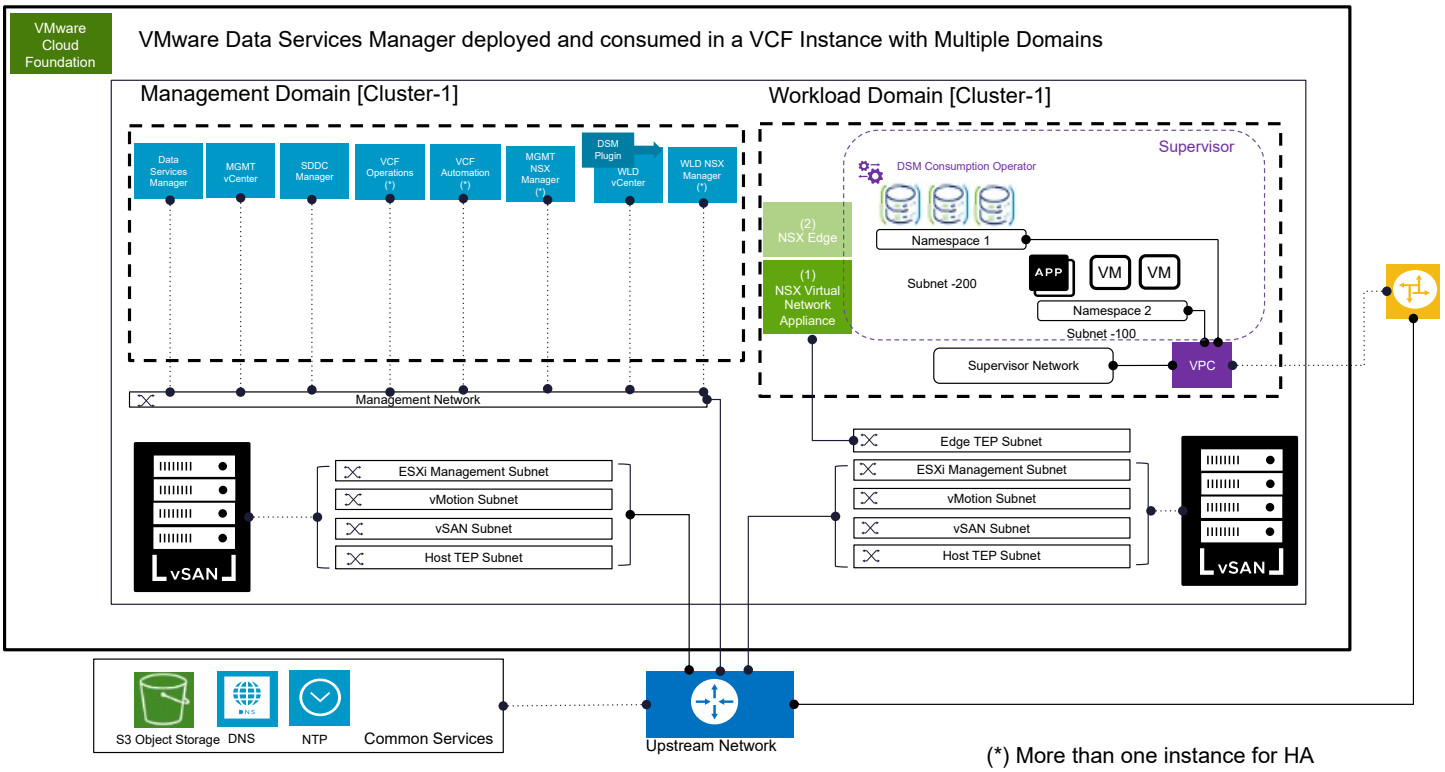
Data Services Manager is deployed in a management domain and database workloads are placed in clusters within the same workload domain.



Benefits	Drawbacks
• Simple deployment • Dedicated VCF Instance for a Tenant or a Line of Business • Near Edge Deployments Smaller Environments and no plan of growth • Smaller hardware footprint • VCF Automation HA is an option	• Limited scalability compared to a multiple domain architecture • Management Overhead to manage each VCF Instance for a dedicated Tenant • VCF Automation HA deployment consumes more infrastructure resources

Figure 283: VMware Data Services Manager is deployed in a management domain with workloads deployed to a separate workload domain.

VMware Data Services Manager is deployed in a management domain and database workloads are placed in clusters within a separate workload domain.



Benefits	Drawbacks
• Simplified integrated management. • independent scaling of compute, storage, and networking components. • Distributed architecture.	• Larger hardware footprint. • Larger cost footprint.

VMware Data Services Attributes

Attribute	Detail
Data Services Provider Virtual Appliance	Deployed as a photon virtual appliance (OVA), it serves as the control plane that coordinates the deployment, scaling, and lifecycle of all database instances
Infrastructure Policy	Allows vSphere administrators to define and set limits to specific compute, storage, and network resources that VMware Data Services Manager database workloads can use. For example, a VCF-A Organization admin creates a "Project" where they attach an Infrastructure Policy to that project. Developers assigned to that project only see the database options and resource limits that are defined by that specific policy
Data Services Manager (DSM) Consumption Operator	Is a software component deployed on VMware Supervisor Cluster, where customers with VCF Automation can integrate with DSM APIs
Infrastructure Policy	Defines the compute, storage, and networking resources for the database to be deployed, which the infrastructure administrator defines
Data Service Policy	Data Service Polices combines with infrastructure policies to define settings for database engines such as allowed versions, maintenance, and backup policy. Data Service policies also offer multi-tenancy by which users/organizations can provision specific database types and versions.
Database VM	A virtual machine that hosts a data service which is requested by the VMware Data Services Manager user. The Database VM is a Kubernetes node which runs the data service modules. The Kubernetes nodes are of dual purpose. A single Database VM acts as the control plane Kubernetes node and a Kubernetes worker node.
Database Topology and Resiliency	VMware Data Services Manager supports one Kubernetes node for standalone Database VMs and three nodes for replicated databases. • Single-Server Database topology Infrastructure resiliency is provided vSphere High Availability and underlying storage redundancy • Three-Node Database topology and Single vSphere Cluster This topology includes one primary database, one database-level synchronous replica and one monitor node. All database nodes share a vSphere Cluster and underlying shared storage • Three-Node Database topology and three vSphere Clusters. This topology includes one primary database, one database-level synchronous replica and one monitor node. The nodes are distributed in three vSphere clusters, providing enhanced data durability and availability **Important:** In a three vSphere cluster topology, all vSphere clusters must share a distributed vSphere switch
Data Services Manager Roles	Three personas interact with DSM, each with distinct responsibilities. • DSM Admin - manages the DSM platform, integrates with vCenter, defines infrastructure policies, and configures the resource supply. • Tenant Admin - focuses on VCF Automation, managing resources at that layer. • Tenant User - application developers or DBAs provision databases through the DSM portal or API, consuming resources within the guardrails established by the DSM Admin.
Data Services Namespaces	Maps users to the infrastructure policies that they can consume.

VMware Data Services Deployment Options

Design Area	Options
Appliance Form Factor	Data services VM deployment appliance is offered as a single OVA form factor, comprising of: • 8 vCPUs • 16-GB memory • 595-GB virtual disk storage
Appliance Shared Datastore Options	Can be deployed on any shared storage that supports vSphere features such as vSphere HA and vSphere DRS.
Appliance Virtual Disk Formats	Options are • Thin-provisioned • Thin-provisioned lazy-zeroed • Thin-provisioned eager-zeroed The capabilities of the underlying storage determine the virtual disk format.
Appliance Virtual Networking	• A Provider VM's Management network can connect to a Standard or vSphere Distributed Switch port group. • Database VM workload network can connect to vSphere Distributed Switch port group or can be assigning to an NSX VPC subnet. • NSX VPC can be centralized or distributed.
Appliance Subnet Reservation	DSM uses two internal networks, Docker networks and Workload cluster networks. These networks can be customized from defaults and overridden: • The docker network can be overridden during ovf deploy. • The workload cluster networks can be overridden by updating the Data Services system using the API or the DSM console.

VMware Data Services Design Objectives

Design Objective	Description
Main Objectives	• Provide a self-service platform for database as service in a VCF private cloud architecture. • Standardized life-cycle management applied to database workloads, including.provisioning, backup, cloning, scaling, patching, and upgrades.
VCF Architecture Integration	Single VCF instance architecture • Management domain for control plane component (DSM appliance). See Management Domain Model • Workload domain for workload consumption (Database workloads). See Workload Domain Model
VCF Automation Integration	• Enable service catalog blueprints for self-service provisioning. • Allow for tenant uses to create and consume databases under a Supervisor namespace using VMware Cloud Foundation Automation. See VCF Automation Models
Private AI Database support	Enable Data service to allow provisioning of PostgreSQL databases to facilitate vector database integration for generative AI applications. See Private AI Platform Detailed Design
Directory Authentication	Authentication support for role-based access control
Backup and Recovery	Control Plane and Database to Create multiple database backup locations to segregate control plane data • Provider VM backup and recovery with S3-compatible storage • Database workload backup and recovery with S3 compatible storage
Certificate Management	Using and managing trusted Certificates in VMware Data Services Manager
VCF Operations Integration	Database Metrics forwarding to VCF Operations, see VCF Operations Detailed Design
Log Management Integration	Configure log forwarding to Log management, see High Availability Log Management Model

Implementation Steps for VMware Data Services Manager

Product Grouping	Procedure	Details
VCF Fleet deployment	VCF Fleet in a Single Site with Minimal Footprint	Follow the steps as designated on implementation guidance of a VCF Fleet
Deploying VMware Data Services appliance	• Download the VMware Data Services Manager plugin • Install the plugin using the vSphere Client plugin wizard.	Download the OVA appliance and use the client plugin to upload and deploy to VMware Data services appliance on the vSphere cluster
Configure the VMware Data Services appliance	Configure a local administrator account	Create a local DSM administrator account. Configure an external LDAPS identity source. Configure trusted root certificates. Configure SMTP settings for system notifications. Create a namespace infrastructure policy in the vSphere client
	Configure an external identity source for VMware Data Services Manager.	
	Configure trusted Root Certificates in VMware Data Services Manager.	
	Configuring SMTP Settings in VMware Data Services Manager	
	Create a vSphere Namespace Infrastructure Policy in VMware Data Services Manager	
Configure S3 Storage for DSM and Database backups	Configure Storage Settings in VMware Data Services Manager	Configure and deploy S3-compatible storage • A bucket for Provider Logs • A bucket for Provider Backups • A bucket for Database backup storage
Activate Data Services	Activate Data Services in VMware Data Services Manager	Activate the data services in VMware Data Services Manager to provision databases.
Deploy DSM Consumption Operator	Install the VMware Data Services Manager Supervisor Service	Before tenant users can consume Supervisor resources from the VCF Automation Organization portal, you must install the VMware Data Services Manager Consumption Operator as a Supervisor service in vCenter.
Integrate VCF Automation with DSM	Configure VMware Data Services Manager in VCF Automation	VCF Automation Provider Admin Establishes Connection to DSM. to create and assign data service policies.
Configure metrics forwarding for VCF Operations	Configure Metrics Forwarding in VMware Data Services Manager	Metrics can be to be exported to VCF Operations
Configure VMware DSM for Log Management	Configure Log Forwarding for VMware Data Services Manager	Configure log forwarding for the system services and the data services.

Design Elements for VMware Data Services Manager

Table 563: Data Services Manager Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-PC-DSM-REQD-001	Deploy Data Services Manager in the management domain and register the plugin with a separate workload domain vCenter instance.	• Provides clear separation between management and workload domains. • The Data Services Manager appliance itself deploys in the management domain which typically hosts core VCF components: vCenter, NSX Managers, and VCF Automation. • The Data Services Manager plugin connects to the vCenter instance managing a separate workload domain. This is where applications and DSM-provisioned databases are deployed. • Data Services Manager appliance maintains a 1:1 relationship to a vCenter instance.	Requires management domain and separate workload domain.
VCF-PC-DSM-REQD-002	Protect the Data Services Manager appliance using vSphere HA.	Ensures availability of Data Services Manager appliance without requiring manual intervention during an ESX host failure event.	None.
VCF-PC-DSM-REQD-003	Use shared storage on management domain cluster to place Data Services Manager provider VM.	Ensure that the Data Services Manager appliance can support vSphere vMotion, vSphere DRS, and vSphere HA.	Requires shared storage and enough storage capacity to satisfy the minimum requirements of the appliance.
VCF-PC-DSM-REQD-004	Allocate a single IPv4 address for the Data Services Manager management IP on the same network as core VCF management components.	Simplifies deployment insuring the same management vlan is used for all management components on VCF.	Requires the IP address to be registered with the DNS server for both forward and reverse lookup.
VCF-PC-DSM-REQD-005	Enable vSphere Supervisor on workload clusters for VCF Automation.	vSphere Supervisor enables a Kubernetes-based IaaS control plane on vSphere. Data Services Manager uses a Consumption Operator (installed as a Supervisor Service) to listen for requests from VCF Automation.	Requires vSphere Supervisor be enabled on the targeted workload cluster. For more information, see vSphere Supervisor Models.
VCF-PC-DSM-REQD-006	Use VPC networking for databases (centralized or distributed) provisioned by Data Services Manager.	With NSX VPC, each project or tenant has its own networking policies and security controls, which allows for tenants to avail of • Simplified Management: Database owners can now manage their VPC subnets. • Automate VPC deployment through VMware Cloud Foundation Automation.	• Requires enablement of vSphere Supervisor on the target workload domain. • Requires deployment of NSX Edge cluster for centralized VPC or Virtual Network Appliance cluster for distributed VPC.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-PC-DSM-REQD-007	Size VPC subnet for the workload databases for desired database topology and maintenance operations.	• While a database cluster operates within a topology, it permanently requires one IP address for each node, and two extra cluster-level IP addresses for the database and control plane access. • If a VPC subnet segment is exhausted, you can no longer create database clusters. Scaling or upgrading existing clusters might lead to database unavailability.	Requires configuring a VPC subnet for network resources within a supervisor deployment. For more information, see Workload Connectivity Models
VCF-PC-DSM-REQD-008	Configure a local Data Services Manager administrator user.	If the primary Identity Provider system is down, allows for local access and a break glass account.	• The local user must be in an email address format and stored locally on the appliance. • Dedicate a unique persona that has a valid email address that is associated with the user ID.
VCF-PC-DSM-REQD-009	Configure Active Directory or OpenLDAPS as a Directory Service for Data Services Manager.	• Data Services Manager provides role-based access for two roles. – DSM Admin - A super user that manages the platform. – DSM User creates and maintains individual databases. • This centralizes user management, enhances security through standardized, group-based permissions. **Note:** VCF Automation provisioned data services are managed by VCF Automation independently.	• Data Services Manager requires Lightweight Directory Access Protocol over TLS (LDAPS) as an external identity source. • The servers certificate must be self-signed, or signed by a Certificate Authority (CA).

Design Requirement ID	Design Requirement	Justification	Implication
VCF-PC-DSM-REQD-010	Configure a S3-compatible object store with TLS enabled.	• Data Services Manager's backup architecture requires S3-compatible object storage—external to vSphere datastores. • S3 provider repositories are used for core VMware Data Services Manager storage, backup, logs, and database backups • Create multiple database backup locations to segregate data. • Data Services Manager requires at a minimum — A bucket for Provider Logs. — A bucket for Provider Backups. — Database backup storage. • Use policies to control which bucket each database uses. This provides granular control over data protection and compliance requirements.	• The S3 compatible storage should be external to VCF infrastructure where Data Services Manager appliances and workload are deployed to. • S3 endpoints must be resolvable by the Data Services Manager management IP • For the Provider VM to be able to access S3-compatible storage, you must configure Identity and Access Management (IAM) policies at the S3 storage provider end.
VCF-PC-DSM-REQD-011	Configure an SMTP server for system emails.	• Email notifications for Database operations and alerts are not possible without a configured SMTP server. • Data Services Manager facilitates sending an email to a system user when the user initiates a Forgot Password password reset. To enable this functionality, you must configure an SMTP server for the installation.	• Requires connectivity to SMTP server. • To receive email notifications, directory service users must have a valid email address attribute (`mail` or `rfc822mailbox` in the LDAPS server).
VCF-PC-DSM-REQD-012	Deploy the Data Services Manager Consumption Operator service.	Development teams can leverage VCF Automation's integration to automate database creation and modification to define database resources. The Consumption Operator processes these requests and provisions databases through Data Services Manager.	You must download and install the Data Services Manager Consumption Operator service on vSphere Supervisor.

Table 564: Data Services Manager Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-PC-DSM-RCMD-001	Configure VCF SSO and vCenter Groups on VCF Operations to deploy DSM appliance.	• This allows a VCF SSO role assigned multiple vCenters to deploy DSM appliance. Appropriate permissions can be assigned to SSO role for DSM Deployment and administration. • This allows DSM installer access to both management domain vCenter for appliance placement and Workload vCenter for DSM plugin registration.	• Requires VCF SSO and vCenter Groups configured within the VCF Operations console. • This allows a VCF SSO role that is assigned to multiple vCenters to deploy DSM appliance. • For more information please review. • Identity and Access Management
VCF-PC-DSM-RCMD-002	Configure Metrics forwarding to VCF Operations.	• With metrics forwarding enabled, platform teams gain a single operational view that correlates VM-level resource metrics that are already collected by VCF Operations with database-specific metrics published by Data Services Manager. • Data Services Manager forwards metrics for all three supported database engines: PostgreSQL, MySQL, and Microsoft SQL Server.	Requires the cloud proxy component certificate to establish connectivity.
VCF-PC-DSM-RCMD-003	Configure log forwarding to Log Management.	• Allows you to configure Log Forwarding for VMware Data Services Manager provider VM and provisioned databases. • Logs can be analyzed to improve productivity of the system, deduce trends, or to troubleshoot, as required.	Requires Log managementdeployed within a VCF Fleet.
VCF-PC-DSM-RCMD-004	For air-gapped deployments, provide access to a private container registry.	• For air-gap deployments, a private container registry must be available to host DSM Consumption Operator images. • Data Services Manager does not ship with Microsoft SQL Server images. • MS SQL Server container images must be uploaded to a private container image registry	Requires deployment of a private container image registry that supports the OCI (Open Container Initiative).

Workload Connectivity Detailed Design

The workload connectivity models present different tradeoffs between simplicity, performance, service availability, and scalability. The corresponding design library elements explain the architecture overview, key benefits, limitations, and specific deployment considerations for each model.

Workload Connectivity Models

The following workload connectivity models are supported:

- Centralized Connectivity Model - Supports VCF Automation All Apps Orgs, vSphere Supervisor, and all VPC network and stateful services.
- Distributed VLAN Connectivity Model - Supports VCF Automation All Apps Orgs, vSphere Supervisor, and most VPC network and stateful services.
- Distributed VXLAN Connectivity Model for Version 9.1.0 - Supports EVPN-VXLAN fabric integration.
- Distributed VXLAN Connectivity Model for Version 9.1.1- Supports EVPN-VXLAN fabric integration and VCF Automation All Apps Orgs, vSphere Supervisor, and most VPC network and stateful services.
- NSX Segment Connectivity Model - Supports NSX Federation with VM mobility across VCF Instances.
- VLAN Networking Connectivity Model- Legacy connectivity model without the benefits of private cloud network virtualization.

Centralized Connectivity Model

The *Centralized Connectivity Model* uses an NSX Edge cluster with Tier-0 gateway to provide VPC-compatible workload domain with network connectivity established via a Centralized Transit Gateway.

Centralized Connectivity Model Overview

The model uses a primary (Tier-0) gateway configured in Active/Active or Active/Standby model. The Tier-0 gateway HA mode is independent of the Transit Gateway HA mode, this allows the Tier-0 gateway to have an Active/Active mode while a Transit Gateway can use Active/Standby. Individual tenants can connect to the Tier-0 gateway through their own Centralized Transit Gateways (CTGWs). This workload networking model is the base building block to support all the VCF Automation tenancy models that are described in VCF Automation Provider Tenant Models and Multi-Tenancy Consumption Patterns. It can also be used in single tenant deployments without the integration with VCF Automation or for vSphere Supervisor deployments without VCF Automation.

Centralized Connectivity Model Attributes

A *Centralized Connectivity Model* has the following attributes.

Attribute	Detail
Network Consumption Models	• VCF Automation All Apps Orgs Consumption Model Consider Adding Supervisor Only Consumption Model • Virtual Private Cloud (VPC) Consumption Model
Transit Gateway Type	Centralized
Node Type	NSX Edge node required
Tier-0 Gateway HA Mode	• Active/Active • Active/Standby
Transit Gateway HA Mode	• Active/Active • Active/Standby – Required for VCF Automation All Apps Orgs and vSphere Supervisor
VPC Gateway Services	• VPC Gateway without North/South Services • VPC Gateway with North/South Services – Required forVCF Automation All Apps Orgs and vSphere Supervisor
Load Balancing Options	• NSX Edge load balancer • Avi Load Balancer
NSX Tenancy Constructs	NSX Projects, VPCs, Multiple Transit Gateways

Attribute	Detail
Consumption Options	• VCF Automation All Apps Orgs • vSphere Supervisor (i.e. VM Service, VKS, and more) • vCenter UI/API • NSX UI/API
External IP Blocks	• Multiple (up to 5)
Provider Gateway Type	• Tenant Dedicated Tier-0 Gateway • Tenant Shared Tier-0 Gateway • Tenant Dedicated VRF Gateway

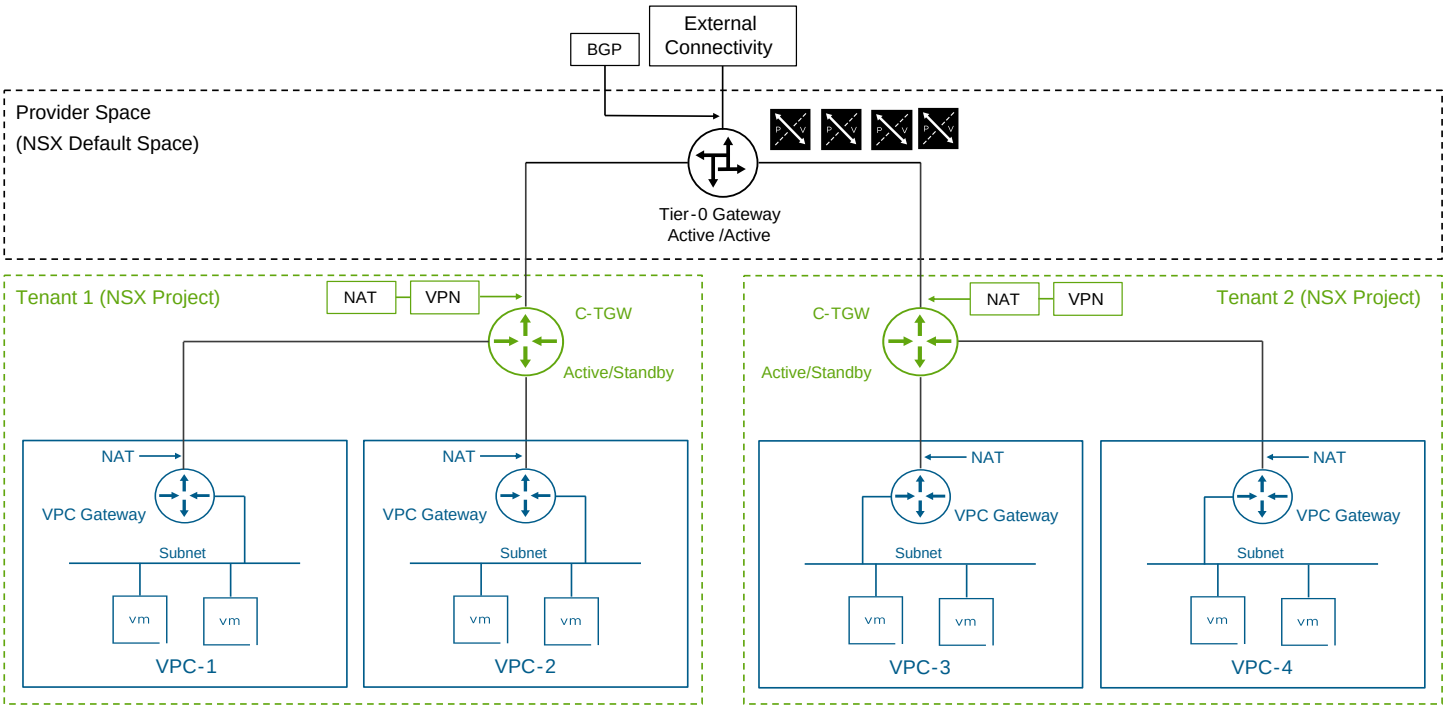
Centralized Connectivity Model Options

Design Area	Options
External Connections	• Single Tier-0/VRF Gateway • Multiple Tier-0/VRF Gateways
Tier-0 Gateway/VRF Deployment Options	• Single Tenant • Shared across multiple tenants • Dedicated, one Tier-0 gateway for each tenant
NSX Edge Cluster	• Shared across Tier-0 Gateway, Transit Gateway, and VPCs • Dedicated clusters for Tier-0 Gateway, Transit Gateway, and VPCs
VRF (Only applies if the Gateway is shared)	Yes or No

Centralized Connectivity Model with Shared Tier-0 Gateway Per Tenant

Multiple tenants can share a single Tier-0 gateway and its NSX Edge node resources. When used with VPC Networking for outbound connectivity the NAT process happens when VPC subnets are set to "Private TGW" or "Private VPC" access model with Auto-SNAT enabled.

Figure 284: Shared Tier-0 Gateway



Centralized Connectivity Model Design Requirements and Recommendations for Shared Tier-0 Gateway per Tenant

Table 565: Centralized Connectivity Model Common Design Requirements

Use this list of general requirements for the *Centralized Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-CTGW-REQD-CFG-001	A minimum of two (2) NSX Edge nodes in the NSX edge cluster. Select one of the presented NSX Edge Cluster Models.	Two (2) or more NSX Edge nodes are required to support a Tier-0 gateway with high availability.	Additional resources must be available to support the NSX edge nodes.
VCF-WLDCON-CTGW-REQD-CFG-002	Use Centralized Connectivity for the connectivity of the Transit Gateway.	This model requires the centralized services that are provided by the NSX Edge nodes.	Additional resources must be available to support the NSX edge node, following one of the models described in the NSX edge design section here NSX Edge Cluster Detailed Design
VCF-WLDCON-CTGW-REQD-CFG-003	When using stateful services in the VPC Connectivity Profile enable N-S Services.	N-S services are required for VPC's Default Outbound NAT (Auto-SNAT), SNAT, and DNAT services.	Any traffic between different VPCs traverse the NSX Edge node.
VCF-WLDCON-CTGW-REQD-CFG-004	In the VPC Connectivity Profile, assign an IP range of enterprise-wide routable IPs to the external IP block (a /24 CIDR or larger is preferred).	This IP range is used for any tenant outbound and inbound traffic. It means that any NAT IP, Public subnet range, or load balancer VIP will be assigned from this range.	If the IP range is exhausted, additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.

Table 566: Centralized Connectivity Model Common Design Recommendations

Use this list of general recommendations for the *Centralized Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-001	Configure the Tier-0 gateway in Active/Active mode.	• A Tier-0 gateway in Active/Active mode supports scale-out up to eight (8) NSX Edge nodes, increasing north-south throughput. • North-south traffic is distributed across all NSX Edge nodes within the same NSX Edge cluster attached to the Tier-0 gateway.	Only Tier-0 gateway in Active/Standby mode support stateful network services.
VCF-WLDCON-CTGW-RCMD-CFG-002	Use a unique private BGP Autonomous System Number (ASN) for the Tier-0 gateway.	• It prevents advertisements from being dropped by the BGP loop detection check. If the peering devices only advertise the default route, having a unique BGP AS may not be necessary. • It is assumed that the Tier-0 gateway is peering to internal devices and it's not an Internet-facing router, in which case a public AS number is required.	A unique BGP Autonomous System Number (ASN) must be available for the Tier-0 gateway.
VCF-WLDCON-CTGW-RCMD-CFG-003	Each of the NSX Edge nodes where the Tier-0 gateway runs will peer with two separate physical network devices.	It provides redundancy at the physical network layer, without requiring an NSX Edge node failover.	• If the NSX Edge node is deployed in a virtual machine form factor, the ESX host where it runs must be connected to two (2) physical switches running BGP. • If the NSX Edge node is a bare-metal NSX Edge, the server where the bare-metal NSX Edge software is installed must be connected to two (2) physical switches running BGP.
VCF-WLDCON-CTGW-RCMD-CFG-004	The physical network devices peering to the Tier-0 gateway are physically connected to the ESX host where the NSX Edge nodes run, or to the bare-metal NSX Edge.	• The Layer 3 peering is not dependent on any Layer 2 technology, making it easier to map a physical failure to a routing convergence. • Peering over a VLAN transported over multiple physical switches and over potentially multiple redundant paths increases the operational complexity of the design.	The physical switches connected to the ESX host where the NSX Edge node is running or those connected to a bare-metal NSX Edge must be BGP capable.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-005	If using NSX Edge VMs, use two (2) interfaces for data path fastpath traffic.	• Two (2) data plane fastpath interfaces provide redundancy for both VLAN and Overlay traffic. Using the additional two (2) data plane fastpath interfaces does not increase the performance of the NSX Edge VM in the majority of situations, in particular when the additional vNics leverage the same set of pNICs. • When the ESX hosts have four (4) pNICs available to the NSX Edge appliances, we recommend the use of multiple NSX Edge nodes connected to different sets of pNICs.	Two (2) physical NICs must be available on the ESX hosts where the NSX Edge node in VM form factor is running.
VCF-WLDCON-CTGW-RCMD-CFG-006	If using bare-metal NSX Edge nodes, use four (4) pNIC interfaces for data path traffic	Bare-metal NSX Edge nodes are generally limited by the speed of their interfaces. In most cases, increasing the number of interfaces in use will linearly improve the throughput that can be achieved.	• Four (4) data plane fastpath pNIC must be available on the bare-metal NSX Edge server. • Using two (2) 100G pNICs is a good option for bare-metal NSX Edge nodes when four (4) pNICs are not available.
VCF-WLDCON-CTGW-RCMD-CFG-007	Each NSX Edge node data path interface transports both VLAN and TEP traffic.	Two (2) data path interfaces are sufficient to provide redundancy for both types of traffic.	Inbound and outbound traffic from the virtual network are mixed on the same vNICs, making monitoring less intuitive.
VCF-WLDCON-CTGW-RCMD-CFG-008	Every NSX Edge node shares the same VLAN to peer to a specific physical network device. With two (2) physical network devices, it means two (2) peering VLANs are needed.	Adding or replacing additional NSX Edge nodes to the NSX Edge node cluster doesn't require creating new VLANs on the physical network fabric.	Peering subnet ranges should accommodate NSX Edge cluster growth.
VCF-WLDCON-CTGW-RCMD-CFG-009	Use /28 or larger uplink peering subnets are recommended.	Allow to scale the Tier-0 gateway to eight (8) NSX Edge nodes when in an Active/Active configuration.	For an Active/Standby Tier-0, the peering subnets can be smaller as the Tier-0 gateway is limited to two (2) NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-010	Peering VLANs are transported on trunk port groups connected to the edge fp-eth0 and fp-eth1 interfaces.	• Trunk port groups allow to transport both peering VLAN and TEP VLAN traffic on the same NSX Edge node vNIC. • Trunk port groups allow seamless integration of VRFs and service interfaces for VLANs.	• Peering VLANs and TEP VLAN are transported on the same peering port groups but they have different NIC teaming requirements. • Peering VLANs do not need to failover between uplinks, the TEP VLAN does. • The teaming policy for port group must be active/ standby to satisfy the TEP VLAN requirement. • Physical switches should prune the peering VLAN from their interfaces where they are not required and should avoid transporting them on the inter-switch link, or via a physical overlay extension (e.g., VXLAN in an EVPN fabric)
VCF-WLDCON-CTGW-RCMD-CFG-011	Set the MTU of the Tier-0 gateway external interfaces to the MTU of the peering interface on the physical network switches (e.g., 9000).	Some network devices set the DF bit to "Do not Fragment" on the BGP peering session, a mismatch in the MTU may cause peering issues.	MTU needs to match the physical network
VCF-WLDCON-CTGW-RCMD-CFG-012	Set the global gateway MTU to the value of the external interfaces of the Tier-0 gateway minus 200 (i.e., 8800 = 9000 -200).	Avoid fragmentation at the NSX layer when packets larger than the default 1500 bytes are exchanged between workloads in the virtual network and workloads on the physical network.	Must be set post deployment in NSX as a global setting.
VCF-WLDCON-CTGW-RCMD-CFG-013	Enable BFD between the Tier-0 gateway and the physical network devices.	Provide fast failure detection and failover with minimal CPU overhead. It permits leaving the BGP timers to their default.	Physical switches need to support BFD.
VCF-WLDCON-CTGW-RCMD-CFG-014	Use the default BGP timers, 60/180 seconds.	When BFD is enabled, there is no need to lower the default BGP timer settings increasing NSX Edge node CPU overhead.	• BFD is required for fast failure detection and failover. • If BFD is not available on the physical switches, BGP timers can be lowered to allow a faster failover.
VCF-WLDCON-CTGW-RCMD-CFG-015	On the Tier-0 gateway, enable redistribution of the following network types into BGP: • Transit Gateway Static	In this model, all workloads are connected to transit gateways. This setting allows for VPC public subnets, NAT IPs, and load balancer VIPs to be advertised to the physical network.	All VPC public subnets, NAT IPs, and load balancer VIPs will be advertised to all BGP peers. If filtering is required, route filters should be configured.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-016	• Create a ingress route filter allowing the advertisement of the default route only from the physical network to the Tier-0 gateway. • A corresponding summarization from the physical network is recommended.	• In most cases, the virtual network is a stub network with a single exit point. The filtering reduces the Tier-0 gateway routing table, simplifying troubleshooting and potentially reducing converge time. • It also prevents RFP checks from being triggered by asymmetric routing advertisements from the physical switches.	• If the virtual network has multiple exit points, for example, one to shared services and one as a private interconnection to the tenant's remote data center, this recommendation cannot be implemented. • In such cases reducing the routing table size via summarization is still recommended.
VCF-WLDCON-CTGW-RCMD-CFG-017	Leave the RPF check enabled (set to strict) on all four external interfaces of the Tier-0 gateway.	RPF check mitigates IP Address spoofing attacks.	• Asymmetric routing advertisements (Not asymmetric routing, which is normal in any ECMP configuration andsupported by the RFP check) from the physical switches will trigger the RPF check and drop nonmalicious traffic. • RPF check should be enabled only if consistent routing advertisement from all BGP peers can be guaranteed via route summarization and filtering.
VCF-WLDCON-CTGW-RCMD-CFG-018	Set BGP Graceful Restart to Helper-only.	BGP Graceful Restart may slow down the BGP convergence by waiting for the peer device control plane to recover. Helper-only ensures that routing convergence is delayed only when a peer device negotiates BGP Graceful Restart. The helper-only setting will allow the Tier-0 gateway to adapt to the peer device behavior.	If the peer device doesn't support BGP Graceful Restart or if BGP Graceful Restart is not desirable it may be better to disable BGP Graceful Restart altogether.
VCF-WLDCON-CTGW-RCMD-CFG-019	Set the Forwarding Up timer to 5 seconds (Default).	This will delay traffic from being forwarded to the NSX Edge node from the virtual network before the routing table is fully populated.	
VCF-WLDCON-CTGW-RCMD-CFG-020	In the VPC Connectivity Profile, assign an IP range of non-routable IPs to the private-TGW IP block (a /23 CIDR or larger is preferred).	This IP range will be used for inter-VPC connectivity within the tenant boundary. It will be used to assign CIDRs for Private TGW VPC Subnets.	Workloads on private-TGW subnets will be reachable from any other tenant workloads without requiring NAT, but no connectivity from workloads outside of the tenant boundary will be available without exposing the workloads via a NAT Rule, or a load balancer VIP, or an External IP (stateless 1:1 NAT).

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-021	To allow for granular application placement restrict the network span of VPC subnets to one or more vSphere clusters within an NSX instance. This is typically required for remote vSphere clusters or DMZ deployments.	• Provides isolation of VPC subnets and applications in different vSphere clusters. • The network span is defined on the Centralized TGW and applies to all connected VPCs and their associated VPC subnets. • vSphere clusters can be shared across multiple network spans.	Exclusive network span allows defining a dedicated network span for a specific TGW and the associated VPCs, preventing it from being shared or overlapping with other network spans.

Table 567: Centralized Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Requirements

Use this list of additional requirements when adopting the *Centralized Connectivity Model* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-CTGW-REQD-CFG-005	In the Default VPC Connectivity Profile of the Default Project enable N-S Services.	N-S Services are required for VPC's Default Outbound NAT (Auto-SNAT), SNAT, and DNAT services, which are necessary for vSphere Supervisor and VCFA All Apps Orgs. N-S Services is required for vSphere Supervisor and VCF-A All Apps Orgs.	Any traffic between different VPCs will traverse the NSX Edge node.
VCF-WLDCON-CTGW-REQD-CFG-006	In the Default VPC Connectivity Profile of the Default Project enable Default Outbound NAT (Auto-SNAT).	Default Outbound NAT (Auto-SNAT) is necessary for vSphere Supervisor and VCFA All Apps Orgs.	Any traffic between different VPCs will traverse the NSX Edge node.
VCF-WLDCON-CTGW-REQD-CFG-007	In the Default VPC Connectivity Profile of the Default Project, assign an IP range of non-routable IPs to the private-TGW IP block (a /23 CIDR or larger is preferred).	This private-TGW IP range will be used for inter-VPC connectivity within the tenant boundary. It will be used to assign CIDRs for private-TGW VPC Subnets.	• Workloads on private-TGW subnets will be reachable from any other tenant workloads without requiring NAT, but no connectivity from workloads outside of the tenant boundary will be available without exposing the workloads via a NAT Rule, or a Load Balancer VIP, or an External IP (stateless 1:1 NAT). • If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.

Table 568: Centralized Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Recommendations

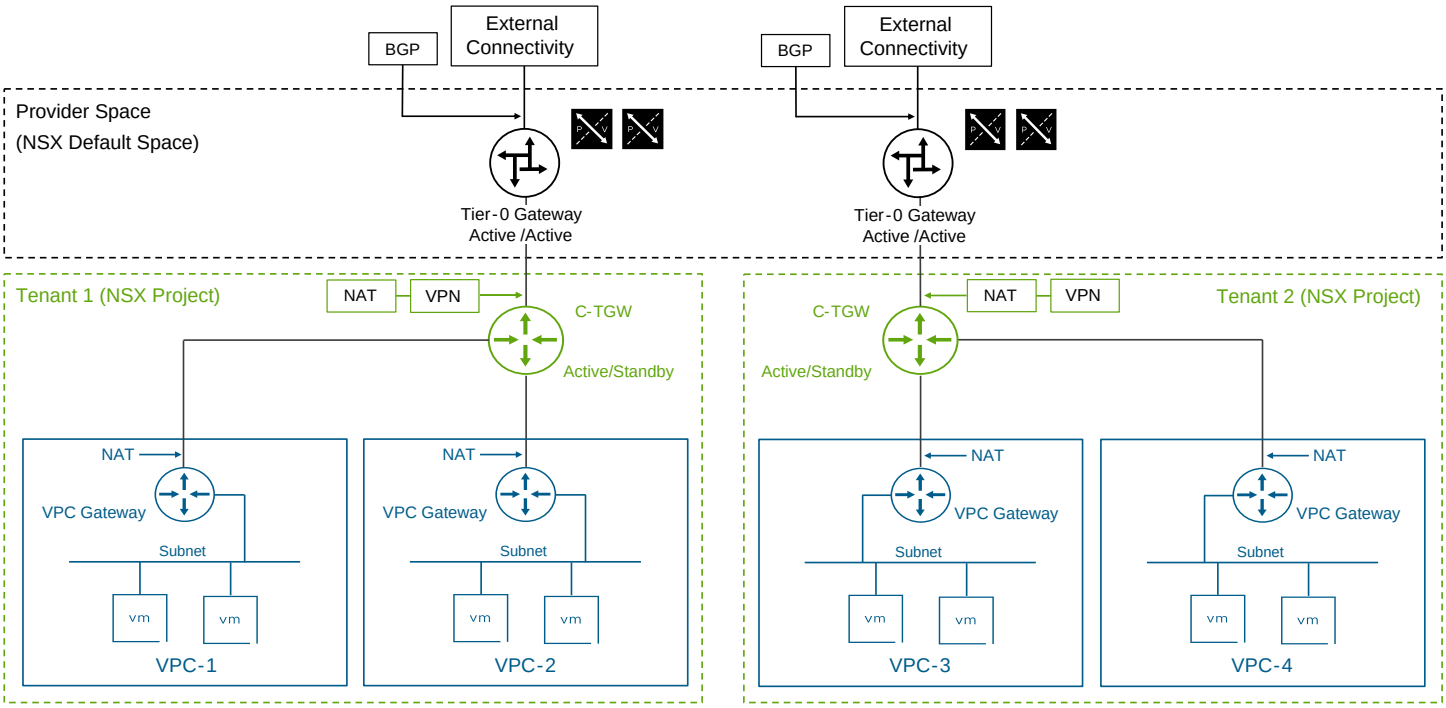
Use this list of additional recommendations when adopting the *Centralized Connectivity Model* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-022	Use large or X-Large nodes if using NSX Edge appliance form factor, or use bare-metal NSX Edge.	Supports better scale for Supervisor cluster load balancing services to be instantiated on the NSX Edge nodes.	Additional resources must be available to support the NSX Edge nodes.
VCF-WLDCON-CTGW-RCMD-CFG-023	Use the Default Project for the vSphere Supervisor namespace's VPCs.	• When VCFA All Apps Orgs are in use, in each Organization will have its dedicated NSX Project. • The Default space will not be used and it will be available to be consumed by system VPCs to support the IaaS control plane. • The WLD creation workflow follows this model when the new WLD has a new dedicated NSX Manager.	• This approach is not recommended if the Default Project is also consumed via other means, for example via the vCenter UI, which allows the configuration of VPCs in the Default Project. • In general, when VCF Automation is in use, it's not recommended to mix another network consumption vehicle in the same workload domain.
VCF-WLDCON-CTGW-RCMD-CFG-024	In the Default VPC Connectivity Profile of the Default Project set the NSX Edge cluster as the same where the Tier-0 gateway is deployed.	This choice will lead to the VPC centralized services running on the same NSX Edge nodes as the Tier-0 gateway and TGW.	• NSX Edge nodes will leverage additional CPU cycles to serve Tier-0 gateway, TGW, and VPC centralized services. Monitor the CPU utilization via one of the following methods: – NSX Edge CLI – NSX API/UI – VCF Ops • Do not use the vCenter UI graphs to monitor the NSX Edge CPU utilization as it will provide unreliable results. • If maximum performance is required consider using dediacted NSX Edge clusters for the Transit Gateway and Tier-0 gateway.

Centralized Connectivity Model with Dedicated Gateway per Tenant

Each tenant can have its own dedicated Tier-0 Gateway. This supports direct routing of tenant-specific IP ranges to the Tier-0 Gateway. This is achieved by creating a dedicated centralized external connection for each tenant Centralized Transit Gateway.

Figure 285: Dedicated Gateway per Tenant Option



Centralized Connectivity Model Design Requirements and Recommendations for Dedicated Gateway per Tenant

Table 569: Centralized Connectivity Model with Dedicated Gateway per Tenant Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-CTGW-REQD-CFG-005	Two (2) or more NSX Edge nodes must be available for each tenant adhering to the dedicated gateway deployment model.	• Each tenant will have exclusive access to the resources provided by the NSX Edge nodes in the NSX Edge node cluster. This will limit resource contention and will allow the multi-tenant design to scale. • Also, the provider will be able to configure tenant services on a tenant-dedicated Tier-0 gateway, for example VPNs, leveraging tenant-specific IP addresses for local and remote networks.	Additional ESX hosts or pNICs must be available to support the additional NSX Edge nodes. In most cases, connecting multiple edge nodes to the same pair of pNICs will not achieve the desired scalability objective.
VCF-WLDCON-CTGW-REQD-CFG-006	A Tier-0 gateway in Active/Active or Active/Standby mode must be deployed for each tenant adhering to the dedicated gateway deployment model.	Required for the dedicated gateway deployment model.	Additional NSX Edge clusters need to be deployed, one for each tenant.

Table 570: Centralized Connectivity Model Common Design Requirements

Use this list of general requirements for the *Centralized Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-CTGW-REQD-CFG-001	A minimum of two (2) NSX Edge nodes in the NSX edge cluster. Select one of the presented NSX Edge Cluster Models.	Two (2) or more NSX Edge nodes are required to support a Tier-0 gateway with high availability.	Additional resources must be available to support the NSX edge nodes.
VCF-WLDCON-CTGW-REQD-CFG-002	Use Centralized Connectivity for the connectivity of the Transit Gateway.	This model requires the centralized services that are provided by the NSX Edge nodes.	Additional resources must be available to support the NSX edge node, following one of the models described in the NSX edge design section here NSX Edge Cluster Detailed Design
VCF-WLDCON-CTGW-REQD-CFG-003	When using stateful services in the VPC Connectivity Profile enable N-S Services.	N-S services are required for VPC's Default Outbound NAT (Auto-SNAT), SNAT, and DNAT services.	Any traffic between different VPCs traverse the NSX Edge node.
VCF-WLDCON-CTGW-REQD-CFG-004	In the VPC Connectivity Profile, assign an IP range of enterprise-wide routable IPs to the external IP block (a /24 CIDR or larger is preferred).	This IP range is used for any tenant outbound and inbound traffic. It means that any NAT IP, Public subnet range, or load balancer VIP will be assigned from this range.	If the IP range is exhausted, additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.

Table 571: Centralized Connectivity Model Common Design Recommendations

Use this list of general recommendations for the *Centralized Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-001	Configure the Tier-0 gateway in Active/Active mode.	• A Tier-0 gateway in Active/Active mode supports scale-out up to eight (8) NSX Edge nodes, increasing north-south throughput. • North-south traffic is distributed across all NSX Edge nodes within the same NSX Edge cluster attached to the Tier-0 gateway.	Only Tier-0 gateway in Active/Standby mode support stateful network services.
VCF-WLDCON-CTGW-RCMD-CFG-002	Use a unique private BGP Autonomous System Number (ASN) for the Tier-0 gateway.	• It prevents advertisements from being dropped by the BGP loop detection check. If the peering devices only advertise the default route, having a unique BGP AS may not be necessary. • It is assumed that the Tier-0 gateway is peering to internal devices and it's not an Internet-facing router, in which case a public AS number is required.	A unique BGP Autonomous System Number (ASN) must be available for the Tier-0 gateway.
VCF-WLDCON-CTGW-RCMD-CFG-003	Each of the NSX Edge nodes where the Tier-0 gateway runs will peer with two separate physical network devices.	It provides redundancy at the physical network layer, without requiring an NSX Edge node failover.	• If the NSX Edge node is deployed in a virtual machine form factor, the ESX host where it runs must be connected to two (2) physical switches running BGP. • If the NSX Edge node is a bare-metal NSX Edge, the server where the bare-metal NSX Edge software is installed must be connected to two (2) physical switches running BGP.
VCF-WLDCON-CTGW-RCMD-CFG-004	The physical network devices peering to the Tier-0 gateway are physically connected to the ESX host where the NSX Edge nodes run, or to the bare-metal NSX Edge.	• The Layer 3 peering is not dependent on any Layer 2 technology, making it easier to map a physical failure to a routing convergence. • Peering over a VLAN transported over multiple physical switches and over potentially multiple redundant paths increases the operational complexity of the design.	The physical switches connected to the ESX host where the NSX Edge node is running or those connected to a bare-metal NSX Edge must be BGP capable.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-005	If using NSX Edge VMs, use two (2) interfaces for data path fastpath traffic.	• Two (2) data plane fastpath interfaces provide redundancy for both VLAN and Overlay traffic. Using the additional two (2) data plane fastpath interfaces does not increase the performance of the NSX Edge VM in the majority of situations, in particular when the additional vNics leverage the same set of pNICs. • When the ESX hosts have four (4) pNICs available to the NSX Edge appliances, we recommend the use of multiple NSX Edge nodes connected to different sets of pNICs.	Two (2) physical NICs must be available on the ESX hosts where the NSX Edge node in VM form factor is running.
VCF-WLDCON-CTGW-RCMD-CFG-006	If using bare-metal NSX Edge nodes, use four (4) pNIC interfaces for data path traffic	Bare-metal NSX Edge nodes are generally limited by the speed of their interfaces. In most cases, increasing the number of interfaces in use will linearly improve the throughput that can be achieved.	• Four (4) data plane fastpath pNIC must be available on the bare-metal NSX Edge server. • Using two (2) 100G pNICs is a good option for bare-metal NSX Edge nodes when four (4) pNICs are not available.
VCF-WLDCON-CTGW-RCMD-CFG-007	Each NSX Edge node data path interface transports both VLAN and TEP traffic.	Two (2) data path interfaces are sufficient to provide redundancy for both types of traffic.	Inbound and outbound traffic from the virtual network are mixed on the same vNICs, making monitoring less intuitive.
VCF-WLDCON-CTGW-RCMD-CFG-008	Every NSX Edge node shares the same VLAN to peer to a specific physical network device. With two (2) physical network devices, it means two (2) peering VLANs are needed.	Adding or replacing additional NSX Edge nodes to the NSX Edge node cluster doesn't require creating new VLANs on the physical network fabric.	Peering subnet ranges should accommodate NSX Edge cluster growth.
VCF-WLDCON-CTGW-RCMD-CFG-009	Use /28 or larger uplink peering subnets are recommended.	Allow to scale the Tier-0 gateway to eight (8) NSX Edge nodes when in an Active/Active configuration.	For an Active/Standby Tier-0, the peering subnets can be smaller as the Tier-0 gateway is limited to two (2) NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-010	Peering VLANs are transported on trunk port groups connected to the edge fp-eth0 and fp-eth1 interfaces.	• Trunk port groups allow to transport both peering VLAN and TEP VLAN traffic on the same NSX Edge node vNIC. • Trunk port groups allow seamless integration of VRFs and service interfaces for VLANs.	• Peering VLANs and TEP VLAN are transported on the same peering port groups but they have different NIC teaming requirements. • Peering VLANs do not need to failover between uplinks, the TEP VLAN does. • The teaming policy for port group must be active/ standby to satisfy the TEP VLAN requirement. • Physical switches should prune the peering VLAN from their interfaces where they are not required and should avoid transporting them on the inter-switch link, or via a physical overlay extension (e.g., VXLAN in an EVPN fabric)
VCF-WLDCON-CTGW-RCMD-CFG-011	Set the MTU of the Tier-0 gateway external interfaces to the MTU of the peering interface on the physical network switches (e.g., 9000).	Some network devices set the DF bit to "Do not Fragment" on the BGP peering session, a mismatch in the MTU may cause peering issues.	MTU needs to match the physical network
VCF-WLDCON-CTGW-RCMD-CFG-012	Set the global gateway MTU to the value of the external interfaces of the Tier-0 gateway minus 200 (i.e., 8800 = 9000 -200).	Avoid fragmentation at the NSX layer when packets larger than the default 1500 bytes are exchanged between workloads in the virtual network and workloads on the physical network.	Must be set post deployment in NSX as a global setting.
VCF-WLDCON-CTGW-RCMD-CFG-013	Enable BFD between the Tier-0 gateway and the physical network devices.	Provide fast failure detection and failover with minimal CPU overhead. It permits leaving the BGP timers to their default.	Physical switches need to support BFD.
VCF-WLDCON-CTGW-RCMD-CFG-014	Use the default BGP timers, 60/180 seconds.	When BFD is enabled, there is no need to lower the default BGP timer settings increasing NSX Edge node CPU overhead.	• BFD is required for fast failure detection and failover. • If BFD is not available on the physical switches, BGP timers can be lowered to allow a faster failover.
VCF-WLDCON-CTGW-RCMD-CFG-015	On the Tier-0 gateway, enable redistribution of the following network types into BGP: • Transit Gateway Static	In this model, all workloads are connected to transit gateways. This setting allows for VPC public subnets, NAT IPs, and load balancer VIPs to be advertised to the physical network.	All VPC public subnets, NAT IPs, and load balancer VIPs will be advertised to all BGP peers. If filtering is required, route filters should be configured.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-016	• Create a ingress route filter allowing the advertisement of the default route only from the physical network to the Tier-0 gateway. • A corresponding summarization from the physical network is recommended.	• In most cases, the virtual network is a stub network with a single exit point. The filtering reduces the Tier-0 gateway routing table, simplifying troubleshooting and potentially reducing converge time. • It also prevents RFP checks from being triggered by asymmetric routing advertisements from the physical switches.	• If the virtual network has multiple exit points, for example, one to shared services and one as a private interconnection to the tenant's remote data center, this recommendation cannot be implemented. • In such cases reducing the routing table size via summarization is still recommended.
VCF-WLDCON-CTGW-RCMD-CFG-017	Leave the RPF check enabled (set to strict) on all four external interfaces of the Tier-0 gateway.	RPF check mitigates IP Address spoofing attacks.	• Asymmetric routing advertisements (Not asymmetric routing, which is normal in any ECMP configuration andsupported by the RFP check) from the physical switches will trigger the RPF check and drop nonmalicious traffic. • RPF check should be enabled only if consistent routing advertisement from all BGP peers can be guaranteed via route summarization and filtering.
VCF-WLDCON-CTGW-RCMD-CFG-018	Set BGP Graceful Restart to Helper-only.	BGP Graceful Restart may slow down the BGP convergence by waiting for the peer device control plane to recover. Helper-only ensures that routing convergence is delayed only when a peer device negotiates BGP Graceful Restart. The helper-only setting will allow the Tier-0 gateway to adapt to the peer device behavior.	If the peer device doesn't support BGP Graceful Restart or if BGP Graceful Restart is not desirable it may be better to disable BGP Graceful Restart altogether.
VCF-WLDCON-CTGW-RCMD-CFG-019	Set the Forwarding Up timer to 5 seconds (Default).	This will delay traffic from being forwarded to the NSX Edge node from the virtual network before the routing table is fully populated.	
VCF-WLDCON-CTGW-RCMD-CFG-020	In the VPC Connectivity Profile, assign an IP range of non-routable IPs to the private-TGW IP block (a /23 CIDR or larger is preferred).	This IP range will be used for inter-VPC connectivity within the tenant boundary. It will be used to assign CIDRs for Private TGW VPC Subnets.	Workloads on private-TGW subnets will be reachable from any other tenant workloads without requiring NAT, but no connectivity from workloads outside of the tenant boundary will be available without exposing the workloads via a NAT Rule, or a load balancer VIP, or an External IP (stateless 1:1 NAT).

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-021	To allow for granular application placement restrict the network span of VPC subnets to one or more vSphere clusters within an NSX instance. This is typically required for remote vSphere clusters or DMZ deployments.	• Provides isolation of VPC subnets and applications in different vSphere clusters. • The network span is defined on the Centralized TGW and applies to all connected VPCs and their associated VPC subnets. • vSphere clusters can be shared across multiple network spans.	Exclusive network span allows defining a dedicated network span for a specific TGW and the associated VPCs, preventing it from being shared or overlapping with other network spans.

Table 572: Centralized Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Requirements

Use this list of additional requirements when adopting the *Centralized Connectivity Model* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-CTGW-REQD-CFG-005	In the Default VPC Connectivity Profile of the Default Project enable N-S Services.	N-S Services are required for VPC's Default Outbound NAT (Auto-SNAT), SNAT, and DNAT services, which are necessary for vSphere Supervisor and VCFA All Apps Orgs. N-S Services is required for vSphere Supervisor and VCF-A All Apps Orgs.	Any traffic between different VPCs will traverse the NSX Edge node.
VCF-WLDCON-CTGW-REQD-CFG-006	In the Default VPC Connectivity Profile of the Default Project enable Default Outbound NAT (Auto-SNAT).	Default Outbound NAT (Auto-SNAT) is necessary for vSphere Supervisor and VCFA All Apps Orgs.	Any traffic between different VPCs will traverse the NSX Edge node.
VCF-WLDCON-CTGW-REQD-CFG-007	In the Default VPC Connectivity Profile of the Default Project, assign an IP range of non-routable IPs to the private-TGW IP block (a /23 CIDR or larger is preferred).	This private-TGW IP range will be used for inter-VPC connectivity within the tenant boundary. It will be used to assign CIDRs for private-TGW VPC Subnets.	• Workloads on private-TGW subnets will be reachable from any other tenant workloads without requiring NAT, but no connectivity from workloads outside of the tenant boundary will be available without exposing the workloads via a NAT Rule, or a Load Balancer VIP, or an External IP (stateless 1:1 NAT). • If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.

Table 573: Centralized Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Recommendations

Use this list of additional recommendations when adopting the *Centralized Connectivity Model* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-022	Use large or X-Large nodes if using NSX Edge appliance form factor, or use bare-metal NSX Edge.	Supports better scale for Supervisor cluster load balancing services to be instantiated on the NSX Edge nodes.	Additional resources must be available to support the NSX Edge nodes.
VCF-WLDCON-CTGW-RCMD-CFG-023	Use the Default Project for the vSphere Supervisor namespace's VPCs.	• When VCFA All Apps Orgs are in use, in each Organization will have its dedicated NSX Project. • The Default space will not be used and it will be available to be consumed by system VPCs to support the IaaS control plane. • The WLD creation workflow follows this model when the new WLD has a new dedicated NSX Manager.	• This approach is not recommended if the Default Project is also consumed via other means, for example via the vCenter UI, which allows the configuration of VPCs in the Default Project. • In general, when VCF Automation is in use, it's not recommended to mix another network consumption vehicle in the same workload domain.
VCF-WLDCON-CTGW-RCMD-CFG-024	In the Default VPC Connectivity Profile of the Default Project set the NSX Edge cluster as the same where the Tier-0 gateway is deployed.	This choice will lead to the VPC centralized services running on the same NSX Edge nodes as the Tier-0 gateway and TGW.	• NSX Edge nodes will leverage additional CPU cycles to serve Tier-0 gateway, TGW, and VPC centralized services. Monitor the CPU utilization via one of the following methods: – NSX Edge CLI – NSX API/UI – VCF Ops • Do not use the vCenter UI graphs to monitor the NSX Edge CPU utilization as it will provide unreliable results. • If maximum performance is required consider using dediacted NSX Edge clusters for the Transit Gateway and Tier-0 gateway.

Centralized Connectivity Model with VRF Integration

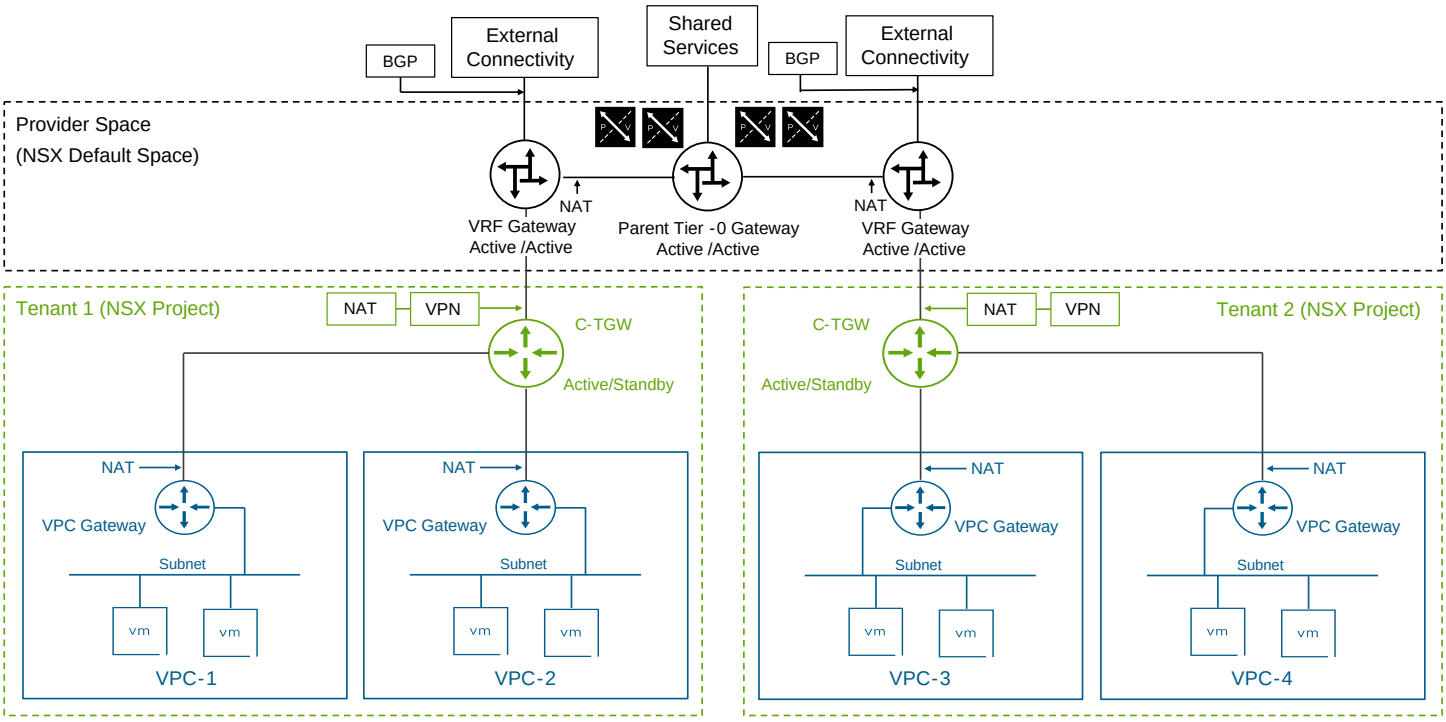
When tenants need dedicated external network access but dedicated NSX Edge nodes are impractical, NSX VRFs (Virtual Routing and Forwarding) offer a solution:

- VRFs provide similar capabilities to dedicated Tier-0 Gateways
- Multiple VRFs can share a single edge node with their parent Tier-0 Gateway
- A Centralized Transit Gateway can also have multiple gateway connections, either VRF or Tier-0.
  - This can provide different communication paths through different external connections, e.g. Datacenter and Internet.
  - Outbound route adveristment of the External IP block and filtering of inbound routes can be used to achieve path selection for traffic to and from the Transit Gaetway.
- VRF limitations in this topology

- Gateway firewall and VPN services are restricted to uplink interfaces

VRF Gateways can be shared by multiple tenants. This variation may be desirable when multiple tenants, but not all, need access to the same set of upstream resources, such as VPNs, private WAN circuits, or VRFs in the physical network. When all tenants need access to the same upstream resources, a shared Tier-0 gateway is the most appropriate choice.

Figure 286: Dedicated VRF Gateway Option



Centralized Connectivity Model Design Requirements and Recommendations for VRF Integration

Table 574: Centralized Connectivity Model with VRF Integration Design Requirements

Use this list of additional requirements when adopting the *Centralized Connectivity Model* to configure virtual networking for a environment with VRF Integration for tenants that need dedicated external network access without dedicated NSX Edge nodes per tenant.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-CTGW-REQD-CFG-010	The tenant VRFs must be associated with the Tier-0 gateway (Parent Tier-0 gateway)	NSX VRF gateway only supported with a Parent Tier-0 gateway	The VRF Gateway will inherent the HA mode of the parent Tier-0 gateway, therefore it is recommended to create the Parent Tier-0 gateway with Active/Active for best performance.
VCF-WLDCON-CTGW-REQD-CFG-011	Each VRF uses two dedicated peering VLANs	VRFs sharing the same NSX Edge node cannot share the same VLAN ID for any of their external interfaces	New VLANs must be created any time a new VRF is provisioned. In highly dynamic environments incorporating inline EVPN in the design will remove this design requirement.

Table 575: Centralized Connectivity Model Design Recommendations for VRF Integration

Use this list of additional recommendations when adopting the *Centralized Connectivity Model* to configure virtual networking for a environment with VRF Integration for tenants that need dedicated external network access without dedicated NSX Edge nodes per tenant.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-025	Use external BGP (EBGP) as the routing protocol between the VRF gateway and the physical network	• BGP is a scalable solution to interconnect the virtual network environment to the physical network fabric.New dynamically created networks in the VCF environment are automatically advertised to the physical network. • BGP is a standard protocol supported by the majority of the datacenter switches. • EBGP simplifies the interconnection compared to internal BGP (IBGP) by eliminating the need for full-mesh peering and creating a clear boundary between the administrative domains of VCF and the physical network.	Physical switches must support EBGP
VCF-WLDCON-CTGW-RCMD-CFG-026	Use a unique private BGP Autonomous System Number (ASN) for the VRF gateway.	• It prevents advertisements from being dropped by the BGP loop detection check. If the peering devices only advertise the default route, having a unique BGP ASN may not be necessary. • It is assumed that the Tier-0 gateway is peering to internal devices and it's not an Internet-facing router, in which case a public AS number is required.	A unique BGP Autonomous System Number (ASN) must be available for each VRF gateway.
VCF-WLDCON-CTGW-RCMD-CFG-027	The VRF will peer to the same physical network devices as the Parent Tier-0 gateway.	• VRF HA status is dependent on the HA status of the parent Tier-0 gateway. For this reason, the VRF and the Parent Tier-0 gateway must share the same type of physical failures and react to them consistently. • Specifically, the design should avoid the VRF having an NSX Edge node in the "All BGP peers down" state while the parent Tier-0 gateway does not. More details are in the VCF networking reference guide, VRF Lite section.	• Designs where each VRF peers with a different upstream device incorporate additional risk. The risk can be avoided via the dedicated gateway model, which is not affected by this behavior as each Tier-0 gateway maintains its own HA state. • If VRFs with different upstream gateways are in use, the risk can be mitigated by peering each VRF NSX Edge node to multiple upstream devices to limit the chance of a VRF having an NSX Edge node in the "All BGP peers down" state while the Parent Tier-0 gatewaydoes not.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-CTGW-RCMD-CFG-028	Enable BFD between the VRF gateway and the physical network devices.	Provide fast failure detection and failover with minimal CPU overhead. It permits leaving the BGP timers to their default.	Physical switches need to support BFD
VCF-WLDCON-CTGW-RCMD-CFG-029	On the VRF, enable redistribution of the following network types into BGP: • Transit Gateway Static	In this model, all workloads are connected to transit gateways. This setting allows for VPC public subnets, NAT IPs, and load balancer VIPs to be advertised to the physical network.	All VPC public subnets, NAT IPs, and Load Balancer VIPs will be advertised to all BGP peers. If filtering is required, route filters should be configured.
VCF-WLDCON-CTGW-RCMD-CFG-030	• Create a ingress route filter allowing the advertisement of the default route only from the physical network to the VRF Gateway. • A corresponding summarization from the physical network is recommended.	• In most cases, the virtual network is a stub network with a single exit point. The filtering reduces the Tier-0 gateway routing table, simplifying troubleshooting and potentially reducing converge time. • It also prevents RFP checks from being triggered by asymmetric routing advertisements from the physical switches.	• If the virtual network has multiple exit points, for example, one to shared services and one as a private interconnection to the tenant's remote data center, this recommendation cannot be implemented. • In such cases reducing the routing table size via summarization is still recommended.

Distributed VLAN Connectivity Model

The *Distributed VLAN Connectivity Model* enables easy adoption of VPC network virtualization without requiring NSX Edge deployment. It uses a Distributed Transit Gateway (DTGW) that connects via external VLANs in the physical network fabric, utilizing a distributed forwarding model at the ESX host level.

Distributed VLAN Connectivity Model Overview

This model allows customers to adopt VPC network virtualization with minimal changes to the physical network fabric, requiring only the addition of at least a single external VLAN. The external VLAN is defined in the physical network infrastructure through the Distributed External Connection, which also specifies a default gateway for each subnet on that external VLAN. East-west traffic between workloads is handled by GENEVE-based virtual networking. The Distributed Transit Gateway is instantiated on each ESX host without requiring NSX Edge nodes, enabling near line rate physical NIC performance for traffic that does not require stateful network services. vSphere Supervisor, VKS workloads and VCF Automation All Apps Orgs require load balancing capabilities, this is achieved using either VNA load balancer or AVI Load Balancer.

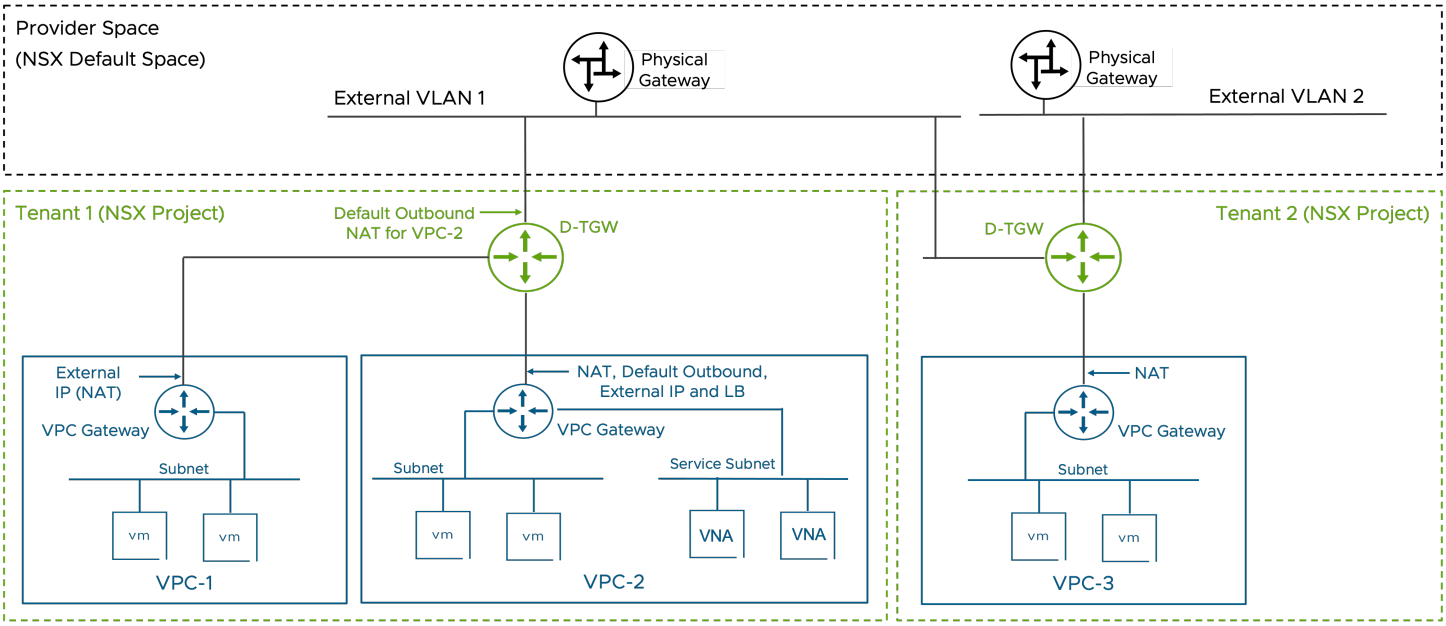
Note:

The *Distributed VLAN Connectivity Model* supports Avi Load Balancer. When only Avi Load Balancer is used, with no other stateful network services such as Default Outbound NAT (Auto-SNAT) or custom NAT, a VNA cluster does not need to be deployed, nor do N-S Services need to be enabled. More details, please refer to section Avi Load Balancer Model.

Note:

Ensure the external VLAN is available on the VDS and pNICs where the host TEPs are connected.

Figure 287: Distributed VLAN Connectivity Model in VMware Cloud Foundation



Distributed VLAN Connectivity Model Attributes

A *Distributed VLAN Connectivity Model* has the following attributes.

Attribute	Detail
Network Consumption Model	• VCF Automation All Apps Orgs Consumption Model • Virtual Private Cloud (VPC) Consumption Model
Transit Gateway Type	Distributed
Node Type	Virtual Network Appliance (optional for North/South Services)
Transit Gateway HA Mode	Distributed
VPC Gateway Services	• VPC Gateway without North/South Services • VPC Gateway with North/South Services – Required for VCF Automation All Apps Orgs and vSphere Supervisor – Instantiated on Virtual Network Appliance cluster
Load Balancing Options	• VNA load balancer • Avi Load Balancer
NSX Tenancy Constructs	NSX Projects, VPCs, Multiple Transit Gateways

Attribute	Detail
Consumption Options	• VCF Automation All Apps Orgs • vSphere Supervisor (i.e. VM Service, VKS, and more) • vCenter UI/API • NSX UI/API
External IP Blocks	Multiple (up to 5)
Provider Gateway Type	Physical Network VLAN Gateway

Distributed VLAN Connectivity Model Options

Design Area	Options
External Connections	• Single VLAN • Multiple VLANs

Distributed VLAN Connectivity Model Design Requirements and Recommendations

Table 576: Distributed VLAN Connectivity Model Design Requirements

Use this list of general requirements for the *Distributed VLAN Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVLAN-REQD-CFG-001	In the physical network fabric, create and extend at least one external VLAN and its corresponding CIDR across the required span of the VPC workloads, which may include one or more workload domains or vSphere clusters.	The external VLAN provides connectivity for VPC public subnets, External IPs (stateless 1:1 NAT), translated NAT IPs, load balancer VIPs and the VPC public subnet gateway IP when the DHCP relay for external DHCP server is enabled.	• Physical network fabric must meet this requirement. If meeting this requirement is impossible, an alternative solution incorporating another Workload Connectivity Models should be implemented. • It is the administrator's responsibility to ensure external VLAN IP gateway reachability for every host. • When deploying a VNA cluster for stateful network services, it must reside within the external VLAN span used by the Distributed TGW. • DHCP relay for an external DHCP server is supported only from version 9.1.1 onward. • When a workload VM or any other device is connected to the external VLAN and its associated CIDR, traffic flow between this VM respective other device and VPC IPs allocated from the External IP block may be asymmetric.
VCF-WLDCON-DTGWVLAN-REQD-CFG-002	Make the external VLAN available on the VDS and pNICs where the host TEPs are connected.	VPC public subnets, External IPs (stateless 1:1 NAT), translated NAT IPs, Load Balancer VIPs and the VPC public subnet gateway IP when the DHCP relay for external DHCP server is enabled, are presented to the external VLAN via the uplinks where host TEPs are enabled.	• Physical network fabric must be configured to allow the external VLANs on the correct ESX interfaces. • It is the administrator's responsibility to ensure external VLAN IP gateway reachability for every host. • When deploying a VNA cluster for stateful network services, it must reside within the external VLAN span used by the Distributed TGW. • DHCP relay for an external DHCP server is supported only from version 9.1.1 onward.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVLAN-REQD-CFG-003	Create one or more External Connection of type Distributed VLAN Connection.	• The External Connection requires an external VLAN and its associated CIDR gateway to provide connectivity to the physical network. • When stateful network services are required, assign a VNA cluster to the Distributed Transit Gateway associated with this External Connection.	Physical network fabric must meet this requirement.
VCF-WLDCON-DTGWVLAN-REQD-CFG-004	Assign at least one range of enterprise-wide routable IPs as part of the External IP block in the VPC Connectivity Profile, sourced from the external VLAN CIDR.	This range, or these ranges, cover all inbound and outbound VPC traffic, including VPC public subnets, External IPs (stateless 1:1 NAT), translated NAT IPs, load balancer VIPs and the VPC public subnet gateway IP when the DHCP relay for external DHCP server is enabled, from it.	• If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only. • DHCP relay for an external DHCP server is supported only from version 9.1.1 onward.
VCF-WLDCON-DTGWVLAN-REQD-CFG-005	Ensure that IPs used by physical network devices or any VM or device on the external VLAN are excluded from the External IP block that is assigned to VPCs.	IP Address conflicts may arise if the exclusion is not configured.	• The external VLAN gateway IP is excluded by default. • Additional IPs can be excluded by defining an exclude IP range for the External IP block.
VCF-WLDCON-DTGWVLAN-REQD-CFG-006	When stateful network services are required, assign a VNA cluster to the VPC Connection Profile. Select one of the presented Virtual Network Appliance Cluster Models.	A VNA cluster is required for stateful network services when using a Distributed TGW with external VLAN.	• Additional resources are required for the VNAcluster. • Only traffic that is sourced from VPC private or private TGW subnets is forwarded to the VNA cluster to perform stateful services such as NAT.
VCF-WLDCON-DTGWVLAN-REQD-CFG-007	When deploying a VNA cluster for stateful network services, ensure that N-S Services is enabled; otherwise, disable it.	N-S Services are only available when a VNA cluster is assigned to the VPC Connection Profile used with the Distributed TGW with external VLAN.	• A VNA cluster is required to enable N-S Services — see Virtual Network Appliance Cluster Models for deployment details. • When only Avi Load Balancer is used, with no other stateful network services such as Default Outbound NAT (Auto-SNAT) or custom NAT, a VNA cluster does not need to be deployed, nor do N-S Services need to be enabled.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVLAN-REQD-CFG-008	When Default Outbound NAT (Auto-SNAT) or custom NAT is required, ensure a VNA cluster is assigned and N-S Services is enabled.	Default Outbound NAT (Auto-SNAT) or custom NAT requires N-S Services to be enabled, which in turn requires a VNA cluster to be assigned to the VPC Connection Profile used by the Distributed TGW with external VLAN.	• A VNA cluster is required to enable Default Outbound NAT (Auto-SNAT) or custom NAT — see Virtual Network Appliance Cluster Models for deployment details. • Default Outbound NAT (Auto-SNAT) and custom NAT is only supported for traffic matching VPC private subnets or private-TGW subnets as the source.
VCF-WLDCON-DTGWVLAN-REQD-CFG-009	When DHCP relay for external DHCP servers is required, ensure a VNA cluster is deployed and N-S Services is enabled.	DHCP relay for external DHCP servers requires N-S Services to be enabled, which in turn requires a VNA cluster to be assigned to the VPC Connection Profile used by the Distributed TGW with external VLAN.	• A VNA cluster is required to enable DHCP relay for external DHCP servers — see Virtual Network Appliance Cluster Models for deployment details. • DHCP relay for external DHCP servers is only supported for workload VMs connected to VPC public subnet. • When the DHCP server is connected to a VPC subnet (of any type), a VNA cluster is not required for DHCP relay. • DHCP relay for an external DHCP server is supported only from version 9.1.1 onward.
VCF-WLDCON-DTGWVLAN-REQD-CFG-010	When a VNA load balancer is required, ensure a VNA cluster is deployed and N-S Services is enabled.	The VNA load balancer requires N-S Services to be enabled, which in turn requires a VNA cluster to be assigned to the VPC Connection Profile used by the Distributed TGW with external VLAN.	• A VNA cluster is required to enable VNA load balancer — see Virtual Network Appliance Cluster Models for deployment details. • Please see Virtual Network Appliance Load Balancer Model for details.

Table 577: Distributed VLAN Connectivity Model Design Recommendations

Use this list of general recommendations for the *Distributed VLAN Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-DTGWVLAN-RCMD-CFG-001	Assign a range of non-routable IPs as part of the private-TGW IP block in the VPC Connectivity Profile.	The private-TGW IP block is used to carve out private-TGW subnets. These subnets enable No-NAT communication between VPCs and allow traffic to destinations northbound of the Distributed TGW using either an External IP (stateless 1:1 NAT), Default Outbound NAT (Auto-SNAT) or custom NAT.	• If private-TGW subnets are not required, this step can be omitted. • If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.
VCF-WLDCON-DTGWVLAN-RCMD-CFG-002	To allow for granular application placement restrict the network span of VPC subnets to one or more vSphere clusters within an NSX instance. This is typically required for remote vSphere clusters or DMZ deployments.	• Provides isolation of VPC subnets and applications in different vSphere clusters. • The network span is defined on the Distributed TGW and applies to all connected VPCs and their associated VPC subnets. • vSphere clusters can be shared across multiple network spans. • Allows the external VLAN to be restricted to a subset of vSphere clusters.	• Exclusive network span allows defining a dedicated network span for a specific TGW and the associated VPCs, preventing it from being shared or overlapping with other network spans. • It is the administrator's responsibility to ensure external VLAN IP gateway reachability for every ESX ESX host.

Table 578: Distributed VLAN Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Requirements

Use this list of additional requirements when adopting the *Distributed VLAN Connectivity Model* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVLAN-REQD-CFG-011	Assign a VNA cluster to the VPC Connection Profile. Select one of the presented Virtual Network Appliance Cluster Models.	A VNA cluster is required for stateful network services when using a Distributed TGW with external VLAN.	• Additional resources are required for the VNA cluster. • Only traffic that is sourced from VPC private or private TGW subnets is forwarded to the VNA cluster to perform stateful services such as NAT.
VCF-WLDCON-DTGWVLAN-REQD-CFG-012	In the Default VPC Connectivity Profile of the Default Project, assign an IP range of non-routable IPs to the private-TGW IP block (a /23 CIDR or larger is preferred).	This private-TGW IP range will be used for inter-VPC connectivity within the tenant boundary. It will be used to assign CIDRs for private-TGW VPC Subnets.	• Workloads on private-TGW subnets will be reachable from any other tenant workloads without requiring NAT, but no connectivity from workloads outside of the tenant boundary will be available without exposing the workloads via a NAT Rule, or a load balancer VIP, or an External IP (stateless 1:1 NAT). • If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.
VCF-WLDCON-DTGWVLAN-REQD-CFG-013	In the Default VPC Connectivity Profile of the Default Project enable N-S Services.	N-S Services is required for vSphere Supervisor and VCF-A All Apps Orgs.	A VNA cluster is required to enable N-S Services — see Virtual Network Appliance Cluster Models for deployment details.
VCF-WLDCON-DTGWVLAN-REQD-CFG-014	In the Default VPC Connectivity Profile of the Default Project enable Default Outbound NAT (Auto-SNAT).	Default Outbound NAT (Auto-SNAT) is necessary for vSphere Supervisor and VCFA All Apps Orgs.	• A VNA cluster is required to enable Default Outbound NAT (Auto-SNAT) — see Virtual Network Appliance Cluster Models for deployment details. • Default Outbound NAT (Auto-SNAT) is only supported for traffic matching VPC private subnets or private-TGW subnets as the source.

Table 579: Distributed VLAN Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Recommendations

Use this list of additional recommendations when adopting the *Distributed VLAN Connectivity Model* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-DTGWVLAN-RCMD-CFG-003	Use the Default Project for the vSphere Supervisor namespace's VPCs.	• When VCFA All Apps Orgs are in use, in each Organization will have its dedicated project. • The default space will not be used and it will be available to be consumed by system VPCs to support the IaaS control plane. • The WLD creation workflow follows this model when the new WLD has a new dedicated NSX Manager.	• This approach is not recommended if the Default Project is also consumed via other means, for example via the vCenter UI, which allows the configuration of VPCs in the Default Project. • In general, when VCF Automation is in use, it's not recommended to mix another network consumption vehicle in the same workload domain.
VCF-WLDCON-DTGWVLAN-RCMD-CFG-004	Assign at least one range of enterprise-wide routable IPs as part of the External IP block in the Default VPC Connectivity Profile, sourced from the external VLAN CIDR within the Default Project (a /24 CIDR or larger is preferred).	This range, or these ranges, cover all inbound and outbound VPC traffic, including VPC public subnets, External IPs (stateless 1:1 NAT), translated NAT IPs, load balancer VIPs and the VPC public subnet gateway IP when the DHCP relay for external DHCP server is enabled, from it.	• If the range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only. • DHCP relay for an external DHCP server is supported only from version 9.1.1 onward.

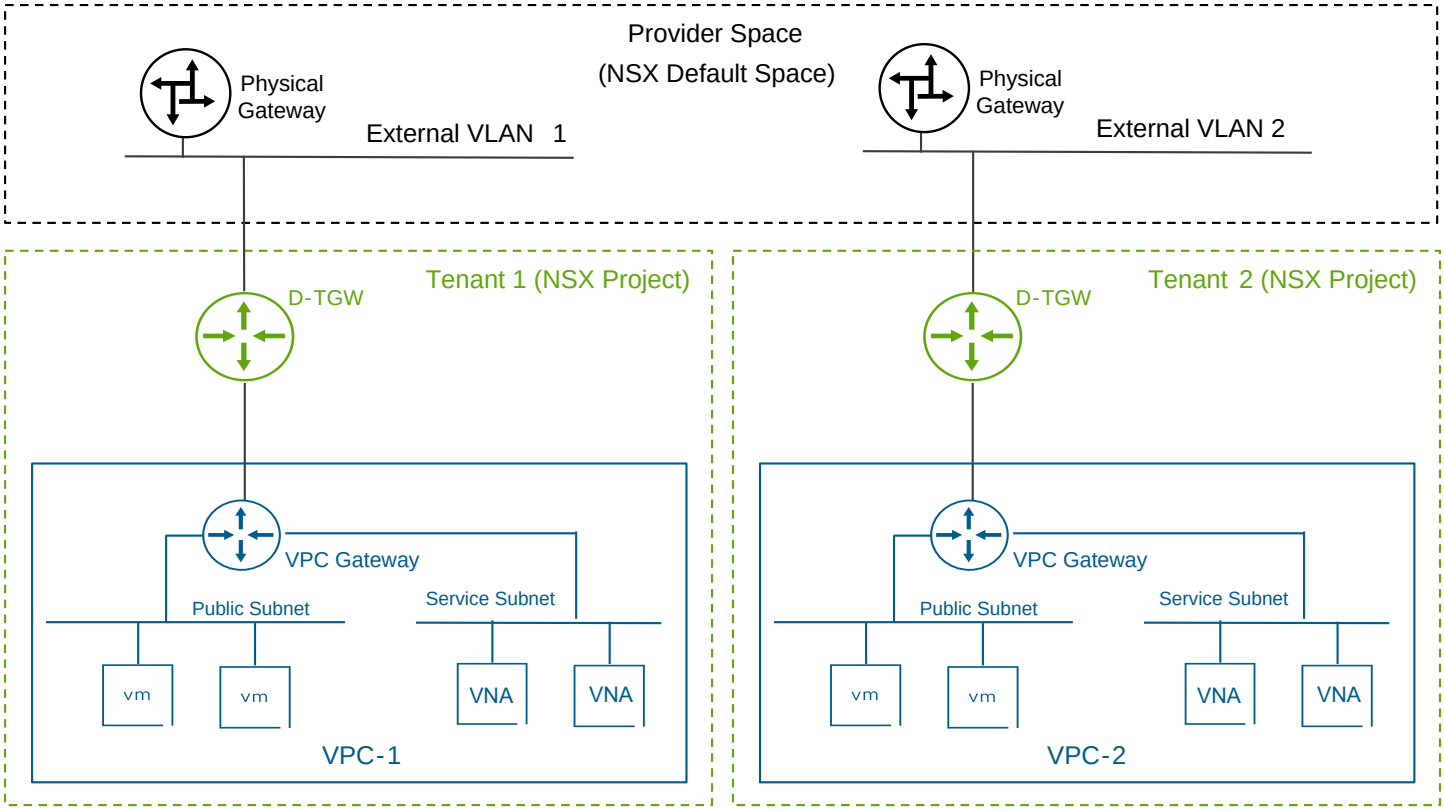
VLAN Backed VPC Connectivity Model

The *VLAN Backed VPC Connectivity Model* introduced in version 9.1.1 enables easy adoption of VPC network virtualization without requiring NSX host TEPs. It uses a Distributed Transit Gateway (DTGW) that connects via external VLANs in the physical network fabric, utilizing a distributed forwarding model at the ESX host level.

VLAN Backed VPC Connectivity Model Overview

This model allows customers to adopt VPC network virtualization with the least number of changes to the physical network fabric, requiring one or more external VLAN. The external VLAN is defined in the physical network infrastructure through the Distributed External Connection, which also specifies a default gateway for each subnet on that external VLAN. East-west traffic between workloads running on different ESX hosts is handled by the fabric in the VLAN physical networking. The Distributed Transit Gateway is instantiated on each ESX host without requiring NSX Edge nodes, enabling near line rate physical NIC performance for traffic that does not require stateful network services. vSphere Supervisor, VKS workloads and VCF Automation All Apps Orgs require load balancing capabilities, this is achieved using either VNA Load Balancer or Avi Load Balancer.

Figure 288: VLAN Backed VPC Connectivity Model in VMware Cloud Foundation



VLAN Backed VPC Connectivity Model Attributes

A *VLAN Backed VPC Connectivity Mode* has the following attributes.

Attribute	Detail
Network Consumption Model	• VCF Automation All Apps Orgs Consumption Model • Virtual Private Cloud (VPC) Consumption Model
Transit Gateway Type	Distributed
Node Type	Virtual Network Appliance (required for Supervisor)
Transit Gateway HA Mode	Distributed
VPC Gateway Services	• VPC Gateway without North/South Services. – Always disabled when created in vCenter. • VPC Gateway with North/South Services. – Required for VCF Automation All Apps Orgs and vSphere Supervisor. – Instantiated on Virtual Network Appliance cluster. – Enabled in VCF Automation All Apps Orgs and vSphere Supervisor.
Load Balancing Options	• VNA Load Balancer • Avi Load Balancer
NSX Tenancy Constructs	NSX Projects, VPCs, Multiple Transit Gateways
Consumption Options	• VCF Automation All Apps Orgs • vSphere Supervisor (i.e. VM Service, VKS, and more) • vCenter UI/API • NSX UI/API
External IP Blocks	Multiple (up to 5)

Attribute	Detail
Provider Gateway Type	Physical Network VLAN Gateway

VLAN Backed VPC Connectivity Model Options

Design Area	Options
Supported Subnet Types	vCenter UI/API • L2 Only • Public Subnet vSphere Supervisor • Public Subnet VCF Automation All Apps • Public Subnet created at the Org level • Public Subnet created at the Provider level

VLAN Backed VPC Connectivity Model Design Requirements and Recommendations

Table 580: VLAN Backed VPC Connectivity Model Design Requirements

Use this list of general requirements for the *VLAN Backed VPC Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-VLANVPC-REQD-CFG-001	In the physical network fabric, create and extend at least one external VLAN and its corresponding CIDR across one or more vSphere clusters.	The external VLAN provides connectivity for VPC public subnets and load balancer VIPs.	• Physical network fabric must meet this requirement. If meeting this requirement is impossible, an alternative solution incorporating another Workload Connectivity Models should be implemented. • It is the administrator's responsibility to ensure external VLAN IP gateway reachability for every host. • If deploying a VNA cluster it must reside within the external VLAN span used by the Distributed TGW.
VCF-WLDCON-VLANVPC-REQD-CFG-002	Create a new network span for the first cluster in a workload domain. Each new workload domain will have it's own span, this also applies to a shared NSX Instance.	Controls the network span to a workload domain boundary, if this needs to be extended the network span can be added to clusters in other workload domains.	Only the first cluster in a workload domain is added to a network span, additional clusters need to be manually added to this network span or another custom network span.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-VLANVPC-REQD-CFG-002	Make the external VLAN available on the VDS prepared for NSX.	VPC public subnet and load balancer VIPs are presented to the external VLAN via the uplinks from the vDS where NSX is enabled.	• Physical network fabric must be configured to allow the external VLANs on the correct ESX interfaces. • It is the administrator's responsibility to ensure external VLAN IP gateway reachability for every host. • When deploying a VNA cluster for load balancing, it must reside within the external VLAN span used by the Distributed TGW.
VCF-WLDCON-VLANVPC-REQD-CFG-003	For an L2 Only subnet created in vCenter create an External Connection of type Distributed VLAN Connection without a Gateway IP.	The External Connection requires an external VLAN.	• The External connection is created automatically. • Physical network fabric must meet this requirement.
VCF-WLDCON-VLANVPC-REQD-CFG-004	For a VPC public subnet created in vCenter create an External Connection of type Distributed VLAN Connection with a Gateway IP.	The External Connection requires an external VLAN and a Gateway IP.	• The External connection is created automatically. • Physical network fabric must meet this requirement.
VCF-WLDCON-VLANVPC-REQD-CFG-005	Assign at least one range of enterprise-wide routable IPs as part of the External IP block in the VPC Connectivity Profile, sourced from the external VLAN CIDR.	This range, or these ranges, cover all inbound and outbound VPC traffic, including VPC public subnets and Load Balancer VIPs.	If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.
VCF-WLDCON-VLANVPC-REQD-CFG-006	Ensure that IPs used by physical network devices or VM on the external VLAN are excluded from the External IP block that are assigned to VPC public subnets.	IP Address conflicts may arise if the exclusion is not configured.	• The external VLAN gateway IP for a Public Subnet is excluded by default. • Additional IPs can be excluded by defining an exclude IP range for the External IP block.

Table 581: VLAN Backed VPC Connectivity Model Design Recommendations

Use this list of general recommendations for the *VLAN Backed VPC Connectivity Model* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-VLANVPC-RCMD-CFG-001	To allow for granular application placement restrict the network span of VPC subnets to one or more vSphere clusters within an NSX instance.	• Provides isolation of VPC subnets and applications in different vSphere clusters. • The network span is defined on the Distributed TGW and applies to all connected VPCs and their associated VPC subnets. • vSphere clusters can be shared across multiple network spans. • Allows the external VLAN to be restricted to a subset of vSphere clusters.	• Exclusive network span allows defining a dedicated network span for a specific TGW and the associated VPCs, preventing it from being shared or overlapping with other network spans. • It is the administrator's responsibility to ensure external VLAN IP gateway reachability for every ESX host. • The second vSphere cluster created in a workload domain is not automatically added to a network span, this needs to be done manually.

Table 582: VLAN Backed VPC Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Requirements

Use this list of additional requirements when adopting the *VLAN Backed VPC Connectivity Model*to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-VLANVPC-REQD-CFG-007	Deploy a VNA cluster. Select one of the presented Virtual Network Appliance Cluster Models.	A VNA cluster is required for stateful network services when using a Distributed TGW with external VLAN.	• Additional resources are required for the VNA cluster. • When Avi load balancer is deployed a VNA cluster is required to provide DHCP addresses for Avi Service Engines.
VCF-WLDCON-VLANVPC-REQD-CFG-008	For the deployment of vSphere Supervisor assign at least one range of enterprise-wide routable IPs as part of the External IP block in the VPC Connectivity Profile, sourced from the external VLAN CIDR.	This range covers all inbound and outbound traffic to the Supervisor and Supervisor Services VPC public subnets.	This is configured during the vSphere Supervisor deployment.
VCF-WLDCON-VLANVPC-REQD-CFG-009	For the deployment of VCF Automation All Apps Org assign at least one range of enterprise-wide routable IPs as part of the External IP block in the VPC Connectivity Profile, sourced from the external VLAN CIDR.	This range covers all inbound and outbound traffic to the Org VPC public subnets.	This is configured in VCF Automation in the Provider Management before creating a new Org/Tenant.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-VLANVPC-REQD-CFG-010	For the deployment of vSphere Supervisor create an External Connection of type Distributed VLAN Connection.	The External Connection requires an external VLAN.	• Physical network fabric must meet this requirement. • The External connection is created automatically when deploying a workload domain with vSphere Supervisor enabled. If Supervisior is deployed day 2 this configuration is done manually before deployment..
VCF-WLDCON-VLANVPC-REQD-CFG-011	For the deployment of VCF Automation All Apps Org create an External Connection of type Distributed VLAN Connection for the Org/Tenant.	The External Connection requires an external VLAN.	• Physical network fabric must meet this requirement. • The External connection must be manually created in VCF Automation for the Org/Tenant.
VCF-WLDCON-VLANVPC-REQD-CFG-012	Assign a VNA cluster to the VPC Connectivity Profile.	• A VNA cluster is required in the VPC Gateway configuration to enable the deployment of Supervisor. • If Avi load balancer is deployed it uses the VPC Services subnet to connect the Avi Service Engines to the VPC.	• Additional resources are required for the VNA cluster. • The is configured automatically when deploying a workload domain with vSphere Supervisor enabled. If Supervisior is manully deployed day 2 manual configuration is requiired.
VCF-WLDCON-VLANVPC-REQD-CFG-013	When using the NSX Default Project for the vSphere Supervisor namespace's VPCs, in the Default VPC Connectivity Profile enable N-S Services.	N-S Services is required for deployment of vSphere Supervisor.	A VNA cluster is required to enable N-S Services — see Virtual Network Appliance Cluster Models for details.
VCF-WLDCON-VLANVPC-REQD-CFG-014	For the NSX Project created for the VCF Automatio Org, in the Default VPC Connectivity Profile enable N-S Services.	N-S Services is required for vSphere Supervisor and VKS based workloads deployed in VCF Automation All Apps Org	A VNA cluster is required to enable N-S Services, this can be shared with the Default Project or be dedicated to the Org — see Virtual Network Appliance Cluster Models for details.
VCF-WLDCON-VLANVPC-REQD-CFG-015	In a multi-zone vSphere Supervisor deployment add all zones to the network span.	Allows the vSphere Supervisor namespaces connect to the VPC Public subnets across zones.	When new zones are added to vSphere Supervisor the network span is automatically adjusted to include the new zone.

Table 583: VLAN Backed VPC Connectivity Model with vSphere Supervisor or VCF-A All Apps Org Design Recommendations

Use this list of additional recommendations when adopting the *VLAN Backed VPC Connectivity Model* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-VLANVPC-RCMD-CFG-002	Use the Default Project for the vSphere Supervisor namespace's VPCs.	• When VCFA All Apps Orgs are in use, in each Organization will have its dedicated project. • The default space will not be used and it will be available to be consumed by system VPCs to support the IaaS control plane. • The WLD creation workflow follows this model when the new WLD has a new dedicated NSX Manager.	• This approach is not recommended if the Default Project is also consumed via other means, for example via the vCenter UI, which allows the configuration of VPCs in the Default Project. • In general, when VCF Automation is in use, it's not recommended to mix another network consumption vehicle in the same workload domain.

Distributed VXLAN Connectivity Model for Version 9.1.0

The *Distributed VXLAN Connectivity Model for Version 9.1.0* integrates NSX with the physical EVPN-VXLAN network fabric, unifying virtual workloads, and physical endpoints into a single routing domain.

Distributed VXLAN Connectivity Model for Version 9.1.0 Overview

The *Distributed VXLAN Connectivity Model for Version 9.1.0* leverages a decoupled Multiprotocol BGP-EVPN control plane and VXLAN data plane to connect Virtual Private Clouds (VPCs) to the physical EVPN-VXLAN network fabric, with Distributed Transit Gateways providing connectivity to tenant Virtual Routing and Forwarding (VRF) instances within the physical network fabric.

Ethernet VPN (EVPN) is a widely adopted data center technology that serves as the distributed control plane for the physical network fabric, using BGP to advertise both Layer 2 MAC (not supported in the current EVPN-VXLAN integration with NSX) and Layer 3 IP reachability across the network. VXLAN acts as the corresponding data plane encapsulation protocol, the counterpart to the GENEVE protocol used for encapsulation and tunneling within the NSX overlay environment.

With the *Distributed VXLAN Connectivity Model for Version 9.1.0*, the NSX control plane managed by the NSX Manager appliances continues to use GENEVE for east-west traffic between ESX hosts, while leveraging VXLAN for north-south routing with the physical EVPN-VXLAN network fabric to reach physical endpoints.

> **Note:**
>
> The *Distributed VXLAN Connectivity Model for Version 9.1.0*, unlike the *Distributed VLAN Connectivity Model*, does not require an external VLAN spanning across the Distributed Transit Gateways. Connectivity between each ESX host and the physical EVPN-VXLAN network fabric is established through the ESX host VTEP interface. This *Distributed VXLAN Connectivity Model for Version 9.1.0* supports, for example, Multi-Rack Layer 3 vSphere Cluster Model for workloads without compromising rack isolation through a Layer 3 boundary. However, it should be noted that in VMware Cloud Foundation 9.1.0, Active/Standby is the only supported mode for EVPN Route Controller nodes, requiring Layer 2 adjacency on the Multiprotocol BGP-EVPN interface.

> **Note:**
>
> The ESX VMkernel Tunnel Endpoint (TEP) interface can handle both GENEVE and VXLAN traffic simultaneously. In this document, TEP (GENEVE) and VTEP (VXLAN) are used interchangeably when referring to ESX hosts.

Egress data plane traffic sourced from a VM connected to a VPC subnet and hosted on an ESX host is forwarded by the Distributed Transit Gateway (DTGW) via the ESX TEP interface to the EVPN gateway in the physical EVPN-VXLAN

network fabric. The EVPN gateway role is configured either on all spine switches or on a subset of VXLAN leaf switches. Configuring the EVPN gateway role on all spine switches achieves optimal traffic patterns. The EVPN gateway, often referred to as the border gateway, must advertise its VTEP IP address as the next-hop to the EVPN Route Controller via EVPN Route Type 5. This information is then propagated to each individual ESX host, ensuring every ESX host knows the path to physical endpoints outside the VCF infrastructure by installing one or more default routes pointing to the VTEP IP address on the EVPN gateway.

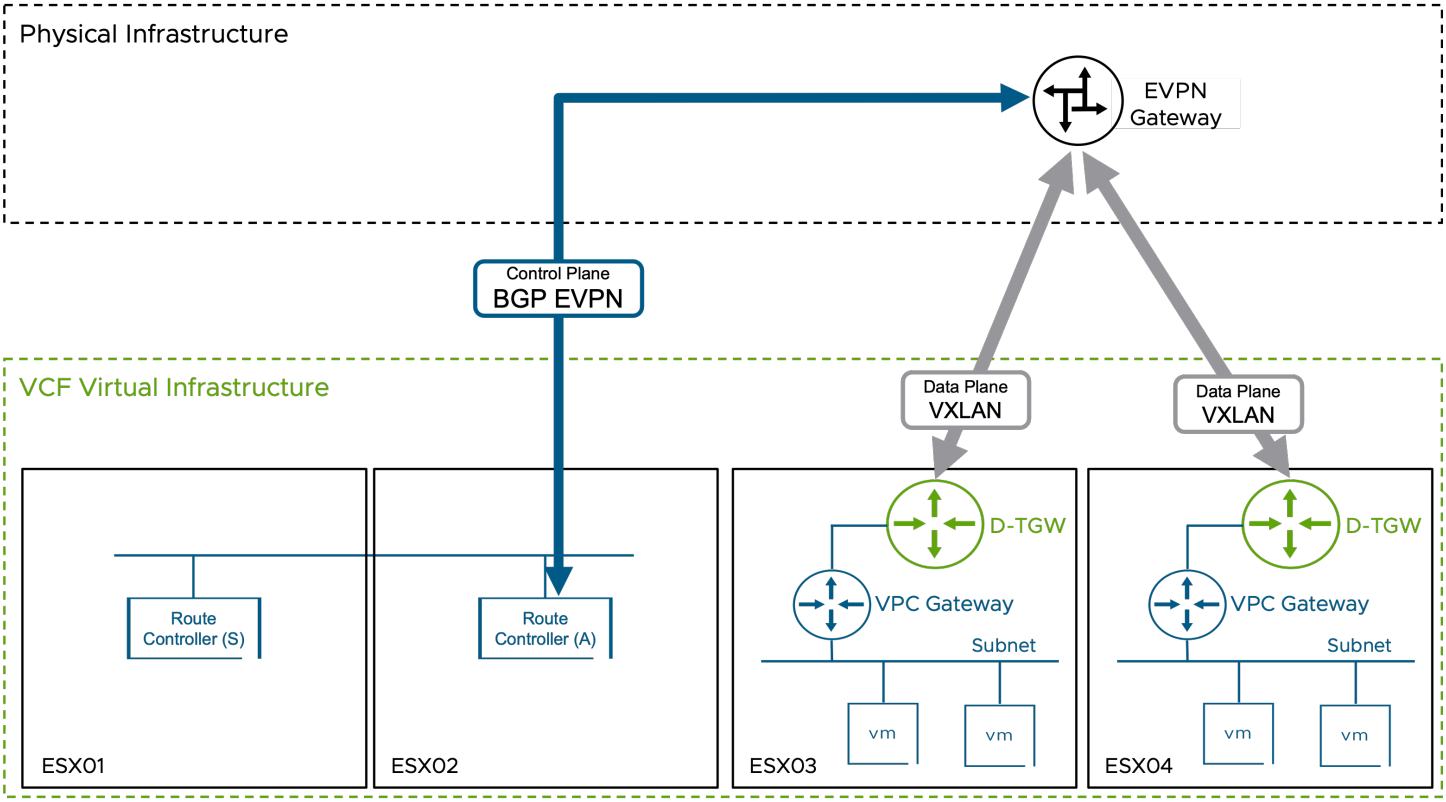
Ingress data plane traffic sourced from physical endpoints connected to a VXLAN leaf switch is initially forwarded to the VTEP of the EVPN gateway using VXLAN encapsulation mapped to the configured Layer 3 VNI. Upon arrival at the EVPN gateway, full topology visibility is provided by the /32 prefix list for each VM behind each ESX host VTEP, learned via the EVPN Route Controller through EVPN Route Type 5. The EVPN gateway then forwards the traffic using VXLAN encapsulation mapped to the same Layer 3 VNI to the ESX host VTEP where the destination VM is located.

Note:

To align Virtual Private Cloud (VPC) in NSX with the VRF tenants in the physical EVPN-VXLAN network fabric, the Distributed Transit Gateway is mapped through the External Distributed VXLAN Connection to a corresponding Layer 3 VNI in the physical EVPN-VXLAN network fabric.

The EVPN Route Controller unifies both environments, establishing a common control plane across the virtual and physical network fabric by peering directly with EVPN gateways using Multiprotocol BGP-EVPN. More details, please refer to section Distributed VXLAN Connectivity Model for Version 9.1.0 EVPN Route Controller Overview

Figure 289: Distributed VXLAN Connectivity Model for Version 9.1.0 Logical View



Distributed VXLAN Connectivity Model for Version 9.1.0 Attributes

A *Distributed VXLAN Connectivity Model for Version 9.1.0* has the following attributes.

Attribute	Detail
Network Consumption Model	Virtual Private Cloud (VPC) Consumption Model
Transit Gateway Type	Distributed
Node Type	EVPN Route Controller is required - Distributed VXLAN Connectivity Model for Version 9.1.0 EVPN Route Controller Overview
Transit Gateway HA Mode	Distributed
VPC Gateway Services	VPC Gateway without North/South Services
NSX Tenancy Constructs	NSX Projects, VPCs, Multiple Transit Gateways
Consumption options	• vCenter UI/API • NSX UI/API
External IP Blocks	Multiple (up to 5)
Provider Gateway Type	Physical EVPN-VXLAN network fabric with EVPN gateway

Distributed VXLAN Connectivity Model for Version 9.1.0 Options

Design Area	Options
EVPN Route Controller: Multiprotocol BGP-EVPN interface per EVPN Route Controller Node	• One • Two

Distributed VXLAN Connectivity Model for Version 9.1.0 Design Requirements and Recommendations

Table 584: Distributed VXLAN Connectivity Model for Version 9.1.0 Design Requirements

Use this list of general requirements for the *Distributed VXLAN Connectivity Model for Version 9.1.0* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-001	The Distributed VXLAN Connectivity Model for Version 9.1.0 requires an EVPN Route Controller cluster.	The EVPN Controller cluster establishing a common control plane across the VCF virtual infrastructure and physical EVPN-VXLAN network fabric by peering with EVPN gateways using Multiprotocol BGP-EVPN.	vSphere cluster resources must be available to support the EVPN Route Controller node(s).
VCF-WLDCON-DTGWVXLAN-REQD-CFG-002	The EVPN Route Controller cluster must belong to the same NSX instance as the Distributed TGW with EVPN-VXLAN integration.	The EVPN Route Controller and the Distributed TGW with EVPN-VXLAN integration must share the same internal overlay segment for data plane notifications.	An EVPN Route Controller cannot be shared across different NSX instances.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-003	Create one of more External Connection of type Distributed VXLAN Connection.	The External Connection requires an EVPN Route Controller cluster to be attached and defines the Multiprotocol BGP-EVPN attributes, such as Layer 3 VNI, Route Distinguisher (RD), and Route Target (RT), used by the EVPN Route Controller to establish peering with the physical EVPN-VXLAN network fabric.	The EVPN Route Controller cluster must be configured prior to creating the External Connection.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-004	Define a single Layer 3 VNI for each External Connections.	• A 1:1 mapping is required between a Distributed TGW and its associated Layer 3 VNI defined in the External Connection and the corresponding VRF/tenant in the physical EVPN-VXLAN network fabric. • The Layer 3 VNI, along with the Route Distinguisher (RD), Route Target (RT) and others, are specific Multiprotocol BGP-EVPN attributes carried as EVPN Route Type 5 between the EVPN Route Controller and the EVPN gateway in the physical EVPN-VXLAN network fabric.	For each VRF/tenant a dedicated Distributed TGW and its associated Layer 3 VNI defined in the External Connection is required.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-005	Define a single Route Distinguisher (RD) for each External Connection.	• A unique Route Distinguisher (RD), prepended to each prefix by the EVPN Route Controller, ensures EVPN routes remain distinguishable across VRFs/tenants when advertised to the EVPN gateway in the physical EVPN-VXLAN network fabric. • The Route Distinguisher (RD) is unique per VRF/tenant and maps directly to a single Distributed TGW and its associated Layer 3 VNI defined in the External Connection.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-006	Define at least one pair of Route Targets (RTs), consisting of an import and an export RT, for each External Connection. These Route Targets (RTs) must correspond to the Route Targets (RTs) defined on the EVPN gateways.	• The Route Target (RT) and Layer 3 VNI together identify the VRF/tenant and is defined in the External Connection. • Route Targets (RTs) are required to control the import and export policy, determining which VRFs/ tenants accept and install the advertised routes. • A corresponding pair of Route Targets (RTs) is required, configured on both the External Connection and the EVPN gateways, to import and install VRF routes.	None.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-007	Assign at least one range of VRF/tenant specific routable IPs as part of the External IP block in the VPC Connectivity Profile.	This range, or these ranges, covers all inbound and outbound VPC traffic for a specific VRF/tenant. It includes public subnets, and External IPs (stateless 1:1 NAT) assigned from it.	If the range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-008	N-S Services must be disabled in the VPC Connectivity Profile.	• N-S Services are not available when using Distributed TGW with EVPN-VXLAN integration. • When assigning a Distributed TGW with EVPN-VXLAN integration to the VPC Connection Profile, the N-S Services configuration option is omitted.	If N-S Services are required, consider using External IP (stateless 1:1 NAT) as an alternative option; otherwise, the Centralized Connectivity Model or Distributed VLAN Connectivity Model should be implemented.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-009	Default Outbound NAT (Auto-SNAT) must be disabled in the VPC Connectivity Profile.	• Default Outbound NAT (Auto-SNAT) is not available when using the Distributed TGW with EVPN-VXLAN integration, as it requires N-S Services, which are not enabled. • When assigning a Distributed TGW with EVPN-VXLAN integration to the VPC Connection Profile, the Default Outbound NAT (Auto-SNAT) configuration option is omitted.	If N-S Services are required, consider using External IP (stateless 1:1 NAT) as an alternative option; otherwise, the Centralized Connectivity Model or Distributed VLAN Connectivity Model should be implemented.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-010	Ensure that the VTEP interface on each ESX host has IP connectivity to the EVPN gateway VTEP interfaces to establish VXLAN tunnels for data plane traffic.	Data plane traffic requires a VXLAN tunnel between the ESX host VTEP interface and the EVPN gateway VTEP, which resides in the global routing table of the physical EVPN-VXLAN network fabric	When the ESX host VTEP interface belongs to the tenant routing space and the EVPN gateway VTEP interfaces reside in the global routing table, connectivity between them requires VRF leaking.

Table 585: Distributed VXLAN Connectivity Model for Version 9.1.0 Design Recommendations

Use this list of general recommendations for the *Distributed VXLAN Connectivity Model for Version 9.1.0* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-001	Assign a range of non-routable IPs as part of the private-TGW IP block in the VPC Connectivity Profile.	The private-TGW IP block is used to carve out private-TGW subnets. These subnets enable No-NAT communication between VPCs and allow traffic to destinations northbound of the Distributed TGW using External IP (stateless 1:1 NAT).	• If private-TGW subnets are not required, this step can be omitted. • If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-002	To allow for granular application placement restrict the network span of VPC subnets to one or more vSphere clusters within an NSX instance. This is typically required for remote vSphere clusters or DMZ deployments.	• Provides isolation of VPC subnets and applications in different vSphere clusters. • The network span is defined on the Distributed TGW and applies to all connected VPCs and their associated VPC subnets. • vSphere clusters can be shared across multiple network spans.	Exclusive network span allows defining a dedicated network span for a specific TGW and the associated VPCs, preventing it from being shared or overlapping with other network spans.

Distributed VXLAN Connectivity Model for Version 9.1.0 EVPN Route Controller Overview

The EVPN Route Controller establishes a common control plane across the virtual and physical network fabric by peering directly with EVPN gateways in the physical EVPN-VXLAN network fabric using Multiprotocol BGP-EVPN.

Although the EVPN Route Controller is not involved in data plane forwarding, its role is crucial and requires careful consideration regarding High Availability, including for example, rack fault tolerance with independent storage per node. It is therefore strongly recommended to deploy the EVPN Route Controller as a two-node Active/Standby cluster, where one node is active while the other remains on standby. Only the active EVPN Route Controller establishes Multiprotocol BGP-EVPN sessions with the EVPN gateways in the physical EVPN-VXLAN network fabric, while the standby node remains in cold standby without an established Multiprotocol BGP-EVPN session. The standby node retains all configuration and BGP routes in its local RIB, but the BGP connection to the EVPN gateway is not established until a failover occurs.

A single EVPN Route Controller cluster is sufficient to support multiple VRFs/tenants, each identified by the Layer 3 VNI configured in the External Connection. Each External Connection supports only a single Layer 3 VNI, with a strict 1:1 mapping between the External Connection and the Distributed Transit Gateway.

The active EVPN Route Controller advertises a /32 host route for each workload VM, including the VTEP of the ESX host where the VM is running, to the EVPN gateway in the physical EVPN-VXLAN network fabric via Multiprotocol BGP-EVPN. These /32 host routes, which represent either public IPs or External IPs (stateless 1:1 NAT), along with the VTEP information, are carried as EVPN Route Type 5 together with the Layer 3 VNI and Router MAC (RMAC), to identify the VRF/tenant represented by the Layer 3 VNI configured in the External Connection. The EVPN Route Controller learns this information about each workload VM, represented as a /32 host route and VTEP, via the regular Central Control Plane (CCP) channel from the individual ESX hosts through the NSX Manager.

The EVPN gateways in the physical EVPN-VXLAN network fabric advertise prefixes from physical endpoints, networks or a default route to the EVPN Route Controller, including the next-hop information configured on each EVPN gateway. To minimize the forwarding table on each ESX host where a Distributed Transit Gateway is instantiated, the EVPN Route Controller advertises only the default route and the EVPN gateway next-hop to each ESX host via NSX Manager through the Central Control Plane (CCP) channel.

Important:

Centralized password management via VCF Operations is the only supported option; local credentials cannot be configured during the EVPN Route Controller node deployment. VCF Operations supports the password management task 'update', which requires the current password before a new one can be configured. A fresh EVPN Route Controller node deployment generates randomized passwords, which can be, for example, retrieved via the SDDC Manager API (GET /v1/credentials) and subsequently used to update the passwords using VCF Operations.

Note:

The EVPN Route Controller is built on top of the Virtual Network Appliance, and this section applies exclusively to its use as an EVPN Route Controller. Throughout this section, the term EVPN Route Controller refers to the EVPN Route Controller function inclusive of the underlying Virtual Network Appliance nodes. For Virtual Network Appliance nodes deployed for stateful network services such as NAT, refer to the Virtual Network Appliance Cluster Detailed Design section.

Note:

Each EVPN Route Controller node exposes up to three configurable interfaces, each connected to a dedicated distributed Virtual Port Group: a management interface and one or two interfaces for Multiprotocol BGP-EVPN peering. An additional internal interface, not user-configurable, handles notification messages during vMotion events.

Distributed VXLAN Connectivity Model for Version 9.1.0 EVPN Route Controller Design Requirements and Recommendations

Table 586: Distributed VXLAN Connectivity Model for Version 9.1.0 EVPN Active/Standby Route Controller Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-001	Deploy two (2) EVPN Route Controllers nodes belonging to the same EVPN Route Controller cluster in Active/Standby mode.	• EVPN Route Controller nodes are required to establish a common EVPN control plane between the physical EVPN-VXLAN network fabric and the Distributed TGW environment. • Two (2) EVPN Route Controller nodes are required to support High Availability.	• vSphere cluster resources must be available to support the two EVPN Route Controller nodes. • Only the active EVPN Route Controller node establishes a Multiprotocol BGP-EVPN session with the EVPN gateway in the physical EVPN-VXLAN network fabric.
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-002	Select the EVPN Route Controller node size medium.	• The medium EVPN Route Controller node size is required for production deployments, supporting enterprise-grade prefix scale. • The small size is limited to Proof-of-Concept and test environments.	You must provide sufficient compute resources to support the chosen EVPN Route Controller node size.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-003	Create at least a new single dedicated vSphere Distributed Port Group (dvPG) for the EVPN Route Controller node Multiprotocol BGP-EVPN interface and make the same vSphere dvPG available to both EVPN Route Controller nodes.	• The active and standby EVPN Route Controller nodes share a common floating IP for Multiprotocol BGP-EVPN peering per vSphere dvPG, requiring Layer 2 adjacency between both nodes connected to this dedicated vSphere dvPG. • Each dedicated vSphere dvPG carries the Multiprotocol BGP-EVPN sessions between the EVPN Route Controllers and the EVPN gateway in the physical EVPN-VXLAN network fabric. • The VLAN and CIDR assigned per dedicated vSphere dvPG must provide IP connectivity between the EVPN Route Controller Multiprotocol BGP-EVPN interface and the EVPN gateway loopback interfaces, which are part of the global routing table in the physical EVPN-VXLAN network fabric. • A second vSphere dvPG for the Multiprotocol BGP-EVPN interface provides VLAN-level redundancy intended for corner cases only. • Traffic segregation from other vSphere dvPGs allows organizations to implement and enforce network security policies.	• Each EVPN Route Controller node supports up to two Multiprotocol BGP-EVPN interfaces, each mapped to a dedicated vSphere dvPG. • The EVPN Route Controller deployment workflow allows only a single vSphere dvPG to be configured for the Multiprotocol BGP-EVPN interface during initial deployment; the second vSphere dvPG must be added post-deployment. • Additional configuration effort is required in the physical EVPN-VXLAN network fabric. • For vSphere clusters deployed across racks, whether hosting workloads or EVPN Route Controller nodes, where the physical EVPN-VXLAN network fabric is required to extend Layer 2 for the ESX host TEP network or the Multiprotocol BGP-EVPN interface, VRF leaking may be required to reach the EVPN gateway VTEP IP address and Multiprotocol BGP-EVPN loopback IP addresses in the global routing table.
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-004	Assign a unique IPv4 address and subnet mask to each Multiprotocol BGP-EVPN interface of each EVPN Route Controller node.	To support the BGP floating IP (VIP), each EVPN Route Controller node requires a dedicated IPv4 address on each Multiprotocol BGP-EVPN interface.	Only IPv4 addresses are supported for the Multiprotocol BGP-EVPN interface.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-005	To support High Availability for Multiprotocol BGP-EVPN peering, configure a BGP floating IP (VIP) for each vSphere dvPG assigned to the Multiprotocol BGP-EVPN interfaces.	• To support High Availability for Multiprotocol BGP-EVPN peering, a floating IP (VIP) is configured per vSphere dvPG as the shared BGP source address for the Multiprotocol BGP-EVPN interfaces. • Each EVPN Route Controller node has its own dedicated IPv4 address per Multiprotocol BGP-EVPN interface and vSphere dvPG, however only the active EVPN Route Controller uses the floating IP, or floating IPs if two vSphere dvPGs are configured, for peering with the EVPN gateways in the physical EVPN-VXLAN network fabric.	Upon an EVPN Route Controller failover, a new Multiprotocol BGP-EVPN session must be established with the new active EVPN Route Controller, as the Multiprotocol BGP-EVPN connections remain administratively down on the standby EVPN Route Controller.
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-006	Configure a unique local Autonomous System Number (ASN) for Multiprotocol BGP-EVPN.	• To establish a Multiprotocol BGP-EVPN session between the EVPN Route Controller and the EVPN gateways in the physical EVPN-VXLAN network fabric, a local ASN is required. • A unique local ASN enables native AS_PATH-based loop detection through eBGP.	A unique BGP ASN must be available for the EVPN Route Controller cluster.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-007	Configure at least two Multiprotocol BGP-EVPN peerings from the EVPN Route Controller cluster to the EVPN gateways in the physical EVPN-VXLAN network fabric.	• Multiprotocol BGP-EVPN peering with multiple EVPN gateways in the physical EVPN-VXLAN network fabric provides redundant Multiprotocol BGP-EVPN paths, ensuring High Availability. • The EVPN gateway role is configured either on all spine switches or on two VXLAN leaf switches in the physical EVPN-VXLAN network fabric. • Multiprotocol BGP-EVPN peering with VXLAN leaf switches not designated as EVPN gateways in the physical EVPN-VXLAN network fabric is neither required nor recommended. • The EVPN gateway, often referred to as the border gateway, must advertise its VTEP IP address as the next hop to the EVPN Route Controller via EVPN Route Type 5, which is then propagated to each individual ESX host to ensure every ESX host knows the path to physical destinations via the default route pointing to VTEP IP address of the EVPN gateway.	Only IPv4-based Multiprotocol BGP-EVPN peering is supported.
VCF-WLDCON-DTGWVXLAN-RC-REQD-CFG-008	Configure the BGP maximum hop limit to match the number of Layer 3 hops between the Multiprotocol BGP-EVPN peering source and destination.	Since Multiprotocol BGP-EVPN peering in a Distributed Transit Gateway environment with EVPN-VXLAN integration is typically not established between directly connected peering endpoints, the maximum hop limit must be adjusted accordingly.	By default, the Multiprotocol BGP-EVPN maximum hop limit is set to 1, meaning peering endpoints must be directly connected.

Table 587: Distributed VXLAN Connectivity Model for Version 9.1.0 EVPN Active/Standby Route Controller Design Recommendations

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-001	Use the default memory reservation with 100% for EVPN Route Controller nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a EVPN Route Controller node, ensuring this full memory amount remains available to the EVPN Route Controller node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-002	Use the default CPU reservation priority in NSX set to high shares for EVPN Route Controller nodes.	Setting CPU shares to "High" ensures that EVPN Route Controller nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are generally less critical than high memory reservations.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-003	Create at new vSphere Distributed Port Group for the EVPN Route Controller node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel.	Traffic segregation enables organizations to implement and enforce network security policies.	• For the management domain, a dedicated vSphere Distributed Port Group for the EVPN Route Controller node management interface already exists — this is the same vSphere Distributed Port Group Port Group used by the management stack. • Depending on the selected EVPN Route Controller cluster design, a EVPN Route Controller node management vSphere Distributed Port Group is required for each vSphere cluster, or rack.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-004	Select static IP allocation for the EVPN Route Controller node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-005	Use the central node profile configuration for EVPN Route Controller nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all EVPN Route Controller nodes.	• EVPN Route Controller nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual EVPN Route Controller node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-006	Deploy the two EVPN Route Controller nodes on two different vSphere clusters across two separate racks to improve availability.	Deploying each of the two EVPN Route Controllers on an independent vSphere cluster in a separate rack provides rack fault tolerance and independent storage when using vSAN.	• Requires additional resources to support multiple vSphere clusters deployed across different racks with independent storage (vSAN). • Layer 2 adjacency for the EVPN Route Controller Multiprotocol BGP-EVPN interface is required across the racks where the two EVPN Route Controller reside.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-007	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the two EVPN Route Controller nodes when both are running on the same vSphere cluster.	Prevents the two EVPN Route Controller nodes from running on the same ESX host and risking the High Availability of the EVPN Route Controller cluster.	A minimum of three ESX hosts is required in the vSphere cluster when deploying two EVPN Route Controller nodes, to allow placing an ESX host into maintenance mode.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-008	Use the default MTU for the EVPN Route Controller node Multiprotocol BGP-EVPN interface.	Since the EVPN Route Controller node is not involved in the data plane forwarding path, the MTU setting is less critical and would only impact the fragmentation of large BGP UPDATE messages.	• If a large BGP UPDATE packet exceeds the MTU of a link along the multi-hop path and fragmentation is blocked due to the DF (Don't Fragment) bit being set, the packet will be silently dropped. • To avoid BGP UPDATE fragmentation issues, ensure Path MTU Discovery is enabled on both BGP peering endpoints and that ICMP is unblocked along the entire path between them.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-009	Ensure BGP Graceful Restart is enabled on the EVPN gateway in the physical EVPN-VXLAN network fabric.	Given the currently supported Active/Standby mode of the EVPN Route Controller and the requirement to re-establish a new Multiprotocol BGP-EVPN session upon a node failover, which can result in extended failover times, it is recommended to configure BGP Graceful Restart to maintain the Multiprotocol BGP-EVPN session.	BGP Graceful Restart is enabled by default on the EVPN Route Controller, requiring no additional configuration.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-010	Do not enable BFD between the EVPN Router Controller nodes and the EVPN gateway in the physical EVPN-VXLAN network fabric.	• Given the currently supported Active/Standby mode of the EVPN Route Controller, disabling BFD is recommended as a new Multiprotocol BGP-EVPN session must be re-established upon a node failover, which can result in extended failover times, and BFD would prevent BGP Graceful Restart from protecting traffic forwarding during this process. • BFD should be disabled to avoid false positive failures caused by ECMP hashing sending BFD and BGP packets on different paths through the physical EVPN-VXLAN network fabric, as BFD would immediately tear down the Multiprotocol BGP-EVPN session without allowing BGP Graceful Restart to protect traffic forwarding.	None.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-011	Use fast, non-default BGP timers, 1/3 seconds.	Since BFD is not recommended for EVPN Route Controller nodes in Active/Standby mode with BGP Graceful Restart enabled, fast BGP timers are required to ensure rapid failover detection.	• Fast BGP timers are required for rapid failure detection and failover in the absence of BFD. • Non-default BGP timers cannot be configured during the EVPN Route Controller deployment workflow, and must be adjusted post-deployment.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-012	Enable Anycast VTEP on the EVPN gateways in the physical EVPN-VXLAN network fabric.	Configuring Anycast VTEP on the EVPN gateways in the physical EVPN-VXLAN network fabric reduces traffic blackholing in the event of an EVPN gateway node failure, as no default route withdrawal on each ESX host is required.	Additional configuration effort is required on all EVPN gateway nodes in the physical EVPN-VXLAN network fabric.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-013	Advertise only the default route from the EVPN gateways in the physical EVPN-VXLAN network fabric to the EVPN Route Controller cluster.	The EVPN Route Controller cluster typically requires no more than the default route to program the egress path on each ESX host, ensuring every host knows the path to physical destinations via the EVPN gateways.	Additional configuration effort is required to configure an outbound route filter on all EVPN gateway nodes in the physical EVPN-VXLAN network fabric.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-014	Enable ECMP on the EVPN Route Controller cluster unless Anycast VTEP is already configured on the EVPN gateways in the physical EVPN-VXLAN network fabric.	• ECMP allows multiple default routes to be installed on each ESX host where a Distributed TGW is instantiated, supporting individual data plane paths to each EVPN gateway. • If Anycast VTEP is enabled, only a single default route needs to be installed on each ESX host, making ECMP unnecessary.	ECMP is only beneficial when BGP multipath criteria are met, which is typically the case in a Clos-based spine-leaf physical EVPN-VXLAN network fabric.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-015	Do not enable multipath relax unless the ECMP multipath criteria are not met due to differing BGP AS_PATH values caused by different ASNs in the AS_PATH.	By default, ECMP BGP multipath requires all criteria to be met, including identical AS_PATH length and identical Autonomous System Numbers (ASNs) in the AS_PATH. When EVPN gateways in the physical EVPN-VXLAN network fabric belong to different ASNs, ECMP BGP multipath cannot be achieved unless multipath relax is enabled.	None.
VCF-WLDCON-DTGWVXLAN-RC-RCMD-CFG-016	Configure a BGP password for each Multiprotocol BGP-EVPN session.	An MD5-based BGP password secures the TCP connection for Multiprotocol BGP-EVPN peering between the EVPN Route Controller and the physical EVPN-VXLAN network fabric.	None.

Distributed VXLAN Connectivity Model for Version 9.1.1

The *Distributed VXLAN Connectivity Model for Version 9.1.1* integrates NSX with the physical EVPN-VXLAN network fabric, unifying virtual workloads, and physical endpoints into a single routing domain.

Distributed VXLAN Connectivity Model Comparison

Attribute	Version 9.1.0	Version 9.1.1
Stateful network services when using Virtual Network Appliance	Not supported	Supported
Avi Load Balancer	Not supported	Supported
EVPN Route Controller HA mode	Active/Standby	• Active/Standby • Active/Active with scale-out capability of up to eight nodes
EVPN-VXLAN Direct Prefix Routing	Not supported	Supported
DHCP relay on VPC public subnets to support external DHCP server connectivity	Not supported	Supported
Host group affinity using the compute policy type Best Effort Restart (BER)	• Not supported for EVPN Route Controller nodes • Not supported for Virtual Network Appliance nodes for stateful network services	• Supported for EVPN Route Controller nodes • Supported for Virtual Network Appliance nodes for stateful network services

Distributed VXLAN Connectivity Model for Version 9.1.1 Overview

The *Distributed VXLAN Connectivity Model for Version 9.1.1* leverages a decoupled Multiprotocol BGP-EVPN control plane and VXLAN data plane to connect Virtual Private Clouds (VPCs) to the physical EVPN-VXLAN network fabric, with Distributed Transit Gateways providing connectivity to tenant Virtual Routing and Forwarding (VRF) instances within the physical network fabric.

Ethernet VPN (EVPN) is a widely adopted data center technology that serves as the distributed control plane for the physical network fabric, using BGP to advertise both Layer 2 MAC (not supported in the current EVPN-VXLAN integration with NSX) and Layer 3 IP reachability across the network. VXLAN acts as the corresponding data plane encapsulation protocol, the counterpart to the GENEVE protocol used for encapsulation and tunneling within the NSX overlay environment.

With the *Distributed VXLAN Connectivity Model for Version 9.1.1*, the NSX control plane managed by the NSX Manager appliances continues to use GENEVE for east-west traffic between ESX hosts, while leveraging VXLAN for routing with the physical EVPN-VXLAN network fabric to reach physical endpoints attached to the physical EVPN-VXLAN network fabric or external network.

The *Distributed VXLAN Connectivity Model for Version 9.1.1* supports multiple VRFs/tenants. Each VRF/tenant is identified by a Layer 3 VNI configured in the External Connection, which maps 1:1 to a Distributed Transit Gateway.

> **Note:**
>
> The *Distributed VXLAN Connectivity Model for Version 9.1.1*, unlike the *Distributed VLAN Connectivity Model*, does not require an external VLAN spanning across the Distributed Transit Gateways. Connectivity between each ESX host and the physical EVPN-VXLAN network fabric is established through the ESX host VTEP interface. This *Distributed VXLAN Connectivity Model for Version 9.1.1* supports, for example, Multi-Rack Layer 3 vSphere Cluster Model for workloads without compromising rack isolation through a Layer 3 boundary.

> **Note:**
>
> The ESX VMkernel Tunnel Endpoint (TEP) interface can handle both GENEVE and VXLAN traffic simultaneously. In this document, TEP (GENEVE) and VTEP (VXLAN) are used interchangeably when referring to ESX hosts.

The EVPN gateway in the physical EVPN-VXLAN network fabric plays a crucial role in the *Distributed VXLAN Connectivity Model for Version 9.1.1* as it can be considered the central hub for both the data plane, handling VXLAN tunnels, and the control plane BGP-EVPN session with the EVPN Route Controller. The EVPN gateway is configured either on all spine switches or on a subset of VXLAN leaf switches in the physical EVPN-VXLAN network fabric. The EVPN gateway, often referred to as the border gateway, advertises its VTEP IP address to the EVPN Route Controller. This VTEP IP, along with the default route advertised by the EVPN gateway, is then installed on each ESX host, with the EVPN gateway as next-hop. This enables traffic to reach destinations outside of NSX and physical endpoints that are connected to VXLAN leaf switches via the default route.

> **Note:**
>
> The *Distributed VXLAN Connectivity Model for Version 9.1.1* requires that the EVPN gateway advertise the default route to the EVPN Route Controller, regardless of whether EVPN-VXLAN Direct Prefix Routing is enabled or disabled. The default route can either originate on the EVPN gateway itself or be advertised from any other physical network device, such as a firewall.

The *Distributed VXLAN Connectivity Model for Version 9.1.1* supports EVPN-VXLAN Direct Prefix Routing. This feature optimizes the traffic forwarding path and reduces packet processing overhead on the EVPN gateway. When EVPN-VXLAN Direct Prefix Routing is disabled, which is the default setting, egress traffic from a VM connected to a VPC subnet is forwarded by the Distributed Transit Gateway (DTGW) through the ESX VTEP to the EVPN gateway VTEP leverage VXLAN encapsulation. At the EVPN gateway, the VXLAN tunnel is decapsulated for a route lookup to identify the physical endpoint destination. The lookup is based on the prefix learned via EVPN Route Type 5 from the VXLAN leaf switch where the physical endpoint is connected, or from outside the EVPN-VXLAN fabric via the routing protocol in use. The traffic is then either re-encapsulated into a new VXLAN tunnel to reach that physical endpoint within the EVPN-VXLAN network fabric, or forwarded without further VXLAN encapsulation to another physical endpoint outside the EVPN-VXLAN network fabric.

Ingress data plane traffic from a physical endpoint connected to a VXLAN leaf switch towards a VM connected to a VPC subnet is initially forwarded again to the EVPN gateway VTEP using VXLAN encapsulation. At the EVPN gateway, the

VXLAN tunnel is decapsulated for a route lookup to identify the destination VM on a VPC subnet. The route lookup is based on the /32 host prefix learned via the EVPN Route Controller using EVPN Route Type 5. The traffic is then re-encapsulated into another VXLAN tunnel toward the ESX VTEP hosting that VM. Regardless of traffic direction, when EVPN-VXLAN Direct Prefix Routing is disabled, and whether it originates from the ESX host or the VXLAN leaf switch, VXLAN encapsulation is always removed and re-added on the physical switch acting as an EVPN gateway.

When EVPN-VXLAN Direct Prefix Routing is enabled, a VXLAN tunnel is established directly between the ESX VTEP and the VXLAN leaf switch VTEP to reach the physical endpoint. This feature can be enabled per External Connection on the Distributed VXLAN Connection as an add-on to the default forwarding behavior. In this mode, selected traffic flows use EVPN-VXLAN Direct Prefix Routing, optional with enabled filters configured on the EVPN gateways and VXLAN leaf switches, bypassing the EVPN gateway without its VXLAN decapsulation and encapsulation. Other traffic flows continue using the path through the EVPN gateway with the additional VXLAN encapsulation. This feature should only be enabled when an optimized datapath is required for latency-sensitive traffic flows, such as storage protocols like NFS and S3/object storage.

Note:

When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN Route Controller receives either EVPN Route Type 5 or Route Type 2 to resolve routes to /32 host prefix destinations, depending on the physical network fabric vendor and their recommendation. Regardless of the received Route Type, the EVPN Route Controller always advertises only EVPN Route Type 5 toward the EVPN gateways.

Note:

EVPN-VXLAN Direct Prefix Routing requires IP connectivity for VXLAN encapsulated traffic between each ESX VTEP and the VTEPs of all EVPN gateways and VXLAN leaf switches. This is typically achieved either through VRF leaking or by connecting the ESX VTEP interfaces to the global routing table where the physical switch VTEPs reside. Without EVPN-VXLAN Direct Prefix Routing, IP connectivity for VXLAN encapuslated is only required between each ESX VTEP and the VTEPs of all EVPN gateways.

The *Distributed VXLAN Connectivity Model for Version 9.1.1* supports stateful network services such as Default Outbound NAT (Auto-SNAT), custom NAT, DHCP relay for external DHCP servers, and load balancing. These stateful network services require a Virtual Network Appliance (VNA) cluster. vSphere Supervisor, VKS workloads and VCF Automation All Apps Orgs require load balancing capabilities, this is achieved using either VNA load balancer or AVI Load Balancer.

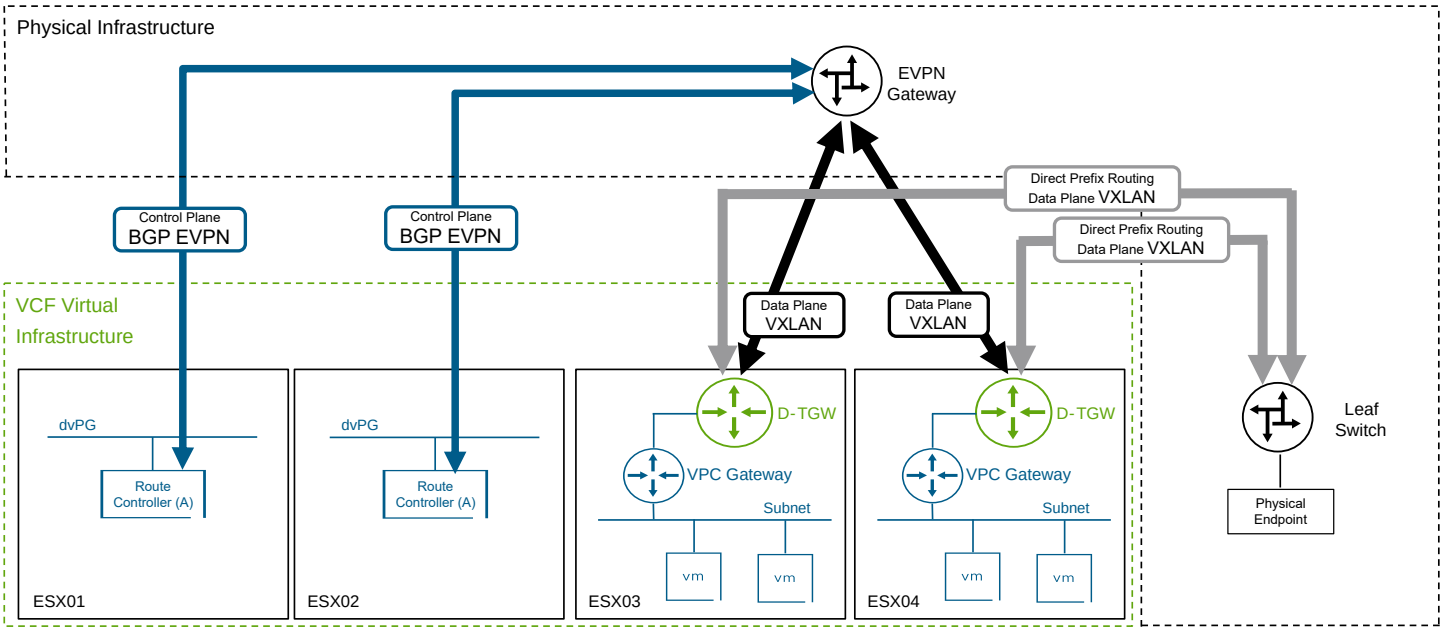
The VNA nodes used for stateful network services are distinct from the VNA used for the EVPN Route Controller. When stateful network services are required in the *Distributed VXLAN Connectivity Model for Version 9.1.1* the deployment results in two separate VNA clusters at least one for the EVPN Route Controller and at least one for stateful network services. For VNA nodes deployed for stateful network services such as NAT, refer to the Virtual Network Appliance Cluster Detailed Design section.

Note:

The *Distributed VXLAN Connectivity Model for Version 9.1.1* supports Avi Load Balancer. When only Avi Load Balancer is used, with no other stateful network services such as Default Outbound NAT (Auto-SNAT) or custom NAT, a VNA cluster does not need to be deployed, nor do N-S Services need to be enabled. More details, please refer to section Avi Load Balancer Model. More details, please refer to section Avi Load Balancer Model.

The EVPN Route Controller unifies both environments, establishing a common control plane across the virtual and physical network fabric by peering directly with EVPN gateways using Multiprotocol BGP-EVPN. More details, please refer to section Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Route Controller Overview.

Distributed VXLAN Connectivity Model for Version 9.1.1 Logical View with Active/Active EVPN Route Controller



Distributed VXLAN Connectivity Model for Version 9.1.1 Attributes

A *Distributed VXLAN Connectivity Model for Version 9.1.1* has the following attributes.

Attribute	Detail
Network Consumption Model	• VCF Automation All Apps Orgs Consumption Model • Virtual Private Cloud (VPC) Consumption Model
Transit Gateway Type	Distributed
Node Type	• EVPN Route Controller is required - Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Route Controller Overview • Virtual Network Appliance (optional for North/South Services)
Transit Gateway HA Mode	Distributed
VPC Gateway Services	• VPC Gateway without North/South Services • VPC Gateway with North/South Services – Required for VCF Automation All Apps Orgs and vSphere Supervisor – Instantiated on Virtual Network Appliance cluster
Load Balancing Options	• VNA load balancer • Avi Load Balancer
NSX Tenancy Constructs	NSX Projects, VPCs, Multiple Transit Gateways
Consumption options	• VCF Automation All Apps Orgs • vSphere Supervisor (i.e. VM Service, VKS, and more) • vCenter UI/API • NSX UI/API
External IP Blocks	Multiple (up to 5)
Provider Gateway Type	Physical EVPN-VXLAN network fabric with EVPN gateway

Distributed VXLAN Connectivity Model for Version 9.1.1 Options

Design Area	Options
EVPN Route Controller High Availability Model	• Active/Active (up to 8 nodes) • Active/Standby (up 2 nodes)
EVPN Route Controller: Multiprotocol BGP-EVPN interface per EVPN Route Controller Node	• One • Two (only supported with Active/Standby mode)
EVPN-VXLAN Direct Prefix Routing	• EVPN gateway only • Direct Prefix Routing (requires version 9.1.1)

Distributed VXLAN Connectivity Model for Version 9.1.1 Design Requirements and Recommendations

Table 588: Distributed VXLAN Connectivity Model for Version 9.1.1 Design Requirements

Use this list of general requirements for the *Distributed VXLAN Connectivity Model for Version 9.1.1* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-011	The Distributed VXLAN Connectivity Model for Version 9.1.1 requires an EVPN Route Controller cluster.	The EVPN Controller cluster establishing a common control plane across the VCF virtual infrastructure and physical EVPN-VXLAN network fabric by peering with EVPN gateways using Multiprotocol BGP-EVPN.	vSphere cluster resources must be available to support the EVPN Route Controller node(s).
VCF-WLDCON-DTGWVXLAN-REQD-CFG-012	The EVPN Route Controller cluster must belong to the same NSX instance as the Distributed TGW with EVPN-VXLAN integration.	The EVPN Route Controller and the Distributed TGW with EVPN-VXLAN integration must share the same internal overlay segment for data plane notifications.	An EVPN Route Controller cannot be shared across different NSX instances.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-013	Create one or more External Connection of type Distributed VXLAN Connection.	The External Connection requires an EVPN Route Controller cluster to be attached and defines the Multiprotocol BGP-EVPN attributes, such as Layer 3 VNI, Route Distinguisher (RD), and Route Target (RT), used by the EVPN Route Controller to establish peering with the physical EVPN-VXLAN network fabric.	The EVPN Route Controller cluster must be configured prior to creating the External Connection.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-014	Define a single Layer 3 VNI for each External Connections.	• A 1:1 mapping is required between a Distributed TGW and its associated Layer 3 VNI defined in the External Connection and the corresponding VRF/tenant in the physical EVPN-VXLAN network fabric. • The Layer 3 VNI, along with the Route Distinguisher (RD), Route Target (RT) and others, are specific Multiprotocol BGP-EVPN attributes carried as EVPN Route Type 5 between the EVPN Route Controller to the EVPN gateway in the physical EVPN-VXLAN network fabric. **Note:** When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN Route Controller may receive EVPN Route Type 2 in addition to EVPN Route Type 5 from the EVPN gateway.	For each VRF/tenant a dedicated Distributed TGW and its associated Layer 3 VNI defined in the External Connection is required.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-015	Define a single Route Distinguisher (RD) for each External Connection.	• A unique Route Distinguisher (RD), prepended to each prefix by the EVPN Route Controller, ensures EVPN routes remain distinguishable across VRFs/tenants when advertised to the EVPN gateway in the physical EVPN-VXLAN network fabric. • The Route Distinguisher (RD) is unique per VRF/tenant and maps directly to a single Distributed TGW and its associated Layer 3 VNI defined in the External Connection.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-016	Define at least one pair of Route Targets (RTs), consisting of an import and an export RT, for each External Connection. These Route Targets (RTs) must correspond to the Route Targets (RTs) defined on the EVPN gateways.	• The Route Target (RT) and Layer 3 VNI together identify the VRF/tenant and is defined in the External Connection. • Route Targets (RTs) are required to control the import and export policy, determining which VRFs/ tenants accept and install the advertised routes. • A corresponding pair of Route Targets (RTs) is required, configured on both the External Connection and the EVPN gateways, to import and install VRF routes.	None.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-017	Advertise a default route from the EVPN gateway to the EVPN Route Controller.	• A default route advertisement is required to ensure there is always network path from each ESX host to the EVPN gateway, regardless of whether the EVPN-VXLAN Direct Prefix Routing feature is enabled or disabled. • The default route can either originate on the EVPN gateway itself, or be advertised through the EVPN gateway from any other physical network device, such as a firewall.	None
VCF-WLDCON-DTGWVXLAN-REQD-CFG-018	Assign at least one range of VRF/tenant specific routable IPs as part of the External IP block in the VPC Connectivity Profile.	This range, or these ranges, covers all inbound and outbound VPC traffic for a specific VRF/tenant. It including VPC public subnets, External IPs (stateless 1:1 NAT), translated NAT IPs, load balancer VIPs and the VPC public subnet gateway IP when the DHCP relay for external DHCP server is enabled.	If the range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-019	When stateful network services are required, assign a VNA cluster to the VPC Connection Profile. Select one of the presented Virtual Network Appliance Cluster Models.	A VNA cluster is required for stateful network services when using a Distributed TGW with EVPN-VXLAN integration.	• Additional resources are required for the VNAcluster. • Only traffic that is sourced from VPC private or private TGW subnets is forwarded to the VNA cluster to perform stateful services such as NAT. • To enable editing of the VPC Connection Profile, which is required, for example, to assign the VNA cluster or enable N-S Services, ensure all ESX hosts have the correct minimum OS version configured by leveraging the feature opt-in API. For more details, see Enable Advanced Networking Services for Distributed VXLAN Connections.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-020	When deploying a VNA cluster for stateful network services, ensure that N-S Services is enabled; otherwise, disable it.	N-S Services are only available when a VNA cluster is assigned to the VPC Connection Profile used with the Distributed TGW with EVPN-VXLAN integration.	• A VNA cluster is required to enable N-S Services — see Virtual Network Appliance Cluster Models for details. • To enable editing of the VPC Connection Profile, which is required, for example, to assign the VNA cluster or enable N-S Services, ensure all ESX hosts have the correct minimum OS version configured by leveraging the feature opt-in API. For more details, see Enable Advanced Networking Services for Distributed VXLAN Connections. • When only Avi Load Balancer is used, with no other stateful network services such as Default Outbound NAT (Auto-SNAT) or custom NAT, a VNA cluster does not need to be deployed, nor do N-S Services need to be enabled.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-021	When Default Outbound NAT (Auto-SNAT) or custom NAT is required, ensure a VNA cluster is assigned and N-S Services is enabled.	Default Outbound NAT (Auto-SNAT) or custom NAT requires N-S Services to be enabled, which in turn requires a VNA cluster to be assigned to the VPC Connection Profile used by the Distributed TGW with EVPN-VXLAN integration.	• A VNA cluster is required to enable Default Outbound NAT (Auto-SNAT) or custom NAT — see Virtual Network Appliance Cluster Models for deployment details. • Default Outbound NAT (Auto-SNAT) and custom NAT is only supported for traffic matching VPC private subnets or private-TGW subnets as the source. • To enable editing of the VPC Connection Profile, which is required, for example, to assign the VNA cluster or enable N-S Services, ensure all ESX hosts have the correct minimum OS version configured by leveraging the feature opt-in API. For more details, see Enable Advanced Networking Services for Distributed VXLAN Connections.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-022	When DHCP relay for external DHCP servers is required, ensure a VNA cluster is deployed and N-S Services is enabled.	DHCP relay for external DHCP servers requires N-S Services to be enabled, which in turn requires a VNA cluster to be assigned to the VPC Connection Profile used by the Distributed TGW with EVPN-VXLAN integration.	• A VNA cluster is required to enable DHCP relay for external DHCP servers — see Virtual Network Appliance Cluster Models for deployment details. • DHCP relay for external DHCP servers is only supported for workload VMs connected to VPC public subnet. • When the DHCP server is connected to a VPC subnet (of any type), a VNA cluster is not required for DHCP relay. • To enable editing of the VPC Connection Profile, which is required, for example, to assign the VNA cluster or enable N-S Services, ensure all ESX hosts have the correct minimum OS version configured by leveraging the feature opt-in API. For more details, see Enable Advanced Networking Services for Distributed VXLAN Connections.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-023	When a VNA load balancer is required, ensure a VNA cluster is deployed and N-S Services is enabled.	The VNA load balancer requires N-S Services to be enabled, which in turn requires a VNA cluster to be assigned to the VPC Connection Profile used by the Distributed TGW with EVPN-VXLAN integration.	• A VNA cluster is required to enable VNA load balancer — see Virtual Network Appliance Cluster Models for deployment details. • Please see Virtual Network Appliance Load Balancer Model for details.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-024	Without EVPN-VXLAN Direct Prefix Routing, connectivity to the EVPN gateway VTEP interfaces from each ESX host is sufficient to establish VXLAN tunnels for data plane traffic. When EVPN-VXLAN Direct Prefix Routing is enabled, ensure that the VTEP interface on each ESX host has IP connectivity to both the EVPN gateway VTEP interfaces and the VXLAN leaf switches VTEPs.	Data plane traffic requires VXLAN tunnels between the ESX host VTEP interface and the EVPN gateway VTEPs, and the VXLAN leaf switch VTEPs when EVPN-VXLAN Direct Prefix Routing is enabled. **Note:** All physical VTEP interfaces reside in the global routing table of the physical EVPN-VXLAN network fabric.	The ESX host VTEP interface must either be placed into the global routing table, or, if the ESX host VTEP interface belongs to the tenant routing space, VRF leaking is required.

Table 589: Distributed VXLAN Connectivity Model for Version 9.1.1 Design Recommendations

Use this list of general recommendations for the *Distributed VXLAN Connectivity Model for Version 9.1.1* to provide workload connectivity for the Virtual Private Cloud (VPC) Consumption Model and the VCF Automation All Apps Orgs Consumption Model.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-003	Assign an IP range of non-routable IPs to the private-TGW IP block in the VPC Connectivity Profile (a /23 CIDR or larger is preferred).	This private-TGW IP range will be used for inter-VPC connectivity within the tenant boundary. It will be used to assign CIDRs for private-TGW VPC Subnets.	• If private-TGW subnets are not required, this step can be omitted. • Workloads on private-TGW subnets will be reachable from any other tenant workloads without requiring NAT, but no connectivity from workloads outside of the tenant boundary will be available without exposing the workloads via a NAT Rule, or a load balancer VIP, or an External IP (stateless 1:1 NAT). • If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-004	To allow for granular application placement restrict the network span of VPC subnets to one or more vSphere clusters within an NSX instance. This is typically required for remote vSphere clusters or DMZ deployments.	• Provides isolation of VPC subnets and applications in different vSphere clusters. • The network span is defined on the Distributed TGW and applies to all connected VPCs and their associated VPC subnets. • vSphere clusters can be shared across multiple network spans.	Exclusive network span allows defining a dedicated network span for a specific TGW and the associated VPCs, preventing it from being shared or overlapping with other network spans.
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-005	For latency-sensitive traffic flows, such as storage protocols like NFS and S3/object storage, enabled Direct Prefix Routing.	With Direct Prefix Routing selected, traffic flow is forwarded between the VXLAN leaf switch and the ESX host, bypassing the EVPN gateway and avoiding its VXLAN decapsulation and encapsulation.	Enable EVPN-VXLAN Direct Prefix Routing only for a limited number of latency-sensitive physical endpoints, rather than at the VXLAN leaf switch level, by applying additional granular prefix filtering for these endpoints, to minimize the routing table on each ESX host.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-006	When using the EVPN-VXLAN Direct Prefix Routing feature, define at least one additional pair of Route Targets (RTs), consisting of an import and an export Route Target (RT), for the VXLAN leaf switches and the External Connection.	• The Route Target (RT) and Layer 3 VNI together identify the VRF/tenant and is defined in the External Connection. • Route Targets (RTs) are required to control the import and export policy, determining which VRFs/ tenants accept and install the advertised routes. • A corresponding pair of Route Targets (RTs) is required, configured on both the External Connection and the VXLAN leaf switches, to import and install VRF routes. • Only the VXLAN leaf switches connected to latency-sensitive endpoints require these additional Route Targets (RTs).	To minimize the routing table on each ESX host, apply granular prefix filtering to these additional Route Targets (RTs) for this limited number of latency-sensitive physical endpoints.

Table 590: Distributed VXLAN Connectivity Model for Version 9.1.1 with vSphere Supervisor or VCF-A All Apps Org Design Requirements

Use this list of additional requirements when adopting the *Distributed VXLAN Connectivity Model for Version 9.1.1* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-025	Assign a VNA cluster to the VPC Connection Profile. Select one of the presented Virtual Network Appliance Cluster Models.	A VNA cluster is required for stateful network services when using a Distributed TGW with VXLAN integrations.	• Additional resources are required for the VNA cluster. • Only traffic that is sourced from VPC private or private TGW subnets is forwarded to the VNA cluster to perform stateful services such as NAT.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-026	In the Default VPC Connectivity Profile of the Default Project, assign an IP range of non-routable IPs to the private-TGW IP block (a /23 CIDR or larger is preferred).	This private-TGW IP range will be used for inter-VPC connectivity within the tenant boundary. It will be used to assign CIDRs for private-TGW VPC Subnets.	• Workloads on private-TGW subnets will be reachable from any other tenant workloads without requiring NAT, but no connectivity from workloads outside of the tenant boundary will be available without exposing the workloads via a NAT Rule, or a Load Balancer VIP, or an External IP (stateless 1:1 NAT). • If a range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-REQD-CFG-027	In the Default VPC Connectivity Profile of the Default Project enable N-S Services.	N-S Services is required for vSphere Supervisor and VCF-A All Apps Orgs.	A VNA cluster is required to enable N-S Services — see Virtual Network Appliance Cluster Models for details.
VCF-WLDCON-DTGWVXLAN-REQD-CFG-028	In the Default VPC Connectivity Profile of the Default Project enable Default Outbound NAT (Auto-SNAT).	Default Outbound NAT (Auto-SNAT) is necessary for vSphereSupervisor and VCFA All Apps Orgs.	• A VNA cluster is required to enable Default Outbound NAT (Auto-SNAT) — see Virtual Network Appliance Cluster Models for details. • Default Outbound NAT (Auto-SNAT) is only supported for traffic matching VPC private subnets and private-TGW subnets as the source.

Table 591: Distributed VXLAN Connectivity Model for Version 9.1.1 with vSphere Supervisor or VCF-A All Apps Org Design Recommendations

Use this list of additional recommendations when adopting the *Distributed VXLAN Connectivity Model for Version 9.1.1* to configure virtual networking for a environment with vSphere Supervisor with VPC networking or VCF Automation with All Apps Org(s).

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-007	Use the Default Project for the vSphere Supervisor namespace's VPCs.	• When VCFA All Apps Orgs are in use, in each Organization will have its dedicated project. • The default space will not be used and it will be available to be consumed by system VPCs to support the IaaS control plane. • The WLD creation workflow follows this model when the new WLD has a new dedicated NSX Manager.	• This approach is not recommended if the Default Project is also consumed via other means, for example via the vCenter UI, which allows the configuration of VPCs in the Default Project. • In general, when VCF Automation is in use, it's not recommended to mix another network consumption vehicle in the same workload domain.
VCF-WLDCON-DTGWVXLAN-RCMD-CFG-008	Assign at least one range of VRF/tenant specific routable IPs as part of the External IP block in the Default VPC Connectivity Profile within the Default Project (a /24 CIDR or larger is preferred).	This range, or these ranges, covers all inbound and outbound VPC traffic for a specific VRF/tenant. It including VPC public subnets, External IPs (stateless 1:1 NAT), translated NAT IPs, load balancer VIPs and the VPC public subnet gateway IP when the DHCP relay for external DHCP server is enabled.	If the range is exhausted additional CIDRs can be added up to a total of 5. Ranges can be added in a CIDR format only.

Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Route Controller Overview

The EVPN Route Controller establishes a common control plane across the virtual and physical network fabric. The EVPN Route Controller achieves this by peering directly with EVPN gateways in the physical EVPN-VXLAN network fabric using Multiprotocol BGP-EVPN.

Although the EVPN Route Controller is not involved in data plane forwarding, its role is crucial. Careful High Availability consideration is required, including rack fault tolerance with independent storage per node. Deploying it as a minimum two-node cluster is recommended.

The *Distributed VXLAN Connectivity Model for Version 9.1.1* supports EVPN Route Controller in Active/Active cluster mode alongside the previously introduced Active/Standby mode. Active/Active supports up to eight nodes; more than two nodes are useful, for example, when there is a need to spread them across multiple zones or failure domains. Active/Active is the only mode allowing a Layer 3 boundary between the EVPN Route Controller nodes on all interfaces, including the Multiprotocol BGP-EVPN interface. Active/Standby, by contrast, requires Layer 2 adjacency on the Multiprotocol BGP-EVPN interface to establish the Multiprotocol BGP-EVPN session from the VIP source interface.

In Active/Active mode, each EVPN Route Controller maintains an active and independent Multiprotocol BGP-EVPN interfaces and establishes sessions with all EVPN gateways in the physical EVPN-VXLAN network fabric. This substantially reduces the data plane impact of EVPN Route Controller or Multiprotocol BGP-EVPN session failures. In Active/Standby mode, only the active EVPN Route Controller node establishes Multiprotocol BGP-EVPN sessions with the EVPN gateways. The standby node remains in cold standby, retaining its configuration but without an established Multiprotocol BGP-EVPN session. Its Multiprotocol BGP-EVPN connection is established only after a failover, resulting in some minimal data plane impact.

Table 592: Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Route Controller Cluster High Availability Mode Comparison

Attribute	Active/Standby Mode	Active/Active Mode
Minimum required supported Version	9.1.0	9.1.1
EVPN Route Controller HA model	One node active, one in idle standby	All nodes active simultaneously
Minimal required EVPN Route Controller Nodes	1	2
Maximal supported EVPN Route Controller Nodes	2	8
Number of supported Multiprotocol BGP-EVPN interfaces per EVPN Route Controller node; this interface is used as for Multiprotocol BGP-EVPN peering	2	1
Multiprotocol BGP-EVPN interfaces Design Requirement	The Multiprotocol BGP-EVPN interfaces require Layer 2 adjacency between the two EVPN Route Controller nodes to support the BGP-EVPN VIP source interface	Independent Multiprotocol BGP-EVPN interfaces per EVPN Route Controller node; any Multiprotocol BGP-EVPN interfaces can be separated by a Layer 3 boundary
Multiprotocol BGP-EVPN peering	Only the active EVPN Route Controller node peers with the EVPN gateways in the physical network fabric	Every EVPN Route Controller node peers with the EVPN gateways in the physical network fabric
Traffic disruption duration following an active EVPN Route Controller failure	A few seconds is achievable.	Sub-second is achievable

A single EVPN Route Controller cluster is sufficient to support multiple VRFs/tenants, each VRFs/tenants identified by the Layer 3 VNI configured in the External Connection. Each External Connection supports only a single Layer 3 VNI, with a strict 1:1 mapping between the External Connection and the Distributed Transit Gateway.

Each active EVPN Route Controller advertises a /32 host routes to the EVPN gateways in the physical EVPN-VXLAN network fabric via Multiprotocol BGP-EVPN. These /32 host routes are carried as EVPN Route Type 5, together with the Layer 3 VNI and Router MAC (RMAC) to identify the VRF/tenant of the External Connection. These /32 host routes represent one of the following address types:

- public IPs of workload VMs
- External IPs used for stateless 1:1 NAT
- Stateful network service IPs, such as the Default Outbound NAT (Auto-SNAT) translated IP or load balancer VIP
- DHCP relay on VPC public subnets to support external DHCP server connectivity. The VPC public subnet gateway IP is advertised alongside the workload VM /32 host routes from the EVPN Route Controller to the EVPN gateway, enabling reachability to the external DHCP server

Each /32 host route also includes VTEP information to identify the ESX host where the workload VM, or the Virtual Network Appliance (VNA) with its IP for stateful network services, resides.

The EVPN Route Controller learns /32 host route and VTEP information for workload VMs through the Central Control Plane (CCP) channel from individual ESX hosts through NSX Manager. IP addresses used by the Virtual Network Appliance (VNA) for stateful network services are learned through a separate internal notification channel.

Note:

Each EVPN Route Controller node exposes up to three configurable interfaces, each connected to a dedicated distributed Virtual Port Group: a management interface and one or two interfaces for Multiprotocol BGP-EVPN peering (EVPN Route Controller in Active/Active mode supports only one Multiprotocol BGP-EVPN interface). An additional internal interface, not user-configurable, handles notification messages during vMotion events and advertises Service IP addresses used for stateful network services from the Virtual Network Appliance.

Without EVPN-VXLAN Direct Prefix Routing, each EVPN gateway advertises prefixes representing physical endpoints attached to the VXLAN leaf switches, or prefixes belonging to networks outside the EVPN-VXLAN network fabric. The EVPN gateway must also advertise a default route configured by the network fabric administrator in the physical network. The default route can either originate on the EVPN gateway itself or be advertised from any other physical network device, such as a firewall.

Each EVPN gateway uses its VTEP as the next-hop toward the EVPN Route Controller, advertised via EVPN Route Type 5. To minimize the forwarding table on each ESX host where a Distributed Transit Gateway is instantiated, the EVPN Route Controller installs only the default route (ECMP) and the corresponding EVPN gateway next-hop.

When the EVPN-VXLAN Direct Prefix Routing feature is enabled, it requires additional Route Targets (RT) to be configured on both the physical network fabric and the EVPN Route Controller. Once configured, each physical endpoint within the EVPN-VXLAN network fabric is advertised either as a Route Type 5 with /32 prefix or as a Route Type 2 host route (MAC/IP), including the VTEP of its attached VXLAN leaf switch. With EVPN-VXLAN Direct Prefix Routing enabled, more granular filtering of specific physical endpoints is recommended. Additional filters can be applied to these Route Targets (RTs).

These advertised /32 host routes, representing the selected physical endpoints, are installed on each ESX host through the Central Control Plane (CCP) channel. They are added alongside the routes already installed and received from the EVPN gateway. These /32 prefix entries use the physical VXLAN leaf switch VTEP as next-hop. This finally enables a direct VXLAN tunnel between the ESX host and the physical VXLAN leaf switch, bypassing the EVPN gateway and its VXLAN encapsulation overhead for latency-sensitive traffic.

Important:

When enabled, EVPN-VXLAN Direct Prefix Routing increases the forwarding table size on each ESX host and the number of VXLAN tunnels from each physical VXLAN leaf switch toward ESX hosts. Network administrators must ensure that supported scale limits are not exceeded.

Important:

Centralized password management via VCF Operations is the only supported option; local credentials cannot be configured during the EVPN Route Controller node deployment. VCF Operations supports the password management task 'update', which requires the current password before a new one can be configured. A fresh EVPN Route Controller node deployment generates randomized passwords, which can be, for example retrieved via the SDDC Manager API (GET /v1/credentials) and subsequently used to update the passwords using VCF Operations.

Note:

The EVPN Route Controller is built on top of the Virtual Network Appliance, and this section applies exclusively to its use as an EVPN Route Controller. Throughout this section, the term EVPN Route Controller refers to the EVPN Route Controller function inclusive of the underlying Virtual Network Appliance nodes. For Virtual Network Appliance nodes deployed for stateful network services such as NAT, refer to the Virtual Network Appliance Cluster Detailed Design section.

Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Route Controller Design Requirements and Recommendations

Table 593: Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Active/Standby Route Controller Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-001	Deploy two (2) EVPN Route Controllers nodes belonging to the same EVPN Route Controller cluster in Active/Standby mode.	• EVPN Route Controller nodes are required to establish a common EVPN control plane between the physical EVPN-VXLAN network fabric and the Distributed TGW environment. • Two (2) EVPN Route Controller nodes are required to support High Availability.	• vSphere cluster resources must be available to support the two EVPN Route Controller nodes. • Only the active EVPN Route Controller node establishes a Multiprotocol BGP-EVPN session with the EVPN gateways, often referred to as border gateways, in the physical EVPN-VXLAN network fabric. • Only the active EVPN Route Controller node establishes a Multiprotocol BGP-EVPN session with every EVPN gateway, often referred to as border gateways, in the physical EVPN-VXLAN network fabric
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-002	Select the EVPN Route Controller node size medium.	• The medium EVPN Route Controller node size is required for production deployments, supporting enterprise-grade prefix scale. • The small size is limited to Proof-of-Concept and test environments.	You must provide sufficient compute resources to support the chosen EVPN Route Controller node size.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-003	Create at least a new single dedicated vSphere Distributed Port Group (dvPG) for the EVPN Route Controller node Multiprotocol BGP-EVPN interface and make the same vSphere dvPG available to both EVPN Route Controller nodes.	• The active and standby EVPN Route Controller nodes share a common floating IP for Multiprotocol BGP-EVPN peering per vSphere dvPG, requiring Layer 2 adjacency between both nodes connected to this dedicated vSphere dvPG. • Each dedicated vSphere dvPG carries the Multiprotocol BGP-EVPN sessions between the EVPN Route Controllers and the EVPN gateways in the physical EVPN-VXLAN network fabric. • The VLAN and CIDR assigned to the dedicated vSphere dvPortGroup must provide IP connectivity between the EVPN Route Controller's Multiprotocol BGP-EVPN interface and the EVPN gateway loopback interfaces, which terminate the Multiprotocol BGP-EVPN session and reside in the global routing table of the physical EVPN-VXLAN network fabric. • A second vSphere dvPG for the Multiprotocol BGP-EVPN interface provides VLAN-level redundancy intended for corner cases only. • Traffic segregation from other vSphere dvPGs allows organizations to implement and enforce network security policies.	• Each EVPN Route Controller node supports up to two Multiprotocol BGP-EVPN interfaces, each mapped to a dedicated vSphere dvPG. • The EVPN Route Controller deployment workflow allows only a single vSphere dvPG to be configured for the Multiprotocol BGP-EVPN interface during initial deployment; the second vSphere dvPG must be added post-deployment. • Additional configuration effort is required in the physical EVPN-VXLAN network fabric. • When the EVPN Route Controller nodes and their Multiprotocol BGP-EVPN interfaces belong to the tenant routing space and the EVPN gateway VTEP interfaces reside in the global routing table, connectivity between them requires VRF leaking.
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-004	Assign a unique IPv4 address and subnet mask to each Multiprotocol BGP-EVPN interface of each EVPN Route Controller node.	To support the BGP floating IP (VIP), each EVPN Route Controller node requires a dedicated IPv4 address on each Multiprotocol BGP-EVPN interface.	Only IPv4 addresses are supported for the Multiprotocol BGP-EVPN interface.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-005	To support High Availability for Multiprotocol BGP-EVPN peering, configure a BGP floating IP (VIP) for each vSphere dvPG assigned to the Multiprotocol BGP-EVPN interfaces.	• To support High Availability for Multiprotocol BGP-EVPN peering, a floating IP (VIP) is configured per vSphere dvPG as the shared BGP source address for the Multiprotocol BGP-EVPN interfaces. • Each EVPN Route Controller node has its own dedicated IPv4 address per Multiprotocol BGP-EVPN interface and vSphere dvPG, however only the active EVPN Route Controller uses the floating IP, or floating IPs if two vSphere dvPGs are configured, for peering with the EVPN gateways in the physical EVPN-VXLAN network fabric.	Upon an EVPN Route Controller failover, a new Multiprotocol BGP-EVPN session must be established with the new active EVPN Route Controller, as the Multiprotocol BGP-EVPN connections remain administratively down on the standby EVPN Route Controller.
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-006	Configure a unique local Autonomous System Number (ASN) for Multiprotocol BGP-EVPN.	• To establish a Multiprotocol BGP-EVPN session between the EVPN Route Controller and the EVPN gateways in the physical EVPN-VXLAN network fabric, a local ASN is required. • A unique local ASN enables native AS_PATH-based loop detection through eBGP.	A unique BGP ASN must be available for the EVPN Route Controller cluster.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-007	Configure at least two Multiprotocol BGP-EVPN peerings from the EVPN Route Controller cluster to the EVPN gateways in the physical EVPN-VXLAN network fabric.	• Multiprotocol BGP-EVPN peering with multiple EVPN gateways in the physical EVPN-VXLAN network fabric provides redundant Multiprotocol BGP-EVPN paths, ensuring High Availability. • The EVPN gateway role is configured either on all spine switches or on two VXLAN leaf switches in the physical EVPN-VXLAN network fabric. • Multiprotocol BGP-EVPN peering with VXLAN leaf switches not designated as EVPN gateways in the physical EVPN-VXLAN network fabric is neither required nor recommended. • The EVPN gateway must advertise its VTEP IP address as the next hop to the EVPN Route Controller via EVPN Route Type 5, which is then propagated to each individual ESX host to ensure every ESX host knows the path to physical destinations via the default route pointing to VTEP IP address of the EVPN gateway. • When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN gateway must additionally advertise the VTEP IP addresses of the VXLAN leaf switches with the /32 host routes of their connected physical endpoints to the EVPN Route Controller as EVPN Route Type 5, ensuring that every ESX host knows the path to physical endpoint via the /32 host routes pointing to VTEP IP address of the VXLAN leaf switch. **Note:** When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN Route Controller may receive EVPN Route Type 2 in addition to EVPN Route Type 5 from the EVPN gateway, based on the network fabric vendor's preference. Regardless of the received EVPN Route Type, the EVPN Route Controller installs only the relevant route	Only IPv4-based Multiprotocol BGP-EVPN peering is supported.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-REQD-CFG-008	Configure the BGP maximum hop limit to match the number of Layer 3 hops between the Multiprotocol BGP-EVPN peering source and destination.	Since Multiprotocol BGP-EVPN peering in a Distributed TGW environment with EVPN-VXLAN integration is typically not established between directly connected peering endpoints, the maximum hop limit must be adjusted accordingly.	By default, the Multiprotocol BGP-EVPN maximum hop limit is set to 1, meaning peering endpoints must be directly connected.

Table 594: Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Active/Standby Route Controller Design Recommendations

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-001	Use the default memory reservation with 100% for EVPN Route Controller nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a EVPN Route Controller node, ensuring this full memory amount remains available to the EVPN Route Controller node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-002	Use the default CPU reservation priority in NSX set to high shares for EVPN Route Controller nodes.	Setting CPU shares to "High" ensures that EVPN Route Controller nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are generally less critical than high memory reservations.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-003	Create at new vSphere Distributed Port Group for the EVPN Route Controller node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel.	Traffic segregation enables organizations to implement and enforce network security policies.	• For the management domain, a dedicated vSphere Distributed Port Group for the EVPN Route Controller node management interface already exists — this is the same vSphere Distributed Port Group Port Group used by the management stack. • Depending on the selected EVPN Route Controller cluster design, a EVPN Route Controller node management vSphere Distributed Port Group is required for each vSphere cluster, or rack.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-004	Select static IP allocation for the EVPN Route Controller node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-005	Use the central node profile configuration for EVPN Route Controller nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all EVPN Route Controller nodes.	• EVPN Route Controller nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual EVPN Route Controller node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-006	Deploy the two EVPN Route Controller nodes on two different vSphere clusters across two separate racks to improve availability.	Deploying each of the two EVPN Route Controllers on an independent vSphere cluster in a separate rack provides rack fault tolerance and independent storage when using vSAN.	• Requires additional resources to support multiple vSphere clusters deployed across different racks with independent storage (vSAN). • In Active/Standby mode, Layer 2 adjacency for the Multiprotocol BGP-EVPN interface is required across the racks where the two EVPN Route Controller nodes reside.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-007	Use the default MTU for the EVPN Route Controller node Multiprotocol BGP-EVPN interface.	Since the EVPN Route Controller node is not involved in the data plane forwarding path, the MTU setting is less critical and would only impact the fragmentation of large BGP UPDATE messages.	• If a large BGP UPDATE packet exceeds the MTU of a link along the multi-hop path and fragmentation is blocked due to the DF (Don't Fragment) bit being set, the packet will be silently dropped. • To avoid BGP UPDATE fragmentation issues, ensure Path MTU Discovery is enabled on both BGP peering endpoints and that ICMP is unblocked along the entire path between them.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-008	Ensure BGP Graceful Restart is enabled on the EVPN gateway in the physical EVPN-VXLAN network fabric.	Given the selected Active/Standby mode of the EVPN Route Controller and the requirement to re-establish a new Multiprotocol BGP-EVPN session upon a node failover, which can result in extended failover times, it is recommended to configure BGP Graceful Restart to maintain the Multiprotocol BGP-EVPN session.	BGP Graceful Restart is enabled by default on the EVPN Route Controller, requiring no additional configuration.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-009	Do not enable BFD for BGP between the EVPN Router Controller nodes in Active/Standby and the EVPN gateway in the physical EVPN-VXLAN network fabric.	• Given the selected Active/Standby mode of the EVPN Route Controller, disabling BFD is recommended as a new Multiprotocol BGP-EVPN session must be re-established upon a node failover, which can result in extended failover times, and BFD would prevent BGP Graceful Restart from protecting traffic forwarding during this process. • BFD should be disabled to avoid false positive failures caused by ECMP hashing sending BFD and BGP packets on different paths through the physical EVPN-VXLAN network fabric, as BFD would immediately tear down the Multiprotocol BGP-EVPN session without allowing BGP Graceful Restart to protect traffic forwarding.	None.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-010	Use fast, non-default BGP timers, 1/3 seconds.	Since BFD is not recommended for EVPN Route Controller nodes in Active/Standby mode with BGP Graceful Restart enabled, fast BGP timers are required to ensure rapid failover detection.	• Fast BGP timers are required for rapid failure detection and failover in the absence of BFD. • Non-default BGP timers cannot be configured during the EVPN Route Controller deployment workflow, and must be adjusted post-deployment.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-011	Enable Anycast VTEP on the EVPN gateways in the physical EVPN-VXLAN network fabric when the EVPN Route Controller operates in Active/Standby mode and BFD for BGP is disabled.	Configuring Anycast VTEP on the EVPN gateways in the physical EVPN-VXLAN network fabric reduces traffic blackholing in the event of an EVPN gateway node failure, especially in the absence of BFD for BGP, as no route withdrawal on each ESX host is required.	Additional configuration effort is required on all EVPN gateway nodes in the physical EVPN-VXLAN network fabric.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-012	When EVPN-VXLAN Direct Prefix Routing is disabled, advertise only the default route from the EVPN gateways to the EVPN Route Controller cluster. When EVPN-VXLAN Direct Prefix Routing is enabled, advertise the default route plus all /32 host routes of the latency-sensitive physical endpoints connected to the VXLAN leaf switches.	• When EVPN-VXLAN Direct Prefix Routing is disabled, the EVPN Route Controller cluster typically requires no more than the default route to program the egress path on each ESX host, ensuring every ESX host knows the path to physical destinations via the EVPN gateways. • When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN Route Controller requires the default route plus all /32 host routes of the latency-sensitive physical endpoints connected to the VXLAN leaf switches. These routes are used to program the egress path on each ESX host, ensuring it can reach either the EVPN gateway or the VXLAN leaf switches directly.	Additional configuration effort is required to configure an outbound route filter on all EVPN gateway nodes in the physical EVPN-VXLAN network fabric.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-013	Enable ECMP on the EVPN Route Controller cluster unless Anycast VTEP is already configured on the EVPN gateways in the physical EVPN-VXLAN network fabric.	• ECMP allows multiple routes to be installed on each ESX host where a Distributed TGW is instantiated, supporting individual data plane paths to each EVPN gateway or a VXLAN leaf switch when VPN-VXLAN Direct Prefix Routing is enabled. • If Anycast VTEP is enabled, only a single route needs to be installed on each ESX host, making ECMP unnecessary.	ECMP is only beneficial when BGP multipath criteria are met, which is typically the case in a Clos-based spine-leaf physical EVPN-VXLAN network fabric.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-014	Do not enable multipath relax unless the ECMP multipath criteria are not met due to differing BGP AS_PATH values caused by different ASNs in the AS_PATH.	By default, ECMP BGP multipath requires all criteria to be met, including identical AS_PATH length and identical Autonomous System Numbers (ASNs) in the AS_PATH. When EVPN gateways in the physical EVPN-VXLAN network fabric belong to different ASNs, ECMP BGP multipath cannot be achieved unless multipath relax is enabled.	None.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-015	Configure a BGP password for each Multiprotocol BGP-EVPN session.	An MD5-based BGP password secures the TCP connection for Multiprotocol BGP-EVPN peering between the EVPN Route Controller and the physical EVPN-VXLAN network fabric.	None.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-016	When a compute policy type Best Effort Restart (BER) is not configured, apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the two EVPN Route Controller nodes when both are running on the same vSphere cluster.. Otherwise, proceed with configuring a compute policy of type Best Effort Restart (BER).	Prevents the two EVPN Route Controller nodes from running on the same ESX host and risking the High Availability of the EVPN Route Controller cluster.	A minimum of three ESX hosts is required in the vSphere cluster when deploying two EVPN Route Controller nodes, to allow placing an ESX host into maintenance mode.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-017	When configuring a compute policy of type Best Effort Restart (BER), create for every ESX host where an EVPN Route Controller node will run an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the EVPN Route Controller workflow.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-018	When configuring a compute policy of type Best Effort Restart (BER), create for every EVPN Route Controller node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The EVPN Route Controller workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-019	When configuring a compute policy of type Best Effort Restart (BER), assign to each individual EVPN Route Controller node the previously configured host group.	Each EVPN Route Controller node has a 1:1 affinity with the underlying ESX host.	• The EVPN Route Controller workflow automatically assign to each individual EVPN Route Controller node the previously configured host group. • If the assignment was not set through the EVPN Route Controller workflow, it can be added post-deployment.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-020	When configuring a compute policy of type Best Effort Restart (BER), create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the EVPN Route Controller node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each EVPN Route Controller node are required to enable the compute policy type Best Effort Restart (BER).	The EVPN Route Controller workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-021	When configuring a compute policy of type Best Effort Restart (BER), create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each EVPN Route Controller node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the EVPN Route Controller nodes together.	The EVPN Route Controller workflow automatically configures the vCenter tag with a category and assign to each EVPN Route Controller node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AS-RCMD-CFG-022	When configuring a compute policy of type Best Effort Restart (BER), create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful EVPN Route Controller node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer EVPN Route Controller node readiness during host upgrades through multiple pre-checks before initiating EVPN Route Controller node failover and ensures the EVPN Route Controller node shutdown process begins gracefully only after successful pre-checks.	• The EVPN Route Controller workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the EVPN Route Controller node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the EVPN Route Controller node rather than performing vMotion. • User-initiated ESX host maintenance mode shuts down the hosted EVPN Route Controller node regardless of whether a healthy peer EVPN Route Controller node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Table 595: Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Active/Active Route Controller Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-001	Deploy at least two (2) EVPN Route Controllers nodes belonging to the same EVPN Route Controller cluster in Active/Active mode.	• EVPN Route Controller nodes are required to establish a common EVPN control plane between the physical EVPN-VXLAN network fabric and the Distributed TGW environment. • At least two (2) and up to eight (8) EVPN Route Controller nodes are supported within the same EVPN Route Controller cluster in Active/Active mode to provide High Availability. • Each EVPN Route Controller node in Active/Active mode within the same EVPN Route Controller cluster operates as an independent EVPN Route Controller, improving resiliency across multiple zones.	• vSphere cluster resources must be available to support the two EVPN Route Controller nodes. • Each active EVPN Route Controller node establishes a Multiprotocol BGP-EVPN session with every EVPN gateway, often referred to as border gateways, in the physical EVPN-VXLAN network fabric. • The EVPN Route Controller deployment workflow allows only the deployment of a two (2) node cluster during the initial deployment; additional nodes must be added post-deployment. • Additional EVPN Route Controller nodes in Active/Active mode within the same EVPN Route Controller cluster do not increase the prefix scale.
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-002	Select the EVPN Route Controller node size medium.	• The medium EVPN Route Controller node size is required for production deployments, supporting enterprise-grade prefix scale. • The small size is limited to Proof-of-Concept and test environments.	You must provide sufficient compute resources to support the chosen EVPN Route Controller node size.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-003	Create at least a new single dedicated vSphere Distributed Port Group (dvPG) for each EVPN Route Controller node Multiprotocol BGP-EVPN interface.	• Each active EVPN Route Controller node can have its own dedicated vSphere Distributed Port Group (dvPG) for the EVPN Route Controller node Multiprotocol BGP-EVPN interface, Layer 2 adjacency between the active nodes is not required. • Each dedicated vSphere dvPG carries the Multiprotocol BGP-EVPN sessions between the EVPN Route Controllers and the EVPN gateways in the physical EVPN-VXLAN network fabric. • The VLAN and CIDR assigned to the dedicated vSphere dvPortGroup must provide IP connectivity between the EVPN Route Controller's Multiprotocol BGP-EVPN interface and the EVPN gateway loopback interfaces, which terminate the Multiprotocol BGP-EVPN session and reside in the global routing table of the physical EVPN-VXLAN network fabric. • Traffic segregation from other vSphere dvPGs allows organizations to implement and enforce network security policies.	• Additional configuration effort is required in the physical EVPN-VXLAN network fabric. • When the EVPN Route Controller nodes and their Multiprotocol BGP-EVPN interfaces belong to the tenant routing space and the EVPN gateway VTEP interfaces reside in the global routing table, connectivity between them requires VRF leaking.
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-004	Assign a unique IPv4 address and subnet mask to each Multiprotocol BGP-EVPN interface of each EVPN Route Controller node.	• Required to establish a Multiprotocol BGP-EVPN session from each EVPN Route Controller node's Multiprotocol BGP-EVPN interface to every EVPN gateway in the physical EVPN-VXLAN network fabric. • A BGP floating IP (VIP) is neither required nor configurable in an EVPN Route Controller deployment using Active/Active mode.	Only IPv4 addresses are supported for the Multiprotocol BGP-EVPN interface.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-005	Configure a unique local Autonomous System Number (ASN) for Multiprotocol BGP-EVPN.	• To establish a Multiprotocol BGP-EVPN session between the EVPN Route Controller and the EVPN gateways in the physical EVPN-VXLAN network fabric, a local ASN is required. • A unique local ASN enables native AS_PATH-based loop detection through eBGP.	A unique BGP ASN must be available for the EVPN Route Controller cluster.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-006	Configure at least two Multiprotocol BGP-EVPN peerings from each EVPN Route Controller to the EVPN gateways in the physical EVPN-VXLAN network fabric.	• Multiprotocol BGP-EVPN peering with multiple EVPN gateways in the physical EVPN-VXLAN network fabric provides multiple redundant Multiprotocol BGP-EVPN paths, ensuring High Availability. • The EVPN gateway role is configured either on all spine switches or on two VXLAN leaf switches in the physical EVPN-VXLAN network fabric. • Multiprotocol BGP-EVPN peering with VXLAN leaf switches not designated as EVPN gateways in the physical EVPN-VXLAN network fabric is neither required nor recommended. • The EVPN gateway must advertise its VTEP IP address as the next hop to the EVPN Route Controller via EVPN Route Type 5, which is then propagated to each individual ESX host to ensure every ESX host knows the path to physical destinations via the default route pointing to VTEP IP address of the EVPN gateway. • When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN gateway must additionally advertise the VTEP IP addresses of the VXLAN leaf switches with the /32 host routes of their connected physical endpoints to the EVPN Route Controller as EVPN Route Type 5, ensuring that every ESX host knows the path to physical endpoint via the /32 host routes pointing to VTEP IP address of the VXLAN leaf switch. **Note:** When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN Route Controller may receive EVPN Route Type 2 in addition to EVPN Route Type 5 from the EVPN gateway, based on the network fabric vendor's preference. Regardless of the received EVPN Route Type, the EVPN Route Controller installs only the relevant route	Only IPv4-based Multiprotocol BGP-EVPN peering is supported.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-007	Do not configure Multiprotocol BGP-EVPN peerings between the active EVPN Route Controllers themselves; instead, ensure each EVPN Route Controller has the same number of Multiprotocol BGP-EVPN peerings with every EVPN gateway in the physical EVPN-VXLAN network fabric.	A consistent number of Multiprotocol BGP-EVPN peerings with the EVPN gateways on each EVPN Route Controller is required to ensure all nodes maintain an identical route table.	None.
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-008	Configure the BGP maximum hop limit to match the number of Layer 3 hops between the Multiprotocol BGP-EVPN peering source and destination.	Since Multiprotocol BGP-EVPN peering in a Distributed TGW environment with EVPN-VXLAN integration is typically not established between directly connected peering endpoints, the maximum hop limit must be adjusted accordingly.	By default, the Multiprotocol BGP-EVPN maximum hop limit is set to 1, meaning peering endpoints must be directly connected.
VCF-WLDCON-DTGWVXLAN-RC-AA-REQD-CFG-009	For each EVPN Route Controller, select the corresponding BGP source address for each Multiprotocol BGP-EVPN peering.	The BGP source address is required to ensure each Multiprotocol BGP-EVPN peering is established from the correct EVPN Route Controller.	None.

Table 596: Distributed VXLAN Connectivity Model for Version 9.1.1 EVPN Active/Active Route Controller Design Recommendations

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-001	Use the default memory reservation with 100% for EVPN Route Controller nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a EVPN Route Controller node, ensuring this full memory amount remains available to the EVPN Route Controller node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-002	Use the default CPU reservation priority in NSX set to high shares for EVPN Route Controller nodes.	Setting CPU shares to "High" ensures that EVPN Route Controller nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are generally less critical than high memory reservations.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-003	Create at new vSphere Distributed Port Group for the EVPN Route Controller node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel.	Traffic segregation enables organizations to implement and enforce network security policies.	• For the management domain, a dedicated vSphere Distributed Port Group for the EVPN Route Controller node management interface already exists — this is the same vSphere Distributed Port Group Port Group used by the management stack. • Depending on the selected EVPN Route Controller cluster design, a EVPN Route Controller node management vSphere Distributed Port Group is required for each vSphere cluster, or rack.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-004	Select static IP allocation for the EVPN Route Controller node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-005	Use the central node profile configuration for EVPN Route Controller nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all EVPN Route Controller nodes.	• EVPN Route Controller nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual EVPN Route Controller node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-006	Deploy at least two (2) EVPN Route Controller nodes on two different vSphere clusters across two separate racks to improve availability.	• When deploying two (2) EVPN Route Controllers, ensure each is placed on an independent vSphere cluster in a separate rack, providing rack-level fault tolerance and independent storage when using vSAN. • When deploying more than two (2) EVPN Route Controllers, ensure the nodes are distributed across at least two independent vSphere clusters in separate racks, providing the same rack-level fault tolerance and independent storage when using vSAN.	Requires additional resources to support multiple vSphere clusters deployed across different racks with independent storage (vSAN).

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-007	Use the default MTU for the EVPN Route Controller node Multiprotocol BGP-EVPN interface.	Since the EVPN Route Controller node is not involved in the data plane forwarding path, the MTU setting is less critical and would only impact the fragmentation of large BGP UPDATE messages.	• If a large BGP UPDATE packet exceeds the MTU of a link along the multi-hop path and fragmentation is blocked due to the DF (Don't Fragment) bit being set, the packet will be silently dropped. • To avoid BGP UPDATE fragmentation issues, ensure Path MTU Discovery is enabled on both BGP peering endpoints and that ICMP is unblocked along the entire path between them.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-008	Use the default BGP timers, 60/180 seconds.	When BFD is enabled, there is no need lower the default BGP timers settings increasing EVPN Router Controller CPU overhead.	• BFD is required for fast failure detection and failover. • If BFD is not available on the EVPN gateway in the physical EVPN-VXLAN network fabric, BGP timers can be lowered to allow faster failure detection and failover. • Non-default BGP timers cannot be configured during the EVPN Route Controller deployment workflow, and must be adjusted post-deployment.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-009	Enable BFD for BGP between the EVPN Router Controller nodes and the EVPN gateways in the physical EVPN-VXLAN network fabric.	Provide faster failure detection and failover with minimum CPU overhead. It permits leaving the BGP timers to their default	• The EVPN gateway in the physical EVPN-VXLAN network fabric need to support BFD for BGP. • BFD and BGP packets may travel on different paths through the physical EVPN-VXLAN network fabric due to different ECMP hashing results, potentially causing false positive failures, for example when a link between the spine and leaf switches goes down.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-010	Depending on the physical network fabric design, Anycast VTEP can optionally be enabled on the EVPN gateways in the physical EVPN-VXLAN network fabric when the EVPN Route Controller operates in Active/Active mode, even if BFD for BGP is enabled.	Configuring Anycast VTEP on the EVPN gateways in the physical EVPN-VXLAN network fabric reduces traffic blackholing in the event of an EVPN gateway node failure, as no route withdrawal on each ESX host is required.	Additional configuration effort is required on all EVPN gateway nodes in the physical EVPN-VXLAN network fabric.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-011	When EVPN-VXLAN Direct Prefix Routing is disabled, advertise only the default route from the EVPN gateways to the EVPN Route Controller cluster. When EVPN-VXLAN Direct Prefix Routing is enabled, advertise the default route plus all /32 host routes of the latency-sensitive physical endpoints connected to the VXLAN leaf switches	• When EVPN-VXLAN Direct Prefix Routing is disabled, the EVPN Route Controller cluster typically requires no more than the default route to program the egress path on each ESX host, ensuring every ESX host knows the path to physical destinations via the EVPN gateways. • When EVPN-VXLAN Direct Prefix Routing is enabled, the EVPN Route Controller requires the default route plus all /32 host routes of the latency-sensitive physical endpoints connected to the VXLAN leaf switches. These routes are used to program the egress path on each ESX host, ensuring it can reach either the EVPN gateway or the VXLAN leaf switches directly.	Additional configuration effort is required to configure an outbound route filter on all EVPN gateway nodes in the physical EVPN-VXLAN network fabric.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-012	Enable ECMP on the EVPN Route Controller cluster unless Anycast VTEP is already configured on the EVPN gateways in the physical EVPN-VXLAN network fabric.	• ECMP allows multiple routes to be installed on each ESX host where a Distributed TGW is instantiated, supporting individual data plane paths to each EVPN gateway or a VXLAN leaf switch when VPN-VXLAN Direct Prefix Routing is enabled. • If Anycast VTEP is enabled, only a single route needs to be installed on each ESX host, making ECMP unnecessary.	ECMP is only beneficial when BGP multipath criteria are met, which is typically the case in a Clos-based spine-leaf physical EVPN-VXLAN network fabric.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-013	Do not enable multipath relax unless the ECMP multipath criteria are not met due to differing BGP AS_PATH values caused by different ASNs in the AS_PATH.	By default, ECMP BGP multipath requires all criteria to be met, including identical AS_PATH length and identical Autonomous System Numbers (ASNs) in the AS_PATH. When EVPN gateways in the physical EVPN-VXLAN network fabric belong to different ASNs, ECMP BGP multipath cannot be achieved unless multipath relax is enabled.	None.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-014	Configure a BGP password for each Multiprotocol BGP-EVPN session.	An MD5-based BGP password secures the TCP connection for Multiprotocol BGP-EVPN peering between the EVPN Route Controller and the physical EVPN-VXLAN network fabric.	None.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-015	When a compute policy type Best Effort Restart (BER) is not configured, apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to all EVPN Route Controller nodes when they are running on the same vSphere cluster.. Otherwise, proceed with configuring a compute policy of type Best Effort Restart (BER).	Prevents the EVPN Route Controller nodes from running on the same ESX host and risking the High Availability of the EVPN Route Controller cluster.	A minimum of (N+1) ESX hosts is required in the vSphere cluster when deploying (N) EVPN Route Controller nodes to put the ESX host in maintenance mode.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-016	When configuring a compute policy of type Best Effort Restart (BER), create for every ESX host where an EVPN Route Controller node will run an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the EVPN Route Controller workflow.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-017	When configuring a compute policy of type Best Effort Restart (BER), create for every EVPN Route Controller node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The EVPN Route Controller workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-018	When configuring a compute policy of type Best Effort Restart (BER), assign to each individual EVPN Route Controller node the previously configured host group.	Each EVPN Route Controller node has a 1:1 affinity with the underlying ESX host.	• The EVPN Route Controller workflow automatically assign to each individual EVPN Route Controller node the previously configured host group. • If the assignment was not set through the EVPN Route Controller workflow, it can be added post-deployment.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-019	When configuring a compute policy of type Best Effort Restart (BER), create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the EVPN Route Controller node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each EVPN Route Controller node are required to enable the compute policy type Best Effort Restart (BER).	The EVPN Route Controller workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-020	When configuring a compute policy of type Best Effort Restart (BER), create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each EVPN Route Controller node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the EVPN Route Controller nodes together.	The EVPN Route Controller workflow automatically configures the vCenter tag with a category and assign to each EVPN Route Controller node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Recommendations ID	Design Requirement	Justification	Implication
VCF-WLDCON-DTGWVXLAN-RC-AA-RCMD-CFG-021	When configuring a compute policy of type Best Effort Restart (BER), create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful EVPN Route Controller node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer EVPN Route Controller node readiness during host upgrades through multiple pre-checks before initiating EVPN Route Controller node failover and ensures the EVPN Route Controller node shutdown process begins gracefully only after successful pre-checks.	• The EVPN Route Controller workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the EVPN Route Controller node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the EVPN Route Controller node rather than performing vMotion. • User-initiated ESX host maintenance mode shuts down the hosted EVPN Route Controller node regardless of whether a healthy peer EVPN Route Controller node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

NSX Segment Connectivity Model

This model employs a Tier-0 gateway, where Tier-1 gateways are directly connected to it rather than using a Centralized Transit Gateway (CTGW) with VPCs. It leverages virtual networking based on GENEVE segments for workload connectivity including full support of NSX Federation and centralized advanced security features, but lacks the simplified consumption model of a VPC-based topology.

NSX Segment Connectivity Model Overview

This model offers strong north-south throughput scalability, with a single Tier-0 gateway supporting multiple workload environments, each mapped to individual Tier-1 gateways. VRF support can be incorporated into this model when tenants require dedicated external network access or when the VRF separation of the physical network fabric must be extended into the virtual network. Best performance is achieved by deploying a stateless Active/Active Tier-0 gateway combined with Tier-1 gateways running in DR-only mode, maximizing distributed routing benefits.

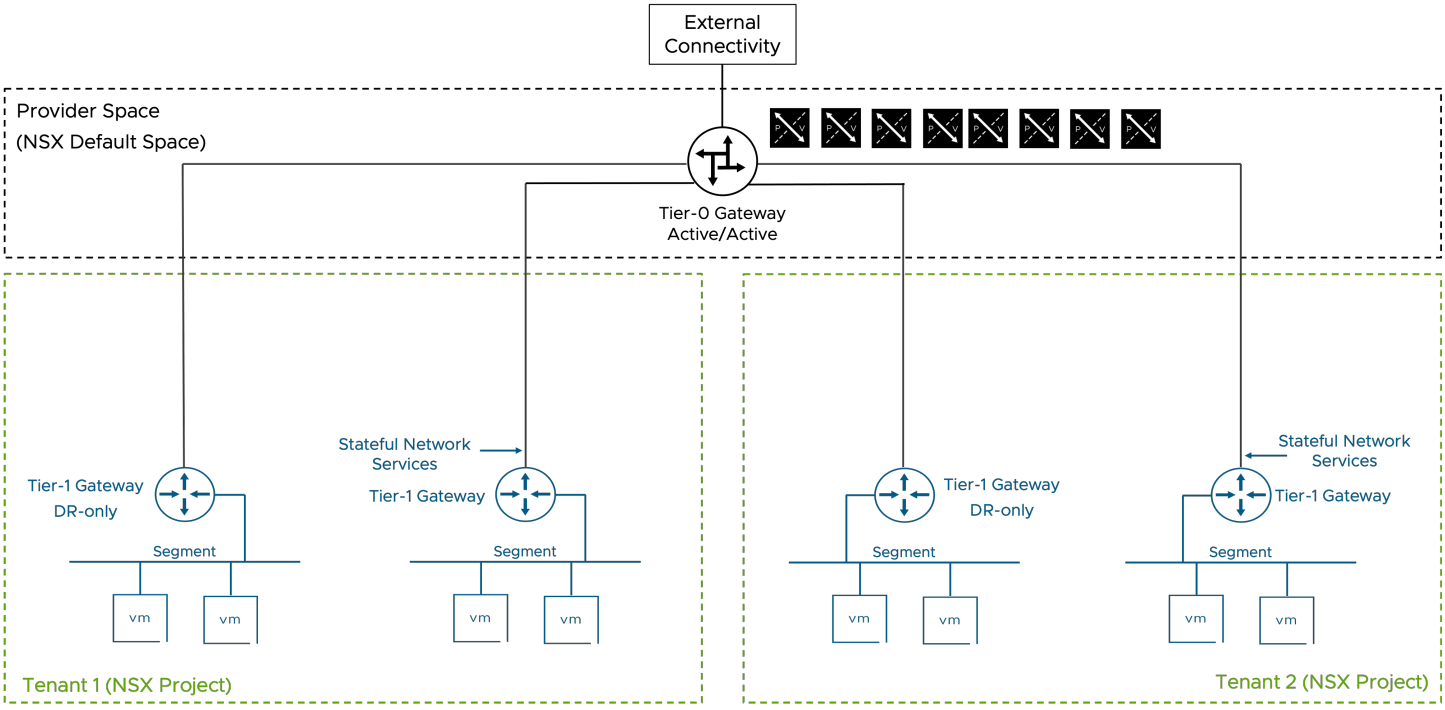
This model lacks the simplified consumption model of a VPC-based topology and does not support certain network services required for specific use cases, such as VCFA All Apps Orgs.

Enabling centralized network services on specific Tier-1 gateways by associating them with an NSX Edge cluster is supported, but should be limited to cases where those services are explicitly required. Such services include stateful NAT, centralized NSX Edge node-based firewall requirements at Layer 4 or Layer 7, and advanced NSX Edge node security services such as Identity Firewall, URL Filtering, FQDN Analysis, IDS/IPS, Malware Detection, and TLS Inspection. VPN support is available on Tier-1 gateways, though it is not supported when running in DR -only mode. Mixing Tier-1 gateways with and without centralized network services enabled is supported. This model supports VRF can be incorporated into the topology when tenants need dedicated external network access or when the VRF separation of the physical fabric must be extended into the virtual network.

Note:

Enabling centralized network services affects communication between workloads connected to different Tier-1 gateways by routing all traffic through the NSX Edge node. It also limits the north-south throughput of all the workloads connected to the Tier-1 gateway to the capacity of a single NSX Edge node.

Figure 290: NSX Segment Connectivity Model in VMware Cloud Foundation



NSX Segment Connectivity Model Attributes

An *NSX Segment Connectivity Model* has the following attributes.

Attribute	Detail
Network Consumption Model	NSX Segment Consumption Model
Node Type	NSX Edge Node Required
Tier-0 Gateway HA Mode	• Active/Active • Active/Standby
Tier-1 HA Mode	• Active/Standby • Distributed
Tier-0 Gateway Services	• Stateless network services (NAT) in Active/Active mode • Stateful network services in Active/Standby mode
Tier-1 Gateway Services	Stateful network services in Active/Standby mode

Attribute	Detail
Load Balancing Options	• NSX Edge load balancer • Avi Load Balancer
NSX Tenancy Constructs	NSX Projects only
Consumption options	NSX UI/API
Provider Gateway Type	• Tenant dedicated Tier-0 gateway • Tenant shared Tier-0 gateway • Tenant dedicated VRF gateway

NSX Segment Connectivity Model Options

Design Area	Options
Gateway Deployment Options	• Single Tenant • Shared across multiple tenants • Dedicated, one Gateway for each tenant
Segment options	• Global Segment (NSX Federation required) • Local Segment
VRF Adoption (Only applies if the Gateway is shared)	Yes or No

NSX Segment Connectivity Model with NSX Federation

NSX Federation provides a single point of management for workload networking and security across NSX instances, regardless if those NSX instances are associated to workload domains, within, or across VMware Cloud Foundation instances. The NSX Federation capabilities are:

- Centrally manages security policy and groups across NSX instances
- Layer 2 and Layer 3 network topology extension across NSX instances

NSX Federation allows you to extend the *NSX Segment Connectivity Model* across VCF workload domains, part of the same or different VCF instances. Each workload domain assigned to a dedicated NSX instance maintains independent control plane availability, making NSX Federation the preferred solution for disaster recovery use cases, while also supporting an independent software lifecycle. Cross NSX instance traffic traverses the NSX Edge nodes and supports any MTU above 1280 bytes (preferred is at least 1700 bytes for workload VMs using MTU 1500 bytes to avoid fragmentation, or 9000 bytes for workload VM above 1500 bytes) via TCP automatic MSS adjustment. For non-TCP traffic such as UDP, no MSS mechanism is available and fragmentation occurs. NSX Federation support local egress.

Note:

When extending the network topology across workload domains via NSX Federation, it's not possible to leverage the NSX multi-tenancy framework, so all configurations must be performed centrally by the provider admin in the NSX default Project.

Note:

See an *NSX Segment Connectivity Model with NSX Federation* use case in the VCF management domain here VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model.

NSX Federation uses an Active/Standby model for NSX Global Managers, where Active and Standby NSX Global Manager clusters can be deployed in the same or different VCF instances. Only Layer 3 connectivity is required between the Active and StandbyNSX Global Manager clusters, with no Layer 2 connectivity required. Please refer to specific NSX Global Managers content in NSX Manager and Control Plane Models for more details.

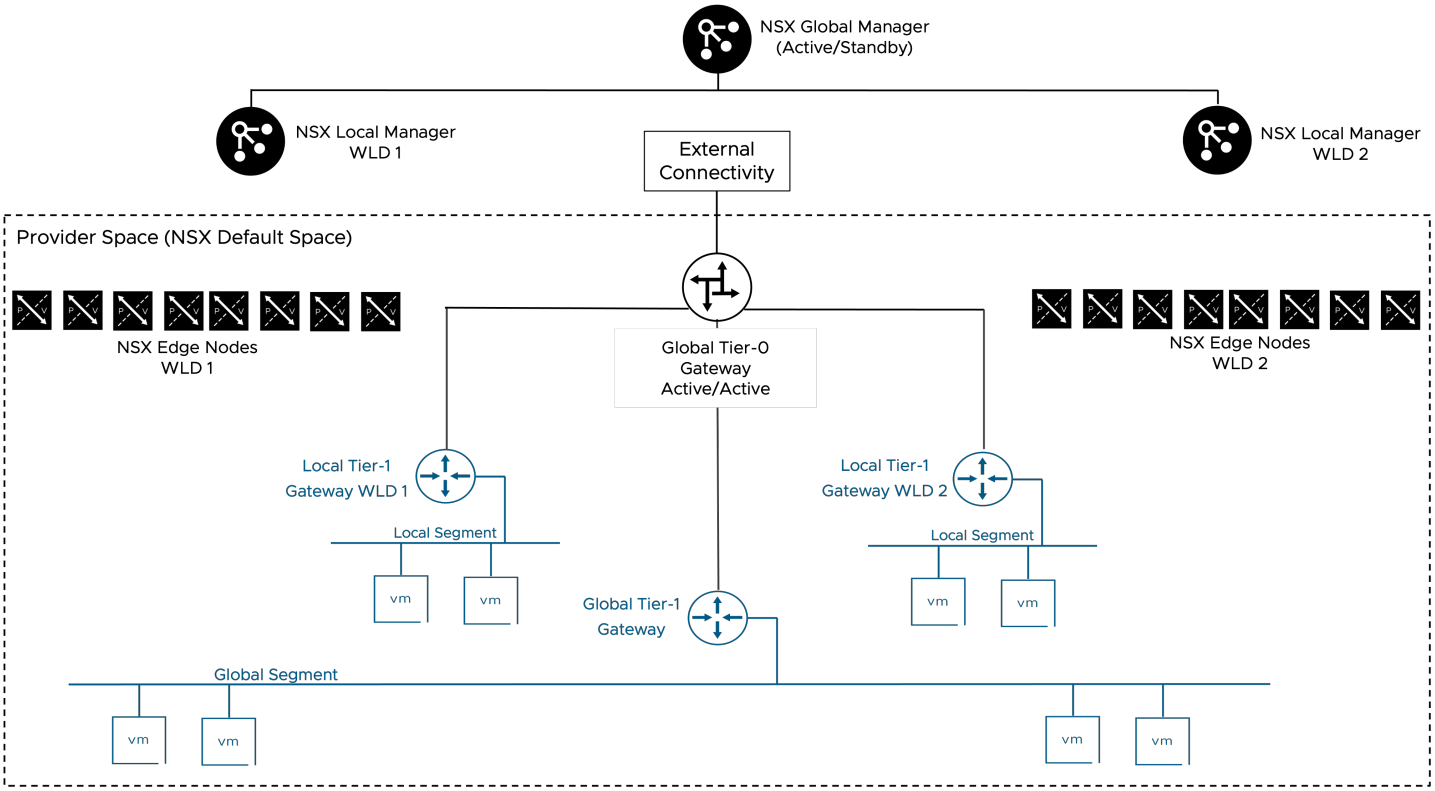
Note:

NSX Federation can coexist with local VPC consumption, please refer to NSX Federation Capabilities in the Different Network Consumption Models

Note:

Tenant (NSX Project) administrators, and VPC administrators do not have the ability to configure global groups and security policies spanning multiple NSX instances, nor managing local groups and security policies for multiple NSX instances via the NSX Global Manager.

Figure 291: NSX Segment Connectivity Model with NSX Federation in VMware Cloud Foundation



NSX Segment Connectivity Model Design Requirements and Recommendations

Table 597: NSX Segment Connectivity Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-NSXSEG-REQD-CFG-001	A minimum of two (2) NSX Edge nodes must be available. Select one of the presented NSX Edge Cluster Models.	• Two NSX Edge nodes are the minimum to provide High Availability. • This model requires the centralized services provided by the NSX Edge nodes. • All NSX Edge nodes from a NSX Edge cluster will contribute to north-south throughput. • The NSX Edge cluster can be scaled up to eight (8) NSX Edge nodes for Tier-0 gateways in Active/Active mode.	• Resources must be available for the NSX Edge nodes. • When scaling out the NSX Edge cluster, additional ESX hosts or pNICs must be available to support the additional NSX Edge nodes. In most cases, connecting multiple NSX Edge nodes to the same pair of pNICs will not achieve the desired performance objective.

Table 598: NSX Segment Connectivity Model Design Recommendations

Design Recommendations ID	Design Recommendation	Justification	Implication
VCF-WLDCON-NSXSEG-RCMD-CFG-001	Configure the Tier-0 gateway in Active/Active mode.	• A Tier-0 gateway in Active/Active mode supports scale-out up to eight (8) NSX Edge nodes, increasing north-south throughput. • North-south traffic is distributed across all NSX Edge nodes within the same NSX Edge cluster attached to the Tier-0 gateway.	Only Tier-0 gateway in Active/Standby mode support stateful network services.
VCF-WLDCON-NSXSEG-RCMD-CFG-002	Use external BGP as the routing protocol between the Tier-0 gateway and the physical network.	• BGP is a scalable solution to interconnect the virtual network environment to the physical network fabric. • New dynamically created networks in the VCF environment are automatically advertised to the physical network. • BGP is a standard protocol supported by the majority of the datacenter switches. • External BGP simplifies the interconnection compared to internal BGP by eliminating the need for full-mesh peering and creating a clear boundary between the administrative domains of VCF and the physical network.	Physical switches must support external BGP.

Design Recommendations ID	Design Recommendation	Justification	Implication
VCF-WLDCON-NSXSEG-RCMD-CFG-003	Use a unique private BGP Autonomous System Number (ASN) for the Tier-0 gateway	• It prevents advertisements from being dropped by the BGP loop detection check. • If the peering devices only advertise the default route, having a unique BGP ASN may not be necessary. • It is assumed that the Tier-0 gateway is peering to internal devices and it's not an Internet-facing router, in which case a public ASN is required.	A unique BGP ASN must be available for the Tier-0 gateway.
VCF-WLDCON-NSXSEG-RCMD-CFG-004	Each of the NSX Edge nodes where the Active/Active Tier-0 gateway runs will peer with two separate physical network devices.	It provides redundancy at the physical network layer, without requiring an NSX Edge node failover.	• If the NSX Edge node is a appliance, the ESX host where it runs must be connected to two (2) physical network switches running BGP. • If the NSX Edge node is a bare-Metal NSX Edge , the server where the bare-Metal NSX Edge software is installed must be connected to two (2) physical network switches running BGP.
VCF-WLDCON-NSXSEG-RCMD-CFG-005	The physical network devices peering to the Tier-0 gateway are physically direclty connected to the ESX host where the NSX Edge node runs, or to the bare metal NSX Edge node.	• The Layer 3 peering is not dependent on a Layer 2 technology, making it easier to map a physical failure to a routing convergence. • Peering over a VLAN transported over multiple physical network switches and over potentially multiple redundant paths increases the operational complexity of the design.	The physical switches connected to the ESX host where the NSX Edge VM is running or those connected to a bare metal NSX Edge must be BGP capable.
VCF-WLDCON-NSXSEG-RCMD-CFG-006	If using NSX Edge virtual appliance form factor, use two (2) data plane fastpath interfaces.	• Two (2) data plane fathpath interfaces provide redundancy for both VLAN and Overlay traffic. • Using the additional two (2) data plane fastpath interfaces does not increase the performance of the NSX Edge VM in the majority of situations, in particular when the additional vNics leverage the same set of physical NICs. When the ESX hosts have four (4) pNICs available to the NSX Edge VMs, the recommendation is to use multiple NSX Edge nodes connected to different sets of physcial NICs.	Two (2) physical NIC must be available on the ESX hosts where the NSX Edge VM is running.

Design Recommendations ID	Design Recommendation	Justification	Implication
VCF-WLDCON-NSXSEG-RCMD-CFG-007	If using NSX Edge in bare metal form factor, use four (4) data plane fastpath interfaces.	NSX Edge in bare metal form factor are generally limited by the speed of their physical NICs. In most cases, increasing the number of physical NICs in use will linearly improve the throughput that can be achieved.	• Four (4) data plane fastpath physcial NIC must be available on the NSX Edge in bare metal form factor. • Using two (2) 100G physcial NICs is a good option for NSX Edge in bare metal form factor when four (4) physical NICs are not available.
VCF-WLDCON-NSXSEG-RCMD-CFG-008	Each NSX Edge data plane fastpath interface transports both VLAN and TEP traffic.	Two data plane fastpath interfaces are sufficient to provide redundancy for both types of traffic.	Inbound and outbound traffic from the virtual network are mixed on the same NSX Edge vNICs, making monitoring less intuitive.
VCF-WLDCON-NSXSEG-RCMD-CFG-009	Every NSX Edge node shares the same VLAN to peer to a specific physical network device. With two physical network devices, it means two peering VLANs are needed.	Adding or replacing additional NSX Edge nodes to the NSX Edge cluster doesn't require creating new VLANs on the physical network fabric.	Peering subnet ranges should accommodate NSX Edge cluster growth.
VCF-WLDCON-NSXSEG-RCMD-CFG-010	Use /28 or larger uplink peering subnets are recommended.	Allow to scale the Tier-0 gateway to eight (8) NSX Edge nodes when in an Active/Active configuration.	For an Active/Standby Tier-0 gateway, the peering subnets can be smaller as the Tier-0 gateway is limited to two (2) NSX Edge nodes.
VCF-WLDCON-NSXSEG-RCMD-CFG-011	Peering VLANs are transported on trunk port groups connected to the NSX Edge fp-eth0 and fp-eth1 interfaces.	• Trunk port groups allow to transport both peering VLAN and TEP VLAN traffic on the same NSX Edge vNIC. • Trunk port groups allow seamless integration of VRFs and service interfaces for VLANs.	• Peering VLANs and TEP VLAN are transported on the same peering port groups but they have different NIC teaming requirements. • Peering VLANs do not need to failover between uplinks, the TEP VLAN does. • The teaming policy for port group must be active/standby to satisfy the TEP VLAN requirement. • Physical switches should prune the peering VLAN from their interfaces where they are not required and should avoid transporting them on the inter-switch link, or via a physical overlay extension (e.g., VXLAN in an EVPN fabric)
VCF-WLDCON-NSXSEG-RCMD-CFG-012	Set the MTU of the Tier-0 gateway external Layer 3 interfaces to the MTU of the peering interface on the physical network switches (e.g., 9000).	Some network devices set the DF bit to "Do not Fragment" on the BGP peering session, a mismatch in the MTU may cause peering issues.	None.

Design Recommendations ID	Design Recommendation	Justification	Implication
VCF-WLDCON-NSXSEG-RCMD-CFG-013	Set the global gateway MTU to the value of the external Layer 3 interfaces of the Tier-0 gateway minus 200 (i.e., 8800 = 9000 -200).	Avoid fragmentation at the NSX layer when packets larger than the default 1500 bytes are exchanged between workloads in the virtual network and workloads on the physical network.	Must be set post deployment in NSX as a global setting.
VCF-WLDCON-NSXSEG-RCMD-CFG-014	Enable BFD between the Tier-0 gateway and the physical network devices.	Provide fast failure detection and failover with minimal CPU overhead. It permits leaving the BGP timers to their default.	Physical switches need to support BFD.
VCF-WLDCON-NSXSEG-RCMD-CFG-015	Use the default BGP timers, 60/180 seconds.	When BFD is enabled, there is no need to lower the default BGP timer settings increasing NSX Edge CPU overhead.	• BFD is required for fast failure detection and failover • If BFD is not available on the physical switches, BGP timers can be lowered to allow a faster failure detection and failover.
VCF-WLDCON-NSXSEG-RCMD-CFG-016	On the Tier-0 gateway, enable redistribution of the following network types into BGP: • Tier-1 connected networks • Tier-1 NAT routes	• In this model, all workloads are connected to Tier-1 gateway segments. Those segments must be advertised for direct workload reachability. • NAT routes advertisement is required by workloads connected to segments not advertised by the Tier-1 gateways.	All Tier-1 gateway connected segments will be advertised to the physical network unless they are not advertised by the Tier-1 gateway. If filtering is required, route filters should be configured.
VCF-WLDCON-NSXSEG-RCMD-CFG-017	• Create an ingress route filter allowing the advertisement of the default route only from the physical network to the Tier-0 gateway. • A corresponding summarization from the physical network is recommended.	• In most cases, the virtual network is a stub network with a single exit point. The filtering reduces the Tier-0 gateway routing table, simplifying troubleshooting and potentially reducing converge time. • It also prevents RFP checks from being triggered by asymmetric routing advertisements from the physical switches.	• If the virtual network has multiple exit points, for example, one to shared services and one as a private interconnection to the tenant's remote data center, this recommendation cannot be implemented. • In such cases reducing the routing table size via summarization is still recommended.
VCF-WLDCON-NSXSEG-RCMD-CFG-018	Leave the RPF check enabled (set to strict) on all four external interfaces of the Tier-0 gateway.	RPF check mitigates IP Address Spoofing Attacks.	• Asymmetric routing advertisements (Not asymmetric routing, which is normal in any ECMP configuration and supported by the RFP check) from the physical switches will trigger the RPF check and drop nonmalicious traffic. • RPF check should be enabled only if consistent routing advertisement from all BGP peers can be guaranteed via route summarization and filtering.

Design Recommendations ID	Design Recommendation	Justification	Implication
VCF-WLDCON-NSXSEG-RCMD-CFG-019	Set BGP Graceful Restart to Helper-only.	BGP Graceful Restart may slow down the BGP convergence by waiting for the peer device control plane to recover. Helper-only ensures that routing convergence is delayed only when a peer device negotiates BGP Graceful Restart. The helper-only setting will allow the Tier-0 gateway to adapt to the peer device behavior.	If the peer device doesn't support BGP Graceful Restart or if BGP Graceful Restart is not desirable it may be better to disable BGP Graceful Restart altogether.
VCF-WLDCON-NSXSEG-RCMD-CFG-020	Set the Forwarding Up timer to 5 seconds (Default).	This will delay traffic from being forwarded to the NSX Edge from the virtual network before the routing table is fully populated.	None.
VCF-WLDCON-NSXSEG-RCMD-CFG-021	Configure Tier-1 gateways in DR-only mode.	• The goal is to have all Tier-1 gateways in DR-only mode unless required for centralized network services. • North-south traffic from Tier-1 gateways running in DR-only mode is distributed across all NSX Edge nodes used by the Tier-0 gateway via ECMP hashing, increasing the scalability of the solution.	Centralized network services will not be available for the Tier-1 gateway unless the user associates the Tier-1 gateway to an NSX Edge cluster to make it Active/Standby.
VCF-WLDCON-NSXSEG-RCMD-CFG-022	Only for specifics Tier-1 gateway select the Active/Standby mode and associate them to the NSX Edge cluster where the Tier-0 gateway runs.	Provide the option to consume centralized network services for specific Tier-1 gateways.	• Any east-west traffic between different Tier-1 gateways will traverse the NSX Edge node if at least one of them is in Active/Standby mode. • NSX Edge nodes will leverage additional CPU cycles to serve Tier-0 gateway, and Tier-1 gateways centralized services. Monitor the CPU utilization via one of the following methods: – NSX Edge CLI – NSX API/UI – VCF Ops • Do not use the vCenter UI graphs to monitor the NSX Edge CPU utilization as it will provide unreliable results.

Design Recommendations ID	Design Recommendation	Justification	Implication
VCF-WLDCON-NSXSEG-RCMD-CFG-023	In a multi-tenant design leveraging NSX Projects, assign a range of routable IPs to each tenant and create a corresponding project route filter.	Avoid IP conflicts between tenants.	• It is not possible to share routable ranges across tenants (the VPC model allows to share the same range across tenants while avoiding individual IP conflicts). • Project route filters must be created via the NSX API. Please refer to Route Filtering in NSX Projects

VLAN Networking Connectivity Model

The *VLAN Networking Connectivity Model* uses traditional VLANs with vSphere Distributed Switch port groups or NSX VLAN segments as the method of connecting workloads to the network.

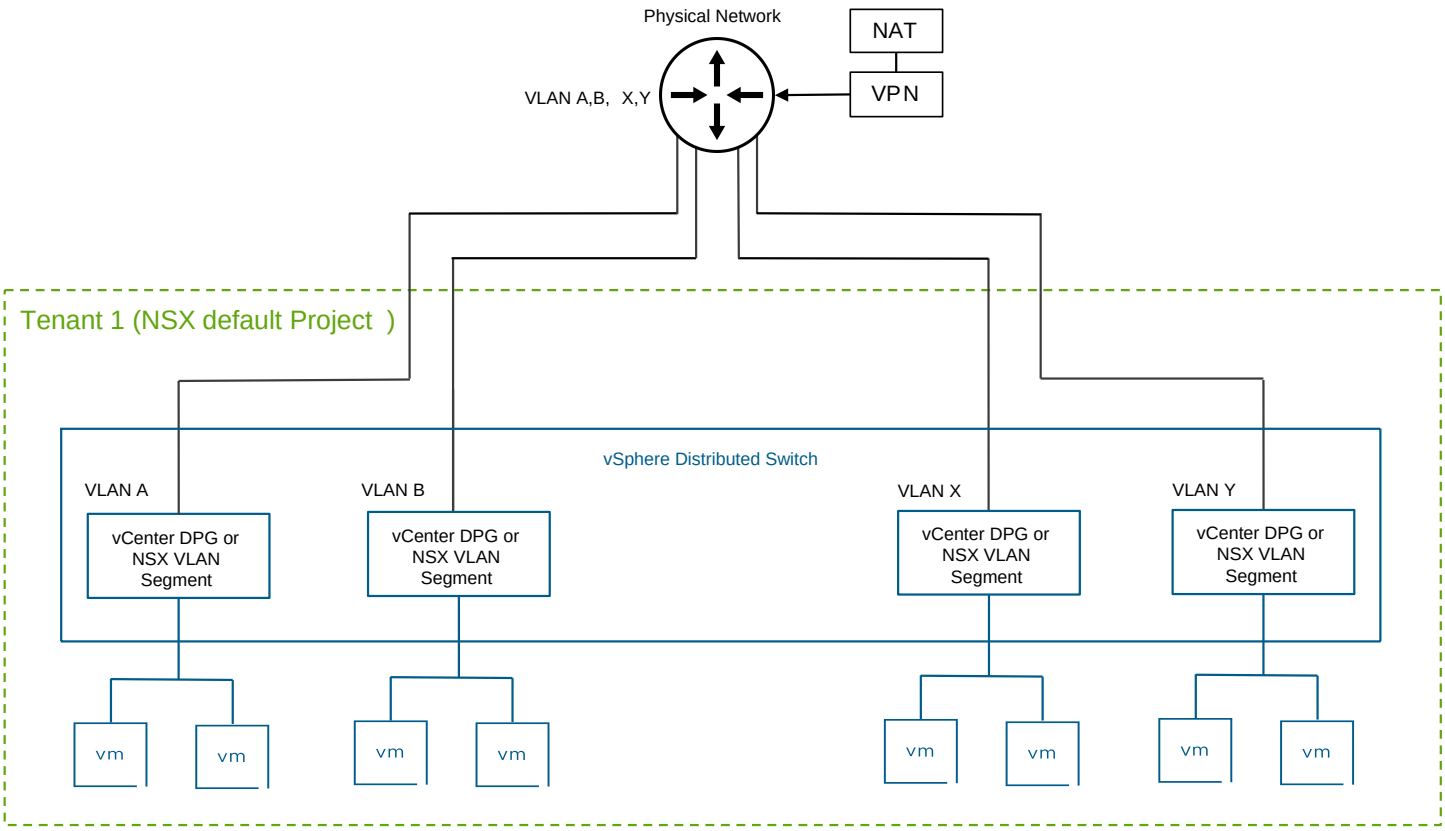
VLAN Networking Model Overview

In this model, the only VMware Cloud Foundation component participating in the networking functionalities is the vSphere Distributed Switch, responsible for switching the traffic between the VMs running in the hypervisor and the physical uplinks. Once the packets are delivered to the upstream physical switch, it is the responsibility of the physical network fabric to deliver them to the appropriate destination. VMs are connected to vSphere Distributed Port Groups or NSX VLAN segments.

This model lacks the benefits that are provided by the NSX network virtualization (VPCs or NSX overlay segments) which decouples the physical network from the topology requirements of the applications and services that run on top of it. With NSX network virtualization, the physical network fabric can be designed and operated based on a simpler and more stable model that does not require following the application's lifecycle. The role of adapting to the ever-changing application requirements is offloaded to the NSX network virtualization layer while the physical network fabric design can be centered around maximizing throughput and high availability, minimizing latency, and optimizing operational efficiency.

The *VLAN Networking Connectivity Model*is supported in VMware Cloud Foundation and may be appropriate for specific use cases, but it is not recommended for the majority of scenarios as it lacks the benefits provided by network virtualization.

Figure 292: VLAN Connectivity Model in VMware Cloud Foundation



VLAN Networking Connectivity Model Attributes

A *VLAN Networking Connectivity Model* has the following attributes.

Attribute	Detail
Network Consumption Model	VLAN Networking Consumption Model
Layer 3 Routing	Performed on the Physical network fabric
Additional Layer 3 Services (NAT, VPN, etc.)	Performed on the Physical network fabric
Load Balancing Options	• NSX Edge load balancer • Avi Load Balancer • Foundation Load Balancer
Tenancy	Achieved using Physical Fabric - VLAN/VRF
Consumption options	• vCenter UI/API for vSphere Distributed Port Groups • NSX UI/API for NSX VLAN Segments

VLAN Networking Connectivity Model Options

Design Area	Options
Network Type	• vSphere Distributed Port Group – Created directly in vCenter • NSX VLAN Segments – Created in NSX Manager
NSX Services Support	• vSphere Distributed Port Group – Supported via "NSX on dvpgs" feature • NSX VLAN Segment – Natively supported
Workload Placement	• VMs connected to vCenter-managed vSphere Distributed Port Groups • VMs connected to NSX-managed vSphere Distributed Port Groups (auto-created)
Consumption	• vSphere Distributed Port Group – Individual configuration of vSphere Distributed Port Groups for each vSphere Distributed Switch. • NSX VLAN Segment – Auto-creates corresponding vSphere Distributed Port Groups on all vSphereDistributed Switches belong to the same VLAN Transport Zone

Note:

In VCF 9.1.0 from a consumption perspective, using NSX VLAN Segments are better suited for environments with a high number of vSphere Distributed Switches, vCenters, and vSphere Distributed Port Groups. When creating a VLAN Segment in NSX, a corresponding vSphere Distributed Port Group is automatically created on any vSphere Distributed Switch part of the corresponding NSX, VLAN Transport Zone. This provides simpler lifecycle management of VLAN networks via NSX Manager in environments with numerous vSphere Distributed Switches, across single or multiple vCenters. This consideration is relevant for VMware Cloud Foundation where each vSphere cluster has a dedicated vSphere Distributed Switch, increasing the number of vSphere Distributed Port Groups to be managed.

Comparison VLAN Networking versus Network Virtualization (VPC or NSX Segment)

The *VLAN Networking Connectivity Model* may be appropriate for specific use cases, but it is not recommended for the majority of scenarios as it lacks the benefits provided by network virtualization. The benefits of network virtualization are summarized in the table below.

Attribute	VLAN Networking	Network Virtualization (VPC or NSX Segment)
VM Mobility	VM mobility is limited to the span of a VLAN (usually one rack)	VM mobility across Layer 3 boundaries – any topology
Rack Availability	Achieving rack availability via vSphere HA is not possible without extending VLANs between racks	Easy to achieve rack availability via vSphere HA for compute workloads regardless of the network fabric topology
Resource Pooling	Resource pooling is limited by the VLAN span	Heterogenous islands of computing are easily integrated. vSphere Clusters can be distributed across N racks regardless of the network fabric topology
Infrastructure Agility	Slow application deployment dependent on physical network reconfigurations	The physical network is abstracted. No changes are required to support the application lifecycle
Change Management	Network changes require the reconfiguration of multiple devices	Single point of management for the entire logical network. The network can embrace an Infrastructure as Code (IaC) approach, where network constructs follow the application lifecycle
Disaster Recovery	Disaster recovery requires re-IP or extending networks across sites via the physical network	Easy disaster recovery without re-IP via NSXFederation (NSX segment only) or multi availabilty zone design
Layer 2 Scale	Fundamentally tied to MAC learning and Layer 2 discovery over different types of underlay technology (e.g., traditional Layer 2, EVPN)	Programmed from vSphere, and thus no MAC learning and abstracted from the physical network, reducing the leaf switch scale requirements
Layer 2 Security	Physical fabric is directly exposed to Layer 2 frames generated by the VMs (e.g., STP BPDU)	NSX Overlay isolates the physical fabric from the Layer 2 frames generated by the endpoints
Support of Advanced Networking (i.e., NAT, Firewall)	Tied to physical ASIC and limited by switch vendor	Software-defined allows scaling of the networking and security stack

VLAN Networking Connectivity Model Design Requirements and Recommendations

Table 599: VLAN Connectivity Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLDCON-VLAN-REQD-CFG-001	Create VLANs for each workload network on the phyical network fabric.	Required for VLAN connectivity for each tenant network.	• Network admin needs to be involved for new network creation. • Slow application deployment dependent on physical network reconfigurations.
VCF-WLDCON-VLAN-REQD-CFG-002	Add the workload VLANs to the trunk switch host ports in Trunk mode using 802.1q tagging.	Required for VLAN connectivity for each tenant network.	• Network admin needs to be involved for new network creation. • Slow application deployment dependent on physical network reconfigurations.

Table 600: VLAN Connectivity Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLDCON-VLAN-RCMD-CFG-001	Create a dedicated vSphere Distributed Switch for workloads.	Isolate workload traffic from other system traffic to maximise performance.	Additional vSphere Distributed Switch required.
VCF-WLDCON-VLAN-RCMD-CFG-002	Optimize NIC Teaming and Failover.	Prevents single point of failure if one physial NIC fails	• For vSphere Distributed Switch port groups this needs to be configured for each new port group • For NSX VLAN segments use named teaming policies on the VLAN transport zone.
VCF-WLDCON-VLAN-RCMD-CFG-003	Enable NSX for DVPGs for Distributed Firewall for vSphere Distributed Switch port groups in vCenter.	Enables the vDefend Distributed Firewall (available as an advanced service), spoofguard, and any other services that can be applied via NSX segment profiles.	Must be manually enabled from NSX manager post vSphere cluster deployment.

Load Balancer Detailed Design

A VCF Instance can include one or more supported *Load Balancer Models*. This section describes the options, requirements, and recommendations for each model *Load Balancer Model*.

- Avi Load Balancer Model
- Virtual Network Appliance Load Balancer Model
- NSX Edge VPC Load Balancer Model
- NSX Edge Tier-1 Load Balancer Model
- Foundation Load Balancer Model

Avi Load Balancer Model

The *Avi Load Balancer Model* provides load balancing for vSphere Supervisor and VMware vSphere Kubernetes Service (VKS) workloads, it can also be used for NSX segment and traditional VLAN-based networking.

The Avi Load Balancer provides load-balancing capabilities for virtualized workloads including Kubernetes (VKS) workloads, it provides web application security and automatic scale. It fully integrates with VCF Automation and enables automated application deployment with Avi Load Balancer. The Avi controllers are deployed and life cycled using VCF Operations, this provides an integrated experience for management of service accounts, password rotation, and certificate management. A single Avi Load Balancer controller cluster can be deployed with NSX deployed in : many configuration where one NSX instance is shared with multiple workload domains. Alternatively, if NSX is deployed 1:1 with one workload domain per NSX instance, then a dedicated Avi Controller cluster can be deployed per NSX Instance. Service Engines can be deployed to any workload domain cluster.

Figure 293: Avi Load Balancer with NSX 1:Many Workload Domain Deployment in VMware Cloud Foundation

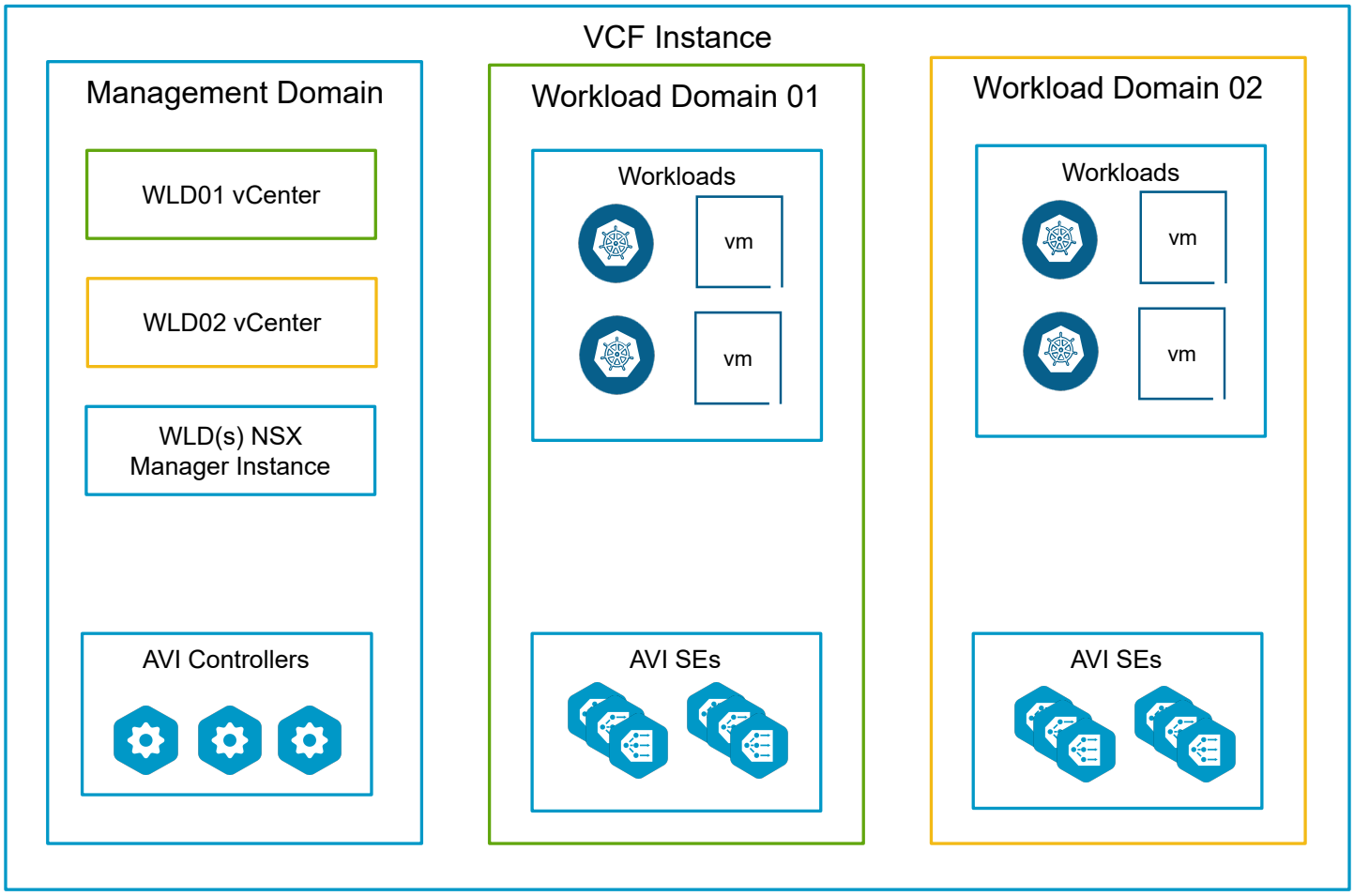
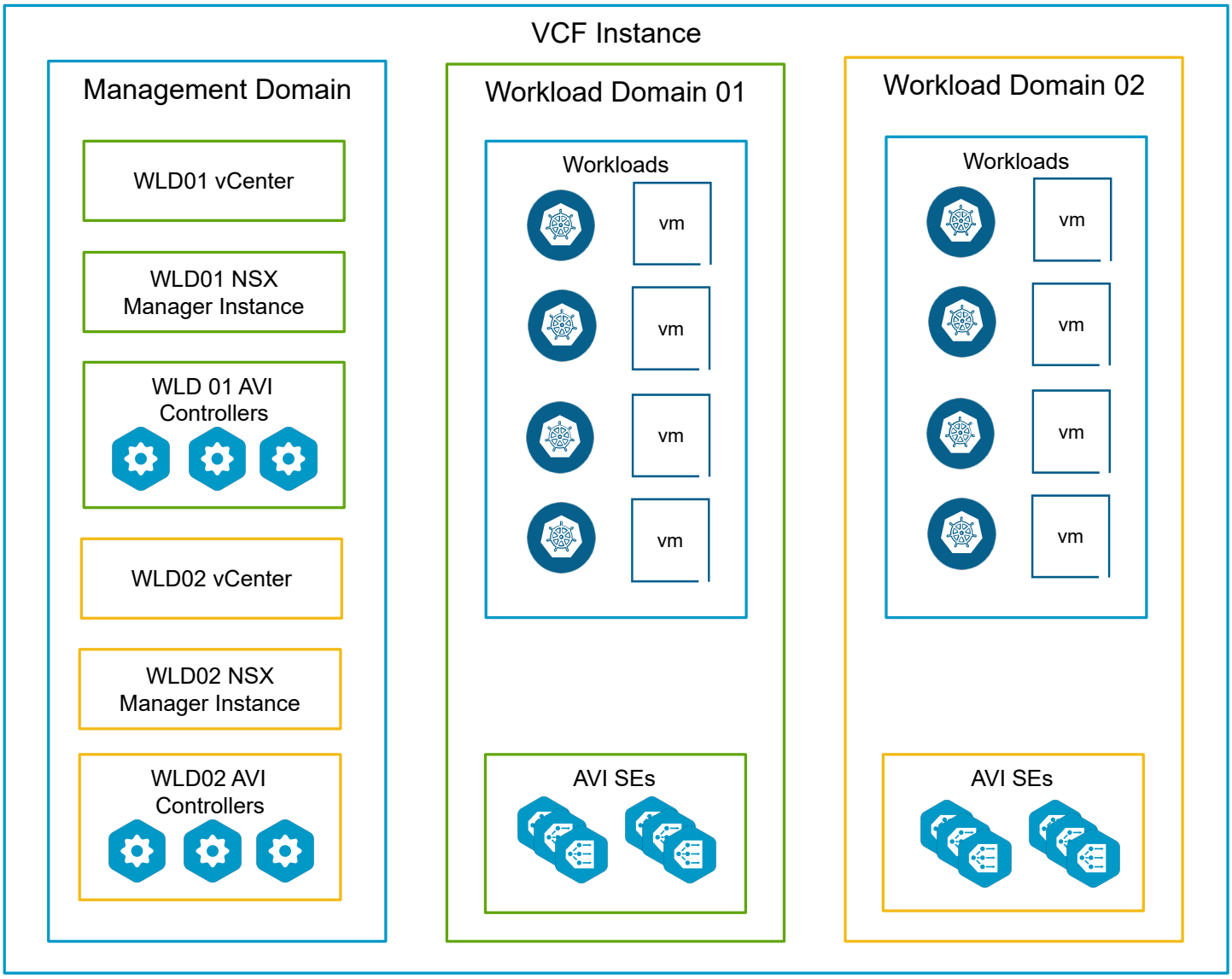


Figure 294: Avi Load Balancer with NSX One to One Workload Domain Deployment in VMware Cloud Foundation



Avi Load Balancer Model Attributes

An *Avi Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Capability	Layer 4 and Layer 7
Networking Models Supported	VPC Networking, NSX Segment Networking, VLAN Networking
Entitlement	Additional entitlement required
Availability	Control Plane - 3 node cluster Data Plane - Service Engine Groups
Security	• Web Application Firewall (WAF) • Denial of service protection • Bot identification and management • IP reputation filtering • Layer 7 security policies
Scalability	Control Plane - Small, Large, X-large Data Plane - Service Engine Group Expansion
Avi Controller Deployment Model	Deployed to a management domain using VCF Operations.
Avi Service Engines Deployment Model	Deployed to a workload domain clusters automatically via Avi controller orchestration.

Avi Load Balancer Model Deployment Options

Design Area	Options
Number of Avi Controllers	• Single – Available as an option in the UI in 9.1.1 – In 9.1 only small controller size and requires KB article to deploy. • Three
NSX Manager to Workload Domain	• One to One – Every workload domain will have a dedicated NSX manager and Avi Controller cluster. • One to Many – Multiple workload domains can share an NSX manager, and Avi Controller cluster.
Controller Size	• Small • Large • X-Large
NSX Cloud - Transport Zone	• VLAN Transport Zone – Support VLAN segment for Service Engine Management Network and data traffic. • Overlay Transport Zone – Support NSX Overlay segment for management and data traffic. • NSX Edge cluster is required. – Support VPC subnet for management traffic.
NSX Cloud - Service Engine Management Connectivity	• NSX VLAN Segment • NSX VPC Subnet – VPC Public subnet – VPC Private Subnet • NSX Overlay Segment – NSX Edge cluster is required. – Tier-1 gateway is required.

Design Area	Options
NSX Cloud - Service Engine Data Network Connectivity	• NSX VLAN Segment — Only supported in non-VPC deployments. • NSX Overlay Segment — NSX Edge cluster required. — Only required in non-VPC deployments. • NSX VPC Subnet — Default when Avi is deployed using VCF Operations — VPC Services Subnet used for communication with VPC **Note:** When Avi is deployed by VCF Operations VPC mode is enabled by default.
Avi Service Engine HA Mode	• N+M (One Virtual Server per Service Engine) (default mode) • Active/Active • Active/Standby
Transit Gateway Support	• Centralized Transit Gateway • Distributed Transit Gateway

Avi Load Balancer integration with VPC

The Avi Load Balancer can be fully integrated with VPC to provide load balancing for Supervisor and VKS-based workloads that can be deployed using VCF Automation. This can be achieved by setting VPC mode in the NSX Cloud account on the Avi Controller cluster. Communication between the VPC and the Avi Service Engines happens over the VPC Services subnet which allows VIP traffic to the service engines. When the Avi controller cluster is deployed using VCF Operations, this setting is configured by default.

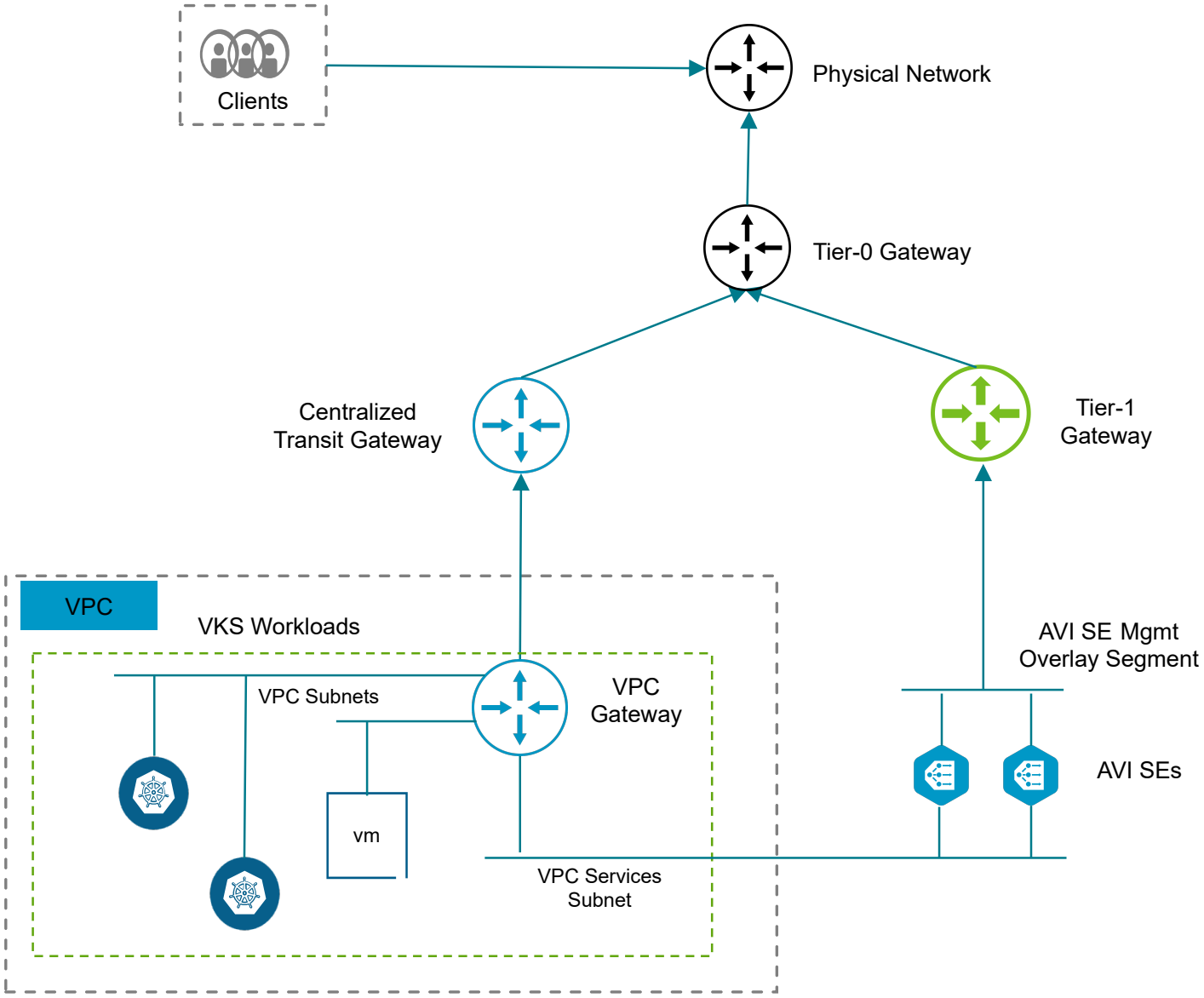
Note:

The Avi Load Balancer also supports traditional VMs connected to VPC subnets without vSphere Supervisor.

Avi Load Balancer with Centralized Transit Gateway for vSphere Supervisor

The Avi Load Balancer can be fully integrated with a Centralized Transit Gateway that is used for VPC connectivity for vSphere Supervisor. This requires an NSX edge cluster and a Tier-0 gateway for external connectivity. A Tier-1 gateway can provide management connectivity between the Avi Controllers and the Avi Service Engines via an NSX segment. Alternatively, an NSX VLAN segment could be used for the Avi Service Engine management connectivity.

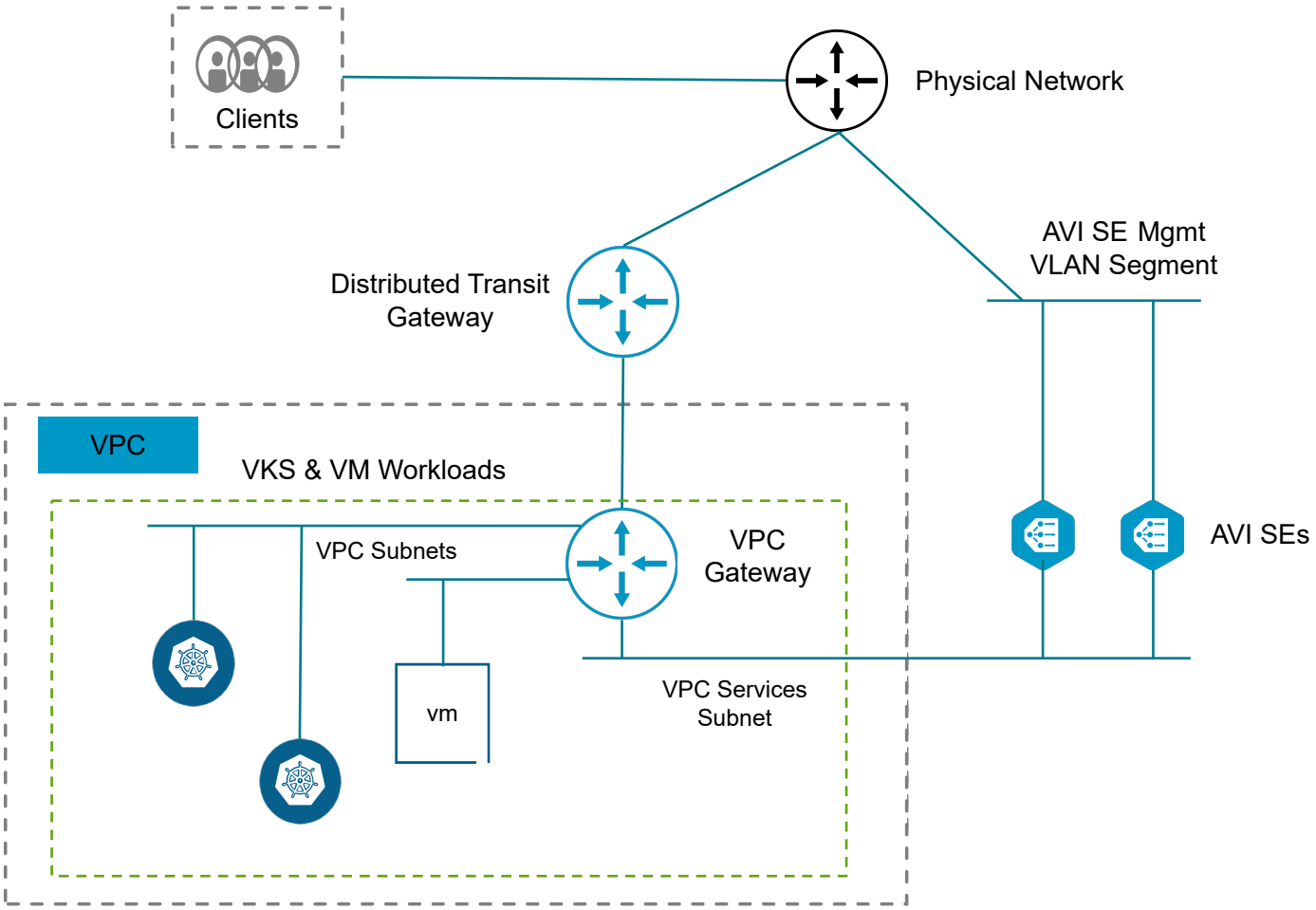
Figure 295: Avi Load Balancer with Centralized Transit Gateway in VMware Cloud Foundation



Avi Load Balancer with Distributed Transit Gateway for vSphere Supervisor

The Avi Load Balancer also integrates with a Distributed Transit Gateway for VPC connectivity used for vSphere Supervisor deployments. In this case, there is no requirement for an NSX Edge cluster for connectivity. Connectivity between the Avi Service Engines and Avi Controllers in the Management domain can be achieved using an NSX VLAN segment rather than an NSX overlay segment.

Figure 296: Avi Load Balancer with Distributed Transit Gateway in VMware Cloud Foundation



Avi Load Balancer Model Design Requirements and Recommendations

Table 601: Avi Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-AVI-LB-REQD-CFG-01	Configure NSX Cloud when using Avi as load balancer for VPC.	Required for integration with NSX and VPC.	Created by default during deployment from VCF Operations.
VCF-AVI-LB-REQD-CFG-02	Register Avi as an endpoint with NSX.	Allows NSX to delegate Layer 4-7 traffic management to the Avi Service Engines.	Configured during deployment from VCF Operations.
VCF-AVI-LB-REQD-CFG-03	Avi must be deployed with a 1:1 relationship with NSX Manager.	Product requirement.	Additional Avi Controller Clusters required if more than one NSX instance is deployed.
VCF-AVI-LB-REQD-CFG-04	Avi Load Balancer must be integrated with NSX before Supervisor activation.	If Avi Load Balancer integration is not detected, Supervisor defaults to utilizing NSX load balancer.	vSphere Supervisor should not be deployed as part of the workload creation, after the workload domain is created, Avi must be deployed before vSphere Supervisor activation.

Table 602: Avi Load Balancer Design Recommendations

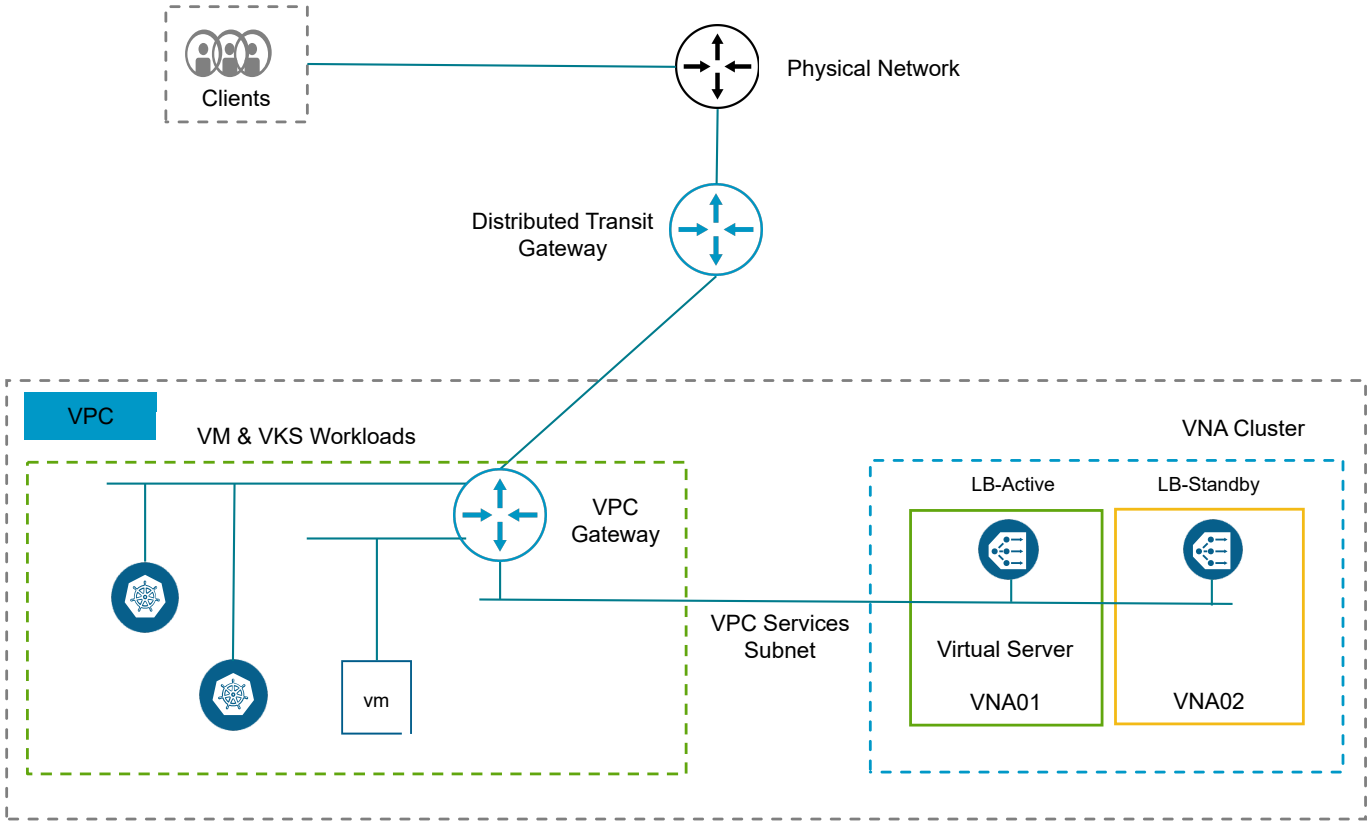
Design Requirement ID	Design Requirement	Justification	Implication
VCF-AVI-LB-RCMD-CFG-01	Deploy a 3 node controller cluster.	Provides a higher level of availability, single node failure won't cause the cluster to go down.	Management domain needs sufficient resources for the Controller VMs.
VCF-AVI-LB-RCMD-CFG-02	Deploy the controller cluster in the management domain primary cluster using VCF Operations.	• Lifecycle of the Controllers can be done using VCF Operations • Service account management from VCF Operations.	• The Avi controllers are always placed in the management domain. • The Avi VM names are fixed during deployment.
VCF-AVI-LB-RCMD-CFG-03	Create DRS anti-affinity rule for the Avi Controllers.	Prevents the Avi controller cluster being impacted by a single ESX host failure.	Created automatically when deployed from VCF Operations.
VCF-AVI-LB-RCMD-CFG-04	Enable VPC mode in the NSX Cloud Account.	• Allows the Avi Controller to discover NSX projects and their corresponding VPCs. • Needed for Supervisor based workloads using VPC subnets with Avi as the load balancer.	Configured automatically during deployment from VCF Operations.
VCF-AVI-LB-RCMD-CFG-05	Enable Service Engine DHCP IP addressing.	Automates the provisioning and assignment of IP addresses for Virtual Services (VIPs) and Service Engines. The Service Engines receive their IPs from the VPC Services Subnet.	• Configured automatically during deployment from VCF Operations. • Required for the Service Engine data network when integrated with NSX VPC.

Virtual Network Appliance Load Balancer Model

The *Virtual Network Appliance Load Balancer Model* can be used to provide load balancing for vSphere Supervisor and VMware vSphere Kubernetes Service (VKS) workloads, it supports VPC-based networking.

A Virtual Network Appliance cluster is required with one or more Virtual Network Appliance nodes; two are recommended to form an Active/Standby pair for the VPC load balancer. The Virtual Network Appliance-based load balancer supports only VPC workloads.

Figure 297: Virtual Network Appliance Load Balancer Model in VMware Cloud Foundation



Virtual Network Appliance Load Balancer Model Attributes

A *Virtual Network Appliance Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Load Balancing for VCF infrastructure components (i.e. VCF appliances) • Layer 4 load balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services)
Networking Models Supported	VPC Networking Required
Entitlement	Included with VMware Cloud Foundation
Availability	Active/Standby
Scalability	• Scale up Virtual Network Appliance nodes in the Virtual Network Appliance cluster • Scale out a single Virtual Network Appliance Cluster • Create additional Virtual Network Appliance clusters
Deployment Model	Virtual Network Appliance Cluster with two or more Virtual Network Appliances. See Virtual Network Appliance Cluster Detailed Design for further details.

Virtual Network Appliance Load Balancer Model Options

Design Area	Options
Number of Virtual Network Appliances per cluster	1 to 10
Deployment Options	• Deploy with a new workload domain • Deploy Day 2 from NSX Manager • Deploy Day 2 from vCenter
Virtual Network Appliance Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger scale depoloyments where a large number of load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

Virtual Network Appliance Load Balancer Design Requirements and Recommendations

Table 603: Virtual Network Appliance Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-LB-REQD-CFG-01	Create a Virtual Network Appliance cluster with two or more Virtual Network Appliance nodes.	Provides availability for a load balancer instance.	None.
VCF-VNA-LB-REQD-CFG-02	Configure the VPC connectivity profile to use Virtual Network Appliance cluster in the VPC Service Gateway Configurations.	Enables the VPC to use the Virtual Network Appliance cluster for load balancer instances.	Virtual Network Appliance cluster can be shared between VPCs.
VCF-VNA-LB-REQD-CFG-03	Create a load balancer instance for the VPC using the API	Supervisor and VKS-based workloads, load balancer creation is automated.	No UI available to configure load balancer instances.
VCF-VNA-LB-REQD-CFG-04	Create a load balancer configuration for VPC using the API.	Supervisor and VKS-based workloads, load balancer configuration is automated.	No UI available to configure load balancer instances.

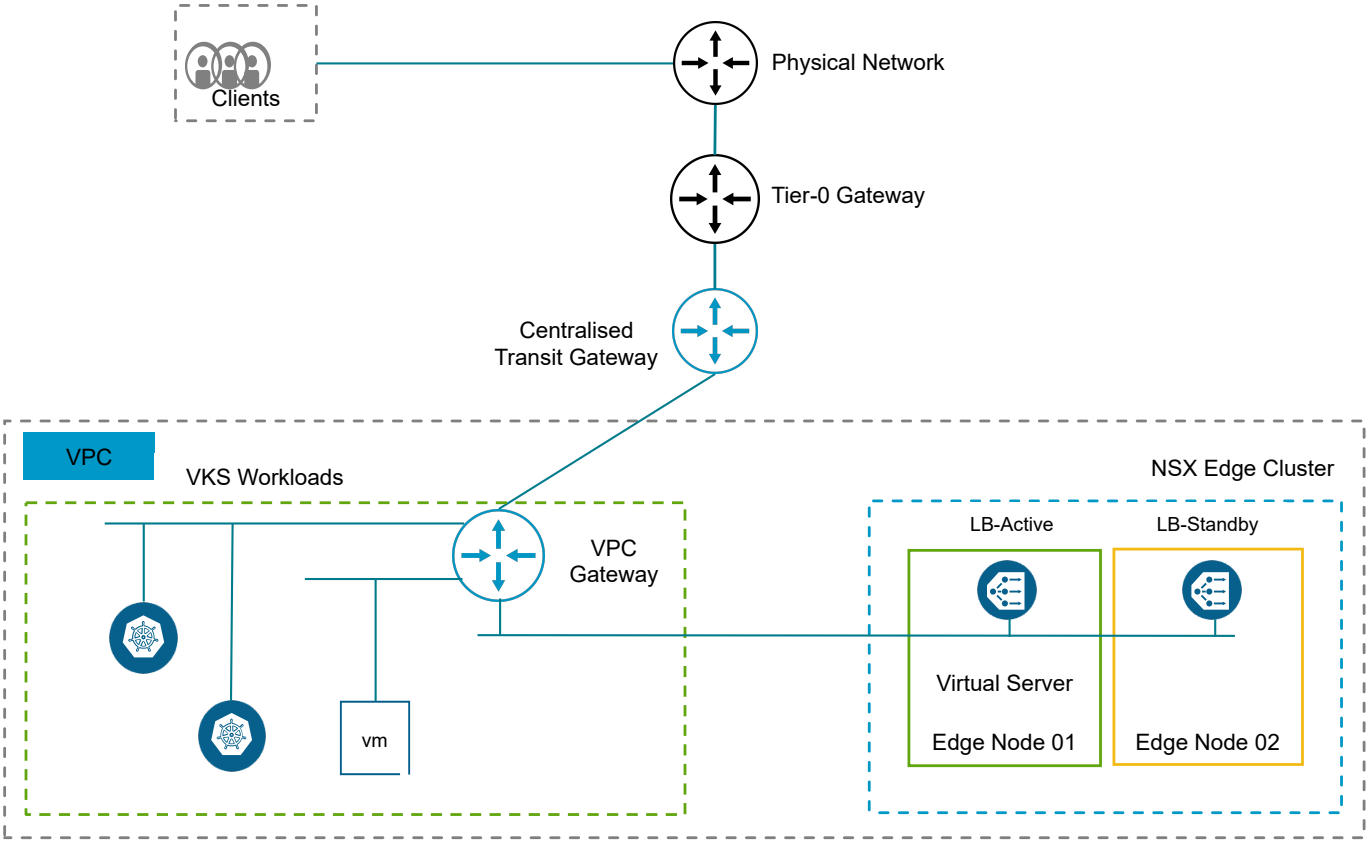
Table 604: Virtual Network Appliance Load Balancer Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-LB-RCMD-CFG-01	Configure failures domain for Virtual Network Appliance clusters with more than two Virtual Network Appliance nodes in the cluster.	Ensure VPC load balancer availability by controlling Actve/Standby placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-VNA-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure VPC load balancer availability by controlling Actve/Standby placement.	Manually configured from vCenter UI.

NSX Edge VPC Load Balancer Model

The NSX Edge VPC load balancer can be used to provide load balancing for vSphere Supervisor and VMware vSphere Kubernetes Service (VKS) workloads, this is based on the VPC Networking model. The NSX Edge VPC load balancer can only be configured through API and is especially suited for vSphere Supervisor and VKS-based workloads.

Figure 298: NSX Edge - VPC Load Balancer



NSX Edge VPC Load Balancer Model Attributes

An *NSX Edge VPC Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Layer 4 Load Balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services) • Load Balancing for VCF Infrastructure Components (i.e. VCF appliances)
Networking Models Supported	VPC Networking
Entitlement	Included with VCF
Availability	Active/Standby
Scalability	• Scale up NSX Edge node size • Scale out NSX Edge Cluster • Create additional NSX Edge clusters
Deployment Model	NSX Edge Cluster with two or more NSX Edge nodes for high availability

NSX Edge VPC Load Balancer Model Options

Design Area	Options
Networking Model	VPC Networking
NSX Edge Cluster Requirements	• Minimum one NSX edge node • Recommended two or more NSX edge nodes for high availability
Load Balancer Deployment Mode	• Deployed using NSX API
NSX Edge Node Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger-scale deployments where many load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

NSX Edge VPC Load Balancer Design Requirements and Recommendations

Table 605: NSX Edge VPC Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-EDGE-VPC-LB-REQD-CFG-01	Create a Centralized Transit Gateway with two or more edge nodes.	Provides availability and routing for a load balancer instance.	None.
VCF-EDGE-VPC-LB-REQD-CFG-02	Create a centralized external connection with Tier-0 Gateway.	Allows external connectivity to the load balancer Virtual Servers.	Tier-0 requires static or dynamic routing configured.
VCF-EDGE-VPC-LB-REQD-CFG-03	If using BGP routing enable Tier-0 route redistribution to include Transit Gateway Static	Allows load balancer Virtual Server IPs to be advertised to the datacenter network	Enabled by default when Centralized Transit Gateway is deployed from vCenter or NSX Manager.
VCF-EDGE-VPC-LB-REQD-CFG-04	Configure the VPC connectivity profile to use NSX Edge cluster in the VPC Service Gateway Configurations.	Enables the VPC to use the NSX Edge cluster for load balancer instances.	NSX edge cluster can be shared between VPCs.
VCF-EDGE-VPC-LB-REQD-CFG-05	Create a load balancer Instance for the VPC using API.	Supervisor and VKS-based workloads, load balancer creation is automated.	No UI available to configure load balancer Instances.
VCF-EDGE-VPC-LB-REQD-CFG-06	Create load balancer configuration for VPC using API.	Supervisor and VKS-based workloads load balancer configuration is automated.	No UI available to configure load balancer Instances.

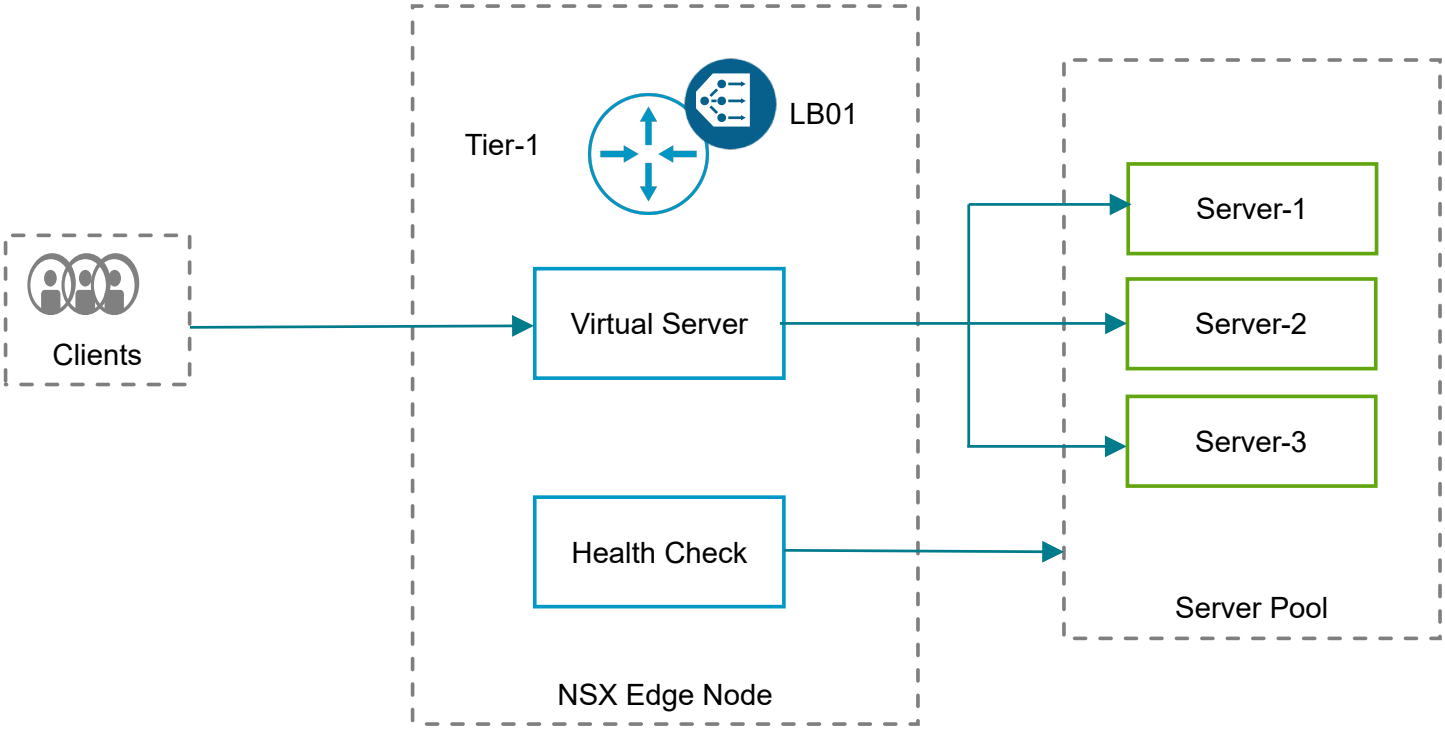
Table 606: NSX Edge VPC Load Balancer Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-EDGE-VPC-LB-RCMD-CFG-01	Configure failures domain for edge clusters with more than two NSX Edge nodes in the cluster.	Ensure VPC load balancer availability by controlling Actve/Standy placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-EDGE-VPC-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure Tier-1 load balancer availability by controlling Actve/Standy placement.	Manually configured from vCenter UI.
VCF-EDGE-VPC-LB-RCMD-CFG-03	NSX Edge nodes should be deployed with a minimum of large form factor.	NSX load balancers have fixed resource allocations on NSX Edge nodes. A large form factor is required to accommodate basic system and workload needs.	Large form factor resources required. If additional load balancers are required, NSX Edges can be scaled up or out.

NSX Edge Tier-1 Load Balancer Model

The NSX Edge Tier-1 load balancer can be used to provide load balancing for workloads connected to NSX overlay and VLAN segments or traditional vSphere distributed port groups. It can be used to provide load balancing for VCF Operations in the VCF management platform, but also vSphere Supervisor if the Supervisor has been deployed with classic NSX mode.

Figure 299: NSX Edge - Tier-1 Load Balancer



NSX Edge Tier-1 Load Balancer Model Attributes

An *NSX Edge Tier-1 Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Support	• Load Balancing for VCF Infrastructure Components (i.e. VCF appliances) • Layer 4 Load Balancing for vSphere Supervisor (including VM Service, vSphere PODs, VKS Clusters, and Supervisor Services)
Networking Models Supported	• NSX Segment Networking, VLAN Networking
Entitlement	Included with VCF
Availability	Active/Standby
Scalability	• Scale Up Edge cluster by adding nodes – Additional capacity for Tier-1 gateways for load balancing pusposes • Scale number of Tier-1 Gateways – Alows additional Load balancers to be created
Deployment Model	NSX Edge Cluster with two or more NSX Edge nodes for high availability

NSX Edge Tier-1 Load Balancer Model Options

Design Area	Options
Networking Model	• NSX Segment Networking • VLAN Networking
NSX Edge Cluster Requirements	• Minimum one • Recommended two or more for high availability
Deployment Options	Deploy Day 2 from NSX Manager.
NSX Edge Node Size	• Small – Not recommened even for POC as it only supports a single load balancer instance • Medium – More suited to POC as scale is limited • Large – Recommended minimum size for vSphere Supervisor to allow for load balancer instance scale • X-Large – Recommended for larger-scale deployments where many load balancer instances will be deployed. Please refer to the Configuration Maximums for details for the number of load balancer instances supported for each size above https://configmax.broadcom.com/home
Load Balancer Size	• Small • Medium • Large • X-Large Please refer to the Configuration Maximums for details https://configmax.broadcom.com/home

NSX Edge Tier-1 Load Balancer Design Requirements and Recommendations

Table 607: NSX Edge Tier-1 Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-EDGE-T1-LB-REQD-CFG-01	Create an NSX Edge cluster with two or more NSX Edge nodes.	Provides availability for a load balancer instance.	None.
VCF-EDGE-T1-LB-REQD-CFG-02	Create a Tier-1 Gateway with HA Mode as Active/Standy.	Only Active/Standby is supported for load balancer.	All load balancer traffic must pass through a single edge node.
VCF-EDGE-T1-LB-REQD-CFG-03	Configure an appropriate NSX Edge pool allocation size.	Ensures that the correct scale can be achieved for load balancers.	Incorrect initial sizing limits scale.
VCF-EDGE-T1-LB-REQD-CFG-04	Create a service interface on the Tier-1 to connect to an NSX Overlay or VLAN segment.	Allows connectivity to the load balancer.	None.
VCF-EDGE-T1-LB-REQD-CFG-05	Create a default static route for the load balancer traffic.	Allows traffic to be routed from the load balancer to external networks.	None.

Table 608: NSX Edge Tier-1 Load Balancer Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-EDGE-T1-LB-RCMD-CFG-01	Configure failures domain for NSX Edge clusters with more than two NSX Edge nodes in the cluster.	Ensure Tier-1 load balancer availability by controlling Actve/Standy placement.	Manually configured from NSX Manager UI or vCenter UI.
VCF-EDGE-T1-LB-RCMD-CFG-02	Use DRS groups and rules to ensure Active/Standby VMs are not on the same host or rack.	Ensure Tier-1 load balancer availability by controlling Actve/Standy placement.	Manually configured from vCenter UI.

Foundation Load Balancer Model

The Foundation Load Balancer is a basic load balancer that can be used to provide load balancing for vSphere Supervisor with VLAN networking. The Foundation Load Balancer requires vSphere Distributed port groups, it does not support VPC Networking or NSX Segment networking.

You can deploy a Foundation Load Balancer during Supervisor activation in vCenter. It can be deployed as a single appliance or in high availability mode using two appliances in active/standby mode. The Foundation Load Balancer can be deployed in a one-arm or two-arm mode. In one-arm mode the Virtual Server Network is used both for incoming application traffic and outgoing forwarded application traffic to the Workload Networks. In two-arm mode, these networks are separated and the appliance uses a separate network adapter for each traffic type.

Figure 300: Foundation Load Balancer One-arm mode

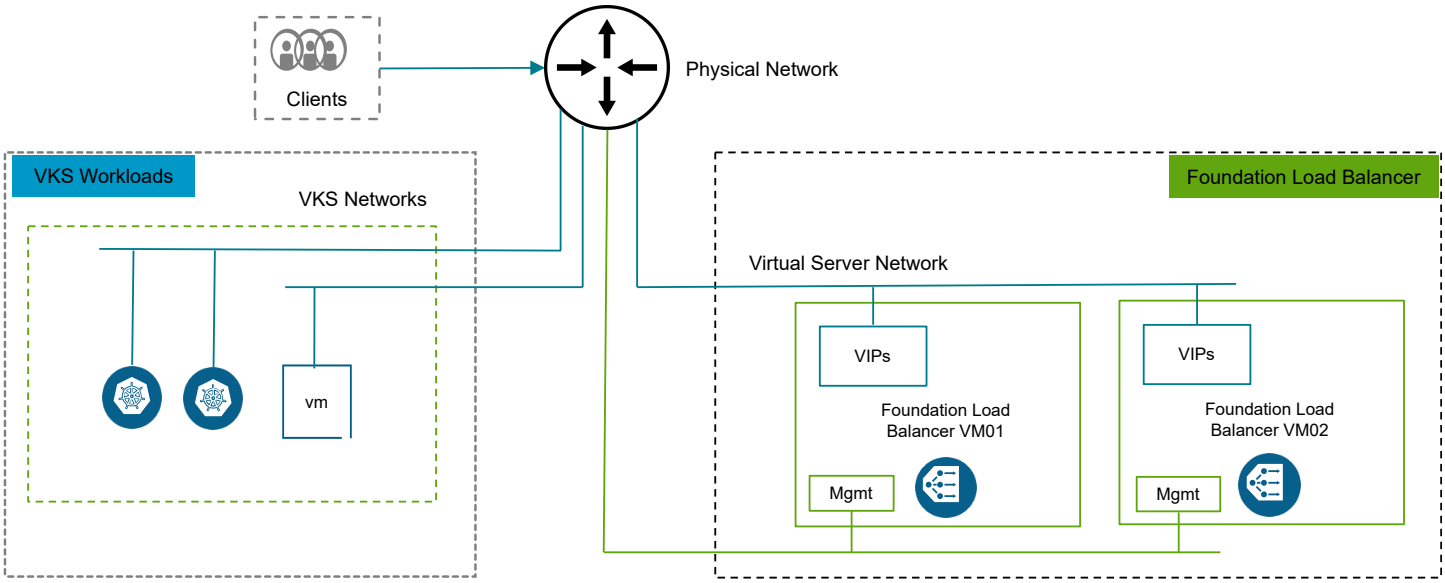
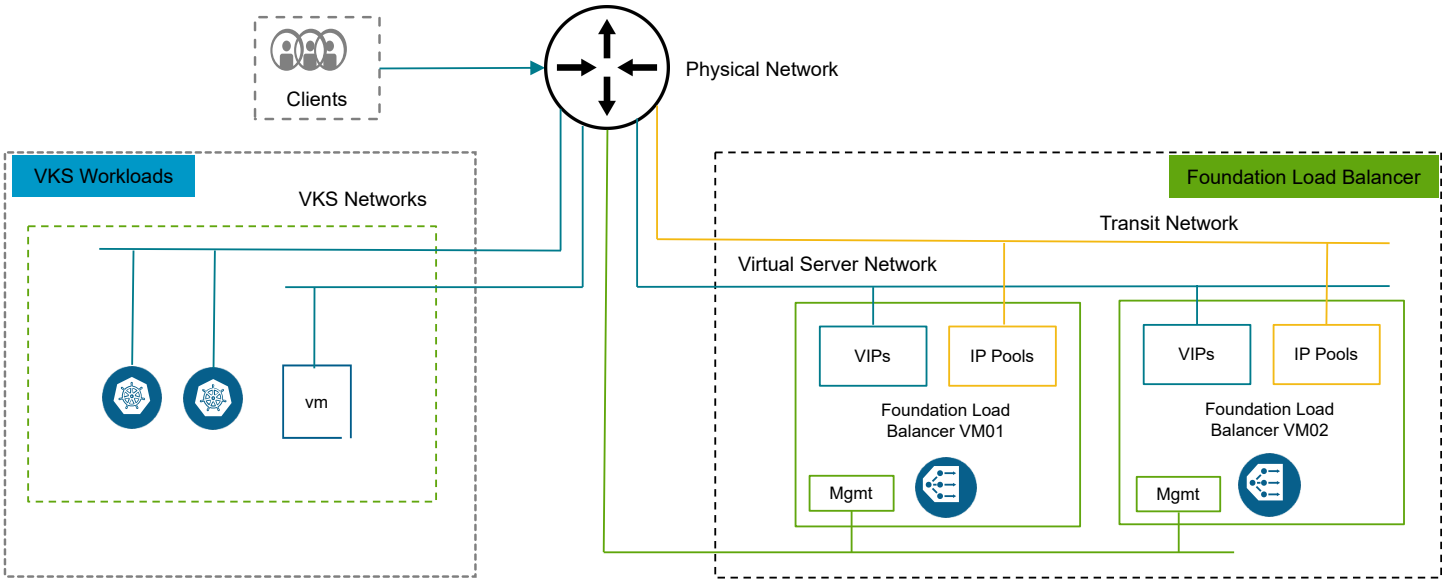


Figure 301: Foundation Load Balancer Two-arm mode



Foundation Load Balancer Model Attributes

A *Foundation Load Balancer Model* has the following attributes.

Attribute	Detail
Load-Balancing Capability	Layer 4
Networking Models Supported	VLAN Networking
Entitlement	Included with VCF
Scalability	Maximum of two appliances
Availability	Active/Standy
Deployment Model	Foundation Load Balancer appliances deployed by vCenter

Foundation Load Balancer Model Options

Design Area	Options
Number of Load Balancer Appliances	• Single • Dual
Network Topology	• One-arm • Two-arm
Deployment	• During Supervisor Activation • Manually from vCenter

Foundation Load Balancer Design Requirements and Recommendations

Table 609: Foundation Load Balancer Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-FLB-LB-REQD-CFG-01	Supervisor deployment with vDS networking.	Does not support other types of network connectivity.	VPC Networking not supported.
VCF-FLB-LB-REQD-CFG-02	Management port group available on the vSphere Distributed Switch.	Required for Management connectivity to Foundation load balancer.	Port Group must be created on vSphere Distributed Switch before deployment.
VCF-FLB-LB-REQD-CFG-03	Virtual Server Network port group available on the vSphere Distributed Switch.	Required for incoming traffic from clients to Foundation load balancer Virtual IPs (VIPs)	Port Group must be created on vSphere Distributed Switch before deployment.
VCF-FLB-LB-REQD-CFG-04	In a Two-arm deployment mode Transit Network port group available on the vSphere Distributed Switch.	Required outgoing traffic from Foundation load balancer forwarded to one of the workload IP pool members and then to the Workload Networks.	Port Group must be created on vSphere Distributed Switch before deployment.

Table 610: Foundation Load Balancer Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-FLB-LB-RCMD-CFG-01	Deploy two Foundation load balancer VMs.	Ensure a higher level of availability, if the active VM fails or an ESX hosts fails the passive becomes active.	Requires more resources.
VCF-FLB-LB-RCMD-CFG-02	Configure DRS anti-affinity rules for the two Foundation load balancer VMs	Ensure a higher level of availability if an ESX host in the cluster fails.	Must be created manually in vCenter.

VCF Management Services Component Detailed Design

The components deployed as part of VCF management services differs depending on whether the VCF Instance being deployed is the first VCF Instance in a VCF fleet or an additional VCF Instance.

- VCF Management Services in the First VCF Instance Model
- VCF Management Services in an Additional VCF Instance Model

VCF Management Services in the First VCF Instance Model

The first VCF Instance in a VCF fleet runs a VCF management services cluster with a combination of fleet level and instance level components for managing all VCF Instances in the fleet.

VCF management services provides a common framework for running VCF management components. Individual VCF management components are deployed at either fleet level or at instance level.

- **Fleet level components** exist in one VCF Instance and are used to manage multiple VCF Instances.
- **Instance level components** exist in each VCF Instance and are used to manage that instance only.

Figure 302: VCF Management Services in the First VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.

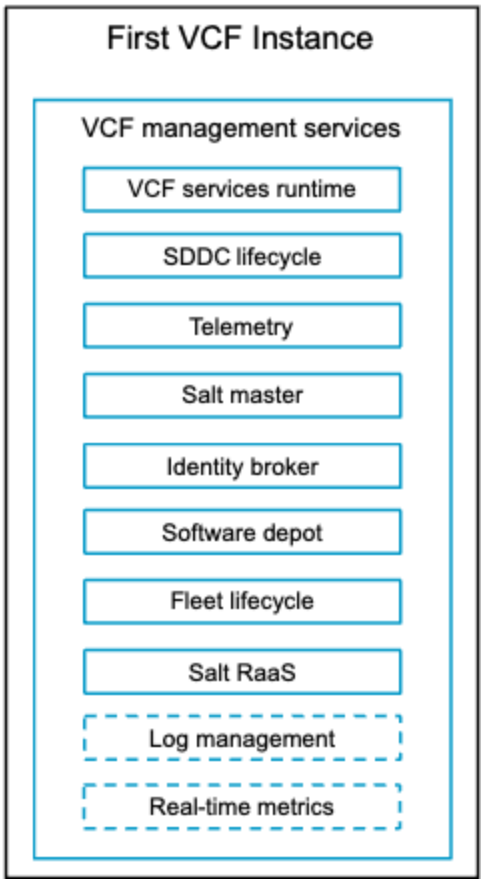


Table 611: VCF Management Services in the First VCF Instance Model Components

The following components run in the first VCF Instance in the VCF fleet.

Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment

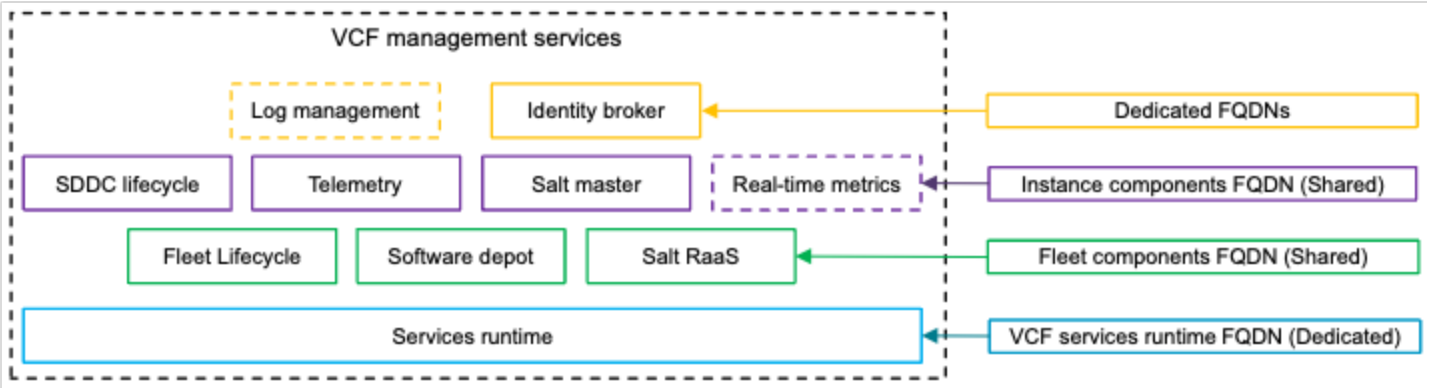
Service	Description	Deployment Time
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations across the fleet.	
Fleet lifecycle	Orchestrates fleet-wide life-cycle operations, including the planning and sequencing of upgrades and patches across multiple VCF Instances.	
Salt RaaS	Defines and manages desired state configurations and policies (Recipe as a Service).	
Log management	Collects logs from all fleet components	Day-N Components
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

* Can be in one or many VCF Instances

VCF Management Services in the First VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 303:



VCF Management Services in the First VCF Instance Model Design Requirements

Table 612: VCF Management Services in the First VCF Instance Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-FIRST-001	Create DNS entries for the following shared FQDNs: • Fleet components FQDN • Instance components FQDN	Required for deploying VCF management services fleet and instance level components in the first VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-FIRST-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN • Identity broker FQDN	Required for deploying VCF management services in the first VCF Instance.	You must create the DNS entries before starting the deployment.

VCF Management Services in an Additional VCF Instance Model

Additional VCF Instances in a fleet run a VCF management services cluster with instance level components for managing that instance only.

VCF management services provides a common framework for running VCF management components. Individual VCF management components are deployed at either fleet level or at instance level.

- **Fleet level components** exist in one VCF Instance and are used to manage multiple VCF Instances.
- **Instance level components** exist in each VCF Instance and are used to manage that instance only.

Figure 304: VCF Management Services in an Additional VCF Instance Model in VMware Cloud Foundation

The diagram below shows the various components for this model. The dotted line components are deployed day-N.

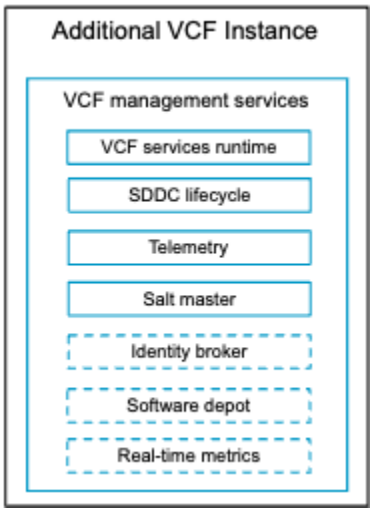


Table 613: VCF Management Services in an Additional VCF Instance Model Components

The following components run in each additional VCF Instance in the VCF fleet.

Service	Description	Deployment Time
VCF services runtime	Hosts and orchestrates VCF management services.	Initial deployment
SDDC lifecycle	Performs local life-cycle tasks (install, update, patch) received from the fleet lifecycle component.	

Service	Description	Deployment Time
Telemetry	Aggregates and transmits diagnostic and usage data.	
Salt master	Enforces configuration states on local infrastructure resources.	
Identity broker *	Provides centralized identity and access management services, handling authentication and token exchange for VCF components.	Day-N Components
Software depot *	Hosts and provides installation binaries, upgrade bundles, and metadata required for life-cycle operations within this VCF Instance.	
Real-time metrics *	Collects and streams high-frequency performance data for monitoring.	

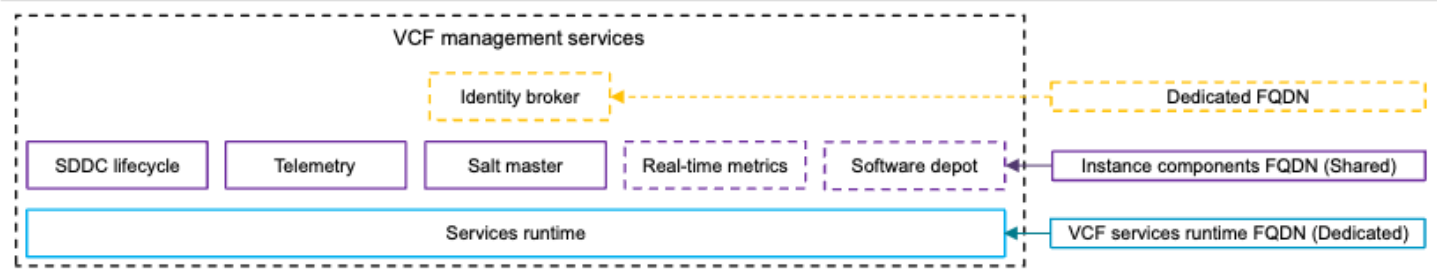
* Can be in one or many VCF Instances

VCF Management Services in an Additional VCF Instance Model FQDN Mappings

VCF management services use DNS entries for the services that are exposed from the cluster. Some services share the same FQDN, and others use dedicated FQDNs.

Figure 305: VCF Management Services FQDN Mapping in an Additional VCF Instance

The diagram below shows the various components for this model. The dotted line components are optional, and are deployed day-N.



VCF Management Services in an Additional VCF Instance Model Design Requirements

Table 614: VCF Management Services in an Additional VCF Instance Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-ADTL-001	Create DNS entries for the following shared FQDNs: • Instance components FQDN	Required for deploying VCF management services in the additional VCF Instance.	You must create the DNS entries before starting the deployment.
VCF-MS-REQD-ADTL-002	Create DNS entries for the following dedicated FQDNs: • VCF services runtime FQDN	Required for deploying VCF management services in the additional VCF Instance.	You must create the DNS entries before starting the deployment.

VCF Management Services Detailed Design

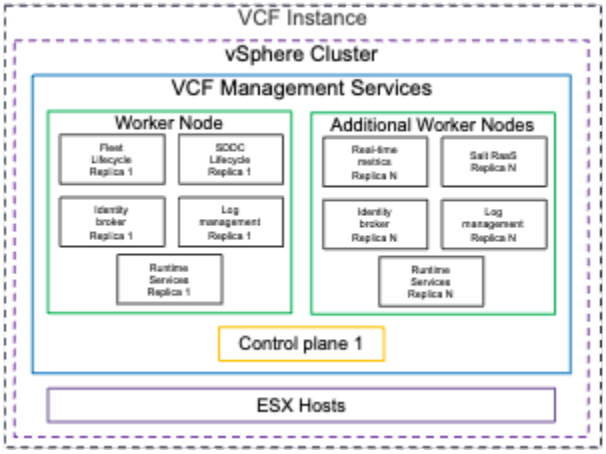
Based on the needs of a VCF Instance, you may choose to deploy VCF management services in a simple or highly available manner.

- Simple VCF Management Services Model
- High Availability VCF Management Services Model

Simple VCF Management Services Model

The *Simple VCF Management Services Model* deploys a single control plane node and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Figure 306: Simple VCF Management Services Model in VMware Cloud Foundation



Simple VCF Management Services Model Attributes

Attribute	Detail
Recommended Usage	• Proof of Concept • Lab deployments
Deployment Size	Small **Note:** Refer to VCF Fleet Sizing Models for more details on sizing.
Availability	• Relies on vSphere HA. • Loss of the single control plane node would result in a need to redeploy the cluster and restore from backup.
Scalability	Scale out to a medium or large high availability deployment model is supported.
Placement	Hosted on a single vSphere cluster.

Simple VCF Management Services Model Options

According to your objectives, the *Simple VCF Management Services Model* can have options from the following design areas.

Design Area	Options
Infrastructure IP Address Model	• IPv4 Addressing Model • Dual Stack IP Addressing Model
VCF Management Network Model	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model

Simple VCF Management Services Model Design Requirements

Table 615: Common VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-001	Allocate a sufficient IP range for the required number of control plane and worker nodes for your chosen deployment type.	Required for initial deployment.	Adding additional components may require auto-scale out of the VCF management services.
VCF-MS-REQD-COM-002	Use *Kubernetes Cluster API Provider vSphere (CAPV)* node anti-affinity to ensure the nodes run on different ESX hosts.	Using *CAPV* prevents the VCF management services nodes from running on the same ESX host and risking the high availability of the cluster.	Distributed Resource Scheduler (DRS) anti-affinity rules are not required.
VCF-MS-REQD-COM-003	Protect the VCF management services nodes by using vSphere HA.	Ensures availability of the VCF management services nodes without requiring manual intervention during an ESX host failure event.	None.

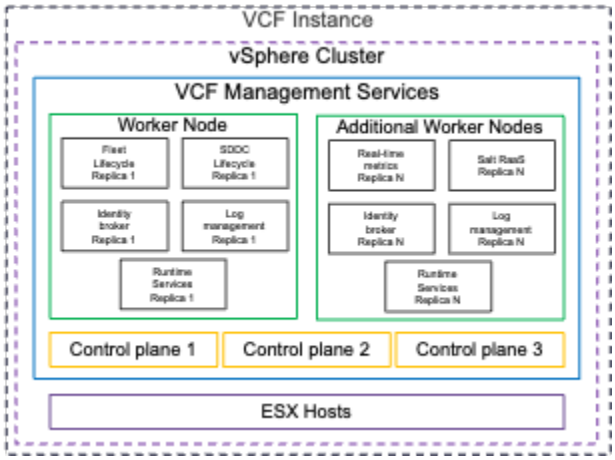
Table 616: Simple VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-SIM-001	Deploy VCF management services with a single control plane node and three worker nodes in the default management vSphere cluster.	• Supports the simple deployment objective. • VCF management services supports scale-out and up to a high availability model.	Loss of the single control plane node would result in a need to redeploy and restore from backup.

High Availability VCF Management Services Model

The *High Availability VCF Management Services Model* deploys multiple control plane nodes and multiple worker nodes, and uses vSphere HA as an availability mechanism.

Figure 307: High Availability VCF Management Services Model in VMware Cloud Foundation



High Availability VCF Management Services Model Attributes

Attribute	Detail
Availability	Relies on vSphere HA
Scalability	Scale out/up worker nodes
Placement	Hosted in a single vSphere cluster

High Availability VCF Management Services Model Options

According to your objectives, the *High Availability VCF Management Services Model* can have options from the following design areas.

Design Area	Options
Infrastructure IP Address Model	• IPv4 Addressing Model • Dual Stack IP Addressing Model
Deployment Size	• Medium • Large Refer to VCF Fleet Sizing Models for more details on sizing.
VCF Management Network Model	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model

High Availability VCF Management Services Model Design Requirements

Table 617: Common VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-COM-001	Allocate a sufficient IP range for the required number of control plane and worker nodes for your chosen deployment type.	Required for initial deployment.	Adding additional components may require auto-scale out of the VCF management services.
VCF-MS-REQD-COM-002	Use *Kubernetes Cluster API Provider vSphere (CAPV)* node anti-affinity to ensure the nodes run on different ESX hosts.	Using *CAPV* prevents the VCF management services nodes from running on the same ESX host and risking the high availability of the cluster.	Distributed Resource Scheduler (DRS) anti-affinity rules are not required.
VCF-MS-REQD-COM-003	Protect the VCF management services nodes by using vSphere HA.	Ensures availability of the VCF management services nodes without requiring manual intervention during an ESX host failure event.	None.

Table 618: High Availability VCF Management Services Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MS-REQD-HA-001	Deploy VCF management services in high availability mode with three (3) control plane nodes and the corresponding number of worker nodes in the default management vSphere cluster.	• Supports the high availability deployment objective. • VCF management services supports scale-out and up from small to medium or large.	Requires additional resources.

VCF Management Network Detailed Design

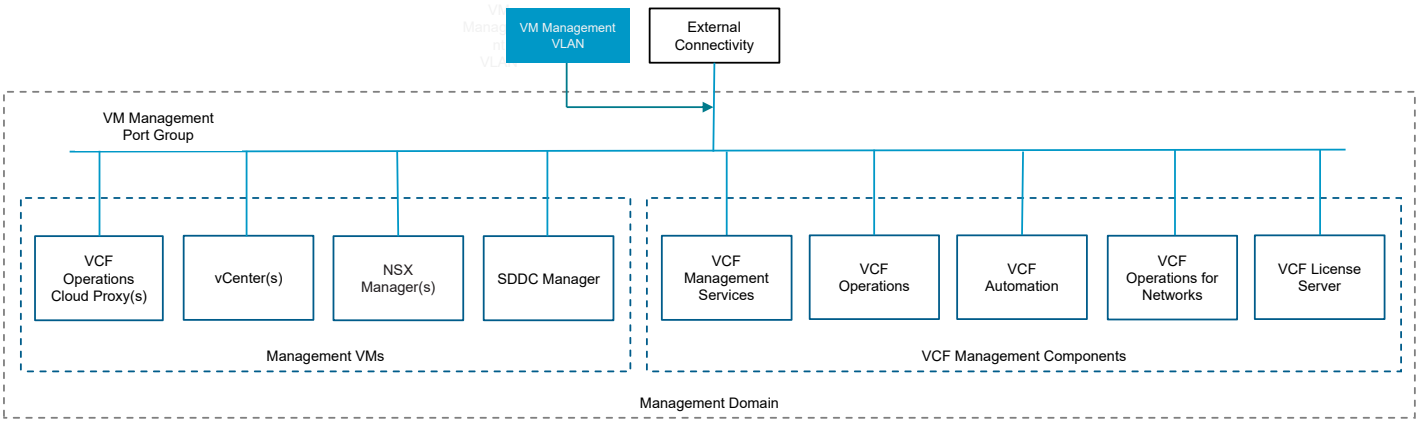
The *VCF Management Network Models* have several connectivity options depending on the requirements. This section describes a number of options, requirements, and recommendations for each *VCF Management Network Models*.

- VCF Management Shared VLAN Network Model
- VCF Management Dedicated VLAN Network Model
- VCF Management Dedicated VLAN and NSX Overlay Segment Network Model
- VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model

VCF Management Shared VLAN Network Model

VCF management components can be deployed to the same vSphere Distributed Switch port group used for the VCF Instance level management components such as (vCenter, NSX Manager, and SDDC Manager). This can be achieved using VCF Installer during initial deployment of the VCF fleet or more instances in the fleet.

Figure 308: VCF Management Shared VLAN Network Model in VMware Cloud Foundation



VCF Management Shared VLAN Network Model Attributes

A *VCF Management Single VLAN Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	VLAN-backed vSphere distributed port group.
Network Isolation	No; Sharing VLAN with the VCF management domain components.
NSX Edge Cluster Required	No; Optional to provide NSX load balancer for VCF Operations. (requires additional NSX VLAN segment for Load balancer connectivity)
Stretching Network Between Availability Zones	• **VM management network** must be stretched in the physical network fabric between availability zones. • Provides automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• Not suited for a disaster recovery use case as the network is shared with management domain components. • Failover testing is not supported.

VCF Management Shared VLAN Network Model Design Requirements and Recommendations

Table 619: VCF Management Shared VLAN Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-SV-NET-REQD-001	Use the management domain vCenter VM management distributed port group for the VCF fleet level components.	No additional network required.	• No logical network isolation. • Done automatically during VCF Installer deployment.
VCF-MGMT-SV-NET-REQD-002	Use the same VLAN for VCF management components and the VCF Instance management virtual machines. e.g. vCenter and NSX Managers.	No additional VLAN required.	• Separate physical firewall rules may not be possible with a single VLAN. NSX Distributed Firewall may be an alternative solution. • Not suitable for Fleet Disaster Recovery use case with single VLAN.

Table 620: VCF Management Shared VLAN Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-SV-NET-RCMD-001	Deploy the VCF management components during initial deployment using VCF Installer.	Simplified deployment using VCF Installer UI.	None.

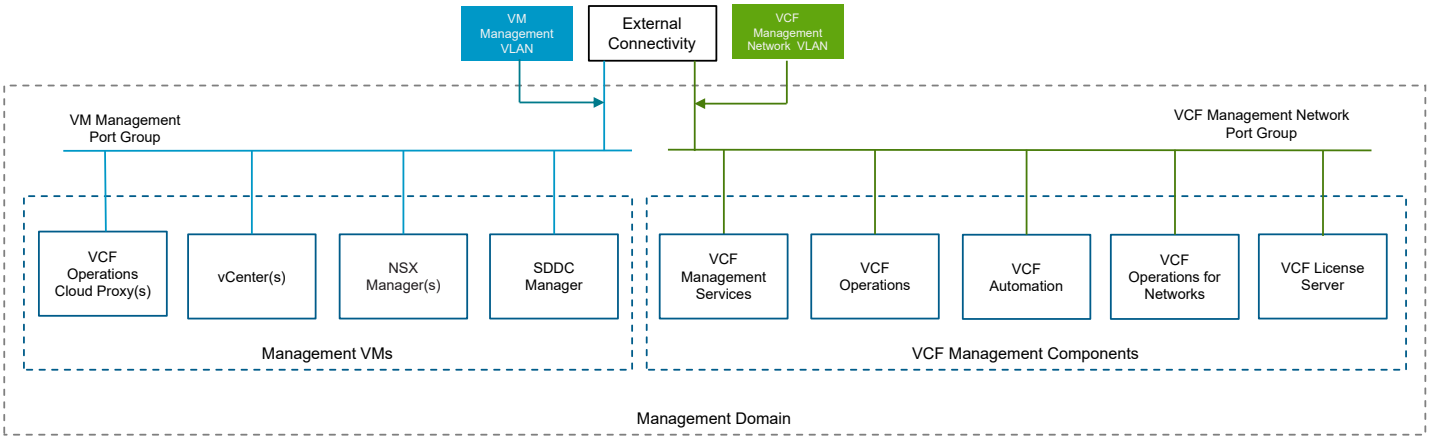
VCF Management Dedicated VLAN Network Model

VCF management components can be deployed to the dedicated vSphere Distributed Switch port group and only used for all the VCF management components.

Note:

The cloud proxy is a VCF Instance level management component and is always deployed to the **VM management network** by VCF Installer.

Figure 309: VCF Management Dedicated VLAN Network Model in VMware Cloud Foundation



VCF Management Dedicated VLAN Network Model Attributes

A *VCF Management Dedicated VLAN Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	VLAN-backed vSphere distributed port group.
Network Isolation	Yes; Isolated using dedicated VLAN and vSphere distributed port group.
NSX Edge Cluster Required	No; Optional to provide NSX load balancer for VCF Operations. (requires additional NSX VLAN segment for Load balancer connectivity)
Multi Availability Zone Network Configuration	• **VCF management network** must be stretched in the Physical Network fabric between availability zones. • Provides automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	There are two options: 1. Option 1: Create a separate **VCF management network**in the second region to deploy the new VCF Automation and VCF management services instances that are used for restore purposes. – The **VCF management network** in the second region needs to be routable to the first region **VCF management network**. – The **VCF management network** in the first region must be available at second region in the case of Fleet disaster recovery to provide IP mobility for VCF Operations and optionally VCF Operations for Networks. – During the restore proces the VIP DNS entries for VCF Automation and VCF management services need to be updated to point to the new network. 2. Option 2: Use a single **VCF management network** and stretch the network between regions.

VCF Management Dedicated VLAN Network Model Design Requirements and Recommendations

Table 621: VCF Management Dedicated VLAN Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-DV-NET-REQD-001	Use a dedicated VLAN for VCF management components. • VCF management services • VCF Operations • VCF Operations for networks • VCF Automation • License Server	• Logical network isolation can be achieved for the VCF Management components traffic. • A physical firewall can be used to secure access to this network.	• Another VLAN and IP subnet required. • For stretched management vSpherecluster, VLAN must be stretched between sites to provide site high availability. • For Fleet Disaster Recovery to provide IP mobility for VCF Operations and optionally VCF Operations for networks , the same VCF Management VLAN must be available at the second Region. • For Fleet Disaster Recovery in this model theVCF management services and VCF Automation uses the same VCF Management VLAN, however it is possible to use a different network at the recovery site.

Table 622: VCF Management Dedicated VLAN Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-DV-NET-RCMD-001	Create a dedicated distributed port group on the management vCenter management vSphere Distributed Switch.	Used for VCF Management component connectivity.	• Automatically created during VCF Installer deployment day 0.
VCF-MGMT-DV-NET-RCMD-002	Use a dedicated port group on the management vCenter management vSphere Distributed Switch for all the VCF management components.	Logically isolates VCF Management components traffic from the management domain VM management traffic.	Does not provide additional isolation of VCF management services from VCF Automation and VCF Operations
VCF-MGMT-DV-NET-RCMD-003	Use the VM management network for the cloud proxy, they are considered VCF Instance level components, and can reside on the same network as the instance components.	The cloud proxy typically communicate with the management components, keeping both in the same network can reduce the latency between components.	Using VCF Installer UI cloud proxy is always deployed to VM management network.
VCF-MGMT-DV-NET-RCMD-004	Use the VCF Installer UI to deploy all VCF management components day 0.	• Provides an automated workflow to deploy the components to the dedicated network. • API not required.	None.

VCF Management Dedicated VLAN and NSX Overlay Segment Network Model

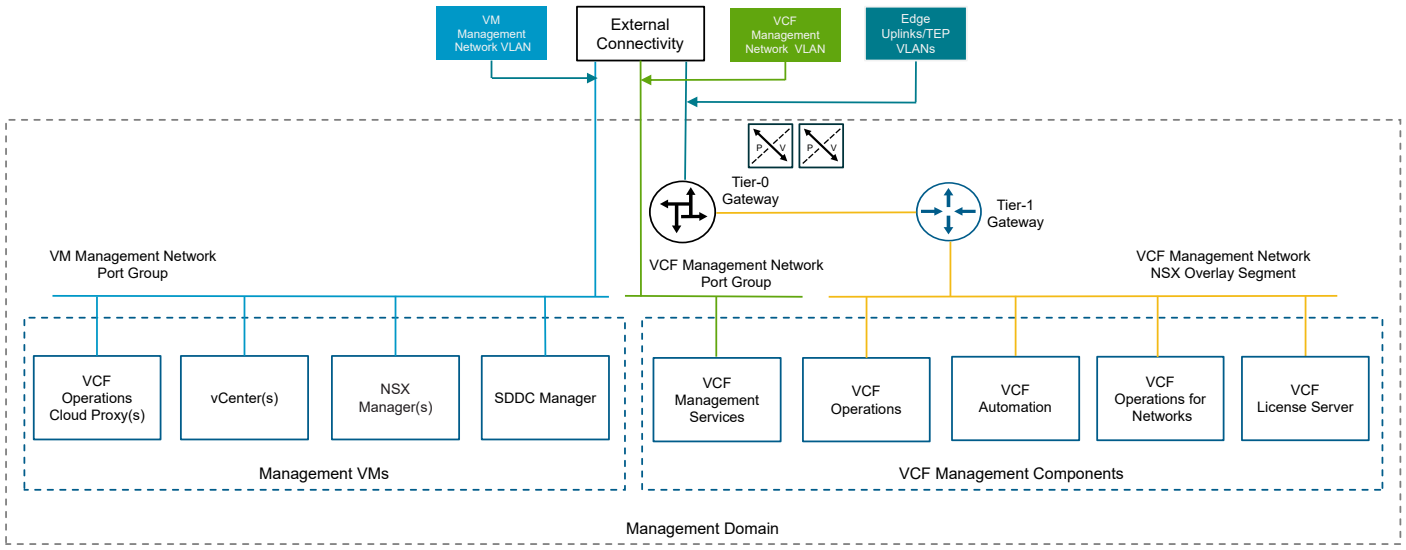
In the *VCF Management Dedicated VLAN and NSX Overlay Segment Network Model*, the VCF management components are deployed to two separate networks, VLAN and NSX overlay segment.

The VCF management services are deployed to a VLAN during initial deployment using VCF Installer. Additional components VCF Operations and VCF Automation are deployed to an NSX overlay segment. This is done as a day 2 operation after VCF Installer has deployed the management domain. An NSX edge cluster with a Tier-0 gateway, Tier-1 gateway, and overlay segment must be created manually in NSX before you deploy the additional VCF management components. Additionally VCF Operations for networks can be deployed to the NSX overlay segment. As the License Server is deployed along with VCF Operations using VCF Installer it will also get deployed to the NSX overlay segment.

Note:

The cloud proxy is a VCF Instance level management component and is always deployed to the **VM management network** when using VCF Installer UI. If cloud proxy must be deployed to another network this can be achieved day2 using VCF Installer API.

Figure 310: VCF Management Dedicated VLAN and NSX Overlay Segment Network Model in VMware Cloud Foundation



VCF Management Dedicated VLAN and NSX Overlay Segment Network Model Attributes

A *VCF Management Single VLAN and NSX Overlay Segment Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	• NSX Overlay segment networking. • VLAN-backed vSphere distributed port group.
Network Isolation	Yes; Isolated using dedicated VLAN and port group and NSX segment.
NSX Edge Cluster Required	Yes; Provides NSX Segment Connectivity and can optionally provide NSX load balancer for VCF Operations.
Stretching Networks Between Availability Zones (Not shown in the diagram)	• NSX segment is stretched using NSX Overlay segment available on hosts in the vSphere stretched cluster connected to the same NSX overlay transport zone. • VCF management services network must be stretched in the Physical Network fabric between availability zones. • Both networks provide automatic network connectivity at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• NSX Federation can be used to stretch the NSX segment used for VCF Operations,VCF Automation, VCF Operations for networks , and the License Server. • For the VCF management services network there are two options: Option 1: Create a separate **VCF management network**in the second region to deploy the new VCF management services instance that is used for restore purposes. – The **VCF management network** in the second region needs to be routable to the first region **VCF management network**. – During the restore process the VIP DNS entries for VCF management services need to be updated to point to the network in the second region. Option 2: Use a single **VCF management network** and stretch the network between regions in which case no DNS updates are needed.

VCF Management Dedicated VLAN and NSX Overlay Segment Network Model Design Requirements and Recommendations

Table 623: Common VCF Management Dedicated VLAN and NSX Overlay Segment Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-DVOVL-NET-REQD-001	Create a dedicated distributed port group on the management vCenter management vSphere Distributed Switch for VCF management services.	Used only for VCF management services connectivity.	Created during VCF Installerdeployment.
VCF-MGMT-DVOVL-NET-REQD-002	Use a dedicated port group on the management vCenter management vSphere Distributed Switch for the VCF management services.	Logically isolatesVCF management services traffic from the management domain VM management traffic.	Connected during VCF Installer deployment.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-DVOVL-NET-REQD-003	Use a dedicated VLAN for VCF management services.	• Logical network isolation can be achieved for the VCF management services traffic. • A physical firewall can be used to secure access to this network. • Traffic originating from services running in VCF management servicesmay be sourced from any one of the IP addresses assigned to the IP Pool for VCFMS, and so any firewall rules or ACLs that might affect that traffic should always include all pool IPs as a source.	• Another VLAN and IP subnet required. • For stretched management cluster, VLAN must be stretched between sites to provide site high availability. • For disaster recovery, this VLAN does not need to be stretched between VCF Instances or sites.
VCF-MGMT-DVOVL-NET-REQD-004	Create an NSX Edge cluster in the management domain.	Provides Tier-0 gateway routing to the physical network.	BGP Routing configuration needed on the physical switches.
VCF-MGMT-DVOVL-NET-REQD-005	Create a Tier-1 gateway and link to the Tier-0 gateway.	Provides connectivity to other networks for the NSX segment used for the VCF Management components.	Manually created in NSX Manager UI.
VCF-MGMT-DVOVL-NET-REQD-006	On the Tier-1 gateway, advertise all connected segments and service ports.	Advertises the segment subnet to the Tier-0 gateway.	Manually created in NSX Manager UI.
VCF-MGMT-DVOVL-NET-REQD-007	On the Tier-0 gateway, advertise all connected segments and service ports.	Advertises the segment subnet to the physical network.	Configured automatically during Centralized Transit Gateway deployment.
VCF-MGMT-DVOVL-NET-REQD-008	Create an NSX overlay segment on management domain overlay transport zone for VCF management components. • VCF Operations • VCF Automation • VCF Operations for networks • License Server	Allows for these critical components to be failed over in a disaster recovery scenario.	Manually created in NSX Manager UI.
VCF-MGMT-DVOVL-NET-REQD-009	Attach segment to management domain overlay transport zone for VCF management components.	• NSX Overlay Segment can be stretched between VCF Instances using NSX Federation. • Provides IP Mobility if the NSX Overlay segment is stretched between sites.	Manually configured in NSX Manager UI.

Table 624: Common VCF Management Dedicated VLAN and NSX Overlay Segment Network Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-DVOVL-NET-RCMD-001	Place the cloud proxy nodes onto the **VM management** network, they are considered VCF Instance level management components, and can reside on the same network as the core management components.	• The cloud proxy nodes typically communicate with the core management components, keeping both on the same network can reduce the latency between components. • If a disaster recovery solution is layered on top only a single Overlay network must be failed over to the recovery site.	If disaster recovery solution is layered on top the cloud proxy nodes are not failed over to a recovery site.
VCF-MGMT-DVOVL-NET-RCMD-002	Use the VCF Installer UI to deploy the VCF management components day 2 using the deferred deployment option.	• Provides the automated workflow to deploy the components to the dedicated networks. • Reduces the complexity of using the SDDC Manager API.	None

VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model

In the *VCF Management Dedicated VLAN and NSX Strethced Overlay Segment Network Model*, the VCF management components are deployed to two separate networks, a VLAN and an NSX overlay stretched segment between two regions to provide fleet disaster recovery.

The VCF management services are deployed to a VLAN during initial deployment using VCF Installer. Additional components VCF Operations and VCF Automation are deployed to an NSX overlay segment and the segment is then stretched between sites using NSX Federation. This is done as a day 2 operation after VCF Installer has deployed the management domain. An NSX edge cluster with a Tier-0 gateway, Tier-1 gateway, and overlay segment must be created manually in NSX before you deploy the VCF management components. The NSX stretched overlay segment will provide IP mobility for VCF Operations in the case of a failover from one region to another. Additionally VCF Operations for networks can be deployed to the NSX stretched overlay segment as it also requires IP mobility.

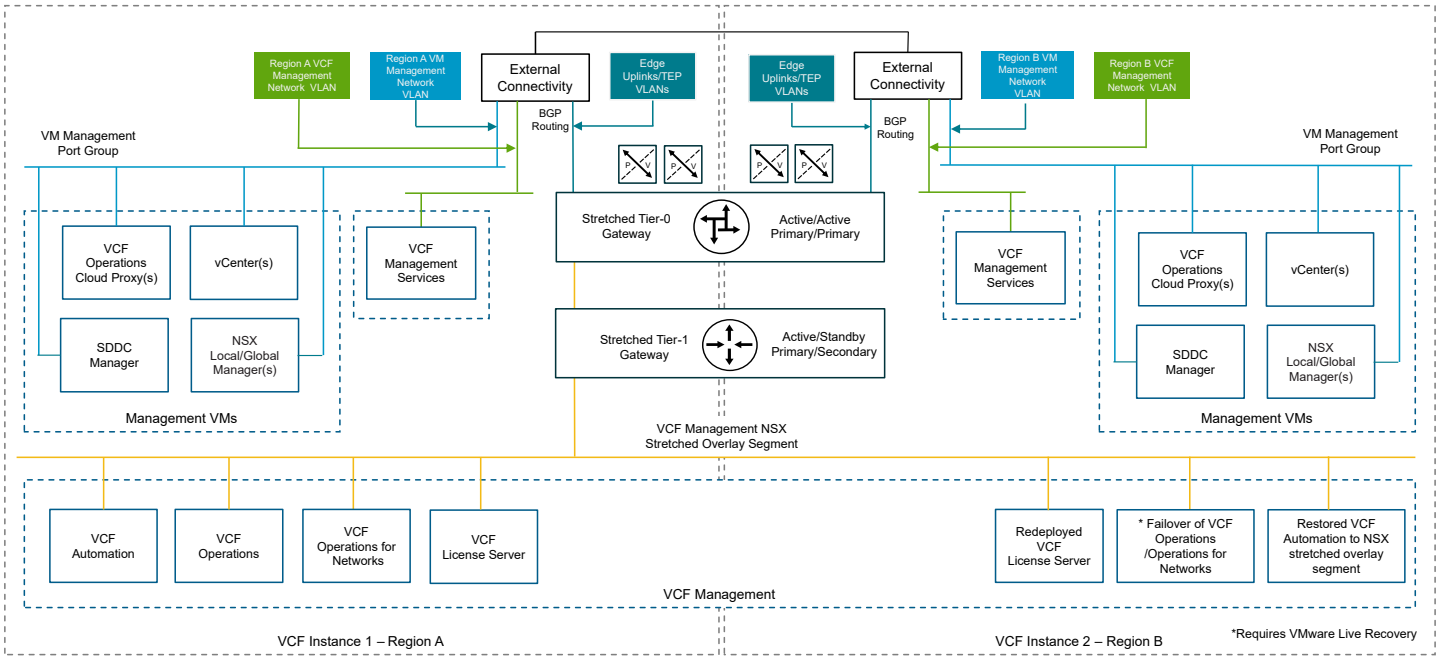
Note:

VCF Automation does not require IP mobility when restored to a second region. But as part of the restore procedure if the same NSX stretched overlay segment is used, no DNS repointing is needed for the VIPs during the restore process.

Note:

The cloud proxy is a VCF Instance level management component and is always deployed to the **VM management network** when using VCF Installer UI. If cloud proxy must be deployed to another network this can be achieved day2 using VCF Installer API.

Figure 311: VCF Management Dedicated VLAN and NSX Stretched Overlay Network Model in VMware Cloud Foundation



VCF Management Dedicated VLAN and NSX Stretched Overlay Network Model Attributes

A *VCF Management Dedicated VLAN and NSX Stretched Overlay Network Model* has the following attributes.

Attribute	Detail
Connectivity Type	• NSX Overlay segment networking. • VLAN networking.
Network Isolation	Yes, dedicated NSX segment.
NSX Edge Cluster Required	Yes; Provides NSX Segment Connectivity and can optionally provide NSX load balancer for VCF Operations.
Stretching Network Between Availability Zones (Not shown in the diagram)	• NSX segment is stretched using NSX Overlay segment available on hosts in the vSphere stretched cluster connected to the same NSX overlay transport zone. • VCF management services network must be stretched in the Physical Network fabric between availability zones. • Allows HA restart of all components at secondary availability zone if a failure occurs at the primary availability zone.
Multi Region Network Configuration	• NSX Federation can be used to stretch the NSX segment used for VCF Operations,VCF Automation, VCF Operations for networks , and the License Server. • For the VCF management services network there are two options: Option 1: Create a separate **VCF management network**in the second region to deploy the new VCF management services instance that is used for restore purposes. — The **VCF management network** in the second region needs to be routable to the first region **VCF management network**. — During the restore process the VIP DNS entries for VCF management services need to be updated to point to the network in the second region. Option 2: Use a single **VCF management network** and stretch the network between regions in which case no DNS updates are needed.

VCF Management Dedicated VLAN and NSX Stretched Overlay Network Model Design Requirements and Recommendations

Table 625: Common VCF Management Dedicated VLAN and NSX Overlay Segment Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-DVOVL-NET-REQD-001	Create a dedicated distributed port group on the management vCenter management vSphere Distributed Switch for VCF management services.	Used only for VCF management services connectivity.	Created during VCF Installerdeployment.
VCF-MGMT-DVOVL-NET-REQD-002	Use a dedicated port group on the management vCenter management vSphere Distributed Switch for the VCF management services.	Logically isolatesVCF management services traffic from the management domain VM management traffic.	Connected during VCF Installer deployment.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-DVOVL-NET-REQD-003	Use a dedicated VLAN for VCF management services.	• Logical network isolation can be achieved for the VCF management services traffic. • A physical firewall can be used to secure access to this network. • Traffic originating from services running in VCF management servicesmay be sourced from any one of the IP addresses assigned to the IP Pool for VCFMS, and so any firewall rules or ACLs that might affect that traffic should always include all pool IPs as a source.	• Another VLAN and IP subnet required. • For stretched management cluster, VLAN must be stretched between sites to provide site high availability. • For disaster recovery, this VLAN does not need to be stretched between VCF Instances or sites.
VCF-MGMT-DVOVL-NET-REQD-004	Create an NSX Edge cluster in the management domain.	Provides Tier-0 gateway routing to the physical network.	BGP Routing configuration needed on the physical switches.
VCF-MGMT-DVOVL-NET-REQD-005	Create a Tier-1 gateway and link to the Tier-0 gateway.	Provides connectivity to other networks for the NSX segment used for the VCF Management components.	Manually created in NSX Manager UI.
VCF-MGMT-DVOVL-NET-REQD-006	On the Tier-1 gateway, advertise all connected segments and service ports.	Advertises the segment subnet to the Tier-0 gateway.	Manually created in NSX Manager UI.
VCF-MGMT-DVOVL-NET-REQD-007	On the Tier-0 gateway, advertise all connected segments and service ports.	Advertises the segment subnet to the physical network.	Configured automatically during Centralized Transit Gateway deployment.
VCF-MGMT-DVOVL-NET-REQD-008	Create an NSX overlay segment on management domain overlay transport zone for VCF management components. • VCF Operations • VCF Automation • VCF Operations for networks • License Server	Allows for these critical components to be failed over in a disaster recovery scenario.	Manually created in NSX Manager UI.
VCF-MGMT-DVOVL-NET-REQD-009	Attach segment to management domain overlay transport zone for VCF management components.	• NSX Overlay Segment can be stretched between VCF Instances using NSX Federation. • Provides IP Mobility if the NSX Overlay segment is stretched between sites.	Manually configured in NSX Manager UI.

Table 626: VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-DVSOVL-NET-REQD-001	Allocate a separate VLAN for edge RTEP overlay that is different from the NSX Edge TEP VLAN.	The Edge TEP and RTEP networks must use separate subnets/VLANs to provide routing and traffic isolation for intra and inter data-centre traffic. It also provides support for configuring different MTU size per network.	You must allocate another VLAN in the data center infrastructure.
VCF-MGMT-DVSOVL-NET-REQD-002	Extend the Tier-0 gateway to the NSX instance used for the management domain in the second VCF Instance.	• Supports ECMP north-south routing on all nodes in the NSX Edge cluster. • Enables support for cross-instance Tier-1 gateways and cross-instance network segments.	Manually done through NSX Global Manager UI.
VCF-MGMT-DVSOVL-NET-REQD-003	Set the Tier-0 gateway as primary for both NSX Instances in both VCF Instances.	• In NSX Federation, a Tier-0 gateway lets egress traffic from connected Tier-1 gateways only in its primary locations. • Local ingress and egress traffic is controlled independently at the Tier-1 level. No segments are provisioned directly to the Tier-0 gateway. • A mixture of network spans (local to a VCF Instance or spanning multiple instances) is enabled without requiring additional Tier-0 gateways and hence NSX Edge nodes. • If an NSX failure in a VCF Instance occurs, the local-instance networking in the other instance remains available without manual intervention.	Manually done through NSX Global Manager UI.
VCF-MGMT-DVSOVL-NET-REQD-004	For the global Tier-0 gateway, establish BGP neighbor peering to the ToR switches connected to the second NSX Instance in management domain of the second VCF Instance.	• Enables the learning and advertising of routes in the second NSX Instance in the management domain of the second VCF Instance. • Facilitates a potential automated failover of networks from the first to the second VCF Instance.	Manually done through NSX Global Manager UI at the second location.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-MGMT-DVSOVL-NET-REQD-005	Use a stretched Tier-1 gateway and connect it to the Tier-0 gateway for cross-instance Fleet level networking.	• NSX network segments follow the span of the gateway that they are attached to. • Enables network span between the NSX Instance in the management domain of both VCF Instances because • Creates a two-tier routing architecture.	Manually done through NSX Global Manager UI.
VCF-MGMT-DVSOVL-NET-REQD-006	Assign the NSX Edge cluster in each NSX Instance in the management domain of each VCF Instance to the stretched Tier-1 gateway. Set the first NSX Instance as primary and the second instance as secondary.	• Enables cross-instance network span between the first and second NSX Instances at each location. • Enables deterministic ingress and egress traffic for the cross-instance network. • If a VCF Instance failure occurs, enables deterministic failover of the Tier-1 traffic flow. • During the recovery of the inaccessible VCF Instance, enables deterministic failback of the Tier-1 traffic flow, preventing unintended asymmetrical routing. • Eliminates the need to use BGP attributes in the first and second VCF Instances to influence location preference and failover.	You must manually fail over and fail back the cross-instance network from the standby NSX Global Manager.
VCF-MGMT-DVSOVL-NET-REQD-007	Extend the stretched NSX overlay segment used to deploy the Fleet level management components to the second VCF Instance.	• Enables IP mobility without a complex physical network configuration. • The VCF fleet level management network must be available at both VCF Instances.	Manually done through NSX Global Manager UI.

Table 627: Common VCF Management Dedicated VLAN and NSX Overlay Segment Network Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-DVOVL-NET-RCMD-001	Place the cloud proxy nodes onto the **VM management** network, they are considered VCF Instance level management components, and can reside on the same network as the core management components.	• The cloud proxy nodes typically communicate with the core management components, keeping both on the same network can reduce the latency between components. • If a disaster recovery solution is layered on top only a single Overlay network must be failed over to the recovery site.	If disaster recovery solution is layered on top the cloud proxy nodes are not failed over to a recovery site.
VCF-MGMT-DVOVL-NET-RCMD-002	Use the VCF Installer UI to deploy the VCF management components day 2 using the deferred deployment option.	• Provides the automated workflow to deploy the components to the dedicated networks. • Reduces the complexity of using the SDDC Manager API.	None

Table 628: VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-DVSOVL-NET-RCMD-001	Use Tier-1 gateway to control the span of VCF management network and ingress and egress traffic in the VCF Instances.	Enables spanning multiple VCF Instances without requiring additional Tier-0 gateways and hence NSX Edge nodes.	To control location span, a Tier-1 gateway must be assigned to an NSX Edge cluster and hence has the Tier-1 SR component.

VCF Operations Detailed Design

VCF Operations is a mandatory component for each VCF fleet and is the central console for operating your VCF fleet, providing dashboards for monitoring, log analysis, reporting, and licensing. With VCF Operations offers fleet management capabilities such as lifecycle management, certificate, and password management. This section describes the options, requirements, and recommendations for each *VCF Operations Model*.

The initial management domain hosts essential components, including SDDC Manager, vCenter, NSX Manager, VCF Operations, and VCF Automation. These components work together to ensure smooth operations and orchestration across the VCF fleet.

This streamlined and scalable approach provides centralized management, allowing a unified framework to oversee VCF Instances through a single VCF Operations instance delivering a strong foundation to expand to multiple VCF Instances while maintaining centralized operations and automation.

To enable more capabilities, VCF Operations can be further extended through the deployment of log management, real-time metrics, andVMware Cloud Foundation Operations for networks. These components support their own deployment models, this section describes the options, requirements, and recommendations for each.

- Simple VCF Operations Model
- High Availability VCF Operations Model
- Continuous Availability VCF Operations Model

VCF Operations Placement

VCF Operations placement for an entire VCF fleet plays a critical role in monitoring, analyzing, licensing, life-cycle management, optimizing the performance, and capacity of the VMware Cloud Foundation platform. Proper placement of the VCF Operations components is essential to ensure high availability, scalability, and efficient resource utilization.

Attribute	Details
Resource Allocation	Ensure that sufficient compute, memory, and storage resources are allocated to the VCF Operations appliances to handle the expected workload. This is particularly important in environments with large-scale deployments or high data ingestion rates.
Network Connectivity	VCF Operations should be placed in a network segment with low latency and high bandwidth to ensure efficient communication with vCenter, NSX, and other monitored components.
Scalability	Plan for future growth by deploying VCF Operations in a manner that allows for easy scaling as the VCF environment expands. This includes considering the placement of additional nodes if required. By carefully planning the placement of VCF Operations, organizations can ensure optimal performance, reliability, and scalability of their VCF environment while gaining actionable insights to improve operations.

VCF Operations Network Placement

Model	Details	Benefits	Implications
Default VM management distributed port group	VCF Operations components are deployed on the same network as the default VM management network or the ESX management network if configured with the same VLAN.	• Simplified deployment with no additional network configuration required. • Immediate integration with the existing management infrastructure. • Suitable for smaller environments or those with limited network segmentation requirements.	• Shared traffic between management and VCF Operations components, potentially impacting performance in larger environments. • Limited isolation for VCF Operations traffic, which may not meet security or compliance requirements in some organizations.
Dedicated distributed port group	VCF Operations components are deployed on a dedicated distributed port group.	• Enhanced security and traffic isolation for VCF Operations components • Improved performance by separating VCF Operations traffic from other management traffic. • Greater flexibility for scaling and integrating with advanced network architectures.	• Automatically done by VCF Installer.
NSX segment	VCF Operations components are deployed to an NSX segment.	• Enhanced security and traffic isolation for VCF Operations components • Improved performance by separating VCF Operations traffic from other management traffic. • Greater flexibility for scaling and integrating with advanced network architectures.	• Requires additional planning and configuration during deployment. • VCF Operations deployment must be skipped during the initial deployment (Day 0) and configured as a Day 2 activity after the network is set up.

VCF Operations Licensing Modes

VCF Operations uses the VMware Cloud Foundation Business Services console to license VCF Instances and report license usage. It is recommended that you use the *Connected Mode* whenever possible to simplify license management.

Mode	Details	Benefits	Implications
Connected Mode	VCF Operations maintain a live connection to the VCF Business Services console via the internet and regularly reports license usage.	Reduced operational overhead of manual license management operations.	An outbound internet connection to VCF Business Services console is required for Connected Mode.
Disconnected Mode	Offline registration of VCF Operations with the VCF Business Services console and license usage reporting.	Internet connection is not required for VCF Operations instances.	Operational overhead of manual license management tasks, such as license usage reporting.

Table 629: Deployment Path

There are two distinctive paths for VCF Operations components deployment - during or after the deployment of the management domain with VCF Installer

Refer to the VCF Operations design decisions for the justifications and implications of each deployment path.

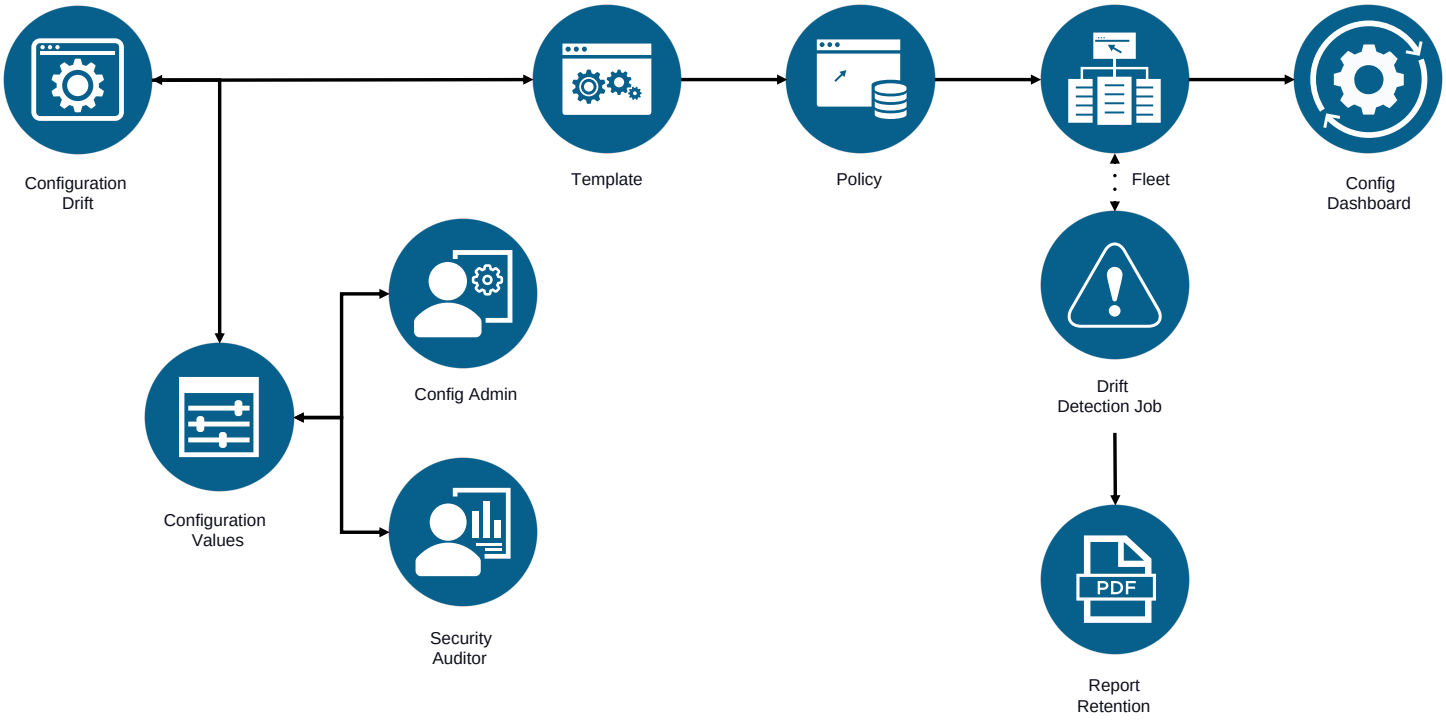
Component Name	Deployment Path
VCF Operation	Must be deployed either: • During the installation of the management domain with VCF Installer. **Important:** All components will be deployed with the management domain. • After the installation of the management domain as a separate workflow. **Important:** VCF Installer should be run second time to install deferred components.
Cloud Proxy	Mandatory component: • During the installation of the management domain with VCF Installer. **Important:** Will be deployed with the management domain. • After the installation of the management domain as a separate workflow. **Important:** VCF Installer should be run second time to install deferred components. • With each new VCF Instance that will be part of the existing VCF fleet. **Important:** Will be deployed in the management domain of each VCF Instance.
Fleet lifecycle(running as a component intoVCF management services)	Must be deployed either: • During the installation of the management domain with VCF Installer. **Important:** Will be deployed with the management domain. • After the installation of the management domain as a separate workflow. **Important:** VCF Installer should be run second time to install deferred components.

Configuration Drift

Using a desired state approach, fleet management can be managed through a configuration drift capability for vCenter and vSphere Clusters. This approach is based on a template style management. A single template can be applied against each component, or multiple templates without overlapping configuration values can be applied to each component.
Due to the range of possible configuration drift values and their impact on security, operations should invite security and compliance teams to participate in the process. Balancing the need to manage operations with the need for appropriate security can alter the approach taken.

Figure 312: Configuration Drift

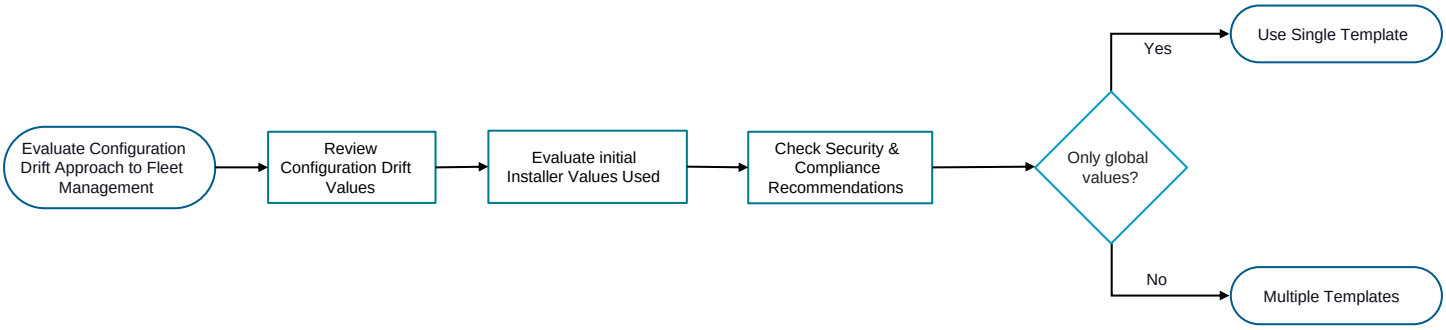
End to end configuration drift components assembled to support fleet management.


Configuration Drift
Template
Policy
Fleet
Config Dashboard
Config Admin
Drift Detection Job
Configuration Values
PDF
Security Auditor
Report Retention

The process for evaluating and confirming values should be balanced with the use of one, or more templates. There are some use cases where global values and a single template to enforce them is possible, while in other use cases this requires a different approach. The range of configuration drift values and how the VCF instance is used should select the best approach to fleet management.

Figure 313: Configuration Drift Values and Templates

Process for evaluating configuration drift values and determining the template structure to use.

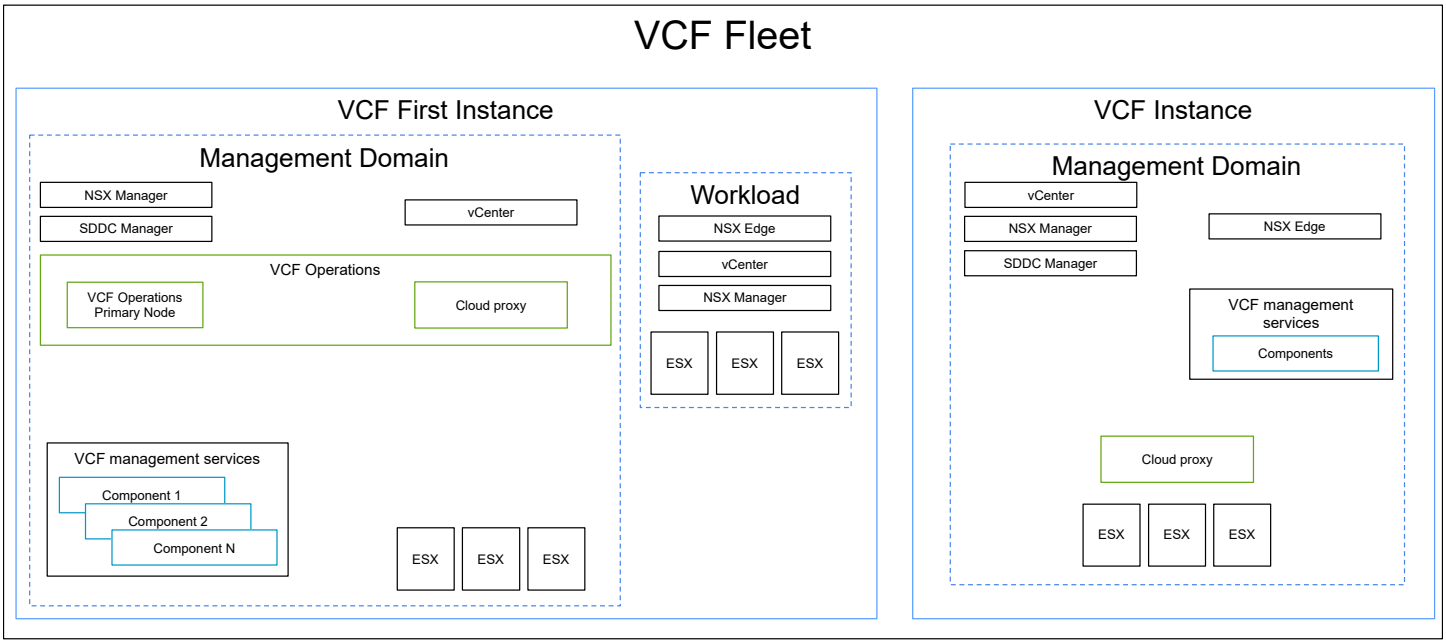

Evaluate Configuration Drift Approach to Fleet Management
Review Configuration Drift Values
Evaluate initial Installer Values Used
Check Security & Compliance Recommendations
Only global values?
Yes
Use Single Template
No
Multiple Templates

Simple VCF Operations Model

The *Simple VCF Operations Model* deploys a single node of all the VCF Operations components and uses vSphere HA as an availability mechanism.

The *Simple VCF Operations Model* includes three single-node components.

Figure 314: Simple VCF Operations Model in VMware Cloud Foundation



Simple VCF Operations Model Attributes

A *Simple VCF Operations Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out. **Note:** Scale-out is supported for the VCF Operations data nodes and cloud proxy nodes.
Recoverability	Restore through a backup.

Simple VCF Operations Model Options

A *Simple VCF Operations Model* can leverage options from the design areas that are listed below, depending on objectives.

Design Area	Options
Infrastructure IP Address Model	• IPv4 Addressing Model • Dual Stack IP Addressing Model
VCF Management Network Model	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model • VCF Management Dedicated VLAN and NSX Overlay Segment Network Model • VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model
License Server Model	• Single License Server Model • Multiple License Server Model
Licensing Mode	• Connected Mode • Disconnected Mode

Simple VCF Operations Model Design Requirements

Table 630: Simple VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-SIM-001	Deploy VCF Operations as a single node in the default management vSphere cluster.	• Supports the simple deployment objective. • Provides the capacity that is required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out to a high availability model.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • No application level high availability.

Table 631: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

Simple VCF Operations Model Design Recommendations

Table 632: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

VCF Operations - Networking and Internet Access Design Choices

There are several options regarding connectivity to the external resources and network placement for the VCF Operations components. You should choose one option from each of the tables below.

Table 633: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on the default distributed port group created by VCF Installer and reserved for management components.	Ensures that the VCF Operations nodes are deployed adjacent to the other components on the same Layer-2 network.	None.

Table 634: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on a dedicated distributed port group **other** than default port group reserved for management components by VCF Installer.	Ensures that the VCF Operations nodes and VCF management services are deployed in a dedicated Layer-2 network for easier restore in a different VCF Instance.	• The VCF Operations and VCF management serviceswill be installed in a dedicated distributed port group. • All the configurations are done automatically by the VCF Installer. • You must ensure that VLAN is provisioned in each availability zone if stretched clusters are used.

Table 635:

Table 636: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on an NSX segment.	Allow portability of the VCF Operations nodes and VCF management services to another VCF Instance in the VCF fleet without the need for reconfiguration of the IP addressing.	• You must skip the deployment of the VCF Operations and VCF management servicesin the VCF Installer. • You must perform configuration to support the networking configuration as a post-deployment task before deploying the VCF Operations andVCF management services. • You must perform deployment using the VCF Installer option to install deferred components.

Table 637: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 638: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

Table 639: VCF Operations Diagnostics Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-DIAG-001	Manually upload Skyline Health Diagnostics and Advisory signatures.	If your policy does not permit access to internet for management components.	Signatures should be downloaded manually and then uploaded into VCF Operations.

Table 640: VCF Operations Diagnostics Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-DIAG-001	Download Skyline Health Diagnostics and Advisory signatures directly from Broadcom's repository.	Ensures always up to date information about potential issues and mitigation steps in your environment.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 641: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Connect to the online depot.	Ensures always up to date information about bundles for Install, Update, Patch and compatibility information.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 642: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Connect to an offline depot.	If your policy does not permit access to internet for management components you may configure an internal depot.	The bundles should be downloaded to a system with access to the Broadcom's depot and then uploaded to the internal depot by following your internal procedures.

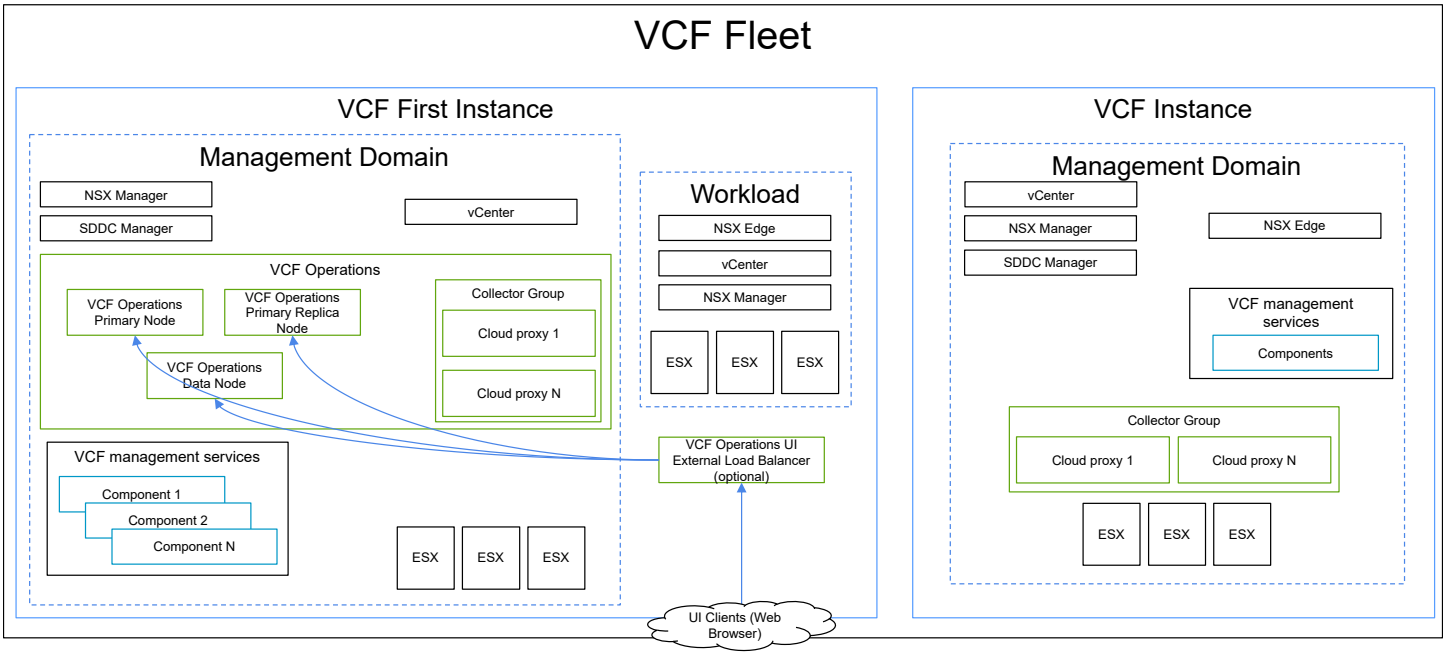
Table 643: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Upload files manually.	Removes the overhead of installing and managing an additional appliance for internal depot.	You should download and upload all the files to the respective storage. A lot of manual labor and error prone process.

High Availability VCF Operations Model

The *High Availability VCF Operations Model* deploys a three node cluster for the analytic engine, and single node cloud proxy appliance. Fleet lifecycle is deployed as a single instance component in the VCF management services. To achieve high availability after the initial deployment, additional cloud proxy nodes can be deployed and then combined in collector groups.

Figure 315: High Availability VCF Operations Model in VMware Cloud Foundation



High Availability VCF Operations Model Attributes

A *High Availability VCF Operations Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure. Additional level of availability for the components in: • VCF Operations cluster • VCF Operations collector groups
Scalability	Supports scale-up and scale-out. **Note:** Scale-out is supported for the VCF Operations data nodes and cloud proxynodes.
Recovery	Restore through a backup.
Load balancing	Optional external load balancer for the VCF Operations cluster UI.

High Availability VCF Operations Model Options

A *High Availability VCF Operations Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Infrastructure IP Address Model	• IPv4 Addressing Model • Dual Stack IP Addressing Model
VCF Management Network Model	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model • VCF Management Dedicated VLAN and NSX Overlay Segment Network Model • VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model
License Server Model	• Single License Server Model • Multiple License Server Model
Licensing Mode	• Connected Mode • Disconnected Mode

High Availability VCF Operations Model Design Requirements

Table 644: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

Table 645: High Availability VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-001	Deploy VCF Operations as a cluster of three nodes in the default management vSphere cluster.	• Supports the high availability objectives by deploying VCF Operations as a cluster. • Provides the capacity that is required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out with additional data nodes.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • Create an anti affinity rule (Separate Virtual Machines) for the VMs in the cluster after the deployment.
VCF-OPS-REQD-HA-002	Deploy a single Unified VCF Operations collector node in the default management vSphere cluster.	Removes the load from the VCF Operations cluster from collecting metrics from local-instance applications.	You may need to scale-up the VCF Operations collector nodes to a greater size based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-HA-003	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations cluster nodes.	Using vSphere DRS prevents the VCF Operations cluster nodes from running on the same ESX host and risking the high availability of the cluster.	• You must perform an additional configuration to set up an anti-affinity rule. • You must have a minimum of four ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode. • If additional data nodes are added, you must update the anti-affinity rule and ensure there are enough ESX hosts in the default management vSphere cluster to ensure you can put an ESX host in maintenance mode.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-HA-004	The certificate for VCF Operations must include the FQDNs for all cluster nodes in the SAN (subject alternative name) attribute. **Note:** The SAN does not need to include the FQDN of VCF Operations collector nodes.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted.	Before scale out, the certificate for VCF Operations must be changed to also include the FQDN of the additional node in the SAN (subject alternative name) attribute.
VCF-OPS-REQD-HA-005	If using an optional external load balancer in the high availability deployment model, the certificate for VCF Operations **must** also include the FQDN of the load balancer in the SAN (subject alternative name) attribute.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted when using an external load balancer with the high availability model.	None.

Table 646: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

VCF Operations - Networking and Internet Access Design Choices

There are several options regarding connectivity to the external resources and network placement for the VCF Operations components. You should choose one option from each of the tables below.

Table 647: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on the default distributed port group created by VCF Installer and reserved for management components.	Ensures that the VCF Operations nodes are deployed adjacent to the other components on the same Layer-2 network.	None.

Table 648: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on a dedicated distributed port group **other** than default port group reserved for management components by VCF Installer.	Ensures that the VCF Operations nodes and VCF management services are deployed in a dedicated Layer-2 network for easier restore in a different VCF Instance.	• The VCF Operations and VCF management serviceswill be installed in a dedicated distributed port group. • All the configurations are done automatically by the VCF Installer. • You must ensure that VLAN is provisioned in each availability zone if stretched clusters are used.

Table 649:

Table 650: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on an NSX segment.	Allow portability of the VCF Operations nodes and VCF management services to another VCF Instance in the VCF fleet without the need for reconfiguration of the IP addressing.	• You must skip the deployment of the VCF Operations and VCF management servicesin the VCF Installer. • You must perform configuration to support the networking configuration as a post-deployment task before deploying the VCF Operations andVCF management services. • You must perform deployment using the VCF Installer option to install deferred components.

Table 651: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 652: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

Table 653: VCF Operations Diagnostics Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-DIAG-001	Manually upload Skyline Health Diagnostics and Advisory signatures.	If your policy does not permit access to internet for management components.	Signatures should be downloaded manually and then uploaded into VCF Operations.

Table 654: VCF Operations Diagnostics Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-DIAG-001	Download Skyline Health Diagnostics and Advisory signatures directly from Broadcom's repository.	Ensures always up to date information about potential issues and mitigation steps in your environment.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 655: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Connect to the online depot.	Ensures always up to date information about bundles for Install, Update, Patch and compatibility information.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 656: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Connect to an offline depot.	If your policy does not permit access to internet for management components you may configure an internal depot.	The bundles should be downloaded to a system with access to the Broadcom's depot and then uploaded to the internal depot by following your internal procedures.

Table 657: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Upload files manually.	Removes the overhead of installing and managing an additional appliance for internal depot.	You should download and upload all the files to the respective storage. A lot of manual labor and error prone process.

VCF Operations High Availability Deployment Model Design Recommendations

In your VCF Operations design, you can apply certain best practices as part of post deployment workflows.

Table 658: High Availability VCF Operations Model Design Recommendations

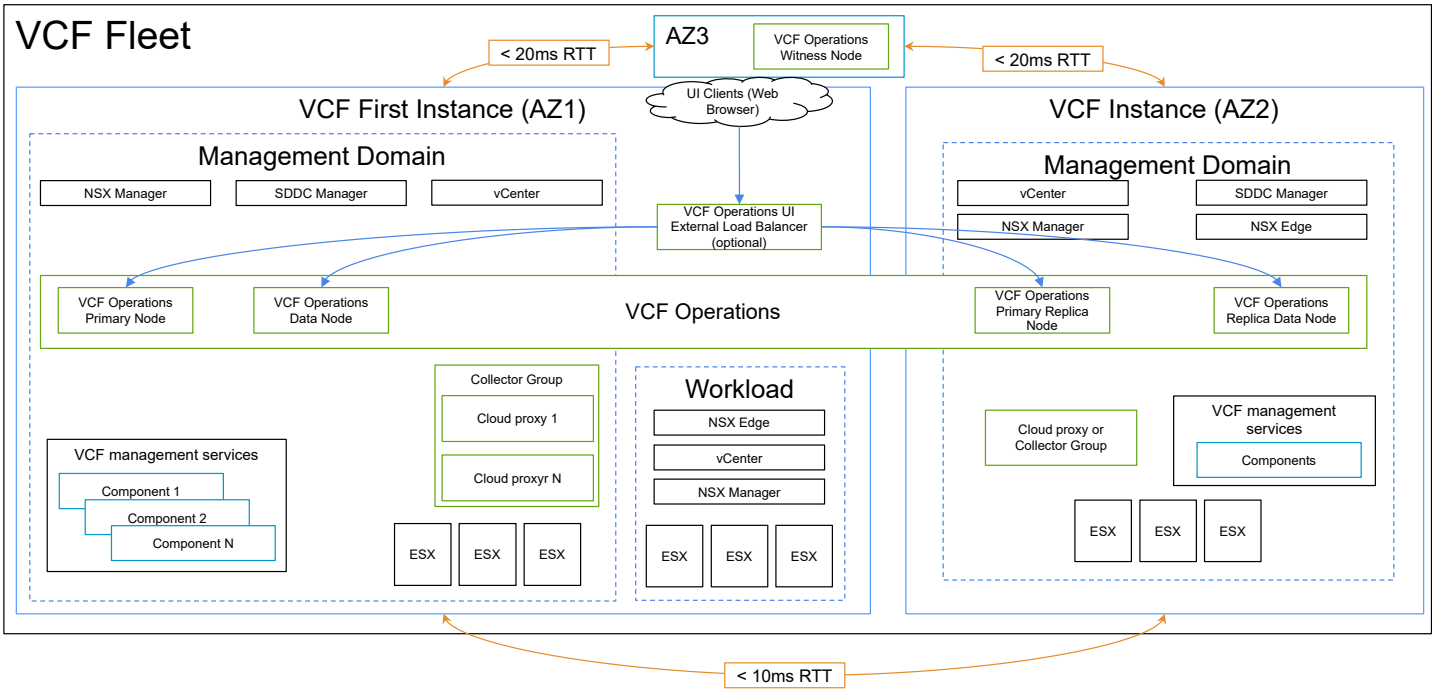
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-HA-001	Deploy VCF Operations with an external load balancer to load balance the connections across the VCF Operations cluster nodes.	Required to deploy a VCF Operations cluster deployment with distributed user interface access across members.	You must configure the external load balancer configuration to support this VCF Operations configuration.
VCF-OPS-RCMD-HA-002	Create a collector group in VCF Operations for each VCF Instance and assign the default VCF Operations collector appliance.	Provides increased availability to metric collection across the VCF Operations collector appliances that are assigned to the collector group.	Manual reconfiguration of the VMware Cloud Foundation adapter required to utilize multiple VCF Operations collector appliances with the group.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-HA-003	Deploy an additional VCF Operations collector node in the default management vSphere cluster.	Extends the availability of the VCF Operations collectors by balancing the load across the collectors when added to a collector group.	• You must deploy the collector node after the deployment of the VCF Operations. • Additional collector node should be deployed for all VCF Instances that requires a collector group.
VCF-OPS-RCMD-HA-004	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations collector nodes.	Using vSphere DRS prevents the VCF Operations collectors from running on the same ESX host and risking the high availability of the collectors.	You must perform manual configuration to set up an anti-affinity rule.
VCF-OPS-RCMD-HA-005	Reconfigure the default VMware Cloud Foundation Adapter in VCF Operations to use a local-instance collector group for a VCF Instance.	Increases the availability of metric collection.	Must be reconfigured post-deployment.

Continuous Availability VCF Operations Model

The *Continuous Availability VCF Operations Model* separates the VCF Operations cluster into two fault domains, stretching across two vSphere instances in two separate availability zones, and protects the analytics cluster against the loss of an entire fault domain.

Figure 316: Continuous Availability VCF Operations Model in VMware Cloud Foundation



VCF Operations uses a robust architecture to ensure data availability and fault tolerance across analytics nodes. Data is stored in pairs of analytics nodes, distributed across two fault domains, ensuring continuous availability and synchronization. Each object is stored in two nodes (primary and replica) across fault domains, providing redundancy and resilience in case of failures.

A fault domain consists of one or more analytics nodes, grouped according to their physical location in the data center. The two fault domains enable VCF Operations to tolerate failure of an entire physical location and failures from resources dedicated to a single fault domain. This design requires a VCF Operations witness in a third site.

To deploy the VCF Operations with continuous availability, you first deploy the initial VCF Operations instance by using the Simple VCF Operations Model or the High Availability VCF Operations Model. You then create a second VCF Instance in a separate site/availability zone where you deploy the VCF Operations witness.

Continuous Availability VCF Operations Model Attributes

A Continuous Availability VCF Operations Model has the following attributes.

Attribute	Detail
Availability	Implements design with two VCF instances, across two availability zones to protect data from an availability zone failure. Inside the availability zone, relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale out. **Note:** Scale-out is supported for the VCF Operations data nodes and VCF Operations collector nodes.
Recovery	Redeploy of a component in one of the availability zones in case of availability zone failure. Restore through a backup for the whole deployment, in case of both availability zones failure.
Load balancing	Optional external load balancer for the VCF Operations cluster UI.

Continuous Availability VCF Operations Model Options

A Continuous Availability VCF Operations Model can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Infrastructure IP Address Model	• IPv4 Addressing Model • Dual Stack IP Addressing Model
VCF Management Network Model	• VCF Management Shared VLAN Network Model • VCF Management Dedicated VLAN Network Model • VCF Management Dedicated VLAN and NSX Overlay Segment Network Model • VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model
License Server Model	• Single License Server Model • Multiple License Server Model
Licensing Mode	• Connected Mode • Disconnected Mode

Continuous Availability VCF Operations Model Design Requirements

Table 659: Common VCF Operations Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-COM-001	Deploy the fleet management component in the VCF management services in default management cluster.	Provides lifecycle management operations for a VCF fleet.	VCF management services must be installed.
VCF-OPS-REQD-COM-002	Protect through the fleet management VCF management services configuration.	Ensures availability of fleet management appliance by inheriting the VCF management servicesprotection, without requiring manual intervention during an ESX host failure event.	The VCF management services availability configuration directly translates to the fleet management.
VCF-OPS-REQD-COM-003	Deploy a single cloud proxy appliance in the default management vSphere cluster.	• Removes the load from the VCF Operations analytic node(s) from collecting metrics from local-instance applications. • Could reuse cloud proxy to gather both logs and metrics across the VMware Cloud Foundation platform. • Centralized management of logs and metrics, reducing administrative overhead.	You may need to scale-up the cloud proxy nodes to a greater size based on the known or forecast virtual machine and object count across each connected VCF Instance to adhere to the scale limits.
VCF-OPS-REQD-COM-004	Protect cloud proxy appliance by using vSphere HA.	Ensures availability of cloud proxy appliance without requiring manual intervention during an ESX host failure event.	None.

Table 660: Continuous Availability VCF Operations Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-CA-001	Deploy VCF Operations in the default management vSphere cluster.	• Supports the high availability objectives by deploying VCF Operations as a cluster. • Provides the capacity required for monitoring virtual machines or objects within the monitoring scale limits. • Supports scale-out with additional data nodes.	• You must select a size for the VCF Operations cluster nodes based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits. • Create an anti affinity rule (Separate Virtual Machines) for the VMs in the cluster after the deployment.
VCF-OPS-REQD-CA-002	Deploy a single Unified VCF Operations collector node in the default management vSphere cluster.	Removes the load from the VCF Operations cluster from collecting metrics from local-instance applications.	You may need to scale-up the VCF Operations collector nodes to a greater size based on the known or forecasted virtual machine and object count across each connected VCF Instance to adhere to the scale limits.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-OPS-REQD-CA-003	The certificate for VCF Operations must include the FQDNs for all cluster nodes in the SAN (subject alternative name) attribute. **Note:** The SAN does not need to include the FQDN of VCF Operations collector nodes.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted.	Before scale out, the certificate for VCF Operations must be changed to also include the FQDN of the additional node in the SAN (subject alternative name) attribute.
VCF-OPS-REQD-CA-004	Deploy a second VCF Instance in a separate availability zone.	A second instance is needed to provide a separate SDDC Manager and fully independent infrastructure.	
VCF-OPS-REQD-CA-005	Satisfy latency and networking requirements for different components.	VCF Operations Continuous Availability requires: • 10 ms with peaks up to 20 ms during 20-sec intervals between the cluster nodes. • 30 ms between the witness and the cluster nodes. • L2 or L3 Connectivity between the VCF Operations nodes in different availability zones. • Communication between the different VCF Operations nodes should be allowed through corporate firewall, based on the Ports and Protocols documentation.	Availability zones should be close and connected with enterprise grade connection.
VCF-OPS-REQD-CA-006	Third location (Separate Availability zone) for witness node deployment.	The witness node ensures cluster health and determines which fault domain remains online during communication failures.	
VCF-OPS-REQD-CA-007	Reconfigure a primary replica cluster node to a data node if VCF Operations was deployed with High Availability.	During the configuration of VCF Operations Continuous Availability, we deploy a replication partner of the primary node in the second availability zone.	
VCF-OPS-REQD-CA-008	Deploy through vSphere: • Replication partners to the nodes in the primary AZ in the secondary AZ. • Witness node in the designated third location.	Replica nodes for all components should be deployed from a template and later configured in VCF Operations.	Manual workflow spanning across vSphere and VCF Operations.
VCF-OPS-REQD-CA-009	If using an optional external load balancer, the certificate for VCF Operations **must** also include the FQDN of the load balancer in the SAN (subject alternative name) attribute.	Ensures that the communication to the VCF Operations UI and API, and cross-component, is encrypted when using an external load balancer with the high availability model.	None.

VCF Operations - Networking and Internet Access Design Choices

There are several options regarding connectivity to the external resources and network placement for the VCF Operations components. You should choose one option from each of the tables below.

Table 661: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on the default distributed port group created by VCF Installer and reserved for management components.	Ensures that the VCF Operations nodes are deployed adjacent to the other components on the same Layer-2 network.	None.

Table 662: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on a dedicated distributed port group **other** than default port group reserved for management components by VCF Installer.	Ensures that the VCF Operations nodes and VCF management services are deployed in a dedicated Layer-2 network for easier restore in a different VCF Instance.	• The VCF Operations and VCF management serviceswill be installed in a dedicated distributed port group. • All the configurations are done automatically by the VCF Installer. • You must ensure that VLAN is provisioned in each availability zone if stretched clusters are used.

Table 663:

Table 664: Network Placement for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-NET-001	Deploy the VCF Operations nodes on an NSX segment.	Allow portability of the VCF Operations nodes and VCF management services to another VCF Instance in the VCF fleet without the need for reconfiguration of the IP addressing.	• You must skip the deployment of the VCF Operations and VCF management servicesin the VCF Installer. • You must perform configuration to support the networking configuration as a post-deployment task before deploying the VCF Operations andVCF management services. • You must perform deployment using the VCF Installer option to install deferred components.

Table 665: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Connected licensing mode".	• Provides easy license management and up to date states in overall licensing portal. • No manual interaction required.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 666: Licensing for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-LIC-001	Use VCF Operations in "Disconnected licensing mode".	If your policy does not permit access to internet for management components.	• You need to upload usage statistics from VCF Operations to Broadcom's licensing portal and then download and apply entitlements to all VCF Operations. • Usage statistics has be uploaded to the licensing portal every 180 days.

Table 667: VCF Operations Diagnostics Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-DIAG-001	Manually upload Skyline Health Diagnostics and Advisory signatures.	If your policy does not permit access to internet for management components.	Signatures should be downloaded manually and then uploaded into VCF Operations.

Table 668: VCF Operations Diagnostics Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-DIAG-001	Download Skyline Health Diagnostics and Advisory signatures directly from Broadcom's repository.	Ensures always up to date information about potential issues and mitigation steps in your environment.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 669: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Connect to the online depot.	Ensures always up to date information about bundles for Install, Update, Patch and compatibility information.	VCF Operations will require outbound connection to internet directly or through a proxy server.

Table 670: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Connect to an offline depot.	If your policy does not permit access to internet for management components you may configure an internal depot.	The bundles should be downloaded to a system with access to the Broadcom's depot and then uploaded to the internal depot by following your internal procedures.

Table 671: Binary Management for VCF Operations Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implications
VCF-OPS-RECD-BIN-001	Upload files manually.	Removes the overhead of installing and managing an additional appliance for internal depot.	You should download and upload all the files to the respective storage. A lot of manual labor and error prone process.

Continuous Availability VCF Operations Model Design Recommendations

Table 672: Common VCF Operations Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-001	Enable data persistence on each cloud proxy appliance.	Provides the ability to store data in case of connectivity issues.	Storage availability on each cloud proxy must be monitored.
VCF-OPS-RECD-COM-002	Configure VCF Operations to use a notifications configuration to route notifications for system events.	Integrates VCF Operations system events notifications to platform engineers, providing an enhanced user experience.	You must configure and maintain the notification integration.
VCF-OPS-RECD-COM-003	Add a physical data center object to the VCF Operations configuration for the geographic location of each VCF Instance.	Provides organization of the cloud proxy nodes in VCF Operations inventory.	VCF Operations must have connection to the Internet to view and set the precise geographic coordinates in the UI.
VCF-OPS-RECD-COM-004	Activate the Ping integration in VCF Operations.	Provides metrics on the availability of endpoints.	You must activate the integration manually.
VCF-OPS-RECD-COM-005	Configure the currency in the VCF Operations global options based on your organization requirements.	To receive benefits of the VCF Operations cost engine, you must select a currency.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RECD-COM-006	Deploy real-time metrics in each VCF Instance.	Provides the ability to show real-time data in the observability workbench for the selected inventory object	• Real-time metrics store component is automatically deployed during the installation. • Data is stored in the real-time metrics store and displayed in VCF Operations on demand. • Real-time metrics is configured automatically to collect supported data from the whole VCF Instance during its installation. Configuration uses a "all or nothing" model at the VCF Instance level. • Once enabled the data collection could be stopped for the whole VCF Instance by stopping real-time metrics components in VCF management services.

Table 673: Continuous Availability VCF Operations Model Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-CA-001	Deploy VCF Operations with an external load balancer to load balance the connections across the VCF Operations cluster nodes.	Required to deploy a VCF Operations cluster deployment with distributed user interface access across members.	You must configure the external load balancer configuration to support this VCF Operationscon figuration.
VCF-OPS-RCMD-CA-002	Create a collector group in VCF Operations for each VCF Instance and assign the default VCF Operations collector appliance.	Provides increased availability to metric collection across the VCF Operations collector appliances assigned to the collector group.	Manual reconfiguration of the VMware Cloud Foundation adapter required to utilize multiple VCF Operations collector appliances with the group.
VCF-OPS-RCMD-CA-003	Reconfigure the default VMware Cloud Foundation Adapter in VCF Operations to use a local-instance collector group for a VCF Instance.	Increases the availability of metric collection.	Must be reconfigured post-deployment.
VCF-OPS-RCMD-CA-004	Deploy a second VCF Operations collector appliance in the default management vSphere cluster and assign it to the collector group.	Increases the availability of metric collection.	None.
VCF-OPS-RCMD-CA-005	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VCF Operations collector nodes in the same group.	Using vSphere DRS prevents the VCF Operations collectors from running on the same ESX host and risking the high availability of the collectors.	You must perform manual configuration to set up an anti-affinity rule.

SDDC Manager Detailed Design

SDDC Manager provides fleet management capabilities for the underlying virtual infrastructure and uses binaries to deploy new workload domains, and to patch and upgrade existing ones.

SDDC Manager Model Options

A SDDC Manager can leverage options from the design areas that are listed below, depending on objectives.

Design Area	Options
Infrastructure IP Address Model	• IPv4 Addressing Model • Dual Stack IP Addressing Model

Table 674: SDDC Manager Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-SDDCM-REQD-CFG-001	Deploy a new SDDC Manager with each new VCF Instance.	The VCF Installer automatically installs a new SDDC Manager with each new management domain.	None.
VCF-SDDCM-REQD-CFG-002	Deploy an SDDC Manager system in the first availability zone of the management domain.	SDDC Manager is required to perform VMware Cloud Foundation capabilities, such as provisioning workload domains, deploying solutions, patching, upgrading, and others.	None.
VCF-SDDCM-REQD-CFG-003	Place the SDDC Manager appliance on the VM management network.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Log Management Detailed Design

A VCF fleet can include one of the supported *Log Management Models*. This section describes the options, requirements, and recommendations for each *Log Management Model*.

- Simple Log Management Model
- High Availability Log Management Model
- Multi-Instance Log Management Model

Log Management Deployment Models

Table 675: Common Log Management Deployment Model Design Attributes

Attribute	Details
Centralized log management	Provides a centralized log management solution for vCenter, ESX, VCF Automation, NSX, and components running on VCF management services (e.g. log management, identity broker). The use cases that are listed are an example, but they are not limited to this subset, for a high availability / cluster log management instance to manage: • Fleet level deployments. • Default deployment model for all management and workload environments. • High availability environments. The use cases listed are an example, but they are not limited to this subset, for a Simple / Standard *log management* instance to manage: • Test/Dev environments only. The use cases that are listed are an example, but they are not limited to this subset, for a multi instance log management model to manage: • Environments that exceed the scale limits of a single log management instance. • Environments that cannot send full logging traffic between locations/regions.
Centralized log source management	Log sources for the VCF fleet can be managed in a single location under VCF Operations. By default, all components in the VCF management services that support log collection are enabled for logging to the VCF log management instance for the Fleet. Other components require activation of log collection on the VCF Operations Integrations page. When activating the log collection, there is the option to: • Send logs directly to the log management instance. • Send logs to the local cloud proxy or proxy group for optimization/aggregation before sending to log management. Once activated, configuration is performed on the Log Collection page under Configurations, including setting custom FQDN/VIP, and managing the types of logs that are collected. **Important:** When log collection for a source uses the cloud proxy, logs are balanced across all available VIPs in a round-robin manner. If a specific VIP is selected under Log Collection configuration, the collection reverts to the direct path.
Initial deployment	Deployed after initial deployment of infrastructure and operations, from the *Build* section of VCF Operations.
Architecture flexibility	Can be deployed in a Simple or Highly Available Cluster deployment. The recommended deployment model is an HA Cluster deployment. Simple deployment is recommended for test/dev environments and not recommended for production use cases. A deployment can be scaled up (increase replica size), scaled out (add replicas) or can increase storage (add storage to every replica) up to the scale limits. There is no provision for scaling back in once a scale-up or scale-out is performed.
Security & Compliance	FIPS mode is enabled by default: • Only FIPS-approved providers, keystores, ciphers, and algorithms are used. • Non-FIPS algorithms are disabled. • Certificates and keystores are in BCFKS format. • End-to-end TLS communication uses supported ciphers only. • Supports hosts with hardware acceleration for FIPS ciphers.

Attribute	Details
VCF management services dependency	Log management is deployed to the VCF services runtime, in the VCF fleet's first instance. Therefore the cluster where VCF management services for the fleet is deployed must be provisioned with sufficient resources to support the anticipated compute and storage requirements of log management. VCF management services automatically scales out additional management services nodes when required by log management, but if nodes require scaling up (increase in size) that change should be scheduled in a maintenance window as it can impact other services running on VCF management services.

Log Management Sizing Considerations

There are three main considerations for log management sizing:

- Events per second (EPS)
- Connections (http & tcp)
- Storage capacity

The number of EPS can be approximated by the number of ESX hosts in the fleet multiplied by 250. Certain configurations may require adjustment, for example, logging of firewall rules can generate a significant volume of EPS that must be taken into account. Connections are generally related to the number of hosts and endpoints (more hosts and endpoints require more connections and generate more logs). Storage capacity is a factor of the retention period and the number of log messages received.

Log management relies on an integrated load balancer that receives and forwards messages to one of the log processing and data replicas. There is one load balancer per VIP/FQDN, and the size of the load balancer is based on the log management instance size profile (small & medium share the same size load balancer). Using an external load balancer is not recommended as it would introduce an additional hop between the log source and log management, since the integrated load balancer is still a required part of the traffic path.

Additional VIPs/FQDNs must be manually created on the integrated load balancer when the number of EPS or connections will exceed the capacity of a single gateway. Log sources must be configured to send to different FQDNs to keep the load on any individual FQDN within the limits shown in the table below.

The number of replicas (which process and store the log messages) can be scaled out independently of the instance's size profile, up to a defined limit. The total capacity for log processing, storage, and queries of the instance is determined by the number of replicas and the size of the instance shown in the table below.

> **Important:**
>
> To support a given EPS load, both the number of VIPs/FQDNs and the number and size of replicas must both be sufficient to handle the total load. While load is automatically distributed by the load balancer across deployed replicas, the traffic sent to each FQDN must be managed by the customer unless the cloud proxy is being used. Log events sent through the cloud proxy will be balanced across all available VIPs using round-robin distribution.
>
> In order for multiple VIPs to provide additional capacity, the VCF management services must be deployed with the HA deployment model.
>
> Capacity utilization and performance of the log management instance is best observed on the *Log Management Health Overview* dashboard in VCF Operations.

Each replica has a minimum of 500 GB of disk, which can be scaled up to 9 TB per replica. All nodes scale to the same size.

Once log management is scaled up to a number of replicas, it cannot be scaled back down to fewer replicas even if the size profile is increased. Likewise the size profile cannot be reduced once scaled up. For this reason, we recommend scaling up the size to medium once you exceed 5-6 small replicas, or large once you get to 6-8 medium replicas. If you deploy a large number of replicas in a smaller size profile, future scaling up to a larger size profile will result in a large jump in resources and capacity (e.g. 12 small to 12 medium replicas).

To provide a higher level of resilience, plan for an extra replica or two (N+1 or N+2) to provide sufficient log processing capacity in case a replica (or the host it is running on) fails. Also consider whether you want to provision for peak loads,

or support an average load with the understanding that some logs may be lost during peak rates if the load exceeds the cluster's capacity.

Log Management does not consume resources directly from the management cluster hosts, but runs on VCF management services worker nodes. VCF management services must have sufficient IP addresses in its pool to support the number of worker nodes required for log management. Allow for 6 IPs for an initial log management installation plus 1 IP for every small or medium replica and 2 IPs for every large replica. The cluster VCF management services is running on requires sufficient CPU and memory capacity for these worker nodes. Each VCF management services worker node consumes 12 vCPU / 24 GB memory for small size log management profile and 24 vCPU / 48 GB memory for medium profile. The large size log management profile uses the same size worker nodes but requires 2 per replica.

The datastore used by the VCF management services will need to support all log management nodes' storage requirements, and as all VCF management services nodes run in a common vSphere cluster on a common datastore, all log management replicas will also run on that common cluster & datastore.

The resources (CPU and Memory) shown in the table below are consumed by log management from the VCF services runtime worker nodes. Depending on how much capacity is available in the existing VCF services runtime footprint, additional worker nodes may be deployed, or worker nodes may be scaled up as required for the log management profile selected.

> **Notice:**
>
> The actual number and size of VCF management services worker nodes is determined by all the components that are deployed and their sizing and availability settings. The VCF services runtime automatically scales the size and number of worker nodes to accommodate the selected size and number of log management replicas and VIPs.

In addition to EPS and the storage/sharding requirements described, there are some other factors that can impact log processing capacity and impact sizing decisions:

- A high complexity & quantity of log filtering rules can decrease log processing capacity
- Disabling event clustering (machine learning-based grouping) can increase log processing capacity
- Running many queries (using automation or multiple administrators accessing the system) decreases available log processing capacity
- Hosts that do not support FIPS-compliant ciphers have reduced capacity; scale numbers are based on hardware-accelerated hosts.

Log ManagementSize Profile (max. hosts supported)	Per VIP/FQDN			Per Replica		
	Resources (vCPU/Memory)	Events Per Second	Connection Limit (http / tcp [syslog])	Resources (vCPU/Memory)	Events Per Second**	Number of replicas
Small (1,500)	4/2GB	200K	20k / 50k*	8/16GB	20K	1-19
Medium (3,000)	4/2GB	200K	20k / 50k*	16/32GB	40K	3-19
Large *** (4,560)	6/4GB	300K	30k / 50k*	32/64GB	60K	3-19

* HTTP & TCP (syslog) connection limits are independent; both limits can be achieved simultaneously.

** These figures are based on a typical load profile. If the load profile is smaller (e.g. fewer queries, smaller log messages, etc.) the observed performance may be higher, up to a hard cap of 40K/70K/120K EPS respectively for Small/Medium/ Large size replicas.

*** Large-size log management replicas are spread across 2 VCF management services worker nodes.

Table 676: Cloud Proxy sizing for Log Forwarding

Cloud Proxy Size	Resources (vCPU/ Memory)	EPS	Connections	Network Latency
Small	4 / 16GB	200K (syslog), 35K TCP (HTTP/API)	300	<500ms, < 2% loss
Standard	8 / 48GB	400K (syslog), 75K TCP (HTTP/API)	600	<500ms, < 2% loss

Log Management Virtual IPs and FQDNs

Log management starts with a default Fully Qualified Domain Name (FQDN) & IP that is configured during initial deployment. Multiple Virtual IPs (VIPs) can be configured post-deployment under the Integrated Load Balancer tab on the Log Collection configuration panel. Some reasons for creating additional VIPs:

Attribute	Description
Scale	Each VIP has a separate Load Balancer with its own event-per-second (EPS) and connections processing capacity. By adding VIPs you can increase the total EPS & connections capacity of the log management instance, up to the limit of the total for all replicas in the instance.
Separation	Every VIP has its own FQDN and the ability to add a key/value pair to every log message that it processes. This can be used to separate logs to a different partition or for easy filtering when searching logs. **Restriction:** Fields can only be added to logs sent using the HTTP (API) protocol, and not those sent by syslog. For example, ESX hosts send syslog data, which will not have any fields added by the VIP.

Tip:

The SSL certificate used by log management must include all the FQDNs for VIPs for the VIP creation to proceed.

Using the Cloud Proxy with log management

VCF deploys a cloud proxy by default in every workload domain. This proxy serves to collect metrics, forward logs, and run the log assist agent. To achieve high-availability or greater scale at a site, multiple cloud proxies can be deployed and added to a collector group.

It provides three primary benefits for log collection:

- Optimizes network traffic to improve stability and efficiency of transmission over high latency connections.
- Buffers logs for a period of time in case of an unreliable network connection, to help minimize the loss of log messages.
- Centralizes all log traffic to a single source/destination pair, making it easier to manage firewall rules.

Tip:

Log message buffering to disk is enabled by default, and is not dependent on the "Data Persistence" setting being enabled. Once the buffer fills up, the oldest logs start to be removed to make room for new logs. The buffer holds about 20GB of logs.

When using the cloud proxy in multi-VIP environments, all logs sent through the cloud proxy are always sent in round-robin rotation to all VIPs configured in log management. When fields are added to log messages processed by a VIP, logs must be sent directly to the VIP rather than the cloud proxy so only the desired logs are processed.

The cloud proxy is available in two sizes: Standard supports 500K EPS and 600 connections. Small size supports about half the throughput and connections of the standard proxy. Both sizes can tolerate up to 300ms network latency to the log

management instance. As latency increases, throughput will also decrease, so expect lower maximum EPS rates for sites with high latency.

Log Retention and Archive Design for log management

Your organization might require specific retention periods for logs due to compliance, auditing, or for troubleshooting purposes. You must factor in the quantity of storage that is required to retain logs in the active index, and the operational impact and space requirements for archived logs. Archived logs are not directly searchable but can be imported to an archive partition for searching and analysis.

By default, logs are split into two different partitions: Audit & Logs. Up to 9 additional partitions can be created to store specific log messages. Each partition can have a custom retention period defined, as well as a different archive location.

Table 677: Log Archiving Attributes

Attribute	Description
Retention period	Logs are stored in a searchable index in the partition for the length of the retention period. Each partition has its own index, and within that index, the logs are divided into shards. A shard corresponds to one day of one replica's portion of the index. Each day of retention therefore consumes one shard per index on each replica. Because the number of shards per replica is fixed (700, 800 and 900 for small, medium, and large size profiles respectively), the retention period has an upper limit. As an example, a medium size profile with 8 partitions would have capacity for at most 100 days of retention. If your retention requirement is longer, you must adjust the size or number of replicas, and/or the number of partitions to accommodate it. Once the retention period for a log message is exceeded, it will be removed from the index and discarded during the next log index cycle. If there is insufficient disk space or insufficient shard capacity to honor the full retention period, logs may be discarded earlier. The status can be monitored on the *Log Management Health Dashboard*. If no retention period is specified, the system will manage it to maintain disk space and shard capacity at an acceptable level. Log indexes are rotated at least every 24 hours, which also runs the archival process. This rotation can occur more frequently if the log index size, or the total size of log messages exceeds a system-defined threshold. Each partition runs on an independent indexing cycle.
Archive location	The archive location must be on an NFS version 3 (NFSv3) shared storage, or an S3-compatible endpoint. The archive location must provide write access, always be available, and have enough capacity to accommodate the archives. Pruning or rotation of archive files is not within the scope of log management, and must be handled by an external solution if required. S3 endpoints must support chunked transfer encoding, and the bucket must be in an AWS-equivalent region (e.g. US_EAST_1). NFS paths should be entered in all lower-case when adding external storage to Log Management. Logs are sent to the archive on the first index refresh cycle, meaning that log messages will reach the archive in at most 24 hours (but possibly less). If using the archive as a recovery solution, this defines the Recovery Point Objective (RPO) for the log data. **Important:** The storage connection will be initiated from the VCF management services node's IPs. When configuring firewall rules or ACLs on storage devices, the entire IP range allocated to the VCF management services should be allowed access to the S3 or NFS endpoint(s).

Attribute	Description
Partition	A partition represents a distinct index for storing log messages. Incoming logs are directed using filters applied to the partition, and the order of processing (logs will go into the first partition they match). Audit is always the first partition processed on ingestion, and default is always the last, and acts as a catch-all for logs that don't match any filters. Any new partitions created can be re-ordered between the two system-defined partitions. When searching logs, queries can also be restricted to specific partitions.

Log Collection

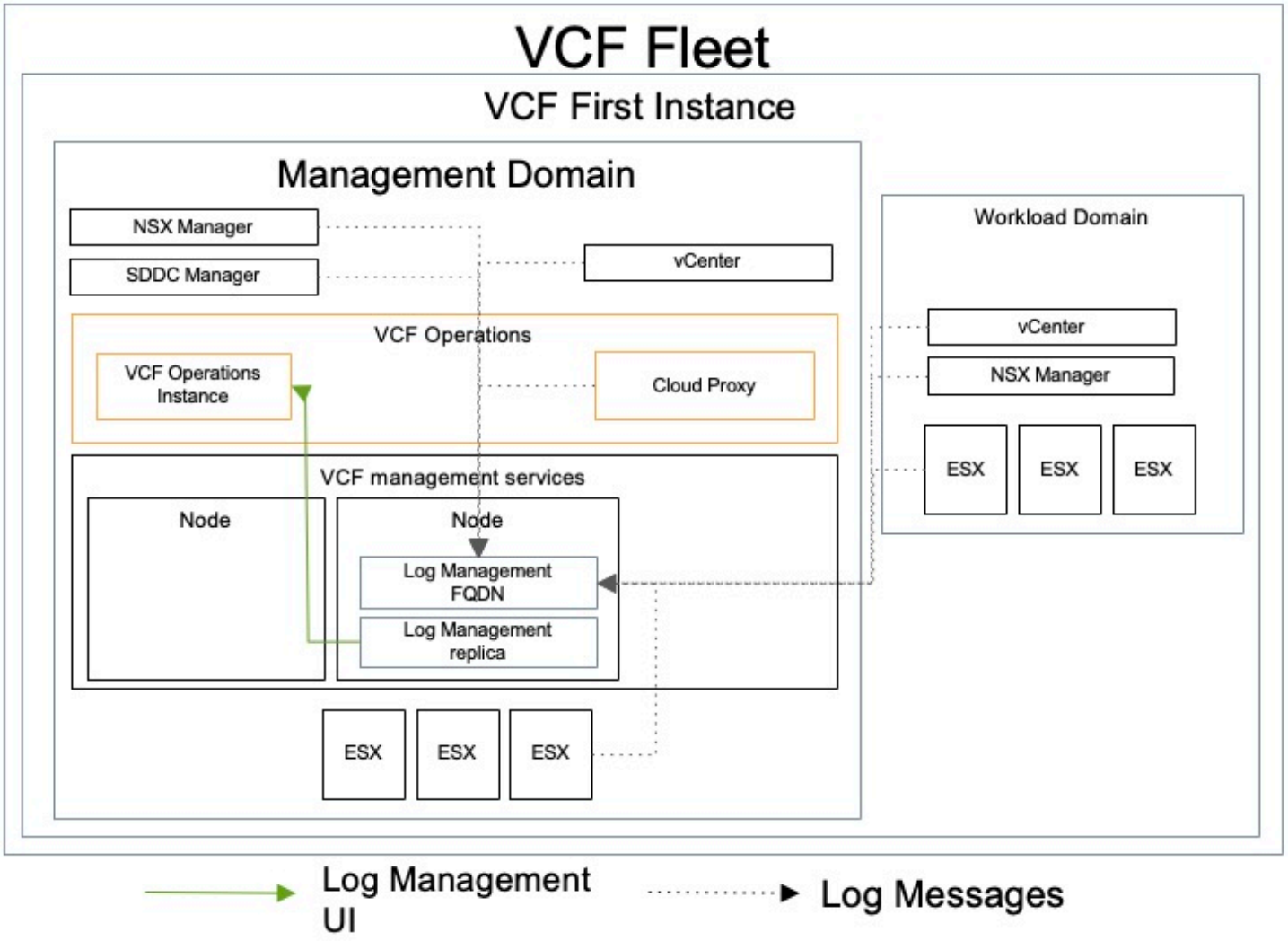
Once log management has been installed, components must send their logs to the log management instance for ingestion. Depending on the component this may be done automatically, or may require manual configuration. The following table provides the method required for each component:

Attribute	Description
vCenter Server, NSX Manager	Log collection is not enabled by default. To activate log collection for a vCenter Server (including the ESX hosts manged by it) or NNSX Manager in a VMware Cloud Foundation domain enable it on the adapter for that component under VCF Operations -> Administration -> Integrations -> VMware Cloud Foundation. For each individual component you can Activate Log Collection, and select whether the logs should be sent directly to the *log management*instance, or use a cloud proxy.
VCF management services components (including VCF Automation, *log management* itself, identity broker)	Log collection for VCF management services is enabled automatically when *log management*is deployed. If a component running on VCF management services supports log forwarding then logs will be forwarded to the *log management* instance for the fleet. Log collection for VCF management services components cannot be disabled nor configured from the *Log Collection* tile in VCF Operations.

Simple Log Management Model

A basic log management deployment in the VCF fleet's primary VCF Instance management domain. Availability is provided by the VCF management services whose nodes are in turn protected by vSphere HA within a single vSphere cluster. The goal of the simple model is to consume minimal resources and require minimal configuration.

Figure 317: Simple Log Management Model in VMware Cloud Foundation



Simple Log Management Model Attributes

A *Simple Log Management Model* has the following attributes.

Attribute	Detail
Centralized Management	Provides a centralized log management framework for vCenter, ESX, NSX, VCF Operations, VCF Automation, and identity broker.
Availability	Relies on VCF management services and vSphere HA to restart nodes in the event of an ESX host or log management component failure, resulting in potential outages for up to 5 minutes depending on the failure.

Attribute	Detail
Scalability	Supports scale-up, storage scale up, and scale out. **CAUTION:** Once a quantity (size or number of replicas) is scaled up, it cannot be scaled back down.
Recoverability	Restore configuration through a backup, restore logs from archives.

Simple Log Management Model Options

A *Simple Log Management Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
None	No options are provided in the simple model. Moving up to the High Availability model is recommended if additional configuration options are needed.

Simple Log Management Model Design Requirements and Recommendations

Table 678: Simple Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-SIM-001	Deploy log management as a single small-size replica.	• Supports the simple deployment objective. • Supports scale-out to a high availability model.	None.

Table 679: Common Log Management Design Recommendations for All Models

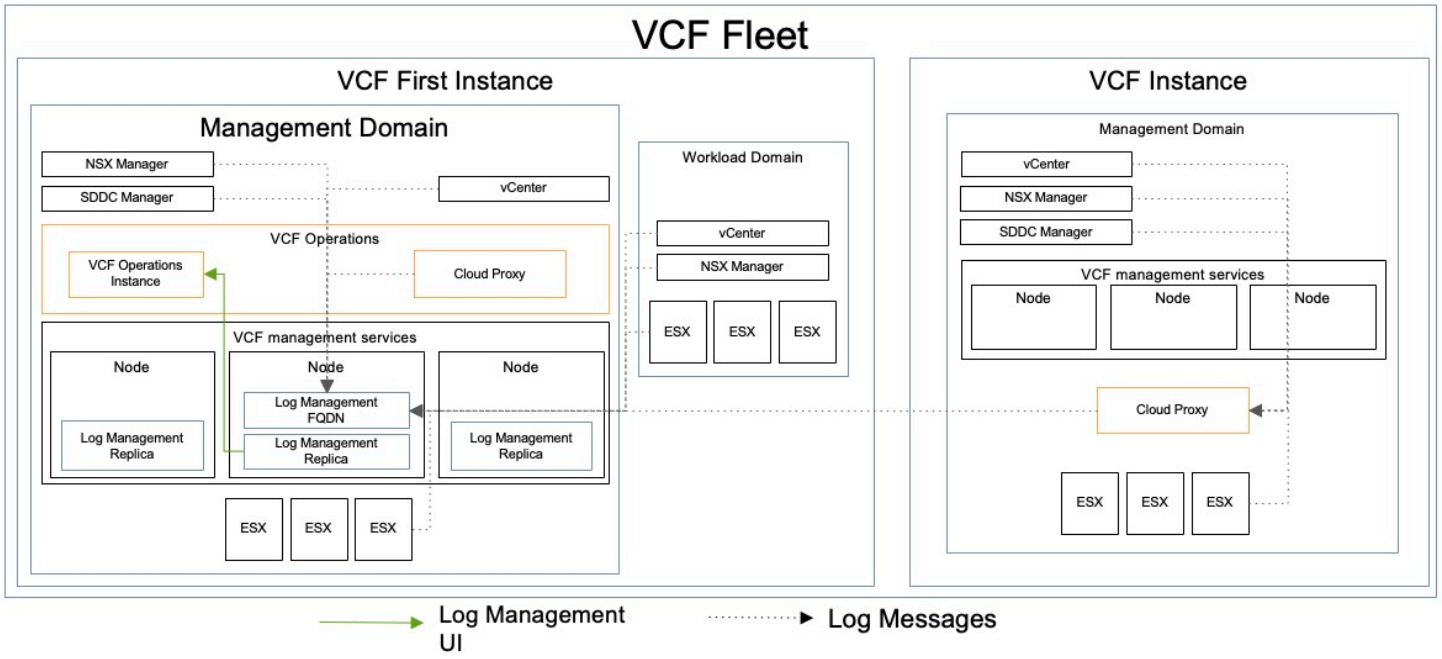
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

High Availability Log Management Model

The *High Availability Log Management Model* consists of a three-node deployment behind an internal load balancer. This model provides resiliency at both the application level through the multi-replica deployment, VCF management services to restart replicas, and vSphere HA to restart nodes in the event of an ESX host or log management component failure.

Figure 318: High Availability Log Management Model in VMware Cloud Foundation



High Availability Log Management Model Attributes

A *High Availability Log Management Model* has the following attributes.

Attribute	Detail
Centralized management	Provides a centralized log management framework for vCenter, ESX, NSX, VCF Operations, VCF Automation, and identity broker.
Availability	Relies on cluster technology in the event of an outage or VCF management services and vSphere HA to restart nodes in the event of an ESX host failure.

Attribute	Detail
Scalability	Supports scale-up, storage scale up, and scale out.
Recoverability	Restore configuration through a backup, restore logs from archives.

High Availability Log Management Model Options

A *High Availability Log Management Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Small • Medium • Large

High Availability Log Management Model Design Requirements and Recommendation

Table 680: High Availability Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-HA-001	Deploy log management with at least 3 replicas.	• Provides high availability. • Using the integrated load balancer prevents a single point of failure. • Using the integrated load balancer simplifies the log management deployment and subsequent integration.	• You must deploy a minimum of three appropriately sized replicas. • All replicas are sized identically. • If the capacity of your log management cluster must expand, identical capacity is added to each node.
VCF-LOGS-REQD-HA-002	Protect all VCF Management service nodes by using vSphere High Availability.	Supports the availability objectives for log management without requiring manual intervention during an ESX host failure event.	None.

Table 681: Common Log Management Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

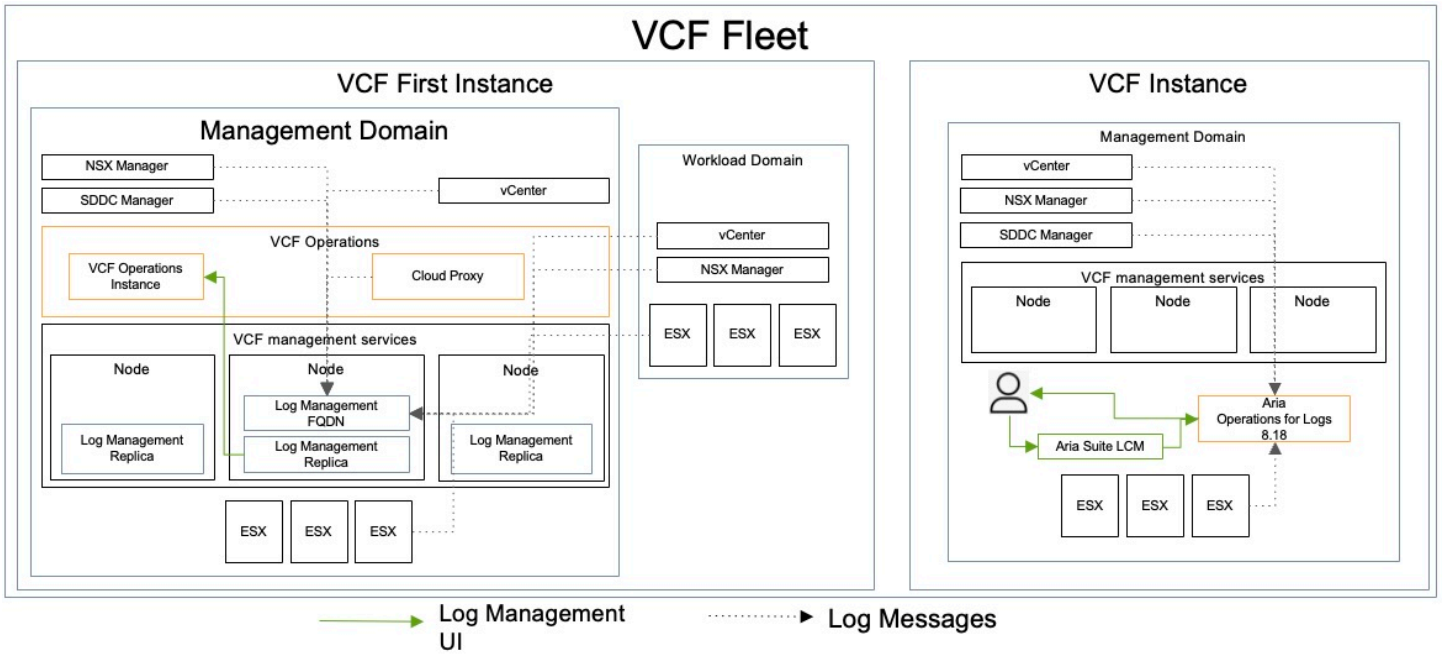
Table 682: High Availability Log Management Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-HA-001	Enable HA on the VCF management services control plane. (Instance deployed with the High Availability deployment model, or scale up VCF services runtime to medium or large.)	To support the high availability objective, the control plane for the VCF management services should be configured for HA, as log management depends upon VCF management services.	Requires additional nodes and capacity in the vSphere management cluster.

Multi-Instance Log Management Model

The *Multi-Instance Log Management Model* consists of a single, VCF Operations integrated log management instance, and multiple non-integrated, Aria Operations for Logs 8.18 instances. These non-integrated instances are deployed in separate VCF Instances within the fleet, and logs from components in that instance are manually configured to go to this non-integrated instance. This model provides increased scale and log segregation at the expense of higher management overhead. Where connectivity, policy and scale permit, a subset of logs can be forwarded to the integrated instance for a centralized view.

Figure 319: Multi-Instance Log Management Model in VMware Cloud Foundation



Multi-Instance Log Management Model Attributes

A *Multi-Instance Log Management Model* has the following attributes.

Attribute	Detail
Distributed management	Provides multiple management interfaces for viewing and managing logs in the VCF fleet.
Availability	Each instance can support different availability requirements by using a simple or high availability log management model for the instance.
Scalability	Each instance supports independent scale-up, storage scale up, and scale-out.
Recoverability	Restore configuration through a backup, restore logs from archives.

Multi-Instance Log Management Model Options

A *Multi-Instance Log Management Model* can apply options from the design areas below according to your objectives.

Design Area	Options
Appliance Size	• Small • Medium • Large

Multi-Instance Log Management Model Design Requirements and Recommendations

Table 683: Multi-Instance Log Management Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LOGS-REQD-MI-001	Deploy the integrated log management for the Fleet with at least 3 replicas through VCF Operations. Deploy additionallog management instances as a three node cluster - one primary and two worker nodes with an integrated load balancer, in the default management vSphere cluster for the instance, using Aria Operations for Logs 8.18 (latest patch release).	• Provides high availability. • Using the integrated load balancer prevents a single point of failure. • Using the integrated load balancer simplifies the log management deployment and subsequent integration.	• You must deploy a minimum of three appropriately sized nodes. • You must size all nodes identically. • If the capacity of your log management cluster must expand, identical capacity must be added to each node.
VCF-LOGS-REQD-MI-002	Protect all log management cluster nodes by using vSphere HA.	Supports the availability objectives for log management without requiring manual intervention during an ESX host failure event.	None.
VCF-LOGS-REQD-MI-003	Apply a vSphere DRS anti-affinity rule to the log management cluster nodes for the non-integrated Aria Operations for Logs instances.	Using vSphere DRS prevents the log management cluster nodes from running on the same ESXhost and risking the high availability of the cluster.	None.

Table 684: Common Log Management Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-001	Create additional partitions to accommodate different retention policies, filtering, or archive location.	Split logs into separate partitions to apply different policies	
VCF-LOGS-RCMD-COM-002	Configure an SFTP server or S3 endpoint for log archives	Define a location to store archived log messages for extended near-line retention	For long-term log archival when capacity on the instance is insufficient. Enables recovery of log messages from the archive location.
VCF-LOGS-RCMD-COM-003	Configure a retention period for every log management partition and enable Archive on it to store logs for recovery or for extended duration.	Supports recovery of log messages and offline availability.	None.
VCF-LOGS-RCMD-COM-004	Define additional VIPs/FQDNs when required for scale or log separation, and set log collection to send to the desired FQDN.	Support higher scale for the instance, or log tagging based on the FQDN used.	The SSL certificate for log management must have all FQDNs included in the SAN field.
VCF-LOGS-RCMD-COM-005	Review and favorite the Log Management Health Overview Dashboard in VCF Operations.	Provides alerts and warnings that impact the health, performance, or availability of the log management environment	To receive alerts from the Log Management dashboard, ensure that there is a notification method that is configured in VCF Operations -> Administration: Configurations -> Alerts: Notifications
VCF-LOGS-RCMD-COM-006	Activate log collection for vCenter and NSX.	To gather logs from vCenter and NSX, the VMware Cloud Foundation adapter must be configured to activate log collection, and must select the path (direct or through the cloud proxy). Collection is not enabled by default.	Requires manual enablement per adapter.
VCF-LOGS-RCMD-COM-007	Install and configure the log agent on SDDC Manager.	Supports collecting logs from SDDC Manager, which includes some audit logs.	Follow: https://knowledge.broadcom.com/external/article/396643/deploy-and-configure-vcf-operations-for.html
VCF-LOGS-RCMD-COM-008	Implement an archive retention/ maintenance policy.	Define a retention policy for archives and set up a process to remove archives that exceed the retention period.	Manage the storage consumption of the archive disk.
VCF-LOGS-RCMD-COM-009	Configure an SFTP server for configuration backups.	Log Management configuration is backed up through VCF management services	Required to recover the environment if there is issue
VCF-LOGS-RCMD-COM-010	Use a cloud proxy for logs being sent from remote sites.	A cloud proxy can optimize for throughput and latency to make long-distance log forwarding more efficient.	Log collection through the cloud proxy will always use all VIPs (in round-robin selection).
VCF-LOGS-RCMD-COM-011	Run log management on hosts that support hardware FIPS-compliant cipher acceleration.	Achieve expected performance numbers. Scale capacity is reduced on hosts that do not support acceleration.	

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-COM-012	Configure external agents or log sources to use a separate VIP.	The VIP can be moved to different log management instances in the future without having to manually reconfigure each individual source.	Same as for having multiple VIPs/FQDNs.

Table 685: Multi Instance Log Management Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-LOGS-RCMD-MI-001	Deploy Aria Operations for Logs 8.18 instances using a dedicated Aria Suite Lifecycle instance for each one to facilitate ongoing Lifecycle Management (LCM)	Aria Suite Lifecycle supports scaling up and out nodes and adding storage capacity for the non-integrated instances.	None.
VCF-LOGS-RCMD-MI-002	Configure local components to send their logs to the non-integrated log instances through the components themselves.	As these Aria Operations for logs instances are not integrated with VCF Operations, central log management is not available.	Components running in VCF management services are not configurable and will use the VCF fleet integrated log management instance.

VCF Operations for Networks Detailed Design

A VCF fleet can include one of the supported *VCF Operations for Networks Models*. This section describes the options, requirements and recommendations for each *VCF Operations for Networks Model*.

- Simple VCF Operations for Networks Model
- High Availability VCF Operations for Networks Model

VCF Operations for Networks - Logical Design

VCF Operations for networks can be deployed as a cluster or a single node. This cluster processes the collected data and presents it using a dashboard. VCF Operations for Networks also uses collector nodes to collect data from the data sources, such as vCenter server and NSX Manager which then sends the data to the platform cluster for processing.

The platform cluster capacity is based on the deployment size of each node and the number of nodes composing the cluster. For more information about how to decide the platform brick (node) size and number of platform bricks (nodes), see the VCF Operations for Networks documentation.

The collector capacity is based on the deployment size. The number of data sources that you can add to a specific collector is based on the documented capacity of the collector size (accounting for the number of VMs and the number of flows).

> **Important:**
>
> VCF Operations for networks is fully integrated with VCF Operations UI for viewing and analyzing flows, and with VCF Operations Lifecycle for deployment and management of the installation. VCF Operations for Networks is always deployed in a 1:1 ratio with VCF Operations, i.e. one VCF Operations for networks instance per VCF fleet.

VCF Operations for Networks - Scaling

VCF Operations for networks scales in two dimensions: the platform and the collectors. The platform can only be clustered if using appliances of at least a large size. Medium deployments cannot be scaled into a cluster without first scaling up to large.

The scaling of a platform to a cluster model does not provide additional availability to the platform. Scaling must be used when maximums for a single node exceed.

The overall dimensioning for collector scaling is based on various factors including:

- Number of vCenter instances
- Number of VMs
- Number of flows to track

To determine when to scale the platform and collector nodes, see the VCF Operations for networks sizing below. When collecting over 10,000 VMs and 4 million active flows, the platform must be upgraded from a single node to a clustered deployment. Similar scale-up and scale-out requirements exist for the Collectors.

Design Recommendations IDs	Design Recommendation	Justifications	Implications
VCF-NECE-RCMD-01	Single Platform Node Size based upon sizing document and best practices.	• Smallest footprint. • Can be scaled-out to a platform cluster. • Supports 10,000 virtual machines or objects.	• Requires a CPU and RAM reservation of 100% of the CPU and RAM assigned. • No, high availability • Requires the single node be deployed as an Extra Large size
VCF-NECE-RCMD-02	Platform Cluster • Three node clusters. Consisting of 3 platform instances. Can scale up to 15 XL nodes. • Cluster node sizing starts at a Large Platform Node Size and go up to Extra Large.	• Highly Performant. • Highly scalable.	• Cluster does not provide high availability. It is for performance. • Requires higher CPU and memory reservations to support additional nodes.
VCF-NECE-RCMD-03	Collector A single VM used to monitor remote VCF deployments and to send data back to the primary cluster.	• Highly Performant. • Highly scalable.	• Limitation of total flows is based on the capacity of the platform. • Requires understanding the amount of VMs, Flows, and Edges will be sending data to the collector.

Understanding how to size VCF Operations for Networks

VCF Operations for Networks provides network visibility across virtual and physical devices. The key considerations are:

- Number of VMs monitored
- Flows Per Day
- Total Flows
- Flow Planning
- Number of Edges

To understand how to properly size the environment see the Capacity for Non-Cluster and Clustered environments below.

- The reservation for the CPU speed and RAM for each node must be 100% of the value specified above.
- If the disk in any of the platform nodes exceeds 95% of the capacity, VCF Operations for Networks UI will not be accessible.

To match your setup to all the specifications, you might have to add the resources (RAM, Disk, CPU). See https://kb.vmware.com/s/article/53550and Increase the Brick Size of Your Setup

Non-Cluster Sizing

Table 686: Single Node Sizing

Brick Size	Cores required for 2.1 GHz CPU	Cores required for 2.3 GHz CPU	Cores required for 2.6 GHz CPU	RAM	Disk
Medium	10	9	8	32 GB	1 TB
Large	15	14	12	48 GB	1 TB
Extra Large	20	18	16	64 GB	2 TB

Table 687: Flow and Network Rule Sizing

Brick Size	Number of VMs (K = Thousand)	Flows per Day (M = Million)	Total Flows (M = Million)	Flow Planning (M = Million)	Number of Network Rules for Network Map and Intents (K = Thousand)	Number of Edges for VeloCloud (K = Thousand)	Number of VMs for Flow-Based Application Discovery	Number of UI Entities in the Network Map Topology
Medium	4K	1M	4M	2M	Not Applicable	2K	Not Supported	Not Supported
Large	6K	2M	8M	4M	Not Applicable	2K	Not Supported	Not Supported
Extra Large	10K	2M	8M	4M	170K	4K	3K	5K

Cluster Sizing

Brick Size	Cluster Size	Number of VMs (K = Thousand)	Flows per Day (M = Million)	Total Flows (M = Million)	Flow Planning (M = Million)	Number of Network Rules for Network Map and Intents (K = Thousand)	Number of Edges for VeloCloud (K = Thousand)	Number of VMs for Flow-Based Application Discovery	Number of UI Entities in the Network Map Topology
Large	3	10K	2M	8M	4M	Not Applicable	4K	Not Supported	5K
Extra Large	3	18K	6M	24M	6M	170K	6K	3K	5K
Extra Large	5	30K	10M	40M	10M	170K	10K	3K	5K
Extra Large	7	58K	12M	48M	10M	170K	10K	3K	5K
Extra Large	10	100K	15M	60M	10M	170K	10K	3K	5K
Extra Large	15	150K	20M	80M	10M	170K	10K	3K	5K

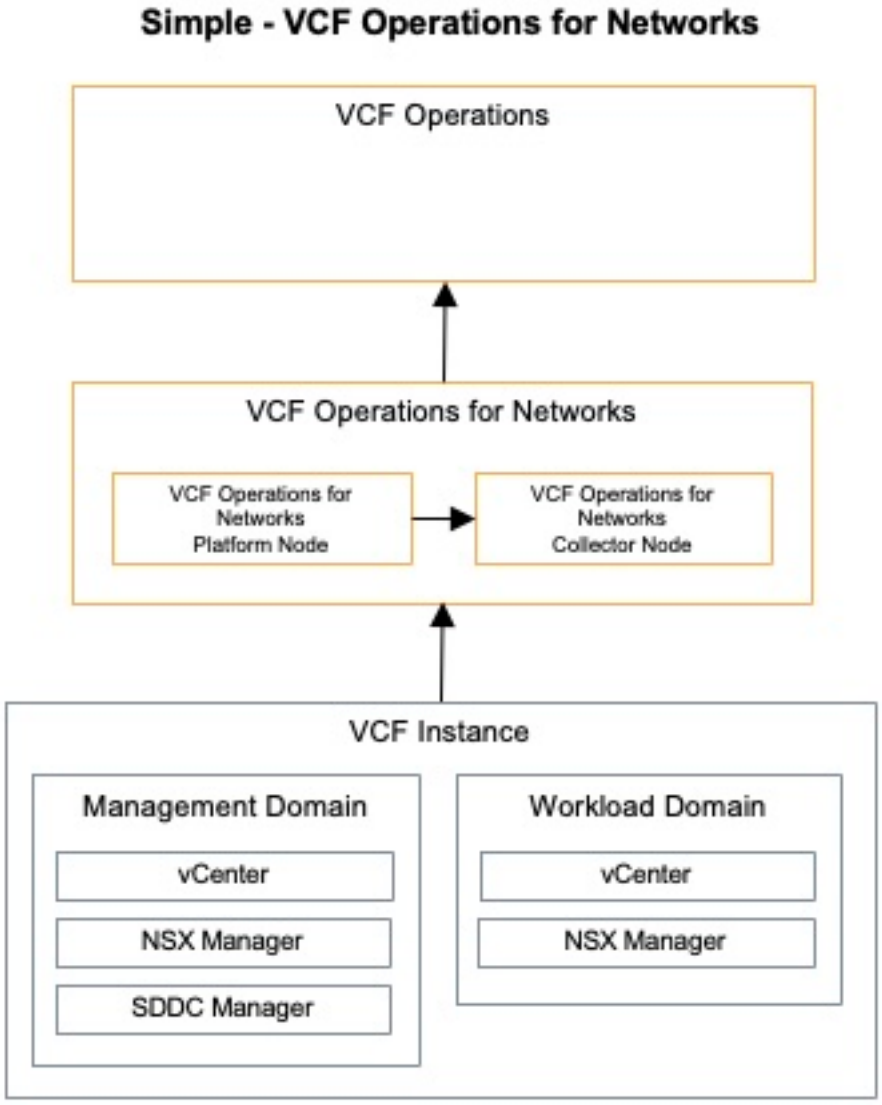
Recommendation for the Collector Deployment

Collector Size	Number of VMs (K = Thousand)	Flows per Day (M = Million)	Flow count in 4 days (M = Million)	Number of Edges for VeloCloud (K = Thousand)
Medium	4K	2.5M	3.25M	4K
Large	10K	5M	6.5M	6K
Extra Large	35K	10M	13M	10K
Extra Extra Large	45K	17M	22M	10K

Simple VCF Operations for Networks Model

The *Simple VCF Operations for Networks Model* consists of a single VCF Operations for networks node deployed in a vSphere cluster. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Figure 320: Simple VCF Operations for Networks Model in VMware Cloud Foundation



Simple VCF Operations for Networks Model Attributes

A *Simple VCF Operations for Networks Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out.
Recoverability	Restore through a backup.

Simple VCF Operations for Networks Model Options

A *Simple VCF Operations for Networks Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Medium • Large • Extra Large

Simple VCF Operations for Networks Model Design Requirements

Table 688: Simple VCF Operations for Networks Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-SIM-001	Deploy VCF Operations for networks as a non-clustered, single platform node in the default management vSphere cluster.	• Provides capacity for monitoring of up to 10,000 virtual machine or objects. • Supports scale-out with additional VCF Operations for networks platform nodes to a platform cluster.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks platform node will be applied.
VCF-NET-REQD-SIM-002	Deploy a single VCF Operations for networks collector node in the default management vSphere cluster.	A minimum of one VCF Operations for networks collector node is required to provide the VCF Operations for networks platform node with network monitoring data.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks collector node will be applied.

Simple VCF Operations for Networks Model Design Recommendations

Table 689: Simple VCF Operations for Networks Model Design Recommendations

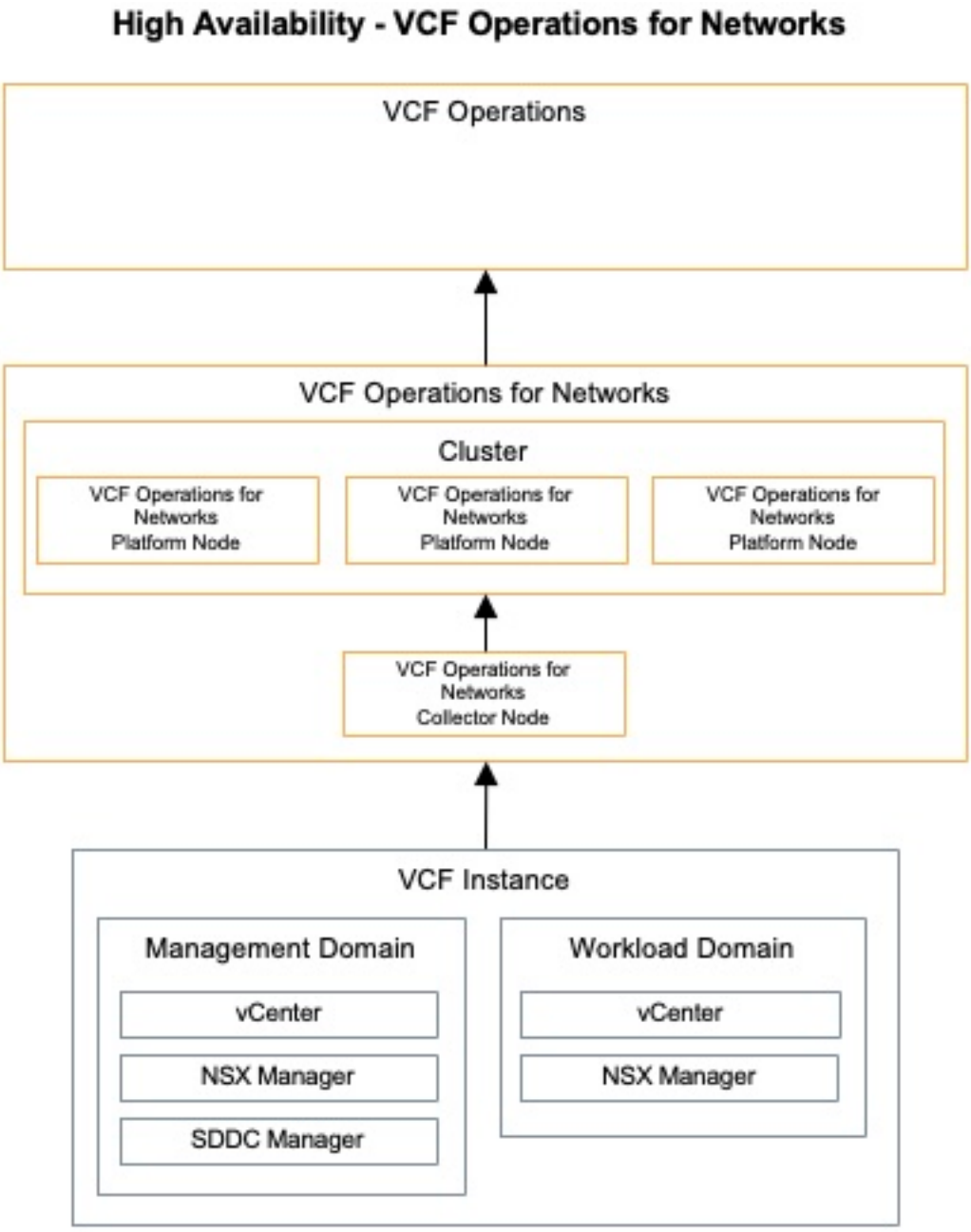
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-SIM-001	Deploy an appropriately sized VCF Operations for networksplatform node for your needs.	Ensures resource availability and usage efficiency.	None.
VCF-NET-RCMD-SIM-002	Protect all VCF Operations for networks nodes by using vSphere High Availability.	Supports the availability objective for VCF Operations for networks without requiring manual intervention during an ESX host failure event.	You must ensure sufficient spare capacity for vSphere High Availability failover operations.
VCF-NET-RCMD-SIM-003	In an environment with multiple VCF Instances, deploy the VCF Operations for networks collector node(s) in the default management vSphere cluster in each VCF Instance by usingVCF Operations fleet management.	• Ensures a consistent deployment in each VCF Instances regarding network data collection. • It is assumed that each VCF Instances introduces a sufficiently high number of virtual machines, so that the deployment of dedicated VCF Operations for networks collector nodes is justified.	Each VCF Operations for networks collector node must be registered with a VCF Operations for networks platform node.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-SIM-004	If the number of virtual machines exceeds 10,000 or if any other object scale limitations are reached, scale out to a VCF Operations for networks platform cluster deployment by adding additional platform nodes by usingVCF Operations fleet management.	Ensures that the VCF Operations for networks platform cluster has enough capacity to meet the SDDC object growth.	You must consider the additional resources required by VCF Operations for networks platform nodes in the management vSphere cluster when scaling out to a platform cluster.
VCF-NET-RCMD-SIM-005	For each vCenter data source, enable NetFlow on each vSphere Distributed Switch within the workload domain.	Provides the collection of network flows via the IPFIX protocol.	For the VCF domains, VCF Operations for networks will automatically update the NetFlow settings for each cluster's vSphere Distributed Switch.
VCF-NET-RCMD-SIM-006	For each NSX data source, enable IPFIX for the distributed firewall.	Provides the collection of network flows via the IPFIX protocol.	The distributed firewall service must be enabled on the NSX Local Manager for the VCF workload domain.
VCF-NET-RCMD-SIM-007	For each NSX Manager data source, enable latency metric collection.	Provides the collection of latency metrics from NSX Transport Nodes.	Any firewall rule sets from all ESX hosts to the VCF Operations for networks collector must allow traffic on TCP 1991.
VCF-NET-RCMD-SIM-008	Use the default retention period of one month for the VCF Operations for networks platform node.	Keeping the default value for retention minimizes required storage resources administrative efforts.	None.
VCF-NET-RCMD-SIM-009	Use a CA-Signed certificate containing the fully qualified domain names (FQDNs) of each VCF Operations for networks platform and collector node in the SAN attributes, when deploying VCF Operations for networks.	Configuring a CA-Signed certificate ensures that the communication to the externally facing web UI and API VCF Operations for networks for is encrypted.	• Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered. • Each time a node is added the certificate must be replaced to include the fully qualified domain name of the additional node.
VCF-NET-RCMD-SIM-010	Use an SHA-2 or higher algorithm when signing certificates.	The SHA-1 algorithm is considered less secure and has been deprecated.	Not all certificate authorities support SHA-2.

High Availability VCF Operations for Networks Model

The *High Availability VCF Operations for Networks Model* consists of a three-node deployment. This model provides increased performance and scalability. Availability is provided by the native vSphere HA functionality within the vSphere cluster.

Figure 321: High Availability VCF Operations for Network Model in VMware Cloud Foundation



High Availability VCF Operations for Networks Model Attributes

A *High Availability VCF Operations for Networks Model* has the following attributes.

Attribute	Detail
Availability	Relies on vSphere HA to restart the nodes in the event of an ESX host failure.
Scalability	Supports scale-up and scale-out.
Recoverability	Restore through a backup.

High Availability VCF Operations for Networks Model Options

A *High Availability VCF Operations for Networks Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Appliance Size	• Large • Extra Large

High Availability VCF Operations for Networks Model Design Requirements

Table 690: High Availability VCF Operations for Networks Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-HA-001	Deploy threeVCF Operations for networksplatform nodes in a cluster.	• Highly Performant. • Highly scalable, can scale up to 15 Extra Large nodes.	• Cluster does not provide high availability. It is for performance. • Requires higher CPU and memory reservations to support additional nodes.
VCF-NET-REQD-HA-002	Deploy a single VCF Operations for networks collector node in the default management vSphere cluster.	A minimum of one VCF Operations for networks collector node is required to provide the VCF Operations for networks platform node with network monitoring data.	A 100% reservation for CPU and RAM for the deployed VCF Operations for networks collector node will be applied.

High Availability VCF Operations for Networks Model Design Recommendations

Table 691: High Availability VCF Operations for Networks Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-001	Deploy appropriately sized VCF Operations for networksplatform nodes for your needs.	Ensures resource availability and usage efficiency.	None.
VCF-NET-RCMD-HA-002	Protect all VCF Operations for networks nodes by using vSphere High Availability.	Supports the availability objective for VCF Operations for networks without requiring manual intervention during an ESX host failure event.	You must ensure sufficient spare capacity for vSphere High Availability failover operations.
VCF-NET-RCMD-HA-003	In an environment with multiple VCF Instances, deploy the VCF Operations for networkscollector node(s) in the default management vSphere cluster in each VCF Instance by using VCF Operations fleet management.	• Ensures a consistent deployment in each VCF Instances regarding network data collection. • It is assumed that each VCF Instances introduces a sufficiently high number of virtual machines, so that the deployment of dedicated VCF Operations for networks collector nodes is justified.	Each VCF Operations for networks collector node must be registered with a VCF Operations for networks platform node.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-HA-004	For each vCenter data source, enable NetFlow on each vSphere Distributed Switch within the workload domain.	Provides the collection of network flows via the IPFIX protocol.	For the VCF domains, VCF Operations for networks will automatically update the NetFlow settings for each cluster's vSphere Distributed Switch.
VCF-NET-RCMD-HA-005	For each NSX data source, enable IPFIX for the distributed firewall.	Provides the collection of network flows via the IPFIX protocol.	The distributed firewall service must be enabled on the NSX Local Manager for the VCF workload domain.
VCF-NET-RCMD-HA-006	For each NSX Manager data source, enable latency metric collection.	Provides the collection of latency metrics from NSX Transport Nodes.	Any firewall rule sets from all ESX hosts to the VCF Operations for networks collector must allow traffic on TCP 1991.
VCF-NET-RCMD-HA-007	Use the default retention period of one month for the VCF Operations for networks platform node.	Keeping the default value for retention minimizes required storage resources administrative efforts.	None.
VCF-NET-RCMD-HA-008	Use a CA-Signed certificate containing the fully qualified domain names (FQDNs) of each VCF Operations for networks platform and collector node in the SAN attributes, when deploying VCF Operations for networks.	Configuring a CA-Signed certificate ensures that the communication to the externally facing web UI and API VCF Operations for networks for is encrypted.	• Using CA-signed certificates from a certificate authority might increase the deployment preparation time as certificate requests are generated and delivered. • Each time a node is added the certificate must be replaced to include the fully qualified domain name of the additional node.
VCF-NET-RCMD-HA-009	Use an SHA-2 or higher algorithm when signing certificates.	The SHA-1 algorithm is considered less secure and has been deprecated.	Not all certificate authorities support SHA-2.

License Server Detailed Design

License server appliances serve as a repository to store VCF component licenses.

License server does not always need to be online or connected to VCF Operations. It is required for the following license operations:

- Licensing new ESX hosts or vCenters (they remain in evaluation mode (90 days) until expiration or the license server can be replaced).
- Generate updated usage files.
- Import new license files.

A VMware Cloud Foundation platform can include one of the two supported *License Server Models*. This section describes the design requirements, recommendations, and choices for each model.

- Single License Server Model
- Multiple License Server Model

Single License Server Model

The *Single License Server Model* is the default configuration for VMware Cloud Foundation platform, which is designed to provide centralized license management for the entire fleet. It's suitable for most typical deployments where reduced operational overhead is prioritized.

Table 692: Single License Server Model Attributes

Attribute	Detail
Use Cases	This is the typical license server deployment model for VMware Cloud Foundation and is suitable for most VCF deployments.
Deployment mechanism	One license server per VCF fleet. A license server is automatically deployed as part of the installation or upgrade to VMware Cloud Foundation 9.1.
Backup and restore	Backup is not required. If the original license server becomes unavailable, deploy a new license server instance and migrate the licenses within 90 days**.**
Lifecycle	The upgrade and patch management of the license server is managed through VCF Operations.
High Availability	High availability is provided by vSphere HA.
License Scope	A single license server provides licensing management for the whole VCF fleet.

Single License Server Model Design Requirements

Table 693: Single License Server Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LIC-REQD-LS-001	Register the default license server in VCF Business Services console.	Registration allows the VCF Business Services console to add license allocations to the license server.	None.

Multiple License Server Model

The *Multiple License Server Model* supports complex environments requiring administrative segregation for multi-tenancy or geographic distribution. This model ensures local license availability for regional locations or edge sites that lack reliable connectivity to the central management domain.

Table 694: Multiple License Server Model Attributes

Attribute	Detail
Use Cases	• Supports licensing for workload domains in edge sites that lack reliable connectivity to the central management domain. • Supports distributed physical locations to ensure local availability and compliance with regional data residency requirements. • Provides dedicated license servers for different customers or internal business units within a single VCF fleet in a multi-tenant environment. • The deployment size exceeds the maximum configuration that is supported by a single license server.
Deployment mechanism	Multiple license servers per VCF fleet. Additional license server must be deployed using an OVA template in a vSphere environment in the VCF fleet.
Lifecycle	The upgrade and patch management of the license server is managed through VCF Operations.
Backup and restore	Backup is not required. Deploy a new license server and migrate the licenses, or migrate the licenses to an existing license server in VCF Business Services console within 90 days. When licenses migrate between license server, license reassignment is not required.
High Availability	High availability is provided by vSphere HA.
License Scope	Each license server manages the licensing for one or multiple workload and management VCF domains.

Multiple License Server Model Design Requirements

Table 695: Multiple License Server Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-LIC-REQD-LS-002	Register all license servers in VCF Business Services console.	Registration allows the VCF Business Services console to add license allocations to the license server.	None.

Identity Broker Detailed Design

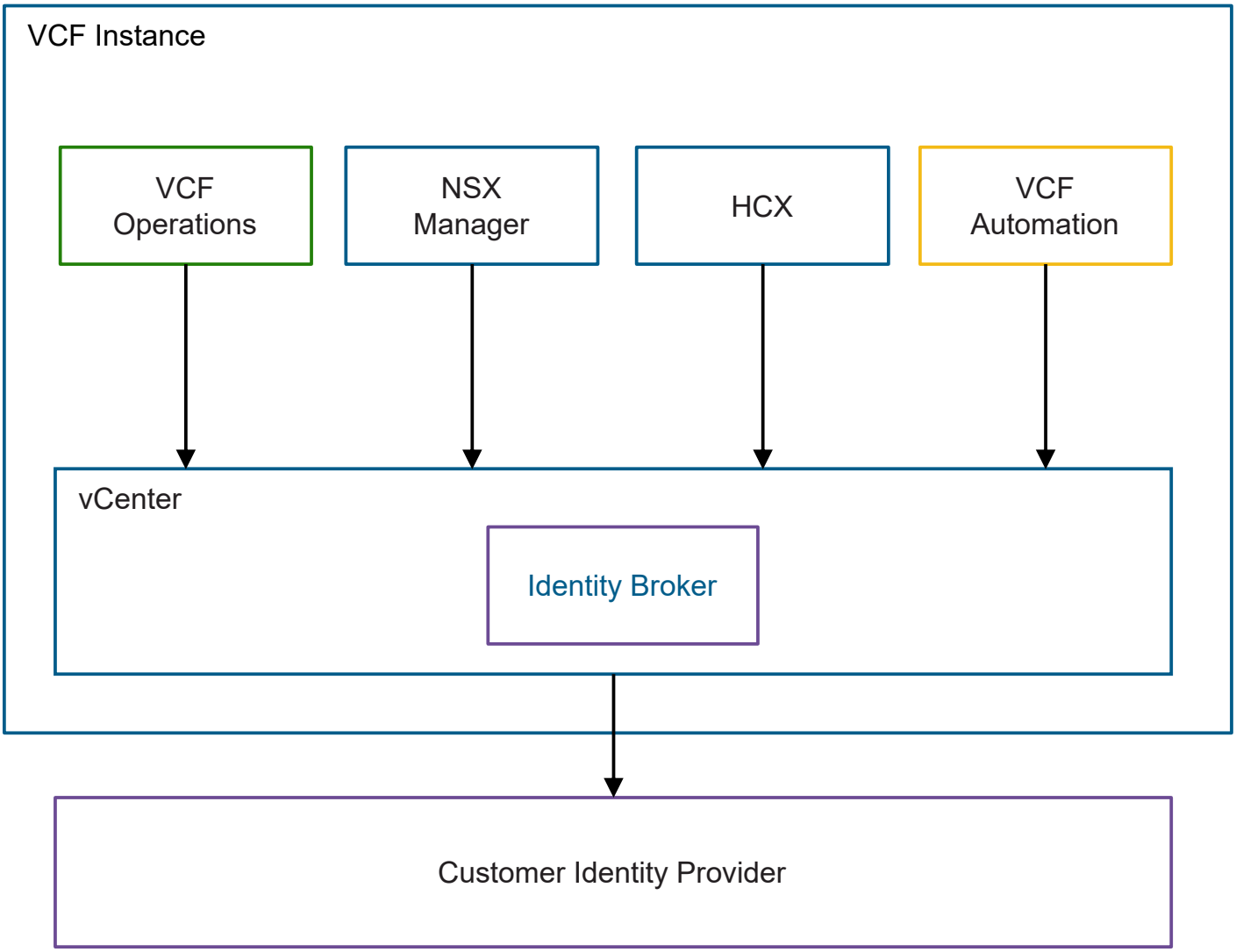
A VMware Cloud Foundation platform can include one of the two supported *Identity Broker Models*. This section describes the design requirements, recommendations, and choices for each *Identity Broker Model*.

- Embedded Identity Broker Model
- Instance Identity Broker Model

Embedded Identity Broker Model

The *Embedded Identity Broker Model* is the simplest form of configuring VCF Single Sign-On for VMware Cloud Foundation. This deployment model is intended for VMware Cloud Foundation lab environments or proof-of-concept (POC) use cases.

Figure 322: Embedded Identity Broker Model in VMware Cloud Foundation



Embedded Identity Broker Model Attributes

The *Embedded Identity Broker Model* has the following attributes.

Attribute	Detail
Service Location	Identity broker runs as a service within the management domain vCenter.
Deployment mechanism	As an embedded service running on vCenter, no additional deployment is required.
Number of Nodes	1 (As part of vCenter)
Scale considerations	One identity broker per VCF Instance.
Supported services	For support directories and IdPs, see Identity Providers and Protocols Supported for Single Sign-On.
Backup and restore	As part of the vCenter backup and restore process.
Resiliency	Resiliency is provided by vSphere HA.
Operational Impacts	As this service run as part of vCenter, maintenance tasks performed on the vCenter instance also impact your ability to authenticate to the VMware Cloud Foundation components.
Migration path	Migration from embedded to the instance model is performed via a streamlined wizard in VCF Operations. This process preserves the existing VCF Single Sign-On configuration and user permissions, eliminating the need for a VCF Single Sign-On configuration reset.

Embedded Identity Broker Model Design Requirements

Table 696: Embedded Identity Broker Model Design Requirements

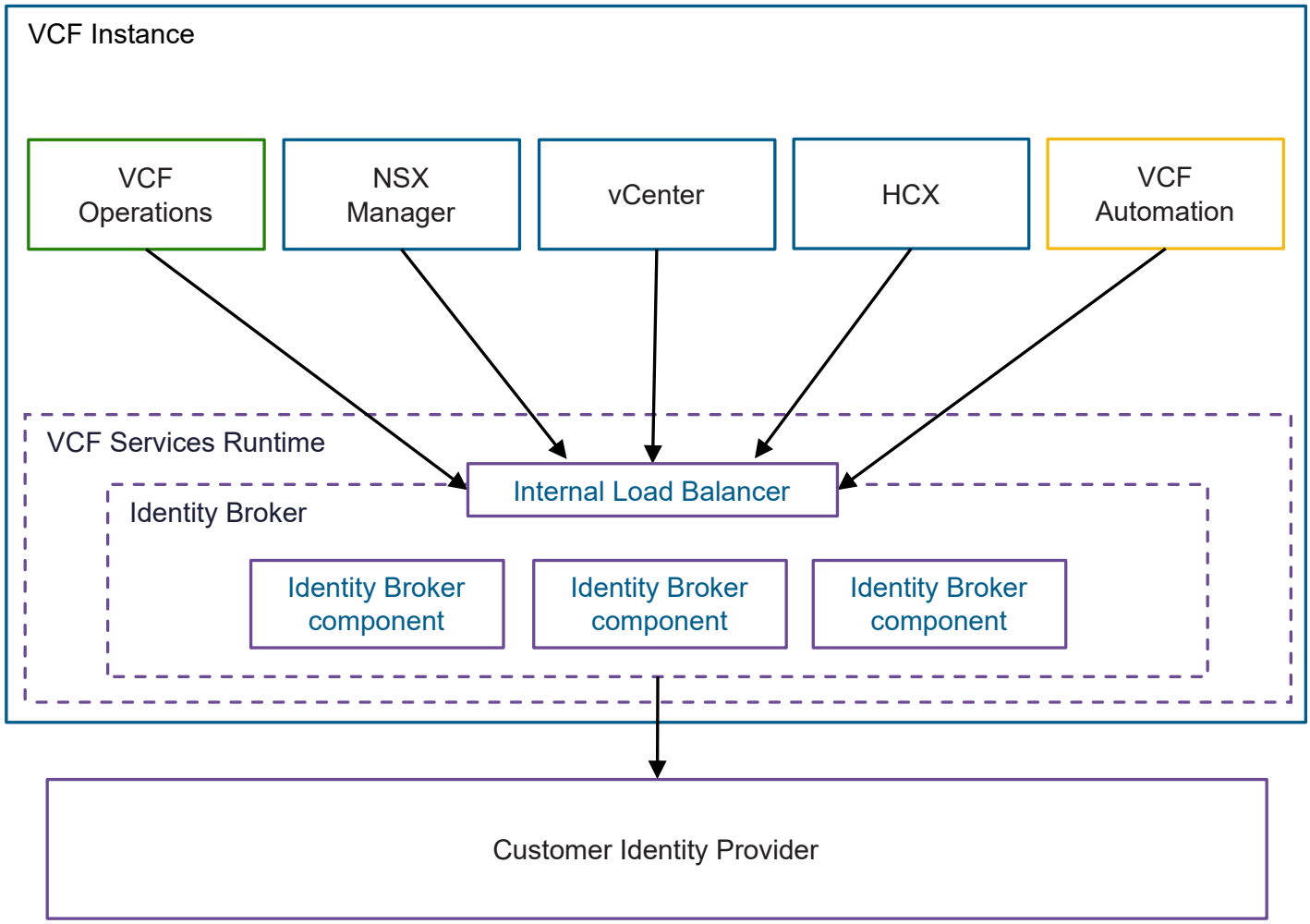
Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-EMB-001	Use the embedded identity broker in the first management domain vCenter.	Simplifies the operational overhead.	• Limits the scale that is supported by the embedded model. • Limits the availability of the authentication service in a VMware Cloud Foundation platform.
VCF-SSO-REQD-EMB-002	Connect the embedded identity broker to your corporate IdP.	Provides directory service (users and group) and user authentication service to the identity broker service.	None.

Instance Identity Broker Model

To avoid a single point of failure for your authentication services, you can use the *Instance Identity Broker Model*. This option is recommended for a production VMware Cloud Foundation deployment.

> **Important:**
>
> The *Instance Identity Broker Model* deployment includes a built-in internal load balancer. An additional load balancer is not required or recommended.

Figure 323: Instance Identity Broker Model in VMware Cloud Foundation



Instance Identity Broker Model Attributes

The *Instance Identity Broker Model* has the following attributes.

Attribute	Detail
Service Location	Identity broker runs as a service on VCF Management Services.
Deployment mechanism	An identity broker is automatically deployed during the installation or upgrade to VMware Cloud Foundation 9.1 in the first management domain. Additional identity brokers can be deployed in each VCF Instance as part of a day 2 operation.
Number of Nodes	multi-node cluster.
Scale considerations	• Scale up: Identity broker supports small, medium, and large sizes to match different configuration demands. • Scale out: Support for multiple identity broker instances to increase redundancy and throughput. • Recommendations: – Scaling up is preferred over scaling out when a single large-size identity broker meets security, resilience, and performance standards, as it reduces operational complexity and simplifies SSO management. – Schedule scale-up operations within a maintenance window, as these activities can impact active services.
Supported services	For support directories and IdPs, see Identity Providers and Protocols Supported for Single Sign-On.
Backup and restore	Use VCF management services backup and restore process.
Resiliency	Resiliency is provided by a distributed multiple-node deployment running each node on separate vSphere hosts.
Operational Impacts	As an isolated, external service, operational tasks performed on vCenter do not impact the authentication stack of the VMware Cloud Foundation components.
Migration path	Migrating from the appliance to embedded requires a reset of the VCF Single Sign-On configuration. Users must be re-added to components.

Instance Identity Broker Model Design Requirements

Table 697: Instance Identity Broker Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-EXT-001	Deploy additional identity broker instances as a Day N operation as needed.	• Provides high availability to the authentication source for a VCF Instance. • Provides additional scale.	Increases the operational overhead of managing the identity broker.
VCF-SSO-REQD-EXT-002	Connect the identity broker service to your corporate IdP.	Provides directory services (users and group) and user authentication service to the identity broker service.	None.

VCF Single Sign-On Detailed Design

VMware Cloud Foundation supports for multiple VCF Single Sign-on Models. This section describes the options, requirements, and recommendations for each VCF Single Sign-on Models. For each of these VCF Single Sign-on Models, both *Identity Broker Models* are supported (embedded and instance), however it is recommended to use the Instance Identity Broker Model for multi-VCF Instance Single Sign-On deployments due to availability and scale.

SSO Realm

Each Identity broker operates within its own SSO realm in a strict 1:1 mapping. This SSO realm defines the logical boundary within which user authentication, identity tokens, and single sign-on sessions are shared and trusted.

An SSO realm can span across multiple VCF instances. However, all components belonging to a single VCF instance must belong to the same SSO realm:

- Workload Domain Components (e.g., vCenter , NSX Manager): Must connect to the same Identity broker instance within their VCF instance. Split-SSO configurations—where different components within the same VCF instance point to separate identity brokers for SSO—are not supported.
- Fleet Management Components (e.g., VCF Operations, VCF Automation): Can connect to any Identity broker instance within the fleet. However, connecting to the Identity broker running on their local VCF instance is strongly recommended.

Instance Identity Broker Failure Recovery Considerations

When planning for VCF SSO failure and disaster recovery (DR), options for recovery and supportability depend on the specific failure scenario and SSO realm boundaries.

Failure Scenario 1: VCF management services Cluster Failure

If a VCF management services cluster failure causes the identity broker to stop functioning while the underlying VCF instance remains operational, you can perform an in-place restoration of the identity broker onto the restored VCF management services cluster using the latest Identity broker configuration backup.

Failure Scenario 2: VCF Instance Failure

When a VCF instance hosting an Identity broker experiences a failure, recovery depends on whether the VCF instance itself is being restored:

- Restoration option 2a: VCF Instance Restoration Required

When VCF Instance restoration is required, the impacted Identity broker can be restored as part of the VCF instance restoration workflow using the latest Identity broker configuration backups.

- Restoration option 2b: VCF Instance Restoration Not Required

When VCF instance restoration is not required, the impacted Identity broker can be restored to another existing VCF instance within the same SSO realm, or to a non-SSO enabled VCF instance, using the latest Identity broker configuration backup. Please note that restoring an identity broker to a VCF instance in a different SSO realm is not supported.

Failure Scenario 3: VCF Fleet Failure

When a VCF fleet experiences a widespread failure, any impacted Identity broker can be restored as a part of the overall VCF Fleet restoration process using the Identity broker configuration backups.

During all failure scenarios, local accounts and VCF SSO emergency access client (for API access) remain functional.

Supported Single Sign-On Model

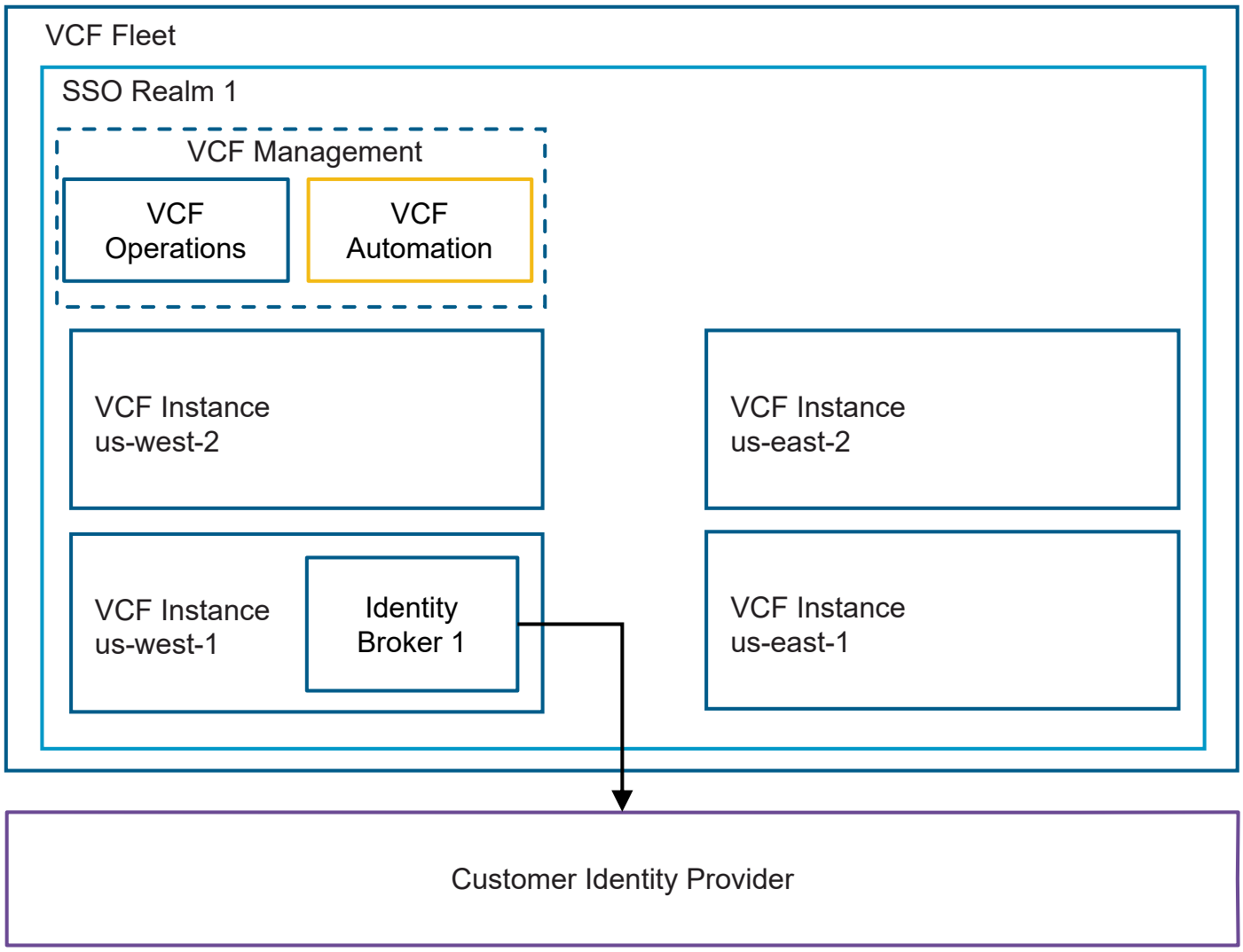
- VCF Fleet-Wide Single Sign-On Model
- Cross VCF Instance Single Sign-On Model
- Single VCF Instance Single Sign-On Model

VCF Fleet-Wide Single Sign-On Model

With the *VCF Fleet-Wide Single Sign-On Model* each VCF fleet contains a single identity broker deployment. A user can log in once and gain VCF Single Sign-On capabilities for all components across all VCF Instances in the same VCF fleet.

For each of these *VCF Single Sign-On Models*, both *Identity Broker Models* are supported (embedded and appliance), however it is recommended to use the Instance Instance Identity Broker Model for multi-VCF Instance Single Sign-On deployments due to availability and scale.

Figure 324: VCF Fleet-Wide Single Sign-On Model in VMware Cloud Foundation



VCF Fleet-Wide Single Sign-On Model Attributes

The *VCF Fleet-Wide Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**Low Scale -** A single identity broker services all VCF Instances in your VCF fleet.
Resources requirements	**Low Resource Footprint** - A single identity broker deployment is required for one VCF fleet.
Operational Overhead	**Low Operational Overhead** - An identity broker per VCF fleet will need to be managed separately for operational tasks such as backup and restore.
Availability	**Low Availability per Private Cloud** - A single identity broker servicing all VCF Instances in your VCF fleet.
Blast Radius	**Large Blast Radius** - An outage of one identity broker instance will impact logins for a single VCF fleet.
Single Sign-On Realm	**Large SSO Realm** - Users do not need to re-authenticate when moving between different VCF Instances within the same VCF fleet.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a only): Because the identity broker is hosted on the primary VCF instance, VCF instance restoration is required; Option 2b is not applicable. • Failure Scenario 3

VCF Fleet-Wide Single Sign-On Model Design Recommendations

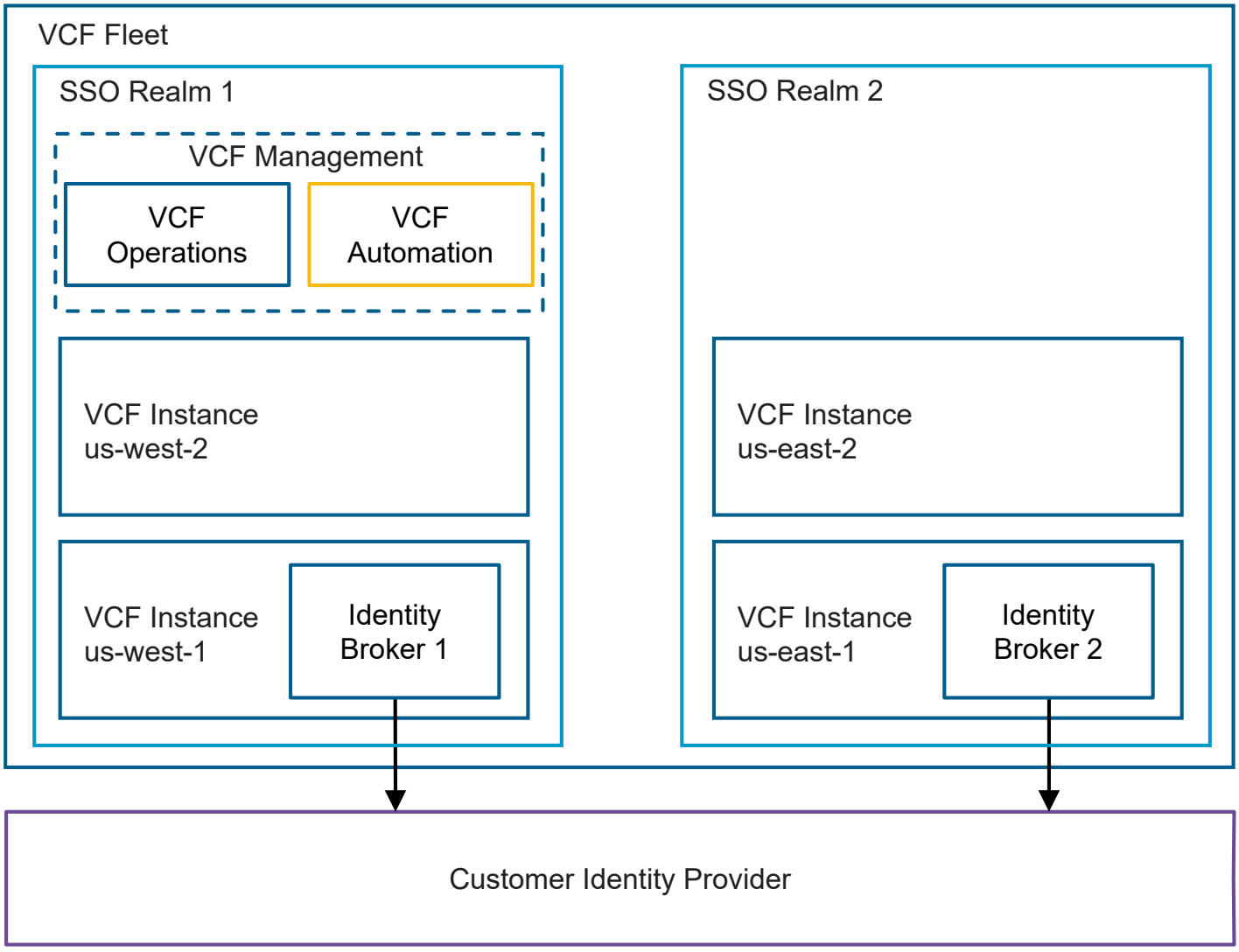
Table 698: VCF Fleet-Wide Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VFL-001	Configure additional VCF Instances in the VCF fleet to consume the same identity broker configured in the first VCF Instance.	• Provides an authentication source for all components in the VCF fleet. • Provides VCF Single Sign-On capabilities to all components in the VCF fleet.	An outage to the identity broker service impacts the VCF fleet.

Cross VCF Instance Single Sign-On Model

The *Cross VCF Instance Single Sign-On Model* includes multiple identity broker instances. Each identity broker instance serves a defined set of VCF instances in the same VCF fleet. VCF management components, such as VCF Operations and VCF Automation, can connect to only one of identity broker instances for Single Sign-On.

Figure 325: Cross VCF Instance Single Sign-On Model in VMware Cloud Foundation



Cross VCF Instance Single Sign-On Model Attributes

The *Cross VCF Instance Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**Balanced Scale -** A single identity broker services a set of VCF Instances in your VCF fleet.
Resources requirements	**Balanced Resource Footprint** - Additional identity broker deployments will require additional compute / memory / storage.
Operational Overhead	**Balanced Operational Overhead** - Additional identity broker will need to managed separately for operational tasks such as backup and restore.
Availability	**Balanced Availability per Private Cloud** - A single identity broker servicing a set of VCF Instances in the VCF fleet.
Blast Radius	**Balanced Blast Radius** - An outage of your identity broker deployment will impact authentication across a set of VCF Instances connected to that identity broker.
Single Sign-On Realm	**Balanced SSO Realm** - With one identity broker deployment for a set of VCF Instances, users authentication will carry across instances in the same fleet requiring no additional re-authentication.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a and Option 2b): Restoring to another VCF instance in the same SSO realm is supported • Failure Scenario 3

Cross VCF Instance Single Sign-On Model Design Requirements

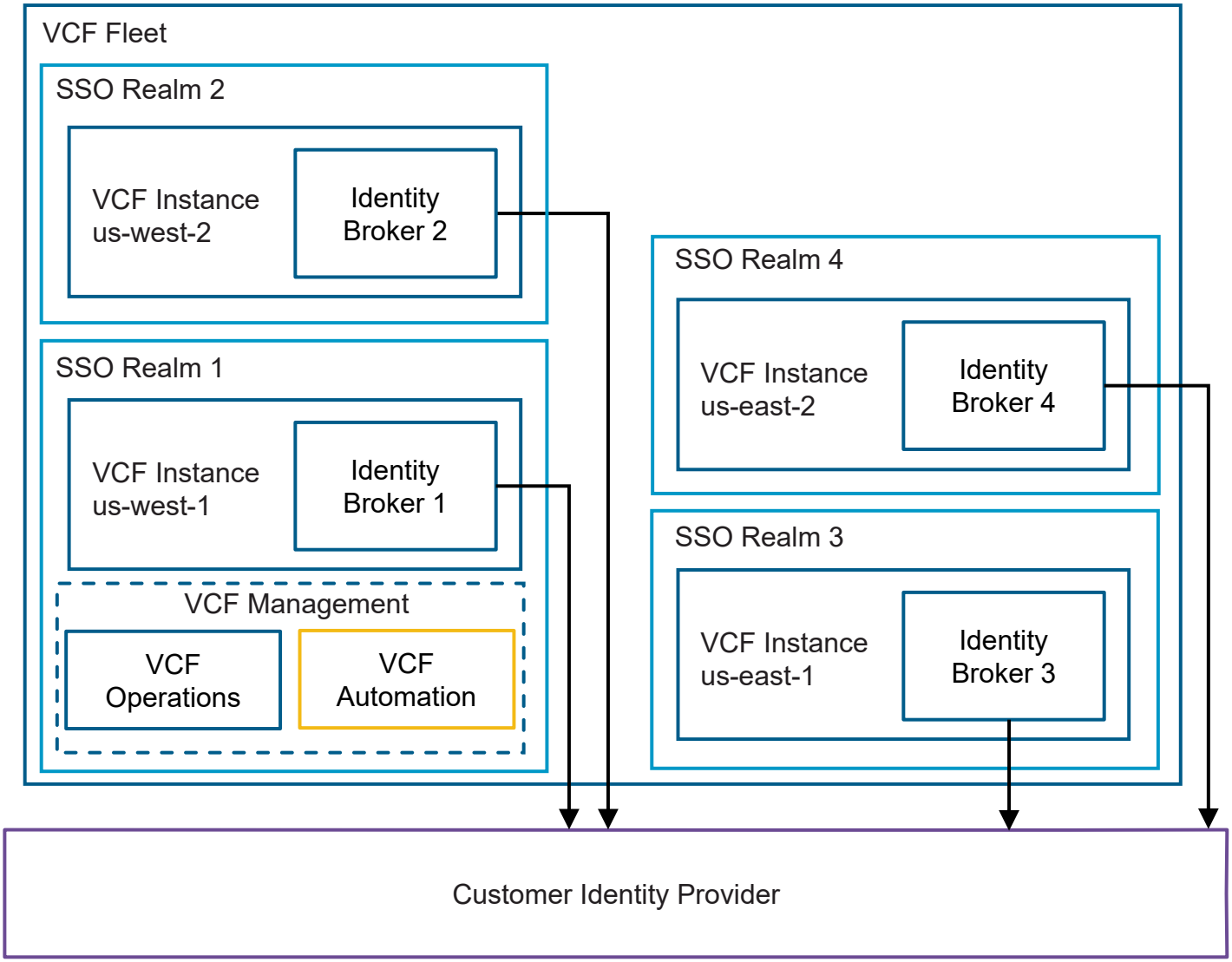
Table 699: Cross VCF Instance Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VCI-001	Deploy additional identity broker in the management domain of a VCF Instance.	• Provides an authentication source for the VCF Instance. • Provides VCF Single Sign-On capabilities to the VCF Instance.	All VMware Cloud Foundation components connected to the additional identity broker use the same authentication source.
VCF-SSO-REQD-VCI-002	Configure additional VCF Instances in the same VCF fleet to consume the additional identity broker.	• Provides an authentication source for all components in the configured VCF Instances. • Provides VCF Single Sign-On capabilities to all components in the configured VCF Instances.	Any outage to the identity broker service impacts all VCF Instances that are configured to use it.

Single VCF Instance Single Sign-On Model

With the *Single VCF Instance Single Sign-On Model* each VCF Instance will utilize its own dedicated identity broker. The VCF Single Sign-On realm will be limited to the VCF Instance. VCF management components, such as VCF Operations and VCF Automation, can connect to only one of identity broker instances for Single Sign-On.

Figure 326: Single VCF Instance Single Sign-On Model in VMware Cloud Foundation



Single VCF Instance Single Sign-On Model Attributes

The *Single VCF Instance Single Sign-On Model* has the following attributes.

Attribute	Detail
Scale	**High Scale -** A single identity broker service for each VCF Instance.
Resources requirements	**Large Resource Footprint** - Each identity broker deployment will require additional compute / memory / storage resources.
Operational Overhead	**High Operational Overhead** - Each identity broker will need to be managed separately for operational tasks such as backup and restore.
Availability	**High Availability per Private Cloud** - A single identity broker servicing each VCF Instance.
Blast Radius	**Small Blast Radius** - An outage of one identity broker instance will only impact logins for the VCF Instance associated with it.
Single Sign-On Realm	**Small SSO Realm** - For each identity broker deployment users will have an additional Single Sign-On realm. Moving across instances will require re-entering credentials.
Disaster Recovery	Supported DR scenarios: • Failure Scenario 1 • Failure Scenario 2 (Restoration Option 2a only): Option 2b is not a viable option because restoring across different SSO realms is not supported. • Failure Scenario 3 Note: local accounts and VCF SSO emergency access client (for API access) continue to work during outages.

Single VCF Instance Single Sign-On Model Design Requirements

Table 700: Single VCF Instance Single Sign-On Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implications
VCF-SSO-REQD-VSI-001	Deploy identity broker in the management domain of the first VCF Instance.	• Provides an authentication service for the first VCF Instance. • Provides VCF Single Sign-On capabilities to the VCF Instance.	All workload domains in a VCF Instance use the same authentication source as the management domain.
VCF-SSO-REQD-VSI-002	Configure VCF Instances to consume the identity broker installed in its management domain.	• Provides an authentication source for all components in the configured VCF Instance. • Provides VCF Single Sign-On capabilities to all components in the configured VCF Instance.	Any outage to the identity broker service impacts all components in the VCF Instance.

Information Security Detailed Design

You design management of access controls, certificates, and accounts for VMware Cloud Foundation according to the requirements of your organization.

- Access Management Design
- Account Management Design
- Certificate Management Design

Access Management Design

Each component in a VMware Cloud Foundation platform has specific access methods. From a security perspective, you must be aware of these access methods to ensure you only grant access to each component to the right users.

Table 701: Access Methods for each VCF Component

Component	Access Method	Additional Information
SDDC Manager	• UI • API • SSH	SSH is active by default. **root** user access is not permitted.
NSX Local Manager	• UI • API • SSH	SSH is deactivated by default.
NSX Edge	• API • SSH	SSH is deactivated by default.
NSX Global Manager	• UI • API • SSH	SSH setting is defined during deployment.
vCenter	• UI • API • SSH	SSH is active by default.
ESX	• Direct Console User Interface (DCUI) • ESX Shell • SSH • VMware Host Client	SSH and ESX shell are deactivated by default.
VCF Operations	• UI • API • SSH	SSH is deactivated by default.
Cloud Proxy	• API • SSH	SSH is deactivated by default.
Log Management	• UI • API	UI is via VCF Operations.
VCF Operations for networks	• UI • API • SSH	SSH is active by default. **root** user access is not permitted.
VCF Automation	• UI • API • SSH	SSH is active by default.
Identity Broker	• UI • API	UI is via VCF Operations.
vSphere Supervisor	• UI • API • CLI • SSH	• UI is via vCenter • CLI requires Kubernetes CLI Tools • SSH is active by default.
VCF Services Runtime	• API • SSH	SSH is active by default.

Account Management Design

You design account management for your VMware Cloud Foundation platform according to industry standards and the requirements of your organization.

Account Management

VMware Cloud Foundation comprises multiple types of accounts: OS accounts, application accounts, and inter-component accounts. Each account has different attributes and is managed according to the specific methods outlined in the table.

Table 702: Account and Password Management in VMware Cloud Foundation

Component	User Account	Password Management	Additional Information
SDDC Manager/VCF Installer	admin@local	• Update by using the VCF Operations UI or API. • Default Expiry: Never	• Local appliance account. • API access (break-glass account).
	vcf	• Update by using the VCF Operations UI or API. • Default Expiry: 365 days	• Local appliance account. • OS level access.
	root	• Update by using the VCF Operations UI or API. • Default Expiry: 365 days	• Local appliance account. • OS level access.
	backup	• Update by using the VCF Operations UI or API. • Default Expiry: 365 days	• Local appliance account. • OS level access.
	svc-sddclcm-sddcm-uuid (when firstly created) svc-vcf-sddc-lcm-SDDC_MANAGER-*uid* (after first rotation)	• System managed. • Automatically rotated every 30 days by default. • Default Expiry: 90 days	Service account between SDDC-LCM and SDDC Manager
	svc-opssr-sddcm-*uid*	• System managed. • Automatically rotated every 30 days by default. • Default Expiry: 90 days	Service account between VCF Operations and SDDC Manager
	svc-vcfa-sddcm-*uid*	• System managed. • Automatically rotated every 30 days by default. • Default Expiry: 90 days	Service account between VCF Automation and SDDC Manager
	administrator@vsphere.local	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• vCenter Single Sign-On account. • Application and API access.
NSX Local Manager	admin	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level, API, and application access.
	root	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access.
	audit	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access. • Read-only application level access.

Component	User Account	Password Management	Additional Information
	svc-*sddc-manager-hostname-nsx-vip-hostname-uid*	• System managed. • Automatically rotated every 30 days by default. • Default Expiry: 90 days	Service account between SDDC Manager and NSX.
	svc-ops-nsx-*uid*	• System managed.	Service account between VCF Operations and NSX.
	svc-ni-nsx-*uid*	• System managed.	Service accounts between VCF Operations for networks and NSX.
	svc-vcfa_*uid*	• System managed.	Service account between VCF Automation and NSX.
	svc-*hcx-hostname*-nsx-hostanme-hcx-uuid	• System managed.	Service account between HCX and NSX.
NSX Edge	admin	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level, API, and application access.
	root	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access.
	audit	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access. • Read-only application level access.
NSX Global Manager	admin	• Manual by using the NSX Global Manager UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level, API, and application access.
	root	• Manual by using each NSX Global Manager appliance. • Default Expiry: 90 days	• Local appliance account. • OS level access.
	audit	• Manual by using the NSX Global Manager UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access. • Read-only application level access.
vCenter	root	• Update or remediate by using the VCF OperationsUI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access. • VAMI access.
	administrator@vsphere.local	• Update by using the VCF Operations UI or API. • Default Expiry: Never	• vCenter Single Sign-On account. • Application and API access. • Relevant to isolated workload domain.
	svc-*sddc-manager-hostname-vcenter-hostname*@vsphere.local	• System managed.	• Service account between SDDC Manager and vCenter. • Administrator role.
	nsxt_cm-*uid*	• System managed.	• Service account between NSX Manager and vCenter. • Administrator role.

Component	User Account	Password Management	Additional Information
	svc-ops-vc-*vcf-ops-cluster-uid*@vsphere.local	• System managed.	• Service account between VCF Operations and vCenter. • Administrator role.
	svc-vcfa_*uid*@vsphere.local	• System managed.	• Service account between VCF Automation and vCenter. • Administrator role.
	svc-vcfsp-vc-*uid*@vsphere.local	• System managed.	• Service account between VCF Services Runtime and vCenter. • VCF Services Runtime role.
	svc-vcfsp-vc-admin-*uid*@vsphere.local	• System managed.	• Service account between VCF Services Runtime and vCenter. • VCF Services Runtime Admin role.
	svc-*hcx-hostname-vc-hostname-hcx-uuid*	• System managed.	• Service account between HCX and vCenter. • Administrator role.
ESX	root	• Update by using the VCF Operations UI or API. • Default Expiry: 99999 (never)	Local root account.
	svc-vcf-*esx-hostname*	• Update by using the SDDC Manager UI or API.	Service account between SDDC Manager and the ESX host.
VCF Operations	admin	• Update by using the VCF Operations UI or API. • Default Expiry: Never	API and application access.
	root	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access.
	svc-salt-ops-*uid*	• System managed	Service Account between Salt RaaS and VCF Operations.
	svc-ops-li-ops-*uid*	• System managed	Service Account between Log Management and VCF Operations.
	svc-ops-ni-ops-*uid*	• System managed	Service Account between VCF Operations for Networks and VCF Operations.
VCF Operations Collectors	root	• Update by using the VCF Operations UI or API. • Default Expiry: 90 days	• Local appliance account. • OS level access.
VCF Services Runtime	vmware-system-user	• Update by using the VCF Operations UI or API. • Default Expiry: 365 days	• Local appliance account. • OS level access.
	admin@vsp.local	• Update by using the VCF Operations UI or API. • Default Expiry: 365 days	• VCF Services Runtime API access
VCF Automation	admin	• Update by using the VCF Operations UI or API. • Default Expiry: never	• Admin account for the VCF Automation system org.

Component	User Account	Password Management	Additional Information
	vmware-system-user	• Update by using the VCF Operations UI or API. • Default Expiry: 365 days	• Local appliance account. • OS level access.
	admin@vsp.local	• Update by using the VCF Operations UI or API. • Default Expiry: 365 days	• VCF Services Runtime API access
Log Management	svc-ops-ops-li-*uid*	• System managed	Service Account between VCF Operations and Log Management.
VCF Operations for Networks	admin@local	• Update by using the VCF Operations UI or API. • Default Expiry: Never	API and application access.
	root	• Manual by using the OS. • Default Expiry: 99999 (never)	• Local appliance account. • OS level access.
	support	• Update by using the VCF Operations UI or API. • Default Expiry: 99999 (never)	• Local appliance account. • OS level access.
	consoleuser	• Update by using the VCF Operations UI or API. • Default Expiry: 99999 (never)	• Local appliance account. • cli access.
	svc-ops-ops-ni-*uid*	• System managed	Service Account between VCF Operations for Networks and VCF Operations.
VCF Operations for Networkscollector	support	• Update by using the VCF Operations UI or API. • Default Expiry: 99999 (never)	• Local appliance account. • OS level access.
	consoleuser	• Update by using the VCF Operations UI or API. • Default Expiry: 99999 (never)	• Local appliance account. • cli access.
Salt RaaS	svc-ops-salt-*uid*	• System managed	Service Account between VCF Operations and Salt RaaS.
vSphere Supervisor	vmware-system-user	Encoded in the control plane namespace secret.	• Local appliance account. • OS level access.
HCX	root	• Update by using the VCF Operations UI or API. • Default Expiry: Never	• Local appliance account. • OS level access.
	admin	• Update by using the VCF Operations UI or API. • Default Expiry: Never	• Local appliance account. • HCX Manager Appliance Management UI access.
	svc-*sddc-manager-hostname*-*hcx-hostname*-*uuid*	• System managed	Service account between SDDC Manager and HCX.
Avi Load Balancer	admin	• Update by using the VCF Operations UI or API. • Default Expiry: Never	API, application, and OS access.
	vcfops-admin	• Update by using the VCF Operations UI or API. • Default Expiry: Never	Used for VCF Operations and Avi Load Balancer integration.

For more information on account and password management, see the VCF Fleet Management Blueprint Account and Password Management.

Account Management Design Recommendations

In your account management design, you can apply certain best practices.

Table 703: Account and Password Management Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SEC-RCMD-ACT-001	Regularly update credentials as per customer password standard.	Reduces the risk of credential compromise and limits the window of opportunity for attackers.	None.
VCF-SEC-RCMD-ACT-002	Store all VCF credentials in an Enterprise Vault system when using API or manual processes.	Ensure that the vault remains the authoritative source of truth for all secrets.	An enterprise vault system is required.
VCF-SEC-RCMD-ACT-003	Monitor and alert on credential retrieval attempts from SDDC Manager via API or CLI access using VCF Operations for Logs	Direct retrieval of break-glass credentials is commonly considered a security and compliance risk in highly regulated environments.	None.
VCF-SEC-RCMD-ACT-004	Use VCF Operations built-in alerts and notifications mechanism to actively monitor password expiration.	Automated monitoring ensures that administrators receive advanced warning of pending password expirations, allowing for timely rotation and preventing unexpected downtime.	Requires integrations with the notification plugin in VCF Operations for outbound alerting.
VCF-SEC-RCMD-ACT-005	Use Audit Records in VCF Operations to centrally trace the password lifecycle (generation, rotation, retrieval) and correlate operations with associated actors and timestamps.	Ensures a verifiable audit trail for compliance and accountability for all credential lifecycle events.	None.
VCF-SEC-RCMD-ACT-006	Configure Logs Management to retain all credential-related events in accordance with regulatory or organizational requirements.	These logs are critical for historical audit validation and compliance reviews.	Requires Logs Management be deployed.
VCF-SEC-RCMD-ACT-007	Assign Password "Manage" and "View" permissions to specific user roles within based on their job requirements.	Ensures users only have the access they need, reducing the risk of unauthorized changes or accidental errors.	None.
VCF-SEC-RCMD-ACT-008	Once the policy is active, update your current password immediately to ensure compliance.	VCF Password Policy complexity settings (password length and character limits) don't apply to the existing passwords.	None.
VCF-SEC-RCMD-ACT-009	Periodically review the password policy settings (complexity, lockout, expiration, history).	Ensure that password policies remain aligned with evolving internal security standards and external regulatory requirements.	None.
VCF-SEC-RCMD-ACT-010	Establish a Standard Operating Procedure (SOP) to handle password policy configuration drift detected by VCF Operations.	Provides the team with a clear playbook to quickly identify an unauthorized change and remediate to a known good state.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SEC-RCMD-ACT-011	Use VCF-managed inter-component accounts for integration between components whenever possible.	Reduces operational overhead and minimizes human error as the platform natively handles account creation and ongoing password management.	None.
VCF-SEC-RCMD-ACT-012	Do not change inter-component account passwords or permissions directly on VCF components.	Manual changes may break the integration between VCF components and may cause unnecessary service outages.	None.
VCF-SEC-RCMD-ACT-013	Use password auto-rotation for inter-component accounts managed by SDDC Manager.	Reduces operational overhead and minimizes human error by ensuring security compliance without manual intervention.	The rotation frequency is limited to 30 days, 60 days, or 90 days.

Certificate Management Design

You design certificate management for VMware Cloud Foundation according to industry standards and the requirements of your organization.

Access to all management component interfaces must be over a Transport Layer Security (TLS) connection. During deployment, each component is assigned a certificate from a default signing CA. Replace the default certificate with a trusted enterprise CA-signed certificate to provide secure access to each component.

Table 704: Certificate Management in VMware Cloud Foundation

Component	Default Signing CA	Life cycle for CA-Signed Certificates
SDDC Manager	Management domain VMCA	VCF Operations
NSX Local Manager	Management domain VMCA	VCF Operations
NSX Edges	Self-Signed	Manual
NSX Global Manager	Self-Signed	Manual
vCenter	Workload domain VMCA (Non-ELM) Management domain VMCA (ELM)	VCF Operations
ESX	Workload domain VMCA	VCF Operations
VCF Operations	Management domain VMCA	VCF Operations
License Server	Default license server: Management domain VMCA Additional license server: self-signed	VCF Operations
VCF Automation	Management domain VMCA	VCF Operations
Log Management	Management domain VMCA	VCF Operations
VCF Operations for Networks	Management domain VMCA	VCF Operations
VCF Management Services	Management domain VMCA	VCF Operations
Identity Broker	Management domain VMCA	VCF Operations
vSphere Supervisor	Workload domain VMCA	Automatic / Manual
VCF Operations HCX	Management domain VMCA	VCF Operations
Avi Load Balancer	Management domain VMCA	VCF Operations

Certificate Management Recommendations

In your account management design, you can apply certain best practices.

Table 705: Certificate Management Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-SEC-RCMD-CERT-001	Perform certificate management for all supported components exclusively through VCF Operations.	The practice aligns with the fleet management model and provides a unified console and APIs.	None.
VCF-SEC-RCMD-CERT-002	Avoid using the SDDC Manager for certificate management; use SDDC Manager APIs only for backward compatibility during the transition to VCF Operations.	The SDDC Manager UI is deprecated in VCF 9.0. Restricting its API use to legacy automation ensures a structured migration to the modern VCF Operations fleet management API.	None.
VCF-SEC-RCMD-CERT-003	Do not use vCenter (API, UI, CLI) to manage ESX host or vCenter certificates once those certificates are being managed by VCF Operations.	Direct management at the component level creates orphaned CSRs in vCenter and results in a "Source of Truth" conflict.	None.
VCF-SEC-RCMD-CERT-004	Default CA-signed certificates on VCF management components should be replaced with certificates that are issued by a trusted enterprise Certificate Authority before going to production.	Using trusted certificates establishes a consistent trust baseline and reduces operational friction during the lifecycle management and compliance activities.	None.
VCF-SEC-RCMD-CERT-005	Use VCF Operations built-in alerts and notifications mechanism to actively monitor certificate expiration.	Automated monitoring ensures that administrators receive advanced warning of pending certificate expirations, allowing for timely rotation and preventing unexpected downtime.	Requires integrations with the notification plugin in VCF Operations for outbound alerting.
VCF-SEC-RCMD-CERT-006	Configure Log Management to retain all certificate-related events in accordance with regulatory or organizational requirements.	These logs are critical for historical audit validation and compliance reviews.	Requires Log Management be deployed.
VCF-SEC-RCMD-CERT-007	Use Audit Records in VCF Operations to centrally trace the certificate lifecycle operations.	Provides a forensic audit trail for compliance verification and incident response across all certificate lifecycle events.	None.
VCF-SEC-RCMD-CERT-008	Assign Certificate "Manage" and "View" permissions to specific user roles within VCF Operations based on their job requirements.	Ensures that users only have the access that they need, reducing the risk of unauthorized changes or accidental errors.	None.

For more information, see the VCF Fleet Management Blueprint Certificate Management.

VCF Domain Detailed Design

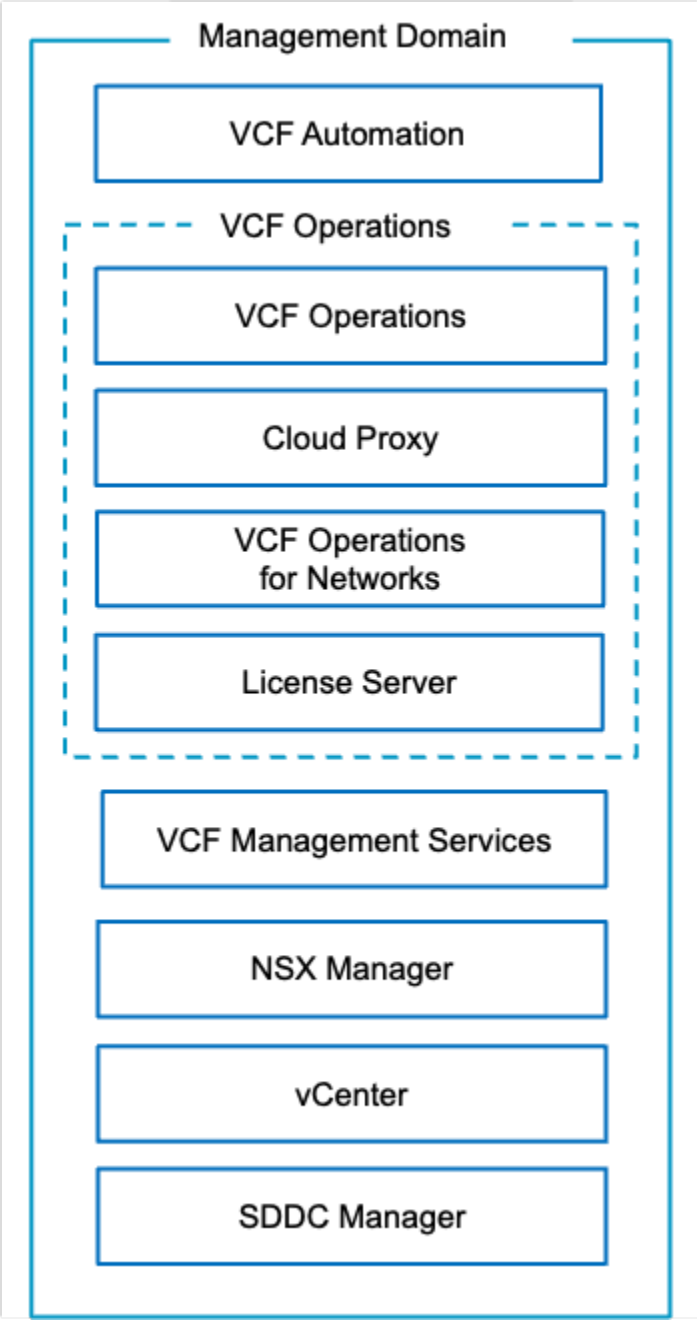
A VCF fleet can include one or more VCF domains that are implemented according to a supported *VCF Domain Model*. Each *VCF Domain Model* has options, requirements, and recommendations.

- Management Domain Model
- Workload Domain Model

Management Domain Model

The management domain in a VCF fleet hosts the management components of the VCF Instance where it is deployed and the management components of the fleet.

Figure 327: Management Domain Model in VMware Cloud Foundation



Management Domain Model Attributes

The *Management Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• VCF fleet management components. • VCF Instance management components. • NSX Edge nodes. • Can also run business workloads.
Management Domain in the First VCF Instance	• The management domain in the first VCF Instance contains the following management components: – VCF management services (fleet and instance level components) – VCF Operations – Cloud Proxy – VCF Operations for Networks – VCF Automation – ESX hosts – vCenter – NSX Manager – SDDC Manager
Management Domains in Additional VCF Instances	• Management domains in additional VCF Instances contain the following instance-level management components: – VCF management services (instance level components) – Cloud Proxy – ESX hosts – vCenter – NSX Manager – SDDC Manager

Management Domain Model Options

A *Management Domain Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
vSphere Cluster Models	• First cluster: – Single-Rack vSphere Cluster Model – Multi-Rack Layer 2 vSphere Cluster Model – Stretched vSphere Cluster Model (Day 2) • Additional clusters: – All cluster models.
VCF management services Models	• Simple VCF Management Services Model • High Availability VCF Management Services Model • VCF Management Services in the First VCF Instance Model • VCF Management Services in an Additional VCF Instance Model
VCF Operations Models	• Simple VCF Operations Model • High Availability VCF Operations Model
VCF Operations for Networks Models	• Simple VCF Operations for Networks Model • High Availability VCF Operations for Networks Model
VCF Automation Models	• Simple VCF Automation Model • High Availability VCF Automation Model
NSX Manager Models	• Simple NSX Manager Model • High Availability NSX Manager Model

Management Domain Model Design Requirements and Recommendations

Table 706: Management Domain Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-WLD-REQD-CFG-001	Deploy a management domain to host all management components for the VCF Instance.	• Guarantees sufficient resources for management components. • Enables the use of specific hardware to meet only the needs of the management components. • Enables the use of dedicated physical compute, network, and storage separately from those used for additional workloads. • Enables separate lifecycle management of management and workload components.	• You must carefully size the management domain to accommodate planned deployments of additional workload domains and management components. • Hardware might not be fully utilized until full-scale deployment has been reached. • The management components for all greenfield VCF domains are deployed to the same cluster, and share the same subnet.

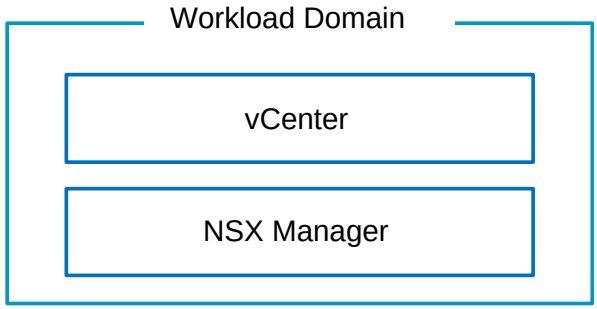
Table 707: Management Domain Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-MGMT-RCMD-NSX-001	Use a dedicated NSX Manager instance for the management domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Managers to manage and patch.

Workload Domain Model

As you expand your VCF fleet, you add workload domains to a VCF Instance. Each workload domain is assigned a vCenter and NSX Manager.

Figure 328: Workload Domain Model in VMware Cloud Foundation



Workload Domain Model Attributes

A *Workload Domain Model* has the following attributes.

Attribute	Details
Resource Allocation	• NSX Edge nodes. • Business workloads.
vCenter Single Sign-On Domain	Dedicated.

Workload Domain Model Options

A *Workload Domain Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
vSphere Cluster Models	• Single-Rack vSphere Cluster Model • Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model • Stretched vSphere Cluster Model
NSX Manager Models	• Simple NSX Manager Model • High Availability NSX Manager Model
NSX Consumption Models	• Dedicated NSX instance. • Shared NSX instance with another workload domain including the management domain
Workload Network Type	• IPv4 • IPv6

Workload Domain Model Recommendations

Table 708: Workload Domain Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-WLD-RCMD-CFG-002	Use workload domains for customer workloads.	• Aligns with the VMware best practice of separating management workloads from customer workloads. • Provides better long-term flexibility and expansion options.	Requires additional hardware.
VCF-WLD-RCMD-NSX-002	Use a dedicated or shared NSX Manager instance for the workload domain.	• Decouples the lifecycle management of the management stack from workload domains. • Isolates workload networking configuration errors or resource exhaustion from impacting the management domain. • Ensures that management operations are not affected by high churn or load in workload domains.	• Requires additional resources to host separate NSX Manager instances for workload domains. • Increases the number of NSX Manager to manage and patch.

vSphere Detailed Design

A VCF Instance includes vCenter and ESX, this section describes the options, requirements, and recommendations for each both.

- vCenter Model
- ESX Model

vCenter Model

The vCenter design considers the location, size, high availability, and identity domain isolation of the vCenter instances for the VCF domains in a VCF fleet.

vCenter Model Attributes

A *vCenter Model* has the following attributes.

Design Area	Attributes
vCenter Deployment Type	• One vCenter instance for the management domain that manages the management components of the VCF Instance. • Optionally, additional vCenter instances for workload domains to support workloads. • vSphere HA protecting all vCenter appliances.
vCenter Single Sign-On	• Each vCenter has a distinct vCenter Single Sign-On domain.

Methods for Protecting vCenter

VMware Cloud Foundation supports multi-rack management vSphere clusters, vSAN stretched clusters, or 3rd party synchronous replication solutions such as metro clusters that are designed to provide disaster recovery and/or high availability for the entire management stack (vCenter, NSX Manager, etc.) as opposed to a single vCenter.

Table 709: Methods for Protecting vCenter in VMware Cloud Foundation

High Availability Method	Supported in VMware Cloud Foundation	Considerations
vSphere High Availability	Yes	You must ensure sufficient resources in the cluster to allow virtual machines to restart on another ESX host in the event of an ESX host failure.
vCenter High Availability (vCenter HA)	No	• vCenter services must fail over to the passive node so there is no continuous availability. • Recovery time can be up to 15 mins. • You must meet additional networking requirements for the private network. • vCenter HA requires additional resources for the passive and witness nodes. • Lifecycle management is complicated because you must manually delete and recreate the standby virtual machines during a lifecycle management operation.
vSphere Fault Tolerance (vSphere FT)	No	• The vCPU limit of vSphere FT vCPU would limit vCenter appliance size to medium. • You must provide a dedicated network.

vCenter Model Design Requirements and Recommendations

Table 710: vCenter Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-CFG-001	Deploy a dedicated vCenter appliance for the management domain of the VCF Instance.	• Isolates vCenter failures to management or customer workloads. • Isolates vCenter operations between management and business workloads. • Supports a scalable vSphere cluster design where you can reuse the management components as more customer workloads are added to the VCF Instance. • Simplifies capacity planning for customer workloads because you do not consider management workloads for the workload domain vCenter. • Improves the ability to upgrade the vSphere environment and related components by enabling for explicit separation of maintenance windows: — Management workloads remain available while you are upgrading the tenant workloads. — Workloads remain available while you are upgrading the management nodes. • Supports clear separation of roles and responsibilities to ensure that only administrators with granted authorization can control the management workloads. • Facilitates quicker troubleshooting and problem resolution. • Simplifies disaster recovery operations by supporting a clear separation between recovery of the management components and tenant workloads. • Provides isolation of potential network issues by introducing network separation of the vSphere clusters in the VCF Instance.	The vSphere cluster must have sufficient resources.
VCF-VCS-REQD-NET-001	Place all workload domain vCenter appliances on the VM management network in the management domain.	• Simplifies IP addressing for management VMs by using the same VLAN and subnet. • Provides simplified secure access to management VMs in the same VLAN network.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VCS-REQD-SSO-001	Create all vCenter instances within a VCF Instance in their own unique vCenter Single Sign-On domain.	• Sharing vCenter Single Sign-On domains is not supported. • Enables isolation at the vCenter Single Sign-On domain layer for increased security separation.	• Each vCenter instance is managed through its own pane of glass using a different set of administrative credentials. • You must manage password rotation for each vCenter Single Sign-On domain separately.

Table 711: vCenter Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VCS-RCMD-CFG-001	Deploy an appropriately sized vCenter appliance for each workload domain.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate compute resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-002	Deploy a vCenter appliance with the appropriate storage size.	Ensures resource availability and usage efficiency per workload domain.	You must ensure that adequate storage resources are available to run all vCenter appliances.
VCF-VCS-RCMD-CFG-003	Protect workload domain vCenter appliances by using vSphere HA.	vSphere HA is the only supported method to protect vCenter availability in VCF.	vCenter becomes unavailable during a vSphere HA failover.
VCF-VCS-RCMD-CFG-004	In vSphere HA, set the restart priority policy for the vCenter appliance to high.	vCenter is the management and control plane for physical and virtual infrastructure. In a vSphere HA event, to ensure that the rest of the SDDC management stack comes up faultlessly, the workload domain vCenter must be available first, before the other management components come online.	If the restart priority for another virtual machine is set to highest, the connectivity delays for the management components are longer.

ESX Model

In the design of the ESX host configuration for your VMware Cloud Foundation platform, consider the resources, networking, and security policies that are required to support the virtual machines in each VCF domain vSphere cluster.

ESX Hardware and Sizing Considerations

For more information on host sizing, refer to the VCF Planning and Preparation Workbook.

Table 712: ESX Hardware and Sizing Considerations

Hardware Element	Considerations
BIOS/Firmware	Ensure the ESX host hardware is on the Hardware Compatability List (HCL) and that you are running the latest supported BIOS and firmware.
CPU	• Total CPU requirements for the workloads that are running in the cluster. • Host failure and maintenance scenarios. • CPU over-commitment ratio guidelines – Management Domain • To ensure optimal performance and maintain resilience during host failures or maintenance, keep the management domain vCPU-to-pCPU over-commitment ratio at or below 2:1. **Note:** While higher ratios such as 3:1 or 4:1 are supported, you must closely monitor CPU ready times to ensure that the management cluster has enough capacity to run the workloads. – Workload domains • CPU over-commitment ratios for workload domains should be calculated on a case by case basis depending on the CPU intensity of the workloads. You must closely monitor CPU ready times to ensure optimal performance. • Number of physical cores, not logical cores. – Simultaneous multi-threading (SMT) in CPUs improve CPU performance however the performance enhancement will not be equivalent to 100% more CPU power.
Memory	• Total memory requirements for the workloads that are running in the cluster. • When sizing memory for the ESX hosts in a cluster, to reserve the resources of one host for failover or maintenance, set the admission control setting to N+1, which reserves the resources of one host for failover or maintenance. • *vSAN* OSA (Original Storage Architecture). – To support the maximum number of disk groups, you must provide 32 GB of RAM. • *vSAN* ESA (Express Storage Architecture). – You must provide a minimum of 128GB of RAM.
Storage	• Use high-endurance devices such as a hard drive or SSD for boot device. • Use a 128GB or greater boot device to maximize the space available for ESX-OS Data. • Use hosts with homogeneous configuration.
Network	• Use multiple physical NICs connected to multiple ToR switches to ensure redundancy. • 25GbE is recommended • Use physical NICs of the same speed, vendor, and model across all host within a cluster.
Power	• Use redundant power supplies, preferably connected to separate circuits to ensure availability. • Consider the impact of enabling Power Management on your ESX hosts. In order to allow ESX to control CPU power-saving features you need to set power management in the BIOS to *OS Controlled Mode*.

ESX Model Design Requirements and Recommendations

In your ESX design, you apply certain best practices.

Table 713: ESX Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-ESX-REQD-CFG-001	Install no less than the minimum number of ESX hosts required for the vSphere cluster model being deployed.	• Ensures that availability requirements are met. • If one of the ESX hosts is not available because of a failure or maintenance event, the CPU over-commitment ratio becomes 2:1.	None.
VCF-ESX-REQD-CFG-002	Ensure each ESX host matches the required CPU, memory, and storage specification.	Ensures that workloads will run without contention even during failure and maintenance conditions.	Assemble the server specification and number according to the sizing in the VCF Planning and Preparation Workbook which is based on projected deployment size.
VCF-ESX-REQD-SEC-001	Regenerate the certificate of each ESX host after assigning the ESX host an FQDN.	Establishes a secure connection with the VCF Installer during the initial deployment and prevents man-in-the-middle (MiTM) attacks.	You must manually regenerate the certificates of the ESX hosts before the deployment.

Table 714: ESX Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-001	When using vSAN as primary storage in the management domain, use vSAN ReadyNodes for each ESX host.	Your management domain is fully compatible with vSAN at deployment. **Note:** For information about the models of physical servers that are vSAN-ready, see vSAN Compatibility Guide for vSAN ReadyNodes.	• Hardware choices might be limited. • If you plan to use a server configuration that is not a vSAN ReadyNode, your CPU, disks, and I/O modules must be listed on the VMware Compatibility Guide under *CPU Series* and *vSAN Compatibility List* aligned to the ESX version.
VCF-ESX-RCMD-CFG-002	Allocate ESX hosts with uniform configuration across the default management vSphere cluster.	A balanced vSphere cluster has these advantages: • Predictable performance even during hardware failures. • Minimal impact of resynchronization or rebuild operations on performance.	You must apply vendor sourcing, budgeting, and procurement considerations for uniform server nodes on a per vSphere cluster basis.
VCF-ESX-RCMD-CFG-003	When sizing CPU, do not consider multi-threading technology and associated performance gains.	Although multi-threading technologies increase CPU performance, the performance gain depends on running workloads and differs from one case to another.	Because you must provide more physical CPU cores, costs increase and hardware choices become limited.
VCF-ESX-RCMD-CFG-004	Install and configure all ESX hosts in the default management vSphere cluster to boot using a 128-GB device or larger.	Provides ESX hosts that have large memory, that is, greater than 512 GB, with enough space for the scratch partition when using vSAN.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-ESX-RCMD-CFG-005	Use the default configuration for the scratch partition on all ESX hosts in the default management vSphere cluster.	• If a failure in the vSAN cluster occurs, the ESX hosts remain responsive and log information is still accessible. • If using vSAN storage its not possible to use a vSAN datastore for the scratch partition.	None.
VCF-ESX-RCMD-CFG-006	For workloads running in the default management vSphere cluster, save the virtual machine swap file at the default location.	Simplifies the configuration process.	Increases the amount of replication traffic for management workloads that are recovered as part of the disaster recovery process.
VCF-ESX-RCMD-NET-001	Place the ESX hosts in each management domain vSphere cluster on the ESX management network that is separate from the VM management network.	• Enables the separation of the physical VLAN between ESX hosts and the other management components for security reasons. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Increases the number of VLANs required.
VCF-ESX-RCMD-NET-002	Place the ESX hosts in each workload domain on a separate ESX host management VLAN-backed network.	Enables the separation of the physical VLAN between the ESX hosts in different workload domains for security reasons.	Increases the number of VLANs required. For each workload domain, you must allocate a separate management subnet.
VCF-ESX-RCMD-NET-003	Use dedicated VLANs for each traffic type.	Enables the separation of the physical VLAN for vSAN, vMotion, NFS & backup traffic.	Increases the number of VLANs required. For each traffic type you must allocate a separate VLAN and subnet.
VCF-ESX-RCMD-SEC-001	Set the advanced setting `UserVars.SuppressShellWarning` to `0` across all ESX hosts in the management domain.	• Ensures compliance with the *vSphere Security Configuration Guide* and with security best practices • Enables the warning message that appears in the vSphere client every time SSH access is activated on an ESX host.	You must turn off SSH enablement warning messages manually when performing troubleshooting or support activities.

vSphere Cluster Detailed Design

A VCF Instance can include one or more of the supported *vSphere Cluster Models*. This section describes the options, requirements, and recommendations for each *vSphere Cluster Model*.

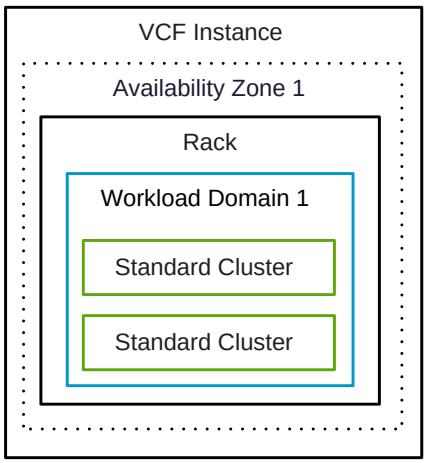
- Single-Rack vSphere Cluster Model
- Multi-Rack Layer 2 vSphere Cluster Model
- Multi-Rack Layer 3 vSphere Cluster Model
- Stretched vSphere Cluster Model

Single-Rack vSphere Cluster Model

The *Single-Rack vSphere Cluster Model* is the simplest form of protection where workloads are deployed to a vSphere cluster in the same physical rack, in a single availability zone.

The *Single-Rack vSphere Cluster Model* relies on vSphere HA to protect workloads against host failures.

Figure 329: Single-Rack vSphere Cluster Model in VMware Cloud Foundation



Single-Rack vSphere Cluster Model Attributes

A *Single-Rack vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter.
Physical Rack Mapping	All cluster hosts exist in one physical rack.
Resilience	vSphere HA protects against host failures.

Single-Rack vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere which supports your storage type and VCF domain considerations.

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
Minimum number of ESX hosts	vSAN	Simple	Three (3)	• 33% CPU and memory • Tolerates one host failure	Three (3)	• 33% CPU and memory • Tolerates one host failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Attribute	Storage Type	VCF Installer Deployment Model	Management Domain (Default Cluster)	Reserved capacity for handling ESX host failures per cluster	Management Domain (Additional Clusters) or Workload Domain (All Clusters)	Reserved capacity for handling ESX host failures per cluster
	NFS or FC	Simple	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure	Two (2)	• 50% CPU and memory • Tolerates one availability zone failure
		High Availability	Four (4)	• 25% CPU and memory • Tolerates one availability zone failure		

Important:

vSphere clusters are configured allowing for the failure of a single ESX host while still having enough resources to run all vSphere cluster workloads.

Attention:

* Designs should be completed in consultation with your storage vendor.

Single-Rack vSphere Cluster Model Options

A *Single-Rack vSphere Cluster Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Storage Models (principal)	• Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model • Single-Rack vSAN Storage Cluster Storage Model • Single-Rack vSAN Compute Cluster Storage Model • Fibre Channel Storage Model • NFS Storage Model

Single-Rack vSphere Cluster Model Design Requirements and Recommendations

Table 715: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Table 716: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Table 717: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Table 718: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX

host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Multi-Rack vSphere Cluster Detailed Design

A VCF Instance can support one or more supported *Multi-Rack vSphere Cluster Models*. This section describes the options, requirements, and recommendations for each *Multi-Rack vSphere Cluster Model*.

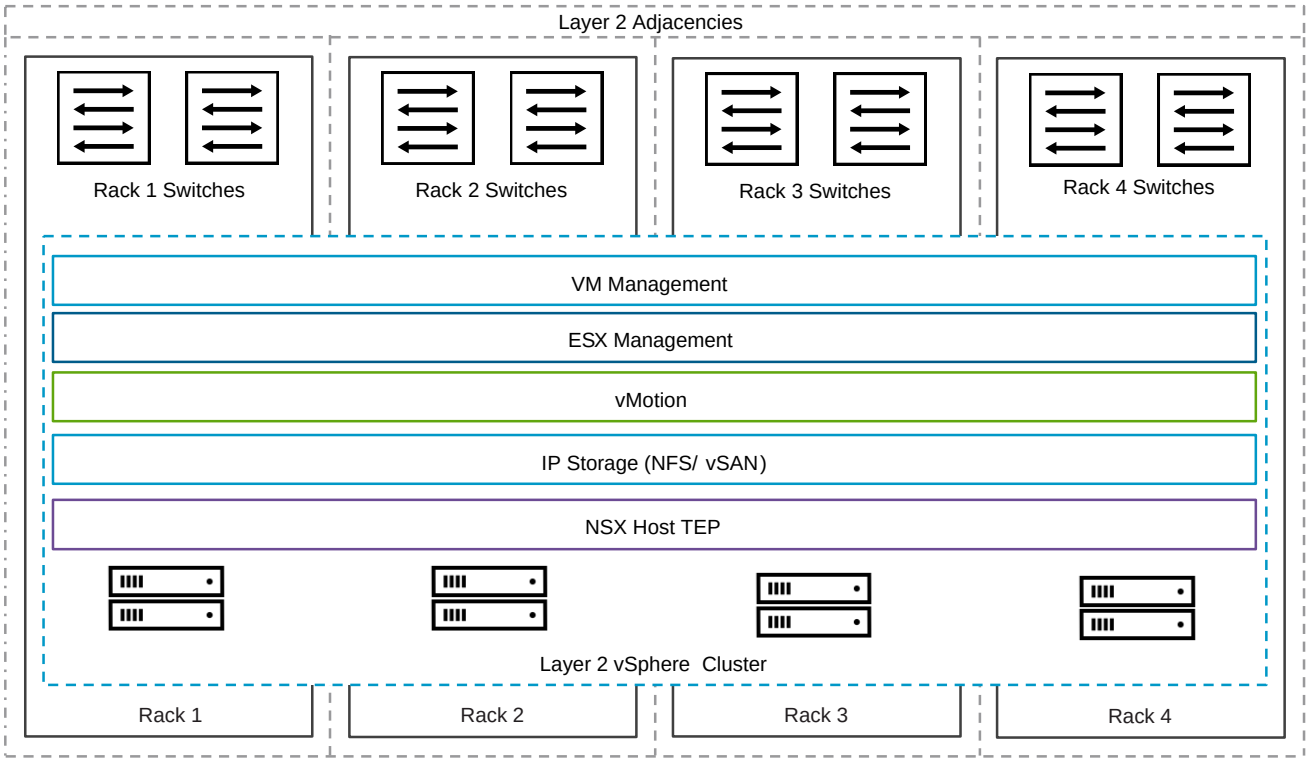
- Multi-Rack Layer 2 vSphere Cluster Model
- Multi-Rack Layer 3 vSphere Cluster Model

Multi-Rack Layer 2 vSphere Cluster Model

The *Multi-Rack Layer 2 vSphere Cluster Model* allows you to protect workloads on vSphere clusters in your environment against a failure of a single fault domain. You implement vSphere cluster across multiple fault domains within a single availability zone. The networks are stretched across racks using a single broadcast domain for each network.

The *Multi-Rack Layer 2 vSphere Cluster Model* incorporates multiple fault domains in the design can help reduce the blast radius of a failure which can increase application availability. In this design the infrastructure networks are stretched across racks, this reduces the number of VLANs and subnets needed for the multi-rack deployment but also uses a single broadcast domain for each network.

Figure 330: Multi-Rack Layer 2 vSphere Cluster Model in VMware Cloud Foundation



Multi-Rack Layer 2 vSphere Cluster Model Attributes

A *Multi-Rack Layer 2 vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter
Cluster Rack Mapping	Hosts are installed horizontally across different physical racks.
Management Domain Clusters	Primary and additional clusters supported.
Resilience	• vSphere HA protects workloads against host failures. • Multiple fault domains protect workloads in the event of a failure of one or more fault domains/racks.
Availability Zones	Cluster can be stretched across availability zones.
Networking	Layer 2 (Including VXLAN/EVPN Fabric) - single broadcast domain for each network.
Storage	Storage capable of surviving the failure of a single fault domain at the storage level.

Multi-Rack Layer 2 vSphere Cluster Model VLANs and Subnets

Function	VLANs and Subnets
VM management	• VLAN stretched across the racks. • Single subnet for each network. • Highly available gateway available across racks.
VCF management	
ESX management	
vSphere vMotion	
vSAN	
Host overlay	
NFS	
NSX Edge Uplink01	See NSX Edge node design for details Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster
NSX Edge Uplink02	
NSX Edge overlay	

Multi-Rack Layer 2 vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere cluster which supports your storage type and workload domain considerations.

Attribute	Storage Type	Management Domain (First Cluster) Simple Deployment of Management Components	Management Domain (First Cluster) HA Deployment of Management Components	Management Domain (Additional Clusters) Workload Domain (All Clusters)
Minimum number of ESX hosts	vSAN	Three (3)	Four (4)	Three (3)
	VMFS on FC* NFS*	Two (2)	Four (4)	Two (2)

Important:

vSphere clusters are configured allowing for the failure of a single ESX host while still having enough resources to run all vSphere cluster workloads.

Attention:

* Designs should be completed in consultation with your storage vendor.

Multi-Rack Layer 2 vSphere Cluster Model Options

A *Multi-Rack Layer 2 vSphere Cluster Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Storage Models (principal)	• Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model • Fibre Channel Storage Model • NFS Storage Model

Multi-Rack Layer 2 vSphere Cluster Model Design Requirements and Recommendations

Table 719: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Table 720: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Table 721: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Table 722: Common Multi-Rack vSphere Cluster Model Design Recommendations

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-MR-RCMD-CFG-001	Use multiple vSphere Distributed Switches.	Provides traffic separation for storage, NSX, and vSphere vMotion traffic.	A minimum of two (2) physical NICs are required per vSphere Distributed Switch.
VCF-CLS-MR-RCMD-CFG-002	Use vSAN ESA as the principal storage.	Provides best performance for workloads.	vSAN ESA-compatible hardware is required.

Table 723: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX

host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

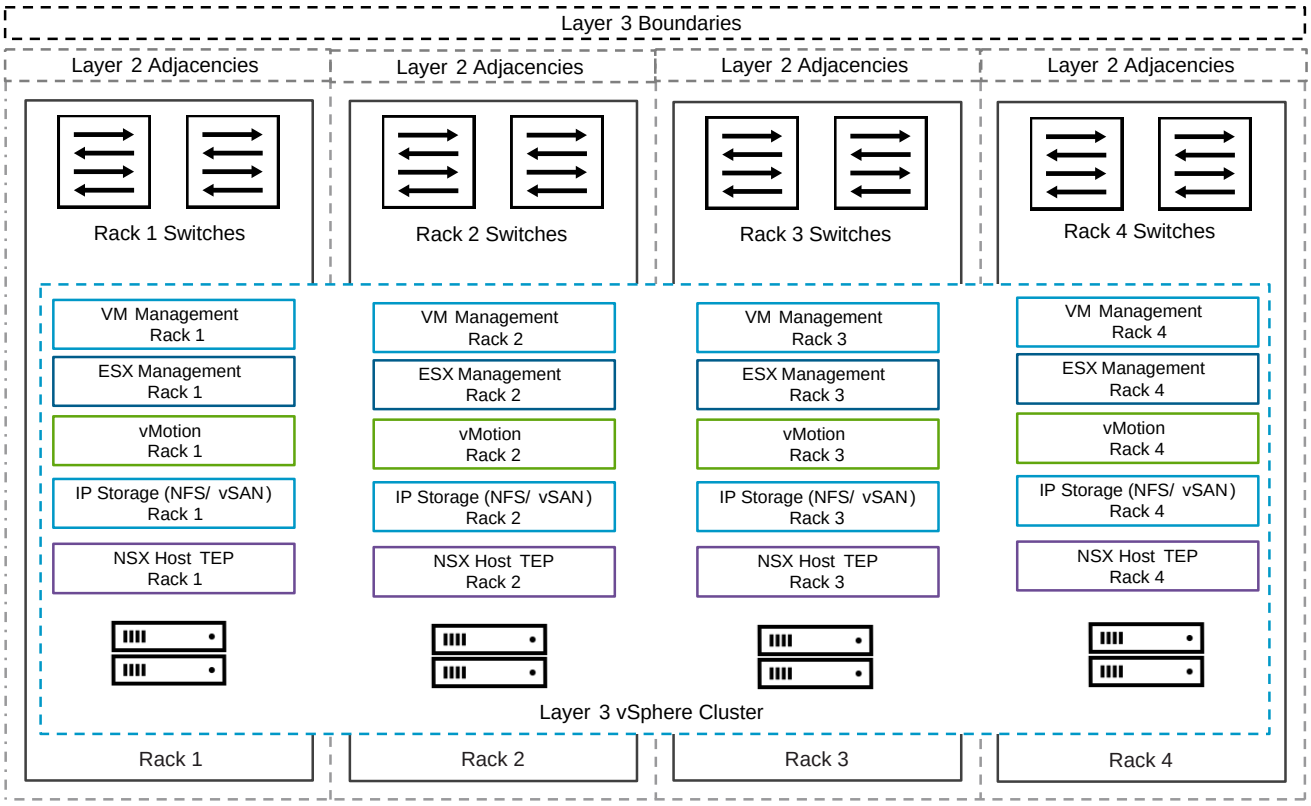
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Multi-Rack Layer 3 vSphere Cluster Model

The *Multi-Rack Layer 3 vSphere Cluster Model* allows you to protect workloads on vSphere clusters in your environment against a failure of a single fault domain by implementing your vSphere cluster across multiple fault domains within a single availability zone. The networks are isolated to a rack using a separate broadcast domain for each network per rack.

The *Multi-Rack Layer 3 vSphere Cluster Model* incorporates multiple fault domains in the design that can help reduce the blast radius of a failure which can increase application availability. In this design the VCF infrastructure networks are not stretched across racks, instead each rack has a separate VLAN and subnet. The network traffic is routed between the racks, this increases the number of VLANs and subnets needed for the layer 3 multi-rack deployment but also uses a single broadcast domain for each network thus reducing the impact of a layer 2 network issue across racks.

Figure 331: Multi-Rack Layer 3 vSphere Cluster Model in VMware Cloud Foundation



Multi-Rack Layer 3 vSphere Cluster Model Attributes

A *Multi-Rack Layer 3 vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter
Cluster Rack Mapping	Hosts are installed horizontally across different physical racks.
Management Domain Clusters	Not supported for primary cluster, additional clusters supported.
Resilience	• vSphere HA protects workloads against host failures. • Multiple fault domains protect workloads in the event of a failure of one or more fault domains/racks.
Availability Zones	Single availability zone only.
Networking	Layer 3 routed networks with isolated broadcast domains per rack.
Storage	Storage capable of surviving the failure of a single fault domain at the storage level.

Multi-Rack Layer 3 vSphere Cluster Model VLANs and Subnets

Function	Network Attributes
VM management	• VLAN local to the top of rack switches per rack. • Separate subnet per rack. • Highly available gateway at the top of rack switches. • Routable network across racks. • *Depends on NFS storage fabric network design.
ESX management	
vSphere vMotion	
vSAN	
Host overlay	
NFS*	
NSX Edge Uplink01	• See NSX Edge node design for details Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster
NSX Edge Uplink02	
NSX Edge overlay	

Multi-Rack Layer 3 vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere cluster which supports your storage type and workload domain considerations.

Attribute	Storage Type	Management Domain (Additional Clusters) Workload Domain (All Clusters)
Minimum number of ESX hosts	vSAN	Three (3)
	VMFS on FC* NFS*	Two (2)

Important:

vSphere clusters are configured allowing for the failure of a single ESX host while still having enough resources to run all vSphere cluster workloads.

Multi-Rack Layer 3 vSphere Cluster Model Options

A *Multi-Rack Layer 3 vSphere Cluster Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Storage Models (principal)	• Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model • Fibre Channel Storage Model • NFS Storage Model

Multi-Rack Layer 3 vSphere Cluster Model Design Requirements and Recommendations

Table 724: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Table 725: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Table 726: Multi-Rack Layer 3 vSphere Cluster Model Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements for a layer 3 multi-rack cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-L3MR-REQD-CFG-001	Create an uplink profile for each rack.	Enables a Layer 3 boundary at the top-of-rack switches for host TEP traffic.	Requires additional subnet per rack.
VCF-VNI-L3MR-REQD-CFG-002	Create a host TEP IP pool for each rack with a subnet that is allocated per rack and a gateway for the subnet.	• Provides the host TEP IP addressing for each rack • Enables a Layer 3 boundary at the top-of-rack switches for host TEP traffic. • Provides routing capabilities for the host TEPs between racks.	Requires additional subnet per rack.
VCF-VNI-L3MR-REQD-CFG-003	Create NSX host sub transport node profiles for additional racks.	Enables NSX host transport node configuration per rack.	None.

Table 727: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Table 728: Common Multi-Rack vSphere Cluster Model Design Recommendations

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-MR-RCMD-CFG-001	Use multiple vSphere Distributed Switches.	Provides traffic separation for storage, NSX, and vSphere vMotion traffic.	A minimum of two (2) physical NICs are required per vSphere Distributed Switch.
VCF-CLS-MR-RCMD-CFG-002	Use vSAN ESA as the principal storage.	Provides best performance for workloads.	vSAN ESA-compatible hardware is required.

Table 729: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Table 730: Multi-Rack Layer 3 vSphere Cluster Model Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations for a layer 3 multi-rack cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

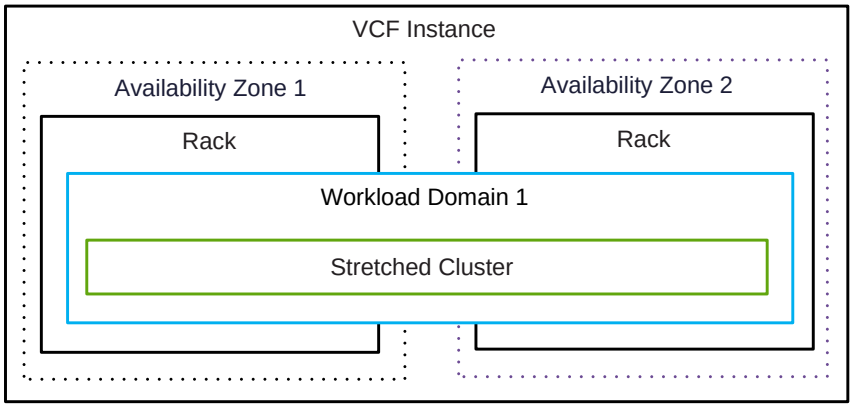
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-L3MR-REQD-CFG-001	Use a separate host TEP transport VLAN ID per rack.	Enables a Layer 3 boundary at the top-of-rack switches for host TEP traffic.	Requires additional host TEP VLAN ID per rack.

Stretched vSphere Cluster Model

The *Stretched vSphere Cluster Model* allows you to protect workloads on vSphere clusters in your environment against a failure of a single fault domain by implementing a stretched vSphere cluster across multiple availability zones.

Incorporating multiple availability zones in your design can help reduce the blast radius of a failure and can increase application availability.

Figure 332: Stretched vSphere Cluster Model in VMware Cloud Foundation



Stretched vSphere Cluster Model Attributes

A *Stretched vSphere Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Multiple datacenters
Cluster Rack Mapping	• VCF domain cluster with multiple availability zones, each zone in a single-rack vSphere cluster. • VCF domain cluster with multiple availability zones, with Layer 2 multi-rack vSphere clusters in each zone.
Resilience	• vSphere HA protects workloads against host failures. • Multiple availability zones protect against availability zone failures.
Bandwidth	• When using vSAN stretched clusters, the bandwidth between the zones must be at least 10 Gbps and the round-trip latency must be less than 5 ms. • When using storage array based stretched clusters, you must work with your storage vendor to define the bandwidth and latency requirements.
Management Domain Requirements	Having the management domain on a stretched cluster is a prerequisite to configure and implement stretched cluster in a workload domain. In this way, the critical management components managing the workload domains are still available if an availability zone failure occurs.
Availability Zones	• You can have up to two availability zones associated with a given stretched vSphere cluster. – When using vSAN stretched clusters, a third independent fault domain is required to host the vSAN witness appliance.

Stretched vSphere Cluster Model VLANs and Subnets

Function	VLANs and Subnets
VM management	Shared across availability zones
VCF management	Shared across availability zones
ESX management	Unique per availability zone
vSphere vMotion	Unique per availability zone
vSAN*	Unique per availability zone
Host overlay	Unique per availability zone
NFS*	Depends on stretched storage design

Note:

*Optional based on the principal storage choice for the cluster.

Stretched vSphere Cluster Model Sizing Considerations

You choose the number of ESX hosts per vSphere cluster which supports your storage type and workload domain considerations.

Storage Type	VCF Installer Deployment Model	Management Domain (First Cluster)	Management Domain (Additional Clusters) Workload Domain (All Clusters)
vSAN	Simple	Six (6)	Six (6)

Storage Type	VCF Installer Deployment Model	Management Domain (First Cluster)	Management Domain (Additional Clusters) Workload Domain (All Clusters)
	Highly Available	Eight (8)	
NFS, FC *	Simple	Refer to KB417356	**Minimum:** • One (1) host in Site A • One (1) host in Site B **Total: 2 ESX hosts** **Recommended:** • Two (2) hosts in Site A • Two (2) hosts in Site B **Total: 4 ESX hosts**
	Highly Available	Refer to KB417356	

Important:

vSphere clusters are configured allowing for the failure of a single availability zone while still having enough resources to run all vSphere cluster workloads.

Attention:

* Storage and networking design requirements should be completed in consultation with your storage vendor.

Stretched vSphere Cluster Model Options

A *Stretched vSphere Cluster Model* can apply options from the design areas that are listed below, depending on your objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Storage Models (principal)	• Stretched vSAN Storage Model • Stretched vSAN Storage Cluster Storage Model • Stretched vSAN Compute Cluster Storage Model • Fibre Channel Storage Model

Stretched vSphere Cluster Model Design Requirements and Recommendations

Table 731: Common vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-001	Create a vSphere cluster in each VCF domain for the initial set of ESX hosts.	• Simplifies configuration by isolating management from customer workloads. • Ensures that customer workloads have no impact on the management stack.	Management of multiple vSphere clusters and vCenter instances increases operational overhead.
VCF-CLS-REQD-CFG-002	Allocate the minimum number of ESX hosts according to the vSphere cluster type being deployed.	Ensures correct level of redundancy to protect against host failure in the vSphere cluster.	To support redundancy, you must allocate extra ESX host resources.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-003	Use vSphere Lifecycle Manager images as the lifecycle management method for all vSphere clusters. **Note:** Imported workload domains may be using vSphere Lifecycle Manager baselines. It is recommended to transition them to use vSphere Lifecycle Manager images.	vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons manually.	A vSphere Lifecycle Manager cluster image is required during workload domain or vSphere cluster deployment.
VCF-CLS-REQD-CFG-004	Use vSphere HA to protect all virtual machines against failures.	vSphere HA supports a robust level of protection for both ESX host and virtual machine availability.	You must provide sufficient resources on the remaining ESX hosts so that virtual machines can be restarted on those hosts in the event of an ESX host outage.

Table 732: Common vSphere Cluster Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-REQD-CFG-001	Configure all ESX hosts in the workload domain as transport nodes in NSX.	Enables distributed routing, logical segments, and distributed firewall.	None.
VCF-VNI-REQD-CFG-002	Configure each ESX host as a transport node using transport node profiles.	• Enables the participation of ESX hosts and the virtual machines running on them connected to VPCs, NSX overlay segments, and VLAN networks. • Transport node profiles can only be applied at the cluster level.	None.
VCF-VNI-REQD-CFG-003	Create a single overlay transport zone in the NSX instance for all overlay traffic across the host and NSX Edge transport nodes of the workload domain or multiple workload domains using a shared NSX instance.	Ensures that all VPCs and Overlay segments are available to all ESX hosts and NSX Edge nodes that are configured as transport nodes.	All clusters in all workload domains that share NSX Manager share overlay transport zone.

Table 733: Stretched vSphere Cluster Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-CLS-REQD-CFG-005	Enable the **Override default gateway for this adapter** setting on the vSAN VMkernel adapters on all ESX hosts.	Enables routing the vSAN data traffic through the vSAN network gateway rather than through the management gateway.	vSAN networks across availability zones must have a route to each other.
VCF-CLS-REQD-CFG-006	Create a host group for each availability zone and add the ESX hosts in the zone to the respective group.	Makes it easier to manage which virtual machines run in which availability zone.	None.

Table 734: Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Requirements

Virtual networking infrastructure requirements for a stretched cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNI-SC-REQD-CFG-001	Create an uplink profile for each availability zone.	Enables different VLAN ids for Host TEP in each availability zone.	Requires additional Host TEP VLAN in availability zone 2.
VCF-VNI-SC-REQD-CFG-002	Create NSX host sub transport node profile for second availability zone.	Enables NSX host transport node configuration per availability zone.	None.

Table 735: Common vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-001	Configure admission control for one (1) ESX host failure and percentage-based failover capacity.	• Using the percentage-based reservation works well in situations where virtual machines have varying and sometimes significant CPU or memory reservations. • vSphere automatically calculates the reserved percentage according to the number of ESX host failures to tolerate and the number of ESX hosts in the vSphere cluster.	In a vSphere cluster of four (4) ESX hosts, the resources of only three (3) ESX hosts are available for use.
VCF-CLS-RCMD-CFG-002	Enable VM Monitoring for each vSphere cluster.	• VM Monitoring provides in-guest protection for most virtual machines. • The application or service running on the virtual machine must be capable of restarting successfully after a reboot or the virtual machine restart is not sufficient.	None.
VCF-CLS-RCMD-CFG-003	Enable vSphere DRS on all vSphere clusters, using the default fully automated mode with medium threshold.	Provides the best trade-off between load balancing and unnecessary migrations with vSphere vMotion.	If a vCenter outage occurs, the mapping from virtual machines to ESX hosts might be difficult to determine.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-004	Enable Enhanced vMotion Compatibility on all vSphere clusters in the management domain.	Supports vSphere cluster upgrades without virtual machine downtime.	• You must enable EVC only if the vSphere clusters contain ESX hosts with CPUs from the same vendor. • You must enable EVC on the default management domain vSphere cluster during bring-up using the API and a JSON spec.
VCF-CLS-RCMD-CFG-006	Set the Enhanced vMotion Compatibility mode on a vSphere cluster to the highest available baseline that is supported for the lowest CPU architecture of the ESX hosts.	Supports vSphere cluster upgrades without virtual machine downtime.	None.

Table 736: Common vSphere Cluster Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations describe the design recommendations for the configuration of NSX host transport nodes in the cluster with Host TEPs to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-RCMD-CFG-001	Use static IP pools to assign IP addresses to the host TEP interfaces.	• Removes the need for an external DHCP server for the host overlay VLANs. • You can use NSX Manager to verify static IP pool configurations.	Not required for VLAN backed VPC clusters.
VCF-VNI-RCMD-CFG-002	Create an uplink profile with a load balance source teaming policy with all active uplinks for ESX hosts.	For increased resiliency and performance, supports the concurrent use of more than one physical NICs on the ESX hosts that are configured as transport nodes.	Not required for VLAN backed VPC clusters.

Table 737: Stretched vSphere Cluster Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-008	Increase admission control percentage to half of the ESX hosts in the vSphere cluster.	Allocating only half of a stretched cluster ensures that all VMs have enough resources if an availability zone outage occurs.	• In a vSphere cluster of eight (8) ESX hosts, the resources of only four (4) ESX hosts are available for use. • If you add more ESX hosts to the default management vSphere cluster, add them in pairs, one per availability zone.
VCF-CLS-RCMD-CFG-009	Create a virtual machine group for each availability zone and add the virtual machines in the zone to the respective group.	Ensures that virtual machines are located only in the assigned availability zone to avoid unnecessary vSphere vMotion migrations.	You must add virtual machines to the allocated group manually.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-CLS-RCMD-CFG-010	Create a should-run-on-hosts-in-group VM-Host affinity rule to run each group of virtual machines on the respective group of hosts in the same availability zone.	Ensures that virtual machines are located only in the assigned availability zone to avoid unnecessary vSphere vMotion migrations.	You must manually create the rules.

Table 738: Stretched vSphere Cluster Model Virtual Networking Infrastructure Design Recommendations

Virtual networking infrastructure recommendations for a stretched cluster describe the design requirements for the configuration of NSX host transport nodes in the cluster to provide virtual networking capabilities including VPC networking and NSX segment networking.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNI-SC-RCMD-CFG-001	Create a host TEP IP pool for each availability zone with a subnet that is allocated per availability zone and a gateway that is configured for the network.	• Provides the host TEP IP addressing for each availability zone. • Enables a Layer 3 routed TEP traffic between availability zones. • Using an external DHCP server for the host overlay VLANs in both availability zones is not required.	Requires additional subnet per availability zone.
VCF-VNI-SC-RCMD-CFG-002	Configure an NSX sub-transport node profile for hosts in availability zone 2.	• The NSX transport node profile can remain attached when using two separate VLANs for host TEPs at each availability zone. • You can use static IP pools for the host TEPs in each availability zone.	Changes to the host transport node configuration are done at the vSphere cluster level.

VCF Network Detailed Design

This section covers all aspects of network design for vSphere clusters, NSX Edge clusters, VPC, and consumption models for workloads deployed in the VCF Instance.

- Distributed Switch Detailed Design
- VMkernel Detailed Design
- Network I/O Control Detailed Design

Distributed Switch Detailed Design

A VCF Instance can support one or more supported *Distributed Switch Models*. This section describes the options, requirements, and recommendations for each *Distributed Switch Model*.

- Single Distributed Switch Model
- Storage Separation Distributed Switch Model
- Workload Separation Distributed Switch Model
- Storage and Workload Separation Distributed Switch Model
- Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model
- vSAN Storage Cluster Separation Distributed Switch Model

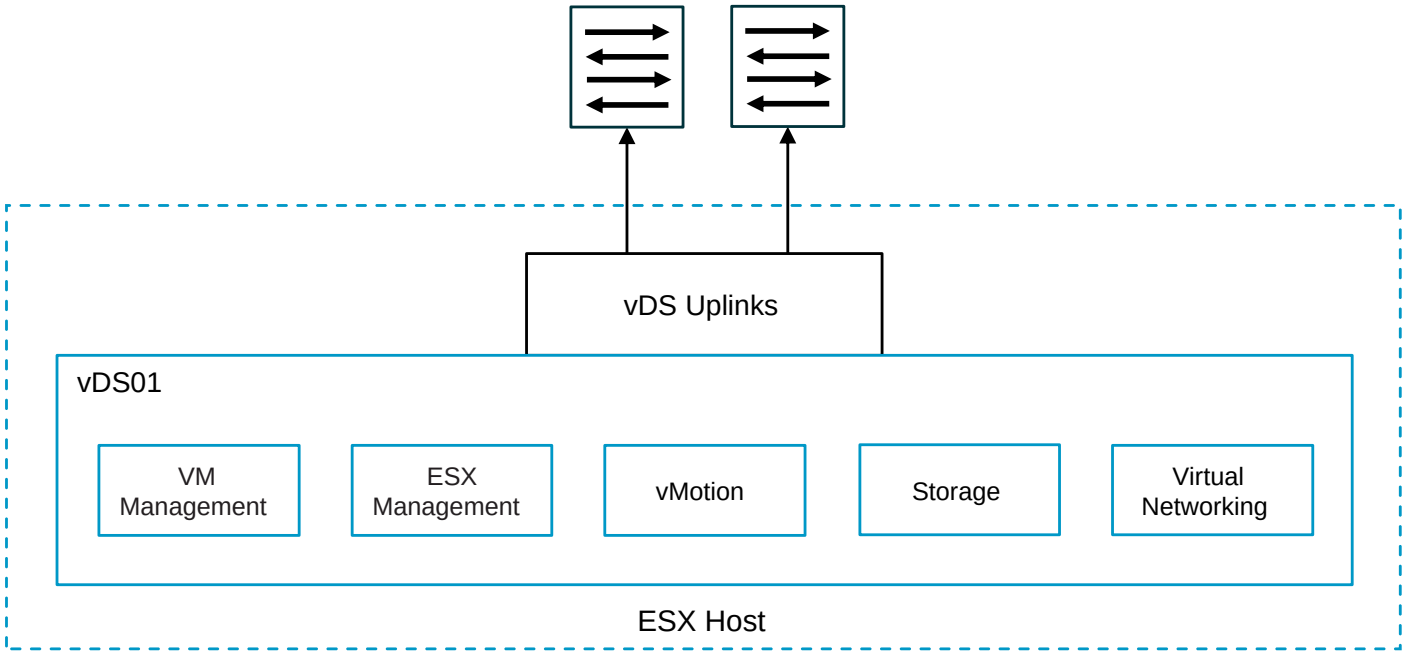
Single Distributed Switch Model

A *Single Distributed Switch Model* in VMware Cloud Foundation provides a minimal deployment of networking using a single vSphere distributed switch for all network services such as management, vSphere vMotion, IP-based storage and, NSX.

A *Single Distributed Switch Model* provides connectivity using the lowest number of physical NICs for each ESX host in the vSphere cluster while still providing availability from a single physical NIC failure, the network services running on the vSphere cluster will share the bandwidth provided by two physical NICs.

The *Single Distributed Switch Model* can be suited to vSphere clusters where the number of physical NICs must be reduced and the overall bandwidth requirements does not exceed that provided by two physical NICs. The bandwidth of the ESX host physical NICs will determine the network performance that can be achieved for the vSphere cluster.

Figure 333: Single Distributed Switch Model in VMware Cloud Foundation



Single Distributed Switch Model Attributes

A *Single Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Single
Number of Physical NICs per ESX Host	Two (2)
Bandwidth	Shared for all network traffic
Network I/O Control	Enabled by default

Single Distributed Switch Model Options

A *Single Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Physical NIC Level Redundancy (NLR)	• Use a single 2-port NIC (no NLR). • Use two separate 2-port NICs. – Each traffic type is spread across two separate NIC to avoid single point of failure. – Results in unused ports on NIC.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Single Distributed Switch Model Design Requirements and Recommendations

Table 739: Common Distributed Switch Design Requirements for VCF 9.1.0

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-COM-REQD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain.	• For each vSphere cluster, you manage more vSphere Distributed Switches. • Does not apply from version 9.1.1 onwards because Shared vSphere Distributed Switch is supported for greenfield deployments.

Table 740: Single Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-SDS-REQD-CFG-001	Create the following distributed port groups on the vSphere Distributed Switch: • VM management (mgmt domain only) • ESX management • vMotion • Storage	Provides the required network services for the vSphere cluster.	VM management distributed port group must be manually created on additional vSphere clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-SDS-REQD-CFG-002	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switch for each additional rack: • ESX management • vMotion • Storage (vSAN/NFS)	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack	Additional port groups are created for each additional rack.
VCF-VDS-SDS-REQD-CFG-003	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switch for availability zone 2 • ESX management • vMotion • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2.

Table 741: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Table 742: Single Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SDS-RCMD-DPG-001	Use the **Route based on physical NIC load** teaming algorithm for any storage distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-SDS-RCMD-NIO-001	Enable Network I/O Control on vSphere Distributed Switch (default setting).	Increases resiliency and performance when network traffic shares uplinks.	Network I/O Control might impact network performance for critical traffic types if misconfigured.
VCF-VDS-SDS-RCMD-NIO-002	Set the share value for management traffic to Normal (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than vSAN traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-SDS-RCMD-NIO-003	Set the share value for vMotion traffic to Low (default setting).	During times of network contention, vMotion traffic is not as important as virtual machine or storage traffic.	During times of network contention, vMotion takes longer than usual to complete.
VCF-VDS-SDS-RCMD-NIO-004	Set the share value for virtual machines to High (default setting).	Virtual machines are the most important asset in a VCF Instance. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-SDS-RCMD-NIO-005	Set the share value for principal IP-based storage traffic to High (default setting).	During times of network contention, storage traffic needs guaranteed bandwidth to support virtual machine performance.	None.
VCF-VDS-SDS-RCMD-NIO-006	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

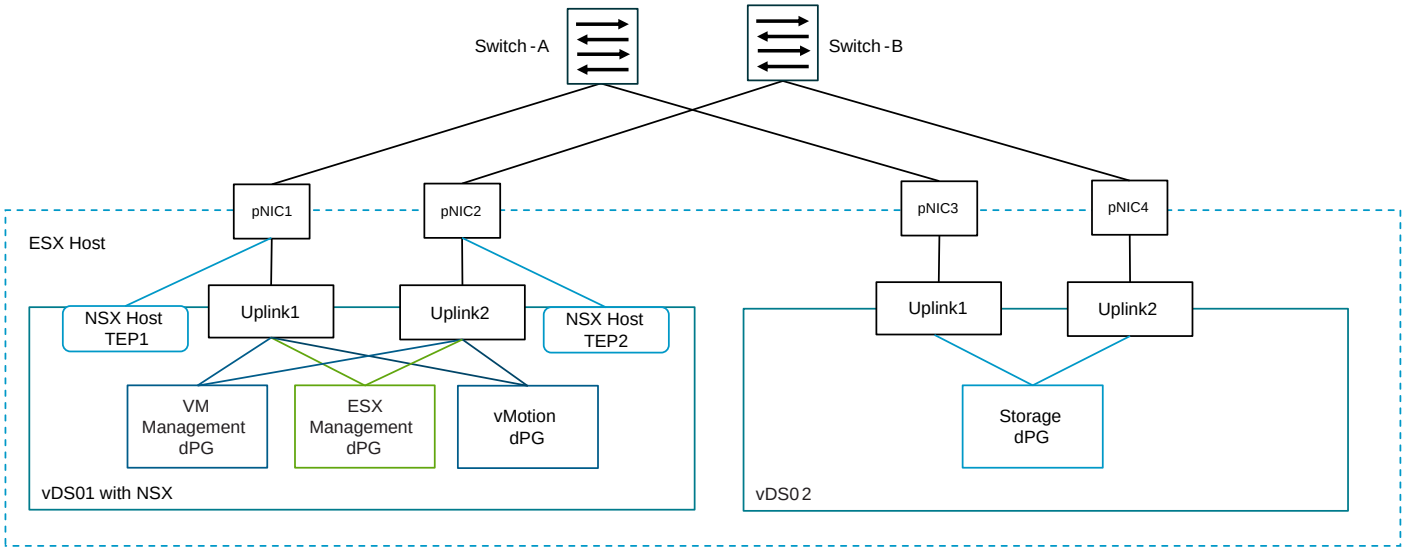
Storage Separation Distributed Switch Model

A *Storage Separation Distributed Switch Model* in VMware Cloud Foundation provides separation of storage traffic from all other network services running on the vSphere cluster.

A *Storage Separation Distributed Switch Model* requires two vSphere Distributed Switches to achieve this level of traffic separation where one vSphere Distributed Switch will be dedicated for storage traffic and the other will be used for all other network services, e.g. management, workload, vSphere vMotion, etc. Each vSphere Distributed Switch has two uplinks requiring four physical NICs on the ESX hosts.

The *Storage Separation Distributed Switch Model* can be suited to clusters where dedicated network bandwidth is required for storage traffic or if this traffic needs to be isolated to its own network fabric. This can be especially suited to vSphere clusters where dedicated bandwidth with no contention is required for storage traffic.

Figure 334: Storage Separation Distributed Switch Model With Single Network Fabric in VMware Cloud Foundation



Storage Separation Distributed Switch Model Attributes

A *Storage Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Shared for Management, vSphere vMotion, NSX workloads. • Dedicated for IP storage.

Storage Separation Distributed Switch Model Options

A *Storage Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (storage traffic isolated to dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two or more network cards. • Each traffic type is spread across separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Storage Separation Distributed Switch Model Design Requirements and Recommendations

Table 743: Common Distributed Switch Design Requirements for VCF 9.1.0

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-COM-REQD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain.	• For each vSphere cluster, you manage more vSphere Distributed Switches. • Does not apply from version 9.1.1 onwards because Shared vSphere Distributed Switch is supported for greenfield deployments.

Table 744: Storage Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-SS-REQD-CFG-001	Use two vSphere Distributed Switch per vSphere cluster: • vDS01 — Management — vMotion — NSX • vDS02 — Storage	• Separates storage traffic onto a dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for storage traffic (NFS or vSAN) with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switch required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than the simple model.
VCF-VDS-SS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion	Provides the required network services for the vSphere cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-SS-REQD-CFG-003	Create the following distributed port groups on vDS02: • Storage	Provides connectivity for the hosts in the cluster to the storage network.	vSphere Distributed Switch storage traffic separation profile provides the required configuration.
VCF-VDS-SS-REQD-CFG-004	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switches for each additional rack: vDS01: • ESX management • vMotion vDS02: • Storage	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack
VCF-VDS-SS-REQD-CFG-005	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switches for availability zone 2 vDS01: • ESX management • vMotion vDS02: • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2

Table 745: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Table 746: Storage Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-DPG-001	Use the **Failover Order** teaming algorithm for the storage port group.	Provides a consistent traffic flow through a single physical ToR switch without needing to traverse Inter Switch Link during normal operations.	It must be manually configured during deployment using vSphere Distributed Switche custom profile option.
VCF-VDS-SS-RCMD-NIO-001	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than Principal storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-SS-RCMD-NIO-002	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as virtual machine traffic.	During times of network contention, vMotion takes longer than usual to complete.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SS-RCMD-NIO-003	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in the SDDC. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-SS-RCMD-NIO-004	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

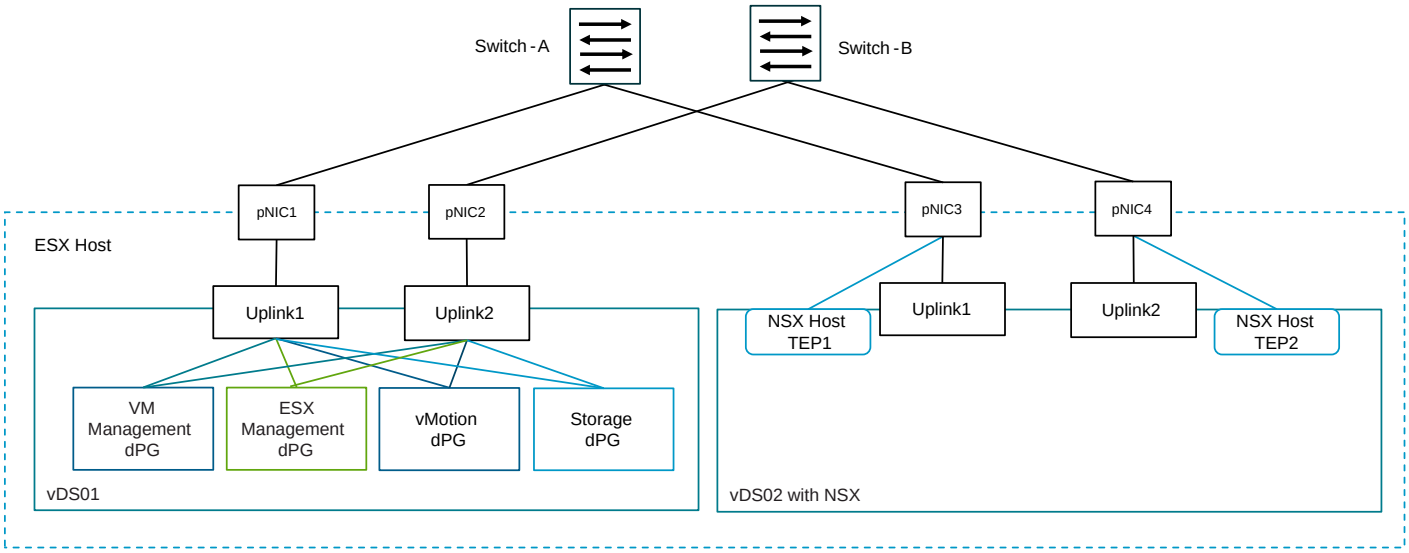
Workload Separation Distributed Switch Model

A *Workload Separation Distributed Switch Model* in VMware Cloud Foundation provides a separation of workload traffic running on NSX segments from all other network services running on the vSphere cluster.

A *Workload Separation Distributed Switch Model* requires two vSphere Distributed Switches to achieve this level of traffic separation where one vSphere Distributed Switch will be dedicated for workload traffic and the other will be used for all other network services, e.g management, vSphere vMotion, etc. Each vSphere Distributed Switch has two uplinks so a minimum of four physical NICs is needed on the ESX hosts.

The *Workload Separation Distributed Switch Model* model can be suited to clusters where dedicated network bandwidth is required for workloads that cant be achieved with a shared network design using just two physical NICs. It also provides the option to isolate workload traffic to a separate network fabric if this is a requirement for the design.

Figure 335: Workload Separation Distributed Switch Model in VMware Cloud Foundation



Workload Separation Distributed Switch Model Attributes

A *Workload Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Shared for Management, vSphere vMotion, vSAN. • Dedicated for workloads on NSX.

Workload Separation Distributed Switch Model Options

A *Workload Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (workload traffic using NSX and VPC networks isolated to a dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two separate 2-port network cards. • Each traffic type is spread across two separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Workload Separation Distributed Switch Model Design Requirements and Recommendations

Table 747: Common Distributed Switch Design Requirements for VCF 9.1.0

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-COM-REQD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain.	• For each vSphere cluster, you manage more vSphere Distributed Switches. • Does not apply from version 9.1.1 onwards because Shared vSphere Distributed Switch is supported for greenfield deployments.

Table 748: Workload Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-WS-REQD-CFG-001	Use two (2) vSphere Distributed Switches per vSphere cluster: • vDS01 — ESX Management — vMotion — Storage • vDS02 — NSX	• Separates NSX workload traffic onto dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for workload traffic with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switches required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than the simple model.
VCF-VDS-WS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion • Storage	Provides the required network services for the cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain clusters.
VCF-VDS-WS-REQD-CFG-003	Configure vDS02 for NSX.	Provides NSX networking capabilities for the workloads on the cluster.	None.
VCF-VDS-WS-REQD-CFG-004	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switch for each additional rack: • ESX management • vMotion • Storage (vSAN/NFS)	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack.
VCF-VDS-WS-REQD-CFG-005	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switch for availability zone 2: • ESX management • vMotion • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2.

Table 749: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Table 750: Workload Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-WS-RCMD-DPG-001	Use the **Route based on physical NIC load** teaming algorithm for the storage distributed port group on vDS01 (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-WS-RCMD-NIO-001	Enable Network I/O Control on vSphere Distributed Switch vDS01 (default setting).	Increases resiliency and performance when network traffic shares uplinks.	Network I/O Control might impact network performance for critical traffic types if misconfigured.
VCF-VDS-WS-RCMD-NIO-002	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than principal storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-WS-RCMD-NIO-003	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as storage traffic.	During times of network contention, vMotion takes longer than usual to complete.

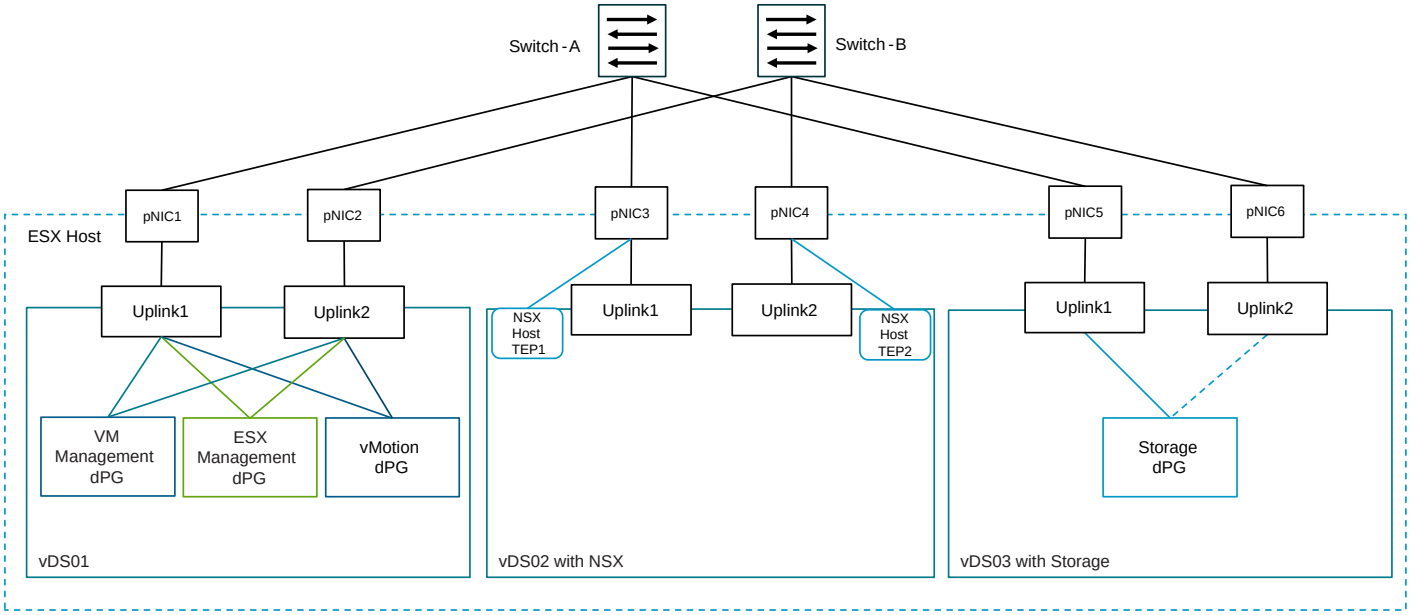
Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-WS-RCMD-NIO-004	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in an VCF Instance. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-WS-RCMD-NIO-005	Set the share value for principal IP-based Storage traffic to High for vDS01 (default setting).	During times of network contention, storage traffic needs guaranteed bandwidth to support virtual machine performance.	None.
VCF-VDS-WS-RCMD-NIO-006	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

Storage and Workload Separation Distributed Switch Model

The *Storage and Workload Separation Distributed Switch Model* in VMware Cloud Foundation provides for full separation of storage and workload traffic from all other network traffic.

The *Storage and Workload Separation Distributed Switch Model* requires three vSphere Distributed Switches to achieve this level of traffic separation where both storage and workload have dedicated vSphere Distributed Switches and a third vSphere Distributed Switch will be used for all other network services, e.g. management, vSphere vMotion, etc. Each vSphere Distributed Switch has two uplinks requiring six physical NICs on the ESX hosts in the vSphere cluster connecting to each vSphere Distributed Switch. This basically combines the previous two network models providing isolation and high performance for storage and production workloads running on NSX segments or VPCs.

Figure 336: Storage and Workload Separation Distributed Switch Model in VMware Cloud Foundation



Storage and Workload Separation Distributed Switch Model Attributes

A *Storage and Workload Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Three (3)
Number of Physical NICs per ESX Host	Six (6)
Bandwidth	• Shared for Management, vSphere vMotion. • Dedicated for vSAN. • Dediacted for NSX workload.

Storage and Workload Separation Distributed Switch Model Options

A *Storage and Workload Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (all traffic on same fabric). • Dual (storage traffic isolated to dedicated fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across different network cards. NLR: • Use two or more network cards. • Each traffic type is spread across separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.
Shared vDS (VCF 9.1.1 onwards for Greenfield Deployments)	• No - Dedicated vDS per cluster • Yes - Shared between clusters

Storage and Workload Separation Distributed Switch Model Design Requirements and Recommendations

Table 751: Common Distributed Switch Design Requirements for VCF 9.1.0

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-COM-REQD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain.	• For each vSphere cluster, you manage more vSphere Distributed Switches. • Does not apply from version 9.1.1 onwards because Shared vSphere Distributed Switch is supported for greenfield deployments.

Table 752: Storage and Workload Separation Distributed Switch Model Design Requirements

Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-SWS-REQD-CFG-001	Use three (3) vSphere Distributed Switch per vSphere cluster: • vDS01 — Management — vMotion • vDS02 — Storage • vDS03 — NSX Workloads	• Separates NSX workload traffic onto dedicated vSphere Distributed Switches. • Separates storage traffic onto dedicated vSphere Distributed Switches. • Can provide maximum bandwidth and performance for storage traffic (NFS or vSAN) with no contention from other traffic on the host physical NICs. • Can provide maximum bandwidth and performance for workload traffic with no contention from other traffic on the host physical NICs.	• Additional vSphere Distributed Switches required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switches with two (2) physical NICs per ESX host. • A higher number of ESX host physical NICs required, a minimum of six (6) with redundancy.
VCF-VDS-SWS-REQD-CFG-002	Create the following distributed port groups on vDS01 and attach ESX VMkernels: • VM management • ESX management • vMotion	Provides the required network services for the vSphere cluster.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-SWS-REQD-CFG-003	Create the following distributed port groups on vDS02: • Storage	Provides connectivity for the ESX hosts in the vSphere cluster to the storage network.	None.
VCF-VDS-SWS-REQD-CFG-004	Configure vDS03 for NSX traffic.	Provides NSX networking capabilities for the workloads on the cluster.	None.
VCF-VDS-SWS-REQD-CFG-005	For Layer 3 multi-rack cluster deployments create the following distributed port groups on the vSphere Distributed Switches for each additional rack: vDS01: • ESX management • vMotion vDS02: • Storage	• Provides separate networks per rack in a multi-rack deployment. • Storage traffic has a separate isolated layer 2 broadcast domain per rack.	Additional port groups are created for each additional rack
VCF-VDS-SWS-REQD-CFG-006	For stretched cluster deployments create the following distributed port groups on the vSphere Distributed Switches for availability zone 2 vDS01: • ESX management • vMotion vDS02: • vSAN	• Provides separate networks per rack in a multi-rack deployment. • vSAN traffic has a separate isolated layer 2 broadcast domain per availability zone.	Additional port groups are created for availability zone 2

Table 753: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Table 754: Storage and Workload Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SWS-RCMD-DPG-001	Use the **Failover Order** teaming algorithm for the storage (vSAN or NFS) distributed port group.	Only one uplink active which provides a consistent traffic flow through a single physical top of rack switch without needing to traverse the Inter Switch Link during normal operations.	It must be manually configured during deployment using vSphere Distributed Switch custom profile option.
VCF-VDS-SWS-RCMD-NIO-001	Enable Network I/O Control on vSphere Distributed Switch vDS01 (default setting).	Increases resiliency and performance of the network when traffic shares uplinks.	Network I/O Control might impact network performance for critical traffic types if misconfigured.
VCF-VDS-SWS-RCMD-NIO-002	Set the share value for management traffic to Normal for shared vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion. Management traffic is important because it ensures that the hosts can still be managed during times of network contention	None.

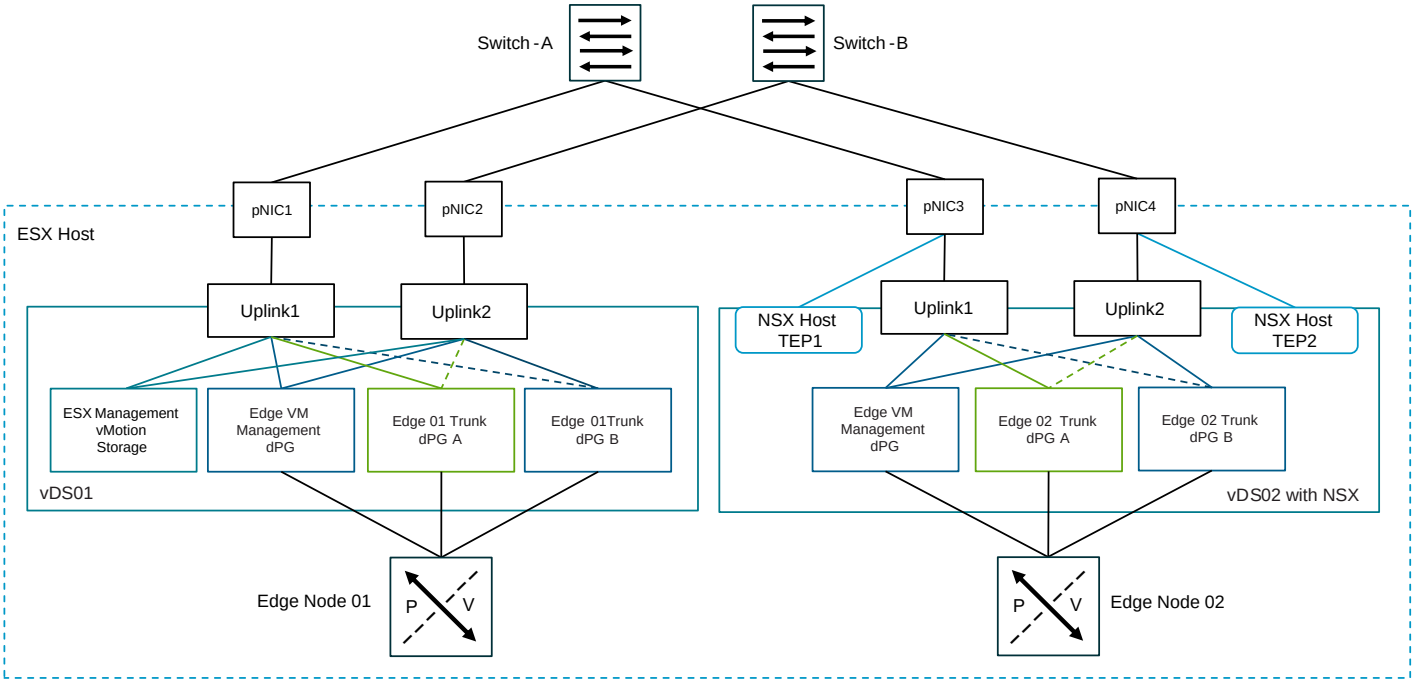
Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-SWS-RCMD-NIO-003	Set the share value for vMotion traffic to Low for shared vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as management traffic.	During times of network contention, vMotion takes longer than usual to complete.
VCF-VDS-SWS-RCMD-NIO-004	Set the share value for virtual machines to High for DS01 (default setting).	Virtual machines are the most important asset in a VCF Instance. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-SWS-RCMD-NIO-005	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model

A *Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model* in VMware Cloud Foundation provides the network design for the deployment of NSX Edge nodes to the vSphere cluster. This vSphere cluster is dedicated to running only NSX Edge to provide for a large-scale, high-performance NSX Edge cluster deployment. To see more details please refer to NSX Edge Cluster Detailed Design.

The *Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model* uses two vSphere Distributed Switches with two uplinks to provide the connectivity for two NSX Edge nodes to run on each ESX host in the vSphere cluster.

Figure 337: Dedicated vSphere Cluster for NSX Edges Distributed Switch Model in VMware Cloud Foundation



Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model Attributes

A *Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Each NSX Edge node has dedicated Uplink/TEP traffic per ESX host pNIC. • No vSphere vMotion traffic as each NSX Edge node is pinned to an ESX host as per the design. • vSAN traffic should be minimal as only two NSX Edge nodes deployed per ESX host. • Host TEP traffic should be minimal as recommended to have no workload virtual machinces deployed using NSX overlay.

Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model Options

A *Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	Single (all traffic on same fabric).
Physical NIC Level Redundancy (NLR)	No NLR: • Don't spread network traffic types across separate network cards. NLR: • Use two or more network cards. • Each traffic type is spread across separate network cards to avoid single point of failure.
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. (only option for NSX Edge node uplinks). • Route based on IP hash.
Data Path Mode	• Standard. • Enhanced Datapath Standard (default). • Enhanced Datapath Dedicated.

Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model Design Requirements and Recommendations

Table 755: Common Distributed Switch Design Requirements for VCF 9.1.0

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-COM-REQD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain.	• For each vSphere cluster, you manage more vSphere Distributed Switches. • Does not apply from version 9.1.1 onwards because Shared vSphere Distributed Switch is supported for greenfield deployments.

Table 756: Dedicated vSphere Cluster for NSX Edges Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication

VCF-VDS-DE-REQD-CFG-001	Use two (2) vSphere Distributed Switches per vSphere cluster:
	• vDS01
	  – NSX Edge VM management
	  – vMotion
	  – Storage
	  – Edge Uplink
	• vDS02
	  – NSX Edge VM management
	  – NSX - Host
	  – Edge TEP
	  – Edge Uplink
		• Separates storage traffic onto a dedicated vSphere Distributed Switch.
		• Can provide maximum bandwidth and performance for storage traffic (NFS or vSAN) with no contention from other traffic on the host physical NICs.
			• Additional vSphere Distributed Switch required increasing management overhead.
			• A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two physical NICs per ESX host.
			• Higher physical switch port density is required than the simple model.

VCF-VDS-DE-REQD-CFG-002	Create the following distributed port groups on vDS01: • NSX Edge management • ESX management • vMotion • Storage • NSX Edge Uplinks	Provides the required network services for the cluster including NSX Edge connectivity.	• vSphere Distributed Switch workload traffic separation profile provides the required configuration for the ESX host. • Distributed port groups for NSX Edge nodes must be created manually.
VCF-VDS-DE-REQD-CFG-003	Create the following distributed port groups on vDS02: • NSX Edge management • NSX Edge Uplinks	Provides the required connectivity for NSX Edge management and NSX Edge TEP and NSX Edge uplink traffic.	• vSphere Distributed Switch workload traffic separation profile provides the required configuration. • Distributed port groups for NSX Edge nodes must be created manually.

Table 757: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Table 758: Dedicated vSphere Cluster for NSX Edges Distributed Switch Model Design Recommendations

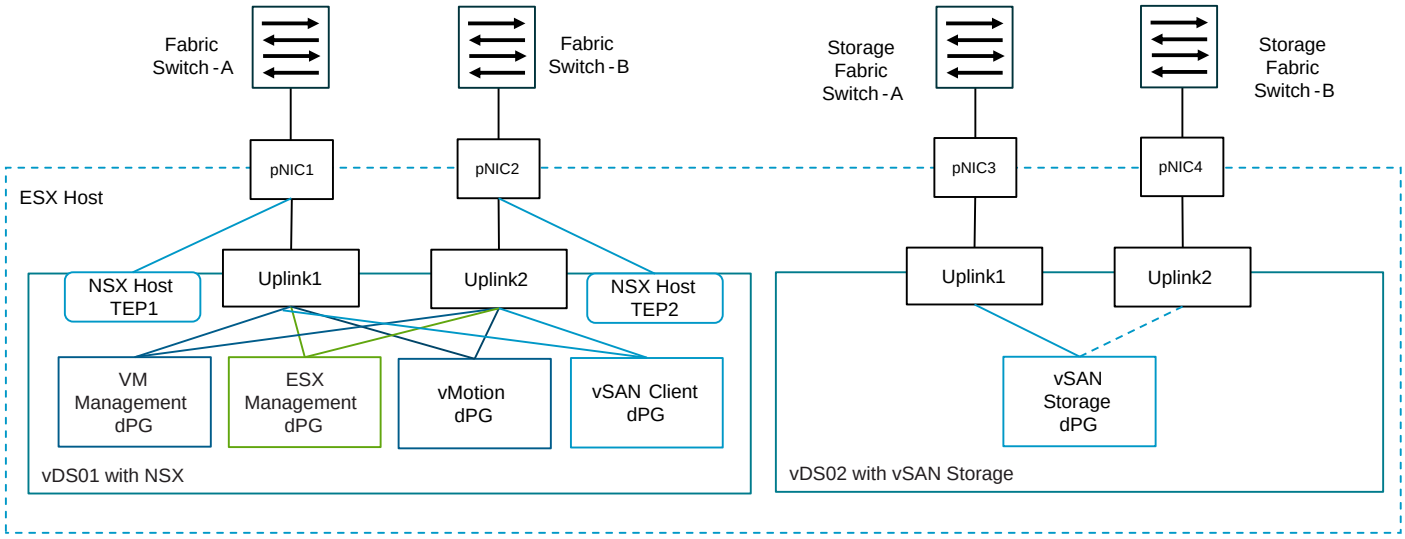
Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-DE-RCMD-DPG-001	Use the **Route based on physical NIC load** teaming algorithm for the storage port group on vDS01 (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	None.
VCF-VDS-DE-RCMD-NIO-001	Disable Network I/O Control on vSphere Distributed Switch vDS01.	Provides maximum performance for NSX Edge when Network I/O control is disabled.	Must be disabled manually.
VCF-VDS-DE-RCMD-NIO-002	Disable Network I/O Control on vSphere Distributed Switch vDS02.	Provides maximum performance for NSX Edge when Network I/O control is disabled.	Must be disabled manually.

vSAN Storage Cluster Separation Distributed Switch Model

A *vSAN Storage Cluster Separation Distributed Switch Model* in VMware Cloud Foundation is used for storage clusters where the vSAN storage traffic is separated from all other traffic on the vSphere cluster.

A *vSAN Storage Cluster Separation Distributed Switch Model* requires two vSphere Distributed Switches to achieve a design where vSAN storage traffic requires its own dedicated physical NICs for guaranteed bandwidth and optional vSAN storage traffic isolation through a separate storage fabric within a rack.

Figure 338: vSAN Storage Cluster Separation Distributed Switch Model With Optional Dedicated Storage Fabric in VMware Cloud Foundation



vSAN Storage Cluster Separation Distributed Switch Model Attributes

A *vSAN Storage Cluster Separation Distributed Switch Model* has the following attributes.

Attribute	Detail
Number of vSphere Distributed Switches	Two (2)
Number of Physical NICs per ESX Host	Four (4)
Bandwidth	• Dedicated for the vSAN storage traffic local to the cluster/rack. • vSAN client traffic serving compute clusters is shared with management traffic.

vSAN Storage Cluster Separation Distributed Switch Model Options

A *Storage Separation Distributed Switch Model* can leverage options from the design areas listed below, depending on objectives.

Design Area	Options
Network Fabric	• Single (All traffic on same fabric) • Dual (vSAN Storage Cluster traffic isolated to dedicated fabric)
Physical NIC Level Redundancy (NLR)	No NLR: • Dont spread network traffic types across different network cards. NLR: • Use two or more network cards. • Each traffic type is spread across separate network cards to avoid single point of failure
Load Balancing Option	• Route based on physical NIC load. • Route based on the originating virtual port. • Use explicit failover order. • Route based on IP hash.
LAG	• Enabled – Requires LACP configuration on the physical switches • Disabled
Data Path Mode	• Standard • Enhanced Datapath Standard (default) • Enhanced Datapath Dedicated

vSAN Storage Cluster Separation Distributed Switch Model Design Requirements and Recommendations

Table 759: Common Distributed Switch Design Requirements for VCF 9.1.0

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-COM-REQD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain.	• For each vSphere cluster, you manage more vSphere Distributed Switches. • Does not apply from version 9.1.1 onwards because Shared vSphere Distributed Switch is supported for greenfield deployments.

Table 760: vSAN Storage Cluster Separation Distributed Switch Model Design Requirements

Design Requirements ID	Design Requirements	Justification	Implication
VCF-VDS-VSCS-REQD-CFG-001	Use two (2) vSphere Distributed Switch per cluster: • vDS01 — Management — vMotion — NSX — vSAN Client • vDS02 — vSAN Cluster local storage	• Separates vSAN cluster storage traffic onto a dedicated vSphere Distributed Switch. • Can provide maximum bandwidth and performance for storage traffic with no contention from other traffic on the ESX host physical NICs.	• Additional vSphere Distributed Switch required increasing management overhead. • A higher number of ESX host physical NICs is required than using a single vSphere Distributed Switch with two (2) physical NICs per ESX host. • Higher physical switch port density is required than using single vSphere Distributed Switch.
VCF-VDS-VSCS-REQD-CFG-002	Create the following distributed port groups on vDS01: • VM management • ESX management • vMotion • vSAN Client traffic	Provides the required network services for the cluster, including vSAN client traffic.	VM management distributed port group must be manually created on additional clusters in the management domain and any workload domain vSphere clusters.
VCF-VDS-VSCS-REQD-CFG-003	Create the following distributed port groups on vDS02: • vSAN	Provides connectivity for the hosts in the cluster to the storage network.	None.

Table 761: Common Distributed Switch Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-001	Do not share a vSphere Distributed Switch across different vSphere clusters.	• Enables independent lifecycle management of vSphere Distributed Switch per vSphere cluster. • Reduces the size of the fault domain. • Supported for stretched clusters. • Supported for layer 3 multi-rack clusters.	• For each vSphere cluster, you manage more vSphere Distributed Switches.
VCF-VDS-COM-RCMD-CFG-002	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	• Supports the MTU size that is required by system traffic types. • Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VDS-COM-RCMD-CFG-003	For NSX enabled vSphere Distributed Switch with Overlay or VLAN transport zone configure Enhanced Datapath Standard.	Provides the best performance for bandwidth and packets per second for the NSX Edge nodes running on the vSphere cluster.	The physical NIC must support the Enhanced Datapath - Interrupt Mode feature.

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-COM-RCMD-CFG-004	Use physical NICs with the same link speed, hardware model, and performance capabilities for all uplinks connected to the same vSphere Distributed Switch.	• Ensures consistent throughput and latency regardless of which uplink traffic traverses. • Ensures that hardware offloads are consistently available on all uplinks. • Prevents scenarios where fail over to a slower NIC causes congestion or dropped packets. • Required for effective use of dynamic teaming policies which assume equal capability across all links.	Restricts the ability to mix different models of network cards within the same vSphere Distributed Switches of hosts in a vSphere cluster.
VCF-VDS-COM-RCMD-CFG-005	Configure the vSphere Distributed Switch to use Uplinks instead of LAG.	• No Link Aggregation configuration required on the physical switches. • Less complex configuration on the physical switches. • Can be easier to troubleshoot network issues. • NSX Edge BGP peering simplified with deterministic traffic flows to the TOR switches.	Can improve performance and bandwidth whereby one host or VMkernel port can communicate with other hosts using different load-balancing options.
VCF-VDS-COM-RCMD-DPG-001	Use ephemeral port binding for the VM management distributed port group (default setting).	• Using ephemeral port binding provides the option to recover the vCenter instance that is managing the vSphere Distributed Switch. • The VM management network is not required for a multi-rack compute-only cluster in a workload domain.	Port-level permissions and controls are lost across power cycles, and no historical context is saved.
VCF-VDS-COM-RCMD-DPG-002	Use static port binding for all non-management distributed port groups (default setting).	Static binding ensures that a virtual machine connects to the same port on the vSphere Distributed Switch. This allows for historical data and port-level monitoring.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-003	Use the **Route based on physical NIC load** teaming algorithm for the VM management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-004	Use the **Route based on physical NIC load** teaming algorithm for the ESX management distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.
VCF-VDS-COM-RCMD-DPG-005	Use the **Route based on physical NIC load** teaming algorithm for the vMotion distributed port group (default setting).	Reduces the complexity of the network design, increases resiliency, and can adjust to fluctuating workloads.	Traffic not always deterministic, can be configured to another teaming policy during or after deployment if necessary.

Table 762: vSAN Storage Cluster Separation Distributed Switch Model Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-VDS-VSCS-RCMD-DPG-001	Use the **Failover Order** teaming algorithm for the vSANdistributed port group.	Provides a consistent traffic flow through a single physical top of rack switch without needing to traverse the Inter Switch Link during normal operations.	It must be manually configured during deployment using the vSphere Distributed Switch custom profile option.
VCF-VDS-VSCS-RCMD-NIO-001	Disable Network I/O Control on vSphere Distributed Switch vDS02 dedicated for vSAN cluster storage traffic.	Provides the best performance for vSAN cluster traffic when Network I/O control is disabled and there is no sharing of traffic.	You must deactivate Network I/O Control manually after the cluster is deployed.
VCF-VDS-VSCS-RCMD-NIO-002	Set the share value for management traffic to Normal for vDS01 (default setting).	By keeping the default setting of Normal, management traffic is prioritized higher than vMotion but lower than storage traffic. Management traffic is important because it ensures that the hosts can still be managed during times of network contention.	None.
VCF-VDS-VSCS-RCMD-NIO-003	Set the share value for vMotion traffic to Low for vDS01 (default setting).	During times of network contention, vMotion traffic is not as important as virtual machine traffic.	During times of network contention, vMotion takes longer than usual to complete.
VCF-VDS-VSCS-RCMD-NIO-004	Set the share value for virtual machines to High for vDS01 (default setting).	Virtual machines are the most important asset in the SDDC. Leaving the default setting of High ensures that they always have access to the network resources they need.	None.
VCF-VDS-VSCS-RCMD-NIO-005	Set the share value for vSAN Client traffic to High for vDS01 (default setting)	During times of network contention, storage traffic needs guaranteed bandwidth to support virtual machine performance.	None.
VCF-VDS-VSCS-RCMD-NIO-006	All other traffic types are set using default settings.	Other traffic types are only added during day 2 operations.	Can be manually adjusted on day 2 if necessary.

VMkernel Detailed Design

The VMkernel networking layer provides connectivity to ESX hosts and handles the system traffic for management, vSphere vMotion, vSphere HA, vSAN, and NSX Host TEPs.

VMkernel Network Adapter Design

The following are the VMkernels created on the ESX hosts used by clusters in VMware Cloud Foundation.

Table 763: VMkernel Network Adapter Configuration

VMkernel Adapter Service	Connected Distributed Port Group	Activated Services	Recommended MTU Size (Bytes)
ESX Management	ESX Management Port Group	Management Traffic	1500 (Default)
vSphere vMotion	vSphere vMotion Port Group	vSphere vMotion Traffic	9000
vSAN	vSAN Port Group	vSAN	9000
NFS	NFS Port Group	NFS	9000

VMkernel Adapter Service	Connected Distributed Port Group	Activated Services	Recommended MTU Size (Bytes)
Host TEPs	Not applicable	Not applicable	9000

Network I/O Control Detailed Design

vSphere Network I/O Control version 3 is enabled by default on the vSphere Distributed Switches created in VMware Cloud Foundation and provides a mechanism to reserve bandwidth for system traffic based on the capacity of the physical adapters on an ESX host. This can be particularly useful when different traffic types share the same physical adapters on the hosts.

vSphere Distributed Switch Network I/O Control Options

Network I/O Control supports different mechanisms for resource management of system traffic related to infrastructure services, such vSAN, vMotion, etc and VM traffic. For system traffic shares are used to ensure a dedicated percentage of the bandwidth at a time when there is contention on the physical adapter. Network I/O Control provisions bandwidth to the network adapters of virtual machines by using constructs of shares, reservation, and limit. In VMware Cloud Foundation the default shares are applied to the system traffic, the table below shows the default values.

Traffic Type	Shares	Shares Value
Management Traffic	Normal	50
Fault Tolerance (FT) Traffic	Low	25
vMotion Traffic	Low	25
VM Traffic	High	100
iSCSI Traffic	Low	25
NFS Traffic	Low*	25
vSphere Replication (VR) Traffic	Low	25
vSAN Traffic	High**	100
vSphere Data Protection Backup Traffic	Low	25
vSphere Backup NFC Traffic	Normal	50
NVMe over TCP	Normal	50
Provisioning Traffic	Normal	50

Depending on the vSphere Distributed Switch configuration and the type of system traffic on the vSphere Distributed Switch these value can be modified post deployment to meet the requirements of the environment.

Note:

* High if NFS storage is used as principal storage for the vSphere cluster.

** Low if vSAN is not used as principal storage for the vSphere cluster.

Note:

In some cases vSphere Network I/O Control needs to be disabled, for vSphere cluster hosting edge clusters, it is recommended to disable the feature.

Storage Detailed Design

A VCF Instance can include one or more supported *Storage Models*. This section describes the options, requirements, and recommendations for each *Storage Model*.

- vSAN Storage Models
- NFS Storage Model
- Fibre Channel Storage Model

vSAN Storage Models

The following section covers multiple *vSAN Storage Models* that provide different vSAN deployment models such as single-rack, multi-rack, stretched vSAN cluster, or disaggregated vSAN storage and compute clusters. This section also covers vSAN specific networking requirements and recommendations.

Single-Rack vSAN Storage Models

- Single-Rack vSAN ESA Storage Model
- Single-Rack vSAN OSA Storage Model
- Single-Rack vSAN Storage Cluster Storage Model
- Single-Rack vSAN Compute Cluster Storage Model

Multi-Rack vSAN Storage Models

- Multi-Rack vSAN Storage Model
- Multi-Rack vSAN Compute Cluster Storage Model

Stretched vSAN Storage Models

- Stretched vSAN Storage Model
- Stretched vSAN Storage Cluster Storage Model
- Stretched vSAN Compute Cluster Storage Model

vSAN Host Network Models

When designing ESX host networking for vSAN, determining the network configuration for vSAN traffic has an impact on the *Distributed Switch Model* that is chosen for a given vSphere cluster. See Distributed Switch Detailed Design.

Figure 339: vSAN Host Network Model - 2 pNIC

vSAN Host Network Model with two (2) physical network adapters where traffic is shared with other services. See Single Distributed Switch Model.

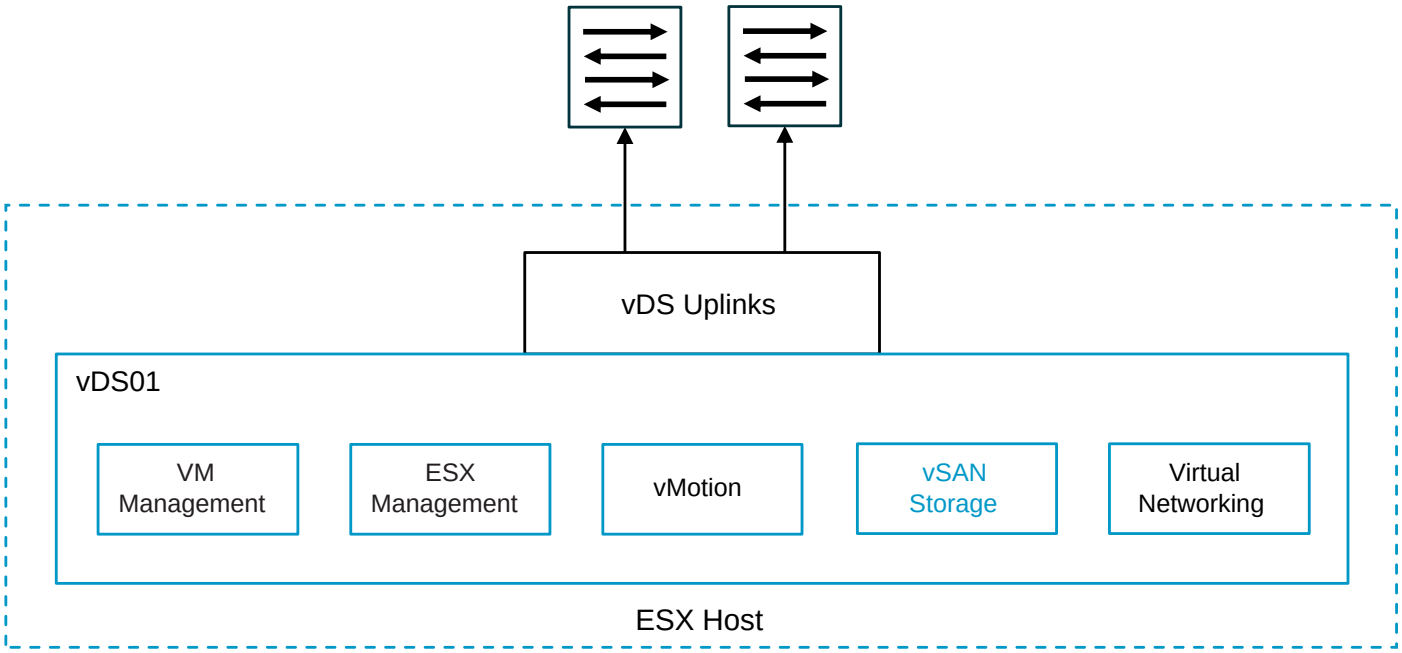
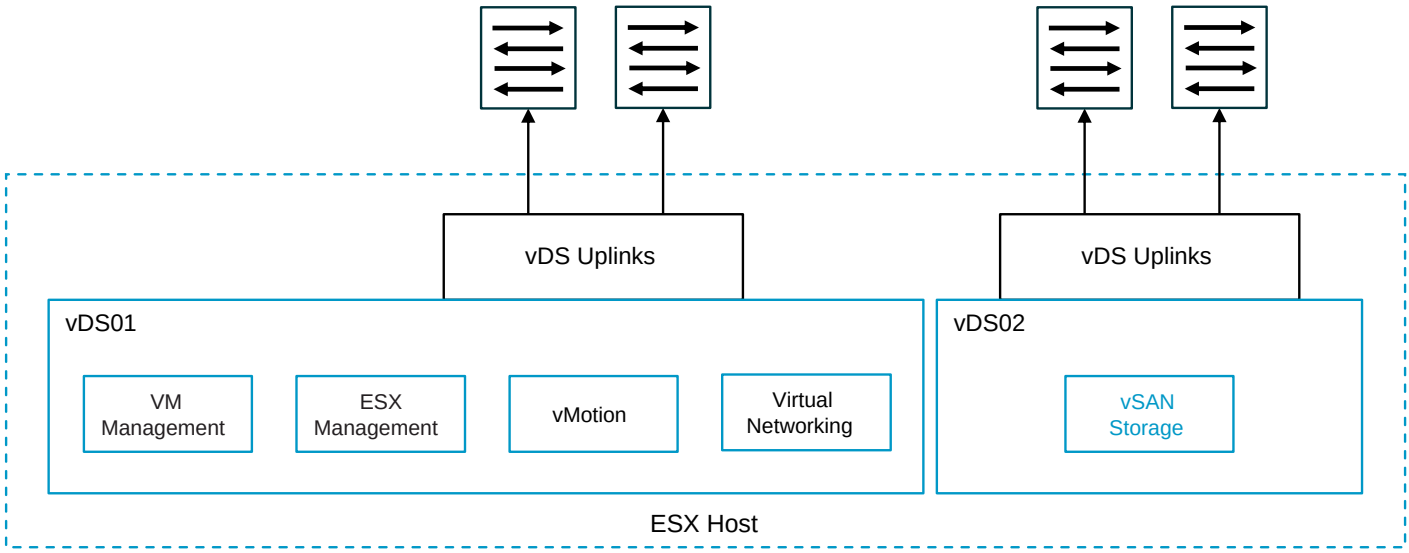


Figure 340: vSAN Host Network Model - 4 pNIC

vSAN Host Network Model with four (4) physical network adapters where vSAN traffic is isolated on a dedicated fabric. See Storage Separation Distributed Switch Model.



vSAN Host Network Model Attributes

A *vSAN Host Network Model* has the following attributes.

Attribute	Detail
vSAN VMkernel	Dedicated VMkernel ports for vSAN traffic communication.
VLAN	VLANs are used for vSAN storage traffic to be isolated on a dedicated VLAN.
Physical Networking	Dedicated or shared physical adapters for storage traffic.
vSphere Distributed Switches	vSphere Distributed Switches provide network services and connectivity, see Distributed Switch Detailed Design.
vSAN Traffic Type	vSAN uses different traffic types for vSAN traffic that are tagged to VMkernel adapters: • vSAN Traffic- vSAN HCI and Storage Cluster traffic. • vSAN Witness Traffic - vSAN Witness node traffic. • vSAN Storage Cluster Client Traffic - vSAN client cluster traffic for datastore storage clusters.
vSAN Network Ports	vSAN communicates between each host in the cluster by using TCP / UDP. For the list of all supported vSAN ports and protocols, see the VMware Ports and Protocols portal at https://ports.broadcom.com/home/vSAN.
MTU Size	vSAN must use a consistent MTU for all vSAN traffic types.

Table 764: vSAN Host Network Model Options

Design Area	Options
vSAN Host Networking	• 10GbE • 25GbE or higher **Important:** vSAN ESA Ready node profile, **vSAN-HCI-SM**, is the only vSAN ESA ready node that supports 10GbE. All other vSAN ESA HCI ready node profiles require 25GbE or greater. vSAN ESA Storage Ready Node profile **vSAN-SC-SM** can use minimum 25GbE for vSAN Storage Traffic (E/W) **and** 10GbE for vSAN Storage Cluster Client Traffic (N/S) https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance
Routing	Layer 2 connectivity between all vSAN hosts sharing the IP space or subnet. Can be achieved by: • Deploying ESX hosts in a physical rack that shares Layer 2 broadcast domain for vSAN network traffic. • Deploying ESX hosts horizontally across different physical racks while sharing Layer 2 broadcast domain for vSAN network traffic.
	Layer 3 connectivity between vSAN hosts using Layer 3 routing: • Multiple physical racks can have a unique set of Layer 2 broadcast domains that are routed through top of rack switches. • ESX hosts deployed across multiple physical racks can consume different routed VLANS. See Multi-Rack Layer 3 vSphere Cluster Model.
Distributed Switch Model	• Single Distributed Switch Model • Workload Separation Distributed Switch Model • Storage Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
MTU	1500 - 9000
NIC Teaming and Load Balancing	vSAN relies on vSphere Distributed Switches to determine load balancing and teaming. Design choices are listed here with justification and implications, see Distributed Switch Detailed Design.

vSAN Host Network Model Design Requirements

Table 765: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

vSAN Host Network Model Design Recommendations

Table 766: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

Single-Rack vSAN Storage Models

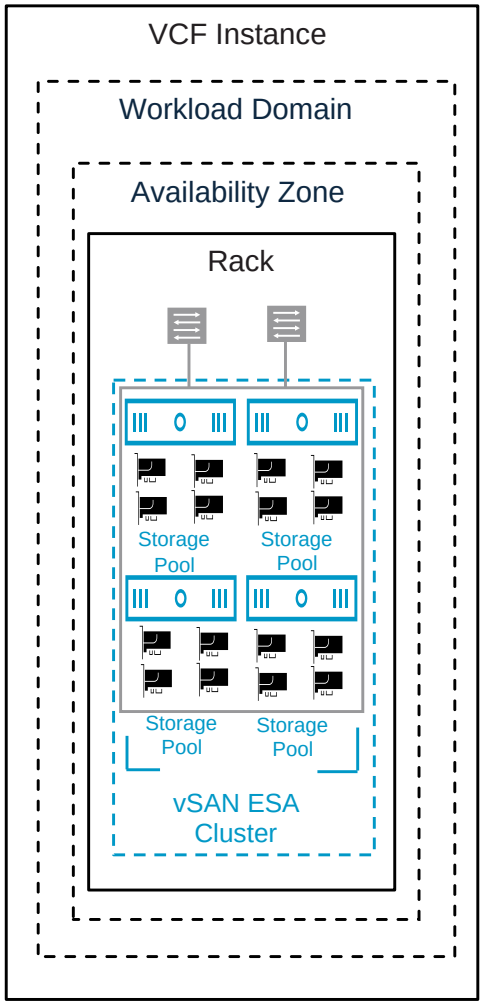
A *Single-Rack vSAN Storage Model* refers to a vSAN cluster deployed within a single physical rack or rack enclosure. This approach uses the local vSAN compatible storage devices within the ESX hosts housed in that rack to create a shared, distributed datastore accessible to all ESX hosts in a vSphere cluster. The following sections describe *Single-Rack vSAN Storage Models*. A *Single-Rack vSAN Storage Model* can either be based on a vSAN ESA or vSAN OSA architecture, and can be deployed as a vSAN Storage Cluster or a vSAN Compute Cluster.

- Single-Rack vSAN ESA Storage Model
- Single-Rack vSAN OSA Storage Model
- Single-Rack vSAN Storage Cluster Storage Model
- Single-Rack vSAN Compute Cluster Storage Model

Single-Rack vSAN ESA Storage Model

vSAN ESA uses a single-tier architecture that uses high-performance flash storage devices. vSAN ESA supports only NVMe high-performance storage devices.

Figure 341: Single-Rack vSAN ESA Storage Model in VMware Cloud Foundation



Single-Rack vSAN ESA Storage Model Attributes

A *Single-Rack vSAN ESA Storage Model* has the following attributes.

Attribute	Detail
Single Tier	All data is stored on the same type of storage device, regardless of access frequency or importance.
Storage Pool	All NVMe Flash Storage.
Data Services	• Compression (enabled by default). • Encryption. (in transit and at rest) • vSAN Global Deduplication **Restriction:** vSAN Global Deduplication is not currently compatible with vSAN Stretched Clusters and Two Node vSAN Clusters. • Integrated vSAN File Services. • vSAN iSCSI Target Service. • vSAN Storage Clusters. • Stretched Cluster. • Fault domains. • Kubernetes persistent storage services (block and file). • Native scalable snapshots
Virtual Machine Storage policies	vSAN storage policies (also known as Virtual Machine Storage Policies or VM Storage Policies) are rules defining virtual machine (VM) storage requirements that vSAN automatically enforces across a cluster. vSAN 9.1 automatically configures vSAN Auto-RAID policy on new vSAN ESA clusters. vSAN Auto-RAID policy manages the resilience for data residing in a cluster, regardless of the size or topology of the vSAN cluster. vSAN Auto-RAID creates a single policy within the vCenter server,

Single-Rack vSAN ESA Storage Model Options

A *Single-Rack vSAN ESA Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Deduplication	vSAN Global Deduplication is supported with VMware Cloud Foundation 9.1 and can be enabled as a day 2 operation.
Data In Transit Encryption	Data In Transit is supported to be enabled as day 0 or day 2 operation.
Data at Rest encryption	Data at Rest Encryption is supported as a day 2 operation.
vSAN Networking	vSAN ESA can utilize 10GbE , 25GbE or higher. See Table 764: vSAN Host Network Model Options.

Single-Rack vSAN ESA Storage Model Design Requirements and Recommendations

Table 767: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Table 768: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

Table 769: vSAN ESA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-ESA-REQD-CFG-001	Ensure vSAN certified Drives used for vSAN ESA clusters.	• When using VMware Cloud Foundation automated workflows, vSAN ESA cluster creation maybe blocked unless vSAN certified drives are detected. • vSAN Auto-RAID will be enabled when vSAN certified components are detected. This allows vSAN to generate an optimal default storage policy, based on the cluster topology and the number of ESX hosts.	None.
VCF-VSAN-ESA-REQD-CFG-002	1.6 TB drive size or higher for vSAN ESA devices for storage pools.	Minimum supported vSAN ESA drive size is 1.6TB.	vSAN ESA supports up to 24 drives per node. Smaller capacity drives may constrain raw capacity of a vSAN node.
VCF-VSAN-ESA-REQD-CFG-003	16 Cores or higher per vSAN node. vSAN ESA has a minimum value of 16 cores per vSAN Node	Minimum vSAN ESA core count per node is 16 for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	Smaller number of cores per node may require more vSAN nodes to support targeted workloads.
VCF-VSAN-ESA-REQD-CFG-004	Memory size per vSAN Ready node 128GB or higher	Minimum vSAN ESA memory is 128GB for the smallest vSAN Ready node, **vSAN-HCI-SM** or **vSAN-SC-SM** vSAN ESA Storage Cluster ReadyNode.	• Smaller memory footprint per node will limit number of vSAN drives per node. • Smaller memory footprint per node for vSAN ESA will limit VM workloads being placed on vSAN nodes.

Table 770: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Table 771: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

Table 772: vSAN ESA Storage Model Design Recommendations

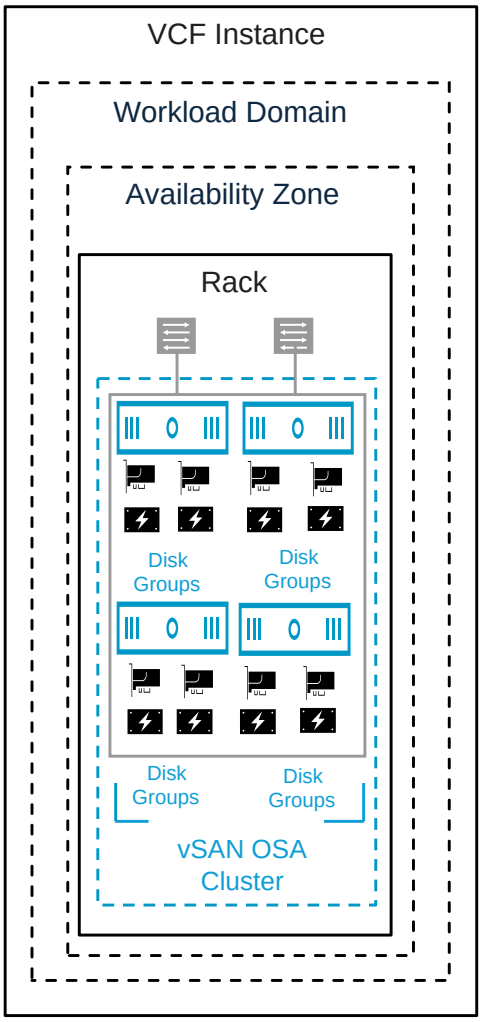
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-001	• Select a specific size of drives used according to the requirements of target workloads. • Use multiple storage devices per vSAN node • NVMe TLC devices with a performance Class F (100000 - 349999) or higher with endurance class 3 DWPD or higher.	• The number of storage devices has an impact on the performance of vSAN ESA. • NVMe drives with mixed-use endurance and performance class typically classified on the vSAN HCL as "3DWPD" (3 Drive Writes per Day) will give a balance between performance and durability.	• Smaller drive sizes will impact overall capacity of vSAN ESA. Large drive sizes will introduce larger capacity fault domain in case of failure. • Sizing vSAN Nodes requires determining the workload capacity requirements prior to deployment. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-002	Use vSAN ready nodes with high core count.	vSAN Express Storage Architecture (ESA) is designed to be highly parallel and leverages multiple CPU cores for improved performance, making it multithreaded, and optimized for NVMe- TLC based flash devices.	There will be a correlation between target workload, number of cores and NVMe devices per vSAN Node. Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.
VCF-VSAN-ESA-RCMD-CFG-003	Use vSAN ready nodes with high memory capacity per vSAN node.	When enabled vSAN has required memory overheads and must have adequate resources account for VM workloads.	Refer to vSAN Sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-ESA-RCMD-CFG-004	At least four (4) nodes for vSAN cluster size.	This ensures in the event of a fault domain failure, vSAN objects can be re-protected to maintain policy compliance. In conjunction with auto policy management, four (4) node vSAN ESA clusters will allow vSAN to automatically create a default storage policy that is capacity optimized.	None. **Note:** New vSAN clusters in VCF 9.1 will use clusters will automatically be configured to use Auto-RAID.
VCF-VSAN-ESA-RCMD-CFG-005	• Select the same minimum requirements for vSAN Storage cluster nodes as vSAN ESA HCI nodes. • Review vSAN Storage minimum node requirements	• vSAN Storage Cluster Nodes use the same architecture as regular vSAN ESA nodes. • vSAN ESA nodes can be re-purposed for vSAN Storage Clusters or vSAN HCI Clusters	None.

Single-Rack vSAN OSA Storage Model

vSAN OSA uses a two-tier storage architecture that utilizes distinct cache and capacity tier storage devices within a disk group construct. vSAN OSA can support a combination of flash and magnetic disks that can be a combination of SATA, SAS, and NVMe storage devices.

Figure 342: Single-Rack vSAN OSA Storage Model in VMware Cloud Foundation



Single-Rack vSAN OSA Storage Model Attributes

A *Single-Rack vSAN OSA Storage Model* has the following attributes.

Attribute	Detail
Two Tier	vSAN OSA architecture consists of two tiers: a cache tier for the purpose of write cache, and a capacity tier for persistent storage.
vSAN Storage Devices	vSAN OSA can support of flash and magnetic disks that can be a combination of SATA, SAS, and NVMe storage devices.
Data Services	• Deduplication and compression at cluster / disk group level. • Encryption. • Integrated file services. • vSAN iSCSI target service. • Stretched cluster. • Fault domains. • Kubernetes persistent storage services (block and file).
Virutal Machine Storage polices	vSAN storage policies (also known as Virtual Machine Storage Policies or VM Storage Policies) are rules defining virtual machine (VM) storage requirements that vSAN automatically enforces across a cluster. vSAN OSA will use a vSAN Default Storage policy with Raid-1 protection if a VM Storage Policy policy is not specified

Single-Rack vSAN OSA Storage Model Options

A *Single-Rack vSAN OSA Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Deduplication and Compression	vSAN OSA can be deployed with duplication and compression either enabled or disabled at a vSAN cluster level. **Note:** Enabling compression only data service is not allowed during VCF cluster deployment.
Data In Transit Encryption	Data In Transit is supported to be enabled as day 0 or day 2 operation.
Data at Rest encryption	Data at Rest Encryption is supported as a day 2 operation.
vSAN Networking	vSAN OSA can utilize 10GbE , 25GbE or higher. See Table 764: vSAN Host Network Model Options.

Single-Rack vSAN OSA Storage Model Design Requirements and Recommendations

Table 773: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Table 774: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

Table 775: vSAN OSA Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-OSA-REQD-CFG-001	vSAN OSA requires one storage device for caching and at least one storage device for capacity to create a valid disk group.	A valid vSAN OSA disk group comprises of one caching tier device (flash) and one capacity tier device (magnetic or flash).	Check all hardware against official hardware compatibility guide.
VCF-VSAN-OSA-REQD-CFG-002	Each vSAN OSA node requires a minimum of one vSAN disk group.	• Each node in a vSAN OSA cluster contributes one or more disk groups to the cluster.	Disk groups can contribute to vSAN node memory overhead https://vcf.broadcom.com/tools/vsansizer/home for more prescriptive sizing guidance.
VCF-VSAN-REQD-CFG-003	Do not use the storage I/O controllers that are running vSAN disk groups for another purpose.	Running non-vSAN filesystem, for example, VMFS, on a storage I/O controller that shares devices making up a vSAN disk group can impact vSAN performance and stability.	If non-vSAN disks are required in ESX hosts, you must have an additional storage I/O controller in the host.

Table 776: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Table 777: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

Table 778: vSAN OSA Storage Model Design Recommendations

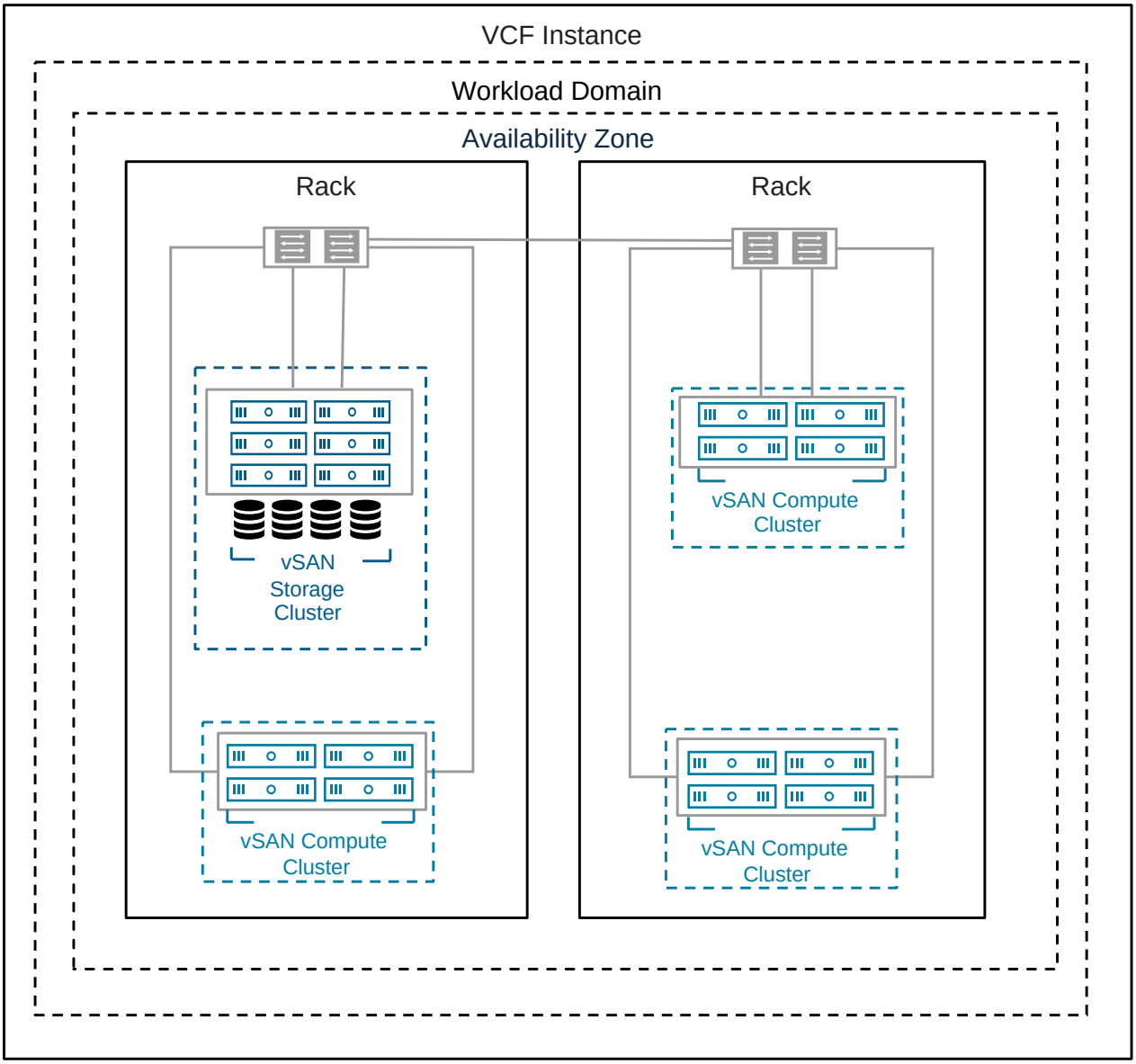
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-OSA-RCMD-CFG-001	Minimum two (2) disk-groups per vSAN node.	For optimal vSAN performance and availability , minimum of two (2) disk-groups per vSAN node is recommended	This requires two (2) cache disks that do not contribute to storage capacity.
VCF-VSAN-OSA-RCMD-CFG-002	Use higher base CPU clock speed for ESX hosts is recommended.	vSAN OSA can be sensitive to CPU clock speed for scaling aggregate performance.	This may restrict certain hardware choices.

Single-Rack vSAN Storage Cluster Storage Model

VMware vSAN storage clusters (previously known as vSAN Max™) is a dis-aggregated storage solution for vSphere clusters. A vSAN Storage cluster provides storage resources to one or more vSAN Compute clusters. vSAN Storage Clusters are based on vSAN ESA architecture. vSAN Storage Clusters implements a de-converged network approach. A "de-converged storage network" (or non-converged storage network) refers to a network architecture where vSAN storage traffic is separated from vSAN compute client traffic. This facilitates using dedicated storage network adapters or physical ports, rather than sharing the same infrastructure.

Figure 343: Single-Rack vSAN Storage Cluster Storage Model in VMware Cloud Foundation

This image depicts a vSAN Storage Cluster with multiple vSAN Client Clusters (vSphere Clusters) consuming storage resources on a vSAN Storage Cluster.



Single-Rack vSAN Storage Cluster Storage Model Attributes

A *Single-Rack vSAN Storage Cluster Storage Model* has the following attributes.

Attributes	Detail
vSAN Storage Cluster Architecture	vSAN Storage Clusters are based on Single-Rack vSAN ESA Storage Model architecture
vSAN Compute Cluster	A vSAN Compute Cluster (vSphere cluster) that can consume vSAN Storage Cluster capacity
vSAN Storage Cluster Client Traffic	• vSAN Storage Clusters introduce a new vSAN networking traffic type to enable a de-converged storage network. • This allows internal vSAN traffic, such as data replication, resync and rebalance to be logically separated from vSAN client cluster traffic. • vSAN client cluster traffic is used for vSAN compute client traffic and can be described as a second vSAN network type configured on a vSAN Storage Cluster.

Single-Rack vSAN Storage Cluster Storage Model Options

A *Single-Rack vSAN Storage Cluster Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
vSAN Storage Cluster	• **Dis-aggregated vSAN Storage traffic** vSAN Storage traffic and vSAN VM IO traffic will be separated. If selected a new traffic type called **vSAN Storage Cluster Client traffic.** This will require an additional VMkernel interface and a separate network pool with an allocated IPv4 IP subnet and VLAN. • **Converged vSAN Storage Traffic** If selected vSAN traffic and vSAN VM IO traffic share the same physical resources. • A vSAN Storage Cluster can be stretched or non stretched.
Distributed Switch Models	• Single Distributed Switch Model • Storage Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
MTU	1500 - 9000
vSAN Networking	• vSAN Storage clusters can use minimum 25GbE or higher for vSAN Storage traffic • With dis-aggregated vSAN traffic, vSAN Storage client traffic can use minimum 10GbE or higher See Table 764: vSAN Host Network Model Options.

Single-Rack vSAN Storage Cluster Storage Model Design Requirements

Table 779: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Table 780: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

Table 781: vSAN Storage Cluster Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-STG-CLU-REQD-CFG-001	It is required to use vSAN ESA compatible hardware.	vSAN Storage Clusters are supported on vSAN ESA architecture	None.
VCF-VSAN-STG-CLU-REQD-CFG-002	Provide at least four (4) ESX hosts. The maximum is thirty-two (32) ESX hosts in a vSAN Storage Cluster.	A vSAN storage cluster must contain at least four (4) ESX hosts.	Do not enable "Host Rebuild Reserve" capacity management mechanism in a vSAN storage cluster that consist of only 4 hosts. When paired with the Auto-Policy Management feature, this will prevent vSAN storage clusters from being able to use space-efficient RAID-5
VCF-VSAN-STG-CLU-REQD-CFG-003	The minimum number of vSAN storage devices per ESX host is two (2).	Please Reference https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance and KB90343	
VCF-VSAN-STG-CLU-REQD-CFG-004	The minimum number of CPU Cores per ESX host is sixteen (16)	Please Reference https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance and KB90343	None.
VCF-VSAN-STG-CLU-REQD-CFG-005	The minimum supported memory per ESX host is 128GB.	Please Reference https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance and KB90343	None.
VCF-VSAN-STG-CLU-REQD-CFG-006	The minimum supported vSAN Node storage capacity is 20TB.	Please Reference https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance and KB90343	None.
VCF-VSAN-STG-CLU-REQD-CFG-007	A vSAN Storage Cluster can serve up to ten (10) vSAN Client clusters.	Please reference VMware Configuration Maximums	None.
VCF-VSAN-STG-CLU-REQD-CFG-008	A vSAN Storage Cluster datastore can mounted by up to 104 vSAN client nodes.	Please reference VMware Configuration Maximums	This is the aggregate max number of hosts in a storage & client cluster

Table 782: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Table 783: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

Table 784: vSAN Storage Cluster Storage Model Design Recommendations

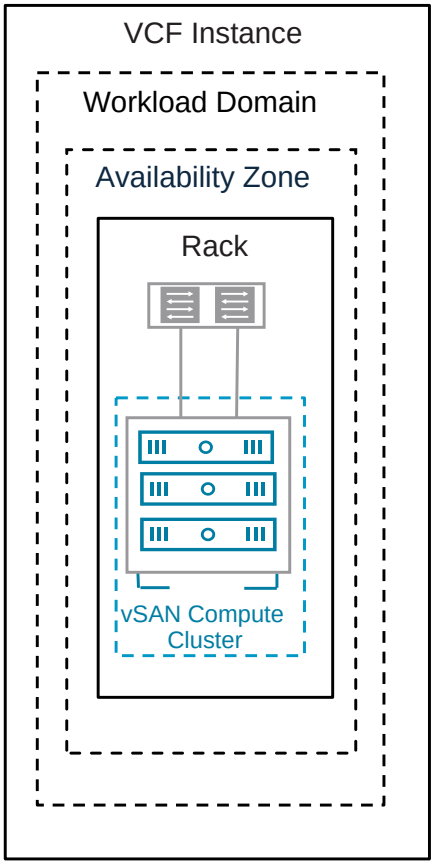
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-STG-CLU-RCMD-CFG-001	Use storage traffic separation for vSAN storage cluster traffic.	• Storage traffic separation will help guarantee dedicated physical host network resources for vSAN cluster traffic. • Storage traffic separation will help isolate and secure vSAN Storage traffic from vSAN Client Cluster traffic	Implementing storage traffic separation may require additional physical and logical networking resources.
VCF-VSAN-STG-CLU-RCMD-CFG-002	Depending on the vSAN client cluster traffic and vSAN storage cluster traffic requirements, use separate physical host networking with different capabilities, such as bandwidth and latency.	vSAN storage replication traffic might require have higher data rate than vSAN client traffic. For example, a 25 GbE Ethernet might satisfy the requirements for vSAN client cluster traffic, while 100 GbE Ethernet offers a significantly higher data transfer rate (100 Gbps) which might satisfy vSAN storage traffic requirements.	Sizing of vSAN client traffic and vSAN storage traffic may have to be evaluated depending on the requirements of the workload Using higher bandwidth hardware can be more expensive due higher data transfer rate and the need for more complex equipment.
VCF-VSAN-STG-CLU-RCMD-CFG-003	Design for 7-node vSAN storage clusters.	A 7-node vSAN storage cluster ensures that the cluster can support FTT=2 using RAID-6. While FTT=2 using RAID-6 only requires 6 hosts, a sustained failure of a host in a cluster consisting of just 6 hosts would result in an insufficient number of hosts to regain its prescribed level of resilience. 7 hosts would be able to automatically regain their prescribed level of resilience upon a sustained host failure	None

Single-Rack vSAN Compute Cluster Storage Model

A *Single-Rack vSAN Compute Cluster Storage Model* is a cluster of ESX hosts configured to mount a remote vSAN datastore provided by a vSAN Storage Cluster in the same VCF domain.

Figure 344: Single-Rack vSAN Compute Cluster Storage Model in VMware Cloud Foundation

A vSAN Compute Cluster is a vSphere Cluster with VMkernel interface tagged with vSAN traffic type to communicate to an existing vSAN Storage Cluster in order to mount a vSAN remote datastore.



Single-Rack vSAN Compute Cluster Storage Model Attributes

A *Single-Rack vSAN Compute Cluster Storage Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter
Cluster Rack Mapping	Hosts are deployed within a single physical rack or rack enclosure.
Resilience	• vSphere HA protects workloads against host failures.
Availability Zones	• Single availability zone only.
Networking	• Layer 2 networking
vSAN Compute Cluster	• A vSAN Compute cluster is a vSphere Cluster Models with a #unique_611/unique_611_Connect_42_id-6b5ac9d1-3397-4f8c-b56e-4453c86757bb defined • The primary function of a vSAN Compute cluster is to provide compute resources (CPU, memory) for running virtual machines, vSAN Compute clusters do not contribute to storage. • A vSAN Compute cluster can act as a client to a vSAN Storage cluster. • A vSAN Compute cluster does not have to be a certified vSAN Ready node, thus does not require certified vSAN local storage devices for vSAN use.

Attribute	Detail
vSAN Network Traffic	• vSAN Client cluster have dedicated VMkernel interfaces tagged with standard vSAN traffic type. • The vSAN Compute cluster nodes can optionally communicate with the vSAN Storage Cluster Client Traffic type defined on the vSAN Storage cluster. (if configured). • On the vSAN Storage cluster, vSAN Storage Cluster Client Traffic can be tagged on a dedicated VMkernel interface on the vSAN Storage cluster to have a disaggregated network design to separate vSAN compute traffic from vSAN storage traffic.
vSAN Remote Datastore	vSAN Compute Cluster will mount a remote datastore from vSAN Storage cluster resources.

Single-Rack vSAN Compute Cluster Storage Model Dependencies

A *Single-Rack vSAN Compute Cluster Storage Model* has the following dependencies.

Design Area	Dependency
vSAN Host Network Traffic Type	Route to vSAN storage network on vSAN Storage cluster.
	Route to vSAN Storage Cluster Client Traffic network on vSAN Storage cluster. (if configured)

Single-Rack vSAN Compute Cluster Storage Model Design Requirements

Table 785: vSAN Compute Cluster Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-COMP-CLU-REQD-CFG-001	Minimum two (2) vSAN compute nodes is required for VCF provisioned clusters	Deploying compute clusters for VCF requires minium 2 (nodes) or greater	none
VCF-VSAN-COMP-CLU-REQD-CFG-001	vSAN networking must be configured on hosts in the Compute cluster.	A vSAN Compute cluster is a vSphere cluster with a vSAN networking element that enables it to mount a vSAN Storage Cluster	Requires planning for vSAN networking requirements to support planned VM workload.
VCF-VSAN-COMP-CLU-REQD-CFG-002	vSAN Compute nodes must be commissioned as vSAN Compute Cluster type	This allows creation of vSAN Compute Clusters with vSAN compute node types	Requires new hosts to be commissioned to an exising network pool that has the correct networking attributes
VCF-VSAN-COMP-CLU-REQD-CFG-003	A vSAN Client Cluster must be deployed within same VCF Workload domain as vSAN Storage Cluster is deployed	A vSAN Client cluster is not supported to be cross mounted across vCenter instances from a VCF workflow perspective	Requires vSAN client clusters to be deployed within a VCF workload domain.
VCF-VSAN-COMP-CLU-REQD-CFG-004	All objects that make up a VM must reside on the same datastore.	This is currently the supported method for VMs deployed on vSAN Client Clusters	VM performance characteristics and fail-over responses will be consistent
VCF-VSAN-COMP-CLU-REQD-CFG-005	A vSAN Compute cluster can mount up to 5 remote datastores	This is a supported maximum configuration	This will require careful planning from compute and networking sizing perspective
VCF-VSAN-COMP-CLU-REQD-CFG-006	Configure vSphere HA failure response for Datastore with APD to be set to Power off and restart VMs.	This ensures Virtual machines respond appropriately to a HA event if a remote vSAN datastore is unavailable on a ESX host	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-COMP-CLU-REQD-CFG-007	Ensure vSAN Compute Cluster latency to vSAN Storage Cluster is less than 5 milliseconds	vSAN compute clusters perform a precheck and require network latency between the client clusters and the server cluster of 5ms or less.	Ensure a high speed low latency network between vSAN compute clusters and vSAN Storage Clusters.

Multi-Rack vSAN Storage Models

Multi-Rack vSAN Storage Models is a configuration of vSAN where the ESX hosts contributing storage to the vSAN cluster are physically located in different racks. This design is primarily used for rack awareness and to provide resilience against the failure of an entire rack. By spreading data across multiple racks, the architecture can tolerate the failure of an entire rack, as workloads can automatically restart affected virtual machines on ESX hosts in other racks. The following sections describe *Multi-Rack vSAN Storage Models*. A *Multi-Rack vSAN Storage Model* can either be based on a vSAN ESA or vSAN OSA architecture, or a vSAN Compute cluster.

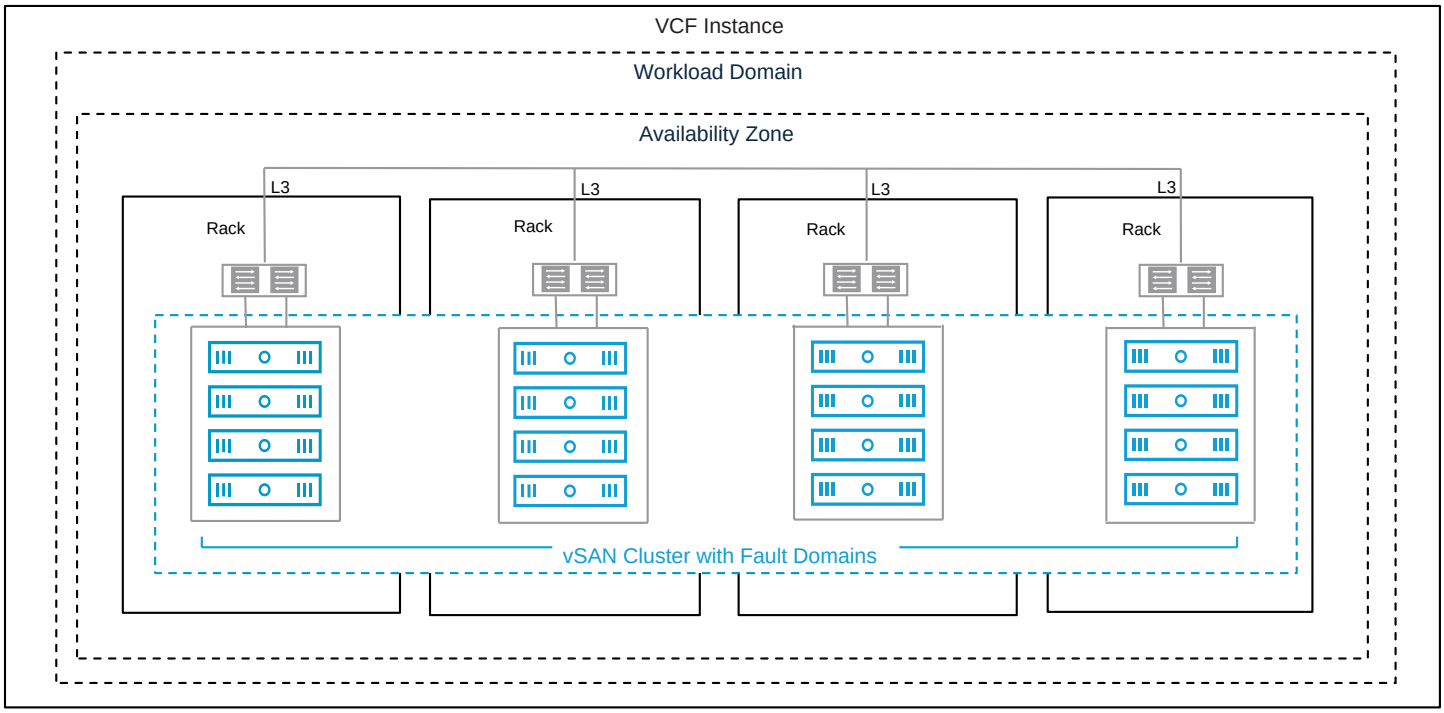
- Multi-Rack vSAN Storage Model
- Multi-Rack vSAN Compute Cluster Storage Model

Multi-Rack vSAN Storage Model

The *Multi-Rack vSAN Storage Model* is based on vSAN fault domains and either the Multi-Rack Layer 2 vSphere Cluster Model or Multi-Rack Layer 3 vSphere Cluster Model. This offers rack-level protection for workloads. This design protects virtual machines and vSAN objects against rack level failures.

Figure 345: Multi-Rack vSAN Storage Model in VMware Cloud Foundation

This figure depicts a vSAN cluster that is distributed across four (4) racks. Each rack can be logically associated to a vSAN fault domain.



Multi-Rack vSAN Storage Model Attributes

A *Multi-Rack vSAN Storage Model* has the following attributes.

Attribute	Detail
vSAN HCI Cluster	A cluster that provides both compute and storage to virtual machines.
vSAN Fault Domain	A logical grouping of hosts based on their physical location such as a rack that allows vSAN to protect against failures of an entire physical location.
vSAN Datastore	A distributed datastore that supports features that require shared storage.
vSAN Physical Networking	Host network interfaces can be dedicated or can be shared with other traffic types.
vSAN Availability	Workloads are protected by synchronously replicating objects in a vSAN cluster.
Data Services	• Deduplication and/or compression. • Encryption. • Integrated file services. • vSAN iSCSI Target Service. • Kubernetes persistent storage. services (block and file).

A *Multi-Rack vSAN Storage Model* shares many of the same attributes as a Multi-Rack vSphere Cluster Detailed Design. One of the key choices is the network topology that a multi-rack deployment consumes.

Multi-Rack vSAN Storage Model Options

A *Multi-Rack Cluster Model* can be deployed using the following networking types.

Multi-Rack Cluster Type	Description
Layer 2	• Hosts are installed horizontally across different physical racks. • Each physical rack shares the same set of Layer 2 broadcast domains for each VLAN.
Layer 3	• Hosts are installed horizontally across different physical racks. • Each physical rack has a unique set of Layer 2 broadcast domains for each VLAN.

vSAN HCI Multi-Rack Storage Model Attributes

A *Multi-Rack Cluster Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter
Cluster Rack Mapping	Hosts are installed horizontally across different physical racks.
Resilience	• vSphere HA protects workloads against host failures. • Multiple fault domains protect workloads in the event of a failure of one or more fault domains/racks.
Availability Zones	• Layer 2 cluster type can be stretched across availability zones. • Layer 3 cluster type is single availability zone only.
Networking	• Layer 2 (Including VXLAN/EVPN Fabric). or • Layer 3
Storage	Storage capable of surviving the failure of a single fault domain at the storage level.

Multi-Rack vSAN Storage Model VLANs and Subnets

Function	Layer 2 VLANs and Subnets	Layer 3 VLANs and Subnets
VM management	• Shared across racks. • Highly available gateway at the ToR switch.	• Required per rack. • Highly available gateway at the ToR switches or leaf nodes in the rack.
ESX management		
vSphere vMotion		
vSAN		
Host overlay		
NFS	Not supported.	Not supported.
NSX Edge Uplink01	• Shared across racks • Highly available gateway at the ToR switch.	Depends on NSX Edge design.
NSX Edge Uplink02		
NSX Edge overlay		

Multi-Rack vSAN Storage Model Design Requirements and Recommendations

Table 786: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Table 787: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

Table 788: Multi-Rack vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-L3MR-REQD-CFG-001	For Layer 3 multi-rack clusters, configure the IP address of the vSAN Network Gateway for each rack accessible over Layer 3 as the isolation address for the nodes in that in the vSphere Cluster.	Enables vSphere HA to validate if an ESX host is isolated from the vSAN Network.	The IP address of the vSAN network gateway(s) must be highly available and reply to ICMP requests.
VCF-VSAN-L3MR-REQD-CFG-002	Use a minimum of three (3) racks for the cluster.	Provides support for re-protecting vSAN objects if a single-rack failure occurs.	• Requires a minimum of 3 hosts in a cluster, distributed across three (3) racks. • Will provide redundancy, but will not provide automatic rebuild of vSAN components.
VCF-VSAN-L3MR-REQD-CFG-003	Configure vSAN fault domains and place the nodes of each rack in their fault domain.	Allows workload VMs to tolerate a rack failure by distributing copies of the data and witness components on nodes in separate racks.	Configure fault domains manually in vCenter after cluster expansion.
VCF-VSAN-L3MR-REQD-CFG-004	Deactivate vSAN ESA auto policy management.	This feature is not supported with vSAN fault domains.	To align with the number of vSAN fault domains, it maybe necessary create a default storage policy manually.
VCF-VSAN-L3MR-REQD-CFG-005	Do not enable reserved Capacity in a vSAN HCI multi-rack cluster.	vSAN capacity reserve is not supported in conjunction with the following vSAN cluster topologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains	This will require manually calculating vSAN slack space. Refer to the vSAN sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

Table 789: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Table 790: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

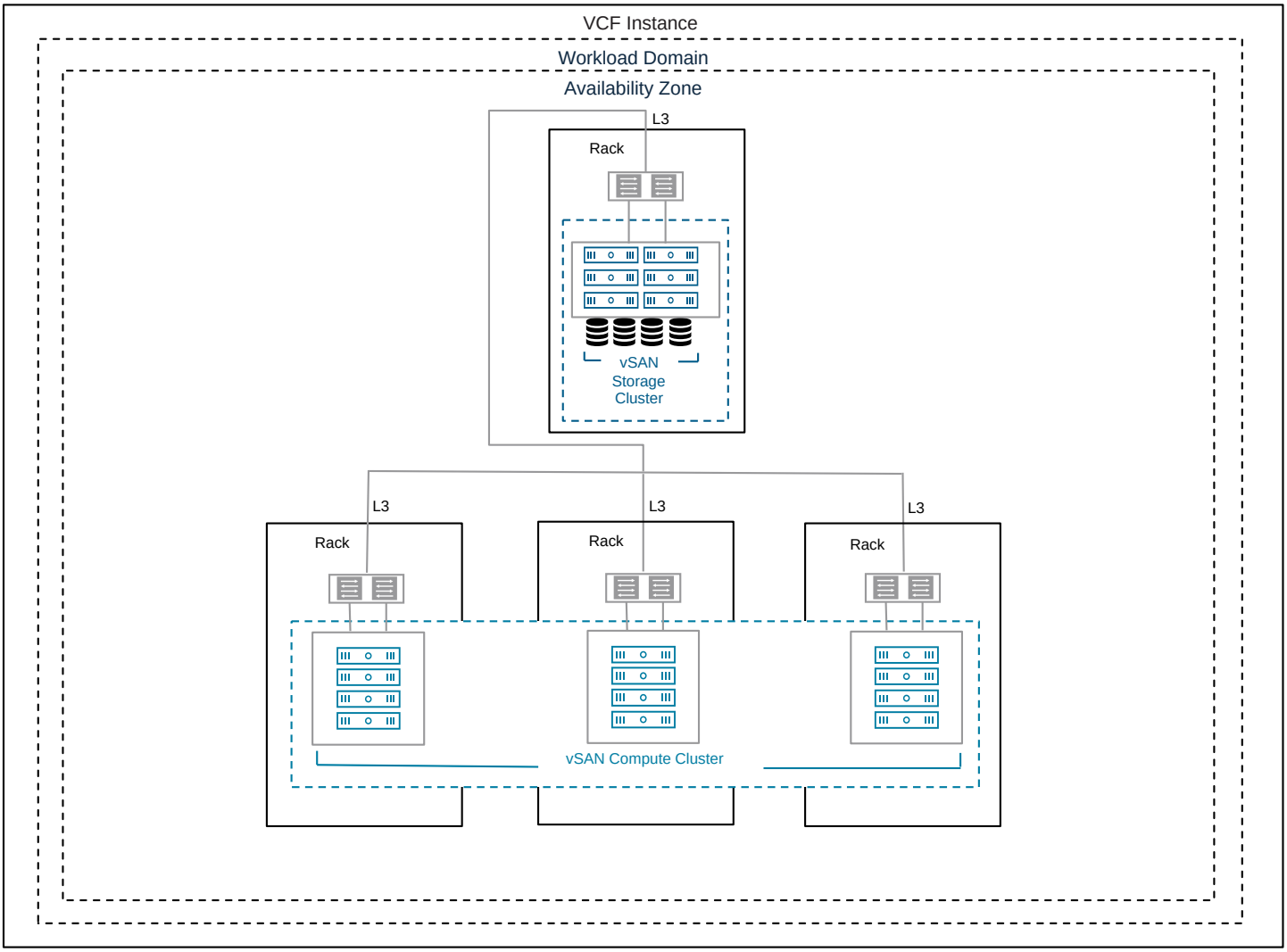
Table 791: Multi-Rack vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-L3MR-RCMD-CFG-001	Use four (4) Fault domains for vSAN HCI multi-rack	Provides support for re-protecting vSAN objects if a single-rack failure occurs.	• Requires a minimum of four(4) ESX hosts added to a cluster, distributed across four (4) racks.
VCF-VSAN-L3MR-RCMD-CFG-002	Use Layer 3 networking between each fault domain (physical rack).	Provides fault isolation per fault domain or physical rack.	• Requires multiple IPv4 or IPv6 subnets per rack. • Requires gateway per IPv4 or IPv6 subnet. • Requires layer 3 routing between racks.

Multi-Rack vSAN Compute Cluster Storage Model

A *Multi-Rack vSAN Compute Cluster Storage Model* consumes storage from a vSAN Storage cluster deployed in the same VCF domain. A vSAN Compute cluster can be spread over multiple racks routed via Layer 3 networking. Similar to the Multi-Rack Layer 3 vSphere Cluster Model this allows protection of workloads on vSphere clusters in your environment against a failure of a single fault domain by implementing a vSAN Compute cluster across multiple fault domains within a single availability zone. The networks are isolated to a rack using a separate broadcast domain for each network per rack.

Figure 346: Multi-Rack vSAN Compute Cluster Storage Model in VMware Cloud Foundation



Multi-Rack vSAN Compute Cluster Storage Model Attributes

A *Multi-Rack vSAN Compute Cluster Storage Model* has the following attributes.

Attribute	Detail
Datacenters	Single datacenter
Cluster Rack Mapping	Hosts are deployed horizontally across different physical racks.

Attribute	Detail
Resilience	• vSphere HA protects workloads against host failures. • Multiple fault domains protect workloads in the event of a failure of one or more fault domains/racks.
Availability Zones	Single availability zone only.
Networking	Layer 3 routed networks with isolated broadcast domains per rack.
vSAN Compute Cluster	• A vSAN Compute cluster is a vSphere Cluster Models with a #unique_611/unique_611_Connect_42_id-6b5ac9d1-3397-4f8c-b56e-4453c86757bb defined • The primary function of a vSAN Compute cluster is to provide compute resources (CPU, memory) for running virtual machines, not storage. • A vSAN Compute-only cluster can act as a client to a vSAN Storage cluster. • A vSAN Compute cluster does not have to be a certified vSAN Ready node, thus does not require certified vSAN local storage devices for vSAN use.
vSAN Network Traffic	• vSAN client clusters have dedicated VMkernel interfaces tagged with standard vSAN traffic type. • The vSAN Compute cluster nodes can optionally communicate with the vSAN Storage cluster client traffic type that is defined on the storage cluster. (if configured) • On the vSAN Storage cluster, vSAN Storage Cluster Client Traffic can be tagged on a dedicated VMkernel interface on the vSAN Storage cluster to have a disaggregated network design to separate vSAN compute traffic from vSAN storage traffic.
vSAN Remote Datastore	vSAN Compute cluster mount as a remote datastore from vSAN Storage cluster.

Multi-Rack vSAN Compute Cluster Storage Model Dependencies

A *Multi-Rack vSAN Compute Cluster Storage Model* has the following dependencies.

Design Area	Dependency
vSAN Host Networking Traffic Type	Route to vSAN storage network on vSAN Storage cluster.
	Route to vSAN Storage Cluster Client Traffic network on vSAN Storage cluster. (if configured)

Multi-Rack vSAN Compute Cluster Storage Model Design Requirements and Recommendations

Table 792: vSAN Compute Cluster Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-COMP-CLU-REQD-CFG-001	Minimum two (2) vSAN compute nodes is required for VCF provisioned clusters	Deploying compute clusters for VCF requires minium 2 (nodes) or greater	none
VCF-VSAN-COMP-CLU-REQD-CFG-001	vSAN networking must be configured on hosts in the Compute cluster.	A vSAN Compute cluster is a vSphere cluster with a vSAN networking element that enables it to mount a vSAN Storage Cluster	Requires planning for vSAN networking requirements to support planned VM workload.
VCF-VSAN-COMP-CLU-REQD-CFG-002	vSAN Compute nodes must be commissioned as vSAN Compute Cluster type	This allows creation of vSAN Compute Clusters with vSAN compute node types	Requires new hosts to be commissioned to an exising network pool that has the correct networking attributes
VCF-VSAN-COMP-CLU-REQD-CFG-003	A vSAN Client Cluster must be deployed within same VCF Workload domain as vSAN Storage Cluster is deployed	A vSAN Client cluster is not supported to be cross mounted across vCenter instances from a VCF workflow perspective	Requires vSAN client clusters to be deployed within a VCF workload domain.
VCF-VSAN-COMP-CLU-REQD-CFG-004	All objects that make up a VM must reside on the same datastore.	This is currently the supported method for VMs deployed on vSAN Client Clusters	VM performance characteristics and fail-over responses will be consistent

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-COMP-CLU-REQD-CFG-005	A vSAN Compute cluster can mount up to 5 remote datastores	This is a supported maximum configuration	This will require careful planning from compute and networking sizing perspective
VCF-VSAN-COMP-CLU-REQD-CFG-006	Configure vSphere HA failure response for Datastore with APD to be set to Power off and restart VMs.	This ensures Virtual machines respond appropriately to a HA event if a remote vSAN datastore is unavailable on a ESX host	None.
VCF-VSAN-COMP-CLU-REQD-CFG-007	Ensure vSAN Compute Cluster latency to vSAN Storage Cluster is less than 5 milliseconds	vSAN compute clusters perform a precheck and require network latency between the client clusters and the server cluster of 5ms or less.	Ensure a high speed low latency network between vSAN compute clusters and vSAN Storage Clusters.

Table 793: Multi-Rack vSAN Compute Cluster Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-COMP-L3MR-RCMD-CFG-001	Use Layer 3 networking between each fault domain (physical rack).	Provides fault isolation per fault domain or physical rack.	• Requires multiple IPv4 or IPv6 subnet per rack. • Requires gateway per IPv4 or IPv6 subnet. • Requires layer 3 routing between racks.

Stretched vSAN Storage Models

Stretched vSAN Storage Models provide a highly resilient and available storage architecture by distributing data and services across multiple locations, leveraging synchronous replication over a low-latency inter-site link. *Stretched vSAN Storage Models* span two distinct physical locations, acting as active-active sites for data storage. This means that data is synchronously replicated between the sites, providing redundancy in case of a site failure. The following vSAN cluster models can be deployed in a stretched cluster configuration, vSAN HCI Cluster (vSAN ESA or vSAN OSA architectures), a vSAN Storage cluster, or vSAN Compute cluster.

- Stretched vSAN Storage Model
- Stretched vSAN Storage Cluster Storage Model
- Stretched vSAN Compute Cluster Storage Model

Sample VLAN and IP Subnet Requirements for Stretched vSAN Cluster in a VCF Environment

In an environment stretched across multiple availability zones, Layer 2 networks must be stretched between the availability zones by the physical infrastructure. You must also provide a Layer 3 gateway that is highly available between availability zones. The method for stretching these Layer 2 networks and providing a highly available Layer 3 gateway is vendor-specific.

This section displays a sample configuration for an environment with a stretched vSAN cluster across two (2) availability zones.

For NSX Edge designs that have a requirement to be stretched, refer to NSX Edge Cluster Models for Dual Availability Zones.

Table 794: Stretched vSAN Cluster VLAN and Subnet Sample Requirements and Recommendations

VLAN Function	Stretched Layer 2 VLAN	Availability Zone 1	Availability Zone 2	HA Layer 3 Gateway	Recommended MTU
VM Management	Yes	Yes	Yes	Yes	1500
Host Management VLAN (AZ1)	No	Yes	No	Yes	1500
vMotion VLAN (AZ1)	No	Yes	No	Yes	9000
vSAN VLAN (AZ1)	No	Yes	No	Yes	9000
NSX Host Overlay (AZ1)	No	Yes	No	Yes	9000
Host Management VLAN (AZ2)	No	No	Yes	Yes	1500
vMotion VLAN (AZ2)	No	No	Yes	Yes	9000
vSAN VLAN (AZ2)	No	No	Yes	Yes	9000
NSX Host Overlay (AZ2)	No	No	Yes	Yes	9000

Important:

Networks that are not stretched can have the same or different VLAN IDs on either availability zone.

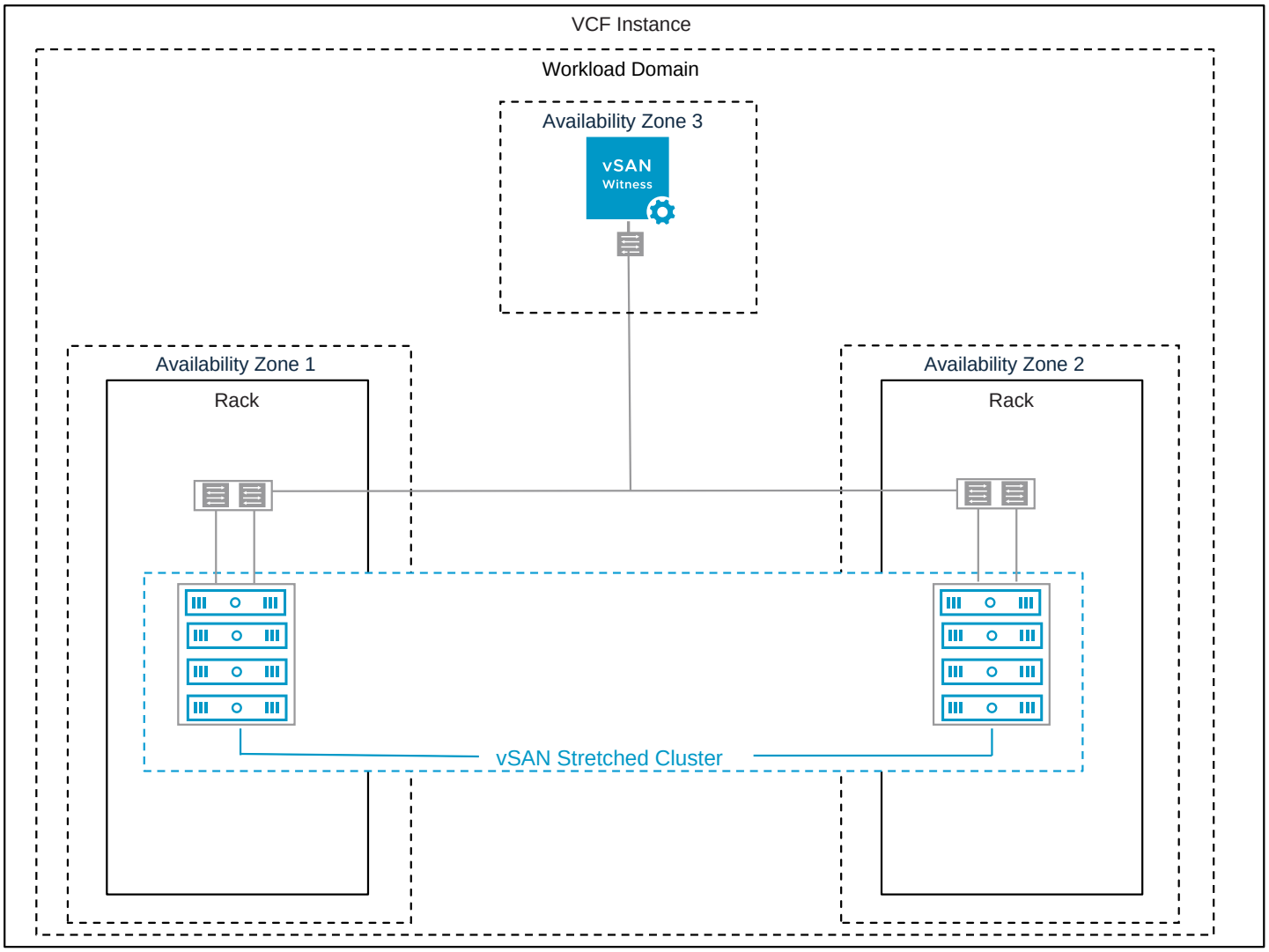
Stretched Layer 2 networks should have the same VLAN ID on either availability zone.

Stretched vSAN Storage Model

A *Stretched vSAN Storage Model* is a cluster configuration where the ESX hosts, that comprise a single vSAN cluster, reside in two availability zones.

Figure 347: Stretched vSAN Storage Model in VMware Cloud Foundation

Stretched vSAN Storage Model depicting three availability zones for vSAN. Two availability zones are designated for vSAN data nodes and a third availability zone is designated for a vSAN witness node.


VCF Instance
Workload Domain
Availability Zone 3
vSAN
Witness
Availability Zone 1
Rack
Availability Zone 2
Rack
vSAN Stretched Cluster

Stretched vSAN Storage Model Attributes

A *Stretched vSAN Storage Model* has the following attributes.

Attribute	Detail
Fault Domain / Avilability zone	• Stretched vSAN clusters comprise of three (3) Fault domains. A fault domain comprises of one or more vSAN nodes. • A fault domain or availability zone can represent a physical location. Fault domains / availability zones can either be two distinct data centers in a metro distance, or two safety or fire sectors (data halls) in the same large-scale datacenter.
vSAN Traffic Type	• Stretched vSAN clusters utilize vSAN Traffic type for data traffic. • Stretched vSAN clusters utilize vSAN Witness Traffic type for vSAN meta-data traffic. **Note:** Stretched vSAN Storage clusters (ESA) can be configured to use vSAN Storage Cluster Client traffic for vSAN compute client clusters. See Stretched vSAN Storage Cluster Storage Model.
vSAN Witness Appliance	A witness node is a pre-configured appliance that runs ESX and is distributed as an OVA file.

Stretched vSAN Storage Model Options

A *Stretched vSAN Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Storage Models	• Single-Rack vSAN ESA Storage Model • Single-Rack vSAN OSA Storage Model
Witness Traffic Separation	• Witness traffic separation allows you to use a distinct VMkernel adapter for vSAN witness traffic communication with the vSAN witness appliance. • There are two choices when stretching a vSAN clusters. – When enabled, the ESX management network is used to communicate with the vSAN witness node via the management network. – When disabled, The vSAN data traffic network is used to communicate with the vSAN witness node. This requires routing the vSAN data network VLAN to the vSAN witness network.

Stretched vSAN Storage Model Design Requirements and Recommendations

Table 795: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Table 796: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

Table 797: Stretched vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-001	Two (2) availability zones and a third site for the witness appliance location.	Fault domains are mapped to availability zones to provide logical and physical ESX host separation and ensure a copy of vSAN data is always available even when an availability zone goes offline.	• This requires both availability zones to have an equal number of ESX hosts residing in each site, and are connected using a high bandwidth/low latency connection known as an inter-site link, or ISL. • The third site should host the witness appliance and must not have a logical or physical dependency on either availability zone.
VCF-VSAN-SC-REQD-CFG-002	The minimum latency required between the data sites hosting virtual machine objects should not exceed 5 milliseconds (ms) in round trip time (RTT), or 2.5ms one way.	The latency minimums exist largely to ensure that applications can write data resiliently across two sites in a timely manner.	Stretched cluster network latency requirements will dictate the realistic geographic distance between sites.
VCF-VSAN-SC-REQD-CFG-003	The minimum latency required between witness site and the data sites should not exceed 200ms RTT, or 100ms one way.	The witness site only stores small amounts of metadata, and is not a part of the vSAN data path.	None.
VCF-VSAN-SC-REQD-CFG-004	vSAN Stretched Cluster Data Site to Data Site Bandwidth minimum is no less than 10GbE. • The vSAN Stretched Cluster Bandwidth Sizing document can be used for general guidance in sizing for vSAN OSA and ESA architectures.	The network requirements for the data-site to data site in a stretch cluster remain the same for both vSAN ESA and OSA.	• VM workloads will ultimately determine the network bandwidth demand between data-sites. • Please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing
VCF-VSAN-SC-REQD-CFG-005	vSAN Stretched Cluster Data Site to Witness Site Bandwidth is based on number of VMs that is protected by a stretch cluster storage policy.	• Network communication from the data sites to the witness host appliance is comprised entirely of metadata and this is wholly dependent on the number of VM and components protected by vSAN Stretch Cluster polices. • There is no minimum bandwidth defined as estimating the bandwidth required for the witness site is based on the number of VM components.	For accurate sizing please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing
VCF-VSAN-SC-REQD-CFG-006	Management Domain initial cluster Minimum of eight (8) nodes for vSAN stretched cluster for high availability (HA) install	Management domain initial clusters deployed in high availability (HA) mode have a requirement for 4 nodes in a non-stretched deployment. Stretching a cluster is a day two operation and will require 4 additional hosts on the second availability zone.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-007	Workload domain clusters (except for management Domain initial cluster) Minimum of six (6) nodes for vSAN stretched clusters	This will provide minimum requirements for site availability and site local availability, tolerating a site failure and one failure on each availability zone. This will give a balance between a host failure on a local availability zone and a full availability zone failure.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure with minimum three (3) nodes per site. However with three nodes per site, vSAN will not automatically rebuild objects that are impacted by a local failure.
VCF-VSAN-SC-REQD-CFG-008	For vSAN OSA architecture Modify the default Storage Policy as follows • **Site disaster tolerance:** Site mirroring • **Failure to tolerate :** One (1) failure RAID-1 (Mirroring)	Provides the minimum necessary protection for VMs in each availability zone, with the ability to recover from an availability zone outage. **Remember:** • For vSAN ESA , Auto Policy configuration will automatically alter the default storage policy to reflect the number of hosts and underlying vSAN topology, when the cluster is converted to a stretch cluster. • With 9.1 vSAN Auto-RAID will be configured by default to all new vSAN ESA clusters and automticlly configure the optimal policy based on the cluster topology	• Additional Storage is necessary to satisfy the storage policy requirements in both availability zones. • Additional policies can be configured for specific workloads depending on their individual performance or availability requirements which may differ from what the default vSAN stretched cluster policy.
VCF-VSAN-SC-REQD-CFG-009	Deploy a vSAN witness appliance in a location that is not co-located to the ESX hosts in any of the availability zones.	Ensures availability of vSAN witness appliance in the event of a failure of one of the availability zones.	You must provide a third availability zone that is logically and physically independent of the data sites The third site must be able to provide a environment capable to host the witness appliance.
VCF-VSAN-SC-REQD-CFG-010	Deploy a witness appliance that corresponds to vSAN Architecture, (ESA or OSA).	Ensures the witness appliance deployed is compatible to the correct vSAN Storage architecture	None.
VCF-VSAN-SC-REQD-CFG-011	Connect the first VMkernel (vmk0) adapter of the vSAN witness appliance to the management network in the witness site.	The ESX host management networks in both availability zones must be routed to the vSAN witness host management network in the third witness site.	When using witness traffic separation, the second interface on vSAN witness host (vmk1) can be removed.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-012	Configure vSAN Witness host identity as per standard ESX host recommendations.	• Allows vSAN Witness host to be managed identically as a ESX host. • Ensures host identity, DNS and NTP is consistently configured	Apply the requirements for External Services Design
VCF-VSAN-SC-REQD-CFG-013	Enable vSphere Lifecycle Manager images on the vSAN witness host.	• VMware Cloud Foundation 9.0 requires vSphere Lifecycle Manager images for ESX hosts. • This will prevent warnings or vSAN health check alarms.	This requires either deploying or converting a vSAN witness host as a vSphere Lifecycle Manager image managed entity.

Table 798: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Table 799: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

Table 800: Stretched vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-RCMD-001	Use Layer 3 (routed) networking between the two data sites, and the witness site. • The data sites will only be able to communicate with each other using the ISL • The witness host should only be able to communicate with hosts in each data site directly, and not through another data site.	Using Layer 3 helps avoid spanning tree protocol (STP) from redirecting traffic across an undesirable link, such as the more bandwidth constrained link to the witness site	Ensure Layer 3 gateways are highly available such that it tolerates the failure of an entire availability zone.
VCF-VSAN-SC-RCMD-002	Use witness traffic separation when configuring a stretched vSAN cluster.	• Network Flexibility. You can configure different VLANs or subnets for vSAN data and witness traffic, allowing for better network segmentation and resource allocation. • Mixed MTU. This will allow vSAN witness traffic type to use a different MTU size than the vSAN data traffic. This avoids configuring static routes between vSAN networks. • Simplified Troubleshooting. Identifying and resolving network issues related to witness traffic becomes easier when it is isolated from the main vSAN data traffic.	This requires the vSAN witness network to be routable to both the ESX host management networks on either availability zone as well to VM management network if configured.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-RCMD-CFG-003	Configure the vSAN witness host to use the first VMkernel adapter (vmk0), that is the management interface, for vSAN witness traffic.	• Removes the requirement to have two IP addresses allocated to the vSAN witness host. • Removes the requirement to have static routes in the witness as the witness traffic is routed over the management networks	• The management networks in both availability zones must be routed to the management witness site. • Ensure to use the same MTU size as the host management interfaces. **Note:** Default MTU size is 1500 for management interfaces
VCF-VSAN-SC-RCMD-CFG-004	For vSAN OSA clusters, configure vSAN storage policies with **Force Provisioning**.	vSAN objects are created if an availability zone becomes unavailable because of a site maintenance operation or a failure.	This would require modifying vSAN OSA policies used as part of the stretched cluster. **Tip:** **Force Provisioning** is no longer an applicable storage rule in a stretched cluster using Auto-RAID. Objects will continue to be created in the event of site mainteance.
VCF-VSAN-SC-RCMD-CFG-005	Ensure DRS Host Groups, VM Groups, and VM/Host Rules are in conjunction with vSAN storage policies.	• VM and vSAN data resilience desired outcomes are satisfied. • Suggests or forces a behavior of VM placement in a vSAN stretched cluster while in steady state. • vSphere HA will trigger the desired behavior in response to a failure.	• Requires creation of additional DRS Host groups, VM Groups, and VM/Host DRS Rules. • The placement rules should match the desired failure response.
VCF-VSAN-SC-RCMD-CFG-006	vSphere HA Admission control fail-over capacity is configured to reserve 50% of the CPU and memory capacity of the entire stretched cluster.	This ensures that VMs can fail between AZs and have adequate resources.	Requires adequate capacity guarantee successful fail-over of protected workloads in the event of failure of an entire availability zone.
VCF-VSAN-SC-RCMD-CFG-007	Do not enable reserved Capacity in a stretched vSAN cluster.	vSAN capacity reserve is not supported in conjunction with the following vSAN cluster topologies: • Three (3) node clusters. • Stretched vSAN clusters. • vSAN clusters using fault domains.	This will require manually calculating vSAN slack space. Refer to the vSAN sizer for recommended capacity at https://vcf.broadcom.com/tools/vsansizer/home.

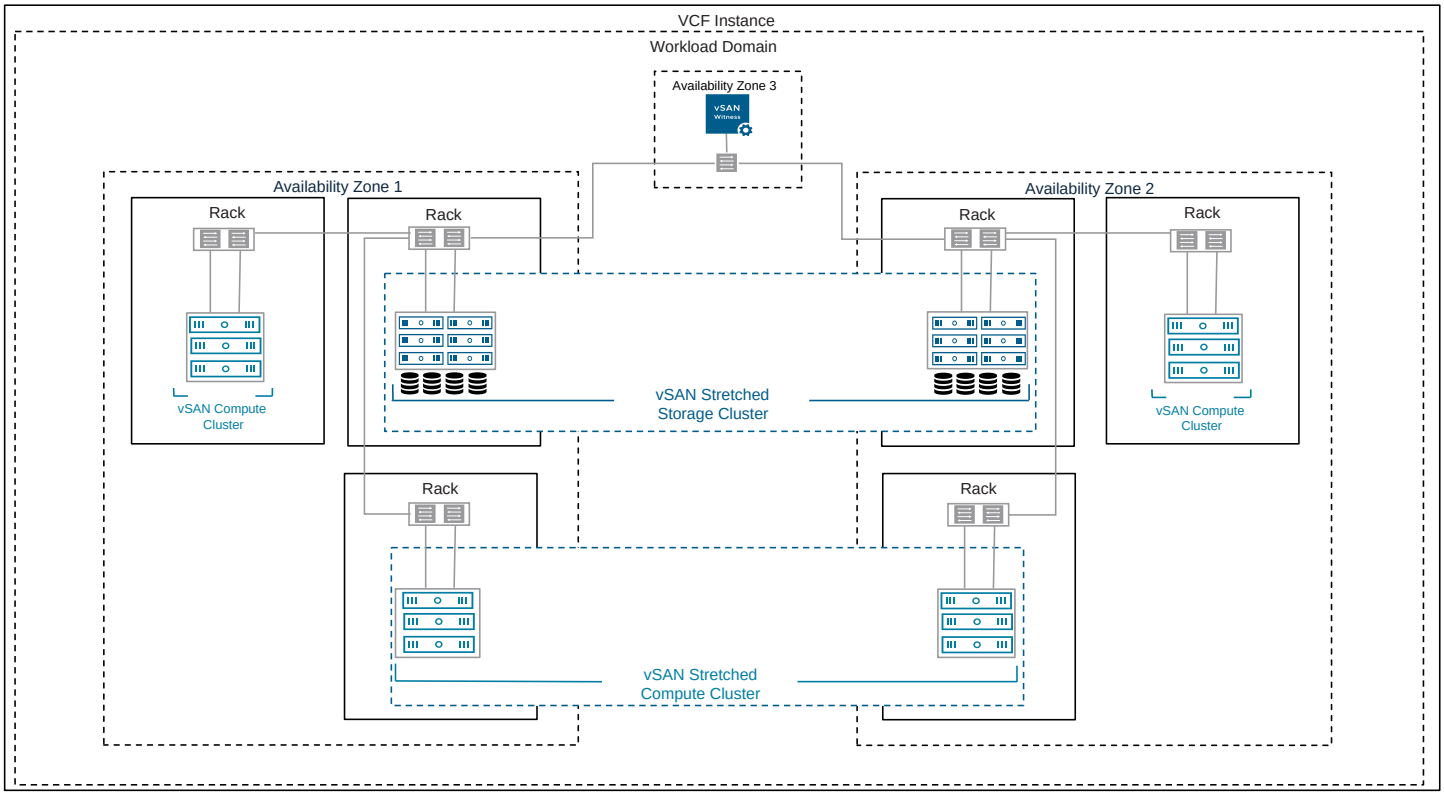
Stretched vSAN Storage Cluster Storage Model

The *Stretched vSAN Storage Cluster Storage Model* is identical from a deployment perspective to a stretched vSAN cluster deployment. This allows for a vSAN Compute Cluster to mount a Stretched vSAN Storage Cluster to avail of higher availability. A vSAN Compute Cluster can also be stretched.

Figure 348: Stretched vSAN Storage Cluster Storage Model in VMware Cloud Foundation

This figure illustrates a stretched vSAN Storage Cluster with vSAN Compute Clusters consuming storage capacity.

A vSAN Compute Cluster can be stretched or non-stretched. A non-stretched vSAN Compute Cluster can be deployed on either availability zone (VCF api only).



Tip:

Stretched vSAN Storage Clusters are stretched vSAN clusters and should inherit the same attributes.

Stretched vSAN Storage Cluster Storage Model Attributes

A *Stretched vSAN Storage Cluster Storage Model* has the following attributes.

Attribute	Detail
Fault Domain / Avilability zone	• Stretched vSAN clusters comprise of three (3) Fault domains. A fault domain comprises of one or more vSAN nodes. • A fault domain or availability zone can represent a physical location. Fault domains / availability zones can either be two distinct data centers in a metro distance, or two safety or fire sectors (data halls) in the same large-scale datacenter.
vSAN Traffic Type	• Stretched vSAN clusters utilize vSAN Traffic type for data traffic. • Stretched vSAN clusters utilize vSAN Witness Traffic type for vSAN meta-data traffic. **Note:** Stretched vSAN Storage clusters (ESA) can be configured to use vSAN Storage Cluster Client traffic for vSAN compute client clusters. See Stretched vSAN Storage Cluster Storage Model.
vSAN Witness Appliance	A witness node is a pre-configured appliance that runs ESX and is distributed as an OVA file.

Stretched vSAN Storage Cluster Storage Model Options

A *Stretched vSAN Storage Cluster Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Witness Traffic Separation	• Witness traffic separation allows you to use a VMkernel adapter for vSAN witness traffic that is different from the adapter for vSAN data traffic. • There are two choices when stretching a vSAN clusters. – When enabled, the ESX management network is used to communicate with the vSAN witness node. – When disabled, The vSAN data traffic network is used to communicate with the vSAN witness traffic.
vSAN Storage Cluster Client Traffic	• **Dis-aggregated vSAN Storage traffic** - vSAN storage traffic and vSAN VM IO traffic will be separated. If selected a new traffic type called vSAN Storage Cluster Client traffic can be selected. This will require an additional VMkernel interface and a separate network pool with an allocated IPv4 Segment. • C**onverged vSAN Storage Traffic** - If selected vSAN traffic and vSAN VM IO traffic share the same physical resources.
Distributed Switch Models	• Single Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model • vSAN Storage Cluster Separation Distributed Switch Model
MTU	1500 - 9000

Stretched vSAN Storage Cluster Storage Model Design Requirements and Recommendations

Table 801: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Table 802: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

Table 803: Stretched vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-001	Two (2) availability zones and a third site for the witness appliance location.	Fault domains are mapped to availability zones to provide logical and physical ESX host separation and ensure a copy of vSAN data is always available even when an availability zone goes offline.	• This requires both availability zones to have an equal number of ESX hosts residing in each site, and are connected using a high bandwidth/low latency connection known as an inter-site link, or ISL. • The third site should host the witness appliance and must not have a logical or physical dependency on either availability zone.
VCF-VSAN-SC-REQD-CFG-002	The minimum latency required between the data sites hosting virtual machine objects should not exceed 5 milliseconds (ms) in round trip time (RTT), or 2.5ms one way.	The latency minimums exist largely to ensure that applications can write data resiliently across two sites in a timely manner.	Stretched cluster network latency requirements will dictate the realistic geographic distance between sites.
VCF-VSAN-SC-REQD-CFG-003	The minimum latency required between witness site and the data sites should not exceed 200ms RTT, or 100ms one way.	The witness site only stores small amounts of metadata, and is not a part of the vSAN data path.	None.
VCF-VSAN-SC-REQD-CFG-004	vSAN Stretched Cluster Data Site to Data Site Bandwidth minimum is no less than 10GbE. • The vSAN Stretched Cluster Bandwidth Sizing document can be used for general guidance in sizing for vSAN OSA and ESA architectures.	The network requirements for the data-site to data site in a stretch cluster remain the same for both vSAN ESA and OSA.	• VM workloads will ultimately determine the network bandwidth demand between data-sites. • Please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing
VCF-VSAN-SC-REQD-CFG-005	vSAN Stretched Cluster Data Site to Witness Site Bandwidth is based on number of VMs that is protected by a stretch cluster storage policy.	• Network communication from the data sites to the witness host appliance is comprised entirely of metadata and this is wholly dependent on the number of VM and components protected by vSAN Stretch Cluster polices. • There is no minimum bandwidth defined as estimating the bandwidth required for the witness site is based on the number of VM components.	For accurate sizing please review https://www.vmware.com/docs/vmw-vsan-stretched-cluster-bandwidth-sizing
VCF-VSAN-SC-REQD-CFG-006	Management Domain initial cluster Minimum of eight (8) nodes for vSAN stretched cluster for high availability (HA) install	Management domain initial clusters deployed in high availability (HA) mode have a requirement for 4 nodes in a non-stretched deployment. Stretching a cluster is a day two operation and will require 4 additional hosts on the second availability zone.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-007	Workload domain clusters (except for management Domain initial cluster) Minimum of six (6) nodes for vSAN stretched clusters	This will provide minimum requirements for site availability and site local availability, tolerating a site failure and one failure on each availability zone. This will give a balance between a host failure on a local availability zone and a full availability zone failure.	vSAN stretched cluster will operate optimally with a balanced ESX host configuration on each availability zone and be able to tolerate a local site failure with minimum three (3) nodes per site. However with three nodes per site, vSAN will not automatically rebuild objects that are impacted by a local failure.
VCF-VSAN-SC-REQD-CFG-008	For vSAN OSA architecture Modify the default Storage Policy as follows • **Site disaster tolerance:** Site mirroring • **Failure to tolerate :** One (1) failure RAID-1 (Mirroring)	Provides the minimum necessary protection for VMs in each availability zone, with the ability to recover from an availability zone outage. **Remember:** • For vSAN ESA , Auto Policy configuration will automatically alter the default storage policy to reflect the number of hosts and underlying vSAN topology, when the cluster is converted to a stretch cluster. • With 9.1 vSAN Auto-RAID will be configured by default to all new vSAN ESA clusters and automticlly configure the optimal policy based on the cluster topology	• Additional Storage is necessary to satisfy the storage policy requirements in both availability zones. • Additional policies can be configured for specific workloads depending on their individual performance or availability requirements which may differ from what the default vSAN stretched cluster policy.
VCF-VSAN-SC-REQD-CFG-009	Deploy a vSAN witness appliance in a location that is not co-located to the ESX hosts in any of the availability zones.	Ensures availability of vSAN witness appliance in the event of a failure of one of the availability zones.	You must provide a third availability zone that is logically and physically independent of the data sites The third site must be able to provide a environment capable to host the witness appliance.
VCF-VSAN-SC-REQD-CFG-010	Deploy a witness appliance that corresponds to vSAN Architecture, (ESA or OSA).	Ensures the witness appliance deployed is compatible to the correct vSAN Storage architecture	None.
VCF-VSAN-SC-REQD-CFG-011	Connect the first VMkernel (vmk0) adapter of the vSAN witness appliance to the management network in the witness site.	The ESX host management networks in both availability zones must be routed to the vSAN witness host management network in the third witness site.	When using witness traffic separation, the second interface on vSAN witness host (vmk1) can be removed.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-REQD-CFG-012	Configure vSAN Witness host identity as per standard ESX host recommendations.	• Allows vSAN Witness host to be managed identically as a ESX host. • Ensures host identity, DNS and NTP is consistently configured	Apply the requirements for External Services Design
VCF-VSAN-SC-REQD-CFG-013	Enable vSphere Lifecycle Manager images on the vSAN witness host.	• VMware Cloud Foundation 9.0 requires vSphere Lifecycle Manager images for ESX hosts. • This will prevent warnings or vSAN health check alarms.	This requires either deploying or converting a vSAN witness host as a vSphere Lifecycle Manager image managed entity.

Table 804: Stretched vSAN Storage Cluster Storage Model Additional Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-STG-CLU-REQD-CFG-001	Use four (4) vSAN Nodes per availability zone (total of 8 vSAN data nodes).	This satisfy the requirement to have at least four nodes for a vSAN Storage cluster. Having four (4) vSAN nodes per availability zone will provide a balance of capacity and availability	This is an additional requirement for a standard vSAN stretch cluster and will require additional vSAN Nodes.

Table 805: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Table 806: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

Table 807: Stretched vSAN Storage Cluster Storage Model Additional Design Recommendations

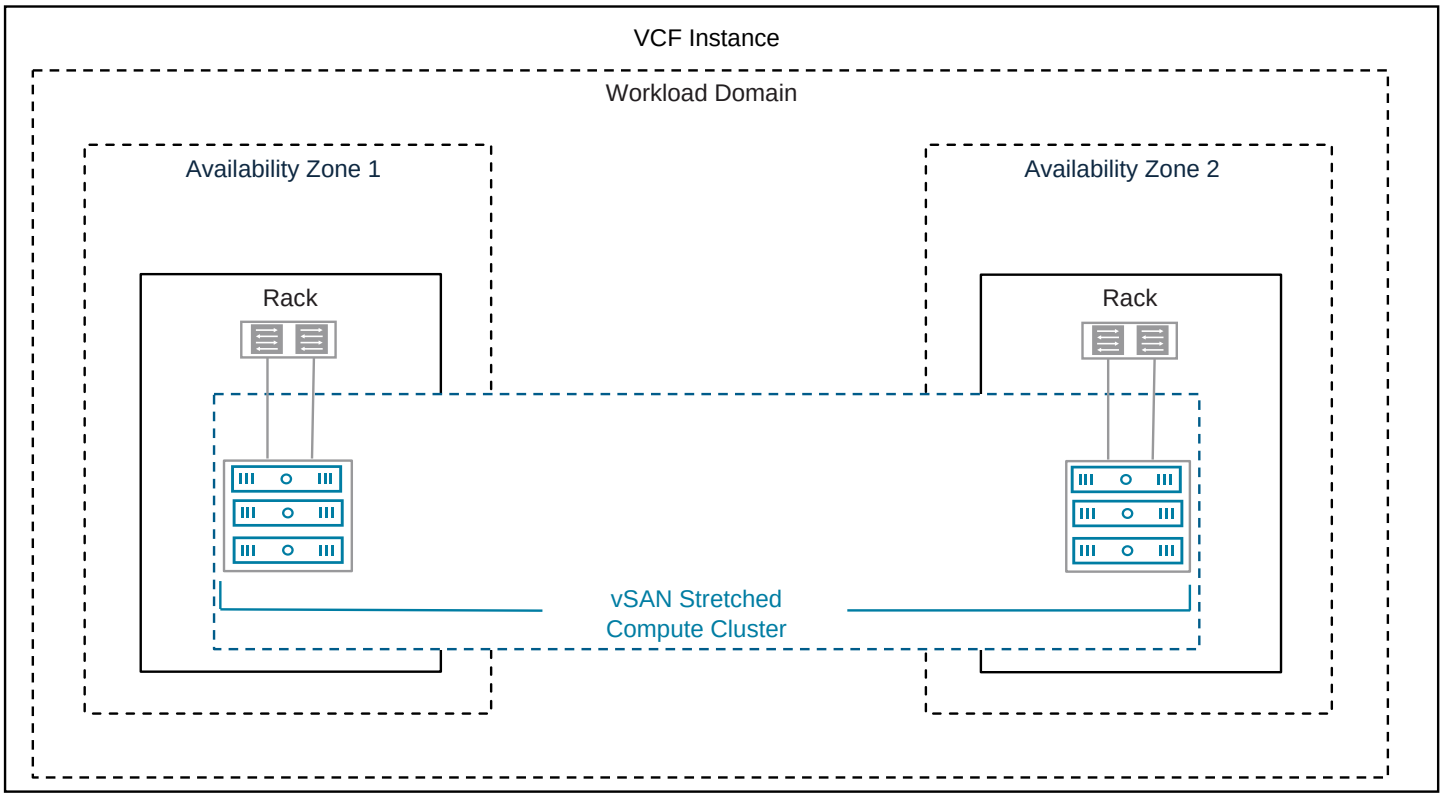
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-STG-CLU-RCMD-CFG-001	Use seven (7) hosts in each availability zone (total of 14 vSAN data nodes).	This configuration ensures that a cluster could support a storage policy with a secondary level of resilience of FTT=2 using RAID-6 erasure coding, and allows vSAN to regain its prescribed level of resilience in the event of a sustained host outage.	An additional host beyond the required minimum allows vSAN to reconstruct the stripe and parity in the most efficient way, with the fewest performance implications.
VCF-VSAN-SC-STG-CLU-RCMD-CFG-002	Use Witness Traffic Separation	Removes the requirement for the backend data node network to route with the vSAN Witness Host vmkernel interface Removes the requirement of static routes between vSAN data node traffic	Simpler network design avoids complex routing between vSAN data nodes and vSAM Witness host

Stretched vSAN Compute Cluster Storage Model

Stretched vSAN Compute Cluster Storage Model is a cluster of ESX hosts deployed over multiple availability zones, configured to mount a remote vSAN datastore provided by Stretched vSAN Storage Clusters in the same VCF domain.

Figure 349: Stretched vSAN Compute Cluster Storage Model in VMware Cloud Foundation

A Stretched vSAN Compute Cluster is a vSphere Cluster, deployed over multiple availability zones. Each VMkernel interface tagged with vSAN traffic type which communicates to an existing Stretched vSAN Storage Cluster to mount a vSAN remote datastore.



Note:

Stretched vSAN Compute Clusters are essentially vSAN Compute Clusters that's stretched over two availability zones and inherit the same attributes. Additional attributes are outlined below.

Stretched vSAN Compute Cluster Storage Model Attributes

A *Stretched vSAN Compute Cluster Storage Model* has the following attributes.

Attributes	Details
Fault Domain / Availability Zone	• Stretched vSAN Compute Clusters comprise of two (2) fault domains. • A fault domain or availability zone can represent a physical location. Fault domains / availability zones can either be two distinct data centers within a metro distance, or two safety or fire sectors (data halls) in the same large-scale datacenter.
vSAN Storage Cluster Client Traffic	vSAN Storage Cluster Client Traffic must be tagged on a VMkernel interface on the storage cluster. When enabled, the vSAN compute cluster nodes will use the vSAN client traffic (on the storage cluster) to communicate with storage cluster nodes for VM IO traffic.
vSAN Remote Datastore	This is a datastore shared from a vSAN storage cluster.

Stretched vSAN Compute Cluster Storage Model Options

A *Stretched vSAN Compute Cluster Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
vSAN Storage Cluster Client VMkernel MTU Size	1500 - 9000
vSAN Compute Site Coupling	**vSAN Compute Site Coupling** This is the concept of configuring and establishing site affinity to the vSAN objects when mounting a vSAN client cluster to a stretched cluster to take advantage of the most optimal path to the data. Compute Cluster Site Coupling can be configured to either availability zone, depending on the characteristics of the network fabric. This addresses scenarios where networking between availability zones has significant differences, This is defined this as asymmetrical and symmetrical network connectivity. **Asymmetrical network connectivity** would describe a topology where the network capabilities (latency & bandwidth) between availability zones could be less than the network capabilities between a local vSAN Compute Cluster and vSAN Storage Cluster fault domain within the same physical location E.g. A stretched cluster spanning across two physical sites with site local compute clusters mounting the shared datastore. **Symmetrical network connectivity** would describe a topology where the network capabilities connecting the availability zones would be the same as the network capabilities between the client cluster and server cluster within each availability zones. Connectivity across each availability zone is equal across all connectivity possibilities. Eg. A stretched cluster spanning across multiple rooms within a physical data-center.
vSAN Storage Cluster Client Traffic	• When disabled vSAN client cluster will share the same network with vSAN Storage Cluster traffic
	• When enabled, the vSAN compute cluster client nodes will communicate via the vSAN client traffic on the storage cluster nodes for vSAN client cluster traffic

Stretched vSAN Compute Cluster Storage Model Design Requirements and Recommendations

Table 808: vSAN Compute Cluster Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-COMP-CLU-REQD-CFG-001	Minimum two (2) vSAN compute nodes is required for VCF provisioned clusters	Deploying compute clusters for VCF requires minium 2 (nodes) or greater	none
VCF-VSAN-COMP-CLU-REQD-CFG-001	vSAN networking must be configured on hosts in the Compute cluster.	A vSAN Compute cluster is a vSphere cluster with a vSAN networking element that enables it to mount a vSAN Storage Cluster	Requires planning for vSAN networking requirements to support planned VM workload.
VCF-VSAN-COMP-CLU-REQD-CFG-002	vSAN Compute nodes must be commissioned as vSAN Compute Cluster type	This allows creation of vSAN Compute Clusters with vSAN compute node types	Requires new hosts to be commissioned to an exising network pool that has the correct networking attributes
VCF-VSAN-COMP-CLU-REQD-CFG-003	A vSAN Client Cluster must be deployed within same VCF Workload domain as vSAN Storage Cluster is deployed	A vSAN Client cluster is not supported to be cross mounted across vCenter instances from a VCF workflow perspective	Requires vSAN client clusters to be deployed within a VCF workload domain.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-COMP-CLU-REQD-CFG-004	All objects that make up a VM must reside on the same datastore.	This is currently the supported method for VMs deployed on vSAN Client Clusters	VM performance characteristics and fail-over responses will be consistent
VCF-VSAN-COMP-CLU-REQD-CFG-005	A vSAN Compute cluster can mount up to 5 remote datastores	This is a supported maximum configuration	This will require careful planning from compute and networking sizing perspective
VCF-VSAN-COMP-CLU-REQD-CFG-006	Configure vSphere HA failure response for Datastore with APD to be set to Power off and restart VMs.	This ensures Virtual machines respond appropriately to a HA event if a remote vSAN datastore is unavailable on a ESX host	None.
VCF-VSAN-COMP-CLU-REQD-CFG-007	Ensure vSAN Compute Cluster latency to vSAN Storage Cluster is less than 5 milliseconds	vSAN compute clusters perform a precheck and require network latency between the client clusters and the server cluster of 5ms or less.	Ensure a high speed low latency network between vSAN compute clusters and vSAN Storage Clusters.

Table 809: Stretched vSAN Compute Cluster Storage Model Additional Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-SC-COMP-CLU-REQD-CFG-001	Minimum two (2) nodes for a stretched compute cluster. This allows for a one (1) plus one (1) configuration where the compute nodes are distributed equally between both availability zones.	This satisfies the constraint of minimum two (2) nodes for a vSAN compute cluster and provides equal resources on both availability zones.	None.
VCF-VSAN-SC-COMP-CLU-REQD-CFG-002	Ensure DRS Host Groups, VM Groups, and VM/Host Rules in conjunction with vSAN Storage Toplogies and policies are aligned.	• This ensures that VM and vSAN data resilience desired outcomes are satisfied. • This will suggest or force a behavior of VM placement that reflects the storage topology backing the Virtual machine data.	• Requires creation of additional DRS Host groups, VM Groups, and VM/Host DRS Rules. • The Placement rules should match the desired failure response.

Table 810: Stretched vSAN Compute Cluster Storage Model Additional Design Recommendations

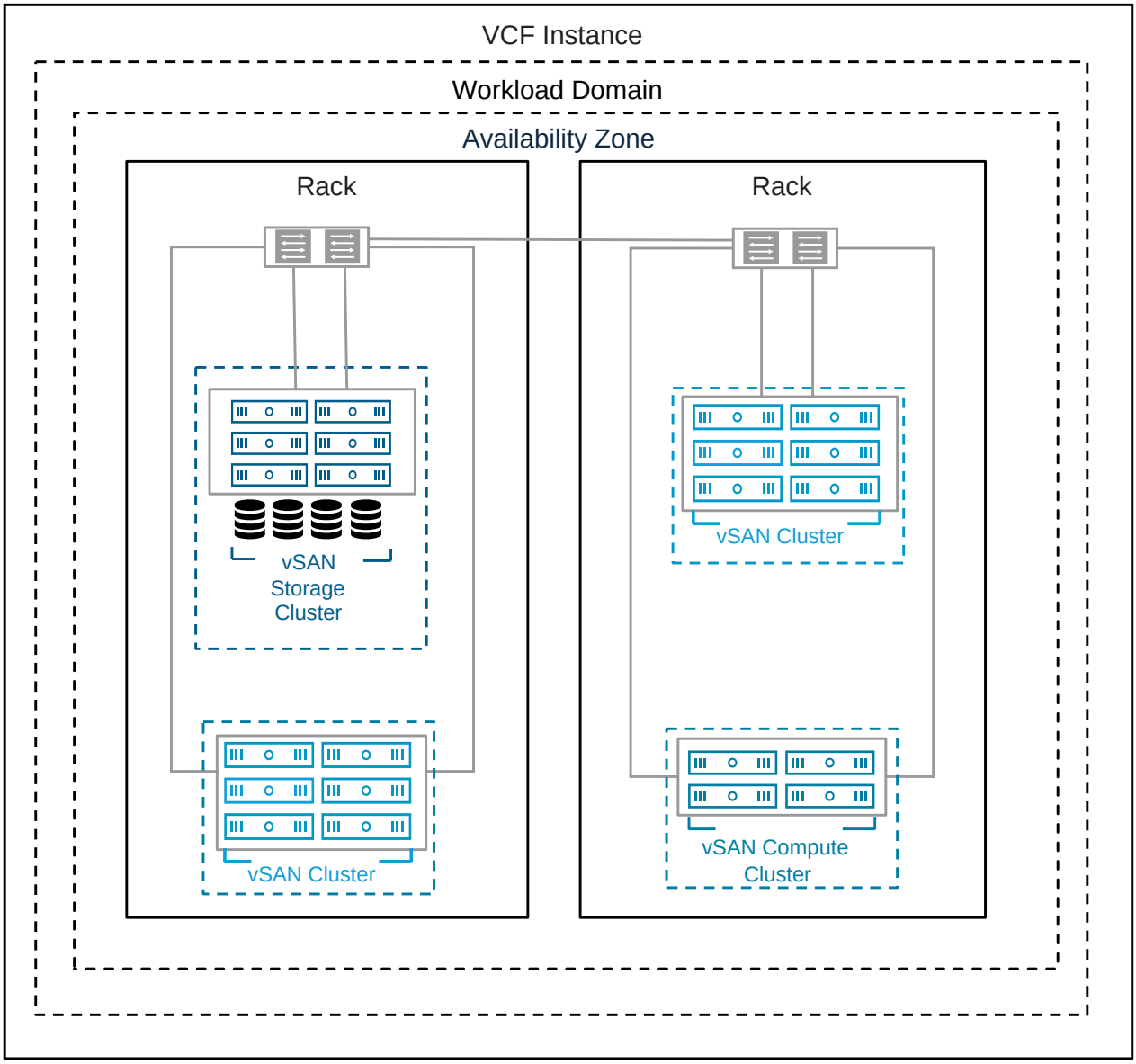
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-SC-COMP-CLU-RCMD-CFG-001	Configure site coupling between the vSAN client cluster and the server cluster so they are 'affinitized' or “coupled” to prevent vSAN Cluster client IO traversing over a sub optimal network route.	• If there is a defined disparity in connectivity, (**asymmetric network connectivity**) Configuring **site coupling** will help maintain data affinity from an availability zone perspective between the client cluster and the server cluster fault domains. • Establishing site affinity (e.g. inter-site coupling) assists with both read and write operations. • For writes, it ensures that a write operation would only need to perform a single round trip across an inter-site link (ISL), and not two. • For reads the IO path would remain local between the client and server cluster	It is possible to have a client stretched cluster use a server stretched cluster that do not share any physical site. As a result, this may change coupling or network affinity, as there may not necessarily be a preferred path, and may be deemed a symmetric connection. **Note:** Site coupling can be changed non-disruptively without any need to unmount or remount a datastore.

Datastore Sharing vSAN Storage Model

The *Datastore Sharing vSAN Storage Model* allows multiple independent vSAN HCI Clusters to consume storage of adjacent vSAN storage resources within the same VCF domain. Also known as “cross-cluster capacity sharing” this represents the highly flexible cross-cluster capacity sharing capabilities between vSAN clusters. This feature was formerly referred to as “HCI Mesh.”

Figure 350: Datastore Sharing vSAN Storage Model in VMware Cloud Foundation

The diagram depicts mounting remote vSAN datastores on vSAN clusters or vSAN Compute clusters and vSAN datastore sharing.



Datastore Sharing vSAN Storage Model Attributes

A *Datastore Sharing vSAN Storage Model*has the following attributes.

Attributes	Detail
vSAN Cluster	Allows multiple independent vSAN clusters to consume storage of adjacent vSAN storage resources within the same workload domain. • vSAN cluster can be either a client cluster or a server cluster. • vSAN clusters are based on Single-Rack vSAN ESA Storage Model or Single-Rack vSAN OSA Storage Model.
vSAN Remote Datastore	vSAN storage that is shared to a client cluster, such as another vSAN HCI Cluster or a vSAN Compute cluster.
vSAN Compute Cluster	A vSAN Compute cluster that can consume vSAN Storage cluster capacity.

Attributes	Detail
Storage Models	• Principle Storage: – vSAN clusters can consume a remote datastore as principle storage. • Supplemental Storage: – vSAN storage can be shared to adjacent clusters within the same VCF domain as supplemental storage using VCF native workflows. – VCF 9.1 supports consuming vSAN storage resources residing in one workload domain by clusters in other workload domains (where they use different vCenter servers). –
vSAN Storage Cluster Client Traffic	• vSAN Storage clusters introduce a new vSAN networking traffic type to enable a de-converged storage network. • It allows internal vSAN traffic, such as data replication, resync, and rebalance to be logically and physically separated from virtual machine / vSAN Storage Cluster Client traffic. • **vSAN** Cluster Client Traffic is used for vSAN client cluster traffic and can be described as a second vSAN network type on a vSAN Storage cluster.

Datastore Sharing vSAN Storage Model Options

A *Datastore Sharing vSAN Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
vSAN Host Networking	• Share vSAN traffic with vSAN Client cluster traffic.
	• Use vSAN Storage Cluster Client Traffic, use dedicated client traffic for VM IO traffic.

Table 811: vSAN Datastore sharing options

This table lists the vSAN datastore sharing options between vSAN clusters

Client Cluster	Server cluster configuration	Outcomes
vSAN HCI clusters (ESA) when it resides in one of the data sites where the vSAN storage cluster resides.	vSAN storage cluster or vSAN HCI cluster (ESA) in a stretched cluster configuration	Provides site level availability of vSAN storage data and VM instances on both client and server cluster
vSAN HCI clusters (ESA) in a stretched cluster configuration.	vSAN storage cluster or vSAN HCI cluster (ESA) in a stretched cluster configuration	Provides resilience of data but no site level availability of running VM instances on non stretched cluster topology
vSphere clusters stretched across two sites using asymmetrical network connectivity.	vSAN storage cluster or vSAN HCI cluster (ESA) in a stretched cluster configuration	Provides site-level resilience of data and VM instances.
vSphere clusters stretched across two sites using symmetrical network connectivity.	vSAN storage cluster or vSAN HCI cluster (ESA) in a stretched cluster configuration	Provides site-level resilience of data and VM instances. Require the same network capabilities (bandwidth and latency) between fault domains defining each site.
Client cluster running vSAN OSA	vSAN storage cluster or vSAN HCI cluster (ESA) in a single site or stretched cluster configuration	support of connectivity between vSAN OSA and vSAN ESA Requires VCF 9.1 for mixed support

Datastore Sharing vSAN Storage Model Design Requirements and Recommendations

Table 812: Common vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-001	Minimum vSAN cluster size.	Minimum valid vSAN HCI cluster is three (3) nodes all contributing to storage.	• Supports simple install of a management domain default vSphere cluster only. • A three (3) node vSAN cluster will not offer automatic rebuild of components in the event of a failure. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster. **Important:** vSAN Storage Clusters require minimum four (4) vSAN Nodes see Single-Rack vSAN Storage Cluster Storage Model for more details.
VCF-VSAN-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the VCF domain vSphere cluster.	Ensures that sufficient resources are present to create the VCF domain vSphere cluster.	Requires determining the workload capacity requirements prior to deployment. **Note:** Refer to the VMware Cloud Foundation Planning and Preparation Workbook for accurate sizing guidance.
VCF-VSAN-REQD-CFG-003	vSphere Lifecycle Management image based management is required for vSAN clusters.	• All new vSAN clusters, irrespective of vSAN ESA or vSAN OSA architectures, are required to use vSphere Lifecycle Manager images. • vSphere Lifecycle Manager images simplify the management of firmware and vendor add-ons.	Imported vSAN OSA clusters may use vSphere Lifecycle Manager baselines, but have to be converted prior to upgrade to VMware Cloud Foundation 9.0.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-REQD-CFG-004	Verify the hardware components used for vSAN are on the vSAN hardware compatibility list.	Helps prevent hardware related issues during deployment and operation.	• Requires validating vSAN hardware to ensure it is on the compatibility list prior to deployment. • Restricts the number of devices that can be used to a certified list. **Note:** Refer to the vSAN hardware compatibility guide for certified hardware https://compatibilityguide.broadcom.com/.
VCF-VSAN-REQD-CFG-005	Do not use NVMe storage devices attached to a Tri-Mode controller for vSAN diskgroups or storage pools.	• Discrete RAID controllers that support SATA/SAS/NVMe are often known as "Tri-mode controllers". • vSAN only supports SAS and SATA devices attached to a Tri-mode controller. • NVMe devices attached to Tri-mode controller are not a supported vSAN configuration. **Note:** Refer to knowledge base article KB314305	• NVMe devices are only supported directly connected to a PCIe slot on the bus. • May require to work with server vendor to verify or modify configurations.
VCF-VSAN-REQD-CFG-006	For vSAN HCI clusters, excluding vSAN Storage clusters, configure the vSAN network gateway IP address as the isolation address for the cluster.	vSphere HA can validate against an IP address on the vSAN network if a host is isolated.	Allocate an additional IP address. Top of rack switches may configure a port for switch virtual interface (SVI) which can be used as the isolation address. See design decsion VCF-VSAN-NET-REQD-CFG-003
VCF-VSAN-REQD-CFG-007	For vSAN HCI clusters, excluding vSAN Storage clusters, set the advanced cluster setting `das.usedefaultisolationaddress` to false.	vSphere HA in vSAN uses the ESX hosts management network default gateway IP for isolation detection when configured with default settings. This setting avoids conflicts if vSphere HA and vSAN triggers different responses when failures occur.	May require to configure a SVI (Switched Virtual Interface) gateway address for the VLAN assigned to the vSAN network.

Table 813: Common vSAN Host Network Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-NET-REQD-CFG-001	vSAN requires a high speed, low latency network for synchronous replication of vSAN components. **Note:** vSAN can use 10GbE , 25GbE or greater.	10GbE is the minimum network required for vSAN (ESA and OSA architectures **CAUTION:** There is limited support for 10GbE and vSAN ESA. For vSAN ESA ready node, **vSAN-HCI-SM** profile, only supports 10GbE. Please refer to https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Using 10GbE for vSAN networking can **severely** limit vSAN performance and can introduce bandwidth constraints combined with high latency. • Larger vSAN ESA nodes require 25GbE. **CAUTION:** Using 10GbE networking on vSAN ESA will trigger a vSAN cluster compliance check. Please review Broadcom Knowledge base KB372309
VCF-VSAN-NET-REQD-CFG-002	• Use a dedicated VLAN for a single rack cluster **Note:** You can share a single VLAN for multiple vSAN clusters within the same rack • For vSAN HCI multi-rack topology, VLAN requirements will depend on vSphere Multi-Rack Cluster Detailed Design if using L2 or L3 design.	• For a single rack cluster using a dedicated VLAN allows for better storage traffic isolation and security.	• For a single rack Cluster a VLAN ID must be allocated for use for a vSAN Cluster. • For vSAN HCI multi-rack topology, VLAN requirements will depend on the deployment, if using L2 or L3 design
VCF-VSAN-NET-REQD-CFG-003	• Use one (1) IPv4 IP or IPv6 address subnet for vSAN traffic in a single rack cluster.	The vSAN VMkernel in a VMware Cloud Foundation platform requires IPv4 or IPv6 addressing. **Notice:** VCF 9.1 does not support dual stack vSAN VMkernel networking for vSAN data traffic. A design must be IPv6 or IPv4 for vSAN VMkernel addressing.	• You require an IPv4 or IPv6 subnet configured with a correctly sized subnet to support the number of free IP addresses for each vSAN node in a single rack vSAN Cluster. • A gateway IP is required.

Table 814: Datastore Sharing vSAN Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VSAN-HCI-DS-REQD-CFG-001	A vSAN client cluster can mount up to 5 remote datastores	This is a supported maximum configuration	This will require careful planning from compute and networking sizing perspective
VCF-VSAN-HCI-DS-REQD-CFG-002	A vSAN HCI Server cluster can serve its local datastore to up to 10 client clusters.	This is a supported maximum configuration	This will require careful planning from compute and networking sizing perspective
VCF-VSAN-HCI-DS-REQD-CFG-003	vSAN networking must be configured on hosts in the Compute cluster.	A vSAN Compute cluster is a vSphere cluster with vSAN networking configured that enables it to mount a vSAN Storage Cluster	If vSAN Compute client network is configured on the vSAN storage cluster it must be able to establish connectivity to vSAN compute cluster.
VCF-VSAN-HCI-DS-REQD-CFG-004	All objects that make up a VM must reside on the same datastore.	This is currently the supported method for VMs deployed on vSAN Client Clusters	VM performance characteristics and fail-over responses will be consistent
VCF-VSAN-HCI-DS-REQD-CFG-005	Configure vSphere HA failure response for Datastore with APD to be set to Power off and restart VMs.	This ensures Virtual machines respond appropriately to a HA event if a remote vSAN datastore is unavailable on a ESX host	None.
VCF-VSAN-HCI-DS-REQD-CFG-006	Ensure vSAN client Cluster latency to vSAN server Cluster is less than 5 milliseconds	vSAN client clusters perform a precheck and require network latency between the client clusters and the server cluster of 5ms or less.	Ensure a high speed low latency network between vSAN compute clusters and vSAN Storage Clusters.

Table 815: Common vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-001	It is recommended to start with four (4) vSAN nodes.	• This ensures vSAN objects will automatically rebuild in the case of a failure or maintenance event. • This allows enabling Reserved Capacity on four (4) node clusters. • This allows space efficient storage polices to be automatically configured when used with vSAN ESA clusters.	This will potentially increase cost and hardware footprint of vSAN cluster.
VCF-VSAN-RCMD-CFG-002	Include TPM (Trusted Platform Module) in the hardware configuration of the ESX hosts.	Ensures that the keys issued to the hosts in a vSAN cluster using Data-at-Rest Encryption are cryptographically stored on a TPM device.	Can limit the choice of hardware configurations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-003	Enable Reserved Capacity on vSAN clusters.	• Reserved Capacity will implement a sliding scale storage capacity resource reservation as the vSAN node count increases per vSAN cluster. For example, the Host Rebuild Reserve for a four (4) node cluster would be 25% while it would be just 8% for a twelve (12) node cluster. This avoids imposing a generic 30% capacity overhead of a vSAN Cluster, thus dynamically optimizing reserved storage capacity on vSAN. • Recommendation is to design for vSphere cluster host counts large enough to lower the percentage needed for Reserved Capacity, This will provide the most efficient, yet agile vSphere cluster design.	• When enabled vSAN will impose two soft limits: • **Operations Reserve** - limits for capacity needed to perform transient storage activities like storage policy changes, re-balancing. • **Host Rebuild Reserve** - will reserve the capacity needed to absorb a sustained failure of a single ESX host in a vSAN cluster – to support an N+1 cluster design strategy. **CAUTION:** vSAN capacity reserve is not supported in conjunction with the following vSAN Cluster typologies: • Three (3) node clusters. • vSAN stretched clusters. • vSAN clusters using fault domains. **CAUTION:** • When vSAN Storage Clusters based 9.0 are configured with "Host Rebuild Reserve" and Auto-Policy in a cluster that consist of only 4 hosts will prevent vSAN using space-efficient and more performant RAID-5 policy. • vSAN Auto-RAID removes this limitaton in vSAN 9.1

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-RCMD-CFG-004	Enable disk Automatic Re-balance.	• When enabled vSAN will attempt to automatically keep the storage consumption of each capacity drive within a default variance threshold of 30%. • This allows vSAN to more evenly distribute the data across the discrete devices to achieve a balanced distribution of resources, and thus, improved performance.	• Adding or scaling out new hosts will trigger a disk rebalance and resynchronization. • When adding or replacing failed resources, automatic rebalance may introduce additional data re-synchronization and as a result additional network traffic utilization • re-synchronization may have an impact when vSAN clusters are network bandwidth constrained.

Table 816: Common vSAN Host Network Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-001	25GbE or higher for vSAN Data traffic is recommended.	• vSAN synchronous replication requires high bandwidth / low latency intra-node communication. While 10GbE is supported for vSAN 25GbE or higher networking is highly recommended to ensure consistent performance.	None.
VCF-VSAN-NET-RCMD-CFG-002	**Small scale workloads:** Two (2) physical NICs per vSAN node for small scale workloads, that utilize smaller vSAN ready node profiles. For vSAN ESA Ready-Node sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	• Simplifies deployment to use a single vSphere Distributed Switch model. • During periods of contention NIOC will distribute shares among competing traffic.	• Requires careful planning to ensure vSAN workload can share resources with other services such as vSphere vMotion and NSX workloads. • For high IO intensive workloads, 100GbE networking might be required for vSAN Ready-Nodes that require two phyiscal nics.
VCF-VSAN-NET-RCMD-CFG-003	**Large scale workloads:** Four (4) physical NICs per vSAN node. For large scale / IO intensive workloads based on bigger vSAN ready node profiles. For vSAN ESA Ready-Nodes sizing guidance review https://compatibilityguide.broadcom.com/pages/vsan-esa-readynode-hardware-guidance	This guarantees storage traffic isolation and full bandwidth of physical network resources on a ESX host.	Additional vSphere Distributed Switches requires increasing management overhead with a higher number of ESX host physical NICs. For more information see Storage and Workload Separation Network Model For high IO intensive workloads, 100GbE networking might be required to satisfy bandwidth and latency requirements.
VCF-VSAN-NET-RCMD-CFG-004	Configure the MTU size of the vSphere Distributed Switch to 9000 bytes for jumbo frames.	Supports the MTU size required by system traffic types. Improves traffic throughput.	When adjusting the MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches, and routers) to support the same MTU packet size.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-NET-RCMD-CFG-005	For Single-Rack vSphere Cluster Model use Layer 2 networking for vSAN.	Layer 2 connectivity between all vSAN hosts sharing the subnet within a single rack simplifies deployment.	None.
VCF-VSAN-NET-RCMD-CFG-006	For Multi-Rack Layer 3 vSphere Cluster Model with Layer 3 networking for vSAN.	Reduces fault domain of network to a specific rack and IP segment.	• Routing must be configured between each network subnet. • Host to rack addressing needs to be planned in advance. • Consider the number of hops and additional latency incurred while the traffic gets routed.
VCF-VSAN-NET-RCMD-CFG-007	Use the Failover Order teaming algorithm for the vSAN storage port group. See VCF-VDS-SS-RCMD-DPG-001	Provides a consistent traffic flow through a single physical ToR switch without need to traverse Inter Switch Link during normal operations.	It needs to be manually configured during deployment using vSphere Distributed Switches custom profile option.

Table 817: Datastore Sharing vSAN Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VSAN-HCI-DS-RCMD-CFG-002	Use vSAN Storage Cluster Client Traffic, use dedicated client traffic for VM IO traffic	Provides network separation and isolation between vSAN replication traffic and vSAN client VM traffic	This can be restrictive to only new clusters within a workload domain, depending on how they were commissioned.

NFS Storage Model

VMware Cloud Foundation can use NFSv3 Storage as the principal storage or supplemental storage for both the management domain and workload domains. NFSv4.1 is only supported as a supplemental storage option. The size of the compute and storage resources for the type of VCF domain will determine the NFS architecture, topology, and how NFS storage is consumed.

NFS Storage Model Attributes

An *NFS Storage Model* has the following attributes.

Attribute	Detail
Storage Model	Principal (NFSv3 only), Supplemental (NFSv3 or NFSv4.1)
Data Services	Supports underlying data services that are provided by the storage array.
Distributed File System	Uses a network protocol to manage data access between the storage server and ESX host.

NFS Storage Model Options

An *NFS Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Workload Separation Distributed Switch Model • Storage and Workload Separation Distributed Switch Model
Cluster Models	• Single-Rack vSphere Cluster Model • Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model **Note:** Multi-rack NFS storage design will be dependant on the Storage vendor design.
Number of Connections to NFS Datastore(s)	• Single NFS VMkernel port (default) • Additional NFS VMkernel ports • nConnect Using multiple connections to NFS datastores can offer increased bandwidth, redundancy, and traffic isolation **Note:** Configuring additional VMkernel ports or nConnect requires manual configuration.
Datastore Clusters	• Enable • Disable **Note:** vSphere Storage DRS is used to manage datastore clusters: A collection of datastores with a shared storage type and a shared management interface.

NFS Storage Model Design Requirements and Recommendations

Table 818: NFS Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NFS-REQD-CFG-001	Minimum vSphere Cluster with NFS storage.	• The minimum number of ESX hosts in a vSphere Cluster with an NFS datastore is two (2). • Ensures that availability requirements are met.	• Supports simple installation of a management domain default vSphere cluster only. • Using a smaller cluster limits the workload that can be placed on a vSAN cluster • If one of the ESX hosts is not available due to failure or maintenance, the CPU commitment ratio becomes 2:1.
VCF-NFS-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the workload domain vSphere cluster.	Ensures that sufficient resources are present to create the workload domain vSphere cluster.	Requires determining the workload capacity requirements before deployment.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NFS-REQD-CFG-003	All physical NICs must be: • Placed in the correct PCIe slot on the ESX host. • On the hardware compatibility list. • The same manufacturer and model. • Have identical firmware and drivers.	Proper configuration ensures that data reliability and performance requirements can be met.	• Restricts the number of NICs that can be used due to hardware compatibility list consideration. • Additional configuration is required to ensure that correct firmware versions are set.

Table 819: NFS Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NFS-RCMD-CFG-001	Start with four (4) ESX hosts.	• Ensures correct level of redundancy to protect against host failure in the vSphere cluster. • Supports the High Availability install of a management domain default vSphere cluster.	Higher costs incurred for additional Hosts, NICs, cables, and network switches.
VCF-NFS-RCMD-CFG-002	Use two (2) or more physical NICs with throughput greater than 10GbE.	The use of multiple NICs and network switches can improve redundancy, increase bandwidth, and can help distribute I/O load.	• Higher costs incurred for additional NICs, cables, and network switches. • Using a single physical NIC does not provide a failover in a failure occurs.
VCF-NFS-RCMD-CFG-003	Configure VMkernel Binding for NFS Datastore	• Enhances performance by isolating a specific VMkernel to handle NFS traffic. • Simplified trouble shooting.	Prevents NFS traffic being redirected via an alternate route in case the NFS VMkernel adapter fails.
VCF-NFS-RCMD-CFG-004	Configure jumbo frames (MTU 9000) for NFS traffic.	Jumbo frames offer benefits such as increased network efficiency, reduced CPU load, and improved throughput.	Physical infrastructure must support jumbo frames end to end.
VCF-NFS-RCMD-CFG-005	Install storage hardware acceleration in vSphere (VAAI).	• Reduced ESX host CPU load. • Reduced storage network bandwidth. • Improved performance.	• Additional configuration is required. • VAAI requires a manual upgrade. Follow vendor guidance during the upgrade procedure.
VCF-NFS-RCMD-CFG-006	Use nConnect to increase the number of connections to NFS datastore(s).	Using multiple connections to NFS datastores can offer increased bandwidth, redundancy, and traffic isolation.	The datastore must be mounted with esxcli command.
VCF-NFS-RCMD-CFG-007	Configure vCenter alarms that monitor disk capacity. Build vCenter alarms or deploy array-specific management packs for VCF Operations to ensure that someone is notified when the volume reaches a pre-defined threshold.	If the NFS datastore becomes full virtual machines are unable to write.	Additional configuration is required.

Fibre Channel Storage Model

VMware Cloud Foundation supports Fibre Channel Storage as principal or supplemental storage for the management domain and workload domains. Review the choices and requirements and follow the recommendations.

Fibre Channel Storage Model Attributes

A *Fibre Channel Storage Model* has the following attributes.

Attribute	Detail
Storage models	Principal, Supplemental
Data Services	Supports underlying data services that are provided by the storage array.
SCSI interface protocol	Uses a fibre channel protocol to manage data access between the storage fabric and ESX host.

Fibre Channel Storage Model Options

A *Fibre Channel Storage Model* can apply options from the design areas that are listed below, depending on objectives.

Design Area	Options
Distributed Switch Models	• Single Distributed Switch Model • Workload Separation Distributed Switch Model **Note:** Fibre Channel does not use the physical network. As a result not all network models are available.
Cluster Models	• Single-Rack vSphere Cluster Model • Multi-Rack Layer 2 vSphere Cluster Model • Multi-Rack Layer 3 vSphere Cluster Model
SAN Fabric	• HBAs – Single – Multiple • Switches – Single – Multiple • Zoning – Single-initiator – Multi-initiator **Note:** Follow Storage Vendor best practices for configuration of the SAN Fabric.
Storage Array	• Storage Provisioning • Host Registration • LUN Masking **Note:** Follow Storage Vendor best practices for configuration of the storage array.
Path Selection Policy	• Round Robin (VMW_PSP_RR) • Fixed • Most Recently Used (MRU) **Note:** Follow Storage Vendor best practices for configuration of the storage array.

Design Area	Options
Datastore Clusters	• Enable • Disable **Note:** vSphere Storage DRS is used to manage Datastore clusters: A collection of datastores with a shared storage type and a shared management interface.
Raw Device Mapping (RDM)	• Physical Compatibility Mode • Virtual Compatibility Mode **Note:** RDM is supported. Customers must make sure to maintain consistent LUN IDs across all hosts in the vSphere Cluster.

Fibre Channel Storage Model Design Requirements and Recommendations

Table 820: Fibre Channel Storage Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-FC-REQD-CFG-001	Minimum vSphere Cluster with Fibre Channel storage.	• The minimum number of ESX hosts in a vSphere cluster with a Fibre Channel datastore is two (2). • Ensures that availability requirements are met.	• Supports simple installation of a management domain default vSphere cluster only. • If one of the ESX hosts is not available due to failure or maintenance, the CPU commitment ratio becomes 2:1.
VCF-FC-REQD-CFG-002	Provide sufficient raw capacity to meet the initial needs of the workload domain vSphere cluster.	Ensures that sufficient resources are present to create the workload domain vSphere cluster.	Requires determining the workload capacity requirements before deployment.
VCF-FC-REQD-CFG-003	Configure one or more HBAs.	At least one (1) HBA is required to access Fibre Channel storage.	Using a single HBA does not provide failover in the event of an HBA or Fibre Channel switch failure.
VCF-FC-REQD-CFG-004	All Fibre Channel HBAs must be: • Placed in the correct PCIe slot on the ESX host. • On the Broadcom hardware compatibility list. • The same manufacturer and model. • Have identical firmware and drivers.	Proper configuration ensures that data reliability and performance requirements can be met.	• Restricts the number of HBAs that can be used due to HCL consideration. • Additional configuration is required to ensure that correct firmware versions are set.

Table 821: Fibre Channel Storage Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-FC-RCMD-CFG-001	Start with four (4) ESX hosts.	• Ensures that availability requirements are met. • Ensures correct level of redundancy to protect against ESX host failure in the vSphere cluster. • Supports the High Availability install of a management domain default vSphere cluster.	Higher costs incurred for additional Hosts, HBAs, cables, Fibre Channel switches.
VCF-FC-RCMD-CFG-002	Use two (2) or more HBAs.	The use of multiple HBAs and multiple Fibre Channel switches can improve redundancy, increase bandwidth, and can help distribute I/O load.	Higher costs incurred for additional HBAs, cables, Fibre Channel switches.
VCF-FC-RCMD-CFG-003	Adjust the queue depth based on storage vendor best practices.	Maintains uniform queue depth across all HBAs in a cluster.	Modifying the HBA queue depth requires manual configuration.
VCF-FC-RCMD-CFG-004	Provision the datastore virtual disk format as thick-provisioned eager-zeroed.	• Allocates the entire disk at creation ensuring data integrity and security. • Offers best performance.	May take longer to format when ESX hardware acceleration is not used. VAAI supports the SCSI operation WRITE SAME which speeds up the format.
VCF-FC-RCMD-CFG-005	Use the VMware Path Selection Policy (PSP) Round Robin (RR).	• Offers increased performance, faster Path Failover Time • Balances IO across your storage array.	Additional configuration of the ESX host.
VCF-FC-RCMD-CFG-006	Configure the Round Robin IOPS limit based on the storage vendor's best practice.	Adjusting the limit can provide a positive impact to performance and is recommended by most storage vendors.	Additional configuration of the ESX host.
VCF-FC-RCMD-CFG-007	Install and register a VASA storage provider in vSphere.	• Provides a complete "end-to-end" view of your infrastructure as the storage system information will be visible in vCenter. • Provides additional storage capabilities.	Additional configuration is required.
VCF-FC-RCMD-CFG-008	Install storage hardware acceleration in vSphere (VAAI).	• Reduced ESX host CPU load. • Reduced storage network bandwidth. • Improved performance.	• Additional configuration is required. • VAAI requires a manual upgrade. Follow vendor guidance during the upgrade procedure.
VCF-FC-RCMD-CFG-009	Private LUNs such as boot LUNs should not be connected to all ESX hosts in the cluster. They should only be connected to the ESX host that they are enabling.	Prevents access to the boot volume from other ESX hosts.	May require additional configuration such as LUN Masking to ensure that the boot LUN is not accessible by other ESX hosts in the vSphere cluster.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-FC-RCMD-CFG-010	Use VMFS version 6.	Use the latest supported VMFS version. VMFS 6 enables advanced format (512e) and Automatic Space Reclamation support.	VMFS version must be selected when the datastore is created.
VCF-FC-RCMD-CFG-011	Configure vCenter alarms that monitor disk capacity. Build vCenter alarms or deploy array-specific management packs for VCF Operations to ensure that someone is notified when the volume reaches a pre-defined threshold.	If the VMFS datastore becomes full virtual machines are unable to write.	Additional configuration is required.

NSX Manager and Control Plane Detailed Design

Apply these design and component principles to size, deploy, and configure NSX Manager as part of your VMware Cloud Foundation platform.

NSX Manager Model

When you deploy NSX Manager appliances, either with a local or global scope, you deploy the appliance with a size and availability suitable for your environment's scale and availability requirements.

- Simple NSX Manager Model
- High Availability NSX Manager Model
- High Availability NSX Manager With External Load Balancer Model

NSX Global Manager Model

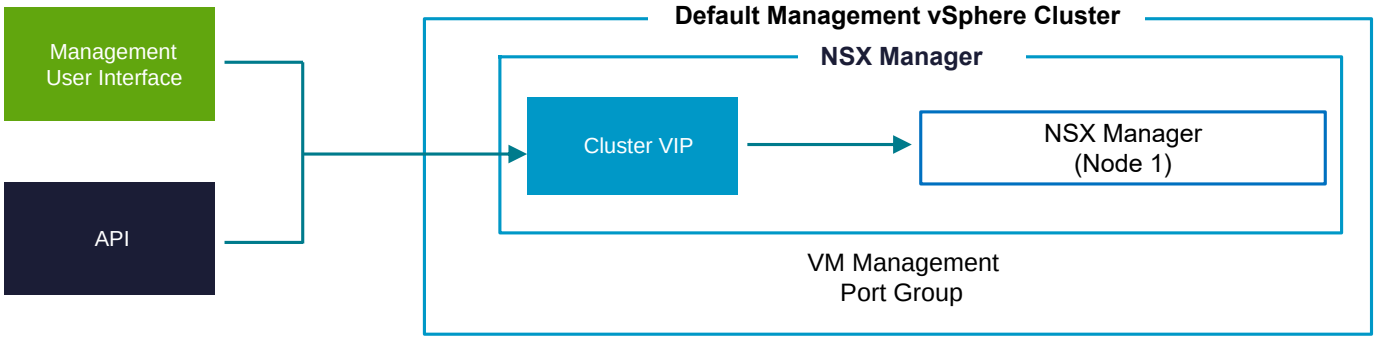
For an environment with multiple VCF Instances, you can use NSX Federation, which requires the manual deployment of NSX Global Manager nodes. Consider the placement requirements for using NSX Global Manager in VCF, and the best practices for having an NSX Global Manager cluster operate in an optimal way, such as the number and size of the nodes and high availability on a standard or stretched management cluster.

- High Availability NSX Global Manager Model
- High Availability with Standby NSX Global Manager Model

Simple NSX Manager Model

The *Simple NSX Manager Model* addresses use cases requiring a minimal infrastructure footprint with lower scale and lower availability requirements.

Figure 351: Simple NSX Manager Model in VMware Cloud Foundation



Simple NSX Manager Model Attributes

The *Simple NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Single NSX Manager Appliance with Internal VIP.
Availability	• vSphere HA protects the node applying high restart priority. • Slower recovery from node failure.
Infrastructure Impact during Upgrade	• Management and control plane impacted during upgrade operations. • Deployment/Power-on/Restart/vMotion of VMs may be impacted.
Virtual Networking Scale	Reduced scale
API Request Scale	Limited scale

Simple NSX Manager Model Design Requirements and Recommendations

Table 822: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

Table 823: NSX Manager Simple Deployment Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-004	Deploy a single NSX Manager node in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports the objective of a small-scale deployment.	• Slower recovery from node failure. • Management and control plane is impacted during upgrade operations. • Supports limited scale.

Table 824: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

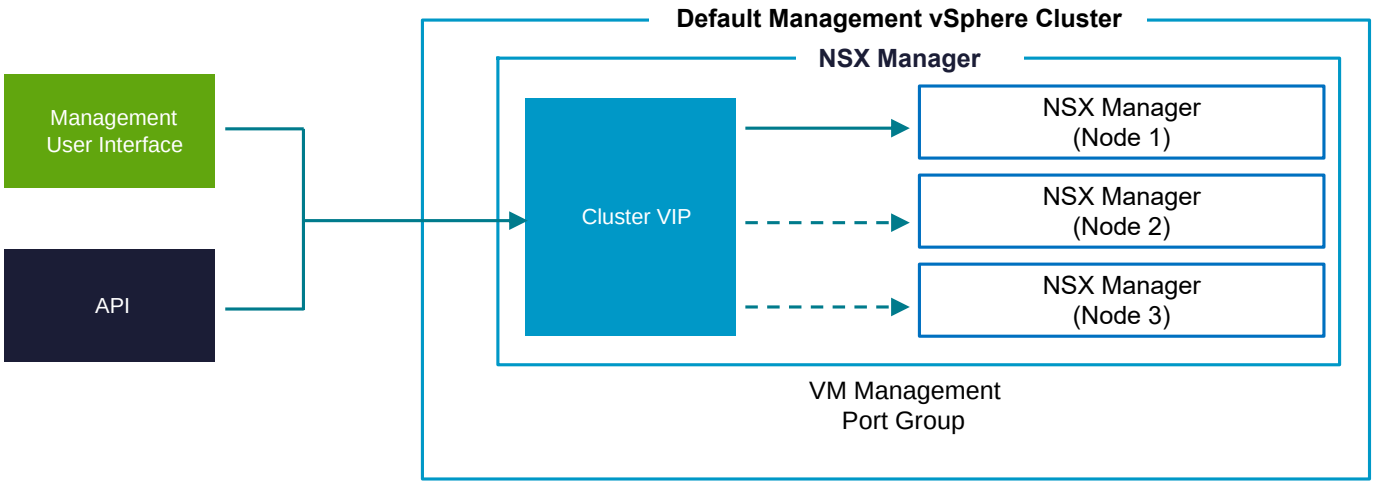
Table 825: Common NSX Manager Design Recommendations for Stretched Clusters for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-002	Add the NSX Manager appliances to the virtual machine group for the first availability zone.	Ensures that the NSX Manager appliances are powered on an ESX host in the primary availability zone.	None.

High Availability NSX Manager Model

The *High Availability NSX Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.

Figure 352: High Availability NSX Manager Model in VMware Cloud Foundation



High Availability NSX Manager Model Attributes

The *High Availability NSX Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Three-node cluster of NSX Manager appliances with internal virtual IP
Availability	• Highly available cluster of virtual IP NSX Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.
API Request Scale	Internal virtual IP directs API requests to a single NSX Manager appliance.

High Availability NSX Manager Model Design Requirements and Recommendations

Table 826: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

Table 827: High Availability NSX Manager Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-005	Deploy three (3) NSX Manager nodes in the management domain default vSphere cluster for configuring and managing the network services for the VCF domain.	Supports high availability of the NSX Manager cluster.	You must have sufficient resources in the default cluster of the management domain to run three NSX Manager nodes.

Table 828: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

Table 829: Common NSX Manager Design Recommendations for Stretched Clusters for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-002	Add the NSX Manager appliances to the virtual machine group for the first availability zone.	Ensures that the NSX Manager appliances are powered on an ESX host in the primary availability zone.	None.

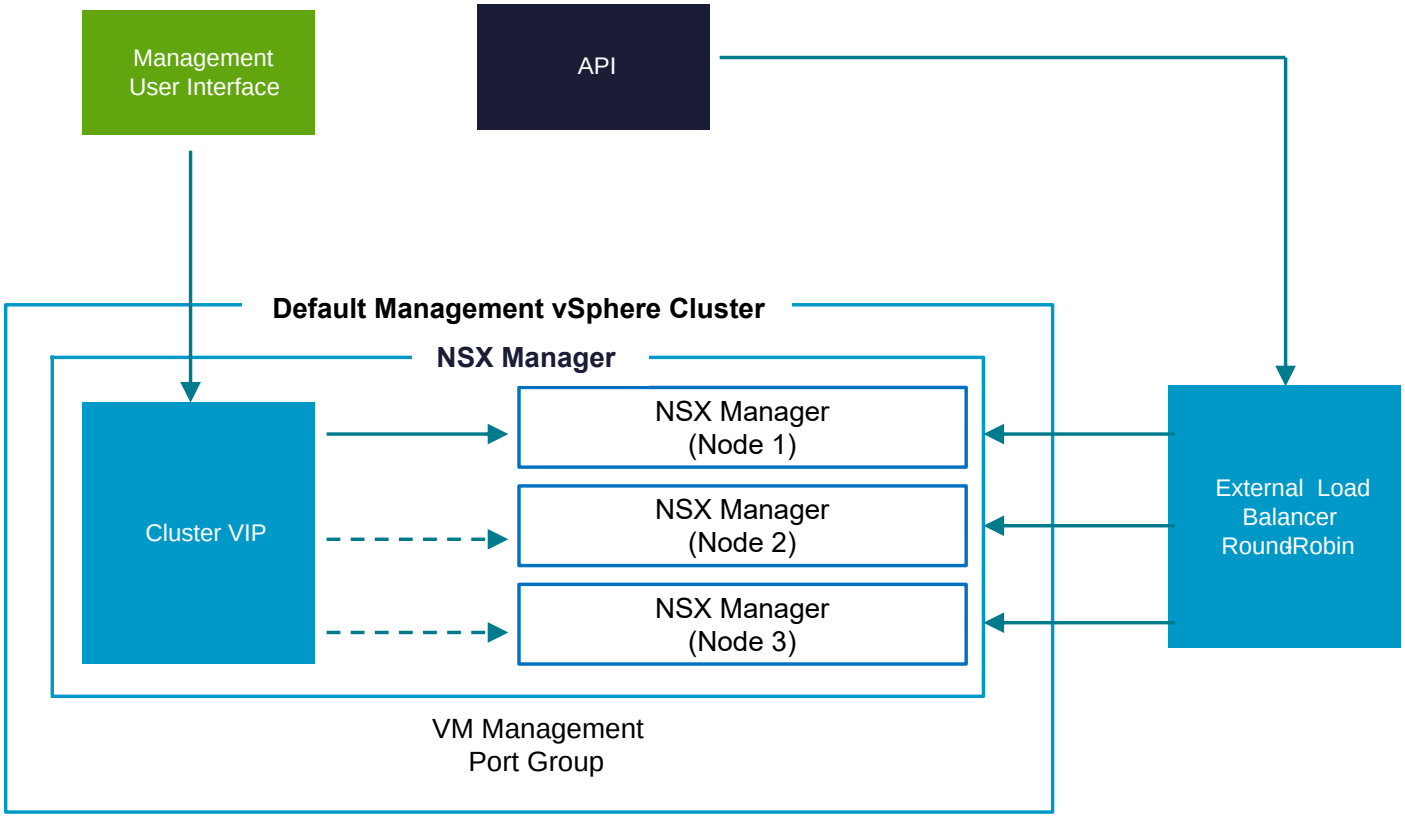
Table 830: High Availability NSX Manager Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-003	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Manager appliances.	Keeps the NSX Manager appliances running on different ESX hosts for high availability.	You must allocate at least four physical hosts so that the three NSX Manager appliances continue running if an ESX host failure occurs.

High Availability NSX Manager With External Load Balancer Model

The *High Availability NSX Manager With External Load Balancer Model* addresses use cases requiring improved API performance while maintaining highest scale, optimized operations, and maximum availability.

Figure 353: High Availability NSX Manager With External Load Balancer Model in VMware Cloud Foundation



High Availability NSX Manager With External Load Balancer Model Attributes

The *High Availability NSX Manager With External Load Balancer Model* has the following attributes.

Attribute	Detail
Deployment Size	• Three-node cluster of NSX Manager appliances with internal virtual IPs for management. • Additional virtual IP on external Load Balancer for API requests.
Availability	• Highly available cluster of NSX Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX Configuration Maximums.
API Request Scale	• Increase API based management scale for high-churn environments. • External virtual IP distributes API requests across all NSX Manager nodes.

High Availability NSX Manager Model Design Requirements and Recommendations

Table 831: Common NSX Manager Design Requirements for All Models

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-001	Place the appliances of the NSX Manager cluster on the VM management network in the management domain.	• Simplifies IP addressing for management components by using the same VLAN and subnet. • Provides simplified secure access to management components in the same VLAN network.	None.
VCF-NSX-LM-REQD-CFG-002	Create a virtual IP (VIP) address for the NSX Manager cluster for the workload domain.	• Used for the user interface and API of NSX Manager. • In a clustered configuration, provides high availability. • In a non-clustered configuration, prepares for future expansion.	• In a clustered configuration, the VIP address feature provides high availability only. It does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Manager nodes must be deployed on the same Layer 2 network.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-003	In vSphere HA, set the restart priority policy for each appliance to high.	• NSX Manager implements the control plane for virtual network segments. vSphere HA restarts the NSX Manager appliances first so that other virtual machines that are being powered on or migrated by using vSphere vMotion while the control plane is offline lose connectivity only until the control plane quorum is re-established. • Setting the restart priority to high reserves the highest priority for adding services that must be started before NSX Manager.	If the restart priority for another management appliance is set to highest, the connectivity delay for management appliances is longer.

Table 832: High Availability NSX Manager With External Load Balancer Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-LM-REQD-CFG-006	Deploy three (3) NSX Manager nodes in the management domain default vSphere cluster for configuring and managing the network services for the workload domain.	Supports high availability of the NSX Manager cluster.	You must have sufficient resources in the default cluster of the management domain to run three NSX Manager nodes.

Table 833: Common NSX Manager Model Design Recommendations for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-001	Deploy appropriately sized nodes in the NSX Manager cluster for the workload domain. **Note:** Check VMware Configuration Maximums to select the right NSX Manager form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Manager nodes.

Table 834: Common NSX Manager Design Recommendations for Stretched Clusters for All Models

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-002	Add the NSX Manager appliances to the virtual machine group for the first availability zone.	Ensures that the NSX Manager appliances are powered on an ESX host in the primary availability zone.	None.

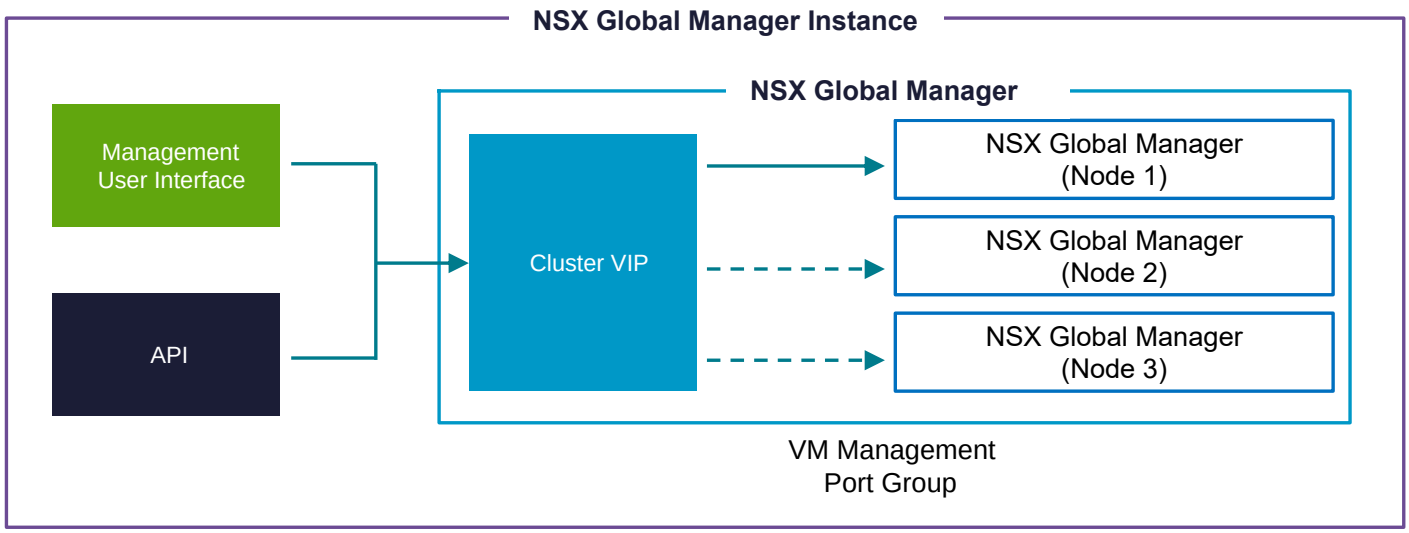
Table 835: High Availability NSX Manager With External Load Balancer Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-LM-RCMD-CFG-003	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Manager appliances.	Keeps the NSX Manager appliances running on different ESX hosts for high availability.	You must allocate at least four physical hosts so that the three NSX Manager appliances continue running if an ESX host failure occurs.
VCF-NSX-LM-RCMD-CFG-004	Create an additional virtual IP (VIP) address for the NSX Manager cluster for the workload domain and assign it to an externally provided load balancer.	• Provides separate VIP for stateless API requests. • Internal VIP is reserved for session persistent management UI usage.	Requires additional IP address.
VCF-NSX-LM-RCMD-CFG-005	Configure an external load balancer to round-robin traffic without source persistence between the three NSX Manager node IPs.	Ensures that stateless API request traffic is balanced across all available NSX Manager appliances.	Requires externally managed HTTP load balancer.
VCF-NSX-LM-RCMD-CFG-006	Configure an external load balancer to monitor health of NSX Manager nodes.	Ensures continued availability of API endpoint during upgrade or outage of a single NSX Manager appliance.	Requires externally managed HTTP load balancer.

High Availability NSX Global Manager Model

The *High Availability NSX Global Manager Model* addresses use cases requiring the highest scale, optimized operations, and maximum availability requirements.

Figure 354: High Availability NSX Global Manager Model in VMware Cloud Foundation



High Availability NSX Global Manager Model Attributes

The *High Availability NSX Global Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	Three-node cluster of NSX Manager appliances with internal virtual IP for management.
Availability	• Highly available cluster of NSX Global Manager appliances with rapid recovery of a single node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.

High Availability NSX Global Manager Model Design Requirements and Recommendations

Table 836: Common High Availability NSX Global Manager Design Recommendation

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-GM-RCMD-CFG-001	Place the appliances of the NSX Global Manager cluster on the VM management network in each VCF Instance.	• Simplifies IP addressing for management VMs. • Provides simplified secure access to all management VMs in the same VLAN network.	None.
VCF-NSX-GM-RCMD-CFG-002	Deploy appropriately sized nodes in the NSX Global Manager cluster for the workload domain. **Note:** Check Configuration Maximums to select the right NSX Global Managers form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	Incorrectly sized NSX Global Managers might impact the ability to scale according to the requirements.
VCF-NSX-GM-RCMD-CFG-003	Create a virtual IP (VIP) address for the NSX Global Manager cluster for the workload domain.	Provides high availability of the user interface and API of NSX Global Manager.	• The VIP address feature provides high availability only, it does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Global Manager nodes must be deployed on the same Layer 2 network.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-GM-RCMD-CFG-004	In vSphere HA, set the restart priority policy for each NSX Global Manager appliance to medium.	• NSX Global Manager implements the management plane for global segments and firewalls. NSX Global Manager is not required for control plane and data plane connectivity. • Setting the restart priority to medium reserves the high priority for services that impact the NSX control or data planes.	• Management of NSX global components are unavailable until the NSX Global Manager virtual machines restart. • The NSX Global Manager cluster is deployed in the management domain, where the total number of virtual machines is limited and where it competes with other management components for restart priority.
VCF-NSX-GM-RCMD-SEC-001	Establish an operational practice to capture and update the thumbprint of the NSX Local Manager certificate on NSX Global Manager every time the certificate is updated by using SDDC Manager.	• Ensures secured connectivity between the NSX Manager instances. • Each certificate has its own unique thumbprint. NSX Global Manager stores the unique thumbprint of the NSX Local Manager instances for enhanced security. • If an authentication failure between NSX Global Manager and NSX Local Manager occurs, objects that are created from NSX Global Manager are not propagated to the software-defined network.	The administrator must establish and follow an operational practice by using a runbook or automated process to ensure that the thumbprint is up to date.
VCF-NSX-GM-RCMD-CFG-005	Deploy three (3) NSX Global Manager nodes for the workload domain to support NSX Federation across VCF Instances.	Provides high availability for the NSX Global Manager cluster.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Global Manager nodes.
VCF-NSX-GM-RCMD-CFG-006	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Global Manager appliances.	Keeps the NSX Global Manager appliances running on different ESX hosts for high availability.	You must allocate at least four (4) physical hosts so that the three (3) NSX Manager appliances continue running if an ESX host failure occurs.

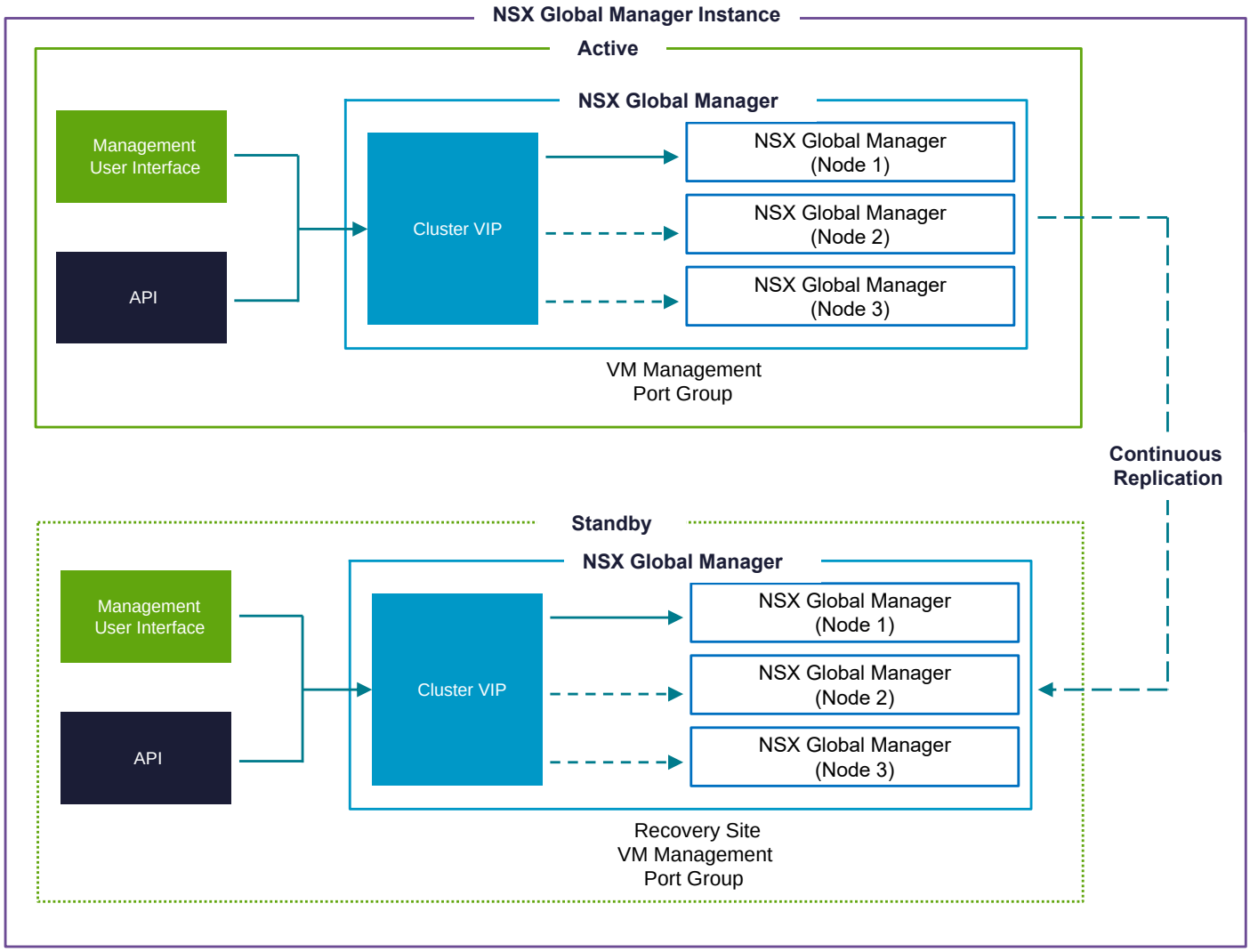
Table 837: Common High Availability NSX Global Manager for Stretched Clusters Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-GM-RCMD-CFG-007	Add the NSX Global Manager appliances to the virtual machine group for the first availability zone.	Ensures that the NSX Global Manager appliances are powered on a host in the primary availability zone.	Must be manually configured.

High Availability with Standby NSX Global Manager Model

The *High Availability with Standby NSX Global Manager Model* ensures enhanced business continuity by implementing a standby NSX Global Manager Cluster.

Figure 355: High Availability with Standby NSX Global Manager Model in VMware Cloud Foundation



High Availability with Standby NSX Global Manager Model Attributes

The *High Availability with Standby NSX Global Manager Model* has the following attributes.

Attribute	Detail
Deployment Size	• Two NSX Global Manager clusters – Three-node active cluster of NSX Manager appliances with internal virtual IP for management – Three-node standby cluster of NSX Manager appliances with internal virtual IP for management.
Availability	• Highly available cluster of NSX Global Manager appliances with rapid recovery of a single-node failure. • vSphere HA recovers a node applying high restart priority. • Anti-affinity rule to keep cluster nodes on different ESX hosts. • Replication from active cluster to standby cluster. • Ability to promote standby cluster to active.
Infrastructure Impact during Upgrade	No impact to control plane during upgrade operations.
Virtual Networking Scale	Enterprise scale up to NSX configuration maximums.

High Availability with Standby NSX Global Manager Model Design Requirements and Recommendations

Table 838: Common High Availability NSX Global Manager Design Recommendation

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-GM-RCMD-CFG-001	Place the appliances of the NSX Global Manager cluster on the VM management network in each VCF Instance.	• Simplifies IP addressing for management VMs. • Provides simplified secure access to all management VMs in the same VLAN network.	None.
VCF-NSX-GM-RCMD-CFG-002	Deploy appropriately sized nodes in the NSX Global Manager cluster for the workload domain. **Note:** Check Configuration Maximums to select the right NSX Global Managers form factor for your scale needs.	Ensures resource availability and usage efficiency per workload domain.	Incorrectly sized NSX Global Managers might impact the ability to scale according to the requirements.
VCF-NSX-GM-RCMD-CFG-003	Create a virtual IP (VIP) address for the NSX Global Manager cluster for the workload domain.	Provides high availability of the user interface and API of NSX Global Manager.	• The VIP address feature provides high availability only, it does not load-balance requests across the cluster. • When using the VIP address feature, all NSX Global Manager nodes must be deployed on the same Layer 2 network.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-GM-RCMD-CFG-004	In vSphere HA, set the restart priority policy for each NSX Global Manager appliance to medium.	• NSX Global Manager implements the management plane for global segments and firewalls. NSX Global Manager is not required for control plane and data plane connectivity. • Setting the restart priority to medium reserves the high priority for services that impact the NSX control or data planes.	• Management of NSX global components are unavailable until the NSX Global Manager virtual machines restart. • The NSX Global Manager cluster is deployed in the management domain, where the total number of virtual machines is limited and where it competes with other management components for restart priority.
VCF-NSX-GM-RCMD-SEC-001	Establish an operational practice to capture and update the thumbprint of the NSX Local Manager certificate on NSX Global Manager every time the certificate is updated by using SDDC Manager.	• Ensures secured connectivity between the NSX Manager instances. • Each certificate has its own unique thumbprint. NSX Global Manager stores the unique thumbprint of the NSX Local Manager instances for enhanced security. • If an authentication failure between NSX Global Manager and NSX Local Manager occurs, objects that are created from NSX Global Manager are not propagated to the software-defined network.	The administrator must establish and follow an operational practice by using a runbook or automated process to ensure that the thumbprint is up to date.
VCF-NSX-GM-RCMD-CFG-005	Deploy three (3) NSX Global Manager nodes for the workload domain to support NSX Federation across VCF Instances.	Provides high availability for the NSX Global Manager cluster.	You must have sufficient resources in the management domain default vSphere cluster to run three (3) NSX Global Manager nodes.
VCF-NSX-GM-RCMD-CFG-006	Apply VM-VM anti-affinity rules in vSphere DRS to the NSX Global Manager appliances.	Keeps the NSX Global Manager appliances running on different ESX hosts for high availability.	You must allocate at least four (4) physical hosts so that the three (3) NSX Manager appliances continue running if an ESX host failure occurs.

Table 839: Common High Availability NSX Global Manager for Stretched Clusters Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-GM-RCMD-CFG-007	Add the NSX Global Manager appliances to the virtual machine group for the first availability zone.	Ensures that the NSX Global Manager appliances are powered on a host in the primary availability zone.	Must be manually configured.

Table 840: Highly Available with Standby NSX Global Manager Cluster Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-GM-RCMD-CFG-008	Deploy an additional NSX Global Manager Cluster in the second VCF Instance.	Enables recoverability of NSX Global Manager in the second VCF Instance if a failure in the first VCF Instance occurs.	Requires additional NSX Global Manager nodes in the second VCF Instance.
VCF-NSX-GM-RCMD-CFG-009	Set the NSX Global Manager cluster in the second VCF Instance as standby.	Enables recoverability of NSX Global Manager in the second VCF Instance when a failure in the first instance occurs.	None

NSX Edge Cluster Detailed Design

This section outlines various *NSX Edge Cluster Models* for virtualized data center environments, including the physical rack placement of NSX Edge nodes. Each design is tailored to address specific requirements for high availability, scalability, capabilities, and infrastructure footprint.

NSX Edge Cluster Models for Single Availability Zone

- Host Fault Tolerant NSX Edge Cluster Model
- Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters
- Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster
- Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster

NSX Edge Cluster Models for Dual Availability Zones

- Dual Availability Zone Based on NSX Edge Node HA Model
- Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model

NSX Edge Cluster Models for Single Availability Zone

NSX Edge cluster designs deployed in a single availability zone have their infrastructure components typically installed in close proximity to each other. For example, when using multiple racks, they are placed in the same room or adjacent rooms in the same building. Asymmetric traffic is not a concern in such single availability zone designs.

Consider the main design attributes of the available models.

Table 841: Host Fault Tolerant NSX Edge Cluster Model Attributes

Attribute	Detail
Compute Availability Model	Host level only
Storage Availability Model	Single shared storage
Physical Network Fabric Supportability	Layer 2 adjacency is required between the ToR switches within the rack
Physical vSphere Cluster Layout	Single vSphere cluster deployed within a single rack
Physical Footprint	Low
Scalability	• Adding more hosts to the existing vSphere cluster to deploy more NSX Edge nodes. • Adding more NSX Edge nodes to existing hosts that have free CPU and memory resources requires additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node. A ratio of two physical NICs for each NSX Edge node is recommended
NSX Edge Node Mobility	Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.

Table 842: Rack Fault Tolerant NSX Edge Cluster Model Comparison

Attribute	Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters	Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster	Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster
Compute availability model	Rack and host level		
Storage availability model	Independent storage for each rack when using vSAN	Single shared storage across racks	
Physical network fabric supportability	• Layer 2 adjacency is required between the ToR switches within the rack. • Layer 3 between racks is supported.	Layer 2 adjacencies for VLANs spanning racks are required.	• Layer 2 adjacency is required between the ToR switches within the rack. • Layer 3 between the racks is supported.
Physical vSphere cluster layout	Multiple vSphere clusters deployed vertically, each vSphere cluster in separate dedicated rack	Single vSphere cluster deployed horizontal across multiple racks	
Physical footprint	Medium to high		
Scalability	• Adding more hosts to the existing vSphere cluster to deploy more NSX Edge nodes • Adding more NSX Edge nodes to existing hosts that have free CPUs and memory resources requires additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node. A ratio of two physical NICs for each NSX Edge node is recommended.		
NSX Edge Node Mobility	• Mobility is not supported across racks • Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.	Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during host maintenance by temporarily disabling the VM/host rule across racks, but not recommended in order to maintain BGP peering local to each rack.	• Mobility is not supported across racks. • Mobility within the same rack is not required and the configuration of a compute policy type Best Effort Restart (BER) is required. Optional during host maintenance within the same rack when multiple ESX hosts are available, by temporarily disabling the VM/host rule.

Note:

A Layer 2 vSphere cluster refers to a vSphere cluster implementation where the host relevant host networks, including vMotion, ESX management, and storage traffic, reside within a single Layer 2 broadcast domain.

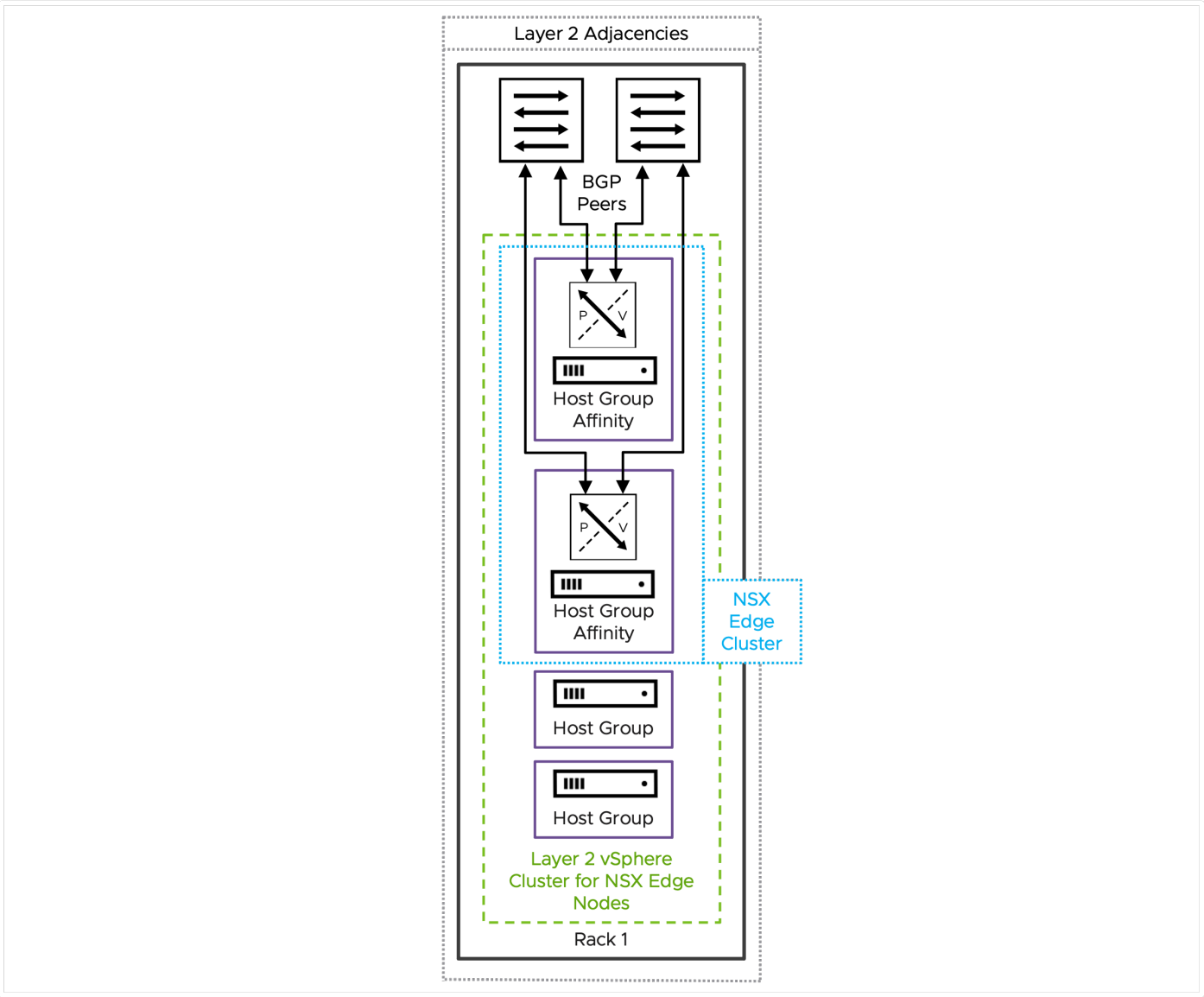
Note:

A Layer 3 vSphere cluster refers to a vSphere cluster implementation where for each individual ESX host or a sub-set of ESX hosts from a vSphere cluster, the relevant host networks, including vMotion, ESX management, and storage traffic, are assigned to individual Layer 2 broadcast domain, separated through Layer 3 boundaries.

Host Fault Tolerant NSX Edge Cluster Model

A Single-Rack vSphere Cluster Model for hosting NSX Edge nodes offer a minimalist footprint while maintaining basic host-level availability.

Figure 356: Host Fault Tolerant NSX Edge Cluster Model in VMware Cloud Foundation



Host Fault Tolerant NSX Edge Cluster Model Attributes

The *Host Fault Tolerant NSX Edge Cluster Model* has the following attributes.

Attribute	Details
Availability Model	• Host level redundancy only. • Single shared storage. • This design cannot ensure continuous service availability for north-south communication and centralized network services during a complete rack failure.
NSX Edge Node Mobility	Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same rack that hosts the NSX Edge node.
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.

Attribute	Details
North-South Bandwidth Growth	• Add hosts to the single existing vSphere cluster or preferably transition to a rack fault tolerant design with two or more vertical Layer 2 clusters in separate racks. • When deploying more NSX Edge nodes on existing hosts, ensure each host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Host Fault Tolerant NSX Edge Cluster Model Design Requirements and Recommendations

Table 843: Common NSX Edge Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMSAZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric Settings.

Table 844: Host Fault Tolerant NSX Edge Cluster Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-HFTSR-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Table 845: Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMSAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMSAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-NSX-EDGE-RCMD-COMSAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs. • When deploying two NSX Edge nodes per host, each NSX Edge node requires two dedicated physical NICs configured for optimal performance.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMSAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface already exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMSAZ-008	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow.
VCF-NSX-EDGE-RCMD-COMSAZ-009	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-NSX-EDGE-RCMD-COMSAZ-010	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESXhost.	If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment.
VCF-NSX-EDGE-RCMD-COMSAZ-011	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-NSX-EDGE-RCMD-COMSAZ-012	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-013	Create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.
VCF-NSX-EDGE-RCMD-COMSAZ-014	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload virtual machines and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-015	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMSAZ-016	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-017	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSX on dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMSAZ-018	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two data plane fastpath interfaces gets an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP)	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMSAZ-019	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (Physical link state up but not forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMSAZ-020	Limit the span of NSX Edge node networks to the local racks.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two (2) NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP session.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-021	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard for the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.
VCF-NSX-EDGE-RCMD-COMSAZ-022	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

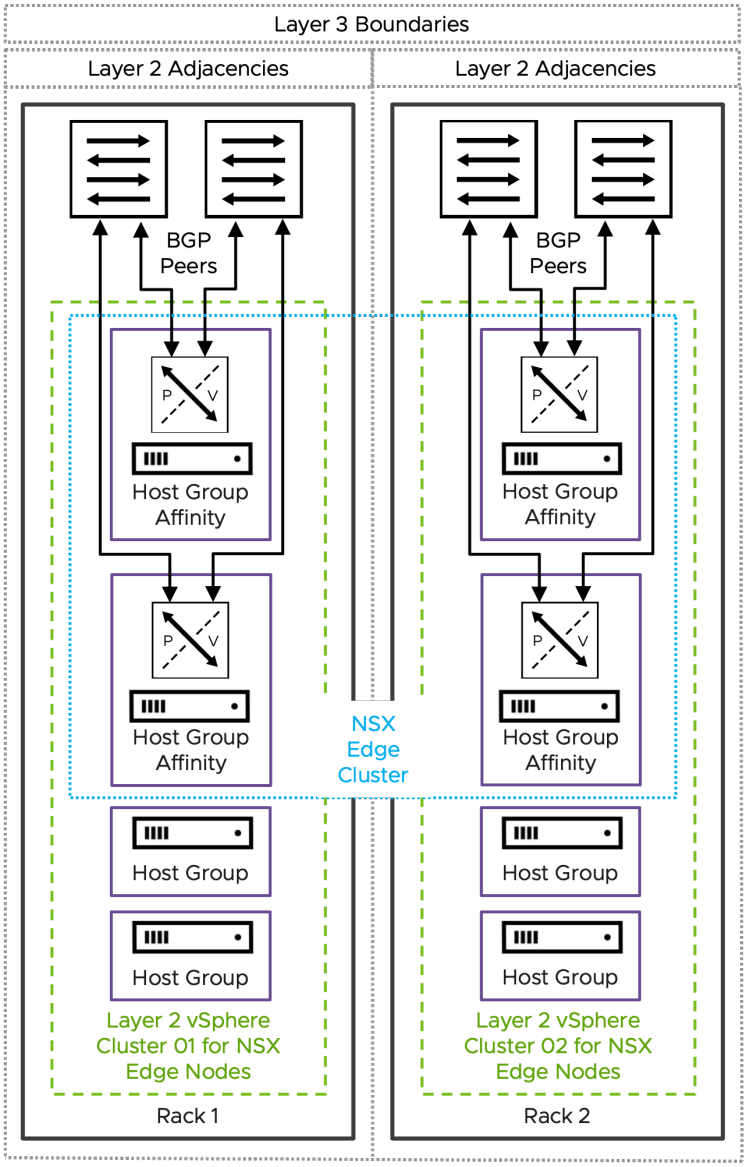
Table 846: Host Fault Tolerant NSX Edge Cluster Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-HFTSR-001	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across different hosts.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-NSX-EDGE-RCMD-HFTSR-002	Configure an NSX failure domain for each host where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability if there was a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters

Using two or more instances of the Single-Rack vSphere Cluster Model to host NSX Edge nodes from the same NSX Edge cluster provides both host-level and rack-level availability, with each rack maintaining independent storage when using vSAN. Each vSphere cluster's infrastructure is contained entirely within a single rack.

Figure 357: Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters in VMware Cloud Foundation



Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters Attributes

The *Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters* has the following attributes.

Attribute	Details
Availabilit Model	• Host and rack-level redundancy. • Independent shared storage per vSphere cluster when using vSAN. • This design ensures continuous service availability for north-south communication and centralized network services during a complete rack failure.
NSX Edge Node Mobility	• Mobility is not supported across racks • Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same rack that hosts the NSX Edge node.

Attribute	Details
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Expand all vSphere clusters simultaneously, ensuring NSX Edge nodes remain evenly distributed across vSphere clusters. This differs from the single vSphere cluster approach where scaling could be limited to one vSphere cluster and one rack. • When deploying more NSX Edge nodes on existing ESX hosts, ensure each ESX host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters Design Requirements and Recommendations

Table 847: Common NSX Edge Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMSAZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric Settings.

Table 848: Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-RFTMC-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Table 849: Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMS AZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMS AZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMS AZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMS AZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs. • When deploying two NSX Edge nodes per host, each NSX Edge node requires two dedicated physical NICs configured for optimal performance.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMSAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface already exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMSAZ-008	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow.
VCF-NSX-EDGE-RCMD-COMSAZ-009	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-NSX-EDGE-RCMD-COMSAZ-010	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESXhost.	If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-011	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-NSX-EDGE-RCMD-COMSAZ-012	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually.
VCF-NSX-EDGE-RCMD-COMSAZ-013	Create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-014	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload virtual machines and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.
VCF-NSX-EDGE-RCMD-COMSAZ-015	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMSAZ-016	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-017	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSX on dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMSAZ-018	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two data plane fastpath interfaces gets an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP)	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMSAZ-019	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (Physical link state up but not forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-020	Limit the span of NSX Edge node networks to the local racks.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two (2) NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP session.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.
VCF-NSX-EDGE-RCMD-COMSAZ-021	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard for the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.
VCF-NSX-EDGE-RCMD-COMSAZ-022	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

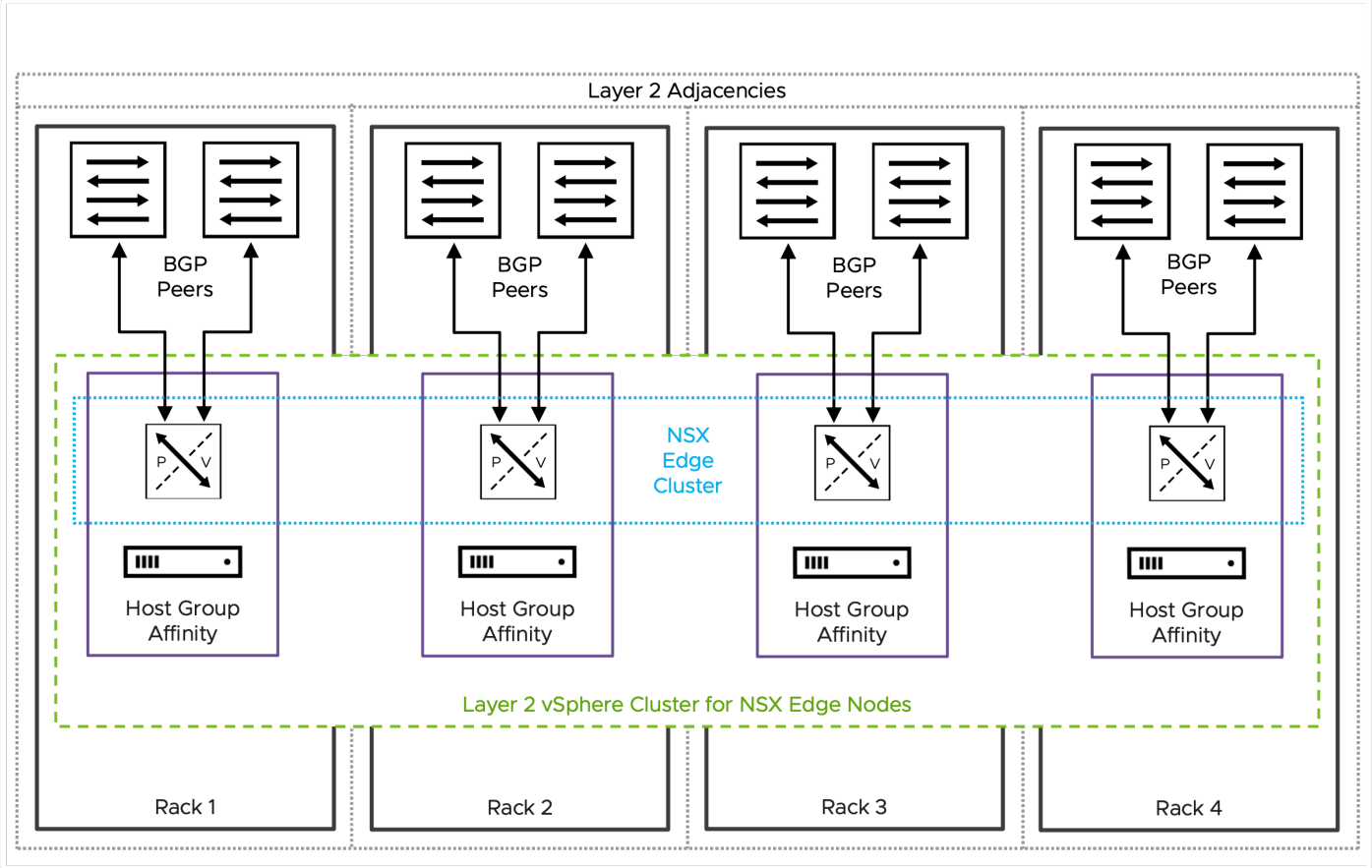
Table 850: Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-RFTMC-001	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across different vSphere clusters.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-NSX-EDGE-RCMD-RFTMC-002	Configure an NSX failure domain for each separate vSphere cluster where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability in case of a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster

A single horizontally deployed vSphere cluster for hosting NSX Edge nodes provides host-level and rack-level availability but without providing independent storage for each rack. The vSphere cluster is deployed with one or more hosts per rack across two or more racks, unlike the other designs with separated vertical vSphere clusters where each vSphere cluster is contained within a single rack. This design distributes ESX hosts from the same vSphere cluster within the same Layer 2 domain between racks rather than across Layer 3 boundaries. The physical network fabric must support Layer 2 adjacencies (e.g., through VXLAN encapsulation) to enable connectivity between ESX hosts across different racks.

Figure 358: Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster in VMware Cloud Foundation



Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster Attributes

The *Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster* has the following attributes.

Attribute	Details
Availability Model	• Host and rack-level redundancy. • Single shared storage. • This design ensures continuous service availability for north-south communication and centralized network services during a complete rack failure, similar to the multiple vertical vSphere cluster design.
NSX Edge Node Mobility	Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during host maintenance by temporarily disabling the VM/host rule across racks, but not recommended in order to maintain BGP peering local to each rack.
BGP Peering	Because BGP peering over extended networks may be problematic in some physical network fabrics, it's recommended to establish BGP peering only with local ToR switches to eliminate this dependency, even when Layer 2 adjacency is available across racks.
NSX Edge Node and Link Failover mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Add hosts to existing racks or extend the vSphere cluster to span additional racks, with NSX Edge nodes distributed evenly across the racks. • When deploying more NSX Edge nodes on existing ESX hosts, ensure each ESX host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster Design Requirements and Recommendations

Table 851: Common NSX Edge Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMSAZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric Settings.

Table 852: Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-RFTL2-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High-Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Table 853: Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMSAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMSAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-NSX-EDGE-RCMD-COMSAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs. • When deploying two NSX Edge nodes per host, each NSX Edge node requires two dedicated physical NICs configured for optimal performance.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMSAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface already exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMSAZ-008	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow.
VCF-NSX-EDGE-RCMD-COMSAZ-009	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-NSX-EDGE-RCMD-COMSAZ-010	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESXhost.	If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment.
VCF-NSX-EDGE-RCMD-COMSAZ-011	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-NSX-EDGE-RCMD-COMSAZ-012	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-013	Create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.
VCF-NSX-EDGE-RCMD-COMSAZ-014	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload virtual machines and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-015	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMSAZ-016	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-017	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSX on dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMSAZ-018	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two data plane fastpath interfaces gets an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP)	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMSAZ-019	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (Physical link state up but not forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMSAZ-020	Limit the span of NSX Edge node networks to the local racks.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two (2) NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP session.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-021	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard for the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.
VCF-NSX-EDGE-RCMD-COMSAZ-022	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

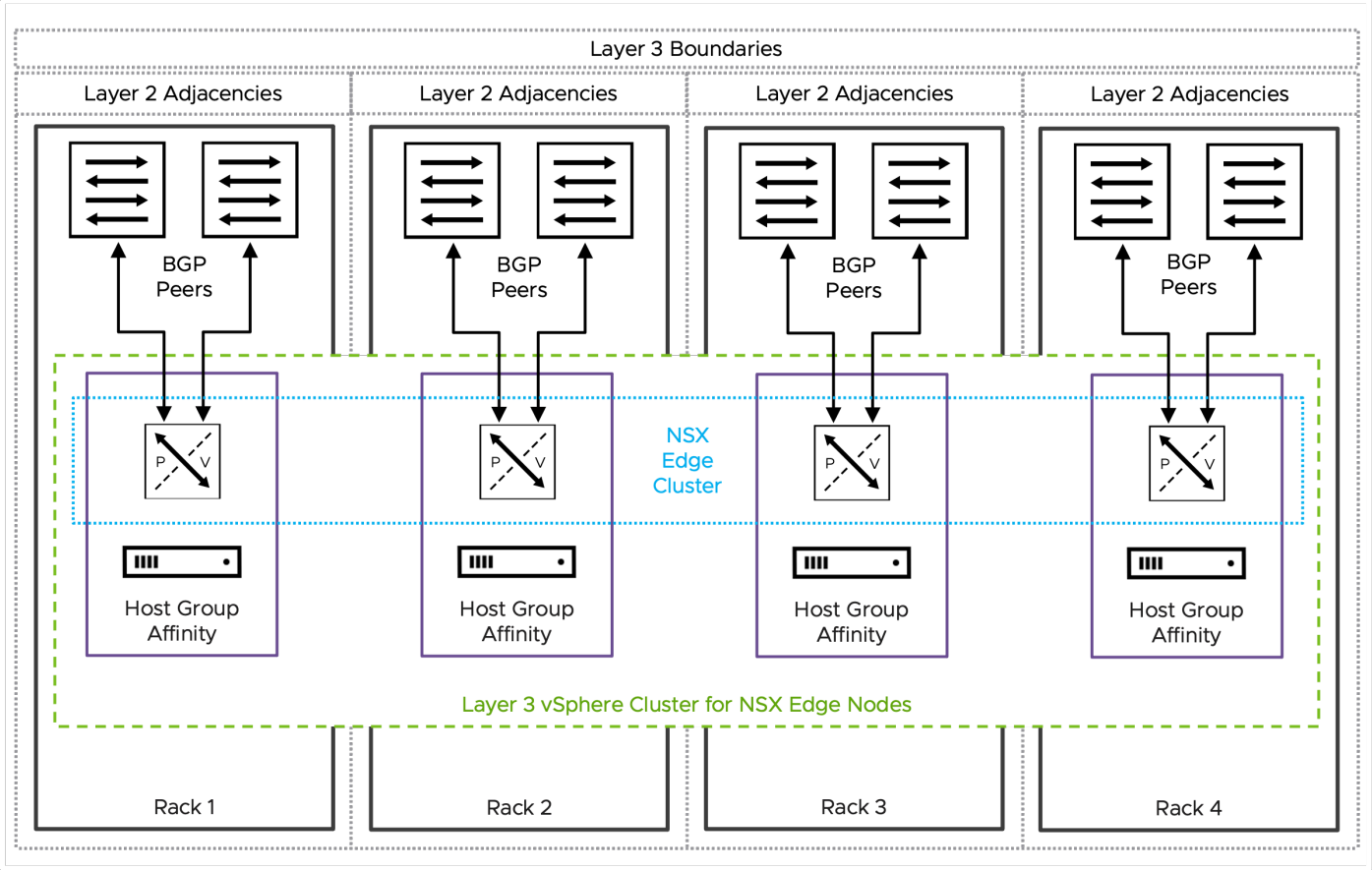
Table 854: Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-RFTL2-001	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across different racks.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-NSX-EDGE-RCMD-RFTL2-002	Configure an NSX failure domain for each rack where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability in case of a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster

A single horizontally deployed vSphere cluster for hosting NSX Edge nodes provides host-level and rack-level availability but without providing independent storage for each rack. The vSphere cluster is deployed with one or more hosts per rack across two or more racks, unlike other designs with separated vertical vSphere clusters where each vSphere cluster was contained within a single rack. This design distributes ESX hosts from the same vSphere cluster across multiple racks, with Layer 3 network boundaries providing separation between each rack. This design has no dependencies on specific Layer 2 capabilities (such as VXLAN) of the physical network fabric and works with a simple Layer 3 physical network fabric.

Figure 359: Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster in VMware Cloud Foundation



Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster Attributes

The *Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster* has the following attributes.

Attribute	Properties
Availability Model	• Host and rack-level redundancy. • Single shared storage. • This design ensures continuous service availability for north-south communication and centralized network services during a complete rack failure, similar to the multiple vertical vSphere cluster design.
NSX Edge Node Mobility	• Mobility is not supported across racks. • Mobility within the same rack is not required and the configuration of a compute policy type Best Effort Restart (BER) is required. Optional during host maintenance within the same rack when multiple ESX hosts are available, by temporarily disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same rack that hosts the NSX Edge node.
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Add hosts to existing racks or extend the vSphere cluster to span additional racks, with NSX Edge nodes distributed evenly across the racks. • When deploying more NSX Edge nodes on existing ESX hosts, ensure each ESX host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster Design Requirements and Recommendations

Table 855: Common NSX Edge Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMSAZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric Settings.

Table 856: Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-RFTL3-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Table 857: Common NSX Edge Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMSAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMSAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-NSX-EDGE-RCMD-COMSAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs. • When deploying two NSX Edge nodes per host, each NSX Edge node requires two dedicated physical NICs configured for optimal performance.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMSAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface already exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMSAZ-008	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow.
VCF-NSX-EDGE-RCMD-COMSAZ-009	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-NSX-EDGE-RCMD-COMSAZ-010	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESXhost.	If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment.
VCF-NSX-EDGE-RCMD-COMSAZ-011	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-NSX-EDGE-RCMD-COMSAZ-012	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-013	Create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.
VCF-NSX-EDGE-RCMD-COMSAZ-014	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload virtual machines and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-015	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMSAZ-016	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMSAZ-017	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSX on dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMSAZ-018	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two data plane fastpath interfaces gets an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP)	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMSAZ-019	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (Physical link state up but not forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMSAZ-020	Limit the span of NSX Edge node networks to the local racks.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two (2) NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP session.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMSAZ-021	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard for the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.
VCF-NSX-EDGE-RCMD-COMSAZ-022	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Table 858: Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-RFTL3-001	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across different racks.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-NSX-EDGE-RCMD-RFTL3-002	Configure an NSX failure domain for each rack where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability in case of a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

NSX Edge Cluster Models for Dual Availability Zones

NSX Edge deployments across two availability zones require careful consideration of network characteristics, even though these zones are located in close proximity to each other. The latency (RTT) requirements vary based on the underlying vSphere cluster topology: a maximum of 5ms latency (RTT) is required when deploying on stretched vSAN clusters, while up to 10ms latency (RTT) is acceptable when running on independent vSphere clusters in each availability zone. When designing for these deployments, organizations must make a binary decision regarding asymmetric traffic flows between availability zones—they are either acceptable or not acceptable, usually based on the presence of a stateful device in the path or not. This decision, along with considerations about network failover speed and north-south throughput requirements, will determine the appropriate design model to implement.

NSX Edge Cluster for Dual Availability Zones Models Comparison

Attribute	One dedicated vSphere cluster per availability zone based on NSX Edge node HA (4 routing options: A, B, C, and D)	Single stretched vSAN cluster across availability zones based on NSX Edge node HA (4 routing options: A, B, C, and D)	NSX Edge node design for a single stretched vSAN cluster across availability zones based on vSphere HA recovery
Max Latency (RTT) between availability zones	10 ms	5 ms	
Availability zone failover	Fast and automatic for routing option A, B, and C; availability zone failover is based on NSX Edge node High Availability detection. Manual failover for routing option D; availability zone failover requires an automated script for Tier-1 gateway switchover. **Note:** See next section below for details around the routing options A, B, C, and D.		Automatic but slower, based on vSphere HA.
Compute availability model	Availability zone and host level		
Storage availability model	Independent storage for each availability zone when using vSAN	Single shared storage across availability zones	
Physical network fabric dependency and fault isolation	• No dependency • Maintain optimal fault isolation between availability zones		• Requires stretched VLAN across availability zones. • Limited error isolation/fault isolation between availability zones.
Physical vSphere cluster layout	Two vSphere clusters each contained within an availability zone.	Single stretched vSAN cluster across the two availability zones	
Physical footprint	Medium to high		
Scalability	• Adding more hosts to the existing vSphere cluster to deploy more NSX Edge nodes • Adding more NSX Edge nodes to existing ESX hosts that have free CPU and memory resources requires additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node. A ratio of 2 to 1 physical NICs for each NSX Edge node is recommended.		
NSX Edge node mobility	• Mobility is not supported across racks/availability zones when using the multiple instances of the Single-Rack vSphere Cluster Model • Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.	• Mobility is not supported across availability zones when using the Stretched vSphere Cluster Model • Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.	Mobility is required across availability zones for NSX Edge nodes to failover and restart at the secondary availability zone.
Routing peering failover	Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.		BGP failover without Bidirectional Forwarding Detection (BFD)

Routing Options for Dual Availability Zone Edge Cluster Models based on NSX Edge Node High Availability

When the NSX Edge nodes are pinned to specific availability zones, the design supports different routing design choices based on the requirement to provide symmetric traffic in and out of the virtual network. Deterministic symmetric traffic

forwarding is required when external stateful services are implemented in the network path connecting the two availability zones.

Choice ID	Attributes	Benefits	Implications
A - Active/Active Tier-0 gateway with asymmetric routing	• Single Tier-0 gateway across availability zones in active/active mode. • Applies to Centralized Connectivity Model and NSX Segment Connectivity Model.	• Scalable throughput. • Fast and automatic failover across availability zones.	Asymmetric traffic, not suitable when a stateful device is in the path between availability zone.
B - Active/Active Tier-0 gateway with symmetric routing	• Single Tier-0 gateway across availability zones in active/active mode. • To direct ingress traffic to a specific availability zone, the Tier-0 gateway must advertise BGP routes with AS-Path prepending to the physical network fabric in the secondary availability zones. • To direct egress traffic to a specific availability zone, the physical network fabric must advertise more specific routes to the NSX Edgenodes in the primary availability zone. • Applies to Centralized Connectivity Model and NSX Segment Connectivity Model.	• Scalable throughput • Fast and automatic failover across availability zones. • Symmetric traffic in and out of the virtual network. • Suitable when a stateful device is in the path between availability zones.	• Physical network fabric configuration requirement (Advertisement of more specific routes in the primary availability zone). • Some outbound flows will traverse the NSX Edge nodes in the secondary availability zone before being redirected to the NSX Edge nodes in the primary availability zone.
C - Active/Standby Tier-0 gateway with symmetric routing	• Single Tier-0 gateway across availability zones in active/standby mode. • In the Tier-0 gateway settings, designate as preferred the NSX Edge node located in the availability zone that manages incoming and outgoing traffic. • Applies to Centralized Connectivity Model and NSX Segment Connectivity Model.	• Fast, automatic failover across availability zones • Symmetric traffic in and out of the virtual network. • Suitable when a stateful device is in the path between availability zones.	• Throughput limited by a single NSX Edge node capacity. • NSX Edge node failover (e.g., for an upgrade) requires availability zone failover.
D - Two Active/Active Tier-0 gateway with manual failover	• Independent Tier-0 gateways per availability zone in active/active mode. • Connect the Tier-1 gateway to the Tier-0 gateway deployed in the primary availability zone that manages egress and ingress traffic flows. • Applies to Centralized Connectivity Model and NSX Segment Connectivity Model.	• Scalable throughput. • Symmetric traffic in and out of the virtual network. • Suitable when a stateful device is in the path between availability zones. • Controlled network failover across availability zone via manual intervention.	Manual network failover across availability zone.

Dual Availability Zone Based on NSX Edge Node HA Model

NSX Edge nodes can be hosted in two ways: either using multiple instances of the Single-Rack vSphere Cluster Model, or using a single Stretched vSphere Cluster Model across two availability zones with half of the NSX Edge nodes from the NSX Edge cluster restricted to the primary availability zone and the other half to the secondary availability zone. Note that NSX Edge nodes cannot move between availability zones in these two design options. Each availability zone can use separate VLANs for NSX Edge node uplink traffic (BGP sessions, north-south traffic, NSX Edge node TEP traffic, and NSX Edge node management interface), improving failure isolation. These two designs supports adaptable Tier-0 gateway routing configurations that prevent asymmetric traffic issues.

Figure 360: Dual Availability Zone Design Based on NSX Edge Node HA Model (Dedicated vSphere Cluster) in VMware Cloud Foundation

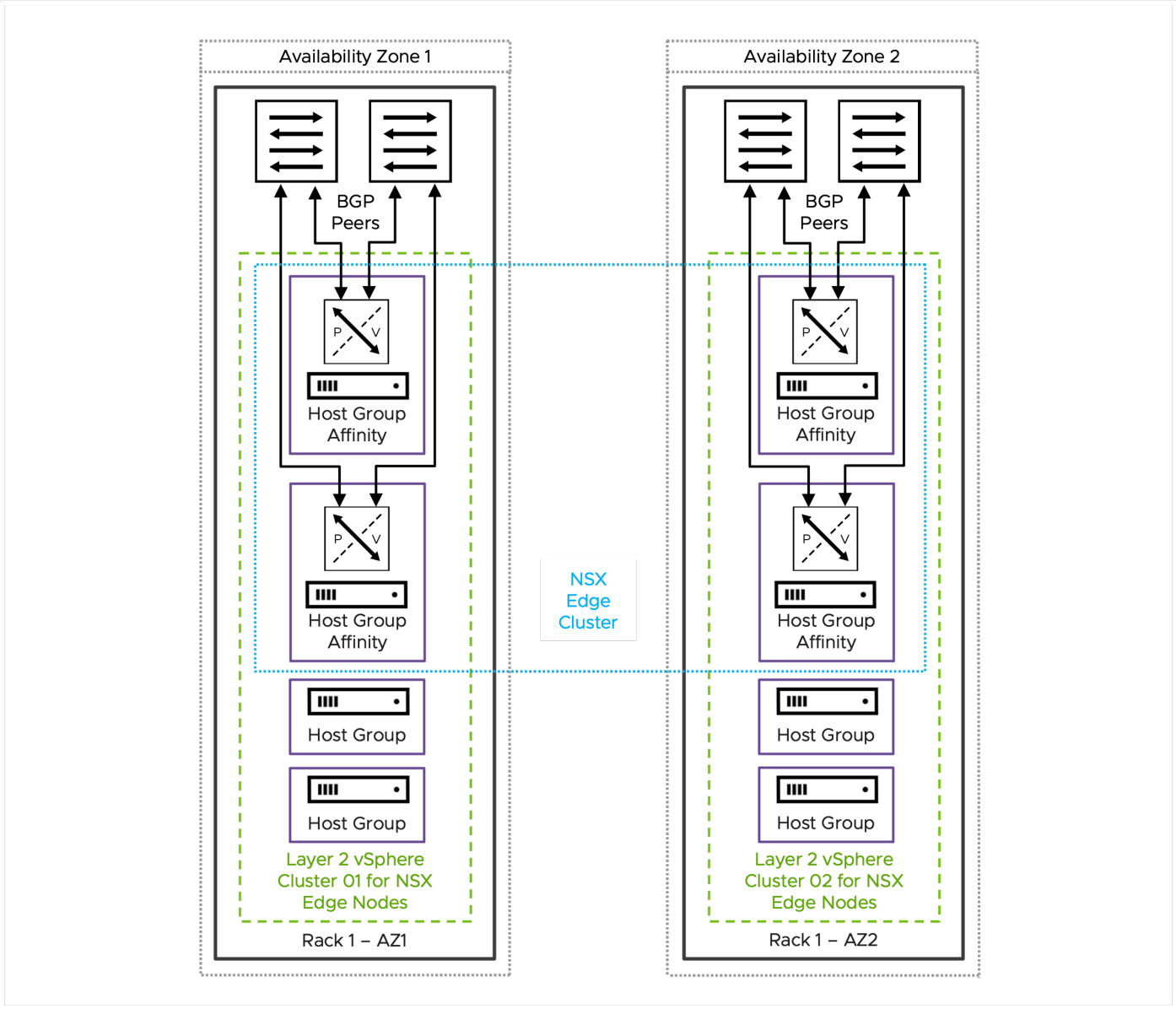
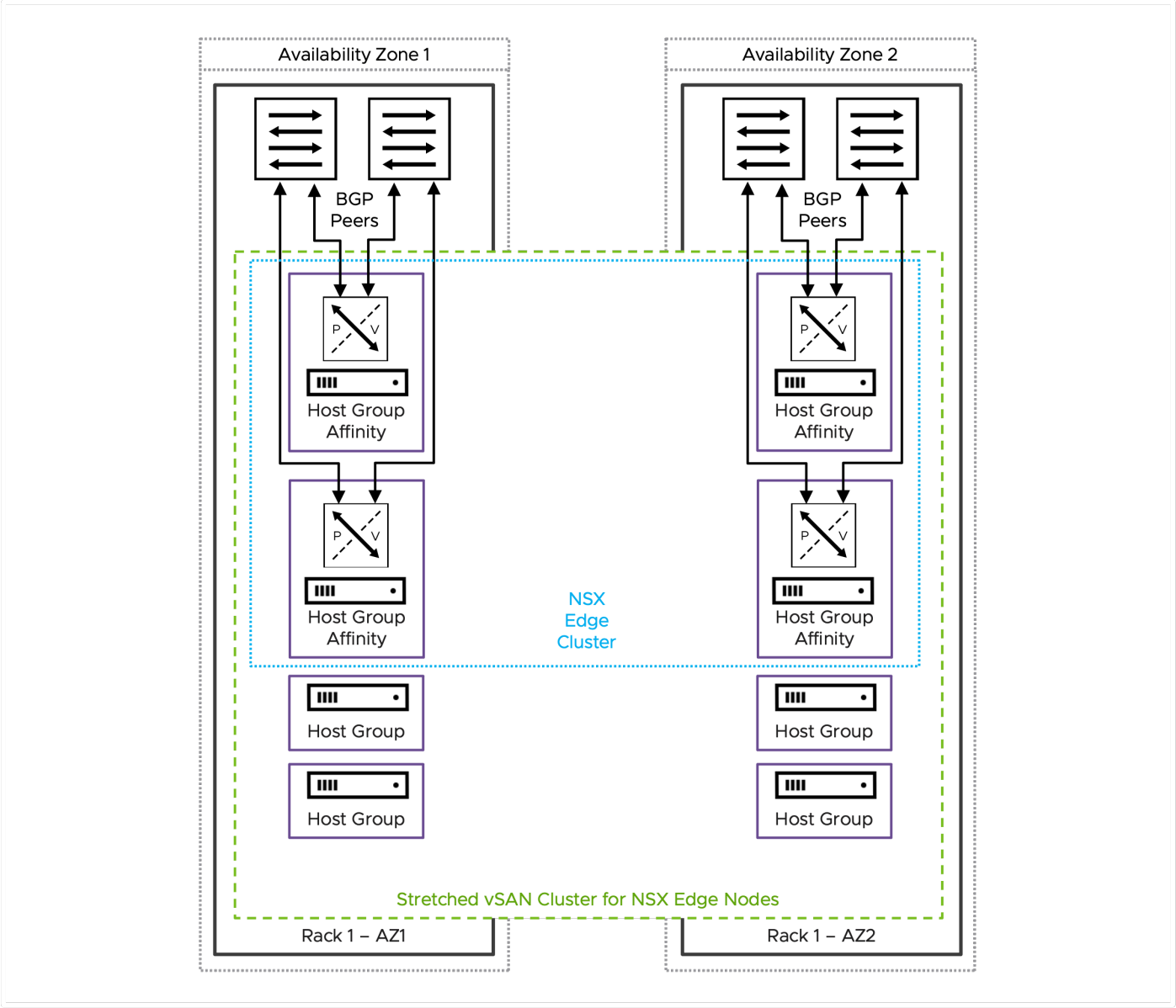


Figure 361: Dual Availability Zone Design Based on NSX Edge Node HA Model (Stretched vSAN) in VMware Cloud Foundation



Dual Availability Zone Design Based on NSX Edge Node HA Model Attributes

The *Dual Availability Zone Design Based on NSX Edge Node HA Model* has the following attributes.

Attribute	Details
Availability Model	• Host and availability zone redundancy. • Independent shared storage per vSphere cluster when using vSAN or single shared storage. • This design ensures continuous north-south communication and centralized network services even if one availability zone completely fails.
NSX Edge Node Mobility	• Mobility is not supported across racks/availabilty zones when using multiple instances of the Single-Rack vSphere Cluster Model. Mobility is not supported across availability zones when using the Stretched vSphere Cluster Model. • Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
BGP Peering	Establish BGP peering solely with the local ToR switches located within the same availability zones that hosts the NSX Edge node.
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Add more hosts to both vSphere clusters simultaneously or scale out the stretched vSphere cluster with additional hosts in each availability zone. • When deploying more NSX Edge nodes on existing ESX hosts, ensure each ESX host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Dual Availability Zone Design Based on NSX Edge Node HA Model Design Requirements and Recommendations

Table 859: Common NSX Edge Cluster for Dual Availability Zones Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMD AZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric settings.

Table 860: Dual Availability Zone Design based on NSX Edge Node HA Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-DAZE HA-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High-Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.

Table 861: Common NSX Edge Cluster for Dual Availability Zones Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node ESX hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMDAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or ESX hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload virtual machines have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart after ESX host failures if it cannot meet their memory reservations on surviving ESX hosts. • Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMDAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are generally less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMDAZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.
VCF-NSX-EDGE-RCMD-COMDAZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMDAZ-008	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across two dedicated vSphere clusters respective across two availability zones.	Maintain High Availability and prevent single points of failure.	• Additional servers are required. • This design recommendation is not applicable to the design model based on vSphere HA Recovery
VCF-NSX-EDGE-RCMD-COMDAZ-009	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload VMs and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-010	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMDAZ-011	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMDAZ-012	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge nodes and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge nodes, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSXon dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMDAZ-013	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two (2) data plane fastpath interfaces get an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP).	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMDAZ-014	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (physical link state up but no forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-015	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard or the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-016	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Table 862: Dual Availability Zone Design based on NSX Edge Node HA Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZEHA-001	Limit the span of NSX Edge node networks to the local rack.	• VLANs including NSX Edge node Tunnel Endpoint (TEP), the NSX Edge node management, and the two NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic are limited to the rack. • BGP will establish peering connections exclusively with Top-of-Rack (ToR) switches located within the same rack, ensuring a deterministic alignment between physical components (ToRs and cabling) and the corresponding BGP sessions.	• Limit the span of networks to reduce failure propagation across racks. • Avoid BGP peering across racks.
VCF-NSX-EDGE-RCMD-DAZEHA-002	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	• Host groups must be configured through the administrator before initiating the Network Connectivity wizard or NSX Edge cluster workflow. • This design recommendation is not applicable with routing option C - Active/Standby Tier-0 gateway with symmetric routing.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZE HA-003	Create for every NSX Edge node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	• The Network Connectivity wizard or the NSX Edge cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-004	Assign to each individual NSX Edge node the previously configured host group.	Each NSX Edge node has a 1:1 affinity with the underlying ESX host.	• Increased configuration effort. • If the assignment was not set through the Network Connectivity wizard or NSX Edge cluster workflow, it can be added post-deployment. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-005	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the NSX Edge node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each NSX Edge node are required to enable the compute policy type Best Effort Restart (BER).	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-006	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each NSX Edge node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the NSX Edge nodes together.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the vCenter tag with a category and assign to each NSX Edge node when the host group affinity option is selected. Do not create the vCenter tag and category manually. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-DAZE HA-007	Create a compute policy type Best Effort Restart (BER) and assign the previously created vCenter tag and category to it.	• The compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful NSX Edge node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer NSX Edge node readiness during host upgrades through multiple pre-checks before initiating NSX Edge node failover and ensures the NSX Edge node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or NSX Edge cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the NSX Edge node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the NSX Edge node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted NSX Edge node regardless of whether a healthy peer NSX Edge node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained. • This design recommendation is not applicable with routing option C - Active/ Standby Tier-0 gateway with symmetric routing.
VCF-NSX-EDGE-RCMD-DAZE HA-008	Configure an NSX failure domain for each availability zone where the NSX Edge node can run - this is only valid when you have more than two NSX Edge nodes; map the NSX Edge nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee service availability in case of a failure affecting multiple NSX Edge nodes. • Active and standby instance of a VPC gateway or Tier-1 gateway always run in different NSX failure domains.	• Additional manual configuration. • This applies only for designs incorporating a VPC gateway or Tier-1 gateway in active/standby mode.

Table 863: Dual Availability Zone Design Based on NSX Edge Node HA Model BGP Routing Design Requirements

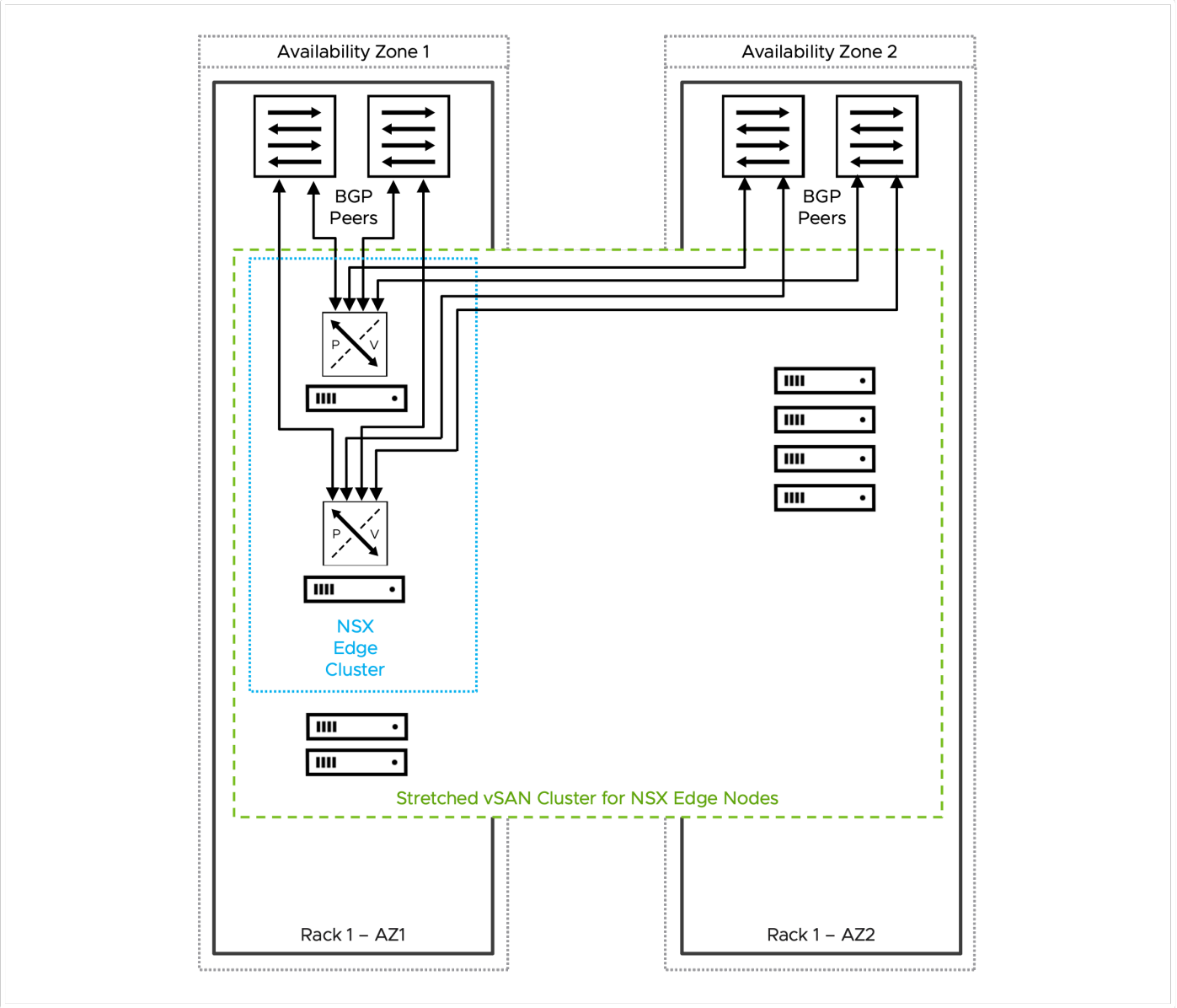
These BGP routing design requirements are specific requirements for the *Dual Availability Zone Design Based on NSX Edge Node HA Model* for routing option B. Routing option B implements a High Availability design using Tier-0 gateways configured in active/active mode and ensures symmetric traffic in and out of the virtual network. For routing options A, C, and D, these specific recommendations do not apply; only the general BGP routing design requirements and rrecommendations that are foundin the Workload Connectivity Modelssection are relevant.

Design requirement ID	Design Requirement	Justification	Implication
VCF-NSX-BGP-REQD-DAZEHA-001	Create a route-map that prepends the BGP ASN path using the local Tier-0 gateway ASN and matches a new configured IP prefix list set to "Any". Apply this route-map as an outbound filter to all BGP neighbors configured with the top of rack switches in the second availability zone.	The Tier-0 gateway uses BGP ASN path prepending in BGP route advertisements to the second availability zone's top of rack switches, ensuring ingress traffic preferentially flows through the primary availability zone, as BGP selects paths with a shorted AS path.	• Additional manual configuration. • The Tier-0 gateway will route north-south traffic through the second availability zone only if the connection to their BGP neighbors in the primary availability zone is lost, for example, if a failure of the top of the rack switch pair or in the availability zone occurs.
VCF-NSX-BGP-REQD-DAZEHA-002	The top of rack switches in the primary availability zone must advertise 0.0.0.0/1 and 128.0.0.0/1 to the NSX Edge nodes in the primary availability zone and the top of rack switches in the second availability zone must advertise 0.0.0.0/0 to the NSX Edge nodes in the second availability zone.	The NSX Edge nodes in the primary availability zone will receive more specific routes compared to the NSX Edge nodes in the second availability zone, ensuring the egress traffic preferentially flows through the primary availability zone.	• Additional manual configuration on the top of rack switches. • The origin of 0.0.0.0/1 and 128.0.0.0/1 depends on the routing topology, but in many cases it is the same device where the default route is injected — for example, the internet-facing firewall.

Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model

NSX Edge nodes are hosted using a stretched vSAN cluster across two availability zones and support NSX Edge node mobility between zones (unlike the previous design where mobility was not supported). The NSX Edge node uplink VLANs (carrying BGP sessions and north-south traffic) and NSX Edge node management VLAN must be stretched across both availability zones, which provides only limited fault isolation between zones. This approach is different from the previous dual availability zone design where separate VLANs improved isolation. The design supports Tier-0 gateway configuration in active/active mode with symmetric flows and availability zone awareness based on NSX Edge node location. This approach is different from the previous adaptable Tier-0 gateway routing design.

Figure 362: Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model (Stretched vSAN) in VMware Cloud Foundation



Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model Attributes

The *Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model* has the following attributes.

Attribute	Properties
Availability Model	• Host and availability zone redundancy. • Single shared storage. • This design ensures continuous north-south communication and centralized network services even if one availability zone completely fails.
NSX Edge Node Mobility	Mobility is required across availability zones for NSX Edge nodes to failover and restart at the secondary availability zone.
BGP Peering	BGP peering over stretched networks across both availability zone is required.

Attribute	Properties
NSX Edge Node and Link Failover Mechanisms	• Fast NSX Edge node High Availability detection for NSX Edge node failures with the same availability zone. • A complete availability zone outage, the recovery depends on vSphere High Availability, which may take several minutes. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for BGP.
North-South Bandwidth Growth	• Scale out the stretched vSAN cluster with additional hosts in each availability zone. • When deploying more NSX Edge nodes on existing ESX hosts, ensure each ESX host has sufficient CPU, memory, and importantly, additional physical NICs (pNICs) to maintain maximum packet forwarding rates (PPS) for each NSX Edge node.

Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model Design Requirements and Recommendations

Table 864: Common NSX Edge Cluster for Dual Availability Zones Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-COMD AZ-001	Align the Tunnel Endpoint (TEP) MTU for NSX Edge nodes to the same MTU as the vSphere Distributed Switch MTU.	Consistent MTU within the same NSX Overlay Transport Zone is required.	Requires manual configuration of the Tunnel Endpoint (TEP) MTU value in NSX Global Fabric settings.

Table 865: Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-DAZV HA-001	Deploy two (2) or more NSX Edge nodes belonging to the same NSX Edge cluster.	• NSX Edge nodes are required to support centralized network connectivity for the virtual network. • At least two NSX Edge nodes are required to support NSX Edge High-Availability.	• vSphere cluster resources must be available to support the NSX Edge nodes. • Additional NSX Edge nodes may be required to satisfy increased bandwidth requirements.
VCF-NSX-EDGE-REQD-DAZV HA-002	Do not enable the compute policy type Best Effort Restart (BER) for each NSX Edge node.	• In a stretched vSAN cluster deployment supporting NSX Edge node mobility across the two Availability Zone the compute policy type Best Effort Restart (BER) cannot be used. • NSX Edge node mobility requires "should" VM-to-host groups to place the NSX Edge nodes into the primary Availability Zone to support deterministic egress and ingress traffic flows, whereas Best Effort Restart (BER) compute policy requires "must" VM/host rule.	None.
VCF-NSX-EDGE-REQD-DAZV HA-003	Create a host group to include all hosts from the primary availability zone.	This host groups is required to enable the NSX Edge node affinity with the primary availability zone.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NSX-EDGE-REQD-DAZV HA-004	Create a VM group to include all NSX Edge nodes from a specific NSX Edge cluster.	This VM group is required to enable the NSX Edge node affinity with the primary availability zone.	None.
VCF-NSX-EDGE-REQD-DAZV HA-005	Create "should" VM/host rule for the primary availability zones.	Deterministic ingress and egress traffic path within the primary availability zone is achieved by associating the VM group containing the NSX Edge nodes with the host group from the primary availability zone through the creation of a VM/host "should" rule.	Supports external stateful services that are implemented in the network path connecting two availability zones.
VCF-NSX-EDGE-REQD-DAZV HA-006	Extend the NSX Edge node networks across both availability zones.	• NSX Edge nodes can fail over between availability zones, ensuring continuous connectivity to the top of rack switches in both availability zones regardless of which zone the NSX Edge nodes are currently hosted in • NSX Edge node networks must be stretched across both availability zones, including the NSX Edge node Tunnel Endpoint (TEP) VLAN, the NSX Edge node management VLAN, and the two NSX Edge Uplink VLANs that carry BGP sessions and north-south traffic.	Stretched VLANs between the availability zones must be configured in the physical network infrastructure.

Table 866: Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model BGP Routing Design Requirements

These BGP routing design requirements are specific requirements for the *Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model* across the availability zones. General BGP routing design requirements and recommendations and can be found in the Centralized Connectivity Model and NSX Segment Connectivity Model.

Design requirement ID	Design Requirement	Justification	Implication
VCF-NSX-BGP-REQD-DAZVHA-001	Do not use Bidirectional Forwarding Detection (BFD) for BGP.	Bidirectional Forwarding Detection (BFD) failure detection for BGP sessions can trigger connectivity failures during vMotion migrations, use instead a lower BGP timer (4/12 seconds)	Slower link failure detection.
VCF-NSX-BGP-REQD-DAZVHA-002	Create a route-map that prepends the BGP ASN path using the local Tier-0 gateway ASN and matches a new configured IP prefix list set to "Any". Apply this route-map as an outbound filter to all BGP neighbors configured with the top of rack switches in the secondary availability zone.	The Tier-0 gateway uses BGP ASN path prepending in BGP route advertisements to the secondary availability zone's top of rack switches, ensuring ingress traffic preferentially flows through the primary availability zone, as BGP selects paths with a shorted AS path.	• Additional manual configuration. • The Tier-0 gateway will route north-south traffic through the secondary availability zone only if the connection to their BGP neighbors in the primary availability zone is lost, for example, if a failure of the top of the rack switch pair or in the availability zone occurs.
VCF-NSX-BGP-REQD-DAZVHA-003	Create a route-map that set a lower BGP local preference, for example, 90 and matches a new configured IP prefix list set to "0.0.0.0/0". Apply this route-map as an inbound filter to all BGP neighbors configured with the top of rack switches in the secondary availability zone.	The Tier-0 gateway sets a lower BGP local preference for the 0.0.0.0/0 route (assuming the physical network advertises only the default route across both availability zones) received from the secondary availability zone's top of rack switches, ensuring egress traffic preferentially flows through the primary availability zone, as BGP selects paths with higher local preference values (default 100).	• Additional manual configuration. • The Tier-0 gateway routes north-south traffic through the secondary availability zone only if the connection to their BGP neighbors in the primary availability zone is lost, for example, if a failure of the top of the rack switch pair or in the availability zone occurs.

Table 867: Common NSX Edge Cluster for Dual Availability Zones Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-001	Dedicated vSphere cluster to exclusively host NSX Edge nodes.	• NSX Edge nodes use dedicated and exclusive host resources (CPU, memory, pNIC) to provide the best packet forwarding performance, reliability, and operational simplification. • Enables the selection of an independent vSphere Distributed Switch design separate from the vSphere cluster hosting workload virtual machines. • Enable specific hardware selection for NSX Edge node ESX hosts.	Additional servers are required.
VCF-NSX-EDGE-RCMD-COMDAZ-002	Limit the number of NSX Edge nodes per host to the number of available physical NICs.	Each NSX Edge node should be connected to a pair of physical NICs to ensure maximum packet forwarding rates (PPS).	Additional physical NICs or ESX hosts, or both are required to support the deployment of a higher number of NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-003	Use the default memory reservation with 100% for NSX Edge nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a NSX Edge node, ensuring this full memory amount remains available to the NSX Edge node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload virtual machines have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart after ESX host failures if it cannot meet their memory reservations on surviving ESX hosts. • Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-NSX-EDGE-RCMD-COMDAZ-004	Use the default CPU reservation priority in NSX set to high shares for NSX Edge nodes.	Setting CPU shares to "High" ensures that NSX Edge nodes receive higher relative priority during CPU contention scenarios, allowing critical network functions to maintain performance when host resources are constrained.	High CPU reservations are generally less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMD AZ-005	For each NSX Edge node, assign a dedicated physical NIC to the data plane fastpath interfaces fp-eth0 and fp-eth1.	• Each NSX Edge node leverages only two of the four data plane fastpath interfaces, which is sufficient for most deployment scenarios. • When two NSX Edge nodes are deployed per host, each NSX Edge node should use a dedicated pair of physical NICs.	The two remaining data plane fastpath interfaces per NSX Edge node are not used.
VCF-NSX-EDGE-RCMD-COMD AZ-006	Map data plane fastpath interface fp-eth0 with physical NIC 1 as active and physical NIC 2 as standby; map data plane fastpath interface fp-eth1 with physical NIC 1 as standby and physical NIC 2 as active, creating complementary failover paths.	Enables redundant paths for both data plane fastpath interfaces on each NSX Edge node, preventing single points of failure.	None.
VCF-NSX-EDGE-RCMD-COMD AZ-007	Create a vSphere Distributed Port Group for the NSX Edge node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected NSX Edge cluster model, an NSX Edge node management vSphere Distributed Port Group is required for each vSphere cluster, rack, or availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the NSX Edge node management interface exists — this is the same vSphere Distributed Port Group that is used by the management stack.
VCF-NSX-EDGE-RCMD-COMD AZ-008	Ensure that NSX Edge nodes belonging to the same NSX Edge cluster are evenly deployed across two dedicated vSphere clusters respective across two availability zones.	Maintain High Availability and prevent single points of failure.	• Additional servers are required. • This design recommendation is not applicable to the design model based on vSphere HA Recovery
VCF-NSX-EDGE-RCMD-COMD AZ-009	Select the NSX Edge node size large or x-large.	• The size of an NSX Edge node depends on the throughput requirements and any configured centralized network services. • For most deployments, particularly when NSX Edge nodes are used solely for north-south traffic forwarding, the large NSX Edge node size provides an optimal balance between performance and resource utilization.	• You must provide sufficient compute resources to support the chosen NSX Edge node size. • In vSphere clusters with a high density of NSX Edge nodes, such as multi-tenancy environments with high oversubscription ratios or collapsed vSphere clusters where both workload VMs and NSX Edge nodes run on the same hosts, the resource requirements of large or extra large NSX Edge nodes occasionally create resource constraints.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-010	Select the auto-generation password option for NSX Edge nodes.	Auto-generation password option enables centralized password management in VCF Operations.	The auto-generated password must be retrieved via SDDC Manager to gain SSH access and subsequently updated in VCF Operations.
VCF-NSX-EDGE-RCMD-COMDAZ-011	Select static IP allocation for the NSX Edge node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-NSX-EDGE-RCMD-COMDAZ-012	• Configure separate Tunnel Endpoint (TEP) VLANs for NSX Edge nodes and ESX hosts. • It is possible to use a single TEP VLAN across host and NSX Edge node by enabling "NSX on dvPG".	• Simplifies Tunnel Endpoint (TEP) IP management by separating IP allocation methods between ESX hosts and NSX Edge nodes, for example, using static IP assignment for NSX Edge nodes while utilizing DHCP-based allocation for ESX hosts. • No need to enable "NSXon dvPG".	• NSX Edge nodes and host Tunnel Endpoint (TEP) require separate VLANs with routing configured between them to enable proper overlay network communication. • Requires the configuration of an additional VLAN for NSX Edge node Tunnel Endpoint (TEP) traffic.
VCF-NSX-EDGE-RCMD-COMDAZ-013	Configure IP Pool allocation for the NSX Edge node Tunnel Endpoint (TEP) interfaces.	• IP Pool allocation eliminates dependency on external DHCP services while reducing configuration overhead compared to static IP assignment. • Each of the two (2) data plane fastpath interfaces get an IP address assigned to enable NSX Edge node Tunnel Endpoint (TEP).	Requires the configuration of one or more IP Pools depending on the vSphere cluster model.
VCF-NSX-EDGE-RCMD-COMDAZ-014	Enable TEP Groups for NSX Edge nodes.	• Higher throughput achievable by a single NSX Edge node. • Granular flow base load balancing of NSX Edge node TEP traffic. • Resiliency to partial link failures (physical link state up but no forwarding).	TEP Groups must be enabled through the NSX API globally for all NSX Edge nodes.
VCF-NSX-EDGE-RCMD-COMDAZ-015	Use a single Edge Switch in the NSX Edge nodes.	• Simplifies deployment of NSX Edge nodes. • Supports multiple Tunnel Endpoint (TEP) interfaces. • Carries both overlay and uplink traffic. • A single Edge Switch is sufficient for most deployment scenarios.	The Network Connectivity wizard or the NSX Edge node workflow in vCenter automatically configures always a single Edge Switch.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NSX-EDGE-RCMD-COMDAZ-016	Use the central node profile configuration for NSX Edge nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all NSX Edge nodes.	• NSX Edge nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual NSX Edge node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Virtual Network Appliance Cluster Detailed Design

This section outlines various *Virtual Network Appliance Cluster Models* for virtualized data center environments, including the physical rack placement of *Virtual Network Appliance* nodes. Each design is tailored to address specific requirements for high availability, scalability, capabilities, and infrastructure footprint.

Virtual Network Appliance Cluster Models for Single Availability Zone

- Host Fault Tolerant Virtual Network Appliance Cluster Model
- Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters
- Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster
- Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster (only support with Distributed VXLAN Connectivity Model for Version 9.1.1)

Virtual Network Appliance Cluster Models for Dual Availability Zones

- Dual Availability Zone Based on Virtual Network Appliance Node HA Model
- Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model

Virtual Network Appliance Cluster Models for Single Availability Zone

Virtual Network Appliance cluster designs deployed in a single availability zone have their infrastructure components typically installed in close proximity to each other. For example, when using multiple racks, they are placed in the same room or adjacent rooms in the same building.

Consider the main design attributes of the available models.

Table 868: Host Fault Tolerant Virtual Network Appliance Cluster Model Attributes

Attribute	Detail
Support Workload Connectivity Models	• Distributed VLAN Connectivity Model • Distributed VXLAN Connectivity Model for Version 9.1.1
Compute Availability Model	Host level only
Storage Availability Model	Single shared storage
Physical Network Fabric Supportability	Layer 2 adjacency is required between the ToR switches within the rack
Physical vSphere Cluster Layout	Single vSphere cluster deployed within a single rack.
Physical Footprint	Low
Scalability	• Adding more hosts to the existing vSphere cluster to deploy more VNA nodes when free CPU and memory resources are unavailable. • Adding more VNA nodes to existing vSphere cluster that have free CPU and memory resources.

Attribute	Detail
VNA Node Mobility	• For version before 9.1.1: Mobility is supported and required during ESX host maintenance. • Starting with version 9.1.1: Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.

Table 869: Rack Fault Tolerant Virtual Network Appliance Cluster Model Comparison

Attribute	Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters	Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster	Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster NEW 9.1.1
Support Workload Connectivity Models	• Distributed VLAN Connectivity Model • Distributed VXLAN Connectivity Model for Version 9.1.1		Distributed VXLAN Connectivity Model for Version 9.1.1
Compute Availability Model	Rack and host level		
Storage Availability Model	Independent storage for each rack when using vSAN.	Single shared storage across racks.	
Physical Network Fabric Supportability	When using the Distributed VLAN Connectivity Model, Layer 2 adjacency for the Distributed Transit Gateway external VLAN is required across all racks where VNA nodes from the same VNA cluster reside.	Layer 2 adjacencies for VLANs spanning racks is required.	• Layer 2 adjacency is required between the ToR switches within the rack. • Layer 3 boundary between the racks is supported.
Physical vSphere Cluster Layout	Multiple vSphere clusters deployed vertically, each vSphere cluster in separate dedicated rack.	Single vSphere cluster deployed horizontal across multiple racks.	
Physical Footprint	Medium to high		
Scalability	• Adding more hosts to the existing vSphere clusters to deploy more VNA nodes when free CPU and memory resources are unavailable. • Adding more VNA nodes to existing vSphere cluster that have free CPU and memory resources.		
VNA Node Mobility	• Mobility is not supported across racks. • For versions before 9.1.1: Mobility is supported and required during ESX host maintenance within the same rack/vSphere cluster. • Starting with version 9.1.1: Mobility is not required within the same rack/vSphere cluster when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.	• For versions before 9.1.1: Mobility is supported and required during ESX host maintenance within the same rack. • Starting with version 9.1.1: Mobility is not required when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.	• Starting with version 9.1.1 with multiple ESX hosts per rack and when a compute policy type Best Effort Restart (BER) is configured: Mobility is not required. Optional during ESX host maintenance by disabling the VM/host rule. • Starting with version 9.1.1 with a single ESX host per rack: A compute policy type Best Effort Restart (BER) is required; mobility is not supported.

Note:

A Layer 2 vSphere cluster refers to a vSphere cluster implementation where the host relevant host networks, including vMotion, ESX management, and storage traffic, reside within a single Layer 2 broadcast domain.

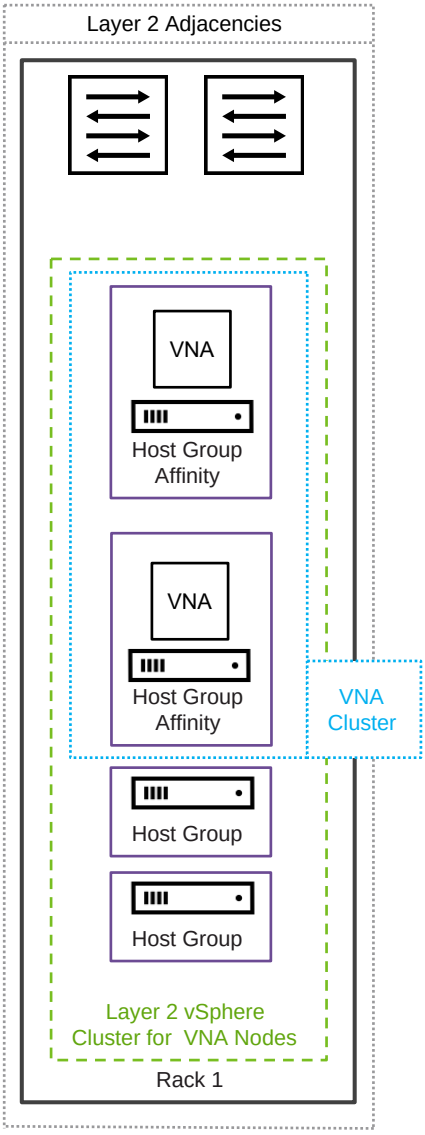
Note:

A Layer 3 vSphere cluster refers to a vSphere cluster implementation where for each individual ESX host or a sub-set of ESX hosts from a vSphere cluster, the relevant host networks, including vMotion, ESX management, and storage traffic, are assigned to individual Layer 2 broadcast domain, separated through Layer 3 boundaries.

Host Fault Tolerant Virtual Network Appliance Cluster Model

A Single-Rack vSphere Cluster Model for hosting a *Virtual Network Appliance* cluster offers a minimalist footprint while maintaining basic host-level availability.

Figure 363: Host Fault Tolerant Virtual Network Appliance Cluster Model in VMware Cloud Foundation



Host Fault Tolerant Virtual Network Appliance Cluster Model Attributes

The *Host Fault Tolerant Virtual Network Appliance Cluster Model* has the following attributes.

Attribute	Details
Availability Model	• Host level redundancy only. • Single shared storage. • This design cannot ensure continuous stateful network service availability for redirected traffic from the distributed VPC gateway to the VNA cluster during a complete rack failure.
VNA Node Mobility	• For versions before 9.1.1: Mobility is supported and required during ESX host maintenance within the same rack/vSphere cluster. • Starting with version 9.1.1: Mobility is not required within the same rack/vSphere cluster, when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
VNA Node and Link Failover Mechanisms	• Fast VNA node High Availability detection for VNA node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for data plane failures.
Supported Distributed Connectivity Models	• Distributed VLAN Connectivity Model • Distributed VXLAN Connectivity Model for Version 9.1.1

Host Fault Tolerant Virtual Network Appliance Cluster Model Design Requirements and Recommendations

Table 870: Common Virtual Network Appliance Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-COMSAZ-001	Deploy two (2) or more VNA nodes belonging to the same VNA cluster.	• VNA nodes are required to support stateful network services within a distributed VPC environment. • At least two (2) VNA nodes are required to support VNA High Availability.	• vSphere cluster resources must be available to support the VNA nodes. • Additional VNA nodes may be required to meet increased stateful network service demands.

Table 871: Host Fault Tolerant Virtual Network Appliance Cluster Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-HFTSR-001	When a compute policy type Best Effort Restart (BER) is not configured, apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VNA nodes. Otherwise, proceed with configuring a compute policy of type Best Effort Restart (BER).	Prevents the VNA nodes from running on the same ESX host and risking the High Availability of the VNA cluster.	A minimum of (N+1) ESX hosts is required in the vSphere cluster when deploying (N) VNA nodes to put the ESX host in maintenance mode.

Table 872: Common Virtual Network Appliance Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-001	Use the default memory reservation with 100% for VNA nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a VNA node, ensuring this full memory amount remains available to the VNA node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-VNA-RCMD-COMSAZ-002	Use the default CPU reservation priority in NSX set to high shares for VNA nodes.	Setting CPU shares to "High" ensures that VNA nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-VNA-RCMD-COMSAZ-003	Create a vSphere Distributed Port Group for the VNA node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected VNA cluster model, a VNA node management vSphere Distributed Port Group is required for each vSphere cluster, or rack.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the VNA node management interface exists — this is the same vSphere Distributed Port Group used by the management stack.
VCF-VNA-RCMD-COMSAZ-004	Select the VNA node size large or x-large.	• The size of an VNA node primarily depends on the stateful network service that is deployed and the volume of redirected traffic that is processed by the VNA node. • For most deployments, particularly when using stateful NAT services, the large node size provides an optimal balance between performance and resource utilization.	You must provide sufficient compute resources to support the chosen VNA node size.
VCF-VNA-RCMD-COMSAZ-005	Select static IP allocation for the VNA node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-006	Use the central node profile configuration for VNA nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all VNA nodes.	• VNA nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual VNA node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Table 873: Host Fault Tolerant Virtual Network Appliance Cluster Model Design Recommendations

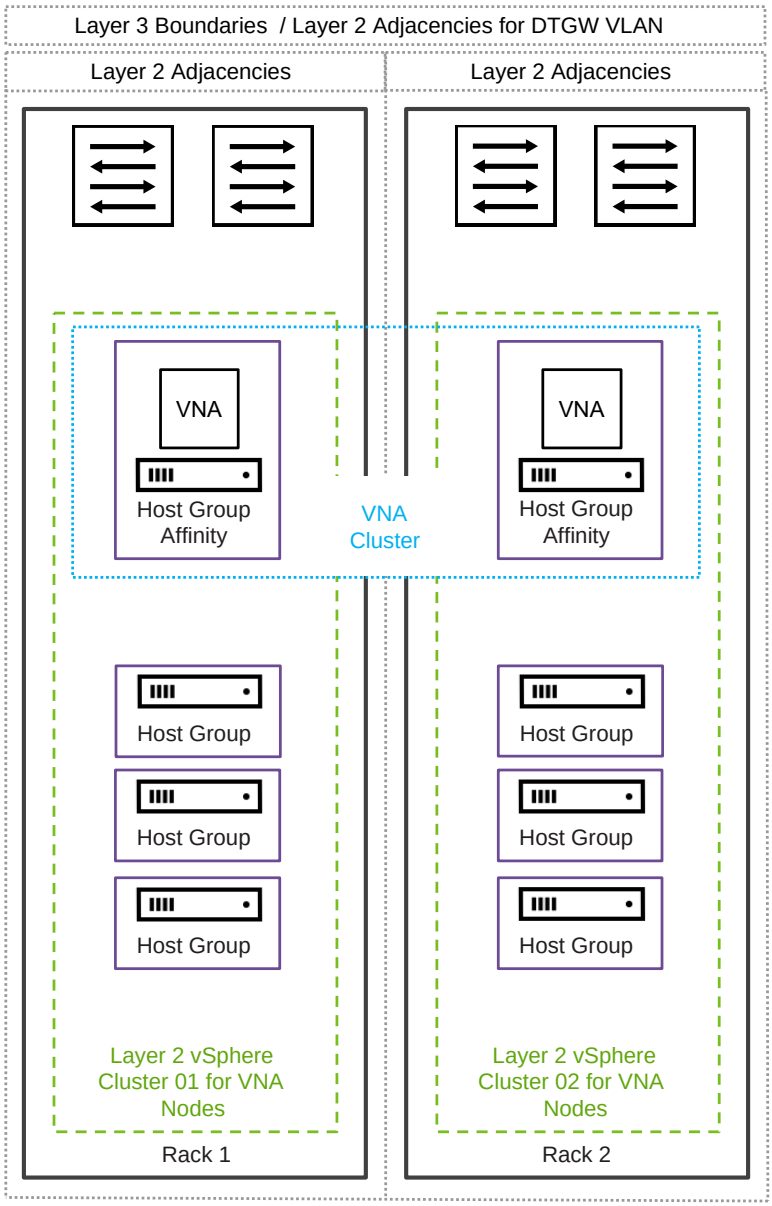
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-HFTSR-001	When configuring a compute policy of type Best Effort Restart (BER), create for every ESX host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or VNA cluster workflow.
VCF-VNA-RCMD-HFTSR-002	When configuring a compute policy of type Best Effort Restart (BER), create for every VNA node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the VNA cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-VNA-RCMD-HFTSR-003	When configuring a compute policy of type Best Effort Restart (BER), assign to each individual VNA node the previously configured host group.	Each VNA node has a 1:1 affinity with the underlying ESX host.	• The Network Connectivity wizard or the VNA cluster workflow automatically assign to each individual VNA node the previously configured host group. • If the assignment was not set through the Network Connectivity wizard or VNA cluster workflow, it can be added post-deployment.
VCF-VNA-RCMD-HFTSR-004	When configuring a compute policy of type Best Effort Restart (BER), create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the VNA node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each VNA node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or VNA cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-VNA-RCMD-HFTSR-005	When configuring a compute policy of type Best Effort Restart (BER), create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each VNA node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the VNA nodes together.	The Network Connectivity wizard or VNA cluster workflow automatically configures the vCenter tag with a category and assign to each VNA node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-HFTSR-006	When configuring a compute policy of type Best Effort Restart (BER), create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful VNA node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer VNA node readiness during host upgrades through multiple pre-checks before initiating VNA node failover and ensures the VNA node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or VNA cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the VNA node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the VNA node rather than performing vMotion. • User-initiated ESX host maintenance mode shuts down the hosted VNA node regardless of whether a healthy peer VNA node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters

Using two or more instances of the Single-Rack vSphere Cluster Model to host *Virtual Network Appliance* nodes from the same *Virtual Network Appliance* cluster provides both host-level and rack-level availability, with each rack maintaining independent storage when using vSAN. Each vSphere cluster's infrastructure is contained entirely within a single rack, with one exception: When using the Distributed VLAN Connectivity Model, the required external VLAN for the Distributed Transit Gateway (DTGW) must be stretched across the two or more instances of the Single-Rack vSphere Cluster Model, where the *Virtual Network Appliance* nodes from the same *Virtual Network Appliance* cluster are running.

Figure 364: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters in VMware Cloud Foundation



Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters Attributes

The *Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters* has the following attributes.

Attribute	Details
Availability Model	• Host level and rack-level redundancy. • Independent shared storage per vSphere cluster when using vSAN. • This design ensures continuous stateful network service availability for redirected traffic from the distributed VPC gateway to the VNA cluster during a complete rack failure.
VNA Node Mobility	• Mobility is not supported across racks. • For versions before 9.1.1: Mobility is supported and required during ESX host maintenance within the same rack/vSphere cluster. • Starting with version 9.1.1: Mobility is not required within the same rack/vSphere cluster when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
VNA Node and Link Failover Mechanisms	• Fast VNA node High Availability detection for VNA node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for data plane failures.
Supported Distributed Connectivity Models	• Distributed VLAN Connectivity Model • Distributed VXLAN Connectivity Model for Version 9.1.1

Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters Design Requirements and Recommendations

Table 874: Common Virtual Network Appliance Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-COMSAZ-001	Deploy two (2) or more VNA nodes belonging to the same VNA cluster.	• VNA nodes are required to support stateful network services within a distributed VPC environment. • At least two (2) VNA nodes are required to support VNA High Availability.	• vSphere cluster resources must be available to support the VNA nodes. • Additional VNA nodes may be required to meet increased stateful network service demands.

Table 875: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-RFTMC-001	When a compute policy type Best Effort Restart (BER) is not configured, apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VNA nodes when more than one VNA node is running on a single-rack vSphere cluster. Otherwise, proceed with configuring a compute policy of type Best Effort Restart (BER).	Prevents the VNA nodes from running on the same ESX host and risking the High Availability of the VNA cluster.	A minimum of (N+1) ESX hosts is required in the vSphere cluster when deploying (N) VNA nodes to put the host in maintenance mode.

Table 876: Common Virtual Network Appliance Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-001	Use the default memory reservation with 100% for VNA nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a VNA node, ensuring this full memory amount remains available to the VNA node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-VNA-RCMD-COMSAZ-002	Use the default CPU reservation priority in NSX set to high shares for VNA nodes.	Setting CPU shares to "High" ensures that VNA nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-VNA-RCMD-COMSAZ-003	Create a vSphere Distributed Port Group for the VNA node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected VNA cluster model, a VNA node management vSphere Distributed Port Group is required for each vSphere cluster, or rack.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the VNA node management interface exists — this is the same vSphere Distributed Port Group used by the management stack.
VCF-VNA-RCMD-COMSAZ-004	Select the VNA node size large or x-large.	• The size of an VNA node primarily depends on the stateful network service that is deployed and the volume of redirected traffic that is processed by the VNA node. • For most deployments, particularly when using stateful NAT services, the large node size provides an optimal balance between performance and resource utilization.	You must provide sufficient compute resources to support the chosen VNA node size.
VCF-VNA-RCMD-COMSAZ-005	Select static IP allocation for the VNA node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-006	Use the central node profile configuration for VNA nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all VNA nodes.	• VNA nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual VNA node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Table 877: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multiple Single-Rack vSphere Clusters Design Recommendations

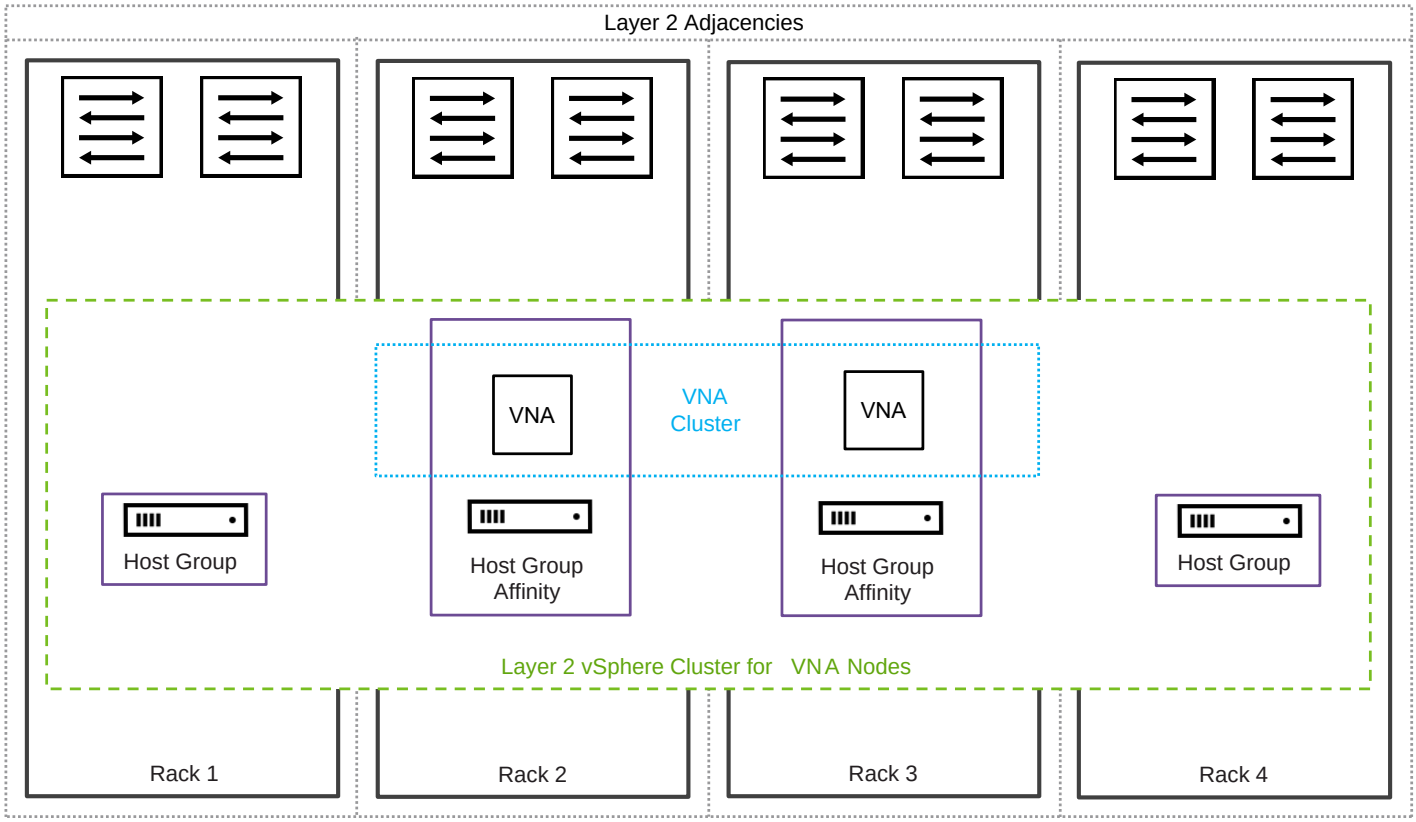
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-RFTMC-001	Ensure that VNA nodes belonging to the same VNA cluster are evenly deployed across different vSphere clusters.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-VNA-RCMD-RFTMC-002	Configure an NSX failure domain for each separate vSphere cluster where the VNA node can run - this is only valid when you have more than two VNA nodes; map the VNA to the NSX failure domains appropriately.	• NSX failure domains guarantee stateful network service availability if there is a failure affecting multiple VNA nodes. • Active and standby instance of a VNA nodes always runs in different NSX failure domains.	Additional manual configuration.
VCF-VNA-RCMD-RFTMC-003	When configuring a compute policy of type Best Effort Restart (BER), create for every ESX host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or VNA cluster workflow.
VCF-VNA-RCMD-RFTMC-004	When configuring a compute policy of type Best Effort Restart (BER), create for every VNA node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the VNA cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-VNA-RCMD-RFTMC-005	When configuring a compute policy of type Best Effort Restart (BER), assign to each individual VNA node the previously configured host group.	Each VNA node has a 1:1 affinity with the underlying ESX host.	• The Network Connectivity wizard or the VNA cluster workflow automatically assign to each individual VNA node the previously configured host group. • If the assignment was not set through the Network Connectivity wizard or VNA cluster workflow, it can be added post-deployment.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-RFTMC-006	When configuring a compute policy of type Best Effort Restart (BER), create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the VNA node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each VNA node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or VNA cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-VNA-RCMD-RFTMC-007	When configuring a compute policy of type Best Effort Restart (BER), create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each VNA node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the VNA nodes together.	The Network Connectivity wizard or VNA cluster workflow automatically configures the vCenter tag with a category and assign to each VNA node when the host group affinity option is selected. Do not create the vCenter tag and category manually.
VCF-VNA-RCMD-RFTMC-008	When configuring a compute policy of type Best Effort Restart (BER), create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful VNA node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer VNA node readiness during host upgrades through multiple pre-checks before initiating VNA node failover and ensures the VNA node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or VNA cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the VNA node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the VNA node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted VNA node regardless of whether a healthy peer VNA node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster

A single horizontally deployed vSphere cluster for hosting *Virtual Network Appliance* nodes provides host-level and rack-level availability but without providing independent storage for each rack. The vSphere cluster is deployed with one or more hosts per rack across two or more racks, unlike the other designs with separated vertical vSphere clusters where each vSphere cluster was contained within a single rack. This design distributes ESX hosts from the same vSphere cluster within the same Layer 2 domain between racks rather than across Layer 3 boundaries. The physical network fabric must support Layer 2 adjacencies (e.g., through VXLAN encapsulation in the physical network fabric) to enable connectivity between ESX hosts across different racks. *Virtual Network Appliance* nodes mobility across racks is not required when a compute policy type Best Effort Restart (BER) is configured.

Figure 365: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster in VMware Cloud Foundation



Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster Attributes

The *Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster* has the following attributes.

Attribute	Details
Availability Model	• Host and rack-level redundancy. • Single shared storage. • This design ensures continuous stateful network service availability for redirected traffic from the distributed VPC gateway to the VNA cluster during a complete rack failure.
VNA Node Mobility	• For versions before 9.1.1: Mobility is supported and required during ESX host maintenance within the same rack and across racks. • Starting with version 9.1.1: Mobility is not required within the same rack and across racks, when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
VNA Node and Link Failover Mechanisms	• Fast VNA node High Availability detection for VNA node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for data plane failures.
Supported Distributed Connectivity Models	• Distributed VLAN Connectivity Model • Distributed VXLAN Connectivity Model for Version 9.1.1

Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster Design Requirements and Recommendations

Table 878: Common Virtual Network Appliance Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-COMSAZ-001	Deploy two (2) or more VNA nodes belonging to the same VNA cluster.	• VNA nodes are required to support stateful network services within a distributed VPC environment. • At least two (2) VNA nodes are required to support VNA High Availability.	• vSphere cluster resources must be available to support the VNA nodes. • Additional VNA nodes may be required to meet increased stateful network service demands.

Table 879: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-RFTL2-001	• When a compute policy type Best Effort Restart (BER) is not configured, apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VNA nodes. Otherwise, proceed with configuring a compute policy of type Best Effort Restart (BER). • When a compute policy type Best Effort Restart (BER) is not configured and when more than one ESX host is installed per rack, apply an additional vSphere Distributed Resource Scheduler (DRS) "should" VM/host rule to prevent multiple VNAnodes from running on the same rack.	Prevents the VNA nodes from running on the same ESX host and risking the High Availability of the VNA cluster.	A minimum of (N+1) ESX hosts is required in the vSphere cluster when deploying (N) VNA nodes to put the host in maintenance mode.

Table 880: Common Virtual Network Appliance Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-001	Use the default memory reservation with 100% for VNA nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a VNA node, ensuring this full memory amount remains available to the VNA node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-VNA-RCMD-COMSAZ-002	Use the default CPU reservation priority in NSX set to high shares for VNA nodes.	Setting CPU shares to "High" ensures that VNA nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-003	Create a vSphere Distributed Port Group for the VNA node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected VNA cluster model, a VNA node management vSphere Distributed Port Group is required for each vSphere cluster, or rack.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the VNA node management interface exists — this is the same vSphere Distributed Port Group used by the management stack.
VCF-VNA-RCMD-COMSAZ-004	Select the VNA node size large or x-large.	• The size of an VNA node primarily depends on the stateful network service that is deployed and the volume of redirected traffic that is processed by the VNA node. • For most deployments, particularly when using stateful NAT services, the large node size provides an optimal balance between performance and resource utilization.	You must provide sufficient compute resources to support the chosen VNA node size.
VCF-VNA-RCMD-COMSAZ-005	Select static IP allocation for the VNA node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-VNA-RCMD-COMSAZ-006	Use the central node profile configuration for VNA nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all VNA nodes.	• VNA nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual VNA node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Table 881: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 2 vSphere Cluster Design Recommendations

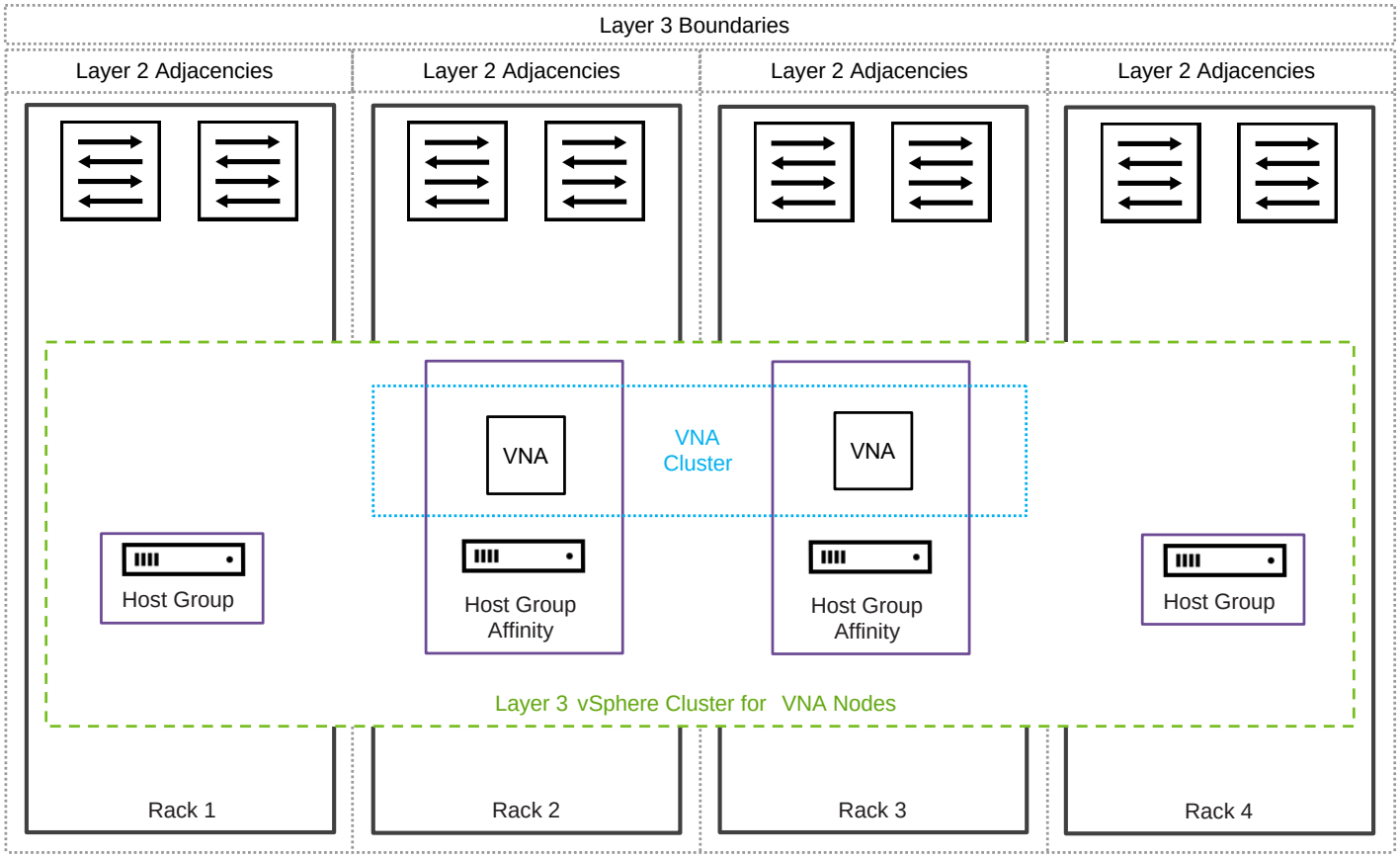
Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-RFTL2-001	Ensure that VNA nodes belonging to the same VNA cluster are evenly deployed across different racks.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-VNA-RCMD-RFTL2-002	Configure an NSX failure domain for each rack where the VNA node can run - this is only valid when you have more than two VNA nodes; map the VNA nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee stateful network service availability if there is a failure affecting multiple VNA nodes. • Active and standby instance of a VNA node always runs in different NSX failure domains.	Additional manual configuration.
VCF-VNA-RCMD-RFTL2-003	When configuring a compute policy of type Best Effort Restart (BER), create for every ESX host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or VNA cluster workflow.
VCF-VNA-RCMD-RFTL2-004	When configuring a compute policy of type Best Effort Restart (BER), create for every VNA node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the VNA cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-VNA-RCMD-RFTL2-005	When configuring a compute policy of type Best Effort Restart (BER), assign to each individual VNA node the previously configured host group.	Each VNA node has a 1:1 affinity with the underlying ESX host.	• The Network Connectivity wizard or the VNA cluster workflow automatically assign to each individual VNA node the previously configured host group. • If the assignment was not set through the Network Connectivity wizard or VNA cluster workflow, it can be added post-deployment.
VCF-VNA-RCMD-RFTL2-006	When configuring a compute policy of type Best Effort Restart (BER), create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the VNA node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each VNA node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or VNA cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-VNA-RCMD-RFTL2-007	When configuring a compute policy of type Best Effort Restart (BER), create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each VNA node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the VNA nodes together.	The Network Connectivity wizard or VNA cluster workflow automatically configures the vCenter tag with a category and assign to each VNA node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-RFTL2-008	When configuring a compute policy of type Best Effort Restart (BER), create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful VNA node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer VNA node readiness during host upgrades through multiple pre-checks before initiating VNA node failover and ensures the VNA node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or VNA cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the VNA node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the VNA node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted VNA node regardless of whether a healthy peer VNA node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster

A single horizontally deployed vSphere cluster for hosting *Virtual Network Appliance* nodes provides host-level and rack-level availability but without providing independent storage for each rack. The vSphere cluster is deployed with one or more hosts per rack across two or more racks, unlike the other designs with separated vertical vSphere clusters where each vSphere cluster was contained within a single rack. This design distributes ESX hosts from the same vSphere cluster across multiple racks, with Layer 3 network boundaries providing separation between each rack. This design has no dependencies on specific Layer 2 capabilities (such as VXLAN) of the physical network fabric and works with a simple Layer 3 physical network fabric. The Distributed VXLAN Connectivity Model for Version 9.1.1 is the only supported model.

Figure 366: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3vSphere Cluster in VMware Cloud Foundation



Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster Attributes

The *Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster* has the following attributes.

Attribute	Details
Availability Model	• Host and rack-level redundancy. • Single shared storage. • This design ensures continuous stateful network service availability for redirected traffic from the distributed VPC gateway to the VNA cluster during a complete rack failure.
VNA Node Mobility	• Mobility is not supported across racks. • The Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster is not supported for versions before 9.1.1. • Starting with version 9.1.1 with multiple ESX hosts per rack and when a compute policy type Best Effort Restart (BER) is configured: Mobility is not required. Optional during ESX host maintenance by disabling the VM/host rule. • Starting with version 9.1.1 with a single ESX host per rack: A compute policy type Best Effort Restart (BER) is required; mobility is not supported.
VNA Node and Link Failover Mechanisms	• Fast VNA node High Availability detection for VNA node failures. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for data plane failures.
Supported Distributed Connectivity Models	Distributed VXLAN Connectivity Model for Version 9.1.1

Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster Design Requirements and Recommendations

Table 882: Common Virtual Network Appliance Cluster for Single Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-COMSAZ-001	Deploy two (2) or more VNA nodes belonging to the same VNA cluster.	• VNA nodes are required to support stateful network services within a distributed VPC environment. • At least two (2) VNA nodes are required to support VNA High Availability.	• vSphere cluster resources must be available to support the VNA nodes. • Additional VNA nodes may be required to meet increased stateful network service demands.

Table 883: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster Design Requirements

Only the Distributed VXLAN Connectivity Model for Version 9.1.1 supports the Multi-Rack Layer 3 vSphere Cluster Model, all other Distributed Connectivity Models dont support this model.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-RFTL3-001	Create for every host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or VNA cluster workflow.
VCF-VNA-REQD-RFTL3-002	Create for every VNA node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The VNA cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.
VCF-VNA-REQD-RFTL3-003	Assign to each individual VNA node the previously configured host group.	Each VNA node has a 1:1 affinity with the underlying ESX host.	• The VNA cluster workflow automatically assign to each individual VNA node the previously configured host group. • If the assignment was not set through the Network Connectivity wizard or VNA cluster workflow, it can be added post-deployment.
VCF-VNA-REQD-RFTL3-004	Create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the VNA node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each VNA node are required to enable the compute policy type Best Effort Restart (BER).	The VNA cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-VNA-REQD-RFTL3-005	Create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each VNA node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the VNA nodes together.	The VNA cluster workflow automatically configures the vCenter tag with a category and assign to each VNA node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-RFTL3-006	Create a compute policy type Best Effort Restart (BER) and assign the previously created vSphere tag and category to it.	• Compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful VNA node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer VNA node readiness during host upgrades through multiple pre-checks before initiating VNA node failover and ensures the VNA node shutdown process begins gracefully only after successful pre-checks.	• The VNA cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the VNA node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the VNA node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted VNA node regardless of whether a healthy peer VNA node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Table 884: Common Virtual Network Appliance Cluster for Single Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-001	Use the default memory reservation with 100% for VNA nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a VNA node, ensuring this full memory amount remains available to the VNA node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-VNA-RCMD-COMSAZ-002	Use the default CPU reservation priority in NSX set to high shares for VNA nodes.	Setting CPU shares to "High" ensures that VNA nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-VNA-RCMD-COMSAZ-003	Create a vSphere Distributed Port Group for the VNA node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected VNA cluster model, a VNA node management vSphere Distributed Port Group is required for each vSphere cluster, or rack.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the VNA node management interface exists — this is the same vSphere Distributed Port Group used by the management stack.
VCF-VNA-RCMD-COMSAZ-004	Select the VNA node size large or x-large.	• The size of an VNA node primarily depends on the stateful network service that is deployed and the volume of redirected traffic that is processed by the VNA node. • For most deployments, particularly when using stateful NAT services, the large node size provides an optimal balance between performance and resource utilization.	You must provide sufficient compute resources to support the chosen VNA node size.
VCF-VNA-RCMD-COMSAZ-005	Select static IP allocation for the VNA node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMSAZ-006	Use the central node profile configuration for VNA nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all VNA nodes.	• VNA nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual VNA node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

Table 885: Rack Fault Tolerant Virtual Network Appliance Cluster Model using Multi-Rack Layer 3 vSphere Cluster Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-RFTL3-001	Ensure that VNA nodes belonging to the same VNA cluster are evenly deployed across different racks.	Maintain High Availability and prevent single points of failure.	Additional servers are required.

Virtual Network Appliance Cluster Models for Dual Availability Zones

Virtual Network Appliance deployments across two availability zones require careful consideration of network characteristics, even though these zones are located in close proximity to each other. The latency (RTT) requirements vary based on the underlying vSphere cluster topology: a maximum of 5ms latency (RTT) is required when deploying on vSAN stretched clusters, while up to 10ms latency (RTT) is acceptable when running on independent vSphere clusters in each availability zone.

Virtual Network Appliance Cluster for Dual Availability Zones Models Comparison

Attribute	One dedicated vSphere Cluster per availability zone based on VNA node HA	Single stretched vSAN Cluster across availability zones based on VNA node HA	Single stretched vSAN cluster across availability zones based on vSphere HA recovery
Max Latency (RTT) Between Availability Zones	10 ms	5 ms	
Availability Zone Failover	Fast Availability zone failover is based on VNA node High Availability detection.		Automatic but slower, based on vSphere HA
Compute Availability Model	Availability zone and host level.		
Storage Availability Model	Independent storage for each availability zone when using vSAN.	Single shared storage across availability zones.	
Physical Network Fabric Supportability	Layer 2 adjacency for the Distributed Transit Gateway external VLAN is required across the availability zones where VNA nodes from the same VNA cluster reside.		Layer 2 adjacencies for VLANs spanning racks are required.
Physical vSphere Cluster Layout	Two vSphere clusters each contained within an availability zone.	Single stretched vSphere cluster across the two availability zones.	
Physical Footprint	Medium to high		
Scalability	• Adding more ESX hosts to the existing vSphere cluster to deploy more VNA nodes when insufficient free CPU and memory resources are available. • Adding more VNA nodes to existing vSphere cluster that have free CPU and memory resources.		

Attribute	One dedicated vSphere Cluster per availability zone based on VNA node HA	Single stretched vSAN Cluster across availability zones based on VNA node HA	Single stretched vSAN cluster across availability zones based on vSphere HA recovery
VNA Node Mobility	• Mobility is not supported across availability zones. • For versions before 9.1.1: Mobility is required within a single availability zone. • Starting with version 9.1.1: Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.	• Mobility is not required across availability zones. • For versions before 9.1.1: Mobility is required within a single availability zone. • Starting with version 9.1.1: Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.	• Mobility is required within a single availability zone. • Mobility is required across availability zones for VNA nodes to fail over and restart at the secondary availability zone.

Dual Availability Zone Based on Virtual Network Appliance Node HA Model

Virtual Network Appliance nodes can be hosted in two ways: either using multiple instances of the Single-Rack vSphere Cluster Model, or using a single Stretched vSphere Cluster Model across two availability zones with half of the *Virtual Network Appliance* nodes from the same *Virtual Network Appliance* cluster restricted to the primary availability zone and the other half to the secondary availability zone. Each availability zone can use a separate VLAN for *Virtual Network Appliance* management interface, improving failure isolation. When using the Distributed VLAN Connectivity Model, the required external VLAN for the Distributed Transit Gateway (DTGW) must be stretched across multiple instances of the Single-Rack vSphere Cluster Model, or across the two availability zones spanning the Stretched vSphere Cluster Model, where the *Virtual Network Appliance* nodes from the same *Virtual Network Appliance* cluster are running

Figure 367: Dual Availability Zone Based on Virtual Network Appliance Node HA Model (Dedicated vSphere Cluster) in VMware Cloud Foundation

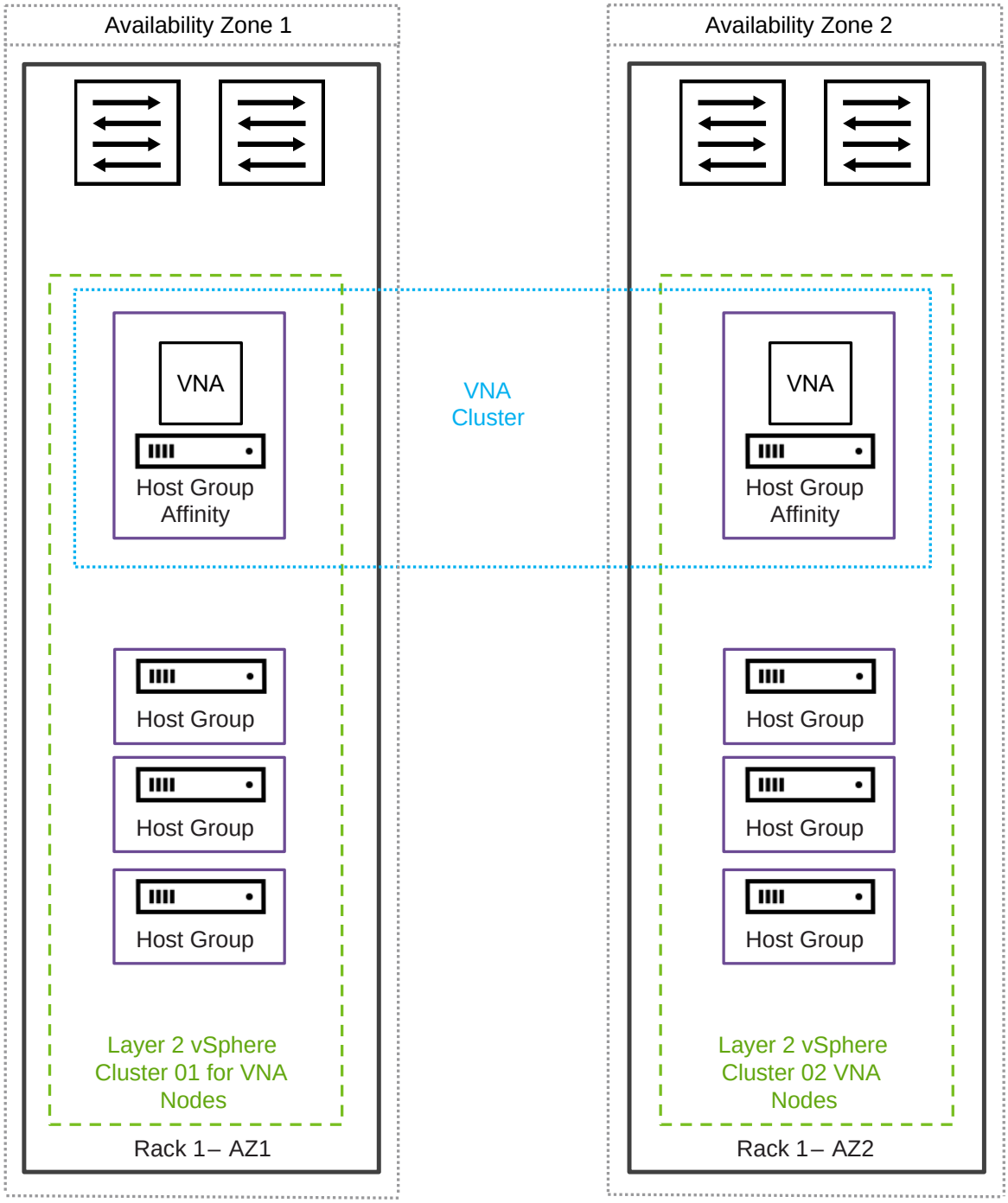
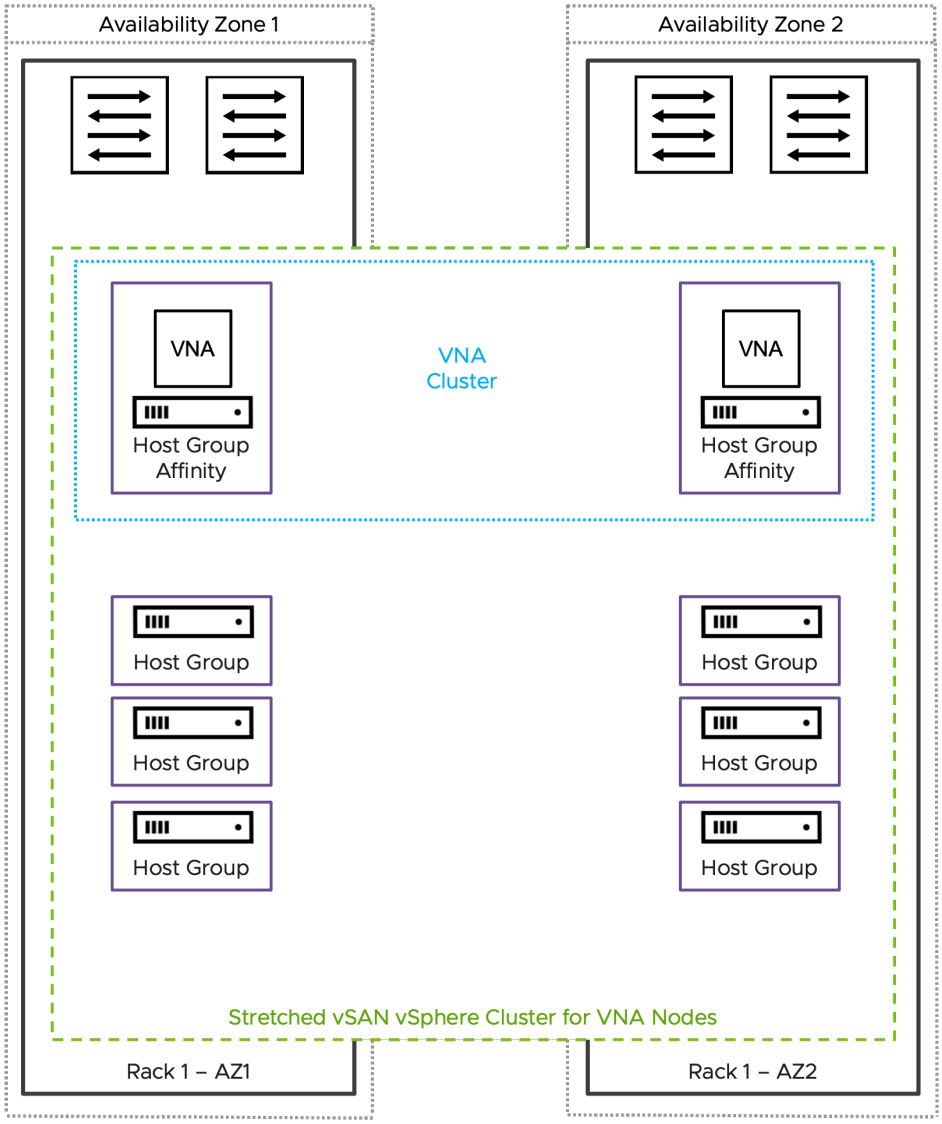


Figure 368: Dual Availability Zone Based on Virtual Network Appliance Node HA Model (Stretched vSAN) in VMware Cloud Foundation



Dual Availability Zone Based on Virtual Network Appliance Node HA Model Attributes

The *Dual Availability Zone Based on Virtual Network Appliance Node HA Model* has the following attributes.

Attribute	Details
Availability Model	• Host and availability zone redundancy. • Independent shared storage per vSphere cluster when using vSAN or single shared storage. • This design ensures continuous stateful network service availability for redirected traffic from the distributed VPC gateway to the VNA cluster even if one availability zone completely fails.
VNA Node Mobility	• Mobility is not supported across racks/availabilty zones when using multiple instances of the Single-Rack vSphere Cluster Model. • Mobility is not required across availability zones when using the Stretched vSphere Cluster Model. • For versions before 9.1.1: Mobility is required within a single availability zone. • Starting with version 9.1.1: Mobility is not required within a single availability zone when a compute policy type Best Effort Restart (BER) is configured. Optional during ESX host maintenance by disabling the VM/host rule.
VNA Node and Link Failover Mechanisms	• Fast VNA node High Availability detection for VNA node failures within and across availability zones. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for data plane failures.

Dual Availability Zone Based on Virtual Network Appliance Node HA Model Design Requirements and Recommendations

Table 886: Common Virtual Network Appliance Cluster for Dual Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-COMDAZ-001	Deploy two (2) or more VNA nodes belonging to the same VNA cluster.	• VNA nodes are required to support stateful network services within a distributed VPC environment. • At least two (2) VNA nodes are required to support VNA High Availability.	• vSphere cluster resources must be available to support the VNA nodes. • Additional VNA nodes may be required to meet increased stateful network service demands.

Table 887: Dual Availability Zone Based on Virtual Network Appliance Node HA Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-DAZVNAHA-001	When a compute policy type Best Effort Restart (BER) is not configured, apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VNA nodes when more than one VNA node is running on a vSphere cluster.	Prevents the VNA nodes from running on the same ESX host and risking the High Availability of the VNA cluster.	A minimum of (N+1) ESX hosts is required in the vSphere cluster when deploying (N) VNA nodes to put the host in maintenance mode.
VCF-VNA-REQD-DAZVNAHA-002	When a compute policy type Best Effort Restart (BER) is not configured and when using a stretched vSAN cluster, create a ESX host group to include all ESX hosts from the primary availability zone and another ESX host group to include all hosts from the secondary availability zone.	These ESX host groups are required to enable the VNA node affinity with the primary availability zone and respective secondary availability zone.	None.
VCF-VNA-REQD-DAZVNAHA-003	When a compute policy type Best Effort Restart (BER) is not configured and when using a stretched vSAN cluster, create a VM group to include the VNA nodes from the primary availability zone and another VM group to include the VNA nodes from the secondary availability zone.	These VM groups are required to enable the VNA node affinity with the primary availability zone and respective secondary availability zone.	None.
VCF-VNA-REQD-DAZVNAHA-004	When a compute policy type Best Effort Restart (BER) is not configured and when using a stretched vSAN cluster, create a "should" VM/host rule for the primary availability zone and another "should" VM/host rule for the secondary availability zone.	For each availability zone, create a VM/host "should" rule associating the VNA node VM group with the corresponding ESX host group to ensure deterministic placement.	None.
VCF-VNA-REQD-DAZVNAHA-005	Ensure that VNA nodes belonging to the same VNA cluster are evenly deployed across two dedicated vSphere clusters respectively across the two availability zones.	Maintain High Availability and prevent single points of failure.	Additional servers are required.
VCF-VNA-REQD-DAZVNAHA-006	When configuring a compute policy of type Best Effort Restart (BER), create for every ESX host an individual host group.	Host groups are required to enable the compute policy type Best Effort Restart (BER).	Host groups must be configured through the administrator before initiating the Network Connectivity wizard or VNA cluster workflow.
VCF-VNA-REQD-DAZVNAHA-007	When configuring a compute policy of type Best Effort Restart (BER), create for every VNA node an individual VM group.	VM groups are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or the VNA cluster workflow automatically configures VM groups when the host group affinity option is selected. Do not create this VM group manually.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-DAZVNAHA-008	When configuring a compute policy of type Best Effort Restart (BER), assign to each individual VNA node the previously configured host group.	Each VNA node has a 1:1 affinity with the underlying ESX host.	• Increased configuration effort. • If the assignment was not set through the Network Connectivity wizard or VNA cluster workflow, it can be added post-deployment.
VCF-VNA-REQD-DAZVNAHA-009	When configuring a compute policy of type Best Effort Restart (BER), create a VM/Host "must" rule by mapping the previously created host group and VM group together, ensuring the VNA node in the VM group must only run on this specific ESX host.	VM/Host "must" rule for each VNA node are required to enable the compute policy type Best Effort Restart (BER).	The Network Connectivity wizard or VNA cluster workflow automatically configures VM/Host rule when the host group affinity option is selected. Do not create this VM/host rule manually.
VCF-VNA-REQD-DAZVNAHA-010	When configuring a compute policy of type Best Effort Restart (BER), create a vCenter tag with a category to identify the compute policy type Best Effort Restart (BER) and assign it to each VNA node.	The vCenter tag with a category is required to map both the compute policy type Best Effort Restart (BER) and the VNA nodes together.	The Network Connectivity wizard or VNA cluster workflow automatically configures the vCenter tag with a category and assign to each VNA node when the host group affinity option is selected. Do not create the vCenter tag and category manually.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-DAZVNAHA-011	When configuring a compute policy of type Best Effort Restart (BER), create a compute policy type Best Effort Restart (BER) and assign the previously created vCenter tag and category to it.	• The compute policy type Best Effort Restart (BER) enables smooth rolling ESX host upgrades leveraging graceful VNA node shutdown to minimize traffic disruption. • The compute policy type Best Effort Restart (BER) verifies peer VNA node readiness during host upgrades through multiple pre-checks before initiating VNA node failover and ensures the VNA node shutdown process begins gracefully only after successful pre-checks.	• The Network Connectivity wizard or VNA cluster workflow automatically configures the compute policy type Best Effort Restart (BER) when the host group affinity option is selected. Do not create this compute policy type Best Effort Restart (BER) manually. • Evacuating the VNA node using vMotion during ESX host upgrades or maintenance is not possible when a compute policy of type Best Effort Restart (BER) is configured. The compute policy type Best Effort Restart (BER) is an override policy, not a fallback for vMotion. Even if vMotion is possible during ESX host maintenance mode operations, the configured compute policy type Best Effort Restart (BER) gives preference to shutting down the VNA node node rather than performing vMotion. • User-initiated host maintenance mode shuts down the hosted VNA node regardless of whether a healthy peer VNA node is available to take over, as readiness pre-checks are not executed during this process; it is the administrator's responsibility to ensure that service availability is maintained.

Table 888: Common Virtual Network Appliance Cluster for Dual Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMDAZ-001	Use the default memory reservation with 100% for VNA nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a VNA node, ensuring this full memory amount remains available to the VNA node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-VNA-RCMD-COMDAZ-002	Use the default CPU reservation priority in NSX set to high shares for VNA nodes.	Setting CPU shares to "High" ensures that VNA nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-VNA-RCMD-COMDAZ-003	Create a vSphere Distributed Port Group for the VNA node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected VNA cluster model, a VNA node management vSphere Distributed Port Group is required for each availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the VNA node management interface exists — this is the same vSphere Distributed Port Group used by the management stack.
VCF-VNA-RCMD-COMDAZ-004	Select the VNA node size large or x-large.	• The size of an VNA node primarily depends on the stateful network service that is deployed and the volume of redirected traffic that is processed by the VNA node. • For most deployments, particularly when using stateful NAT services, the large node size provides an optimal balance between performance and resource utilization.	You must provide sufficient compute resources to support the chosen VNA size.
VCF-VNA-RCMD-COMDAZ-005	Select static IP allocation for the VNA node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMDAZ-006	Use the central node profile configuration for VNA nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all VNA nodes.	• VNA nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual VNA node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

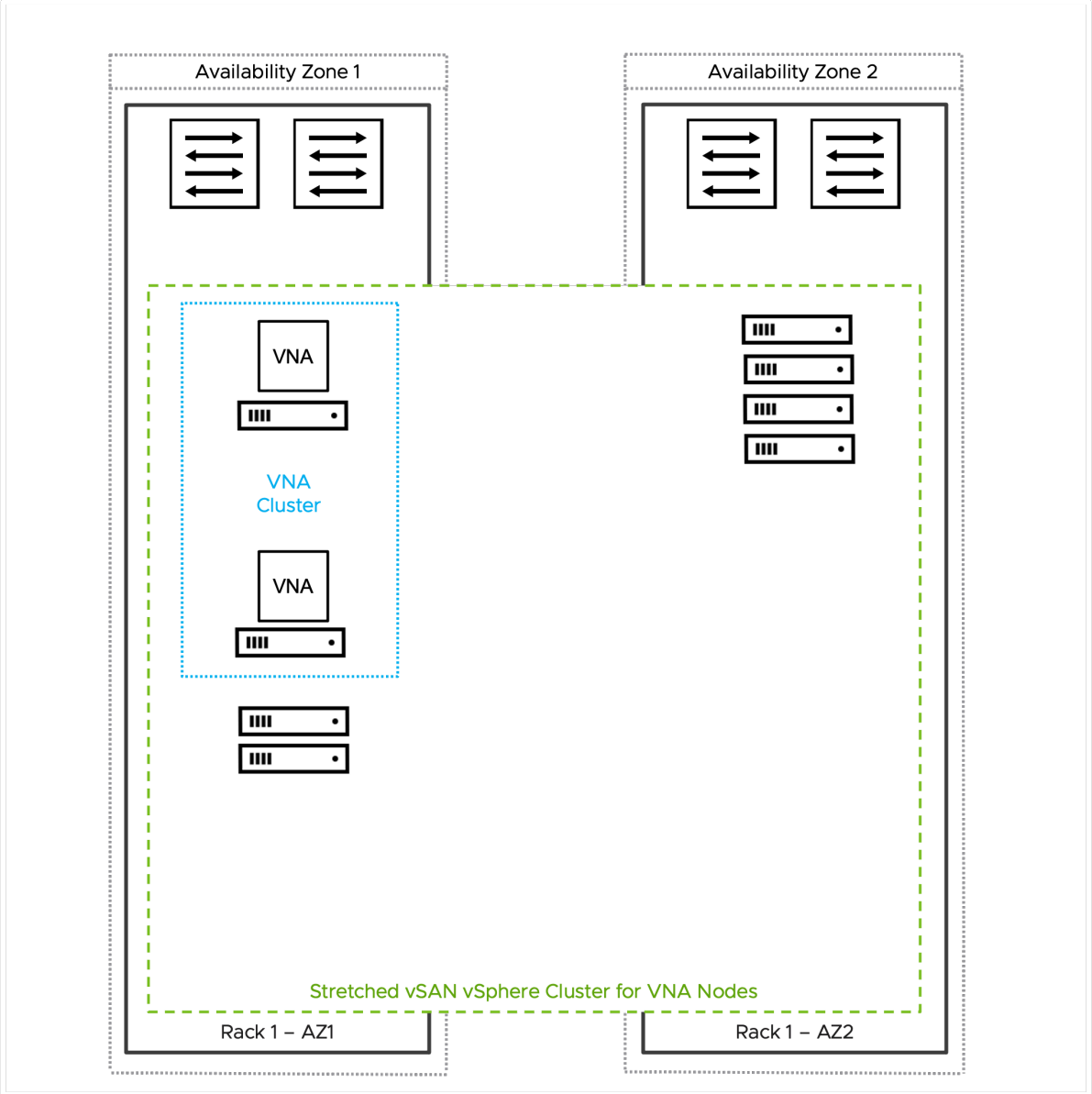
Table 889: Dual Availability Zone Based on Virtual Network Appliance Node HA Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-DAZVNAHA-001	Configure an NSX failure domain for each separate vSphere cluster respective for each availability zone where the VNA node can run - this is only valid when you have more than two VNA nodes; map the VNA nodes to the NSX failure domains appropriately.	• NSX failure domains guarantee stateful network service availability in case of a failure affecting multiple VNA nodes. • Active and standby instance of a VNA node always runs in different NSX failure domains.	Additional manual configuration.

Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model

Virtual Network Appliance nodes are hosted using a stretched vSAN cluster across two availability zones and support *Virtual Network Appliance* node mobility between zones. The *Virtual Network Appliance* node management VLAN must be stretched across both availability zones, which provides only limited fault isolation between zones. This approach is different from the previous dual availability zone design where separate VLANs improved isolation.

Figure 369: Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model (Stretched vSAN) in VMware Cloud Foundation



Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model Attributes

The*Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model* has the following attributes.

Attribute	Details
Availability Model	• Host and availability zone redundancy. • This design ensures continuous stateful network service availability for redirected traffic from the distributed VPC gateway to the VNA cluster even if one availability zone completely fails.
VNA Node Mobility	• Mobility is required within a single availability zone. • Mobility is required across availability zones for VNA nodes to fail over and restart at the secondary availability zone.

Attribute	Details
VNA Node and Link Failover Mechanisms	• Fast VNA node High Availability detection for VNA node failures. • A complete availability zone outage, the recovery depends on vSphere High Availability (vSphere HA), which may take several minutes. • Fast link failure detection with Bidirectional Forwarding Detection (BFD) for data plane failures.

Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model Design Requirements and Recommendations

Table 890: Common Virtual Network Appliance Cluster for Dual Availability Zone Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-COMDAZ-001	Deploy two (2) or more VNA nodes belonging to the same VNA cluster.	• VNA nodes are required to support stateful network services within a distributed VPC environment. • At least two (2) VNA nodes are required to support VNA High Availability.	• vSphere cluster resources must be available to support the VNA nodes. • Additional VNA nodes may be required to meet increased stateful network service demands.

Table 891: Dual Availability Zone Based on Virtual Network Appliance vSphere HA Recovery Model Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-DAZVHA-001	Apply a vSphere Distributed Resource Scheduler (DRS) anti-affinity rule to the VNA nodes.	Prevents the VNA nodes from running on the same ESX host and risking the High Availability of the VNA cluster.	A minimum of (N+1) ESX hosts is required in the vSphere cluster when deploying (N) VNA nodes to put the host in maintenance mode.
VCF-VNA-REQD-DAZVHA-002	Create a ESX host group to include all ESX hosts from the primary availability zone.	This ESX host group is required to enable the VNA node affinity with the primary availability zone.	None.
VCF-VNA-REQD-DAZVHA-003	Create a VM group to include all VNA nodes from a specific VNA cluster which should run in the primary availability zone.	This VM groups is required to enable the VNA node affinity with the primary availability zone.	None.
VCF-VNA-REQD-DAZVHA-004	Create a "should" VM/host rule for the primary availability zone.	Create a VM/host "should" rule associating the VNA node VM group with the corresponding ESX host group to ensure deterministic placement.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-VNA-REQD-DAZVHA-005	Do not enable the compute policy type Best Effort Restart (BER) for each VNA node.	• In a stretched vSAN cluster deployment supporting VNA node mobility across the two Availability Zone the compute policy type Best Effort Restart (BER) cannot be used. • VNA node mobility requires "should" VM-to-host groups to place the VNA nodes into the primary Availability Zone to support deterministic egress and ingress traffic flows, whereas Best Effort Restart (BER) compute policy requires "must" VM/host rule.	None.

Table 892: Common Virtual Network Appliance Cluster for Dual Availability Zone Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMDAZ-001	Use the default memory reservation with 100% for VNA nodes.	The 100% memory reservation configuration is a guaranteed minimum allocation of host physical memory that is entirely dedicated to a VNA node, ensuring this full memory amount remains available to the VNA node even when the host system is in an over-committed state.	In a collapsed vSphere cluster where other workload VMs have memory reservation too: • Virtual machines may fail to power on when memory reservations cannot be guaranteed, even if idle reserved memory exists. • vSphere HA may be unable to restart VMs after host failures if it cannot meet their memory reservations on surviving hosts. Memory reservation locks physical RAM, while CPU reservation guarantees minimum access to a shared resource.
VCF-VNA-RCMD-COMDAZ-002	Use the default CPU reservation priority in NSX set to high shares for VNA nodes.	Setting CPU shares to "High" ensures that VNA nodes receive higher relative priority during CPU contention scenarios, allowing critical stateful network services to maintain performance when host resources are constrained.	High CPU reservations are less critical than high memory reservations.
VCF-VNA-RCMD-COMDAZ-003	Create a vSphere Distributed Port Group for the VNA node management interface, separate from the vSphere Distributed Port Group used for ESX host management VMkernel. Depending on the selected VNA cluster model, a VNA node management vSphere Distributed Port Group is required for each availability zone.	Traffic segregation enables organizations to implement and enforce network security policies.	For the management domain, a dedicated vSphere Distributed Port Group for the VNA node management interface exists — this is the same vSphere Distributed Port Group used by the management stack.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-VNA-RCMD-COMDAZ-004	Select the VNA node size large or x-large.	• The size of an VNA node primarily depends on the stateful network service that is deployed and the volume of redirected traffic that is processed by the VNA node. • For most deployments, particularly when using stateful NAT services, the large node size provides an optimal balance between performance and resource utilization.	You must provide sufficient compute resources to support the chosen VNA size.
VCF-VNA-RCMD-COMDAZ-005	Select static IP allocation for the VNA node management interface.	Static IP allocation removes the dependency on external DHCP services.	None.
VCF-VNA-RCMD-COMDAZ-006	Use the central node profile configuration for VNA nodes to configure timezone, NTP, search domain names, DNS server(s), logging server(s) and SNMP.	Consistent node configuration across all VNA nodes.	• VNA nodes are always deployed using the central node profile configuration; however, additional configuration can be applied at the individual VNA node level — for example, via CLI. • In the event of a configuration conflict, the central node profile configuration takes precedence.

External Services Detailed Design

IP addressing scheme, name resolution, and time synchronization must support the requirements of the VMware Cloud Foundation platform.

External Services Design Requirements for VMware Cloud Foundation

Table 893: Network Time Protocol Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-NTP-001	Configure time synchronization using an internal NTP time source for all management components.	Ensures that all management components are synchronized with a valid time source.	An operational NTP service must be available in the environment.
VCF-NET-REQD-NTP-002	Set the NTP service for all management components to start automatically.	Ensures that the NTP service remains synchronized after you restart a component.	None.

Data Center Network Requirements

This section describes the options, requirements, and recommendations for the data center network requirements.

Due to the large number of hardware specifics between data centers, no one-size-fits-all approach to data center network design exists. The specifics of bandwidth requirements depend heavily on the dynamism of the workloads. Environments with high virtual machine churn, high virtual machine storage I/O, or high inter-virtual machine communication need more network resources than computationally intensive workloads. Moreover, latency-sensitive applications require special consideration even at low bandwidth.

Note:

The terms "performant" and "adequate" describe a network that is sized and configured to meet the requirements for VCF components, considering the expected workloads and traffic flows in the production environment. Network paths must be reliable and stable, as even low levels of packet loss and retransmits result in a substantial decrease in performance and throughput.

Sites, Availability Zones, and Rack Fault Tolerance

Underneath the virtual infrastructure is the physical infrastructure. Servers, storage, and network switches are all physical objects which reside in physical locations. VMware Cloud Foundation uses compute, storage, and network virtualization to abstract consumption from the physical underpinnings.

A key design aspect of a highly available deployment is to ensure that components are distributed into different physical locations to avoid faults from a common source. It is up to each customer to evaluate their own risk tolerance and fault isolation requirements.

The following areas are typical levels of physical fault tolerance that is found within an enterprise.

Sites: A separate site is typically a considerable geographic distance apart. In some instances, connectivity between this site may have higher latency (over 10ms RTT) or lower bandwidth, so synchronous data replication technologies are impractical.

Availability Zones: Availability zones are separate sites that are geographically close with low latency (under 5ms RTT) and high-bandwidth connectivity. They may be physically separate buildings or parts of an existing data center that meet the goals for physical fault tolerance. There may be a common or separate network fabric between the availability zones. However the network must be able to provide a stretch Layer 2 network between these sites to support mobility of management appliances. Availability zones should have independent network paths such that the failure of one availability zone does not impact another.

Note:

With some fabrics, the network vendor may not support BGP peering over the Layer 2 networks that are stretched between sites.

Witness Site: Clustered technologies, such as vSAN, require a third tie-breaker site to enable automated fast recovery. These witness sites may be a separate availability zone or separate site. Network paths to other sites should be independent of each other. Inter-site connections may have higher latency and lower bandwidth connections.

Racks: A rack is a physical location within a site or availability zone that contains physical servers or storage components. Each rack typically has dedicated network switches to which all rack components connect. Typically, connectivity within a rack provides the best network performance to other components within the rack with minimal or no over-subscription. For this design, all servers that are connected to the same physical network switch are considered part of the same rack regardless of where they are physically located within the data center.

Physical Data Center Network Design

This section describes the requirements and recommendations that are related to the physical characteristics of the data center network.

Fabric Options

Table 894: Data Center Rack to Network Switch and ESX Host Co-location

Network switches and ESX hosts are physical objects which are located in physical racks. Typically, ESX hosts within a physical rack are connected only to network switches associated with the physical rack.

Physical Location of hosts and connected switches.	Description	Benefits	Implications
Hosts and switches co-located	Connect ESX hosts only to network switches within the same physical rack.	• Ensures hosts and network switches share a common fate by aligning fault domains to the physical rack boundary. • ESX hosts share the same fate as their connected switches.	None.
Host and switches distributed	ESX hosts connect to network switches in other racks.	• ESX hosts may share separate fate from connected switches. • Allows for different physical resiliency for hosts vs switches.	• More complex cabling. • Fault domain for ESX hosts includes associated network switches.

Table 895: Fabric Type Comparison

Fabric Type	Example	VLAN Segment Presentation
Layer 2 Fabric	Campus-style Network Fabric (Core–Aggregation–Access)	Same across all racks **Note:** Depends on the Layer 3 / Layer 2 demarcation point.
Layer 3 Fabric	CLOS Network Fabric (Leaf-Spine/ Superspine)	Unique per rack.
Layer 3 Fabric with Hardware SDN	• Hardware SDN - Standards Based BGP-EVPN • Hardware SDN - Proprietary Fabric	Configurable per-VLAN: • Same across all racks. • Unique per rack.

Table 896: Host to Physical Switch Mapping Models

Each physical NIC in an ESX host connects to the data center fabric switch port. ESX hosts with multiple physical NICs may connect to one or more switches that are part of the data center fabric.

Number of Switches per ESX Host	Description	Benefits	Implications
Single switch per ESX host	A single ESX host with one (1) or more physical NICs are connected to a single physical switch for all network connectivity.	• Lowest cost. • Simple networking design. • Non-blocking connectivity between ESX hosts on single switch.	• No ESX host level physical network link redundancy. • Switch or link outages isolate ESX hosts. • Switch maintenance may isolate ESX host.
Multiple switches per ESX host	A single ESX host with two (2) or more physical NICs are connected to two (2) or more physical switches for network connectivity.	When properly configured, provides host-level physical network link redundancy.	• An additional switch is needed. • Must provide gateway redundancy. • Inter-switch path must be appropriately sized. • Needs to provide MAC mobility between switches, via inter-switch link or fabric overlay.

Note:

In VMware Cloud Foundation networking, switches utilized for out-of-band connectivity to the ESX host are out of scope.

Table 897: Number of Fabrics per ESX Host

Each physical NIC in an ESX host connects to a single switch. For ESX hosts with multiple NICs, switches may be part of different fabrics providing physical separation of traffic flows.

Number of Connected Fabrics per ESX Host	Description	Benefits	Implications
Single L2/L3 domain for all switches	• All ESX host physical NICs are connected to a common physical fabric. • Physical NICs may have different logical networks (VLANs), but connect to a single physical fabric. Network flows thru the fabric are mixed with all other traffic.	• Lowest cost. • Simple networking design. • Ability to share all fabric bandwidth with all traffic types. • Can utilize vSphere traffic prioritization and bandwidth management (NIOC and teaming policies) to prioritize traffic flows.	• Improperly sized fabric links can result in congestion of specific traffic types. • Relies on traffic prioritization and bandwidth management within the ESX host and fabric to manage congestion and allocated bandwidth.
Separate L2/L3 domain for switches	• ESX host physical NICs are connected to two (2) or more physical fabrics, where each fabric has (one) 1 or more physical NICs connected. • Fabrics are physically separated. • Network flows through one fabric are physically isolated from the other fabric. • Commonly used to physically isolate critical network traffic such as storage or workloads, or to separate infrastructure traffic from workload traffic.	• Provided physical isolation of some traffic types from another. • Traffic types may egress network via different paths. • Allows switches to be optimized for expected traffic patterns including port speed and features.	• Additional cost requiring additional NICs, switches and fabric. • Complex design. • VCF Networking Overlay segments and VPCs will only be available on one fabric.

Table 898: Fabric Subscription Ratio

Highly performant and low latency inter-host communication is the key to a highly performant VCF Instance. Latency within the fabric may be directly seen as latency in workloads. Minimizing over-subscription between network switches connecting ESX hosts in a VCF Instance can reduce network congestion.

Over-subscription Models	Description	Benefits	Implications
Over-subscribed Fabric	• The total aggregate bandwidth of all utilized ESX host ports is less than the total aggregate bandwidth to/through the fabric. • Traffic flows pinned to the same switch are line rate; however, ESX hosts connected to different switches will pass through congested paths in the fabric.	Lowest cost.	• Improperly sized fabric links can result in congestion of specific traffic types. • Congestion of storage or workload traffic impacts workload performance. • Pinning of traffic flow is manual and error prone.
Non-blocking within a vSphere Cluster	• All ESX hosts within a vSphere cluster have non-blocking paths to all other ESX hosts within that vSphere cluster. Paths to ESX hosts outside of the vSphere cluster may traverse oversubscribed fabric links and are subject to congestion. • Typically, all ESX hosts are connected to a single switch or a pair of switches with a high-bandwidth link between them.	Limits congestion for intra-cluster traffic such as vSAN or vMotion traffic.	• Requires additional high bandwidth paths or inter-switch links between all switches connected to ESX hosts in a vSphere cluster. • Impractical with vSphere clusters that span racks.
Non-blocking within VCF Instance	• All ESX hosts within the VCF Instance have a non-blocking path to all other ESX hosts within the vSphere cluster. • Uplinks from the top of the rack switch are sized equal to the aggregate bandwidth of all utilized ESX host switch ports.	Limits congestion through the entire VCF Instance.	• Higher cost.

Routing Models

VMware Cloud Foundation supports the following routing options for the NSX Segment Connectivity Model and Centralized Connectivity Model.

Routing Type	Description	Benefits	Implications
Static routing	• The network administrator manages the routing information, adding routing information to the routing table. • If any change occurs in the network, the administrator has to update the related information in the routing table.	• No dynamic routing protocol is required on the ToR switches. • Might reduce ToR switch license costs.	• You must manually create static routes in NSX Manager on the Tier-0 gateway. • With Active-Standby Tier-0, you must manually create a High Availability VIP in the NSX Manager on the Tier-0 gateway to provide redundancy across ToR switches. • No ECMP support, limiting throughput and redundancy.
Static routing with BFD	• The network administrator manages the routing information, adding routing information to the routing table. • If any change occurs in the network, the administrator has to update the related information in the routing table. • Use of BFD can provide resilience in the event of NSX Edge node or switch failure.	• No dynamic routing protocol is required on the ToR switches. • Might reduce ToR switch license costs. • Support Active/Active Tier-0 Gateway • Supports ECMP for scalable North-South throughput and automated failover.	• You must manually create static routes in NSX Manager on the Tier-0 gateway. • With Active-Standby Tier-0, you must manually create a High Availability VIP in the NSX Manager on the Tier-0 gateway to provide redundancy across ToR switches.
OSPF	• The routing protocol automatically adds and manages the routing information in the routing table. If any change occurs in the network, the routing protocol automatically updates the related information in the routing table. If any new segments or subnets are added in NSX, they are automatically added to the routing table.	• If the physical fabric is running the OSPF routing protocol, using OSPF at the virtual layer might be a simpler approach for the network administrator. • Supports ECMP for scalable North-South throughput and automated failover.	• You must manually configure OSPF inNSX Manager on the Tier-0 gateway. • Not supported with NSX Federation. • Combined use of BGP and OSPF on a single Tier-0 gateway not supported.

Routing Type	Description	Benefits	Implications
eBGP	• BGP is known as an exterior gateway protocol. It is designed to share routing information between disparate networks, known as autonomous systems (ASes). • When multiple BGP-derived paths exist, the protocol chooses a path to send traffic based on certain criteria. • The routing protocol automatically adds and manages the routing information in the routing table. If any new segments or subnets are added in NSX, they are automatically added to the routing table.	• Fully supported by the automated network connectivity workflow in VMware Cloud Foundation. • Fully supported for all VMware Cloud Foundation topologies. • Supports flexible traffic path manipulation through BGP attributes for optimized ingress and egress routing.	• None.

Note:

iBGP is not recommended due to full-mesh peering requirements.

Network Latency

Latency is the total round-trip time that is measured between ESX hosts and/or virtual machines listed below. Ensure that latency requirements are met even under high operational load. See VMware Ports and Protocols for more details.

Latency Between...	Maximum Latency
ESX hosts in a vSphere cluster	5ms
ESX hosts, which contain NSX Edge nodes that are part of the same NSX Edge cluster	10ms
ESX hosts in a vSAN Stretched cluster to the vSAN witness	150ms
vSphere clusters in workload domain	100ms
Workload domains in a VCF Instance	100ms
VCF Instances in a VCF fleet	300ms

Note:

High latency can affect workload performance and VCF Operations. It is important to minimize latency whenever possible.

Table 899: Network Latency Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-LAT-001	Ensure the maximum latency between ESX hosts within a vSphere Cluster is under 5 ms.	With vSAN storage, VM write latency is impacted by latency between ESX hosts.	• Clusters with connections to multiple switches must have appropriately sized paths between all switches—ISL, Fabric paths, etc. • In stretch clusters, all ESXhosts must be within 5 ms of each other. • Network links must be adequately sized to avoid latency spikes caused by congestion.
VCF-NET-REQD-LAT-002	Ensure the maximum latency between ESX hosts which contain NSX Edge nodes that are part of the same NSX Edge cluster is under 10 ms.	NSX Edge nodes use BFD to provide rapid failover of NSX services.	None.
VCF-NET-REQD-LAT-003	Ensure the maximum latency between ESX hosts in a vSAN stretched cluster to the vSAN witness is under 150 ms.	Maximum latency supported for a vSAN stretched cluster witness connectivity.	None.
VCF-NET-REQD-LAT-004	Ensure the maximum latency between vSphere clusters in a workload domain is under 100 ms.	Maximum latency supported for an NSX Manager to transport nodes and vCenter to ESX hosts.	• With remote sites, links must be adequately performant. • Remote sites must be geographically close.
VCF-NET-REQD-LAT-005	Ensure the maximum latency between workload domains in a VCF Instance is under 100 ms.	Maximum latency that is supported between a VCF Operations to ESX hosts.	None.
VCF-NET-REQD-LAT-006	Ensure the maximum latency between VCF Instances in a VCF fleet is under 300 ms.	Maximum latency supported for a VCF Operations to managed entities and NSX Federation to managed NSX Manager.	With remote sites, links must be adequately performant.

Maximum Transmission Unit / Jumbo Frames

Geneve packets between Host TEPs and Edge TEPs have the DF flag set and cannot be fragmented. Other traffic types gain a throughput increase by larger frame types.

Data Center MTU Requirements

MTU Between	Minimum MTU	Recommended MTU
ESX hosts in a vSphere cluster	1700*	9000
ESX hosts which contain NSX Edge nodes that are part of the same NSX Edge cluster	1700*	9000
ESX hosts in a vSAN stretched cluster to the vSAN witness	1500	1500
vSphere clusters in a workload domain	1700*	9000
Workload domains in a VCF Instance	1500	9000
VCF Instances in a VCF fleet	1500	1500-9000

Note:

* 1600 is the minimum required MTU, however 1700 is recommended to address future expansion of the geneve header.

Host to Host Bandwidth

Ensure adequate bandwidth between all ESX host NICs within a cluster. Two ESX hosts that are connected to the same switch typically, will not have any fabric-induced bottleneck. However, ESX hosts that are connected to different switches within the same rack or in different racks may have limited bandwidth due to the over-subscription of links between switches or through the core/spine of the fabric. Considering the actual traffic flows when determining the required bandwidth at each point in the fabric.

For example, a high over-subscription ratio between leaf and spine switches can degrade performance for data-intensive operations, such as vSAN resynchronization or vMotion. Minimize traffic across highly oversubscribed links by properly sizing switch-to-switch interconnects and localizing related traffic types to the same physical switch.

Network Link Aggregation

Link Aggregation within Fabric

Link aggregation within the fabric - between fabric switches, i.e., leaf and spine switches, or ISL between top of rack switches - is supported. Ensure that the fabric vendor supports peering of BGP across link aggregation paths from non-fabric endpoints - such as NSX Edge Node.

Link Aggregation to Host

Link aggregation to the host is supported but not recommended - the default switch independent load-based teaming (LBT) provides benefits similar to those of LACP without added configuration complexity.

The use of LACP-based dynamic link aggregation between the ESX host and the Top of Rack is supported, provided that:

1. The switch port for the first pNIC in each host is configured to enable a link and traffic to flow before receiving LACP PDU - i.e., LACP Failback mode.
2. Ensure that the fabric vendor supports peering of BGP across link aggregation paths from non-fabric endpoints - such as NSX Edge Node.

The use of other link aggregation technologies - static, EtherChannel, etc. - is not supported.

When using link aggregation, ensure the host VMkernels are correctly configured to utilize a single VMkernel per traffic type. NSX Edge nodes should utilize a single uplink design.

Table 900: Link Aggregation in Fabric Design Choices

Design Choice	Justification	Implication
Do not use link aggregation withing the switching fabric	• Utilizes standardized protocols to simplify interoperability between switches from different vendors. • Easier to scale the spine layer compared to MLAG-based fabrics • Uplinks in separate Layer 2 domains provide more robust protection from Layer 2 events, such as broadcast storms or temporary loops. • Preferred by some fabric types	None
Use of link aggregation between switching within the fabric	• Provides additional bandwidth and resilience while presenting as on logical link • Preferred by some fabric types to avoid spanning tree convergence	• If using dynamic routing protocol for NSX Tier 0, the network fabric provider must support routing adjacency to traverse path from for NSX Edge to fabric neighbor. • Requires implementation of proprietary MLAG protocols to support redundancy across devices, may be a challenge in a multi vendor fabric.

Table 901: Link Aggregation to Host Design Choices

Design Choice	Justification	Implication
Use independent uplinks without LACP link aggregation between the fabric switch and the ESX host.	• Simplest Configuration • vSphere Distributed Switch independent load-based teaming provides adequate balancing. • Can support deterministic traffic path using Active/Standby teaming. • Easier to troubleshoot deterministic traffic paths. • Additional VMkernels allow added bandwidth and resilience at the application layer. • vSphere Distributed Switches do not create a network loop when multiple pNICs are connected to the same fabric.	None.
Use LACP link aggregation between the fabric switch and the ESX host.	Aligns with specific customer fabric operations practices.	• Requires specific LACP configuration and switch setup. • Not all vSphere features are supported with LACP. See KB 324555 • If using dynamic routing protocol for NSX Tier 0, the network fabric provider must support routing adjacency to traverse path from for NSX Edge to fabric neighbor. • Non-deterministic traffic flow complicates troubleshooting.

Table 902: Link Aggregation Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-REQD-LAG-001	Ensure each connection between the fabric switch and ESX host operates as an independent uplink.	• While dynamic link aggregation with LACPv2 is supported, it adds additional considerations to the deployment and operation of the SDDC. • Use of independent links provides flexibility and deployment simplicity while providing similar resilience and performance. • Independent uplinks allow for a mix of active/standby and active/active traffic patterns by traffic type. • Independent uplinks allow infrastructure traffic VMKs to independently balance traffic patterns. • Independent uplinks allow storage traffic a deterministic traffic path. • Independent uplinks provide NSX Edge Nodes BGP Peering a predictable fabric path to the BGP Neighbor. • Independent uplinks do not require the configuration of switch-specific aggregation protocols.	None

Number of Connected Top of Rack Switches per Host

Each physical NIC from the ESX host may be connected to the same or different switches. In a properly configured fabric, connecting multiple switches provides connectivity continuity during a switch outage due to failure or maintenance.

Physical NICs from an ESX host may connect to multiple switches provided that:

1. All physical NICs associated with a single vSphere Distributed Switch are connected to the same fabric, with identical access to all pertinent VLANs and gateways.
2. Ensure that the failure of a physical switch does not result in the loss of next-hop gateway availability. The use of VRRP or similar technology is recommended.

Table 903: Top of Rack Switch per Host Choices

Design Choice	Justification	Implication
Connect all ESX hosts to a single top of rack switch per vSphere Distributed Switch. The fabric switch may be dedicated or shared for hosts with multiple vSphere Distributed Switch.	Lowest cost option.	• No redundancy. • Top of Rack switch failure or maintenance may result in isolation of the host.
Connect all ESX hosts to two or more top of rack wsitches. The fabric switch may be dedicated or shared with ESX hosts with multiple vSphere Distributed Switches.	• When properly configured, provides redundant links to the fabric. • Top of Rack switch failure or maintenance may not result in isolation of the host. • Additional fabric bandwidth.	Requires additional switches.

Table 904: Top of Rack Switch per ESX Host Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-TOR-001	Each ESX host must be connected to at least one network switch for IP connectivity.	ESX hosts require IP connectivity to other ESX hosts and management virtual machines for management, control, and data plane communication.	None.

Table 905: Top of Rack Switch per ESX Host Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-TOR-001	Connect each ESX host to two (2) or more top of rack switches. The fabric switch may be dedicated or shared for ESX hosts with multiple vSphere Distributed Switches.	Provides excellent performance and resilience.	Requires additional switches.

Number of connected Fabrics per Hosts

Each vSphere Distributed Switch may only be associated with a single fabric. When ESX hosts are configured with multiple vSphere Distributed Switches, each vSphere Distributed Switch may be associated with a different fabric. For instance, in the Storage Separation Distributed Switch Model option, the physical NICs associated with the vSphere Distributed Switch dedicated to storage traffic may be physically separated from the default data center fabric.

IP connectivity between all physical NICs, regardless of associated fabric, is required. Fabrics that are isolated from each other are not supported for infrastructure traffic (VMKs, TEPs, etc).

Table 906: Fabrics per Host Choices

Design Choice	Justification	Implication
Connect all ESX hosts to a single network fabric.	• Unified fabric is adequate to support VCF Operations and workloads. • Physical (pNICs) and logical (VLANs) provide adequate isolation between traffic types. • Where dedicated bandwidth is required, more physical NICs provide adequate separation.	Network Fabric must be correctly designed and monitored to ensure no congestion points for latency sensitive traffic such as storage.
Connect all ESX hosts to two or more network fabrics.	• Provides physical separation between traffic types. • May offer a different path out of the fabric. • Separates critical traffic types (Workloads, storage, etc) on to physically separate fabric.	Requires additional fabric.

Table 907: Fabrics per ESX Host Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-FBR-001	Each ESX host must be connected to at least one network fabric for IP connectivity.	ESX hosts require IP connectivity to other ESX hosts and management virtual machines for management, control, and dataplane communication.	None.

Table 908: Fabrics per ESX Host Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-FBR-001	Connect all ESX hosts to a single network fabric.	Unified fabric is adequate to support VCF Operations and workloads.	Network fabric must be correctly designed and monitored to ensure no congestion points for latency-sensitive traffic such as storage.

Table 909: Host Switch Port Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-SWT-001	Ensure ESX host switch ports are compatible with ESX host pNIC speed.	Required for adequate throughput from fabric to ESX hosts.	None.
VCF-NET-REQD-SWT-002	Ensure MTU of ESX host switch ports that are used for NSX host or NSX Edge TEP traffic is set to the largest MTU of any traffic type to host, 1700 minimum.	MTU may be set and different points in the data center fabric. MTU of switch port must but adequate to handle larger MTU required for NSX Edge and NSX host TEP traffic.	None.

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-SWT-003	Ensure MTU of ESX host switch ports are set to the largest MTU of any traffic type (excluding NSX host or NSX Edge TEP) to 1500 minimum.	MTU may be set and different points in the data center fabric. MTU of switch port for ESX hosts which do not contain NSX Edgenodes and are not configured as NSX transport nodes may be configured with standard frame size.	None.

Table 910: Host Switch Port Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-SWT-001	Configure switch host ports in Trunk mode using 802.1q tagging.	Enables use of VLANs for logical separation of infrastructure traffic functions.	Additional VLANs IDs are needed for each traffic function.
VCF-NET-RCMD-SWT-002	Configure the trunk ports that are connected to ESX NICs as trunk PortFast.	Reduces the time to transition ports over to the forwarding state.	None.
VCF-NET-RCMD-SWT-003	Enable a Layer 2 network device discovery protocol such as LLDP or CDP.	Simplifies troubleshooting of network connectivity issues.	May not align with organizational security guidelines.
VCF-NET-RCMD-SWT-004	Configure ESX host port MTU of 9000 or larger.	MTU of the host port must be equal to or larger than the largest traffic MTU.	None.

Network Fabric Detailed Design

The data center network fabric supports the VCF private cloud in achieving the desired availability goals. The exact model that is chosen should align with the customer availability and resilience goals.

Network Fabric Models

Three *Network Fabric Models* exist which a vSphere cluster is built upon:

- Single-Rack Network Fabric Model
- Multi-Rack Network Fabric Model
- Multi Availability Zone Network Fabric Model

Common Design Requirements

The data center network fabric provides VLANs to support the operation of the VCF workload domain. These VLANs can be grouped into a few categories of common requirements.

Table 911: VLAN Traffic Types

Traffic Type	Usage	Presented To	Notes
ESX Infrastructure Traffic	Provides connectivity for ESX Management, vMotion, vSAN, etc.	All ESX host PNICs that host VMkernels.	Used for vSAN witness traffic in stretched cluster model.
VM Management Traffic	Connectivity for management appliances. I.e vCenter, NSX Edge nodes, VCF Automation, etc.	• All ESX hosts in first management cluster. • All ESX hosts that contain NSX Edge nodes.	Optional segment that may be shared with ESX management VMkernel Segment.

Traffic Type	Usage	Presented To	Notes
NSX Edge Node Uplink Networks	• North-South data path for traffic to/from NSX overlay segments or VPC subnets. • Utilized for BGP peering from NSX Edge nodes to fabric.	All NSX Edge nodes with Tier-0 configured.	
NSX Host and NSX Edge TEPs	East-West data path for NSX overlay traffic (encapsulated in GENEVE). North-South datapath to physical path for DTGW with EVPN-VXLAN integration (excapsulated in VXLAN)	• All ESX hosts prepared for NSX overlay networking. • All NSX Edge nodes.	
NSX Edge RTEP	East-west data path for NSX cross-instance traffic when using NSX Federation.	NSX Edge nodes prepared for NSX Federation.	Optional.
Workload VM VLAN Segments	Provides connectivity for customer workloads.	ESX hosts based on customer use case need.	
External VLAN for DTGW	Provides connectivity for customer workloads within DTGW-backed VPC.	ESX hosts based on span of DTGW.	

Table 912: VLAN MTUs

Individual VLANs may have different minimum MTUs based on their traffic type. VLAN MTUs are typically equal to, or occasionally smaller than, the vSphere Distributed Switch or network fabric MTU.

Traffic Type	Minimum MTU	Recommended MTU	Notes
ESX Infrastructure Segments	1500	ESX management VMK - 1500* Other VMKs - 9000**	Used for vSAN witness traffic in Multi-AZ cluster model.
VM Management Segments	1500	1500	Optional segment that may be shared with ESX management VMK Segment.
NSX Edge Node Uplink Segments	1500	9000	
NSX Host and Edge TEPs	1700***	9000	
NSX Edge RTEP	1500	9000****	NSX RTEPs are used with NSXFederation.
Workload VM Segments	Based on customer requirements		

* Alignment of this MTU with enterprise inter-site links MTU is recommended. With vSAN Stretch Cluster clusters, the management VMkernel can be used for vSAN witness traffic, which will traverse inter-site links.
** Refer to the storage design section for storage-specific MTU recommendations.
*** To support NSX overlay traffic over Geneve, the recommended MTU is 9000, with a minimum recommended MTU of 1700. Geneve is supported with MTU as low as 1600, but that is not the recommended minimum as it does not provide adequate overhead for future Geneve header expansion.
**** With NSX Federation, the RTEP will traverse inter-site links - a larger MTU is not required, but a larger MTU will minimize frame segmentation for cross-instance traffic flows.

Table 913: MTU Choices

Requirement	Option	Description
MTU	Minimum	The minimum MTU for the VLAN segment. All intermediate devices must be able to forward traffic without fragmentation.
	Preferred	The best practices recommend MTU for the VLAN segment. All intermediate devices must be able to forward traffic without fragmentation.

Table 914: Infrastructure VLAN MTU Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-MTU-001	Ensure all VLAN segment MTU's less than or equal to the fabric MTU.	Segment MTU greater than fabric MTU may result in fragmentation during transport.	None.
VCF-NET-REQD-MTU-002	Ensure the VLAN segment and first hop gateway for ESX management VMKs is configured to 1500 or greater.	A standard ethernet frame size is sufficient.	None.
VCF-NET-REQD-MTU-003	Ensure the VLAN segment and first hop gateway for management component virtual machines is configured to 1500 or greater.	A standard ethernet frame size is sufficient.	None.
VCF-NET-REQD-MTU-004	Ensure the VLAN segment and first hop gateway for ESX VMkernels for IP-based storage or vSAN are configured to 1500 or greater.	A standard ethernet frame size is sufficient.	None.
VCF-NET-REQD-MTU-005	Ensure the VLAN segment and first hop gateway for ESX VMkernels for vMotion are configured to 1500 or greater.	A standard ethernet frame size is sufficient.	None.
VCF-NET-REQD-MTU-006	Ensure the VLAN segment and first hop gateway for ESX VMkernels for host TEP and NSX Edge Node TEP are configured to 1700 or greater.	Geneve overlay requires 1600. However, 1700 provides additional overhead for future Geneve header expansion.	Jumbo frames must be configured end-to-end, including all intermediate segments and gateways.
VCF-NET-REQD-MTU-007	Ensure the VLAN segment utilized for BGP Peering to the edge node is configured to 1500 or greater.	A standard ethernet frame size is sufficient.	None.

Table 915: Infrastructure VLAN MTU Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-MTU-001	Ensure the VLAN segment and first hop gateway for ESX management VMkernels is configured to 1500.	• A standard ethernet frame size is sufficient. • vSAN witness traffic may traverse WAN links that do not support jumbo frames.	None.
VCF-NET-RCMD-MTU-002	Ensure the VLAN segment and first hop gateway for management component VMs is configured to 9000 or greater.	Jumbo frames provide additional throughput and reduced overhead where large data transfers may occur.	Jumbo frames must be configured end-to-end, including all intermediate segments and gateways.
VCF-NET-RCMD-MTU-003	Ensure the VLAN segment and first hop gateway for ESX VMkernels for IP-based Storage or vSAN are configured to 9000 or greater.	Jumbo frames provide additional throughput and reduced overhead where large data transfers may occur.	Jumbo frames must be configured end-to-end, including all intermediate segments and gateways.
VCF-NET-RCMD-MTU-004	Ensure the VLAN segment and first hop gateway for ESX VMkernels for vMotion are configured to 9000 or greater.	Jumbo frames provide additional throughput and reduced overhead where large data transfers may occur.	Jumbo frames must be configured end-to-end, including all intermediate segments and gateways.

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-MTU-005	Ensure the VLAN segment and first hop gateway for ESX VMkernels for Host TEP and Edge Node TEP are configured to 9000 or greater.	Jumbo frames enable larger overlay network segment sizes. Workload Overlay Segments must be 200 bytes smaller than the MTU.	Jumbo frames must be configured end-to-end, including all intermediate segments and gateways.
VCF-NET-RCMD-MTU-006	Ensure the VLAN segment that is utilized for NSX Edge node Uplinks and BGP Peering to the edge node is configured to 9000 or greater.	• NSX Edge node Uplink segments carry workload overlay segment traffic to the data center fabric. • BGP peering occurs over NSX Edge node Uplink segments. Jumbo frames provide additional throughput and reduced overhead where large BGP peer updates may occur.	Jumbo frames must be configured end-to-end, including all intermediate segments and gateways.

NSX overlay segments within an NSX VPC or not, must account for the Geneve encapsulation overhead. The maximum overlay segment MTU equals the Host/Edge TEP MTU minus 200 bytes. i.e., if the Host/Edge TEP MTU is 9000, the maximum overlay segment MTU is 8800.

Table 916: VCF Overlay Segment Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-OVL-001	Ensure that overlay segments utilize an MTU which is 200 less than the ESX host TEP MTU.	• Geneve packets are flagged DF (Do not fragment) and will be dropped by network devices if over MTU. • It is recommended that 200 bytes be reserved for the Gevene IP Header.	Jumbo frames must be configured end-to-end, including all intermediate segments and gateways.

IP Connectivity

All components of the VMware Cloud Foundation platform require IP connectivity to the other components in the platform. In environments with multiple fabrics per cluster, routing for IP connectivity is required.

Table 917: IP Connectivity Choices

Design Choice	Justification	Implication
Provide Internet connectivity to Broadcom services from the VCF Operations, License Manager, Software Depot, and SDDC Manager components.	• Required to download license files and upload license usage. • Required to download new updates. • Used for support bundles.	VCF management components must be able to directly connect to a Broadcom service via Internet.
Provide Internet connectivity to Broadcom services from the VCF Operations, Software depot, License server, and SDDC Manager components via a properly configured HTTP proxy.	• Required to download license files and upload license usage. • Required to download new updates. • Used for support bundles.	Requires setup and configuration of an external proxy server.
Use a manual offline method for license and depot sync.	Useful in dark site without Internet connectivity.	• Requires setup and configuration of depot server for software updates • Require manual operations that would otherwise be automatic when Internet access is available.

Internet Connectivity Requirements

Dark Sites

The management domains require Internet connectivity, directly or through a proxy, to support lifecycle and licensing management. Manual processes are available for dark sites. Refer to Public URLs Required for Online Functionalities for VCF and vSphere Foundationfor details.

Table 918: IP Connectivity Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-IPC-001	Ensure IP connectivity between all management component VMs and ESX hosts within a VCF Instance.	• Management and cluster planes reside in the management component VMs and must be able to access ESX VMs for proper operations. • ESX hosts require IP connectivity within and between clusters to support cluster-level operations such as vMotion, networking, and vSAN Storage.	• Fabric must provide routing between management VLANs and ESX VMKs. • IPs used for VMKs must be routable within/between data centers that contain the VCF fleet.

Table 919: IP Connectivity Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-IPC-001	Provide Internet connectivity to Broadcom services from the VCF Operations and SDDC Manager components.	• Provides automated license management. • Provides the easiest method for software update downloads. • Provides the easiest method for support.	• VMware Cloud Foundation management components must be able to directly connect to a Broadcom service via the Internet. • Appropriate outbound firewall ports must be opened.

First Hop Gateway

A first hop gateway refers to the first router a network packet encounters as it leaves a VLAN segment enroute to another network in the data center.

The data center fabric may provide resilience to the first hop gateway via VRRP or other method. Depending upon the Availability model chosen, and the scope/span of the VLAN segment it is associated with, the gateway will need to tolerate the failure of an individual switch, rack, or entire availability zone.

Table 920: First Hop Gateway Choices

First Hop Gateway Availability	Highly Available - Between AZs	The First Hop Gateway must be provided by a single highly available IP which can fail-over (Active/Standby) between AZs, or be present in each AZ simultaneously (Active/Active).
	Highly Available - Within a Rack	The First Hop Gateway must be provided by a single highly available IP that can fail over (Active/Standby) between the top-of-rack switches or be present in each top of rack simultaneously (Active/Active).

	None	A highly available First Hop Gateways may be provided, but is not required for this VLAN Segment.

Table 921: First Hop Gateway Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-GTW-001	Provide a first hop gateway for all infrastructure VLAN segments.	A data center provided gateway is required for connectivity to other segments within and outside of the data center.	None.
VCF-NET-REQD-GTW-002	Ensure the first hop gateway is configured with the appropriate MTU for the VLAN segment.	• MTU may be set and different points in the data center fabric. • Ensure the gateway MTU is appropriately set for traffic types which will traverse it.	None.

Table 922: First Hop Gateway Design Recommendations

When designing the VLAN and subnet configuration for your VMware Cloud Foundation deployment, consider the following recommendations:

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-GTW-001	Use the IP address of the floating interface for Virtual Router Redundancy Protocol (VRPP) or Hot Standby Routing Protocol (HSRP) as the gateway.	Provides high availability to the first hop gateway.	None.
VCF-NET-RCMD-GTW-002	For network segments which are stretched between availability zones - have a Layer 3 gateway at the first hop that is highly available such that it tolerates the failure of an entire availability zone,	When VMs move between availability zones during a failure, the first hop gateway must be available on the same IP.	Requires network fabric provide ability to failover first hop gateway IP between AZs.

Note:

NSX Edge uplink networks that are used with BGP Peering may receive the next hop gateway and availability via BGP.

Infrastructure VLANs

Note:

Infrastructure networks are used to provide management, control, overlay, and IP Storage communications within a VCF Instance. Workload networks are used to provide IP connectivity to non-management workloads.

The required span of a Layer 2 segment depends on the type of traffic and the design of the vSphere Cluster Models chosen. Using a separate 802.1q-based VLAN for each type of infrastructure network traffic within each cluster is advisable.

VMkernel Segments: Each cluster's network function should be separated into its own VMkernel and Layer 2 domain. For clusters that span multiple racks or availability zones, different VLAN IDs and Layer 2 domains may be utilized for each VMkernel traffic type in each rack.

VM Segments: Any infrastructure VLAN that supports management or workload VMs must be equally accessible to all ESX hosts within the vSphere cluster to support workload mobility via vSphere HA or vMotion. Additionally, the VLAN must be reachable across all racks or availability zones associated with the vSphere Cluster. The VLAN ID must be consistent across all hosts/racks/AZ.

The gateway for these VLANs should also be resilient enough to handle the failure of a single switch, rack, or availability zone. The use of a First Hop Redundancy Protocol (FHRP) is recommended.

Note:

For cluster-level VLAN scope, each ESX host must have the same VLAN ID and VLAN segment presented, including vSphere clusters that span racks or availability zones. The first hop gateway must also be highly available across racks or availability zones.

Requirement	Option	Description
Required VLAN Scope	Global across availability zones	For this traffic type, a single VLAN segment with the same VLAN ID is presented to all ESX hosts in both availability zones.
	Availability Zone Local	A single VLAN segment with the same VLAN ID is presented to all ESX hosts in individual availability zones. ESX hosts in the second availability zone may have different VLAN segments/VLAN IDs for this traffic type.
	Rack Local	A single VLAN segment with the same VLAN ID is presented to the pNICs for an ESX host. Individual ESX hosts may have different VLAN segments/VLAN IDs for this traffic type.

Single-Rack Network Fabric Model

The *Single-Rack Network Fabric Model* describes the VLAN and gateway requirements to support the Single-Rack vSphere Cluster Model.

The data center network fabric provides connectivity to the vSphere clusters. The vSphere Cluster Models and Storage Models chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support a Single-Rack vSphere Cluster Model. Depending upon the storage type chosen, additional VLAN requirements may apply.

Note:

If this rack will contain NSX Edge nodes, additional VLAN Requirements will apply. NSX Edge Cluster Models for Single Availability Zone

Table 923: Infrastructure VLAN Scope and First Hop Gateway Availability for Single Rack

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management VM Appliances*		Rack Local	Highly Available	1500	1500
ESX host	ESX management VMK *				1500
	IP Storage VMKs (vSAN, NFS, iSCSI)				9000
	vMotion VMK				
	Host TEP			1700	
NSX Edge node	Refer to Host Fault Tolerant NSX Edge Cluster Model or Rack Fault Tolerant NSX Edge Cluster Model using Multiple Single-Rack vSphere Clusters				
	NSX Edge Node management	Rack Local	Highly Available	1500	9000

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available Static Routing with BFD - Optional BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across Racks	Highly Available - Between Racks**	1500	1500
	BGP-EVPN Peering	Across Racks	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* When using separate VLAN segments for management component VMs and ESX VMKs.					

Table 924: Workload VLAN Scope and First Hop Gateway Availability for Single Rack

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VLAN Connectivity Model	Rack Local	Highly Available	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model Attributes	Refer to Edges			

Multi-Rack Network Fabric Model

The *Multi-Rack Network Fabric Model* describes the VLAN and gateway requirements to support Multi-Rack Layer 2 vSphere Cluster Model and Multi-Rack Layer 3 vSphere Cluster Model.

The data center fabric provides connectivity to the vSphere clusters. The vSphere Cluster Models and Storage Models chosen will determine which data center availability model is required to support the connectivity and availability within the cluster. This section describes the requirements to support Multi-Rack Layer 2 vSphere Cluster Model and Multi-Rack Layer 3 vSphere Cluster Model. Depending upon the storage type chosen, additional VLAN requirements may apply.

Multi-Rack Network Fabric Model with Layer 2

In this model, all VLANs are stretched across all hosts within the vSphere cluster. This stretch is provided by the data center fabric using its intrinsic capabilities such as EVPN-VXLAN.

Table 925: Infrastructure VLAN Scope and First Hop Gateway Availability for Layer 2 Multi Rack

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management component VMs		Across Racks	Highly Available - Across Racks**	1500	1500
ESX host	ESX management VMK *				1500
	IP Storage VMKs (VSAN, NFS, iSCSI)				9000

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
	vMotion VMK				
	Host TEP			1700	
NSX Edge node	Refer to Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 2 vSphere Cluster				
	NSX Edge Node Management	Across Racks	Highly Available - Across Racks**	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available Static Routing with BFD - Optional BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Across Racks	Highly Available - Across Racks**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNAVNA Node Management	Across Racks	Highly Available - Across Racks**	1500	1500
	BGP-EVPN Peering	Across Racks	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* When using separate VLAN segments for management component VMs and ESX VMKs. ** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.					

Table 926: Workload VLAN Scope and First Hop Gateway Availability for Layer 2 Multi Rack

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Across Racks	Highly Available - Between Racks**	1500	9000
Distributed VLAN Connectivity Model	Across Racks *	Highly Available - Between Racks**	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model Attributes	Refer to Edges			
* Will align with the span of the Distributed Transit Gateway ** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.				

Multi-Rack Network Fabric Model with Layer 3

In this model, VLAN segments are isolated to individual racks within the data center. ESX hosts within the vSphere cluster may come from different racks with different VLAN segments presented to them.

Note:

This model is not applicable to the default management vSphere cluster VCF Instance.

Table 927: Infrastructure VLAN Scope and First Hop Gateway Availability for Layer 3 Multi Rack

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
ESX host	ESX management VMK *	Rack Local	Highly Available - Within a Rack	1500	1500

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
	IP Storage VMKs (VSAN, NFS, iSCSI)				9000
	vMotion VMK				
	Host TEP			1700	
NSX Edge node	Refer to Rack Fault Tolerant NSX Edge Cluster Model using Multi-Rack Layer 3 vSphere Cluster				
	NSX Edge Node Management	Rack Local	Highly Available - Within a Rack	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available Static Routing with BFD - Optional BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Rack Local**	Highly Available - Within Rack	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0 NOTE: Both nodes of the EVPN Route Controllers must reside within a single Layer 2 domain.				
	VNA Node Management	Rack Local	Highly Available -Within Rack	1500	1500
	BGP-EVPN Peering	Rack Local**	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* When using separate VLAN segments for management component VMs and ESX VMKs. ** All Node of a VNA cluster must share a common Layer 2 network.					

Table 928: Optional - Infrastructure VLAN Scope and First Hop Gateway Availability for Stretched VLANs in Multi Rack

In a Multi-Rack Layer 3 Network Fabric Model, you can optionally stretch Layer 2 networks between racks to support management components or clustered services requiring Layer 2 adjacency. This approach enhances fault tolerance and ensures the continuous availability of critical management appliances as they move across different physical racks.

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Single Availability Zone				
	VNA Node Management	Across Racks*	Highly Available - Across Racks**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0 NOTE: Both nodes of the EVPN Route Controller must reside within a single Layer 2 domain.				
	VNA Node Management	Rack Local	Highly Available - Across Racks**	1500	1500
	BGP-EVPN Peering	Across Racks*	Single-Hop BGP - None Multi-Hop BGP - Highly Available	1500	1500
* All Node of a VNA cluster must share a common L2 network. Optional stretching of this VLAN between racks will enable seamless maintenance and recovery across racks. ** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.					

Table 929: Workload VLAN Scope and First Hop Gateway Availability for Layer 3 Multi Rack

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model Attributes	Refer to Edges			

Table 930: Optional - Workload VLAN Scope and First Hop Gateway Availability for Stretched VLANs in Multi Rack

In a Multi-Rack Layer 3 Network Fabric Model, you can optionally stretch a limited number of Layer 2 networks between racks to meet specific customer workload requirements for high availability or mobility. This approach enhances fault tolerance and ensures a seamless maintenance experience for workloads that require Layer 2 adjacency.

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Across Racks	Highly Available - Across Racks*	1500	9000
* Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.				

Multi Availability Zone Network Fabric Model

This section describes the VLAN and gateway requirements to support a vSphere cluster within multiple availability zones.

This design requires stretching a minimum number of VLAN segments between each availability zone. The physical network fabric stretches these segments to provide a single Layer 2 domain across zones, enabling virtual machine mobility during an availability zone outage. For these fabric-stretched VLAN segments, the first-hop gateway must remain highly available across both availability zones. You must give special consideration to VLAN segments for NSX Edge uplinks when using the Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model, as BGP peering from the NSX Edge node to the fabric occurs over these networks.

> **Note:**
>
> VLAN IDs are switch-level numerical labels identifying a Layer 2 domain. In some fabric configurations, the same VLAN ID may identify different domains, or different VLAN IDs may identify the same domain. A distributed port group can be associated with a single VLAN ID. Any distributed port groups supporting virtual machines must have the same VLAN ID and Layer 2 domain mapping across the entire vSphere cluster. You may use different VLAN IDs for separate vSphere clusters that do not share a common vSphere Distributed Switch.

Table 931: Infrastructure VLAN Scope and First Hop Gateway Availability for Multiple AZ

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
Management component VMs*		Global across AZs	Highly Available - Across AZs **	1500	9000
ESX host	ESX management VMK	AZ Local***	Highly Available - Within AZ**		1500
	IP Storage VMKs (VSAN, NFS, iSCSI)		Highly Available - Within AZ **		9000
	vMotion VMK		Highly Available - Within AZ **		

Traffic Type		Required VLAN Scope	First Hop Gateway Availability	MTU	
				minimum	preferred
	Host TEP		Highly Available - Within AZ **	1700	
NSX Edge node	Refer to Dual Availability Zone Based on NSX Edge Node HA Model				
	NSX Edge Node Management	AZ Local	Highly Available - Within AZ **	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available - Within AZ** BGP - None	1700	
	Refer to Dual Availability Zone Based on NSX Edge vSphere HA Recovery Model				
	NSX Edge Node Management	Global across AZs	Highly Available - Across AZs *	1500	9000
	NSX Edge TEP			1700	
	NSX Edge RTEP			1500	
	NSX Edge Uplink		Static Routing - Highly Available - Across AZs ** BGP - None	1700	
VNA - Stateful Services	Refer to Virtual Network Appliance Cluster Models for Dual Availability Zones				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
VNA - EVPN Route Controller	Refer to Distributed VXLAN Connectivity Model for Version 9.1.0				
	VNA Node Management	Across AZs	Highly Available - Across AZs**	1500	1500
	BGP-EVPN Peering	Across AZs	Single-Hop BGP - None Multi-Hop BGP - Highly Available - Across AZs **	1500	1500

* When using separate VLAN segments for management component virtual machines and ESX VMkernels.
** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.
**In this table, ESX host infrastructure VLANs are limited to an individual availability zone. Stretching of these VLANs between availability zones is supported, however the first hop gateway must also be highly available between availability zones.

Table 932: Workload VLAN Scope and First Hop Gateway Availability for Multiple Availability Zones

Traffic Type	Required VLAN Scope	First Hop Gateway Availability	MTU	
			minimum	preferred
VLAN Networking Connectivity Model	Global between AZs	Highly Available - Across AZs**	1500	9000
Distributed VLAN Connectivity Model	Global between AZs*	Highly Available - Across AZs**	1500	9000
Distributed VXLAN Connectivity Model for Version 9.1.0	Same as Host TEP			
Centralized Connectivity Model	Refer to Edges			
NSX Segment Connectivity Model	Refer to Edges			

* All Node of a VNA cluster must share a common L2 network. Optional stretching of this VLAN between racks will enable seamless maintenance and recovery across racks.
** Examples of Highly Available gateways include HSRP, VRRP, AnyCast, etc.

IP Connectivity and Symmetry

Traffic must be routed between availability zones for all VLAN segments that require cross-site communication. For example, vSAN networks in each availability zone must have a route to each other to support synchronous data replication.

For VLAN segments that stretch between availability zones, the physical network fabric determines the ingress and egress paths. In most cases, to simplify troubleshooting, symmetrical routing is recommended.

For workloads, the ingress and egress path depends upon the Network Consumption Models chosen. For the Centralized Connectivity Modelingress and egress paths depend on the specific NSX Edge Cluster Models for Dual Availability Zonesand routing model:

- **Static Routing**: The physical network fabric determines the ingress and egress paths.
- **eBGP**: A combination of the physical fabric and the NSX Edge gateway determines the ingress and egress paths.

For all stretched VLAN network segments (infrastructure networks, VLAN Networking Connectivity Model or Distributed VLAN Connectivity Model), or Distributed VXLAN Connectivity Model for Version 9.1.0, the physical network fabric must provide a highly available Layer 3 gateway at the first hop. This ensures that virtual machines maintain connectivity on the same IP address when moving between availability zones.

Witness Location

For vSAN stretched cluster, an independent witness site is required. This requires independent network paths from each availability zone to the witness site, such that the failure of an individual availability zone does not impact the traffic path from the other availability zones to the witness site.

When configured with vSAN Witness Traffic Separation (WTS), the vSAN witness traffic flows between the ESX management VMkernels and the vSAN Witness node. This traffic uses the MTU of the ESX management VMkernels.

Availability Zone Data Center Interconnect

The data center inter-connect between the availability zones must be appropriately sized to handle all infrastructure, workload, and storage replication traffic. The interconnect is important to the operations of the vSAN Cluster. It should have the appropriate level of resilience to achieve business objectives.

Infrastructure IP Addressing Detailed Design

The infrastructure IP addressing models focus exclusively on the IPv4 and IPv6 unicast addressing schemes used for VCF management components and infrastructure VMkernel interfaces. This section details the supported address assignment methods to ensure reliable, point-to-point communication across the data center fabric for all VCF endpoints.

Table 933: IP Address Family Models

The following table defines the supported *IP Address Family Models* for VMware Cloud Foundation. Each model dictates the addressing scheme used for Management and Infrastructure traffic.

IP Families	Description	Interoperability with Other Components Deployed as		
		IPv4	IPv6	Dual-Stack
IPv4 Only	Traditional deployment standard where components are assigned exclusively IPv4 addresses.	✓		✓
IPv6 Only	Components are assigned exclusively IPv6 addresses.		✓	✓
Dual-Stack	Components are assigned both IPv4 and IPv6 addresses. **Note:** In a dual-stack configuration, communication occurs over IPv6 when both endpoints support it. If an established IPv6 session fails, the system does not fail back to IPv4.	✓	✓	✓

IP Address Assignment Methods

Methods by which IP addresses are allocated and assigned to specific virtual and physical VCF components.

Table 934: IPv4 Address Assignment Methods

IPv4 Assignment Method	Description
Static	Statically assigned by the user via manual configuration prior to workflow execution. Statically assigned for the user from a user provided range during workflow execution.
Network Pools	Statistically assigned from a pre-configured range of addresses.
DHCPv4	Assigned via the customer-provided DHCPv4 server.

Table 935: IPv6 Address Assignment Methods

IPv6 Address Assignment Method	Description
Static	• Statically assigned by the user via manual configuration prior to workflow execution. • Statically assigned for the user from a user provided range during workflow execution.
Network Pools	Statistically assigned from a pre-configured range of addresses.
DHCPv6	• Assigned via the customer-provided DHCPv6 server. • Requires upstream Router Advertisements (RA).
SLAAC	• Automatic address assignment via Stateless Address Autoconfiguration (RFC 4862). • Requires upstream Router Advertisements (RA).

IPv4 Addressing Model

Guideline for IPv4 Infrastructure IP Addressing

All infrastructure networks (management VMkernels, primary storage, vMotion, NSX TEPs, etc) must be IPv4 based. The number of IPs required in each segment varies based on the cluster size, storage type, and VMkernel design for each infrastructure traffic type.

Note:

Workload VLAN and overlay segments may leverage IPv6.

Table 936: IPv4 Addressing Model by Infrastructure Component

Component	Infrastructure Traffic Type	Minimum IPs required Per Host	Supported Addressing Method
ESX Host	Management VMK	1 per host	Static
	vMotion VMK	1 per host	• Network Pool • DHCPv4
	Storage VMKs (VSAN, NFS)	1 per host per VMK	• Network Pool • DHCPv4
	Host TEP	1 per active VDS Uplink per Host (typically 2)	• Static IP List • Network Pool • DHCPv4
NSX Edge Node	Management	1 per node	• Static • DHCPv4

Component	Infrastructure Traffic Type	Minimum IPs required Per Host	Supported Addressing Method
	NSX Edge TEP	1 per node (typically 2)	• Static IP List • Network Pool • DHCPv4
	NSX Edge Tier-0 Gateway Uplinks	1 per node (typically 2)	Static
VNA - Stateful Services	VNA Node Management	1 per node	• Static • DHCPv4
VNA - EVPN Route Controller	VNA Node Management	1 per node	• Static • DHCPv4
	BGP-EVPN Peering	1 per node (+1 for VNA cluster VIP)	Static
vSAN Witness	Management	1 per node	Static
	VSAN Witness	1 per node	Static

IPv4 Addressing Model Design Requirements and Recommendations

Table 937: IPv4 Addressing Model Design Requirements

Design Requirements ID	Design Requirement	Justification	Implication
VCF-NET-REQD-IPA-001	Allocate statically assigned IPv4 addresses and host names for all management components.	Ensures stability across the VCF fleet, and makes it simpler to maintain, track, and implement DNS configuration.	You must provide precise IP address management.
VCF-NET-REQD-IPA-002	Configure forward and reverse DNS records for all management components.	Ensures that all components are accessible by using a fully qualified domain name instead of using IP addresses. It is easier to remember and connect to components across the VCF fleet.	You must provide DNS records for each component.

Table 938: IPv4 Addressing Model Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-IPA-001	Ensure that your subnets are scaled appropriately to allow for expansion.	• Expanding subnets at a later time is disruptive. • VMware Cloud Foundation does not support changing of subnet masks for infrastructure networks.	None.
VCF-NET-RCMD-IPA-002	Use the RFC 1918 IPv4 address space for these subnets and allocate one octet by VCF Instance and another octet by function.	VMware Cloud Foundation management and Infrastructure components do not require public IPs.	None.

Note:

Expansion or shrinkage of the IP subnet for a specific traffic type is not recommended. Adding an additional IP address to an existing IP subnet (without changing the subnet mask) is supported.

Note:

A subset of the RFC 6598 100.64.0.0/10 address block is utilized for allocating Inter-Tier Transit Links for internal connectivity within the NSX SDN between the Tier 0 and Tier 1 components and between the Transit Gateway (TGW) and individual VPC Gateways. This address space is not externally advertised. If this 100.64.0.0/10 block is utilized elsewhere in the enterprise, and alternative address block for the *Inter-Tier Transit Link* may be provided to VCF during deployment time.

Note:

Changing of the Inter-tier transit link after deployment requires manual redeployment of NSX routing components.

Note:

The RFC 2544 198.18.0.0/15 block is reserved for use within the VCF management services. This address space is not externally advertised. If this 198.18.0.0/15 block is utilized elsewhere in the enterprise, and alternative address block for internal cluster IPs may be provided during deployment time via API.

Dual Stack IP Addressing Model

The dual-stack model enables the deployment of VCF components to minimize the consumption of IPv4 address and optimize the usage of IPv6 addresses.

Guideline for Dual-Stack IPv4/IPv6 Infrastructure IP Addressing

Supported IP address families are unique to each component. Each component within the VCF solution has its own IP address support where components may support single-stack IPv4, single-stack IPv6, or dual-stack (IPv4 and IPv6).

VMware Cloud Foundation applies dual-stack networking as a domain-wide configuration for the management domain or workload domain, rather than as a component-level choice. Dual-stack deployment of a new domain configures all dual-stack capable management components with both IPv4 and IPv6 addresses, while it provisions single-stack capable components with IPv4 only. Also, you may select either IPv4 or IPv6 for vMotion and storage (VSAN/NFS) interfaces to best fit your specific storage and mobility requirements.

Note:

Dual-stack networking is applied as a domain-wide configuration rather than a component-level choice.

Note:

This model does not eliminate the need for IPv4 addressing.

IP Address Family Support by Management Component

The following tables summarize the IP address family support for individual VCF components when the dual-stack model is selected.

Table 939: Dual-Stack IP Addressing Model by Management Components

Component	Supported IP Family Models	
	IPv4 Only	Dual-Stack
VCF Operations		✓
Cloud Proxy		✓
VCF Management Services		✓
License Server	✓	
Log Management		✓
Identity Broker		✓

Component	Supported IP Family Models	
	IPv4 Only	Dual-Stack
VCF Automation	✓	
VCF Automation Services	✓	
VCF Operations Orchestrator	✓	
VCF Operations for Networks - Server		✓
VCF Operations for Networks - Collector		✓
VCF Operations HCX	✓	
NSX Global Manager	✓	
Data Services Manager	✓	
VMware Live Recovery	✓	

Table 940: Dual-Stack IP Addressing Model by Instance Management Components

Component	Supported IP Family Models	
	IPv4 Only	Dual-Stack
vCenter		✓
NSX Manager		✓
vSphere Supervisor	✓	
VCF Installer		✓
SDDC Manager		✓

Important:

DNS must be able to resolve FQDN with both IPv4 and IPv6 addresses.

IP Address Family Support by Infrastructure Component

The number of IPs required in each segment varies based on the cluster size, storage type, and VMkernel design for each infrastructure traffic type.

Table 941: Dual-Stack IP Addressing Model by Infrastructure Component

Component	Infrastructure Traffic Type	Minimum IPs required Per Component	IP Family Model			Supported Addressing Method
			IPv4	Dual-Stack	IPv6	
ESX Host	Management VMK	1 per host		✓		Static
	vMotion VMK	1 per host			✓	• VCF Network Pool • DHCPv6 • SLAAC
	Storage VMKs (VSAN, NFS)	1 per host per VMK			✓	• VCF Network Pool • DHCPv6 • SLAAC
	Host TEP	1 per VDS Uplink per Host (typically 2)	✓			• Static • DHCPv4
NSX Edge Node	Management	1 per node		✓		• Static

Component	Infrastructure Traffic Type	Minimum IPs required Per Component	IP Family Model			Supported Addressing Method
			IPv4	Dual-Stack	IPv6	
	Edge TEP	1 per node (typically 2)	✓			• Static - IP List • Static - IP Pool • DHCPv4
Virtual Networking Appliance	Management	1 per node		✓		• Static • DHCPv4
Virtual Networking Appliance (EVPN Route Controller)	Management	1 per node	✓			• Static • DHCPv4
vSAN Witness	Management	1 per node		✓		Static
	VSAN	1 per node			✓	Static

Note:

Expansion or shrinkage of the IP subnet for a specific traffic type is not recommended. Adding an additional IP address to an existing IP Subnet (without changing subnet mask) is supported.

Dual Stack IP Addressing Model Design Requirements and Recommendations

Only Global Unicast Addresses (GUA) are supported for VCF components. Unique Local Addresses (ULA) are not recommended for use within the VCF management components. Link-Local Addresses (LLA) are supported for traditional IPv6 roles such as routing.

Table 942: Dual-Stack Requirements

Design Requirements ID	Design Requirement	Justification	Implication
VCF-NET-REQD-IPA-003	Allocate statically assigned IPv4 addresses for all management components. Additionally, allocate statically assigned IPv6 addresses for all dual-stack capable management components.	Ensures stability across the VCF fleet, and makes it simpler to maintain, track, and implement DNS configuration. Dual-stack address assignment is required for all dual-stack capable management components.	You must provide precise IP address management.
VCF-NET-REQD-IPA-004	Configure forward and reverse DNS records for all management components. Dual-stack components require IPv6 and IPv4 relevant records. IPv4 components must only have IPv4 relevant records.	• Ensures that all components are accessible by using a fully qualified domain name (FQDN) instead of using IP addresses. It is easier to remember and connect to components across the VCF fleet. • FQDNs are required for Day-0 and Day-N VCF workflows. • Correct DNS resolution is required to determine IP address families available for component communication.	• You must provide DNS records for each component. • IPv4 only components must not have IPv6 related DNS records.

Design Requirements ID	Design Requirement	Justification	Implication
VCF-NET-REQD-IPA-005	Use the Global Unicast Address (GUA) space for IPv6 subnets.	• Ensure routable IPv6 address space. • Unique Local Address (ULA) and Link-Local Address (LLA) are not supported for VCF component addressing	Requires a routable IPv6 block.

Table 943: Dual-Stack Addressing Model Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-NET-RCMD-IPA-003	Ensure that your subnets are scaled appropriately to allow for expansion.	• Expanding subnets at a later time is disruptive. • VMware Cloud Foundation does not support changing of subnet masks for infrastructure networks.	None.
VCF-NET-RCMD-IPA-004	Use the RFC 1918 IPv4 address space for these subnets and allocate one octet by VCF Instance and another octet by function.	VMware Cloud Foundation management and Infrastructure components do not require public IPs.	None.
VCF-NET-RCMD-IPA-005	For IPv6, use /64 for subnet length.	Aligns with industry best practices.	None.

Note:

A subset of the RFC 6598 100.64.0.0/10 address block is utilized for allocating Inter-Tier Transit Links for internal connectivity within the NSX SDN between the Tier 0 and Tier 1 components. This address space is not externally advertised. If this 100.64.0.0/10 block is utilized elsewhere in the enterprise, and alternative address block for the *Inter-Tier Transit Link* may be provided to VCF during deployment time.

Note:

Changing of the Inter-tier transit link after deployment requires manual redeployment of NSX routing components.

Note:

A subset of the RFC 2544 198.18.0.0/15 block is reserved for use within the VCF Managed Services Platform. This address space is not externally advertised. If this 198.18.0.0/15 block is utilized elsewhere in the enterprise, and alternative address block for internal cluster IPs may be provided during deployment time via API.

Note:

The ULA FD00::/111 is reserved for use within the VCF Managed Services Platform. This address space is not externally advertised. If this FD00::/111 block is utilized elsewhere in the enterprise, and alternative address block for internal cluster IPs may be provided to VCF during deployment time via API.

Dual-Stack Physical Network Design Considerations

Implementing a dual-stack IP addressing model requires ensuring the underlying physical network fabric is explicitly configured to support both IPv4 and IPv6 traffic concurrently. This configuration extends beyond basic IP allocation to include considerations for Layer 2 multicast handling, routing advertisements, and security policy enforcement.

Furthermore, you must determine the appropriate method for assigning IPv6 addresses to management and infrastructure components. VMware Cloud Foundation supports multiple assignment methods, including Stateless Address Autoconfiguration (SLAAC) and Dynamic Host Configuration Protocol for IPv6 (DHCPv6). Selecting the correct method depends on your organizational requirements for address control, visibility, and automation.

Table 944: Dual-Stack Physical Network Design Requirements

Design Requirement ID	Design Requirement	Justification	Implication
VCF-NET-REQD-DS-001	Configure IPv6 on all physical switch interfaces (SVI) servicing dual-stack infrastructure VLANs (ESX Management, Management VM, vMotion, and Storage).	Ensures that the physical fabric can route and process IPv6 traffic for all primary VCF services.	None
VCF-NET-REQD-DS-002	Use a consistent MTU (e.g., 9000 bytes) for both IPv4 and IPv6 on Dual-Stack VLANs.	Prevents fragmentation issues.	IPv6 has a larger header (40 bytes) than IPv4 (20 bytes), making header overhead slightly higher.
VCF-NET-REQD-DS-003	Update physical firewalls and Access Control Lists (ACLs) to permit bidirectional traffic for both IPv4 and IPv6 address families.	Ensures that security policies are consistently applied and do not inadvertently block required IPv6 management traffic.	Increases the complexity of firewall rule management and auditing.

Table 945: Dual-Stack Physical Network Design Recommendations

Design Recommendations ID	Design Recommendations	Justification	Implication
VCF-NET-RCMD-DS-001	Configure IPv6 Router Advertisements (RA) on the physical interfaces for VLANs using DHCPv6 or SLAAC.	Required for SLAAC and DHCPv6 to enable automatic discovery of gateways and obtain IPv6 address configuration.	None
VCF-NET-RCMD-DS-002	Enable MLD Snooping (Multicast Listener Discovery) on physical Layer 2 switches.	• IPv6 relies heavily on multicast for Neighbor Discovery. • MLD Snooping prevents IPv6 multicast traffic from flooding all ports in the VLAN.	Requires careful physical network configuration including MLD Querier.

Dual-Stack Operational Considerations

When operating a dual-stack VMware Cloud Foundation environment, you must account for the specific constraints regarding workload mobility. Because VCF applies dual-stack networking as a domain-wide configuration rather than a component-level choice, maintaining protocol consistency across the environment is critical for stable operations.

Table 946: Dual-Stack Operational Design Recommendations

Design Recommendation ID	Design Recommendation	Justification	Implication
VCF-OPS-RCMD-DS-001	Restrict vMotion operations to clusters that share a common IP address family for their vMotion VMkernel adapters.	• vMotion is only supported between clusters where the vMotion VMKs reside in the same IP family. If a dual-stack cluster uses IPv6 for vMotion, it cannot migrate workloads to an IPv4-only cluster. • vMotion between a dual-stack and an IPv4 cluster is only possible if both selected IPv4 for the vMotion VMK.	Limits workload mobility between legacy and modern clusters if IPv6 is selected for vMotion. Requires uniform cluster deployment models within a VCF instance to maintain full mobility.
VCF-OPS-RCMD-DS-002	Ensure external infrastructure services (NTP, DNS) are dual-stack capable before activation.	Dual-stack components prioritize communication over IPv6. If primary IPv6 communication fails, there is no automatic failback to IPv4 for these services.	Legacy services that only support IPv4 must be upgraded or replaced to avoid service failures.

Workload IP Addressing Detailed Design

Guideline for Workload IP Addressing

Workload networks—whether VLAN or NSX Overlay—can utilize either IPv4 or IPv6 addressing to meet specific application requirements. The sizing and configuration of these segments are determined by organizational needs and the chosen networking model. While these methods offer broad protocol flexibility, note that the Virtual Private Cloud (VPC) network virtualization model currently provides a prescriptive design optimized for the IPv4 address family.

To optimize IP route management and ensure scalability, all overlay workload segments should originate from one or larger super-net blocks. These blocks are delegated to the NSX software-defined network (SDN) through BGP or static routing. This approach centralizes route propagation and reduces the complexity of managing individual workload segments within the data center fabric. Refer to the Networking Consumption Models section for further details on workload addressing

Network Type	Supported Workload IP Families		
	IPv4	IPv6	Dual-Stack
VLAN Segments	✓	✓	✓
Overlay Segments	✓	✓	✓
VPC-Subnets	✓		

VCF Edge Detailed Design

VMware Cloud Foundation Edge extends the centralized management capabilities of the platform to geographically dispersed edge locations.

This section describes different design options for deploying VCF Edge, covering various needs and complexities.

Feature	1-Node cluster in a VCF Instance	2-Node with external storage (minimum) in a workload domain	3-Node vSAN cluster (minimum) in a workload domain	4-Node VCF Fleet with single domain and one or multiple clusters.
Prime Use Case	Remote / Branch Offices, Edge locations	Remote / Branch Offices, Edge locations	Small to medium-sized deployments	Large-scale, distributed environments or locally managed edge environments
Storage	Local or external NFS or FC	External NFS or FC (No vSAN)	vSAN or External Storage	vSAN or External Storage
Networking	• NSX supported • vSphere Distributed Switches supported	• NSX supported • vSphere Distributed Switches supported	• NSX supported • vSphere Distributed Switches supported	• NSX supported • vSphere Distributed Switches supported
Max Hosts per Cluster	64	64	64	64
Max Workload Domains	Not supported	24	24	24
Life cycle Management	Central or local lifecycle management with vCenter	Central or local lifecycle management with VCF Operation	Central or local lifecycle management with VCF Operation	Central or local lifecycle management with VCF Operation
Latency & Bandwidth Requirements	300ms latency, 10 Mbps bandwidth	300ms latency, 10 Mbps bandwidth	300ms latency, 10 Mbps bandwidth	300ms latency, 10 Mbps bandwidth
VKS Support	Yes	Yes	Yes	Yes
Key Advantage	Lower cost for edge deployments	Lower cost for edge deployments	Simplified management for smaller environments	A highly scalable, resilient architecture, can fit any deployment models such as centrally managed, locally managed and Isolated Edge.

Design Patterns	Description
VCF Edge Option 1: VCF Operation with vCenter for Distributed Edge	Uses centralized vCenter operations to manage distributed edge locations while maintaining operational consistency across sites.
VCF Edge Option 2: Thin Edge with vSphere Supervisor on Two-Node vSAN	Lightweight edge architecture using a compact two-node vSAN cluster with vSphere Supervisor enabled for Kubernetes workloads at the edge.
VCF Edge Option 3: vSAN Witness Placement for Distributed Edge	Defines placement strategies for vSAN Witness appliances supporting resilient two-node edge deployments.
VCF Edge Pattern 1: Centrally Managed Edge Compute Clusters in a Workload Domain	Edge compute clusters are centrally managed within a VCF workload domain to provide consistent lifecycle and governance operations.
VCF Edge Pattern 2: Segregated Edge Deployment Across Multi-Region Workload Domains	Edge environments are separated across regional workload domains to improve isolation, resiliency, and regional autonomy.
VCF Edge Pattern 3: Locally Managed Edge	Edge locations operate with localized management to reduce dependency on centralized infrastructure and improve site autonomy.
VCF Edge Pattern 4: VCF Fleet with Multi-Region Distributed Edge	Multiple distributed edge regions operate under a shared VCF Fleet model with centralized governance and regional isolation.
VCF Edge Pattern 5: Dark Site Edge	Edge deployment that is designed for disconnected or air-gapped environments with minimal dependency on external services.
VCF Edge Pattern 6: vSphere Supervisor at the Edge in a VCF Instance	Extends Kubernetes platform services to edge environments through vSphere Supervisor integration within a VCF instance.
VCF Edge Pattern 7: Supervisor Zonal Separation at the Edge	Uses zonal separation for Supervisor workloads to improve resiliency and fault isolation across edge environments.

Design Patterns	Description
VCF Edge Pattern 8: Single Host vSphere Supervisor at the Edge in a VCF Instance	Minimal-footprint edge deployment using a single host running vSphere Supervisor services for lightweight or non-critical workloads.
VCF Edge Pattern 9: Leveraging Existing VCF Operations for Far Edge Expansion	Extends existing VCF operational models and tooling to support scalable far-edge deployments.
VCF Edge Pattern 10: Colocated vCenter with ESX Host at the Edge	Deploys vCenter alongside ESX hosts at the edge site to provide localized infrastructure management and operational independence.

Design Options Without VCF Full Stack

VCF Edge offers flexibility in management of different VCF components in edge deployments.

Customers also have the flexibility to deploy a reduced infrastructure footprint without implementing the complete VMware Cloud Foundation stack. In smaller environments or edge deployments, customers may choose to manually deploy only selected core components such as:

- VMware vCenter
- VCF Operations components
- VMware License Server

These environments can still be licensed under a VCF Edge model for lightweight or resource-constrained deployments where the full private cloud platform capabilities of VCF are not required.

This approach can reduce:

- Infrastructure footprint
- Hardware requirements
- Deployment complexity
- Operational cost for small-scale environments

However, this simplified deployment model introduces important tradeoffs compared to a full VCF deployment.

Tradeoffs and Considerations

By not deploying the full VCF stack, customers may lose several integrated platform capabilities that enable a consistent private cloud operating model across data center and edge environments.

Potential Limitations

Capability Area	Impact Without Full VCF Stack
Centralized Private Cloud Operations	Reduced ability to manage infrastructure consistently from a centralized platform
Centralized Governance and Policy Enforcement	Limited ability to apply standardized policies and governance across environments
Fleet-Wide Operations	Reduced capability to manage distributed environments from a single operational framework
Automated Infrastructure Provisioning	More manual deployment and operational processes
Integrated Networking and Security Automation	Reduced integration across compute, storage, and networking domains
Standardized Cloud Consumption Experience	Limited self-service and automation consistency across environments
Compliance and Configuration Consistency	Increased risk of operational drift between locations
Scalable Multi-Tenancy Models (IT and OT)	Reduced support for centralized tenant governance and consumption such as IT and OT environments
Coordinated Disaster Recovery and Recovery Automation	More manual recovery orchestration across sites

Design Consideration

While this approach simplifies a single deployment and reduces initial infrastructure overhead, it comes at the cost of managing every site differently. Customers must evaluate whether this isolated simplicity outweighs the loss of centralized governance, as this fragmented approach becomes impossible to manage when scaling to thousands of locations without a complete VCF deployment.

In environments expected to scale over time, operational fragmentation, and manual lifecycle management overhead may increase significantly without the centralized automation and governance capabilities of the full VCF platform.

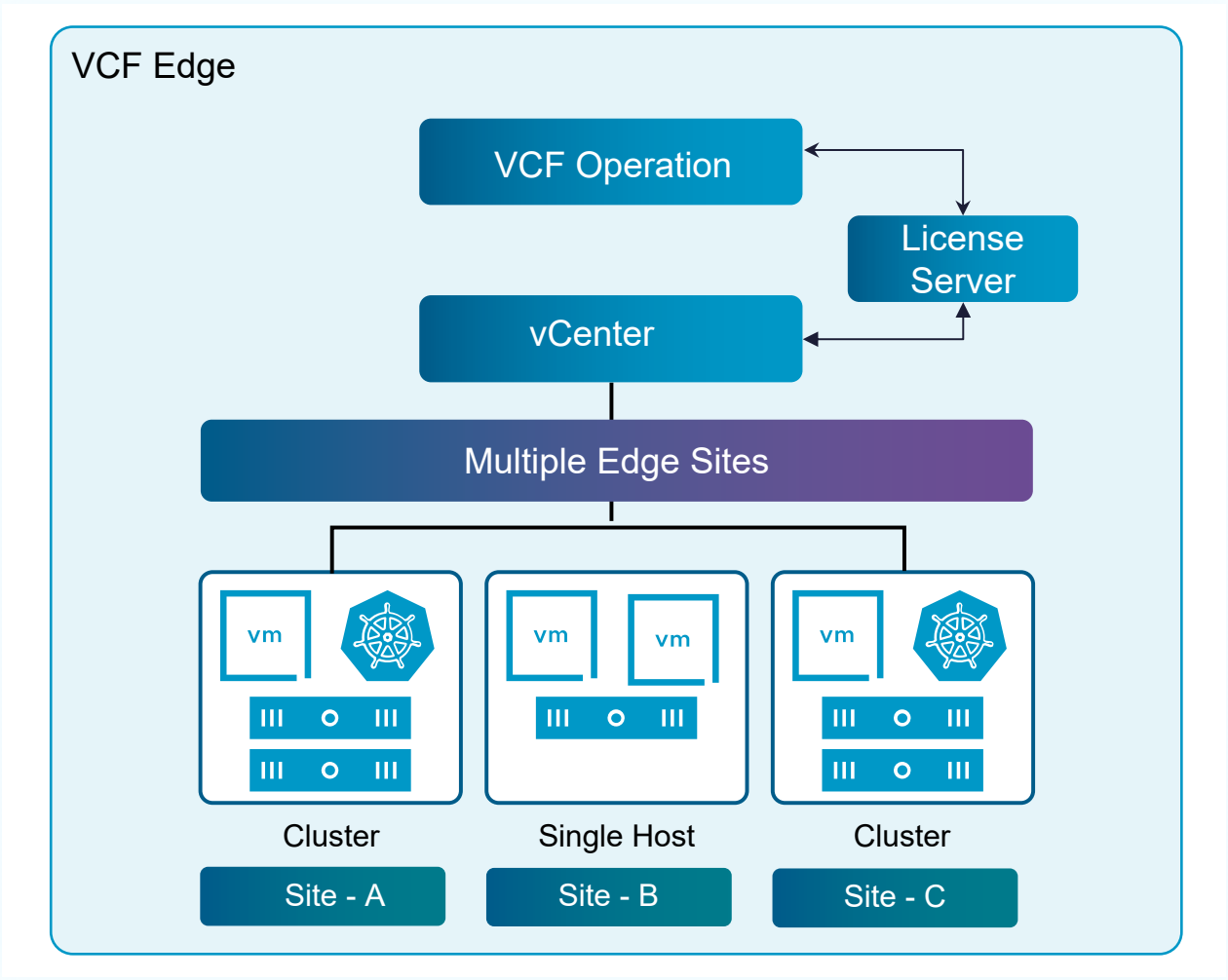
The following design patterns support VCF Edge deployments without requiring the full VMware Cloud Foundation stack.

Design Patterns	Description
VCF Edge Option 1: VCF Operation with vCenter for Distributed Edge	Uses centralized vCenter operations to manage distributed edge locations while maintaining operational consistency across sites.
VCF Edge Option 2: Thin Edge with vSphere Supervisor on Two-Node vSAN	Lightweight edge architecture using a compact two-node vSAN cluster with vSphere Supervisor enabled for Kubernetes workloads at the edge.
VCF Edge Option 3: vSAN Witness Placement for Distributed Edge	Defines placement strategies for vSAN Witness appliances supporting resilient two-node edge deployments.

VCF Edge Option 1: VCF Operations with vCenter for Distributed Edge

A design where a central VCF Operations instance, including one or more centrally located vCenter instances, is used to manage and monitor multiple geographically distributed edge locations.

Figure 370: VCF Edge Option 1a: VCF Operations with vCenter for Distributed Edge



Design Attributes	Details
Life-Cycle Management	Lifecycle management is performed through vCenter.
Hybrid Deployment Models	The design accommodates different deployment scales at the edge, including: • **Multi-Host Clusters:** Providing compute and potentially storage resources across multiple physical hosts for higher availability and capacity (e.g., Site-A, Site-C). • **Single-Host Deployments:** Offering a smaller footprint and potentially lower costs for less demanding edge locations (e.g., Site-B). Only supports for VM-base workload deployments.
Network Connectivity	Latency and Bandwidth: 300 ms for connecting vCenter to ESX hosts. 500 ms required for vSAN witness VM.
Deployment and Licensing	License Server and VCF Operation are mandatory requirements along with vCenter to support this design. **Attention:** Deploy License Server, VCF Operation, and vCenter manually then entitle VCF Edge license with VCF Operation.
Centralized License Check-in	If this design used for a dark site edge, then VCF Operation mandates a periodic license check-in (e.g., every 180 days, even for dark sites via their limited connectivity or alternative methods) to ensure the entire environment, including isolated edge locations, remains compliant with licensing terms and to prevent operational disruptions.

Design Decision – Option 1b: Centralized Operations Extension for Distributed Edge

Option 1b extends the architecture that is introduced in **Option 1a** by adding centralized operational services for distributed edge environments, including centralized logging, password management, and certificate management capabilities.

In this model, **VCF management, operations, licensing, and operational governance services remain centrally managed** across all edge locations, while individual edge sites continue to follow the same hybrid deployment approach defined in Option 1a. This enables organizations to maintain localized edge operations while benefiting from centralized operational visibility, security management, and administrative consistency across distributed environments.

The primary enhancement that is introduced in this design is the addition of centralized operational management services that provide:

- centralized log aggregation and observability
- centralized password and credential management
- centralized certificate lifecycle management
- consolidated operational monitoring and troubleshooting
- unified operational governance across edge environments

When to Choose Option 1b

Option 1b is recommended when organizations require:

- centralized logging and observability across edge sites
- centralized password and credential governance
- centralized certificate lifecycle and trust management
- consolidated operational monitoring and troubleshooting
- centralized security auditing and compliance visibility
- unified operational management for distributed edge environments

This model is particularly beneficial for organizations operating multiple edge sites that require stronger operational governance, centralized security controls, and standardized operational management practices across environments.

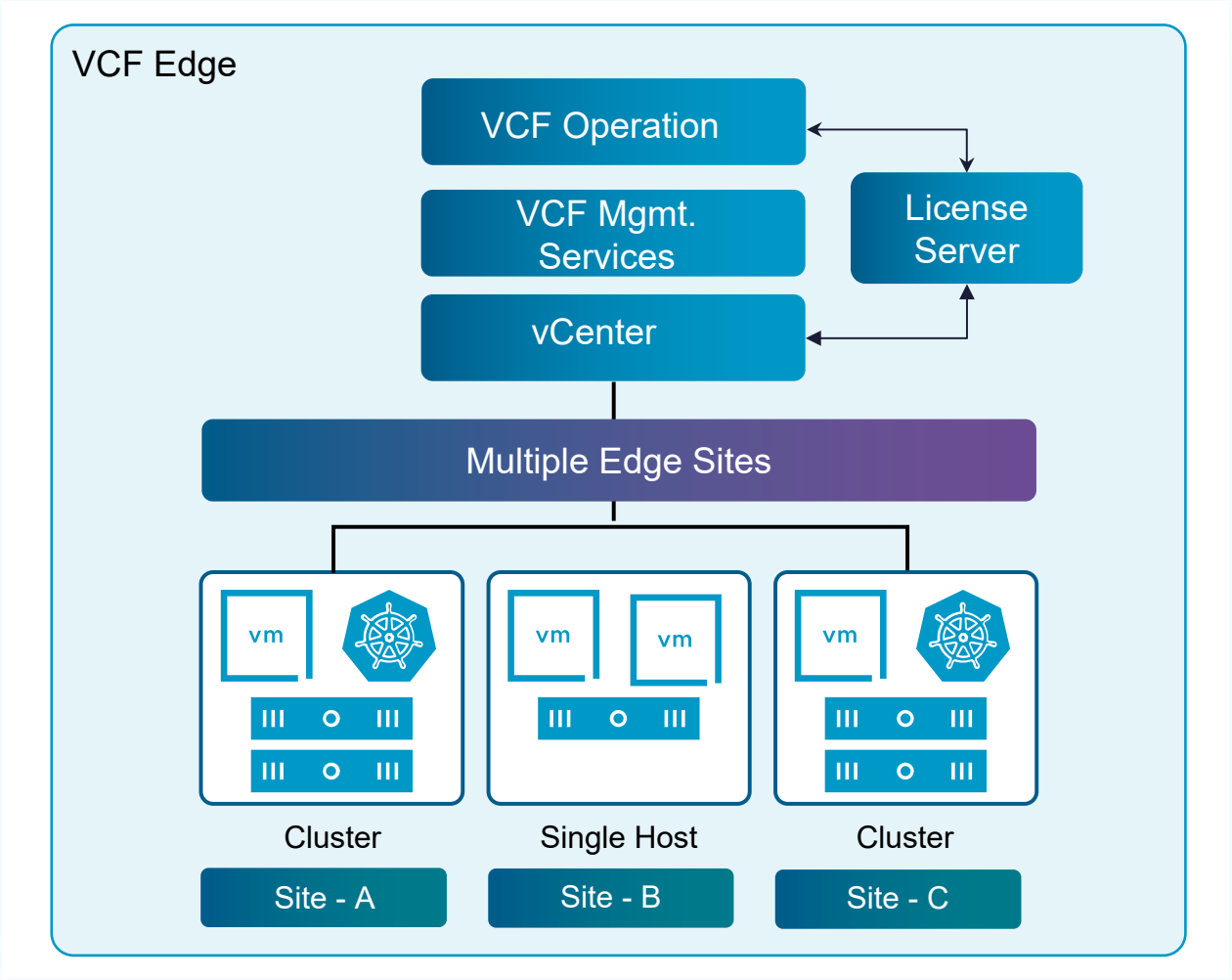
Attention:

Important Design Consideration

Unlike Option 1a, which can be deployed using manually assembled infrastructure components, **Option 1b must be deployed using the VCF installer workflow with VVF** to properly integrate centralized operational services such as logging, password management, and certificate management into the platform architecture.

As a result, this deployment model should not be implemented using the standalone manual deployment approach used in Option 1a.

Figure 371: VCF Edge Option 1b: VCF Operations with vCenter for Distributed Edge with Unified Logging, Password, and Certificate Management

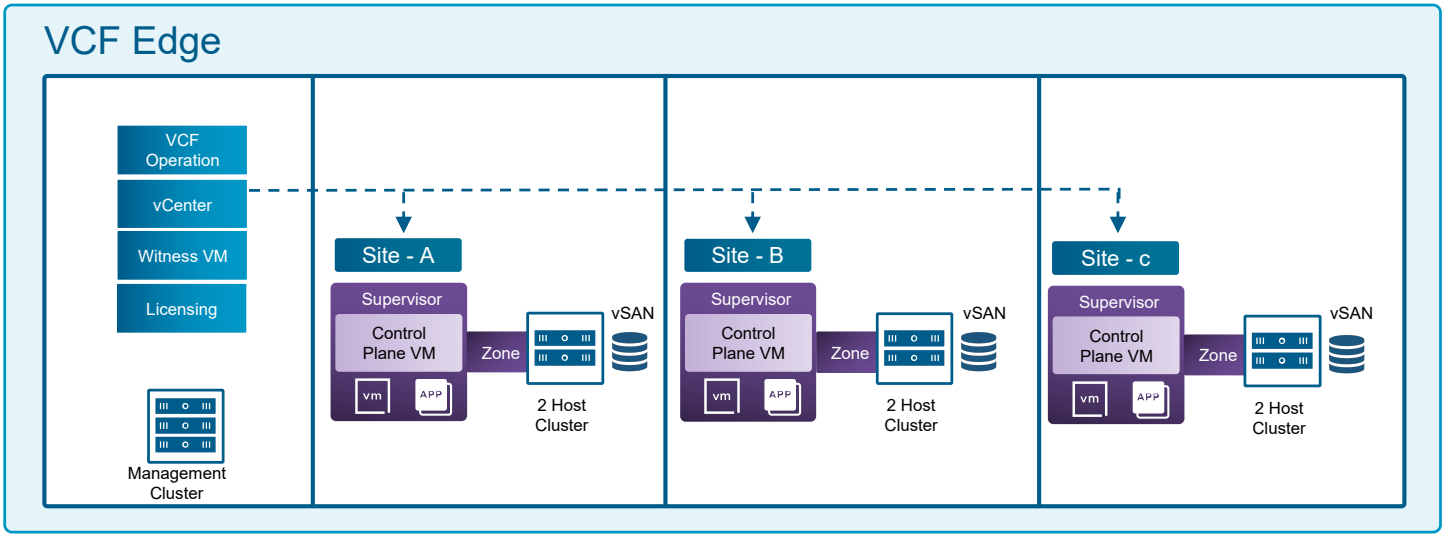


Design Attributes	Details
Lifecycle Management	Lifecycle management is performed through vCenter.
Unified Log Management, Password, and Certificate Management	VCF Management Service provides integration for • Unified log management within VCF Ops providing detailed insights into cluster health, utilization, scale, alongside the health of both VCF and non-VCF component agents.
Hybrid Deployment Models	The design accommodates different deployment scales at the edge, including: • **Multi-Host Clusters:** Providing compute and potentially storage resources across multiple physical hosts for higher availability and capacity (e.g., Site-A, Site-C). • **Single-Host Deployments:** Offering a smaller footprint and potentially lower costs for less demanding edge locations (e.g., Site-B). Only supports for VM-base workload deployments.
Network Connectivity	Latency and Bandwidth: Maximum 300 ms for connecting vCenter to ESX hosts. 500 ms required for vSAN witness VM.
Deployment and Licensing	**Attention:** VCF Management Service is a mandatory requirement along with License Server, vCenter and VCF Ops to support this design. Deploy using VCF installer with VVF option and entitle VCF Edge license with VCF Operation.
Centralized License Check-in	If this design used for a dark site edge, then VCF Operation mandates a periodic license check-in (e.g., every 180 days, even for dark sites via their limited connectivity or alternative methods) to ensure the entire environment, including isolated edge locations, remains compliant with licensing terms and to prevent operational disruptions.

VCF Edge Option 2: Thin Edge with vSphere Supervisor on Two-Node vSAN

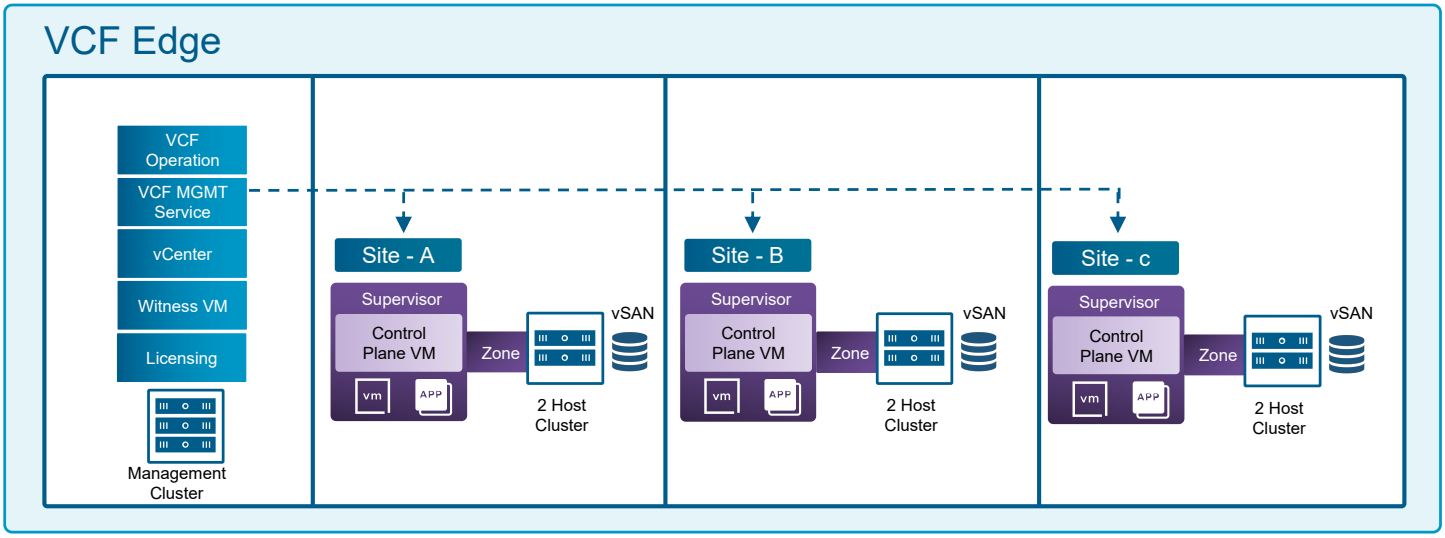
A resource-constrained edge deployment utilizing a small, two-node vSAN cluster to provide local storage for running both traditional VMs and modern containerized applications that are orchestrated by vSphere Kubernetes Service (VKS)

Figure 372: VCF Edge Option 2a: Thin Edge with vSphere Supervisor on Two-Node vSAN



vSphere Supervisor	Supervisor is enabled in the vSphere environment at each edge site (within each 2-node cluster). This enables the deployment and management of Kubernetes workloads alongside traditional VMs.
vSAN for Local Storage	Each 2-node cluster utilizes vSAN for local shared storage.
Control Plane VM at each edge location	Provides the necessary control plane components for managing the Kubernetes environment and potentially other edge-specific services locally at each site. Recommend to have 1 control plane VM on a 2 node vSAN cluster. With 1 CP VM on a two-node cluster, the cluster degrades but functioning if one of the nodes fails. While redundancy is lost, the CP VM remains operational on the surviving node, existing workloads continue running and management operations (though potentially with reduced performance) to still be possible.
Network Connectivity	Latency and Bandwidth: Maximum 300ms / 10Mbps for connection between vCenter to the ESX hosts.
Deployment and Licensing	License Server and VCF Operation are mandatory along with vCenter to support this design. Deploy License Server, VCF Operation, and vCenter manually and then entitle VCF Edge license with VCF Operation.

Figure 373: VCF Edge Option 2b: Thin Edge with vSphere Supervisor on Two-Node vSAN with Unified Software Depot and Logging



Option 2b extends 2a with unified software depot and centralized Logging for edge sites. Design requirements for the VCF managements are identical as 2a. VCF edge services continue to use vSAN for local storage with remote witness and vSphere Supervisor runs on top of the two node vSAN cluster.

Design Requirement	Details
Use Option 2a as a baseline for VCF management and edge infrastructure design	Option 2b builds on top of option 2a by adding a unified depot and logging services for edge
Software Depot, Unified Log Management, password and certificate management	VCF Management Service provides integration for • Unified log management within VCF Ops providing detailed insights into cluster health, utilization, scale, alongside the health of both VCF and non-VCF component agents. • The Fleet Depot service streamlines how VMware binaries from Broadcom are managed. It introduces a standardized authorization method for securely accessing the Broadcom Depot and a single, centralized depot for saving all necessary software. • Password and Certificate management for the Edge vSphere infrastructure.
Network Connectivity	Latency and Bandwidth: Maximum 300ms / 10Mbps for connection between vCenter to the ESX hosts.
Deployment and Licensing	License Server, VCF Operation, VCF Management Services, are mandatory requirements along with vCenter to support this design. Deploy using the VCF Installer with VVF option and then entitle VCF Edge license with VCF Operation.

Note:

A key deviation from Pattern 6 is that this design does not incorporate a full VCF stack, but it does leverage vSAN.

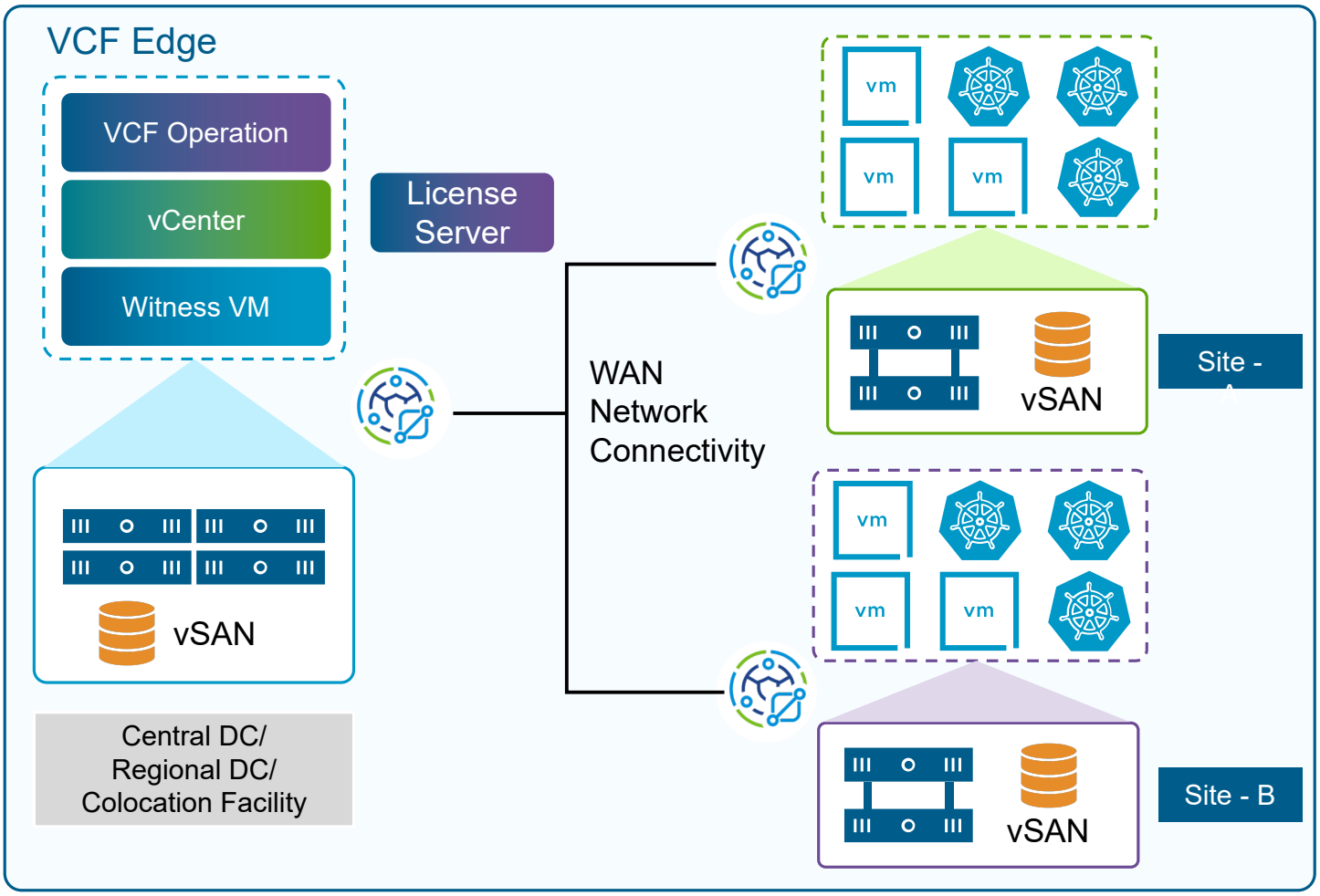
Important:

Based on this design, to deploy and onboard single-host edge sites, including Argo CD integration, customers can use the **VCF Edge at Scale automation script** to simplify and streamline edge deployments.

VCF Edge Option 3: vSAN Witness Placement for Distributed Edge

Strategically locating the vSAN Witness component outside the primary data-bearing edge sites (e.g., in a central data center or in an independent site) to ensure quorum and data availability in 2-node vSAN edge deployments during site failures.

Figure 374: VCF Edge Option 3: vSAN Witness Placement for Distributed Edge



Design Requirement	Details
vSAN Witness VM placement	**Option 1: vSAN Witness VM in the Central DC/Regional DC/ Co-location Facility.** This location offers maximum isolation from the potential failure domains of both edge sites, providing the highest level of resilience. **Option 2: If a Central DC/Regional DC/Co-location Facility is unavailable, implement a cross-site Witness VM configuration.** Hosting Site A's Witness VM on Site B and vice versa provides a degree of resilience against a single site failure when a dedicated third location is not feasible. However, be mindful of potential network connectivity and latency between the sites. **Option 3: Embedded Witness Host - If using embedded hardware specifically designed for a vSAN Witness, consider local hosting at the Edge.** Embedded Witness appliances are often lightweight and designed for remote locations. However, carefully consider the management and potential single point of failure if the entire physical host housing the embedded Witness fails. This option should be evaluated based on specific hardware capabilities and risk tolerance.
Deployment and Licensing	VCF Operation is the mandatory requirement along with vCenter to support this design. Deploy License Server, VCF Operation, and vCenter manually. Alternatively if unified depot service or central logging is desired, using VCF installer with VVF option, entitle VCF Edge license with VCF Operation.
VSAN Witness Licensing	To license vSAN Witness: 1) Attach your vSAN Witness Host or Appliance to the vCenter 2) Assign your vSAN license to the vCenter 3) The Witness host or appliance automatically license. vSAN Witness host cores do NOT consume capacity from your license capacity.
Network Connectivity	Latency and Bandwidth: 300 ms / 10Mbps required for connection between vCenter to the ESX hosts.

Design Options With Full VCF Stack

Customers requiring a fully integrated private cloud operating model across both core data center and edge environments can deploy the complete VMware Cloud Foundation stack. This approach provides a unified operational framework that integrates compute, storage, networking, security, automation, and lifecycle management into a consistent cloud platform architecture.

Importantly, deploying the full VCF stack does not require large-scale infrastructure footprints at every location. Customers still have the flexibility to deploy smaller edge environments using compact configurations such as single-host or two-host edge deployments while continuing to benefit from the broader capabilities of the full VCF platform.

This allows organizations to extend the same operational model, governance framework, and lifecycle management capabilities from centralized data centers to lightweight edge locations without sacrificing consistency or platform functionality.

Key Advantages of Full VCF Stack Deployments

Capability Area	Benefit
Unified Private Cloud Operations	Consistent operational model across core data center and edge environments
Centralized Lifecycle Management	Coordinated patching, upgrades, and lifecycle orchestration across distributed infrastructure
Policy-Driven Governance	Standardized security, networking, and operational policy enforcement
Scalable Edge Flexibility	Supports both large-scale deployments and smaller edge footprints, including single-host and two-host configurations
Centralized Fleet Operations	Enables centralized visibility and management across distributed regions and edge sites
Integrated Networking and Security	Native integration with networking, segmentation, and security services
Multi-Tenant Consumption Models (IT/ OT)	Supports scalable tenant isolation and governance across environments such as IT/ OT.
Unified Observability	Centralized monitoring, logging, telemetry, and operational analytics
Kubernetes and Modern Application Platform Support	Consistent VM and Kubernetes workload operations across core and edge locations
Consistent Operational Experience	Reduces operational drift between data center and edge deployments

Design Consideration

By deploying the full VCF stack, customers gain the flexibility to standardize operations across environments regardless of deployment size. Even in smaller edge deployments with minimal infrastructure footprints, customers can still leverage the broader value of VCF, including centralized governance, lifecycle management, automation, security integration, and unified operations.

This approach enables organizations to maintain a consistent cloud operating model from the core data center to the far edge while adapting infrastructure sizing to the requirements of each location.

The following design patterns support VCF Edge deployments with the full VMware Cloud Foundation stack.

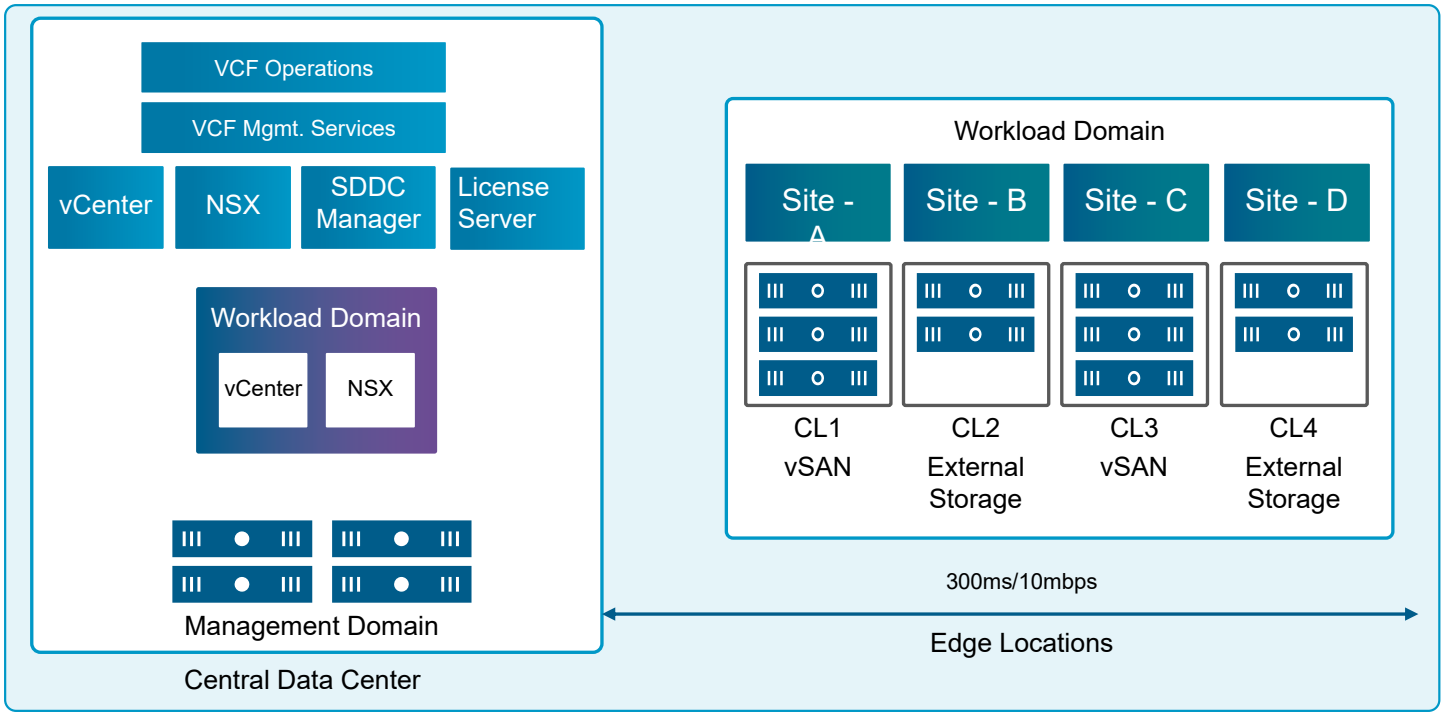
Design Patterns	Description
VCF Edge Pattern 1: Centrally Managed Edge Compute Clusters in a Workload Domain	Edge compute clusters are centrally managed within a VCF workload domain to provide consistent lifecycle and governance operations.
VCF Edge Pattern 2: Segregated Edge Deployment Across Multi-Region Workload Domains	Edge environments are separated across regional workload domains to improve isolation, resiliency, and regional autonomy.
VCF Edge Pattern 3: Locally Managed Edge	Edge locations operate with localized management to reduce dependency on centralized infrastructure and improve site autonomy.
VCF Edge Pattern 4: VCF Fleet with Multi-Region Distributed Edge	Multiple distributed edge regions operate under a shared VCF Fleet model with centralized governance and regional isolation.
VCF Edge Pattern 5: Dark Site Edge	Edge deployment that is designed for disconnected or air-gapped environments with minimal dependency on external services.
VCF Edge Pattern 6: vSphere Supervisor at the Edge in a VCF Instance	Extends Kubernetes platform services to edge environments through vSphere Supervisor integration within a VCF instance.
VCF Edge Pattern 7: Supervisor Zonal Separation at the Edge	Uses zonal separation for Supervisor workloads to improve resiliency and fault isolation across edge environments.

Design Patterns	Description
VCF Edge Pattern 8: Single Host vSphere Supervisor at the Edge in a VCF Instance	Minimal-footprint edge deployment using a single host running vSphere Supervisor services for lightweight or non-critical workloads.
VCF Edge Pattern 9: Leveraging Existing VCF Operations for Far Edge Expansion	Extends existing VCF operational models and tooling to support scalable far-edge deployments.
VCF Edge Pattern 10: Colocated vCenter with ESX Host at the Edge	Deploys vCenter alongside ESX hosts at the edge site to provide localized infrastructure management and operational independence.

VCF Edge Pattern 1: Centrally Managed Edge Compute Clusters in a Workload Domain

VCF Edge manages compute resources centrally for simplified operations and consistent infrastructure across distributed locations.

Figure 375: Centrally Managed VCF Edge Compute Clusters in a Single Workload Domain

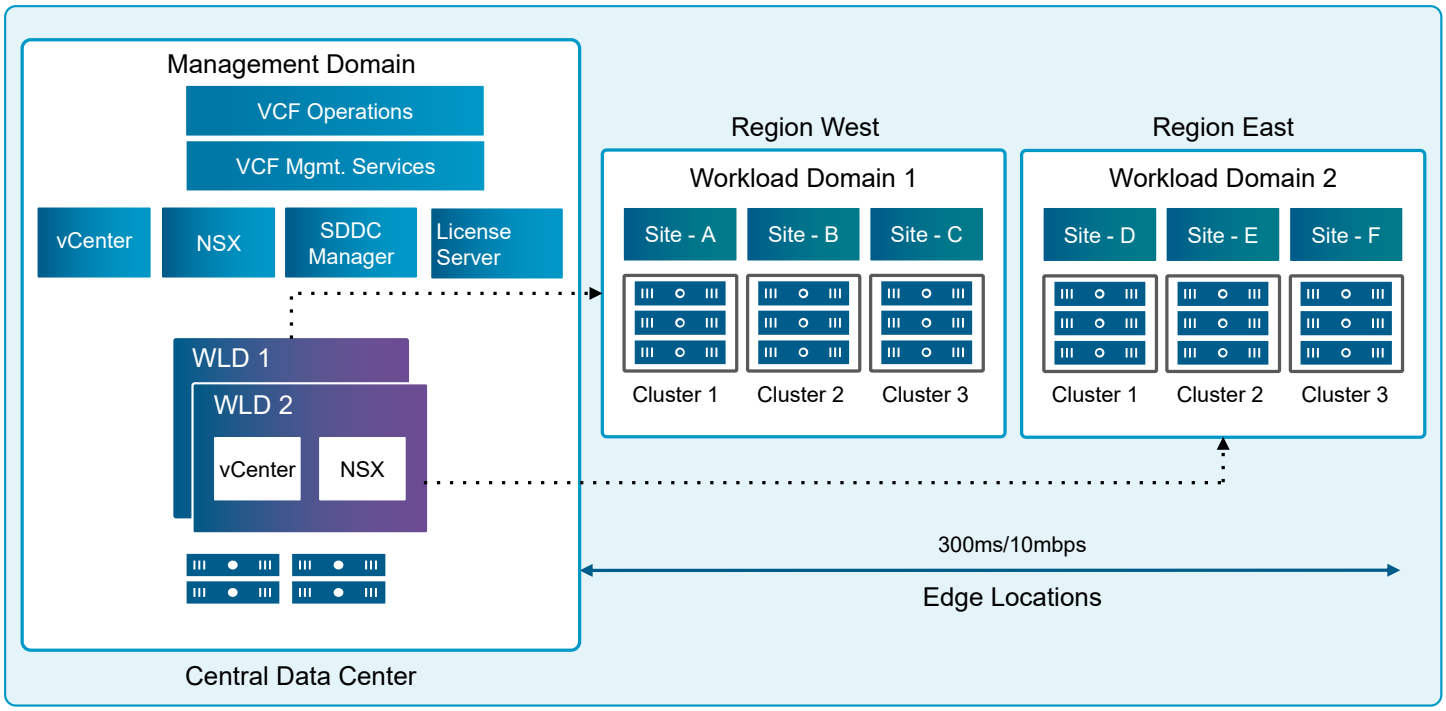


Design Attributes	Details
Storage Options at Edge	• Flexibility to choose vSAN or external storage. • Mixed storage environment with both vSAN and external storage solutions.
Network Connectivity	• 300ms / 10Mbps latency and bandwidth requirement for connecting the central data center to VCF Edge sites. Refer to Latency Details for each component.

VCF Edge Pattern 2: Segregated Edge deployment across Multi-Region Workload Domains

This design describes a multi-region edge architecture with segregated edge sites, each having dedicated workload domains. Each region possesses its own independent vCenter for daily operation.

Figure 376: VCF Edge Pattern 2: Segregated Edge deployment across Multi-Region Workload Domains

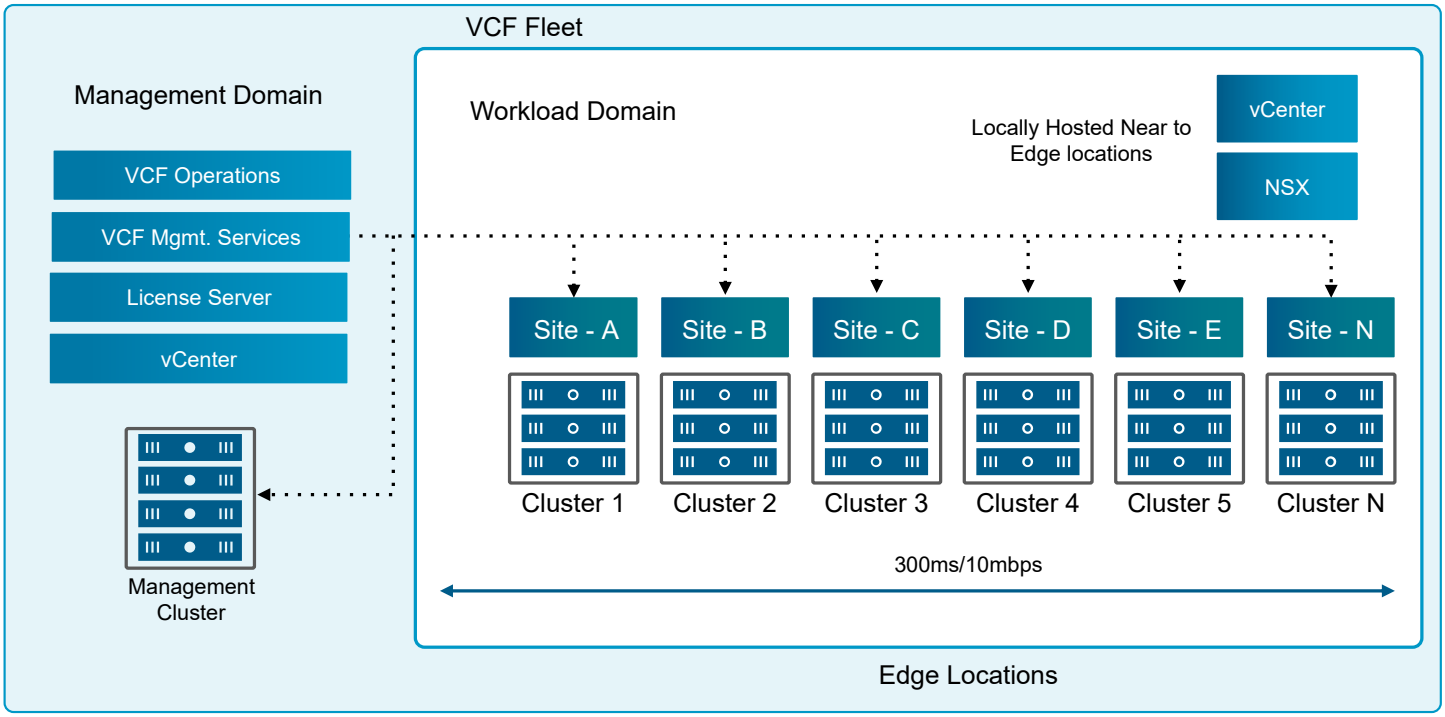


Design Attributes	Details
Compute clusters for multiple sites within regions	Each region is further divided into multiple sites (Site A, B, C in Region West; Site D, E, F in Region East). This suggests a need for localized compute and storage resources at different physical locations.
Network Connectivity	300ms / 10Mbps latency and bandwidth requirement for connecting the central data center to VCF Edge sites. Refer to Latency Details for each component.

VCF Edge Pattern 3: Locally Managed Edge

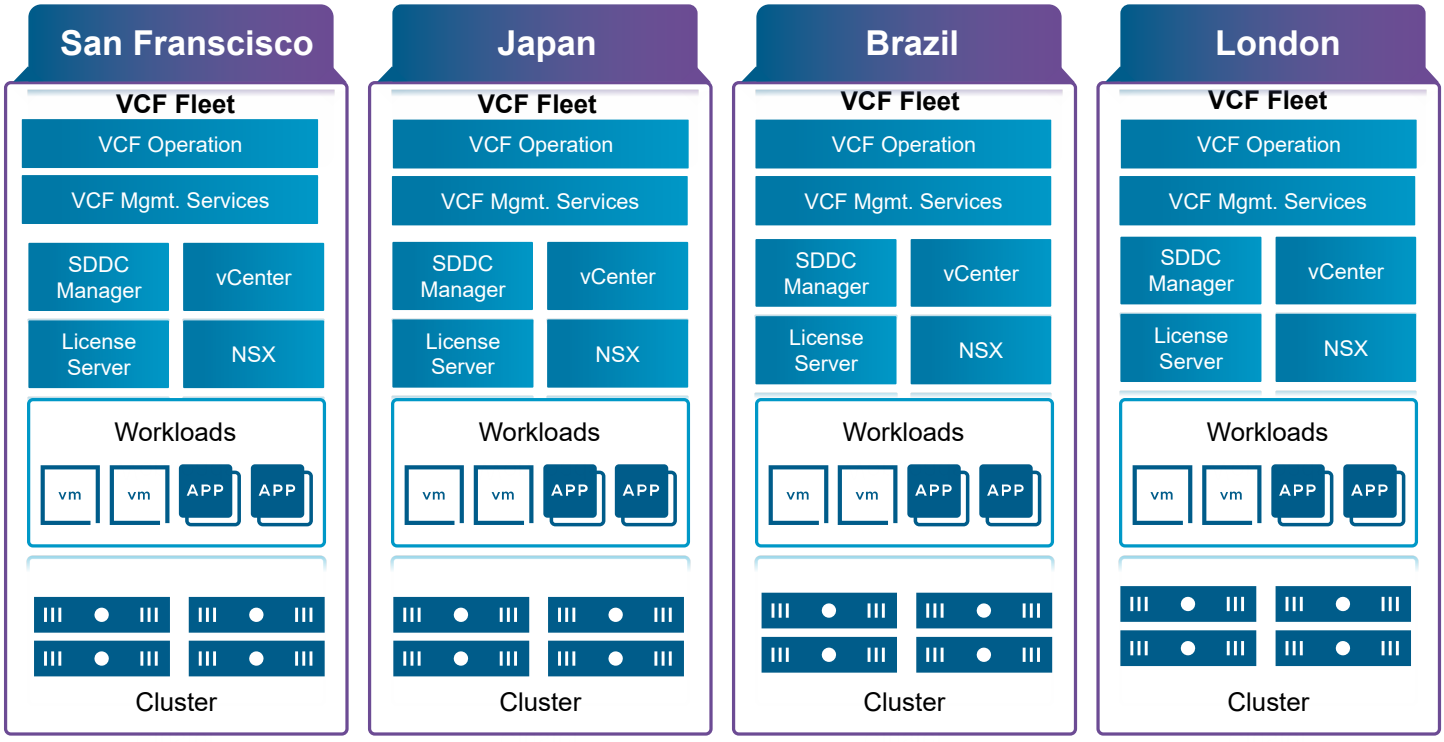
Independent edge compute regions, each with its own local management infrastructure, operating autonomously across a distributed environment.

Figure 377: VCF Edge Pattern 3a: Locally Managed Edge



Design Attributes	Details
Network connectivity	Latency and Bandwidth: Minimum 300ms / 10Mbps, indicating latency and bandwidth requirement for both management and edge workloads. However, due to the close proximity of both management and edge locations, this may not be an issue. If necessary, colocate the vCenter and NSX at the Edge cluster is an option to overcome latency. Refer to Latency Details for each component.

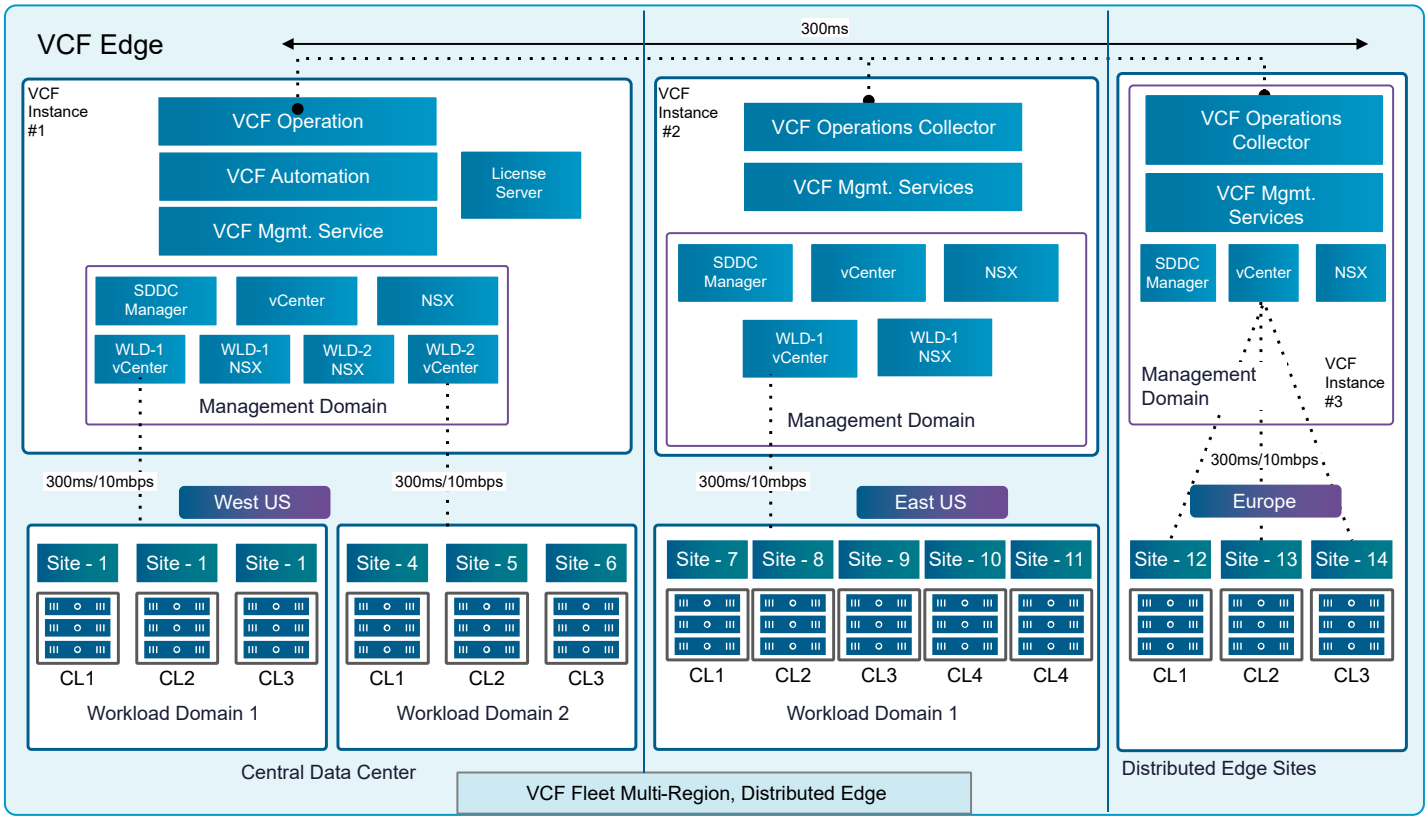
Figure 378: VCF Edge Pattern 3b: Locally Managed Thick Edge



VCF Edge Pattern 4: VCF Fleet with Multi Region, Distributed Edge

A large-scale VCF Edge deployment spanning multiple geographical regions, encompassing numerous distributed edge locations, all managed as a unified VCF fleet for consistent operations and life-cycle management across a wide and diverse infrastructure.

Figure 379: VCF Edge Pattern 4: VCF Fleet with Multi Region, Distributed Edge



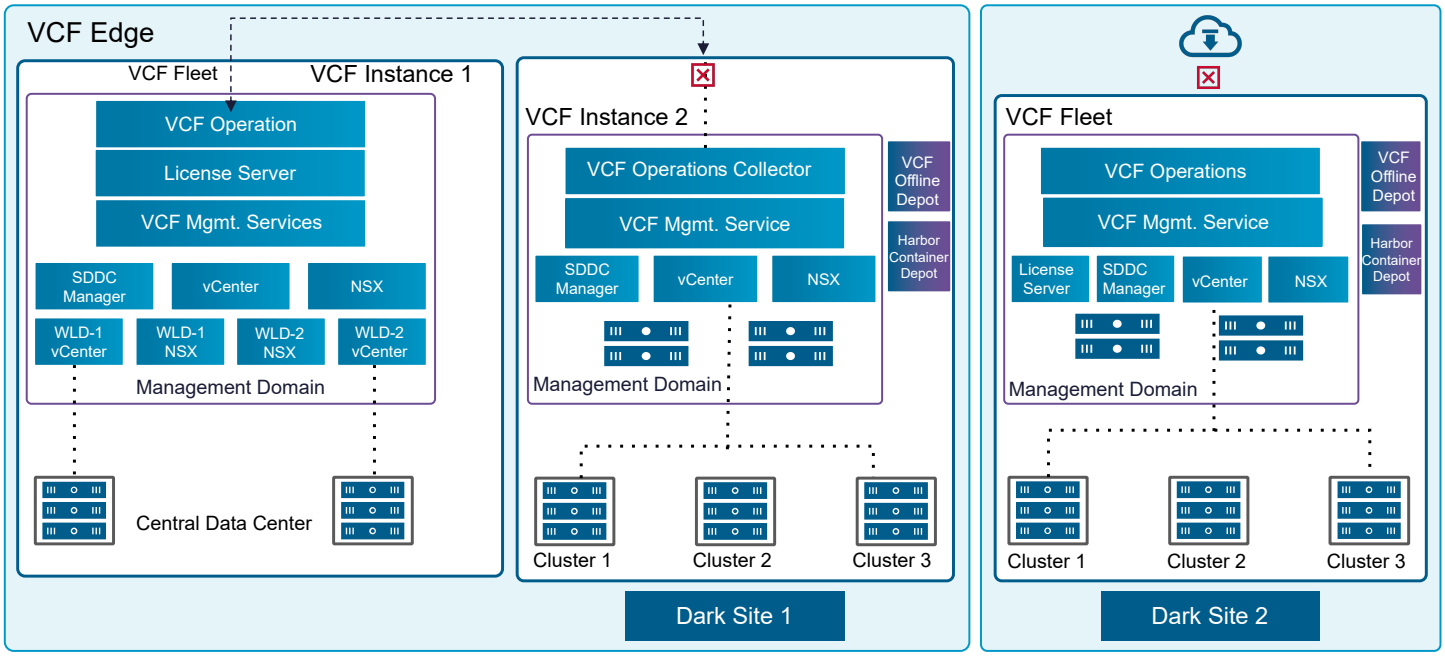
Design Attributes	Justification
Distributed Edge Sites	Extends compute and application capabilities closer to users and data sources in geographically dispersed locations.
VCF Operations Collector (Regional)	Enables localized monitoring and data collection within each region, potentially for performance optimization and troubleshooting, especially given the WAN latency.

Best Fit Scenarios
• Organizations with geographically dispersed users or data sources: This is a fundamental requirement. If your users or data are spread across multiple cities, regions, or countries, VCF Fleet can help you bring applications and services closer to them. • Edge workloads with strict latency requirements: If your applications must respond quickly, VCF Fleet's distributed edge architecture can help minimize latency by deploying workloads closer to the source of the data or the end user. • Requirements for high availability and disaster recovery: VCF Fleet's multi-region capabilities enable you to deploy applications across multiple locations, ensuring that they remain available even if one location experiences an outage. • Need for centralized management and consistent operations at scale: If you have a large number of vSphere deployments to manage, VCF Fleet simplifies management by providing a single point of control and enabling you to enforce consistent configurations and policies across all locations.

VCF Edge Pattern 5: Dark Site Edge

Dark Site Edge describes a VCF Edge deployment that is designed for environments with limited or no reliable external network connectivity.

Figure 380: VCF Edge Pattern 5: Dark Site Edge



Design Attributes	Details
Local Compute Clusters	VCF Instance 2 manages its own compute clusters. These clusters provide the necessary compute resources to run workloads within the isolated dark site. VCF Instance 3 with own VCF fleet, dedicate to the site and manages its own compute clusters.
VCF Operations Collector	In the case of Dark Site 1, Data collected locally during offline mode and synchronized with VCF Operations, when connectivity is established with the central site. here might be periodically exported or analyzed locally. In the case of Dark Site 2, Enables monitoring and potentially local operational insights within the dark site, even without direct connection to an outside world. Data collected here monitored and analyzed locally with a dedicated VCF Operation.
VCF Offline Depot	A local VCF Offline Depot is deployed within the dark site. In the absence of direct Internet connectivity, the Offline Depot serves as a local repository for software bundles, patches, and upgrades required for the VCF Instance and its components. This ensures the lifecycle management of the dark site's VCF environment.
Harbor Bootstrap Registry	For Dark Sites that require vSphere Kubernetes Service, a bootstrap container registry to stage the initial components is deployed. Those components are then used to spin up a production-grade Harbor registry operating as a native Supervisor Service. Bootstrap harbor deployments steps can be based on upstream Harbor or Bitnami Virtual Harbor Appliance. Load Carvel Packages and Push the Harbor Carvel packages and their dependencies into the bootstrap container registry.
License Check-in	VCF Operation mandates a periodic license check-in (e.g., every 180 days, even for dark sites via their limited connectivity or alternative methods) to ensure the entire VCF environment, including isolated edge locations, remains compliant with licensing terms and to prevent operational disruptions.

Note:

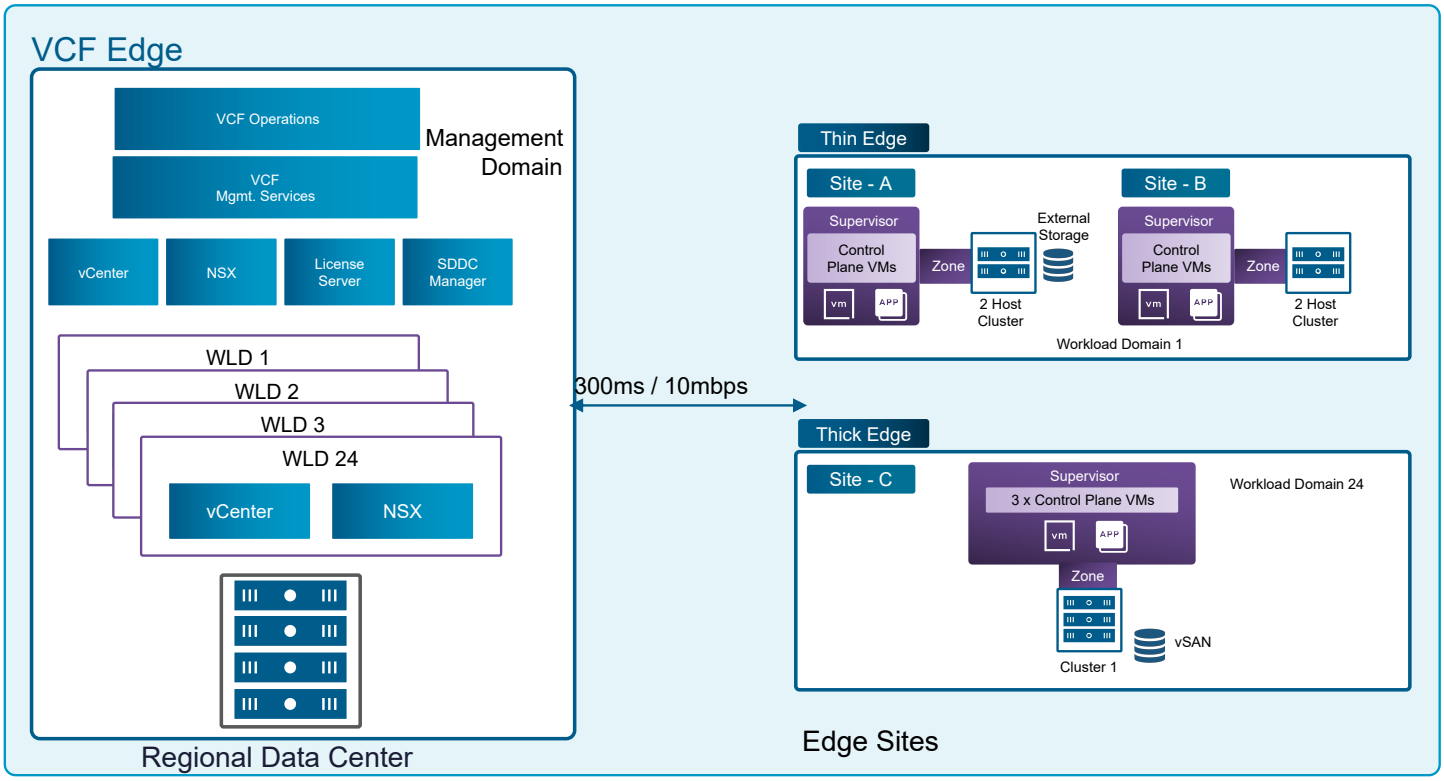
If you are entitled for VCF Edge, you have the option to deploy a dark site edge using a combination of VCF Operations and vCenter only. Refer to the **other flexible design** options available and select the one that best fits your requirements.

Best Fit Scenarios
• Strict Isolation and Security Requirements • Unreliable or Intermittent Network Connectivity • Autonomous Operations Requirement - Situations where the edge location must function independently for extended periods without external dependencies, including management, monitoring, and software updates.

VCF Edge Pattern 6: vSphere Supervisor at the Edge in a VCF Instance with Multiple Domains

This enables running Kubernetes workloads alongside traditional VMs at the edge, leveraging consistent VCF management and infrastructure.

Figure 381: VCF Edge Pattern 6: vSphere Supervisor at the Edge in a VCF Instance with Multiple Domains

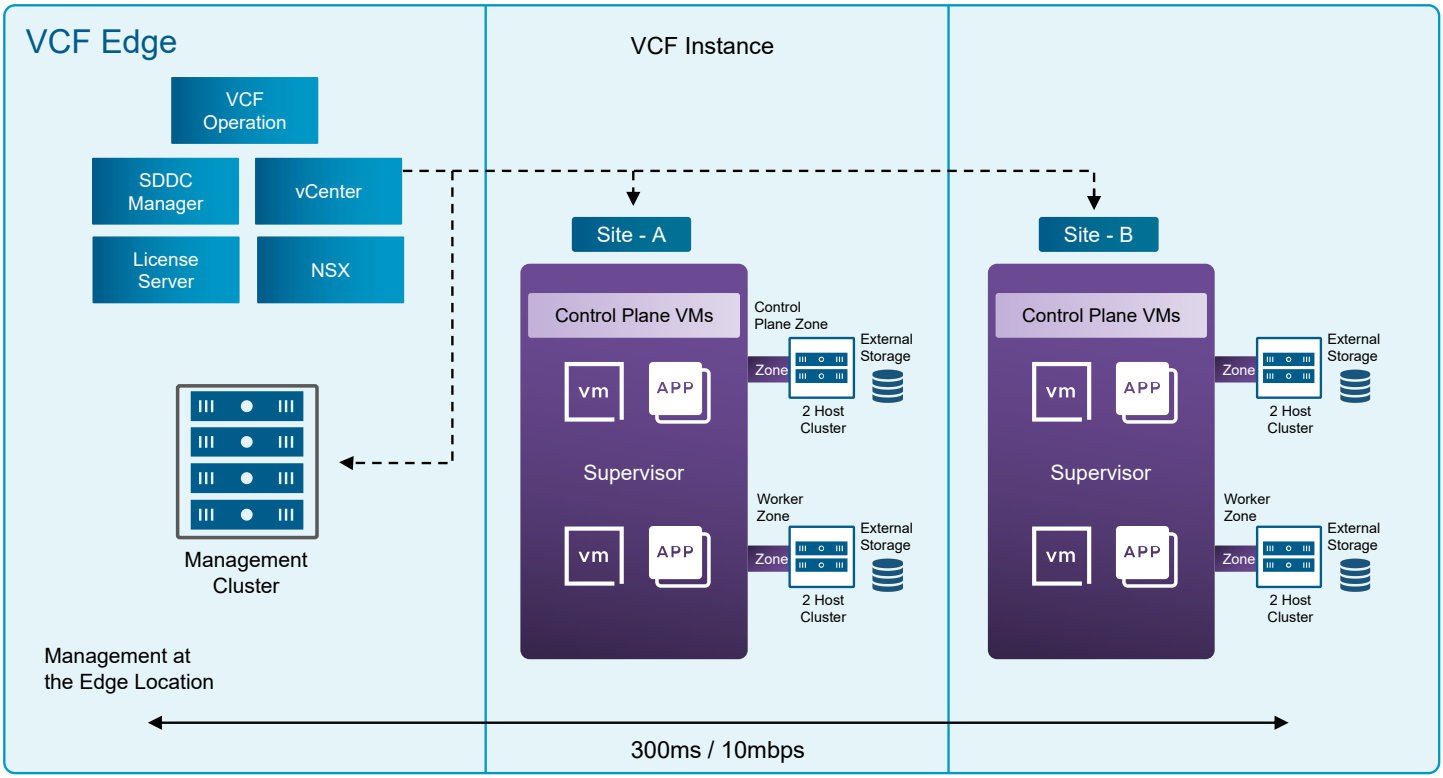


Design Attributes	Details
Thin edge	2-host cluster in a workload domain minimizes the hardware footprint and cost for edge locations with limited resources.
Thick edge	Thick edge suggests a more substantial cluster with more resources. Supports more demanding workloads and provides greater local compute and storage capacity.
Supervisor cluster	Supervisor is enabled in the vSphere environment at each edge site. Enables the deployment and management of workloads in a declarative manner through Supervisor services alongside traditional virtual machines (VMs) directly at the edge, supporting modern, containerized applications.

VCF Edge Pattern 7: Supervisor Zonal separation at the Edge

Designing edge locations with vSphere Supervisor into distinct logical zones (e.g., Control Plane Zone, Worker Zone) for improved resource isolation, security, and resilience of the workloads.

Figure 382: VCF Edge Pattern 7: Supervisor Zonal separation at the Edge



Design Attributes	Details
Supervisor Cluster	Supervisor is enabled in the vSphere environment at each edge site. Enables the deployment and management of Kubernetes workloads alongside traditional virtual machines (VMs) directly at the edge, supporting modern, containerized applications.
Zonal Separation	Each edge site is further divided into Control Plane Zone and Workload Zone. Provides logical isolation and resource separation for critical control plane components and application workloads within each physical site, enhancing stability and security. The Supervisor's Control Plane VMs reside within the dedicated "Control Plane Zone" at each site. Virtual machines and applications running workloads and acting as Kubernetes worker nodes reside within the dedicated "Worker Zone" at each site.
2 Host Cluster with External Storage	Both Control Plane and Worker Zones at each site running on a separate 2-host cluster with external storage.

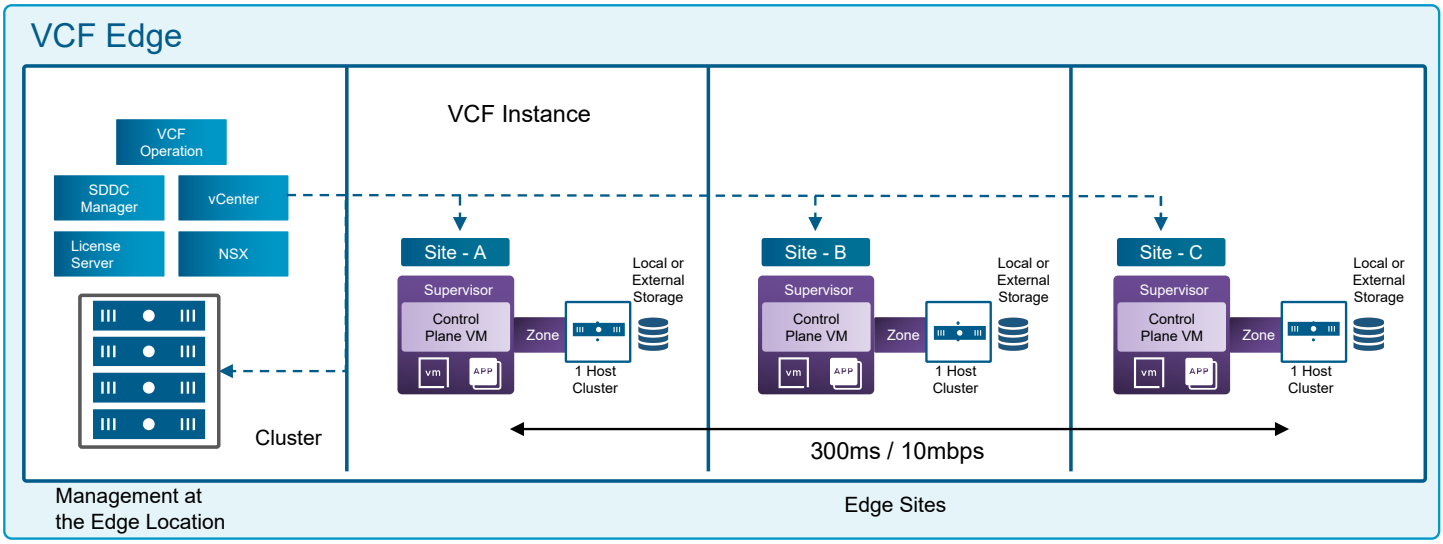
Best Fit Scenarios
Organizations Requiring Strong Local Control and Security: Environments where strict local control over the infrastructure and data is paramount, and minimizing external dependencies is a key requirement.

Important:

This design can be implemented with a VCF Instance with a single workload domain or multiple domain and clusters inside the workload domain as described in Pattern 6 above.

VCF Edge Pattern 8: Single Host vSphere Supervisor at the Edge in a VCF Instance

Figure 383: VCF Edge Pattern 8: Single Host vSphere Supervisor at the Edge in a VCF Instance



This design pattern focuses on deploying a highly efficient and compact vSphere Supervisor footprint at remote or edge locations. A single host leverages vSphere Supervisor to bring Kubernetes capabilities directly to resource-constrained environments, while maintaining centralized management from a core VCF instance.

Important:

Based on this design, to deploy and onboard single-host edge sites, including vSphere Supervisor and ArgoCD, customers can use the **VCF Edge at Scale automation script** to simplify and streamline edge deployments.

Design Attributes	Details
Single ESX Host per Site	Each edge location is built upon a single physical ESX server. Emphasizing a minimal hardware footprint.
vSphere Supervisor	vSphere Supervisor is enabled on the host which enables the deployment and management of Kubernetes workloads directly on the ESX host. This includes the deployment of a "Control Plane VM" for the Supervisor functionality.
Centralized VCF Management	A VCF Fleet (comprising License Server, VCF Operation, VCF Automation, vCenter, and NSX) resides at a central data center or management location. Central vCenter instance manages the distributed edge sites.
Single Cluster Host Deployment	It is a manual process within the vCenter to create a cluster and add the host.
vSphere Supervisor Network	vSphere Distributed Switch (vDS) is used for vSphere Supervisor management and workload network
HA Admission control	vSphere HA is enabled on the cluster however, to make vSphere Supervisor to work, we have disabled HA Admission control in the cluster level in vCenter.
Storage Options	Only shared storage or local storage supported.
Network Connectivity	• Bandwidth**:** Aim for 1 Gbps or higher for efficient VM or container image transfer. A minimum of 10 Mbps might be acceptable for smaller images or less frequent deployments, but it is slow. • Latency: Keep latency below 300 ms for a smooth management experience. While deployments might complete with higher latency, the user experience degrades.

Important:

To support single node vSphere Supervisor deployment, you must disable HA Admission Control on the vSphere Cluster in vCenter.

Best Fit Scenarios
• Numerous Far Edge Locations: Such as retail stores, bank branches, quick-service restaurants, remote industrial sites, or cellular towers, where space and IT staff are limited. • Resource-Constrained Environments: Locations where deploying multiple servers or complex networking is not feasible due to physical limitations or cost. • Applications Requiring Ultra-Low Latency: Workloads like IoT data ingestion and real-time analytics, AI/ML inference at the edge, or local POS systems that demand immediate responses. • High Volume of Small Workloads: Deployments where many small, independent applications must run close to the data source or end user. • Simplified Operations for Distributed Deployments: Organizations looking to standardize and automate the deployment and ongoing management of a large fleet of edge sites from a central location.

Benefits	Implications
Minimized Hardware Footprint: Significantly reduces the amount of physical hardware needed at each edge site. **Lowers CapEx** (capital expenditure) for hardware, power, cooling, and space. **Reduced Operational Overhead:** Simplifies local IT management at the edge due to fewer components. Leverages centralized VCF management for automated lifecycle operations (upgrades, patching), reducing the need for on-site IT staff. **Optimized for Edge Workloads:** Enables deployment of cloud-native (Kubernetes) and traditional VM workloads directly at the edge. Facilitates ultra-low latency processing and data ingestion close to the source.	**Limited High Availability/Resilience:** A single ESX host means a single point of failure. If the host goes down, all workloads at that specific edge site become unavailable. Requires robust strategies for application-level resilience or rapid recovery from a central location. **Resource Constraints:** The single host limits the total compute, memory, and storage capacity available at the edge site. Careful sizing and workload placement are critical to avoid resource contention. **Storage Dependency:** Reliance on local storage or direct-attached storage means storage failure directly impacts the site. **Connectivity Criticality:** The WAN link to the central VCF Management Cluster is crucial for management, monitoring, and lifecycle operations. Latency and bandwidth limitations on the WAN as described in the design requirement **No vMotion for Local HA:** With only one host, vMotion cannot be used for high availability within the edge site itself. If a host fails, VMs cannot automatically migrate to another host.

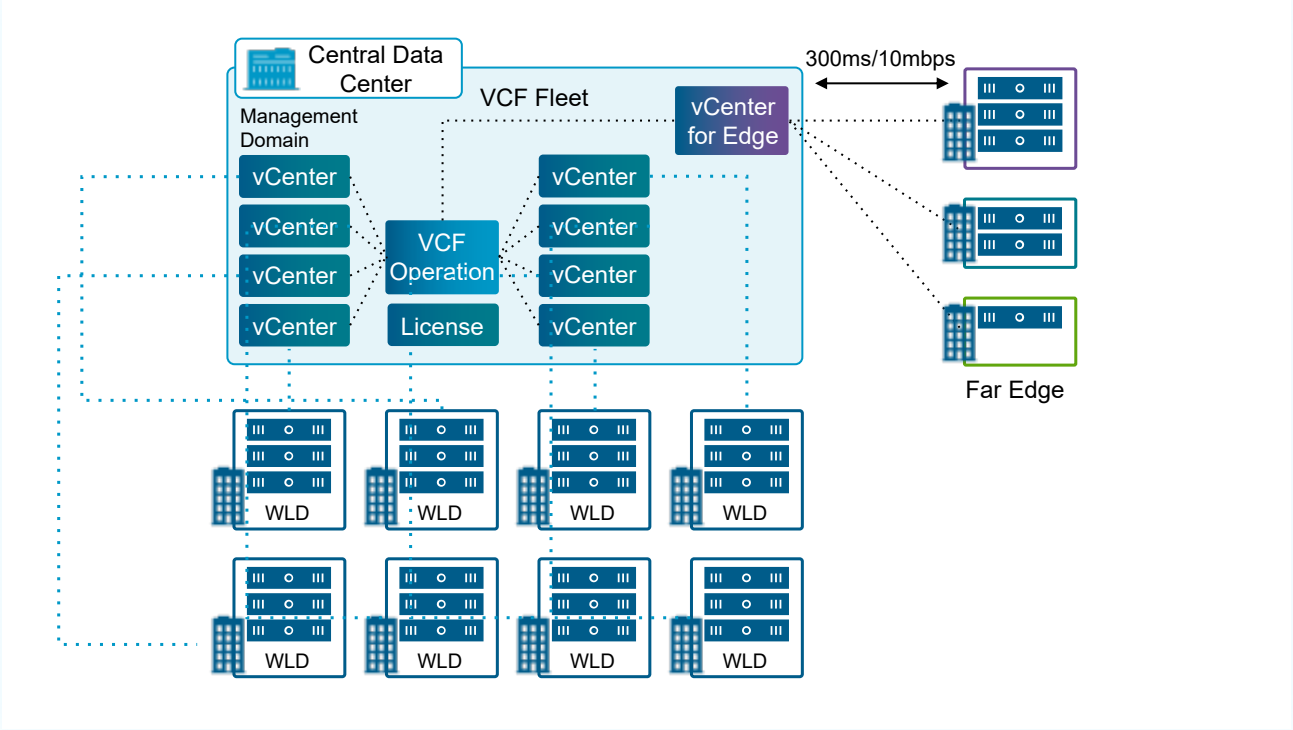
Important:

For multi-node VKS clusters deployed at the edge, the resilience of even localized workloads depends on central vCenter management. Should a Kubernetes pod crashes or its VKS worker node fails, the Kubernetes controller automatically attempts to restart the pod. However, if a pod utilizes a persistent volume and needs to be restarted on an alternative VKS worker node, the underlying vSphere Supervisor must establish communication with vCenter Server to facilitate the necessary volume detach and attach operations between the nodes. **Therefore, maintaining dedicated and stable connectivity to vCenter Server is paramount.** Without this vital connection, the vSphere Supervisor is unable to coordinate with vCenter to provision resources or orchestrate the recovery of the pod, ultimately preventing the affected pod service from coming back online.

VCF Edge Pattern 9: Leveraging Existing VCF Operations for Far Edge Expansion

This design outlines how customers can extend their existing VCF Fleet to far-edge locations with VCF Edge, leveraging VCF Operations for unified license management and monitoring.

Figure 384: VCF Edge Pattern 9: Leveraging Existing VCF Operations for Far Edge Expansion


Central Data Center
VCF Fleet
Management Domain
vCenter
vCenter
vCenter
vCenter
VCF Operation
License
vCenter
vCenter
vCenter
vCenter
vCenter for Edge
300ms/10mbps
Far Edge
WLD
WLD
WLD
WLD
WLD
WLD
WLD
WLD

Design Attributes	Details
vCenter for Far Edge locations	A vCenter can be deployed in an existing data center or co-location facility for far-edge locations and can be connected to VCF Operations for license management. However, VCF Operation will not perform lifecycle management of vCenter and hosts which are connected to this vCenter.
Lifecycle Management of vCenter and Edge Hosts	In this scenario, the lifecycle management of edge site hosts must be handled by vCenter for VCF Edge.

VCF Edge Pattern 10: Colocated vCenter with ESX Host at the Edge

The standard VMware Cloud Foundation deployment model places the Management Domain along with the Workload Domain vCenter and NSX components in a centralized data center, while edge-site ESX hosts are managed remotely from that central location. This model is well suited for environments where latency between the management plane and managed infrastructure remains within supported operational limits and where edge locations do not require independent operational control.

This design addresses deployment scenarios where edge environments require greater operational locality or autonomy. Typical use cases include:

- regional hub architectures where a locally colocated vCenter manages nearby edge sites within supported latency boundaries, while the central data center may reside at higher WAN latency
- environments with data residency, sovereignty, or security requirements requiring localized management control
- intermittently connected or disconnected edge environments requiring local operational continuity during WAN outages
- existing independently managed edge vCenter environments that need to be brought under centralized VCF governance

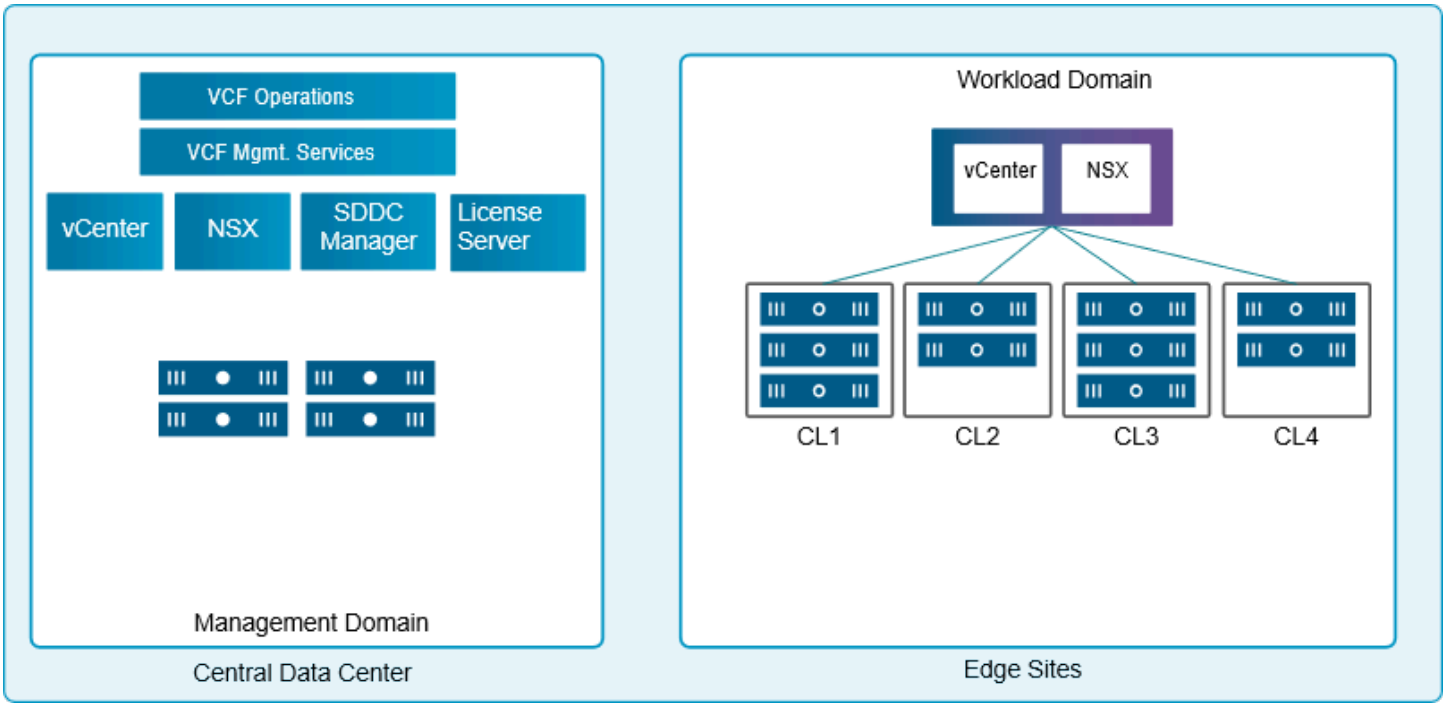
In these scenarios, a locally colocated workload domain vCenter can be deployed at the edge or regional site and imported into VCF as a workload domain. This approach does not eliminate overall VCF latency and support requirements between platform components; rather, it localizes operational control and management functions closer to the managed infrastructure and applications.

This pattern supports two implementation approaches:

- **Option A:** deploy a new vCenter locally at the edge or regional site before performing a brownfield import into VCF
- **Option B:** import an existing locally managed vCenter already operating the edge ESXi hosts

Both approaches result in a VCF-governed workload domain with a locally colocated vCenter. The design also preserves flexibility in networking architecture, allowing customers to either deploy NSX locally at the edge site or continue using centralized NSX management services depending on operational and connectivity requirements.

Figure 385: Colocated vCenter with ESX Host at the Edge



Design Attributes	Details
Centralized Management Domain	A central data center hosts the management components, including vCenter, NSX Manager, VCF Operation Manager, and SDDC Manager. This suggests centralized control and administration of the entire environment.
Workload Domains vCenter at the Edge	• **Local control plane:** each edge site that requires colocated management deploys a separate vCenter Server Appliance (VCSA) on-site, colocated with the edge ESX cluster. This eliminates WAN dependency for all vSphere control plane operations at that site. • **Workload Domain scope:** the edge vCenter manages only the ESX hosts at its local edge site or nearby edge sites where latency is not a concern.
Network Connectivity	Typical latency between SDDC manager and vCener must be less than 100ms. Refer to Latency Details for each component.

Deployment Path Options

Option A — New vCenter (Greenfield)	Option B — Existing vCenter (Brownfield)
• No prior vCenter at the edge site • Deploy a fresh vCenter colocated with edge ESX hosts • Perform brownfield import of the new vCenter into VCF • Add edge ESX hosts to • Recommended for: new edge deployments	• Existing vCenter already manages edge ESX hosts independently of VCF • Import the existing vCenter into VCF 9 as a Workload Domain vCenter • No vCenter redeployment - existing vCenter is retained in place • Existing VM workloads preserved - no maintenance window required for VMs • Recommended for: edge sites with existing production workloads and an independently managed vCenter

Private AI Platform Detailed Design

Selecting the right AI infrastructure deployment model for VMware Private AI Foundation begins with understanding the specific use case requirements. The decision tree starts with what type of AI workload is your organization planning to run. If the use case falls squarely within the realm of large language models, such as building conversational AI applications, implementing retrieval-augmented generation systems, creating data indexing and retrieval solutions, developing MCP servers, or generating content, then Private AI Services is the optimal choice.

Architectural Options

Private AI Services is purpose-built for LLM applications and provides a fully managed platform where application developers can work through an intuitive UI to build knowledge bases, and AI agents leveraging model endpoints made available by the cloud administrator without needing any knowledge of Kubernetes or virtual machine management. The platform abstracts away the complexity of infrastructure management and delivers an end-to-end workflow that spans from model selection through knowledge base integration to agent deployment, while providing production-ready monitoring and governance. However, it's critical to understand that Private AI Services is specialized for LLM and RAG use cases and cannot support workloads like computer vision, time-series analysis, optimization problems, or audio processing models.

For organizations whose AI workloads extend beyond LLMs or require the flexibility to support diverse model types, the choice narrows to either VKS Clusters or Deep Learning VMs. VKS Clusters are Kubernetes-based environments designed for scalable AI and ML workloads where multiple data scientists or ML engineers need to share GPU resources, run distributed training across multiple physical hosts, or deploy production inference services that require high availability and scalability. These clusters support any type of AI workload, from computer vision and time-series forecasting to LLM fine-tuning and audio processing, making them the most versatile option. The trade-off for this versatility is complexity, users need at least a basic understanding of Kubernetes concepts, and while the VCF Automation catalog can simplify deployment by pre-installing the NVIDIA GPU operator and providing form-based configuration, the underlying platform still requires more technical sophistication than other options. VKS Clusters are particularly well suited for ML engineers and platform teams building production-grade AI services, especially when workloads need to scale horizontally or when distributed inference and training capabilities are required.

In contrast, Deep Learning VMs are designed for individual data scientists who need a personal GPU-accelerated workstation for exploratory work, model development, and prototyping. These VMs provide the fastest path to productivity because they come with a pre-built Ubuntu image that includes all the essential components, Docker, ML package managers, popular frameworks like PyTorch and TensorFlow, the NVIDIA Container Toolkit, and monitoring tools like DCGM. The vGPU driver installation and the vGPU licensing happen automatically and matches the host driver version, eliminating a common source of configuration errors. While DLVMs can be deployed through multiple methods, including the VCF Automation catalog, vSphere Client from OVA or content library, or kubectl, the catalog deployment provides the simplest experience, where data scientists simply select their desired software bundle and get a fully configured workstation within minutes. For the vast majority of data science work involving model experimentation, fine-tuning, and development, DLVMs provide the ideal balance of capability and simplicity. It is possible to validate ML models on DLVMs against the security, privacy, and technical requirements of your organization, then upload the models to a dedicated model gallery for use by AI application developers or MLOps engineers for automated CI/CD-based deployment of model runtimes.

Private AI Services Model

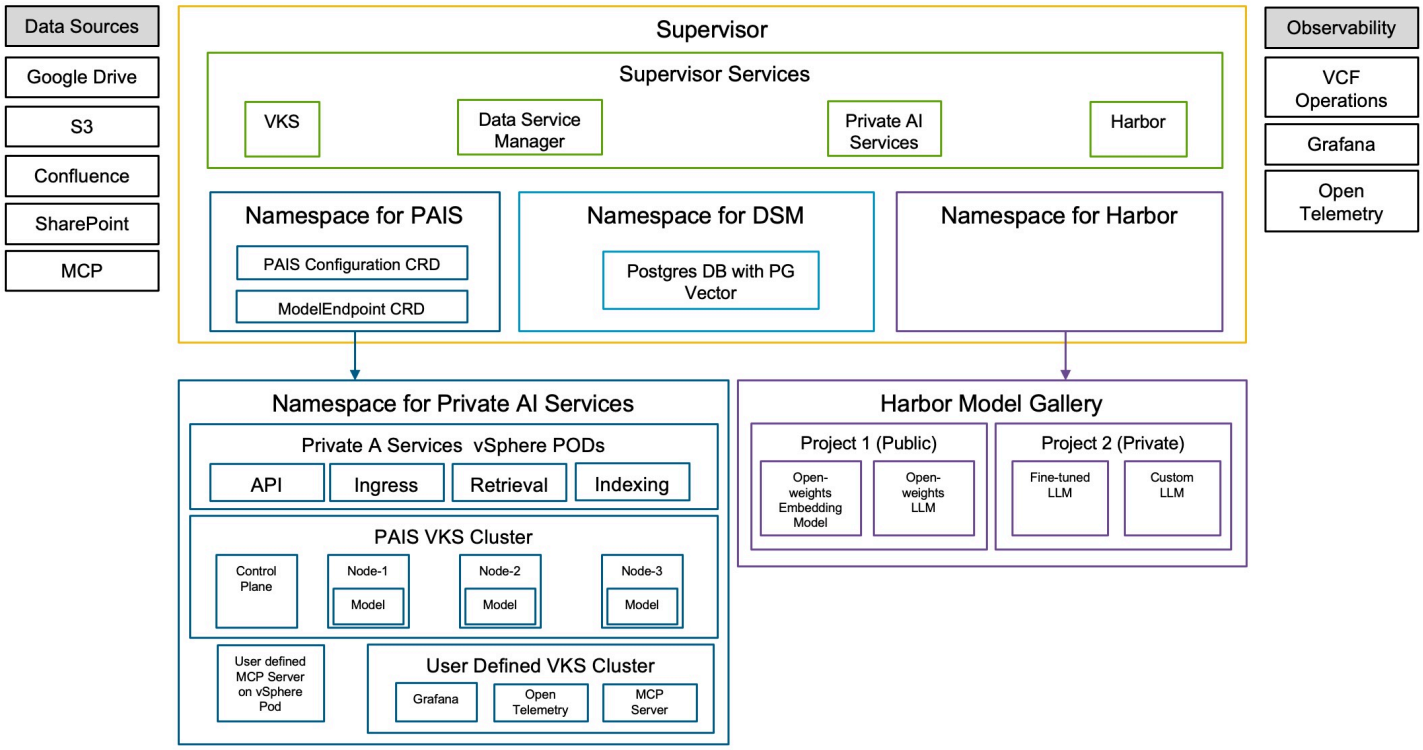
VMware Private AI Services provide an end-to-end platform for deploying generative AI applications within the VMware Private AI Foundation environment.

The architecture consists of four core modules: Model Gallery, Model Runtime, Data Indexing and Retrieval, and Agent Builder.

The Model Gallery utilizes the Harbor container registry as the central repository for AI model artifacts. Models stored in Harbor are consumed by the Model Runtime module, which executes both completion models and embedding models through inference engines such as vLLM, llama.cpp, and Infinity. Model endpoints expose these models via OpenAI-compatible APIs, with completion models generating text responses and embedding models producing vector representations for semantic search operations.

The Data Indexing and Retrieval module connects to an external PostgreSQL database with the pgvector extension, to store and retrieve vector embeddings. This component indexes data from multiple sources including Google Drive, Confluence, and SharePoint, and S3 splitting documents into configurable text chunks using configurable splitting strategies. The embedding model transforms these chunks into vectors that are stored in the external PostgreSQL database, enabling similarity search with configurable top-k retrieval and similarity thresholds. The Agent Builder orchestrates interactions between completion models, knowledge bases, and indexes. Agents can be configured with system instructions, session management policies, and context retrieval parameters, exposing chat completion endpoints that inject relevant document chunks as context for the completion model's responses.

Figure 386: Private AI Services Model in VMware Cloud Foundation



Private AI Services Model Attributes

Attribute	Detail
Platform type	Fully managed Generative AI application platform
User Interaction	• GUI/API to deploy model endpoints, knowledge bases, and agents • OpenAI-compatible API • No Kubernetes/VM knowledge required
Manageability	• PAIS abstracts the majority of infrastructure complexity • MLOPs Engineer operates higher in the stack, dealing with knowledge bases, Tools, and Model Endpoints rather than VMs, Kubernetes, and microservices.
Use cases	• Supports LLM use cases, including Conversational AI, Data Indexing and Retrieval, MCP servers, and Content Generation. • Cannot run: Computer vision, time-series, optimization, audio/speech models
Target Persona	GenAI Applications Developers

Design Options - Consumption Patterns

VMware Private AI Services offers two distinct consumption patterns for accessing AI models: **Direct Model Endpoints** and **Agents**. Direct Model Endpoints provide a stateless, OpenAI-compatible API for maximum flexibility, requiring developers to manage conversation state and implement RAG workflows themselves. In contrast, Agents offer a

stateful, opinionated approach with built-in session management, automated knowledge base integration, and persistent conversation tracking. The choice between these patterns depends on your application's complexity, control requirements, and whether you prefer managing state client-side or server-side. The two approaches can coexist in the same Private AI Services Deployment and share the same Model Endpoints.

The consumption pattern also determines how much of the platform has to be deployed. Starting in Private AI Services version 3.0, the Agent Builder and the Data Indexing and Retrieval modules are optional and can each be disabled in the **paisconfiguration** custom resource, allowing the footprint and the prerequisites of a Private AI Services instance to be aligned to the intended use case. This results in three deployment profiles driven by the desired consumption pattern:

- **Model Endpoints only.** Both Agent Builder and Data Indexing and Retrieval are disabled. In this simplified deployment, no external PostgreSQL database with pgvector is required, removing the dependency on VMware Data Services Manager (or on any other externally provided compatible database) and the associated sizing, and lifecycle work. The result is the fastest path to a governed, OpenAI-compatible inference endpoint, with conversation state and any retrieval logic implemented by the consuming application.
- **Agent Builder without Data Indexing and Retrieval.** Agent Builder is enabled and Data Indexing and Retrieval is disabled. An external PostgreSQL database is still required, because Agent Builder persists agent definitions, sessions, and conversation history server-side. This middle ground delivers an OpenAI Assistants-style capability: pre-configured system instructions, server-side session management with TTL and summarization, and MCP tool integration that lets an agent call external APIs, query databases, and act on live systems — all without knowledge bases, indexes, or data source connectors. Because no embeddings or text chunks are stored, the pgvector extension is not exercised and database sizing is driven only by agent metadata and session data, so it is significantly smaller than a RAG-enabled deployment. This profile fits action-oriented agents, where the value comes from tool invocation against current system state rather than from retrieval over an indexed corpus.
- **Agent Builder with Data Indexing and Retrieval.** Both modules are enabled, adding knowledge bases, data source ingestion, embedding generation, and automated context injection on top of the agent capabilities.

Option	Attributes	Benefits	Implications
Direct Model Endpoints	• Stateless API calls • OpenAI-compatible interface (/chat/completions, /completions, /embeddings) • Per-request model selection • Manual message history management • No session persistence • Return only the standard OpenAI response fields • Full parameter control per request • No external PostgreSQL database required	• Maximum flexibility and control • Simple integration, ideal for prototyping • Can switch models per request • No state overhead • Standard OpenAI client library compatibility • Simplified prerequisites: no DSM or other externally provided database to size, and lifecycle • Smallest platform footprint	• Client must manage conversation history • Manual RAG implementation required • No automatic context window management • Must handle embeddings and vector search separately • Higher client complexity for multi-turn conversations
Agents - Agent Builder only, Data Indexing and Retrieval disabled	• Stateful with persistent sessions • Pre-configured system instructions • Automatic conversation history tracking • Configurable session TTL (session_max_ttl) • Session length management (session_max_length, session_summarization_strategy) • Optional MCP tool integration	• Session management handled automatically • Conversation history persisted server-side • Automatic context window management with summarization • Reduced client-side complexity for chatbots • Automatic session cleanup via TTL • Extends agent capabilities with external tools (APIs, databases, actions) • Database sized for metadata and sessions only, so significantly smaller than a RAG deployment	• Requires upfront agent creation and configuration • Fixed model per agent • Less flexibility in per-request customization • When MCP tools are required, MCP server deployment and configuration is an end-user responsability • MCP tools must be pre-configured and linked to the agent. • No grounding in enterprise data beyond model knowledge, prompt context, and MCP tools
Agents – Agent Builder with Data Indexing and Retrieval	All Agent Builder only attributes, plus: • Built-in Knowledge Base integration • Data Source integration (Google Drive, Confluence, SharePoint, S3, local files) • Configurable chunking, top-k, and similarity thresholds	All Agent Builder only benefits, plus: • Automated RAG workflow (indexing → retrieval → context injection) • Built-in document citations and references • Enterprise data source integration • Agents can both retrieve from knowledge bases and act through MCP tools	All Agent Builder only implications, plus: • Requires Knowledge Base and indexing setup • Opinionated RAG workflow (less control over retrieval strategy) • Database must be to accomodate the source documents volume

Design Options - Authentication Methods

Private AI Services validates every API and UI request against a configured authorization provider. Providers are declared in the **paisconfiguration** custom resource, more than one may be configured on the same instance, and when several are present the UI offers a provider picker at login. Because all three options are OIDC-based, an instance can be brought up with one provider and moved to another with a configuration change rather than a redesign.

Direct OIDC points Private AI Services at the identity provider the organization already operates, authorizing users from the claims in the resulting token. It requires nothing beyond the existing identity infrastructure and is suitable for production. Identity is wired up per deployment and per service, and serving several organizations from one instance requires them to share a single identity provider.

VCFA OIDC places VCF Automation between Private AI Services and the organization's identity solution, federating each tenant's identity provider upstream and issuing the tokens PAIS trusts. One integration covers any upstream identity stack. Tenant entitlement is explicit and delegated to tenant administrators, and long-lived VCFA API tokens work directly as bearer credentials for scripted access. It is production-suitable but requires VCF Auitomation All Apps Orgs to be part of the architecture.

Local Authentication removes the external dependency altogether, running an embedded OIDC issuer with users held in a Kubernetes Secret. It exists so evaluation can proceed where no identity provider is reachable, and it offers self-service tokens for non-interactive clients. It is explicitly not intended for production.

Option	Attributes	Benefits	Implications
Direct OIDC	• Customer identity provider trusted directly • Public client with PKCE and OIDC Discovery • Authorization from token claims, including group membership	• Uses the identity provider already in place • No additional VCF component required, specifically VCF Automation • Enterprise policy, MFA, and audit apply unchanged • Suitable for production	• Fresh integration for each Private AI Services deployment • No static API token; non-interactive clients obtain and refresh their own • Users from multiple VCF Automation organizations consuming the same Private AI Services instance require a single shared identity provider holding all users and entitlements
VCF Automation OIDC	• VCFA is the issuer Private AI Services trusts • Each tenant identity provider is federated upstream • Permitted organizations declared explicitly • Access granted through a VCF Automation role assigned by the tenant administrator • Long-lived VCFA API tokens accepted as bearer credentials	• One integration point for any customer identity stack • Each organization keeps its own identity provider, normalized into a common claim • User onboarding delegated to VCFA Automation Org Admins • Common IDP integration accross Priavte AI Services and VKS • Scripted access and cross-instance model sharing do not need an OIDC flow thanks to VCF Automation long-lived API tokens • Suitable for production	• Requires VCF Automation • Entitlement managed in two places: allowed organizations on the Private AI Services instance, roles in VCFA • Token lifetimes follow VCFA-wide settings
Local Authentication	• Private AI Services is its own issuer via an embedded OIDC server • Users and service accounts that are held in a Kubernetes Secret • Self-service personal access tokens • Configurable token lifetimes	• No identity provider required, enabling evaluation when no IDP is available • Personal access tokens work directly as bearer credentials	• **Not intended for production** • No tenancy model; a single flat user list per instance, maintained by hand • Parallel user store beside the organization's existing directory • No MFA, password reset, or token audit logging • No per-token revocation; revocation removes a user or invalidates all tokens

Design Options - Model Endpoint Locations

Beyond how models are consumed, a Private AI Services deployment must decide where the models it exposes actually run. Private AI Services presents every model through the same inference gateway and the same OpenAI-compatible API. This holds regardless of whether the model is deployed and managed by Private AI Services itself, served by a self-managed inference stack, hosted by another Private AI Services instance, or provided by a public cloud service. Consumers of the data plane (external clients, Agents, and Data Indexing and Retrieval) see a single model catalog, and are unaware of the underlying hosting location.

This abstraction turns model hosting into an infrastructure decision rather than an application one. The choice is made per model, not per deployment: a single Private AI Services instance can expose a locally managed model, a self-managed endpoint, a model borrowed from another instance, and a public cloud model side by side in the same catalog. A model can also be relocated later without any change to the applications consuming it. Where an instance manages none of its own models, the result is Private AI Services acting purely as an AI gateway. It provides authentication, tenancy integration, Agents, and Data Indexing and Retrieval over models that all run elsewhere.

Private AI Services managed model endpoints are the default and the starting point for most deployments. The model is deployed as a ModelEndpoint custom resource in the same namespace as the Private AI Services instance, with its artifacts sourced from the Harbor Model Gallery. Inference is served on the Private AI Services managed VKS cluster. Deployment, scaling, upgrade, and deletion are driven from the Private AI Services UI and API, and the corresponding inference gateway routes are created automatically. Because everything sits inside one namespace, inference traffic introduces no external network dependency, and GPU and inference telemetry are collected natively. High availability is handled through model endpoint replicas, and no credentials have to be exchanged between components. The tradeoff is resource locality. GPU capacity is dedicated to that namespace, so making the same model available to several teams means deploying it several times. The supply of GPUs becomes the practical limit on how many Private AI Services instances an organization can operate.

A **self-managed VKS cluster** is the right choice when the constraint is the serving architecture rather than the model itself. The customer deploys and operates the inference stack directly, and exposes it through a customer-managed ingress using an OpenAI-compatible API. It is then registered with Private AI Services as an InferenceGatewayRoute. This opens up capabilities that sit above the inference engine, such as KV-cache-aware and prefix-aware routing across replicas, and disaggregated inference. Model weights can be held on ReadWriteMany volumes, so that replicas start without pulling multi-gigabyte artifacts. Multi-node inference becomes possible for models too large for a single node, along with the freedom to adopt newer engines on the customer's own cadence. The model still appears in the Private AI Services catalog and is consumed identically by Agents, knowledge bases, and the data plane API. So applications are unaffected by the choice, and remain unaffected if the model is later migrated to a managed endpoint. In exchange, the customer takes back responsibility for lifecycle, scaling, high availability, and telemetry, and governs model artifacts outside Harbor. They gain no resource sharing benefit, since the GPU capacity is still consumed in the same namespace.

Consuming a model from a **different Private AI Services instance** addresses the duplication inherent in the first option. One instance hosts GPU-backed model endpoints on behalf of others, which register those models as remote routes and require no GPU capacity of their own. The serving instance may sit in another namespace on the same Supervisor, on a different Supervisor, or in an entirely different VCF instance. The significant consequence is that isolation no longer has to be traded for utilization. Each team or tenant runs a dedicated Private AI Services instance with its own authentication and application data, while drawing on a shared pool of GPU-backed models. Onboarding an additional team no longer requires an additional GPU allocation. The considerations are those of any shared service. The serving instance must be sized and made highly available for aggregate demand, and its administrator owns model lifecycle exclusively.

A **public cloud model** is hosted entirely by an external provider and consumed through its OpenAI-compatible API, with no compute footprint on premises at all. Google Cloud Gemini Enterprise Agent Platform (formerly Vertex AI) and Google Gemini are also supported. This is the fastest route to proprietary and frontier models that cannot be deployed locally. It requires no capacity planning, and provides elastic headroom that a fixed on-premises GPU pool cannot. Because cloud and private models share the same catalog and the same authentication, applications treat them identically. This makes it

a practical starting point for teams building against a model today and moving it on premises as GPU capacity arrives. The defining consideration is that prompt and completion data leaves the environment.

Options	Attributes	Benefits	Implications
Private AI Services Managed Model Endpoints	Deployed as a **ModelEndpoint** custom resource in the Private AI Services namespace Model artifacts sourced from the Harbor Model Gallery Inference served on the Private AI Services managed VKS cluster using vLLM, Infinity and llama.ccp Full lifecycle (deploy, scale, upgrade, delete) driven from Private AI Services UI/API and supervisor namespace CRDs. Inference gateway routes created and maintained automatically	Simplest operating model Inference traffic does not leave the namespace, and no external network dependency is introduced GPU telemetry (DCGM) and inference metrics (vLLM) are collected natively. Example Grafana dashboards are available separately for customers to import into their own Grafana instance as a starting point. These are provided as examples only. Model artifacts are curated and governed through Harbor High availability handled by Private AI Services through model endpoint replica No credential exchange configuration required	GPU capacity is dedicated to a single Private AI Instance Making the same model available to several teams requires deploying it several times, consuming GPU capacity for each copy Serving topology is limited to the options exposed by Private AI Services Replica startup time is governed by pulling model artifacts from Harbor, and loadinginto GPU, which is significant for large models and affects how quickly capacity can be added or recovered GPU availability becomes the practical limit on how many Private AI Services instances an organization can operate
Self-Managed VKS Cluster	Inference engine deployed and operated directly by the customer on a self-managed VKS cluster Model exposed through a customer-managed ingress using an OpenAI-compatible API Registered through an InferenceGatewayRoute referencing the endpoint URL, with spec.engine set to OpenAI TLS verification configured through the Private AI Services trust bundle Typically the same organization and security boundary as the Private AI Services instance, so no cross-tenant entitlement is involved	Access to advanced serving architectures that sit above the inference engine. These include KV-cache-aware and prefix-aware request routing across replicas, and disaggregated inference, which separates the prefill and decode phases onto independently scaled pools. Model weights can be held on ReadWriteMany volumes and mounted by every replica. So scaling out or recovering a replica does not require pulling multi-gigabyte artifacts from a registry, which materially shortens startup time. Supports multi-node inference for models too large to be served from a single node Alternative or newer inference engines can be adopted on the customer's own cadence rather than waiting for them to be integrated into the platform Custom scheduling, autoscaling, and GPU partitioning policies can be applied to the serving workload The model still appears in the Private AI Services model catalog and is consumed identically by agents, knowledge bases, indexes, and the data plane API Existing self-managed inference stacks can be brought under Private AI Services authentication and governance without re-platforming them Applications require no change if the model is later migrated to	Model lifecycle, scaling, high availability, upgrades, and GPU scheduling become the customer's responsibility Model artifacts are governed outside the Harbor Model Gallery unless the customer integrates it explicitly Inference and GPU telemetry for the model are not collected by the Private AI Services Prometheus stack and must be instrumented separately The endpoint must be registered explicitly; Private AI Services does not discover it or detect subsequent changes to it The ingress CA certificate must be added to the Private AI Services trust bundle for strict TLS verification GPU capacity is still consumed within the namespace, so this option provides serving flexibility but no resource sharing benefit The advanced serving architectures that motivate this option carry their own operational burden: cache-aware routing and disaggregated inference add components that must be deployed, tuned, and debugged by the customer Requires Kubernetes and inference expertise that a managed model endpoint would otherwise abstract

Options	Attributes	Benefits	Implications
Different Private AI Services Instance	A Private AI Services instance hosts GPU-backed model endpoints on behalf of other instances Consuming instances register the remote models as **Inference GatewayRoute** resources Consuming instances require no GPU capacity of their own In multi-tenant environments, entitlement is granted through the VCF Automation auth provider using **allowedOrgs.** In environments without VCF Automation multi-tenancy, entitlement can be granted through the built-in static credentials auth provider The serving instance ingress CA certificate is registered in the trust bundle of each consuming instance The serving instance may reside in another namespace on the same Supervisor, on a different Supervisor, or in an entirely different VCF instance	Scarce GPU capacity is deployed once and consumed by many teams, eliminating duplicate model deployments Isolation no longer has to be traded for utilization: each team or tenant operates a dedicated Private AI Services instance with its own authentication, Agents, knowledge bases, indexes, and application data, while drawing on a common pool of GPU-backed models A team's data, agent configuration, and conversation history remain confined to its own instance Remote models appear in the local catalog exactly like local ones, so Agents and Data Indexing and Retrieval consume them without modification New teams can be onboarded onto namespaces with no GPU allocation, substantially lowering the cost and lead time of standing up an additional instance Model selection and versioning are curated centrally by the administrator of the serving instance Shared models and locally deployed private models can be combined in the same catalog	The serving instance becomes a shared-fate component and must be sized, made highly available, and monitored for its aggregate load; contention between consumers is possible Model lifecycle is owned exclusively by the administrator of the serving instance, and administrators of consuming instances cannot deploy or modify models there On the serving instance, entitlement is granted to a service account rather than to individual users End user identity is not propagated to the serving instance; the consuming instance authorizes its own users API tokens must be issued, distributed, and rotated between the serving instance and each consuming instance The ingress CA certificate must be tracked and rotated before expiry to avoid route failures Remote models must be registered explicitly; changes on the serving instance are not discovered or propagated automatically Network reachability between the instances is required, and inference traffic may traverse a network and trust boundary, adding latency

Options	Attributes	Benefits	Implications
Public Cloud Model	Registered through an **InferenceGatewayRoute** pointing at the provider endpoint, with **spec.engine** set to **OpenAI** for generic OpenAI-compatible services, or **GoogleGemini**/**GoogleVertexAI** for Google-specific API endpoints. Requires outbound connectivity from the namespace to the provider	Access to proprietary and frontier models that cannot be deployed on premises No GPU capacity, capacity planning, or model deployment effort required, the model is available immediately Elastic capacity absorbs bursts that a fixed on-premises GPU pool cannot Cloud and private models sit in the same catalog behind the same authentication, so applications treat them identically Provides a practical starting point: applications can be built against cloud models and moved to on-premises models as GPU capacity becomes available, without application changes	Prompt and completion data leaves the environment, requiring privacy, compliance, and data sovereignty review, and potentially conflicting with the objectives of a private AI deployment Cost shifts to consumption-based token billing. Model versions, deprecations, and behavior changes are controlled by the provider and can change Availability and latency depend on the external network path and the provider's service levels Provider rate limits and quotas apply and must be accounted for in application design Provider credentials must be stored, referenced, and rotated, and egress through proxies or firewalls must be established

Design options - Private AI Services Enablement and Lifecycle

Private AI Services is enabled in a vSphere Supervisor namespace in either design. The difference is how that namespace is created and consumed.

In the first option, the namespace belongs to a VCF Automation All-Apps organization and project, and Private AI Services is enabled through the guided workflow in the Build & Deploy tab. Organizations are the tenancy boundary, so a single instance can be consumed by one organization or by several, and VCFA OIDC becomes available as an authorization provider, allowing each organization to keep its own identity provider federated upstream.

In the second option, the namespace is created in the vCenter UI or on the Supervisor through the CLI. The service is enabled with kubectl or through the Local Consumption Interface. There is no VCF Automation dependency, and any

Supervisor networking option can be used. But tenancy is expressed through namespace segmentation rather than an organization construct, which suits a single operational organization or a centralized infrastructure team.

	Attributes	Benefits	Implications
VCF Automation (single or multiple organizations)	Private AI Services instance enabled in a namespace part of a VCF Automation All-Apps organization and project Guided enablement workflow in the Build & Deploy tab VCFA OIDC available as an authorization provider	No manifest authoring for initial deployment One instance can serve multiple organizations, each retaining its own identity provider User onboarding delegated to organization administrators Databases deployed via VCF Automation are auto-discovered at enablement	Requires VCF Automation All Apps organizations Advanced configuration still requires editing the **paisconfiguration** custom resource with kubectl NSX VPC networking model only
vCenter	Namespace created in the vCenter UI or Supervisor via CLI. Service enabled via kubectl or Local Consumption Interface (LCI) Tenancy expressed through namespace segmentation Suited to a single operational organization or centralized infrastructure team	No VCF Automation dependency Any supervisor networking option is available (VDS or NSX VPC)	Advanced configuration still requires editing the **paisconfiguration** custom resource with kubectl Users from multiple business units on one instance require a single shared identity provider

Design requirements and recommendations

Table 947: AI/ML Platform based on Private AI Services Design Recommendations

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-001	Enable the Private AI Services in a namespace with access to a single vSphere zone	While it is supported to deploy the Private AI Services in a multi-zone namespace, critical components including the VKS cluster control plane are deployed as single instances rather than three instances across supervisor zones, preventing cross-zone fault tolerance. Replicas of the same model endpoint must be located in the same zone.	Private AI Services infrastructure lacks zone-level high availability. A zone failure causes a complete PAIS service outage, including loss of model inference endpoints and control plane functionality. Organizations must implement alternative resilience strategies such as multiple independent single zone deployments.
PAIF-PL-RCMD-PAIS-002	Deploy Private AI Services in a namespace with at least 5 GHz of CPU reservation	Default Private AI Services components are defined with minimal CPU reservations. They will not start in a namespace without enough reserved resources.	Reservation may need to be increased depending on the additional workloads running in the namespace

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-003	Deploy Private AI Services in a namespace with memory reservation that accounts for GPU memory allocation	vSphere reserves the full system memory of VMs that have GPU devices that are attached, ensuring guaranteed memory availability for GPU operations. When configuring VM Classes that include GPUs, the VM's system memory should be set to around 2X the GPU memory to account for the GPU memory allocation plus operating system, vLLM runtime, model loading buffers, and KV cache overhead. For example, a VM with an A100 80GB GPU should be configured with at least 160 GB of system RAM.	This requires advanced capacity planning around the number of model endpoints and their replicas to be hosted in the namespace. Calculate the total namespace memory reservation as: (number of GPU VMs) × (2 × GPU memory per VM). For instance, hosting 3 replicas of models on A100 80-GB GPUs requires planning for at least 480 GB (3 × 160 GB) of namespace memory reservation for GPU workloads alone.
PAIF-PL-RCMD-PAIS-004	Deploy the Harbor Model Registry as a supervisor service.	Harbor provides an OCI compatible registry where an approved model can be stored and consumed. Enabling Harbor as a supervisor service provides the most integrated experience.	The Harbor Supervisor Service is not enabled by default and must be manually enabled.
PAIF-PL-RCMD-PAIS-005	Expose the Harbor Model Registry through the NGINX load balancer service by setting **enableNginxLoadBalancer: true** and **enableContourHttpProxy: false** in the Harbor data values file.	Removes the Contour Supervisor Service from the Private AI Services stack entirely, eliminating one deployment step, and one service to lifecycle-manage. The design already requires manual edits to the same data values file to size the registry volume, so no additional operational step is introduced.	Harbor consumes one Virtual IP from the load balancer IP pool for the lifetime of the service. Harbor cannot share a VIP with other Ingress-based services on the Supervisor. Harbor Supervisor Service **2.11.2 or later** must be used, as earlier versions do not support the NGINX load balancer exposure method
PAIF-PL-RCMD-PAIS-006	Increase the size of the Harbor registry volume before enabling the service.	The registry volume stores all OCI artifacts including model weight files, which for large LLMs can be 50-150GB per model, with multiple versions and formats multiplying storage needs. A collection of 10-20 LLMs with their configurations and multiple versions can easily consume 2-5TB of storage**.**	Manual modification of the data YAML file is required when enabling the harbor supervisor service. The same edit applies the registry volume sizing and the NGINX load balancer flags so both changes are made in a single operation.
PAIF-PL-RCMD-PAIS-007	Replicate the Harbor registry content to a secondary Harbor registry in separate infrastructure.	Provides recoverability for the stored models.	Requires deploying an additional Harbor registry instance with sufficient storage to mirror all OCI artifacts. Harbor replication policies must be configured and monitored.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-008	Configure VCF Automation as the authorization provider for Private AI Services when VCF Automation is part of the deployment.	VCF Automation federates each tenant's upstream identity provider and issues the tokens Private AI Services trusts, so the deployment integrates with VCF Automation rather than separately with whatever identity solution each customer or organization operates	Requires VCF Automatin to be be part of the design The permitted organizations must be declared on the Private AI Services instance and the access role assigned within VCFA, so entitlement is managed in two places.
PAIF-PL-RCMD-PAIS-009	Use VCF Automation API tokens for non-interactive access to the Private AI Services API, rather than implementing an OIDC flow in client tooling.	Long-lived VCFA API tokens work directly as bearer credentials against the Private AI Services API, allowing scripts, CI/CD pipelines, OpenAI SDK clients, and cross-instance model consumption to authenticate without an interactive login or a token refresh implementation.	Long-lived credentials must be stored securely by the consuming client and rotated according to the organization's credential policy.
PAIF-PL-RCMD-PAIS-010	Do not use the local authentication provider for production deployments.	The provider is explicitly scoped to lab, demonstration, and proof-of-concept use. It has no tenancy model and no audit logging for issued tokens.	Deployments that begin with local authentication require a migration to VCF Automation or a customer identity provider before production use, including reissuing any personal access tokens held by non-interactive clients.
PAIF-PL-RCMD-PAIS-011	Integrate the Private AI Services deployment with a Postgres Database deployed via Data Service Manager (DSM).	Data Service Manager provides a simplified way to deploy a compatible vector database.	The DSM appliance must be deployed and integrated with vCenter.
PAIF-PL-RCMD-PAIS-012	Deploy the Postgres database via VCF Automation rather than via the Data Services Manager console.	• The Private AI Services enablement workflow in the VCF Automation UI will automatically discover databases that are deployed via VCF Automation. • Connection details for other compatible databases will have to be provided manually.	• VCF Automation must be part of the design • The Data Services Manager supervisor service must be enabled the supervisor
PAIF-PL-RCMD-PAIS-013	Size the database disk to 5-10X the size of the source documents that will be indexed into knowledge bases.	PostgreSQL with pgvector stores not only vector embeddings but also the original text chunks, metadata, full-text search indexes, and database overhead.	The namespace class storage limit must be sized accordingly to accommodate the storage requirements of the database instance. For example, if planning to index 500 GB of source documents, provision 2.5-5TB of database storage. DSM database is deployed in a different namespace than Private AI Services.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-014	Enable Prometheus metric collection in the PAISConfiguration custom resource. The DCGM exporter and vLLM metric endpoints are automatically deployed and maintained.	Prometheus collects critical metrics across the Private AI Services infrastructure. They provides complete visibility into GPU health, inference performance, and pod status. These metrics include GPU hardware telemetry (temperature, utilization, power, memory, clock speeds, and interconnect bandwidth), vLLM inference performance data (request latencies, token throughput, queue depths, and cache utilization), and Kubernetes pod health indicator.	Enabling Prometheus adds minimal computational overhead from metric scraping and requires storage for time-series data.
PAIF-PL-RCMD-PAIS-015	Use the pre-built Private AI Services Grafana Dashboards to visualize Prometheus metrics.	Grafana transforms raw Prometheus metrics into interactive dashboards that make infrastructure state immediately visible and actionable. The pre-configured dashboards display GPU hardware metrics alongside vLLM inference performance and pod health, enabling operators to correlate issues across layers (such as identifying that high GPU memory usage is causing increased inference latency).	Grafana requires a separate deployment (containerized or standalone), ongoing version management, and user access control configuration. Teams need basic training on dashboard navigation and interpretation.
PAIF-PL-RCMD-PAIS-016	Enable LLM traces in the PAISConfiguration custom resource.	Enabling LLM traces in the PAISConfiguration custom resource sends OpenTelemetry-compliant trace data to observability tools like Phoenix or Langfuse, providing application-level visibility that complements infrastructure metrics. While Prometheus/ Grafana show system-level performance, traces capture the application developer's view: individual request flows, prompt/ completion pairs, token-level timing, model parameters used, and semantic context of each inference call. This enables developers to debug specific user requests, and provides insights that infrastructure metrics alone cannot provide.	Traces contain sensitive application data, requiring careful consideration of where trace data is stored. Organizations must choose between SaaS observability platforms (easier setup, managed infrastructure, but data leaves the premises) or self-hosted deployments (full data control and confidentiality, but requires deploying and managing Phoenix/Langfuse infrastructure on-prem). Trace sampling rates must be configured to balance observability coverage with storage and processing costs, and access controls must restrict trace visibility to authorized developers only.

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-017	Visualize LLM traces in an observability platform that is dedicated to the Private AI Services instance.	Deploying a self-hosted observability platform dedicated exclusively to the Private AI Services ensures that sensitive trace data containing prompts, completions, and model interactions never leaves the on-premises environment, maintaining complete data sovereignty and compliance with privacy requirements. A dedicated deployment simplifies role-based access control (RBAC) by eliminating the complexity of multi-tenant isolation.	A dedicated observability platform requires provisioning separate resources exclusively for tracing data, increasing infrastructure footprint and operational overhead.
PAIF-PL-RCMD-PAIS-018	Configure the Telegraf supervisor service to scrape Private AI Services controller metrics, enabling visualization in VMware Cloud Foundation (VCF) Operations.	While the Private AI Services Grafana dashboards serve application developers focused on inference performance and model behavior, VCF Operations serves VI admins who must correlate Private AI Services workload demands with cluster capacity, identify GPU resource contention across namespaces, and plan infrastructure scaling	Configuring Telegraf to scrape Private AI Services controller metrics requires manual setup.
PAIF-PL-RCMD-PAIS-019	Deploy MCP (Model Context Protocol) servers as POD VMs in the same namespace where the PAIS service is enabled.	Deploying MCP servers as Pod VMs within the same vSphere namespace where PAIS is enabled creates a self-contained system where all components share network locality and administrative boundaries. MCP servers can be exposed via ClusterIP services that remain local to the namespace without consuming routable IP addresses from the broader network infrastructure. Co-locating MCP servers with Private AI Services in the same namespace consolidates the entire LLM application stack into a single namespace, enabling users with namespace-level RBAC permissions to manage both PAIS model endpoints and their associated MCP servers through a unified access control model.	Pod VMs consume compute and memory resources from the same vSphere namespace quota as Private AI Services workloads, requiring careful capacity planning to avoid resource contention between model inference and MCP server operations.
PAIF-PL-RCMD-PAIS-020	Instantiate the Private AI Services VKS control plane as a VMClass XLarge.	It ensures that all Private AI services that are part of this design run smoothly with sufficient compute and memory resources.	VMClass XLarge consumes 4 vCPUs and 32 GB of memory, requiring capacity planning at the namespace level to accommodate both the control plane and workload resources.

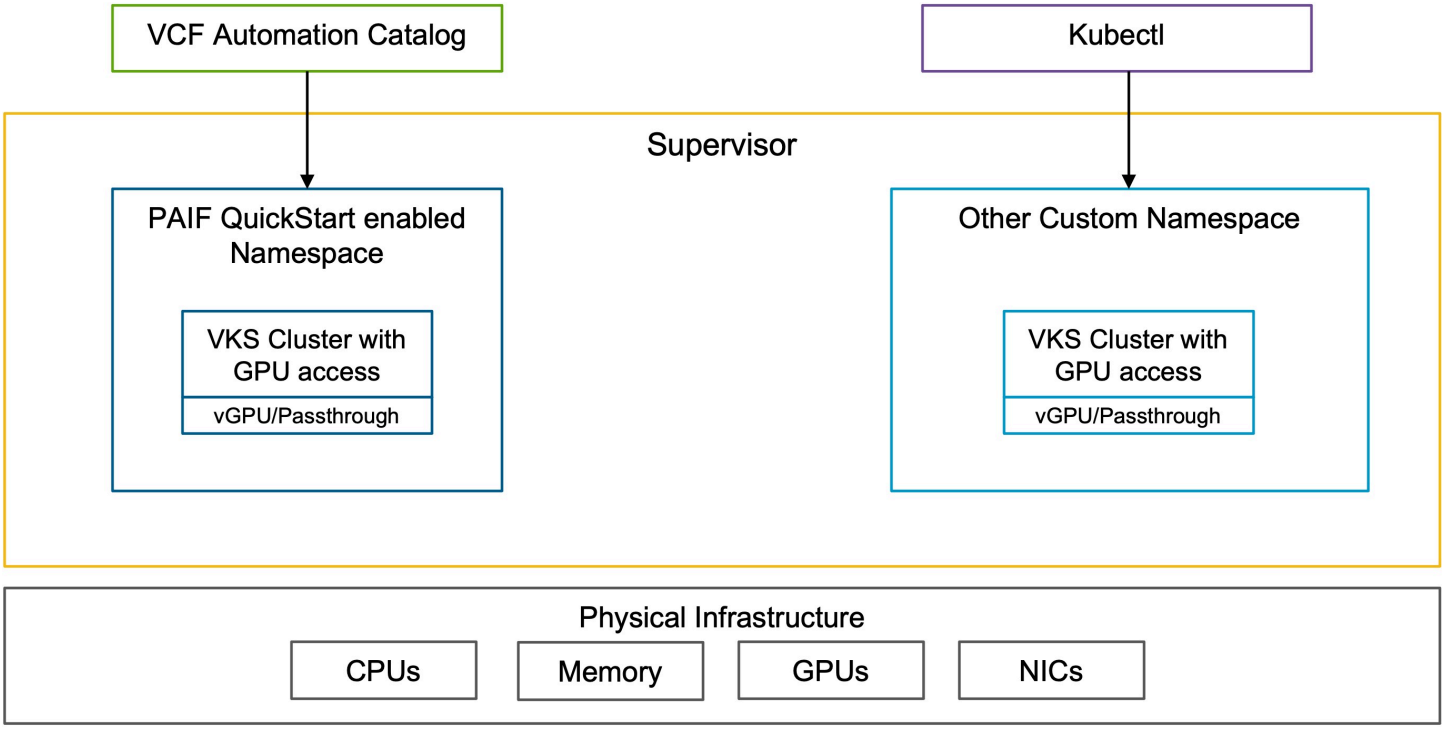
ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-021	Deploy Model endpoints with replica = 2 or higher.	Configuring model endpoints with at least 2 replicas ensures high availability for inference workloads by distributing pods across separate VKS worker nodes. This architecture provides redundancy that is critical for those GPU-accelerated models that cannot leverage vSphere's vMotion capabilities. Models running GPUs in passthrough mode or vGPU requiring Unified Virtual Memory (UVM) must be shut down for host maintenance or failure scenarios, causing service interruptions for single-replica deployments. With 2+ replicas, the Private AI Services inference gateway automatically routes requests to healthy replicas while a worker node undergoes maintenance or recovers from failure, maintaining continuous inference service availability.	Deploying multiple replicas multiplies the resource consumption per model endpoint. This significantly impacts namespace capacity planning and increases infrastructure costs proportionally to the replica count.
PAIF-PL-RCMD-PAIS-022	Deploy a dedicated Private AI Services instance for each team or tenant that requires a distinct set of models or a distinct data boundary.	Access to a Private AI Services instance grants access to everything configured in it: model endpoints, agents, knowledge bases, and MCP tools Access cannot be scoped to a subset of those resources. The enforcement boundary is therefore the instance itself. Consuming models from a shared serving instance makes this practical, as each team's instance can draw on a common pool of GPU-backed models rather than requiring its own GPU capacity.	Each instance requires its own namespace, authentication configuration, agents, knowledge bases, indexes, and vector database. Content and tools needed by more than one team must be configured in each instance.
PAIF-PL-RCMD-PAIS-023	When deploying a Private AI Services instance dedicated to shared model serving, do not co-locate tenant application workloads in the same namespace.	The serving instance determines inference performance for every consumer, and application workloads compete for the same namespace resources as inference. Separation keeps maintenance windows independent.	Requires an additional namespace and instance operated by the provider or platform team.
PAIF-PL-RCMD-PAIS-024	Size model endpoint replicas on a shared serving instance for the aggregate demand of all consuming instances.	A shared model receives the combined load of every consuming team, so replica count must be driven by aggregate concurrency rather than by availability alone.	Capacity planning requires demand estimates from each consuming team before the model is deployed, and must be revisited as teams are onboarded. Replica count multiplies GPU consumption and interacts with the memory reservation calculation

ID	Design Recommendations	Justification	Implication
PAIF-PL-RCMD-PAIS-025	Issue a dedicated service account and API token to each user consuming Private AI Services instance.	A credential shared between consumers cannot be revoked or rotated for one of them without interrupting all of them. Distinct credentials also allow requests arriving at the serving instance to be attributed to their originating instance.	The number of credentials grows with the number of consuming instances.
PAIF-PL-RCMD-PAIS-026	Track the expiry of the ingress CA certificate of the shared serving Private AI Services instance and rotate it before it expires.	Every consuming instance trusts the same certificate, so its expiry causes remote model routes to fail simultaneously across all of them.	The certificate must be recorded in a certificate inventory and monitored. Rotation requires a coordinated trust bundle update on every consuming instance, involving each tenant administrator in a multi-tenant environment.
PAIF-PL-RCMD-PAIS-027	Use strict TLS verification for inference gateway routes that point to model endpoints outside the organization.	Requests to a model endpoint carry prompt content and return completions, both of which routinely contain sensitive data. Both **strict** and **caOnly** validate the endpoint certificate chain against a CA registered in the Private AI Services trust bundle; the difference is the identity check. **strict** additionally verifies that the FQDN or IP address used in the route matches an entry in the certificate Subject Alternative Name, confirming that the endpoint presenting the certificate is the one the route intended to reach. **caOnly** omits that check and should be used only where the endpoint is legitimately reached through an address absent from the SAN. Disabling verification permits plaintext transport and should not be used outside testing.	The CA certificate of each external endpoint must be added to the trust bundle, and the certificate must include the FQDN or IP address used in the route in its SAN. Endpoints reached by an address not present in the SAN require either a reissued certificate or a documented exception using **caOnly**.
PAIF-PL-RCMD-PAIS-028	Deploy Private AI Services instances that consume only remote models in namespaces without GPU capacity	An instance with no local model endpoints does not require GPU-backed VM classes reducing the namespace reservation to what the control plane, gateway, and application components require.	The instance depends entirely on the serving instance and the network path to it for all inference. The CPU reservation floor in PAIF-PL-RCMD-PAIS-003 continues to apply.

AI and ML Platform Based on VMware Kubernetes Services Model

This model enables VKS cluster deployment with direct access to GPU resources, providing a flexible Kubernetes-based platform for running AI/ML workloads.

Figure 387: AI and ML Platform Based on VMware Kubernetes Services Model in VMware Cloud Foundation



Design Attributes

Attribute	Detail
Platform type	Kubernetes platform for AI/ML microservices
User Interaction	• Kubectl • GitOps • VCF Automation service catalog
Manageability	• Kubernetes concepts and tools are required: kubectl, Helm, Operators • Catalog Items simplify day-0 deployment of the VKS clusters
Use cases	• Supports ANY AI workload (Computer Vision, time-series, LLM, audio, optimization) • Suitable for Distributed Inference and Training across multiple physical hosts
Target Persona	ML engineers and platform teams

Design Options

When deploying VKS clusters for AI and ML workloads in VMware Private AI Foundation, organizations have two primary deployment approaches that differ significantly in complexity and control. The choice between deploying through the VCF Automation catalog versus manual kubectl-based deployment depends on your team's Kubernetes expertise, need for customization, and infrastructure-as-code requirements. Understanding these differences helps determine which approach aligns best with your operational model and technical capabilities.

The VCF Automation catalog approach provides a self-service deployment experience that reduces the complexity of standing up GPU-accelerated Kubernetes clusters. When using the catalog, users interact with a form-based interface where they select VM classes, specify node counts, and configure GPU sharing parameters without needing to write any YAML or understand the intricacies of Kubernetes cluster provisioning. The catalog item automates the installation of the NVIDIA GPU operator at deployment time, which eliminates the need to manually download Helm charts from the NVIDIA NGC catalog, install the NGC CLI, or manage GPU operator installation scripts. This approach abstracts away a large part

of the manual steps required for cluster deployment, making it accessible to ML engineers and data scientists who may have only basic Kubernetes knowledge.

Manual VKS cluster deployment via kubectl offers complete control over every aspect of cluster configuration at the cost of increased complexity. This approach requires administrators to create detailed YAML specifications that define the cluster control plane and worker nodes, and manually install the NVIDIA GPU operator using Helm charts downloaded from the NGC catalog. The manual deployment process involves many discrete steps spanning cluster provisioning, GPU operator installation, NIM deployment, and application configuration. However, this investment in complexity yields substantial benefits for organizations with advanced requirements. Manual deployment supports both vGPU and DirectPath I/O GPUs, providing flexibility to choose the GPU virtualization model that best fits specific workload performance characteristics. It also enables the optional installation of the NVIDIA Network Operator, which is essential for distributed inference and training scenarios that require high-performance networking across multiple ESX hosts. Manual deployment provides a true infrastructure-as-code capability where every configuration decision is captured in version-controlled YAML files, enabling GitOps workflows, automated testing, and consistent deployments across multiple environments.

Starting with VCF 9.1, both approaches support both vGPU and DirectPath I/O GPUs, enabling optimal GPU virtualization choices for specific performance requirements.

Option	Attributes	Benefits	Implications
VKS Cluster via VCF Automation Catalog Item	• Self-service AI/ML-ready Kubernetes cluster via the VCF Automation catalog • Pre-installed NVIDIA GPU operator providing NVIDIA vGPU drivers • Pre-configured NIM (NVIDIA Inference Microservices) in the catalog template • Form-based deployment with VM class selection • Both vGPU and DirectPath I/O GPUs are supported	• GPU operator complexity abstracted (no manual Helm charts) • Management complexity reduction vs manual VKS	Requires VCF Automation All Apps organization setup
Manual VKS Cluster Deployment	• Kubectl-based cluster provisioning • Manual YAML cluster specifications • Manual GPU operator installation (Helm, NGC CLI) • Manual NVIDIA NIM/ workload deployment • Both vGPU and DirectPath I/O GPUs are supported • Optional manual installation of the NVIDIA Network Operator for Distributed Inference (Across ESX hosts)	• Complete infrastructure-as-Code capability • Maximum flexibility for custom configurations • Support for Distributed GPU Architectures • No VCF Automation dependency	• Manual management of the GPU operator and NIMs • Error-prone (no validation until deployment) • The user must know or check the host driver version before GPU operator installation • Requires a supervisor-enabled vSphere cluster

Design requirements and recommendations

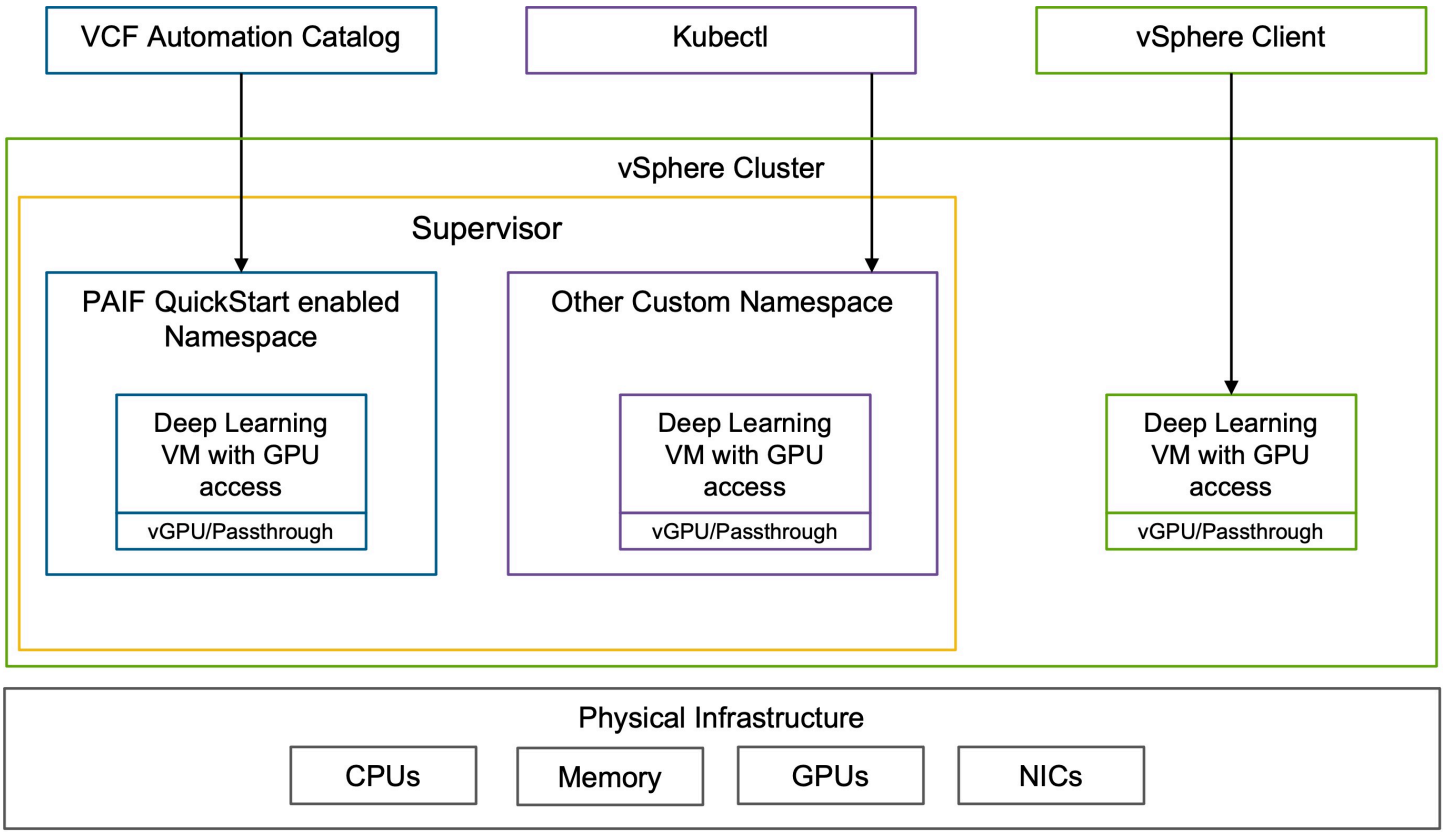
Table 948: AI/ML Platform based on VKS Design Recommendations

ID	Design Recommendation	Justification	Implication
PAIF-PL-RCMD-VKS-001	Deploy VKS clusters via the VCF Automation Catalog as the default option	Minimize management complexity	Not all deployment options are supported
PAIF-PL-RCMD-VKS-002	Deploy VKS clusters manually only when required by the desired customization, or when an IaC approach is desired	Specific configurations may require more flexibility	Management overhead

AI and ML Platform Based on Deep Learning Virtual Machines Model

This model provides pre-configured Deep Learning Virtual Machines that deliver immediate GPU-accelerated compute environments for machine learning workloads, complete with popular AI/ML frameworks and tools from NVIDIA NGC.

Figure 388: AI and ML Platform Based on Deep Learning Virtual Machines Model in VMware Cloud Foundation



AI and ML Platform Based on Deep Learning Virtual Machines Model Attributes

Attribute	Detail
Platform type	Pre-built virtual machines for individual use
User Interaction	• Jupyter Notebooks Web UI • SSH and Docker CLI for advanced/custom use cases

Attribute	Detail
Manageability	• Available as self-service catalog items in VCF Automation • Customizable ready-to-use AI/ML software stack • Also available via typical VI Admin workflows (OVF Deployment)
Use cases	• Individual data science work and experimentation • Training and inference for any AI/ML use case when the VM form factor is preferred
Target Persona	Data Scientists

Design Options

When deploying Deep Learning Virtual Machines in VMware Private AI Foundation, organizations can choose from three distinct approaches based on user roles, technical expertise, and infrastructure requirements.

The VCF Automation catalog approach targets data scientists and ML engineers who need rapid access to GPU-accelerated development environments without infrastructure expertise. Users interact with a form-based interface to customize machine parameters, select AI/ML frameworks, and specify NGC packages. This self-service model abstracts away all YAML and OVF configuration complexity, but requires a fully configured VCF Automation infrastructure, an All Apps organization, and Supervisor-enabled clusters.

The kubectl deployment method serves DevOps engineers with Kubernetes knowledge who need granular control over VM specifications. Administrators create YAML manifests defining all configuration parameters, including networking, storage, and GPU assignments, while managing cloud-init files for workload automation. It requires Kubernetes competency and comfort with kubectl operations, but eliminates dependency on VCF Automation, needing only an enabled Supervisor vSphere cluster. In this case, the supervisor manages the lifecycle of the virtual machine. Interacting with the supervisor is possible via kubectl or by installing the local consumption interface supervisor service, which provides a graphical user interface. Leveraging the standard vSphere Client UI workflows is not supported for supervisor managed virtual machines.

Direct deployment through the vSphere Client suits VI administrators preferring traditional virtualization workflows with complete independence from both VCF Automation and Supervisor infrastructure. Administrators deploy from content libraries directly onto vSphere clusters, manually configuring OVF properties through the deployment wizard. This method provides full configuration control, but demands a detailed understanding of OVF properties and manual cloud-init file management, including base64 encoding.

Starting with VCF 9.1, all approaches support both vGPU and DirectPath I/O GPUs, enabling optimal GPU virtualization choices for specific performance requirements.

Option	Attributes	Benefits	Implications
DLVM via VCF Automation Catalog Item	• Self-service AI/ML-ready workstation via VCF Automation catalog • Form-based deployment (no OVF/YAML knowledge)	No infrastructure knowledge required for data scientists	• Requires VCF Automation, All Apps organization, and Supervisor-enabled vSphere cluster • The administrator must first run the Private AI Foundation Quickstart in VCF Automation
kubectl DLVM Deployment	• Direct deployment kubectl • Same capabilities as the VCF Automation catalog item • Requires manual configuration of a Kubernetes configuration secret • Both vGPUs and DirectPath I/O GPUs are supported	• Infrastructure-as-Code capability • Full control over all configuration parameters • No dependency on VCF Automation catalog setup and VCF Automation All Apps Org.	• Manual configuration is error-prone (no form validation) • Complex cloud-init file management and base64 encoding for advanced setups • Requires a Supervisor-enabled vSphere cluster

Option	Attributes	Benefits	Implications
vSphere DLVM Deployment	• Direct deployment via the vSphere Client • Same capabilities as the VCF Automation catalog item • Requires manual configuration of OVF properties • Both vGPUs and DirectPath I/O GPUs are supported	• Full control over all configuration parameters • No dependency on VCF Automation or Supervisor • Support for compute policies (e.g., BestEffortRestart), providing better ESX LCM when Live vMotion is not supported (DirectPath I/O, or UVM configuration)	• Must understand the purpose of the OVF properties • The kubectl method requires Kubernetes knowledge • Complex cloud-init file management and base64 encoding for advanced setups

Design Requirements and Recommendations

Table 949: AI/ML Platform-based on Deep Learning VMs Design Recommendations

ID	Design Recommendation	Justification	Implication
PAIF-PL-RCMD-DLVM-001	Deploy DLVMs via the VCF Automation Catalog for large Data Science teams	Self-service experience and customizations	Higher infrastructure setup requirements
PAIF-PL-RCMD-DLVM-002	Deploy DLVMs via vSphere for quick experimentation	Lower infrastructure requirements	No self-service experience

Private AI Compute Detailed Design

VMware Cloud Foundation (VCF) offers three distinct compute models to support AI applications.

Overview

vGPU enables multiple virtual machines to share a single physical GPU through either time-sliced execution or hardware-isolated partitions (MIG). It also enables pooling multiple GPUs on the NVIDIA HGX platform. This model balances performance with operational flexibility, supporting live migration and efficient resource pooling, though it requires NVIDIA licensing and introduces an additional software layer.

DirectPath (Passthrough) dedicates one or more entire physical GPUs to a single virtual machine, providing bare-metal performance with minimal overhead. This approach offers the highest performance for GPU-intensive workloads but sacrifices flexibility. VMs cannot be live migrated and each GPU serves only one VM at a time. Pooling multiple GPUs on the NVIDIA HGX platform requires additional software components on the workloads VMs.

CPU-only inference eliminates GPU hardware entirely by running AI workloads on standard CPU cores using optimized runtimes like llama.cpp. While offering complete operational flexibility and zero specialized hardware costs, this approach delivers lower throughput and is best suited for smaller models and less demanding inference tasks.

Accelerator Design Based on AI/ML Workloads Requirements

Foundational Concepts for GPU Sizing

For practical sizing examples, please refer to this blog.

Model Parameters and Memory Format

Large Language Models consist of billions of parameters (weights) that define the model's behavior. These parameters are typically stored in FP16 (16-bit floating point) format, requiring 2 bytes per parameter. An 8 billion parameter model, therefore, requires 16 GB of memory just to store the weights (8B × 2 bytes), while a 70 billion parameter model requires 140 GB. This memory footprint is fixed and must be loaded into GPU memory before the model can perform inference.

The Two Phases of LLM Inference

LLM inference occurs in two distinct phases. The prefill phase processes the user's input prompt, computing attention for all prompt tokens simultaneously in parallel. The generation phase then produces the response one token at a time sequentially, with each new token depending on all previous tokens. This fundamental difference, parallel vs sequential processing, determines which hardware resources bottleneck each phase.

KV Cache

The KV (Key-Value) cache stores intermediate attention results from previously processed tokens, enabling the model to avoid recomputing attention for the entire conversation history when generating each new token. Each user conversation maintains its own separate KV cache containing its unique context, while all users share the same read-only model weights. The KV cache size per token is determined by the model architecture using the formula: 2 matrices × 2 bytes × n_layers × n_kv_heads × d_head. For example, an 8B model with 32 layers and 8 KV heads has a KV cache size of 0.000122 GiB per token. As conversations grow longer or more users connect concurrently, KV cache memory consumption increases proportionally.

Attention Heads

Attention heads are parallel attention mechanisms within each transformer layer that allow the model to focus on different aspects of the input simultaneously. Each attention head learns to attend to different patterns or relationships in the data, for example, one head might focus on syntactic relationships while another captures semantic meaning. In the attention mechanism, the input is projected into Query (Q), Key (K), and Value (V) matrices, and each head performs these projections independently with its own learned weights. The number of attention heads varies by model architecture: smaller models might have 32 heads while larger models can have 64 or more. The dimension per head (d_head) is calculated by dividing the model's hidden dimension by the number of heads. For memory efficiency, many modern LLMs use Grouped Query Attention (GQA), where multiple query heads share the same key-value head, significantly reducing KV cache memory requirements. The n_kv_heads parameter in the KV cache formula refers to these shared key-value heads rather than the total number of query heads.

Compute-Bound vs Memory-Bound Operations

Prefill is compute-bound because it processes many tokens in parallel, reusing the same loaded weights across all tokens, in this case the GPU's computational cores become the bottleneck.

Generation is memory-bound because the GPU must load the entire model (16 GB for an 8B model) from High Bandwidth Memory (HBM) to cache for each token generated. The GPU spends most of its time waiting for this data transfer rather than computing. The time to generate a token is determined primarily by how fast weights can be loaded: generation_time = model_bytes ÷ memory_bandwidth.

GPU Memory Hierarchy

GPUs have a memory hierarchy with vastly different speeds and capacities. HBM (High Bandwidth Memory) is the main GPU memory—large (48-141 GB) but relatively slow (864-3,350 GB/s). L1/L2 cache is on-chip memory—extremely fast (10,000+ GB/s) but tiny (~40-50 MB). Model weights are stored in HBM, but computation happens in GPU cores that access data through cache. Since a 16 GB model cannot fit in 40 MB of cache, weights must be continuously streamed from HBM through cache during inference, with different weight segments constantly loaded and evicted. This streaming from HBM to cache is what creates the memory bandwidth bottleneck during token generation.

Context Window

Context window refers to the total number of tokens stored in the KV cache for a single user's conversation, calculated as prompt_length + generated_response_length. This differs from a model's maximum context capacity—the architectural limit it supports (e.g., 8K, 128K tokens). While a model may support 128K tokens maximum, real applications typically use much shorter contexts (2-6K tokens). Context window directly impacts both memory consumption (longer contexts require more KV cache memory) and latency (longer contexts take longer to process during prefill).

Batching

Batching refers to processing multiple user requests simultaneously on the same GPU. The model weights are loaded once from HBM into the cache and then reused to generate tokens for all users in the batch, while each user maintains their own separate KV cache containing their unique conversation. This amortizes the expensive weight-loading operation across many users. For an 8 billions parameters model, instead of loading 16 GB for each user sequentially, the GPU loads 16 GB once and serves all users. Batching dramatically increases total system throughput (tokens/second across all users) while maintaining similar per-user latency, but is limited by available memory for storing multiple users' KV caches.

Data Parallelism vs Tensor Parallelism

Data parallelism deploys independent, complete copies of the model across multiple GPUs, with a load balancer distributing user requests among replicas. Each GPU operates independently without inter-GPU communication.

Tensor parallelism splits a single model across multiple GPUs, with each GPU holding a fraction of the model's weights and layers. During inference, GPUs must communicate during every forward pass to exchange intermediate results.

Data parallelism is simpler and achieves near-linear scaling but multiplies memory requirements (N replicas need N× the model weight memory). Tensor parallelism is more memory-efficient and necessary when models exceed single-GPU capacity, but it introduces communication overhead that degrades performance and requires high-speed interconnects like NVLink.

Accelerator Design Based on AI/ML Workloads Design Requirements and Recommendations

Table 950: Accelerator Design Recommendations Based on AI/ML Workload Requirements

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-001	Size GPU memory to accommodate model weights plus KV cache for expected concurrent requests at target context length. Sizing GPU memory at 2.5× model weight for production workloads is a general practical guideline.	Model weights require 2 bytes per parameter (FP16) as a fixed baseline. KV cache grows dynamically based on the number of simultaneous requests and their context lengths. Each model has a specific KV cache size per token (calculated from model architecture: 2 × 2 bytes × n_layers × n_kv_heads × d_head). Example: 8B model with 32 layers and 8-KV heads has 0.000122 GiB/token cache size. For 10 simultaneous requests each using 4,000 tokens: need 16 GB (weights) + 4.88 GB (10 × 4,000 × 0.000122) = 21 GB minimum. Formula: Total_Memory ≥ Model_Weights + (concurrent_requests × tokens_per_request × KV_cache_per_token).	Underestimating concurrent load or context length causes OOM failures during production traffic. Overestimating wastes expensive GPU memory that sits idle. Different model families have vastly different KV cache sizes even at the same parameter count.
PAIF-ACC-RCMD-002	Procure GPUs with adequate memory bandwidth to meet token generation throughput requirements	Throughput (tokens per second) measures how fast the model generates output tokens during the streaming response phase. Token generation is memory-bound because the GPU must load all model weights from High Bandwidth Memory (HBM) to the cache for each token, spending the majority of time on memory operations rather than computation. Generation time = model_bytes ÷ memory_bandwidth. Example: 16-GB model on 864-GB/s GPU takes 18.5 ms/token; on 3,350-GB/s GPU takes 4.8 ms/token - nearly 4× faster despite similar memory capacity.	Memory bandwidth cannot be improved through software, as it's hardware-limited. Maximizing throughput encourages premium GPU purchases even when compute capabilities are underutilized.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-003	Procure GPUs with adequate compute (TFLOPS) to meet Time-to-First-Token (TTFT) requirements	Time-to-First-Token (TTFT) measures the latency from submitting a prompt until the first output token appears, which is critical for interactive responsiveness. The prefill phase is compute-bound because it processes many prompt tokens in parallel, reusing loaded weights across tokens. Prefill time = model_params × 2 ÷ GPU_TFLOPS. Example: 8B model on 181 TFLOPS GPU takes 0.088 ms/token prefill; on 989 TFLOPS takes 0.016 ms/token, 5.5× faster.	Some applications tolerate higher TTFT latency, making this requirement unnecessary. Must size based on the expected prompt length distribution in the actual workload. Must validate that the selected GPU meets TTFT targets for expected prompt lengths in the actual workload.
PAIF-ACC-RCMD-004	Target sub-200ms Time-to-First-Token for interactive applications	200 ms represents the human visual reaction time threshold. Delays beyond this value create perceptible lag in a conversational interface. Time-to-first-token equals prompt length multiplied by per-token prefill time: TTFT = prompt_tokens × (model_params × 2 ÷ GPU_TFLOPS). Example: 8B model on mid-range GPU (181 TFLOPS) processes tokens at 0.088 ms each, supporting a maximum of ~2,300 token prompts within a 200-ms TTFT. High-end GPU (989 TFLOPS) at 0.016 ms/token supports ~12,500 tokens. The prefill phase is compute-bound and benefits from higher TFLOPS GPUs.	Severely constrains GPU selection for long-context applications like document analysis or code review. Some applications tolerate higher latency (batch processing, background tasks, non-interactive workflows), making this target unnecessary.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-005	Target minimum tokens-per-second generation rate based on application responsiveness requirements. Plan GPU allocation using a 30 tokens/sec minimum throughput for chat applications.	Generation throughput determines perceived responsiveness during streaming output. Human reading speed averages several words per second; generating significantly faster ensures the user never waits. Memory-bound generation time = model_bytes ÷ memory_bandwidth. Example: 16-GB model on 864-GB/s GPU generates at 54 tokens/sec (18.5 ms/token); on 3,350 GB/s generates at 208 tokens/sec (4.8 ms/token). Real-time chat applications typically require higher throughput than asynchronous workflows. The average reading speed is estimated to be between 200-300 words per minute. In comparison, a model generating 30 tokens per second (approximately 90 words per second) can produce around 1,350 words per minute.	Throughput requirements may eliminate entire GPU classes from consideration. The formula calculates per-request generation speed, which remains relatively constant regardless of batching. However, batching multiple requests allows the GPU to amortize weight loading across many users, dramatically increasing total system throughput (tokens/sec across all users) while maintaining individual user experience.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-006	Size GPU memory for the average context window, not the maximum model capacity	Context window refers to the total tokens in the KV cache per user (prompt + generated response). Models advertise maximum context lengths (8K, 128K, or more), but real workloads use much shorter contexts. Sizing for maximum dramatically reduces concurrent capacity since KV cache memory scales linearly with context length. Formula: Max_concurrent_users = (Total_GPU_Memory - Model_Weights) ÷ (context_window × KV_cache_per_token). Example: 8B model on 48GB GPU sized for 4K average context supports 64 concurrent users; sizing for 8K maximum halves capacity to 32 users; sizing for 128K reduces to 2 users. Typical applications: RAG workflows use prompt (1-2K) + retrieved docs (2-3K) + response (1K) = 4-6K total; chat uses 1-3K context. TTFT also scales with context length: Total_TTFT = context_window × prefill_time_per_token. Longer contexts increase prefill time linearly - an 8K context takes 2× longer to process than 4K. Optimizing for realistic average usage maximizes both concurrent capacity and TTFT performance.	Must monitor actual context length distribution in production to validate sizing assumptions and identify when undersizing causes user impact.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-007	When scaling the GPU infrastructure for a model endpoint, use data parallelism (independent model replicas) over tensor parallelism (model sharding across GPUs) when possible	Data parallelism deploys complete, independent copies of the model on separate GPUs with load balancing, distributing requests across replicas. Each GPU holds full model weights and processes requests independently. Tensor parallelism splits a single model across multiple GPUs - each GPU holds a fraction of the model's layers or weights and must communicate with other GPUs during every forward pass to complete inference. Example: 8B model (16 GB) can run as 4 independent replicas on 4×48GB GPUs (data parallelism) or as 1 model split across 4 GPUs with 4 GB weights per GPU (tensor parallelism). Tensor parallelism is theoretically more memory efficient, pooling memory across GPUs for shared KV cache enables higher concurrent capacity. Example: Single model across 2× 48-GB GPUs has 80 GB for KV cache (after 16-GB weights); 2 independent replicas have only 64-GB total for KV cache. However, data parallelism achieves near-linear throughput scaling without inter-GPU communication overhead, offers simpler deployment and better fault isolation. Single replica failure reduces capacity but does not crash the entire system (vs tensor parallelism, where one GPU failure brings down the model). Tensor parallelism requires high-speed interconnects (NVLink/ NVSwitch) and introduces latency from GPU-to-GPU communication every forward pass.	Data parallelism multiplies total memory requirements. 5 replicas need 5× the model weight storage (80-GB total for 8B model) vs 1× with tensor parallelism (16 GB). Prevents KV cache memory pooling across GPUs, reducing effective concurrent capacity per GPU compared to shared memory in tensor parallelism.
PAIF-ACC-RCMD-008	Provision VM/container RAM at 1-2× total GPU memory for inference workloads	Modern LLM inference frameworks load models directly into GPU memory and should not swap to system RAM during normal operation. VM RAM supports the operating system, inference framework, monitoring tools, and request handling overhead. System RAM below this ratio risks framework instability or slow startup times.	Over-provisioning wastes resources since inference primarily uses GPU memory.

ID	Design Recommendations	Justification	Implication
PAIF-ACC-RCMD-009	Size physical server RAM at 2-3× total GPU memory to accommodate host overhead and ancillary workloads	Physical servers require additional RAM beyond VM allocations for hypervisor overhead and potential non-GPU workloads.	Over-provisioning wastes resources since inference primarily uses GPU memory.
PAIF-ACC-RCMD-010	Provision 5-10× storage capacity relative to source document size for RAG embedding databases	RAG systems multiply your source document storage due to vector embeddings, indexes, and chunk overlap. This multiplier increases with higher-dimensional embeddings, greater chunk overlap, or storing multiple embedding models.	Difficult to estimate precisely without testing actual documents and parameters. The database grows continuously as documents are added.
PAIF-ACC-RCMD-011	Start with 4-8 vCPUs per GPU and monitor CPU utilization. Scale based on actual bottlenecks, not a fixed ratio.	CPU cores handle request preprocessing, tokenization, output post-processing, batching logic, and framework orchestration while the GPU performs model inference. CPU requirements scale with the number of GPUs and expected request concurrency. Insufficient CPU cores bottleneck request processing, even when the GPU is underutilized.	CPU is typically not the bottleneck. Over-provisioning wastes resources and increases costs.

Common Design Requirements and Recommendations for both vGPUs and DirectPath GPUs

Common vGPUs and DirectPath GPUs Design Requirements and Recommendations

Table 951: Common Design Requirements for vGPUs and DirectPath (Passthrough) GPUs Compute Models

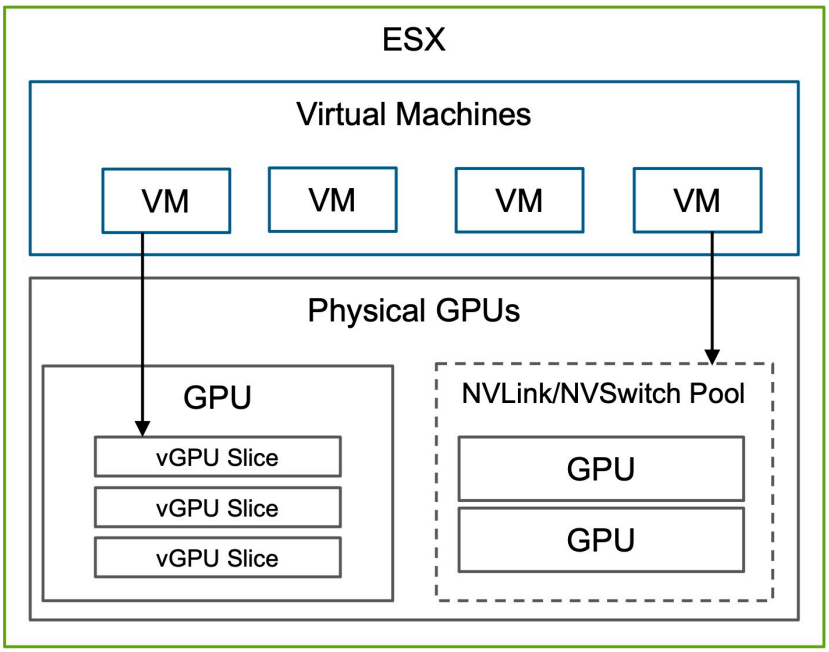
ID	Design Recommendations	Justification	Implication
PAIF-COM-REQD-COM-001	Enable Intel VT-d or AMD I/O VT in the ESX server BIOS	Required for IOMMU functionality that enables Direct Memory Access (DMA) isolation and allows the VMkernel to assign GPU devices to VMs directly. Provides a direct path to hardware while maintaining complete isolation of resources. Without VT-d or AMD-V enabled, passthrough GPUs and vGPUs cannot function	Verify CPU support and BIOS enablement before deployment.
PAIF-COM-REQD-COM-002	Enable Memory Mapping Above 4 GB in the ESX server BIOS	Modern GPUs require larger contiguous MMIO address spaces (beyond the 256-MB limit in the sub-4GB region). Enables the system to map entire GPU BARs above the 4-GB threshold, reducing fragmentation and improving stability. Without this setting, GPUs with BAR sizes exceeding 256 MB may fail to initialize or be prevented from use entirely. The device may have to incorporate multiple BARs during operations, increasing complexity and overhead, negatively impacting performance.	Verify BIOS enablement before deployment.
PAIF-COM-REQD-COM-003	Workloads must use a 64-bit operating system	32-bit operating systems can only address up to 4 GB of memory, which is insufficient for modern GPU memory capacities. A 64-bit OS can address up to 16 million TB, enabling full utilization of GPU memory and large MMIO spaces.	Private AI Services model endpoints VMs, DLVMs, and VKS clusters deployed using VCF Automation catalog use 64-bit OSs. Verify the OS of custom workloads.
PAIF-COM-REQD-COM-004	Workloads using GPUs must be configured with the EFI firmware boot option	BIOS firmware only supports BARs up to 256 MB in the MMIO-low region. GPUs with BAR1 sizes exceeding 256 MB require EFI firmware to map their memory spaces above 4 GB properly. GPUs with BAR1 sizes larger than 256 MB cannot be properly initialized with BIOS firmware.	Firmware cannot be changed after OS installation without reinstalling the guest OS.

ID	Design Recommendations	Justification	Implication
PAIF-COM-REQD-COM-005	Workloads must have all their memory reserved	Memory reservation "pins" VM memory pages to fixed physical addresses, preventing ESXi from swapping or ballooning them during memory pressure. Memory reservation is required because GPU DMA operations use host physical addresses and cannot handle page faults mid-transfer. Without memory reservation, ESX may swap out or may balloon memory pages during DMA operations, causing page faults that cannot be serviced.	Any VM configured with a PCI devices has their full memory that is automatically reserved, no user configuration is required. Memory overcommitment is not possible for GPU-enabled workloads.
PAIF-COM-REQD-COM-006	Set pciPassthru.set.use64bitMMIO = true in the workload VM advanced configuration parameters	Enables 64-bit MMIO addressing for the VM, allowing access to MMIO regions above 4 GB. The default 32GB MMIO space is often insufficient for modern GPUs with large BAR sizes. Without this setting, the VM is limited to the default 32GB MMIO space, which is insufficient for modern GPUs requiring larger MMIO regions. GPU initialization will fail if BAR requirements exceed available MMIO space. https://knowledge.broadcom.com/external/article/312208/vsphere-vmdirectpath-io-and-dynamic-dire.html	This setting is automatically applied to Private AI Services model endpoints VMs, VKS, and supervisor managed DLVM workloads configured with GPUs. Manual configuration is required for DLVM deployed from OVA.
PAIF-COM-REQD-COM-007	Specify a value for the pciPassthru.64bitMMIOSizeGB setting in the workload VM advanced configuration parameters. The value is based on the total memory of the GPUs that the VM has access to. Calculated as: sum of GPU memory capacities, round up to power of 2, then use the next power of 2 value (e.g., 40GB GPU → 64 GB → 128GB MMIO).	Defines the total MMIO address space for all GPUs assigned to the VM. MMIO space must accommodate all GPU BARs with natural alignment, meaning memory addresses must be allocated on boundaries that match their size (e.g., a 64GB BAR must start on a 64-GB boundary). GPUs expose multiple BARs (BAR0 for control, BAR1 for frame buffer, BAR3 for device-specific functions) that must be properly aligned within the MMIO space. Without a properly sized MMIO space, the system cannot naturally align GPU BARs, preventing GPU initialization. Insufficient MMIO allocation will cause the GPU device to fail to start or function incorrectly.	This setting is set to 512 GB to Private AI Services model endpoints VMs, VKS, and supervisor managed DLVM workloads configured with GPUs. Manual configuration is required for DLVM deployed from OVA.

ID	Design Recommendations	Justification	Implication
PAIF-COM-REQD-COM-008	Use non-reserved (best-effort) GPU VM Classes. Avoid reserved GPU VM Classes unless tenant isolation requirements explicitly demand them.	GPU resources remain in a shared pool accessible to all namespaces associated with the vSphere cluster, preventing artificially stranded capacity. When GPU hosts are added to the cluster, their capacity becomes immediately available to existing workloads without requiring the administrator to update zone reservations, VCFA Automation organization quotas, and namespaces.	In environments where organizations or teams require a guaranteed and pre-allocated number of GPU slots reserved, GPU VM classes are the appropriate mechanism. Without reservation, there is no guarantee that GPU capacity will be available when a workload attempts to start.

vGPU Compute Model

Figure 389: vGPU Compute Model in VMware Cloud Foundation



vGPU Compute Model Attributes

Attribute	Detail
Platform type	Flexible GPU allocation, from fractional sharing to multiple dedicated physical GPUs per VM
Resource sharing	GPU sharing is enabled via Time-sliced mode (multiple VMs share GPU resources serially) or MIG mode (hardware-isolated GPU partitions). Dedicating a full GPU or multiple GPUs to a workload is also supported.
Performance	• Near bare-metal performance for dedicated full GPU profiles. • Time-sliced: variable performance based on co-tenant GPU activity. • MIG: consistent performance with hardware isolation and QoS guarantees
Live Migration Support	Fully supported unless UVM is enabled for the VM (UVM is required for multi-GPU profiles)
High Availability	• Full DRS integration for intelligent initial placement and load balancing • HA automatically restarts VMs on alternate hosts with available GPU capacity • Supports fully automated cluster operations like maintenance mode (UVM VMs must be powered off manually)

Attribute	Detail
GPU Pooling	• Native vSphere device group support for NVLink/NVSwitch topologies • Direct high-speed GPU-to-GPU communication for multi-GPU configurations • Automated discovery and presentation of interconnected GPU sets
Licensing Requirements	Requires NVAIE for C-series licensing
Operational complexity	• Additional software layer: requires NVIDIA vGPU Manager installation and maintenance on ESX hosts. vGPU Manager (VIBs) must be added to LCM image for the cluster. • Profile-based capacity planning and management
Use Cases	• Production environments requiring live migration for zero-downtime maintenance • Multi-tenant platforms with GPU sharing requirements for cost efficiency • Environments prioritizing operational flexibility and automation

vGPU Compute Model Options

Option	Attributes	Benefits	Implications
Time-share vGPU	• Multiple VMs share a single physical GPU using time-slicing • Frame buffer memory is exclusively allocated to each vGPU at power-on and cannot be overcommitted between VMs • GPU compute resources (Streaming multiprocessors, encoders, decoders, copy engines) are shared via time-slicing • The GPU scheduler assigns time windows to each vGPU in a round-robin fashion • Active workloads can utilize all compute resources during their time slice	• Dynamic compute utilization: idle vGPUs yield GPU cycles to active workloads • Potential for higher peak performance when co-tenants are idle • Supports vMotion for live migration • Lower barrier to entry - no SR-IOV requirement (except T4)	• Performance varies based on co-tenant activity - no QoS guarantees • Workloads execute serially on GPU compute resources • No hardware-level isolation of compute paths, L2 cache, or memory controllers • Less predictable performance for latency-sensitive inference workloads
MIG vGPU	• Physical GPU partitioned into isolated GPU instances • Hardware-level isolation of frame buffer, streaming multiprocessors (GPU compute cores), L2 cache, memory controllers, and copy engines • Each GPU Instance has dedicated resources that cannot be accessed by other instances • Configurable profiles define the number of GPU compute cores and memory allocation per instance (e.g., for an NVIDIA A100 80 GB, 1g.10gb = 1/7th compute + 10-GB memory, 7g.80gb = full GPU) • Requires SR-IOV enabled in BIOS	• Strict QoS guarantees: one workload cannot impact another • Predictable, consistent performance regardless of co-tenant activity • Hardware-level fault isolation and error containment • Suitable for multi-tenant environments and inference production workloads • Supports vMotion for live migration	• Lower VM density compared to time-sliced vGPU (maximum 7 instances vs. potentially more with time-slicing) • Workloads cannot exceed the allocated slice even if other slices are idle • Limited GPU model support (Ampere and Hopper architecture only) • Cannot mix MIG and time-sliced profiles on the same physical GPU • Only one MIG-backed vGPU profile per VM (no multi-GPU with MIG)

Option	Attributes	Benefits	Implications
Multi-GPU vGPU (NVIDIA HGX Platform)	• Multiple physical GPUs assigned to a single VM as a device group • GPUs connected via NVLink bridges (2 GPUs) or NVSwitch fabric (2, 4, 8 GPUs or more) • High-speed bidirectional bandwidth between GPUs • Device groups automatically discovered by the vGPU host driver and presented as assignable units in vSphere Client	• Massive aggregate GPU memory for large models (e.g., with 8 NVIDIA A100, 8x80GB = 640 GB) • High-bandwidth GPU-to-GPU communication is critical for distributed training gradient synchronization • Enables training and inference of large language models that exceed single GPU memory capacity • Reduced stunt times in VCF 9	• Requires specialized server hardware (HGX baseboard with NVSwitch or PCIe GPUs with NVLink bridge support) • PCIe NVLink bridges are limited to 2 GPU pairs; 4+ GPU full-mesh connectivity requires NVSwitch/HGX systems • Higher cost. Dedicates multiple full GPUs to a single workload with no sharing • Unified Virtual Memory (UVM) may be required for multi-GPU workloads, disabling live vMotion. • vMotion stun times increase with total GPU memory (mitigated in VCF 9.0)

vGPU Compute Model Design Requirements and Recommendations

Table 952: Design Recommendations for vGPU Compute Model

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-001	Configure a high-bandwidth vMotion network (100 GigE recommended for vGPU workloads)	GPU frame buffer memory must be copied during vMotion. VMs can have multiple GPUs assigned (up to 8 in NVSwitch configurations). vMotion stun time is directly proportional to the data volume and inversely proportional to network bandwidth. Insufficient bandwidth causes extended stun times that may exceed the 100-second default timeout, resulting in failed vMotion operations. 10-GigE networks may be inadequate for VMs with large or multiple vGPU profiles.	Higher cost
PAIF-COM-RCMD-VGPU-002	Deploy VCF 9.0+ with supported processors (Intel Sapphire Rapids+, AMD Milan+) for better vMotion performance	ESX 9.0 or later enables vMotion to identify static versus actively changing GPU memory pages. Static data (~70% for LLM inference - model weights/ parameters) is copied during the pre-copy phase while the VM runs. Only changing data (KV cache, activations) is copied during stun time, reducing stun times by 3-4x. Without this capability, all GPU memory is copied during stun time regardless of whether it's actively changing.	VCF 9.0 or later is required

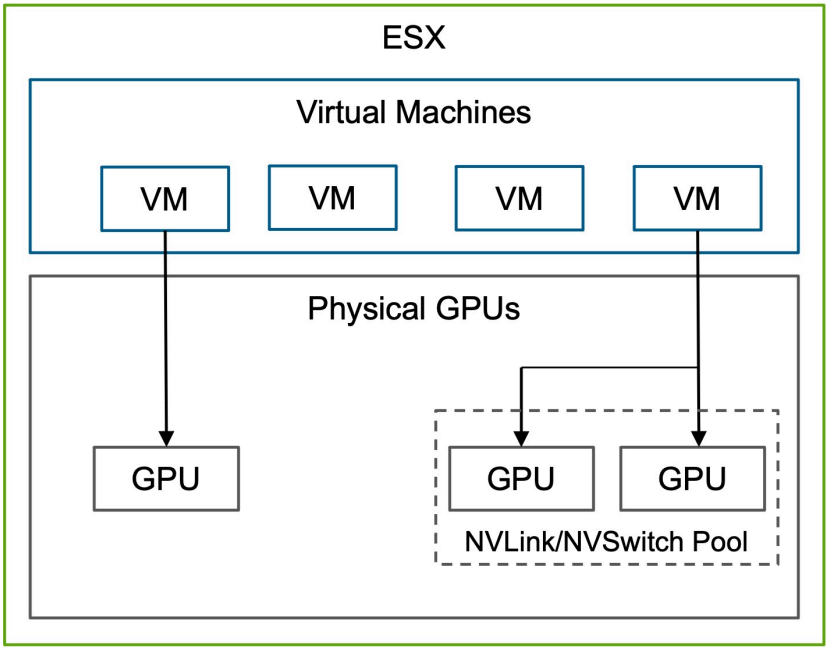
ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-003	Increase vMotion default Stun Time Limit (for large vGPU profiles)	The default vMotion stun time limit is 100 seconds. For VMs with large vGPU profiles (multiple GPUs or GPUs with large frame buffer memory), the time that is required to copy GPU memory during vMotion may exceed this default. vSphere allows administrators to set a custom stun time limit that their workload can tolerate. If the stun time limit is set too low for the VM's vGPU configuration, vMotion operations will time out and fail. The VM remains on the source host, preventing maintenance operations or DRS load balancing. vSphere calculates its own estimate and warns if the allocated time is insufficient. https://knowledge.broadcom.com/external/article?legacyId=88271	ESX evacuation operations during maintenance activities may take more time.

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-004	Enable DRS automation for vGPU VMs by setting PassthroughDrsAutomation=1 and LBMaxVmotionPerHost=1 in the DRS cluster advanced settings	By default, DRS sets Load Balancer recommendations to Manual Remediation for vGPU VMs due to extended vMotion downtimes. Setting PassthroughDrsAutomation=1 enables automatic DRS load balancing and Maintenance Mode evacuations for vGPU VMs, provided the stun time is acceptable according to DRS calculations. Setting LBMaxVmotionPerHost=1 limits DRS to one vMotion per host per scan cycle, reducing the number of concurrent vMotions during consolidation and preventing network saturation from multiple large GPU memory transfers. Without PassthroughDrsAutomation enabled, DRS will not automatically migrate vGPU-enabled VMs for load balancing or maintenance mode evacuation. Administrators must manually approve each recommendation. Without LBMaxVmotionPerHost, DRS may initiate multiple vMotions simultaneously, potentially overwhelming the vMotion network when multiple VMs with large vGPU profiles are migrated concurrently. https://knowledge.broadcom.com/external/article?articleNumber=313479	ESX evacuation operations during maintenance activities may take more time.

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-VGPU-005	Enable VgpuVmConsolidation for GPU Capacity Optimization	When set to 1, this DRS advanced setting causes DRS to tightly pack VMs with smaller vGPU profiles (e.g., single vGPU) onto hosts with extra compatible GPU capacity. This consolidation frees up larger contiguous GPU capacity on other hosts, allowing VMs with multiple full-memory vGPU devices to power on. Without this setting, GPU resources may become fragmented across the cluster. VMs requiring multiple GPUs may fail to power on even when sufficient total GPU capacity exists, because the capacity is scattered across multiple hosts rather than consolidated on fewer hosts. https://knowledge.broadcom.com/external/article/312021/vgpu-vm-placement-may-result-in-suboptim.html	Manual configuration is required. This optional setting may not be beneficial in deployments leveraging homogenous vGPU profiles.
PAIF-COM-RCMD-VGPU-006	Enable vMotion stun time validation at VM power-on by setting PassthroughRequireDrs=1 in the DRS cluster advanced settings	When set to 1, this setting performs checks at VM power-on time to ensure that the VM can be moved using vMotion and that its stun time fits within the estimate calculated by DRS based on vGPU profile sizes and configured vMotion network bandwidth. Without this setting, VMs may be powered on without validation of their vMotion feasibility, resulting in VMs that cannot be migrated during maintenance or load balancing and only discovered when vMotion is attempted. https://knowledge.broadcom.com/external/article/313178/insufficient-resources-fault-when-using.html	VMs may be prevented from being powered on even if the required resources are available on the cluster. It may not be appropriate in deployments leveraging multi-GPU profiles or UVM, where VMs cannot be live-migrated via vMotion.

Direct Path GPUs Compute Model

Figure 390: Direct Path GPUs Compute Model in VMware Cloud Foundation



Direct Path GPUs Compute Model Attributes

Attribute	Detail
Platform type	Dedicated GPU assignment: one or multiple physical GPUs exclusively allocated to one VM
Resource sharing	No sharing: GPU dedicated to a single VM
Performance	Bare-metal equivalent performance with direct hardware access. Predictable and consistent, no co-tenant impact
Live Migration Support	Not supported: VM cannot be live migrated (vMotion unavailable)
High Availability	• DRS can place a VM on any compatible host at power-on • HA can restart the VM on alternate hosts.
GPU Pooling	• To leverage NVLink/NVSwitch functionality, all GPUs and NVSwitches must be configured as passthrough devices and assigned to the workload • Multi-GPU configurations may communicate via the PCIe bus without proper placement and Guest Driver configuration based on GPU topology • NVIDIA GPU fabric manager software must be installed on a workload VM
Licensing Requirements	No additional GPU software licensing required
Operational complexity	Simpler software stack, standard workload GPU drivers only. No hypervisor GPU manager layer.
Use Cases	• Environments where workload requirements are clear, and the infrastructure can be designed accordingly. • Tight collaboration between the infrastructure team and the ML/AI platform team.

Direct Path GPUs Compute Model Design Requirements and Recommendations

Table 953: Design Requirements for DirectPath Compute Model

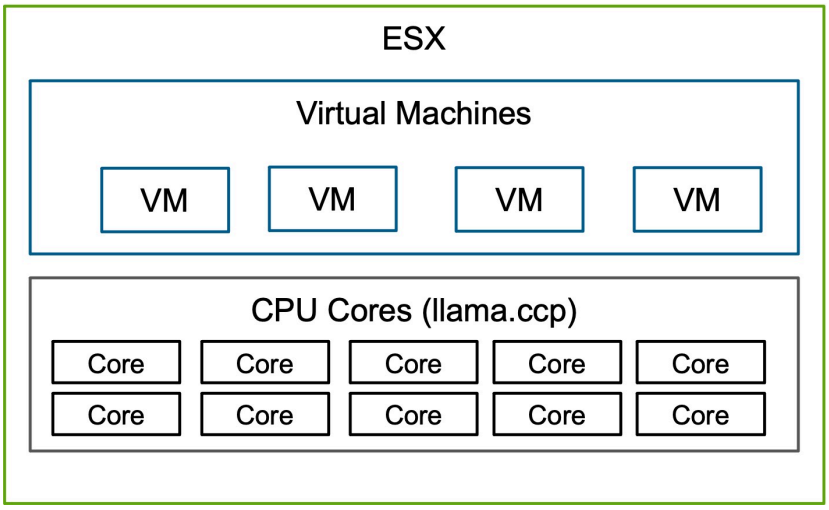
ID	Design Requirements	Justification	Implication
PAIF-COM-REQD-DPIO-001	Use Dynamic DirectPath for GPU assignment instead of static DirectPath	Dynamic DirectPath I/O uses the assignable hardware framework with key-value labels instead of static Segment/ Bus/Device/Function (SBDF) identifiers. This decouples the static relationship between VM and device, enabling DRS initial placement, vSphere HA restart on alternate hosts, and flexible GPU assignment across the cluster. Starting with vSphere 8, Dynamic DirectPath I/O with vendor-generated labels is used by default. Using static DirectPath I/O confines the VM to a specific ESXi host and specific GPU device. If the host fails, vSphere HA cannot restart the VM on another host, it will remain powered off. DRS cannot perform initial placement or load balancing. The VM must be manually reconfigured to use a different GPU.	None, Dynamic DirectPath is the default.

Table 954: Design Recommendations for DirectPath Compute Model

ID	Design Recommendations	Justification	Implication
PAIF-COM-RCMD-DPIO-001	Use custom labels with Dynamic DirectPath to capture operational characteristics beyond hardware specifications	While vendor-generated labels identify GPU model and memory capacity, custom labels enable tagging GPUs with operational attributes such as workload designation (training vs. inference), network topology (RDMA-enabled vs. standard), location constraints (rack, datacenter, compliance zone), or availability tier (production vs. development). This allows placement policies to assign workloads based on operational requirements, not just hardware specifications. Without custom labels, all identical GPU models appear interchangeable to DRS placement logic. Administrators cannot enforce operational policies (e.g., "inference workloads only on GPUs with high-speed networking").	Manual labeling of each GPU device is required.

CPU Only Compute Model

Figure 391: CPU Only Compute Model in VMware Cloud Foundation



CPU Only Compute Model Attributes

Attribute	Detail
Platform type	• Infrastructure design aligns with general VCF principles for VM and VKS workloads. • AI/ML Inference only • No accelerator infrastructure consideration. • For Private AI Services, inference runs on CPU cores using llama.cpp runtime
Resource sharing	• Standard vSphere resource sharing • Models run in workload memory, which should not be overcommitted.
Performance	Significantly lower than GPU acceleration but adequate for latency-tolerant workloads
Live Migration Support	Fully supported, vMotion works without restrictions. No special configuration required
High Availability	• Full DRS support for VM initial placement and load balancing • vSphere HA restarts VMs on alternate hosts • Standard vSphere operational capabilities apply (e.g., ESX evacuation for Maintenance Mode)
GPU Pooling	Not applicable
Licensing Requirements	Not applicable
Operational complexity	No operational overhead compared to generic VCF Deployment
Use Cases	• Embedding generation • Inference with small to medium language models (7B-13B parameters) with relaxed latency requirements • Development, testing, and experimentation environments • Cost-sensitive deployments where GPU ROI is unclear

CPU Only Compute Model Design Requirements and Recommendations

Table 955: Design Recommendations for CPU-only Compute Model

ID	Design Recommendation	Justification	Implication
PAIF-COM-RCMD-CPU-001	Configure full memory reservation for CPU-only inference VMs running llama.cpp	llama.cpp loads the entire quantized model into system RAM and relies on fast, consistent memory access for token generation. Memory reservation prevents ESX from swapping or ballooning VM memory during memory pressure, which would cause severe performance degradation.	Memory reservation reduces cluster memory overcommitment flexibility. Reserved memory cannot be reclaimed by ESX for other VMs, potentially reducing overall cluster density. For development or non-production workloads where performance consistency is less critical, this trade-off may not be justified.